



The Holistic Guide to Wellness:



Herbal Protocols for Common Ailments

Nicole Apelian, Ph. D.

The Holistic Guide to Wellness:

Herbal Protocols for Common Ailments

Nicole Apelian, Ph. D.

Edited and Written by Nicole Apelian, Ph.D.

Copyrighted by Global Brother SRL

© 2023

All rights reserved. No part of this book may be reproduced or used in any manner without the prior written permission of the copyright owner, except for the use of brief quotations in a book review.

Disclaimer Page

This book was created to provide information about a holistic approach for common ailments. This information is made available with the knowledge that the publisher, editor, and author do not offer any legal or otherwise medical advice. You should always consult with your physician or another medical specialist.

This book does not claim to contain and indeed does not contain all the information available on the subject.

While the author, editor, and publisher have gone to great lengths to provide the most useful and accurate information, there may still exist typographical and/or content errors. Therefore, this book should not be used as a medical guide.

The author, editor, and publisher shall incur no liability or be held responsible to any person or entity regarding any loss of life or injury, alleged or otherwise, that happens directly or indirectly as a result of using the information contained in this book. It is your own responsibility and if you want to use a protocol from this book, you should consult with your physician first.

Some of the remedies and treatments found within this book do not comply with FDA guidelines. The information in the book has not been reviewed, tested, or approved by any official testing body or government agency.

The author and editor of this book make no guarantees of any kind, expressed or implied, regarding the final results obtained by applying the information found in this book. Making, using, and consuming any of the products described will be done at your own risk.

The author, editor, and publisher hold no responsibility for the misuse of a plant, lichen, mushroom, or any species when using the contents of this book, or any and all consequences to your health or that of others that may result.

Some names and identifying details have been changed to protect the privacy of the author and other individuals.

By reading past this point you hereby agree to be bound by this disclaimer, or you may return this book within the guarantee time period for a full refund.

Table of Contents

DISCLAIMER PAGE.....	2
TABLE OF CONTENTS.....	3
ABOUT THE AUTHOR.....	4
ACKNOWLEDGEMENTS.....	5
INTRODUCTION AND HOW TO BEST USE THIS BOOK.....	6
AUTOIMMUNE ISSUES	
Ch. 1: Autoimmune Protocol (General)	7
Graves' Disease Protocol (<i>see page 103, under Endocrine System – Thyroid Protocol</i>)	
Hashimoto's Disease Protocol (<i>see page 103, under Endocrine System – Thyroid Protocol</i>)	
Inflammatory Bowel Disease (IBD) Protocol: Crohn's Disease and Ulcerative Colitis (UC) (<i>see page 64, under Digestive System</i>)	
Ch. 2: Lupus (SLE) Protocol (Systemic Lupus Erythematosus).....	14
Ch. 3: Multiple Sclerosis Protocol (also under Nervous System).....	20
Ch. 4: Psoriasis and Eczema Protocol (also under Skin).....	28
Ch. 5: Rheumatoid Arthritis Protocol (also under Skeletomuscular System).....	35
CIRCULATORY SYSTEM	
Ch. 6: Heart Health Protocol (Cholesterol, High Blood Pressure, Arteriosclerosis).....	42
DIGESTIVE SYSTEM	
Ch. 7: Candida Protocol (also under Immune System and Infections)	46
Ch. 8: Food Poisoning Protocol.....	54
Ch. 9: GERD and Acid Reflux Protocol	60
Ch. 10: Inflammatory Bowel Disease (IBD) Protocol: Crohn's Disease and Ulcerative Colitis (UC) (also under Autoimmune Issues)	64
Ch. 11: Leaky Gut, Gastritis and IBS Protocol	70
Ch. 12: Liver Health Protocol.....	76
Ch. 13: Ulcer Protocol.....	83
ENDOCRINE SYSTEM (HYPOTHALAMUS, PITUITARY, PANCREAS, THYROID)	
Ch. 14: Adrenal Fatigue Protocol: The Hypothalamic-Pituitary-Adrenal (HPA) Axis.....	90
Ch. 15: Diabetes Type 2 Protocol.....	96
Graves' Disease Protocol (<i>see page 103</i>)	
Hashimoto's Thyroiditis Protocol (<i>see page 103</i>)	
Ch. 16: Thyroid Protocol – Including Hashimoto's Thyroiditis and Graves' Disease.....	103
IMMUNE SYSTEM AND INFECTIONS	
Ch. 17: Bacterial Infection Protocol.....	113
Candida Protocol (<i>see page 46, under Digestive System</i>)	
Ch. 18: Flu, Viral, and Immune Protocol.....	121
Ch. 19: Fungal Infection Protocol.....	127
Ch. 20: Lyme Disease Protocol.....	133
Ch. 21: Parasite Protocol.....	139
Sinusitis Protocol (<i>see page 228, under Respiratory System</i>)	
NERVOUS SYSTEM	
Ch. 22: Alzheimer's and Dementia Protocol.....	145
Ch. 23: Anxiety and Stress Protocol.....	150
Ch. 24: Brain and Nerve Protocol - Including Stroke and Concussion.....	157
Ch. 25: Depression Protocol.....	162
Multiple Sclerosis Protocol (<i>see page 20, under Autoimmune Issues</i>)	
Ch. 26: Parkinson's Disease Protocol.....	169
Ch. 27: Vertigo and Tinnitus Protocol (also under Other).....	174
REPRODUCTIVE SYSTEM AND SEXUAL HEALTH	
Ch. 28: Menopause Protocol.....	178
Ch. 29: Men's Sexual Health Protocol.....	186
Ch. 30: Prostate Protocol.....	191
Ch. 31: Women's Sexual Health Protocol (PCOS, Endometriosis, Uterine Fibroids, Fertility, Birth Control, STDs).....	198
RESPIRATORY SYSTEM	
Ch. 32: Allergy Protocol.....	211
Ch. 33: Asthma Protocol	217
Ch. 34: Lungs and COPD Protocol.....	223
Ch. 35: Sinusitis Protocol (also under Immune System and Infections)	228
SKELETOMUSCULAR SYSTEM	
Ch. 36: Osteoporosis Protocol.....	234
Ch. 37: Pain Management Protocol (also under Other)	240
Rheumatoid Arthritis Protocol (<i>see page 35, under Autoimmune Issues</i>)	
SKIN	
Ch. 38: Acne Protocol.....	246
Psoriasis and Eczema Protocol (<i>see page 28, under Autoimmune Issues</i>)	
URINARY/EXCRETORY SYSTEM	
Ch. 39: Kidney Protocol.....	250
OTHER	
Ch. 40: Anti-Aging Protocol.....	259
Ch. 41: Energy Protocol.....	268
Ch. 42: Hair Loss Protocol.....	274
Ch. 43: Hemorrhoids Protocol.....	281
Pain Management Protocol (<i>see page 240, under Skeletomuscular System</i>)	
Ch. 44: Sleep Protocol.....	286
Vertigo and Tinnitus Protocol (<i>see page 174, under Nervous System</i>)	
Ch. 45: Weight Loss Protocol.....	292
DIETARY CHAPTERS	
Ch. 46: Anti-Inflammatory Diet.....	297
Ch. 47: Elimination Diet.....	300
Ch. 48: Intermittent Fasting.....	303
Ch. 49: Ketogenic Diet.....	305
Ch. 50: Mediterranean Diet.....	308
Ch. 51: The Wahls Protocol Diet (for MS and Autoimmune Disorders)	310
INDEX.....	312
REFERENCES.....	316

About the Author



Dr. Nicole Apelian is an herbalist, author, biologist, anthropologist, survival consultant, TV celebrity, wilderness skills instructor, foraging expert, and mother. She graduated with a degree in Biology from McGill University in Canada and has her Master's degree in Ecology from the University of Oregon.

She earned her Doctorate through Prescott College while working as an anthropologist and ethnobotanist in Botswana. She is also the author of *The Lost Book of Herbal Remedies: The Healing Power of Plant Medicine*; *A Reference Guide to Surviving Nature: Outdoor Preparation and Remedies*; and *The Forager's Guide to Wild Foods: Edible Plants, Lichens, Mushrooms, and Seaweeds*. For more about Nicole please visit www.nicoleapelian.com and visit her herbal apothecary at www.nicolesapothecary.com.

An unexpected diagnosis of multiple sclerosis in 2000 led Nicole to apply her research skills toward her own personal wellness. Through changes in her lifestyle, recognizing profound mind-body linkages, and making and using her own herbal remedies, Nicole went from bedridden to being fully alive and

from surviving to thriving.

She believes there are many people who need to find their own scientifically backed holistic remedies. This became her life's mission and the main reason for writing this book. In these pages she pairs her experiential and scientific knowledge with peer-reviewed research – doing the research for you so you don't have to. She has helped thousands of people treat themselves naturally by following her holistic wellness protocols and by using herbal remedies.

In 2015 she was among the first women selected for the History Channel's hit TV show "Alone". Despite having MS, she went on to survive solo for 57 days in a remote area of Vancouver Island with little more than her hunting knife and the wild foods and medicines she found there.

She has spent years living in nature with the San Bushmen of the Kalahari Desert, one of the last Indigenous peoples who still live as hunter-gatherers. Today she continues her work with the San through their non-profit, "The Origins Project".

Dr. Nicole Apelian's scientific knowledge was key to this book. Many of the plants, lichens, and mushrooms you'll find here are ones she's used with great results. She has selected the best-researched protocols for people who want to treat themselves naturally. These remedies are a part of a holistic philosophy of being self-reliant - of connecting your mind with your body and your body with nature. Here's to thriving and taking charge of your health.

Dedication

For my mom...

Acknowledgements

Thank you to my family, as this book took a lot of time spent deep in research, writing, and editing. Thank you to my sons Quinn and Colton for the joy and love they bring to my life. Thanks to my dear friend John Friess, for the many nights spent around the fire. Here's to many more.

Huge gratitude to my stepfather, Dr. Herbert Beall, who fueled my passion for scientific research, nature, and independent thinking, and to my mom, Dr. Barbara Apelian Beall, for her continued support, love, and understanding as we grow older together.

Thanks to Diana for her behind-the-scenes work putting this book together. I couldn't have done it without her help. Grateful for Claude and the rest of my incredible publishing team – I am blessed to have each and every one of them in my life.

Thank you to the scientists who continue to do rigorous, peer-reviewed research, especially those looking into natural health and healing. Your work makes my work possible. And deep gratitude to the NDs, MDs, and scientists who consulted on this book – your input and collaboration were invaluable.

A huge shout out to my readers (you!) for your desire to learn more about natural ways of healing. I hope this book helps you and your family on your journey to renewed health.

I trust that this book will help more people realize all that wild medicine and nature connection have to offer us, and I hope that every reader of this book will form a deeper connection to the natural world.

To everyone who has used natural medicine successfully. Keep sharing your stories!

Nicole Apelian

Introduction and How to Best Use This Book

I'm thrilled that you've secured your copy of *The Holistic Guide to Wellness: Herbal Protocols for Common Ailments*. I hope it will be extremely helpful to you and your family as you seek to optimize your health.

When my publishing team and I were deciding to put this book together, we wanted to create a reliable resource of protocols that included evidence-based holistic therapies-interventions that are backed by science, time-tested, and that actually work-and to also include information about current medical practices and the benefits and risks associated with relevant conventional therapies. We incorporated all of these factors so that readers could make informed decisions about treatment options to ultimately find the healing you're in search of-healing that aligns with your values.

Now that I've shared why we created this resource, allow me to briefly explain the layout of this book and how it is intended to be used. Within this book, you'll find 45 research-backed, holistic protocols designed to address various physical and mental health concerns. As you flip through the table of contents, you'll notice that each chapter (protocol) is based on a health condition. When you turn to a protocol, you'll find that the chapter begins with an introduction to the health condition, followed by a weighted pyramid. The pyramids prioritize the interventions. The most important or foundational interventions are located at the base of the pyramid, and the more conventional treatment options are

at the top of the pyramid. We designed it this way so that a person who only has a limited amount of time, finances, or resources could determine with minimal effort which interventions are likely to make the most significant difference given their unique circumstances, adding on more of the protocol as they are able. Following the introduction, pyramid, and detailed explanations of the available treatment and/or management options, you'll find a weekly chart. This chart provides a sample plan that includes the interventions discussed in the chapter and shows one way that they can be incorporated into a weeklong plan of action to address that specific health concern.

In the Dietary section, you'll see our nutritional protocols. Here you'll find detailed instructions on implementing various therapeutic diets, such as the anti-inflammatory diet, intermittent fasting, the elimination diet, the ketogenic diet, and other nutritional protocols. As you reference the holistic protocols, you'll notice that many of them point you back to the nutritional protocol that is most relevant to the health condition you're reading about. You'll read about the basics of the nutritional intervention in the disease chapter, and then refer to the nutritional protocol chapter for the finer details. This will help you tailor the dietary intervention to your unique needs and combine it with the lifestyle, herb, and supplement-related interventions discussed in the protocol. This is how you'll get the most out of this resource.

Ch. 1: Autoimmune Protocol (General)

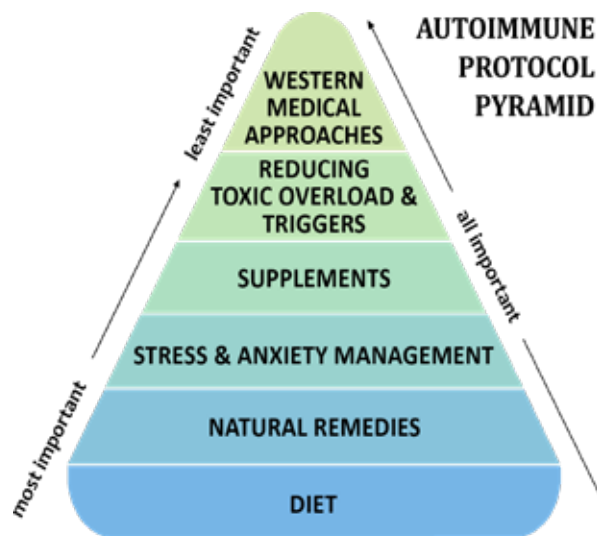
It is estimated that about 50 million people, about 20% of the total population, suffer from an autoimmune disease in the United States alone. This number may be even higher as many people are unaware that they have an autoimmune issue.

Autoimmune diseases are known to affect females more than males (75-80% of those with an autoimmune issue are female), but current males may have higher than reported rates. The autoimmune disease Type-1 Diabetes Mellitus affects an equal number of men and women.

What exactly is an autoimmune disease? Autoimmunity is the presence of antibodies (made by B-lymphocytes) or T Lymphocytes directed against the human body. For those with an autoimmune disease the immune system mistakenly attacks healthy tissue. If left untreated the damage can become extensive.¹

Elevated levels of pro-inflammatory compounds are linked with autoimmune disorders, where the immune system attacks the body. It's crucial to calm this inflammatory cycle. "Herbs that reduce inflammation may help down-regulate the autoimmune response."²

Currently, researchers have discovered more than one hundred different types of autoimmune disorders. Common autoimmune disorders include: multiple sclerosis, Crohn's disease, Graves' disease, Hashimoto's thyroiditis, type 1 diabetes, psoriasis, rheumatoid arthritis, ulcerative colitis, lupus, endometriosis, Addison's disease, celiac disease, Kawasaki disease, interstitial cystitis, and vitiligo.



Autoimmune disorders have a continued upward rising trend. Many believe this is due to our modern diet, leaky gut syndrome, viruses (such as Epstein-Barr), exposure to environmental toxins, an unhealthy lifestyle, increased stress, mold toxicity, and parasites. Autoimmunity likely has a genetic component. Another contributing factor may be better disease detection and understanding of

autoimmunity in the medical community.

If you have an autoimmune disease, following the protocols listed in this book may help prevent flares and control exposure to triggers that make autoimmunity worse. If you have a family history of autoimmune disease, following these protocols may help prevent future issues.

1. Diet: Anti-Inflammatory Foods and Your Microbiome

A healthy, whole-foods anti-inflammatory

diet is. Some foods nourish the immune system by providing our body with necessary vitamins, minerals, antioxidants, and nutrients to thrive. Others specifically feed the good gut bacteria in our microbiome, contributing to a healthy probiotic population in the gut. There is a direct link between our microbiome and our immune system. Other foods are naturally anti-inflammatory and help lower elevated blood sugar levels; *lowering inflammation and blood sugar levels is key for autoimmune issues.*

THE BEST DIETS TO FOLLOW FOR AUTOIMMUNE ISSUES: Anti-Inflammatory and Ketogenic Diets (see pages 295 and 305):

A diet that focuses on an abundance of brightly colored non-starchy fruits and vegetables, healthy fats, and wild-caught fish like salmon, mackerel, and sardines helps calm inflammation, protect brain and nerve health, and reduce autoimmune flares. The Ketogenic/Wahls Paleo Plus Diet switches around the carb-protein-fat ratio. A ketogenic diet doesn't do away with carbs completely. They just represent a much smaller part of your daily caloric intake, which is mostly made up of fats and some protein. The golden ratio of a ketogenic diet is: 70-80% fat from total daily calories, 5-10% carbohydrate, and 10-20% protein. For a 2000-calorie diet, this translates to about 165 grams fat, 40 grams carbohydrate, and 75 grams protein. If you decide on a keto diet your body will start to burn fat instead of carbs for energy. This takes a bit of getting used to and you may experience some headaches and fatigue in the beginning, but it will get easier over time. Visit these sections in this book for more information.

The Wahls Protocol Diet (see page 310): The Wahls Protocol Diet is an excellent autoimmune diet protocol for autoimmune issues with much research to back it up. It was created by Dr. Terry Wahls, MD – a researcher who, like me, had debilitating MS and decided to take her health into her own hands through scientific research. I highly recommend her eating protocol. Visit this section of the book for more information.

Intermittent Fasting (see page 303): Not only does intermittent fasting help reduce inflammation, it also helps regulate immune function. With intermittent fasting, you are not told what foods you should eat—although it is best to keep your diet healthy at all times—but rather when you should eat. Intermittent fasting is not a diet – it is a way of eating. If you're a beginner start off with a simple 16-8 routine. That means that you will consume all your daily calories in an

8-hour timeframe, for example from 12 noon until 8 p.m. Once your eating window closes only drink water, unsweetened coffee, or tea (nothing with calories). Doing so has many health benefits including weight loss and the fact that your body is put into a regenerative state where it repairs DNA damage and clears out toxins. Once you get used to fasting you can try and switch to an 18-6 or a 20-4 daily regimen. The longer you can go without food, the more it will benefit your body. And if you can go for a few days without eating from time to time that will have the greatest impact. Visit this section of the book for more information on how to fast, health benefits, and also who should not intermittently fast.

THE GUT AND IMMUNE SYSTEM CONNECTION:

Gut bacteria help “train” our immune system. From the moment we are born, our body is exposed to bacteria that become part of our microbiome. Over time, our body learns how to distinguish dangerous pathogens from helpful bacteria. Autoimmunity is often linked to an imbalanced microbiome and leaky gut syndrome (herbs for leaky gut are in the herbal section below). For more information on leaky gut, please see page 70.

Some foods, like sugar, hinder the growth of good bacteria in the gut. When our diet is unhealthy, harmful pathogens can thrive. Some foods feed these pathogens, causing a cascade of health issues. We can alter our microbiome by eating the right foods and taking steps to ensure good bacteria have an environment to thrive. One of the best ways to feed the good bacteria in your gut is by consuming prebiotics.

Prebiotic Foods: It is recommended that you eat a few of these foods with every meal. To preserve their nutrients, do not overcook or use high heat. Suggested foods include: leeks, onions, garlic, asparagus, seaweed, flax seeds, chia seeds, apples, bananas, blueberries, oats, burdock root (best as a tea), chicory root (roasted for a coffee substitute), dandelion leaves, spinach, Jerusalem artichokes, jicama root, lentils, split peas, and chickpeas.

When you consume the above foods, you feed the good microbes in your gut. As a result, they produce short-chain fatty acids that fortify the gut barrier, prevent inflammation, and increase insulin sensitivity. The lowering of inflammation as a result of eating prebiotics can significantly reduce autoimmune flares.

Probiotics: Probiotics can help reduce autoimmune flares and prevent autoimmune issues. Probiotics are the actual microbes in our gut, as opposed to prebiotics, which feed them. The best way to encourage these microbes is by eating fermented foods and drinks. When you consume probiotics, you replenish your gut with good bacteria. This helps balance your microbiome, remove bad bacteria that may lead to inflammation, and prevent the triggering of autoimmune flares.

Helpful probiotic foods and drinks include: Kimchi, sauerkraut (not canned), miso, yogurt with live cultures, kefir, tempeh, pickles, natto, and organic

bone broth. Bone broth is especially good when made with healthy vegetables and greens like stinging nettle and dandelion leaves.

Healthy gut flora is linked to less inflammation, lowered risk of autoimmune flares, better digestive health, improved heart health, and even reduced depression. See the herbal section for more gut health help, as well as our Leaky Gut Protocol for more information (see page 70).

Below are things to avoid to maintain a healthy microbiome:

- Processed packaged foods contain high amounts of toxins and preservatives that can trigger inflammation. Avoid most foods that come in a box.
- White sugar is linked to systemic inflammation and numerous diseases.
- Synthetic sugar is linked to inflammation and cancer.
- Caffeine is linked to increased anxiety, which can trigger autoimmune issues. One cup in the morning is probably fine, but more than that may cause issues.
- Alcohol is a known trigger for inflammation, especially in the gut. It also encourages leaky gut syndrome, which can foster inflammation throughout the body due to the release of proinflammatory cytokines.³
- Gluten is a huge autoimmune trigger, and not only for people with Celiac disease. I avoid it completely.
- Certain animal products like dairy milk, cheese, and beef may cause an inflammatory response in the body, which can make autoimmune conditions worse. While healthy fats are an important part of a good diet, I recommend sticking to wild-caught fatty fish, grass-fed organic meats, extra virgin olive oil, avocado, and coconut oil to minimize inflammation.
- Synthetic food colorings are linked to numerous cancers, disease, and behavior issues.
- Antibiotics may be necessary at times, but they can destroy the gut flora so it's important to repopulate the gut with probiotics.
- Pesticides: Buy organic or grow your own garden to avoid foods sprayed with pesticides.

2. Natural Remedies

There are natural remedies for autoimmune diseases that help balance an overstimulated immune system.⁴ Chronic inflammation is correlated with the onset of many autoimmune disorders. Adaptogenic, anti-inflammatory, and immunomodulatory herbs are thus especially good for autoimmune issues. Many medicinal mushrooms have immunomodulatory and anti-inflammatory properties. I take dual-extracted Lion's Mane, Reishi, Turkey Tail, and Cordyceps Mushroom tinctures daily. These mushrooms have the added benefit of being antiviral. Other well-researched herbs for autoimmune issues include Hops, Chinese Skullcap, and Stinging Nettle. Please also see the Leaky Gut Protocol in this book (see page 70).

Reishi (*Ganoderma lucidum*): Reishi mushroom is a formidable anti-inflammatory that helps modulate the immune system. Reishi suppresses certain inflammatory compounds and inhibits histamine release.⁵ Since autoimmune disorders are inflammatory in nature, reishi is an excellent choice for slowing or reversing disease progression. It has immune-modulating and immune-potentiating capabilities that target inflammatory cytokines.⁶ It has neuroprotective qualities and is effective in calming the inflammation and immune response involved with Leaky Gut, which is often associated with autoimmune issues. It also helps to strengthen the body against the negative effects of stress-like inflammation, hormonal imbalances, high cortisol, fatigue, brain fog, and low energy. Suggested dose: 2 dropperfuls of a dual-extracted tincture twice daily.

Lion's Mane (*Hericium erinaceus*): Lion's mane mushroom is neuroprotective and is known to help nervous system degeneration, decrease gut inflammation, boost brain function, lessen stress and anxiety, and help repair damaged nerve cells in the spinal cord and brain.⁷ Studies demonstrate great potential for myelination and regeneration of nerves. Research shows lion's mane mushroom stimulates Nerve Growth Factor (NGF), which is crucial for the repair of the protective, fatty coating surrounding nerve fibers known as the myelin sheath.⁸ It has even been shown to help reduce symptoms of Alzheimer's, Dementia, and Parkinson's⁹ in animal studies. Suggested dose: 2 dropperfuls of a dual-extracted tincture twice daily.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): High in anti-inflammatory compounds, turkey tail mushroom assists in reducing internal inflammation and is an excellent medicinal remedy for a range of chronic diseases. Turkey tail is an immune modulator, making it an effective medicinal herb for autoimmune conditions. It is also a prebiotic and controls both Candida and small intestinal bacterial overgrowth (SIBO) and is a remarkable remedy for helping heal leaky gut. Suggested dose: 2 dropperfuls of a dual-extracted tincture twice daily.

Cordyceps (*Cordyceps* spp.): Cordyceps has been shown to have neuroprotective effects and to reduce neuronal cell death. It has immunomodulatory and anti-inflammatory properties, helps to combat fatigue, and has anti-aging and antioxidant properties. Suggested dose: 2 dropperfuls of a dual-extracted tincture twice daily.

Choose one of the remedies below to use in conjunction with the medicinal mushroom tinctures if you need additional help with inflammation:

Turmeric/Curcumin and Black Pepper for Inflammation: Turmeric is famous for lowering inflammation in the body. This is especially helpful for those with an autoimmune disease, as high levels of inflammation are associated with autoimmune disorders. Decreasing your body's inflammatory response is key.

Combining turmeric with black pepper helps make your tincture more effective, as black pepper contains compounds that enable your body to absorb more curcumin from the turmeric. Additionally, black pepper has its own anti-inflammatory properties that can help with pain and stiffness.

Sarsaparilla Tincture (*Smilax glabra*):¹⁰ This plant helps give root beer its unique flavor. Sarsaparilla is a natural source of plant steroids that help to reduce inflammation. Its anti-inflammatory mechanism is linked to its ability to inhibit TNF-alpha-induced NFkB activation^{11,12}, which is especially important for arthritis. Its saponins emulsify and bind to endotoxins in the gastrointestinal tract¹³. Many illnesses, including RA¹⁴, gout, psoriasis¹⁵, and acne have been associated with increased endotoxin levels¹⁶. This herb is especially helpful for autoimmune diseases that directly affect the skin, like psoriasis. Suggested dose: 2 dropperfuls of a tincture twice daily.

HERBS FOR PREVENTING ILLNESS; PREVENTING ILLNESSES THAT CAN LEAD TO A FLARE-UP: An important thing to keep in mind when you suffer from an autoimmune condition is to keep your immune system from needlessly activating. This means preventing general illnesses so it doesn't cause your immune system to flare up. Since inflammation and immune system dysfunction go hand-in-hand with autoimmune disorders, those with them have to be careful to avoid viral illness. For that reason, I take these herbal remedies when I am around others.

Elderberry (*Sambucus nigra*) and Ursolic Acid: Beyond shortening the duration of colds and the flu, elderberry is helpful for inflammation, chronic fatigue, allergies, nerve pain, heart pain, headaches, dental pain, sciatica, and cancer.¹⁷ Elderberry contains the antioxidant flavonols quercetin, kaempferol, and isorhamnetin. The berries are rich in anthocyanins, which provide strong antioxidant and anti-inflammatory benefits.¹⁸

Elderberry extract reduces the severity and length of colds and influenza. In one study¹⁹, 60 people ill with the flu who took elderberry extract four times a day experienced symptom improvement in 2 to 4 days, while it took the control group 7 to 8 days to improve. A double-blind placebo controlled clinical trial involving 312 air travelers who took 300 mg elderberry extract three times a day²⁰ found that the participants who did fall ill with a cold or the flu recovered faster and had less severe symptoms.

A systematic review found that elderberry is a safe option for treating viral respiratory illness, and there is no evidence that it over-stimulates the immune system²¹ or creates a cytokine storm²². A study published in the medical journal *Autoimmunity Reviews* states that elderberry should be considered at an early course of viral illness.²³ Elderberry tincture can be used by most people who have an autoimmune disorder.

The alcoholic extraction of elderberry contains the

triterpenoid ursolic acid (UA)²⁴, which has been shown to halt damage and spur neuronal repair in multiple sclerosis²⁵, prevent and repair neurons in animal models of MS, reduce further damage to neurons, help rebuild the protective sheaths covering neurons, shows significant potential as an oral anti-inflammatory for MS, and has significant potential as a neural repair agent for MS²⁶. Ursolic acid is also found in apple peels, aronia berries, many kitchen herbs (oregano, thyme, rosemary, sage, basil, tulsi), cranberries, bilberries, and hawthorn. Suggested dose: 1-2 dropperfuls of an alcohol-based elderberry tincture daily for prevention; 2 dropperfuls 3x/day for acute illness.

Usnea Dual Extracted Tincture: Usnea (“Old Man’s Beard”) is my go-to herb (paired with Elderberry) for keeping healthy during the cold and flu season, and when I am traveling or around crowds. Usnea has powerful antibiotic, antifungal, antimicrobial, and antiviral properties. I spray it in the back of my throat to help prevent illness. It can be used as an anti-bacterial, anti-microbial, anti-viral, anti-fungal, for MRSA, respiratory issues, staph infection, strep throat, toenail fungus (topical spray), and urinary disorders. I use 1-3 sprays of Usnea tincture in my throat daily for prevention and more when ill. As with mushroom tinctures, Usnea needs both an alcohol and a water extraction (a dual-extraction).

Eastern Blend Tincture: This is a blend of Chinese skullcap, Japanese honeysuckle, Japanese knotweed, and cordyceps mushroom. These research-backed herbs (see below) address a wide-range of health concerns, including: Lyme disease, viral and bacterial infections, diabetes, cancer, anxiety, insomnia, and cardiovascular issues. This blend has powerful anti-inflammatory action and helps modulate the immune system. I use it for viral prevention. Suggested dose: 1-2 dropperfuls daily for prevention; 2 dropperfuls 3x/day for acute illness.

The baicalin and wogonin content in Chinese skullcap (*Scutellaria baicalensis*) is anti-inflammatory and exhibits strong antiviral activity for severe respiratory illness, the common cold, and influenza.²⁷ It also protects against oxidative stress and has both neuroprotective and cognitive enhancement effects due to its baicalin content.²⁸

Japanese honeysuckle is antiviral,²⁹ antibacterial, anti-inflammatory, and modulates the immune system and significantly increases killer (NK) cells.³⁰

Japanese knotweed is antiviral, antibacterial, and antifungal. It contains high concentrations of trans-resveratrol³¹, is a strong antioxidant, inhibits viral replication, and reduces neuroinflammation and neurodegenerative diseases such as Alzheimer’s and dementia.³²

HERBS FOR LEAKY GUT: Your gut plays an integral role in your immune system. A normal-functioning intestinal tract acts as a sophisticated barrier that controls nutrient absorption into our bloodstream

to nourish our body. Some people have “leaky guts”. This means that the barrier is not functioning properly and allows toxins and microbes, which would normally be discarded as waste, to absorb or “leak” into the surrounding tissues and blood. The effects of a leaky gut result in inflammation and changes in our gut bacteria. These changes may trigger digestive and autoimmune issues, and aggravate a range of health problems. An anti-inflammatory diet helps immensely with Leaky Gut issues, as will fermented foods, probiotics, and herbs such as:

- **Lion’s Mane (*Hericum erinaceus*):** A known antioxidant and gastroprotective, lion’s mane helps tame the inflammatory response and protect the lining of the stomach. It also protects the gut from *H. pylori* and the polysaccharides in lion’s mane protect against gastric ulcers.³³

- **Marshmallow Root (*Althaea officinalis*):** A soothing herbal remedy for the digestive system, marshmallow is an excellent herb for gut issues because it creates a protective layer over intestinal perforations.

- **Plantain (*Plantago major*):** An all-around tonic for the digestive system, plantain is anti-inflammatory and soothes mucous membranes—an ideal combination for relieving the symptoms of a leaky gut.

- **Reishi (*Ganoderma lucidum*):** Reishi exerts anti-inflammatory action that’s helpful for calming intestinal inflammation, it modulates the immune system, and it inhibits *Candida* yeast. It also has neuroprotective qualities. It is an effective herb for calming the inflammation and immune response involved with leaky gut.

- **Slippery Elm (*Ulmus rubra*):** Slippery elm is a demulcent, forming a protective layer in the digestive tract and reducing gut inflammation. It stimulates mucus secretion, helping ease IBS and gastritis.

- **Turkey Tail Mushroom (*Trametes versicolor* or *Coriolus versicolor*):** Turkey tail contains important prebiotics that balance the microbiome, encourages the growth of “good” gut bacteria, and controls *Candida* and small intestinal bacterial overgrowth (SIBO). Also, turkey tail is a remarkable remedy for healing the gut.

3. Stress and Anxiety Management

Managing stress and anxiety is crucial to your overall health. It is even more important if you have an autoimmune disease. Over time, stress takes a toll on the body. It targets the heart and cardiovascular system, and also weakens the immune system. Stress can result in ulcers and is linked to many digestive issues.

Stress can trigger autoimmune flares and makes it difficult to manage autoimmune diseases. It is important to practice stress management to avoid aggravating autoimmune symptoms now and in the future. Please see our Anxiety and Stress Protocol for more detailed information (see page 150).

Tips for managing stress in a healthy way include:

Exercise: Exercise helps to release endorphins in the body that promote a sense of calm happiness and wellbeing. Yoga and tai chi are especially calming. Try to exercise at least thirty minutes each day.

Time in Nature: Studies show that when people spend time in the natural world, their stress levels decrease dramatically. Combine nature and exercise if possible by going on a walk outdoors. Spend at least twenty minutes outside each day.

Aromatherapy: Useful essential oils for stress include lavender, chamomile, cedarwood, patchouli, bergamot, vetiver, and vanilla. Diffuse in the morning, inhale throughout the day, and diffuse again in the evening for best results.

Herbs for Stress: Try one of the following tinctures or teas: lemon balm, ashwagandha, lavender, skullcap, or motherwort. Personally, I use my Anxiety and Stress tincture, a blend of ashwagandha, lemon balm, lion's mane mushroom, and reishi mushroom when I am feeling anxious or stressed.

Meditation: Daily meditation is a useful tool for lowering stress and clearing the mind. Meditation can lower heart rate and blood pressure.

Breathing Exercises: Deep lymphatic breathing can help the body relax, while also promoting the flow of lymphatic fluids. Try inhaling deeply through the nose (slowly) until your lungs are completely full and then exhaling through the mouth slowly. Make sure to fully release all the air from the lungs before taking another deep breath. Do this for fifteen to twenty minutes. Repeat up to three times daily, or as needed.

Massage: Massage releases tension and promotes deep relaxation.

Acupuncture: Acupuncture has pain relief and stress-relieving benefits.

Self-Care: Put time aside (on a regular basis) for self-care.

Vacation/Travel: Vacation and travel time are necessary to reset and get away from the hustle and bustle of daily life. If possible, don't check your phone or email.

Prioritize: If you are feeling overwhelmed, make a list of the most to least important things you need to accomplish. Do the important things first.

Learn to Say "No": If you are overwhelmed, say "no" to new tasks.

Gardening: Many find gardening calming. Plus, you can grow organic produce to give you the healthy foods you need to help with autoimmune issues.

Find a Hobby: A hobby that you love can give you a sense of purpose and relieve stress. Take the necessary steps to immerse yourself in something you enjoy.

Avoid Alcohol, Nicotine, and Drugs: These aggravate anxiety and stress.

Avoid Overeating: If you are stressed, you may find yourself overeating as a form of comfort. Overeating can lead to weight gain and increased anxiety.

Sleep: Good sleep is key. If sleep is an issue for you, please see our Sleep Protocol (see page 286).

4. Supplements Some supplements help reduce inflammation and ease the symptoms of autoimmune disease. Below are supplements I recommend taking daily to manage autoimmune issues.

Vitamin D and K2: Vitamin D plays an important role in immune system function. It helps to regulate inflammatory markers and immune cells. It can be difficult to get enough vitamin D if you do not spend much time outdoors in the sunlight. Get a blood test to measure your levels before starting a daily routine, as if you are very low you will likely take a high dose to get you back to normal levels. Suggested maintenance dose: 2,000-4000 IU daily; reduce the amount you take in the summer months when/if you are spending more time outdoors. It's important to take vitamin K2 alongside vitamin D because they work synergistically to promote cardiovascular and bone health. Suggested dose: 80 mcg vitamin K per day.

Vitamin B-12: Vitamin B12 deficiency is associated with chronic stomach inflammation and may be involved in pernicious anemia, which is an autoimmune malabsorption syndrome. Additionally, levels of B12 tend to be low in MS patients. It's a cofactor in myelin formation and plays a role in regulating the immune system. Suggested dose: Before you supplement with B12, have your general practitioner measure your blood levels. Once you have a baseline, begin with 2,000 mcg of oral vitamin B12 daily, then decrease the dose to 1,000 mcg per day, followed by 1,000 mcg weekly, and finally 1,000 mcg monthly.

Essential Fatty Acids: Chronic inflammatory conditions like autoimmunity respond well to the supplementation of fish oils. Research has shown that these essential fatty acids help to decrease disease activity, reduce pain, and lower the use of anti-inflammatory drugs. Suggested dose: 0.3-9.6 g/day Omega-3 PUFA; EPA: 0.01-4.1 g; DHA: 0.01-2.7 g for autoimmunity (I take 600-1,000 mg daily.)

CoQ10: CoQ10 helps to suppress inflammation, improves fatigue and depression, and has beneficial effects on inflammatory cytokines and oxidative stress in autoimmune patients. Findings suggest that coenzyme Q10 might be useful in treating neurodegenerative diseases.³⁴ Suggested dose: 500 mg/day for MS; 100 mg/day for rheumatoid arthritis; and 1,200 mg/day for Parkinson's disease.

Turmeric/Curcumin: Curcumin is the active medicinal compound in turmeric that lowers inflammation in the body. Suggested dose: 3 mg of curcumin per day for each kilogram of body weight, or between 500-2,000 mg daily.

Resveratrol: This powerful antioxidant can support a healthy metabolism, as well as reduce inflammation, and may help with aging. It is also a great immune modulator. Additionally, resveratrol can help balance the immune system while also inhibiting autoantibody production. Suggested dose: 500-1,500 mg daily (Start with the lowest dosage and increase if needed.).

Glutathione: Glutathione plays a role in immune system health and the body's detoxification process. It is a potent antioxidant and can reduce autoimmune triggers. The body naturally produces glutathione, but most of us have insufficient levels due to poor diet, medications, stress, and other factors. Suggested dose: Take acetylated glutathione – 500 to 1,000 mg. Note that over-the-counter medications like Tylenol deplete glutathione levels in the body.

L-Glutamine: This supplement works to help repair the gut and restore gut health. Healing the gut is a vital step in dealing with autoimmune disease. This supplement can also promote thyroid health and overall immune health. Suggested dose: 2-5 grams twice daily.

5. Reducing Toxic Overload and Triggers

It is vital to reduce your toxic load; every small change you make adds up. As you reduce your toxicity, you may notice fewer autoimmune flares. Below are tips and ways to avoid toxins as you go about your daily life. Try to implement as many as you can.

Food Triggers: Pay close attention to the foods you eat. Many foods and ingredients can trigger autoimmune flares. For example, it is recommended to avoid gluten because it can cause an autoimmune reaction for those with celiac disease or gluten intolerance. Gluten is in a wide-range of products, not just bread. Read labels carefully.

Other foods that may trigger allergies and flares include cow dairy, red meat, eggs, tree nuts, peanuts, and soy. If you want to drink milk, try goat milk, almond milk, coconut milk, or oat milk. It is recommended that you eat eggs from chickens that are free-range. Get a food sensitivity test or try an elimination diet (see page 300) and keep a food diary so that you can pinpoint any correlations between the foods you eat and any autoimmune flares.

Chemicals in Products: Familiarize yourself with common chemicals that are lurking in your everyday bath and beauty items. Check your shampoos, conditioners, body washes, lotions, face creams, and makeup. Below are ingredients to avoid:

Sulfates: Sulfates are used in products like shampoos and body washes as foaming agents. They have been linked to cancer, neurotoxicity, organ toxicity, skin irritation, and endocrine disruption. Using products with sulfates can significantly worsen autoimmune diseases of the skin.

Parabens: Products like moisturizers and hair care items may contain parabens. These are linked to thyroid dysfunction and cancer. Especially avoid these if you have an autoimmune disease that affects the

thyroid.

Tricolsan: The FDA has banned the use of this antibacterial chemical after it was found to be ineffective. However, it was also found to be an endocrine disruptor and potential carcinogen. It may still be found in body washes, soaps, and hand sanitizers.

Dye: If an autoimmune disease affects your skin, pay special attention to products with dyes. Synthetic dyes can cause skin irritation, hives, allergic reactions, and may be carcinogenic. Your skin is your biggest organ and it absorbs the products you use, so dangerous toxins can be absorbed into your bloodstream this way. Skip the cute-colored body lotions, bath bombs, shampoos, and other products.

Fragrance: Synthetic fragrances in products have been linked to skin irritations and allergies. They are also linked to birth defects, cancers, nervous system disorders, and may exacerbate asthma. Steer clear of any products with synthetic fragrances. Even “natural” fragrances can cause issues. Look for products that use pure essential oils instead.

Alcohol: Alcohol is used in many moisturizing products. This seems counterintuitive, but alcohol gives the appearance that your skin is absorbing the lotion quickly. It is very deceiving and often results in skin dryness, irritation, and inflammation. If you have an autoimmune disease that affects the skin, avoid products with alcohol.

Cleaners: Many cleaners contain harmful chemicals. You come into contact with the surfaces you clean regularly, so you are exposing yourself to harmful chemicals more than you probably realize. If you want to clean with a safer alternative, try infusing orange peels in vinegar and use it to clean counters, floors, etc. It is much more environmentally friendly as well.

Plastics: Many plastics contain a cancer-causing chemical called bisphenol A (BPA). There is a possible link between BPA and birth defects, blood pressure, diabetes, and endocrine disorders. Use glass cups and glass dishware. Be mindful of the plastic products you purchase and check to make sure they are BPA-free.

Nonstick Surfaces: Some pots and pans are coated with a nonstick surface to make cooking easier. However, they are linked to serious health problems and may greatly worsen autoimmune disease. Opt for stainless steel, ceramic, or cast iron pots and pans when cooking. Stir whatever you cook using wooden spoons to avoid BPA.

Pesticides: Pesticides are a potential trigger for autoimmune flares if you consume foods that have been sprayed with them. Opt for organic produce or grow your own vegetable garden.

Water: In cities, water is often treated with fluoride. This may be fine for the teeth, but it is not so great for the bones and immune system. If you drink tap water, find out if it has been fluoridated. Opt for spring water or filtered well water. Get your water tested.

Radiation: Be mindful of everyday radiation exposure.

Note where you keep your cell phone throughout the day. If you are shopping for a new home, be mindful of how close it is to power lines and power stations. Charge your phone in a different room at night when you are sleeping.

6. Western Medical Approaches

Low-dose naltrexone is a good option for autoimmune issues. Research has shown that this pharmaceutical is helpful for reducing pain, inflammation, fatigue, and elevating mood. Negative side effects are very rare.

Depending on the autoimmune condition, a doctor may prescribe immune-suppressing medications like tofacitinib, cyclosporine, tacrolimus, sirolimus, everolimus, mycophenolate, or adalimumab. In addition,

corticosteroids like prednisone may be prescribed for inflammation. There are a lot of negative side effects associated with these medications.

Some doctors may suggest over the counter medications like Ibuprofen to tackle pain and inflammation, but this is only a temporary fix for a complex problem.

Research each recommended medication so that you fully understand possible side effects so you can make an informed decision.

Try to find a doctor who takes into consideration your lifestyle and diet in addition to potential medications for autoimmune diseases. A naturopathic doctor is an excellent option.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	4 Mushroom tinctures (Lion's Mane, Reishi, Turkey Tail & Cordyceps dual-extracted tinctures) Deep breathing exercise Vit D, B-12, CoQ10, Resveratrol, Glutathione, and L-Glutamine	Weekly meal plan/ grocery shop Physical exercise Turmeric & Omega 3s Herbs for illness prevention Gut herbs	Time outside/nature walk Meditation	4 mushroom tinctures Gut herbs L-Glutamine	Aromatherapy (Lavender or Chamomile to aid sleep & stress) Deep breathing exercise or epsom salt bath Sleep herbs, if needed
MONDAY	4 Mushroom tinctures Deep breathing exercise Vit D, B-12, CoQ10, Resveratrol, Glutathione, and L-Glutamine	Physical exercise Turmeric & Omega 3s Herbs for illness prevention Gut herbs	Time outside/nature walk Meditation	4 mushroom tinctures Gut herbs L-Glutamine	Aromatherapy Deep breathing exercise or epsom salt bath Sleep herbs, if needed
TUESDAY	4 Mushroom tinctures Deep breathing exercise Vit D, B-12, CoQ10, Resveratrol, Glutathione, and L-Glutamine	Physical exercise Turmeric & Omega 3s Herbs for illness prevention Gut herbs	Time outside/nature walk Meditation	4 mushroom tinctures Gut herbs L-Glutamine	Aromatherapy Deep breathing exercise or epsom salt bath Sleep herbs, if needed
WEDNESDAY	4 Mushroom tinctures Deep breathing exercise Vit D, B-12, CoQ10, Resveratrol, Glutathione, and L-Glutamine	Physical exercise Turmeric & Omega 3s Herbs for illness prevention Gut herbs	Time outside/nature walk Meditation	4 mushroom tinctures Gut herbs L-Glutamine	Aromatherapy Deep breathing exercise or epsom salt bath Sleep herbs, if needed
THURSDAY	4 Mushroom tinctures Deep breathing exercise Vit D, B-12, CoQ10, Resveratrol, Glutathione, and L-Glutamine	Physical exercise Turmeric & Omega 3s Herbs for illness prevention Gut herbs	Time outside/nature walk Meditation	4 mushroom tinctures Gut herbs L-Glutamine	Aromatherapy Deep breathing exercise or epsom salt bath Sleep herbs, if needed
FRIDAY	4 Mushroom tinctures Deep breathing exercise Vit D, B-12, CoQ10, Resveratrol, Glutathione, and L-Glutamine	Physical exercise Turmeric & Omega 3s Herbs for illness prevention Gut herbs	Time outside/nature walk Meditation	4 mushroom tinctures Gut herbs L-Glutamine	Aromatherapy Deep breathing exercise or epsom salt bath Sleep herbs, if needed
SATURDAY	4 Mushroom tinctures Deep breathing exercise Vit D, B-12, CoQ10, Resveratrol, Glutathione, and L-Glutamine	Physical exercise Turmeric & Omega 3s Herbs for illness prevention Gut herbs	Time outside/nature walk Meditation	4 mushroom tinctures Gut herbs L-Glutamine	Aromatherapy Deep breathing exercise or epsom salt bath Sleep herbs, if needed

Ch. 2: Lupus (SLE) Protocol (Systemic Lupus Erythematosus)

Systemic Lupus Erythematosus (SLE) is a chronic autoimmune disease and the most common type of lupus. It occurs when the immune system erroneously attacks the body's tissues, resulting in inflammation, tissue damage, skin issues, joint problems, and even heart, lung, brain, blood cell, and kidney issues. People with autoimmune diseases like lupus experience flares - periods of illness where their malfunctioning immune system causes symptoms. Flares range from mild to severe and don't always have a pattern of predictability.

Risk Factors: Women make up 90% of lupus diagnoses,¹ though men tend to have more serious organ system issues. The majority of people with lupus are diagnosed between the ages of 15 and 45. However, children or older adults may also develop lupus. African Americans tend to suffer from lupus more than Caucasians and lupus is more common in those with American Indian or Asian descent.² Those with family members with lupus or another autoimmune disease are more likely to develop lupus.

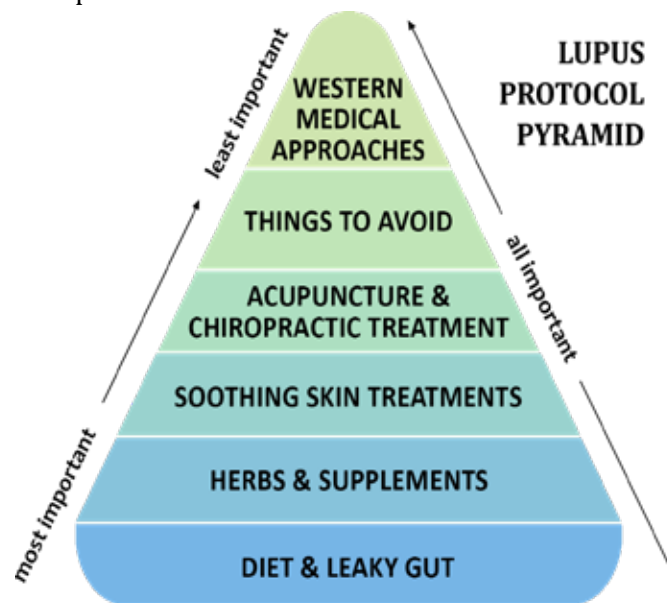
Symptoms: Symptoms of lupus vary from person to person. They can be mild or severe, affect one or multiple areas of the body, change over time, and can come and go without warning. Commonly reported symptoms include:

- Fevers
- Sensitivity to sunlight
- A butterfly rash on the face across the nose and cheeks
- Swollen, painful joints (arthritis) and stiffness in the morning
- Stomach and abdominal pain
- Pain when taking a deep breath or lying down
- Lethargy, fatigue, and/or weakness
- Hair loss
- Skin rashes
- Headaches, dizziness, confusion, seizures, and/or depression
- Swollen glands
- Sores in the nose and mouth
- Fingers and toes change color to a blue-purple, white, or red due to cold and stress. This is called Raynaud's phenomenon.
- Swelling around the eyes or in the legs
- Inflammation that leads to other issues with the heart, lungs, or kidneys.

Inflammation caused by lupus can lead to problems with vital organs. Kidney issues occur in 50% of those with lupus. Lupus nephritis is caused when lupus autoantibodies affect the kidney structures that filter out waste. This leads to kidney inflammation and may cause blood and protein in the urine (often seen as foamy urine), increased blood pressure, edema (swelling), and impaired kidney function. It may even lead to

kidney failure.

When the brain and central nervous system are affected, seizures and memory problems may occur. Sometimes lupus causes heart valve damage, leading to scarring in cardiac tissue. Other heart complications include pericarditis and myocarditis. Blood vessels may also be affected. Vasculitis occurs due to inflammation in the blood vessels, as well as blood clots due to the presence of high levels of autoantibodies. Pleurisy caused by inflammation of the tissue that surrounds the lungs can occur, making it hard to take a deep breath. When the heart and vessels are inflamed chronically, conditions like atherosclerosis and coronary artery disease may develop.



The cause of lupus is still unknown, but what is known is that certain factors can trigger the immune system. These include genetics, influences on the immune system, systemic inflammation, environmental factors like viral infections, stress, certain medications, smoking, diet, hormones, and sunlight exposure. Western treatments for lupus focus on treatment of symptoms as there is no known allopathic treatment for lupus. Read more on allopathic treatments and diagnostic testing for lupus in the "western medical treatment" section of this protocol.

While there are still a lot of unknown factors concerning the causes and treatment of lupus, more is known about how to avoid flares and keep the body in remission. Since lupus is an autoimmune disease, there are steps you can take to control inflammation and live your best life. This protocol outlines what you need to know to manage lupus.

1. Diet and Leaky Gut The foods we consume (and avoid) make a huge difference in our overall health. While this is somewhat common knowledge, some are unaware that their diet plays a large role in keeping autoimmune flares at bay. Many people find that when they eat certain trigger foods, they experience a flare. When they avoid these inflammatory foods, they find that they can go much longer between flares. This section focuses on the best diet for controlling flares and healing leaky gut, which is

strongly tied to autoimmune diseases.

Begin by eliminating all major triggers from your diet. These are foods strongly associated with autoimmune flares. These include: fried foods, trans fats, dairy, gluten, sugar, tree nuts, excess table salt, and processed foods of any kind. Also watch out for “cross-reactive foods.” Cross-reactive foods are foods that the immune system mistakes for gluten.³ They may cause the same autoimmune reaction that gluten causes for many people. Examples of potential cross-reactive foods include cereal grains, corn, yeast, rice, millet, milk, milk chocolate, and instant coffee. Eggs are triggering for some people. However, many people find that eating cage-free organic eggs (in moderation) do not cause them trouble. Get an IGG food sensitivity test or do an elimination diet (see page 300) to be sure.

Peppers and other nightshades rarely trigger allergies. They may (rarely) lead to nightshade intolerance, though there are not any large-scale studies currently supporting this. If concerned, you can reduce their lectin content by soaking grains, nuts, seeds, and legumes and by cooking nightshades, grains, and legumes.⁴

WHAT TO EAT: An Elimination Diet is crucial to helping control flares, but it may leave you wondering what you can eat. Below are healthy and anti-inflammatory foods to incorporate into your diet. Please also see our anti-inflammatory diet section (see page 21), an excellent diet to follow for lupus.

Fresh Vegetables and Fruits: Eat vegetables (including leafy greens) and fruits with each meal. Avoid or cook foods in the nightshade family if they are a trigger for you. Eat vegetables of different colors with each meal and aim for 6-9 servings of vegetables daily.

Tubers: If you are craving something more substantial, try tubers like Jerusalem artichokes, taro, or sweet potatoes. Compounds in Jerusalem artichokes are especially good for controlling inflammation in addition to being prebiotic.

Healthy, Lean Meats: Avoid processed meats; instead eat omega-3 rich fish, wild game, chicken, turkey, and seafood. Choose organic, grass-fed, pasture-raised, or wild caught protein. Avoid lunchmeats, as they often contain excess salt and are processed.

Bone Broth: Organic bone broth is extremely nutritious. It is great on its own, and you can add nutrient-dense ingredients to make a tasty soup. Try adding stinging nettle leaves, kale, spinach, dandelion greens, and other vegetables.

Fermented Foods: It is also important to nourish your gut health by eating fermented foods for their probiotic benefits. Avoid dairy-based fermented foods if dairy is a trigger for you. Incorporate probiotic-rich foods such as kimchi, sauerkraut (not canned), miso, and kombucha. This is an important step for healing leaky gut; try to consume twice a day.

Substitutes for Coffee and Sugar: Processed white sugar is your enemy. Instead try natural sweeteners like stevia, maple syrup, and raw honey in moderation in place of sugar. Instead of coffee, which can be triggering, try green tea. Green tea is rich in antioxidants and can help lower inflammation in the body.

HEALING LEAKY GUT: Leaky gut (increased intestinal permeability) can greatly contribute to autoimmune disease.⁵ It occurs when the lining of the intestines becomes more permeable and toxins and proteins from the foods we eat leach into our body. This causes systemic inflammation and the triggering or worsening of autoimmune diseases. In addition to inflammation, leaky gut can cause digestive upset, diarrhea, pain, allergies, brain fog, skin issues, mood swings, and hormone imbalances. Many people aren't aware that they have a leaky gut. Healing your gut helps reduce flares, helps your body to heal, and helps inflammation to subside. See our Leaky Gut Protocol (see page 70) for more details.

Increased intestinal permeability has been directly linked with the autoimmune disease multiple sclerosis (MS).⁶ The intestinal tissue of mice infected with a MS-like disease showed indicators of leaky gut plus increased inflammation. A surprising revelation was that the intestinal mucous membrane became inflamed even before the mice began displaying symptoms of MS.

Healing the gut is crucial for healing any autoimmune disease. Normal intestinal permeability can be restored by paying closer attention to your diet and the foods you eat. You can also support gut health with herbal remedies. For more on herbal remedies for gut health, see the “herbs and supplements” section of this protocol.

The foods listed above help lower inflammation and nourish the gut. Avoid food triggers, as they can worsen leaky gut and inflammation. Below are other foods to avoid:

- Baked Goods/Pizza (Avoid processed baked goods like muffins, cakes, pies, pastries, cookies, and pizza.)
- Snack and Junk Food (Avoid crackers, pretzels, popcorn, fast food, potato chips, cereals (especially sugary cereals), and candy bars.)
- Gluten-Containing Grains (Gluten usually worsens leaky gut and autoimmune issues. Read labels carefully and avoid foods like barley, rye, seitan, bulgur, and wheat.)
- Wheat-Based Foods (Wheat-based foods like breads, pastas, and cereals can worsen leaky gut.)
- Refined Oils (Refined oils can cause much harm, especially to the digestive tract. Avoid oils like canola oil, vegetable oil, sunflower oil, soybean oil, and peanut oil. Replace with olive and coconut oil.)
- Sauces and Beverages (Sauces hide a lot of salt and sugar. They can also contain refined oils and unhealthy preservatives. These include salad dressings, soy sauce, teriyaki sauce, and hoisin sauce. Avoid alcohol altogether, as well as sugary drinks.)

- Artificial Sweeteners (Avoid using artificial sweeteners such as aspartame, sucralose, and saccharin. Opt instead for raw honey, raw maple syrup, or stevia leaf.)

2. Herbs and Supplements

Supplements and herbs can help control inflammation, immune system malfunctions, and other imbalances that contribute to lupus. There are also remedies for skin issues related to lupus. I also include herbs that target leaky gut.

HERBAL ADAPTOGENS: Adaptogens are one of the best remedies for autoimmune diseases because unlike some herbs that overly “boost” the immune system, adaptogens bring much-needed balance. The herbs below are beneficial for autoimmune diseases and leaky gut. The medicinal mushrooms listed below are all immunomodulatory, anti-inflammatory, anti-viral, anti-cancer, and antioxidant.

Reishi (*Ganoderma lucidum*): This medicinal mushroom is called the “mushroom of immortality.” Its anti-inflammatory properties help calm intestinal inflammation, and its ability to balance and support immune function⁷, hormone levels, and stress makes it an excellent choice for those seeking help with leaky gut and lupus flares. As with most medicinal mushrooms, the best way to get the most from reishi is via a dual-extracted tincture since mushrooms contain both water and alcohol-soluble compounds; a double-extracted tincture accesses all of these medicinal compounds. Suggested dose: two dropperfuls of reishi dual-extracted tincture twice daily.

Lion’s Mane (*Hericum erinaceus*): Lion’s mane helps cool an overactive inflammatory response, has neuroprotective and brain-boosting qualities⁸, and is excellent for those with autoimmune and leaky gut issues. Suggested dose: two dropperfuls of lion’s mane tincture twice daily.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): Turkey tail mushrooms contain prebiotics that help balance the microbiome to control candida overgrowth and heal the gut⁹. They are also known for their immunomodulatory and anti-inflammatory properties. Suggested dose: two dropperfuls of turkey tail dual-extracted tincture twice daily.

Ashwagandha (*Withania somnifera*): The root of this plant has been used for thousands of years to support adrenal function. It also helps support and strengthen the immune system. It calms anxiety¹⁰ and inflammation, eases depression, and helps balance hormone levels. It may protect the brain against the negative consequences of emotional, physical, and chemical stress. Suggested dose: two dropperfuls of ashwagandha tincture twice daily.

American (*Panax*) Ginseng: Used for stamina, energy, and to support overall health, American ginseng is a powerful herb for normalizing our body’s potentially harmful stress response. It also helps with HPA-axis dysfunction (“adrenal fatigue”¹¹), common in those with autoimmune conditions. Suggested dose: two

dropperfuls up to three times daily.

HERBS FOR PREVENTING ILLNESS THAT CAN CAUSE A FLARE-UP: Please see our Autoimmune Protocol (General) for herbs (e.g., *Elderberry*) to prevent viral illness (see page 9).

MORE HERBS FOR LEAKY GUT AND AUTOIMMUNE DYSFUNCTION: **Slippery Elm (*Ulmus rubra*) and Marshmallow Root (*Althaea officinalis*):** Slippery elm bark and marshmallow root have mucilage that acts as a protective layer over intestinal perforations¹² in the digestive tract. Consuming both as a tea can help the gut heal and controls bowel inflammation.¹³ Suggested dose: one to two teaspoons of each infused in a cup of hot water to make tea up to three times daily.

Plantain (*Plantago major*): Broadleaf plantain is of the most common yard “weeds” next to dandelion. It is excellent for the digestive system and has potent anti-inflammatory properties¹⁴ that can soothe mucous membranes, beneficial for those seeking to heal inflammation from leaky gut. Suggested dose: two dropperfuls of plantain tincture up to three times daily or one cup of plantain tea up to three times daily.

Berberine: One well-known contributor to leaky gut is the bacterium *H. pylori*. This bacterium can cause ulcers and make the gut more permeable. If *H. pylori* is the cause of your leaky gut, berberine may be able to eliminate it naturally without the use of harmful antibiotics that can kill off good bacteria in the microbiome.¹⁵ For more on naturally healing leaky gut and ulcers with herbs and other techniques, see the ulcer protocol. Suggested dose: 500mg 2-3 times a day if needed for *H. pylori* (Do not use it as goldenseal, as goldenseal is too stimulating for those with lupus).

Turmeric/ Curcumin: Turmeric is very well studied for its ability to calm inflammation¹⁶. It’s best paired with black pepper/piperine as compounds in black pepper help the body absorb curcumin much more efficiently. Suggested dose: two dropperfuls of turmeric root and black pepper tincture up to four times daily or 500-2,000 mg of a curcumin/piperine supplement.

SUPPLEMENTS: Vitamin D: Studies show that vitamin D may play a beneficial role in immune function for those with autoimmune disease¹⁷. Vitamin D can be created by time in the sunlight, but for those with lupus, too much sun may trigger a flare. Get your levels tested. Suggested dose: 4000 IU daily.

Vitamin K2: If you are avoiding eggs and dairy on an autoimmune diet, you may need supplementation with vitamin K2. This vitamin is responsible for cardiovascular, bone, and dental health. It helps get calcium where it needs to go in the body, while avoiding accumulation in the soft tissues, kidneys, and joints. Suggested dose: 80-120 mcg daily.

Note: Check with your doctor before beginning a vitamin K2 regimen and take with Vit D.

Omega-3: Daily supplementation can reduce inflammation and help bring relief for those with lupus¹⁸. Omega-3s can be obtained through diet through foods like salmon, mackerel, and other fatty fish. You can also take a daily supplement. Suggested dose: 600-2,000 mg daily.

B-12: Vitamin B-12 is associated with energy; it also aids in red blood cell formation and has been shown to help with mood disorders. Vitamin B12 has been found to be lacking in those with autoimmune disease¹⁹. With many autoimmune diseases, malabsorption occurs, which can result in a wide range of nutrient deficiencies. Get your levels tested before supplementing. B12 is found in meat and animal products, so if you are vegan or vegetarian, you may be deficient. Suggested dose: Once you have a baseline blood test, begin with 2,000 mcg of oral vitamin B12 daily, then decrease to 1,000 mcg per day, followed by 1,000 mcg weekly, and finally 1,000 mcg monthly.

Nicotinamide Riboside (Tru Niagen): This is an alternative form of vitamin B3, which is also called niacin. It is converted by the body into nicotinamide adenine dinucleotide, a coenzyme or helper molecule. This fuels a variety of biological processes, such as repairing damaged DNA, converting food into energy, supporting the body's circadian rhythm and internal clock, and fortifying cell's defense systems. As we age, this important nutrient is produced less and less. Because it can help protect DNA damage and oxidative stress, it has potential anti-cancer properties²⁰. It shows promise as an immune-protector and anti-inflammatory agent for those with lupus²¹. Suggested dose: 300 mg daily.

Coenzyme Q10: Coenzyme Q10 has immunomodulatory activity²², suppresses inflammation, improves fatigue and depression, and has beneficial effects on inflammatory cytokines and oxidative stress in autoimmune patients²³. Suggested dose: 300-400 mg daily.

3. Soothing Skin Treatments

Lupus can cause three major types of skin issues: chronic cutaneous lupus, which consists of round, disc-shaped sores on the face and scalp; subacute cutaneous lupus, which can cause a red, scaly rash or ring-shaped sores; and acute cutaneous lupus, appearing as a butterfly-shaped rash across the cheeks and nose. It may also appear on the arms and legs. Other skin-related issues that arise with lupus include calcinosis (hard, white lumps under the skin due to calcium buildup), mucosal ulcers (sores in the mouth, nose, or vagina), and hair loss. For more on hair loss treatment, see the hair loss protocol (see page 274).

Calendula Salve (*Calendula officinalis*): Calendula reduces pain by lowering inflammation. It is antimicrobial, antifungal, and supports cell rejuvenation and wound healing²⁴. It is used for rashes like diaper rash, yeast infections, eczema, and contact dermatitis. You can make your own salve by drying calendula flowers and filling an eight-ounce jar with them. Then completely cover it in a carrier oil like organic jojoba or almond oil and allow this to sit and infuse for six to eight weeks. After 6-8 weeks, strain out the oil and set it aside. Use as is or, to make a

salve, measure your calendula oil and pour into a double boiler. Add one ounce of beeswax pellets for every 4 oz of oil, place on low heat, and allow this to melt. Pour this into tins or jars to cool and use this as needed for skin rashes.

Lavender (*Lavandula angustifolia*): Lavender helps lower inflammation, soothes skin, and reduce redness from rashes. It is also antimicrobial and antifungal. Add five or six drops of lavender essential oil to a teaspoon of carrier oil (like organic jojoba oil) and massage into rashes daily.

Cottonwood Buds (*Populus trichocarpa* and *P. deltoides*): Also known as Balm of Gilead, Cottonwood buds are anti-inflammatory, antibacterial, antifungal, and analgesic²⁵. They provide relief from painful, inflamed rashes and joints, and work even better when paired with calendula. Apply a liberal amount of cottonwood bud oil to rashes, joints, and other swollen, painful areas of the body.

Chickweed Salve (*Stellaria media*): For hot and irritated rashes, chickweed can cool and soothe to reduce irritation, redness, and inflammation. Chickweed is a very common yard "weed". It can be harvested and wilted overnight to infuse in a carrier oil to make a salve. Simply follow the steps above in the calendula salve recipe. Feel free to combine chickweed, calendula, and lavender to make an even more powerful salve.

4. Acupuncture and Chiropractic Treatment

Acupuncture is an ancient Traditional Chinese Medicine approach to healing. It requires triggering specific points on the skin with tiny needles. It is minimally invasive and painless. The needles stimulate nerve-rich areas of the surface of the skin to influence tissues, glands, organs, and bodily functions. Each needle sends a signal to other areas of the body to help stimulate the immune system, promote circulation, heal wounds, and modulate pain.

Many people swear by acupuncture for its ability to relieve stress. It is also credited for helping the following conditions: hypertension, insomnia, allergies, anxiety and depression, chronic pain, menstrual cramps, morning sickness, migraines, osteoarthritis, strokes, and sprains. Because there is clinical evidence that acupuncture has a positive impact on stress, anxiety, and depression²⁶, it may indirectly help with lupus flares. Stress can be a huge trigger for autoimmune flares like lupus.

Chiropractic treatment involves human health and disease processes. Chiropractors address structural, spinal, musculoskeletal, neurological, vascular, emotional, environmental, and nutritional relationships. They may help with diagnostics and therapeutic procedures and they perform adjustments to bring alignment to tissues in and around the spinal column. It is believed that when the spinal column is properly adjusted, the nerves in the body work better, which aids healing. Many people find that regular chiropractic care lowers bodily inflammation²⁷, helps them go longer between flares, and lowers stress and anxiety levels.

5. Things to Avoid

There are some bad habits that significantly contribute to lupus flares. Below are lifestyle changes you may want to make (if you haven't already) to better manage lupus.

Sleep: Most people need 8 hours of good sleep every night. Not getting enough sleep affects many bodily processes; some of the most significant for those with lupus include problems with the immune system, increased inflammation, and higher stress/anxiety, all of which can worsen or trigger flares. Take the time to establish a solid bedtime routine. Avoid blue light and screens, as these send signals to the brain to keep you awake. Dim the lights as bedtime approaches. Take a relaxing bath. Read a good book. It may help to diffuse certain essential oils, like lavender, chamomile, and cedarwood, and to take sleep herbs such as valerian root and hops. For more information on how to get a good night's rest, see the Sleep Protocol (see page 286).

Manage Stress and Anxiety: Chronic stress and anxiety take a major toll on health. For those with an autoimmune disease, chronic stress and anxiety are especially hazardous to health. Make time for self-care, take time off to relax, and set time aside to do the things you love to help keep flares at bay. For more on how to manage stress, anxiety, and depression, see the Anxiety and Stress Protocol (see page 150) and the Depression Protocol (see page 162) if depression is an issue for you.

Stop Smoking: Smoking prevents the oxygenation of blood. Smoking causes too much CO₂ in the blood, which in turn, means that vital organs and tissues are not getting the oxygenated blood they need to stay healthy. Since those with lupus already have issues with organs and tissues, it is important to stop smoking in order for these tissues and organs to stay as healthy as possible. It also damages the lungs and causes cancer.

Don't Overdo Sun Exposure: Too much sun (specifically, ultraviolet) exposure is a known to cause lupus flares. You don't have to smear on the sunscreen when you are outdoors, as sunscreen contains its own share of toxic chemicals. Instead, wear sun-protective clothing like long sleeves, long pants, and hats with a big brim.

Be More Active: Those who lead a sedentary life have a higher risk for lupus complications. It is vital that you stay active and set time aside to exercise every day. Whether you choose swimming, yoga, running, jogging, strength training, or a daily walk, make sure you take at least 20 minutes each day to exercise. Not only can it reduce your chance of a lupus flare-up, it will help you get to a healthy weight, have less stress and depression, and increase energy levels. A recent study found that those with lupus who exercised for at least 20 minutes three times a week experienced less fatigue and more vitality²⁸.

6. Western Medical Approaches

No one test diagnoses lupus. Usually, a series of diagnostic tests ranging from blood and urine tests to imaging tests are needed. Below are common diagnostic tests for lupus.

DIAGNOSING LUPUS: Kidney and Liver

Assessment: You may have tests on your liver and kidneys to see how well they are functioning, as lupus is known to affect these organs.

Complete Blood Count: This is a test to measure the number of red and white blood cells, platelets, and hemoglobin. The results can indicate whether or not you have anemia, a common malady with lupus. It is also not uncommon for those with lupus to have a low white blood cell and platelet count.

Urinalysis: Your urine may be examined to look for increased protein or red blood cells, as this can indicate if lupus is affecting the kidneys.

Erythrocyte Sedimentation Rate: This blood test looks at the rate in which red blood cells settle to the bottom of a tube (within an hour's time). If this happens faster than normal, this may indicate lupus. Note that this increased sedimentation rate can also indicate infections, cancer, and other inflammatory diseases.

ANA Test: ANA stands for Antinuclear Antibody test. If you test positive for these antibodies, you may have lupus. Note that a positive ANA test can indicate other conditions as well.

Echocardiogram: This uses sound waves to produce images of your heart as it is beating in real time. This helps physicians check for problems with the heart, such as valves issues. This test is a way to see if lupus is affecting the heart.

Chest X-Ray: An image of the chest may be taken to look for lupus-associated lung issues like inflammation or fluid in the lungs.

Biopsies: This involves taking a small sample of tissue from an area of the body to look for damage. Oftentimes, biopsies for those with lupus will be conducted on the kidneys to check for damage. Sometimes a skin biopsy may be taken to look for skin conditions associated with lupus.

TREATMENTS: Treatments for lupus are most often designed to help control the symptoms of the disease, as there are no current medical treatments that put lupus into complete remission.

NSAIDs: This acronym stands for Nonsteroidal Anti-Inflammatory Drugs. These are drugs like naproxen, ibuprofen, and Motrin. They can treat pain, swelling, and fever associated lupus, but only temporarily. Some NSAIDs are available by prescription and some are over-the-counter. Taking NSAIDs on a regular basis can cause kidney problems, heart problems, and stomach bleeding, so weigh the risks and benefits carefully.

Corticosteroids: Corticosteroids include drugs like prednisone. These help with inflammation. Sometimes, high doses are prescribed to control serious kidney and brain issues. Side effects may include weight gain, thinning and brittle bones, high blood pressure, diabetes, and an increased risk of infection. The longer you take them, the more at risk for complications you become.

Antimalarial Drugs: Although drugs like hydroxychloroquine are prescribed to treat malaria, they may also be prescribed to decrease the risk of lupus flares. This is because these drugs affect the immune system. One needs regular eye exams if taking an antimalarial drug long-term, as they can cause retinal damage.

Biologics: These are medications that are administered intravenously, like belimumab and rituximab. They may help reduce lupus symptoms for some people, but can also cause nausea, diarrhea, infections, and a worsening of depression.

Immunosuppressants: It is not uncommon for a doctor to prescribe an immunosuppressant for lupus, since this disease is caused by the immune system malfunctioning. They can be helpful for reducing flares. Examples of commonly prescribed immunosuppressants include azathioprine, mycophenolate, methotrexate,

cyclosporine, and leflunomide. These come with side effects, such as an increased risk of infections, an increased risk of cancer, liver damage, and decreased fertility.

LDN: LDN stands for Low Dose Naltrexone, an opioid antagonist that attaches to the opioid receptors in the body. This helps block opioids from binding to receptors. Opioids and their receptors play a role in pain relief and feelings of pleasure. While these are prescribed in higher doses for those with an addiction to opioids and alcohol, they are also prescribed in a low-dose form to those with lupus and other autoimmune issues because they help reduce lupus-associated pain and regulate the immune system. They also help with fatigue and most people do not report any negative side effects. Take at night. You need a prescription from your doctor but they will need to be made in a compounding pharmacy.

	Morning	Noon	Afternoon	Evening	Before Bed
S U N D A Y	Healthy breakfast Vit D, K2, Omega-3, B12, Tru Niagen, & CoQ10 Lion's Mane, Reishi, and Turkey Tail dual-extracted mushroom tinctures	Weekly grocery shopping for lupus-friendly foods Healthy lunch Marshmallow & Slippery Elm tea Turmeric tincture or supplement	Exercise (at least 20 min.) – outdoors, if possible Weekly chiropractic care Herbal salve: calendula, cottonwood buds, lav- ender (if skin issues are present)	Healthy dinner Lion's Mane, Reishi, and Turkey Tail dual-extracted mushroom tinctures Marshmallow, Slippery Elm, and Plantain tea Turmeric supplement or tincture	Bedtime routine Calendula, cottonwood buds, lavender salve (if skin issues are present) Sleep herbs, if needed Ashwagandha or Anxi- ety & stress tincture
M O N D A Y	Healthy breakfast Vit D, K2, Omega-3, B12, Tru Niagen, & CoQ10 Lion's Mane, Reishi, and Turkey Tail tinctures	Healthy lunch Marshmallow & Slippery Elm tea Turmeric tincture or supplement	Exercise (at least 20 min.) Herbal salve, if needed	Healthy dinner Lion's Mane, Reishi, and Turkey Tail tinctures Marshmallow, Slippery Elm, and Plantain tea Turmeric	Bedtime routine Herbal Salve, if needed Sleep herbs, if needed Ashwagandha or Anxi- ety & Stress tincture
T U E S D A Y	Healthy breakfast Vit D, K2, Omega-3, B12, Tru Niagen, & CoQ10 Lion's Mane, Reishi, and Turkey Tail tinctures	Healthy lunch Marshmallow & Slippery Elm tea Turmeric tincture or supplement	Exercise (at least 20 min.) Herbal salve, if needed	Healthy dinner Lion's Mane, Reishi, and Turkey Tail tinctures Marshmallow, Slippery Elm, and Plantain tea Turmeric	Bedtime routine Herbal Salve, if needed Sleep herbs, if needed Ashwagandha or Anxi- ety & Stress tincture
W E D N E S D A Y	Healthy breakfast Vit D, K2, Omega-3, B12, Tru Niagen, & CoQ10 Lion's Mane, Reishi, and Turkey Tail tinctures	Healthy lunch Marshmallow & Slippery Elm tea Turmeric tincture or supplement	Exercise (at least 20 min.) Weekly acupuncture Herbal salve, if needed	Healthy dinner Lion's Mane, Reishi, and Turkey Tail tinctures Marshmallow, Slippery Elm, and Plantain tea Turmeric	Bedtime routine Herbal Salve, if needed Sleep herbs, if needed Ashwagandha or Anxi- ety & Stress tincture
T H U R S D A Y	Healthy breakfast Vit D, K2, Omega-3, B12, Tru Niagen, & CoQ10 Lion's Mane, Reishi, and Turkey Tail tinctures	Healthy lunch Marshmallow & Slippery Elm tea Turmeric tincture or supplement	Exercise (at least 20 min.) Herbal salve, if needed	Healthy dinner Lion's Mane, Reishi, and Turkey Tail tinctures Marshmallow, Slippery Elm, and Plantain tea Turmeric	Bedtime routine Herbal Salve, if needed Sleep herbs, if needed Ashwagandha or Anxi- ety & Stress tincture
F R I D A Y	Healthy breakfast Vit D, K2, Omega-3, B12, Tru Niagen, & CoQ10 Lion's Mane, Reishi, and Turkey Tail tinctures	Healthy lunch Marshmallow & Slippery Elm tea Turmeric tincture or supplement	Exercise (at least 20 min.) Herbal salve, if needed	Healthy dinner Lion's Mane, Reishi, and Turkey Tail tinctures Marshmallow, Slippery Elm, and Plantain tea Turmeric	Bedtime routine Herbal Salve, if needed Sleep herbs, if needed Ashwagandha or Anxi- ety & Stress tincture
S A T U R D A Y	Healthy breakfast Vit D, K2, Omega-3, B12, Tru Niagen, & CoQ10 Lion's Mane, Reishi, and Turkey Tail tinctures	Healthy lunch Marshmallow & Slippery Elm tea Turmeric tincture or supplement	Exercise (at least 20 min.) Herbal salve, if needed	Healthy dinner Lion's Mane, Reishi, and Turkey Tail tinctures Marshmallow, Slippery Elm, and Plantain tea Turmeric	Bedtime routine Herbal Salve, if needed Sleep herbs, if needed Ashwagandha or Anxi- ety & Stress tincture

Ch. 3: Multiple Sclerosis Protocol

I was diagnosed with Multiple Sclerosis in early 2000, but my first undeniable symptoms presented in 1999. I was just thirty years old. I'd been fatigued and had been losing my balance but had brushed it off since I was young, active, and thought that I was healthy.

Then one day I felt something in my eye, which progressed to a tiny black spot in my field of vision. When I woke the next morning my vision in my left eye was almost gone. A very large black spot was in the center, leaving me with only a little peripheral vision.

I went to see an ophthalmologist and then a neurologist who gave me a diagnosis of Multiple Sclerosis, confirmed by my classic MS symptom of optic neuritis and clear brain lesions as seen on an MRI. I was shocked. Nobody ever expects to get so sick, so young. After I left my neurologist's office, I went straight to the clinic's reading room (this was before the days of Internet) and started researching MS. At home I began looking up original research papers on Multiple Sclerosis. As a scientist I wanted statistics and peer-reviewed studies. Later on, I took this same approach to develop the protocol that saved my quality of life and that of countless others affected by this disease.

After years of failed treatments and the complete failure of western medicine, which left me bedbound on my bad days and using a cane and/or a wheelchair on my "good" days, I took my health into my own hands. It has paid off well, as now my Progressive Multiple Sclerosis is no longer progressing. I have had no new MS lesions in the past 20 years and am happy and healthy, even surviving 57 days alone in the wild on the History Channel's hit TV show "Alone". I know that following my protocol is how I've achieved good health.

My success with MS inspired me to research and discover the other powerful protocols in this book, and is what motivated me to open my own apothecary.

Multiple sclerosis is a type of autoimmune disorder, where the immune system mistakenly targets healthy cells in the body instead of foreign bacteria or viruses. In multiple sclerosis, the immune system attacks and causes damage to the nervous system. Specifically, the immune system attacks the protective sheath around our nerves called myelin, resulting in faulty nerve function and demyelination. The course of MS is associated with an exaggerated inflammatory response and increased oxidative stress.

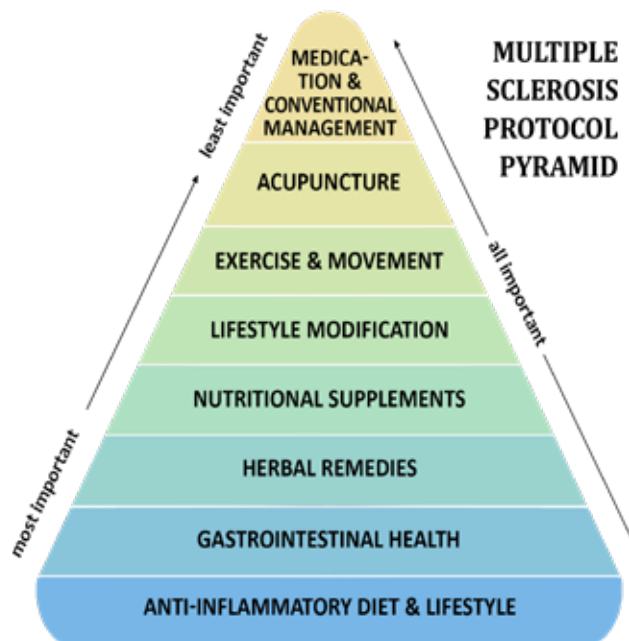
MS affects people differently. Some people experience mild or infrequent symptoms. For others, it is an extremely severe condition since the central nervous system is responsible for controlling many essential activities in the body, like regulating body temperature and heart rate, breathing, control of voluntary movements, learning and memory, balance, elimination, and sight.

In young and middle-aged adults, multiple sclerosis is the most common permanently disabling disorder of the central nervous system. It is commonly diagnosed between the ages of 20 and 50. People living in higher latitudes, smokers, and people with a family history are at increased risk of being diagnosed with multiple sclerosis. Women are three times more likely to be affected than men.

Symptoms of multiple sclerosis include but are not limited to:

- Weakness in the limbs, numbness, and/or tingling
- Poor coordination and balance, gait disturbances
- Dizziness and vertigo
- Double vision, blurred vision, and/or eye pain
- Urination and bowel problems
- Facial weakness, pain and/or sensation loss
- Fatigue
- Depression
- Poor memory and attention
- Nerve pain
- Muscle spasms.

Multiple sclerosis is broken into three subtypes. **Relapsing-remitting MS** accounts for 85-90% of initial cases where people experience discrete symptom flares or attacks over a period of days to weeks that then steadily improve over weeks to months. They may regain normal neurological functioning between attacks. This may progress to secondary progressive MS. **Secondary progressive MS** cases appear like relapsing-remitting at first, but people experience gradual deterioration of nervous system functioning over time. **Primary progressive MS** accounts for 10-15% of cases. People experience symptoms with gradual deterioration of functioning over time without any noticeable periods of attacks or improvement.



The cause of multiple sclerosis is unknown, but thought to be a combination of factors that lead to abnormal and unregulated immune system function. Besides genetic susceptibility, infections are a possible cause. There is

evidence relating a past infection with the Epstein-Barr virus, or infectious mononucleosis^{1 2}, with increased risk of developing multiple sclerosis and that chronic/recurrent viral activity may be responsible for symptom flares³. A proposed hypothesis behind why people in higher latitudes are diagnosed more is decreased sunlight exposure and thus lower levels of Vitamin D. Disturbances in the gut bacterial microbiome and exposure to environmental toxins are other leading areas of ongoing investigation. Our understanding of the disease is always developing, but since it is not well understood, no known cure for multiple sclerosis exists. It is a condition that people live with and manage as best they can.

Multiple sclerosis is diagnosed when there are two episodes of neurological disturbance, lasting a minimum of 24 hours, separated in time, which cannot be attributed to an active infection, fever, or other diagnosis. An MRI of the brain and spinal cord with and without contrast is used to visualize areas of nerve damage or scarring, called 'lesions,' in the nervous system and to confirm diagnosis.

1. Anti-Inflammatory Diet and Lifestyle

Managing inflammation and facilitating healthy immune system function is the first priority to improve multiple sclerosis symptoms. Profound changes can be made with modifications to daily rituals and food choices.

Anti-Inflammatory Diet Choices: *Lowering inflammation and blood sugar levels are key for MS.* A **ketogenic diet** is one that is high in fat and low in carbohydrates (see page 305). This causes a shift in energy metabolism where the body uses fat as regular fuel for energy as opposed to its usual carbohydrate fuel, glucose. This metabolic change is linked to numerous health benefits that pertain to common multiple sclerosis concerns, like providing protective⁴ and anti-inflammatory⁵ influences on the central nervous system. Adopting a ketogenic⁶ and/or **anti-inflammatory diet** (see page 297) has been shown to significantly improve fatigue and depression in people with multiple sclerosis. Common triggers are gluten, dairy, and sugar. An IGG test will help you determine specific triggers.

The Wahls Protocol Diet is an excellent autoimmune diet protocol for MS. It was created by Dr. Terry Wahls, MD – a researcher who, like me, had debilitating MS and decided to take her health into her own hands. I highly recommend her dietary protocol. Visit page 310 in this book for more information.

Using **intermittent fasting** can provide similar benefits as a ketogenic diet, although intermittent fasting has nothing to do with changing what you eat. Intermittent fasting is a style of eating that specifies when you eat and when you do not (see our diet section at page 303). I pair an anti-inflammatory diet with intermittent fasting.

Studies analyzing the effect of intermittent fasting in people with multiple sclerosis have found that it improves quality of life and emotional wellbeing⁷ and decreases

symptoms of depression. One possible reason is the fact that intermittent fasting has immunomodulatory⁸ or balancing effects related to increased diversity of the intestinal microbiome. Intestinal health is an important concern for multiple sclerosis patients, as we will discuss, so intermittent fasting is an attractive option for its ability to improve symptoms while possibly addressing an underlying cause of disease and dysfunction. The therapeutic benefit of ketogenic, anti-inflammatory, and intermittent fasting style diets is best seen when adhered to for at least 6 months.

Mediterranean style and gluten-free diet approaches both have something to offer multiple sclerosis patients looking to improve their condition. While a gluten-free diet is not the same as Mediterranean and vice-versa, one could certainly adopt a Mediterranean diet that is gluten-free. The high fruit, vegetable, and whole grain intake in a Mediterranean diet increases dietary antioxidants and dietary fiber, which assist the immune system and promote healthy microbial balance in the gut. This paired with the natural reduction of inflammatory influences, like some fats, animal products, and simple sugars, allows the Mediterranean diet to have an all-around anti-inflammatory effect, which can benefit autoimmune diseases greatly. It is associated with⁹ decrease in fatigue, increase in mental and physical quality of life, increase in exercise capacity, and improvements in hand and leg function amongst multiple sclerosis patients. It may also improve mood and cognition¹⁰. Make sure you are still eating healthy fats such as monounsaturated (e.g. olive oil) and polyunsaturated (e.g. fish oil), as they are necessary for proper nerve function^{11 12}.

A gluten free diet alone can decrease¹³ MS lesion activity seen on MRIs, lower relapse rates, and improve physical capacity, so it is sensible to at least go gluten-free.

Remove Alcohol and Cigarettes from Daily Use: Although its recreational use is widely accepted, alcohol acts as a toxin to the neurological system, with excessive or chronic alcohol consumption causing significant central nervous system dysfunction, which can be irreversible¹⁴. For these reasons, anyone who is already suffering from deficits in their nervous system function are encouraged not to drink alcohol due to the risk worsening of symptoms. Multiple sclerosis risk increases¹⁵ with increasing alcohol abuse or alcohol dependence. Smoking is harmful to the entire body, not just the lungs. There are thousands of different chemicals present in cigarettes and vaping liquids, all which enter the body through the bloodstream. These chemicals cause a pro-inflammatory cascade that worsens auto-immunity and increases oxidative stress. Smoking increases the risk of being diagnosed with multiple sclerosis and worsens disease progression¹⁶. It causes faster rates of brain atrophy, or shrinking, and a greater disease burden¹⁷. Quitting smoking must be part of addressing multiple sclerosis.

2. Gastrointestinal Health

The intestinal tract, when functioning normally, is a sophisticated barrier that controls nutrient absorption into our bloodstream. It is the

production center for hormones, neurotransmitters, and immune signals, as well as a host to many species of bacteria and fungi that help carry out those functions. When the intestinal tract, or gut, is not functioning normally, it has numerous effects on the body that extend beyond the system itself. Poor gastrointestinal health has been linked to the development of several chronic diseases and there is a direct link between abnormal intestinal barrier function, (commonly termed 'leaky gut') with inflammation and multiple sclerosis¹⁸. Integrity of the intestinal barrier is under current investigation as a possible target for treatment in MS.

It is not just the barrier and its functions that are important. Dysbiosis, an altered imbalance in gut flora with enrichment of some species and depletion of others, is commonly documented in people with multiple sclerosis as compared to healthy populations. These changes affect immune response and communication with intestinal hormones and the nervous system. The current scientific belief is that there is a bidirectional relationship¹⁹, meaning that the altered flora predisposes one to multiple sclerosis and that multiple sclerosis predisposes one to a change in flora.

People with MS can experience the double effect of having an altered microbiome and decreased integrity of the intestinal barrier²⁰, both of which are known to exacerbate the inflammatory response and perpetuate intestinal dysfunction. With this in mind, natural treatments aimed at supporting normal gastrointestinal function, barrier integrity, and microflora balance are logical and practical choices.

Probiotics: Probiotics are living bacteria and yeasts that have health benefits when eaten. They are the organisms found in fermented foods and foods made with bacterial cultures. After eating these foods, these probiotics continue to live in our digestive tract where they serve a variety of functions: they help break down and digest food, promote healthy acidity, regulate the immune system, and help protect against infection. Try to integrate at least one type of the following probiotic-rich foods into your daily diet: yogurt or kefir, sauerkraut or kimchee, tempeh, miso, kombucha, apple cider vinegar, buttermilk or raw milk cheese, cured olives or pickles.

Not only will adding probiotics help your immune system but adding a probiotic supplement²¹ helps balance dysbiosis and reduce systemic inflammation in multiple sclerosis patients. When choosing a probiotic capsule, be sure it contains at least 5 different kinds of bacteria and has at least 10 billion organisms per capsule. These are the minimum amounts needed for therapeutic benefit and one capsule per day, taken with food, will provide a sufficient boost.

L-Glutamine: L-Glutamine is a non-essential amino acid that has nutritive and protective effects on the gastrointestinal system. It acts as an anti-oxidant²² in the gut, it helps intestinal cells to grow and survive²³, it improves intestinal barrier functioning²⁴, and it may help repair damaged intestinal lining. In

experiments, oral doses range from 0.3g to 0.9g/kg body weight per day, equaling roughly 21g- 65g per day for a 72.5 kg (160 pound) adult. Without supervision, suggested dose for routine daily consumption is 2-5 grams.

Fiber: Dietary fiber is a key component of overall gastrointestinal health through its regulation of the gut microbiome and its use as food by beneficial bacteria. The resulting health benefits²⁵ of high fiber intake include reduction of inflammation, improvements in intestinal permeability and function, and decreased incidence of depression - likely through promotion of neurotransmitter production. A fiber-rich diet of foods like steamed vegetables, brown rice, legumes, chia, and flax seeds will build a great foundation. You can also take a fiber supplement, like psyllium, acacia fiber, or beta-glucan to help ensure you are getting optimal amounts. The total recommended intake of fiber is between 25-38 grams daily, and supplemental doses range from 2-6 grams once or twice a day, depending on tolerance.

Collagen Powder and Bone Broth: Daily collagen supplementation can help boost the production of gastric juices, as well as help restore mucosal lining²⁶ and permeability. Collagen is rich in amino acids that help nourish the intestinal cells and there is evidence that collagen rich supplements have an anti-inflammatory effect²⁷ in the GI tract. Collagen powder is usually tasteless and dissolves well in liquids; warm liquids are best. Suggested dose: 10-20grams; this is easily and safely obtained using most over the counter collagen supplements. Daily organic bone broth is another excellent source of bioavailable collagen and gelatin.

Herbs for Gut Health: Lion's mane, Marshmallow root, Slippery elm, and Turkey tail are great for the gut. A known antioxidant and gastroprotective, **Lion's mane** helps tame the inflammatory response and protect the lining of the stomach. It also protects the gut from *H. pylori* and the polysaccharides in lion's mane protect against gastric ulcers²⁸. **Marshmallow and slippery elm** are demulcents and create a protective layer over intestinal perforations. **Reishi's** anti-inflammatory action is helpful for calming intestinal inflammation. It also modulates the immune system, inhibits *Candida* yeast, and is neuroprotective. **Turkey tail** contains prebiotics that balance the microbiome, encourages the growth of "good" gut bacteria, and controls *Candida* and small intestinal bacterial overgrowth (SIBO). For more information on Leaky Gut, please visit page 70 in the book.

3. Herbal Remedies Herbal medicines can provide much needed support to people with multiple sclerosis. There are different herbal approaches one can take for MS, and often, a combination is best. There are herbs that act as anti-inflammatories and immunomodulators to help normalize immune system functions, herbs that act as protective and regenerative agents in the nervous system, herbs with anti-viral properties to address possible underlying causes, and herbs that help with symptoms

caused by a damaged nervous system.

MS HERBS: Medicinal Mushrooms: Lion's Mane, Reishi, Turkey tail, and Cordyceps are great mushrooms for people with MS. *Adaptogenic, anti-inflammatory, and immunomodulatory herbs are especially good for autoimmune issues.* Many medicinal mushrooms have immunomodulatory and anti-inflammatory properties. I take 2 dropperfuls each of dual-extracted Lion's Mane, Reishi, Turkey tail, and Cordyceps mushroom tinctures daily, or 2x/day if feeling "off". These mushrooms have the added benefit of being antiviral. Best used as a dual-extracted tincture to access all of the medicinal compounds (not just a tea).

Lion's Mane (*Hericium erinaceus*) is neuroprotective and is known to help nervous system degeneration, decrease gut inflammation, boost brain function, and lessen stress, anxiety, depression, & insomnia. Studies show that lion's mane helps repair damaged nerve cells in the spinal cord and brain²⁹ and support re-myelination and regeneration of nerves through stimulation of Nerve Growth Factor (NGF). NGF proteins also help to maintain the nerve cells responsible for attention, learning, and memory. Additionally, lion's mane has been shown to help reduce symptoms and brain cell degeneration³⁰ in MS, Alzheimer's, Dementia, and Parkinson's³¹ in animal studies, as well as the progression of dementia. It boosts neuronal function³² and significantly improves cognitive performance³³, making it an excellent treatment choice for people with MS. The dominant components of lion's mane are hericenones and erinacines, which possess significant nerve repair ability.

Reishi (*Ganoderma lucidum*) is a formidable anti-inflammatory that helps modulate the immune system. Since autoimmune disorders are inflammatory in nature, reishi is an excellent choice for slowing or reversing disease progression. Reishi suppresses certain inflammatory compounds³⁴ and is effective in calming the inflammation and immune response involved with leaky gut and intestinal dysbiosis. Research also suggests that it could serve as a regenerative therapeutic agent for the treatment of cognitive decline associated with neurodegenerative diseases³⁵. In addition, reishi mushroom is also known for its adaptogenic qualities - helping to strengthen the body against the negative effects of stress, like inflammation, hormonal imbalances, high cortisol, fatigue, depleted energy levels, brain fog, and low energy.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): is high in anti-inflammatory compounds and assists in reducing internal inflammation. Turkey tail is an immune modulator, making it an effective medicinal herb for autoimmune conditions. It also contains prebiotic compounds that help stimulate the growth and population of probiotic bacteria in the intestines.

Cordyceps (*Cordyceps* spp.) has neuroprotective effects and reduces neuronal cell death³⁶. It has immunomodulatory and anti-inflammatory properties, helps to combat fatigue, improves cognitive health, and

has anti-aging and antioxidant properties. Cordyceps is also anti-microbial, anti-bacterial, anti-viral, anti-fungal, and antiprotozoal making it an effective agent against infection. It improves memory, learning, and reduces oxidative damage to cells^{37 38}. It also improves insulin sensitivity while lowering bad LDL cholesterol, which helps to prevent plaque build-up in the arteries, reducing the risk of heart attack and stroke³⁹.

Chinese Skullcap (*Scutellaria baicalensis*): Chinese skullcap is an ancient herb long known for its anti-inflammatory, anti-viral, and antioxidant properties⁴⁰. It may be useful for multiple sclerosis due to its evidence in reducing neural inflammation and improving motor function through protective action on nervous system cells^{41 42}. Some of this protection may be due to its antioxidant⁴³ functions. Suggested dose is 300-900mg of a supplement or 2 dropperfuls of tincture twice per day. It is also anti-viral (see below), as are the above medicinal mushrooms.

HERBS FOR PREVENTING ILLNESS: *Preventing illness that can lead to a flare-up:* An important thing to keep in mind when you suffer from an autoimmune condition is to keep your immune system from needlessly activating. That means preventing general illnesses so it doesn't cause your immune system to flare up. Since inflammation and immune system dysfunction go hand-in-hand with autoimmune disorders, those with them have to be careful to avoid viral illness. Please see page 9 in the Autoimmune Protocol for more detail and research about the following herbs.

Elderberry (*Sambucus nigra*) and Ursolic Acid: Elderberry shortens the duration of colds and flu, decreases inflammation, chronic fatigue, allergies, nerve pain, heart pain, headaches, dental pain, and sciatica. Elderberry is rich in anthocyanins, which have antioxidant and anti-inflammatory benefits⁴⁴. Elderberry is a safe option for treating viral respiratory illness, and there is no evidence that it over-stimulates the immune system⁴⁵ or creates a cytokine storm⁴⁶. Alcoholic extractions of elderberry contain the triterpenoid ursolic acid (UA)⁴⁷, which has been shown to prevent and repair neurons in animal models of MS, reduce further damage to neurons, help rebuild the protective sheaths covering neurons, shows significant potential as an oral anti-inflammatory for MS, and has significant potential as a neural repair agent for MS⁴⁸.

Usnea: I pair Usnea lichen spray with Elderberry to stay healthy during the cold and flu season, and when I am traveling or around crowds. Usnea has powerful antibiotic, antifungal, antimicrobial, and antiviral properties. I spray 1-3 sprays in the back of my throat to help prevent viral illness, and more if ill. I also coat the inside of my nose with my All-Purpose Salve for prevention.

Eastern Blend Tincture: This is a blend of Chinese skullcap, Japanese honeysuckle, Japanese knotweed, and cordyceps mushroom. I use these research-backed herbs for viral prevention as the baicalin and

wogonin content in Chinese skullcap is anti-inflammatory and has been shown to exhibit strong antiviral activity for severe respiratory illness, the common cold, and influenza⁴⁹. It also protects against oxidative stress and has neuroprotective and cognitive enhancement effects due to its baicalin content⁵⁰. Japanese honeysuckle is antiviral⁵¹, antibacterial, anti-inflammatory, and modulates the immune system and significantly increases killer (NK) cells⁵². Japanese knotweed has high concentrations of trans-resveratrol⁵³, has strong antioxidant activity and inhibits viral replication, and also reduces neuroinflammation and neurodegenerative diseases such as Alzheimer's and dementia⁵⁴.

4. Nutritional Supplements

The following nutrients are

either key to normal neurological functioning or are found to be deficient in patients with multiple sclerosis and other autoimmune conditions. Boosting your body's levels of these vitamins and minerals can help the body normalize nerve function and possibly reduce the likelihood of flares and disease progression.

Vitamin D and K-2: Vitamin D is a hormone that helps with immunomodulation and healthy immune system function. Research shows that Vitamin D helps regulate the release of nerve growth factor (NGF), an essential molecule for the neuronal survival of hippocampal neurons as well as cortical neurons⁵⁵. People with low blood levels of Vitamin D are more likely to be diagnosed with MS and other autoimmune conditions. Low Vitamin D is also associated with more severe disease progression. Additionally, Vitamin D supplementation⁵⁶ in people with MS may help decrease relapse rate and disability, prevent growth of new lesions, and improve cognitive performance⁵⁸.

It is difficult for us to make the amount of Vitamin D needed from sun exposure unless we spend a lot of time outdoors with most of our skin exposed. Get a blood test to measure your levels before starting a daily routine. Suggested dose is 2,000-4000 IU daily, but reduce this amount in the summer months if you are spending more time outdoors. Vitamin D deficient adults can take from 5,000-10,000iu of Vitamin D per day over a period of months to reach normal levels. It's important to take vitamin K2 alongside vitamin D because they work synergistically to promote cardiovascular and bone health. Aim for 80mcg vitamin K2 per day – if you need calcium, you can take a calcium supplement with K2 when you take your Vit D.

Omega-3 Fatty Acids: Chronic inflammatory conditions like MS respond well to Omega-3 fatty acid supplementation. Omega-3s reduce oxidative stress and the inflammatory response in multiple sclerosis specifically⁵⁹. Omega-3s help reduce inflammation in the bowels, help with myelin repair, decrease disease activity, reduce pain, and lower the use of anti-inflammatory drugs. These fatty acids also help maintain brain, nerve, cell membrane, and heart function. Suggested daily dose: 600-3,000mg.

Vitamin B-12 and Folic Acid: These are discussed together because of their reliance on the other's presence to work properly, and their shared fundamental responsibilities in central nervous system function at all ages and stages of development. Deficiency of either vitamin can result in indistinguishable neurological and psychological dysfunction. Multiple sclerosis and vitamin B12 deficiency share common inflammatory and pathophysiological characteristics. Vitamin B12 is thought to have an immune balancing effect on the nervous system, as well as assist with nerve cell growth and myelin formation. Decreased levels^{60 61} of vitamin B12 and folate⁶² have been demonstrated in MS patients. B12 deficiency is thought to inhibit MS recovery⁶³. Supplementation of B12 and folic acid has been shown to improve⁶⁴ physical abilities and quality of life for people with multiple sclerosis. Suggested dose: 1,000mcg of each daily.

Turmeric/Curcumin: Turmeric is a widely studied herb known for its anti-inflammatory and antioxidant actions. These properties make it a good fit for addressing the causes and effects of multiple sclerosis. It blocks inflammation and protects nerve cells^{65 66}. Best taken in tincture or encapsulated form (with piperine). Suggested dose: 500-1000mg per day.

Nutrient Antioxidants: Reactive oxygen species, or free radicals, are unstable molecules that are a part of normal metabolism, but they can cause damage if left unbalanced by other resources in the body. These molecules have been implicated as mediators of demyelination and axonal damage, enhancing inflammation and neurodegeneration seen in multiple sclerosis.

Vitamin C is important in the synthesis of the neurotransmitter dopamine, and protects the brain against oxidative stress. Vitamin C deficiency is associated with increased demyelination⁶⁷ and supplementation may help promote remyelination⁶⁸. Suggested dose: 1,000 mg twice a day.

Zinc is concentrated in high amounts in the forebrain. Lack of zinc is associated with various neurological and psychological impairments and low zinc status is associated with multiple sclerosis⁶⁹ diagnosis. Suggested dose: 30mg per day.

Resveratrol^{70 71} helps support a healthy metabolism, reduce inflammation, and may help with aging. It is also a great immune modulator. There is some evidence supporting the use of resveratrol to assist with motor control, demyelination, and reduced oxidative stress in multiple sclerosis. Suggested dose is 500-1,500 mg daily. Start with the lowest dose and work your way up if needed.

Coenzyme Q10 may be useful in treating neurodegenerative diseases⁷². CoQ10 helps suppress inflammation⁷³ in multiple sclerosis patients, may improve fatigue and depression, and has beneficial effects on oxidative stress in autoimmune patients. Suggested dose for MS: 500 mg/day.

N-Acetylcysteine (NAC)⁷⁴ is a naturally occurring compound in the body that is antioxidant, anti-inflammatory, and anti-viral. It also protects DNA and assists in the production of neurotransmitters like glutamate. Supplementation of N-Acetylcysteine has been associated with improvements in cognition⁷⁵ and attention in multiple sclerosis patients. It may also help reduce fatigue⁷⁶. Suggested dose: 500mg twice a day.

5. Lifestyle Modification The most effective treatment plans are holistic and include healthy self-care habits, a healthy environment, and a positive mind-body connection. Research concludes that several lifestyle factors have an impact in the diagnosis and development of MS.

Stress Management: Research⁷⁷ shows that high levels of stress are related to higher risk of relapse and development of lesions, and that utilizing stress management techniques can help prevent the development of new lesions. Additionally, people with multiple sclerosis with strong coping strategies are better able to self-manage and suffer fewer impacts of the disease overall.

Meditation has a lot to offer people with multiple sclerosis looking for symptom improvement. Meditating once a week or more^{78 79} is associated with improved physical and mental wellbeing, increased energy, decreased pain, decreased depression, decreased anxiety, and fewer sleep problems⁸⁰. You can meditate alone or in a group class environment - both are equally helpful. Giving **gratitude** daily is another proven way to manage anxiety and cultivate a positive mindset.

If meditating doesn't appeal to you, **stress management and emotional regulation techniques** can help. People who have taken stress management classes⁸¹ have better measurements of anxiety, depression, and quality of life than MS patients who have not. Stress self-assessment techniques, when utilized twice a day, have positive effects on cognition and memory⁸². **Spending time in nature** is another important way to help both physical and mental health⁸³, including the lowering of cortisol levels.

Optimize Sleep: People with multiple sclerosis are more likely to suffer from sleep disturbances⁸⁴ and sleep disorders⁸⁵ like insomnia, restless legs syndrome, sleep apnea, and narcolepsy. Not only do these sleep issues have a negative effect on patients' quality of life⁸⁶, but sleep issues increase the risk of developing nervous system, mood, and cardiovascular conditions.

There are several hypotheses behind the link between multiple sclerosis and sleep issues. One important finding is that melatonin, our sleepy hormone, is not released normally⁸⁷ in MS patients, so optimizing melatonin is a potential target for therapy. **Melatonin** supplements improve sleep quality and duration for multiple sclerosis patients⁸⁸ and can easily be found over-the-counter. Suggested dose is 1-3mg before bed. Herbal sleep blends also work well when used regularly. See our Sleep

Protocol on page 286 for herbal suggestions.

While medications may be used in the short-term to help manage sleep disturbances, safe, longer-term solutions are preferable. Underlying causes must be addressed-like neuromuscular training for sleep issues related to nighttime urination, and positive airway pressure machines for those with sleep apnea. For others, exercising, better sleep hygiene, cognitive behavioral therapy, mindfulness, and sleep education⁸⁹ have a positive effect on sleep, physical quality of life, and mental quality of life.

Common sleep hygiene recommendations are keeping the same sleep and wake times, ensuring exposure to bright light in the morning, daily exercise, and having no screen time at least an hour before bed.

Disconnect from Technology: By surrounding ourselves with LED bulbs and televisions, computer monitors, and smart phones, we are exposed to more blue light than ever, especially at night. While the most mild effect of the added blue light is digital eyestrain, a more profound effect is that it disrupts our body's sleep-wake cycle, or circadian rhythm. Harvard researchers found that blue light suppresses our sleepy hormone⁹⁰, melatonin, for twice as long as light from other wavelengths, and there is evidence that links circadian rhythm disruption with depression, diabetes, and cardiovascular problems; risks already associated with multiple sclerosis.^{91 92 93}

Aside from the light that our devices emit, there are also the effects of what we're seeing. Social media⁹⁴ use has been linked with higher levels of concentration problems, fatigue, and loneliness. Several studies⁹⁵ have concluded that negative news reports increase levels of sadness, worries, and anxiety while also decreasing positive emotions; these effects can persist for weeks. Considering that sleep, mood, and cognitive disturbances may already be worse due to MS, it's sensible to remove influences that may exacerbate these symptoms further. Taking time to disconnect from technology, especially before bed, is an important step.

6. Exercise and Movement For many years MS patients were told to avoid exercise because it was thought to exacerbate their symptoms. Recent research shows that exercise dramatically improves overall cognitive function in those with MS.

Cardiovascular Exercise: Cardiovascular exercise or aerobic fitness has an added benefit for people with multiple sclerosis: the improvement of brain function and memory. This effect has been seen in people with multiple sclerosis specifically. Just 12 weeks of a treadmill walking program resulted in improved learning and memory ability⁹⁶, attributed to increased stimulation and flow to the hippocampus. Cardiovascular exercise also results⁹⁷ in improved fatigue, movement capacity, quality of life, and overall mood.

Cardiovascular fitness may also result in better disease outcomes. Researchers observed that

when someone with multiple sclerosis had a high level of fitness, less brain damage and deterioration was detected⁹⁸. MRI scans revealed they had a greater volume of vital gray matter of the brain. Aerobically fit patients had far fewer brain lesions and also scored higher on brain functioning tests.

Other Fitness Approaches: Other types of exercise are also helpful for multiple sclerosis patients. **Yoga** has a beneficial effect⁹⁹ on mood and energy for people with MS. It has also been shown to improve fatigue, balance, and walking coordination¹⁰⁰. Additionally, **resistance training**¹⁰¹ has a significant positive effect on muscle strength and might also help balance, fatigue, quality of life and mood. Some may need a pre-exercise screening prior to beginning an exercise program to assess for safety issues but there is plenty of evidence in support of exercise as an effective treatment for multiple sclerosis.

7. Acupuncture Acupuncture is known to help produce natural opioids and helps with pain management¹⁰². It may reduce depression¹⁰³ and anxiety and improve working memory¹⁰⁴. For multiple sclerosis patients specifically, acupuncture treatments improve many symptoms^{105 106}, including fatigue, pain, gait impairments, and bladder dysfunction. It may also reduce relapses and improve cognitive deficits.

There have even been studies investigating specifically which style of acupuncture treatments and administration yields the best results for multiple sclerosis patients, be it minimal acupuncture, Chinese medical acupuncture¹⁰⁷, electroacupuncture, acupoint injection, shallow needling technique, or hand and foot. The conclusion of these studies is that while some are more effective than others, *all kinds* of acupuncture result in symptom improvement, decreased disability, and overall quality of life¹⁰⁸. One study suggests that in order to effectively see and measure reduced recurrence¹⁰⁹ rates, the course of treatment should be 5 days a week for at least three months.

8. Medication and Conventional Management

Multiple sclerosis is conventionally managed with medications, although additional approaches like physical therapy, speech therapy, and counseling may be recommended. Which medication is chosen depends on what subtype is present, which symptoms, and whether acute flare-ups are being targeted or the disease process as a whole. Note that most current disease-modifying therapies have a lot of negative side effects and are often not as effective as using the prior suggestions in this protocol. LDN is an exception.

Low Dose Naltrexone (LDN): Naltrexone is a drug that blocks opioid receptors and is most widely known as a drug for opioid and alcohol dependency, given in doses of 50 to 150 mg. However, applications of naltrexone at lower than usual doses, from 1-5mg

given orally and sometimes intranasally, exhibit effects like neuroprotection, inflammation reduction, immune system modulation, reduced fatigue, and pain relief^{110 111} without negative side effects. LDN has become an emerging treatment¹¹² for autoimmune diseases like multiple sclerosis and Crohn's disease, as well as pain conditions like complex regional pain syndrome and fibromyalgia. Because of its ability to increase patients' general quality of life, it is also under consideration¹¹³ for treating mood disorders. Side effects, if present, are usually mild and temporary, like vivid dreams and headache. I use LDN every night before bed.

Low-dose naltrexone has not yet been approved by the FDA for the treatment of multiple sclerosis, but it is used unofficially, or off-label. Current evidence points to LDN as an inexpensive, safe, and well-tolerated^{114 115} treatment for MS, alone or alongside other medications. Support for its use is mounting.

Disease Modifying Therapies (DMTs): These are the most commonly prescribed group of medications used for MS. DMTs are drugs that suppress the immune system or change how it works in an attempt to reduce relapse, nerve cell damage, and lesion growth. There are over a dozen different kinds of Disease Modifying Therapies - some are given intravenously, some are injections, and some are taken orally. It is thought that oral and intravenous therapies are the most effective, but they also come with more side effects than injections. Choosing the exact course of treatment can be a complicated process. It depends on several factors, including how easily the patient can show up for intravenous treatments or injections, side effect profile, other diseases the patient has at the same time, and patient tolerance. Personally, I manage MS well without these.

Each medication has its own treatment intervals and frequencies in which they need to be taken. Alemtuzumab, for example, is initially given intravenously on 5 consecutive days in a row, but is not given again for a year, whereas ocrelizumab is given once, then 2 weeks later, then for every 6 months afterwards. Interferon is a common injectable medication for multiple sclerosis and it is administered either three times a week, once a week or every 2 weeks depending on the type.

Side effects and patient tolerance should be considered. Flu-like symptoms, headaches, itching, and nausea are common immediately after treatment. Since these drugs change the way the immune system functions, they are commonly associated with long-term side effects that require frequent monitoring like: low white blood cell count, increased susceptibility to infections, impaired liver function, mouth sores, thinning of hair, high blood pressure or heart disease, other auto-immune disease, depression or suicidal thoughts, and increased risk of developing certain cancers.

Corticocorticoids: Also known as steroids, corticocorticoids are a different class of medications added for acute flares. Methylprednisolone is most typical,

given either intravenously or orally. Steroid medications act as anti-inflammatories and immunosuppressants and are only meant to be used short-term. Side effects of this class of drugs¹¹⁶ include susceptibility to infections, fluid retention, and muscle weakness.

Symptom Relief: Other various medications may be

recommended to deal with specific symptoms on an as-needed basis. When people experience pain, muscle spasms, constipation, depression, or tremors that the DMTs do not improve, they may be given additional targeted medications that won't modify the course of the disease, but will make the person more comfortable.

	Morning	Noon	Afternoon	Evening	Before Bed
S U N D A Y	Collagen/bone broth Meditation & morning gratitude Lion's Mane, Reishi, Turkey Tail & Cordyceps mushroom tinctures Vit D w/ K2 or Vit D + Calcium w/ K2, if Ca needed	Anti-inflammatory diet meal planning and shopping NAC & antioxidants Fiber supplement, as needed B12, Folic acid & Omega-3 oils with lunch	Resistance training Quiet time outside Probiotics	Vitamin C & CoQ10 Chinese skullcap tincture Turmeric Anxiety & stress tincture, or Lion's Mane & Reishi	Restorative yoga Melatonin Avoid devices Sleep herbs, if needed
M O N D A Y	Collagen/bone broth Meditation & gratitude Lion's Mane, Reishi, Turkey Tail & Cordyceps mushroom tinctures Vit D w/ K2	NAC & antioxidants Fiber supplement B12, Folic acid & Omega-3 oils with lunch	30 min. walking or cardio (preferably outside) Probiotics	Vitamin C & CoQ10 Chinese skullcap tincture Turmeric Anxiety & stress tincture, or Lion's Mane & Reishi	Melatonin Avoid devices Sleep herbs, if needed
T U E S D A Y	Collagen/bone broth Meditation & gratitude Lion's Mane, Reishi, Turkey Tail & Cordyceps mushroom tinctures Vit D w/ K2	NAC & antioxidants Fiber supplement B12, Folic acid & Omega-3 oils with lunch	Acupuncture Probiotics	Vitamin C & CoQ10 Chinese skullcap tincture Turmeric Anxiety & stress tincture, or Lion's Mane & Reishi	Restorative yoga Melatonin Avoid devices Sleep herbs, if needed
W E D N E S D A Y	Collagen/bone broth Meditation & gratitude Lion's Mane, Reishi, Turkey Tail & Cordyceps mushroom tinctures Vit D w/ K2	NAC & antioxidants Fiber supplement B12, Folic acid & Omega-3 oils with lunch	30 min. walking or cardio (preferably outside) Probiotics	Vitamin C & CoQ10 Chinese skullcap tincture Turmeric Anxiety & stress tincture, or Lion's Mane & Reishi	Melatonin Avoid devices Sleep herbs, if needed
T H U R S D A Y	Collagen/bone broth Meditation & gratitude Lion's Mane, Reishi, Turkey Tail & Cordyceps mushroom tinctures Vit D w/ K2	NAC & antioxidants Fiber supplement B12, Folic acid & Omega-3 oils with lunch	Acupuncture Probiotics	Vitamin C & CoQ10 Chinese skullcap tincture Turmeric Anxiety & stress tincture, or Lion's Mane & Reishi	Restorative yoga Melatonin Avoid devices Sleep herbs, if needed
F R I D A Y	Collagen/bone broth Meditation & gratitude Lion's Mane, Reishi, Turkey Tail & Cordyceps mushroom tinctures Vit D w/ K2	NAC & antioxidants Fiber supplement B12, Folic acid & Omega-3 oils with lunch	30 min. walking or cardio (preferably outside) Probiotics	Vitamin C & CoQ10 Chinese skullcap tincture Turmeric Anxiety & stress tincture, or Lion's Mane & Reishi	Melatonin Avoid devices Sleep herbs, if needed
S A T U R D A Y	Collagen/bone broth Meditation & gratitude Lion's Mane, Reishi, Turkey Tail & Cordyceps mushroom tinctures Vit D w/ K2	NAC & antioxidants Fiber supplement B12, Folic acid & Omega-3 oils with lunch	Resistance training Probiotics	Vitamin C & CoQ10 Chinese skullcap tincture Turmeric Anxiety & stress tincture, or Lion's Mane & Reishi	Restorative yoga Melatonin Avoid devices Sleep herbs, if needed

Ch. 4: Psoriasis and Eczema Protocol

PSORIASIS is an autoimmune disease that affects the skin. It can cause red, scaly, itchy patches that mostly occur on the knees, elbows, trunk, and scalp. Psoriasis is more complicated than a simple skin rash as it is an autoimmune disorder; the body thinks its own tissues are a type of foreign invader that must be attacked by the immune system. In the case of psoriasis, the immune system attacks skin cells, which causes the flare-ups associated with psoriasis. This also means that unlike many other skin conditions, you can't just treat psoriasis with topical lotions or creams.

Symptoms: Symptoms of psoriasis include red patches of skin that are covered with thick, pink, red, and/or silvery scales on lighter skin, or brown or violet-grey scales on darker-pigmented skin. The patches may be dry and cracked and they often itch or burn. Common symptoms include soreness, thickened, pitted, or ridged nails, and swollen and stiff joints. In children, small scaling spots are commonly seen. Severity can vary. Some psoriasis patches are comprised of a few spots of dandruff-like scales and others are major eruptions that cover large areas of the body. Commonly affected areas include the face, palms, soles of the feet, elbows, knees, legs, and lower back.¹ Psoriasis often emerges in cycles; flaring up for a few weeks or months then subsiding or going into remission. This is typical of autoimmune issues as flare-ups can be triggered by diet, stress, and other factors that affect the immune system.

Diagnosis: When diagnosing psoriasis, a doctor will ask you questions about your health and take a look at your skin, nails, and scalp. They may take a small biopsy of the skin to examine under a microscope. This helps to determine what type of psoriasis you have, and rule out other disorders.

The most common type of **ECZEMA** is atopic dermatitis. This condition makes skin red and itchy. It is very common in children, but can occur at any age. Some types of eczema last a long time, referred to as chronic eczema. Other types flare up from time to time. Sometimes, eczema is accompanied by asthma or hay fever. It likely results from a disturbed skin barrier and aberrant autoimmune reactions.² Like psoriasis, there is no definitive cure for eczema. It is regarded as an overreaction by the immune system strongly driven by specific T-cell responses³ in response to a trigger, such as certain foods.

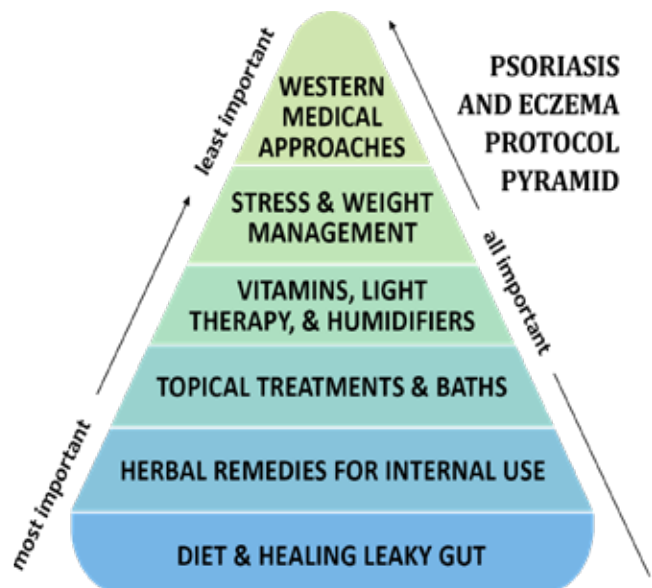
Symptoms: Common symptoms of eczema include dry, itchy skin that can itch severely at times (especially at night), red to brown-gray patches of skin on the hands, feet, ankles, wrists, neck, upper chest, eyelids, and inside the bends of the knees. In infants, it is commonly found on the face and scalp. Other symptoms include small, raised bumps that may leak fluid or crust over when scratched, thickened and cracked skin that may appear scaly, and raw or sensitive skin that is swollen from

scratching.⁴

Diagnosis: A doctor makes a diagnosis by examining the skin and reviewing medical history. Lab tests are not needed but a doctor may use patch testing or other tests to rule out other skin issues and to identify other conditions that may accompany eczema. Make sure to tell your doctor about any food allergies to help in your diagnosis.

While both psoriasis and eczema are thought to be types of autoimmune skin diseases, they are different from each other in the way they manifest. Psoriasis is characterized by thick, dry, scaly patches on the knees, elbows, trunk, and scalp, whereas eczema manifests with red, inflamed, itchy areas on the hands, feet, ankles, wrists, neck, upper chest, eyelids, and inside the bends of the knees. Both skin disorders can come and go and both seem to be worsened or triggered by factors such as diet. However, eczema seems to occur after contact with allergic substances and seems to accompany various allergies. Some common triggers for eczema include soaps, shampoos, dyes, fabrics, animals, and other irritants. Generally, topical treatments do a better job of clearing up eczema than psoriasis.

Since children have more sensitive skin than adults, they are at an increased risk for developing eczema. In some cases, there is overlap between the two and a person will show signs of having both. Stress and attacks on the immune system such as recurring infections, strep throat, or upper respiratory infections can trigger and worsen psoriasis. Psoriasis seems to be less common in people with darker skin pigmentation, though this may be partially due to lack of diagnosis as it presents differently on dark skin. On darker skin both psoriasis and eczema present as brown and as violet with grey scales (often leaving behind discolored skin that can be darker or lighter than usual skin tone) versus pink or red or even silver in tone on lighter skin.



While psoriasis and eczema have no known cures, emerging evidence suggests that changes in diet and lifestyle, as well as healing leaky gut, has an impact on

flares. For many, eating foods like gluten or dairy trigger flares. Upon eliminating these triggers some people are able to go long stretches in remission. In addition, there are ways you can flush toxins from your body, keep your immune system balanced, and treat itchy patches before they become a problem. This protocol aims to help arm you with evidence-based practices to keep you flare-free.

1. Diet and Healing Leaky Gut

It is well known that certain foods trigger inflammatory responses in the body. These foods put your body's immune system into overdrive, and the result is inflammation that can trigger flare-ups. While it is still not yet known exactly why certain foods cause this overreaction, it is hypothesized that they increase inflammation in fat tissues. This adipose (fat) tissue inflammation greatly increases the risk of autoimmune skin issues like psoriasis.

Foods high in saturated fat and certain other foods can make the liver work overtime, as the liver must produce chemicals to metabolize these foods; this leads to inflammation and autoimmune flare-ups. In this section, you will learn about foods to avoid, as well as foods to consume to keep inflammation down and avoid flares. Many foods also harm gut health and lead to or worsen existing leaky gut issues. Diet and healing leaky gut go hand-in-hand.

FOODS (AND BEVERAGES) TO AVOID:

Dairy: Dairy products are often high in saturated fat, which can lead to inflammation. However, a lot of people have sensitivities or allergic reactions to dairy for another reason: casein, a protein in dairy that many people have problems digesting. This can lead to chronic gastrointestinal irritation, inflammation, and leaky gut. A lot of people find relief from psoriasis and eczema simply by cutting dairy from their diet. Some people may not be full-blown lactose intolerant, so they think it is okay to consume dairy, but many people have dairy *sensitivities* (not the same as an intolerance, but can cause significant inflammation and health issues over time) that can trigger an autoimmune flare. Try cutting dairy from your diet or keeping a food journal to track dietary triggers.

Refined Carbohydrates: Foods containing refined carbohydrates are often highly processed. Examples include white breads, white rice, pastas, pastries, and some cereals. Not only are these foods full of sugar, they have also been stripped of fiber and whole grains. They are notorious for causing blood sugar spikes. One of the main reasons those with autoimmune diseases like psoriasis and eczema should avoid refined carbohydrates is their tendency to increase AGEs (advanced glycation end products), which are formed in your bloodstream when fat or protein combines with sugar. These harmful substances in your blood amplify the inflammatory response⁵.

Saturated and Trans Fats: Foods high in saturated and trans fats include cheese, fried foods, margarine, fast foods, red meats, and processed snacks. These increase

low-density lipoprotein (LDL) in the blood and contribute to excess body fat. The link between excess fat and worsening of psoriasis is strong. Note that unsaturated fats, which come from fish and plants, are good for you and help lessen inflammation⁶.

Sugar: Sugar is one of the most addictive substances on the planet. In order to process sugar, our body must produce insulin. However, when we eat too much sugar, our body is forced to store this for extra energy in the form of fat. In addition to triggering inflammation, sugar can also lead to increased inflammatory proteins in the body called cytokines. Artificial sweeteners are no healthier; they are linked to chronic inflammation, as well as cancer. Use sugar sparingly, and when you do, opt for sweet, ripe, whole fruit.

Gluten: Gluten is a protein found in wheat and some grains. Gluten is inflammatory for many people. Those with psoriasis tend to have higher rates of celiac disease, pointing to a strong connection between gluten intake and the worsening of other autoimmune conditions. Due to the rise in conditions like celiac disease, gluten-free substitutes are on the rise. However, keep an eye on the sugar and processed food intake because not every "gluten free" label is healthy.

Alcohol: Alcohol often contains sugar, which causes an inflammatory response in the body. It is a double offender because alcohol also leads to inflammation via the liver, which must work harder to metabolize alcohol. Another negative aspect of alcohol is its effect on gut health. It is notorious for killing off good bacteria while feeding the bad bacteria in the gut. This can lead to gut damage, leaky gut, and ultimately more inflammation.

ANTI-INFLAMMATORY DIET: For information on how to consume more foods that nourish the body while also reducing inflammation, please see the "anti-inflammatory diet" section and the Wahls Protocol diet sections in this book (see pages 297 and 310).

HEALING LEAKY GUT: In addition to diet, addressing a leaky gut is crucial for calming symptoms of both eczema and psoriasis. Leaky gut syndrome is associated with increased intestinal permeability. The waste and other materials that are supposed to go through our intestines find their way into our body due to leaky gut. Evidence points to a link between leaky gut and autoimmune diseases. This is no surprise since waste that was supposed to stay contained in the intestines gets into our bloodstream, triggering inflammation.

Leaky gut symptoms include: brain fog, seasonal allergies, hormone imbalance, skin problems, mood swings, digestive issues, chronic fatigue, and autoimmune disease(s).

There are strategies you can employ to help heal a leaky gut. Start with diet. Eat a whole-foods anti-inflammatory diet that includes fresh fruits and vegetables. See page 297 in this book for details. In addition to clean eating, try incorporating more fermented

foods and beverages into your diet to improve your gut health and nourish the beneficial bacteria in your microbiome. Examples of fermented foods include sauerkraut, kimchi, and kombucha. A daily **probiotic** is recommended. There are herbs you can take that help leaky gut syndrome, including:

Lion's Mane (*Hericium erinaceus*): This mushroom has powerful gastroprotective⁷ antioxidants to help cool the inflammatory response to relieve symptoms of leaky gut. It also helps (as does Reishi) autoimmune issues, as it's an immunomodulator. Suggested dose: 2 dropperfuls twice a day.

Reishi (*Ganoderma lucidum*): Reishi is known as the "mushroom of immortality." This adaptogen possesses strong anti-inflammatory properties that calm intestinal inflammation⁸ and help with leaky gut. The best way to use any medicinal mushroom is to make or purchase a dual extracted reishi tincture. Suggested dose: 2 dropperfuls twice a day.

Slippery Elm (*Ulmus rubra*): This mucilaginous inner bark helps form a protective layer in the digestive tract to coat and soothe inflamed tissues. This helps calm the gut and promote healing⁹. Suggested dose: Infuse two teaspoons of chopped slippery elm bark into a cup of hot water for fifteen minutes and drink this tea up to twice daily or use a tincture.

Marshmallow Root (*Althaea officinalis*): Another mucilaginous plant, this root soothes the digestive tract and helps leaky gut by creating a protective film to cover irritated areas¹⁰ and intestinal perforations. Make an infusion with marshmallow root to sip throughout the day by filling a quart jar with one to three ounces of marshmallow root (chopped). On the stove, heat a quart of water and just before it begins boiling, pour it carefully into the jar with the marshmallow root. Let this steep for up to six hours. Strain and drink up to three cups daily (spaced throughout the day). Refrigerate between uses for up to 3 days.

Plantain (*Plantago major*): This common yard plant soothes gut inflammation and irritated mucous membranes¹¹ to help relieve symptoms of leaky gut. Suggested dose: Infuse two teaspoons into a cup of hot water and drink this tea up to four times daily or use a tincture.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): This common polypore mushroom contains prebiotics that feed the good bacteria in your microbiome and also controls candida overgrowth. Turkey tail mushrooms are a powerful ally for leaky gut as well as for autoimmune issues in general as this mushroom helps modulate the immune system¹². Use a dual-extracted tincture. Suggested dose: 1-2 dropperfuls twice daily.

Berberine: Berberine is a compound found in several medicinal plants, including Oregon grape root and goldenseal. Berberines have been shown to kill *H. pylori* bacteria¹³, which are linked with leaky gut,

ulcers, and more. If your leaky gut is the result of *H. pylori*, you may find berberine useful. Suggested dose: Infuse one tsp of chopped Oregon Grape root into a cup of hot water for ten to fifteen minutes and drink one to three times daily. (Goldenseal is not recommended, as it may be too stimulating for those with an autoimmune condition).

Anti-Parasitic Herbs: Sometimes, leaky gut is caused by parasites. Strong anti-parasitic herbs include wormwood and black walnut hull. Be careful, as too much can cause gastrointestinal distress. Wormwood has a long history of use as an anti-parasitic, especially for livestock¹⁴. In moderation it is also safe for human consumption. Suggested dose: Infuse 1-2 tsp of wormwood leaves in a cup of hot water for ten minutes. Drink one to two cups daily. Black walnut hulls were historically used for the treatment of parasites. You can tincture the chopped green hulls. Suggested dose is one dropperful up to three times daily for two weeks. Start with one dropperful, working your way up to 3x per day.

2. Herbal Remedies for Internal Use Many topical conditions are outward manifestations of something going on inside the body. There are herbs that can help your body recover, reduce inflammation, and flush out toxins to help you heal and stay flare-free. Pair these with a healthy diet and other protocols in this chapter.

IMMUNOMODULATORY HERBS: Since an autoimmune disease occurs when the immune system attacks body tissues that it would normally leave alone, herbs with immunomodulatory properties are a wise addition. Below are some excellent immunomodulatory herbs.

Reishi (*Ganoderma lucidum*): Reishi has already been mentioned for its ability to help with leaky gut. It is also a powerful anti-inflammatory mushroom that helps modulate the immune system to minimize flares. Take reishi tincture in a double extracted form. Suggested dose: 2-3 dropperfuls twice daily.

Sarsaparilla (*Smilax glabra*): Sarsaparilla root has been shown to dramatically improve skin lesions for those with psoriasis¹⁵. It is hypothesized that a steroid in the root called sarsaponin binds to the endotoxins that cause lesions and helps remove lesions from the body. Suggested dose: 1-2 grams daily in tea, capsules, or tincture form (Take with food.).

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): This polypore is an antioxidant-rich powerhouse. In addition, it helps address inflammation that leads to autoimmune issues like psoriasis and eczema. This adaptogen also helps modulate the immune system to prevent flares. In addition to immune modulation¹⁶, compounds in this mushroom, as well as reishi and lion's mane, have been studied for their effects on various cancers. Take a double extracted tincture to access all of its medicinal compounds. Suggested dose: 1-2 dropperfuls twice daily.

Ashwagandha (*Withania somnifera*): Ashwagandha is an adaptogenic herb that helps lower inflammation, as well as help the body respond to stress. Since stress can up your risk of having a flare-up, it is especially important to manage it. In addition, ashwagandha is a great immunomodulator, as are many adaptogens. It helps restore energy, stamina, and even help lift mood¹⁷ (as can Reishi). Suggested dose: 1-2 dropperfuls twice daily.

HERBS FOR PURIFICATION AND CLEANSING: It can be beneficial to cleanse the lymphatic system and organs to help remove toxins and waste that may be leading to inflammation. Below are some excellent purifying herbs to help you flush toxins, lower inflammation, and support the immune system:

Dandelion Root (*Taraxacum officinale*): This common plant boasts a variety of uses, and some of the most potent medicinal properties are found in its root. Dandelion root is a known tonic, blood purifier, and diuretic. It promotes the flushing of waste by helping the body expel water and it helps improve liver function so the liver can do its job filtering toxins. The root also possesses anti-inflammatory¹⁸ properties. Suggested dose: Infuse 2 tsp. chopped roots into a cup of hot water for fifteen minutes and drink this tea up to twice daily. Drink more water when you consume any herb or product with diuretic properties to stay hydrated. You may also use a tincture.

Stinging Nettle (*Urtica dioica*): Stinging nettle helps detoxification, improves circulation, lowers blood pressure, is a diuretic, and significantly reduces inflammation. Stinging nettle can calm allergic reactions and the inflammation that often accompanies eczema and psoriasis¹⁹. It has a long history of traditional use in Germany to treat rheumatoid arthritis. It inhibits inflammation found in autoimmune and allergic reactions. Suggested dose: 2 dropperfuls of stinging nettle tincture once to twice daily.

Burdock Root (*Arctium lappa*): Burdock contains compounds that help flush toxins from the bloodstream. In addition, burdock root boasts anti-inflammatory and antioxidant properties that can greatly benefit those with autoimmune issues like psoriasis and eczema. Burdock root has also been studied for its circulation-enhancing effects that help boost skin health to improve conditions like eczema²⁰. Purchase or create a tincture with the chopped roots to preserve its antioxidants and flavonoids. Suggested dose: 1-2 dropperfuls up to two times daily.

Cleavers (*Galium aparine*): Cleavers (Bedstraw/ Galium) are a wonderful lymphatic cleanser and tonic. Sometimes the lymphatic system can become stagnant or clogged with waste. Cleavers help clear blockages and help the body remove waste more efficiently. The result of a better-functioning lymphatic system is a better functioning immune system²¹. Collect fresh cleavers from a clean area (they are very common throughout North America in the spring) and create a tincture or

add them to a smoothie. Suggested tincture dose: 2 dropperfuls up to three times daily for lymphatic support and purification for 2 weeks.

ANTI-INFLAMMATORY HERBS: Many of the herbs previously mentioned possess anti-inflammatory properties. However, the herbs in this section are known specifically for their powerful anti-inflammatory action. These are especially beneficial for those with autoimmune diseases since inflammation plays a key role in triggering flare-ups.

Turmeric/Curcumin: Turmeric possesses strong anti-inflammatory properties to reduce swelling and inflammation throughout the body²². The main medicinal compound in turmeric (curcumin), is best combined with black pepper for absorption. Tincture chopped turmeric roots with a tablespoon or two of black peppercorns. Suggested dose: 2-3 dropperfuls twice daily, or 500-2,000 mg daily as a supplement.

Ginger (*Zingiber officinale*): Ginger and turmeric have similar anti-inflammatory actions. Ginger is known for its ability to calm pain and swelling when applied topically or taken internally. You can add a few drops of ginger essential oil to a carrier oil and massage this into inflamed areas or tincture chopped ginger to take internally for its anti-inflammatory properties. Ginger is also helpful for gastrointestinal issues like an upset stomach or nausea. It has been studied for its anti-cancer effects in conjunction with its anti-inflammatory properties²³. Suggested dose: 1-2 dropperfuls of ginger tincture twice daily, take a supplement, or infuse a few teaspoons into a cup of hot water for tea up to three times daily.

3. Topical Treatments and Baths **TOPICAL HERBS:**

St. John's Wort (*Hypericum perforatum*): Many people know that St. John's wort is effective for mild to moderate depression and Seasonal Affective Disorder (SAD) when taken internally. It also possesses anti-inflammatory²⁴ and skin-soothing properties that may help those with psoriasis and eczema. You can make your own salve with the fresh flowers by harvesting them and letting them wilt overnight. The next day, fill a jar with the flowers and completely cover the plant material in a carrier oil like organic olive oil. Infuse for four to six weeks or sit the jar in a pan of hot water on the "warm" setting on your stove overnight if you are in a hurry. Strain this out through cheesecloth (make sure to squeeze the oil out of the plant material well) and use this infused oil to massage into affected areas. Follow this technique to make plant-infused oil/massage treatments for other external herbs mentioned in this protocol. St. John's wort also helps with nerve-related pain issues when applied topically.

Calendula (*Calendula officinalis*): Calendula flowers help improve blood flow to areas of inflammation and hydrate the skin. Calendula has been well studied for its skin-healing effects for dermatitis, and may lower your risk of infection²⁵. You can make a salve for eczema or psoriasis by following these steps:

Soothing Calendula and Chickweed Salve (Fast Method). Ingredients: ½ cup organic olive oil, 1/3 cup organic coconut oil, 3 Tbsp. dried calendula flowers, 1/2 Tbsp. dried chickweed flowers, 1 to 2 ounces (28g to 56g) beeswax. In a double boiler, melt the olive oil and coconut oil. Add the flowers and allow the mixture to steep on low for 2 to 3 hours. Strain out the flowers. Return the pan to heat and add the beeswax. Once the wax melts, pour into your containers (adjust the amount of beeswax to get the consistency you want). Allow to cool before use. Salves are portable and less messy than oils.

Chickweed (*Stellaria media*): This common yard “weed” helps redness, itching, irritation, and inflamed areas of skin - perfect for psoriasis and eczema. Studies back up its soothing and anti-inflammatory properties²⁶.

Lavender (*Lavandula angustifolia*): Lavender has been used for centuries to treat wounds, burns, rashes, itching, redness, and many types of skin problems. Not only is it soothing, it helps prevent infection due to its antimicrobial compounds. The calming smell helps relieve anxiety and stress. It has specifically been studied for its anti-psoriatic activity. Infuse into an organic carrier oil to make a massage oil, add a few drops of lavender essential oil to a carrier oil and massage into affected areas, or add into a salve.

Aloe and EO Topical Eczema Treatment: Aloe vera gel is a cooling and anti-inflammatory²⁷ powerhouse for skin affected by psoriasis or eczema. For eczema, it works especially well to calm redness, discoloration, itching, and inflammation. When you add tea tree and lavender essential oil to aloe vera gel, they combine to create a healing remedy to soothe itching, a common complaint for those with eczema and psoriasis. When people itch these areas of skin, they make the problem worse and may cause infections. Try blending two tablespoons of aloe vera gel with 15 drops each of lavender and tea tree essential oils in a bowl and whisk to mix. Pour into in a jar and shake well before massaging affected areas. Store in the refrigerator between uses and shake before each use.

BATHS: Some people find relief by soaking in a warm bath with a cup or two of oatmeal²⁸ and Epsom salts²⁹. You can also make herbal bath “tea” to infuse in the water as you soak. Effective herbs include yarrow, self-heal, lavender, and calendula. Simply fill a muslin cloth bag with a half cup of the chopped herb and let this sit in the bath water with you as you soak. Essential oils can be helpful too. They need to be added in a specific way to be effective and emulsify properly so they don’t just sit on top of the water: Try adding 10-20 drops of lavender, chamomile, rose, or geranium essential oil to a tablespoon of fragrance-free, organic, liquid castile soap and then add this to a cup of Epsom salt. Blend well and add to your bath.

For those with eczema, allergens in many products trigger major flare-ups. Sadly, there are a great number of products that contain synthetic fragrances,

dyes, parabens, sulfates, and more. These ingredients often cause skin reactions. When shopping for laundry soaps, shampoos, lotions, soaps, and anything else that will come into contact with your skin, opt for naturally-derived products that are dye and fragrance-free.

4. Vitamins, Light Therapy, and Humidifiers

VITAMINS:

In addition to the techniques

discussed in this protocol, adding certain vitamins to your daily regimen can help with flare-ups and nourishes the immune system to keep it functioning properly. Below are some potentially helpful vitamins for psoriasis and eczema (see the Autoimmune Protocol at page 7 for a list of additional vitamins that target autoimmune diseases):

Vitamin D: Vitamin D plays an important role in immune system function. It is difficult to get enough vitamin D if you do not spend much time outdoors or live in a temperate zone. Being in the sun is the optimal way to get this vitamin, but many with autoimmune issues are deficient. Vitamin D helps to regulate inflammatory markers and immune cells, so it is important to make sure you have sufficient levels. Vitamin D also plays a role in the proliferation and maturation of keratinocytes. Because of this, it is a great addition for psoriasis³⁰. Get a blood test to measure your levels before starting a daily routine. Suggested dose is 2,000-4000 IU daily, but reduce the amount you take in the summer months when/if you are spending more time outdoors.

B Vitamins: Studies indicate that B-12 deficiency is linked to psoriasis.³¹ Vitamin B-12 deficiency is also associated with chronic stomach inflammation and may be involved in pernicious anemia, which is an autoimmune malabsorption syndrome. Before you supplement with B-12, have your general practitioner measure your blood levels. Suggested dose: Once you have a baseline, begin with 2,000 mcg of oral vitamin B-12 daily, then decrease the dose to 1,000 mcg per day, followed by 1,000 mcg weekly, and finally 1,000 mcg monthly. Vitamin B-7 (biotin) may also be beneficial for those with psoriasis because it is positively associated with skin health. Try taking 30 micrograms daily.

Omega-3 Fatty Acids: Chronic inflammatory conditions like psoriasis often respond well to the supplementation of fish oils like omega-3 fatty acids. Research shows that these essential fatty acids help to decrease disease activity, reduce pain, and lower the use of anti-inflammatory drugs³². Suggested dose: 0.3-9.6 g/day Omega-3 PUFA; EPA: 0.01-4.1 g; DHA: 0.01-2.7 g for autoimmunity

Glucosamine and Chondroitin: Glucosamine and Chondroitin promote cartilage elasticity, help with repair of damaged cartilage, and keep existing cartilage from breaking down. Many with psoriatic arthritis use these to ease symptoms. Glucosamine and Chondroitin have also been studied for the treatment of pain and inflammation associated with osteoarthritis³³. Suggested dose: 1,200 mg daily.

Methylsulfonylmethane (MSM): This sulphur-

containing compound has been studied for its effects on inflammation and joint pain³⁴, which are commonly associated with psoriasis. Suggested dose: 2-4 grams daily.

LIGHT THERAPY: Phototherapy with a special light device is often used for psoriasis. It consists of exposing the skin to controlled amounts of ultraviolet (UV) radiation similar to that from the sun. It is often administered as a supplementary treatment for those with moderate to severe psoriasis that covers large areas of the body³⁵. There are now several different forms of light therapy available. While it is known that exposure to these kinds of light may increase the risk for skin cancers like melanoma, studies suggest the risk is low for those undergoing this therapy due to the duration of exposure.

Sunlight is another option if you wish to go the all-natural route. Spend twenty minutes in the full sun each day to help control flares and heal existing patches of dryness or irritation. Everything in moderation, so make sure to cover up after twenty minutes to avoid overexposure. If you plan on trying phototherapy or spending more time in the sun each day, make sure you are proactive in scheduling yearly exams to check for skin cancers, etc.

USING A HUMIDIFIER: Because psoriasis can result in skin dryness, running a cool mist humidifier in the room when you sleep helps to hydrate skin and prevent flare-ups. This is a good option for those with psoriasis who live in dry areas. Our skin is made up of around 30 percent water and water is essential to skin health.

5. Stress and Weight Management

Autoimmune diseases like psoriasis, as well as many other health conditions, are worsened by stress. **Stress** weakens the immune system. In fact, not only does stress worsen these conditions, it is being studied as a cause. In addition to avoiding food triggers, healing leaky gut, and using herbal remedies and supplements, do what you can to avoid stress in your life.

Start putting time aside to care for your mental, emotional, and spiritual health like you care for your physical health. For a list of helpful tips and strategies to tackle stress and anxiety, see the Anxiety and Stress Protocol in this book (see page 150). If **Sleep** is an issue, please visit our Sleep Protocol in this book (see page 286).

Exercise is not only a great way to reduce stress, it's a great way to live a healthier lifestyle and lose weight if needed. Try exercising outdoors as much as possible to get the stress-relieving benefits of being in nature. Yoga, walking, jogging, hiking, and swimming are all great ways to get the exercise and outdoor time your body needs.

There is an association between being obese and having psoriasis³⁶, so weight management is an important strategy to help you control your flare-ups. For more information on how to lose weight the healthy way, see the **Weight Loss Protocol** (see page 292).

6. Western Medical Approaches

According to the National Psoriasis Foundation³⁷, there are several conventional medical treatments available. These treatments include topical applications, phototherapy (mentioned above), and systemics like biologics and oral medications. They may also include an integrative medicine approach or complimentary therapies (strategies already discussed in this protocol). Below is more information about prescription topical applications and drugs used for psoriasis and eczema. Make sure to ask about side effects if you go this route.

Topical Applications: External corticosteroids control inflammatory responses to lower swelling, inflammation, and irritation for affected areas of skin. While some types of topical steroids are available over the counter, many require a prescription. Some corticosteroids contain a retinoid to help control inflammation and heal the skin.

Make sure to discuss pros and cons of using these topical treatments. Common side effects may include temporary pain and soreness, loss of color in the affected area, infection, and rosacea (if used for prolonged periods on the face).

Other topical treatments include non-steroidal treatments like anthralin, synthetic vitamin D3, and Vitamin A. These are often used in place of steroids to control psoriasis plaques for those who are sensitive to steroids.

Systemics: Systemics are prescription drugs that are sometimes used alongside topical treatments and sometimes used on their own for those who do not have success using topical treatments. There are two main types of systemics for psoriasis: biologics and oral treatments.

Biologics are protein-based drugs given by injection or intravenous infusion. Common biologics for psoriasis include Cimzia, Enbrel, Humira, and Remicade. These do not come without side effects, so ask your doctor about potential side effects. Biologics are also known to increase the risk of infection for some people. Pay attention to your body and signs of infection like fever, cough, or flu-like illness. See a doctor right away if you develop these.

Oral Treatments for psoriasis consist of small molecule medicines. These are used for people who did not have success with topical applications. Oral treatments are taken in pill or liquid form; some are given via injection.

Oral treatments for psoriasis have been around for a little over a decade. The newer treatments selectively target molecules inside immune cells to adjust the inflammatory process inside the cell.

When this happens, a correction in the overactive immune response causing psoriasis inflammation may occur. As a result, a person may notice improvements in the appearance of their skin like reduced redness and

scales. They may also experience less tenderness and swelling in the affected areas.

Methotrexate is a common oral treatment for psoriasis. It is a disease-modifying anti-rheumatic drug that may relieve symptoms and slow/stop joint and tissue damage for those with psoriatic arthritis.

Other oral drugs for psoriasis include Soriatane, Cyclosporine, NSAID's, Otezla, and Xeljanz. These drugs can have side-effects like tingling or numbness of the skin, headaches, joint pain, muscle twitching, and high blood pressure. Because some of these drugs suppress the immune system, they can make you more susceptible to some diseases and infections.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Vitamin D, B vitamins, and Omega-3 fatty acids with breakfast Ginger supplement Reishi, Lion's Mane & Turkey Tail dual-extracted mushroom tinctures	Weekly grocery shopping for psoriasis-friendly, whole foods Sunlight/light therapy (20 min.) Dandelion root/Burdock tea or tincture w/ lunch Stinging nettle tincture External St. John's wort/Calendula/Herbal salve as needed	Daily exercise (at least 20 min.) Leaky gut herbs Probiotic	Sarsaparilla tea or tincture with dinner Turmeric/black pepper supplement Reishi, Lion's Mane & Turkey Tail Dual-extracted mushroom tinctures	Herbal/oatmeal/epsom salt bath soak Run humidifier St. John's wort/Calendula/Herbal salve as needed Stress and/or sleep herbs, if needed
MONDAY	Vitamin D, B vitamins, and Omega-3 fatty acids with breakfast Ginger supplement Reishi, Lion's Mane & Turkey Tail tinctures	Sunlight (20 min.) Dandelion root/Burdock tea or tincture w/ lunch Stinging nettle tincture Herbal salve as needed	Daily exercise (at least 20 min.) Leaky gut herbs Probiotic	Sarsaparilla tea or tincture with dinner Turmeric/black pepper supplement Reishi, Lion's Mane & Turkey tail tinctures	Run humidifier Herbal salve as needed Stress and/or sleep herbs, if needed
TUESDAY	Vitamin D, B vitamins, and Omega-3 fatty acids with breakfast Ginger supplement Reishi, Lion's Mane & Turkey Tail tinctures	Sunlight (20 min.) Dandelion root/Burdock tea or tincture w/ lunch Stinging nettle tincture Herbal salve as needed	Daily exercise (at least 20 min.) Leaky gut herbs Probiotic	Sarsaparilla tea or tincture with dinner Turmeric/black pepper supplement Reishi, Lion's Mane & Turkey tail tinctures	Herbal/oatmeal/epsom salt bath soak Run humidifier Herbal salve as needed Stress and/or sleep herbs, if needed
WEDNESDAY	Vitamin D, B vitamins, and Omega-3 fatty acids with breakfast Ginger supplement Reishi, Lion's Mane & Turkey Tail tinctures	Sunlight / light therapy (20 min.) Dandelion root/Burdock tea or tincture w/ lunch Stinging nettle tincture Herbal salve as needed	Daily exercise (at least 20 min.) Leaky gut herbs Probiotic	Sarsaparilla tea or tincture with dinner Turmeric/black pepper supplement Reishi, Lion's Mane & Turkey tail tinctures	Run humidifier Herbal salve as needed Stress and/or sleep herbs, if needed
THURSDAY	Vitamin D, B vitamins, and Omega-3 fatty acids with breakfast Ginger supplement Reishi, Lion's Mane & Turkey Tail tinctures	Sunlight (20 min.) Dandelion root/Burdock tea or tincture w/ lunch Stinging nettle tincture Herbal salve as needed	Daily exercise (at least 20 min.) Leaky gut herbs Probiotic	Sarsaparilla tea or tincture with dinner Turmeric/black pepper supplement Reishi, Lion's Mane & Turkey tail tinctures	Herbal/oatmeal/epsom salt bath soak Run humidifier Herbal salve as needed Stress and/or sleep herbs, if needed
FRIDAY	Vitamin D, B vitamins, and Omega-3 fatty acids with breakfast Ginger supplement Reishi, Lion's Mane & Turkey Tail tinctures	Sunlight / light therapy (20 min.) Dandelion root/Burdock tea or tincture w/ lunch Stinging nettle tincture Herbal salve as needed	Daily exercise (at least 20 min.) Leaky gut herbs Probiotic	Sarsaparilla tea or tincture with dinner Turmeric/black pepper supplement Reishi, Lion's Mane & Turkey tail tinctures	Run humidifier Herbal salve as needed Stress and/or sleep herbs, if needed
SATURDAY	Vitamin D, B vitamins, and Omega-3 fatty acids with breakfast Ginger supplement Reishi, Lion's Mane & Turkey Tail tinctures	Sunlight (20 min.) Dandelion root/Burdock tea or tincture w/ lunch Stinging nettle tincture Herbal salve as needed	Daily exercise (at least 20 min.) Leaky gut herbs Probiotic	Sarsaparilla tea or tincture with dinner Turmeric/black pepper supplement Reishi, Lion's Mane & Turkey tail tinctures	Herbal/oatmeal/epsom salt bath soak Run humidifier Herbal salve as needed Stress and/or sleep herbs, if needed

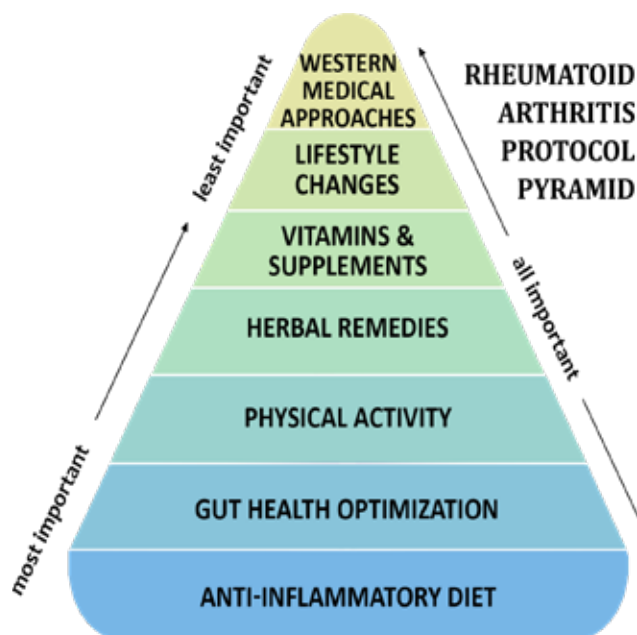
Ch. 5: Rheumatoid Arthritis Protocol

Rheumatoid arthritis (RA) is an autoimmune disease. It occurs when the immune system attacks the joints, leading to chronic inflammation, pain, and joint stiffness. RA can also cause fatigue, reduced cognitive function, secondary Sjogren's syndrome, lung inflammation and scarring (leading to interstitial lung disease/pulmonary fibrosis), skeletal muscle loss and strength (sarcopenia¹), and osteoporosis. RA patients may be at greater risk of cancer, especially blood (hematologic) and kidney cancers².

If not addressed RA can seriously damage joint structures, cartilage, and bones. This damage typically worsens over time and can be irreversible. Cartilage degradation can be seen as joint-space narrowing on radiography³.

Fortunately, there are multiple interventions that can be utilized to construct a comprehensive plan for addressing rheumatoid arthritis. These typically include addressing three important pillars of rheumatoid arthritis-modulating the immune system and curbing inflammation, optimizing gut health, and managing stress.

Joint destruction in rheumatoid arthritis is fueled by a host of pro-inflammatory chemical messengers. The most important of these are tumor necrosis factor alpha (TNF α), interleukin-6 (IL-6), and interferon gamma (IFN- γ). Many conventional treatment options for rheumatoid arthritis target these messengers. Many of the herbal remedies that are beneficial for rheumatoid arthritis target these pro-inflammatory chemicals as well.



The health of your gut plays an important role in the development of autoimmunity. The cells lining your gut are tightly connected and are covered by a protective barrier known as the mucus layer. This layer and the tight connections between cells prevent resident gut bacteria

and microbes from contacting the blood vessels and tissues adjacent to the gastrointestinal tract. In response to various dietary and environmental factors, the mucus layer may be degraded and the tightly arranged gut cells may begin to loosen. When this happens, microbes, microbial products, and inflammatory molecules can pass from the gut into the blood stream. This is referred to as "leaky gut." It leads to systemic inflammation and can lead to the development of autoimmunity.

The development of rheumatoid arthritis has been linked to high levels of stress in early life. The hypothalamic-pituitary-adrenal (HPA) axis controls the chronic stress response within the body. When we experience highly stressful events during childhood, it leads to changes in the way our body handles stress. Because the stress response and the immune response work in conjunction with each other, these changes to the stress response affect the immune system. Research demonstrates that, on average, people who've experienced early stress are twice as likely to be diagnosed with rheumatoid arthritis than people who did not experience significant stressors in early life.⁴ Because of this stress-immune connection, it is important to manage current stress and to mentally process past traumatic events when addressing RA.

In this chapter, we'll discuss several interventions that can be combined to create a comprehensive plan for addressing rheumatoid arthritis.

1. Anti-Inflammatory Diet

In order to effectively bring rheumatoid arthritis under control, you need to significantly reduce inflammation and restore balance to the immune system. An excellent way to do this is by adopting a gluten-free anti-inflammatory diet, specifically one that includes lots of vegetables, fruits, high fiber, bone broth, a moderate amount of protein, plenty of unsaturated fats and omega-3 fatty acids, and minimal saturated and trans fats⁵. This type of diet is especially beneficial in reducing rheumatoid arthritis symptoms when it is combined with fasting (see our Intermittent Fasting Section at page 303). It is similar to the Mediterranean diet with the addition of probiotics. Please visit our section on the Anti-Inflammatory Diet for full details and dietary suggestions (see page 297).

A systematic review and meta-analysis demonstrated that a brief period of fasting followed by a vegetarian or mostly plant-based diet was useful in the treatment of rheumatoid arthritis.⁶ In another study, people with rheumatoid arthritis fasted for 7-10 days and then transitioned to an individually adjusted, gluten-free, vegan diet for 3.5 months. After four months, they were gradually transitioned to a lactovegetarian diet, which they remained on for a year. Researchers compared this group to a group of participants who continued eating their regular omnivorous diets. The group that fasted and consumed the anti-inflammatory diet showed significant improvements in their number of tender or swollen joints, the amount of pain and stiffness they experienced, their grip strength, and blood

work values like erythrocyte sedimentation rate and c-reactive protein (both of which measure inflammation) as well as their white blood cell counts.⁷ The benefits that the fasting and diet group experienced were still present after one year on the diet.

THE MOST EFFICIENT WAY TO TRANSITION TO AN ANTI-INFLAMMATORY DIET: In this section, we'll discuss the most efficient way to transition from a typical Western diet to an anti-inflammatory diet that is individualized to your unique needs. The most efficient way to transition is to start with an elimination-and-challenge protocol. Please read our complete Elimination Diet Section (see page 300); a brief version is below.

Start with a 28-day period and eliminate all processed foods, dairy, eggs, meat, sugar, food coloring, chocolate, caffeine, and alcohol (unless eliminating alcohol poses a withdrawal issue). Only eat whole foods, including fruits and vegetables. You may also eat gluten-free grains and pseudograins, like quinoa, amaranth, and millet; beans and lentils; nuts and seeds; and root crops.

After the 28-day elimination phase, reintroduce the elimination foods one by one (one new food every 2 days for RA) to see how your body responds. Most people experience symptoms within 10 minutes to 12 hours after reintroducing problem foods. The exception is arthritic joint pain, which can take up to 48 hours for symptoms to show. Document any symptoms you experience as you reintroduce these foods.

Eat a large amount of the food on the day you're reintroducing it to determine whether or not the food triggers or worsens your symptoms. Have the food item (e.g., glass of milk) at breakfast. If you experience symptoms after consuming that glass of milk, document them and add milk to the list of trigger foods. If you do not experience symptoms, have milk again at lunch and at dinner. If no symptoms arise, add milk to your list of foods that do not provoke a reaction.

During the testing phase, continue eating the basic anti-inflammatory diet plus the one food that you are testing. Once you've gathered the needed information from your daily test, don't eat that food again until you've finished testing all of the foods on your list. Do not fully reincorporate foods that do not provoke a reaction back into your diet until you have tested all of the foods you plan to test.

Test sugar, food coloring, chocolate, caffeine, and alcohol toward the end of your reintroduction phase, if at all. These foods should only be tested if you plan to continue consuming them; they have a higher likelihood of triggering symptoms. For complete testing details and tips, please see our Elimination Diet Section at page 300.

After completion of your elimination protocol, you'll have valuable information about foods that do and don't provoke a reaction and trigger RA symptoms. Use this information for your individually adjusted anti-inflammatory diet.

2. Gut Health Optimization

Gut health plays a critical role in the development and progression of rheumatoid arthritis. People with RA have different bacteria in their gut than healthy people⁸ and the presence of certain gut bacteria is associated with rheumatoid arthritis. Some research suggests that specific bacterial strains (e.g., *Prevotella copri*) may induce or cause rheumatoid arthritis.⁹

The types of bacteria that are present within the gut can affect the strength, integrity, and function of the intestinal barrier, and if the intestinal barrier is compromised, it can lead to more severe rheumatoid arthritis symptoms. In this section, we'll discuss why to optimize gut health when addressing RA, and ways to do so. For more in-depth instructions on healing leaky gut, see our Leaky Gut Protocol at page 70.

Fertile Field Theory: With advances in technology, our understanding of the relationship between the gut microbiome and the development of rheumatoid arthritis has grown. One hypothesis is the "fertile field theory". This states that viruses, bacteria, and other pathogens normally interact with the human body, but under some circumstances, may colonize or infect the body and interact with the immune system in a way that creates a "fertile field."¹⁰ In other words, the pathogens prime the immune system and set the stage for the development of autoimmunity.

In rheumatoid arthritis, evidence suggests that the gut microbiome changes in response to dietary or environmental factors, and this change leads to the production of pro-inflammatory compounds. These compounds damage the intestinal barrier, causing it to become easier for the bacteria and their products to leave the gut and enter the blood where they travel to joints and cause inflammation.¹¹ People with rheumatoid arthritis have different microbiome compositions, and bacterial products have been found in the joints of people diagnosed with RA.^{12 13} In order to address this, you need to decrease systemic inflammation and restore the integrity of the gut barrier.

RESTORING THE INTEGRITY OF THE GUT BARRIER: Probiotics:

Try to integrate at least one type of the following probiotic-rich foods into your daily diet: yogurt, kefir, sauerkraut, kimchi, tempeh, miso, kombucha, apple cider vinegar, and pickles. Prebiotic foods include apples, cocoa, bananas, garlic, barley, berries, chicory, dandelion greens, flax seed, green vegetables, leeks, legumes (peas and beans), oats, onions, tomatoes, Jerusalem artichoke, and burdock root. These feed the good bacteria in your gut and to help restore balance to the microbiome. Suggested dose, if taking probiotics as a supplement is 50-100 CFU of probiotics per day with meals, in divided doses.

N-Acetyl Glucosamine: N-acetyl glucosamine may decrease symptoms of rheumatoid arthritis by increasing the production of anti-inflammatory chemicals like interleukin-4 and decreasing the production of pro-inflammatory chemicals like TNF- α .^{14 15} Suggested dose:

1 gram of N-acetyl glucosamine per day.

Chondroitin Sulfate: Chondroitin sulfate is marketed as a joint-protective supplement; however, it is not well-absorbed into the blood so very little of it reaches the joints. Research suggests that most of its effects are actually on the gut microbiome.¹⁶ Chondroitin sulfate is used by the gut microbiota leading to positive changes in the gut, which has a positive impact on joint health in both osteoarthritis and rheumatoid arthritis.¹⁷ Suggested dose: 300 mg chondroitin sulfate per day.

Lion's Mane Mushroom (*Hericium erinaceus*): A known antioxidant and gastroprotective, lion's mane helps tame the inflammatory response and protect the lining of the stomach. It also protects the gut from *H. pylori*, and the polysaccharides in lion's mane protect against gastric ulcers.¹⁸ Suggested dose: 2 dropperfuls of a dual-extracted lion's mane tincture, twice daily.

Marshmallow Root (*Althaea officinalis*) and Slippery Elm (*Ulmus rubra*): Marshmallow and slippery elm are demulcents that create a protective layer over the gut, preventing bacterial products from leaving the gut through the damaged barrier and entering circulation. Suggested doses: Take marshmallow as a cold infusion, 1-2 tablespoons of chopped root per cup of water, three times per day. Take 800-1000 mg of slippery elm, three to four times per day as a supplement or tea to start.

Reishi Mushroom (*Ganoderma lucidum*): Reishi's anti-inflammatory action is helpful for calming intestinal inflammation. It also modulates the immune system, inhibits *Candida* yeast, and helps reduce pain in rheumatoid arthritis. Suggested dose: 2 dropperfuls of a dual-extracted reishi tincture, twice daily.

Turkey Tail Mushroom (*Trametes versicolor* or *Coriolus versicolor*): Turkey tail contains prebiotics that balance the microbiome, encourage the growth of "good" gut bacteria, and control *Candida* and small intestinal bacterial overgrowth (SIBO). Suggested dose: 2 dropperfuls of a dual-extracted turkey tail tincture, twice daily.

3. Physical Activity

Physical activity plays an important role in reducing symptoms of rheumatoid arthritis. A study involving women with RA found that those who did water aerobics three times per week over a 16-week span experienced significantly less pain and fewer symptoms; they also had greater functional capacity.¹⁹ Walking is another safe and effective intervention for people with rheumatoid arthritis; people who walked 3-4 times per week over a six-week period showed greater self-efficacy and an improved sense of wellbeing.²⁰ Mobility exercises help increase range of motion in people with RA.²¹ In addition to strengthening muscle, exercise helps maintain range of motion and flexibility.²² Tai Chi is a relaxing and effective remedy for arthritis.²³

4. Herbal Remedies

Herbal medicines provide much needed support to people with rheumatoid arthritis. There are different herbal approaches one can take for RA, and often, a combination is best. There are herbs that act as

anti-inflammatories and immunomodulators to help normalize immune system functions, herbs that act as protective and regenerative agents in the joints, and herbs with anti-microbial and demulcent properties to address possible underlying causes.

MEDICINAL MUSHROOMS: *Adaptogenic, anti-inflammatory, and immunomodulatory herbs are especially helpful for autoimmune issues.* Many medicinal mushrooms have immunomodulatory and anti-inflammatory properties. I take 2 dropperfuls each of my dual-extracted Lion's Mane, Reishi, Turkey Tail, and Cordyceps Mushroom tinctures daily or twice daily if feeling "off". These mushrooms have the added benefit of being antiviral and helping with leaky gut. For their immune-modulating properties, they are best if used as a dual-extracted tincture to access all of the medicinal compounds (not just a tea).

Lion's Mane (*Hericium erinaceus*) is a potent immunomodulator, decreases gut inflammation, and lessens stress, anxiety, depression, & insomnia. Lion's mane helps modulate the immune system by suppressing TNF- α and reducing interleukin-6 production.²⁴ Suggested dose: 2 dropperfuls of a dual-extracted tincture, once to twice daily.

Reishi (*Ganoderma lucidum*) is a formidable anti-inflammatory that helps modulate the immune system.²⁵ Since autoimmune disorders are inflammatory in nature, reishi is considered an excellent choice for slowing or reversing disease progression. In a clinical trial involving people with rheumatoid arthritis, reishi helped reduce pain in people with active RA.²⁶ Reishi suppresses certain inflammatory compounds²⁷ and is effective in calming the inflammation and immune response involved with leaky gut and intestinal dysbiosis. In addition, reishi mushroom is known for its adaptogenic qualities - helping to strengthen the body against the negative effects of stress, like inflammation, hormonal imbalances, high cortisol, fatigue, depleted energy levels, brain fog, and low energy. Suggested dose: 2 dropperfuls of a dual-extracted tincture, once to twice daily.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*) is high in anti-inflammatory compounds and assists in reducing internal inflammation.²⁸ Turkey tail is an immune modulator, making it an effective medicinal herb for autoimmune conditions. It also contains prebiotic compounds that help stimulate the growth and population of probiotic bacteria in the intestines.²⁹ Suggested dose: 2 dropperfuls of a dual-extracted tincture, once to twice daily.

Cordyceps (*Cordyceps* spp.) is immunomodulatory and anti-inflammatory, helps combat fatigue, and has anti-aging and antioxidant properties. Cordyceps is also anti-bacterial, anti-viral, anti-fungal, and antiprotazoal making it an effective agent against infection. One of its active constituents, cordycepin, has been found to modulate inflammation within the joints in a manner that may help reduce connective tissue damage in individuals with rheumatoid arthritis.³⁰ Cordyceps also improves insulin sensitivity while lowering bad

LDL cholesterol, which helps to prevent plaque build-up in the arteries, reducing the risk of heart attack and stroke.³¹ Suggested dose: 2 dropperfuls of a dual-extracted tincture, once to twice daily.

White Button Mushrooms: In a study involving white button mushrooms and shiitake (*Lentinula edodes*) mushrooms, while both types of mushrooms led to a decrease in the pro-inflammatory chemical TNF- α , shiitake led to an increase in the pro-inflammatory chemical interleukin-6.³² This suggests that the effects of shiitake may be mixed but the effects of white button mushrooms on rheumatoid arthritis are more consistent. More research is needed. Cook with white button mushrooms or purchase the powder and take as a supplement.

Black Hoof Mushroom (*Phellinus linteus*): This mushroom reduces rheumatoid arthritis-related joint inflammation in experimental models, specifically by modulating the immune system and reducing pro-inflammatory chemicals like interleukin-12, TNF- α , and interferon gamma.³³ Suggested dose: 500 mg of black hoof mushroom powder, once to twice daily.

OTHER HERBS: Stinging Nettle (*Urtica dioica*): Commonly used in Germany as treatment for RA, stinging nettle helps to reduce pain, halt the cascade of inflammation, and flush toxins from the system³⁴. A 2022 study showed that stinging nettle decreases inflammatory factors, and can improve the symptoms of RA³⁵. An animal study involving stinging nettle extract demonstrated that it was just as effective as celastrol from thunder god vine in reducing inflammation through powerful antioxidant action³⁶. It also suppresses “human leukocyte elastase”, a compound that encourages inflammation³⁷. Another study found that participants who took a stinging nettle extract experienced significantly reduced arthritic pain and inflammation³⁸. It can be used internally and externally. Using the herb as an internal tincture is easiest. It can be applied externally by soaking a cotton pad in nettle tincture, then placing it over the painful joint to bring relief, help restore and repair joint injuries, and reestablish nerve communication. Suggested dose: 2 dropperfuls stinging nettle tincture twice daily.

Thunder God Vine (*Tripterygium wilfordii*): In one study, people with rheumatoid arthritis who were stable on non-steroidal anti-inflammatory drugs and/or corticosteroids were given either sulfasalazine or Thunder god vine extract. After 24 weeks, participants who were given Thunder god vine were significantly more likely to have at least a 20% improvement in RA symptoms (e.g., the number of tender and swollen joints) and significantly lower levels of the pro-inflammatory chemical interleukin-6 (those given sulfasalazine did not have this decrease).³⁹ Suggested dose: 60 mg three times daily.

Licorice Root (*Glycyrrhiza glabra*): Glycyrrhizin and glycyrrhetic acid, active constituents in licorice, help non-steroidal anti-inflammatory drugs and disease-modifying anti-rheumatic drugs work

more effectively and make it less likely that a person taking these drugs will experience unwanted side effects from the drugs.⁴⁰ Furthermore, 18 β -Glycyrrhetic acid reduces inflammation in joints and protect against the side effects of the RA drug methotrexate.⁴¹ Licorice also has demulcent properties, helping protect and heal the gut barrier. Some of the active components of licorice can cause an increase in blood pressure, so people with hypertension should use it cautiously, and ideally under the care of their physician. This blood pressure increase is usually transient; it typically resolves when you stop taking the licorice. Suggested dose is 100 grams of licorice root per day; 1 dropperful of tincture, three times daily (note: not DGL or deglycyrrhized licorice). Talk to your doctor about long-term use; it is intended for shorter-term use.

Frankincense (*Boswellia serrata*): The boswellic acids found in Frankincense exhibit anti-arthritic activity.⁴² They reduce inflammation by inhibiting the activity of white blood cells in arthritic joints. Frankincense also reduces pain and considerably improves joint function.⁴³ Suggested dose is 450-750 mg per day in divided doses or 3-4 dropperfuls tincture, 2 times daily.

Turmeric/Curcumin: The active ingredient in turmeric, curcumin, is well known for decreasing inflammation and curbing joint pain. Scientific evidence supports its efficacy for arthritis. Curcumin effectively suppresses inflammation through the inhibition of pro-inflammatory cytokine production and modulates B and T cell differentiation and function⁴⁵. Suggested dose is 1000 mg/day of curcumin with piperine (black pepper) for maximum absorption⁴⁶.

Ginger (*Zingiber officinale*): Ginger has potent anti-inflammatory properties.⁴⁷ Its active constituents inhibit joint swelling and prevent joint inflammation and joint destruction in experimental animal models of RA.⁴⁸ Clinical research shows ginger reduces pain in people with rheumatoid arthritis by blocking the synthesis of inflammatory prostaglandins and leukotrienes.⁴⁹ Suggested dose: 1-2 teaspoons of chopped ginger per cup of water, three times daily or 1 dropperful of tincture, 2-3 times daily.

Cat's Claw (*Uncaria tomentosa*): In a 52-week long clinical trial, people with rheumatoid arthritis who were taking sulfasalazine or hydroxychloroquine were given cat's claw or placebo. Those who were given cat's claw reported a greater reduction in swollen and painful joints.⁵¹ Because cat's claw can have mixed effects on the immune system, some recommend that it be avoided in people with autoimmune disorders like RA. However, cat's claw has been found to be beneficial in reducing RA symptoms. People taking immunosuppressive drugs such as corticosteroids should discuss with their doctor before taking cat's claw. Please note that the herb *Acacia greggii* is also known as cat's claw; make sure you are using *Uncaria tomentosa*. Suggested dose: 1000 mg per day in divided doses or 2 dropperfuls of tincture, twice daily.

Chinese Skullcap (*Scutellaria baicalensis*): Chinese skullcap has potent anti-inflammatory properties. Chinese skullcap inhibits many pro-inflammatory chemicals, including those that play important roles in rheumatoid arthritis such as prostaglandins, interleukin-6, and TNF- α .⁵² Baicalin, one of its main active constituents, inhibits interleukin-17-mediated joint inflammation in experimental models of arthritis.⁵³ Suggested dose: 300-900mg of a supplement, or 2 dropperfuls of tincture, twice daily.

EXTERNAL HERBS AND BALMS FOR ARTHRITIS: There are a number of plants that are useful for arthritis. I make a Joint and Movement salve that curbs pain and inflammation. It contains the following ingredients: Organic Olive Oil Infused with the following organic herbs: Calendula, Arnica, St. John's Wort, Cottonwood Buds (Balm of Gilead), and Cayenne, plus Organic Beeswax, Organic Helichrysum Essential Oil, Organic Frankincense Essential Oil, and Vitamin E. It is an outstanding remedy for arthritis, muscle and joint pain, carpal tunnel, general aches and pain, nerve pain, and tennis elbow. It is also helpful for diabetic neuropathy pain. It's the best mixture I've found. Apply 3x/day.

Arnica: Anti-inflammatory, stimulates blood flow, and analgesic. Excellent for pain, strains, sprains, bruising, and swelling.

Calendula: Reduces pain by soothing inflammation, encourages collagen production, and curbs collagen degradation. Additionally, calendula softens skin, supports cell rejuvenation, and reduces the appearance of scars. It prevents and heals dry skin, wrinkles, and photodamage. Soothes rashes and bites, and is antifungal and antimicrobial.

Cayenne: Rich in capsaicin to help relieve arthritis, joint, muscle, tendon, and diabetic neuropathy pain. Works best with regular use.

Cottonwood Buds (*Populus trichocarpa* and *P. deltoides*): Anti-inflammatory, antibacterial, antifungal, and analgesic. Rich in salicylic acids for pain relief and to reduce swelling.

Helichrysum and Frankincense (*Boswellia*): Provides effective pain relief through strong anti-inflammatory action and may fight cartilage loss.

St. John's Wort (*Hypericum perforatum*): Consistent, daily use reduces inflammation and pain, including nerve pain.

There are several vitamins and supplements that can be helpful for rheumatoid arthritis. These include vitamin D, quercetin, omega-3 fatty acids, gamma-linolenic acid, avocado-soybean unsaponifiables, and methyl sulfonylmethane.

5. Vitamins and Supplements

Vitamin D: People with low

vitamin D levels tend to have more severe symptoms when diagnosed with rheumatoid arthritis and low vitamin D levels have been linked to musculoskeletal

pain.⁵⁴ Maintaining optimal vitamin D levels may help reduce pain and decrease the severity of disease in people with rheumatoid arthritis. Suggested dose: 5,000-10,000 IU per day until 25-hydroxyvitamin D levels are around 50 ng/dL, then cut back to a maintenance dose of 4,000 IU. Get your blood tested.

Quercetin: Quercetin has potent immune-modulating effects and is beneficial in addressing both acute and chronic immune system dysfunction. In a study conducted on people with rheumatoid arthritis, quercetin supplementation significantly reduced early morning stiffness, morning pain, and pain after engaging in activity.⁵⁵ The number of people with RA classified as having active disease significantly decreased in the group treated with quercetin. Suggested dose: 500 mg quercetin per day.

Omega-3 Fatty Acids: Omega-3 fatty acids like eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) modulate inflammation and the immune system. In clinical trials, omega-3 fatty acids modulated disease activity in rheumatoid arthritis and decreased the number of swollen and tender joints.⁵⁶ People who took omega-3 fatty acids were more likely to be able to decrease the amount of pain medication they were taking.⁵⁷ Suggested dose: 3-4 grams of omega-3 fatty acids (DHA and EPA) per day.

Gamma-Linolenic Acid (GLA): Gamma-linolenic acid is a potent anti-inflammatory. GLA is found in evening primrose oil and borage seed oil. In a study involving people with rheumatoid arthritis and active joint inflammation, gamma-linolenic acid significantly reduced the number of tender joints and the number of swollen joints, with no improvement noted in the placebo group.⁵⁸ Treatment with GLA was effective and well tolerated. Suggested dose: 1.4 grams of gamma-linolenic acid per day.

Avocado-Aoybean Unsaponifiables: Avocado-soybean unsaponifiables (ASUs) are vegetable extracts made from fruits, seeds of avocado, and soybean oil. They have potent anti-inflammatory properties, protect the joints, and have been used effectively in the treatment of various autoimmune disorders.⁵⁹ Research demonstrates that ASUs modulate the immune system and limit the spread of inflammation throughout the joints in rheumatoid arthritis.⁶⁰ Suggested dose: 300 mg ASU per day.

Methyl Sulfonylmethane: Methyl sulfonylmethane (MSM) is a potent anti-inflammatory and antioxidant that is beneficial for RA. It improves range of motion, physical function, and inflammation levels, and protects joints from destruction in rheumatoid arthritis. It is generally recognized as safe and is well tolerated at doses of up to 4 grams daily.⁶¹ Suggested dose: 2-4 grams of MSM daily, in divided doses.

6. Lifestyle Changes

Lifestyle factors like stress levels and sleep quality play important roles in the development and progression of rheumatoid arthritis.

Stress: The connection between stress and

rheumatoid arthritis goes back to childhood. One study found that individuals who experienced two or more types of early life stress (e.g., physical, sexual, or verbal abuse; incarceration of a parent; mental illness or substance abuse in the home; parental divorce or separation; domestic violence; physical or emotional neglect) were twice as likely to be diagnosed with rheumatic conditions as people who did not experience any of these stressors in early life.⁶² Adult stress is also a factor in the progression of rheumatoid arthritis.

Many people with rheumatoid arthritis report that a period of intense stress preceded the onset of their symptoms and others report that higher stress levels often lead to more intense symptoms.⁶³ In order to effectively address symptoms of rheumatoid arthritis, it is important to get stress under control.

This could involve working with a therapist to process past trauma, daily meditation and gratitude, practicing tai chi, and making lifestyle changes and taking herbs to decrease daily stress. See our Anxiety and Stress Protocol for more (see page 105).

Sleep: Sleep plays a critical role in the progression of rheumatoid arthritis. People with rheumatoid arthritis tend to have non-optimal sleep, defined as nightly sleep that does not last for 7-8 hours.⁶⁴ Non-optimal sleep is associated with higher pain scores; people with optimal sleep experience lower pain levels. See our Sleep Protocol for more (see page 286).

7. Western Medical Approaches There are several types of conventional drugs that are used to address rheumatoid arthritis. These include non-steroidal anti-inflammatory drugs, corticosteroids, conventional disease-modifying anti-rheumatic drugs, biologics, and targeted synthetic disease-modifying anti-rheumatic drugs.

Non-Steroidal Anti-Inflammatory Drugs (NSAIDs): NSAIDs like ibuprofen and naproxen are commonly prescribed to treat RA because they help reduce

inflammation and pain. Potential side effects include kidney damage and peptic ulcers, and they may also increase your risk for heart attack, stroke, or high blood pressure.

Corticosteroids: Corticosteroids like prednisone have an immunosuppressive effect; they decrease inflammation and pain quickly and effectively by decreasing the activity of the immune system.

Potential side effects include acne, vision changes, mood changes, hyperglycemia, fluid retention, headache, sleep disturbances, fluid retention, swelling, thinning skin, weight gain, and nausea and/or vomiting. If you do experience side effects while on corticosteroid drugs, it's important not to stop taking the medication without talking to your doctor. Stopping these drugs abruptly can lead to significant and dangerous unwanted effects.

Conventional Disease-Modifying Anti-Rheumatic Drugs (DMARDs): Conventional DMARDs like methotrexate and hydroxychloroquine help prevent your immune system from permanently damaging your joints and other tissues. Potential side effects include liver damage and lung infections, but vary from drug to drug.

Biologics: Biologics are a newer type of DMARDs that help protect against permanent joint and tissue damage. They are usually given alongside conventional DMARDs as they are most effective when used together. They may increase the risk of infections.

Targeted Synthetic Disease-Modifying Anti-Rheumatic Drugs (DMARDs): Targeted synthetic DMARDs include Baricitinib (Brand name: Olumiant®), tofacitinib (Brand name: Xeljanz®) and upadacitinib (Brand name: Rinvoq®). These drugs are typically only used if conventional DMARDs and biologics have not been able to control the progression of the disease because they increase the risk for serious adverse health effects, such as blood clots in the lungs (pulmonary embolism) and cancer.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Dose 1: 4 medicinal mushroom tinctures* (Lion's Mane, Reishi, Turkey Tail & Cordyceps dual-extracted tinctures) Joint & Movement Salve (topical use)*** Vit D, B-12, MSM, Probiotics, & Stinging nettle tincture	Weekly grocery shop for Elimination Protocol Quercetin Omega-3s, Gamma-linolenic acid, Thunder god vine & Turmeric w/ lunch Hydration check (Are you drinking enough water?) Physical activity (30 min.)	Dose 2: 4 mushroom tinctures (Lion's Mane, Reishi, Turkey Tail & Cordyceps dual-extracted tinctures) and Stinging nettle tincture Meditation & time outside/nature walk	Non-mushroom herb of your choice** Thunder god vine & Turmeric ASU (Avocado-soybean unsaponifiables) and MSM Leaky gut herbs	Joint & Movement Salve (topical use)*** Gratitude practice Chinese skullcap and Ginger Sleep herbs, if needed
MONDAY	Dose 1: 4 medicinal mushroom tinctures* (Lion's Mane, Reishi, Turkey Tail & Cordyceps) Joint & Movement Salve (topical use)*** Vit D, B-12, MSM, Probiotics, & Stinging nettle tincture	Quercetin Omega-3s, Gamma-linolenic acid, Thunder god vine & Turmeric w/ lunch Hydration check Physical activity (30 min.)	Dose 2: 4 mushroom tinctures (Lion's Mane, Reishi, Turkey Tail & Cordyceps) and Stinging nettle tincture Meditation	Non-mushroom herb of your choice** Thunder god vine & Turmeric ASU and MSM Leaky gut herbs Outdoor walk after dinner	Joint & Movement Salve (topical use)*** Gratitude practice Chinese skullcap and Ginger Sleep herbs, if needed
TUESDAY	Dose 1: 4 medicinal mushroom tinctures Joint & movement salve Vit D, B-12, MSM, Probiotics, & Stinging nettle tincture	Quercetin Omega-3s, Gamma-linolenic acid, Thunder god vine & Turmeric w/ lunch Hydration check Physical activity (30 min.)	Dose 2: 4 mushroom tinctures (Lion's Mane, Reishi, Turkey Tail & Cordyceps) and Stinging nettle tincture Meditation & time outside/nature walk	Non-mushroom herb of your choice** Thunder god vine & Turmeric ASU and MSM Leaky gut herbs	Joint & Movement Salve (topical use)*** Gratitude practice Chinese skullcap and Ginger Sleep herbs, if needed
WEDNESDAY	Dose 1: 4 medicinal mushroom tinctures* Joint & movement salve Vit D, B-12, MSM, Probiotics, & Stinging nettle tincture	Quercetin Omega-3s, Gamma-linolenic acid, Thunder god vine & Turmeric w/ lunch Hydration check Physical activity (30 min.)	Dose 2: 4 mushroom tinctures (Lion's Mane, Reishi, Turkey Tail & Cordyceps) and Stinging nettle tincture Meditation	Non-mushroom herb of your choice** Thunder god vine & Turmeric ASU and MSM Leaky gut herbs Outdoor walk after dinner	Joint & Movement Salve (topical use)*** Gratitude practice Chinese skullcap and Ginger Sleep herbs, if needed
THURSDAY	Dose 1: 4 medicinal mushroom tinctures* Joint & movement salve Vit D, B-12, MSM, Probiotics, & Stinging nettle tincture	Quercetin Omega-3s, Gamma-linolenic acid, Thunder god vine & Turmeric w/ lunch Hydration check Physical activity (30 min.)	Dose 2: 4 mushroom tinctures (Lion's Mane, Reishi, Turkey Tail & Cordyceps) and Stinging nettle tincture Meditation & time outside/nature walk	Non-mushroom herb of your choice** Thunder god vine & Turmeric ASU and MSM Leaky gut herbs	Joint & Movement Salve (topical use)*** Gratitude practice Chinese skullcap and Ginger Sleep herbs, if needed
FRIDAY	Dose 1: 4 medicinal mushroom tinctures* Joint & movement salve Vit D, B-12, MSM, Probiotics, & Stinging nettle tincture	Quercetin Omega-3s, Gamma-linolenic acid, Thunder god vine & Turmeric w/ lunch Hydration check Physical activity (30 min.)	Dose 2: 4 mushroom tinctures (Lion's Mane, Reishi, Turkey Tail & Cordyceps) and Stinging nettle tincture Meditation	Non-mushroom herb of your choice** Thunder god vine & Turmeric ASU and MSM Leaky gut herbs Outdoor walk after dinner	Joint & Movement Salve (topical use)*** Gratitude practice Chinese skullcap and Ginger Sleep herbs, if needed
SATURDAY	Dose 1: 4 medicinal mushroom tinctures* Joint & movement salve Vit D, B-12, MSM, Probiotics, & Stinging nettle tincture	Quercetin Omega-3s, Gamma-linolenic acid, Thunder god vine & Turmeric w/ lunch Hydration check Physical activity (30 min.)	Dose 2: 4 mushroom tinctures (Lion's Mane, Reishi, Turkey Tail & Cordyceps) and Stinging nettle tincture Meditation & time outside/nature walk	Non-mushroom herb of your choice** Thunder god vine & Turmeric ASU and MSM Leaky gut herbs	Joint & Movement Salve (topical use)*** Gratitude practice Chinese skullcap and Ginger Sleep herbs, if needed

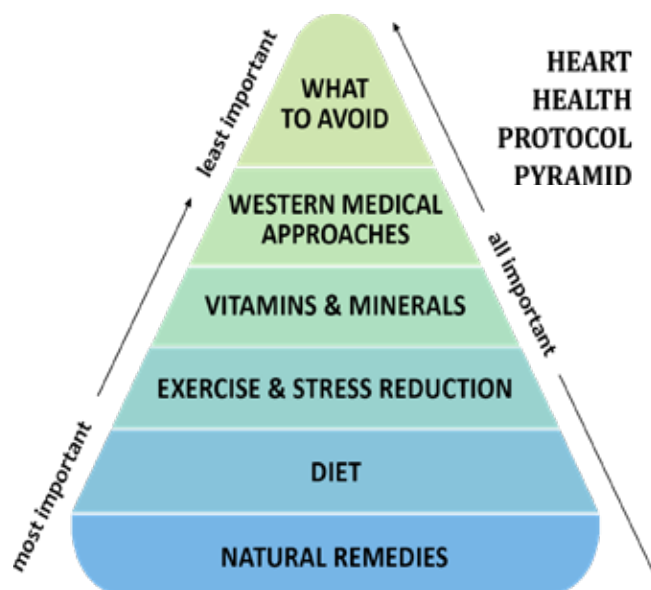
*Medicinal mushroom options: Lion's mane, reishi, turkey tail, cordyceps, white button mushrooms, black hoof mushroom.

**Non-mushroom herbs options: Licorice root, frankincense, cat's claw.

***Joint & Movement salve contains these topical/external herbs: Arnica, calendula, cayenne, cottonwood buds (balm of Gilead), helichrysum, frankincense, and St. John's wort.

Ch. 6: Heart Health Protocol (Cholesterol, High Blood Pressure, Arteriosclerosis)

Heart disease is the number one killer in the world. The heart's job is to pump blood throughout the body; heart disease occurs when the heart can no longer perform this function. According to the Centers for Disease Control and Prevention, one in every four deaths in the United States is the result of heart disease and one person dies every 34 seconds in the United States from cardiovascular disease.¹ Heart disease is an umbrella term for a variety of illnesses affecting the heart, including hypertension, high cholesterol, congestive heart failure, and heart attacks.



While there is no single cause for heart disease, risk factors include high blood pressure, high cholesterol, type 2 diabetes, and unhealthy lifestyle habits.² Despite its high mortality rate, it is estimated that 80% of heart disease is preventable through lifestyle changes, such as eating a healthy diet, regular exercise, and avoiding tobacco.³

Natural therapies can be extremely effective in preventing and treating heart disease. In fact, many pharmaceutical drugs used to treat heart disease (ex. digitoxin and lovastatin) are derived from herbs. Through diet, nutrition, herbal medicine, and lifestyle changes, you can optimize your heart health.

*It's important to note that due to the high rate of heart disease, many people are already medicated with pharmaceutical drugs. These medications can interact with nutritional supplements and herbal medicines. Due to the possibility of interactions, it's important to work with a qualified healthcare professional to ensure your health plan is safe.

1. Natural Remedies

Ask your physician before beginning an herbal

regimen, especially if you are on Western Medications, as there are many interactions. Some common

interactions are listed below.

Cordyceps Mushroom (*Cordyceps* spp.): Cordyceps lowers bad LDL cholesterol, which helps prevent plaque build up in the arteries, thereby reducing the risk of heart attack and stroke⁴. It also improves insulin sensitivity. Suggested dose: 3 dropperfuls dual-extracted cordyceps tincture twice daily.

Fenugreek Seeds (*Trigonella foenum-graecum*): Fenugreek reduces triglycerides and low-density lipoproteins ("bad" LDL cholesterol) and increases high-density lipoprotein ("good" HDL cholesterol) levels.⁵ Through its lipid-altering properties, fenugreek can help prevent heart disease. Suggested dose is 2-4 dropperfuls of tincture daily. Possible interactions: anticoagulant and hypoglycemic medications⁶

Ginkgo Biloba: Ginkgo is best known for increasing blood flow by promoting the dilation of blood vessels, a major factor in coronary artery disease.⁷ Additional research indicates other medicinal benefits due to its antioxidant effects, anti-platelet activity, and ability to lower blood pressure.⁸ Suggested dose is 3-6 dropperfuls of tincture daily. Possible Interactions: anticoagulant, hypoglycemic, and anti-depressant medications⁹

Hawthorn (*Crataegus monogyna*): Hawthorn has a long history of use for heart health. Hawthorn protects against heart disease by controlling high blood pressure and high cholesterol. Hawthorn increases coronary artery blood flow, improves circulation, and lowers blood pressure, important factors in reducing heart disease.¹⁰ Hawthorn is considered a safe and effective treatment for CVD (cardiovascular disease) and ischemic heart disease (IHD)¹¹. Suggested dose is 160 to 1,800 mg daily¹², or use as a tincture (2 dropperfuls 3 times daily). Possible hawthorn interactions: digoxin, beta-blockers, calcium channel blockers (CCBs), phenylephrine, medications for male sexual dysfunction (phosphodiesterase-5 inhibitors), and nitrates.

Hibiscus (*H. sabdariffa*): Hibiscus is known for its antibacterial, anti-oxidant, anti-cholesterol, anti-diabetic, and anti-hypertensive effects.¹³ Regular consumption of Hibiscus tea has beneficial effects in controlling glycemic status and blood pressure, two major risk factors for heart disease.¹⁴ Suggested dose is 3.75 g per day or 2 tsp of dried herb in tea daily. Possible interactions: May potentiate diuretic effects; caution with hydrochlorothiazide.

Holy Basil/Tulsi (*Ocimum tenuiflorum*/*Ocimum sanctum*): Tulsi helps reduce blood glucose, enhance insulin production, improve lipid profiles, prevent weight gain, and protect the liver and kidneys from high blood glucose¹⁵. Its antihypertensive effect is attributed to eugenol, which exerts its effect by blocking calcium channels¹⁶ ¹⁷. It can be taken as a tea, tincture, or supplement. Suggested dose: 2 dropperfuls tincture 2-3 times per day.

Maitake (*Grifola frondosa*): Maitake has become a popular food due to its protein content, dietary fiber, vitamin D, and minerals (K, P, Na, Ca, Mg).¹⁸ It improves

heart health through its nutritional value and its ability to lower blood pressure.¹⁹ Suggested dose: mushroom extract tincture 3-7.0 grams daily, or consume as food²⁰.

Mistletoe (*Viscum alba*): Mistletoe can mediate vasodilation and prevent myocardial injury.²¹ Suggested dose is 1.5 ml (1-2 dropperfuls) in tincture form daily. Possible interactions: immune-modulating medications, hypertensive medications²².

Olive Leaf (*Olea europaea*): Derived from the olive plant, olive leaf extracts have historically been used to treat infections, inflammation, and hypertension.²³ The Olive plant is a staple in Mediterranean cuisine, well known for reducing the risk of heart disease. Olive leaf extract helps prevent low-density lipoproteins (bad) cholesterol from building up in the arteries. This effect helps increase blood flow and lower blood pressure, reducing the risk of heart disease.²⁴ Suggested dose is 500 to 1,000 mg daily.²⁵ Possible interactions: blood pressure medications and insulin.

2. Diet **Mediterranean Diet:** The Mediterranean diet is an ideal diet for cardiovascular health. It is abundant in minimally processed plant-based foods, rich in monounsaturated fat from olive oil, and low in saturated fat, meats, and dairy products.²⁶ People of the Mediterranean region have dramatically lower rates of heart disease. Following a Mediterranean diet is helpful for people with cardiovascular disease, diabetes, high blood pressure, and metabolic syndrome.²⁷ This diet is rich in high-quality, nutrient-dense foods such as fruits and vegetables, olive oil, nuts, legumes, and seafood. For more information on the Mediterranean diet, please see page 308.

DASH Diet: The DASH diet (Dietary Approaches to Stop Hypertension) was created to prevent and treat high blood pressure, a common form of heart disease. It includes foods that are rich in potassium, calcium, and magnesium. These nutrients help control blood pressure. This diet limits foods that are high in sodium, saturated fat, and added sugars. Studies have shown that the DASH diet can lower blood pressure in as little as two weeks.²⁸

SMASH Fish: While it is common knowledge that seafood is great for heart health, "SMASH" fish are the best sources of heart-healthy omega-3s (known to reduce inflammation, heart rhythm disturbances, triglyceride levels, and blood pressure). They are also less likely to be contaminated with toxins when compared to other types of fish.²⁹ SMASH fish are sardines, mackerel, anchovies, salmon, and herring.

Pomegranate Juice: Originally from Iran and India, pomegranates have an antioxidant capacity higher than red wine and green tea³⁰. Multiple studies show that pomegranate juice reduces both systolic and diastolic blood pressure.^{31 32}

3. Exercise and Stress Reduction **Exercise:** Exercise plays a major role in the prevention and treatment of heart disease. The American Heart Association recommends moderate exercise for 30 minutes, 5 times a week.³³ That

said, even a minimal amount of activity is better than nothing.³⁴ Unfortunately, many people fall into the trap of all-or-nothing thinking when it comes to exercise. They often become discouraged if they aren't experiencing results or get derailed due to time constraints. As a result, many people give up. People who live in countries with the lowest rates of heart disease don't report having specific exercise or workout routines.³⁵ Instead, simple movement, such as walking or cycling, is a regular part of life. Research consistently shows that walking regularly significantly reduces cardiovascular disease.³⁶ Integrating some type of movement into your lifestyle is a great way to improve cardiovascular health.

Stress Reduction: Known as a silent killer, stress is a major contributor to illness, especially heart disease. Studies suggest that the high levels of cortisol that result from long-term stress can increase cholesterol, triglycerides, blood sugar, and blood pressure, common risk factors for heart disease.³⁷ Psychological stress levels can predict rates of heart disease.³⁸ Limiting your exposure to stressful situations and finding ways to de-stress is critical for preventing and managing heart disease. Effective stress-reducing strategies include mindfulness meditation, deep lymphatic breathing, journaling, engaging in supportive relationships, and fulfilling hobbies. For more strategies for stress reduction, refer to the Anxiety and Stress Protocol (see page 150).

4. Vitamins and Minerals **Coenzyme Q10:** Also known as CoQ10, this compound helps generate energy in cells. The body produces CoQ10 naturally, but its production tends to decrease with age³⁹ and some medications used to treat heart disease (like statins) reduce CoQ10 levels.⁴⁰ Statins belong to a class of drugs called HMG-coenzyme A reductase inhibitors, and they frequently deplete levels of coenzyme Q10 in the blood and muscle.⁴¹ This is what causes the fatigue, physical weakness, and muscle pain that people who take statins may experience. Taking coenzyme Q10 along with your statin may help alleviate some of these symptoms. Supplementing with CoQ10 also improves heart disease symptoms, reduces the risk of dying from heart problems,⁴² and is associated with a reduction in the frequency of hospitalization in those with heart disease, as well as reduced heart failure and fewer serious complications.⁴³ CoQ10 seems to help treat heart failure by improving heart function, increasing energy through ATP production, and limiting oxidative damage.⁴⁴ Suggested dose: 300mg daily.

Fish Oil: Fish oils contain high amounts of omega-3 essential fatty acids, specifically EPA (eicosapentaenoic acid) and DHA (docosahexaenoic acid). Omega-3s reduce the risk of developing an irregular heartbeat (arrhythmias), slow the buildup of plaque, which hardens and blocks the arteries, and lower blood pressure.⁴⁵ They reduce the likelihood of outcomes associated with heart disease like fatal and non-fatal myocardial infarction, stroke, coronary artery disease (CAD), sudden cardiac death, and all-cause mortality.⁴⁶ Suggested

dose is 1,500mg of EPA and 500mg of DHA daily.⁴⁷ Buy high-quality fish oil and keep in the refrigerator to prevent rancidity. Possible interactions: anti-platelet medications.

L-Arginine: L-arginine is an amino acid that helps the body build protein and is helpful in treating heart conditions.⁴⁸ While the body makes L-arginine it is also found in protein-rich foods and supplements. It improves blood flow in the arteries of the heart, which may improve symptoms of clogged arteries, chest pain (angina), and coronary artery disease.⁴⁹ L-arginine can decrease mild to severe chest pain and lower blood pressure in healthy people as well as those suffering from hypertension. Suggested dose is 3g daily.⁵⁰ Possible interactions: anticoagulants and anti-platelet medications, blood pressure medications, diabetes medications, Isoproterenol (Isuprel), Nitrates, Potassium-sparing diuretics, Sildenafil (Revatio, Viagra).

L-Theanine: L-theanine is an amino acid found in green tea. It is well known for several health benefits including improvements in mood, cognition, and reduction of anxiety-like symptoms.⁵¹ It boosts cardiovascular health by improving vascular function and decreasing the risk of cardiovascular disease. It works by reducing serum cholesterol and increasing artery vasodilation.⁵² Suggested dose is 400mg daily (if possible, 30-minutes before a meal).⁵³

Magnesium: Magnesium plays a role in over 300 enzymatic reactions in the human body.⁵⁴ Magnesium is necessary for maintaining a steady heartbeat and normal blood pressure. It also plays a role in regulating atherogenesis and thrombosis, vascular calcification, and proliferation and migration of endothelial and vascular smooth muscle cells, all factors in the development of a heart attack.⁵⁵ Magnesium has a major influence on the development of cardiovascular disease⁵⁶ and magnesium supplementation significantly lowers blood pressure.⁵⁷ Suggested dose: Magnesium glycinate - 280mg daily for women and 350mg daily for men.

Quercetin: Quercetin is a flavonoid, a compound found in plants. It has an extremely high antioxidant capacity and gives dark leafy greens their color. Emerging research has demonstrated quercetin's positive health benefits, especially for the heart.⁵⁸ It improves blood pressure, reduces low-density lipoprotein, aids in vasodilation of blood vessels, reduces clot formation, and lowers oxidative stress and other inflammatory markers, all of which contribute to heart disease.⁵⁹ Quercetin may also play a supportive role in preventing heart disease through its ability to improve exercise performance and blood sugar management.⁶⁰ Suggested dose is 500mg daily.⁶¹ To optimize absorption, take quercetin with Vitamin C.

Vitamin C: Vitamin C is best known as an antioxidant needed to form blood vessels, cartilage, muscle, and collagen in bones. Due to its important role in blood vessel formation, it plays a key role in supporting the blood vessels of the heart. While it's been well

documented that a deficiency in Vitamin C is associated with a higher risk of death from cardiovascular disease, new research demonstrates its protective effects. Vitamin C may improve endothelial function and lipid profiles,⁶² likely due to its ability to neutralize harmful free radicals, and in turn, reduce oxidative stress. Suggested dose: 500mg daily⁶³.

5. Western Medical Approaches

There are a number of pharmaceutical and surgical treatments used to treat heart disease. Due to its high rates in the general population, many people are currently medicated for heart disease. In fact, nearly half of adults in the United States are taking medication for hypertension, just one form of heart disease.⁶⁴

The most common kinds of medications prescribed for heart disease are anti-coagulants, anti-platelet medications, ACE inhibitors, angiotensin II receptor blockers, beta-blockers, calcium channel blockers, cholesterol-lowering medications, blood sugar lowering medications, digitalis preparations, diuretics, and vasodilators. Research their risks and side effects and learn about alternative treatment options.

6. What to Avoid

A Sedentary Lifestyle:

Characterized by an inactive lifestyle, being sedentary typically means rarely engaging in physical activity. Due to the demands of modern life, it can be difficult to make time to be active. Approximately 60% of all Americans age 18 and older report that they are physically inactive.⁶⁵ "Sitting is the new smoking"; the more you sit, the less you move, and the less you move, the more your risk increases for heart disease.⁶⁶ Even small improvements in your amount of daily walking are better than no walking at all.⁶⁷ By intentionally making an effort to get up and move you reduce your risk of heart disease and other medical issues.⁶⁸

Refined Carbohydrates: The evidence linking carbohydrate-rich foods, especially sugar-sweetened beverages, with heart disease has been steadily increasing.⁶⁹ Research shows that consuming more than 7 servings of refined carbohydrates per day increases the risk of heart disease and death drastically.⁷⁰ Choose complex carbohydrates over simple carbohydrates whenever possible and avoid overindulging in sweets.

Highly Processed Foods: Processed food is any food that has been altered in some way during preparation.⁷¹ Highly processed foods usually have 5 or more ingredients, come from a package, contain a combination of sugar, starch, and unhealthy oils, are often deep-fried, and/or are made with additives and preservatives. While they may be convenient, these foods pose a major risk factor for heart disease and should be avoided.⁷²

High-Stress Levels: Stress plays a major role in heart disease. While stress is an inevitable part of life, reducing your overall stress load plays a key factor in preventing heart disease. Research shows chronic stress predicts the occurrence of heart disease. Employees who experience

work-related stress and individuals who are socially isolated or lonely have an increased risk of cardiovascular disease.⁷³ Making sure you actively take steps to reduce and avoid stress is critical to maintaining a healthy heart.⁷⁴

Smoking: Smoking is a major risk factor for developing heart disease. If you smoke, you are almost three times more likely to die of heart disease.⁷⁵ The chemicals inhaled from smoking can cause permanent damage to the heart and blood vessels.⁷⁶ Even secondhand smoke poses health risks. By quitting or avoiding smoking you dramatically reduce your risk for heart disease, reduce the risk of premature death, and add as many as 10 years

to life expectancy.⁷⁷

Maintaining good cardiovascular health is one of the most important aspects of living a long and healthy life. While heart disease is the leading cause of death worldwide, the good news is that most cases of heart disease are preventable. Simple lifestyle changes can dramatically reduce your risk of getting heart disease. By choosing nourishing and nutrient-dense foods, supporting the body with proper supplementation, utilizing medical herbal medicine, incorporating movement into your daily life, and avoiding risk factors, you can optimize your heart health and lifespan.

	Morning	Noon	Afternoon	Evening	Before Bed
S U N D A Y	Cordyceps and Maitake dual-extracted mushroom tinctures Coenzyme Q10 Ginkgo or Mistletoe	Grocery shop for Mediterranean or Dash diet foods Quercetin, Vitamin C, and Fish Oil after heart-healthy lunch Hawthorn & Olive leaf extracts	Cordyceps and Maitake mushroom tinctures Pomegranate juice Exercise (any form); outdoors, if possible	L-arginine & L-theanine Heart-healthy dinner Stress reduction activity	Hibiscus tea Magnesium
M O N D A Y	Cordyceps and Maitake dual-extracted mushroom tinctures Coenzyme Q10 Ginkgo or Mistletoe	Quercetin, Vitamin C, and Fish Oil post-lunch Hawthorn & Olive leaf extracts	Cordyceps and Maitake mushroom tinctures Pomegranate juice Exercise	L-arginine & L-theanine Heart-healthy dinner After-dinner walk	Hibiscus tea Magnesium
T U E S D A Y	Cordyceps and Maitake dual-extracted mushroom tinctures Coenzyme Q10 Ginkgo or Mistletoe	Quercetin, Vitamin C, and Fish Oil post-lunch Hawthorn & Olive leaf extracts	Cordyceps and Maitake mushroom tinctures Pomegranate juice Exercise	L-arginine & L-theanine Heart-healthy dinner Stress reduction activity	Hibiscus tea Magnesium
W E D N E S D A Y	Cordyceps and Maitake dual-extracted mushroom tinctures Coenzyme Q10 Ginkgo or Mistletoe	Quercetin, Vitamin C, and Fish Oil post-lunch Hawthorn & Olive leaf extracts	Cordyceps and Maitake mushroom tinctures Pomegranate juice Exercise	L-arginine & L-theanine Heart-healthy dinner After-dinner walk	Hibiscus tea Magnesium
T H U R S D A Y	Cordyceps and Maitake dual-extracted mushroom tinctures Coenzyme Q10 Ginkgo or Mistletoe	Quercetin, Vitamin C, and Fish Oil post-lunch Hawthorn & Olive leaf extracts	Cordyceps and Maitake mushroom tinctures Pomegranate juice Exercise	L-arginine & L-theanine Heart-healthy dinner Stress reduction activity	Hibiscus tea Magnesium
F R I D A Y	Cordyceps and Maitake dual-extracted mushroom tinctures Coenzyme Q10 Ginkgo or Mistletoe	Quercetin, Vitamin C, and Fish Oil post-lunch Hawthorn & Olive leaf extracts	Cordyceps and Maitake mushroom tinctures Pomegranate juice Exercise	L-arginine & L-theanine Heart-healthy dinner After-dinner walk	Hibiscus tea Magnesium
S A T U R D A Y	Cordyceps and Maitake dual-extracted mushroom tinctures Coenzyme Q10 Ginkgo or Mistletoe	Quercetin, Vitamin C, and Fish Oil post-lunch Hawthorn & Olive leaf extracts	Cordyceps and Maitake mushroom tinctures Pomegranate juice Exercise	L-arginine & L-theanine Heart-healthy dinner Stress reduction activity	Hibiscus tea Magnesium

Ch. 7: Candida Protocol

Candida is a type of fungus.¹ At normal levels, it aids with nutrient absorption and digestion.² However, it can grow out of control (Candida Overgrowth Syndrome (COS)).³ Symptoms can vary widely depending on where it is overproducing and the initial triggers. If left untreated, it can begin to break down the walls of the intestinal lining and get into the bloodstream. Toxins and other byproducts can also find their way into the bloodstream, resulting in leaky gut syndrome. Candida overgrowth can lead to serious health issues and even the development of new allergies, sensitivities, or food intolerances, especially to corn, eggs, dairy, and gluten.

Candida infections are commonly caused by *Candida albicans*. This is a type of yeast infection found in the mouth, intestinal tract, and/or vagina. It can affect the skin and mucous membranes as well. Although this type of candida infection is rarely serious, it can be if the immune system isn't functioning properly. In this case, candida may migrate to other areas of the body, such as the blood and membranes around the heart and brain. When this happens, it is referred to as invasive candidiasis⁴.

Candida yeasts often grow out of control when the body's natural pH balance is altered. To effectively fight candida, you first need a healthy immune system and plenty of good bacteria in your microbiome. It's a 2-phased approach – first take care of the overgrowth and/or infection and then rebuild and rebalance your beneficial gut bacteria and microflora.

Symptoms of Candida Overgrowth: There are some obvious symptoms of candida overgrowth, such as those that accompany a yeast infection, but other symptoms may not be as apparent. Many people with candida overgrowth experience subtle symptoms and thus don't link their symptoms to the root cause. COS symptoms overlap with other issues, so *it is best to test for candida overgrowth* using a stool, urine, or blood test. Symptoms include:

Chronic Fatigue: If you are feeling wiped out, you may have issues with candida overgrowth. If you are getting enough sleep but still chronically tired, this may be a sign of chronic fatigue syndrome. This syndrome is often diagnosed when a person experiences fatigue for a minimum of six months accompanied by candida symptoms like joint pain and brain fog. Chronic fatigue is one of the most common symptoms of candida overgrowth.

Hormonal Imbalance: Hormonal imbalance is a symptom of many health conditions, including candida overgrowth. Candida infections, if left untreated, can result in early menopause, a decreased sex drive, hormonal migraines, endometriosis, PMS, weight gain, and water retention. Inflammation from candida overgrowth causes more yeast to grow and spread. A byproduct from candida mimics estrogen, which can lead to serious hormonal issues.

Mood Disorders: Candida overgrowth can manifest as anxiety, mood swings, irritability, panic attacks,

and depression. The connection between the gut and the brain is becoming clearer as scientists conduct more research, so it makes sense that issues that begin in the gut have an effect on neurological function.

Recurring Sinus Infections: If you suffer from sinus infections on a regular basis, candida may be the surprising culprit rather than seasonal allergies. Candida affects the sinuses and may present itself as a persistent cough, increased congestion, post-nasal drip, and influenza symptoms.

Recurring Urinary Tract Infections: It isn't uncommon for women to suffer from vaginal infections⁵ or urinary tract infections, but if these become a recurring burden, you may have candida overgrowth. Candida can be sexually transmitted, so partners can spread it back and forth. It may be necessary to treat both partners. Avoid wearing tight-fitting undergarments and taking hot baths when you are experiencing an active infection.

Thrush: Oral thrush is a yeast infection caused by the same candida that is responsible for vaginal yeast infections.⁶ It can spread from person to person via saliva. If left untreated it can spread to other parts of the body, such as the digestive tract, liver, heart valves, and lungs. If you are experiencing thrush on a recurring basis, consider candida overgrowth as a possible root cause and address it as soon as possible using the protocols in this chapter.

Skin and Nail Infections: A warm, moist, and dark environment can trigger fungal infections of the nails, especially the toenails. Fungal infections can spread to other parts of the body such as the skin. A recurring, itchy rash that seems to come out of nowhere may be a candida rash. Candida skin rashes look inflamed and bright red. They may have bumps and stay in regions of the body that may not get as much air, like skin folds. Address the root cause of these fungal infections (candida overgrowth) to get rid of them once and for all.

Brain Fog: Because imbalances in the gut interfere with neurological functions, brain fog is a common symptom of candida overgrowth. Watch for signs of brain fog, such as an inability to focus, difficulty concentrating on tasks, poor memory, and poor physical coordination.

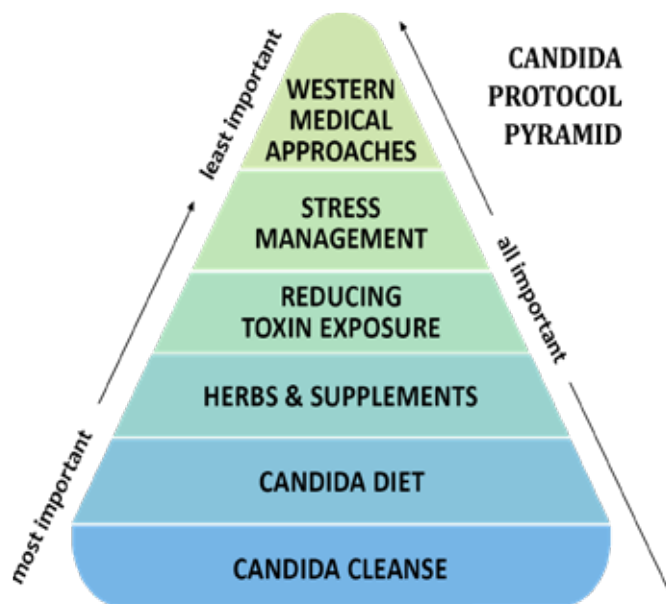
Gastrointestinal Issues: If you experience an increase in intestinal issues, such as persistent gas, bloating, diarrhea, or abdominal cramping, you may have an issue with candida overgrowth in the intestines. A lack of beneficial bacteria in the microbiome can greatly contribute to this as they are responsible for fighting off infection in this region, so when they are diminished, symptoms may manifest. Treating a candida infection should result in a decrease in digestive discomfort and may help heal leaky gut damage.

Other symptoms include cravings for sweets, joint pain, low sex drive, a weakened immune system, and bad breath. Leaky gut symptoms are a big indicator of an underlying candida problem. Watch for autoimmune issues (due to waste penetrating the lining of the gut and getting into the bloodstream). These can range from gas, bloating, diarrhea, and other GI issues to skin rashes,

fatigue, mood changes, and fibromyalgia. A candida overgrowth test is a great first step.

What Causes Candida Overgrowth?: Several factors may contribute to candida overgrowth, including:

- Frequent use of antibiotics kills off all bacteria in the gut, bad and good. The killing off of good bacteria leaves the gut more susceptible to overgrowth of bad bacteria.
- A diet high in sugar is one of the biggest dietary contributions to candida overgrowth. Candida loves to consume sugar because it helps grow and build its cell walls. Sugar also weakens immune defenses, leaving the body less able to fight off candida overgrowth.
- Chronic stress and anxiety weaken immune defenses and can open the door to candida infections if not kept in check.
- Hormonal birth control: When hormones levels are interfered with, candida overgrowth can occur. Estrogen is linked to the growth of yeast.



- Exposure to toxins in your drinking water. Chemicals added to some water, such as fluoride and chlorine, kill beneficial bacteria in the gut and weaken the immune system. Not only are many people drinking water with these substances, they are bathing in it. Do what you can to minimize exposure, such as buying a good filter. Switch to natural products to further lessen your toxic load. Avoid deodorant and other products with aluminum, parabens, and sulfates.
- Diabetics often have high blood sugar levels, and this sugar feeds candida. Many with diabetes have a weakened immune system, allowing candida to spread more easily.

1. Candida Cleanse

To effectively push the “reset” button, a candida cleanse helps rid the body of excess candida by flushing it out through the digestive tract. There are two types of cleanses; choose the one that works best for your needs.

The first type of cleanse is a bit harder to do because it involves sticking to a liquid diet for a few days. It can be very effective at flushing out candida and helping the body heal. The second type of cleanse involves eating gentle foods that help reboot gut health, as well as overall health.

Liquid Candida Cleanse (1-2 days): To start your liquid candida cleanse, make vegetable broth with pure water and organic garlic, celery, kale, onions, and sea salt. Simmer for several hours and strain. Refrigerate when cool. Drink throughout the day and make sure to drink plenty of water during this cleanse in addition to sipping the broth. The additional water will help the body flush out toxins faster. Drink the broth and water for one to two days before going back to solid foods.

When returning to solid food, eliminate sugar, alcohol, processed foods, and anything else that contributes to candida overgrowth. Stick to a diet of organic vegetables, leafy greens, and lean meats like chicken, turkey, and healthy fish (like wild-caught salmon). If your eating habits become unhealthy again and you feel candida overgrowth symptoms beginning to manifest, repeat this cleanse every two weeks or as needed. A good option is to perform the liquid-only candida cleanse as a jumpstart to the gentle food cleanse outlined below.

Gentle Food Cleanse (3-5 days): If a liquid-only cleanse is a bit too much for you, you can target candida with a cleanse that involves gentle foods and the elimination of anything that contributes to candida.

Start by eliminating big candida promoters like sugar, fruit, starches, alcohol, and grains (especially anything containing gluten). Stick to a diet of fresh, organic vegetables that have been steamed. Do not eat starchy vegetables like potatoes, carrots, radishes, or beets. These starches should be avoided during this cleanse because they add to sugar levels when the body processes them. Sugar feeds candida, so this is counterproductive to its elimination.

Drinking water is an integral part of any cleanse. In addition to healthy meals of steamed vegetables, drink at least 72 ounces of pure water each day to help flush candida. Once a day, enjoy leafy greens like kale, chard, wild greens, or spinach in your salad with a small amount of apple cider vinegar, lemon juice, or coconut oil as a dressing.

Detox Baths while Cleansing: Bentonite clay is excellent for helping to pull toxins from the body. Take detox baths every few days, especially before and after a cleanse. Run a hot water bath and add one cup of bentonite clay, one cup of Epsom salts, one tablespoon of liquid castile soap, and 15-20 drops of lavender essential oil and soak.

Supplements to Accompany a Candida Cleanse: There are supplements that support detox pathways, while also aiding in the elimination of candida metabolites. There are two groups of supplement categories. The first are liver-supportive supplements that assist the liver in removing toxins. The second bind themselves to

toxins to help eliminate them. Herbal remedies that deal directly with candida are in the Herbs and Supplements section later on in this protocol.

Milk thistle (*Silybum marianum*) is a liver-supportive herb and contains compounds that help repair liver damage, as well as protect the liver from any toxins released from candida.

Dandelion root (*Taraxacum officinale*) is another excellent liver detox herb. I use a blend of both in tincture form.

Swedish bitters support the digestive system and help regulate acidity to support liver function. Note these are different than digestive bitters.

Molybdenum helps convert acetaldehyde (which is a toxin released by candida) into acetic acid so it can be excreted by the body.⁷

Vitamin C is an important antioxidant known for its ability to support the immune system⁸, but it also helps restore adrenal function and aids in the elimination of toxins.

Probiotics help replenish beneficial bacteria in your gut⁹ and prevent pathogenic fungi from adhering to epithelial cells when invading the body.¹⁰ They also inhibit the formation of fungal biofilms, which protect fungi and prevents anti-fungal drugs from reaching them, making it more difficult to eradicate fungal infections.¹¹ Suggested dose: 50-100 billion colony-forming units per day, in divided doses, with meals.

Other Ways to Assist in a Candida Cleanse: Oil pulling and sauna can also help. Not only is oil pulling great for oral health, it helps pull toxins from the body. Oil pulling is a very simple procedure that is performed right after brushing your teeth. Simply swish a teaspoon or two of organic coconut oil for two to three minutes and spit it out when you are finished. Many cultures and people understand the detoxifying power of saunas. Sweating helps flush out toxins through the pores of the skin when the body is exposed to heat.

2. Candida Diet A candida diet¹² can rid your body of candida, as well as discourage it from coming back. Your chances of success will increase dramatically if you jumpstart this diet with the candida cleanse outlined in the previous section.

To start this diet, you need to give your immune system what it needs to help eliminate candida. You can do this by *eliminating certain foods* from your diet. These foods include sugar, alcohol, and refined carbohydrates. These foods not only weaken the immune system, but they help candida grow.

Replace the foods you eliminate with foods that help boost the immune system and overall health. Leafy greens are packed with nutrients and also contain fiber. This fiber encourages the formation of butyrate, a compound that helps maintain the cells lining the intestinal tract, boosts gut immunity, reduces inflammation, and even reduces the risk of colon cancer¹³. Eat organic

vegetables (skip starchy vegetables like carrots, corn, and potatoes for now), spinach, kale, collard greens, asparagus, and peas. The occasional sweet potato is okay, but no more than one a week. Try to eat a cup of healthy vegetables and leafy greens with each meal. While eating raw greens and vegetables is good for the body, it is recommended to eat steamed or warmed vegetables and greens on the candida diet, as warm foods may heal the gut better for those with candida issues. Also try and include wild greens, which you can learn about in my book, *The Forager's Guide to Wild Foods: Edible Plants, Lichens, Mushrooms, and Seaweeds*.

Pay Attention to Your Body: Consider keeping a food journal. Watch for stomach distress after eating so you know what to avoid in the future. For example, you may not be able to tolerate sweet potato, so you might have to replace it with cauliflower, cucumbers, or mushrooms instead. Consume healthy greens and vegetables such as wild greens, broccoli, onions, and okra.

Grains You Can Eat: Eat grains such as barley, quinoa, sorghum, teff, and amaranth. Avoid gluten, as this seems to be a trigger for many. Stick to one serving of grains per day.

Fruits: Fruit contains sugar, and even though it is natural sugar, it still feeds candida and encourages its growth. While you don't have to eliminate fruit (it contains antioxidants and other essential nutrients), limit fruit to one serving a day. If you want to sweeten something, reach for stevia instead of sugar or synthetic sweeteners. Raw honey is a good natural sweetener, but try to limit your use of this to the occasional teaspoon in tea.

Dairy: Cultured dairy, such as kefir or yogurt with probiotics, is a great choice for the candida diet. For many, dairy can be a trigger, so limit other dairy products like milk and cheese. Instead, opt for coconut milk, oat milk, or almond milk.

Fermented Foods Are Vital: In addition to kefir and yogurt, try to eat probiotic foods or drink probiotic drinks with every meal. Probiotic foods include sauerkraut, kimchi, and other fermented vegetables. Fermented vegetables contain microflora that kill off candida and protect the gut. Eat these daily to help boost your immune system and keep candida at bay. Start with a half-cup of fermented vegetables each day, and work your way up to a half-cup with each meal.

It is essential to replenish beneficial bacteria¹⁴ back into your gut with **daily probiotics** so you can feel better, reduce cravings for refined carbohydrates and sugars, and reduce candida symptoms.

Meat/Protein: Consuming a source of protein with each meal helps control appetite, provides energy, and helps you feel fuller longer. The key to protein with the candida diet is to stick with healthy protein sources like organic chicken, fish (such as wild-caught salmon), free-range eggs, and shellfish. You can also opt for plant-based proteins like beans, legumes, nuts (except peanuts and pistachios), and seeds. *Note that a ketogenic diet is not*

recommended as there are new studies that show that candida thrives on ketones.

Oils: Some oils are great for promoting healthy cell membranes and some oils are counterproductive to health and contribute to health issues like cholesterol and obesity. Choose unrefined, cold-pressed oils if possible. These include coconut oil, olive oil, avocado oil, flax seed oil, and chia seed oil. Add a small amount of healthy oil to a salad and use these oils for cooking. Avoid processed oils like canola oil, peanut oil, corn oil, and vegetable oil.

Yes to Xylitol: Xylitol is a natural sugar alcohol found in many fruits and vegetables. It is frequently used to sweeten chewing gum, toothpaste, and mouthwash. Xylitol has potent anti-fungal effects against *Candida albicans*.¹⁵ Replacing natural and artificial dietary sugars with xylitol could help prevent or address oral thrush.¹⁶

3. Herbs and Supplements **T O P I C A L CANDIDA SKIN**

RASHES: Candida can manifest itself in many ways, including skin rashes. These rashes are often red, inflamed, and itchy. For help with candida rashes, the first line of action is following a strict candida cleanse and diet. Next, try the natural remedies below.

Apple cider vinegar displays significant anti-fungal activity against *Candida albicans*.¹⁷ Breastfeeding mothers may apply apple cider vinegar topically to the nipples and areolas to help kill *Candida* and reduce the likelihood of it causing oral thrush in their babies. Suggested dose: Apply 4% apple cider vinegar topically three times per day or as needed (minimum concentration 2500-10,000 µg/ml); for denture stomatitis soak dentures in apple cider vinegar for 30 minutes or more.

Calendula (*Calendula officinalis*) and Chickweed (*Stellaria media*) in Coconut Oil: Calendula and chickweed help calm irritated and inflamed skin. They also have antifungal properties¹⁸. The best base for each skin remedy in this section is coconut oil because it is antifungal and works well with these herbs to eradicate fungal rashes. One study found coconut oil to have more potent anti-fungal activity against *Candida albicans* than the anti-fungal drug fluconazole.¹⁹ To make a medicinal oil, fill a glass jar with dried calendula flowers and dried chickweed. Melt organic coconut oil by sitting the oil jar in warm water until enough melts to cover the herbs in the glass jar when you pour it in. Once you have completely covered the herbs in the jar with coconut oil, sit this jar in a pan of hot water on the stovetop (on low) for eight plus hours. Strain the plant material through cheesecloth and squeeze the cloth well to get all the oil out of the herbs before discarding. Suggested dose: Apply a liberal amount 3x/day to affected areas.

Antifungal Essential Oils: You can easily create a powerful antifungal skin rash treatment in a few steps with essential oils and coconut oil²⁰. Antifungal essential oils to combat candida include tea tree, lavender²¹, clove, oregano, rosemary, lemongrass, thyme, and geranium. Tea tree (*Melaleuca alternifolia*) oil exhibits potent anti-

fungal activity against 32 strains of *Candida albicans* and other *Candida* species.²²

The best carrier oil is anti-fungal coconut oil. Melt some coconut oil by sitting the jar in warm water. Pour one ounce of coconut oil into a glass jar and add 15-20 drops of two or three of the essential oils mentioned. Blend well and apply to skin rashes. You may add these essential oils to the calendula and chickweed oil above for an antifungal powerhouse. Avoid getting this essential oil blend near open wounds and mucous membranes.

ORAL THRUSH: Oral thrush is a big indicator of a potentially bigger issue with candida. Make sure to follow the candida diet and cleanse in addition to these remedies.

Usnea Lichen: Oral thrush is an overgrowth of candida in the mouth. Alcoholic extractions (tinctures) of Usnea lichen^{23 24 25} are very effective for candida both internally and externally. I prefer using it in an easy-to-use spray bottle for thrush (and other fungal infections). Spray 3x/day on the affected area (for oral thrush). For candida on other parts of the body Usnea Tincture can be paired with the above oil(s) or salves after the Usnea spray dries.

Lemongrass (*Cymbopogon citratus*): Lemongrass is highly antifungal and has a pleasant taste.²⁶ Boil one to two tablespoons of lemongrass in a cup of water until the liquid is reduced by half and cool before straining. Gargle this every few hours (store in the refrigerator between uses). When gargling the thrush remedies in this protocol, gargle for at least one-minute 3x/day. It can take anywhere from one to three days to see an improvement; keep gargling until it is gone. You can pair these gargles with the Usnea Spray (spray after your gargle).

Pau D'arco Decoction: Pau d'arco is a tree found in the rainforest. Its inner bark is antifungal, antiviral, and antimicrobial²⁷. The medicinal qualities of pau d'arco are likely due to its naphthoquinones, specifically lapachol and beta-lapachone. Suggested dose²⁸: Infuse 1 Tbsp. of the bark in one cup of hot water for fifteen minutes and gargle for at least one minute, three times daily. Refrigerate between uses; lasts up to 3 days.

GUT HEALTH AND CANDIDA HELP: Reishi Mushroom (*Ganoderma lucidum*): Reishi is an adaptogenic mushroom that exhibits anti-biofilm activity against both *Candida albicans* and *Candida glabrata* – the two most common pathogenic candida species^{29 30 31}. It is also an excellent anti-inflammatory; and taming inflammation is a crucial step to healing a leaky gut. Take 2-3 dropperfuls of dual-extracted Reishi tincture daily.

Turkey Tail Mushroom (*Trametes versicolor* or *Coriolus versicolor*): Turkey tail is a rich source of prebiotics that support a healthy microbiome by feeding the “good bacteria”, while also controlling candida overgrowth.³² Take 2-3 dropperfuls of dual-extracted Turkey Tail tincture daily.

Lion's Mane Mushroom (*Hericium erinaceus*): Another outstanding medicinal mushroom for

candida and gut health is Lion's Mane. Lion's mane exhibits antifungal activity against *Candida* strains *C. albicans* (the most common form) and *C. neoformans*. It also inhibits biofilm formation of candida^{33 34}. Additionally, it is high in antioxidants and helps with gut inflammation.

Plantain (*Plantago major*), Marshmallow Root (*Althaea officinalis*), and Slippery Elm (*Ulmus rubra*): Known as an all-around tonic for the digestive tract, **plantain** is anti-inflammatory and soothes mucous membranes to help alleviate symptoms of leaky gut. Two additional herbs that support gut healing are **slippery elm** and **marshmallow root**, which form a protective layer that helps the gut to regenerate. All of these ingredients are in my Balanced Gut Tincture, and can be taken as a tea or infusion. Suggested dose: Tincture - 2 dropperfuls 3x/day for 3 weeks, and then one dropperful 3x/day for maintenance. Eat plantain and its seeds and take slippery elm and marshmallow as strong teas (infusions) for best results.

Berberine: Berberine exhibits significant antifungal activity against multiple *Candida* species. It is particularly noted for its use for biofilm infections associated with antifungal resistance³⁵. Berberine, which is yellow in color, is found naturally in plants such as Oregon Grape root, barberry, and goldenseal. Best taken 30 minutes to an hour before meals. Suggested dose: 1500mg/day in 500mg doses 3x/day.

Japanese Honeysuckle (*Lonicera japonica*): New research shows that *Lonicera japonica* extract has very strong antimicrobial activity against *C. albicans* and *C. tropicalis*. It also demonstrates enhanced production of anti-inflammatory cytokines and significant anti-inflammatory activity³⁶. Suggested dose: 2 dropperfuls of tincture/day.

Vitamin B-3/Nicotinamide/Niacinamide (NAM): Nicotinamide, also known as niacinamide, is antifungal, probably through affecting cell wall organization. It also inhibits biofilm formation³⁷. (Note: this is different than niacin). Suggested dose: 100-500mg/day

Caprylic Acid: Caprylic acid extracted from coconut oil exerts potent anti-fungal activity against *Candida albicans*.³⁸, and can be used topically and/or internally. Caprylic acid damages the cell membranes of *Candida albicans* cells and fungal efflux pumps so that the fungi are less able to defend themselves against anti-fungal treatments.³⁹ When caprylic acid is combined with carvacrol and thymol, greater than 83% of membranes are damaged and greater than 95% of efflux pumps are damaged. Suggested dose is 306-612 mg of magnesium caprylate per day, following meals. Note that caprylic acid is not intended for long-term use when taken internally.

Black Walnut (*Juglans nigra*): Black walnut hulls contain a flavanoid called isoquercetin that has potent antifungal properties.⁴⁰ Specifically, isoquercetin damages the membrane of *Candida albicans* cells, increasing the cells' permeability and eventually leading to cell death.⁴¹ Suggested dose: 2 dropperfuls of a 1:3

tincture, twice daily.

Turmeric/Curcumin: Turmeric shows anti-fungal activity against *Candida albicans*. Turmeric contains active constituents that inhibit fungal growth at lower concentrations and directly lead to fungal cell death at higher concentrations.⁴² Suggested dose: 2 dropperfuls of a 1:4 tincture, twice daily.

DIGESTIVE RELIEF: A common symptom of candida overgrowth is digestive discomfort, such as abdominal cramping, pain, diarrhea, gas, bloating, and nausea.

Peppermint (*Mentha piperita*) and Ginger (*Zingiber officinale*) Tea: A combination of peppermint and ginger helps provide relief from upset stomach and nausea that sometimes accompany candida overgrowth. Infuse one teaspoon of each into a cup of hot water for fifteen minutes.

Fennel Tea (*Foeniculum vulgare*): If gas and bloating is an issue, fennel seeds are a very effective stomachic herb that helps relieve severe gas, bloating, and an upset stomach. Infuse one to two teaspoons into a cup of hot water for fifteen minutes. Drink up to three times daily.

Chamomile Tea (*Matricaria chamomilla*): For intestinal spasms and abdominal pain, chamomile tea can provide gentle and calming relief. If your stomach is upset or you feel abdominal pain/discomfort, infuse two teaspoons of chamomile flowers into one cup of hot water for fifteen minutes and drink up to three times daily. In addition, you can use chamomile essential oil externally. Add ten drops to a tablespoon of carrier oil and massage into your abdomen.

Agrimony Tea (*Agrimonia eupatoria*): For those suffering from chronic diarrhea as a result of candida overgrowth, agrimony soothes irritation in the intestinal tract. Infuse two teaspoons into one cup of hot water for fifteen minutes and drink as needed for diarrhea and intestinal relief.

Witch Hazel (*Hamamelis virginiana*) – External Use: The astringent properties of witch hazel help soothe skin for after-effects of diarrhea. For irritated skin in the anal region, add a few cups of witch hazel decoction to your bath water. This helps irritated skin, candida rashes, and even hemorrhoids that result from chronic diarrhea. Make a witch hazel tea to spray on skin rashes by infusing two teaspoons of witch hazel in a cup of hot water for fifteen minutes and adding this to a spray bottle once cool.

IMMUNE SUPPORT: Supporting your immune system can help stop candida overgrowth before it starts. If you suffer from immune system issues that prevent your body from being able to fight off candida, try the remedies below. Note that Reishi and Turkey Tail Mushrooms are also helpful with candida overgrowth, so these medicinal mushrooms are excellent additions as they help in more than one way.

Elderberry (*Sambucus nigra*) and Ursolic Acid: Elderberry is a great addition for immune system support⁴³. It is antiviral and immunomodulatory⁴⁴,

meaning it helps regulate the immune system. It also has ursolic acid, known to inhibit Candida^{45 46}(note: need an alcoholic extraction for triterpenoids). Suggested dose: Take 2 dropperfuls of a dual-extracted elderberry tincture up to twice daily for immune support, and 3-4x/day if you are ill with an infectious disease.

Reishi (*Ganoderma lucidum*): Reishi is an adaptogenic immunomodulatory mushroom that helps to support the immune system and overall health.⁴⁷ It is often referred to as the “mushroom of longevity”. Suggested dose is 2-3 dropperfuls daily as a dual-extracted tincture for full spectrum of medicinal compounds.

Rosehips: Your immune system needs vitamin C to function properly. Many supplements made with synthetic vitamin C are poorly absorbed because the body prefers vitamin C via food. Rosehips contain a high amount of bioavailable vitamin C⁴⁸. You can make a rosehip tincture by blending or mashing your rosehips (so that they infuse better). Then fill a glass jar with rosehips and add alcohol (vodka or brandy work well), leaving ¼ inch at the top. Cover and infuse for six to eight weeks. Strain using cheesecloth and bottle in a dropper bottle. Take 2-3 dropperfuls twice daily.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): Another adaptogenic, antiviral, immunomodulatory mushroom, turkey tail has also been well researched for immune help for those undergoing chemotherapy and radiation. This powerful immune system ally is best taken daily as a dual-extracted tincture.

YEAST INFECTIONS: Like oral thrush, yeast infections are another telltale sign of a bigger issue with candida. The fungus that causes a yeast infection is the same fungus responsible for thrush. The only difference is the location. A yeast infection often affects the vagina and sometimes the outer parts of the vagina, such as the labia and vulva. It can be very irritating and itchy. It isn't uncommon for women to suffer from these, but if they are recurring, this could be a sign of candida overgrowth. Below are some effective remedies to treat yeast infections.

Coconut Oil Bolus: Coconut oil is safe enough to use in the vaginal canal, so it makes the perfect remedy to get rid of various infections in the area, such as bacterial vaginosis and yeast infections.⁴⁹ It is naturally antifungal. To make it even more effective, add a combination of antifungal tea tree and lavender essential oils. Melt 1/2 cup of coconut oil and pour into a wide-mouthed glass jar. Add ten drops each of tea tree and lavender essential oils and blend well. Let cool until the coconut oil has solidified. Take a 1-2” spoonful and roll it into a ball. Insert this into the vagina and lie down for as long as possible (at least one hour), or do this when you go to sleep. Repeat up to three times daily or until your symptoms subside. This remedy can also be applied to external candida rashes, if the coconut oil is too solid, heat it slightly.

Garlic Cloves: Millions of women have personally testified that a garlic clove inserted into the vagina can help get rid of a yeast infection. While there is some skepticism

surrounding this, it is not far-fetched, as garlic is highly antifungal⁵⁰. Instead of inserting a clove into the vagina, try infusing garlic into coconut oil to make a bolus. Melt ½ cup of coconut oil and pour this into a wide-mouthed glass jar. Add a few cloves of finely chopped garlic. Sit the jar in a pan of hot water on the stove on low and let this infuse with gentle heat for eight to ten hours. Remove from heat and strain through cheesecloth. Let cool until the coconut oil has solidified. Take a 1-2” spoonful and roll it into a ball. Insert this into the vagina and lie down for as long as possible (at least one hour). Repeat up to twice daily for best results.

LYMPHATIC CLEANSERS: Another way to help reduce candida and toxins is by supporting the lymphatic system. The lymphatic system plays an important role in the removal of waste from the body. It plays many roles, but some of its most important include protecting the body from potentially dangerous pathogens that invade, maintaining body fluid levels, and absorbing digestive tract fats. It also removes cellular waste. The remedies below help the body flush excess waste through the lymphatic system, while also supporting the immune system:

Cleavers (Bedstraw/*Galium*): This common springtime “weed” is a very effective lymphatic cleanser and stimulates lymphatic flow to flush out toxins. Cleavers can be collected, chopped and infused in alcohol to make a tincture that can be taken daily for immune⁵¹ and lymphatic support. Take two dropperfuls up to twice daily for best results. I also add young cleavers to my smoothies in the springtime. Drink plenty of water.

Dandelion Root (*Taraxacum officinale*): Dandelion root is a great overall detoxifier, diuretic, and helps stimulate lymphatic flow. Infuse a few teaspoons of the chopped roots in a cup of hot water for fifteen minutes and drink to three times daily. Since it is diuretic, it helps the body get rid of excess water to flush the system. Make sure to drink a lot of water daily to aid this process and prevent dehydration.

Lymphatic Massage: Don't forget to nourish the lymphatic system as you are conducting a cleanse. Use a carrier oil (such as coconut oil) and massage down each side of the neck for several minutes. Then work your way down the body, massaging other lymph nodes and pathways, such as under the arms. Gently warming the oil before massaging can help further stimulate lymph flow.

4. Reducing Toxin Exposure

Exposure to toxins can weaken the immune system and make it harder for the body to fight off candida. It may also interfere with digestion, lymphatic flow, and the elimination of waste. Toxins can build up over time and trigger more severe issues like autoimmune disease or even cancer. Reduce your exposure to common toxins. In addition, help your liver and lymphatic system function as they should to eliminate waste by taking daily supplements like cleavers, dandelion root, and milk thistle.

Shampoo, Soap, Lotion, Deodorant, and Makeup:

Research the products you are using on your skin each day to make sure they don't contain toxins⁵² like sulfates, parabens, polyethylene glycol, phthalates, formaldehyde, triclosan, dimethicone, retinyl palmitate, alcohols, dyes, aluminum, and synthetic fragrances. These are common in many toiletries like shampoos, conditioners, deodorants, and lotions. Many makeup products are heavily tainted with toxins and other heavy metals⁵³. When applied to the skin, they are absorbed into the bloodstream and these toxins can bypass your detox organs. Read labels carefully.

Water: Research shows that states like California, Montana, Texas, and Nevada have higher amounts of arsenic in their water. Many cities add chemicals to the water supply such as fluoride. This is good for teeth, but it is not good to ingest (or bathe in) and can interfere with many bodily functions while also degrading bone health. According to a Harvard study⁵⁴, fluoride may also lower IQ. Check your area's water quality by going to the EWG's Tap Water Database. There are water filters that filter out heavy metals and other toxins; this is an excellent choice if you are concerned about the quality of your water.

Food: Try to eat organic, whole foods as much as possible to avoid exposure to pesticides and other toxins. Stick to cage-free eggs, antibiotic-free meats, and grass-fed beef (limit beef to once or twice a week) if you are eating meat. Avoid processed foods; they are full of additives, preservatives, dyes, and other harmful toxins. One of the most common toxins found in foods is mercury. This heavy metal is more common in large fish such as carp, tilefish, swordfish, tuna, and halibut. Avoid these fish to lower your overall heavy metal exposure. Rice can contain the heavy metal arsenic. Arsenic is naturally occurring in soil, but rice plants suck it up from the soil more than other foods. Soaking rice before cooking it reduces the amount of heavy metals, and limit your rice intake to once or twice a week.

Cookware: Avoid aluminum cookware, as aluminum is linked with reduced brain function and cancer. Teflon contains toxic compounds⁵⁵ that are linked to cancer and other health issues. Cook with stainless steel, ceramic, or cast iron (iron is a metal, but many of us are deficient). Use wooden or stainless-steel utensils instead of plastic, as plastic can leach toxins when heated.

Plastics: Many plastics contain chemicals that are linked with cancer and other health issues.⁵⁶ These can interfere with healthy immune function if you are exposed on a regular basis. Since so many things we use are made from plastic, try to limit exposure. Shop for BPA-free plastic and switch out your plastic dinnerware and water bottles with glass or stainless steel.

5. Stress Management

When you are stressed on a regular basis, it affects immune function, thyroid function, waste/toxin elimination, digestion, and your mental state. Continual stress and anxiety makes it harder for your immune system to fight off candida. Stress directly affects gut health and triggers an imbalance of gut flora,

making it easy for the bad bacteria to take over, leaky gut symptoms to emerge, and autoimmune issue to flare. Below are simple steps you can take to reduce stress, reduce your chances of candida overgrowth, and help your body heal from candida overgrowth. Please also see our Anxiety and Stress Protocol at page 70.

Sleep: You need around eight hours of sleep each night for your body to work as it should. Good sleep helps your immune system function better and helps keep candida at bay. Take supplements if you have trouble with sleep, such as magnesium, valerian, hops, passionflower, chamomile, lavender, and California poppy. Establish a solid bedtime routine that helps your body know when it is time for bed. Avoid staring at blue light (from screens like cell phones and computers) at least two hours before bedtime. See our Sleep Protocol at page 286, if sleep is an issue for you.

Essential Oils and Herbs: If you need extra help reducing anxiety and stress, try using essential oils and herbs that have been proven to help⁵⁷. Essential oils that uplift and calm the mind and body include lavender, chamomile, bergamot, frankincense, and vetiver. You don't have to have an essential oil diffuser; simply add a few drops to a cotton ball and place this in a plastic bag. Take the cotton ball out as needed and take a few deep breaths of the oils when you are feeling stressed or anxious. You can also take herbal supplements to help your body manage stress better. Effective natural remedies for stress include adaptogens like ashwagandha, holy basil, and reishi mushroom. Other stress-relieving nervine herbs include lemon balm, lion's mane mushroom, California poppy, lavender, chamomile, and skullcap.

Breathing Exercises: Deep breathing exercises help get the lymphatic system flowing and reduce stress⁵⁸. Put time aside in the morning, afternoon, and evening to practice deep breathing. Just five to fifteen minutes a few times a day can help immensely. Find a quiet place with minimal distractions and take a deep, slow breath in through your nose, making sure to inhale as much air as you can. Slowly exhale through your mouth, controlling the amount of air you release so you don't release too much at once. Make sure to completely exhale all the air from your lungs. Keep breathing like this, focusing on nothing else but your breathing, for five to fifteen minutes. Optimally, do this in the morning when you wake up; it's a great way to start your day mindfully. If you feel the pressures of stress during the day, take some time to practice deep breathing. Before bed is another great time to practice deep breathing as it helps with sleep.

Time for Yourself: Take a look at your schedule and cut out whatever you can. Put time aside to do things you enjoy, such as playing an instrument, reading, a relaxing bath, or self-care of any kind. Do this on a regular basis, not just every now and then. This can significantly impact your mental and physical health.

Exercise Outdoors: Time and again, exercise and nature come up as top contenders for reducing stress. Not only does exercise reduce stress, it can help boost your

immune system to help fight off candida overgrowth. Exercise outdoors when possible⁵⁹, as studies show a direct correlation between time in nature and stress reduction.

6. Western Medical Approaches When you see a physician for a potential candida infection, they will examine the rash and ask questions to get a better idea of your medical history and symptoms. Occasionally a doctor may run a blood panel and take a culture from an infected area for testing and a more accurate diagnosis.

Most often, a doctor will prescribe an antifungal medication⁶⁰. Some antifungal medications are given intravenously and some are given in pill form. Examples

of commonly prescribed antifungal medications include echinocandins like caspofungin, micafungin, and anidulafungin. These are often given intravenously. Other options include fluconazole, amphotericin B, and other antifungal medications. Dosages vary depending on age and weight. The length of antifungal treatment usually lasts two weeks after symptoms have resolved and the candida is no longer in the bloodstream. For more invasive forms of candidiasis, treatment may last longer.

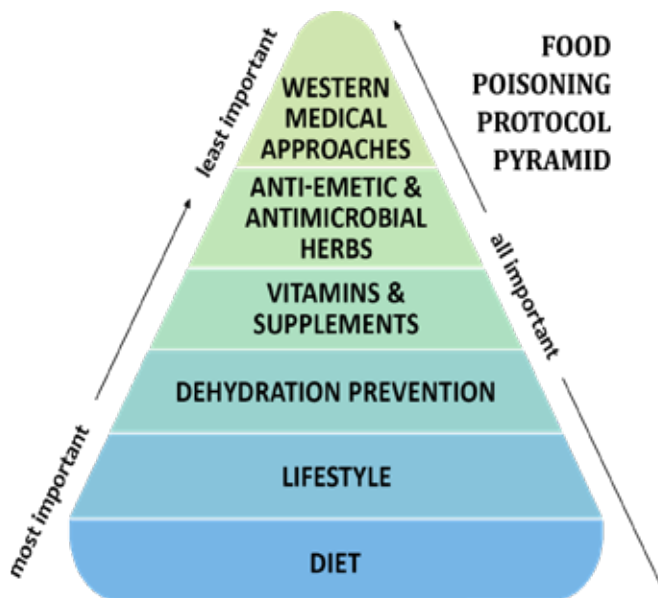
Although serious side effects to antifungal drugs are rare, they do exist. Side effects include liver damage (jaundice), kidney problems, low red blood cell count (anemia), nausea, vomiting, blood pressure changes, and anaphylaxis. Skin reactions like peeling skin or blisters have also been reported.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Mushroom Trio: Reishi, Turkey Tail, & Lion's Mane Dual-extracted mushroom tinctures Vitamin C, Molybdenum, Probiotics, Nicotinamide Candida cleanse broth/gentle solids Oil pulling	Weekly grocery shop for Candida Diet and/or Cleanse Berberine before lunch Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Tincture/Herbs	Candida cleanse broth/gentle solids Outdoor exercise Sauna Japanese Honeysuckle Drink plenty of water Balanced Gut Tincture/Herbs Berberine before dinner	Candida cleanse broth/gentle solids Digestive tea Mushroom Trio Cleavers tincture Probiotics & Turmeric	Berberine Bedtime routine Oil pulling Detox bath Herb-infused coconut oil (for candida rash) Aromatherapy/Sleep herbs, if needed
MONDAY	Mushroom Trio Vitamin C, Molybdenum, Probiotics, Nicotinamide Candida cleanse broth/gentle solids Oil pulling	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs	Candida cleanse broth/gentle solids Outdoor exercise Japanese Honeysuckle Drink plenty of water Balanced Gut Herbs Berberine before dinner	Candida cleanse broth Digestive tea Mushroom Trio Cleavers tincture Probiotics & Turmeric	Berberine Bedtime routine Oil pulling Herb-infused coconut oil (for candida rash) Sleep herbs, if needed
TUESDAY	Mushroom Trio Vitamin C, Molybdenum, Probiotics, Nicotinamide Candida cleanse broth/gentle solids Oil pulling	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs	Fermented food or drink Outdoor exercise Japanese Honeysuckle Drink plenty of water Balanced Gut Herbs Berberine before dinner	Candida diet-friendly dinner Digestive tea Mushroom Trio Cleavers tincture Probiotics & Turmeric	Berberine Bedtime routine Oil pulling Herb-infused coconut oil (for candida rash) Sleep herbs, if needed
WEDNESDAY	Mushroom Trio Vitamin C, Molybdenum, Probiotics, Nicotinamide Candida cleanse broth/gentle solids Oil pulling	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs Lymphatic massage	Fermented food or drink Outdoor exercise Japanese Honeysuckle Drink plenty of water Balanced Gut Herbs Berberine before dinner	Candida dinner Digestive tea Mushroom Trio Cleavers tincture Probiotics & Turmeric	Berberine Bedtime routine Oil pulling Herb-infused coconut oil (for candida rash) Sleep herbs, if needed
THURSDAY	Mushroom Trio Vitamin C, Molybdenum, Probiotics, Nicotinamide Candida cleanse broth/gentle solids Oil pulling	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs	Fermented food or drink Outdoor exercise Japanese Honeysuckle Drink plenty of water Balanced Gut Herbs Berberine before dinner	Candida dinner Digestive tea Mushroom Trio Cleavers tincture Probiotics & Turmeric	Berberine Bedtime routine Oil pulling Detox Bath Herb-infused coconut oil (for candida rash) Aromatherapy/Sleep herbs, if needed
FRIDAY	Mushroom Trio Vitamin C, Molybdenum, Probiotics, Nicotinamide Candida cleanse broth/gentle solids Oil pulling	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs	Fermented food or drink Outdoor exercise Japanese Honeysuckle Drink plenty of water Balanced Gut Herbs Berberine before dinner	Candida dinner Digestive tea Mushroom Trio Cleavers tincture Probiotics & Turmeric	Berberine Bedtime routine Oil pulling Herb-infused coconut oil (for candida rash) Sleep herbs, if needed
SATURDAY	Mushroom Trio Vitamin C, Molybdenum, Probiotics, Nicotinamide Candida cleanse broth/gentle solids Oil pulling	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs Lymphatic massage	Fermented food or drink Outdoor exercise Japanese Honeysuckle Drink plenty of water Balanced Gut Herbs Berberine before dinner	Candida dinner Digestive tea Mushroom Trio Cleavers tincture Probiotics & Turmeric	Berberine Bedtime routine Oil pulling Detox Bath Herb-infused coconut oil (for candida rash) Aromatherapy/Sleep herbs, if needed

* Add caprylic acid daily for short-term/acute use and add oral thrush remedies, if needed.

Ch. 8: Food Poisoning Protocol

Food poisoning, also known as food-borne illness, occurs when we eat food that has been contaminated by a pathogen or toxin. Food-borne infection requires that a live pathogen colonize the gastrointestinal tract after consuming contaminated food. An example is when a live bacterium multiplies in your food and then colonizes your gastrointestinal tract. Food-borne intoxication, on the other hand, occurs when a toxin in the food enters the gastrointestinal tract. Food-borne intoxication is different from food-borne infection because intoxication can occur after ingestion of bacterial toxins, even if a person has not consumed any live bacteria. *Clostridium botulinum* and *Staphylococcus aureus* are two examples of bacteria that produce toxins that can remain in food and cause food poisoning long after the actual bacteria have died. Another example are toxins that naturally exist in poisonous fungi.



Many cases of food-borne illness are self-limiting, which means the immune system can resolve the infection on its own without much assistance; however, extremely severe cases of food-borne infection may require antibiotic treatment. Some types of food-borne intoxication, such as intoxication due to *Clostridium botulinum*, require medical attention because the person needs to be given antitoxin as soon as possible in order to neutralize the toxin and reduce the likelihood of long-term consequences. *Clostridium botulinum* infection does not usually require antibiotics because, as mentioned, the infection is due to the toxin and not the bacterium itself.

Some bacteria are capable of causing both food-borne infection and food-borne intoxication because they invade and colonize the tissues and then they produce damaging toxins. *Shigella dysenteriae* is one such bacterium.

Because food poisoning can be caused by a host of different pathogens and toxins, it can lead to a variety of different symptoms. Common symptoms

are discussed below.

Clostridium Botulinum: *C. botulinum* produces botulinum toxin, which causes neurological symptoms including weakness, vision changes, light sensitivity, difficulty speaking and swallowing, paralysis that spreads downward, and, if left untreated, death. Botulism is most likely to be contracted after eating improperly-canned foods, usually foods that were canned at someone's home (as opposed to commercially available canned goods). It can also occur in infants if they are given honey before age one.

Campylobacter Jejuni: *C. jejuni* is the most common food-borne bacterial pathogen causing gastroenteritis worldwide.¹ It typically causes watery and bloody diarrhea that lasts for 3-7 days, as well as fever and chills. It is mostly self-limiting, but can lead to dehydration, shock, and death in susceptible individuals.

Salmonella Species and Shigella Species: *Salmonella* spp. and *Shigella* spp. cause symptoms that are very similar to those of *Campylobacter jejuni*. They can cause fever, chills, and bloody diarrhea, as well as fatigue, abdominal cramping, and diminished appetite. Most people who experience these types of infections recover on their own within 4-7 days; however, some people may experience more severe symptoms, which can be fatal. Dehydration and an immunocompromised state are major risk factors for severe disease in salmonellosis and shigellosis.

Escherichia Coli: There are several different types of *E. coli* that can cause food poisoning. Symptoms range from mild, watery diarrhea to severe, bloody diarrhea. The strain of *E. coli* that causes diarrhea with large volumes of blood and very little stool is known as *E. coli* O157:H7. Symptoms of *E. coli* O157:H7 can appear up to 3 days after eating contaminated foods, such as under-cooked hamburgers, dried cured salami, unpasteurized fresh-pressed apple cider, yogurt, and cheese made from raw milk.

Amatoxin Poisoning from Mushrooms: *Amanita phalloides* (death cap mushrooms) and other related mushrooms produce a toxin called amatoxin that can cause liver and kidney failure, coma, and death when ingested. Symptoms of amatoxin poisoning include gastrointestinal distress, confusion, vision changes, heart problems, liver and kidney failure, coma, and eventually death.

In this chapter, we'll discuss ways to prevent, treat, and support a person through various types of food poisoning.

1. Diet In the event of food poisoning, a bland diet is best. Bland diets, which are also referred to as soft diets or low-residue diets, are the ideal option because they contain foods that are easier to break down and digest. Because they put less stress on the gastrointestinal tract, bland diets allow the gastrointestinal system to rest and to heal more effectively.² Because they contain minimal fiber, these types of diets minimize the size and quantity of bowel movements. This is especially

beneficial in food poisoning that manifests as diarrhea or runny stool, as it can help to prevent dehydration.

The foods that are typically included on bland diets are soft, easily digestible, low-fiber, usually cooked, only lightly-seasoned, and not spicy. Examples of common foods on this diet include beets, beans, spinach, carrots, broth, fruit juices, cream of wheat, tofu, low-fat dairy products, eggs, pudding, lean meat, and herbal tea. Foods that are typically avoided include fried foods; spices; seeds and nuts; acidic fruits, such as berries, grapes, oranges, lemons, and limes; whole-grain products, such as bread, pasta, and cereals; fatty dairy products, such as whole milk, whipped cream, and ice-cream; non-lean meat, such as beef, fried fish, chicken that includes the skin; dressings and sauces; pickles; alcoholic beverages; strong cheeses; and vegetables that cause excess bloating or flatulence, such as cabbage, cauliflower, onions, and peppers.³

It's important to note that some people find a bland diet difficult to follow, even if only for a few days. This is reportedly because this diet requires food to be only lightly seasoned and because there seems to be lots of restricted and permitted foods to remember on this diet. The lack of seasonings and flavor (compared to what most people are used to) and the extensive list of what can and cannot be eaten make the diet difficult to stick to for some people.

2. Lifestyle

Because some people find the diet difficult to adhere to, some people advocate for an easier to remember variation of the bland diet, one that includes foods that don't necessarily need to be highly seasoned and that only includes a few easy-to-remember foods. This variation is called the BRAT diet, and it consists of bananas, rice, applesauce, and toast (which is what the letters BRAT stand for). These four foods are considered to be low-residue, and a diet consisting of these four foods is typically acceptable short-term while the gastrointestinal tract and body are recovering from food poisoning.

Lifestyle factors such as kitchen cleanliness and personal hygiene play significant roles in preventing and limiting the spread of pathogens that can cause food poisoning.

Clean: To reduce the likelihood of food-borne illness or to prevent its spread once one family member has fallen ill, it is imperative that kitchen counters and other surfaces in the home be thoroughly and frequently cleaned. It's also important for members of the home to wash their hands frequently using soap and warm water for at least 20 seconds. Hands should be washed before and after handling food; after using the bathroom; after changing a diaper; after tending to or caring for a sick person; after blowing one's nose, coughing, or sneezing; after handling uncooked eggs or raw meat, poultry, or fish and their juices; and after touching or being in contact with pets.⁴ Be sure to wash fruits and vegetables thoroughly before eating them.

Separate: In order to prevent cross-contamination, vegetables and raw meat should be chopped or cut on

separate cutting boards, or the cutting boards should be washed with hot, soapy water between uses. You may also sanitize cutting boards with a bleach-containing cleaning solution (one tablespoon of bleach combined with one gallon of water). To sanitize, saturate the cutting board with the solution, allow it to sit for several minutes, and then rinse and dry the board with clean paper towels. Avoid using utensils and plates or dishes that have been in contact with raw meat or food products to serve cooked foods without thoroughly cleansing them. This helps ensure that the food that has been safely cooked is not recontaminated by the uncooked juices of the meat.

Cook: In order to reduce the likelihood of pathogens multiplying in improperly cooked meat, it is important to ensure that foods are cooked to appropriate temperatures. Ground meat and eggs should be cooked to at least 160°F, poultry should be cooked to at least 165°F, and fresh meat steaks, chops, and roasts should be cooked to at least 145°F.⁵ Use a food thermometer to ensure that foods are cooked to appropriate temperatures. Be sure to thoroughly cleanse the food thermometer's probe with hot soapy water between uses to make sure that you are not recontaminating meat that has been appropriately cooked.

Refrigerate: To help ensure that any bacteria remaining on your food is not given the opportunity to multiply and cause food-borne illness, keep your refrigerator colder than 40°F and your freezer below 0°F. Keep foods that are marked "keep refrigerated" in the fridge and refrigerate cooked food promptly. Cooling your cooked food rapidly helps ensure that the food quickly reaches the safe refrigerator storage temperature of 40°F without bacteria being given extensive time to multiply.⁶ To ensure that foods are rapidly cooled, place leftovers in smaller, shallow containers to help them cool faster before placing in the fridge. You can place hot food directly in the refrigerator or you can choose to chill the food rapidly by placing it in an ice- or cold water-bath, and then placing it in the refrigerator.

Play it Safe: To reduce the likelihood of contracting food-borne illness, avoid eating food that was stored in cans that are dented, bulging, or leaking, because this may indicate that the food has been contaminated and is not safe to eat. Additionally, never give honey to children under one year of age, as this can lead to infantile botulism (caused by the bacterium *Clostridium botulinum*), which can be fatal.

3. Dehydration Prevention

Dehydration is a major concern when dealing with food poisoning. The vomiting and diarrhea that can come along with food-borne illness increase your risk for fluid and electrolyte loss.

Signs of dehydration can include thirst; dry lips, tongue, mouth, eyes, or skin; headache; fatigue or weakness; feeling dizzy or lightheaded; lack of focus; nausea; cramping; rapid breathing; and dark yellow or strong-smelling urine.⁷ Many cases of food-borne illness are self-limiting, meaning the immune system is

eventually able to fight off the pathogen or toxin that is causing the illness and return the body to a state of wellness; however, dehydration can lead to even these cases becoming life-threatening emergencies. Prioritize hydration when fighting food-borne illnesses.

In some cases of food poisoning, intense nausea can make it difficult to drink water without regurgitating, except in very small amounts. As a result, it is extremely important to rest and refrain from intense physical exertion, because physical activity can lead to sweating and more fluid loss.

Electrolytes are minerals that the body needs to function optimally. When we lose fluids through vomiting or diarrhea, we also lose electrolytes. Electrolytes need to be replenished. The most important minerals that need to be replenished are sodium, potassium, and magnesium.

If drinking larger volumes of water becomes difficult because of nausea, consider sucking on ice chips or taking sips of water frequently throughout the day. You can replenish water and electrolytes simultaneously by adding powdered electrolyte mixes or electrolyte tablets to your water or by drinking (healthy) sports drinks. In a pinch, you can also make your own homemade electrolyte drink using unsweetened coconut water, Himalayan pink salt, and citrus juice⁸.

4. Vitamins and Supplements

There are a few supplements that may be of benefit during food poisoning. These include activated charcoal, probiotics, and milk thistle.

Activated Charcoal: Activated charcoal is a special form of carbon that is commonly used to treat food poisoning. Activated charcoal works because the charcoal can adhere to poisons or drugs that have been swallowed. After adhering to the toxins, the charcoal then prevents the toxins from being absorbed in the gastrointestinal tract and entering the bloodstream. Activated charcoal is most effective up to one hour after a person consumes a toxin that can be adsorbed by activated charcoal.⁹ It's important to note that oral ingestion of activated charcoal can cause black-colored stool and that activated charcoal should not be given to anyone who is unconscious or whose consciousness is impaired, neither should it be given to anyone whose airway is obstructed or otherwise compromised.¹⁰ Activated charcoal can also interfere with the absorption of certain drugs, vitamins, and minerals if taken together, so it's best to take your medication and supplements 1-2 hours before taking a dose of activated charcoal. Suggested dose is 0.5–1g per kg of body weight in children or 50 g in adults, thoroughly mixed in water and consumed as soon as possible after toxin ingestion.¹¹

Bentonite Clay: Bentonite clay can also absorb toxic organic and inorganic materials in the gastrointestinal tract; and it does so without negatively affecting mineral metabolism and absorption.¹² In one study, orally administered bentonite clay successfully addressed 97% of cases of diarrhea, including diarrhea caused by viral infections and other forms of food poisoning.¹³

Before purchasing or using bentonite clay, however, it's important to note that the United States Food and Drug Administration (FDA) has found elevated levels of lead in some brands, so make sure this isn't the case with the product you purchase if you intend to take Bentonite clay internally.¹⁴ Suggested dose is 0.5 to 1 teaspoon of bentonite clay mixed in water per day.

Probiotics: Food poisoning can disrupt the body's normal gut bacteria. Probiotics are live bacteria that are similar to the bacteria in your gut and can be taken in the form of foods or supplements. Taking probiotics can help restore the balance between good and potentially harmful bacteria in the gut. It can also reduce symptoms of food-borne illness. For example, the probiotic yeast *Saccharomyces boulardii* significantly reduces the duration of watery diarrhea in a randomized, controlled trial.¹⁵ Probiotic bacterial strains such as *Lactobacillus GG* and *Lactobacillus reuteri* are beneficial in addressing symptoms of food-borne illness.¹⁶ Because they help support gut bacteria diversity, taking probiotics regularly may also help reduce the likelihood of you experiencing symptoms of food poisoning if exposed to food-borne pathogens or toxins in the future. Suggested dose: 100 billion CFU while symptoms persist.

Milk Thistle (*Silybum marianum*): Silibinin is a substance derived from the milk thistle plant. It can be used to combat some forms of mushroom poisoning, specifically poisoning with amatoxin-containing mushrooms including *Amanita phalloides* (death cap mushroom) and related species.¹⁷ These mushrooms cause serious liver damage and death. Silibinin is beneficial due to its hepatoprotective properties;¹⁸ it protects the liver by preventing the amatoxins from reaching and damaging liver cells. Silibinin is typically administered intravenously to address mushroom toxicity, because this bypasses the gastrointestinal tract leading to enhanced bioavailability. If dealing with mushroom toxicity, it is best to seek emergent medical care in order to prevent serious liver damage and death; however, if medical care is not available and it becomes necessary to address this concern using milk thistle (and not silibinin), you would need to take very high doses of milk thistle seed in order to see a therapeutic effect. Suggested dose: 13 grams of milk thistle seed per day.

5. Anti-Emetic and Antimicrobial Herbs

Anti-emetic and antimicrobial herbs can be of benefit when addressing food-borne illnesses. These include peppermint, ginger, thyme, clove, pomegranate extract, black walnut, Japanese knotweed, garlic, and cumin. Adding antimicrobial essential oils like thyme, oregano, and pennyroyal oils to food during the food preservation processes may also help reduce the likelihood of those who eat these foods contracting food-borne illnesses.

Peppermint (*Mentha piperita*): Peppermint has anti-emetic properties, meaning it can help reduce nausea and decrease the likelihood of vomiting. It can help to settle the stomach. Inhalation of peppermint essential

oil (aromatherapy) is beneficial in reducing nausea and vomiting.^{19 20 21} Furthermore, peppermint tea exhibits significant antibacterial and antiviral activity.²² Note that long-term internal use of peppermint is discouraged as its high tannin content can lead to impaired iron absorption. Also note that, in many cases of food poisoning, vomiting can be helpful since it helps the body rid itself of the pathogen or toxin that is causing the illness, so anti-emetics like peppermint should be used sparingly and only if vomiting is causing or has the potential to cause significant problems (e.g., if you're at risk for dehydration or mineral imbalance because of repeated vomiting). Suggested dose is 2-5 grams of peppermint leaf per quart of water per day (as an infusion); as a standard extract or glycerin extract, use 5-10 drops of a 1:5 to 1:3 extract per day.

Ginger (*Zingiber officinale*), Thyme (*Thymus vulgaris*), Clove, and Pomegranate Extract: Ginger has anti-emetic properties;²³ it has been used successfully for centuries in the treatment of nausea and vomiting. In addition, ginger, thyme, clove, and pomegranate extract were all found to have antimicrobial activity against multiple strains of bacteria that are known to be responsible for food-borne illnesses.²⁴ These bacterial strains include *Bacillus cereus*, *Staphylococcus aureus*, *Escherichia coli*, *Pseudomonas aeruginosa*, and *Salmonella typhi*. Of the four plants and plant extracts, clove and pomegranate extract showed the most antimicrobial activity. They exhibited both bacteriostatic and bacteriocidal activity, meaning that these plants were able to prevent the bacteria from reproducing and they were also able to kill the bacteria.

Suggested dose of ginger: 1-2 teaspoons of chopped ginger per cup of water, three times daily; 0.5 to 1 dropperful of a 1:3 tincture, 3 times daily.

Suggested dose of thyme: 1-2 teaspoons of dried thyme leaves per cup of water, three times daily; 5 dropperfuls of a 1:5 tincture, 3 times daily.

Suggested dose of clove: 1 dropperful of a 1:3 tincture, 3 times daily.

Suggested dose of pomegranate extract: 750 mg of pomegranate extract per day, in divided doses.

Black Walnut (*Juglans nigra*): As a result of its high tannin content, black walnut has strong antimicrobial activity against *Listeria monocytogenes*, a bacterium that commonly causes food-borne infection.²⁵ An extract from the hull of black walnut has also been found to have antimicrobial activity against *Staphylococcus aureus*,²⁶ a bacterium that is capable of causing food-borne intoxication. Suggested dose: 2 dropperfuls of a 1:3 tincture, 2 times daily.

Japanese Knotweed (*Polygonum cuspidatum*): Japanese knotweed also possesses antimicrobial properties. An ethanolic extract has antimicrobial activity against multiple strains of *Staphylococcus aureus* and *Pseudomonas aeruginosa*,²⁷ both of which can cause food-borne illness and a host of other concerns. Note that

Japanese knotweed should not be taken by individuals who are on blood thinners or during the 10 days prior to surgery. Suggested dose: 1 dropperful of a 1:5 tincture, 3 times daily.

Garlic: Garlic possesses powerful antimicrobial and immune-modulating properties. It is beneficial in addressing food-borne illness due to *Campylobacter jejuni* infection. Garlic essential oil preserves the health of the cells of the colon, reduces the immune response in the large intestine, decreases the amount of pro-inflammatory messengers that are secreted, and reduces the likelihood of *Campylobacter jejuni* moving from the intestine to other organs where it could cause further problems.²⁸ Other forms of garlic, such as garlic concentrates and allicin extracts, also have antimicrobial activity against *Campylobacter jejuni*.²⁹ Aqueous extracts of garlic (teas) exhibit superior antimicrobial activity against multiple drug-resistant *Shigella dysenteriae* 1, *Shigella flexneri* Y, *Shigella sonnei*, and enterotoxigenic *Escherichia coli*.³⁰ Note that high doses of garlic (greater than 5 grams per day) should not be taken by individuals undergoing warfarin therapy, and garlic supplementation should be discontinued 10 days prior to surgery. Suggested dose: 2-6 cloves of garlic per day; 5-12 mg of allicin extract in tablet form per day.

Cumin: The essential oil extracted from cumin seed has an inhibitory effect on various microorganisms that are known to cause food spoilage and food-borne illnesses.³¹ This includes *Klebsiella pneumoniae*, *Vibrio* spp., *Staphylococcus aureus*, *Bacillus cereus*, *Escherichia coli* O157:H7, *Salmonella enteritidis*, and *Listeria monocytogenes*. Other forms of cumin extracts also exhibit antimicrobial activity.³² Suggested dose: 2 dropperfuls of a 1:3 tincture, 2 times daily.

Thyme (*Thymus vulgaris*), Oregano (*Origanum vulgare*), and Pennyroyal Oils: Thyme, oregano, and pennyroyal essential oils show significant antimicrobial activity against a wide range of bacteria, including *Listeria monocytogenes*, *Clostridium perfringens*, *Bacillus cereus*, *Staphylococcus aureus*, *Enterococcus faecium*, *Enterococcus faecalis*, *Staphylococcus epidermidis*, *Salmonella enterica*, *Escherichia coli*, and *Pseudomonas aeruginosa*.³³ Of the three, thyme oil was effective at inhibiting bacteria, even at very low concentrations, leading researchers to conclude that adding these essential oils to food may decrease the likelihood of food spoiling or causing food-borne illness. Note that while these essential oils may be safely added to food in small doses, they should not be taken internally in essential oil form to address food poisoning, because there are compounds in these oils that can cause liver damage and other concerns in high doses.

6. Western Medical Approaches Food-borne illness is frequently self-limiting, meaning that the body is usually able to clear the pathogen or toxin that is causing the illness on its own, without much assistance. This is why treatment for food-borne illness is usually

supportive in nature.

There are a few key factors, however, that can lead to food-borne illness becoming life-threatening, and it is important for you to know and be able to recognize when food poisoning may be life-threatening so that you can seek appropriate care.

Threat of Dehydration: As previously mentioned, dehydration can lead to an otherwise routine course of food-borne illness becoming fatal. This is because dehydration can lead to hypovolemic shock if not corrected, and hypovolemic shock can lead to rapid death.

As a result, it is imperative that you recognize the signs of dehydration (thirst; dry lips, tongue, mouth, eyes, or skin; headache; fatigue or weakness; feeling dizzy or lightheaded; lack of focus; nausea; cramping; rapid breathing; and dark yellow or strong-smelling urine) and take steps to prevent and correct it. In the event that there are factors present that would increase your risk for dehydration, it is important to go to your nearest hospital for intravenous rehydration.

Factors that would increase your risk of dehydration include a fever above 100.4°F that does not abate, diarrhea that persists for longer than 3 days, or nausea and/or vomiting that make it difficult to drink sufficient amounts of water.

Blood in Stool or Vomit: There are multiple strains of bacteria that are known to cause damage to the gastrointestinal tract, leading to bloody diarrhea. Of these, the most common offenders in the developed world are *Campylobacter jejuni*, *Escherichia coli* O157:H7, other Shiga toxin-producing *Escherichia coli*, *Salmonella* species, *Shigella* species, and *Yersinia* species.³⁴

These microbes can wreak havoc on the body, even leading to rapid death. As a result, if you are experiencing food poisoning with blood in your stool or vomit, it is important to seek emergency medical care so that you can receive the medical support you need. In these cases, physicians may prescribe antibiotics in order to kill the pathogen and they will also likely rehydrate your body using IV fluids.

Signs of Dropping Blood Pressure: If you begin to feel lightheaded or experience fainting or blurred vision upon standing, this may be a sign that your blood pressure is dropping. This is of major concern when dealing with food-borne illness, because it may be a sign that your body is entering into a state of hypovolemic shock.

This can happen if the bacterium or toxin that is causing the illness multiplies or spreads systemically, leading to your immune system redoubling its efforts to fight it off. The massive immune response that ensues leads to white blood cells and fluid leaving your blood stream and entering your tissues in order to fight off the invader.

This may sound like it would be a good thing; however, if too much fluid leaves the blood stream, blood pressure drops; the heart is no longer able to pump

enough blood, oxygen, and nutrients to the tissues; and this leads to hypovolemic shock and death.

This is why it's important to seek emergency medical care at the first sign of decreasing blood pressure when dealing with food-borne infection or intoxication.

Neurological Concerns: Neurological concerns such as tingling limbs, blurry vision, or paralysis always require prompt medical attention. This is especially the case if the symptoms arise after having eaten home-canned goods, because neurological concerns when dealing with food-borne illness are frequently a sign of botulism.

Botulism is caused by the bacterium *Clostridium botulinum*. It is a type of food-borne intoxication where the bacterium produces toxins that can cause flaccid paralysis. If you are dealing with botulism, it is important to seek emergency medical care so that doctors can administer antitoxin in order to neutralize the botulinum toxin.

Threat of Liver or Kidney Failure: Mushroom poisoning with amatoxin-containing mushrooms can cause liver failure, kidney failure, coma, and death. It's important to be able to recognize the signs of potentially fatal mushroom poisoning.

Recognizing mushroom poisoning can be tricky, but your life or the life of someone else may depend on it. Here's how mushroom poisoning usually begins: about 6 to 24 hours after ingesting toxic mushrooms, you'll likely begin to experience severe vomiting, abdominal cramping, and diarrhea.

You may also experience dizziness, headache, or coughing. Symptoms may persist for 1-2 days, followed by a period of improvement lasting about 1-3 day(s).

This recovery period frequently leads to people thinking they've gotten over the worst of the poisoning and not seeking medical care; however, it is followed by a return of the symptoms lasting 3-5 days, as well as jaundice (a sign of which is yellowing of the skin and eyes). Coma and death due to liver and kidney failure usually occur 1-2 weeks after eating the mushroom.

To summarize, if you become ill with food poisoning symptoms within 1-2 days of eating wild or foraged mushrooms, do not fall into the trap of thinking you've sufficiently recovered and don't need to seek treatment. It is imperative that you seek emergency medical care as soon as possible in order to reduce the likelihood of coma and death.

As mentioned, it is critical that you be able to distinguish between food poisoning that is likely self-limiting and food poisoning that is progressing to the state where you need emergency medical care.

At the hospital, health care professionals may administer appropriate antibiotics, antitoxin, rehydration therapy, or other supportive therapies in order to help your body recover from the food-borne illness.

	Morning	Noon	Afternoon	Evening	Before Bed
S U N D A Y	<u>With a meal:</u> Pomegranate extract Allicin extract Black walnut or Japanese knotweed tincture*	Activated charcoal** Mid-day hydration & electrolyte check (Are you drinking enough water & getting enough minerals?)	<u>With a meal:</u> Probiotics Milk thistle seed	Early evening hydration & electrolyte check (Are you drinking enough water & minerals?) Bentonite clay***	Unsweetened peppermint tea
M O N D A Y	<u>With a meal:</u> Pomegranate extract Allicin extract Black walnut or Japanese knotweed tincture*	Hydration & electrolyte check Clove tincture	<u>With a meal:</u> Probiotics Milk thistle seed	Hydration & electrolyte check Bentonite clay***	Unsweetened ginger tea
T U E S D A Y	<u>With a meal:</u> Pomegranate extract Allicin extract Black walnut or Japanese knotweed tincture*	Hydration & electrolyte check Clove tincture	<u>With a meal:</u> Probiotics Milk thistle seed	Hydration & electrolyte check Bentonite clay*** Light walking or physical activity	Unsweetened peppermint tea
W E D N E S D A Y	<u>With a meal:</u> Pomegranate extract Allicin extract Black walnut or Japanese knotweed tincture*	Hydration & electrolyte check Thyme tincture	<u>With a meal:</u> Probiotics Milk thistle seed	Hydration & electrolyte check Bentonite clay***	Unsweetened ginger tea
T H U R S D A Y	<u>With a meal:</u> Pomegranate extract Allicin extract Black walnut or Japanese knotweed tincture*	Hydration & electrolyte check Clove tincture	<u>With a meal:</u> Probiotics Milk thistle seed	Hydration & electrolyte check Bentonite clay*** Light walking or physical activity	Unsweetened peppermint tea
F R I D A Y	<u>With a meal:</u> Pomegranate extract Allicin extract Black walnut or Japanese knotweed tincture*	Hydration & electrolyte check Clove tincture	<u>With a meal:</u> Probiotics Milk thistle seed	Hydration & electrolyte check Bentonite clay***	Unsweetened ginger tea
S A T U R D A Y	<u>With a meal:</u> Pomegranate extract Allicin extract Black walnut or Japanese knotweed tincture*	Hydration & electrolyte check Thyme tincture	<u>With a meal:</u> Probiotics Milk thistle seed	Hydration & electrolyte check Bentonite clay*** Light walking or physical activity	Unsweetened peppermint tea

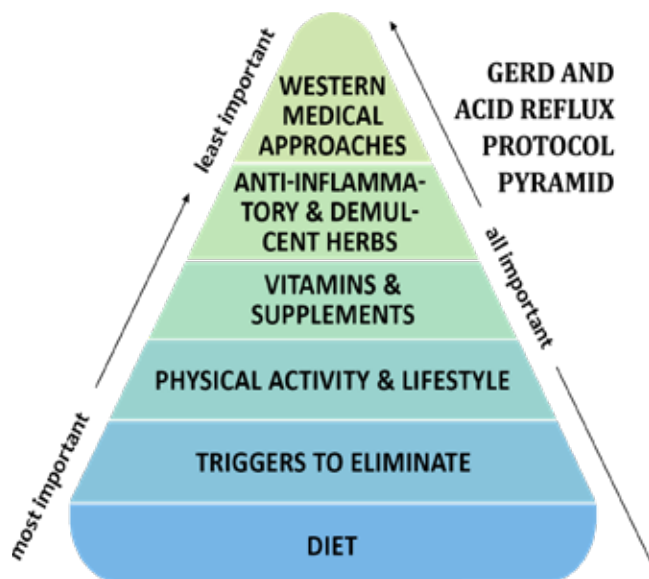
*Avoid Japanese knotweed tincture if on blood thinners or within 10 days of a surgical procedure.

**Activated charcoal: 0.5–1g per kg of body weight (children) or 50g (adults), mixed in water and consumed as soon as possible, after toxin ingestion.

***Bentonite clay: 0.5 to 1 teaspoon of bentonite clay mixed in water per day for 1-2 weeks.

Ch. 9: GERD and Acid Reflux Protocol

Acid reflux occurs when your stomach contents back up into your esophagus. Approximately 25 to 40 percent of Americans suffer from a more severe form of acid reflux: gastroesophageal reflux disease (GERD). GERD is associated with chronic (more than twice a week) reflux of stomach acid into the esophagus, throat, and mouth, and can develop across all age ranges—from babies to the elderly. It often causes a burning sensation or chest pain that may mimic the pain of a heart attack – a symptom called heartburn. GERD tends to worsen when you lie flat, and the acid that travels up the esophagus can cause damage to your tooth enamel. Sore throat, bloating, hiccups, difficulty swallowing, and a lump in the throat are also indicators of GERD.



In babies, symptoms include spitting up repeatedly, discomfort after eating, refusing food, and sleep issues. Nausea and vomiting after eating, trouble breathing, a bitter taste in the mouth, black stools, and difficulty swallowing often indicate GERD in older children or adults. GERD is more likely to occur in individuals who are overweight or pregnant, due to increased pressure on the stomach. Quantity and concentration of stomach acid as well as the presence of localized inflammation have been linked to the development of GERD.

When you swallow, food travels down your esophagus, a muscular tube that connects the mouth to the stomach. At the junction of the esophagus and the stomach, there is a muscular ring called the lower esophageal sphincter. This ring is normally closed; it functions to keep stomach contents from flowing back into the esophagus. About two seconds after you swallow, the muscular lower esophageal sphincter relaxes to let food into the stomach and the ring stays open for six to ten seconds before closing again.¹

In GERD, the lower esophageal sphincter has a harder time closing and remaining closed. This can be due to factors that cause a reduction in the pressure or

tension of the lower esophageal sphincter. It can also be due to factors that lead to an increase in the pressure within or on the stomach, stimulation of receptors that perceive sensations in the esophagus, or changes in the secretion of stomach acid.²

1. Diet High-Fiber Anti-Inflammatory Diets support optimal tension in the lower esophageal sphincter and are associated with fewer episodes of reflux and heartburn in people suffering with GERD.³ Each of the following foods can increase your risk for GERD:^{4 5}

Fatty and Fried Foods: Fatty foods cause the lower esophageal sphincter to relax so that the acidic contents of the stomach are more likely to flow backward into the esophagus, leading to GERD symptoms. Fatty foods also increase the length of time it takes for food to leave the stomach and enter the small intestine. This leads to increased pressure in the stomach for a prolonged period of time and makes GERD symptoms more likely.^{6 7}

Spicy and Sour Foods: Spicy foods contain plant products called alkaloids that stimulate sensory receptors in the esophagus.⁸ This can lead to pain and burning sensations typical of GERD. You are especially likely to feel painful burning sensations if the lining of your esophagus is already inflamed.⁹ Sour foods tend to have a low pH, which is associated with gastroesophageal reflux.¹⁰

Garlic and Onion: Some compounds in garlic can irritate the digestive tract and produce a burning sensation in the abdomen and chest for some people.¹¹ Onions, which are also in the Allium family of plants, have been associated with a higher number of heartburn and reflux episodes, lower pH in the stomach, and more frequent burping in people with GERD.¹² Onion consumption does not produce these effects in healthy individuals.

Tomatoes and Tomato Juice: Raw tomatoes and tomato products such as tomato juice and tomato sauce can exacerbate GERD symptoms,¹³ perhaps due to compounds that increase the secretion of stomach acid. Higher levels of stomach acid lead to increased pressure in the stomach and make it more likely that the stomach acid will flow backward into the esophagus.

Chocolate and Chocolate Products: Regular consumption of chocolate and chocolate-containing products increases GERD symptoms and exposes the lower esophageal sphincter to stomach acid.¹⁴

Carbonated and Caffeinated Beverages: Consumption of carbonated and caffeinated beverages is associated with an increase in GERD symptoms.¹⁵ This may be because carbonated beverages lead to a reduction in pH of the stomach¹⁶ and because both carbonated and caffeinated beverages have been found to cause relaxation of the lower esophageal sphincter, which makes it easier for stomach acid to flow back into the esophagus.

Peppermint Tea (*Mentha piperita*): In one study, people who frequently drank peppermint tea were more likely to experience GERD symptoms than people who did not; however, researchers noted that we should be careful about how we interpret these results since the

study design made it difficult to tell if the GERD symptoms led to the individuals drinking peppermint tea or if the peppermint tea led to the GERD symptoms.¹⁷

It's important to take note of how these foods affect you, so that you can avoid the ones that trigger your symptoms. Making these dietary adjustments can have a significant impact on the frequency and discomfort of GERD. Opt for still water instead of alcohol or carbonated beverages. Eat foods such as ginger, bone broth, fermented and cultured foods, apple cider vinegar, and fiber-rich oatmeal and psyllium¹⁸. Minimize the consumption of high-fat foods, vegetable oils, processed foods, chocolate, caffeine, citrus, tomatoes, spicy food, and mint.

2. Triggers to Eliminate In addition to the foods you eat, there are other factors and habits that can trigger GERD. These factors are related to eating and drinking, weight, and drug and alcohol use.

Avoid Lying Down or Bending Over after Eating: People who suffer from GERD typically have decreased pressure in their lower esophageal sphincter, meaning that the sphincter tends not to close as tightly. Lying down or bending over aggravates symptoms because, instead of moving against gravity (which is the case when in an upright position), the stomach acid can flow with gravity when bending over or reclining and enter the esophagus. Consider taking a relaxing stroll after eating; people who do this regularly have significantly fewer GERD symptoms.¹⁹ If you're not able to go for a walk after your meals, remain in an upright position, as this leads to fewer GERD symptoms than lying down (although people who did this tended to experience more symptoms than people who took a post-meal stroll).

Avoid Overeating/Eat Smaller Meals: Consuming large volumes of food at mealtimes is associated with increased risk for GERD symptoms.²⁰ Keep portion sizes reasonable and do not continue to eat after you're full. Eating at the same times each day may also help. In addition, chew your food thoroughly.

Limit Drinking with Meals: Just like overeating, drinking large amounts of liquids increases the volume of chyme contained within the stomach. This leads to increased pressure on the lower esophageal sphincter, making reflux more likely. Prioritize consuming the majority of your fluids at least 30 minutes before or after your meals.

Avoid Vigorous Exercise after Eating: High-intensity physical activity triggers GERD symptoms in some people. It's important to note that light or short exercise sessions did not increase GERD symptoms.²¹

Increase Your Dinner-to-Bed Time Interval: Having your last meal 3-4 hours or more before bedtime leads to decreased likelihood of experiencing GERD symptoms.^{22 23}

Elevate the Head of Your Bed and Lie on Your Left Side: When preparing for bed, keep in mind that sleeping at an incline or elevating the head of your bed decreases the likelihood of stomach acid flowing back into your esophagus while you are sleeping.²⁴ Because of the anatomical shape of the stomach, lying on your left side

also produces fewer GERD symptoms than lying on your right side for most people.

Maintain a Healthy Weight: To reduce GERD symptoms, it's important to maintain a healthy weight. Weight gain increases risk for GERD symptoms, even if the body-mass index (BMI) remains in the normal range. Additional weight gain, leading to abnormal BMI (e.g., overweight or obesity) significantly increases risk for GERD.²⁵

Avoid Restrictive Clothing: For some people, wearing belts or other tight bands around the waist increases pressure on the stomach, making it more likely that stomach acid will flow backwards into the esophagus.²⁶

Avoid Caffeine, Alcohol, and Tobacco: Caffeinated beverages are associated with an increase in GERD symptoms, likely because they cause the lower esophageal sphincter to relax, making it easier for stomach acid to flow back into the esophagus.²⁷ Alcohol consumption is strongly associated with GERD, especially at higher doses, because it can lead to relaxation of the lower esophageal sphincter.²⁸ This is why people with GERD are typically encouraged to eliminate or significantly reduce their alcohol consumption. Smoking tobacco is associated with increased relaxation of the lower esophageal sphincter and more frequent episodes of symptomatic GERD.^{29 30}

3. Physical Activity and Lifestyle Both high-intensity exercise and being overweight or obese are associated with GERD. Physical activity is an essential part of maintaining a healthy weight, so it's important to ensure that your exercise routine reduces and does not exacerbate your GERD symptoms.

One way to engage in physical activity is taking a walk after meals.³¹ It's also important to prioritize low-to moderate-intensity exercises over high-intensity exercise and to engage more frequently in shorter sessions of physical activity over longer sessions. Instead of exercising for one hour per day, try breaking your exercise session up into four 15-minute sessions throughout the day. Try different combinations of length and intensity to find the combination that works best for you and does not exacerbate your symptoms. Keep in mind that you should have your lowest intensity physical activity session (for example, that post-meal walk) after eating, since vigorous activity in the 2-3 hours after a meal can trigger GERD symptoms.

4. Vitamins and Supplements There are several supplements that are beneficial in addressing gastroesophageal reflux disease. These include melatonin; vitamins A, C, E, B6, folate (B9), and B12; probiotics; and digestive enzymes.

Melatonin: Melatonin has antioxidant properties and protects the most superficial layer of the gastrointestinal tract, including the esophagus, from oxidative damage. Melatonin can help reduce GERD symptoms whether used alone; in combination with other natural supplements like vitamins B6, folate, and B12; or in combination with the proton-pump inhibitor omeprazole.^{32 33 34} Suggested dose: 6 mg once daily at bedtime.

Vitamins A, C, and E: Vitamins A, C, and E are powerful antioxidants. People who consume higher amounts of these vitamins, both through diet and through supplementation, are less likely to report symptoms of GERD.³⁵ Suggested dose: Vitamin A, 700-900 mcg RAE daily; Vitamin C, 100 mg per day, ideally in the form of calcium ascorbate or another buffered form of vitamin C; Vitamin E, 15 mg daily.

Vitamins B6, B9, and B12: At high doses, vitamins B6 and B12 can alleviate acute pain because they inhibit pain signals within the body.³⁶ These vitamins are water-soluble, so your body is able to get rid of what you don't need through your urine if your kidney function is normal. Your body also needs vitamin B9, which is also known as folate, in order to produce S-adenosyl-L-methionine (SAME). SAME may help reduce inflammation and pain without causing damage to the gastrointestinal tract.³⁷ Suggested dose: Vitamin B6, 25 mg per day; Folic acid, 10 mg per day; Vitamin B12, 50 mcg per day.³⁸

Probiotics: We usually think of probiotics as being beneficial to the large intestine and other parts of the lower gastrointestinal tract, but they have positive effects in the upper GI tract as well. Probiotic use is associated with improvements in digestion, reflux, nausea, and gas-related symptoms.³⁹ Suggested dose: 20 billion CFU per day, taken with meals.

Digestive Enzymes: Digestive enzymes such as papain, which is extracted from the papaya plant, help improve constipation and bloating.⁴⁰ More research is needed to form a definitive conclusion about the use of digestive enzymes in GERD; however, preparations containing papain are more effective than placebo in addressing GERD symptoms.⁴¹ Suggested dose: 20 grams of papaya pulp or 25-100 mg of extracted papain per day.

5. Anti-Inflammatory and Demulcent Herbs

If you are suffering from GERD or other digestive issues, including leaky gut, anti-inflammatory and demulcent herbs are an outstanding option for symptom relief and healing. Examples include STW-5 (Iberogast), Ginger, Deglycyrrhized Licorice, Turkey Tail, Reishi, Lion's Mane, Plantain, Marshmallow, and Slippery Elm.

STW-5 (Iberogast): STW-5 (brand name Iberogast) is an herbal preparation consisting of 9 medicinal plants that is effective in the treatment of multiple gastrointestinal disorders.⁴² Research suggests it is highly beneficial in addressing GERD symptoms,⁴³ and this is partly due to its function as a prokinetic. In other words, STW-5 helps move food forward from the stomach into the small intestine, instead of backward toward the esophagus. Its benefits are also due to the presence of demulcent and inflammation-modulating herbs like licorice and chamomile. Suggested dose: 1 ml (20 drops) 3 times per day, immediately before meals.

Ginger (*Zingiber officinale*): Ginger has been used for centuries as a digestive aid because of its anti-inflammatory, carminative, anti-nausea, and anti-emetic properties. It is highly beneficial in addressing symptoms of GERD.⁴⁴ Suggested dose: 3 cups per day (1 tsp. dried ginger or 3-5 grams fresh grated ginger per

cup of water) or 3-5 ml of 1:5 ethanolic extract/tincture, 3 times per day.

Deglycyrrhized Licorice (DGL): DGL is the root of the licorice plant with the active constituent glycyrrhizin removed. Because of its demulcent and anti-inflammatory properties, DGL has been found to be very beneficial in addressing GERD.^{45 46} When consumed before or with a meal, DGL coats the mucosa, which is the layer of the GI tract that comes into contact with food, forming a protective barrier, reducing inflammation and protecting the tissue from damage due to stomach acid. Suggested dose: 450 mg per day in divided doses, with meals

Turkey Tail Mushroom (*Trametes versicolor* or *Coriolus versicolor*): Maintaining a healthy gut flora is crucial when you are dealing with any kind of digestive issue, including GERD. Turkey tail is an excellent prebiotic option as it feeds beneficial bacteria and controls the candida overgrowth; reducing bloating and pressure that can trigger a GERD episode. Suggested dose: 2 dropperfuls of dual-extracted tincture 2 times daily.

Reishi Mushroom (*Ganoderma lucidum*): Reishi mushroom has potent anti-inflammatory properties,⁴⁷ and can be beneficial in soothing irritated and inflamed esophageal mucosa, such as in GERD. Suggested dose: 2 dropperfuls of dual-extracted tincture 2 times daily.

Lion's Mane Mushroom (*Hericium erinaceus*): Lion's Mane has exceptional anti-inflammatory properties and helps tackle leaky gut and gut inflammation. A study published in Evidence-Based Complementary and Alternative Medicine found that lion's mane helps protect the gastrointestinal tract from damage.⁴⁸ Suggested dose: 750 to 1000 mg per day in divided doses (e.g., 250mg three times per day or 500mg twice per day), or 2-3 dropperfuls of lion's mane tincture twice daily.

Plantain (*Plantago major*): Another excellent anti-inflammatory herb, plantain coats and protects the mucous membranes of the throat, esophagus, stomach, and intestines.⁴⁹ Suggested dose: 2 tsp. per cup, three times per day; or 5-15 ml of fresh juice, three times per day; or as a tincture, 3 dropperfuls, 3 times daily.

Marshmallow Root (*Althaea officinalis*): The polysaccharides in marshmallow root help soothe irritated mucus membranes and promote regeneration of the mucosa,⁵⁰ which can be damaged in chronic GERD. Suggested dose: Take as a cold infusion, 1-2 tablespoons of chopped root per cup of water, three times daily.

Slippery Elm (*Ulmus rubra*): Slippery elm is beneficial for addressing GERD,⁵¹ specifically due to its anti-inflammatory and demulcent properties, which enable it to coat the mucosa and prevent and/or reduce inflammation and pain. Suggested dose: 800-1000 mg, three to four times per day as a supplement or tea.

6. Western Medical Approaches

Conventional treatments for gastroesophageal reflux disease include antacids, H2 blockers, proton pump inhibitors, and surgery.

Antacids: Antacids are medications obtained over-the-

counter or prescribed by a doctor. Examples include Tums, Mylanta, and Roloids. Because antacids work by neutralizing stomach acid and our bodies need stomach acid to properly digest our food, antacids are a short-term solution and should not be used on a daily basis. Side effects of antacids can include constipation, for calcium- or aluminum-based antacids; calcium loss, for aluminum-based antacids; diarrhea, for magnesium-based antacids; and swelling for sodium bicarbonate-based antacids. Antacid use can also lead to gas and bloating.

H2 Blockers: H2 blockers work by reducing the amount of acid that is produced by stomach glands. The H2 stands for histamine, but these drugs work differently from typical antihistamines prescribed to treat allergies since those drugs block H1 receptors, not H2 receptors. Examples of common H2 blockers include famotidine and ranitidine. H2 blockers can be purchased over the counter or prescribed by your provider. Side effects can include headache, constipation, diarrhea, dizziness, weakness, or insomnia.

Proton Pump Inhibitors: Proton pump inhibitors (PPIs) are typically prescribed for more severe GERD. Examples include omeprazole and pantoprazole. These drugs work by decreasing the amount of acid the stomach produces. PPIs are very effective at reducing stomach acid production, but long-term PPI use can lead to increased fracture risk, hypomagnesemia, B12 deficiency, pneumonia, *Clostridium difficile* diarrhea, chronic kidney disease, and dementia.⁵²

Surgery: As a last resort, your doctor may recommend surgery if your GERD is not well-controlled using the previously mentioned treatment options. The most common surgery option is fundoplication, when a surgeon sews the top of your stomach around the edge of your esophagus. This effectively increases the pressure of the lower esophageal sphincter and is usually an effective treatment option. Another option is bariatric surgery, which is most relevant to people who have both GERD and obesity. The most common type of bariatric surgery is gastric bypass surgery. These surgeries help people lose weight and address GERD symptoms.

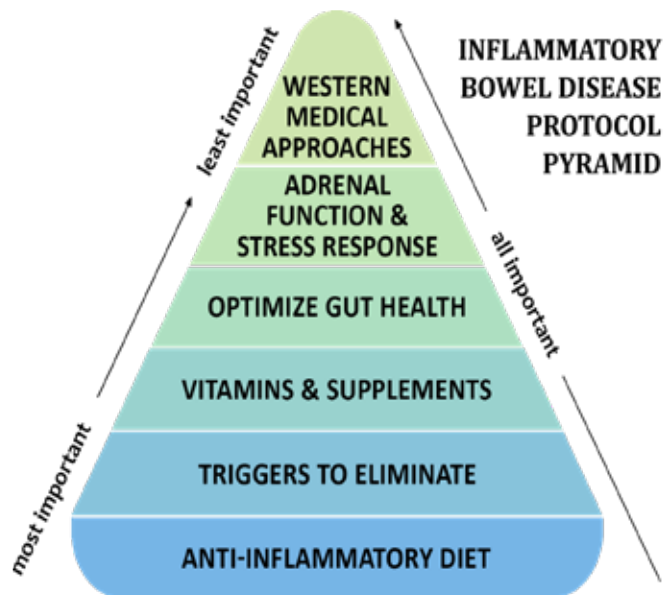
	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Oatmeal, bone broth, or low-fat breakfast with yogurt or other cultured foods Vitamins A, C, E, B6, B9, & B12 Mushroom Trio: Lion's Mane, Reishi, & Turkey Tail dual-extracted mushroom tinctures Post-breakfast walk	Weekly grocery shopping <u>With a meal:</u> digestive enzymes (if needed), Probiotics, & DGL Post-lunch walk	Hydration check (Are you drinking enough water?) Mushroom Trio	Iberogast or your choice of 2 non-mushroom demulcent herbs, 30 min. before afternoon/evening meal Post-dinner walk	Melatonin Bedtime: 3-4 hours after last meal, lie on left side
MONDAY	GERD-friendly breakfast Vitamins A, C, E, B6, B9, & B12 Mushroom Trio Post-breakfast walk	<u>With a meal:</u> digestive enzymes (if needed), Probiotics, & DGL Post-lunch walk	Hydration check Moderate-intensity physical activity Mushroom Trio	Iberogast or your choice of 2 non-mushroom demulcent herbs, 30 min. before afternoon/evening meal Post-dinner walk	Melatonin Bedtime: 3-4 hours after last meal, lie on left side
TUESDAY	GERD-friendly breakfast Vitamins A, C, E, B6, B9, & B12 Mushroom Trio Post-breakfast walk	<u>With a meal:</u> digestive enzymes (if needed), Probiotics, & DGL Post-lunch walk	Hydration check Mushroom Trio	Iberogast or your choice of 2 non-mushroom demulcent herbs, 30 min. before afternoon/evening meal Post-dinner walk	Melatonin Bedtime: 3-4 hours after last meal, lie on left side
WEDNESDAY	GERD-friendly breakfast Vitamins A, C, E, B6, B9, & B12 Mushroom Trio Post-breakfast walk	<u>With a meal:</u> digestive enzymes (if needed), Probiotics, & DGL Post-lunch walk	Hydration check Mushroom Trio	Iberogast or your choice of 2 non-mushroom demulcent herbs, 30 min. before afternoon/evening meal Post-dinner walk	Melatonin Bedtime: 3-4 hours after last meal, lie on left side
THURSDAY	GERD-friendly breakfast Vitamins A, C, E, B6, B9, & B12 Mushroom Trio Post-breakfast walk	<u>With a meal:</u> digestive enzymes (if needed), Probiotics, & DGL Post-lunch walk	Hydration check Moderate-intensity physical activity Mushroom Trio	Iberogast or your choice of 2 non-mushroom demulcent herbs, 30 min. before afternoon/evening meal Post-dinner walk	Melatonin Bedtime: 3-4 hours after last meal, lie on left side
FRIDAY	GERD-friendly breakfast Vitamins A, C, E, B6, B9, & B12 Mushroom Trio Post-breakfast walk	<u>With a meal:</u> digestive enzymes (if needed), Probiotics, & DGL Post-lunch walk	Hydration check Mushroom Trio	Iberogast or your choice of 2 non-mushroom demulcent herbs, 30 min. before afternoon/evening meal Post-dinner walk	Melatonin Bedtime: 3-4 hours after last meal, lie on left side
SATURDAY	GERD-friendly breakfast Vitamins A, C, E, B6, B9, & B12 Mushroom Trio Post-breakfast walk	<u>With a meal:</u> digestive enzymes (if needed), Probiotics, & DGL Post-lunch walk	Hydration check Mushroom Trio	Iberogast or your choice of 2 non-mushroom demulcent herbs, 30 min. before afternoon/evening meal Post-dinner walk	Melatonin Bedtime: 3-4 hours after last meal, lie on left side

* Non-mushroom demulcents: Choose from plantain, marshmallow root, and slippery elm.

Ch. 10: Inflammatory Bowel Disease (IBD) Protocol: Crohn's Disease and Ulcerative Colitis (UC)

Inflammatory bowel disease (IBD) is chronic inflammation of the small and/or large intestines. The two main types of inflammatory bowel disease are Crohn's disease and ulcerative colitis (UC).

Crohn's and UC have many similarities. They can both include extremely debilitating symptoms, such as diarrhea, rectal bleeding, abdominal pain, fatigue, and weight loss. The two conditions also differ in a few key ways. Crohn's disease typically affects the last portion of the small intestine (the ileum) and sometimes affects the large intestine (colon), whereas UC typically affects the large intestine and rectum. Crohn's usually affects the entire thickness of the intestinal wall, whereas UC typically affects only the most superficial mucosal layer of the colon and rectum.



Stool-type also differs. Loose or semi-formed stool is typically seen in Crohn's. On the other hand, because UC leads to ulcers in the mucosa of the colon and rectum, frequent, watery stools with blood and mucus are typical.

Crohn's disease and UC have different long-term effects. Because Crohn's disease affects the small intestine, it can lead to malabsorption and malnutrition, whereas UC typically doesn't. Crohn's can also lead to fistulas, fissures, abscesses, and obstructions, whereas UC can lead to the development of other immune system issues, like ankylosing spondylitis (an inflammatory arthritis that affects the spine and large joints), iritis (inflammation of the colored part of the eye), arthritis (inflammation of the joints), and nephrolithiasis (kidney stones).

The hallmarks of inflammatory bowel disease are inflammation of the intestinal tract, immune system dysregulation, a dysregulated stress response, and

suboptimal gut health.

1. Anti-Inflammatory Diet

A key factor in getting IBD under control is adopting an anti-inflammatory diet, specifically one that includes lots of fruits and vegetables, a moderate amount of protein, and minimal to no saturated fat.¹ Research demonstrates that diets centered on animal-based products like dairy, eggs, cheese, fish and other meat, lead to significant changes in the bacteria that live in the large intestine. This type of diet leads to a shift in the gut microbiome toward bile-tolerant bacteria, like *Bilophila wadsworthia*.²

To understand this, we need to look at human physiology. Our liver makes bile, and our gallbladder stores and concentrates that bile. When we consume fat, our liver and gallbladder work together to secrete bile into the small intestine to emulsify the fat. If you've had your gallbladder removed, then the liver will secrete bile directly into the small intestine. When we habitually consume large amounts of fat, the composition of microbes in our gut changes. This can take place after fewer than five days on a high-fat diet. Bacteria that don't tolerate bile well begin to die off, and bile-loving bacteria like *Bilophila wadsworthia* thrive. *Bilophila wadsworthia* trigger gut inflammation and are associated with increased risk and worse outcomes for IBD.³

In order to address inflammatory bowel diseases like Crohn's and ulcerative colitis, it's important to adopt a mostly or completely plant-based diet that minimizes processed food, includes a moderate amount of protein, and contains minimal to no saturated fat.^{4 5}

THE MOST EFFICIENT WAY TO TRANSITION TO AN ANTI-INFLAMMATORY DIET⁶: If you currently consume a typical Western diet, transitioning to an anti-inflammatory diet may seem daunting. The most efficient way to transition is to start with a 28-day elimination-and-challenge protocol. Eliminate all processed foods, dairy, eggs, meat, sugar, food coloring, chocolate, caffeine, and alcohol (unless eliminating alcohol poses a health hazard by leading to withdrawal) for 28 days. For those 28-days, your diet should consist mostly of whole foods, including fruits and vegetables. You can also eat gluten-free grains and pseudograins, like quinoa, amaranth, and millet; beans and lentils; nuts and seeds; and root crops.

After the 28-day elimination phase, it's very important to reintroduce the foods you've been avoiding into your diet in a very specific way. The "challenge" or reintroduction phase involves reintroducing the foods you had eliminated over the preceding four weeks and seeing how your body responds. Documenting foods you reintroduce and symptoms you experience during this process is critical. Tips include:

Reintroduce Foods One at a Time: In order to know exactly which food is causing a return of your symptoms, reintroduce foods one at a time.

Limit Yourself to Reintroducing (at the Most) One New Food Each Day: Most people experience symptoms

within 10 minutes to 12 hours after reintroducing problem foods. The exception is arthritic joint pain, which can take up to 48 hours for symptoms to show. If arthritic pain is one of your symptoms, reintroduce a new food every two days.

Always Reintroduce Foods in Their Purest Form: Don't use a cheeseburger to reintroduce cheese, because you're also reintroducing gluten (in the bun) and beef. If you were to experience symptoms, it would not be clear if they were due to the cheese, the gluten, or the beef. Instead try cheese alone, beef prepared with minimal sauce or additional flavorings, or whole-wheat crackers.

Eat a Large Amount of the Food on the Day You're Reintroducing it: It's important to consume a large enough amount of the food to provoke a reaction to get accurate information about whether or not it is triggering or worsening your symptoms. In order to do this, have the food item (*e.g.*, glass of milk) at breakfast. If you experience symptoms after consuming that glass of milk, document them and add milk to the list of trigger foods. If you do not experience symptoms, have milk again at lunch and at dinner. If no symptoms arise, then milk can be added to your list of foods that do not provoke a reaction.

Do Not Fully Reincorporate Foods that Do Not Provoke a Reaction Back into Your Diet until You Have Tested All of the Foods You Plan to Test: During the testing phase, continue eating the basic anti-inflammatory diet comprising mainly of fruits and vegetables, plus the one food that you are testing that day. Once you've gathered the needed information from your daily test, don't eat that food again until you've finished testing all of the foods on your list. If you do not plan to continue eating a food, simply cut it out of your diet.

Although Milk and Cheese Are Both Considered Dairy, They Should Be Reintroduced on Separate Days: Some people react to cheese and don't react to milk, and vice versa. You may benefit from reintroducing different types of cheese on different days since some people react to one type of cheese and not another.

Sugar, Food Coloring, Chocolate, Caffeine, and Alcohol Should Be Tested Toward the End of Your Reintroduction Phase, if at All: These foods should only be tested if you plan to continue consuming them, and they should be tested toward the end of your reintroduction journey because of their higher likelihood of triggering symptoms. Test sugar by putting 4 teaspoons of sugar in a drink or mixed with another food. Test food additives by buying a pack of food coloring, mixing 1/2 teaspoon of each color in one cup, adding one teaspoon of the mixture to a glass of water, and drinking it. Alternatively, you may test each color on separate days. Test chocolate with 1-2 tablespoons of pure baker's chocolate or Hershey's cocoa powder. Test caffeine with plain coffee (without milk, non-dairy creamer, or sugar). If you drink caffeinated tea like green or black tea, decaffeinated coffee, or decaffeinated tea, test these on separate days. Test wine, beer, and hard liquor on different days, as reactions may differ. Have 2

drinks per test day, but only if you can afford to not feel well that day and possibly the next day.

Limit Fiber and Pause the Reintroduction Phase as Needed During Flares: If an attempt to reintroduce a food leads to a significant flare, wait until your symptoms resolve before attempting to reintroduce a new food. If the flare is significant, it may be necessary to consume a low-fiber diet for some time to promote healing.

After completion of your elimination and challenge protocol, you should have extremely valuable information about which foods do not provoke a reaction, which foods typically lead to a return of your symptoms and need to be avoided, what and how severe those symptoms are (so you can make informed choices about whether or not you can still have those foods once in a while), and how you can eat a more anti-inflammatory diet that promotes wellness by avoiding the foods that are most likely to aggravate your symptoms.

2. Triggers to Eliminate

When optimizing diet and lifestyle to address inflammatory bowel disease, there are known triggers that should be eliminated. Research demonstrates that these factors, specifically artificial sweeteners, fried foods, and smoking, significantly increase risk for the development and flaring of IBD.

Artificial Sweeteners: Artificial sweeteners like sucralose (Splenda) negatively impact gut health in people with Crohn's disease and ulcerative colitis by exacerbating both gut damage and inflammation.⁷⁻⁸ Artificial sweeteners may also lead to negative changes in our gut bacteria; disruption of the gut barrier, which separates our gastrointestinal tract from the tissues and blood supply beneath it and needs to be intact to prevent development of further immune system reactions; and an increase in digestive enzymes in the feces, which suggests further damage to the gut barrier.⁹ Although the majority of research has been conducted on sucralose, other artificial sweeteners like aspartame, saccharin, and acesulfame potassium lead to similar changes to the gut barrier and gut bacteria.¹⁰ Because these changes can worsen IBD, they should be eliminated. If you need to use a sweetener, some research suggests that erythritol may actually help improve inflammation of the small intestine,¹¹ which may be beneficial in Crohn's disease.

Fried Foods: Fried foods cause inflammation of the colon and lead to disruption of the gut barrier.¹² When the gut barrier is disrupted, it becomes possible for the contents of the gastrointestinal tract (including bacteria and toxic bacterial products) to enter the bloodstream. When this happens, the inflammation that was previously localized to the GI tract spreads to the body, leading to low-grade systemic inflammation.

Smoking: Smoking tobacco worsens the course of Crohn's disease; however, research suggests that it does not have this effect on ulcerative colitis.¹³ This may be because tobacco smoke has specific negative effects on the small intestine, which is typically affected in Crohn's disease but is usually not affected

in ulcerative colitis. Smokers with Crohn's disease experience significant health benefits by quitting or at least cutting back on smoking.

3. Vitamins and Supplements

In addition to making calculated dietary changes, there are a few key supplements that should be considered when addressing inflammatory bowel disease.

Vitamin D: Vitamin D plays a critical role in maintaining the structural integrity of the gastrointestinal barrier.¹⁴ It regulates proteins that keep the space between the cells of the gastrointestinal tract tight. If vitamin D and other protective factors are low, the integrity of the gut barrier is compromised – this is referred to as “leaky gut syndrome.” This leads to increased inflammation and increases the risk for flaring of symptoms in inflammatory bowel disease. Suggested dose: 4000 IU per day, with a goal of a 25-hydroxy vitamin D level of 50 ng/mL. Get your levels tested.

Omega-3 Fatty Acids: Omega-3 fatty acids reduce pro-inflammatory chemicals in the body and have immune-modulating properties.¹⁵ Research specifically demonstrates that increased omega-3 fatty acid intake leads to decreased risk for Crohn's disease and ulcerative colitis.¹⁶ Suggested dose: 1.6 grams per day for men and 1.1 grams per day for women.

Probiotics: It is important to maintain a wide variety of healthy bacterial strains in the gut and to keep abnormal strains of bacteria to a minimum. We'll discuss probiotics for IBD in the next section, “Optimize Gut Health.” Suggested dose (for IBD): at least 100 CFU per day.

4. Optimize Gut Health

Poor gut health can trigger flares of inflammatory bowel disease. The mucus layer of the gut typically protects the blood and submucosal tissues from bacteria and other contents of the GI tract. If this barrier is damaged, bacteria and toxic bacterial products can spread to other tissues, including the blood, and trigger localized and systemic inflammation. This leads to a worsening of IBD.

To heal and restore the gut barrier and optimize the gut microbiome, eliminate trigger foods and add supplements, foods, and herbs that help modulate the immune system and optimize gut health.

RESTORING THE GUT MICROBIOME WITH PROBIOTICS AND PREBIOTICS: People with inflammatory bowel disease tend to have high levels of abnormal bacteria and low levels of beneficial bacteria in the gut; this has been identified as a trigger for both the onset and relapse of IBD.¹⁷ Taking probiotics introduces bacteria that we know are beneficial to gastrointestinal health, so that we can gradually change the composition of the gut microbiome. Having the right forms of bacteria in the gut helps to decrease gut permeability, increase the production of short-chain fatty acids in the gut (which feed healthy bacteria and help improve gut health), decrease the secretion of pro-inflammatory

chemicals in the gut, and increase the secretion of anti-inflammatory chemicals.¹⁸ Suggested dose: at least 100 CFU per day.

Prebiotics are specialized plant fibers that feed gut bacteria and stimulate the growth of healthy gut bacteria. It is recommended to implement dietary changes and take probiotics for some time before beginning prebiotics. This is especially the case in individuals with irritable bowel syndrome, small-intestinal bowel overgrowth (SIBO), or other forms of dysbiosis because abnormal bacteria rapidly ferment prebiotics, leading to bloating and GI upset.

The best method is to consume prebiotics in food form (e.g. Jerusalem artichokes, asparagus, oats, bananas, garlic, and dandelion greens). Incorporate these into your diet gradually, giving your body and gut bacteria time to adjust.

ANTI - INFLAMMATORY SHORT-CHAIN FATTY ACIDS: Short-chain fatty acids like butyric acid are produced by healthy gut bacteria. These have positive effects on gut health, such as enhancing the mucus layer of the gut barrier so that bacteria and bacterial toxins are not able to enter the blood and other tissues,¹⁹ and supporting colon cell health so that they build a stronger intestinal barrier and gut permeability is reduced.²⁰ Suggested dose: 150-300 mg per day²¹.

INFLAMMATION - MODULATING HERBS

AND DEMULCENTS: Slippery Elm (*Ulmus rubra*): Slippery elm is a demulcent, a substance that relieves irritation of the mucus membranes that coats and soothes irritated GI tissue. Slippery elm soothes gastrointestinal mucosa, is anti-inflammatory, promotes the development of the mucosal layer of the GI tract,²² and has antioxidant properties.²³ Suggested dose is 800-1000 mg, three to four times per day as a supplement or tea to start.

Marshmallow Root (*Althaea officinalis*): Marshmallow root forms a protective layer over the gut that covers the damaged mucosal layer, allowing it to heal.²⁴ It has a mild laxative effect that may be beneficial for those with IBS-C and many find it beneficial for ulcerative colitis and Crohn's.²⁵ Suggested dose: Take as a cold infusion, 1-2 tablespoons of chopped root per cup of water, three times per day.²⁶

Aloe Vera: Aloe vera has anti-inflammatory, antioxidant, and demulcent properties. Aloe vera helps reduce inflammation, ulcerations, and tissue damage in ulcerative colitis. Suggested dose: 100 ml twice per day.

Deglycyrrhized Licorice: Licorice root has anti-inflammatory and immune-modulating properties. Its ability to improve ulcers in ulcerative colitis is likely due to its ability to decrease oxidative stress and reduce the quantity of pro-inflammatory chemicals in the body.²⁷ Deglycyrrhized licorice is licorice that has had one of its active constituents, glycyrrhizin, removed. By removing glycyrrhizin, we are able to harness its anti-inflammatory benefits without increasing risk for hypertension, edema, hypokalemia, and hyporeninemia

in people who are susceptible to these. Suggested dose: 450 mg per day in divided doses, with meals.

Lion's Mane Mushroom (*Hericium erinaceus*): Lion's mane is a medicinal mushroom whose benefits are largely due to its antioxidant, anti-inflammatory, and soothing effects on the gut. Lion's Mane protects against worsening of ulcerative colitis by blocking the production of pro-inflammatory chemicals in the body and by modulating the production of compounds that reduce oxidative stress.^{28 29 30} Suggested dose: 750 to 1000 mg per day in divided doses (e.g., 250mg three times per day or 500mg twice per day), or 2-3 dropperfuls of tincture 2x/day.

Turkey Tail Mushroom (*Trametes versicolor* or *Coriolus versicolor*): A number of digestive disorders respond well to Turkey Tail, including Crohn's disease and ulcerative colitis. Turkey Tail supports healthy gut flora through its prebiotic properties.³¹ It feeds beneficial bacteria and controls the overgrowth of pathogenic microbes. It also regulates the pro-inflammatory chemicals that cause intestinal inflammation and damage in inflammatory bowel disease.^{32 33 34} Suggested dose: 2 grams, twice per day, or 2 dropperfuls of dual-extracted tincture 2x/day.

Reishi Mushroom (*Ganoderma lucidum*): This medicinal mushroom is an outstanding option for Crohn's disease and ulcerative colitis because of its immune-modulating properties. Specifically, research demonstrates that reishi improves ulcerative colitis by altering the gut microbiome in the first section of the large intestine and by altering the expression of genes that control the inflammatory response.³⁵ In Crohn's disease, reishi mushroom reduces inflammation and intestinal damage by significantly reducing the production of pro-inflammatory chemicals within the body.³⁶ Suggested dose: 2 dropperfuls of dual-extracted tincture 3x/day.

Plantain (*Plantago major*): Plantain is anti-inflammatory and has a calming effect on mucous membranes, helping enhance the integrity of the gut barrier so that bacteria and their toxic products don't enter the blood and other tissues, resulting in elevated inflammation. For IBD, try drinking Plantain Tea two to three times daily. Eat fresh leaves and seeds (when in season) and drink it juiced. A tincture is also effective, but supplementing with fresh plantain or drinking plantain tea helps the mucilaginous components directly contact the gut lining. Suggested dose: 2 tsp. per cup, three times per day; or 5-15 ml of fresh juice, three times per day; or as a tincture, 3 dropperfuls/ml, three times per day.

5. Adrenal Function and Stress Response

When we experience chronic stress, the hypothalamus, which is a region of the brain, releases corticotropin releasing hormone (CRH). CRH then stimulates the pituitary gland, which is also located within the brain, to secrete adrenocorticotrophic hormone (ACTH). ACTH then stimulates the adrenal glands to secrete cortisol and other glucocorticoids. These

hormones play a critical role in the fight or flight response, and they also play an important role in modulating the immune system. This series of events-the HPA axis-enables us to handle short- and medium-term stress very efficiently, but when stressors don't resolve and our stress becomes chronic, it can have negative effects on our moods and on our immune systems.

It is well-known that adverse life events, depression, and chronic stress increase the likelihood of relapse in people with inflammatory bowel disease. This is likely due to the changes in the hypothalamic-pituitary-adrenal (HPA) axis, which controls our body's response to chronic stress.³⁷ When we are chronically stressed, our bodies secrete higher levels of cortisol. This has an immunosuppressive effect and leads to higher inflammation levels.³⁸ There is a similar increase in inflammation levels in individuals who are depressed.³⁹

In order to decrease the likelihood of IBD flares, it is critical that you work to reduce your current stress levels. This can be helped through working with a therapist, heart rate variability exercises, gratitude journaling, meditation, deep breathing, good sleep, exercise, time in nature, and rearranging your life so that you are under less stress. In addition to intentionally decreasing the amount of stressors you're exposed to on a daily basis, try taking adaptogenic herbs that modulate the stress and immune response.

Ashwagandha (*Withania somnifera*): Ashwagandha has been safely used in Ayurvedic medicine to address gastrointestinal concerns for centuries.⁴⁰ Ashwagandha root safely and effectively improves resilience while under stress and leads to an improved quality of life. Ashwagandha significantly reduces morning cortisol levels, which is evidence of its ability to regulate the body's chronic stress response.⁴¹ We also know that Ashwagandha can modulate the immune system by reducing c-reactive protein levels,⁴² and that the root of this plant offers specific anti-inflammatory benefits that are beneficial for Crohn's disease and ulcerative colitis.⁴³ Suggested dose: 500 mg twice per day as a supplement or tincture.

Medicinal Mushrooms: In addition to being beneficial in addressing inflammatory bowel disease because of their demulcent and inflammation- and immune-modulating properties, Lion's Mane, Turkey Tail, and Reishi mushrooms all have adaptogenic properties and are known to lead to a healthier stress response.^{44 45}

Cordyceps is another medicinal mushroom with adaptogenic and immune-modulating properties. Cordyceps improves the body's ability to handle stressful experiences, protects against oxidative stress, and decreases the production of pro-inflammatory chemicals in the body.⁴⁶ Research specifically shows that Cordyceps has beneficial effects in the prevention and treatment of inflammatory bowel disease.⁴⁷ Suggested dose: 2-3 grams twice per day of CS-4 extract, 3-4 grams of powdered mushroom twice per day, or a dual-

extracted Cordyceps tincture, 2 dropperfuls 2x/day.

Nature/Green Space Exposure: Being exposed to green space significantly reduces cortisol levels and plays a critical role in restoring balance to dysregulated stress and immune responses. People exposed to green spaces feel less stressed, have a decline in cortisol levels,⁴⁸ and have significantly reduced pro-inflammatory chemicals in their body.⁴⁹ Green space recommendation: Prioritize spending at least 20 minutes exposed to nature per day.

6. Western Medical Approaches

Conventional treatments for inflammatory bowel diseases include anti-inflammatory drugs, immune system suppressors, biologics, antibiotics, and surgery.

Anti-Inflammatory Drugs: The anti-inflammatory drugs that are typically prescribed for inflammatory bowel disease fall into two categories: corticosteroids and oral 5-aminosalicylates.

Corticosteroids (including prednisone) act similarly to the hormone cortisol. Cortisol acts to diminish or turn off the immune response, so its effects are mostly anti-inflammatory in this context.

Because they function to turn the immune system off, long-term use of corticosteroids can lead to atrophy of immune-related tissue, which makes you more prone to infections. 5-aminosalicylates also help reduce inflammation, and tend to act in the lining of the intestine or, in some cases, the joints.

Because of this, 5-aminosalicylates tend to be more effective for ulcerative colitis. Remember, in UC, the most superficial lining of the gut is typically affected, whereas the entire thickness of the gut wall is typically affected in Crohn's disease. Aminosalicylates can cause kidney damage in rare cases, so people who have kidney disease should not take it and others should have their kidney function monitored regularly.

Immune System Suppressors: These block or prevent the release of the pro-inflammatory chemicals that cause inflammation and trigger the immune response. Examples include azathioprine, mercaptopurine, and methotrexate. Note that methotrexate removes folate from the body; people experience fewer side effects by supplementing with folate.

Biologics: These drugs are typically administered intravenously or through self-administered injections. They work by neutralizing one of three proteins in the body that have a pro-inflammatory function. The proteins this class of medication blocks are tumor necrosis factor, interleukins, and integrins. Common examples of biologics include adalimumab (Humira), certolizumab (Cimzia), and infliximab (Remicade).

These drugs tend to have fewer side effects than medications that block or turn off the entire immune system, like corticosteroids. Individuals who've had exposure to tuberculosis should receive treatment before starting biologics, because these drugs may make reactivation of tuberculosis more likely. Because these drugs do block normal immune function, those who are already at increased risk for infections may benefit from a different type of treatment.

Antibiotics: Antibiotics are typically prescribed in combination with other medication, specifically when bacterial infection may also be present. This is more common in Crohn's disease because of the possibility for abscess formation. Broad-spectrum antibiotics like ciprofloxacin and metronidazole are typically prescribed. Because these also kill beneficial bacteria, consider taking probiotics.

Surgery: Surgery is typically viewed as a last resort in the management of inflammatory bowel disease. If, after making dietary and lifestyle changes and exhausting the other treatment options, IBD is still not well-controlled, then surgery may be the next step.

For ulcerative colitis, surgery involves complete removal of the large intestine and the creation of an internal pouch (called a J-pouch) that connects the small intestine to the anus to enable defecation. If it's not possible to form a pouch, surgery will create a permanent abdominal opening for stool to pass through to be collected in an attached bag.

Because Crohn's disease does not typically involve the entire ileum (last portion of the small intestine) or colon, surgery is typically limited to removal of the affected regions and then reconnecting the healthy regions. Surgery may also be necessary to drain abscesses that form as a result of the condition.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Morning gratitude and green space/nature exposure (20-30 min.) Probiotics Vitamin D Mushroom Trio (Lion's Mane, Reishi, & Turkey Tail Mushroom Tinctures)	Weekly grocery shopping for Elimination and Challenge protocol Omega-3s w/ food Non-mushroom demulcents* (e.g., Slippery Elm & Marshmallow root infusions)	Hydration check (Are you drinking enough water?) Non-mushroom demulcents* Make time for a hobby you enjoy	Non-mushroom demulcents* Butyric acid Mushroom Trio	Ashwagandha capsules or tincture Document symptoms and food in your log, if you have not done so throughout the day
MONDAY	Morning gratitude and green space/nature exposure Probiotics Vitamin D Mushroom Trio	Omega-3s w/ food Non-mushroom demulcents x2	Hydration check Non-mushroom demulcents*	Non-mushroom demulcents* x2 Butyric acid Evening walk or other physical activity (30 min.) Mushroom Trio	Ashwagandha Document symptoms and food in your log
TUESDAY	Morning gratitude and green space/nature exposure Probiotics Vitamin D Mushroom Trio	Omega-3s w/ food Non-mushroom demulcents x2	Hydration check Non-mushroom demulcents*	Non-mushroom demulcents* x2 Butyric acid Mushroom Trio	Ashwagandha capsules or tincture Document symptoms and food in your log
WEDNESDAY	Morning gratitude and green space/nature exposure Probiotics Vitamin D Mushroom Trio	Omega-3s w/ food Non-mushroom demulcents x2	Hydration check Non-mushroom demulcents*	Non-mushroom demulcents* x2 Butyric acid Mushroom Trio	Ashwagandha capsules or tincture Document symptoms and food in your log
THURSDAY	Morning gratitude and green space/nature exposure Probiotics Vitamin D Mushroom Trio	Omega-3s w/ food Non-mushroom demulcents x2	Hydration check Non-mushroom demulcents*	Non-mushroom demulcents* x Butyric acid Evening walk or other physical activity (30 min.) Mushroom Trio	Ashwagandha capsules or tincture Document symptoms and food in your log
FRIDAY	Morning gratitude and green space/nature exposure Probiotics Vitamin D Mushroom Trio	Omega-3s w/ food Non-mushroom demulcents x2	Hydration check Non-mushroom demulcents*	Non-mushroom demulcents* x2 Butyric acid Mushroom Trio	Ashwagandha capsules or tincture Document symptoms and food in your log
SATURDAY	Morning gratitude and green space/nature exposure Probiotics Vitamin D Mushroom Trio	Omega-3s w/ food Non-mushroom demulcents x2	Hydration check Non-mushroom demulcents*	Non-mushroom demulcents* x2 Butyric acid Mushroom Trio	Ashwagandha capsules or tincture Document symptoms and food in your log

*Non-mushroom demulcents: slippery elm, marshmallow root, aloe vera gel, plantain, and deglycyrrhized licorice. Choose 2-3.

Ch. 11: Leaky Gut, Gastritis, and IBS Protocol

Medical professionals are diagnosing more digestive tract diseases today than ever before. The cause varies, but they're usually linked to an autoimmune disorder, a poor diet, imbalanced gut flora, and an unhealthy lifestyle. Leaky gut, gastritis, and IBS are three common types of digestive disorders.

Leaky Gut/Increased Intestinal Permeability: A normal-functioning intestinal tract acts as a sophisticated barrier that controls nutrient absorption into our bloodstream. Some people have "leaky guts." This means the barrier is hyperpermeable, allowing toxins, undigested food particles, and microbes to absorb or "leak" into the surrounding tissues and blood. This results in increased inflammation and changes in our gut bacteria. These changes may trigger digestive and autoimmune issues, and aggravate a range of health problems. Pay attention to your body and know the signs of leaky gut syndrome. These include: diarrhea, inflammatory bowel disease, allergies, low energy, ulcers, bacterial overgrowth, and a variety of autoimmune diseases.



Gastritis: Gastritis is a condition caused by inflammation in the lining of the stomach. Gastritis and leaky gut often go hand-in-hand. Like leaky gut, there is often a common autoimmune issue at the core of each disease. Both conditions involve inflammation and possible bacterial overgrowth. Each may be triggered by a poor diet and/or overuse of pain pills. Some common symptoms of gastritis include: nausea, vomiting, indigestion and pain in the upper abdomen, and a feeling of fullness in the upper abdomen after eating. If steps aren't taken to fix this issue, it could develop into a serious stomach ulcer.

Irritable Bowel Syndrome (IBS): IBS primarily affects the colon, also known as the large intestine. It is a type of inflammatory bowel disease that can cause a variety of uncomfortable digestive issues like diarrhea, constipation, bloating and gas, cramping, and

abdominal pain. IBS is linked to leaky gut syndrome.

With each of these digestive conditions, you may find significant relief after a change in diet. Many diseases like these are triggered or made worse by eating trigger foods that you may be allergic to (and not know it) or by eating foods that worsen inflammation. The addition of probiotics can also help restore the proper balance of gut flora, helping to relieve symptoms.

There is much evidence supporting the fact that an imbalance in gut flora can trigger or is linked to the digestive conditions above.¹ An overgrowth of *Helicobacter pylori* (*H. pylori*) can attack the lining of the stomach and upper GI tract and can cause leaky gut, gastritis, ulcers, and other digestive issues. *Candida albicans* can also contribute to intestinal diseases, including leaky gut².

1. Diet Diet is the foundation for leaky gut, gastritis, and irritable bowel syndrome. The foods we eat affect the gut flora that reside in our intestinal tract. Be mindful of what you ingest as food plays a major role in your overall health and can alleviate or trigger inflammation. This is especially important for anyone with an autoimmune condition. Try eating an anti-inflammatory diet paired with intermittent fasting. Intermittent fasting and multiday fasts increase microbial diversity, improve gut barrier function, and help reverse intestinal inflammation^{3 4 5}.

Probiotics: Probiotics are essential as they help maintain a proper balance of essential gut flora. Our microbiome contains more than ten times the amount of cells in our body, which in itself has 100 trillion cells. Our microbiome is directly shaped by what we eat. In a healthy person who eats a well-balanced diet, their gut flora is balanced between the good and bad bacteria. When this balance is disturbed, digestive issues ranging from IBS to weight gain can occur.

Getting probiotics from food is the best way to get these beneficial bacteria in your gut. Foods that contain probiotics have usually been fermented in some way to naturally produce these bacteria. Fermented probiotic foods to incorporate into your daily diet include: kombucha, kimchi, miso soup, fresh (not canned) sauerkraut, plain full fat yogurt with live or active cultures, kefir, tempeh, pickles, traditional buttermilk (unless dairy is a trigger food for you), and natto.

Prebiotics: In addition to probiotic foods and drinks, consume prebiotic foods. Prebiotics are what probiotics feed upon. Prebiotic foods that can help keep your existing gut bacteria healthy and balanced⁶ include: leeks, onions, garlic, asparagus, apples, bananas, seaweed, chia and flax seeds, dandelion leaves, burdock root, dark cocoa, jicama root, chicory root (great coffee substitute), lentils, split peas, beans, chickpeas, spinach, and blueberries (preferably wild).

Other prebiotic foods that can help are:

- Jerusalem artichokes (*Helianthus tuberosus*): These tubers are one of the most potent sources known for

prebiotic fiber. They are made up of about seventy six percent inulin. They are easy and fun to grow in the garden.

- **Mushrooms⁷:** All mushrooms are a good source of beta-glucans, prebiotic compounds that help modulate the immune system, improve gastrointestinal health, and lower body weight, inflammation, and insulin resistance. Shiitake, Portobello, Lion's Mane, and white button mushrooms are great for your diet. Reishi, Lion's Mane, and Turkey Tail mushrooms can be used in dual-extracted tincture form.
- **Oats:** Eating a cup of minimally processed oats for breakfast daily can result in a noticeable increase in your good gut bacteria after only three weeks.⁸

Foods that Help Relieve Inflammation: In addition to daily consumption of probiotic and prebiotic foods, eating the following foods can help relieve inflammation in the intestinal tract:

- **Bone broth:** Bone broth is full of valuable nutrients, collagen, and bone-strengthening minerals. It is very nourishing, especially when made with healthy vegetables and greens like stinging nettle and dandelion leaves. It contains compounds that may ease inflammation and keep your gut healthy.
- **Organic olive oil:** Olive oil is rich in protective antioxidants. Avoid other oils like canola and corn oil. They are high in saturated fats, while olive oil is high in omega-3 fatty acids that help lower inflammation.
- **Healthy fish:** Opt for fish like salmon and mackerel. These fish are high in omega-3 fatty acids that target inflammation and protect cells.
- **Fruits:** Fruits like strawberries, blueberries, and raspberries are high in antioxidants that protect cells from oxidative damage and have beneficial phytonutrients.
- **Leafy greens:** Plants like spinach, kale, lamb's quarter, and watercress are packed with vitamins and minerals. They help decrease inflammation and increase energy levels.
- **Vegetables:** Vegetables aren't just good for tackling inflammation; they also contain dietary fiber that is essential to healthy digestion.
- Just as there are foods that can help the body fight inflammation, there are foods that make inflammation worse.

Avoid the following:

- **Processed foods:** These contain high amounts of toxins and preservatives that can trigger inflammation. A good rule of thumb is to avoid anything that comes in a box.
- **Synthetic food coloring:** It has been linked to several types of cancer and other diseases, including behavior issues.
- **Foods sprayed with pesticides:** Try to buy organic

foods whenever possible or grow your own vegetables to ensure you are getting fresh, pesticide-free food.

- **Sugar:** Sugar is linked to inflammation in the body and many diseases.
- **Caffeine:** Caffeine is linked to increased anxiety, which can trigger autoimmune issues.
- **Synthetic sugar:** It may be worse than processed cane sugar. It is linked to inflammation and cancer.
- **Alcohol:** It weakens the immune system and almost always worsens gastritis.
- **Gluten:** Gluten is a major autoimmune trigger and often provokes an inflammatory response.
- **A diet high in animal products:** It may cause an inflammatory response in the body, which may make autoimmune conditions worse.
- **Fried foods:** Fried foods especially harm those with digestive issues as they are usually fried in unhealthy oils.
- **Meats with antibiotics:** Some farmers regularly treat their animals with antibiotics to prevent infections. The meat from these animals may contain high levels of antibiotics. Exposure to antibiotics on a regular basis kills off gut flora and creates antibiotic-resistant infections.

2. Natural Remedies

Following a diet rich in probiotic, prebiotic, and anti-inflammatory foods may provide significant relief for those with digestive and inflammatory bowel issues, especially when paired with the rest of this protocol.

Herbs that help to repair the intestinal tract and restore healthy gut function include lion's mane, reishi, and turkey tail mushrooms, marshmallow root, and slippery elm. All of these are found in my Balanced Gut Tincture. Take 2 dropperfuls 3x/day for a month. Continue 1-2x/day for maintenance.

Lion's Mane (*Hericium erinaceus*)⁹: A known antioxidant and gastroprotective, lion's mane helps tame the inflammatory response and protect the lining of the stomach. It also protects the gut from *H. pylori* and the polysaccharides in lion's mane protect against gastric ulcers¹⁰. Take as a tea or a tincture.

Marshmallow Root (*Althaea officinalis*): Marshmallow root forms a protective layer over the gut that covers the damaged mucosal layer, allowing intestinal perforations to heal.¹¹ It has a mild laxative effect that may be beneficial for those with IBS-C. Suggested dose: Take as a cold infusion, 1-2 tablespoons of chopped root per cup of water, three times per day.

Slippery Elm (*Ulmus rubra*): Slippery elm is a demulcent, forming a protective layer in the digestive tract and reducing gut inflammation. It stimulates mucus secretion, helping ease IBS and gastritis^{12 13}. Suggested dose: 800-1000 mg, three to four times per day as a supplement or tea to start.

Reishi (*Ganoderma lucidum*): Reishi exerts anti-inflammatory action that's helpful for calming intestinal inflammation, it modulates the immune system, and it inhibits Candida yeast. It also has neuroprotective qualities. It is an effective herb for calming the inflammation and immune response involved with leaky gut¹⁴. Suggested dose: 2 dropperfuls of reishi dual-extracted tincture per day.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): Turkey tail calms inflammation, contains important prebiotics that balance the microbiome, encourages the growth of "good" gut bacteria, and controls Candida and small intestinal bacterial overgrowth (SIBO), turkey tail is a remarkable remedy for healing the gut¹⁵. Suggested dose: 2 dropperfuls of a dual-extracted tincture 2x/day.

REMEDIES FOR BACTERIAL OVERGROWTH:

Goldenseal (*Hydrastis canadensis*) and/or Lion's Mane for *H. pylori*: Goldenseal can kill the harmful digestive bacteria *H. pylori*. Lion's mane protects the gut from *H. pylori* and the polysaccharides in lion's mane also protect against gastric ulcers¹⁶ and assist with gut inflammation. *H. pylori* is responsible for serious injury to the lining of the stomach and may lead to gastritis and ulcers. Goldenseal acts like an antibiotic, use mindfully and with caution. It can kill both good and bad bacteria. Make sure that if you drink goldenseal tea to kill *H. pylori*, you also consume plenty of probiotic foods and drinks to replace the good bacteria. *Note that many people with autoimmune issues find goldenseal too stimulating. Lion's Mane is a better option.*

Candida Overgrowth: Please see our Candida Protocol at page 46.

Aloe Vera Juice for Bacterial Overgrowth: Aloe Vera works similarly to marshmallow root and slippery elm and is also thought to help with gut inflammation. Aloe contains compounds that help reduce bacterial overgrowth and infection, while also soothing the GI tract. Aloe Vera is easy to drink and almost flavorless. It can be added to smoothies and juices. Try consuming two to four ounces of this juice throughout the day.

REMEDIES FOR SOOTHING THE GI TRACT:

Aloe Vera for Coating and Soothing: Aloe Vera is also highly effective at coating the digestive tract and stomach to prevent spasms and pain associated with many inflammatory bowel conditions. It comes in juice or gel. Both are equally effective. It is harder to drink the gel by itself than it is to drink the juice, but both blend well into smoothies and juices. Drink up to four ounces spaced out throughout the day.

Marshmallow Root for Coating and Reducing Spasms: Marshmallow root coats and soothes the GI tract. It has been used to help with inflammation and spasms as well. Try drinking it as a tea for daily support in addition to the Balanced Gut Tincture. Add up to three teaspoons of ground marshmallow root to a tea bag and infuse in a cup of hot water for fifteen minutes. Drink up to

three times daily.

Chamomile (*Matricaria chamomilla*) Tea for Spasms and Upset Stomach/GI Tract: Chamomile doesn't just help with stress, it also helps relieve spasms and calms an upset stomach. Chamomile tea is a soothing and relaxing way to enjoy this flower. Simply add two teaspoons of dried chamomile flowers to a tea bag and infuse in a cup of hot water for ten minutes. Drink up to four cups daily.

REMEDIES FOR DIARRHEA: Agrimony (*Agrimonia eupatoria*) for Diarrhea: The tannins in agrimony help to soothe inflammation and irritation in the intestinal tract and helps calm the digestive tract and reduce diarrhea. Agrimony is an especially gentle herb that has been used traditionally for children and adults. For daily maintenance for chronic diarrhea, try agrimony tea. Add two teaspoons of dried agrimony to a tea bag and infuse this in a cup of hot water for ten minutes. Drink up to three cups daily.

Cramp Bark for Intestinal Spasms and Cramping Related to Diarrhea: Cramp bark is known for helping with menstrual issues, but it can also help with digestive spasms associated with irritable bowel disease and other digestive complaints. If you suffer from constant cramping as a result of diarrhea, try making a tincture with cramp bark. Fill a sterile glass jar with well-chopped cramp bark. Completely cover the plant material with at least 80 proof alcohol and tightly cap. Place the jar in a cool, dark location for four to six weeks. Shake the jar daily to help it infuse better. Strain out the liquid and put it in a dropper bottle when ready. Take two dropperfuls at the first sign of cramping and one dropper full an hour later if needed. You may also purchase a ready-made tincture.

REMEDIES FOR ULCERS, ABDOMINAL PAIN, AND GASTRITIS: Fennel Tea for Stomach Pain: If ulcers or gastritis has your stomach in knots, try a cup of fennel tea to soothe and reduce burning. Place two teaspoons of fennel seeds in a tea bag and infuse in a cup of hot water for ten minutes. Drink as needed for stomach pain.

Ginger for Digestive Pain and Inflammation: Because ginger is excellent for reducing inflammation, it is very helpful for inflammatory bowel issues. Inflammation in the body leads to pain, and when the inflammation is reduced, so is the pain. It can also soothe an upset stomach, reduce gas and bloating, and dispel nausea. Place a tablespoon of freshly chopped ginger root in a tea bag and infuse in a cup of hot water for fifteen minutes. Drink up to three cups daily as needed for pain and inflammation. Pair with the Balanced Gut Tincture.

Calendula for Easing Discomfort and Supporting the GI Tract: Calendula helps relieve an upset stomach, gas, bloating, and spasms. Take two dropperfuls of calendula tincture up to four times daily.

REMEDIES FOR CONSTIPATION: Wild Senna for Bowel Movement: Wild Senna is a common and potent remedy for constipation. This plant has a rich

history as a purgative due to powerful laxatives in the leaves and seedpods called anthraquinones. Take this remedy with caution and only if you have a serious need for it. Use as a tea. Add a teaspoon of leaves to a tea bag and infuse in hot water for ten minutes. Drink one cup and wait a few hours. If your bowels do not move, try drinking another cup. Drink plenty of water with any remedy for constipation so you do not lose too much water when it begins to work.

Prunes for Constipation Relief: Prunes (dried plums) are one of the best natural remedies for constipation. Eat five to seven medium sized prunes to start and wait a few hours to see if anything happens. If nothing happens, try eating a few more. Eat a few prunes daily if you suffer from chronic constipation.

3. Stress Relief Stress directly affects the digestive tract. Stress may trigger serious flare-ups with gastritis, irritable bowel disease, and even leaky gut. There is a reason people say: “My stomach is in knots” when they are nervous.

Even occasional stress may trigger symptoms like diarrhea, abdominal cramping, and an upset stomach for those who suffer from chronic inflammatory bowel conditions and digestive issues. Not only is the management of stress vital to keeping your digestive tract healthy, it is vital to maintaining a normal-functioning immune system, a healthy weight, and enjoying sound mental health.

Try to implement some of the following practices to help you manage stress. See our Anxiety and Stress Protocol at page 150 for more details and tips.

Spend More Time Outdoors: When people spend time outdoors in nature, their stress levels decrease dramatically. Combine nature and exercise if possible and go on a walk outdoors. Spend at least twenty minutes outside each day. A bird feeder on your porch is a great way to bring nature to you.

Natural Remedies for Stress: If you need some extra help calming down, herbal remedies may be of assistance. Try one or two of the following tinctures: lemon balm, motherwort, lion's mane, reishi, and ashwagandha. My Anxiety & Stress tincture is a blend containing four of these herbs.

Aromatherapy for Stress: Aromatherapy has been proven to help with stress and anxiety. To effectively use essential oils for stress and anxiety, diffuse them in an ultrasonic diffuser with some water or place a few drops in an aromatherapy inhaler to breathe throughout the day. Useful essential oils for stress include lavender, chamomile, cedarwood, bergamot, vetiver, and vanilla.

Meditation and Daily Gratitude: Daily meditation is a useful tool for lowering anxiety and stress and clearing the mind. Meditation can lower your heart rate and blood pressure. Not a meditator? Try gardening, weaving a basket, or other quiet activities that get you out of your head. A daily gratitude practice (saying thanks for

something external) causes a positive internal shift. It is simple yet effective.

Breathing Exercises: Breathing exercises are very helpful for anxiety and stress. Deep lymphatic breathing can help the body relax, while also promoting the flow of lymphatic fluids. Try inhaling deeply through the nose (slowly) until your lungs are completely full and then exhaling through the mouth slowly. Make sure to fully release all the air from the lungs before taking another deep breath. Do this for fifteen to twenty minutes. Repeat up to three times daily or as needed. While breathing, focus on happy, calm thoughts.

Massage, Acupuncture, Regular Self-Care, and Learning to Prioritize and Say No all help with stress relief. **Avoid alcohol, nicotine, and drugs** as these can greatly worsen anxiety and stress. Please visit our Anxiety and Stress Protocol.

4. Supplements Along with a healthy diet, certain supplements help with inflammatory bowel disorders, like Irritable Bowel Syndrome (IBS), leaky gut, and gastritis.

Omega 3 Fatty Acids: Inflammation is a big deal for those with digestive disorders. Omega 3 fatty acids help reduce inflammation in the bowels, reduce oxidative stress, and help maintain remission in patients with certain inflammatory bowel diseases. Suggested dose: 1.5 grams per day for men and 1 g (100mg) per day for women.

Vitamin D: Vitamin D plays a critical role in maintaining the structural integrity of the gastrointestinal barrier.¹⁷ Vitamin D regulates proteins that keep the space between the cells of the gastrointestinal tract tight. If vitamin D is low, the integrity of the gut barrier is compromised – causing “leaky gut.” Suggested dose is 4000 IU per day, with a goal of a 25-hydroxy vitamin D level of 50 ng/mL. Get your levels tested.

Fiber: A good fiber supplement can make a world of difference for those who have a digestive disorder, especially if constipation is an issue. Your gut flora depends on fiber to live. A high fiber diet rich in foods like steamed vegetables and sprouted chia seeds is a great way to get fiber. Suggested supplement dose: 25-35 grams of fiber daily, accompanied by a healthy diet.

Probiotics: While your best bet for getting enough probiotics is from eating fermented foods and drinks, a probiotic supplement is often necessary. Suggested dose: Start with two to four capsules daily of a high-quality probiotic that has at least 50 billion Colony Forming Units (CFUs) with strain diversity. Oftentimes, the supplements that require refrigeration are better.

Digestive Enzymes: If you have issues with leaky gut, try taking digestive enzyme supplements before and after meals. Digestive enzymes can help the body break down proteins, starches, and complex sugars. When the body is able to better break these down, the chances of intestinal inflammation is significantly reduced.

Look for a full spectrum digestive enzyme supplement.

L-Glutamine: This is another highly anti-inflammatory supplement ideal for those with inflammatory bowel conditions.¹⁸ It may even help repair intestinal lining that has been damaged. Stress depletes glutamine levels, so if you are stressed a lot, this is a good addition. Suggested dose: 2-5 g daily. If you are severely stressed or are extremely active, you can take up to ten grams daily. **Turmeric** is also an excellent addition as a supplement for inflammation.

Collagen Powder: Bone broth is suggested above in the diet section because it is highly nutritious and anti-inflammatory. It also contains collagen. Daily collagen supplementation can help boost the production of gastric juices, as well as help repair the intestinal mucosal lining¹⁹²⁰. Suggested dose: Drink eight to sixteen ounces of bone broth or take four tablespoons of collagen powder daily (or a mix of both).

5. Exercise and What to Avoid

IBS symptoms can make it especially hard to get the daily exercise you need to stay healthy. However, studies show that there is a link between high activity and reduced symptoms.

If you are feeling fatigued frequently, try a lower impact exercise that also works to reduce stress, such as yoga. Yoga helps tone the body and also reinforces a mind-body connection, which has been proven to lower stress levels, which in turn helps with leaky gut, gastritis, and IBS.

Do not feel like you need to work out hard. Intense cardio and strength training may be too much. If you wish, try starting with circuit-based exercises, low-impact strength training exercises, and low to moderate impact intervals.

In the diet section of this protocol, certain foods to avoid were listed. However, there are a few additional things below that you should avoid to prevent triggering a flare:

Unnecessary Antibiotics: Antibiotics are often overprescribed. The result of this is antibiotic-resistant bacteria and damage to our gut health. Antibiotics do have their place but many doctors have stopped

prescribing antibiotics for ailments such as ear infections. In addition, viruses never require antibiotics. Choose a doctor who doesn't overprescribe antibiotics. If you have to take antibiotics, double up on the amount of probiotic foods and drinks you consume each day, and combine with prebiotics.

Environmental Toxins: We expose ourselves to environmental toxins each day. Whatever we can do to lessen that exposure is a good thing. Opt for BPA free plastics, ditch the nonstick cookware and keep your phone away from your body whenever possible. Turn off your Wi-Fi router at night, opt for organic produce and products, and be mindful of body lotions, soaps, and shampoos you use.

Wheat Products and Baked Goods: These products increase inflammation in the body. Furthermore, they contain high amounts of gluten, which may negatively affect the digestive tract.

6. Western Medical Approaches

Many western doctors will advise patients to keep a healthy diet to avoid flare-ups with leaky gut, gastritis, and IBS. For minor cases, this may be the only protocol recommended/needed. Other doctors may prescribe medications to help control the symptoms.

A popular drug for IBS is dicyclomine. Another drug often prescribed is Bentyl. Sometimes certain diagnostic procedures may be administered to see the progression of the disease before prescribing a drug. These procedures range from a colonoscopy to an X-ray or CT scan. Stool tests may also be conducted.

Certain diagnostic tests may be conducted for those with gastritis. These include a test for *H. pylori* bacteria, an endoscopy, and an X-ray. Antibiotics may be prescribed if the test comes back positive for *H. pylori*. Many times, doctors will prescribe antacids and acid blockers to neutralize or block acid production so the lining can heal.

The diagnosis and treatment of leaky gut/increased intestinal permeability is still developing, as this is a somewhat new issue for many western doctors. A change in diet may be suggested, as well as a colonoscopy.

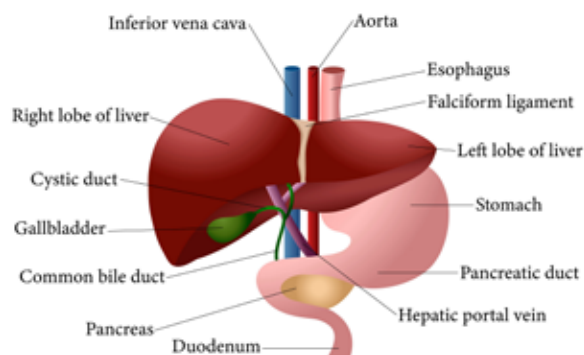
	Morning	Noon	Afternoon	Evening	Before Bed
S U N D A Y	Bone broth Digestive enzymes Lion's Mane, Reishi, & Turkey Tail dual-extracted tinctures (2 dropperfuls of each) Collagen & aloe vera (in a smoothie, if desired) Omega 3, vit. D, fiber, probiotics, & L-glutamine	Weekly meal plan Weekly grocery shop Balanced gut tincture Marshmallow root & slippery elm tea Stress relief activity Daily gratitude	Digestive enzymes Collagen & aloe vera Ginger and fennel tea Yoga	Balanced gut tincture Fiber, probiotic, & turmeric Digestive enzymes	Anxiety & stress tincture Chamomile, slippery elm, & marshmallow root tea Aromatherapy Epsom salt bath
M O N D A Y	Bone broth Digestive enzymes Lion's Mane, Reishi, & Turkey Tail tinctures Collagen & aloe vera (in a smoothie, if desired) Omega 3, vit. D, fiber, probiotics, & L-glutamine	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs	Digestive enzymes Collagen & aloe vera Ginger and fennel tea Yoga	Balanced gut tincture Fiber, probiotic, & turmeric Digestive enzymes	Anxiety & stress tincture Chamomile, slippery elm, & marshmallow root tea Aromatherapy Epsom salt bath
T U E S D A Y	Bone broth Digestive enzymes Lion's Mane, Reishi, & Turkey Tail tinctures Collagen & aloe vera (in a smoothie, if desired) Omega 3, vit. D, fiber, probiotics, & L-glutamine	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs	Digestive enzymes Collagen & aloe vera Ginger and fennel tea Yoga	Balanced gut tincture Fiber, probiotic, & turmeric Digestive enzymes	Anxiety & stress tincture Chamomile, slippery elm, & marshmallow root tea Aromatherapy Epsom salt bath
W E D N E S D A Y	Bone broth Digestive enzymes Lion's Mane, Reishi, & Turkey Tail tinctures Collagen & aloe vera (in a smoothie, if desired) Omega 3, vit. D, fiber, probiotics, & L-glutamine	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs Lymphatic massage	Digestive enzymes Collagen & aloe vera Ginger and fennel tea Yoga	Balanced gut tincture Fiber, probiotic, & turmeric Digestive enzymes	Anxiety & stress tincture Chamomile, slippery elm, & marshmallow root tea Aromatherapy Epsom salt bath
T H U R S D A Y	Bone broth Digestive enzymes Lion's Mane, Reishi, & Turkey Tail tinctures Collagen & aloe vera (in a smoothie, if desired) Omega 3, vit. D, fiber, probiotics, & L-glutamine	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs	Digestive enzymes Collagen & aloe vera Ginger and fennel tea Nature walk	Balanced gut tincture Fiber, probiotic, & turmeric Digestive enzymes	Anxiety & stress tincture Chamomile, slippery elm, & marshmallow root tea Aromatherapy Epsom salt bath
F R I D A Y	Bone broth Digestive enzymes Lion's Mane, Reishi, & Turkey Tail tinctures Collagen & aloe vera (in a smoothie, if desired) Omega 3, vit. D, fiber, probiotics, & L-glutamine	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs	Digestive enzymes Collagen & aloe vera Ginger and fennel tea Yoga	Balanced gut tincture Fiber, probiotic, & turmeric Digestive enzymes	Anxiety & stress tincture Chamomile, slippery elm, & marshmallow root tea Aromatherapy Epsom salt bath
S A T U R D A Y	Bone broth Digestive enzymes Lion's Mane, Reishi, & Turkey Tail tinctures Collagen & aloe vera (in a smoothie, if desired) Omega 3, vit. D, fiber, probiotics, & L-glutamine	Berberine before lunch Candida cleanse broth/gentle solids Drink plenty of water Milk thistle & Dandelion root tincture or tea Balanced Gut Herbs Lymphatic massage	Digestive enzymes Collagen & aloe vera Ginger and fennel tea Nature walk	Balanced gut tincture Fiber, probiotic, & turmeric Digestive enzymes	Anxiety & stress tincture Chamomile, slippery elm, & marshmallow root tea Aromatherapy Epsom salt bath

Ch. 12: Liver Health Protocol

The liver is a large organ located in the upper right portion of the abdominal cavity on top of the stomach, right kidney, and intestines.¹ At any given time, the liver holds around one pint (13%) of the body's blood supply.² The liver is supplied with blood via the hepatic artery, which supplies the liver with oxygenated blood and the hepatic portal vein, which supplies the liver with nutrient-rich blood.

The liver consists of two main lobes. Each lobe is made up of eight segments that contain one thousand small lobules, which are connected to tubes that connect with larger ducts to form the common hepatic duct. The job of this duct is to transport bile made by liver cells to the gallbladder and duodenum through the common bile duct. Bile is made by the liver to break down fats and also helps with digestion, absorption, and excretion; bile is stored in the gallbladder.

STRUCTURE OF THE LIVER



The liver performs more than *five hundred* important functions, including helping regulate chemical levels in the blood and excreting bile. The blood leaving the stomach and intestines also passes through the liver, and the liver helps break down, balance, and create nutrients for this blood.

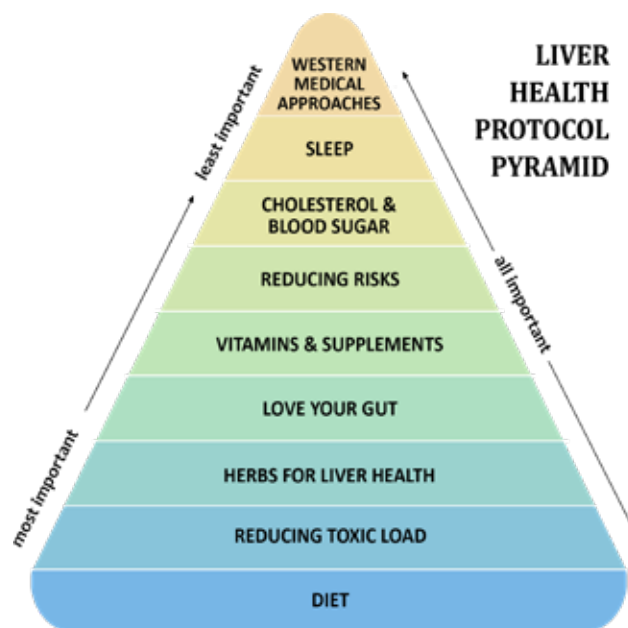
Other functions include the production of proteins for blood plasma, production of cholesterol and other special proteins to help carry fats through the body, the conversion of extra glucose into glycogen for storage, balancing/making glucose as needed, the regulation of amino acids in the blood, the clearing of drugs and other harmful substances in the blood, the processing of hemoglobin for iron storage, the conversion of ammonia to urea, the regulation of blood clotting, and helping to resist infections by creating immune factors and removing bacteria from the bloodstream³. The liver also clears bilirubin.

One common theme among many of these functions is the clearing of toxins and waste products, such as drugs/medications, ammonia, and bilirubin. The breaking down of harmful substances means the

liver must excrete the by-products into the blood. These by-products then enter the intestine and leave the body through excretion. Other blood by-products are filtered out by the kidneys.

Your liver is your biggest and hardest-working detoxification organ. With all the toxins moving through it, the liver can be susceptible to damage and disease. Impaired function of this important organ is becoming increasingly common due to environmental toxins, obesity, diets high in processed foods and saturated fats, viral infections, and autoimmune diseases. Today, it is estimated that one in ten people in the United States have some sort of liver or gallbladder disease.

Common diseases of the liver include fatty liver disease, cirrhosis of the liver, autoimmune hepatitis, viral hepatitis, primary biliary cholangitis, and primary sclerosing cholangitis.⁴ Some liver diseases are inherited but most liver diseases are caused by alcohol use, drugs and medications, poor diet, and autoimmune diseases.



Symptoms of liver disease include: yellowing of the skin and eyes that appear yellow (jaundice); bloating and/or gas; constipation; moodiness and irritability; bruising easily; high blood pressure; weight gain and the inability to lose weight; nausea, vomiting, or loss of appetite; pale stools; dark urine; swelling or pain in the upper right abdomen; chronic fatigue; swelling in the ankles and/or legs; itchy soles of the feet and/or palms of the hands.

If you have a family history of liver cancer or liver disease, you may be at an increased risk of liver complications. There are steps you can take today to support your liver, help it filter waste more efficiently, and reduce your toxic load.

1. Diet Your liver must process all the toxins that enter your body – and one of the most common ways those toxins enter is in the foods you eat. In the first part of this section, you will learn about foods you can incorporate into your diet to help your liver do its job more efficiently. The second half will cover foods

and beverages that harm the liver.

BENEFICIAL FOODS (AND BEVERAGES):

Eat Potassium-Rich Foods: Focus on foods like sweet potatoes, spinach, bananas, tomatoes, and blackstrap molasses to help cleanse your liver. Dandelion is a good source of potassium (more on this below). Potassium assists with balancing sodium levels so that you are not retaining excess water, thereby helping to successfully flush out toxins.

Eat More Berries: Antioxidant-rich berries like blueberries, aronia berries, and cranberries contain anthocyanins that protect the liver from damage while also increasing immune cell response and antioxidant enzymes. Antioxidants in blueberries may help to slow the development of lesions and fibrosis in the liver^{5 6} and blueberry extract was shown to stop the growth of human liver cancer cells in vitro⁷. Try incorporating one half to one cup of berries with a meal or two each day.

Cruciferous Vegetables Are Great for Your Liver:

Vegetables like broccoli, mustard greens, brussel sprouts, watercress, and collard greens contain a lot of fiber plus they contain compounds that may help to increase the levels of detoxification enzymes that help protect the liver from damage. Broccoli may inhibit the growth of tumors and fatty liver disease⁸. These benefits remain even when cooked. Try lightly steaming cruciferous vegetables and eating a cup with each meal.

Nuts: Nuts are full of nutrients and antioxidants that benefit the liver. People who eat more nuts have a decreased risk of nonalcoholic fatty liver disease⁹.

Omega 3-Rich Foods: Omega 3s lower liver fat and triglycerides for those with nonalcoholic fatty liver disease or nonalcoholic steatohepatitis^{10 11}. Consume omega 3-rich fish (wild caught not farm raised) to help support your liver instead of omega 6 fats, which can harm the liver.

Beet-Root Juice: You can eat beets or juice them for their liver protective effects. Beet-root juice has high levels of nitrates, which are altered to nitric oxides, which have a vasodilatory effect – reducing blood pressure and increasing oxygen and nutrient delivery¹². They also have high levels of antioxidants, which reduces oxidative damage and inflammation in the liver^{13 14}. Opt for organic produce to protect your liver from harmful pesticides and toxins.

Coffee: There is an association between coffee and a lower risk of death for those with liver disease. Coffee may protect the liver, lower the risk of cirrhosis and liver damage, and reduce the risk of developing a certain type of liver cancer. Coffee may also have positive effects on liver inflammation. Coffee can help prevent the buildup of fat and collagen, two markers of liver disease. Those who drink coffee may have increased levels of the antioxidant glutathione to neutralize free radicals that damage liver cells. Choose organic coffee, as other types of coffee have been exposed to harmful chemicals that the liver will have to process. Don't forget to drink plenty of water when drinking coffee and other diuretic beverages that

flush water from the body, and best to limit your intake to the morning only.

Grapefruit: These antioxidant-rich citrus fruits reduce inflammation and protect liver cells. Additionally, antioxidants in grapefruits can reduce the development of hepatic fibrosis and even reduce the amount of fat in the liver. Other studies link grapefruit with the ability to metabolize alcohol and counteract its negative effects on the liver.

Grapes: Organic red and purple grapes are great sources of resveratrol, a compound shown to benefit liver health. They are potent sources of antioxidants and have been shown to benefit the liver by lowering inflammation, preventing liver damage, and increasing antioxidant levels to fight free radicals that could lead to issues. Supplementing with grapeseed extract may help improve liver function¹⁵.

Green Tea (and Other Teas): Green tea improves blood markers for liver health and improves liver enzyme levels and reduces oxidative stress and fat deposits in those with nonalcoholic fatty liver disease¹⁶. Those who drink green tea are less likely to develop liver cancer. It may also help to reverse the negative effects of a poor diet on the liver¹⁷. Drink a cup a day, along with plenty of water, for best results. Avoid green tea in large amounts, concentrated forms, or extract forms, as too much may have the opposite effect.

Lemon Water: Drinking a full 16-ounce glass of purified water with the juice of half a lemon first thing in the morning helps to cleanse the liver, hydrate you, and flush away toxins that have accumulated from the previous day. Make sure to drink enough water throughout the day to support detoxification¹⁸.

Prickly Pear Cactus: This edible cactus produces liver-protective fruits and juice that help liver disease. Prickly pear has been shown to help reduce hangovers, which demonstrates its ability to help the liver better detox potentially harmful substances. Prickly pear can help normalize liver enzymes and cholesterol levels¹⁹. The fruits can be consumed or the juice can be added to smoothies, etc.

FOODS TO AVOID (AND BEVERAGES): Alcohol:

Excessive alcohol consumption contributes to three different types of liver disease: cirrhosis, fatty liver, and alcoholic hepatitis. Of all the things you should avoid for liver health, alcohol should be at the top of the list.

Fried Foods: Fried foods are especially bad for the liver. Saturated and trans fats are harder for the liver to process and they end up accumulating in the liver. This can lead to inflammation and an increased risk of cirrhosis. Saturated fats contribute to fatty liver disease because they also increase blood cholesterol levels.

Limit Intake of Sugar: Sugar is strongly linked with many chronic diseases such as obesity, heart disease, cancer, and diabetes²⁰. Consuming high amounts of sugar and processed foods puts excessive strain

on the liver and kidneys - your two most important detoxification organs. High blood sugar levels contribute to the accumulation of fat in the liver.

Non-Organic Foods: Whenever possible, choose organic foods to avoid harmful toxins that result from the spraying of pesticides, herbicides, and other chemicals. These force the liver to work harder to filter out these toxins and they can accumulate in the liver.

Processed Meats: Deli meats, sausage, and other processed meats are high in sodium, saturated fat, and other toxins that are harmful to the liver. The liver breaks down and processes protein at a much slower rate than other compounds. If saturated fats and sodium are allowed to stay at high concentrations in the body, the liver struggles to function at its normal rate.

Processed Wheat Flour: Processed wheat flour is used to make pasta, breads, pancakes, cake, cookies, and sauces. Refined wheat flour can damage the liver because the body converts it into sugar, which increases blood sugar levels. Additionally, these excess sugars remain in the body as energy reserves that become harmful if they are not burned through exercise or digestion. Not only are these processed flours potentially harmful for the liver, they also contain very little fiber.

Sodium: Sodium is much higher in processed and canned foods. Too much sodium can result in fluid retention and inhibits the toxin elimination process. Limit sodium to less than 1,500 milligrams a day.

2. Reducing Toxic Load

The liver's job is to filter toxins. In this day and age, it is confronted with a constant barrage of toxins in the form of drugs, bad habits, chemicals in everyday products, and pesticides and herbicides in various foods. In this section, you will learn what you can do to minimize your toxic load and what you can use to replace certain toxic offenders.

Medications and Other Drugs: It is the liver that is responsible for drug uptake, distribution, and metabolism. Thus the liver and kidneys are often the first to be affected by medications²¹. Watch your intake of over-the-counter pain relievers/fever reducers like acetaminophen (safe for the liver in lower doses), ibuprofen, and naproxen. Non-steroidal anti-inflammatory drugs (NSAIDs) are not safe for those with cirrhosis and none are safe when taken with alcohol. Prolonged use of certain antibiotics can harm the liver. Research the medications you take and try to tackle the root cause of illnesses rather than the symptoms to prevent unnecessary medication consumption.

Cleaning Products and Aerosols: Aerosol sprays such as room sprays contain toxins that are inhaled into the body and get into the bloodstream²². They are especially harmful for the liver. Cleaning products may absorb into the skin or be inhaled into the lungs. Replace room fragrance sprays with natural essential oils by diffusing them in an ultrasonic diffuser to freshen

a room. Replace cleaning products with natural, vinegar-based products.

Cookware: Cookware matters. Avoid aluminum cookware, as it is linked with reduced brain function and cancer. Teflon contains extremely toxic compounds that are linked to cancer and other health issues. Cook with stainless steel or cast iron (iron is a metal, but most of us are deficient and we need iron to be healthy)²³.

Makeup: It is estimated that women use an average of 12 makeup products on their skin every day. Many makeup products are heavily tainted with heavy metals and harmful toxins²⁴. When these toxins are applied to the skin, they are absorbed into the bloodstream. Research the makeup you use to make sure your makeup is safe.

Plastics: Many plastics contain chemicals that are linked with cancer and other health issues²⁵. Opt for BPA free plastic and switch out your plastic dinnerware and water bottles with glass or stainless steel. Never microwave plastics, as this encourages toxins to leach into your food.

Toiletries: Research the products you are using on your skin each day to make sure they don't contain toxins like sulfates, parabens, polyethylene glycol, phthalates, formaldehyde, triclosan, dimethicone, retinyl palmitate, alcohols, dyes, aluminum²⁶, and synthetic fragrances. These are common in many toiletries like shampoos, conditioners, deodorants, and lotions²⁷. The skin, our largest organ, absorbs these and they get into the bloodstream. Purchase from reputable companies that exclude these ingredients or make your own.

Water: Heavy metals and other toxins occur naturally in soil, so it isn't uncommon for water to contain these potentially harmful compounds²⁸. Research shows that states like California, Montana, Texas, and Nevada have higher amounts of arsenic in water. Check your area's water quality by going to the EWG's Tap Water Database. There are water filters you can buy that do a good job of filtering out heavy metals and other toxins.

3. Herbs for Liver Health Along with diet and the elimination of toxins in your life, there are herbs you can incorporate to support your liver, help eliminate toxins, and protect the liver from damage.

Reishi (*Ganoderma lucidum*): Reishi is the most studied mushroom with respect to hepatoprotective (liver-protecting) effects²⁹. Reishi helps balance liver enzymes and improve liver function. The mechanisms suggested include its antioxidant and radical scavenging activity, anti-fibrotic and antiviral activity, modulation of hepatic phase I and II enzymes, inhibition of b-glucuronidase, modulation of nitric oxide production, maintenance of hepatocellular calcium homeostasis, and its immunomodulatory effects^{30 31}. Like most mushrooms, it should be taken as a dual extracted tincture to get the most benefits. Mushrooms contain both water and alcohol soluble compounds. A double extraction helps you obtain all medicinal compounds, including the water-soluble

polysaccharides and the alcohol-soluble triperpenoids, which are hepatoprotective perhaps due to the promotion of scavenging enzymes for hepatic free radicals³². Suggested dose: 2 dropperfuls of a dual extracted tincture twice daily.

Cordyceps (*Cordyceps* spp.): This powerful adaptogenic mushroom supports cardiovascular and liver health. Research shows Cordyceps may also inhibit the growth of cancer cells in the lungs, colon, and liver.^{33 34} Suggested dose: 2 dropperfuls of a dual extracted tincture twice daily.

Dandelion (*Taraxacum officinale*): Dandelion root can help detox the liver, gallbladder, and kidneys. It is a diuretic and blood purifier, helps the liver flush out toxins easier, reduces lipid accumulation and liver inflammation, and is hepatoprotective^{35 36}. In addition, dandelion root helps replace potassium and other minerals. Since it is also a source of inulin, it also supports healthy gut flora. Dandelion root tincture makes a great addition to a liver cleanse, especially when combined with milk thistle. Suggested dose: Drink two to three cups of dandelion tea for two weeks, or use 2 dropperfuls of a dandelion/milk thistle tincture twice daily.

Milk Thistle Seeds (*Silybum marianum*): Milk thistle is one of the most popular liver-supporting remedies available. Its active component is silymarin, composed of silybin, silydianin, and silychristin³⁷. It helps detoxify and support the liver by clearing environmental toxins, chemotherapy toxins, poison, and harmful drugs from the body. It helps with cirrhosis, fatty liver disease, jaundice, and hepatitis³⁸. Best used as a daily tincture along with dandelion root. Suggested dose: 2 dropperfuls of a dandelion/milk thistle tincture or supplement twice daily.

Cat's Claw (*Uncaria tomentosa*): This herb can detoxify the body and cleanse blood and lymphatic fluid by aiding in the removal of toxins, drugs, heavy metals, and other chemicals. It enhances the function of the kidneys, spleen, pancreas, and digestive system, shows promise for tumor reduction for liver cancer³⁹, and combats oxidative stress and liver damage in animal models⁴⁰. (NOTE: It may be too stimulating for some people with an autoimmune disorder – ask your ND).

Chickweed (*Stellaria media*): Chickweed shows great promise against the Hepatitis-B virus (HBV). It is currently under early development as a potential anti-HBV drug candidate⁴¹.

Cleavers (*Galium aparine*): Cleavers (Bedstraw/*Galium*) is known for cleansing the lymphatic system. They help promote the unclogging of ducts and get lymphatic fluids moving to detox the body, helping the liver do its job more efficiently. Cleavers are most effective when prepared fresh, whether in a decoction, tea or tincture.

Herbs to Avoid for Liver Health: Some herbs can put a heavier load on the liver. Avoid using the following

herbal remedies and supplements internally if you have liver issues: Kava kava, comfrey, angelica, tansy, cascara sagrada, ephedra, cloves (in large amounts only), golden ragwort, gotu kola, pennyroyal, mugwort, rue, sassafras, thuja, wormwood, and chaparral. Avoid ingesting any essential oil, as these are very concentrated plant isolates that can put a heavy load on the liver and kidneys.

4. Love Your Gut

Gut health is crucial for proper detoxification. Our gut cells have a system in place for detoxifying and excreting toxins; this protects our body from chemicals, heavy metals, and other harmful substances. If the gut becomes compromised, this can manifest as leaky gut and the toxins that should go through the gut and be excreted are instead reabsorbed into the bloodstream.

HERBAL REMEDIES FOR GUT HEALTH AND REPAIR: The following botanicals are outstanding for the gut and stomach. They are a great place to begin your journey to gut restoration.

Lion's Mane (*Hericium erinaceus*): A known antioxidant and gastroprotective, Lion's Mane helps to tame the inflammatory response and protect the lining of the stomach and gut. It also helps protect the gut from *H. pylori*.

Marshmallow Root (*Althaea officinalis*) and Slippery Elm (*Ulmus rubra*): Marshmallow root and slippery elm are demulcents, forming a protective layer over intestinal perforations. This can help prevent the absorption of waste into the bloodstream from the gut.

Plantain (*Plantago major*): Plantain is anti-inflammatory and soothes mucous membranes.

Reishi (*Ganoderma lucidum*): Reishi exerts anti-inflammatory action that's helpful for calming intestinal inflammation and inhibits *Candida* yeast.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): Containing important prebiotics that balance the microbiome, encourage the growth of "good" gut bacteria, and control *Candida* overgrowth, Turkey Tail is a remarkable remedy for the gut.

In addition to the herbal remedies above, the *Leaky Gut Protocol* (see page 70) in this book can give you many resources to help you restore and heal your gut so you can better care for your liver. The Leaky Gut Protocol also discusses probiotics and how you can get more in your diet (via foods and beverages, as well as supplements) to enhance and restore gut health.

5. Vitamins and Supplements

Some vitamins and minerals can become depleted when the liver is not functioning properly. If we aren't getting enough of these in our diet, it makes it even harder for the liver to function. As with any supplement, make sure your body needs the nutrient before adding it to your daily supplement regimen. Have bloodwork and other tests performed to see if you are deficient. A daily multivitamin might be

sufficient to cover your daily intake of certain important vitamins and minerals needed for overall wellness. The following vitamins and minerals are needed for proper liver function:

Vitamin D: The best way to get this vitamin is through sun exposure – those of us who live in temperate zones most likely need supplementation. The liver and kidneys convert vitamin D into its active form when the body is exposed to sunlight⁴². Vitamin D has a long list of health benefits such as immune system support, lowered risk of inflammation, and a reduced risk of chronic disease. Vitamin D deficiency is associated with chronic liver disease⁴³, and vitamin D levels are inversely related to the severity of CLD⁴⁴. Suggested dose: 2000 units of Vitamin D drops daily.

Vitamin B12: The liver stores around 90% of your body's vitamin B12, which is metabolized in the liver; its by-products are removed by the kidneys. Too much B-12 may cause damage to liver cells, leading to inflammation and scar formation⁴⁵. However, chronic vitamin B12 deficiency might lead to decreased damage repair, diminished hepatic detoxification, and even to chronic liver disease, like liver cirrhosis⁴⁶. Vitamin B12 plays a vital role in the production of methionine (an amino acid that is converted into S-adenosylmethionine). Best to get your levels tested to see if you need supplementation.

Vitamin E: Vitamin E is an antioxidant and can thus neutralize free radicals. It has been studied for its ability to treat chronic hepatitis C, alcoholic hepatitis, and it reduces oxidative stress in nonalcoholic fatty liver disease⁴⁷. Take vitamin E only when recommended by a doctor, as too much can have negative effects in the body. If you want a natural source of vitamin E, you can add more to your diet in the form of foods like almonds, sunflower seeds, and spinach.

Vitamin C: Vitamin C is a powerful antioxidant that can improve liver health and glucose metabolism in patients with nonalcoholic fatty liver disease^{48 49 50}. It is thought to be beneficial for liver health because it helps maintain healthy tissues and eliminate free radicals that cause damage. Vitamin C helps regenerate tissues and heal wounds. Suggested dose: 1000mg/day.

6. Reducing Risks

Sometimes the best defense is a good offense. Make sure you are aware of viruses that can cause liver damage and take steps to avoid getting these. In addition, take steps now to avoid fatty liver disease that can lead to liver damage:

Hepatitis can be alcohol-induced, caused by medications, and even some autoimmune diseases. However, it can also come from a virus like hepatitis B and C. Various forms of hepatitis can lead to significant liver damage and even death over a period of time (depending on the severity). Both hepatitis B and C are spread by bodily fluids.

Some doctors suggest getting a vaccine to prevent hepatitis; do your research and make an informed decision by weighing the pros and cons. A majority of the time, hepatitis C is caused by sharing or using

dirty drug needles and having unprotected sex. Practice safe sex and avoid drugs, which are bad for your liver and overall health. Other ways to avoid viral hepatitis include washing your hands after using the restroom and when you come into contact with an infected person's body fluids. Avoid unclean food and water. Avoid alcohol, as it can worsen hepatitis.

Fatty liver disease is one of the most common causes of liver damage⁵¹. It is often caused by obesity, high blood pressure, diabetes, excessive alcohol consumption, the overproduction of fatty acids, medications that interfere with insulin function (like anabolic steroids), or diseases that affect how substances are produced or removed from the body (like hemochromatosis).

Since a common cause of fatty liver disease is an unhealthy lifestyle and a diet that leads to obesity and high blood pressure, there is a strong chance you can reverse liver damage and allow the liver to heal by losing weight and making dietary changes⁵². Weight loss leads to lower blood pressure, cholesterol, and more stabilized blood sugar. Along with a healthy diet focused on weight loss (if you are overweight), more physical activity can further prompt weight loss and promote optimal health. Avoid fatty foods like fried foods, processed foods, too many carbohydrates, sugar, and bad oils. Make sure you are getting probiotics in your diet through fermented foods and drinks like sauerkraut and kombucha. Consider kicking off your weight loss with a cleanse to promote the flushing of waste from the body and create avenues for the elimination of fat cells in the liver. A simple two-day cleanse that involves drinking lots of lemon water and a healthy broth made with cruciferous vegetables (or sticking to vegetables, berries, and small portions of lean meat like chicken if you can't tolerate broth for two days) is a great way to kick start your journey to health. See the **Candida Protocol** chapter (page 46) for a detailed food cleanse option.

7. Cholesterol and Blood Sugar

For a healthy liver, one must keep cholesterol levels in check. Cholesterol levels and the liver are connected⁵³. One of the liver's many functions is the metabolism of fat, carbohydrates, and protein. The liver also produces and clears cholesterol. Cholesterol isn't necessarily a bad thing - cholesterol is necessary for the creation of hormones, vitamin D, and certain enzymes needed for healthy digestion. Cholesterol is carried throughout the body by bundles of lipoproteins: high-density lipoproteins (HDL) and low-density lipoproteins (LDL). The body needs both of these in the correct proportion to function. HDL is often called the "good cholesterol" and LDL the "bad cholesterol." The problem is, with the current modern-day diet, many are getting far too much LDL cholesterol in their diet and this is leading to a stressful load on the liver.

The liver's ability to produce and clear cholesterol is hindered by poor diet, obesity, diabetes, and other factors⁵⁴. A spike in cholesterol can damage the liver. To keep cholesterol in check, eat a diet with more cholesterol-lowering foods such as organic oats, nuts, okra, fatty

fish, berries, and citrus fruits. Fiber supplements may also help with cholesterol levels. Go easy on meat-based protein and opt for non-animal sources of protein, such as nuts, seeds, and protein rich vegetables like peas. See our **Heart Health Protocol** at page 42 for more details.

Those with high blood sugar levels and diabetes are at an increased risk of liver disease, as the liver plays a large role in blood sugar regulation. It can store sugar, release sugar, and even create sugar. The liver works with the pancreas in a precise system to control blood sugar. Additionally, the majority of the body's glucose, protein, and fat metabolism occur in the liver. Glucose from our diet is taken to the liver where it is fed into various metabolic pathways. The liver and pancreas communicate to aid in the secretion of insulin and glucagon, two hormones involved in blood glucose regulation⁵⁵. Insulin lowers blood glucose by allowing sugar to enter cells. Glucagon signals for the breakdown of glycogen (these are glucose molecules store in the liver) and this helps raise blood glucose levels. See our **Diabetes Type-2 Protocol** at page 96 for more detail.

Below are some herbs recommended for blood sugar management:

Bilberry (*Vaccinium myrtillus*): Both the berries and leaves of this plant lower blood sugar. It is believed that bilberries prevent the breakdown and absorption of carbohydrates in your gut, acting similarly to certain blood sugar-lowering medications⁵⁶. Bilberry can be taken in supplement form, tea, or as a tincture. Bilberries can also be eaten fresh or dried.

Cordyceps (*Cordyceps* spp.): Cordyceps mushrooms contain a compound that acts like insulin to help lower blood glucose levels. It has hypoglycemic, anticholesterol, and antihypertriglyceridemic actions. Cordyceps may improve insulin sensitivity while simultaneously lowering bad cholesterol (LDL), making them doubly beneficial for those looking to improve liver health^{57 58}. Suggested dose: 2 dropperfuls dual-extracted Cordyceps tincture twice daily.

Reishi (*Ganoderma lucidum*): Reishi mushroom helps lower high blood pressure, high blood sugar, and high cholesterol, and it protects against insulin resistance^{59 60 61 62}. Suggested dose: 2 dropperfuls twice daily⁶³ of a dual-extracted tincture.

Fenugreek Seeds: Fenugreek has been studied for its effects on blood sugar levels and may help lower blood glucose while leaving serum insulin levels unaffected. In addition to being helpful for blood sugar regulation, fenugreek may also improve blood pressure⁶⁴. Suggested dose: a dose of 1 gram daily of an extract of fenugreek seeds or fenugreek powder starting at 5mg per day (add it to your daily meals), or fenugreek tincture.

Hawthorn (*Crataegus monogyna*): This berry is known for its positive effects on blood pressure and cardiovascular health, allowing for better circulation and blood flow to the liver. Another benefit of this berry is

its ability to help control blood sugar levels⁶⁵. Suggested dose: 2 dropperfuls of hawthorn tincture twice daily.

Holy Basil/Tulsi: This adaptogenic herb stabilizes blood sugar (glucose) levels, enhances insulin production, improves lipid profiles, helps with insulin resistance, prevents weight gain, and protects the liver and kidneys from high blood glucose⁶⁶. Because tulsi (holy basil) helps the body respond better to stress, it is also associated with an improvement in blood pressure. Suggested dose: The median effective supplement dose is 30 mg/kg or take as a tincture twice daily⁶⁷.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): This polypore has been proven effective at reducing blood glucose and managing blood sugar levels for those with diabetes. They also help lower cholesterol and blood pressure. Suggested dose: 2 dropperfuls dual-extracted turkey tail tincture twice daily.

8. Sleep In order to support successful detoxification, you must get good deep sleep. Deep sleep aids in the restoration and healing of tissues, detoxification, and immune function. When we sleep, our body is able to focus on removing potentially dangerous toxins and toxic byproducts that have accumulated in our system - including those of the brain called beta-amyloid, which has been shown to increase your risk of Alzheimer's disease⁶⁸. Please see our **Sleep Protocol** at page 286 for more information. Quick tips include establishing a regular bedtime routine, avoiding blue light a few hours before bedtime; diffusing sleep-friendly essential oils, and trying sleep herbs like valerian and hops in tincture-form before bed.

9. Western Medical Approaches Sometimes those with liver disease must rely on medical treatment like pharmaceuticals and surgery to help their liver function better. Both gastroenterologists and hepatologists are qualified to treat liver diseases. For most people seeking treatment, the first step in the process is testing and diagnostics to better determine the cause of the issue⁶⁹. These tests may include bloodwork to check liver function, imaging tests to look for liver damage, and perhaps taking a tissue sample to get a better understanding of the condition of the liver.

Medical treatment of liver disease depends on the disease itself. Some liver issues may be treated with lifestyle modifications like avoiding alcohol or losing weight. Others may be treated with medication⁷⁰. For serious liver disease, some people may need a liver transplant or surgery. Transplants for liver disease are often performed when a person has liver failure due to the extent of their disease.

If you are concerned with your liver health, see a specialist like a gastroenterologist or a hepatologist to get a better understanding of your disease, your options, and how to heal. Schedule yearly exams and checkups to stay on top of your liver health.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Lemon water Liver-friendly breakfast Cordyceps and reishi dual-extracted mushroom tinctures Leaky gut herbs Multivitamin Vit. D and C	Weekly grocery shopping for liver-friendly foods Liver-friendly lunch Stay hydrated – drink plenty of water Milk thistle & dandelion root supplement	Exercise at least 20 min. Lemon water Leaky gut herbs Cordyceps and reishi mushroom tinctures	Liver-friendly dinner Avoid screens/blue light two hours before bed Milk thistle & dandelion root supplement	Pre-bed aromatherapy and/or Epsom salt bath if needed for sleep Bedtime routine Sleep herbs, if needed
MONDAY	Lemon water Liver-friendly breakfast Cordyceps and reishi dual-extracted mushroom tinctures Leaky gut herbs Multivitamin Vit. D and C	Liver-friendly lunch Stay hydrated – drink plenty of water Milk thistle & dandelion root supplement Yearly check-up with hepatologist	Exercise at least 20 min. Lemon water Leaky gut herbs Cordyceps and reishi mushroom tinctures	Liver-friendly dinner Avoid screens/blue light two hours before bed Milk thistle & dandelion root supplement	Aromatherapy Bedtime routine Sleep herbs, if needed
TUESDAY	Lemon water Liver-friendly breakfast Cordyceps and reishi dual-extracted mushroom tinctures Leaky gut herbs Multivitamin Vit. D and C	Liver-friendly lunch Stay hydrated – drink plenty of water Milk thistle & dandelion root supplement	Exercise at least 20 min. Lemon water Leaky gut herbs Cordyceps and reishi mushroom tinctures	Liver-friendly dinner Avoid screens/blue light two hours before bed Milk thistle & dandelion root supplement	Aromatherapy Bedtime routine Sleep herbs, if needed
WEDNESDAY	Lemon water Liver-friendly breakfast Cordyceps and reishi dual-extracted mushroom tinctures Leaky gut herbs Multivitamin Vit. D and C	Liver-friendly lunch Stay hydrated – drink plenty of water Milk thistle & dandelion root supplement	Exercise at least 20 min. Lemon water Leaky gut herbs Cordyceps and reishi mushroom tinctures	Liver-friendly dinner Avoid screens/blue light two hours before bed Milk thistle & dandelion root supplement	Aromatherapy Bedtime routine Sleep herbs, if needed
THURSDAY	Lemon water Liver-friendly breakfast Cordyceps and reishi dual-extracted mushroom tinctures Leaky gut herbs Multivitamin Vit. D and C	Liver-friendly lunch Stay hydrated – drink plenty of water Milk thistle & dandelion root supplement	Exercise at least 20 min. Lemon water Leaky gut herbs Cordyceps and reishi mushroom tinctures	Liver-friendly dinner Avoid screens/blue light two hours before bed Milk thistle & dandelion root supplement	Aromatherapy Bedtime routine Sleep herbs, if needed
FRIDAY	Lemon water Liver-friendly breakfast Cordyceps and reishi dual-extracted mushroom tinctures Leaky gut herbs Multivitamin Vit. D and C	Liver-friendly lunch Stay hydrated – drink plenty of water Milk thistle & dandelion root supplement	Exercise at least 20 min. Lemon water Leaky gut herbs Cordyceps and reishi mushroom tinctures	Liver-friendly dinner Avoid screens/blue light two hours before bed Milk thistle & dandelion root supplement	Aromatherapy and/or Epsom salt bath Bedtime routine Sleep herbs, if needed
SATURDAY	Lemon water Liver-friendly breakfast Cordyceps and reishi dual-extracted mushroom tinctures Leaky gut herbs Multivitamin Vit. D and C	Liver-friendly lunch Stay hydrated – drink plenty of water Milk thistle & dandelion root supplement	Exercise at least 20 min. Lemon water Leaky gut herbs Cordyceps and reishi mushroom tinctures	Liver-friendly dinner Avoid screens/blue light two hours before bed Milk thistle & dandelion root supplement	Aromatherapy Bedtime routine Sleep herbs, if needed

* Add in Cholesterol and/or Blood Sugar herbs if needed.

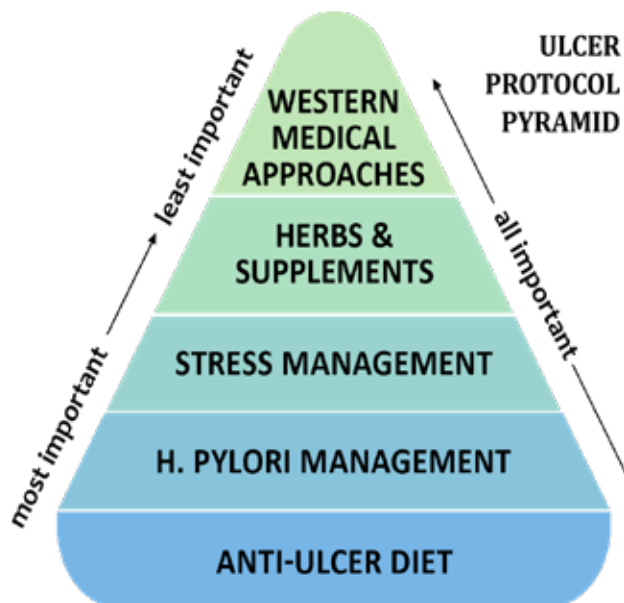
Ch. 13: Ulcer Protocol

Gastric/stomach ulcers are open sores in the stomach lining¹. Duodenal ulcers occur in the upper part of the small intestine. They occur when the layer that protects the stomach lining breaks down, penetrating through the muscularis mucosa. When these breaks occur, it can trigger changes in the gastric mucosa that over time result in erosion and ulceration.

These open sores are both types of peptic ulcer, meaning they are acid-related. The stomach contains acids that can cause the worsening of erosion and damage to the affected area².

Symptoms include³:

- Stomachache and/or severe pain and burning in stomach
- Nausea and/or Vomiting (especially after eating)
- Blood in stool (stool sometimes appearing black or tarry) or vomiting blood (in severe cases) – *consult your physician immediately if you have these symptoms.*
- Feelings of fullness, bloating, or belching
- Heartburn
- Intolerance of fatty foods
- Unexplained weight loss and/or appetite changes
- Trouble breathing
- Feeling faint
- Shortness of breath, fatigue and pale skin can be signs of anemia due to a bleeding ulcer.



Of all the symptoms of peptic ulcers, the most common symptom is severe stomach pain⁴. Peptic ulcers are often the result of one of several factors⁵.

- **A Bacterium:** *Helicobacter pylori* (*H. pylori*) is a common bacterium responsible for stomach ulcers. It lives in the mucous layer that covers and protects tissues that line the stomach and small intestine. Many people have this bacterium in their body, but it isn't enough to cause issues. However, when it is allowed

to thrive, it causes inflammation in the stomach lining, which can result in an ulcer.

- **Regular Use of NSAIDs:** Consistent use of over-the-counter pain reliever/fever reducers called Non-steroidal Anti-Inflammatory Drugs (NSAIDs) irritate and inflame the lining of the stomach and small intestine. These drugs include aspirin, Ibuprofen, Advil, Motrin, and Aleve. Drugs containing acetaminophen (such as Tylenol) aren't associated with ulcers.

- **Certain Medications:** Some medications such as steroids, anticoagulants, serotonin reuptake inhibitors, alendronate, and risedronate increase your chances of developing an ulcer, especially when taking these medications along with NSAIDs.

- **Unhealthy Habits:** Smoking and regular consumption of alcohol are associated with an increased risk for ulcers.

- **Untreated Stress:** Stress is a trigger of existing ulcers caused by other factors.

- **Spicy Foods:** Spicy foods are linked to worsening the symptoms of existing ulcers. Spicy foods cause an increase in stomach acid that worsens ulcers and leads to stomach pain. Taking regular NSAIDs in addition to eating spicy foods can exacerbate ulcers.

When ulcers are left untreated, they can cause significant damage. Untreated ulcers are known to result in stomach cancer, internal bleeding, a hole in the wall of the stomach, and blockage of material in the digestive tract. All of these things can lead to serious health complications or death as a result of not treating an ulcer when you suspect you have a problem. *See your physician right away if you suspect you have an ulcer or vomit blood.*

If you have an ulcer, the first things to do are to quit smoking, stop drinking alcohol, and stop taking NSAIDs. Then continue with the protocol below.

1. Anti-Ulcer Diet

A diet rich in antioxidants and flavonoids can help reverse and soothe ulcers⁶.

Antioxidants and Polyphenols: Help your body heal by focusing on a diet that is rich in fruits, vegetables, and whole grains. Start by purchasing more foods that contain antioxidants. Antioxidant-rich foods with polyphenols have been shown to help protect and heal ulcers⁷. Some foods to include are berries like blueberries, raspberries, strawberries, blackberries, and elderberries (cooked). Elderberries contain the flavonoid quercetin, which is linked with lower systemic inflammation. It is important to eat them in moderation, as too many berries can increase acid. Try one half to one cup of berries with a meal or as a snack daily. In addition to berries, try incorporating black olives, rosemary, flaxseed, and oregano into your diet.

Flavonoids: Foods that are rich in flavonoids can help treat stomach ulcers⁸. Flavonoids are found in

several foods such as legumes, red grapes, kale, broccoli, apples, berries, and green tea. Many fruits and vegetables contain flavonoids; incorporate plenty of these into your daily meals. Flavonoids also help fight *H. pylori*, which contributes to or causes ulcers. These gastroprotective compounds help defend the lining of the stomach and allow ulcers to heal.

Other Healthy Foods to Eat: While sugary foods are not good for ulcers, using raw honey in place of sugar may help⁹. Raw honey contains both polyphenols and antioxidants that can soothe ulcers, as well as manage potential ulcer triggers like *H. pylori*. Garlic is another helpful food. It can be quite strong, so eat it in moderation. Garlic inhibits the growth of *H. pylori*. Bananas are great for peptic ulcers because they help treat overacidity that leads to irritation in the stomach and small intestine. Cabbage can relieve pain and irritation from ulcers¹⁰. It is best taken as a juice or boiled in water. Almond milk can help bind excess acid in the stomach. It is a great alternative to dairy and doesn't trigger allergies like many dairy products. Try juicing cucumbers, carrots, and spinach and drinking an eight-ounce glass each day to help negate stomach overacidity.

Foods to Avoid: Some foods contribute to acidity and cause irritation to the stomach and intestines. This can cause a worsening of ulcers and acid reflux. Here are some foods to limit or avoid, as they can increase acid: spicy foods like chilies and hot peppers, salty foods, coffee and other beverages with caffeine, chocolate, fried foods, soda and other carbonated drinks, processed foods, citrus fruits, tomatoes (high acidity), pizza, chili, alcohol, red meat, refined vegetable oils, corn oil and canola oil, sugary foods. Consider limiting dairy and gluten, as these foods can cause an increase in acid. In addition, they are common triggers for allergies and inflammation. Go easy on starchy foods, sauces, and seafood as well.

Don't Overeat: Overeating or eating close to bedtime significantly increases your chances of experiencing discomfort. Many people with acidity issues have trouble sleeping at night when they eat too close to bedtime because the acid comes back up into their esophagus (acid reflux).

Keep a Food Journal: Keep track of the foods you eat and your symptoms with a food journal. This can help you determine which foods affect you the most. Everyone is different, so foods that may bother some people may not bother you as much. You'll be able to narrow down your triggers.

Don't Go without Food for Too Long: When you go long stretches with no food in your body, you may notice that your stomach becomes upset. This can also cause the stomach acid to worsen and affect your ulcer by triggering more pain and discomfort, as well as gas and bloating. Consider eating small healthy meals multiple times a day rather than three large meals a day.

many people's stomach and small intestine. When it is allowed to flourish, it can cause ulcers and even lead to stomach cancer if left untreated.

2. H. Pylori Management

Some symptoms of *H. pylori* overgrowth include an aching or burning in the abdomen, abdominal pain that worsens when the stomach is empty, nausea, loss of appetite, bloating, gas, and unintentional weight loss¹². Seek medical help if your symptoms worsen to include severe or persistent abdominal pain, difficulty swallowing, bloody or tarry stools, or bloody or black vomit (or vomit that looks like coffee grounds).

If you suspect you may have an overgrowth of this bacterium, you can have tests performed to see if this is the case. Common tests include a urea breath test, a stool test, a biopsy of the digestive tract, and blood tests.

Treating an *H. pylori* infection isn't that easy, especially if you go the allopathic route. Doctors often prescribe two different kinds of antibiotics to kill off the bacteria. Taking these antibiotics will kill off all gut flora, both good and bad. This will leave you with virtually no microbiome. If you go this route, it is important to do everything you can to rebuild your microbiome with probiotic and prebiotic foods and drinks, as well as strong probiotic supplements.

You have other options when it comes to tackling *H. pylori* that may not kill off the microbiome so absolutely. Natural plant-based remedies like goldenseal and lion's mane mushroom help kill *H. pylori*¹³. A study at the University of Chicago found that goldenseal was highly active against *H. pylori* due to the presence of a compound in the plant called berberine. If you use goldenseal, make sure to also consume plenty of probiotic foods and drinks to replace the good bacteria. *Note that people with autoimmune issues may find goldenseal too stimulating.* Medicinal mushrooms such as Lion's Mane mushroom or other sources of berberine may be a better option (see herbal section below).

Some risk factors that contribute to *H. pylori* are antibiotic use (see above), an unhealthy diet, crowded living conditions, and unclean water.

A diet full of processed and unhealthy foods can contribute to *H. pylori* complications. Just as healthy foods feed good bacteria in the microbiome, unhealthy foods feed the bad bacteria, giving *H. pylori* a chance to thrive. Sugary foods and alcohol are bad for the microbiome and feed bad bacteria. Avoid these foods and stick to healthy fruits, vegetables, leafy greens, and probiotics¹⁴.

Because *H. pylori* is extremely contagious and can spread from person to person rather easily, crowded areas can increase your chances of contracting it. It can spread through contact with saliva, fecal matter, or vomit. Make sure to wash your hands and avoid touching your mouth or nose. If someone you know is infected, avoid contact until they have had a chance to eliminate the bacteria.

Sometimes this bacterium is found in contaminated food and water. Most people with a reliable source of

clean, running water will not experience any issues with *H. pylori* spreading this way. However, in developing countries this bacterium spreads faster due to lack of access to clean water.

3. Stress Management Stress affects the defenses of the mucosa in the stomach and can trigger the emergence of ulcers¹⁵. It is well known that stress affects the body, and it is especially linked to ulcers and the worsening of ulcers. Stress weakens the body's immune system, which results in potential overgrowth of bacteria in the stomach and intestines that can lead to ulcers. Because the gut and brain interact with each other closely, it is no coincidence that stress leads to the manifestation of ulcers in the stomach and digestive tract.

There is a type of ulcer called a "stress ulcer," characterized by physical stress. Some examples of physical stress that lead to ulcers include a serious long-term illness, physical trauma, surgery, or injury to the central nervous system. In addition to physical stress, psychological stress commonly affects ulcers. Learn to get your stress under control to avoid serious complications. Learning to manage your stress can also help you improve your immune function, which in turn will help your body heal. Below are some ways you can manage stress to aid in healing and relief. Please also see the Anxiety and Stress Protocol in this book.

Herbs for Stress Management: Try a daily adaptogenic herb or mushroom like ashwagandha, reishi, and/or cordyceps to help your body respond to stress in a healthier way. Adaptogens help regulate the body's stress response. Other great herbs for stress management include lion's mane, motherwort, lavender, lemon balm, valerian, skullcap, and hops. Consider taking capsules, tinctures, or drinking tea with one of these herbs every day or as needed for stress management. Some, like valerian and hops, are best taken at night. See our Sleep Protocol for more information.

Find a Hobby: Finding a hobby you enjoy releases endorphins that help you feel happier and more at peace. Whether it is learning how to play an instrument, woodworking, fishing, or gardening, do something that makes you happy and make time for it each week.

Meditate and Practice Deep Breathing: Meditation significantly reduces stress and helps one adapt to stressful situations better. Even taking fifteen minutes to sit in a quiet place and practice deep breathing can make a significant difference¹⁶. Try sitting or lying down in a quiet place. Focus on your breathing and take deep, slow breaths that fill your lungs. Breathe in through your nose and out through your mouth slowly. Exhale all the air in your lungs until you feel there is no more to exhale. Then repeat this process for five to ten minutes. Focus only on your breathing.

Stop the Full Schedule: Keeping a jam-packed schedule can trigger a major stress response, the release of cortisol, and eventually adrenal fatigue. You may not even feel like

you are that stressed, but running from place to place all the time will eventually take its toll on your health.

Put Time Aside for Self-Care: Society may regard self-care as a luxury, but it is a necessity. The attitude that taking time to care for your mental and emotional needs is a luxury is one of the reasons why many are suffering from mental and emotional issues. Daily self-care is necessary to help keep stress and anxiety levels in check. Self-care looks different for each individual. It could be taking a nightly bath, getting a facial, booking a weekly massage, or setting time aside to do something you love.

Exercise: Daily exercise has a positive effect on both physical and mental health¹⁷. Not only does it get the blood pumping to oxygenate your cells; it also helps you feel better about yourself. Take time to exercise every day for at least twenty minutes. Choose a form of exercise that appeals to you. There are many ways you can get your blood pumping, such as swimming, cardio, jogging, daily walks, or yoga. Yoga is especially helpful for calming the mind and reducing stress.

Spend Time Outdoors: Studies show that spending time in nature significantly reduces stress and anxiety. Even twenty minutes out in the sunshine can help improve your mental and emotional health. In addition, sunshine on the skin helps the body create vitamin D, which improves immune function.

Find Someone to Talk to: It isn't healthy to bottle up emotions, and this can lead to worsened anxiety and stress. Whether it is a trusted friend or a professional counselor, consider finding someone to talk to on a regular basis.

The bottom line is that stress greatly contributes to ulcers and other health issues. If it is left unchecked, it can take a toll on your physical and mental health. We live in a busy, rushed world, and this can make it difficult to stop and take time out for yourself. However, as the old saying goes, "if you don't make time for your wellness, you will be forced to make time for your illness."

4. Herbs and Supplements In addition to a healthy diet of low-acid foods, not smoking, not drinking alcohol, no NSAIDs, stress management, and controlling *H. pylori*, there are various herbs and supplements you can take to manage the pain, discomfort, bloating, and negative effects of ulcers, as well as promote the healing of the stomach and small intestine.

HERBS FOR ULCERS: Aloe Vera: It is one of the best natural remedies for ulcers, as well as minimizing pain and discomfort. Aloe vera helps coat the stomach and intestinal lining and provides soothing relief¹⁸. It is also antimicrobial and anti-inflammatory, helping to reduce inflammation that comes with ulcers and killing potentially harmful bacteria like *H. pylori*. Suggested dose: Two to four capsules daily or purchase food grade aloe vera juice or gel and add one or two ounces to a smoothie each day.

Marshmallow Root (*Althaea officinalis*): It helps to coat, soothe, and protect the stomach and small intestine^{19 20}. The root forms a healing, mucilaginous coating when it is ingested. It can be taken in capsules, as a tea, as an infusion, or as a tincture. A nourishing infusion can be made by boiling a quart of water on the stove and pouring the water into a mason jar with a ½ cup of chopped marshmallow root. Allow to sit and infuse for up to ten hours before straining and refrigerating the liquid. Suggested dose: two cups of marshmallow nourishing infusion daily

Plantain (*Plantago* spp.): It is often overlooked as it is such a common yard “weed.” This plant contains anti-inflammatory and soothing properties that help reduce acid, inflammation, and stomach pain for those with ulcers²¹. The mucilage gel-like juice inside the leaves can be extracted by boiling a half cup of the leaves in two cups of water for fifteen minutes and then straining out the liquid. Suggested dose: Drink four to eight ounces of plantain decoction daily or take two dropperfuls of plantain tincture up to three times daily.

Slippery Elm (*Ulmus rubra*): Slippery Elm forms a protective coating that helps to reduce inflammation and pain and heal ulcers²². The bark from the slippery elm tree can be taken in capsules, as a tea, or as a tincture. Suggested dose: two dropperfuls of slippery elm tincture up to twice daily.

Calendula: Calendula tea or tincture can help reduce the symptoms of peptic ulcers, while also helping promote healing^{23 24}. It assists with digestion, and also protects the lining of the stomach and repairs the gut. Suggested dose: Drink one to two cups of calendula flower tea daily or take 2 dropperfuls of tincture daily.

Ginger: Ginger root is a popular natural remedy that spans many regions and cultures. Ginger is highly anti-inflammatory, helping to settle the inflamed lining of the stomach and small intestine. Additionally, ginger is great for settling an upset stomach and other digestive issues^{25 26}. It can also reduce pain as it has analgesic properties. Suggested dose: one to two cups of ginger root tea daily.

Lemon Balm (*Melissa officinalis*): Lemon Balm is great for ulcers for several reasons. First, as a member of the mint family, lemon balm has stomach-calming properties that can help relieve symptoms of ulcers like gas, bloating, and discomfort. It also has a protective effect on stomach ulcers in animal studies²⁷. Second, lemon balm is an effective nervine. Nervine herbs help to calm the nervous system. If stress or anxiety makes your ulcer symptoms worse, consider adding lemon balm tea or tincture to your daily treatment regimen. Because it is a nervine, it may also help reduce nerve-related pain that can sometimes accompany ulcers. Suggested dose: two dropperfuls of lemon balm tincture twice daily.

Licorice Root (*Glycyrrhiza glabra*): It is one of the most popular herbs for ulcer treatment. Licorice can help the body better respond to stress and bring things into balance where balance is needed²⁸.

Licorice contains a compound called glycyrrhizinic acid that is potentially harmful, but you can purchase deglycyrrhized licorice. This herb treats stomach ulcers by reducing inflammation, reducing harmful bacteria, and stimulating the production of mucus in the stomach to help coat the gastric walls and reduce pain from ulcers²⁹. Suggested dose: two dropperfuls of licorice root tincture twice daily.

Mastic: Mastic is the sap from a mastic tree. It is collected to treat a wide range of digestive conditions from ulcers to inflammatory bowel disease (IBD). It can help reduce stomach acid and protect the lining of the stomach and small intestine. It is antimicrobial and can help to kill bacteria as well³⁰. Take this resin in capsules or extracts as directed on the bottle. Suggested dose: Take up to four mastic gum capsules (250 mg each) daily.

Yarrow (*Achillea millefolium*): Yarrow leaves and flowers are antibacterial and also helps soothe the stomach lining to reduce discomfort. It is also styptic, so it can help staunch bleeding for those with severe ulcers³¹. Suggested dose: two dropperfuls of yarrow tincture up to three times daily or 1-3 cups of yarrow tea daily.

Digestive Bitters: Digestive bitters stimulate healthy digestion, while also alleviating bloating, nausea, indigestion, heartburn, and constipation. Bitters stimulate bitter receptors in the tongue, stomach, gallbladder, and pancreas. This promotes the production of healthy digestive juices such as bile and enzymes that aid in the breaking down of food. They also help the body absorb nutrients better. Bitters can be consumed by themselves (such as adding dandelion leaves to a salad) or you can powder them and add them to capsules or smoothies, or use as a tincture. Some other examples of digestive bitters include gentian, bitter melon, chamomile, milk thistle, and yarrow. Suggested dose: Drink one to two cups of digestive bitter tea or take a bitters tincture daily.

HERBS FOR *H. PYLORI* AND FOR LEAKY

GUT: Goldenseal (*Hydrastis canadensis*): Goldenseal acts like an antibiotic, so it should be used mindfully and with caution as it can kill both good and bad bacteria. If you drink goldenseal tea to kill *H. pylori*, also consume plenty of probiotic foods and drinks to replace the good bacteria. *Note that people with autoimmune issues may find goldenseal too stimulating.* Medicinal mushrooms (see below) or other sources of berberine may be a better option. Suggested dose: Make a tea using two teaspoons of chopped and dried goldenseal leaves and roots and drink one to two cups daily or take a dropperful of goldenseal root and leaf tincture twice daily with food for up to three weeks.

The following herbs may all be found in the Balanced Gut Tincture. Take 2 dropperfuls 3x/day for a month. Continue 1-2x/day for maintenance.

Lion's Mane (*Hericium erinaceus*): It is high in antioxidants, protects the lining of the stomach, and helps with gut inflammation. Lion's mane protects the gut from *H. pylori* and the polysaccharides in lion's mane protect

against gastric ulcers³². Suggested dose: two dropperfuls of lion's mane mushroom tincture twice daily.

Marshmallow Root (*Althaea officinalis*), Plantain (*Plantago* spp.), and Slippery Elm (*Ulmus rubra*): These herbs help form a protective layer in the gut, help with inflammation, and help the gut to regenerate. Suggested dose: two cups of a blended infusion daily or two dropperfuls blended tincture 2-3 times daily.

Reishi (*Ganoderma lucidum*): It helps leaky gut through its anti-inflammatory action. It is also high in beneficial beta-glucans, glycoproteins, and triterpenes to support gut health. Best taken as a dual-extracted mushroom tincture. Suggested dose: two dropperfuls of reishi dual extracted tincture twice daily.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): Turkey Tail calms inflammation and is an excellent source of prebiotics, which encourage a healthy microbiome by controlling the overgrowth of candida, while also feeding the “good” bacteria in the gut. Best taken as a dual-extracted mushroom tincture. Suggested dose: two dropperfuls of turkey tail dual extracted tincture twice daily.

SUPPLEMENTS AND REMEDIES FOR

ULCERS: L-Glutamine: This nonessential amino acid naturally occurs in some foods, such as chicken, eggs, beef, dairy, and some fruits and vegetables. However, you can make sure you are getting enough glutamine by taking a daily supplement³³. L-glutamine is beneficial for healing stomach ulcers because it can protect the lining of the stomach from damage caused by *H. pylori*, which is a common culprit of ulcer damage. It may even help repair intestinal lining that has been damaged. Suggested dose: 14 grams of l-glutamine capsules daily.

Probiotics: While it is possible to get probiotics from foods and beverages, it may be necessary to get the probiotics your gut needs from a good supplement. Make sure the supplement you choose contains *Lactobacillus* and *Bifidobacterium*, *Saccharomyces boulardii*, and soil-based probiotic blends (often *Bacillus* species). This will ensure you are getting a wide range of beneficial bacteria. Probiotics are especially helpful for those with gut damage due to *H. pylori* bacteria³⁴. Taking probiotics can help control its spread. If you have had *H. pylori* in the past, consider taking probiotics for prevention. In addition to restoring balance in the gut to promote health, daily probiotic supplements can improve immune function and mood (because of a strong connection between the gut and brain/nervous system). Suggested dose: Dosages will vary depending on the brand you purchase, so follow directions on the bottle.

Quercetin: Quercetin is a flavonoid that is found in some foods, such as elderberries and capers³⁵. In supplement form, you are able to get higher amounts of quercetin, so the effects may be more pronounced. This strongly anti-inflammatory supplement can help reduce the inflammation in the lining of the stomach and small intestine that cause pain and discomfort for those with

ulcers. It can also help reduce histamine and autoimmune reactions that may trigger worsening of ulcers. Suggested dose: 250 mg up to three times daily with meals.

Sodium Bicarbonate: Commonly called baking soda, sodium bicarbonate is highly alkaline³⁶. For those with overacidity that contributes to ulcers, a teaspoon or two of baking soda in water can reduce the acidity and bring relief from pain and inflammation. If you suffer from an upset stomach on a regular basis or have chronic pain, this common kitchen staple might bring relief. Look for baking soda that has been extracted via a simple process that doesn't require the use of chemicals. Suggested dose: 25 to 2000 mg orally one to four times a day.

Bismuth: Bismuth subcitrate is a common treatment for peptic ulcers and has a long history of positive outcomes in patients taking it to help aid in healing³⁷. It is also associated with a lower relapse rate. Bismuth is commonly found in a variety of over-the-counter treatments for digestive discomfort because it helps to settle the digestive system for those with gastrointestinal discomfort. Suggested dose is 524 mg orally every 30 to 60 minutes as needed; do not exceed eight doses in any 24-hour period.

Zinc: Zinc sulfate may aid in the healing of gastric ulcers by promoting wound healing and upholding the integrity of mucous membranes³⁸. It may also be taken to prevent ulcers if you are prone to them or at risk for developing one. Suggested dose: 8-10 mg daily.

Digestive Enzymes: Digestive enzymes are natural proteins that stimulate and accelerate biological reactions in the body. They can be taken in supplement form for help with a wide range of issues, including digestive complaints and ulcers³⁹. These enzymes help aid in the recovery of peptic ulcers in several ways. They help the body digest meals for the first 40-60 minutes of pre-digestion. This in turn makes the body require less acid in the digestion phase. Less acid is good for those with ulcers. In addition to lower acid levels in the stomach, digestive enzymes help break down food in the pre-digestion phase so that the body doesn't have to work as hard to digest food. Over time, the body adjusts to this and may also allow for an increased reserve capability. Even if you suffer from chronic low levels of acid, digestive enzymes can help by aiding in digestion. Suggested dose: Dosages will vary depending on the brand and size of each supplement, so follow directions on the bottle for best results.

5. Western Medical Approaches

For people dealing with peptic ulcers, it can help to see a healthcare professional to properly diagnose and treat this condition⁴⁰. Oftentimes, seeing a medical professional will begin with a discussion on your medical history. They may ask you questions about symptoms, medications you take, and if you take any NSAIDs. The use of NSAIDs is of special concern since it is a common contributor to ulcers and stomach issues.

A physical exam may be conducted. Your doctor

might check for bloating in the abdomen, listen to the sounds in the abdomen using a stethoscope, tap on the abdomen to check for any tenderness or pain, or take blood for testing.

If your blood is taken for a lab test, they may check for *H. pylori* infection and also check for other disorders and infections. Another common test that may be performed is a urea breath test. This test involves drinking a special drink that contains urea. This is a waste product the body makes as it breaks down protein. If you have an *H. pylori* infection, the bacteria will change the waste into carbon dioxide (a harmless gas that normally appears in the breath as you exhale). A breath sample is taken where you breathe into a bag that is sent for testing. If your breath happens to contain high levels of carbon dioxide, this is a good indication that you have an *H. pylori* infection in the stomach or small intestine. You may also have a stool test for *H. pylori*.

In some cases, your doctor may order further testing, such as an upper gastrointestinal (GI) endoscopy or biopsy. These tests are more invasive, but they can assist in understanding what is going on in the stomach and intestines. This procedure often takes place at an outpatient center or hospital. To begin, an intravenous needle is placed in the arm to administer a sedative to help you stay relaxed and comfortable during the procedure. However, it can also be performed without sedation. A liquid anesthetic is also given to numb the back of the throat where the endoscope goes in. The endoscope is carefully placed down the esophagus and into the stomach and duodenum. A tiny camera at the end of the endoscope helps the surgeon see the lining on a monitor. The endoscope also pumps small amounts of air into the stomach and duodenum to make things easier to see. Sometimes, a biopsy may be taken. This involves taking a small piece of tissue from the lining of the esophagus to test in a lab.

Another diagnostic procedure that may be conducted is an upper GI series. This involves performing an X-ray on the upper GI tract to see the shape and try to detect any other issues. An X-ray is a minimally invasive procedure, although you may need to drink barium to help the technician see the organs more clearly. Barium helps to coat the esophagus, stomach, and small intestine and illuminate them on an X-ray. X-rays do not require the use of anesthesia. You may be asked to fast beforehand to keep the area clear.

They may also do a CT scan. This involves a combination of X-rays and computer technology that helps create clear images of the area. For a CT scan, you may be asked to drink a solution or have an injection of dye that helps make things easier to see. You will lie on a table and slide into a tunnel-like device that takes X-rays. This procedure doesn't require anesthesia and is often performed at an outpatient center or hospital. This scan will help aid in the diagnosis of a peptic ulcer, especially if it has done enough damage to put a hole in the lining of the stomach or small intestine.

MEDICATION: Treatment varies depending on the cause and severity of the ulcer. Below are some common ulcer treatments that physicians prescribe and the reason they may be prescribed.

Proton Pump Inhibitors (PPIs): These can help reduce stomach acid and also protect the lining of the stomach and duodenum from further damage. They cannot kill *H. pylori* but can help fight the infection. Some common PPIs prescribed for ulcers include Nexium, Dexilant, Prevacid, Prilosec, Zegerid, Protonix, and AcipHex. These do not come without side effects, so make sure you research every drug you are prescribed to understand what to look for and how to manage any negative effect you may experience.

Histamine Receptor Blockers: These are prescribed to help block chemicals (called histamines) in the body that signal the stomach to produce acid. These are commonly prescribed for those with overacidity issues. Some commonly prescribed histamine receptor blockers include Tagamet, Pepcid, Zantac, and Axid.

Antibiotics: This may seem to be counterproductive, and sometimes it is. But for severe *H. pylori* infections, antibiotics may be prescribed. It is important to get *H. pylori* under control for the ulcer to heal. Different antibiotics are prescribed for the treatment of this bacterium and they can be very effective at eliminating *H. pylori* when it is the cause of peptic ulcers. Antibiotics clear both the bad and good bacteria in the gut, so make sure to take plenty of probiotic supplements if you are taking antibiotics. Failure to take probiotics with antibiotics can result in an imbalance of gut bacteria that may affect overall health.

Protectants: Your doctor may think you need extra protection from ulcers, and may prescribe something to coat and protect the ulcer against acid and enzymes that inhibit healing. A commonly prescribed protectant is Carafate.

Antacids: These help provide temporary relief from pain caused by ulcers. Antacids do not kill *H. pylori* and may affect the function of any antibiotics you are taking. Some antacids are made with heavy metals that can affect the function of other organs in the body. They are not something that should be taken on a regular basis.

Other Therapies: Sometimes, doctors will attack an ulcer on several different fronts. This may include a treatment plan that utilizes both antibiotics and proton pump inhibitors, or even proton pump inhibitors, bismuth, and antibiotics. This treatment plan is often utilized for those with *H. pylori*-induced ulcers. Be mindful of potential side effects when undergoing any treatment, especially treatments that involve a combination of the drugs discussed above. Some side effects of a combined therapy may include nausea, altered taste, diarrhea, a darkened tongue, darkened stools, headaches, yeast infections, and temporary reddening of the skin when drinking alcohol. Be sure to communicate with your doctor if you have any concerns about your treatment and be honest about any

negative affects you may be experiencing.

SURGICAL TREATMENT: The most common surgery for a peptic ulcer is an emergency surgery. While elective surgeries for ulcers have declined over the years, emergency ulcer surgery has not. Therefore, the likelihood for ulcer surgery increases if you allow your symptoms to persist without intervention. The type of surgery you may need will depend on the type of ulcer, location, and severity. Three common types of ulcer surgery include a laparotomy, a laparoscopic surgery, or an endoscopic surgery.

A laparotomy is the most invasive procedure because it

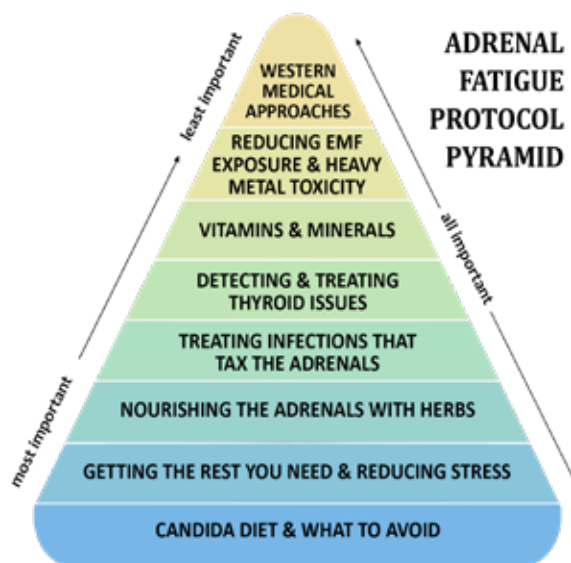
involves a large abdominal incision to repair the ulcer. A laparoscopic surgery is less invasive because it involves a small incision and camera-equipped devices that aid in visualization of the ulcer to help repair it. The least invasive type of ulcer surgery is an endoscopic procedure that consists of a tube inserted down the esophagus and down to the stomach to repair the ulcer. Laparoscopic and endoscopic surgeries have a faster recovery time and likely involve less pain afterwards. Whatever treatment you decide to pursue, make sure you have thoroughly researched the pros and cons and understand the potential side effects to look for as you begin your healing journey.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Light breakfast L-Glutamine, Quercetin, Probiotics & Digestive Enzymes Adaptogenic herbs Balanced gut tincture	Weekly grocery shopping for ulcer-friendly foods Light lunch Goldenseal root tea with raw honey (for <i>H. pylori</i>) Aloe vera and/or marshmallow root supplement or tea	Small (ulcer- friendly) snack Exercise 20-30 min Adaptogenic herbs for stress management Balanced gut tincture Outdoor time	Early dinner (try to eat dinner around 5pm) Goldenseal root tea with raw honey (for <i>H. pylori</i>) Aloe vera and/or slippery elm supplement or tea Self-care/hobby time Quercetin	Calendula & ginger Deep breathing and/ or meditation (for 15-20 min.) Lemon Balm & bal- anced gut tincture Zinc
MONDAY	Light breakfast L-Glutamine, Quercetin, Probiotics & Digestive Enzymes Adaptogenic herbs Balanced gut tincture	Light lunch Goldenseal root tea (for <i>H. pylori</i>) Aloe vera and/or marshmallow root supplement or tea	Small (ulcer- friendly) snack Exercise 20-30 min Adaptogenic herbs Balanced gut tincture Outdoor time	Early dinner (~5pm) Goldenseal root tea Aloe vera and/or slippery elm supplement or tea Self-care/hobby time Quercetin	Calendula & ginger Deep breathing and/ or meditation Lemon Balm & bal- anced gut tincture Zinc
TUESDAY	Light breakfast L-Glutamine, Quercetin, Probiotics & Digestive Enzymes Adaptogenic herbs Balanced gut tincture	Light lunch Goldenseal root tea Aloe vera and/or marshmallow root supplement or tea	Snack Exercise 20-30 min Adaptogenic herbs Balanced gut tincture Outdoor time	Early dinner (~5pm) Goldenseal root tea Aloe vera and/or slippery elm supplement or tea Self-care/hobby time Quercetin	Calendula & ginger Deep breathing and/ or meditation Lemon Balm & bal- anced gut tincture Zinc
WEDNESDAY	Light breakfast L-Glutamine, Quercetin, Probiotics & Digestive Enzymes Adaptogenic herbs Balanced gut tincture	Light lunch Goldenseal root tea Aloe vera and/or marshmallow root supplement or tea Weekly counselling session	Snack Exercise 20-30 min Adaptogenic herbs Balanced gut tincture Outdoor time	Early dinner (~5pm) Goldenseal root tea Aloe vera/slippy elm Self-care/hobby time Quercetin	Calendula & ginger Deep breathing and/ or meditation Lemon Balm & bal- anced gut tincture Zinc
THURSDAY	Light breakfast L-Glutamine, Quercetin, Probiotics & Digestive Enzymes Adaptogenic herbs Balanced gut tincture	Light lunch Goldenseal root tea Aloe vera and/or marshmallow root supplement or tea	Snack Exercise 20-30 min Adaptogenic herbs Balanced gut tincture Outdoor time	Early dinner (~5pm) Goldenseal root tea Aloe vera/slippy elm Self-care/hobby time Quercetin	Calendula & ginger Deep breathing and/ or meditation Lemon Balm & bal- anced gut tincture Zinc
FRIDAY	Light breakfast L-Glutamine, Quercetin, Probiotics & Digestive Enzymes Adaptogenic herbs Balanced gut tincture	Light lunch Goldenseal root tea Aloe vera and/or marshmallow root supplement or tea	Snack Exercise 20-30 min Adaptogenic herbs Balanced gut tincture Outdoor time	Early dinner (~5pm) Goldenseal root tea Aloe vera/slippy elm Self-care/hobby time Quercetin	Calendula & ginger Deep breathing and/ or meditation Lemon Balm & bal- anced gut tincture Zinc
SATURDAY	Light breakfast L-Glutamine, Quercetin, Probiotics & Digestive Enzymes Adaptogenic herbs Balanced gut tincture	Light lunch Goldenseal root tea Aloe vera and/or marshmallow root supplement or tea	Snack Exercise 20-30 min Adaptogenic herbs Balanced gut tincture Outdoor time	Early dinner (~5pm) Goldenseal root tea Aloe vera/slippy elm Self-care/hobby time Quercetin	Calendula & ginger Deep breathing and/ or meditation Lemon Balm & bal- anced gut tincture Zinc

Ch. 14: Adrenal Fatigue Protocol: The Hypothalamic-Pituitary-Adrenal (HPA) Axis

Adrenal fatigue is a term used to describe a set of symptoms that include fatigue, nervousness, sleep issues, brain fog, and digestive issues. However, what many people call “adrenal fatigue” is actually the body in a state of constant “fight or flight” response. This can lead to many problems, including immune system burnout. Those suffering from this condition often complain of feeling constantly tired and fatigued but energetic at night. They experience chronic stress and anxiety and tend to have body aches, hair loss, bouts of lightheadedness, and gastrointestinal problems.

The adrenal glands¹ sit on top of our kidneys and their job is to secrete certain hormones, including cortisol. When we are constantly experiencing stress², these glands secrete more cortisol. Cortisol is often called the “stress hormone” because it is what the body releases to help deal with stress. The problem is that too much cortisol can lead to health issues like severe weight gain (especially around the midsection). Excess weight can lead to heart and cardiovascular issues. Because the gastrointestinal tract is especially sensitive to cortisol, this can lead to problems like irritable bowel syndrome (IBS), nausea, cramping, diarrhea, and heartburn. When the adrenal glands are overproducing cortisol, this can lead to low cortisol in the glands, essentially depleting them.



Many allopathic practitioners have yet to fully acknowledge “adrenal fatigue” or accept it as a conventional diagnosis. It is better classified as hypothalamic-pituitary-adrenal (HPA) axis dysfunction, and it plays a significant role in the body’s response to stress³. In short, HPA-D⁴ can be defined as the body’s stress response system. The term “adrenal fatigue” may be incorrect terminology, but sometimes this is actually the case for those who suffer from conditions like Addison’s disease (if you suspect this is the case, get

the help you need to address it as soon as possible). If you are meeting with a Western practitioner, using the term HPA axis dysfunction instead of “adrenal fatigue” may be best.

More Symptoms of Adrenal Fatigue/HPA-D⁵: If you find that you are always exhausted, depressed⁶, gaining weight, craving sweet or salty foods, and experiencing brain fog on a regular basis, you may have an adrenal issue with the HPA axis⁷. The following list comprises more symptoms to watch for if you suspect an issue:

- Brain fog
- Chronic fatigue
- Hair loss
- Depression
- Lightheadedness
- Irritability and mood swings
- Sleep issues
- Craving sweet and salty foods
- Weight gain
- Tired during the day, wired at night.

Possible Causes of HPA-D: There are many ways your body gets into a constant state of “fight or flight.” Anything that puts an undue stress on the body can affect it in a negative way. Below are some common causes of adrenal fatigue (HPA-D):

- Lack of or poor-quality sleep
- Relentless stress
- Surgery
- Emotional trauma
- Poor diet
- Negative thinking
- EMF exposure
- Heavy metal toxicity
- Thyroid problems
- Caffeine, nicotine or other stimulant use
- Chronic or acute infections, especially respiratory infections (influenza, bronchitis, or pneumonia).
- Candida can play a big part as well.

HPA-D and Autoimmunity - A Potential Link: HPA axis dysfunction is often found in people with multiple sclerosis and other autoimmune issues. It is not fully understood why this occurs, but researchers believe it may be due to general hypothalamic-pituitary-adrenal-axis hyperactivity, disordered orexin neurotransmission, and abnormal cortisol secretion. Dysregulation of extra-adrenal glucocorticoid production may play a key role in a variety of autoimmune disease like multiple sclerosis (MS), lupus, rheumatoid arthritis, and inflammatory conditions of the skin like psoriasis and atopic dermatitis⁸.

If you are struggling with HPA dysfunction, it can seem like you will never regain your health or vitality. But don’t lose hope. By embracing positive lifestyle habits, revamping your diet, reducing exposure to harmful substances, keeping stress and anxiety at bay, and tapping into the power of medicinal herbs and other supplements, you can, once again, live a vibrant life.

1. Candida Diet and What to Avoid

Because candida fungal overgrowth can play a big role in HPA dysfunction by putting extra stress on the body, it is important to keep your gut health in check. Candida tends to get out of control when we eat poorly, consume too much sugar, take antibiotics, and practice unhealthy habits. To learn more about how

to keep candida at bay, as well as how to encourage more healthy bacteria in the gut, please see the Candida Protocol.

If you are prone to drinking coffee, energy drinks, or sugary drinks every day, these may be sabotaging your body and worsening your adrenal issue. Stimulants like these can trigger a worsening of symptoms and continue the cycle of the body in a “fight or flight” state. Swap the coffee, energy drink, etc. for a cup of green tea. It is full of beneficial antioxidants and can give you a nice substitution to the unhealthy drinks.

Maintaining stable blood sugar levels is important. Eating more protein like wild-caught fish, free range eggs, and grass-fed, organic meats will help to encourage sustained energy throughout the day without sugar crashes. High-quality fats such as coconut, avocado, and nuts are smart choices. Avoid carb-rich meals and refined carbohydrates, including: sugar, breads, pastas, cookies, and cakes. Instead focus on non-starchy vegetables and leafy greens. Make sure to eat small, regular meals.

Sugar is all-around unhealthy, but especially for those concerned about HPA-D. Cut sugar from your diet as best you can and if you must sweeten something, try a small amount of a natural sweetener like organic stevia leaf. Read labels carefully for added sugar. When reading labels, look for foods that do not contain preservatives, dyes, and other unnatural substances that can adversely affect health. Choose organic whole foods whenever possible.

2. Getting the Rest You Need and Reducing Stress

Chronic stress and anxiety are the leading causes of adrenal fatigue and related disorders. The body is equipped to handle stress here and there, but when we are constantly stressed, our body overproduces hormones like cortisol that end up doing more harm than good. For this reason, it is imperative that you do everything you can to put your needs first and focus on eliminating stress from your life. While there are many lifestyle factors that cause stress, one of the most obvious issues that can lead to stress and a worsening of various bodily processes is a lack of sleep. We all know what it is like to experience a lack of sleep at one time or another. A lack of sleep greatly affects our mood and behavior. It can lead to depression and severe anxiety. Being in this state too often triggers a stress response in the body that will keep you in an unhealthy cycle of hormonal imbalance. For helpful information on how to get the sleep you need, as well as how to successfully lower stress levels, see both the Anxiety and Stress Protocol and the Sleep Protocol.

The Way You Think Matters: Don't underestimate the power of lifestyle choices in healing HPA dysfunction. Keeping a positive outlook and avoiding negative thinking and negative people is crucial. When we are exposed to negativity, whether through our own thinking or by those surrounding us, it triggers a stress response in the body that adversely affects health. It is entirely possible to “reprogram” your brain and the way you think by forming positive-thinking habits. In time, you

will automatically begin thinking more positively (if you can manage to stay consistent with positive thinking). A morning gratitude practice is a great place to begin. Giving thanks for something external causes an internal positive shift in our mind-body connection.

Stress-relieving practices and gentle exercise help calm HPA dysfunction and assist with balancing your system and helping you feel more positive. Yoga, meditation, singing, massage, gratitude journaling, spending time in nature, crafts and hobbies, and gardening are all excellent options. In addition to maintaining a regular sleep schedule and resting during the day when fatigued, laughter really is the best medicine for stress and negativity. Find things that make you laugh and bring you joy.

3. Nourishing the Adrenals with Herbs

Herbal supplements can be a great way to nourish your body, as well as protect it from harmful stress responses. They can help bring balance to hormones, as well as heal damage already done. There are herbs that help lessen anxiety. You can read more about nervines in the Anxiety and Stress Protocol. However, when it comes to herbal supplements that can protect the body from the effects of stress, adaptogens are your best friend. Adaptogens are herbs and mushrooms that help our bodies deal with the negative aspects of stress, such as inflammation, hormonal imbalances, increased cortisol levels, fatigue, and low energy levels. They help with quality of sleep, adrenal fatigue, and with immune function. An excellent review published in the journal of Chinese Medicine⁹ goes into great detail about the benefits of adaptogens¹⁰ if you would like to learn more. Some of the best adaptogens for HPA-D include ashwagandha, reishi, cordyceps, and lion's mane.

Ashwagandha (*Withania somnifera*): The root of this plant has been used for centuries in Ayurvedic medicine. It is best known for its adaptogenic properties relating to stress-related hypertension. Ashwagandha root reduces the impact of stress in the body that cause spikes in blood pressure and high cortisol levels, as well as stabilize hormones, improve thyroid function, and support immune system function.¹¹ It is also known for improving energy levels, making it very helpful for those suffering from chronic fatigue. Many people taking this adaptogen report improvement in mental acuity, memory, and processing speed. It may also be useful for neurodegenerative diseases such as Alzheimer's and Parkinson's. Suggested dose: one to two dropperfuls of ashwagandha tincture up to three times daily.

Reishi (*Ganoderma lucidum*): When reishi mushroom is taken regularly, it can dramatically improve a sense of wellbeing and quality of life. It also helps to regulate blood sugar and improves blood flow and oxygenation¹². I personally use it every day as it's an exceptional botanical for autoimmune disorders, cancer, liver function, stress, brain fog, gut health, hormonal imbalances, and cardiovascular health. Suggested dose: two dropperfuls of dual-extracted reishi tincture two to three times daily.

Lion's Mane (*Hericium erinaceus*): This anti-inflammatory brain-nourishing mushroom is well-known for its ability to protect and regenerate the brain not only from the ravages of chronic or acute stress, but also from damage caused by brain and neurological disorders¹³, including Alzheimer's, Parkinson's, and multiple sclerosis. Suggested dose: two dropperfuls of lion's mane tincture two to three times daily.

Cordyceps (*Cordyceps* spp.): Cordyceps mushroom improves circulation, balances blood sugar, and increases energy and stamina. It is an outstanding herbal remedy for chronic fatigue. What's more, it improves overall brain function, memory, and learning¹⁴. It's an excellent remedy for colds, the flu, respiratory ailments, and kidney issues, and has been traditionally used to prevent and treat cancer and tumor growth. Suggested dose: two dropperfuls of cordyceps dual extracted tincture one to three times daily.

Ginseng: Ginseng is an adaptogen known for its ability to enhance energy levels, stamina, and vitality. It has long been used for this purpose by Indigenous people of North America. It has been studied for its effects on cardiovascular health¹⁵, as well as its neuromodulatory, vasomodulatory, and immunomodulatory effects. Suggested dose: one dropperful of ginseng tincture up to three times daily, though not at night.

Licorice Root (*Glycyrrhiza glabra*): Licorice root has been found to help the body regulate cortisol more efficiently¹⁶. This is great for those with adrenal issues. This can give adrenals the break they need, in addition to helping protect the body from harmful stress responses. Suggested dose: one dropperful of licorice root tincture twice daily.

4. Treating Infections that Tax the Adrenals

A bacterial infection left untreated in the body can significantly impact adrenal health. Infections in the body are a major cause of adrenal exhaustion because they cause the body to be in a state of stress. It is crucial to learn how to detect and treat an infection as soon as you suspect an issue so your body can heal and recover. In this section, you will learn how to detect an infection, as well as speed healing so you can take the pressure off your adrenals.

If you experience any of the following symptoms, you may have an infection:

- **Fever:** this is one of the first signs of an infection. However, you need to rule out a viral infection, which also causes fevers.
- **Inflammation:** look for redness, swelling, painful, and irritated areas of the body.
- **Pus:** this is a sign of an infection, as it means white blood cells have congregated to an area.
- **Heat:** a hot, red, and swollen area on the skin is a sign of an infection.
- **Malaise:** general discomfort and feeling tired or "off" could mean you have an infection. Malaise on its own may not indicate an infection, but when you

experience this, along with symptoms above, this could indicate a problem.

- **Tightness in chest/respiratory issues:** if you experience this, you may have a respiratory infection. Look for additional symptoms such as fever to confirm.

If you suspect you have an infection, it is important to get treatment right away. When infections are left untreated, they not only tax the body, they may also continue to grow and can lead to severe health issues. For some infections that are more serious, see a healthcare professional for the antibiotics you need to fight the infection fast. Make sure to eat plenty of probiotic-rich foods to offset the negative effects of antibiotics (they can destroy gut health).

For skin and topical infections, try the remedies below:

Garlic: Garlic has strong antibiotic and antifungal properties. It has been studied for its effectiveness for antibiotic-resistant strains of bacteria¹⁷. Fresh garlic can be applied to wounds to speed healing. Simply mash up a clove of fresh garlic and apply this poultice to the wound (after it has been cleaned with soap and water). Cover this with a bandage to keep it in place. Reapply as needed.

Raw Honey: Raw honey is an antimicrobial¹⁸ powerhouse. It has been used for centuries to treat all kinds of infections, especially when applied topically. Make sure you use raw, unprocessed honey. Other kinds of honey have been pasteurized, which kills any benefits it has. It can be messy, but apply a liberal amount to an infected area (after thoroughly cleaning it) and cover it with gauze or a bandage. Re-apply as needed.

Oregano: Oregano is another plant studied for its antimicrobial effects¹⁹ when it comes to treating antibiotic-resistant strains of bacteria. It can kill bacteria like *E. coli* and *Pseudomonas aeruginosa*. It is available in essential oil form for ease of use and a stronger potency. When using oregano essential oil, it is important to dilute it in a carrier oil or it can burn the skin. Add two drops to a teaspoon of a carrier oil like grapeseed or olive oil. You can also add two drops to a teaspoon or two of raw honey for a doubly-potent remedy. Massage this into the cleaned wound and reapply as needed.

To help speed healing for a respiratory infection, try these strategies:

Herbal Steam Inhalation: Try this to help open up the airways, kill bacteria in the respiratory tract, and speed healing. Simply boil water in a pot on the stove and add antimicrobial herbs. Some great herbs to add include oregano, thyme, rosemary, eucalyptus, and peppermint. Add a half cup or more of the herb and bring the pot to a boil. Carefully place your face over the pot to inhale the steam (don't get too close or you will burn yourself). Turn off the burner. Place a towel over your head to trap in the steam and take deep breaths of this for up to fifteen minutes. Do this twice daily for best results.

Usnea: Usnea is basically nature's version of the antibiotic doxycycline. It is a powerful ally against respiratory infections, as well as a variety of other infections²⁰. Make a tincture with this common lichen (sometimes referred to as "old man's beard"). A dual extraction is the perfect way to get the most out of this powerful lichen for treating respiratory infections. Suggested dose: one to three dropperfuls of usnea dual extraction up to three times daily.

Mullein (*Verbascum thapsus*): Mullein is great for opening up the airways to assist in respiratory healing during an infection. It can also help the body get rid of mucus that causes issues and prolongs infections.²¹ Use mullein alongside usnea and herbal steam inhalation treatments as part of a protocol for respiratory infections. Mullein leaves are the part of the plant used to help with respiratory complaints. These common leaves can be harvested and used to make a tincture. Suggested dose: one to three dropperfuls of mullein tincture up to three times daily.

5. Detecting and Treating Thyroid Issues

Don't forget to seek medical help for an infection that doesn't seem to go away or gets worse with treatment.

Your adrenals and thyroid are connected, so when you have issues with one, it may worsen issues with the other. Not only this, but many people confuse the symptoms of adrenal fatigue with thyroid problems and vice versa. Always rule out thyroid conditions and make sure your doctor performs thyroid bloodwork to confirm that this isn't the root cause of your issue.

When we are stressed, our adrenals secrete hormones like cortisol to help our body deal with it. Issues can arise when we are chronically stressed. Being constantly stressed means the adrenals are pushing out cortisol in large quantities most of the time. This leads to adrenal fatigue and thyroid health complications. When sudden, large amounts of cortisol and other stress hormones are being produced and then stop producing (due to adrenal fatigue), the thyroid can become imbalanced.

When the thyroid becomes imbalanced as a result of adrenal fatigue, it can halt the production of stress hormones, as well as T3 and T4 production. This interferes with other metabolic processes in the body and can result in symptoms of hypothyroidism.

Because of the strong link between the adrenals and thyroid, taking care of your adrenals means taking care of the thyroid. This entails keeping a close eye on symptoms of thyroid imbalance. There are two major thyroid issues called hypothyroidism (the thyroid is causing the metabolism to slow down too much) and hyperthyroidism (the thyroid is causing the metabolism to speed up too much). These include, but are not limited to, the following symptoms:

Hypothyroidism: Common symptoms include: fatigue and tiredness, constipation, weight gain, depression,

muscle and joint aches, slow heart rate, feeling cold all the time, fertility issues, menstrual irregularities, reduced sweating, and dry hair and skin, as well as brittle nails.

Hyperthyroidism: Common symptoms include: diarrhea, weight loss, fast heart rate, feeling hot all the time, insomnia, restlessness, increased sweating, anxiety and nervousness, shaky hands, and muscle weakness.

Since the symptoms thyroid issues can look a lot like the symptoms of adrenal fatigue, make sure to see your doctor so they can perform a more detailed analysis of symptoms as well as bloodwork. For more detailed information on how to care for the thyroid, as well as naturally treat a variety of thyroid issues, see the Thyroid Protocol.

6. Vitamins and Minerals **Vitamin B complex:**

Our body needs B vitamins to protect the heart, as well as support a healthy nervous and digestive system. These vitamins may also help to improve mood²² and energy production. For this reason, they may be especially helpful for those concerned with adrenal fatigue. There are many brands and dosages to choose from when purchasing a vitamin B complex, so make sure to follow the dosage directions on the bottle and take them regularly for best results.

Selenium: This mineral is important for proper cell function and also helps maximize thyroid and adrenal performance.²³ It has also been studied for its anti-cancer properties. Those who are deficient in selenium may exhibit signs of fatigue and weakness. Suggested dose: 200-400 mcg daily.

Fish Oil: Fish oil has shown promise for helping to prevent adrenal activation elicited by mental stress.²⁴ This makes it an attractive option for those wishing to protect adrenal and thyroid health. Suggested dose: 250-500 mg daily.

Vitamin D: There is a link between low vitamin D levels and high cortisol. Studies confirm that vitamin D supplementation may help prevent adrenal disease²⁵. Spending around 20 minutes each day in the sunlight is great for helping your body absorb what it needs to make its own vitamin D. You can also take a daily vitamin D supplement. Suggested dose: 400-800 IU daily.

Vitamin C: The production of cortisol and other hormones depends on the body's vitamin C supply. This is why taking enough of this powerful antioxidant is crucial for adrenal health²⁶. The body doesn't absorb this well when taken in supplement form, so just because you are taking a daily pill, doesn't mean your body is benefiting significantly. Eat plenty of vitamin C-rich foods in addition to taking a daily supplement. These include citrus fruits, rosehips, bell peppers, guava, papaya, strawberries, kiwis, pineapples, kale, and broccoli. Suggested dose: 1,000 to 7,000 mg divided into two doses daily.

Magnesium: This mineral is crucial for managing stress and energy production²⁷, so it is no wonder it may help for those with adrenal fatigue. Magnesium can

be obtained in foods like bananas, almonds, avocados, spinach, Swiss chard, and dark chocolate. It can also be taken in supplement form. Suggested dose: 420 mg daily.

Probiotics: While it is advisable to eat plenty of probiotic-rich foods like sauerkraut, kimchi, and kombucha, you can also take a daily probiotic. As previously mentioned, candida²⁸ can play a part in adrenal fatigue. It is important to allow the good bacteria in your microbiome to thrive while deterring the bad bacteria with healthy eating habits and daily probiotic use. In turn, this can help control candida and keep it from causing more issues in the body. Suggested dose: one to ten billion CFU of a broad-spectrum probiotic daily.

7. Reducing EMF Exposure and Heavy Metal Toxicity

As already demonstrated via the thyroid and adrenal connection, taking care of the thyroid means taking care of the adrenals. One way to do this is to reduce your exposure to heavy metals and harmful radiation. A comprehensive list of ways to cut your exposure can be found in the Thyroid Protocol. In addition, this protocol discusses what you can swap potentially harmful everyday substances with to reduce exposure and live a healthier life. When it comes to Electromagnetic Frequency (EMF) exposure, there are little things you can do to help:

- Buy shungite panels to place on your phone or phone case. Shungite is a mineral (heavily composed of carbon) that may help block EMF radiation. You can purchase protective panels in different sizes to stick on your phone or phone case.
- Don't store your cellphone in your bra or pockets. Keep it away from your body.
- Consider storing your phone in a faraday bag to block EMF. You can take it out to use as needed.
- Don't sit your laptop on your lap to work. Keep it away from your body.
- Consider purchasing a protective barrier to sit your laptop on. This helps block radiation from affecting your body.
- Avoid wearing Bluetooth earbuds.

In times of nuclear fallout, many of those affected experienced changes in the thyroid because it is especially sensitive to radiation. Do what you can to protect your thyroid and adrenals from everyday radiation exposure to be on the safe side. While we aren't exposed to levels common with nuclear radiation, we are exposed constantly in the form of technology we now use all the time.

8. Western Medical Approaches

There are several options a physician may choose when determining medical treatment for adrenal fatigue. Many of these will depend on the results of various tests²⁹. Sometimes a simple diet change, coupled with orders to reduce stress may be all a doctor advises, but when these fail to work, a doctor may prescribe more invasive treatments like hormone replacement and certain medications.

Hormone Replacement: Hormone replacement entails replacing depleted cortisol with a corticosteroid. Most often, this corticosteroid is hydrocortisone. Most people are advised to take these two to three times daily by mouth. Sometimes a doctor may prescribe a corticosteroid like prednisone or dexamethasone (especially if you have a negative reaction to hydrocortisone). There are side effects to taking corticosteroids, such as bone loss, so make sure to watch for any issues that may arise.

Another type of hormone that may be administered is fludrocortisone. This is often given to patients that are unable to make their own aldosterone. This can help balance the amount of sodium and other fluids in the body. Those with secondary adrenal insufficiency can make enough aldosterone, and often do not need this medication. Hormonal treatments for adrenal crisis (which is more severe) may include prompt intravenous injections of corticosteroids and a large amount of saline solution with dextrose.

Pharmaceuticals: Most pharmaceuticals prescribed for adrenal insufficiency include various types of corticosteroids like hydrocortisone, prednisone, or dexamethasone (for hormone replacement). Other drugs administered include fludrocortisone. These may be given by mouth or intravenously, depending on the severity of your condition.

Surgery Precautions: Those with adrenal insufficiency who need to have surgery for any reason may have to have IV treatments with corticosteroids and saline prior to surgery. In addition, the treatment may need to last through surgery and after surgery (and will often entail the medications being given by mouth as soon as the patient can take them). This helps to counter any stress the body experiences during the surgery.

Illness: During illness, the body may experience more stress than usual, which can take a toll on those who already have issues with adrenal insufficiency. When this happens, a doctor may adjust the dose of corticosteroids you normally take to compensate for this. Most often, if a patient is ill with a high fever, they will need an increased dose of corticosteroids. If you are unable to keep medication down due to illness, you may need to be treated in the emergency room because those conditions may lead to an adrenal crisis.

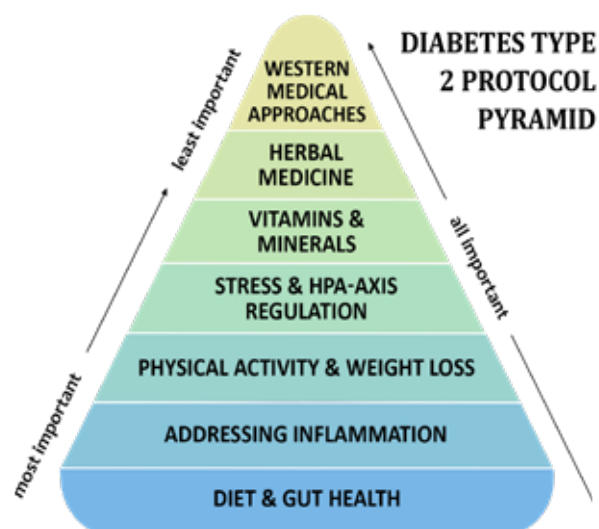
Other Situations that May Require Increased Doses: Surgery and illness aren't the only conditions that may require you to increase your dose of corticosteroids. Any condition that causes the body to become stressed can cause your doctor to up the dose. For example, if you experience a traumatic event or an injury you may need a higher dose of corticosteroids. Other situations that may call for this include passing out, shock, or severe mental or emotional trauma. Oftentimes, once the stress-inducing trauma has subsided, a doctor will adjust the dose back to the original dose you were taking prior to the incident.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Healthy breakfast Vitamin B complex, vit. C, vit. D, and Selenium Adaptogenic tinctures (ex. Reishi, Lion's Mane, and Cordyceps)	Weekly grocery shopping for adrenal and thyroid-friendly foods Healthy lunch Probiotic supplements Fish oil	Time in sunlight (at least 20 min. for vitamin D) Exercise (at least 20 min.) Adaptogenic tinctures (ex. Reishi, Lion's Mane, and Ashwagandha)	Healthy dinner Adaptogenic herbs Magnesium	Avoid screens and blue light at least one hour before bed Aromatherapy (lavender, cedarwood, or chamomile) and herbs as needed for sleep (valerian, hops)*
MONDAY	Healthy breakfast Vitamin B complex, vit. C, vit. D, and Selenium Adaptogenic tinctures (ex. Reishi, Lion's Mane, and Cordyceps)	Healthy lunch Probiotic supplements Fish oil	Time in sunlight (at least 20 min. for vitamin D) Exercise (at least 20 min.) Adaptogenic tinctures	Healthy dinner Adaptogenic herbs Magnesium	Avoid screens and blue light at least one hour before bed Aromatherapy/sleep herbs
TUESDAY	Healthy breakfast Vitamin B complex, vit. C, vit. D, and Selenium Adaptogenic tinctures (ex. Reishi, Lion's Mane, and Cordyceps)	Healthy lunch Probiotic supplements Fish oil Set time aside for a hobby or something you enjoy (stress management)	Time in sunlight (at least 20 min. for vitamin D) Exercise (at least 20 min.) Adaptogenic tinctures	Healthy dinner Adaptogenic herbs Magnesium	Avoid screens and blue light at least one hour before bed Aromatherapy/sleep herbs
WEDNESDAY	Healthy breakfast Vitamin B complex, vit. C, vit. D, and Selenium Adaptogenic tinctures (ex. Reishi, Lion's Mane, and Cordyceps)	Healthy lunch Probiotic supplements Fish oil	Time in sunlight (at least 20 min. for vitamin D) Exercise (at least 20 min.) Adaptogenic tinctures	Healthy dinner Adaptogenic herbs Magnesium	Avoid screens and blue light at least one hour before bed Aromatherapy/sleep herbs
THURSDAY	Healthy breakfast Vitamin B complex, vit. C, vit. D, and Selenium Adaptogenic tinctures (ex. Reishi, Lion's Mane, and Cordyceps)	Healthy lunch Probiotic supplements Fish oil Set time aside for a hobby or something you enjoy (stress management)	Time in sunlight (at least 20 min. for vitamin D) Exercise (at least 20 min.) Adaptogenic tinctures	Healthy dinner Adaptogenic herbs Magnesium	Avoid screens and blue light at least one hour before bed Aromatherapy/sleep herbs
FRIDAY	Healthy breakfast Vitamin B complex, vit. C, vit. D, and Selenium Adaptogenic tinctures (ex. Reishi, Lion's Mane, and Cordyceps)	Healthy lunch Probiotic supplements Fish oil	Time in sunlight (at least 20 min. for vitamin D) Exercise (at least 20 min.) Adaptogenic tinctures	Healthy dinner Adaptogenic herbs Magnesium	Avoid screens and blue light at least one hour before bed Aromatherapy/sleep herbs
SATURDAY	Healthy breakfast Vitamin B complex, vit. C, vit. D, and Selenium Adaptogenic tinctures (ex. Reishi, Lion's Mane, and Cordyceps)	Healthy lunch Probiotic supplements Fish oil Set time aside for a hobby or something you enjoy (stress management)	Time in sunlight (at least 20 min. for vitamin D) Exercise (at least 20 min.) Adaptogenic tinctures	Healthy dinner Adaptogenic herbs Magnesium	Avoid screens and blue light at least one hour before bed Aromatherapy/sleep herbs

*See our Sleep Protocol (page 286) for more information.

Ch. 15: Diabetes Type 2 Protocol

Type 2 diabetes is driven by the function of insulin, a hormone responsible for regulating the amount of glucose in the bloodstream. When we eat, our blood glucose levels rise, which signals the release of insulin from the pancreas. Impaired insulin sensitivity or reduced insulin can lead to high blood glucose, which can damage the body over time. With type 2 diabetes, insulin isn't recognized by the cells within the body. The progression of the disease starts with elevated insulin levels, which causes cells within the body to lose sensitivity to insulin. As a result, the body creates less insulin, which leads to insulin deficiency. While patients with Type 1 diabetes rely on insulin to survive, those with type 2 diabetes may find that they can avoid the use of insulin and instead correct their insulin resistance with diet, exercise, and weight loss.



Environmental factors like diet and exercise play a crucial role in the development of type 2 diabetes. The majority of individuals with this disease are overweight or obese, with abdominal obesity being a key-contributing factor in the progression of altered glucose tolerance. This impaired function of glucose uptake leads to insulin resistance, hyperinsulinemia, and cardiovascular complications¹. High levels of cortisol are another mechanism in the development of insulin resistance. By increasing blood glucose and promoting oxidative stress, the upregulation of cortisol in times of high stress has been shown to induce the pathways that trigger diabetes. Chronically high levels of cortisol not only increase glucose and inhibit glucose uptake, it also encourages gluconeogenesis, which is the process of forming glucose from amino acids, fats, and other compounds within the body. Gluconeogenesis is regulated by hormones like insulin, but with insulin resistance, the body can't effectively manage this process and the liver is encouraged to make glucose from these non-carbohydrate sources. The increased production of glucose is a consistent feature of type 2 diabetes, which is a result of the increased gluconeogenesis caused by insulin resistance².

Factors that contribute to type 2 diabetes include: a sedentary lifestyle, a diet rich in saturated fat (namely trans fats), high intake of refined carbohydrates, insufficient dietary fiber and omega-3 fatty acids, excess alcohol consumption, and stress.

Risk factors include: history of cardiovascular disease or diabetes, age (risk increases over the age of 45), being overweight or obese, hypertension, history of gestational diabetes, PCOS, and previously identified impaired glucose tolerance or fasting glucose.

Testing for diabetes includes blood glucose analysis, urinalysis, blood pressure, and measuring BMI and waist circumference. Since type 2 diabetes is incredibly multifactorial, the combined results of testing and physical assessment provide the most conclusive analysis. Recommended pathology testing for diabetes includes fasting glucose, fasting insulin, and GTT (oral glucose tolerance test). HbA1c (glycated hemoglobin) is another marker that measures how well diabetics are managing their condition. It is comprised of a glucose molecule attached to the oxygen-carrying hemoglobin. The higher the glucose levels in the blood, the more they attach to hemoglobin and thus, the higher the HbA1c level. The HbA1c test differs from a fasting blood glucose test in that it measures blood sugar levels over the previous few months, versus assessing numbers in the moment³.

1. Diet and Gut Health

Simple dietary changes can cause significant results. Weight loss and reducing visceral fat are crucial in reversing insulin resistance and regulating glucose control. Weight loss results in a significant reduction of risk factors including lower blood pressure, improved insulin sensitivity, and a more favorable lipid profile. Together with physical exercise, implementing the below changes can effectively slow the progression and development of type 2 diabetes.

Low Carbohydrate Diet: The debate surrounding the best diets for the management of diabetes continues to evolve, but the influence of a low glycemic or low carbohydrate diet has been shown to be superior in multiple studies. With the treatment goal of managing blood sugar control, avoiding foods with a high glycemic index is crucial in regulating blood glucose and improving insulin sensitivity. The glycemic index is a measure to determine how different foods influence blood glucose after their consumption. Foods with a high glycemic index are typically rich in carbohydrates, and have a dramatic impact on blood sugar levels. Avoiding these foods prevents spikes in blood sugar after meals, helping to manage blood sugar levels over the course of the day⁴.

High Protein: The gluconeogenesis that occurs as a result of elevated cortisol commonly leads to a reduction in lean body mass, which is why it's important to focus on consuming an adequate amount of protein. Many studies support the use of high protein diets for patients with diabetes; most recommend that protein should make up ~30% of total daily energy⁵.

Fiber: Fiber slows the digestion of carbohydrates, steadying the release of glucose into the bloodstream. It binds to fats and causes delayed gastric emptying, which enhances satiety and reduces cravings or hunger after meals, assisting with weight loss. Studies highlight the benefits of functional fibers like psyllium husk and flaxseed for their ability to reduce fasting blood glucose and cholesterol⁶.

MUFAs: Monounsaturated fats derived from olive oil, avocados, nuts, and seeds form the foundations of the Mediterranean diet. One of the largest intervention studies⁷ evaluated the effects of a Mediterranean diet on cardiovascular health and prevention. Rich in protective monounsaturated fats, the Mediterranean diet significantly reduces cardiovascular complications including obesity, insulin resistance, and HbA1c. The benefits of omega-3 fatty acids are also well-researched, particularly when it comes to reducing cardiovascular disease risk factors. See our section on the Mediterranean Diet (page 308) for more information.

Specific anti-diabetic foods are:

Flaxseed: As a rich source of fiber, flaxseed slows the digestion of carbohydrates and works to manage blood glucose. Flaxseeds are low in calories, rich in heart healthy omega-3 fatty acids, and many studies highlight their cardiovascular benefits. In one study, patients with diabetes who consumed 10 grams of ground flaxseed powder daily reduced fasting blood glucose by over 19%⁸.

Apple Cider Vinegar: Apple cider vinegar may have a significant influence on the progression of diabetes by supporting appetite, weight management, and regulating blood glucose. Drinking apple cider vinegar before meals significantly improves insulin sensitivity. One study found that drinking 20 grams of apple cider vinegar before a meal rich in carbohydrates improved insulin sensitivity by over 30%⁹.

Fatty Fish: The prevalence of type 2 diabetes is much lower in populations that consume a large amount of fatty fish, like salmon. Rich in omega-3 fatty acids and protein, fatty fish may aid in reducing inflammation and other risk factors of cardiovascular disease. A 2017 study¹⁰ found that 750 grams of fatty fish each week significantly improved glucose regulation in those with type 2 diabetes.

Okra: Okra helps stabilize blood glucose by slowing the rate at which sugar is absorbed from the intestinal tract^{11 12}. Suggested dose: Eat five okra pods, medium sized, per day.

SUPPORTING THE GUT MICROBIOME: While obesity was once speculated to be mostly caused by living a sedentary lifestyle, new evidence¹³ shows there are many factors at play. The gut microbiome is a recent focus for the progression of many diseases and studies now identify a link between our gut bacteria and the development of type 2 diabetes. A diet rich in inflammatory foods causes

poor microbial health, throwing the delicate balance of our gut bacteria out of equilibrium. This imbalance shifts the gut into a state of dysbiosis, which may result in insulin resistance, inflammation, and further metabolic dysregulation. There's a clear link between dysbiosis and the progression of type 2 diabetes. One large study¹⁴ identified approximately 60,000 type 2 diabetes-associated markers among over 300 individuals with diabetes - all of which were correlated with dysbiosis.

The microbiome has a significant influence over insulin and other markers of metabolic health. Metformin is commonly prescribed for type 2 diabetes management, where it functions to improve insulin sensitivity and reduce the production of glucose. Metformin also alters the gut microbiome. One study identified that metformin¹⁵ increased a specific strain of gut bacteria by 30%, demonstrating the powerful influence this drug has on the gut.

These studies suggest that increasing microbial diversity and working to restore the gut microbiome may influence the risk and development of type 2 diabetes¹⁶.

Eating whole foods helps preserve the interstitial microbiota, which has beneficial effects on metabolic health. In addition to eating a whole foods diet, emerging research continues to reveal various functional foods and supplements that benefit microbial diversity.

Fermented Foods: Fermented foods were originally created for preservation purposes. The fermentation process creates microorganisms and enzymes that transform the nutritional profile of foods and influences the gut microbiome. The fermentation process produces short-chain fatty acids, which act as functional forms of fiber. These reduce inflammation and support metabolic health. Fermented foods such as kimchi, kefir, yogurt, sauerkraut, tempeh, miso, and pickles are a wonderful way to support microbial diversity.

Probiotics: Probiotics are widely researched for their beneficial effects. These bacteria are live microorganisms that influence digestive health, and evidence suggests that probiotics influence the nervous system, hormonal and neurotransmitter function, skin health, and immunity. Probiotics also have beneficial effects for diabetes. Specific strains of probiotics improve glucose control and reduce insulin resistance and body weight in patients with type 2 diabetes¹⁷. For more on gut health, please see our Leaky Gut Protocol (page 70).

2. Addressing Inflammation

As a metabolic disorder that largely involves the immune system, low-grade inflammation is common in type 2 diabetes. Cardiovascular complications like heart disease, metabolic syndrome, and type 2 diabetes are all characterized by increased cytokines as a result of inflammation. The role of inflammation in the pathophysiology of type 2 diabetes is of growing interest, and it's proven to be an effective intervention in the prevention and control of the disease.

Inflammation is exacerbated by the intake of various fatty acids within the diet, including polyunsaturated fats, or omega-6 fatty acids (14). The excess consumption of omega-6 fatty acids in the diet (trans fats) plays a role in activating pro-inflammatory cytokines, contributing to systemic inflammation. The intake of omega-3 and omega-6 fatty acids has changed dramatically over the last few decades. Omega-6 fatty acid consumption has significantly increased with the convenience and affordability of vegetable oils like corn and canola. In contrast to the Western diet that is typically rich in omega-6 fatty acids, some of the healthiest continents eat a proportionally higher intake of omega-3 fatty acids derived from fish. Those with type 2 diabetes often have low levels of the heart healthy omega-3 fatty acid, linolenic acid. This fatty acid is formed by enzymes that require the production of insulin to be activated, suggesting that those with diabetes may benefit from omega-3 rich fish oil as a supplement to bypass these enzymatic pathways.

3. Physical Activity and Weight Loss

The positive influence of exercise on blood sugar regulation is undeniable.

Exercise and physical activity not only benefit body composition and promote weight management, but also facilitate the uptake of glucose into cells and lower insulin resistance. Regular movement reduces the storage of fatty acids, which leads to a reduction in triglycerides and improved glucose control. Although a simple increase in movement throughout the day does wonders for glucose regulation, strength training has superior benefits when it comes to body composition. Our muscles utilize glucose in order to produce energy; therefore low muscle mass is suggested to contribute to the progression of the disease. Strength training is an effective way to reduce diabetic risk factors, while maintaining lean body mass. Lifestyle interventions like physical activity have long been proven to reduce the development of diabetes. Just thirty minutes of exercise five days each week reduces the incidence of diabetes by 58%¹⁸.

4. Stress and HPA-Axis Regulation

We've discussed the role cortisol has in visceral obesity, insulin resistance, and impaired glycemic control. High cortisol wreaks havoc for many metabolic processes. It encourages fat storage, promotes gluconeogenesis, and leads to increased carbohydrate cravings, which perpetuates the fat storage cycle. Chronic stress can over-activate the HPA-axis, causing an increase in the production and release of hormones that cause the dysregulation of insulin, including cortisol and adrenaline. These stress hormones also lead to immune suppression and can aggravate autoimmune pathways¹⁹. Psychological stress is a risk factor for diabetes, and perceived stressful events may influence the onset of prediabetes.

Stress has long been associated with abdominal obesity, and it's linked with changes in eating habits, dietary preferences, food intake, and weight. The hypothalamic-pituitary-adrenal axis, commonly

known as the HPA-axis, is responsible for the body's stress response, so regulating the HPA-axis and supporting the body's adaptation to stress is crucial for breaking the stress cycle and promoting a healthy weight.

Signs of HPA-axis dysfunction ("adrenal fatigue") include fatigue, poor sleep, difficulty coping with stress, craving salt, increased anxiety, brain fog or poor concentration, and digestive disturbances.

REDUCING STRESS AND SUPPORTING THE HPA-AXIS FOR TYPE 2 DIABETES: Eat regular meals:

Poor blood sugar control is a perceived stress on the body. Supporting blood sugar control by eating frequent meals helps reduce stress and regulate the HPA-axis. Eating less and moving more is a necessary approach in diabetic management for those who need to lose weight. However, skipping meals can exacerbate poor blood sugar control as it encourages longer fasting periods; this can have detrimental effects on blood glucose. People who skip breakfast have higher blood glucose levels and are much less likely to have glycemic control. Studies have also shown that skipping meals isn't necessarily associated with a lower total energy intake, as it often causes people to eat a greater amount of food later in the day. Eating regular meals that contain a balance of all macronutrients (carbohydrates, protein, and fats) is the best way to balance blood sugar levels and reduce stress on the body²⁰.

Reducing Caffeine and Stimulants: Caffeine is a powerful stimulant. It increases the release of cortisol in those at rest, and exacerbates the production of stress hormones in people who are already experiencing ongoing stress. Avoiding caffeine, or reducing daily consumption to one per day before 10am, is recommended.

Support the Adrenals: Along with eating regular meals and reducing exposure to stimulants, it's important to focus on key nutrients that support the adrenals. The adrenal glands play a crucial role in blood sugar regulation. The blood sugar lows that occur as a result of poor blood glycemic control are perceived as a significant stress to the body, causing the body to enter into a state of fight or flight. Low blood sugar or hypoglycemia causes the adrenals to produce epinephrine, norepinephrine, and cortisol in an attempt to protect the body from a perceived danger or threat. Help relieve this adrenal load by feeding them with nutrients and cofactors that support a healthy stress response. *Specific herbs and nutrients for the adrenals include adaptogenic herbs, vitamins B and C, magnesium, and adequate protein.* Please see our Adrenal Fatigue/HPA-Axis Protocol and our Anxiety and Stress Protocol for more detailed information.

5. Essential Vitamins and Minerals

Alpha Lipoic Acid (ALA):

This antioxidant promotes the effective uptake of glucose by muscle cells at a rate that is comparable to insulin^{21 22}. Clinical studies have investigated the effects of ALA in diabetes, with one study showing an improvement in glucose clearance by 27% in diabetic patients²³. ALA supplementation results in significant improvements in

insulin sensitivity and diabetic complications such as diabetic neuropathy and polyneuropathy. Food sources of alpha lipoic acid are poorly absorbed, making it difficult to obtain a therapeutic dose through diet. Suggested dose: 600-1200mg per day.

B Vitamins: B vitamins are involved in the metabolism of food.²⁴ Vitamin B5 may be useful for weight management and reducing LDL cholesterol and triglycerides. Thiamine or B1 plays a specific role in carbohydrate metabolism, and animal studies show it may reduce the microvascular complications of diabetes²⁵. Food sources of B vitamins include eggs, organ meat, nuts, and whole grains. Suggested dose: Look for a daily B-complex vitamin with these approximate dosages: B1: 300mg, B2: 10-200mg, B5: 20-500mg, B6: 5-25mg, and B12²⁶: 1000-2000 mcg.

Chromium: This mineral plays a role in the breakdown of macronutrients (carbs, protein, and fats), and is required for the uptake of insulin into cells. Chromium improves the ability of insulin to bind to cells and enhances beta-cell activity, which increases insulin sensitivity²⁷. The body has higher requirements of chromium under certain circumstances, such as type 2 diabetes. Supplementation of chromium in type 2 diabetes results in improved glucose and lipid profiles, and reduced insulin and HbA1c. By supporting carbohydrate metabolism, chromium may have positive effects on body composition. Food sources of chromium include grape juice, brewer's yeast, cheese, tomato juice, and organ meats. Suggested dose: Supplements within the range of 200-1000 mcg per day are considered safe for glycemic control, with the most benefit seen in higher doses. The best absorbed form is chromium picolinate²⁸.

Essential Fatty Acids: Omega-3 fatty acids play a role in energy and glucose regulation. By increasing beta oxidation and glycogen synthesis, essential fatty acids like omega-3 positively influence fat burning and prevent fat storage. Omega-3 fatty acids reduce triglycerides, reduce insulin resistance, and reverse organ damage in diabetes²⁹. Diets rich in essential fatty acids reduce cardiovascular risk factors including reduced LDL cholesterol, improved endothelial function, and reduced oxidative damage. Suggested dose: For cardiovascular disease risk factors including elevated triglycerides and hypertension a dose of 1-4 grams of fish oil is recommended daily. For weight reduction and improved insulin sensitivity, a dose of 660 mg of EPA and 440 mg of DHA per day.

Magnesium: Magnesium is involved in over 300 enzymatic reactions within the body. Magnesium deficiency is related to poor glycemic control and impaired insulin secretion. In reverse, insulin resistance has been shown to interfere with magnesium uptake - creating a vicious cycle. Magnesium deficiency is strongly associated with diabetes and hypertension, with magnesium deficiency present in up to 38% of all diabetic patients³⁰. Foods rich in magnesium include nuts and seeds, dark green leafy vegetables, avocados, bananas, and cacao. Suggested dose: 400mg daily for increased glucose tolerance and

insulin sensitivity.

Vitamin C: This water-soluble vitamin is an essential nutrient, as we can't manufacture it on our own. Vitamin C is involved in metabolizing cholesterol, collagen and neurotransmitter synthesis, immune function, and it acts as an important cofactor for many reactions within the body. The many actions of vitamin C provide the basis for many uses in diabetes, including blood glucose regulation, modulating insulin activity, and fighting free radicals. Vitamin C is found in fresh fruits and vegetables, with the richest sources being bell peppers, leafy greens, rosehips, blueberries, kiwi, strawberries, papaya, and oranges. Suggested dose: 0.5 - 3g daily.

Vitamin D: This fat-soluble vitamin is mainly responsible for the regulation of calcium and phosphorus, as well as bone mineralization and immunomodulation. Vitamin D is essential for the normal secretion of insulin, and vitamin D deficiency puts populations at a higher risk of developing insulin resistance. Vitamin D supplementation can improve insulin secretion in response to glucose in both non-diabetics and those with mild type 2 diabetes³¹. Because it directly supports the immune system, vitamin D is beneficial for conditions that have an autoimmune pathology. Food sources include salmon, eggs, tuna, eggs and cod liver oil. Suggested dose is 1000-4000 IU per day. Get your blood levels tested.

Selenium: Selenium reduces oxidative stress, and studies have identified a complex relationship between selenium and diabetes. Selenium appears to be involved in beta-cell function and increases insulin secretion. Diabetic patients tend to have lower levels of selenium. Food sources of selenium include Brazil nuts (two Brazil nuts provide the daily recommended intake of selenium), organ meats, and seafood. Suggested dose is 2 Brazil nuts or 60-100mcg per day. Do not exceed 150 mcg/day.

Zinc: Zinc plays a key role in endocrine function. Zinc supplementation may be protective against the development of both type 1 and type 2 diabetes. Animal studies show that zinc supplementation may enhance insulin sensitivity and antioxidant status in insulin resistance models³². In addition to influencing insulin and improving glucose control, zinc plays an important role in preventing diabetic complications such as diabetic retinopathy and poor wound healing. Foods rich in zinc include oysters, red meat, and pumpkin seeds. Suggested dose: 25-30 mg of zinc acetate per day³³.

Fiber: Mix one teaspoon of psyllium powder with water and take daily before bedtime. In addition to containing fiber, psyllium (both the husk and the powder) stabilizes blood sugar, reduces cholesterol, and also has a mild sedative effect.

6. Herbal Medicine There are many ways to support the pathology of diabetes with herbal medicine, from reducing stress on the body (both physically and psychologically) to cooling inflammation. Herbs are an incredible tool for diabetics to aid in managing blood sugar levels,

support the HPA-axis, and reduce cravings. *Ask your physician before beginning an herbal regimen, especially if you are on Western Medications.*

Ashwagandha (*Withania somnifera*): This adaptogenic ayurvedic herb reduces inflammation and stress and supports the immune system. Ashwagandha reduces the risk of diabetes by lowering triglycerides, increasing heart-healthy HDL cholesterol, and improving glucose tolerance and insulin sensitivity³⁴. One study compared the blood glucose regulating effects of ashwagandha to metformin³⁵. Suggested dose: 2-7 dropperfuls (ml) of tincture (fluid extract) per day or 3-6 grams of the dried root per day³⁶.

Berberine: Berberine reduces both fasting blood glucose and postprandial blood glucose in patients with diabetes. Berberine reduces HbA1c levels similarly to metformin and Rosiglitazone³⁷ without the unpleasant side effects. In addition to its hypoglycemic actions, berberine reduces triglycerides and total cholesterol³⁸. Suggested dose: 1000-1500mg (500 mg 2-3 times a day) powdered extract daily³⁹.

Bilberry (*Vaccinium myrtillus*): Bilberries are used to lower blood sugar levels in people with type 2 diabetes. The berries prevent the breakdown and absorption of carbs in your gut, similar to some blood sugar-lowering medications. Bilberry can be taken as a supplement, eaten fresh or dried, or taken in tea or a smoothie. Suggested dose⁴⁰: 3 dropperfuls bilberry tincture 2-3 times per day.

Bitter Melon: Bitter melon lowers blood sugar levels. Over one hundred studies have investigated the hypoglycemic effects of bitter melon, which is attributed to a range of insulin-like constituents found within the concentrated fruit. One study⁴¹ revealed that the fresh fruit extract reduced fasting blood glucose by an impressive 50%. Bitter melon has also shown promising results in the prevention and delay of diabetic complications, including neuropathy, cataracts, and insulin resistance⁴². Suggested dose is 100g of bitter melon fruit juice mixed with 100ml of extract daily or 2 grams of oven-dried fruit pulp daily⁴³. Adapt dosage depending on how your blood sugar levels respond.

Cinnamon: Cinnamon effectively lowers blood glucose levels and reduces associated cardiovascular complications. Cinnamon reduces fasting insulin, glucose, total cholesterol, and LDL cholesterol and enhances insulin sensitivity⁴⁴ in those with elevated blood glucose^{45 46}. "Cinnamon improves diabetes... by enhancing insulin sensitivity and insulin secretion; regulating the enzyme activity involved in glucose; regulating glucose metabolism in the liver; adipose tissue and muscle; ameliorating oxidative stress and inflammation to protect islet cells; and improving diabetes complications⁴⁷." When combined with insulin the effects are greater, suggesting a synergistic effect⁴⁸. Suggested dose: Start at 1g (1000mg) of powdered cinnamon per day, increasing to 6 grams only if needed.

Cordyceps (*Cordyceps spp.*): Cordyceps lowers blood glucose levels by increasing insulin sensitivity and "possesses strong hypoglycemic, anticholesterolemic, and anti-hypertriglyceridemic actions."⁴⁹ It contains two unique compounds, d-mannitol cordycepin and 3'-deoxyadenosine, which help lower blood glucose^{50 51} levels by mimicking the activity of insulin. Cordyceps improves insulin sensitivity while also lowering bad LDL cholesterol, which helps to prevent plaque build up in the arteries, thereby reducing the risk of heart attack and stroke⁵³. Cordyceps helps to improve kidney function, an important consideration as kidney disease often accompanies diabetes⁵⁴. Cordyceps is also effective in addressing the immune dysfunction and low-grade chronic inflammation of diabetes⁵⁵. Suggested dose: 3 dropperfuls dual-extracted cordyceps tincture twice daily.

Fenugreek Seeds: Fenugreek has powerful anti-inflammatory and hypoglycemic properties. Several studies have identified the anti-diabetic effects of fenugreek seeds, and the fiber in the seeds slows digestion and reduces the intestinal absorption of glucose. This results in a positive effect on postprandial blood glucose, increased sensitivity to insulin, and the ability to enhance glucose uptake. Suggested dose: 50-100g of seeds taken daily (doses divided throughout the day with each meal) or 1 gram daily of an extract/tincture of fenugreek seeds.

Gymnema sylvestre: Gymnema is an effective Ayurvedic herb for blood sugar dysregulation. Gymnema reduces sugar absorption by the small intestine, reducing blood sugar levels⁵⁶. Gymnema also regenerates pancreatic Beta cells, which make and secrete insulin. Other benefits include decreased cholesterol and triglycerides and lowered HbA1c levels⁵⁷. HbA1c levels are directly proportional to blood glucose levels and measures the average level of blood glucose over a period of 2-3 months⁵⁸. Gymnema also suppresses the desire and taste (and thus intake) of sweet foods⁵⁹. Suggested dose: For diabetes: 3.6-11mls of alcoholic extract (tincture) once a day. For reducing the perception of sweet foods: 1-2 dropperfuls of tincture dropped on the tongue and rinsed off. Repeat every 2-3 hours as required.

Lemon Balm (*Melissa officinalis*): Lemon balm safely and effectively improves lipid profiles, glycemic control, and inflammation⁶⁰. When taken regularly, lemon balm helps control blood glucose levels and protects tissues against the oxidative stress associated with diabetes⁶¹. Moreover, it safeguards the heart by regulating the electrical pulses that cause palpitations, arrhythmias, and tachycardia⁶². Suggested dose: 2 dropperfuls lemon balm tincture twice daily.

Licorice Root (*Glycyrrhiza glabra*): Licorice reduces inflammation and protects mucus membranes. It has anti-diabetic and insulin altering effects; it improves glucose tolerance and insulin sensitivity. In addition to regulating blood glucose, licorice lowers triglycerides, reduces LDL cholesterol and encourages weight loss⁶³. Suggested dose is 2-4ml of the fluid extract/tincture,

three times daily. Note do not use DGL-licorice as it does not contain glycyrrhizic acid. Consult with your doctor or ND about glycyrrhizic acid use long-term.

Milk Thistle (*Silybum marianum*)⁶⁴: Milk thistle decreases insulin resistance and improves the glycemic profile in people with type 2 diabetes⁶⁵. The silymarin in milk thistle improves the glycemic and lipid profiles in T2DM patients⁶⁶. Suggested dose: 140 mg of silymarin three times a day.

Korean/American Ginseng (*Panax ginseng*): This adaptogenic herb influences Beta cells within the pancreas, *Panax* significantly reduces fasting glucose levels, improves insulin sensitivity, and reduces other cardiovascular risk factors like high blood pressure and triglycerides⁶⁷. *Panax* also has an immunomodulatory effect⁶⁸. Suggested dose: 1.5 - 2 grams of powdered extract, three times daily.

Reishi Mushroom (*Ganoderma lucidum*): Reishi is an adaptogenic mushroom that helps support and regulate the immune system. Reishi helps lower blood sugar and protects against insulin resistance^{69 70}.

It also reduces “bad” LDL cholesterol and fatty acids in the blood, while increasing “good” HDL cholesterol. Additionally, reishi increases oxygenation of the blood, regulates the heartbeat, and increases blood flow while normalizing blood pressure.

It improves heart function and lowers the risk of cardiovascular disease. Suggested dose: 2-3 dropperfuls of a dual-extracted tincture, twice daily⁷¹.

Tulsi (*Holy Basil*)⁷²: Tulsi is an outstanding herb for managing blood sugar and insulin resistance. It can reduce blood glucose, enhance insulin production, improve lipid profiles, prevent weight gain, and protect the liver and kidneys from high blood glucose⁷³.

It also reduces diabetic symptoms in those with type 2 diabetes⁷⁴. It can be taken as a tea, tincture, or in supplement form. Suggested dose: 2 dropperfuls tincture 2-3 times per day.

Turkey Tail Mushroom (*Trametes versicolor* or *Coriolus versicolor*): Known to help lower glucose

levels in the bloodstream, turkey tail is an outstanding medicinal mushroom for managing diabetes^{75 76}.

It also helps reduce LDL cholesterol and hypertension. It lowers the risk of heart disease, often an issue for diabetics. Suggested dose: 2 dropperfuls of a dual-extracted tincture two times per day.

7. Western Medical Approaches

Traditional Western

medical treatment aims to gain control of diabetes by regulating blood sugar. Biguanides are traditionally prescribed to reduce gluconeogenesis and increase insulin sensitivity.

Metformin is the first in-line treatment for most patients who struggle to manage blood glucose with lifestyle interventions alone.

Lifestyle interventions are widely recommended as part of diabetic management for high-risk populations and those with type 2 diabetes. Following a balanced diet and regular exercise (30 minutes each day) are the chief recommendations when it comes to lifestyle interventions.

Sulfonylureas are recommended as second to biguanides, used to increase insulin secretion by binding to potassium channels of beta-cells in the pancreas. Care must be taken by those with metabolic syndrome or obesity, as weight gain is a common side effect of taking sulfonylureas.

Insulin is considered to be the last-call of treatment in type 2 diabetes. As the primary treatment of type 1 diabetes, insulin is now used as a viable treatment in addition to metformin in type 2 diabetes. Insulin is effective in reducing blood glucose and HbA1C⁷⁷.

As a lifestyle disease, patients must be actively involved in supporting their treatment. In order to achieve glycemic control and reduce the risk of diabetic complications, patients must be willing to cooperate and stay focused on their lifestyle goals, and practitioners must do their best to encourage these important recommendations.

In order to make lasting change, cultural preferences and the emotional and social support from friends and family is crucial within a holistic treatment plan⁷⁸.

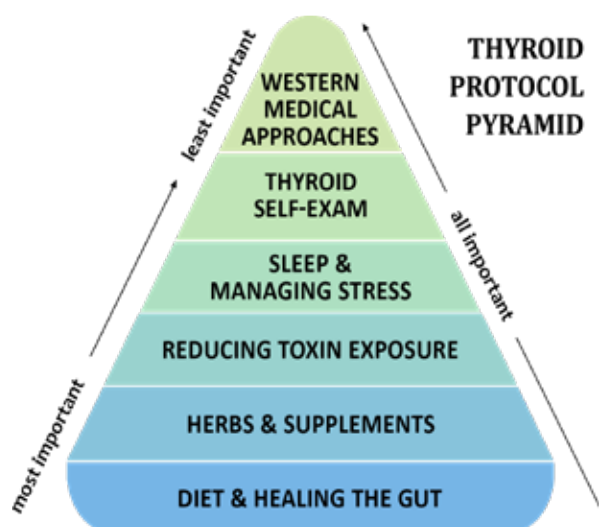
	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Healthy breakfast Alpha-lipoic acid, Chromium, vitamins B, C, D Herbal tinctures: Dose #1 of your herbal choices (ex. Gymnema, Cinnamon, Lemon Balm, Bitter Melon, Berberine, Ashwagandha, Bilberry, Fenugreek, Tulsi *)	Probiotics Healthy lunch Zinc and Omega-3 fatty acids after lunch Cordyceps & Turkey Tail dual-extracted mushroom tinctures	Blood sugar stabilising snack 20-30 min. of movement (e.g. walking, preferably outside) Herbal tinctures: Dose #2 of your herbal choices (ex. Gymnema, Cinnamon, Lemon Balm, Bitter Melon, Berberine, Ashwagandha, Bilberry, Fenugreek, Tulsi *)	Cordyceps & Reishi mushroom tinctures Healthy dinner	Enjoy a protein rich snack before bed (ex. Greek yoghurt with berries) Magnesium Psyllium husk
MONDAY	Healthy breakfast Alpha-lipoic acid, Chromium, vitamins B, C, D Herbal tinctures: Dose #1 of your herbal choices (ex. Gymnema, Cinnamon, Lemon Balm, Bitter Melon, Berberine, Ashwagandha, Bilberry, Fenugreek, Tulsi *)	Probiotics Healthy lunch Zinc and Omega-3s Cordyceps & Turkey Tail tinctures	Blood sugar stabilising snack 20-30 min. of movement (e.g. walking, preferably outside) Herbal tinctures: Dose #2 of your herbal choices (ex. Gymnema, Cinnamon, Lemon Balm, Bitter Melon, Berberine, Ashwagandha, Bilberry, Fenugreek, Tulsi *)	Cordyceps & Reishi mushroom tinctures Healthy dinner	Protein rich snack Magnesium Psyllium husk
TUESDAY	Healthy breakfast Alpha-lipoic acid, Chromium, vitamins B, C, D Herbal tinctures: Dose #1 of your herbal choices*	Probiotics Healthy lunch Zinc and Omega-3s Cordyceps & Turkey Tail tinctures	Blood sugar stabilising snack 20-30 min. of movement (e.g. walking, preferably outside) Herbal tinctures: Dose #2 of your herbal choices*	Cordyceps & Reishi mushroom tinctures Healthy dinner	Protein rich snack Magnesium Psyllium husk
WEDNESDAY	Healthy breakfast Alpha-lipoic acid, Chromium, vitamins B, C, D Herbal tinctures: Dose #1 of your herbal choices*	Probiotics Healthy lunch Zinc and Omega-3s Cordyceps & Turkey Tail tinctures	Blood sugar stabilising snack 20-30 min. of movement (e.g. walking, preferably outside) Herbal tinctures: Dose #2 of your herbal choices*	Cordyceps & Reishi mushroom tinctures Healthy dinner	Protein rich snack Magnesium Psyllium husk
THURSDAY	Healthy breakfast Alpha-lipoic acid, Chromium, vitamins B, C, D Herbal tinctures: Dose #1 of your herbal choices*	Probiotics Healthy lunch Zinc and Omega-3s Cordyceps & Turkey Tail tinctures	Blood sugar stabilising snack 20-30 min. of movement (e.g. walking, preferably outside) Herbal tinctures: Dose #2 of your herbal choices*	Cordyceps & Reishi mushroom tinctures Healthy dinner	Protein rich snack Magnesium Psyllium husk
FRIDAY	Healthy breakfast Alpha-lipoic acid, Chromium, vitamins B, C, D Herbal tinctures: Dose #1 of your herbal choices*	Probiotics Healthy lunch Zinc and Omega-3s Cordyceps & Turkey Tail tinctures	Blood sugar stabilising snack 20-30 min. of movement (e.g. walking, preferably outside) Herbal tinctures: Dose #2 of your herbal choices*	Cordyceps & Reishi mushroom tinctures Healthy dinner	Protein rich snack Magnesium Psyllium husk
SATURDAY	Healthy breakfast Alpha-lipoic acid, Chromium, vitamins B, C, D Herbal tinctures: Dose #1 of your herbal choices*	Probiotics Healthy lunch Zinc and Omega-3s Cordyceps & Turkey Tail tinctures	Blood sugar stabilising snack 20-30 min. of movement (e.g. walking, preferably outside) Herbal tinctures: Dose #2 of your herbal choices*	Cordyceps & Reishi mushroom tinctures Healthy dinner	Protein rich snack Magnesium Psyllium husk

*Choose 2-3 that best fit your needs.

Ch. 16: Thyroid Protocol - Including Hashimoto's Thyroiditis and Graves' Disease

The thyroid is a small 2-inch long butterfly-shaped gland located in the front of your neck. This little gland plays a big role in our health and wellbeing. It is an important part of the endocrine system; producing hormones that control a number of bodily functions.

Sitting just below the voice box near the base of the neck, the thyroid is constantly producing hormones that regulate metabolism, control how fast the heart beats, control how deep we breathe, body temperature, cholesterol levels, menstrual cycles, and hair growth, and affects energy levels, mood, brain function, and weight loss or gain¹. The main hormones are T3 (thyroxine) and T4 (triiodothyronine)². They are released into the bloodstream and travel to nearly every cell, triggering changes at the cellular level. The pituitary gland (located in the head just under the brain and behind the eyes) tells the thyroid how much of these hormones the body needs by sending it signals.



Your body is constantly monitoring conditions and triggering the thyroid to produce hormones it thinks are necessary. Changes such as stress, anxiety, and lack of sleep can significantly impact thyroid hormone production. Even changes you may regard as insignificant may have a large impact on your thyroid. For example, if you go on a crash diet and significantly reduce the number of calories you consume, your body tells the thyroid to produce more hormones to slow your metabolism and hold on to fat. It is a survival mechanism designed to keep you alive in emergency scenarios.

The thyroid is one of the first parts of the body to be affected by environmental toxins, heavy metals, and stress. As a vulnerable part of the endocrine system, the thyroid can and does malfunction for many people. There are many things that can cause the thyroid to underperform or over perform.

Hypothyroidism: This is the most common type of thyroid malfunction. Hypothyroidism is characterized by low thyroid production³. The thyroid is unable to produce the hormones the body needs to stay healthy. Those with hypothyroidism may experience weight gain, fatigue, sensitivity to cold, constipation, dry skin, thinning hair, a puffy face, elevated blood cholesterol, slowed heart rate, impaired cognition or brain fog, depression, heavy or irregular menstrual cycles, pain and stiffness in the joints, muscle weakness, and an enlarged thyroid (also called a goiter). See below for specifics about Hashimoto's thyroiditis.

Hyperthyroidism: This is the opposite of hypothyroidism. Hyperthyroidism results when the thyroid produces too many thyroid hormones⁴. If left untreated, it can significantly impact heart health and even lead to death. An overproduction of thyroid hormones can cause metabolism and heart rate to speed up. When the heart is working overtime for an extended period, this can lead to damage. Other symptoms of hyperthyroidism include weight loss, heart palpitations and irregular heartbeat, increased appetite, anxiety, sweating, menstrual changes, frequent bowel movements, enlarged thyroid gland (goiter), fatigue and muscle weakness, difficulty sleeping, thinning skin, and fine hair. See below for specifics about Graves' disease.

Autoimmune Diseases: Sometimes the body's immune system turns on the thyroid and attacks it. Autoimmune diseases that target the thyroid are becoming increasingly common⁵. While the exact cause is unknown, they are thought to be triggered by a combination of diet, inflammation, stress, genetics, and environmental factors. Two autoimmune disorders of the thyroid are Graves' disease and Hashimoto's thyroiditis, both more common in women.

Hashimoto's Thyroiditis (HT), also called chronic lymphocytic thyroiditis, is an autoimmune disease in which the immune system attacks the thyroid. When these antithyroid antibodies attack, they kill the thyroid's hormone-producing cells, resulting in a decline in hormone production (=hypothyroidism or underactive thyroid) and inflammation. Risk factors for Hashimoto's Disease include being female, genetics, having another type of autoimmune disease, high stress, food sensitivities, leaky gut, having too much iodine, and radiation exposure. The thyroid is especially sensitive to radiation. Most people are diagnosed when they are middle-aged (35-55).

Symptoms may include fatigue, an enlarged thyroid gland (goiter), cold sensitivity, weight gain, puffy face, constipation, dry skin, brittle nails, hair loss, menstrual changes, mood changes, memory and concentration issues, joint pain, and muscle weakness⁶. If left untreated it can cause heart and cholesterol issues, reduced libido, ovulation and menstrual issues, poor pregnancy outcomes, and enlarged goiter. In rare cases, it can cause a medical endocrine emergency called myxedema (severely advanced hypothyroidism), a life-threatening condition that can lead to seizures,

heart failure, coma, and death⁷.

Graves' Disease is an autoimmune disease where the body overproduces Thyroid Stimulating Immunoglobulin antibodies. These immune cells connect with receptors on the thyroid and mistakenly activate it, causing the thyroid to grow and release more hormones than the body needs. It is the most common reason for hyperthyroidism, an overactive thyroid. It is also sometimes called *diffuse toxic goiter* because of how the thyroid grows in response to the excess immune stimulation. Because thyroid hormones help the body control several different processes, Graves' disease can cause a wide variety of symptoms, including: unexplained weight loss, nervousness or irritability, anxiety, insomnia, sweating, increased heart rate, frequent stools and/or diarrhea, menstrual irregularities, muscle weakness or trembling, hair loss, and fatigue.

Diagnosis of Graves' disease is based on symptoms paired with certain physical exam abnormalities, like an enlarged thyroid, increased heart rate, proptosis (bulging eyes), tremors, or exaggerated reflexes. Tests are used to confirm the presence of excess thyroid hormones, thyroid antibodies, or thyroid enlargement and overactivity.

In addition to enlarging the thyroid, disease antibodies can sometimes cause enlargement of the tissues behind the eyes, causing the eyes to bulge or protrude. This symptom of Graves' disease is called proptosis or *Graves' ophthalmopathy*. Other symptoms of Graves' ophthalmopathy are dry eyes, puffy eyelids, redness, eye pain, and/or eye irritation. In rare cases, Graves' causes *pretibial dermopathy* - a thickening and discoloration of the skin around the front of the shins and feet.

Thyroid Nodules: Most of the time, nodules do not interfere with thyroid hormone production.⁸ They are quite common and are characterized by an abnormal growth of thyroid cells that form a lump where the thyroid is. Sometimes they can be felt and other times they cannot. The majority of these lumps are benign and not cancerous.

Thyroiditis: Thyroiditis is characterized by inflammation in the thyroid gland from an infection, immune attack, or trauma to the area⁹. Hashimoto's thyroiditis is one common reason for thyroiditis, but inflammation of the thyroid can also be caused by postpartum thyroiditis. Certain drugs and infections can trigger this inflammation as well.

Thyroid Cancer: This is the least common cause of thyroid issues, and usually thyroid cancer doesn't affect thyroid hormone production.¹⁰ Although it is one of the least common forms of thyroid trouble, it is the most common form of endocrine system cancer. It has a five-year survival rate of around 97%. Early intervention is key; see a physician right away if you have any lumps, tenderness, or swelling in the thyroid area.

If you suspect you have thyroid issues, it may take a while to ascertain the cause, but figuring out the root cause is the first step to being able to eliminate or lessen the damage. See an endocrinologist. In addition, take

this assessment at www.hrt.org/thyroid-assessment to help narrow down your issue.

1. Diet and Healing the Gut For people with Hashimoto's and hypothyroidism an Anti-Inflammatory Diet (see page 297), the Wahl's Protocol Diet (see page 310), or the Mediterranean Diet¹¹ (see page 308) is recommended, making sure there is adequate intake of vitamin D, iodine, and selenium, which are found in plant products rich in polyphenols, antioxidants, and omega-3 fatty acids¹². These healthy diets paired with removing known triggers (such as gluten¹³) allows for decreased inflammation and thus a diminished autoimmune response. People with HT need a healthy intake of protein, dietary fiber, and unsaturated fatty acids. They often need to eliminate lactose because of intolerance and interactions with levothyroxine, and eliminate gluten because of possible interactions with thyroid antigens¹⁴.

People with **Graves' disease and hyperthyroidism** often suffer from digestive upset, increased hunger, frequent stools, and diarrhea because of how the excess hormones speed up the body's processes. This can cause people to lose weight unintentionally. Eating small frequent meals and grazing throughout the day can help reduce gastrointestinal upset and combat weight loss.

The Mediterranean Diet is often recommended for those with Graves' with the added caveat of cutting out gluten. It prioritizes eating whole grains, beans, fruits, and vegetables than proteins, fats, and sweets. The high fruit, vegetable, and whole grain intake increases dietary anti-oxidants and dietary fiber, which assist the immune system and promote healthy microbial balance in the gut. This paired with the natural reduction of inflammatory influences, like some fats, animal products, and simple sugars, has an all-around anti-inflammatory effect, which benefits autoimmune disorders.

Omnivores have higher incidences of **hyperthyroidism** than vegetarians and vegans. This could be due to their increased vegetable and anti-oxidant intake or due to the absence of animal protein and animal products that can be inflammatory for some people. The reason is not well understood, but the relationship is. A trial period of 3-6 months on a vegetarian or vegan diet is recommended to look for positive effects on symptoms, thyroid hormones, and antibody activity.

The connection between the thyroid and gut health is well documented.¹⁵ Those with thyroid issues often complain of symptoms like trouble swallowing, constipation, gas, bloating, nausea and vomiting, diarrhea, indigestion, heartburn, and Irritable Bowel Syndrome (IBS). People with autoimmune thyroid diseases are even more at risk for gut issues and vice versa. One reason for this is because gut issues like leaky gut trigger the immune system to go into overdrive. When the immune system is overactive, it worsens existing autoimmune conditions like Hashimoto's and Graves'. Getting leaky gut under control can significantly reduce symptoms of these autoimmune diseases.

Leaky gut is caused when the intestinal barrier becomes compromised, allowing substances from the gut to pass into the bloodstream where they don't belong. The body's immune system detects these substances and tries to fight them. This results in inflammation as well as digestive discomfort. When the immune system is busy trying to fight off various antigens that have escaped into the body from the gut, this can trigger an autoimmune response¹⁶ for those with existing autoimmune conditions. They may experience constant issues with their thyroid function, even with treatment. Gut health must be addressed to fully heal.

In addition to addressing leaky gut, those with thyroid issues need to focus on a diet rich in foods that nourish the thyroid rather than harm it. There are foods and herbs that can help heal the microbiome, help the gut function more effectively, and nourish the thyroid all at the same time¹⁷. There are also foods that can harm the thyroid.

Studies in people who have an autoimmune thyroid condition have shown a strong link between bacterial overgrowth¹⁸ and their condition. Those with hypothyroidism often have a slowed metabolism, which makes the digestive system slow down¹⁹. Food takes longer to digest and this creates a problem with bacterial overgrowth in the intestines. This can turn into a vicious cycle if you don't take measures to get your thyroid working properly and nourish your gut with probiotic and prebiotic foods.

The microbiota in your gut plays a role in the conversion of thyroid hormones.²⁰ Specifically, they help with the conversion of T4 to T3 (the more potent hormone of the two). With such a strong connection between gut health and thyroid function, it is vital to address possible bacterial overgrowth and leaky gut. Start by eating the right foods and by reading our Leaky Gut Protocol.

PROBIOTIC FOODS: Probiotic foods contain live probiotics that promote gut health. The more good probiotics you have in your gut, the less bad bacteria can thrive. Probiotics help to reduce markers of intestinal permeability, regulate intestinal permeability, repair the intestines, and promote healing after infections of the intestinal tract. Adding probiotics has a beneficial effect²¹ on thyroid function and hormones.

Sometimes the bad bacteria in the microbiome can take over (bacterial overgrowth) due to factors such as poor diet and antibiotic use. This is called gut dysbiosis. This overgrowth can make autoimmune diseases worse by triggering leaky gut. Probiotics help restore balance so good bacteria can thrive. Consider eating/drinking these probiotic foods daily:

Kimchi: Kimchi is a fermented Korean dish with cabbage as the main ingredient. It is usually spicy and often contains red chili peppers. Try a one-cup serving with lunch.

Kefir: Kefir is a type of fermented milk that contains even more probiotics than yogurt. Kefir seems to be well

tolerated by those with lactose intolerance.

Kombucha: Kombucha is a fermented probiotic tea that comes in a variety of flavors. Drink one to two cups daily.

Sauerkraut: It is made by fermenting shredded cabbage with lactic acid bacteria. It is rich in probiotics and fiber. Avoid purchasing canned sauerkraut, as the canning process destroys the probiotics. Eat a cup with lunch or dinner.

Natto: Natto is a traditional Japanese fermented soybean product, often served with rice for breakfast. It supports bone health and contains probiotics.

Tempeh: Tempeh is an Indonesian dish made by fermenting soybeans and then forming them into a patty. This is a great option for vegetarians looking for a natural probiotic.

Yogurt: Yoghurt is a popular source of probiotics. Make sure to purchase yogurt that has live or active cultures, as pasteurization kills probiotics. Eat 1-2 cups daily.

Miso: Miso is a type of soup popular in Japan. It is made using a special seasoning blended with fermented soybeans. This fermented soybean paste is processed into the soup.

PREBIOTIC FOODS: Prebiotics are starchy fibers that feed probiotics (good gut bacteria), keeping the gut balanced to prevent bacterial overgrowth. The best way to get prebiotics is through diet. A fiber-rich diet includes fruits, vegetables, whole grains, legumes, nuts, and seeds. Some excellent prebiotic foods include asparagus, garlic, leeks, onions, chia seeds, flax seeds, seaweed, burdock root, chicory root, cocoa, coconut meat and flour, dandelion leaves, Jerusalem artichokes, apples, bananas, blueberries, carrots, jicama, chickpeas, beans, lentils, split peas, radishes, spinach, tomatoes, yams, barley, and oats.

Try eating a serving of one of the above foods with each meal. For snacks between meals, try a cup of plain yogurt with apple slices, bananas, blueberries, or nuts sprinkled on top. Another healthy and beneficial snack for prebiotic nourishment is hummus (a spread made from chickpeas). Try dipping vegetables like carrots or snap peas in hummus.

OTHER FOODS TO NOURISH THE THYROID: Thyroid-nourishing foods include: seaweeds (they are a great source of iodine—a mineral the thyroid depends on to function). Seaweeds are NOT for people with an over-functioning thyroid (hyperthyroid) as iodine can make hyperthyroidism and Graves' disease worse and may further enlarge the thyroid, causing goiter. There have been cases when iodine is helpful for balancing Graves', but there is no way to know without repeated monitoring, so iodine is best avoided. Other good foods to eat include: fish (opt for wild-caught instead of farm-raised), vegetables, leafy greens, and fruit. Try to stick with organic foods, as they haven't been sprayed with potentially damaging substances. The thyroid

is especially sensitive to contaminants and is usually the first gland affected when the body is exposed to harmful substances (like radiation). Be mindful of the toxins you put in your body.

FOODS THAT CAN HARM THE THYROID

(INFLAMMATORY FOODS): There are foods that can damage the thyroid or trigger inflammation that leads to autoimmune thyroid issues. Avoid gluten, most dairy, processed foods, fried foods, canola and vegetable oils, sugar, caffeine, and alcohol. All of these have been shown to trigger inflammation and weaken the immune system (potentially making autoimmune disease worse).

ADDRESSING POSSIBLE GUT INFECTIONS:

Sometimes leaky gut can be caused by an infection, overgrowth of certain bacteria, or intestinal parasites. A common cause of gastrointestinal issues that also triggers leaky gut is the bacterium *H. pylori*. This bacterium lives in most of our guts, but not in large amounts. When it is allowed to grow (it can proliferate when you eat unhealthy foods and/or take antibiotics on a regular basis) it can cause severe problems in the gut such as stomach pain, diarrhea, and can even lead to stomach cancer if not addressed. To address *H. pylori* naturally try herbs like lion's mane mushroom or goldenseal. Lion's mane protects the gut from *H. pylori* and the polysaccharides in lion's mane also protect against gastric ulcers²² and assist with gut inflammation. The root of the goldenseal plant kills *H. pylori*. If you see a physician, they will likely prescribe strong antibiotics. Sometimes these antibiotics can do as much harm as they do good. If you choose to try the natural route, infuse two teaspoons of goldenseal root in a cup of hot water for fifteen minutes and drink once or twice daily for three weeks or use two dropperfuls of lion's mane tincture daily. *Note that many people with autoimmune issues find goldenseal too stimulating.* Medicinal mushrooms, like lion's mane, are a better option if you have an autoimmune condition.

Another common cause of leaky gut is a parasitic infection. Herbs that kill parasites include wormwood, pumpkin seeds, and black walnut hulls. Wormwood has a steroid-sparing effect and can boost mood for patients with autoimmune bowel diseases²³. Take in capsule form or in a liquid extract (tincture). Follow the dosage directions on the bottle. Most protocols will call for people to take the capsule, tincture, etc. a certain number of times per day for one to three weeks.

An infection of the GI tract may result in a more permeable intestinal tract. Watch for signs of infection and get treatment as soon as possible. If you have a fever, diarrhea, fatigue, and GI issues, see a physician to determine if you have an infection.

HERBS FOR LEAKY GUT: If you have thyroid issues, make sure that you have a healthy, balanced gut. Herbs that help to repair the intestinal tract and restore healthy gut function include lion's mane, reishi, and turkey tail mushrooms, marshmallow root, and slippery elm.

All of these are found in my Balanced Gut Tincture.

Take 2 dropperfuls 3x/day for a month. Continue 1-2x/day for maintenance.

Lion's Mane (*Hericium erinaceus*): Lion's Mane is high in antioxidants, protects the lining of the stomach, and helps with gut inflammation. Take as a tea or a tincture. Lion's mane also protects the gut from *H. pylori* and the its polysaccharides protect against gastric ulcers²⁴.

Marshmallow Root (*Althaea officinalis*) and Slippery Elm (*Ulmus rubra*): These herbs help form a protective layer in the gut and help the gut to regenerate.

Reishi (*Ganoderma lucidum*): Reishi helps leaky gut through its anti-inflammatory action. It is also high in beneficial beta-glucans, glycoproteins, and triterpenes to support gut health. Best taken as a dual-extracted mushroom tincture.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): Turkey Tail calms inflammation and is an excellent source of prebiotics, which encourage a healthy microbiome by controlling the overgrowth of candida, while also feeding the "good" gut bacteria. Best taken as a dual-extracted mushroom tincture.

2. Herbs and Supplements Many with thyroid issues discover that they are deficient in minerals and nutrients responsible for thyroid function. There is a concrete link between mineral deficiency²⁵ and poor thyroid function²⁶. Before beginning an herbal or vitamin supplementation routine, see a physician who can perform blood work to determine any deficiencies.

The herbs and supplements in this section help with thyroid performance and with detoxing from heavy metals and other substances that cause harm to the thyroid.

HERBS FOR THYROID HEALTH (HYPOTHYROIDISM AND HASHIMOTO'S):

Ashwagandha (*Withania somnifera*): Ashwagandha helps reduce the impact of stress in the body; it is strongly associated with thyroid health and stabilization for hypothyroidism; it helps normalize serum thyroid indices^{27 28}. This adaptogenic herb can help with several thyroid-related issues, such as fatigue and focus, immune function, and reducing inflammation. Suggested dose: two dropperfuls of tincture 2-3 times daily.

Bacopa: Bacopa helps counteract hypothyroidism-related focus and concentration issues, as well as inflammation that makes thyroid conditions worse. Bacopa may help enhance memory and contains antioxidants.

Cilantro: For heavy metal detox, cilantro contains substances that prevent lead deposition in the body²⁹. In addition to detoxing lead, this herb has the potential to detox other heavy metals. Take in capsule or tincture form and follow the directions on the bottle. Avoid taking this herb on a regular basis if you have mercury fillings, as it is suspected to pull the mercury out into the body in

an attempt to cleanse it from heavy metals.

Cordyceps (*Cordyceps* spp.)³⁰: Cordyceps is beneficial for hypothyroidism and hyperthyroidism. Cordyceps improves adrenal function and helps restore thyroid balance by increasing helper T cells and cytotoxic T cells. Cordyceps helps lower levels of thyroid antibodies by over 50% while reducing Hashimoto's immune-inflammatory processes³¹. Cordyceps is also rich in selenium. Suggested dose: 2 dropperfuls twice a day of a dual-extracted Cordyceps tincture.

Evening Primrose (*Oenothera biennis*) Oil: Evening primrose seeds are processed into an oil that contain beneficial amino acids that help nourish the thyroid gland. Evening primrose oil can also boost overall thyroid function to reduce symptoms of a sluggish thyroid. It is best known for helping with dry skin, hair loss, and heavy menstrual cycles that accompany some thyroid conditions. Another benefit is its ability to curb inflammation. Take daily in capsule form.

Ginger (*Zingiber officinale*): Ginger is one of the best herbs for easing nausea and an upset stomach, and it is also a great herb for thyroid health. It is a source of thyroid-nourishing nutrients like zinc, magnesium, and potassium. Ginger reduces inflammation, beneficial for thyroid-triggering inflammatory issues. It is also thought to stimulate the metabolism. You can add ginger to your diet, infuse it in water to make tea, or take ginger capsules or a ginger extract/tincture.

Guggul: Guggul has been shown to potentially increase T4 output from the thyroid gland and helps the body convert T4 into T3. Those who take guggul may notice a decrease in hypothyroidism symptoms³². Other benefits include normalizing thyroid stimulating hormone (TSH) and helping the liver convert thyroid hormones. This herb can interact with certain medications, so make sure you research interactions before taking guggul. Suggested dose: 750mg/day until thyroid hormone levels normalize.

Licorice Root (*Glycyrrhiza glabra*): Licorice root can help bring balance and stability to thyroid hormones and help other endocrine glands to function more efficiently. Licorice contains a triterpenoid called glycyrrhetic acid that can help prevent the development of invasive thyroid cancer cells.

Panax Ginseng Root: This popular adaptogen is known for its ability to boost energy and stamina levels. Since many suffering from an underactive thyroid experience fatigue and lethargy, panax ginseng is a good option for not only lifting energy levels, but bringing balance to thyroid function due to its adaptogenic properties. Studies with this herb have shown a favorable effect on the production of thyroid hormones. It may also help with thyroid cancer³³. Best taken in the morning.

Reishi (*Ganoderma lucidum*): This adaptogenic mushroom helps with the negative effects of stress — such as increased inflammation, depleted energy levels, and various types of hormonal imbalances. Studies³⁴

have repeatedly shown that reishi mushrooms have antioxidant abilities that strengthen the body's defenses against cancer³⁵, autoimmune diseases, heart disease, allergies, and infections. It has neuro-protective effects and is good for treating Leaky Gut, a common cause of inflammation and immune response, and chronic fatigue. Suggested dose: 2 dropperfuls of reishi dual-extracted tincture per day.

Seaweed (Bladderwrack and Irish Sea Moss): Use only if you have an iodine-deficiency. Bladderwrack and sea moss contain iodine. Iodine is necessary for the efficient running of the thyroid gland, and many people with thyroid issues are iodine deficient³⁶. If you need more iodine, add sea moss gel to health drinks and smoothies. Bladderwrack can be tinctured, ground into powder for capsules, or infused in water to make tea. You can even cook with it. *Make sure that your iodine levels need supplementation before beginning an iodine treatment.*

Stinging Nettle (*Urtica dioica*): Inflammation in the thyroid contributes to both hypothyroidism and hyperthyroidism. Stinging nettle's anti-inflammatory effects work with both underactive and overactive thyroid. For those with an underactive thyroid, nettle can reverse iodine deficiency to help the thyroid function more optimally. Try adding stinging nettle leaves to bone broth to boost nutrient content. I use it as an easy-to-take tincture. Stinging nettle is also great for urinary health, arthritis, allergies, and boasts a variety of vitamins and minerals. Suggested dose: Drink two cups of stinging nettle tea or broth daily or use 2 dropperfuls of tincture daily.

HERBS FOR THYROID HEALTH (HYPERTHYROIDISM AND GRAVES')

There are herbs that should be avoided if you have hyperthyroidism and Graves' disease. The majority of thyroid and hormone support supplements are designed to increase function and hormones for those with hypothyroidism. This is the opposite of what someone with Graves' disease needs. The following herbs are known to stimulate the thyroid and increase thyroid hormones, so avoid these: Bladderwrack (*Fucus vesiculosus*)³⁷, Coleus (*Coleus forskohlii*)³⁸, Guggul (*Commiphora mukul*)³⁹, and Ashwaganda (*Withania somnifera*)⁴⁰. The following herbs are recommended:

Bugleweed (*Lycopus virginicus*/*Lycopus europeus*)⁴¹

⁴² ⁴³ ⁴⁴: Bugleweed helps improve symptoms for mild hyperthyroidism and Graves' disease; it acts against the antibodies that cause Graves'. Studies show it works in a variety of ways: it blocks auto-antibody stimulation at the thyroid, it blocks stimulation of the thyroid from the brain, it prevents the thyroid hormone T4 from becoming active⁴⁵, and it may help the body get rid of excess T4⁴⁶. It decreases T3 levels, most likely due to inhibiting the conversion of T4 to T3⁴⁷. Suggested dose: tea - 1 tsp. dried herb in one cup hot water three times a day; tincture (extract) form - take 2 dropperfuls (ml) twice a day.

Cordyceps (*Cordyceps* spp.)⁴⁸: Cordyceps is beneficial for hyperthyroidism and hypothyroidism. It improves adrenal function, helps restore thyroid

balance by increasing helper T cells and cytotoxic T cells, and helps lower levels of thyroid antibodies in both Graves' and Hashimoto's patients. Suggested dose: 2 dropperfuls twice a day of a dual-extracted Cordyceps tincture.

Lemon Balm (*Melissa officinalis*)⁴⁹: Lemon balm helps normalize an overactive thyroid. Extracts from both bugleweed and lemon balm inhibit stimulation of thyroid hormone production by both TSH and Graves' antibodies⁵⁰. Lemon balm can reduce excessive thyroid activity and inhibits TSH, which in turn helps to reduce the excessive secretion of thyroid hormone. Lemon balm prevents circulating thyroid hormones from becoming active and acts on the nervous system to reduce symptoms like anxiety, palpitations, and hypertension. Suggested dose: Tea - 1-2 tsp. herb per cup hot water three times a day; as an extract or tincture - 3 ml/dropperfuls twice a day (my preference for ease of use). If encapsulated as a dried herb, 300-600 mg three times a day.

Motherwort (*Leonurus cardiaca*)⁵¹: Motherwort is known for its effectiveness for cardiac conditions such as increased heart rate, palpitations, and irregular heart rhythm, making it useful for people with Graves' disease with these issues. Motherwort also directly blocks stimulation of the thyroid and reduces circulating thyroid hormones, making it an excellent addition for hyperthyroidism⁵². Suggested dose: tea - 1 tsp. dried motherwort per cup daily; tincture - 3-4 ml/dropperfuls twice a day or 250mg three times a day in capsule form.

Skullcap, Chinese and American (*Scutellaria baicalensis* and *S. lateriflora*): Skullcap acts as a mild sedative and anti-spasmodic. It improves insomnia, anxiety, cardiac irritability, muscle tension and spasms. In addition, Chinese Skullcap (*S. baicalensis*) improves thyroid hormones, adrenergic activity, and lipid metabolism in rats with LT4-induced hyperthyroidism. This may be mediated by the modulation of thyroid hormone-regulated gene expression⁵³. Suggested dose: tea - 1 tsp.-1 Tbsp. per cup three times a day or at night; tincture - 2-3ml/dropperfuls twice a day, not to exceed 100ml per week. In encapsulated form take up to 1 gram per day.

Stoneseed/Gromwell (*Lithospermum officianalis* and *Lithospermum rudemale*): The gromwell plant is thought to work^{54 55} through some of the same mechanisms as lemon balm and bugleweed. It is used in combination with them to reduce thyroid activation from the brain, block activation of the thyroid receptors, and block activation of thyroid hormones already in circulation. Suggested dose: 3ml/dropperfuls of herbal tincture twice a day.

Stinging Nettle (*Urtica dioica*): Inflammation in the thyroid contributes to both hyperthyroidism and hypothyroidism. Stinging nettle is an excellent anti-inflammatory for both. Suggested dose: 2 cups of stinging nettle tea or broth daily or 2 dropperfuls of stinging nettle tincture daily.

MEDICINAL MUSHROOMS^{56 57 58 59} **FOR AUTOIMMUNE HEALTH AND INFLAMMATION:** **Cordyceps, Lion's Mane, Reishi, and Turkey Tail Mushrooms** all show evidence of reducing autoimmunity, down-regulating overactive immune responses, and acting as anti-inflammatory antioxidants in the body. They are also anti-viral. Medicinal mushrooms are best taken as a dual-extracted tincture to access the full spectrum of medicinal compounds. Suggested dose: 2ml/dropperfuls of *each tincture* daily for those with Hashimoto's or Graves' disease.

SUPPLEMENTS FOR THYROID HEALTH:

Selenium: Those who are deficient in selenium are at a higher risk for thyroid disease. The thyroid contains a higher amount of selenium than any other mineral⁶⁰. It plays a direct role in the production of thyroid hormones and some scientists believe that selenium deficiency is a key reason why people develop autoimmune thyroid diseases⁶¹. You can get your daily selenium with a supplement or eat foods that contain this mineral, such as chicken, turkey, spinach, eggs, and Brazil nuts. Cordyceps Mushroom is also rich in selenium. Suggested dose: 60-200 mcg daily in supplement form. Brazil nuts are a very easy way to supplement selenium (Eat 2-3 brazil nuts a day).

Vitamin D: Low levels of Vitamin D are associated with autoimmune-related thyroid disease such as Hashimoto's thyroiditis and Graves' disease⁶². Vitamin D decreases the level of thyroid antibodies^{63 64} and treatment with Vitamin D helps normalize thyroid levels⁶⁵. Supplementation with vitamin D has a favorable effect on autoimmunity. Many people are Vitamin D deficient; make sure you have adequate levels by getting plenty of time in the sunlight, eating mushrooms such as maitake, and adding in a daily vitamin D supplement. Suggested dose: 2000 units/day. Upwards of 5000 IU/day may be necessary. To accurately monitor your levels, have your health practitioner perform a quarterly blood test.

Copper: Copper is not produced by the body; we get it via diet. Copper is necessary for adequate thyroid function and those who are deficient can experience adverse health effects like thyroid disease. Those with hyperthyroidism often have high levels of copper in their system and those with hypothyroidism are often deficient⁶⁶. Consider taking a supplement if you are deficient. Get tested before supplementing.

Magnesium: People deficient in magnesium can have issues with thyroid vascularization and inflammation in the body. Some doctors prescribe a magnesium supplement to take with thyroid medication. Not only is magnesium necessary for thyroid health, it is necessary for other important functions such as the formation of bone, muscle health, and nerve health. It also helps with sleep and stress. Suggested dose: 300-400 mg daily at night.

Zinc: Zinc supports immune function, acts as an anti-inflammatory and an antioxidant, and plays a role in supporting the thyroid hormone T3. Daily supplementation can help the thyroid produce more necessary hormones⁶⁷. Zinc has also been shown to

reduce cortisol levels, thus reducing the impact of stress on the thyroid. Suggested dose: 30mg/day.

L-Tyrosine: The body makes this amino acid from another amino acid called phenylalanine. L-tyrosine is a precursor to many proteins and neurotransmitters. The hormones that L-tyrosine helps produce assist the body in handling stress better and in turn, the impact of stress on the thyroid is reduced. If you suffer from brain fog due to thyroid issues, supplementation of L-tyrosine may help improve concentration and focus. Suggested dose: 150-300mg/day.

Iron: There is a correlation between iron deficiency and thyroid dysfunction⁶⁸. People with hypothyroidism often have low iron levels. Natural iron supplements are available (such as supplements made with beet root) or eat more iron-rich foods like lentils, grass fed beef, and black beans. Cooking with cast iron may also improve iron levels. Suggested dose: 150-200mg daily.

Manganese: Manganese is thought to affect thyroid hormones by regulating the selenium-containing deiodinase enzymes that help with activation and deactivation of the thyroid hormones. Suggested dose: Take a daily multivitamin that contains this mineral or eat more manganese-rich foods such as nuts, legumes, leafy greens, and fish/shellfish.

L-Glutamine: Low levels of this combination of three amino acids are correlated with poor gut health and a compromised immune system. Those with autoimmune-related thyroid issues need to make sure they are getting enough to support their immune system. Since thyroid disease and gut health are strongly connected, L-glutamine may be beneficial to those with inflammatory bowel diseases and leaky gut. Suggested dose: 5g twice daily.

Digestive Enzymes: Digestive enzymes can help thyroid disease by helping with leaky gut. Healing leaky gut can lessen stress on the thyroid, help with inflammation, and help reduce autoimmune responses.

Probiotics: In addition to getting your daily probiotics from fermented foods and drinks consider taking a good probiotic supplement that contains a variety of beneficial bacteria. Probiotic supplementation can have a positive impact on the thyroid, especially if your thyroid disease is worsened or triggered by leaky gut. Suggested dose: Start with ten to fifteen billion CFU daily.

Turmeric (*Curcuma longa*)⁶⁹: Turmeric has proven anti-inflammatory, immunomodulatory, and anti-oxidant effects. It is also helpful for metabolic disorders and cardiovascular disease. Suggested dose is 500-1000mg daily of a supplement that also contains piperine.

3. Reducing Toxin Exposure

After the fallout from atomic bombs dropped in Japan during WWII, there was a drastic spike in thyroid issues due to the radiation. There are several reasons why the thyroid is more susceptible to environmental toxins like radiation, heavy metals⁷⁰,

and other contaminants. First, many contaminants have a structural similarity to hormones the thyroid produces, so it tends to attract them. Second, the thyroid gland has an affinity for halogens and metals. The thyroid draws iodine (a halogen) and selenium (a metalloid) so it can produce the hormones it needs to keep the body running. However, this easy avenue for drawing in halogens and metalloids makes it draw in harmful halogens and metalloids. There are four major groups of toxins that may affect thyroid health. These include heavy metals, toxins in everyday products, pesticides and herbicides, and industrial chemicals. These 'endocrine disruptors' interfere with normal hormone levels and metabolism. To reduce your daily chemical exposure to endocrine disruptors^{71 72} and assist thyroid health, adjust your choices using the following guidelines:

- Buy organic produce whenever possible.
- Drink filtered/purified water.
- Use an air filter/purifier in the home.
- Avoid using cookware with Teflon or other industrial non-stick surfaces.
- Avoid using plastics in the home⁷³; switch to glass or ceramic containers and bottles and switch to BPA-free plastics.
- Avoid products with flame-retardants, which contain polybrominated diphenyl ethers (PBDE). Flame-retardants are found in mattresses, furniture, carpet padding, synthetic textiles, and computer screens.
- Use personal care products and cleaning products made with natural ingredients⁷⁴.
- Use naturally sourced pesticides, herbicides, and fungicides in the home and garden.
- Avoid lead products. Lead is linked with reduced thyroid function and elevated thyroid stimulating hormone (TSH)⁷⁵. Common sources include children's toys, inexpensive metal jewelry, and old paint.

4. Sleep and Managing Stress

Your thyroid and adrenal glands work together⁷⁶. The adrenal glands release cortisol into the body when you encounter stress⁷⁷. However, under constant stress, the adrenals can produce too much of this hormone and cause damage to the body, including the thyroid. They can even stop working correctly, resulting in adrenal fatigue.

Managing Stress: There is a direct correlation with stress and interference with thyroid hormone production. Patients reporting higher stress⁷⁸ levels have greater auto-antibody activity and relapse rates than patients reporting lower levels of stress. Stress is a trigger for autoimmune thyroid diseases like Graves' disease and Hashimoto's thyroiditis⁷⁹. Stress is additionally linked with a sluggish metabolism since it affects thyroid function.

Taking steps to eliminate potential stress triggers can help ensure proper thyroid function. The following techniques may result in lower perceived stress and increased mood, and therefore, better thyroid health outcomes. Please see our Anxiety and Stress

Protocol at page 150 for more tips on how to mitigate stress. Choose two or three techniques that suit you and integrate them into your daily and weekly routines to help counteract the effects of stress.

Getting Enough Sleep: Eight solid hours of sleep each night is optimal for a healthy-functioning thyroid⁸⁰. Getting proper sleep also helps lower stress levels that interfere with thyroid hormone production. Please see our Sleep Protocol for tips on how to get a good night's sleep. A few helpful tips include establishing a bedtime routine, avoiding blue light before bed, a warm bath, aromatherapy, and taking sleep herbs such as valerian root, hops, passionflower, California poppy, chamomile, lavender, and magnesium. A combination of valerian root⁸¹ and hops is quite powerful – both are in my Sleep Blend Tincture. If a racing mind is more of an issue than a restless body, try herbs like lion's mane, reishi mushroom, or motherwort as a tincture or tea.

5. Thyroid Self-Exam It is important to learn how to perform a thyroid self-exam.⁸² It is possible to perform this at-home to help with early detection of thyroid issues like Graves' disease or thyroid cancer. A self-exam checks for lumps, enlargements, or other abnormalities. While a thyroid self-exam is a good way to find glaring abnormalities, they miss issues below the surface. If you suspect you have a serious issue with your thyroid, seek help from a medical professional and get your hormone levels tested.

If you feel a lump, nodule, or swelling, try not to worry. This doesn't mean you have thyroid cancer or a thyroid disorder. Make an appointment with an endocrinologist to take a closer look.

Follow the steps below to do a thyroid self-exam:

1. Stand up facing a mirror. Focus on the lower-front portion of your neck and remove anything that obstructs this view.
2. Tilt your head back slightly, making sure your chin is pointed toward the ceiling while still able to see into the mirror.
3. Place your finger on the tip of your chin and then slide it down the midline. The first structure you feel is the top of the thyroid cartilage. Keep moving your finger down to the Adam's apple. Just below this you should feel the cricoid cartilage. Continue sliding your finger down. Below the cricoid ring are the first two rings of the trachea. On top of this lies that thyroid gland.
4. Take a sip of water, keeping your neck extended backward. Swallow the water.
5. Look for lumps as you swallow with your neck extended back. Examine the neck for any lumps, irregularities, or protrusions.
6. Feel the area around the thyroid for any abnormalities, enlargements or lumps. Nodules often appear round and move or roll with the gland as you swallow. A goiter is a lump or swollen protrusion that can appear on one or both sides of the gland. Men need to feel for bumps around the Adam's apple as

well.

7. Repeat the process again. Perform the tilt, swallow, and look process two or three times every few months.

If you detect any irregularities, talk to your doctor to determine the best course of action. Again, the discovery of a lump doesn't automatically indicate you have cancer or a thyroid disorder. Your doctor may perform diagnostic tests such as an ultrasound, blood work, or a computerized tomography (CT) scan.

6. Western Medical Approaches

Hashimoto's Thyroiditis: One of the most common treatments for Hashimoto's thyroiditis is the drug levothyroxine. It is used to restore normal function of the thyroid.

The "watch and wait" approach is often used with Hashimoto's. The hope is that medication will be enough to control the effects of this autoimmune disease. However, sometimes it is not enough and other measures are considered, such as surgery. It all depends on how much damage the thyroid has sustained due to the progression of the disease.

Thyroid surgery often involves a thyroidectomy. This removes the main thyroid gland, but leaves the parathyroid. In this case, the patient needs to continue with thyroid hormone replacement therapy throughout their lifetime. Every year, or several times a year, the patient needs to have bloodwork done in order to monitor their thyroid hormone levels. Medications are then adjusted as needed.

Graves' Disease: Common conventional treatment for Graves' disease and the resulting hyperthyroidism is to stop the body from following the instructions given by the abnormal thyroid hormone levels. These treatments do not change the hormone levels. Other treatments are aimed at relieving common side effects of hyperthyroidism or permanently altering the structure of the thyroid gland or, if affected, the eye socket.

Anti-Thyroid Medications: The anti-thyroid medications methimazole and carbimazole work by blocking the thyroid from making excess hormones. These drugs do not remove existing hormones from circulation; they only prevent new hormones from being made, so they can take 4-6 weeks to have an effect. People may opt to take these medications over the long-term to help manage Graves' disease or for the short-term if they plan on having surgical/definitive treatment of the thyroid. Common side effects are joint pain, fever, changes in taste, nausea, vomiting, itching, hives, and other skin rashes.

If hyperthyroidism occurs in early pregnancy, propylthiouracil may be recommended. While it has a decreased risk to the fetus and causes fewer birth defects than methimazole and carbimazole, it has more risks for maternal liver damage.

Medications for Symptom Management: Beta-blockers, like propranolol and atenolol, are a class of

drugs that act on the heart and nervous system and may be recommended to address Graves' symptoms like elevated heart rate, tremor, anxiety, and nervousness.

For people struggling with the side effects of Graves' ophthalmopathy, various kinds of eye drops may be recommended in addition to topical or systemic steroid drugs to decrease inflammation and suppress the immune system.

DEFINITIVE/IRREVERSIBLE TREATMENT:

More permanent solutions to Graves' disease involve making irreversible structural changes to the thyroid or structures of the eye socket.

Radioactive Iodine Ablation: This is a treatment for Graves' and hyperthyroidism that ultimately destroys all

or portions of the thyroid gland. The procedure involves introducing radioactive iodine in liquid or capsule form into the body for the thyroid to absorb. It takes between 6-16 weeks for the radioactivity to kill the thyroid cells. Some people may require a second dose of radioiodine to achieve more normal thyroid hormone levels. While the amount of radiation used is very small, is thought to be safe, and is not known to cause birth defects or cancers, radioactive iodine ablation is known to worsen ophthalmopathy.

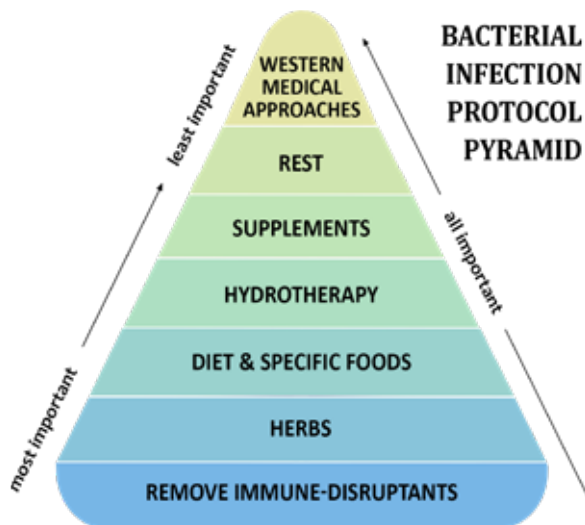
Surgery: For people with severe eye symptoms or those unwilling to take radioiodine, surgical removal of parts of the thyroid may be recommended. Eye surgery may be recommended to correct severe proptosis that does not resolve with other treatments.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	L-tyrosine before eating Meditation/gratitude Balanced gut tincture & Cordyceps, Reishi, & Turkey Tail dual-extracted tinctures Selenium, vit. D, & Iron	Grocery shop for the week for anti-inflammatory & thyroid-nourishing foods Lunch with probiotics and prebiotics (+ supplement if needed) Sea moss smoothie only if needed for iodine Stinging nettle tincture L-glutamine & turmeric	Walk in nature/daily exercise Ashwagandha & Guggul or Ginger Thyroid self-exam Balanced gut tincture	Thyroid-nourishing dinner with probiotics & prebiotics Turn off screens/blue light L-glutamine supplements Cordyceps & Lion's Mane dual-extracted tinctures	Bedtime routine (bath, book, dim lights, etc) Meditation/deep breathing exercise Zinc and magnesium Sleep tincture, if needed Ashwagandha tincture
MONDAY	L-tyrosine before eating Meditation/gratitude Balanced gut tincture & Cordyceps, Reishi, & Turkey Tail dual-extracted tinctures Selenium, vit. D, & Iron	Lunch with probiotics and prebiotics Stinging nettle tincture L-glutamine & turmeric	Walk in nature/daily exercise Ashwagandha & Guggul or Ginger Balanced gut tincture	Thyroid-nourishing dinner with probiotics & prebiotics Turn off screens/blue light L-glutamine Cordyceps & Lion's Mane tinctures	Bedtime routine Meditation/deep breathing exercise Zinc and magnesium Sleep tincture, if needed Ashwagandha tincture
TUESDAY	L-tyrosine before eating Meditation/gratitude Balanced gut tincture & Cordyceps, Reishi, & Turkey Tail dual-extracted tinctures Selenium, vit. D, & Iron	Lunch with probiotics and prebiotics Stinging nettle tincture L-glutamine & turmeric	Yoga or Tai Chi Guggul or Ginger Balanced gut tincture	Thyroid-nourishing dinner with probiotics & prebiotics Turn off screens/blue light L-glutamine Cordyceps & Lion's Mane tinctures	Bedtime routine Meditation/deep breathing exercise Zinc and magnesium Sleep tincture, if needed Ashwagandha tincture
WEDNESDAY	L-tyrosine before eating Meditation/gratitude Balanced gut tincture & Cordyceps, Reishi, & Turkey Tail dual-extracted tinctures Selenium, vit. D, & Iron	Lunch with probiotics and prebiotics Stinging nettle tincture L-glutamine & turmeric	Walk in nature/daily exercise Ashwagandha & Guggul or Ginger Balanced gut tincture	Thyroid-nourishing dinner with probiotics & prebiotics Turn off screens/blue light L-glutamine Cordyceps & Lion's Mane tinctures	Bedtime routine Meditation/deep breathing exercise Zinc and magnesium Sleep tincture, if needed Ashwagandha tincture
THURSDAY	L-tyrosine before eating Meditation/gratitude Balanced gut tincture & Cordyceps, Reishi, & Turkey Tail dual-extracted tinctures Selenium, vit. D, & Iron	Lunch with probiotics and prebiotics Stinging nettle tincture L-glutamine & turmeric	Yoga or Tai Chi Guggul or Ginger Balanced gut tincture	Thyroid-nourishing dinner with probiotics & prebiotics Turn off screens/blue light L-glutamine Cordyceps & Lion's Mane tinctures	Bedtime routine Meditation/deep breathing exercise Zinc and magnesium Sleep tincture, if needed Ashwagandha tincture
FRIDAY	L-tyrosine before eating Meditation/gratitude Balanced gut tincture & Cordyceps, Reishi, & Turkey Tail dual-extracted tinctures Selenium, vit. D, & Iron	Lunch with probiotics and prebiotics Stinging nettle tincture L-glutamine & turmeric	Walk in nature/daily exercise Ashwagandha & Guggul or Ginger Balanced gut tincture	Thyroid-nourishing dinner with probiotics & prebiotics Turn off screens/blue light L-glutamine Cordyceps & Lion's Mane tinctures	Bedtime routine Meditation/deep breathing exercise Zinc and magnesium Sleep tincture, if needed Ashwagandha tincture
SATURDAY	L-tyrosine before eating Meditation/gratitude Balanced gut tincture & Cordyceps, Reishi, & Turkey Tail dual-extracted tinctures Selenium, vit. D, & Iron	Lunch with probiotics and prebiotics Stinging nettle tincture L-glutamine & turmeric	Yoga or Tai Chi Guggul or Ginger Balanced gut tincture	Thyroid-nourishing dinner with probiotics & prebiotics Turn off screens/blue light L-glutamine Cordyceps & Lion's Mane tinctures	Bedtime routine Meditation/deep breathing exercise Zinc and magnesium Sleep tincture, if needed Ashwagandha tincture

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Meditation & gratitude Bugleweed & Stoneseed tinctures Lemon Balm tincture Selenium & probiotics Vit. D and multivitamin	Weekly meal plan Weekly grocery shop Cordyceps, Reishi, & Turkey Tail dual-extracted mushroom tinctures Stinging Nettle tincture	Bugleweed & Stoneseed tinctures 30 min. walking in nature and 10 min. breathwork Lemon Balm tincture	Bugleweed & Stoneseed tinctures Probiotic food or supplement Turmeric Chinese Skullcap tincture Lion's Mane & Cordyceps dual-extracted tinctures	Motherwort tea Restorative yoga Zinc and magnesium Sleep blend tincture (with Valerian & magnesium), if needed
MONDAY	Meditation & gratitude Bugleweed & Stoneseed tinctures Lemon Balm tincture Selenium & probiotics Vit. D and multivitamin	Cordyceps, Reishi, & Turkey Tail tinctures Stinging Nettle tincture 30 min. walking/exercise	Bugleweed & Stoneseed 10 min. breathwork Lemon Balm tincture	Bugleweed & Stoneseed Probiotic food or supplement Turmeric Chinese Skullcap tincture Lion's Mane & Cordyceps	Motherwort tea Tai chi or qi gong Zinc and magnesium Sleep blend tincture
TUESDAY	Meditation & gratitude Bugleweed & Stoneseed tinctures Lemon Balm tincture Selenium & probiotics Vit. D and multivitamin	Cordyceps, Reishi, & Turkey Tail tinctures Stinging Nettle tincture 30 min. walking/exercise	Bugleweed & Stoneseed 10 min. breathwork Lemon Balm tincture	Bugleweed & Stoneseed Probiotic food or supplement Turmeric Chinese Skullcap tincture Lion's Mane & Cordyceps	Motherwort tea Restorative yoga Zinc and magnesium Sleep blend tincture
WEDNESDAY	Meditation & gratitude Bugleweed & Stoneseed tinctures Lemon Balm tincture Selenium & probiotics Vit. D and multivitamin	Cordyceps, Reishi, & Turkey Tail tinctures Stinging Nettle tincture 30 min. walking/exercise	Bugleweed & Stoneseed 10 min. breathwork Lemon Balm tincture	Bugleweed & Stoneseed Probiotic food or supplement Turmeric Chinese Skullcap tincture Lion's Mane & Cordyceps	Motherwort tea Restorative yoga Zinc and magnesium Sleep blend tincture
THURSDAY	Meditation & gratitude Bugleweed & Stoneseed tinctures Lemon Balm tincture Selenium & probiotics Vit. D and multivitamin	Cordyceps, Reishi, & Turkey Tail tinctures Stinging Nettle tincture 30 min. walking/exercise	Bugleweed & Stoneseed 10 min. breathwork Lemon Balm tincture	Bugleweed & Stoneseed Probiotic food or supplement Turmeric Chinese Skullcap tincture Lion's Mane & Cordyceps	Motherwort tea Tai chi or qi gong Zinc and magnesium Sleep blend tincture
FRIDAY	Meditation & gratitude Bugleweed & Stoneseed tinctures Lemon Balm tincture Selenium & probiotics Vit. D and multivitamin	Cordyceps, Reishi, & Turkey Tail tinctures Stinging Nettle tincture 30 min. walking/exercise	Bugleweed & Stoneseed 10 min. breathwork Lemon Balm tincture	Bugleweed & Stoneseed Probiotic food or supplement Turmeric Chinese Skullcap tincture Lion's Mane & Cordyceps	Motherwort tea Restorative yoga Zinc and magnesium Sleep blend tincture
SATURDAY	Meditation & gratitude Bugleweed & Stoneseed tinctures Lemon Balm tincture Selenium & probiotics Vit. D and multivitamin	Cordyceps, Reishi, & Turkey Tail tinctures Stinging Nettle tincture	Bugleweed & Stoneseed 30 min. walking in nature and 10 min. breathwork Lemon Balm tincture	Bugleweed & Stoneseed Probiotic food or supplement Turmeric Chinese Skullcap tincture Lion's Mane & Cordyceps	Motherwort tea Restorative yoga Zinc and magnesium Sleep blend tincture

Ch. 17: Bacterial Infection Protocol

Our skin and mucus membranes serve as our main lines of defense against invading pathogens like bacteria. When our skin integrity becomes compromised, such as if we fall and get scratched or cut, or if bacteria are able to penetrate our initial defenses, then we become infected. It is now the job of our immune systems to identify, wall off, fight, or otherwise block or destroy those bacteria. If our immune systems are not able to retard or destroy them, it can spread to other areas of our body. For example, bacteria in a superficial wound on the toe can spread to the bone and the infection can spread up the foot and toward the leg if our circulation is compromised and our immune cells are not able to address the infection in a timely manner. If the bacterial infection spreads to the blood, this can lead to sepsis and even death. This is why it is critical that we support our immune system and address bacterial infections promptly.



There are several different types of infections that we are susceptible to; these include urinary tract infections, ear infections, skin infections, throat infections, respiratory infections, vaginal infections, and gastrointestinal infections. Risk for bacterial infections varies from person to person. For example:

- Children are at increased risk for middle ear infections because of the shape of their eustachian tubes. Because their eustachian tubes are shorter and more narrow and horizontal than adult tubes, air and fluid do not move in them as readily and bacteria can become trapped, leading to middle ear infections.
- Both children and older individuals are at greater risk for infection. Children's immune systems are still developing and immune system function begins to wane as we age.
- People with disorders that lead to impaired circulation and vessel and organ damage (like diabetes and peripheral artery disease) have wounds that heal more slowly. Since the integrity of the skin remains

compromised for longer periods of time after injury, bacteria have a greater opportunity to colonize wounds.

- Being diagnosed with multiple chronic diseases and/or taking multiple medications can decrease the rate of healing and lead to an increased susceptibility to infection.
- Our bodies require nutrients to repair damaged tissue. Being malnourished increases our risk for infection and increases healing time.
- Fecal or urinary incontinence and being bed-bound increase risk of infection. Fecal and urinary incontinence increase the likelihood of skin breakdown if the person is not changed in a timely manner, and the bacteria in feces or in the environment can then infect the wounds that arise when skin breaks down. Similarly, being confined to a bed increases the risk for decubitus ulcers (bed sores), which can then become infected.
- In females, wiping from back to front instead of from the urethra and vagina toward the anus significantly increases risk for urinary tract infections as colonic bacteria can colonize the urethra and ascend up the urinary tract.
- Cooking foods to appropriate temperatures and washing produce help get rid of potentially pathogenic bacteria. Eating raw or undercooked foods or unwashed fruits and vegetables can increase your risk for gastrointestinal infections.
- Being hospitalized for a prolonged period of time increases your risk for nosocomial or hospital-acquired infections. Using indwelling catheters, feeding tubes, ventilators, and other medical devices can further increase this risk.
- Cognitive impairment can potentially lead to increased infection risk as this can compromise people's abilities to remember proper hygiene practices.

WHAT TO WATCH FOR IN DIFFERENT TYPES OF INFECTIONS:

Gastrointestinal Infections: Watch for pain in the stomach or abdomen, nausea and vomiting, frequent bowel movements, diarrhea, pressure to go to the bathroom but nothing comes out, loss of appetite, and fever.

Respiratory Infections: Watch for congestion; pain in the facial region; sore throat; post-nasal drip; mucus that is green, yellow, or bloody; runny nose; cough; chest pain; shortness of breath; rapid and shallow breathing; fatigue; and fever.

Vaginal Infections: Watch for discharge, itching, burning, pain in the vagina, burning when urinating, or a strong unpleasant odor.

Ear Infections: Watch for pain and tenderness inside and around the ear, ear itching, fever, drainage, pressure in the ear, fatigue, and hearing loss.

Throat Infections: Watch for sore throat, pain when swallowing, red or white dots in the back of the mouth or throat, swollen tonsils, swollen lymph nodes, and fever.

Skin Infections: Watch for redness, swelling, fever, pus, drainage, blisters, pitted areas, fatigue, blisters, and sores.

Urinary Tract Infections: Watch for pain during urination, the feeling of a full bladder even after urinating, lower back pain, abdominal pain, pain in the pelvic area, fatigue, chills, urine leakage, and foul smelling urine that may be cloudy.

Blood Infections: Watch for fever and chills, clammy skin, excessive sweating, confusion and disorientation, pain and discomfort, shortness of breath, and an increased heart rate.

Spinal or Brain Infections: Watch for a headache, stiff neck, fever, confusion, vomiting, and sensitivity to light.

Pay attention to your body and watch for signs and symptoms of worsening infection, especially if the symptoms are increasing in intensity. If you are feeling “off,” do not ignore it. Make sure you have the right tools on hand to monitor your symptoms, such as a thermometer, blood pressure monitor, oxygen saturation monitor, and pulse oximeter. An otoscope is helpful to look into the ear canal and throat for signs of infection. If at any time you feel like an infection is not resolving with treatment, seek emergency medical care.

In this chapter, we'll discuss several interventions that can be helpful in supporting optimal immune function and addressing bacterial infections.

1. Remove Immune-Disruptants

There are many factors that can disrupt normal immune function making us more susceptible to bacterial and other infections. These include sugar, artificial sweeteners, dairy, alcohol, tobacco, and processed foods.

Sugar: The immune system needs glucose to perform its basic functions; however, when we consume too much glucose (sugar), our immune system function becomes impaired and our risk for infection and disease increases.¹ Glucose plays a major role in regulating the immune system; however, when glucose levels are higher than they should be, it leads to increased production of pro-inflammatory cytokines, low-grade inflammation, and immune system dysregulation. Limit your intake of added sugar to reduce your risk for infection and to increase your immune system's likelihood of overcoming them.

Artificial Sweeteners: Artificial sweeteners such as saccharine, sucralose, aspartame, and acesulfame potassium cause changes to the gut microbiome that are similar to the negative changes seen with antibiotics. Specifically, research suggests that these sweeteners may promote the spread of antibiotic-resistant genes among bacteria.² Removing artificial sweeteners from your diet may reduce the likelihood of infectious bacteria in your body transferring antibiotic-resistant genes amongst themselves, which would make your

bacterial infections more difficult to treat.

Dairy: Dairy consumption has been associated with immune system dysfunction. A high intake of dairy has been correlated with allergies, type-1 diabetes, multiple sclerosis, and other immune-related concerns.^{3 4 5 6} Dairy products like milk have been associated with more severe respiratory tract infections⁷ and with increased incidence of middle ear infections. In fact, there has been a recommendation for over 40 years that children with recurrent ear infections strictly avoid cow's milk for at least 3 months, while other well-respected pediatricians recommend life-long avoidance of dairy in children who are prone to ear infections⁸.

Alcohol: Alcohol disrupts immune pathways in such a way that it impairs the body's ability to defend itself against infection.⁹ Many of these changes to the immune system are due to alcohol's effects on the gut microbiome. Beneficial gut bacteria play a critical role in helping the immune system mature and function optimally and alcohol disrupts the communication that normally exists between the gut microbiome and the immune system. It also damages epithelial cells and two types of immune cells: T cells and neutrophils¹⁰.

Tobacco: Smoking negatively affects the function of immune cells of the innate and adaptive immune system, including T helper cells, regulatory T cells, B cells, memory T and B lymphocytes, dendritic cells, macrophages, and natural killer cells.¹¹ Even exposure to tobacco smoke is associated with increased risk for infections.

Processed Foods: Highly processed foods such as refined flour and sugar, juices, and other simple carbohydrates disrupt the microbiome and have pro-inflammatory effects.¹² This may impair the body's ability to combat bacterial infections and other pathogens.

2. Herbs

There are several herbs that exhibit antimicrobial activity against various strains of bacteria, support optimal immune function and the production of new infection-fighting white blood cells, and modulate inflammation. In this section, we'll discuss a few multi-purpose antibacterial herbs, and then we'll discuss them along with a few others in subsections, based on the types of infections they are most commonly used to address. If you have an autoimmune condition best to skip Echinacea, Astragalus, and Goldenseal unless working with a well-trained herbalist or ND.

MULTI-PURPOSE ANTIBACTERIAL AND IMMUNE-MODULATING HERBS: **Astragalus (*Astragalus membranaceus*):** Astragalus leaves and root demonstrate antibacterial activity against a variety of bacteria, including *Salmonella* spp., *Bacillus subtilis*, and *Escherichia coli*, while astragalus leaf tea shows antibacterial activity against *Staphylococcus aureus*, *Salmonella* spp., *Bacillus subtilis*, and *E. coli*¹³. Astragalus also upregulates various elements of immune system function. Note that astragalus is may overstimulate the immune system and therefore should only be used with caution or under the guidance of a trained herbalist or naturopathic doctor in individuals with autoimmune

conditions. Suggested dose: 3-9 grams of dried root daily or 4 dropperfuls of a 1:5 tincture three times daily, or, as a decoction (preferred), 1 tablespoon per cup of water, 2-3 times daily.

Echinacea (*Echinacea purpurea*, *Echinacea angustifolia*): Monocytes are a type of white blood cell that normally circulates in the blood. When tissue becomes damaged or infected, monocytes travel to the tissues and become macrophages - white blood cells that phagocytose (eat) bacteria and other pathogens. Echinacea upregulates macrophage function and may also promote the initial activation of macrophages.¹⁴ Echinacea extract shows antibacterial activity against multidrug-resistant *Klebsiella pneumoniae*,¹⁵ *Streptococcus pyogenes*, *Haemophilus influenzae*, *Legionella pneumophila*, *Propionibacterium acnes*, *Clostridium difficile*, and other bacterial strains.¹⁶ Note that echinacea may overstimulate the immune system and therefore should only be used with caution or under the guidance of a trained herbalist or naturopathic doctor in individuals with autoimmune conditions. Suggested dose: 3 dropperfuls of a 1:5 tincture, 3 times daily.

Goldenseal (*Hydrastis canadensis*): Goldenseal is commonly used to treat skin infections, and it is especially beneficial in the treatment of methicillin-resistant *Staphylococcus aureus* (MRSA).¹⁷ Specifically, goldenseal leaf extract interrupts bacterial communication, blocks MRSA's ability to produce toxins, and prevents it from causing damage to skin cells. Interestingly, the main known active constituents in goldenseal (alkaloids known as berberine, hydrastine, and canadine) do not have this effect when used in isolation, suggesting that a synergistic effect takes place when all of goldenseal's active constituents are used together. Note that goldenseal may overstimulate the immune system and therefore should only be used with caution or under the guidance of a trained herbalist or ND in individuals with autoimmune conditions. Suggested dose: 2 dropperfuls of a 1:3 tincture, 2 times daily.

Olive Leaf (*Olea europaea*): Phenolic compounds (e.g. oleuropein) found in olive leaf extract have strong antimicrobial activity against Gram-negative bacteria, Gram-positive bacteria, and mycoplasma. Olive leaf extract inhibits *Staphylococcus aureus*, *Salmonella enteritidis*, *Bacillus cereus*, *Klebsiella pneumoniae*, and *Escherichia coli*.¹⁸ Another study demonstrated that olive leaf extracts inhibit *Campylobacter jejuni*, *Helicobacter pylori*, and methicillin-resistant *Staphylococcus aureus* (MRSA).¹⁹ Suggested dose: 2 dropperfuls of a 1:5 tincture, 2 times daily.

Oregano (*Origanum vulgare*): The phenolic compounds in oregano have potent antibacterial properties. Oregano has antibacterial activity against *Escherichia coli* ATCC 25922, *Escherichia coli* O157:H7, *Salmonella enterica* subsp. *serovar Enteritidis*, *Bacillus cereus*, *Listeria monocytogenes*, and *Staphylococcus aureus*.²⁰ Greek oregano (*Origanum vulgare* L. subsp. *Hirtum*) was found to have more potent antibacterial properties than common oregano. Suggested dose: 2 dropperfuls of a 1:3

tincture, 2 times daily.

Thyme (*Thymus vulgaris*): Thyme exhibits potent antibacterial activity. Thyme essential oil shows antibacterial activity against *Haemophilus influenzae*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Streptococcus pyogenes*,^{21 22} and multi-drug resistant *Staphylococcus aureus*.²³ Alcoholic extracts of thyme exhibit antibacterial activity against *Escherichia coli*, *Klebsiella pneumoniae*, *Yersinia enterocolitica*, *Staphylococcus aureus*, *Listeria monocytogenes*, and *Enterococcus faecalis*.²⁴ Suggested dose: 5 dropperfuls of a 1:5 tincture, 2 times daily.

Turmeric (*Curcuma longa*): In addition to its anti-inflammatory properties, turmeric rhizome displays antibacterial properties against *Staphylococcus epidermidis*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, and *Escherichia coli*.²⁵ Methanolic extracts of curcumin inhibit *Bacillus subtilis* and *Staphylococcus aureus*.²⁶ Suggested dose: 2 dropperfuls of a 1:4 tincture, 2 times daily.

Usnea/Old Man's Beard (*Usnea barbata*): The usnic acid in the usnea lichen displays antibacterial activity against multiple strains of bacteria including *Pseudomonas aeruginosa*, *Staphylococcus aureus*, *Enterococcus casseliflavus*, *Streptococcus pyogenes*, and *Streptococcus pneumoniae*.²⁷ Suggested dose: 2 dropperfuls of a 1:5 tincture, up to 3 times daily.

Yarrow (*Achillea millefolium*): Yarrow displays antibacterial activity against *Bacillus cereus*, *Bacillus subtilis*, *Staphylococcus aureus*, *Salmonella typhimurium* and *Salmonella agona*.²⁸ Externally, it is one of the best antibacterial wound care herbs available. Not only does it make a great cleanser for a wound, it helps stop bleeding while promoting healing. I recommend that you have yarrow in your first aid kit. You can infuse yarrow in water to make a strong tea and let this cool to wash a wound regularly. You can also grind dried yarrow into powder to make a styptic powder to treat bleeding (note that bleeding does help cleanse the wound). Infuse yarrow in a carrier oil or salve to apply to wounds as they heal. I always carry my all-purpose salve, which has yarrow as its first ingredient. Suggested dose: Internally: 2 dropperfuls of a 1:5 tincture, up to 3 times daily. Externally, apply a salve 3x daily.

MEDICINAL MUSHROOMS: Medicinal mushrooms modulate the immune system and many have antibacterial properties.

Reishi (*Ganoderma lucidum*): Reishi exhibits antibacterial activity against *Bacillus cereus*, *Enterobacter aerogenes*, *Staphylococcus aureus*, *Escherichia coli*, and *Pseudomonas aeruginosa*. Reishi also inhibits *Pseudomonas aeruginosa*, *Proteus vulgaris*, *Enterococcus faecalis*, *Salmonella typhimurium*, *Klebsiella pneumoniae*, and *Streptococcus mutans*.²⁹ Suggested dose: 2 dropperfuls of a dual-extracted tincture, up to 3 times daily.

Cordyceps (*Cordyceps militaris* and *C. sinensis*): Cordyceps modulates the immune system and

potentiates the activity of PA-824, an antibiotic that is used in the treatment of tuberculosis (*Mycobacterium tuberculosis* infection).³⁰ Suggested dose: 2 dropperfuls of a dual-extracted tincture, up to 3 times daily.

Lion's Mane (*Hericium erinaceus*): In addition to its immune-modulating properties, lion's mane displays antibacterial activity against *Helicobacter pylori*, a bacterium that can cause peptic ulcers.³¹ Suggested dose: 2 dropperfuls of a dual-extracted tincture, up to 3 times daily.

METHICILLIN-RESISTANT STAPHYLOCOCCUS AUREUS (MRSA): To address methicillin-resistant *Staphylococcus aureus* infections, consider usnea lichen (*Usnea barbata*), hops (*Humulus lupulus*), elecampane (*Inula helenium*), and Oregon grape (*Mahonia aquifolium*) root.

In addition to its antibacterial effects against other drug-sensitive and multi-drug resistant bacteria, usnea exhibits inhibitory activity against MRSA.³² These effects may be due to its many active constituents, such as usnic acid and related compounds, depsides, depsidones, depsones, dibenzofurans, chromanones, monocyclic phenols, xanthenes, napthaquinones, anthraquinones, and pulvic acid derivatives. The beta acids found in hops strobiles exhibit potent anti-MRSA activity for addressing MRSA-infected wounds.³³ Elecampane has a high content of sesquiterpene lactones and volatile oils, which contribute to its anti-MRSA effects. Elecampane inhibits methicillin-resistant *S. aureus* as well as other bacteria.³⁴ Oregon grape root shows potent anti-MRSA activity; it blocks MRSA by acting as an efflux pump inhibitor, meaning that it prevents bacteria from excreting the drugs so that they are no longer resistant to them.³⁵ In another study, berberine, an active constituent in Oregon grape root, enhanced the efficacy of the pharmaceutical antibiotics azithromycin and levofloxacin.³⁶

Suggested dose: 2 dropperfuls of a 1:5 usnea tincture, up to 3 times daily.

Suggested dose: 4 dropperfuls of a 1:4 hops tincture, 2 times daily.

Suggested dose: 2 dropperfuls of a 1:5 elecampane tincture, 3 times daily.

Suggested dose: 3 dropperfuls of a 1:4 Oregon grape root tincture, 2 times daily.

EAR INFECTIONS: GARLIC AND MULLEIN FLOWERS FOR EAR INFECTIONS: Garlic is a potent antimicrobial agent, and when combined with mullein flowers, it provides relief from the pain and inflammation typical of middle ear infections. Make an oil infusion with these two plants by placing half cup of mullein flowers and a few cloves of chopped garlic in a glass jar. For a severe infection, you can also add yarrow (flowers and leaves) to the mixture. Completely cover the herbs with olive oil and place the jar in a pan of hot water. Make sure the water is very hot, but never boiling. Let the herbs and oil sit and infuse for six

to eight hours before removing the jar and straining out the oil. Discard the dried herbs, and keep the oil in the refrigerator between uses. When you are ready to use it, gently warm the oil by placing the bottle in a pan of warm water until it reaches a comfortable temperature for putting in the ear or use at room temperature. Put three to four drops in the ear and keep the head tilted so that the oil does not pour out. Alternatively, gently place a cotton ball in the ear to hold it in. Repeat as needed up to three times per day until the pain, redness, and infection is gone. You may also massage this oil around the ear. Note: do not place anything (including oil) in the ear if the eardrum is ruptured.

RESPIRATORY INFECTIONS: For respiratory infections, try inhalation of antimicrobial plants that help to open the airways in addition to an internal herbal antibiotic tincture (like yarrow and lungwort lichen tinctures) or modern antibiotics. You can also add mullein to help open the airways.

URINARY TRACT INFECTIONS (UTIS): For urinary tract infections, herbal medicine works best at the first sign of an infection or as a preventative. When addressing urinary tract infections, I usually use more than one herb because they have different mechanisms of action and I've found that using multiple herbs leads to a better response.

My go to herbs for UTIs are as follows: Barberry; Uva ursi; Kinnikinnick leaf; berberine-containing herbs like Oregon grape root and Goldenseal; Juniper leaves; Stinging Nettle; Goldenrod; Usnea; Marshmallow Root; Corn Silk (as a tea); and Cranberry extract, which contains d-mannose, fructose, and proanthocyanidins.³⁷
³⁸ ³⁹ I also add D-Mannose in supplement form because it helps prevent *E. coli*, the most common cause of UTIs, from sticking to the walls of the urethra.

HERBS FOR CLEANSING/WASHING, DRESSING, AND HEALING EXTERNAL WOUNDS: Cleansing external wounds is vital to prevent or keep an infection at bay. First assess the damage. Wash it well to see how deep it is and to get all visible matter out of the wound. If it will not stop bleeding with pressure, keep applying pressure and get to the nearest hospital. If the wound isn't bleeding badly and will not need stitches, cleansing the wound to prevent infection is critical.

Proper cleaning of an external wound helps prevent the development or worsening of a skin infection. Our skin is our first line of defense against foreign invaders in the body. It often takes the brunt of the bacteria to which we are exposed. As soon as possible after sustaining a wound or scratch and assessing it to ensure you don't need emergent care, clean the wound and then add a topical antibiotic to protect against infection.

To properly cleanse and protect an external wound, take the following steps:

- Irrigate the wound thoroughly (ideally with saline

and then an herbal wash – like a yarrow tea). You can make your own saline by mixing 2 teaspoons of non-iodized salt to 1 quart (4 cups) of water. Do not use alcohol or hydrogen peroxide on the wound as it harms the healthy tissue. Warm soapy water also works well. For a very dirty wound/scrape that is too painful to scrub, raw honey helps draw out the dirt so you can treat it.

- Apply an antibiotic salve or ointment to the wound and cover it with a bandage, removing the bandage at least once per day to expose the wound to air. If you don't have an antibiotic salve with you, yarrow (my go-to for external wounds) and usnea are good fresh herbs to use on a wound. If you don't have a salve but have access to other materials, the yarrow or lavender hydrosol described below also works well.
- Assess the wound every few hours to make sure it is not worsening. Apply the antibiotic salve or ointment or the hydrosol 2-3 times per day.
- You will need to remove the bandage daily, clean the wound if needed, and reapply the ointment or hydrosol.

OTHER HERBS FOR USE WITH EXTERNAL WOUNDS: Witch Hazel (*Hamamelis virginiana*):

Witch hazel makes a great wound wash for minor wounds. It is astringent and reduces inflammation and redness. Infuse witch hazel bark into hot water for ten minutes to make a tea and let cool. When cool, pour the infusion over the wound to wash it.

Lavender (*Lavandula angustifolia*): Try spraying lavender hydrosol on a wound every hour. Lavender is naturally antimicrobial and reduces redness. It is especially helpful for burn wounds, as is raw honey.

3. Diet and Specific Foods

There are specific foods that help support optimal immune system function and reduce the likelihood of us succumbing to various types of infections. These include garlic; onion; honey; apple cider vinegar; ginger; vitamin C-containing foods; green, leafy vegetables; and organic bone broth.

Garlic: Garlic exhibits potent antibacterial properties, including inhibiting methicillin-resistant *Staphylococcus aureus* infection.⁴⁰ Garlic ointment prevents bacterial pathogens in burn wounds (*Staphylococcus aureus*, *Staphylococcus epidermidis*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, and *Klebsiella pneumoniae*) from forming biofilms, which make the bacteria more difficult to destroy.⁴¹ Garlic ointment also partially disrupts biofilms formed by *Enterococcus faecalis*. The allicin in garlic exhibits anti-microbial activity against methicillin-resistant *Staphylococcus aureus*.⁴² Fresh garlic extract increases the efficacy of antibiotics against bacterial infections.⁴³ This is especially relevant in cases of multiple drug-resistant bacterial strains.

Onion: Infection leads to increased free radicals and oxidative stress within the body. Red and yellow onions are some of the most potent natural sources of quercetin, a powerful antioxidant. Onions also have

direct antibacterial properties. Onion extract exhibits antibacterial activity against pathogenic bacteria of the mouth, such as *Streptococcus mutans* and *Streptococcus sobrinus*, which are the main causes of dental cavities, and *Porphyromonas gingivalis* and *Prevotella intermedia*, which are the main causes of adult periodontitis.⁴⁴ Aged onions show antibacterial activity against *E. coli*, *Pseudomonas aeruginosa*, *Staphylococcus aureus*, and *Bacillus cereus*.⁴⁵ Onion also possesses antibacterial activity against *Vibrio cholera*, and enhances the effectiveness of conventional antibiotics.^{46 47}

Raw Honey: Honey has been used topically for years in wound care because it has antibacterial properties, helps keep wounds moist, and creates a protective barrier.⁴⁸ Honey speeds up healing by stimulating the production of white blood cells and increasing levels of pro-inflammatory chemical messengers including tumor necrosis factor-alpha (TNF- α), interleukin-1-beta (IL-1 β), and interleukin-6 (IL-6).^{49 50} Because honey exerts these effects when applied topically, it may be antibacterial within the gastrointestinal tract when eaten.

Apple Cider Vinegar: Apple cider vinegar shows antibacterial activity against *E. coli*, *Staphylococcus aureus*,⁵¹ methicillin-resistant *Staphylococcus aureus* (MRSA), and antibiotic-resistant *E. coli*.⁵²

Ginger (*Zingiber officinale*): An active constituent in ginger called zingerone and its derivatives exhibit antibacterial properties.⁵³ Ginger root extracts and gingerol inhibit the growth of *H. pylori*, a bacterium that causes peptic ulcers.⁵⁴ Ginger displays antimicrobial activity against *E. coli*, *Salmonella typhi*, and *Bacillus subtilis* and an ethanolic extract of ginger was shown to inhibit *Salmonella typhi*.⁵⁵ Preliminary research suggests that constituents in ginger may inhibit *Mycobacterium avium* and *Microbacterium tuberculosis*, the bacterium that causes tuberculosis.⁵⁶

Foods High in Vitamin C: Vitamin C has immunoprotective effects and helps the body fight infection.⁵⁷ Foods rich in vitamin C include citrus, kiwi, bell peppers, strawberries, tomatoes, and cruciferous vegetables (cabbage; bok choy; broccoli; kale; cauliflower; Brussels sprouts; mustard greens; and other green, leafy vegetables).

Organic Bone Broth: Bone broth is nourishing and healing. It helps reduce inflammation, soothes the digestive tract, strengthens bones, increases energy, and aids in wound healing.⁵⁸ Bone broth feeds healthy gut bacteria, leading to the production of healthy immune cells.

4. Hydrotherapy

There are several water therapies that can help support optimal immune system function and help the body to clear various types of infections. These include using simple water and salt water gargles, Neti pot, steam inhalation with essential oils, and cold wet socks and cold wet shirt treatments.

Simple Water Gargle: Gargling with water is an effective means of preventing upper respiratory

tract infections in healthy people.⁵⁹

Salt Water Gargle: Salt is bacteriostatic; it prevents some bacteria from growing and reproducing.⁶⁰ It can also be bacteriocidal, causing lysis of the bacterial capsid thus killing bacteria. Salt accumulates at the site of bacterial infection in the skin, where it has an antimicrobial effect.⁶¹ Gargling with salt water may help with removal of bacteria from gums, reducing halitosis and gingivitis, and it may help reduce the risk of upper respiratory tract infections.⁶²

Neti Pot/Saline Nasal Irrigation: Irrigating the upper respiratory mucosa with salt water, (e.g. using a Neti pot) is beneficial in addressing acute upper respiratory infections.⁶³

Steam Inhalation with Essential Oils: Many essential oils have antibacterial properties.⁶⁴ Steam inhalation with essential oils can help address concerns associated with upper respiratory infections. Eucalyptus oil (*Eucalyptus globulus*) exhibits antibacterial activity against various bacteria, including *Staphylococcus aureus*, which can cause staphylococcal pneumonia; *Pseudomonas aeruginosa*, which can cause pneumonia; and *Streptococcus pyogenes*, which causes strep throat.⁶⁵ Peppermint oil exhibits bactericidal activity against methicillin-resistant *Staphylococcus aureus* (MRSA) and methicillin-sensitive *S. aureus*.⁶⁶

Cold Wet Socks and Cold Wet Shirt Treatments: Cold wet socks and cold wet shirt treatments operate under a similar premise. Both therapies require a person to put on a cold, damp, thin garment and allow the person's body heat to warm and dry the article of clothing. Because the effectiveness of the therapy depends on the body's ability to warm the socks or the shirt, it is important to use thin, cotton socks or a thin, cotton T-shirt and to make sure the garments are only slightly damp (not dripping wet). The wet material is usually covered with something that will help retain heat. For example, cold wet sock treatments typically require the damp socks to be covered with another pair of dry socks that are made with a warm material like wool. The wet t-shirt procedure typically requires the person to cut holes in a large trash bag for the arms and head, and wear that over the T-shirt so that the body heat is trapped and able to warm the shirt. Both treatments are usually done before bed, and the person sleeps with the cold garment. Upon rising, the garment should be warm and dry. These treatments stimulate the immune system to fight infection, enhance circulation, and reduce upper respiratory tract inflammation.

5. Supplements There are several supplements that can help support the immune system and help the body eradicate infection. These include vitamins C, D, and E; and probiotics.

Vitamin C: Vitamin C plays an important role in immune system function. This is partly because infections increase oxidative stress and vitamin C is a potent antioxidant. Similar to infections, physical activity also increases oxidative stress. Research demonstrates that vitamin C supplementation cuts the number of colds in

physically active people in half.⁶⁷ Additionally, people who took vitamin C regularly had reduced duration of upper respiratory infections. Vitamin C was also found to benefit pneumonia patients and people with tetanus. Suggested dose: 6-8 grams of buffered vitamin C per day if ill.

Vitamin D: Vitamin D plays a crucial role in immune function. It modulates both the innate and the adaptive immune systems, regulates the inflammatory response, and influences the production of antimicrobial proteins within the body.⁶⁸ Vitamin D deficiency is associated with increased risk of developing infections and more severe infections. Suggested dose: 5,000-10,000 IU per day until 25-hydroxyvitamin D levels are around 50 ng/dL, then cut back to a maintenance dose of 4,000 IU. Get your levels tested.

Vitamin E: Another potent antioxidant, vitamin E also plays a regulatory role in the immune system. Vitamin E modulates the function of T cells. Vitamin E supplementation enhances immune system function and reduces the risk for infectious respiratory infections like pneumonia.⁶⁹ Suggested dose: 800 mg per day.

Probiotics: Probiotics help modulate the immune system and regulate the function of intestinal epithelial cells.⁷⁰ They produce antimicrobial compounds⁷¹ and play an important role in preventing and treating gastrointestinal infections. This is because beneficial bacteria inhabit the gut and prevent other transient and potentially harmful bacteria from taking up residence in the small and large intestines. Suggested dose: 80-100 billion CFU per day with meals, in divided doses.

6. Rest Sleep has a profound effect on the immune system. Adequate sleep is critical in order for our bodies to fight off bacterial and other infections. Sleep helps immune cells called T cells to leave the blood vessels and travel to the lymph nodes, where they can interact with another type of immune cell called antigen-presenting cells.⁷² These interactions are required for our immune systems to efficiently fight off pathogens.

Chronic sleep deprivation causes the body to enter into a state of physiological stress. It leads to a low-grade, persistent, systemic inflammation and increases our risk for various chronic diseases. It also makes us more susceptible to infections.^{73 74} Adopting healthy sleep hygiene practices, incorporating physical activity into your daily routine, appropriately managing your stress, and using hypnotic and sedative herbs as well as vitamins and supplements can help you improve your sleep and harness the immune system-related benefits associated with optimal sleep. For additional instructions on incorporating these interventions into your current routine in order to help you reduce your risk for bacterial infections and combat existing bacterial infections, refer to the Sleep Protocol (page 286).

7. Western Medical Approaches Oftentimes people don't realize they have an infection until very late, when systemic modern antibiotics may be necessary. In our household we utilize natural remedies/herbs as our #1 go to,

rarely defaulting to Western medicine interventions like antibiotics. This decreases our risk of developing bacterial infections that are resistant to modern antibiotics and it also keeps our gut healthy. That said, modern antibiotics are necessary at times.

Before life-saving antibiotics like penicillin were invented, people died of what we consider minor wounds today. Scientists have discovered a variety of antibiotic medications that target different types of infections. These work by targeting various qualities of the bacteria, and either preventing them from multiplying so that the immune system can kill them (bacteriostatic drugs) or killing the bacteria directly (bacteriocidal drugs).

If you and your provider determine that the benefits associated with taking antibiotics outweigh the risks in your unique situation, there are a few things you should know about taking these medications. First of all, once you begin taking antibiotics, it's important to finish the prescribed course. Stopping the medication before you finish simply because you feel better can lead to the infection coming back worse than before. In some cases, it can also lead to the bacteria developing resistance to the drug. In some cases, forgetting to take your antibiotics for more than a day may result in having to start over with treatment. If you forget to take them for over twenty-four hours, call your doctor immediately to figure out what steps to take next. If you happen to spill or lose your antibiotic prescription, call your doctor immediately to get it refilled so you do not miss a dose or risk stopping taking them too early.

There are many types of antibiotics. Some are specific to a certain type of bacteria, while others are effective against a wider range of infections. Like all other drugs, antibiotics are divided into various drug classes based on how they work. Some common antibiotic classes include beta-lactam antibiotics, tetracyclines, macrolides, lincosamides, dihydropyrimidines, sulfonamides, fluoroquinolones, and nitroimidazoles. Below is a short list of ten common antibiotics prescribed today and their uses.

Amoxicillin: Amoxicillin (Brand names: Moxatag, Amoxil, Larotid) is one of the most frequently prescribed antibiotics.⁷⁵ It exhibits activity against a variety of gram-positive bacteria and it also covers more gram-negative bacteria than penicillin. Because amoxicillin is a beta-lactam antibiotic and is similar to penicillin, it can trigger allergic reactions in people who are allergic to penicillin. Amoxicillin is frequently prescribed to treat bacterial infections such as chest infections (including pneumonia), dental abscesses, and ear and sinus infections.

Amoxicillin/Clavulanic Acid: Amoxicillin/clavulanic acid (Brand names: Augmentin, Augmentin ES-600, AugmentinXR) is a combination medication that can tackle bacteria that are resistant to amoxicillin alone. Clavulanic acid is a beta lactamase inhibitor, meaning that it helps prevent some bacteria from inactivating and becoming resistant to the amoxicillin. This combination

medication is commonly used to treat bronchitis, pneumonia, urinary tract infections, ear infections, and skin infections.

Cephalexin: Cephalexin (Brand names: Keflex, Daxbia) are a type of beta-lactam antibiotic known as first-generation cephalosporins. This antibiotic exhibits activity against many gram-positive cocci and gram-negative rods.⁷⁶ It is given to treat urinary tract, respiratory, skin, and other infections.⁷⁷

Doxycycline: Doxycycline (Brand names: Monodox, Doxy-100, Targadox) is in a class of antibiotics known as tetracycline antibiotics. It exhibits bacteriostatic activity against both gram-positive and gram-negative bacteria, meaning it prevents these bacteria from growing without directly killing the bacteria. It is used to treat rosacea, adult acne, gastrointestinal infections, urinary tract infections, respiratory infections, gum disease, eye infections, and a variety of sexually transmitted infections.

Azithromycin: Azithromycin (Brand names: Zithromax, Zithromax Z-Pak, Zmax) is a macrolide antibiotic. It exhibits bacteriostatic activity against many gram-positive bacteria. It works by blocking bacterial protein synthesis. Azithromycin is commonly used to treat atypical pneumonia and other respiratory infections, sexually transmitted infections, skin infections, eye infections, and ear infections. It is sometimes referred to as a "Z pack."

Clindamycin: Clindamycin (Brand names: Cleocin, Cleocin T, Evoclin, and Clindesse) is a lincosamide antibiotic. It prevents bacteria from making the proteins they need to survive. This drug shows bacteriostatic activity against gram-positive and gram-negative bacteria. It is a broad-spectrum antibiotic, and is used to treat more serious infections. It is typically prescribed to treat methicillin-resistant *Staphylococcus aureus* (MRSA) infections and infections of the lungs, skin, blood, female reproductive organs, and internal organs. Diarrhea is a known side-effect of clindamycin.

Trimethoprim / Sulfamethoxazole: Trimethoprim / Sulfamethoxazole (Brand names: Bactrim DS, Sulfatrim, Bactrim) is a combination medication. It is comprised of a dihydropyrimidine and a sulfonamide antibiotic. This combination is usually prescribed to treat ear infections, urinary tract infections, bronchitis, traveler's diarrhea, and shigellosis.

Levofloxacin: Levofloxacin (Brand name: Levaquin) is a fluoroquinolone antibiotic and it exhibits bacteriocidal activity against a wide variety of gram-positive and gram-negative bacterial strains. This drug is a very broad-spectrum antibiotic. Its use is typically reserved for severe or otherwise concerning bacterial infections (e.g., in cases where the person is immunocompromised). Levofloxacin is prescribed to treat exposure to anthrax, tuberculosis, plague, pneumonia, skin infections, and some sexually transmitted infections. Fluoroquinolone antibiotics carry an increased risk of tendinitis or tendon rupture, both during treatment and for

several months after treatment has ended.

Ciprofloxacin: Ciprofloxacin (Brand names: Cetraxal, Cipro XR, Otiprio) is also a fluoroquinolone with widespread bacteriocidal activity. This powerful antibiotic might be given to those exposed to anthrax, typhoid, or the plague. It is usually a last resort if other antibiotics do not work since it can have unwanted side effects. Fluoroquinolone antibiotics carry an increased risk of tendinitis or tendon rupture, both during treatment and for several months after treatment has ended.

Metronidazole: Metronidazole (Brand names: Metrogel, MetroCream, Nuversa) is a nitroimidazole antimicrobial; it arrests bacterial growth by preventing bacteria from making nucleic acids that they need to survive. Metronidazole can be prescribed to treat parasitic or bacterial infections. When prescribed to treat bacterial infections, it is usually prescribed to treat skin infections, rosacea, and mouth infections, including infected gums and dental abscesses. It can also treat bacterial infections of the vagina, stomach, liver, skin, joints, lungs, brain and spinal cord, blood, or heart.

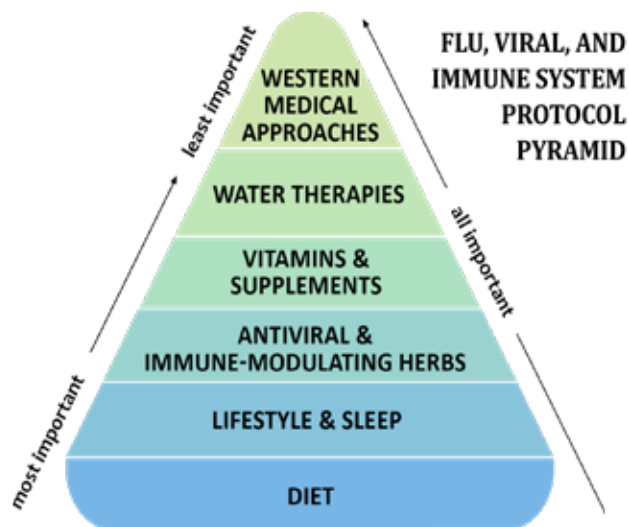
	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Probiotics Vitamins D & E Dose 1: Medicinal mushroom(s) of your choice* Bone broth	Weekly grocery shop Dose 1: non-mushroom herbs (2-3)** Hydration check (Are you drinking enough water?)	Usnea, garlic, & yarrow tinctures Dose 2: Medicinal mushroom(s) of your choice* Vitamin C Probiotics	Usnea, garlic, & yarrow tinctures Dose 2: non-mushroom herbs (2-3)**	Steam inhalation with essential oil OR Neti pot irrigation, if needed Sleep herbs, if needed (see our Sleep Protocol)
MONDAY	Probiotics Vitamins D & E Dose 1: Medicinal mushroom(s) of your choice* Bone broth	Dose 1: non-mushroom herbs** Hydration check	Usnea, garlic, & yarrow tinctures Dose 2: Medicinal mushroom(s)* Vitamin C Probiotics	Usnea, garlic, & yarrow tinctures Dose 2: non-mushroom herbs (2-3)**	Steam inhalation OR Neti pot irrigation Sleep herbs, if needed
TUESDAY	Probiotics Vitamins D & E Dose 1: Medicinal mushroom(s) of your choice* Bone broth	Dose 1: non-mushroom herbs** Hydration check	Usnea, garlic, & yarrow tinctures Dose 2: Medicinal mushroom(s)* Vitamin C Probiotics	Usnea, garlic, & yarrow tinctures Dose 2: non-mushroom herbs (2-3)**	Steam inhalation OR Neti pot irrigation Sleep herbs, if needed
WEDNESDAY	Probiotics Vitamins D & E Dose 1: Medicinal mushroom(s) of your choice* Bone broth	Dose 1: non-mushroom herbs** Hydration check	Usnea, garlic, & yarrow tinctures Dose 2: Medicinal mushroom(s)* Vitamin C Probiotics	Usnea, garlic, & yarrow tinctures Dose 2: non-mushroom herbs (2-3)**	Steam inhalation OR Neti pot irrigation Sleep herbs, if needed
THURSDAY	Probiotics Vitamins D & E Dose 1: Medicinal mushroom(s) of your choice* Bone broth	Dose 1: non-mushroom herbs** Hydration check	Usnea, garlic, & yarrow tinctures Dose 2: Medicinal mushroom(s)* Vitamin C Probiotics	Usnea, garlic, & yarrow tinctures Dose 2: non-mushroom herbs (2-3)**	Steam inhalation OR Neti pot irrigation Sleep herbs, if needed
FRIDAY	Probiotics Vitamins D & E Dose 1: Medicinal mushroom(s) of your choice* Bone broth	Dose 1: non-mushroom herbs** Hydration check	Usnea, garlic, & yarrow tinctures Dose 2: Medicinal mushroom(s)* Vitamin C Probiotics	Usnea, garlic, & yarrow tinctures Dose 2: non-mushroom herbs (2-3)**	Steam inhalation OR Neti pot irrigation Sleep herbs, if needed
SATURDAY	Probiotics Vitamins D & E Dose 1: Medicinal mushroom(s) of your choice* Bone broth	Dose 1: non-mushroom herbs** Hydration check	Usnea, garlic, & yarrow tinctures Dose 2: Medicinal mushroom(s)* Vitamin C Probiotics	Usnea, garlic, & yarrow tinctures Dose 2: non-mushroom herbs (2-3)**	Steam inhalation OR Neti pot irrigation Sleep herbs, if needed

*Medicinal mushrooms: Reishi, Cordyceps, Lion's mane (choose 1-2).

**Non-mushroom herbs: Astragalus, Echinacea, Goldenseal, Olive leaf, Oregano, Thyme, Turmeric, (choose 2-3).

Ch. 18: Flu, Viral, and Immune System Protocol

Viruses are responsible for the flu, the majority of incidences of the common cold, and a host of other conditions. Unlike bacteria, viruses cannot function independently. Viruses are organic material comprised of, at minimum, a protein coat and genetic material. The genetic material can be deoxyribose nucleic acid (DNA) or ribonucleic acid (RNA), and it can be either single-stranded or double-stranded.



Viruses enter into a host cell in order to replicate, taking control of the cell's DNA-making and protein-making capability. They then replicate the virus' genetic material and create more virus particles. The various parts of the virus assemble and whole viruses are created. These viruses exit the cell, infect and take control of other cells, and replicate their genetic material to produce more viruses. As this process continues, we begin to experience symptoms of viral illness.

There are several ways we can make it less likely that our bodies will be infected by viruses, limit the extent of viral illness once infected, and, in many cases, help our bodies clear the infection more quickly. We can support optimal immune system function, which enables our immune system to better attack, identify, and eliminate the viruses from our bodies; we can take antiviral herbs that specifically prevent the virus from entering or exiting our cells; and we can disrupt the process of viral replication by denaturing enzymes that the virus needs to replicate its genome.

In this chapter, we'll discuss various therapies that have been shown to have antiviral and immune-modulating properties against influenza, SARS-CoV-2, herpes simplex 1 and 2, and other viruses.

1. Diet The foods you consume significantly impact your risk for viral illnesses. Specifically, foods that suppress or otherwise impair your immune system function, like sugar and refined carbohydrates, increase your risk of viral infection. These foods lead to increased inflammation and abnormal function of the innate

immune system, which is the body's initial immune response to pathogens like viruses and bacteria.¹ See our dietary chapter for help with food choices.

Diets rich in **whole foods**, like nutrient-rich fruits and vegetables, support optimal immune system function and protect against viral infection.² The immune-related benefits of whole foods have been attributed to their high concentrations of polyphenols, terpenoids, flavonoids, alkaloids, sterols, pigments, unsaturated fatty-acids, and micronutrient vitamins and minerals.

Garlic: Garlic exhibits potent antiviral properties. Integrate it into your meals to support your immune system and reduce your risk of viral infection. The active organosulfur compounds in garlic block viruses from entering host cells and inhibit enzymes that viruses need to replicate like RNA polymerase and reverse transcriptase.³

Probiotic Foods: Our gut health and immune system are directly linked. Make sure you are eating enough probiotic foods each day to keep your good gut bacteria happy. Try eating probiotic foods or drinking probiotic drinks with each meal. Some great choices include kombucha, sauerkraut (not canned), yogurt (with live cultures), kimchi, and miso soup.

Limit or avoid alcohol and sugar; they weaken the immune system and make you more susceptible to illness.

Organic Bone Broth: Bone broth is a highly nourishing and beneficial broth you can easily make at home with scrap bones from fish, chicken, turkey, beef, or lamb. Simply throw the bones into a stockpot and cover them in water. Add healthy, immune-boosting herbs and plants like garlic, onion, rosemary, oregano, stinging nettle, and thyme to the broth and let this simmer for ten to twelve hours. This process pulls the beneficial collagen and gelatin out of the bones. Bone broth can help restore gut health, support immune system function, and promote a healthy inflammatory response. Use as a soup base or drink it by itself daily.

Fasting: In addition to consuming a diet rich in whole foods, limiting refined carbohydrates and sugars, and eating probiotic foods, fasting decreases your risk of viral infection and helps the body to clear the virus once it has invaded the body. Fasting for three days stimulates the immune system and leads to the formation of new immune cells.⁴ Even a less restrictive form of fasting-time-restricted eating, which is a form of intermittent fasting-leads to similar effects on the immune system.⁵ Time-restricted eating involves limiting the window of time when you eat to a 6- to 8-hour daytime window and fasting or refraining from eating during the remaining 16 to 18 hours. Time restricted eating primes the immune response and activates autophagy, which is the body's way of cleaning out damaged cells to make room for new, functioning cells.⁶

In order to begin incorporating time-restricted eating into your daily routine, first choose the window in which you will consistently eat. For example, does an 8 am to 2 pm window make sense for you, or does a 10 am

to 6 pm make more sense with your schedule and family obligations? Keep in mind that it's best to consume most of your calories and carbohydrates earlier in the day, and for your last meal to be lighter or less calorie dense.⁷ It's also ideal for there to be a 3- to 6-hour period between meals. If you're new to time-restricted eating, consuming unsweetened, non-caffeinated herbal teas outside of your eating period can make the transition easier.

2. Lifestyle and Sleep Lifestyle factors such as sleep, exercise, hygiene, and hydration play important roles in preventing and addressing viral illnesses.

Sleep: Chronic sleep deprivation is a form of unspecific stress. As a form of stress, sleep deprivation negatively impacts immune system function and health in general, making you more susceptible to viruses and other pathogens.⁸ The body requires at least 8 hours of sleep for optimal immune system function. One study demonstrated that people who went from sleeping 8 hours nightly to sleeping 6 hours per night experienced more immune system dysregulation, impaired performance, and significant sleepiness.⁹ Please see our Sleep Protocol at page 286 for help with sleep, if this is an issue for you.

Exercise: As you age, the function of your innate immune system wanes. This leads to a decrease in your body's ability to protect you from viral and bacterial infections and it makes long-term complications more likely. Regular physical activity has an anti-inflammatory effect on the body, promotes healthier aging, and reduces your likelihood of death by any cause (a concept referred to in medicine as all-cause mortality).¹⁰ In order to harness these immune-system benefits, moderate-intensity physical activity (64–76% of the maximal heart rate) for at least 30 minutes per day, five or more days per week is recommended.¹¹

Hygiene: In order to make viral infection less likely, refrain from sharing utensils or toothbrushes. Wash your hands after potential exposure to viruses, such as after shaking hands with others or after sneezing, coughing, or using the bathroom. Wash and change your linens frequently.

Hydration: Even slight dehydration has negative effects on the immune system. Research demonstrates that modest dehydration leads to a decrease in three types of antimicrobial proteins that are typically found in the saliva-secretory IgA, alpha amylase, and lysozyme secretion rates.¹² Proper hydration is also important to the immune system because this system needs the nutrients in our blood stream in order to function optimally, and proper hydration makes the transport of these nutrients possible. For most people, taking your body weight (in pounds) and dividing it by 2 gives you a good ballpark figure for the ounces of water you should drink on a daily basis. Keep in mind that if you are more active than the average person or if you live in a hotter climate, you may need to drink more in order to replace the fluids you've lost through sweating.

3. Antiviral and Immune-Modulating Herbs

There are several herbs with antiviral and immune-modulating properties that can be used for prevention and treatment of viral infection. These include Astragalus, Chinese Skullcap, Cordyceps, Echinacea, Elderberry, Goldenseal, Japanese Honeysuckle, Lemon Balm, Licorice, Lion's Mane, Myrrh, Reishi, and Usnea. Note that some of these herbs may not be appropriate for people with autoimmune conditions, such as Astragalus, Echinacea, and Goldenseal. Luckily there are plenty of other effective herbal options.

Astragalus: Astragalus has antiviral and immune-stimulating effects.¹³ It promotes immune cell proliferation and increases the release of inflammatory chemical messengers called cytokines.¹⁴ These properties allow the plant to inhibit viruses and to enhance immunity in the event of viral infection. Its immune-stimulating properties mean that people with autoimmune disorders like systemic lupus erythematosus and rheumatoid arthritis should opt for immune-modulating herbs instead or use extreme caution when using this herb. Suggested dose: 3-9 grams of dried root daily or 4 dropperfuls of tincture three times daily, or as a decoction: 1 tablespoon per cup of water, 2-3 times daily.

Chinese Skullcap (*Scutellaria baicalensis*): The baicalin and wogonin content in Chinese skullcap exhibit strong antiviral activity and are beneficial in addressing severe respiratory illness, the common cold, and influenza virus. This herb demonstrates great potential for acute lung injury induced by influenza and other respiratory viruses. In addition to being active against both influenza A and B, this herb also demonstrates antiviral activity against hepatitis B virus, hepatitis C virus, herpes simplex virus 1 and 2, and Epstein-Barr virus.¹⁵ Suggested dose: 3 dropperfuls Chinese skullcap tincture, 3 times daily.

Cordyceps (*Cordyceps militaris* and *C. sinensis*): Cordyceps can help your body better respond to everyday stress and regulate your immune system as it is an immunomodulator. It has strong antiviral properties and inhibits pro-inflammatory cytokines.¹⁶ Cordycepin from *C. militaris* is thought to inhibit viral entry and viral replication into the host body's cells and is being heavily researched for SARS-CoV-2^{17 18}. *C. militaris* also has anti-HIV, anti-influenza, and anti-HCV (Hepatitis-C) effects^{19 20}. Suggested dose: 2 dropperfuls of a dual-extracted cordyceps tincture, up to three times daily for illness, and 2 dropperfuls daily for maintenance.

Echinacea: Echinacea enables macrophages to better engulf pathogens, increases the production of cytokines, enhances natural killer cell activity, and improves lymphocyte and monocyte cell counts and the antibody response.²¹ While *Echinacea purpurea*, *Echinacea angustifolia*, and *Echinacea pallida* are all used medicinally, *Echinacea purpurea* exhibits more consistent results in supporting optimal immune function.²² Like Astragalus, Echinacea's immune-stimulating properties mean that people with autoimmune disorders

should opt for immune-modulating herbs instead or use extreme caution when using this herb. It's also important to note that this medicinal plant may interact with immunosuppressive drugs, so individuals taking prednisone or other immunosuppressants should avoid Echinacea or use it only under the care of a knowledgeable provider. Suggested dose: 3 dropperfuls echinacea tincture, three times daily.

Elderberry: Elderberry shortens the duration of colds and the flu. In a randomized controlled trial involving influenza A and B virus infections, people treated with elderberry syrup experienced relief from symptoms four days earlier, on average, than people who were treated with a placebo.²³ Elderberry is a safe option for treating viral respiratory illness and does not over-stimulate the immune system or create a 'cytokine storm'.²⁴ In addition, "the use of elderberry supplements should be considered at an early course of [severe respiratory] disease".²⁵ I use it preventatively and curatively. Suggested dose: curative: 15 mL elderberry syrup, four times daily, or 3 dropperfuls of elderberry tincture, 3 times daily; preventatively: 1-2 dropperfuls of tincture daily.

Goldenseal (*Hydrastis canadensis*): Goldenseal inhibits growth of the H1N1 influenza A virus, and inhibits the production of inflammatory mediators.²⁶ Note that goldenseal should not be used by pregnant or breastfeeding women or by people with kidney disease. People with autoimmune disorders should opt for immune-modulating herbs instead or use extreme caution when using this herb. Suggested dose: 2 dropperfuls goldenseal tincture, twice daily.

Japanese Honeysuckle: Japanese honeysuckle has strong antiviral properties.²⁷ It alleviates viral entry and replication and suppresses pro-inflammatory cytokines, including IL-6 and TNF- α , which are part of the cytokine storm found in respiratory distress syndrome²⁸ from SARS-CoV-2. Suggested dose: 2 dropperfuls Japanese honeysuckle tincture, three times daily.

Lemon Balm (*Melissa officinalis*): Lemon balm shows promising results in the treatment of cold sores, genital herpes, and shingles, both internally and topically.^{29 30 31} It also exhibits antiviral activity against the common cold and influenza.³² Suggested dose: 2 dropperfuls lemon balm tincture, three times daily.

Licorice Root (*Glycyrrhiza glabra*): Licorice exhibits antiviral properties against several viruses, including influenza, hepatitis C, herpes simplex virus, enterovirus 71, coxsackievirus A16, and human immunodeficiency virus (HIV).³³ Licorice also shows promising inhibitory effects as an antiviral botanical against SARS-CoV-2.³⁴ Note that individuals with hypertension, liver cirrhosis, congestive heart failure, or renal failure should not use licorice. Suggested dose: 1 dropperful tincture, three times daily (note: not DGL or deglycyrrhizinated licorice).

Lion's Mane (*Hericium erinaceus*): Lion's mane supports optimal nervous and immune system function, reduces inflammation, is immunomodulatory, and

exhibits strong antibacterial and antiviral activity.³⁵ Suggested dose: 2 dropperfuls of a double-extracted lion's mane tincture, up to 3 times daily for illness, and 2 dropperfuls daily for maintenance.

Myrrh: Myrrh extract exhibits antiviral activity against influenza A virus by inhibiting viral replication.³⁶ This botanical also exhibits antiviral activity against herpes simplex virus type 1, the type that most commonly causes cold sores.³⁷ Suggested dose: 2 dropperfuls myrrh tincture twice daily.

Reishi (*Ganoderma lucidum*): In traditional Chinese medicine, this anti-inflammatory adaptogenic mushroom is called "The Mushroom of Immortality." Reishi is a well-researched antiviral, and has immune modulating³⁸ and immune-potentiating capabilities that target inflammatory cytokines³⁹. *Ganoderma lucidum* triterpenoids (soluble in alcohol, not water) demonstrate anti-HIV-1 protease activity^{40 41} and antiviral activity against pathogenic viruses such as herpes simplex virus types 1 (HSV-1 and HSV-2), influenza A⁴², Epstein-Barr Virus (EBV)⁴³, vesicular stomatitis virus (VSV), human immunodeficiency virus (HIV), and possibly Dengue virus (DENV)⁴⁴. Suggested dose: 2 dropperfuls of a dual-extracted reishi tincture, up to three times daily for illness, and 2 dropperfuls daily for maintenance. Note that reishi tea is not enough to access all of the medicinal compounds. A dual-extracted tincture is recommended.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): Turkey tail has been shown to fight the following viral infections: herpes, shingles, HPV, EBV, colds, and influenza. It also inhibits human immunodeficiency virus type 1 (HIV-1) reverse transcriptase and protease, two enzymes necessary for HIV to replicate⁴⁵. Suggested dose: 2 dropperfuls of a dual-extracted turkey tail tincture, up to three times daily for illness, and 2 dropperfuls daily for maintenance.

Usnea (*Usnea barbata*): Also known as Old Man's Beard, Usnea has powerful antibiotic, anti-fungal, antimicrobial, and antiviral properties.⁴⁶ I use it as a throat spray to prevent illness. Suggested dose is 2 dropperfuls of Usnea tincture, 3 times daily. As a throat spray, 5 sprays three times daily are best.

HERBAL REMEDIES TO COMBAT

SYMPTOMS: It is inevitable that at some point, a virus will begin to affect your body. You may notice upper respiratory symptoms such as a cough or congestion. In addition, it is common to have a fever, body aches, or fatigue when you are suffering from a viral infection. The remedies below can help combat these symptoms and help bring much needed relief:

Elecampane (*Inula helenium*) and/or Yerba Santa (*Eriodictyon californicum*) for mucus: Elecampane root is used to tackle stubborn coughs that are especially phlegmy. Yerba Santa leaves dry up mucus quickly, and are a good symptom-reliever for colds, allergies, and bronchial issues.⁴⁷ Suggested dose: 2 dropperfuls of tincture as needed, up to 3x/day for symptom relief. 123

Mullein (*Verbascum thapsus*) for Mucus/Inflammation/Airways: Mullein is famous as an expectorant to rid airways from excess mucus and for soothing mucous membranes.⁴⁸ It is perfect for help with a cough, cold, or bronchitis.⁴⁹ It helps to open the airways and relieve inflammation that may be hindering breathing.⁵⁰ Suggested dose is 2 dropperfuls mullein tincture or tea as needed, up to 3 times daily. For maximum effect, I combine this with Lungwort Lichen into a Bronchial Blend tincture, which has an anti-inflammatory effect on the respiratory system.⁵¹

Yarrow (*Achillea millefolium*) and Boneset (*Eupatorium perfoliatum*) for fever: Boneset has febrifuge properties, meaning it can help to bring down a high fever. Yarrow is a febrifuge plant that works by inducing sweating to bring the body temperature down. Yarrow also has antiviral and antibacterial properties.⁵²
^{53 54} Fevers should be left alone to do their job whenever possible. However, there are times when treating a fever is necessary to help bring relief. Suggested dose: 2 dropperfuls yarrow tincture as needed, up to three times daily.

4. Vitamins and Supplements

There are several vitamins and supplements that have antiviral properties and support optimal immune system function. These include vitamins C, D, and A; zinc; n-acetyl cysteine; and nicotinamide adenine dinucleotide (NAD).

Vitamin C: Vitamin C has been found to be effective in the prevention of the common cold.⁵⁵ In another study, cold and flu symptoms decreased by 85% in individuals who were taking megadoses of vitamin C.⁵⁶ In high doses, vitamin C can cause diarrhea and/or nausea. To limit these effects, take a buffered form of vitamin C, such as calcium ascorbate or magnesium ascorbate. Suggested dose: hourly doses of 1000 mg of Vitamin C for the first 6 hours and then 3 times daily thereafter.

Vitamin D: Vitamin D helps reduce the risk associated with respiratory tract infections by lowering the rate at which viruses are able to replicate, increasing the concentration of anti-inflammatory chemical messengers in the body, and reducing the concentration of pro-inflammatory chemical messengers, which can lead to lung damage and pneumonia. Research suggests that vitamin D supplementation may reduce the risk of influenza and COVID-19 infection and death.⁵⁷ Suggested dose: 10,000 IU per day of vitamin D3 for 2-4 weeks, then a maintenance dose of 5000 IU per day with a goal of raising 25-hydroxyvitamin D levels to 40-60 ng/mL (Get your vitamin D levels tested to know where you stand.).

Vitamin A: Vitamin A plays critical roles in both the innate immune response (your body's less specific immune response, which comes into play in the first few days after encountering a pathogen) and the adaptive immune response (your body's more specific response, which is tailored to the specific pathogen it's fighting).

Vitamin A also has an anti-inflammatory effect on

the respiratory tract. It prevents fibrosis of the lungs and promotes healing of lung tissue.⁵⁸ Suggested dose is 10,000 IU three times daily. Do not take if pregnant or if you may become pregnant, as high-dose vitamin A can lead to fetal malformation.

Zinc: Zinc lozenges in the form of zinc gluconate have been found to significantly reduce the duration of symptoms in the common cold⁵⁹ such as cough, headache, hoarseness, nasal drainage, nasal congestion, and sore throat. A similar study involving zinc acetate lozenges also led to a decrease in the duration and severity of cold symptoms.⁶⁰ Suggested dose: one 12-14 mg zinc lozenge every 2-3 waking hours while upper respiratory infection symptoms persist. Do not exceed 1-2 weeks of zinc supplementation without balancing your zinc intake with copper.

N-Acetyl Cysteine (NAC): N-acetyl cysteine is a powerful antioxidant that can help protect the lungs and other organs from damage in the event of a viral infection. NAC may help reduce symptom severity and improve outcomes in individuals with acute respiratory distress syndrome and acute lung injury.⁶¹ Suggested dose: 500 mg daily.

Nicotinamide Adenine Dinucleotide (NAD): Research demonstrates that the antiviral effects of specific enzymes in the body depend on the presence of nicotinamide adenine dinucleotide to function optimally, and that supplementing with NAD may lead to increased function of the innate immune system, specifically enhancing immunity to coronaviruses.⁶² Suggested dose: To enhance NAD levels, consider intravenous administration of NAD or take an NAD precursor such as nicotinamide riboside (300 mg per day) orally (e.g. the product Tru Niagen).

5. Water Therapies Water therapies, including salt water gargles, hot-to-cold showers, steam inhalation with or without aromatherapy, and Neti pots, can be beneficial in addressing viral infections.

Hot-to-Cold Showers: There are benefits associated with ending a hot shower with a 30 second to 1-minute-long cold shower. In a randomized controlled trial, routine hot-to-cold showers led to a 29% reduction in self-reported sickness-related absence days,⁶³ indicating that hot-to-cold showers may reduce illness severity in people who habitually take them. The viruses considered in this study were influenza virus A and B, respiratory syncytial virus, enterovirus, and rhinovirus, and that the study was conducted in the Netherlands during the 2014/2015 Influenza epidemic, which was the longest lasting influenza epidemic in the region for the preceding 40 years. The benefits associated with hot-to-cold showers have been attributed to the short-term effects of cold exposure, specifically immune system modulation.⁶⁴

Steam Inhalation with and without Aromatherapy: Two 20-minute sessions of steam inhalation alleviate cold symptoms and increase airflow into and out of the

nostrils.⁶⁵ Adding drops of essential oils to this water treatment may be beneficial, specifically if the oils exhibit antiviral properties. *Melaleuca alternifolia*, also known as tea tree, inhibits the replication of the influenza virus.⁶⁶ *Eucalyptus globulus* essential oil prevents viral infection by binding to free viruses and directly inactivating them.⁶⁷ ⁶⁸ *Mentha piperita*, peppermint, exhibits beneficial effects within the respiratory tract and beyond. Within the respiratory tract, the menthol in peppermint helps to make breathing easier by increasing nasal patency and it functions as a decongestant.⁶⁹ Peppermint oil also exhibits antiviral activity against herpes simplex viruses 1 and 2.⁷⁰

Salt Water Gargle and Nasal Irrigation Using the Neti Pot: Salt water is beneficial in addressing symptoms of respiratory viral illness, whether by gargling or by using it to irrigate the nasal passages. In this context, the salt water draws fluid out of inflamed tissues.

This reduces swelling and pain, making breathing easier. The salt water can also help break up mucus so that it is more easily expelled from the body. Mucus expulsion makes it possible for the body to get rid of pathogens and other irritants. In one study, salt water nasal irrigation and gargling led to shorter duration of cold symptoms.⁷¹

6. Western Medical Approaches

Treatment for rhinovirus, the virus that most frequently causes the common cold, is symptomatic and focuses on keeping the person comfortable.

Conventional treatments for influenza include oseltamivir, zanamivir, peramivir, and baloxavir. Acyclovir is an antiviral drug that is commonly used to address infection with herpes simplex virus.

Oseltamivir, Zanamivir, and Peramivir: Oseltamivir, zanamivir, and peramivir are approved to treat both influenza A and influenza B infections. They are neuraminidase inhibitors, meaning that they prevent the

virus from leaving the infected cell and infecting more cells.

The major difference between the three drugs is the route of administration and whether it is used for treatment only or for treatment and prevention of influenza. Oseltamivir (brand name Tamiflu®) can be administered orally or rectally and is typically given to people in the outpatient setting or to hospitalized patients for both prevention and treatment.

Zanamivir (brand name Relenza®) is administered by oral inhalation and is typically not given to people who have underlying respiratory diseases like asthma or chronic obstructive pulmonary disease (COPD).

It can be used for both prevention and treatment. Peramivir (brand name Rapivab®) is administered intravenously and is typically only used for treatment. It carries a risk of hypersensitivity reaction, so it may not be the best choice for individuals who have a history of reacting to medication.

Baloxavir: Baloxavir (brand name Xofluza®) is the fourth drug that is approved for treatment of influenza. It works differently from the first three drugs because it blocks viral replication in both influenza A and influenza B infection. It is administered orally and it can be used for both treatment and for prevention in people age 12 and older.

Acyclovir: Acyclovir is an antiviral drug that can be used to treat herpes infections, including oral and genital herpes, and to prevent oral herpes outbreaks in individuals who have previously been infected with the virus.

This drug is also used to address other viral disorders, including shingles. Acyclovir works by blocking viral DNA polymerase, an enzyme that the virus needs in order to replicate its genome.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Organic bone broth or probiotic-containing meal 2 dual-extracted medicinal mushroom*** tinctures of your choice (Cordyceps, Reishi &/or Turkey Tail) Nicotinamide riboside, NAC, & zinc	Weekly grocery shop for healthy foods Hydration check One symptom-based herbal remedy* (if needed) and 1-2 anti-viral herbs With food: Vit. A, C, & D	Medicinal mushroom tinctures (choose 2) Probiotics Outdoor walk	Optional third meal in the early evening (within 6-8 hour eating window if doing intermittent fasting) One symptom-based herbal remedy* (if needed) and 1-2 anti-viral herbs**	Hot-to-cold shower or steam inhalation with aromatherapy Medicinal mushroom*** tinctures (choose 2) Sleep Protocol for sleep help, if needed
MONDAY	Organic bone broth or probiotic-containing meal Medicinal mushrooms*** (2) Nicotinamide riboside, NAC, & zinc	Hydration check Physical activity outside One symptom-based herbal remedy and 1-2 anti-viral herbs With food: Vit. A, C, & D	Neti pot (use daily if needed for congestion) Medicinal mushroom tinctures (choose 2) Probiotics	Optional third meal in the early evening (within 6-8 hour eating window if doing intermittent fasting) One symptom-based herbal remedy* (if needed) and 1-2 anti-viral herbs**	Hot-to-cold shower or steam inhalation with aromatherapy Medicinal mushroom*** tinctures (choose 2) Sleep Protocol for sleep help, if needed
TUESDAY	Organic bone broth or probiotic-containing meal Medicinal mushrooms*** (2) Nicotinamide riboside, NAC, & zinc	Hydration check Physical activity outside One symptom-based herbal remedy and 1-2 anti-viral herbs With food: Vit. A, C, & D	Medicinal mushroom tinctures (choose 2) Probiotics	Optional third meal in the early evening (within 6-8 hour eating window if doing intermittent fasting) One symptom-based herbal remedy* (if needed) and 1-2 anti-viral herbs**	Hot-to-cold shower or steam inhalation with aromatherapy Medicinal mushroom*** tinctures (choose 2) Sleep Protocol for sleep help, if needed
WEDNESDAY	Organic bone broth or probiotic-containing meal Medicinal mushrooms*** (2) Nicotinamide riboside, NAC, & zinc	Hydration check Physical activity outside One symptom-based herbal remedy and 1-2 anti-viral herbs With food: Vit. A, C, & D	Medicinal mushroom tinctures (choose 2) Probiotics	Optional third meal in the early evening (within 6-8 hour eating window if doing intermittent fasting) One symptom-based herbal remedy* (if needed) and 1-2 anti-viral herbs**	Hot-to-cold shower or steam inhalation with aromatherapy Medicinal mushroom*** tinctures (choose 2) Sleep Protocol for sleep help, if needed
THURSDAY	Organic bone broth or probiotic-containing meal Medicinal mushrooms*** (2) Nicotinamide riboside, NAC, & zinc	Hydration check Physical activity outside One symptom-based herbal remedy and 1-2 anti-viral herbs With food: Vit. A, C, & D	Neti pot Medicinal mushroom tinctures (choose 2) Probiotics	Optional third meal in the early evening (within 6-8 hour eating window if doing intermittent fasting) One symptom-based herbal remedy* (if needed) and 1-2 anti-viral herbs**	Hot-to-cold shower or steam inhalation with aromatherapy Medicinal mushroom*** tinctures (choose 2) Sleep Protocol for sleep help, if needed
FRIDAY	Organic bone broth or probiotic-containing meal Medicinal mushrooms*** (2) Nicotinamide riboside, NAC, & zinc	Hydration check Physical activity outside One symptom-based herbal remedy and 1-2 anti-viral herbs With food: Vit. A, C, & D	Medicinal mushroom tinctures (choose 2) Probiotics	Optional third meal in the early evening (within 6-8 hour eating window if doing intermittent fasting) One symptom-based herbal remedy* (if needed) and 1-2 anti-viral herbs**	Hot-to-cold shower or steam inhalation with aromatherapy Medicinal mushroom*** tinctures (choose 2) Sleep Protocol for sleep help, if needed
SATURDAY	Organic bone broth or probiotic-containing meal Medicinal mushrooms*** (2) Nicotinamide riboside, NAC, & zinc	Hydration check Physical activity outside One symptom-based herbal remedy and 1-2 anti-viral herbs With food: Vit. A, C, & D	Medicinal mushroom tinctures (choose 2) Probiotics	Optional third meal in the early evening (within 6-8 hour eating window if doing intermittent fasting) One symptom-based herbal remedy* (if needed) and 1-2 anti-viral herbs**	Hot-to-cold shower or steam inhalation with aromatherapy Medicinal mushroom*** tinctures (choose 2) Sleep Protocol for sleep help, if needed

*Symptom-based herbal remedies: Elecampane (mucus), Yerba Santa (mucus), Mullein (mucus/inflammation/airways), Yarrow (fever), Boneset (fever).

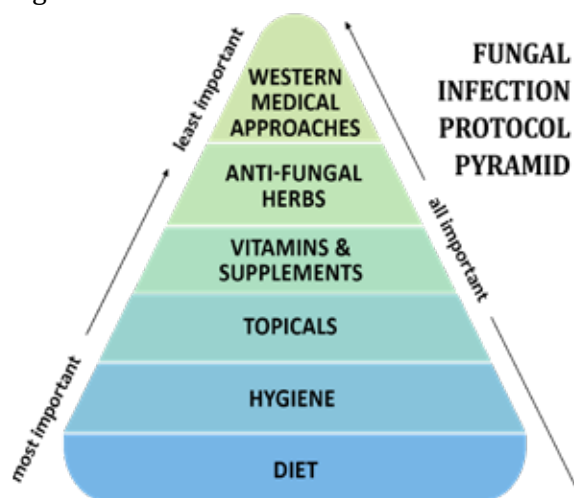
**Anti-viral herbs: Astragalus, Chinese skullcap, Echinacea, Elderberry, Goldenseal, Japanese Honeysuckle, Lemon Balm, Licorice, Myrrh, Usnea.

***Medicinal mushrooms: Cordyceps, Lion's Mane, Reishi, and Turkey Tail.

Ch. 19: Fungal Infection Protocol

Of the millions of species of fungi in existence on this planet, there are several that are known to infect humans. Some of these fungi are more likely to infect us at varying stages of our lives, such as oral thrush in babies or vaginal yeast infections during pregnancy. Given the fact that some of these fungi are highly contagious and are likely to infect you at some point in your life, it's important for you to know the symptoms that typically accompany these fungal infections as well as how to effectively treat them. We'll consider symptoms of the most common fungal infections in this introductory section.

Seborrheic Dermatitis: Seborrheic dermatitis is a fungal infection of the top layer of the skin in areas that contain a lot of sebaceous glands. This includes the face, scalp, chest, and back. Researchers believe the condition is due to infection by *Malassezia* yeast.¹ Symptoms include greasy and flaking skin, mostly on the scalp; redness; and peeling.



Oral Thrush: Oral thrush is a fungal infection of the mouth that is caused by *Candida albicans*. When a person has oral thrush, you can usually see white, slightly raised patches in the mouth or on the tongue. This condition is common in infants, but can also occur in immunocompromised individuals. People who use steroid sprays for asthma are more likely to be diagnosed with oral thrush. Oral thrush can be contagious in people who are at risk for the condition. Note there are other conditions that can lead to white patches in the mouth, but the white patches are not usually easily wiped off like the ones that occur in oral thrush. If you're unsure if you're dealing with oral thrush or another condition, see a healthcare provider. Please also see our Candida Protocol on page 46.

Tinea Versicolor: Tinea versicolor is a common fungal infection of the skin. It typically leads to small, lighter, or otherwise discolored patches of skin. It is not usually contagious and is due to an overgrowth of the yeast *Malassezia* on the skin. This yeast typically lives on our skin but can get out of control in response to changing internal or external conditions, such as

hot or humid weather; increased sweating, oily skin, immunocompromised state, or hormonal changes. Tinea versicolor mostly affects teenagers and young adults.

Tinea Corporis/Ringworm: Ringworm is the most well-known tinea infection. Symptoms include a ring-shaped rash with raised edges, with some clearing in the middle of the circle; redness; itching; hair loss in the affected area; and enlarged lymph nodes in the affected region. There are some conditions that look like tinea corporis, such as nummular eczema and pityriasis rosea, but these conditions do not have clearing at the center and pityriasis rosea is typically not itchy. Tinea corporis is highly contagious, while nummular eczema and pityriasis rosea are not.

Tinea Cruris/Jock Itch: Tinea cruris causes redness and itching in the areas of the body that are usually warm and moist. This includes the genital region and inner thighs. Tinea cruris is most common in athletes, and especially those who wear tight or non-breathable clothing. It is highly contagious.

Tinea Pedis/Athlete's Foot: Athlete's foot is a fungal infection that affects the skin of the feet. It usually begins between the toes and can look like a peeling or scaly rash. Itching and burning are common. Tinea pedis typically occurs in people who sweat on their feet a lot, especially if their shoes are tight and non-breathable. Tinea pedis is highly contagious.

Tinea Barbae: Tinea barbae is one of a group of conditions that is caused by dermatophytes that infect the top layer of the skin, hair, and hair follicles.² It is a rarer form of tinea infection and is mostly seen in the area surrounding men's beards. Symptoms can include pimples or blistering in the affected area, redness, swelling, inflammation, and crusting and/or hair loss around the hairs of the affected area.

Vaginal Yeast Infections: Vaginal yeast infections are caused by the same fungi that cause oral thrush, *Candida albicans*. Please also see our Candida Protocol.

Onychomycosis/Nail Fungus: Onychomycosis is a fungal infection of the nails that affects ~10% of the population; risk increases with age.³ It is most common on the feet and people with poor circulation are more likely to develop onychomycosis. Onychomycosis typically causes yellowing and thickening of the nails, leading to separation of the nail from the nail bed. It is not usually itchy or painful; any concerns are usually cosmetic.

1. Diet In order to reduce the likelihood of developing a fungal infection and to promote an internal environment that is conducive to healing fungal infections, there are a few dietary changes that are necessary. Please also see our Candida Protocol for the Candida Diet if *Candida* is causing your issue.

Avoid Sugar: Diets that contain a lot of added sugar negatively affect our immune function. Specifically, diets high in sugar impair our white blood cells' abilities to engulf pathogens, lead to higher inflammation and immune cell death, and cause other unwanted

effects.⁴ Consuming just 75 grams of sugar (the amount of sugar in less than 2 cans of soda) can lead to immune system function being suppressed for over five hours.⁵ These changes to the immune system can lead to our cells being less able to ward off pathogens, such as invading fungi, and can make healing from these infections much more difficult.

These changes can occur on a high sugar diet even if your blood sugar level remains within the normal range. If your condition progresses to diabetes, however, the negative effects on the immune system are compounded. Uncontrolled blood sugar levels lead to even more immune system dysregulation, including greater susceptibility to infection, even more inflammation, and wounds taking longer to heal.⁶

Avoid Artificial Sweeteners: Artificial sweeteners like aspartame, saccharin, and sucralose have negative effects on immune system function. These artificial sweeteners suppress the immune cells' ability to secrete interleukin-6, a chemical messenger that the body typically produces in response to infection as a part of its defenses.⁷ Sucralose also suppresses the production of another chemical messenger, interleukin-10.

Interleukin-10 plays a role in limiting the immune response and preventing the body from being damaged by a dysregulated immune system; it helps to keep the immune response in balance. In essence, these artificial sweeteners can throw the body's normal immune function out of balance, making it harder for the body to appropriately defend itself against pathogens.

Avoid Refined Flour: Like sugar, refined flour products such as white breads, pastries, and many breaded products have high glycemic indices. This means that they cause spikes in blood sugar shortly after eating them. In addition to disrupting immune cell function on a cellular level, these spikes in blood sugar can promote inflammation, which can lead to impairment of immune function and a worsening of fungal infections.

Additionally, research suggests that when you remove the bran and germ from whole grains in order to produce refined flours, it leads to negative effects upon the immune system.⁸ Specifically, consuming refined flours is associated with lower concentrations of bacteria that produce anti-inflammatory short-chain fatty acids and higher concentrations of pro-inflammatory bacteria. Refined flour consumption is also associated with negative changes to the body's innate immune response, which is the part of your immune system that springs into action when you first encounter a pathogen like a fungus.

Yeast: There aren't many reliable clinical studies that look at the effect of dietary yeast, such as is found in breads, cheeses, beer, and other foods made by yeast and fungi; however, people who have a difficult time controlling seborrheic dermatitis, which can be caused by a fungal infection, frequently report improvement after eliminating yeast and yeast products from their diets.⁹

Xylitol: Xylitol is a natural sugar alcohol found in many fruits and vegetables. It is frequently used to sweeten chewing gum, toothpaste, and mouthwashes. Xylitol has potent anti-fungal effects against *Candida albicans* and oral thrush.¹⁰ Replacing natural and artificial dietary sugars with xylitol could be beneficial in preventing or addressing oral thrush in people who are at risk.¹¹

2. Hygiene

Hygiene plays an important role in limiting the spread of fungal infections.

Keep the Affected Area Clean, Dry, and Cool: Fungi thrive in warm, damp environments. Uncleanliness, moisture, and heat promote the spread of fungal infections. While keeping the infected area dry is important, avoid vigorously rubbing affected areas with towels, as this can damage the integrity of the skin and promote spread of the infection.

Avoid Sharing Personal Items: Many fungal infections are highly contagious, and, under the right conditions, sharing personal items like towels, sheets, toothbrushes, sporting equipment, shoes, or clothing can promote the spreading of infections. Sharing sheets or towels can lead to the spread of ringworm or jock itch; sharing toothbrushes can lead to the spread of oral thrush, particularly in immunocompromised individuals; and sharing shoes or sporting equipment can lead to the spread of athlete's foot.

Wear Slippers at Home and Shoes in Public: To avoid spreading fungal infections like athlete's foot to others in your home, wear slippers indoors if you have athlete's foot (tinea pedis). The dermatophytes that cause athlete's foot can live for extended periods in spore form on carpet, in shoes, or on other surfaces. Similarly, you can contract athlete's foot if you walk barefoot in public places, such as in public showers, at public swimming pools, or on locker room floors. Wear shoes in public places.

Wear Breathable Fabrics: In order to recreate an environment where your body can heal from fungal skin infections, it's important to wear breathable fabrics. If your clothing is not breathable, it is more likely to trap heat and retain moisture. Heat and moisture can promote the spread of fungal infections. Fabrics made with cotton, nylon, polyester, rayon, linen, and silk are preferable while healing from fungal skin infections.

Wash Your Hands after Playing with Pets: Wash your hands with soap and water after playing with pets. Some pets (for example, dogs) can contract fungal infections like ringworm and spread them to humans. If you think your pet may have given you a fungal infection, take your pet to a veterinarian for treatment.

Breastfeeding: If you have an infection of *Candida albicans* on your breasts, you can transfer the yeast to your baby during breastfeeding and cause oral thrush. Signs of *Candida* infecting your breasts include excessive soreness, redness, and severe pain in the breast area. To reduce the likelihood of you and your baby passing the infection back and forth, wash your nipples and allow them to dry thoroughly after each breastfeeding session.

Kissing: It is possible to pass *Candida* to a partner through kissing; however, your partner likely already has *Candida* in his or her mouth. Kissing typically only leads to oral thrush in individuals who are immunocompromised or have other risk factors for *Candida albicans* infection, so if neither you nor your partner have immune-related dysfunction, this is likely not a concern for you.

3. Topicals There are multiple compounds that can be applied topically in order to address fungal infections. These include apple cider vinegar, coconut oil, aloe vera gel, tea tree oil, lavender essential oil, honey, *Eucalyptus pauciflora*, *Usnea barbata*, caprylic acid, undecylenic acid, and bitter orange essential oil.

Apple Cider Vinegar: It displays significant anti-fungal activity against *Candida albicans*.¹² *Candida albicans* causes both oral thrush and denture stomatitis (redness, swelling, and soreness of the gums, usually due to fungal infection in people who wear dentures). Breastfeeding mothers may apply apple cider vinegar topically to the nipples and areolas to help kill *Candida* and reduce the likelihood of it causing oral thrush in their babies.

In cases of denture stomatitis, soak dentures in apple cider vinegar for 30 minutes or more to kill *Candida* and reduce the likelihood of oral thrush spreading and fueling *Candida* overgrowth in the mouth.¹³ Suggested dose: Apply 4% apple cider vinegar topically three times per day or as needed (minimum concentration 2500-10,000 µg/ml); soak dentures in apple cider vinegar for 30 minutes or more.

Coconut Oil: Coconut oil also displays anti-fungal activity against *Candida albicans*. One study found coconut oil to have more potent anti-fungal activity against *Candida albicans* than the anti-fungal drug fluconazole.¹⁴ The researchers concluded that, in view of the fact that many species of *Candida* are becoming increasingly resistant to the currently available anti-fungal drugs, coconut oil should be used to treat *Candida albicans* infections.

Furthermore, an extract from the coconut shell was found to exhibit anti-fungal activity against multiple fungi that cause conditions like ringworm, jock-itch, and athlete's foot in humans, including *Aspergillus niger*, *Microsporum canis*, *Microsporum gypseum*, *Aspergillus flavus*, *Trichophyton rubrum*, *Aspergillus fumigatus* and *Trichophyton vercosum*.¹⁵ Suggested dose: Apply virgin coconut oil topically three times per day or as needed (minimum concentration 1:4).

Aloe Vera Gel: Aloe vera exhibits significant anti-fungal activity against *Aspergillus niger*, a fungus known to infect the outer ear. Alcohol-based extracts of Aloe vera are more potent than water-based extracts, although both are effective.¹⁶ Some studies show favorable effects of *Aloe vera* against *Candida albicans*, while others did not find it to be effective.^{17 18}

Studies that did note anti-fungal activity of *Aloe vera* against *Candida albicans* used higher concentrations and specifically used alcohol-based extracts. Suggested dose: Apply Aloe vera ethanolic extract topically three

times per day or as needed (minimum concentration 6.25 g/mL).

Tea Tree Oil: Tea tree (*Melaleuca alternifolia*) oil exhibits potent anti-fungal activity against 26 strains of dermatophyte species (fungi known to cause skin infections in humans), 54 yeasts (including 32 strains of *Candida albicans* and other *Candida* species), and 22 different *Malassezia* species (including *Malassezia furfur*, a species that causes *Tinea versicolor*).¹⁹ Suggested dose: Apply tea tree oil diluted in coconut oil topically three times per day (minimum concentration 4450 mcg/mL).

Lavender Essential Oil: Lavender (*Lavandula angustifolia*) essential oil has anti-fungal activity and may be an effective treatment against *Candida albicans* infections.²⁰ This study specifically used *Candida albicans* isolated from the vaginal canals of women with vaginal yeast infections. The aim was to compare the anti-fungal effects of lavender to those of clotrimazole in the treatment of yeast infections; it was conducted due to the unwanted side effects of the anti-fungal clotrimazole.

Lavender and tea tree oil, when used in combination, exhibited even more potent anti-fungal activity than when each was used alone.²¹ In this study, the oils were used against *Trichophyton rubrum* and *Trichophyton mentagrophytes* var. *interdigitale*, fungi that cause tinea (fungal infection of the skin) and onychomycosis (fungal infection of the nail) and were found to be highly effective, both alone and even more so in combination. Suggested dose: Apply lavender essential oil topically three times per day.

Raw Honey: Various types of honey are effective at treating *Tinea* infections (fungal skin infections), seborrheic dermatitis, and diaper dermatitis of fungal origin.²² Some types of honey exhibit potent anti-fungal activity against *Candida* species, while others exhibit minimal to no anti-fungal activity.^{23 24} The difference in efficacy and potency depends on the botanical origin of the honey.

A study that evaluated the anti-fungal activity of various types of honeys against *Trichophyton mentagrophytes*, which causes tinea barbae in humans and *Trichophyton rubrum*, which causes athlete's foot, as well as *Candida albicans* found that mono-floral Agastache honey had the most notable anti-fungal activity compared to other commercially available brands.²⁵ Suggested dose: Apply raw honey topically three times per day or as needed.

Eucalyptus Pauciflora: The essential oil of *Eucalyptus pauciflora* exhibits potent anti-fungal activity. It has strong anti-fungal activity against *Epidermophyton floccosum*, *Microsporum audouinii*, *Microsporum canis*, *Microsporum gypseum*, *Microsporum nanum*, *Trichophyton mentagrophytes*, *Trichophyton rubrum*, *Trichophyton tonsurans*, and *Trichophyton violaceum* at concentrations that did not lead to any adverse effects when applied to the skin.²⁶ This particular study involved people with tinea corporis (ringworm), tinea pedis (athlete's foot), or tinea cruris (jock itch). *Eucalyptus*

pauciflora essential oil was applied with 60% of patients recovering completely and 40% showing significant improvement. For those who recovered completely, there was no evidence of relapse when they were re-checked 2 months after the end of treatment. Suggested dose: Apply *Eucalyptus pauciflora* oil topically three times per day or as needed (minimum concentration 1 microliter/mL).

Usnea barbata: See Anti-Fungal Herbs.

Caprylic Acid: See Vitamins and Supplements.

Undecylenic Acid: See Vitamins and Supplements.

Bitter Orange: See Vitamins and Supplements.

4. Vitamins and Supplements

There are various supplements that exert anti-fungal activity. These include caprylic acid, undecylenic acid, bitter orange, and probiotics.

Caprylic Acid: Caprylic acid extracted from coconut oil exerts potent anti-fungal activity against *Candida albicans*.²⁷ and can be used topically and/or internally. Caprylic acid damages the cell membranes of *Candida albicans* cells and fungal efflux pumps so that the fungi are less able to defend themselves against anti-fungal treatments.²⁸

When caprylic acid is combined with carvacrol and thymol, greater than 83% of membranes are damaged and greater than 95% of efflux pumps are damaged. Suggested dose is 306-612 mg of magnesium caprylate per day, following meals. Note that caprylic acid is not intended for long-term use when taken internally.

Undecylenic Acid: Undecylenic acid is an unsaturated fatty acid derived from the castor plant. It has potent anti-fungal activity.²⁹ Although it is most commonly applied topically, it is also used internally in its zinc undecylenate form. In a double-blind, placebo-controlled clinical trial, a formulation including 2% undecylenic acid and 20% zinc undecylenate was applied topically twice per day to affected areas in patients who had *Trichophyton rubrum*- or *Trichophyton mentagrophytes*-induced tinea pedis (athlete's foot).³⁰

At the end of the four-week trial, 80% of those treated with undecylenic acid were free from the fungus. Redness and scaling had also improved significantly, and no side effects were reported. Suggested dose: Topically, apply undecylenic acid-zinc undecylenate to the affected area twice daily for four weeks; for internal use, 120-240 mg of zinc undecylenate per day. Note that undecylenic acid is not intended for long-term use when taken internally.

Bitter Orange: Bitter orange is an anti-fungal supplement that can be used either topically as an essential oil or internally in capsule or tincture form. Bitter orange essential oil exhibits potent anti-fungal activity when applied topically for tinea corporis (ringworm), tinea cruris (jock itch), and tinea pedis (athlete's foot).³¹ Participants taking a diluted 25% bitter orange essential oil did not report any side effects and 100% of participants in this group finished the study (note those taking

a non-diluted form had some skin irritation so best to dilute to 25%).

In this treatment group, 80% of participants were cured after 1-2 weeks of applying the diluted bitter orange oil to the affected areas and the remaining 20% were cured in 2-3 weeks. Suggested dose: Topically, apply bitter orange oil topically (diluted in coconut oil at a concentration of 25%) one to two times daily for 3 weeks or as needed; or 500 mg of bitter orange per day, with meals; or 2 ml of a 1:3 bitter orange tincture 2 times daily.

Probiotics: Probiotics such as *Saccharomyces cerevisiae* can prevent filamentation (a process that usually leads to a pathogenic fungus becoming more harmful to the human) in disease-causing fungi, including *Candida tropicalis* and *Candida parapsilosis*.³² *Lactobacillus acidophilus* and *Lactobacillus rhamnosus* inhibit filamentation in *Candida albicans*.^{33 34}

Probiotics prevent pathogenic fungi from adhering to epithelial cells when invading the body.³⁵ They also inhibit the formation of fungal biofilms, which usually protect fungi and prevents anti-fungal drugs from reaching them, making it more difficult to eradicate fungal infections.³⁶

In the early stages of fungal infection, probiotics such as *Lactobacillus rhamnosus*, *Lactobacillus acidophilus*, and *Lactobacillus casei* inhibit the formation of fungal biofilms, although they were less successful in later stages of biofilm formation (such as after the biofilms reached maturation, approximately 48 hours after infection).^{37 38} Suggested dose: 50-100 billion colony forming units of probiotics per day, in divided doses, with meals.

5. Anti-Fungal Herbs

There are several herbs that exhibit anti-fungal activity and that can be used to address fungal infections. These include Pau d'arco, garlic, usnea, black walnut, and turmeric.

Pau d'Arco: Pau d'arco is a tree native to the Amazonian rain forest. Its inner bark exhibits potent anti-fungal activity.⁴⁰ The medicinal qualities of pau d'arco are likely due to its naphthoquinones, specifically lapachol and beta-lapachone. The inner bark of the tree is usually taken as a decoction.⁴¹ Suggested dose: as a decoction, 1 tablespoon of pau d'arco per cup of water, 3 times daily; or 2 dropperfuls of a 1:5 tincture, 3 times daily.

Garlic: Garlic extract demonstrates anti-fungal activity against *Meyerozyma guilliermondii* and *Rhodotorula mucilaginosa*, both of which can cause onychomycosis (fungal infection of the toenail).⁴² Its anti-fungal properties are attributed to garlic's main active constituent, allicin. In this experiment, garlic led to changes to the cell structure of these fungi and, ultimately, cell death. Suggested dose: Apply garlic extract topically once daily or as tolerated (minimum concentration 120 mg/mL); 5 mL of a 1:3 tincture, 3 times daily.

Usnea barbata: The usnic acid in *Usnea barbata* demonstrates anti-fungal activity against multiple types of fungi, including *Aspergillus flavus*, *Aspergillus fumigatus*, *Candida albicans*, *Penicillium purpurescens*, and

Penicillium verrucosum.⁴³ Acetone and methanol extracts of *Usnea barbata* also inhibited the growth of various fungi, including *Alternaria alternaria*, *Aspergillus niger*, *Mucor hiemalis*, *Penicillium notatum*, and *Schizophyllum commune*.⁴⁴ Suggested dose: Apply *Usnea barbata* extract topically three times per day or as needed (minimum concentration 12.5 mg/mL – an *Usnea* spray works well); or 2 dropperfuls of a 1:5 tincture, 3 times daily.

Black Walnut (*Juglans nigra*): Black walnut hulls contain a flavanoid called isoquercetin that has potent anti-fungal properties.⁴⁵ Specifically, isoquercetin damages the membrane of *Candida albicans* cells, increasing the cells' permeability and eventually leading to cell death.⁴⁶ Suggested dose: 2 dropperfuls of a 1:3 tincture, twice daily.

Turmeric: Turmeric also shows anti-fungal activity against *Candida albicans*. Turmeric contains active constituents that inhibit fungal growth at lower concentrations and directly lead to fungal cell death at higher concentrations.⁴⁷ Curcumin and turmeric oil also had an anti-fungal effect against *Fusarium solani*, a fungus that causes fungal illness, particularly in immunocompromised people.⁴⁸ Suggested dose: 2 dropperfuls of a 1:4 tincture, twice daily.

6. Western Medical Approaches

Conventional treatment for fungal infections typically includes the anti-fungal drugs miconazole, clotrimazole, and terbinafine.

Miconazole: Miconazole (brand names: Micatin®, Zeasorb®, Triple Paste AF®) is an anti-fungal drug. It is typically used to treat fungal or yeast infections of the skin or the vaginal tract, and it works by preventing

yeast and fungi from producing ergosterol, which is a necessary component of their cell membranes.

Without this, the fungal cells become more permeable and cellular components begin to leak out of the cell.⁴⁹ Miconazole is not usually well-absorbed and it is rapidly excreted from the body, so it is mostly used as a topical treatment for superficial and localized infections. Miconazole is usually fairly safe, but in some cases, it can cause redness or itching.

Clotrimazole: Clotrimazole (brand name: Lotrimin®) works similarly to miconazole, because it also blocks ergosterol production and makes fungal cells more permeable.⁵⁰ Clotrimazole is better absorbed than miconazole so it can be used topically (including intravaginally) or orally.

Side effects can include redness, itching, irritation, nausea, and vomiting. If taking clotrimazole orally, side effects can also include liver function abnormalities or liver damage, so it's important to have liver function monitored while taking oral clotrimazole.

Terbinafine: Terbinafine [brand names: Antifungal (terbinafine)®, Jock Itch (terbinafine)®, Athlete's Foot (terbinafine)®] also inhibits ergosterol production, specifically by inhibiting the enzyme squalene epoxidase.⁵¹ It is usually taken orally, and side effects can include headaches, gastrointestinal symptoms, and rashes.

Other less common symptoms include visual disturbances, altered taste, and changes to liver function. Rarely, terbinafine can also lead to liver failure and autoimmune disorders.^{52 53}

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Apply your choice from Topical List A* Usnea &/or Black Walnut tincture Probiotics	Weekly grocery shop for foods compliant w/ the <i>Candida</i> / Anti-fungal Diet, whichever using) Turmeric & Garlic extract w/ a meal Caprylic acid**** post-meal Apply your choice from Topical List B** or C***	Hydration check (Are you drinking enough water?) Usnea &/or Black Walnut tincture Apply choice from Topical List A*	Probiotics Turmeric & Garlic extract w/ a meal After-dinner walk	Pau d'arco unsweetened decoction or tincture Apply your choice from Topical List B** or C***
MONDAY	Apply your choice from Topical List A* Usnea &/or Black Walnut tincture Probiotics	Turmeric & Garlic extract w/ a meal Caprylic acid**** post-meal Apply your choice from Topical List B** or C***	Hydration check Usnea &/or Black Walnut tincture Physical activity (30-45 minutes) Apply choice from Topical List A*	Probiotics Turmeric & Garlic extract w/ a meal	Pau d'arco Apply your choice from Topical List B** or C***
TUESDAY	Apply your choice from Topical List A* Usnea &/or Black Walnut tincture Probiotics	Turmeric & Garlic extract w/ a meal Caprylic acid**** post-meal Apply your choice from Topical List B** or C***	Hydration check Usnea &/or Black Walnut tincture Apply choice from Topical List A*	Probiotics Turmeric & Garlic extract w/ a meal After-dinner walk	Pau d'arco Apply your choice from Topical List B** or C***
WEDNESDAY	Apply your choice from Topical List A* Usnea &/or Black Walnut tincture Probiotics	Turmeric & Garlic extract w/ a meal Caprylic acid**** post-meal Apply your choice from Topical List B** or C***	Hydration check Usnea &/or Black Walnut tincture Physical activity (30-45 minutes) Apply choice from Topical List A*	Probiotics Turmeric & Garlic extract w/ a meal	Pau d'arco Apply your choice from Topical List B** or C***
THURSDAY	Apply your choice from Topical List A* Usnea &/or Black Walnut tincture Probiotics	Turmeric & Garlic extract w/ a meal Caprylic acid**** post-meal Apply your choice from Topical List B** or C***	Hydration check Usnea &/or Black Walnut tincture Apply choice from Topical List A*	Probiotics Turmeric & Garlic extract w/ a meal After-dinner walk	Pau d'arco Apply your choice from Topical List B** or C***
FRIDAY	Apply your choice from Topical List A* Usnea &/or Black Walnut tincture Probiotics	Turmeric & Garlic extract w/ a meal Caprylic acid**** post-meal Apply your choice from Topical List B** or C***	Hydration check Usnea &/or Black Walnut tincture Physical activity (30-45 minutes) Apply choice from Topical List A*	Probiotics Turmeric & Garlic extract w/ a meal	Pau d'arco Apply your choice from Topical List B** or C***
SATURDAY	Apply your choice from Topical List A* Usnea &/or Black Walnut tincture Probiotics	Turmeric & Garlic extract w/ a meal Caprylic acid**** post-meal Apply your choice from Topical List B** or C***	Hydration check Usnea &/or Black Walnut tincture Apply choice from Topical List A*	Probiotics Turmeric & Garlic extract w/ a meal After-dinner walk	Pau d'arco Apply your choice from Topical List B** or C***

*Topical List A: Tea tree oil, lavender oil, *Eucalyptus* essential oil, bitter orange essential oil + topical Usnea spray.

**Topical List B: Undecylenic acid, caprylic acid, raw honey.

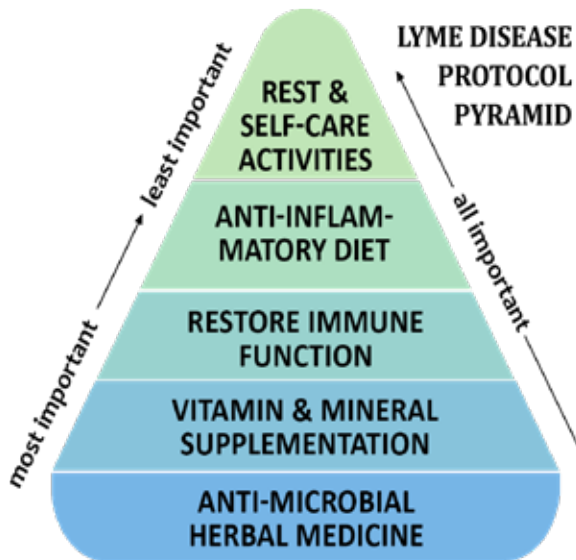
***Topical List C: Apple cider vinegar, coconut oil, aloe vera gel.

****Caprylic acid is not for long-term use.

Ch. 20: Lyme Disease Protocol

Lyme disease is a dangerous and rapidly growing health problem. Since being identified in the 1970s, its incidence has increased at an alarming rate. According to the CDC, Lyme is the number one vector-borne disease in the United States.¹ While its primary geographic distribution was initially localized to the Northeastern United States, it's now present in all states, Canada, Europe, and Asia. The number of reported cases has quadrupled since 1992.² Despite its escalating infection rate, Lyme disease is often overlooked and misdiagnosed.

Lyme disease is caused by the bite of a tick infected with *Borrelia burgdorferi*, a spiral-shaped bacterium (spirochete). Spirochetes can burrow into or between cells and hide as a cyst, making Lyme difficult to treat.



The tick species carrying this bacterium, *Ixodes scapularis*, primarily live near or in wooded areas. An infection typically occurs after a person has been hiking or camping.³ Following a bite from an infected tick, the bacterium enters the bloodstream, where it can infiltrate other parts of the body. While some mechanisms of this bacterium are still unclear, once it enters the body it can cause significant health issues.

STAGES: After a person has contracted Lyme disease, they can progress through three stages.

Stage 1 - Acute or Early Localized Lyme: This stage typically occurs a few days to weeks after being bitten by an infected tick. The infection is localized (it hasn't spread to other parts of the body). A person may experience various symptoms that resemble a mild viral illness, such as a stiff neck, fever, weakness, fatigue, aches, and pains.⁴ The characteristic feature of this stage is the "bulls-eye rash", known medically as erythema chronicum migrans. It is estimated that this rash occurs in 70-80% of people infected.⁵ However, this statistic is widely contested and may be exaggerated. The rash usually forms a bulls-eye pattern⁶ with an expanding red area that is sometimes clears in the center but may be solid. The rash isn't painful or itchy and often goes unnoticed. Typically, patients

have the best outcomes when the infection is diagnosed and treated with antibiotics in this stage.

Stage 2 - Early Disseminated Lyme: This stage typically occurs a few weeks to months following the initial infection. Since Lyme is often unnoticed or untreated in the first stage, the infection can spread. This stage is characterized by more severe symptoms affecting the skin, nervous system, and heart. Multiple rashes may develop. Heart block is another potential manifestation of this stage; it is caused by myocarditis and occurs when the electrical signal that controls the heartbeat is partially or entirely blocked.⁷ Neurological symptoms such as numbness, tingling, shooting pain, or weakness in the arms or legs may occur. Facial nerve palsy is a hallmark of this stage, where the muscles of the face droop due to paralysis of the facial nerves.⁸

Stage 3 - Chronic Lyme, Late Persistent Lyme, Post Treatment Lyme Disease: This is the most severe stage and can occur months to years following the initial infection. Damage to the joints, nerves, and brain may be present. Symptoms can emerge in the presence or absence of having received antibiotic treatment. Two signs of this stage include arthritis of large joints, like the knees and elbows, and arthritis of other joints that moves from joint to joint, known as *migratory arthritis*.⁹ Various neurological manifestations can occur at this stage. Depending on the part of the brain affected, symptoms can present as memory difficulty, inability to process information, cognitive slowing known as "brain fog," migraines, vertigo, and insomnia.¹⁰ Symptoms of this stage may be so debilitating that a person cannot live an ordinary life.

CONVENTIONAL TREATMENT: If a person has been bitten by a tick in a high-risk area but is unsure whether they have contracted the disease, the CDC recommends a single prophylactic dose of 200-mg of doxycycline.¹¹ Note that a 2020 study from Johns Hopkins University found that certain herbal medicines may have more potent activity against *Borrelia burgdorferi* than some antibiotics.¹² See our herbal section below.

Once a person has been diagnosed with Lyme disease, treatment should be started as soon as possible. The current recommended conventional treatment is an initial course of doxycycline antibiotics for 2-4 weeks. A common sentiment among the medical community is that treatment only requires one dose of antibiotics. However, 10-20% of patients continue to suffer from persisting symptoms after the initial antibiotic treatment.¹³

Early disseminated Lyme disease and Chronic Lyme disease are treated with oral or intravenous antibiotics (typically Ceftriaxone) for two or more weeks. Oral antibiotics are used for Lyme arthritis, while intravenous antibiotics are used for neurological issues. Under certain circumstances, antibiotics may be prescribed for several months and often a combination of antibiotics are given.¹⁴ Some physicians believe the cyst form is impervious to antibiotics, though some have had success with metronidazole (Flagyl) and tinidazole

plus grapefruit seed extract (one 250 mg pill 2 times a day¹⁵). Research around grapefruit seed extract is mixed¹⁶.

Often people will improve slowly, but it can take months and even years to feel well again.¹⁷ Unfortunately, even with aggressive antibiotic treatment, some people don't recover. Chronic Lyme disease is controversial among medical professionals due to the vagueness of symptoms and lack of established diagnostic criteria.¹⁸

MICROBIOLOGY: Lyme bacteria, *Borrelia burgdorferi*, are stealthy, and can hide in the body. It can survive inside, between, and outside of cells. While inside a cell, this bacterium can conceal itself from your immune system by manufacturing a barrier called biofilm. This ability to evade the host's immune system is an integral part of its survival¹⁹.

Once it has depleted the cell's available resources, it can resurface into the bloodstream and find other cells to infect.²⁰ The bacterium is thought to have an affinity for the interstitial system (the tiny, narrow spaces between tissues or parts of an organ).²¹ These areas are less accessible to the immune system. The bacterium's ability to hide in the body can make testing, diagnosis, and treatment difficult.

PREVENTION: The following steps can prevent exposure to Lyme disease:

Avoid Tick Habitats: The *Ixodes scapularis* tick usually lives in wooded, bushy areas with long grass. In North America, the tick is most active in May, June, and July.²² Spending time in nature is an essential component of health, so try to choose open spaces that are well maintained and free of brush and long grass, and do a thorough tick check post-hike.

Clothing: Wear shoes, long pants tucked into socks or pant legs, and long sleeves when outside in wooded areas.²³ Wearing white clothes will make it easier to see attached ticks. Stop and check for ticks whenever you brush up on vegetation. After returning home, place your clothes directly in a dryer for a minimum of 6 minutes on high heat to kill ticks attached to clothing.²⁴

Tick Checks: Early identification of attached ticks is crucial. Be vigilant about checking for ticks. Thoroughly check your entire body. Pay special attention to the back of the neck, scalp, groin, and legs where ticks commonly attach—taking a shower after a tick check is recommended.²⁵ Pets often bring ticks home. Regularly check pets for ticks and ensure they receive tick prevention medication.²⁶ Typically, ticks need to be attached for roughly 24-36 hours to become infective, giving you time to identify and remove them.²⁷ If you find a tick save it as there are places you can mail it to to see if it is infected.

Insect Repellent: Conventional recommendations encourage insect repellent with 20%-30% DEET around the ankles, other areas of bare skin, and clothes.²⁸

However, there is controversy around the safety of DEET. Some research suggests DEET has neurotoxic

activity with the potential to exacerbate current neurological issues.²⁹ Neurological issues are a common symptom of Chronic Lyme disease so DEET alternatives may be a safer option.

Essential oil formulations can be effective tick repellents. It's important to note that they don't last as long as chemical alternatives and should be re-applied frequently. The following essential oils have demonstrated tick repellent properties: citronella, lemon eucalyptus, lavender, juniper, oregano, clove, and geranium.³⁰

Permethrin can be applied to clothing before going outdoors. Permethrin is an insecticide that targets the insect nervous system.³¹ While it displays low toxicity to mammals, permethrin has demonstrated health risks when applied directly to the skin.³² Use on clothing only.

PREVENTATIVE HERBAL MEDICINE: Research suggests certain herbal medicines may be effective in preventing tick bites; these include:

Wild Teasel (*Dipsacus fullonum*)^{33 34}: Wild Teasel is an herb with a long history of use for its antibacterial properties. It is popularly used for Lyme disease (I personally use it as a tincture for prevention). While it's unclear how Wild Teasel works, it has been suggested that it can flush the bacteria out of hard-to-reach tissues in the body and back into the bloodstream, where the immune system can attack it. A new 2022 study pinpointed two main bioactive compounds - sylvestrosides III and IV - that showed considerable anti-Borrelia activity³⁵. Once this spirochete is flushed out using teasel root, antibiotics can be more effective. Suggested dose: (for prevention only – treatment dose is very different): Take one 330-mg capsule, 1 time per day or use a teasel root tincture 2x/day post-exposure or when in a known Lyme region.

Anamu (*Petiveria alliacea*): Anamu contains chemical compounds that may possess broad-spectrum antimicrobial properties against bacteria, viruses, yeast, and fungi. Anamu is immune enhancing and has potent anti-inflammatory properties (inhibits COX-1). It increases cellular immunity and increases NK cells.³⁶ Suggested dose: Take one 1000-mg capsule, 2 times per day. Note that it will give urine and feces a distinct odor.

Garlic (*Allium sativum*): Throughout history, garlic has been used to prevent and treat many diseases. The active chemical in garlic, allicin, has antibacterial, antiviral, antifungal, and antiparasitic properties.³⁷ Allicin is a volatile compound and breaks down soon after the plant is crushed, making its medicinal qualities difficult to absorb. However, through extraction techniques, allicin can be stabilized in the form of a supplement that retains its active compounds.³⁸ Suggested dose: Take one 180-1200-mg capsule, three times daily.

1. Anti-Microbial Herbal Medicine According to a 2020 study from Johns Hopkins University, certain herbal medicines have more potent activity against *Borrelia burgdorferi* than some antibiotics, including doxycycline.³⁹ The following herbal medicines are suggested for treating Lyme disease:

Japanese Knotweed (*Polygonum cuspidatum*): Japanese Knotweed is an herb that has been used for centuries in Traditional Chinese Medicine. Renowned for its potent antioxidant, antimicrobial, and anti-inflammatory properties, Japanese Knotweed contains resveratrol, a popular supplement. This compound shows activity against glycation, oxidative stress, inflammation, neurodegeneration, several types of cancer, and aging.⁴⁰ It crosses the blood-brain barrier and is neuroprotective.

Japanese knotweed has strong activity against both growing *B. burgdorferi* and non-growing stationary phase *B. burgdorferi*.^{41 42} It may also eradicate co-infections associated with Lyme. Suggested dose: Take one 200-800-mg capsule or a tincture 2-3 times per day, for six weeks. Take with food. Caution if taken with anticoagulants as resveratrol has blood-thinning properties.

Cryptolepis (*Cryptolepis sanguinolenta*): A plant used by traditional healers in West Africa, *Cryptolepis* has a long history of use for infectious diseases, fever, and parasitic infections. Research demonstrates *Cryptolepis* has systemic antibacterial properties against *B. burgdorferi*,⁴³ as well as anti-inflammatory (blocks COX2 and inflammatory cytokines) and anticancer properties.⁴⁴ This herb has antimicrobial activity against Lyme co-infections, such as *Bartonella* and *Babesiosis*.⁴⁵ In subculture studies, only 1% *Cryptolepis sanguinolenta* extract caused complete eradication of *B. burgdorferi*. Suggested dose: Take one 420-mg capsule once per day or 2 dropperfuls of an alcohol-extracted tincture 2-3 times daily.

Cat's Claw (*Uncaria tomentosa*): Grown in the rainforest of South and Central America, Cat's claw has a long history of use by indigenous people of the Amazon for inflammatory conditions. Cat's claw is considered an immunomodulator, which means it calms an overactive immune system and reduces cytokine cascades while enhancing how well the immune system works. It contains chemicals that increase WBCs, including B and T lymphocytes, natural killer (NK) cells, and granulocytes. Cat's claw enhances a specific type of natural killer cell called CD57, commonly deficient in people with Lyme disease.⁴⁶ Suggested dose: Take two to three 400-800-mg capsules daily or 2 dropperfuls of tincture twice per day with food. Acid-blocking drugs reduce effectiveness. May have an antagonizing effect on immunosuppressive drugs.

Chinese Skullcap (*Scutellaria baicalensis*): Chinese Skullcap is popular for the treatment of various diseases, especially hepatitis, diarrhea, vomiting, viral infections, and high blood pressure.⁴⁷ Research shows it has multiple biological activities, including anti-inflammatory, antiviral, antitumor, antioxidant, and antibacterial effects.⁴⁸ Suggested dose: Take one 400-1,000mg capsule or 2 dropperfuls of tincture 2-3 times per day with food.

Sweet Wormwood (*Artemisia annua*): Used in Traditional Chinese Medicine for over 2,000 years, Sweet Wormwood is considered a potent anti-parasitic herb.⁴⁹ Artemisinin, a chemical compound found in Sweet

Wormwood, is a primary treatment for malaria. The isolation of artemisinin was awarded the Nobel Prize in 2015. Artemisinin's antimicrobial effect is thought to be related to its ability to generate free radicals that damage proteins.⁵⁰ Research shows that high-dosage extracts of *Artemisia annua* exhibited better activity against stationary phase *B. burgdorferi* as compared to the control antibiotics.⁵¹ Wormwood is also indicated in the treatment of Babesiosis, caused by the protozoan *Babesia*, which is the most common co-infection in Lyme disease. Suggested dose: Take one 6-12-g capsule once per day, for 7 days. Take with food. Should not be taken for more than seven days. The dose can be repeated after three weeks. Do not use for more than 1-2 months.

Ashwagandha (*Withania somnifera*): Native to India and Africa, Ashwagandha is a well-known herb in Ayurveda, the traditional medicine of India. Ashwagandha is a calming adaptogen, particularly useful in balancing the HPA axis in the brain (the control center for stress regulation). Through this pathway, it is thought to improve stress resistance, allow for improved sleep, reduce brain fog and fatigue, and has anti-inflammatory, antioxidant, and immune-enhancing properties.⁵² Suggested dose: Take one 450-mg capsule or tincture 2 times per day. Take 20-30 minutes before meals. May stimulate thyroid function (used for hypothyroidism); caution in those with hyperthyroidism.

Rhodiola (*Rhodiola rosea*): Primarily sourced in Siberia, Rhodiola is used to improve resistance to stress and tolerance at high altitudes. Research suggests it may increase oxygen delivery to tissues, especially the heart. Rhodiola is a popular herb amongst Russian athletes to decrease fatigue, increase alertness, and improve memory. Rhodiola enhances cardiovascular function and immune function and protects nerve and brain tissue. Rhodiola has also been clinically shown to help lessen anxiety and stress (fatigue/burnout) and possibly improve concentration and focus.⁵³ Suggested dose: Take one 100-200-mg capsule 2 times per day. Take with food. Can be stimulating; insomnia can occur if taken in the evening.

Wild Teasel (*Dipsacus fullonum*) Leaves and Root Tincture^{54 55}: Spirochetes like *B. burgdorferi* bore into human tissue and then seal themselves with biofilm to create a barrier against antibiotics. While teasel isn't an antibiotic, it acts as a powerful agent to flush the bacteria into the bloodstream so that antibiotics can more easily target the infection. When dealing with Chronic Lyme Disease or in situations where an antibiotic protocol wasn't effective, adding teasel into your treatment regime may help. A new 2022 study pinpointed two main bioactive compounds - sylvesterosides III and IV - that showed considerable anti-Borrelia activity⁵⁶. Note that it still may still take up to six months or longer to clear the infection.

Suggested dose: It's important to be very careful with internal dosing of teasel in the case of Lyme. If you begin with too high of a dose it can cause a Jarisch-Herxheimer ("Herx") reaction. This is when

cytokines are released as the Lyme bacteria die, which may make you feel worse before feeling better.

The standard tincture protocol is: For the first day, take one drop in the morning. On the second day, increase the dose to one drop in the morning and one in the evening. On the third day, take one drop in the morning, afternoon, and evening. Proceed to increase the dose by one drop each day until you reach the maximum of 9 drops/day (three in the morning, three in the afternoon, and three in the evening).

I also use teasel as a Lyme preventative. I take a dropperful of wild teasel tincture 2x/day when I'm in a known Lyme region to keep the *Borrelia burgdorferi* spirochetes from burrowing in.

2. Vitamin and Mineral Supplementation

Vitamin D: Healthy vitamin D levels are required for optimal health – get your levels tested.⁵⁷ Vitamin D is frequently low in patients with persistent Lyme.⁵⁸ Daily sun exposure and the consumption of vitamin D-rich foods, including fatty cold-water fish and pastured egg yolks, help maintain Vitamin D levels. However, supplementation is often necessary.⁵⁹ The recommended form of vitamin D is vitamin D3 or cholecalciferol. Suggested dose: Take 1,000 IU daily with food; more if your levels are low.

Vitamin C: Vitamin C aids in the production of white blood cells that battle infections. It may help people with lowered immunity, such as Lyme disease patients, by improving their ability to fight off further infection.⁶⁰ Additionally, Vitamin C is involved in the formation of collagen, which is frequently destroyed in Lyme disease.⁶¹ Suggested dose: Take one 500-1,000-mg capsule daily with food.

Zinc: Zinc is crucial for the development and function of cells that regulate innate and adaptive immunity. Zinc inhibits complement activity, a part of the immune system that is excessively activated by Lyme infection and contributes to inflammation.⁶² Zinc deficiency impairs the ability of immune cells to generate compounds that kill invading pathogens, like *B. burgdorferi*.⁶³ Suggested dose: Take one 30-mg capsule once daily with food.

N-Acetylcysteine (NAC): N-acetylcysteine (NAC) is a supplement that comes from the amino acid L-cysteine. It aids in replenishing glutathione, a powerful antioxidant in the body that is often depleted by inflammation, common in Lyme disease. Research shows that NAC alleviates chronic inflammatory pathologies.⁶⁴ Suggested dose: Take two 1,000-mg capsules, 2 times daily. Take with juice, away from food, either 30-minutes before a meal, or 1.5 hours after a meal.

Turmeric (*Curcuma longa*): Turmeric is a popular spice in India. It's well known for its anti-inflammatory compounds. It blocks the activation of NF-κB, a pro-inflammatory signaling pathway in Lyme-induced inflammation⁶⁵. It's best absorbed when consumed with fat and black pepper. Suggested dose: 500mg 2-3x daily.

3. Restore Immune Function

Lyme disease can have devastating effects on the immune system. By altering the function of immune cells, it can cause chronic inflammation, leading to pain and illness.⁶⁶ Under normal circumstances, inflammation is an important part of a healthy immune system; it triggers immune cells to attack pathogens or heal damaged tissue.⁶⁷ However, chronic inflammation can be dangerous and plays a major role in many serious health conditions.⁶⁸ It may manifest as joint discomfort, stiffness, aches and pains, fatigue, low stamina, brain fog, slow mental activity, and depressed mood.⁶⁹ When treating Lyme, reducing chronic inflammation helps optimize immune function and restore the body's capacity to heal. The following supplements support immune restoration.

Reishi (*Ganoderma lucidum*): A popular medicinal mushroom used in Traditional Chinese Medicine, Reishi (*Ganoderma lucidum*) is used for its immune-modulating properties.⁷⁰ Through its effect on the immune system, Reishi reduces destructive inflammatory processes while enhancing the ability of the immune system to deal with harmful microbes, like *B. burgdorferi*.⁷¹ Suggested dose: Take 2 dropperfuls of a dual-extracted reishi tincture twice daily.

Omega-3 Fatty Acids: Known as “essential molecules”, Omega-3 fatty acids can't be made by our bodies and must be consumed via diet. Omega-3 fatty acids help support the structure and function of our cells. They also help the innate and adaptive immune system.⁷² Research shows Omega 3s promote brain health by protecting cognitive function.⁷³ Regularly consuming wild-caught fish is an excellent way of increasing your levels of Omega 3s, however, fish oil supplements may be a more convenient and cost-effective way to obtain the benefits of dietary omega 3's. Suggested dose: Take 1,500-mg of EPA and 500-mg of DHA daily. Keep in the refrigerator to avoid rancidity.

Multi-Strain Probiotic: Many individuals with Lyme disease have gone through multiple rounds of antibiotic treatments. While antibiotics may offer significant benefits to Lyme sufferers, they also reduce levels of beneficial microorganisms in the GI tract, enabling opportunistic and pathogenic organisms to proliferate. The resulting imbalance in the gut microbiota can compromise immune function, digestion, and nutrient absorption. Multi-strain probiotics restore the microbial balance of the GI tract by increasing the amount of beneficial bacteria in the small and large intestine.⁷⁴ Research shows probiotics improve digestion, absorption of nutrients, and regulate immune response through anti-inflammatory cytokines.⁷⁵ Suggested dose: Take enough pills daily to obtain 10-12 billion colony forming units.

Organic Bone Broth: Bone broth is a slow-cooked broth made from boiling animal bones and connective tissues in water. It is gelatinous and rich in glycine and proline, amino acids that help create a healthy gut. Lyme disease can cause extensive damage to collagenous tissues in the body, such

as skin, tendons, and ligaments.⁷⁶ Bone broth may help rebuild those tissues by replacing the collagen damaged by Lyme disease. Suggested dose: Drink 1 cup per day.

4. Anti-Inflammatory Diet A healthy and balanced diet is essential for the optimal function of every part of our health, including the immune system.⁷⁷ Certain foods create chronic inflammation in the body. Inflammatory foods trigger the release of chemicals and immune cells that attack surrounding tissues; causing widespread damage. An anti-inflammatory diet consists of removing foods that initiate inflammation and consuming nutrient-dense, whole foods that support the body's health.

Please see our dietary section for more information on an Anti-Inflammatory Diet (see page 297), including foods to avoid, what to eat, and healthy recipes.

Begin by removing pro-inflammatory foods, such as processed food, refined carbohydrates, gluten, sugar, industrial seed oils, and deep-fried foods, and adding in anti-inflammatory foods, such as organic fruits and vegetables, nuts and seeds, beans and legumes, fruits, fermented foods, hormone & antibiotic-free protein, wild-caught seafood, cold-pressed oils, and spices and herbs. Choose fresh whole foods that are organic and GMO-free to support your body's health while reducing your exposure to toxic inflammatory compounds.⁷⁸

Fruits and vegetables are rich in vitamins, minerals, and antioxidants. Additionally, they are rich in Vitamin C, a cofactor for enzymes involved in collagen synthesis. Optimizing vitamin C intake can offset the destruction of collagen that commonly presents with Lyme Disease.

Whole grains such as oatmeal, brown rice, quinoa, and other unrefined grains are high in fiber and support the growth of beneficial gut bacteria, which positively impact our immune function and digestion.

Eat healthy proteins like meat, poultry, fish, and eggs. The brain is severely affected in many cases of chronic Lyme disease, referred to as "neuroborreliosis."⁷⁹ Lyme infection promotes brain inflammation and impairs energy production in neurons, making it difficult to think clearly and maintain a balanced mood. Vitamin B12, Omega 3's, and essential amino acids found in protein are important building blocks in brain function.

Fermented foods such as sauerkraut, kimchi, beet kvass, and yogurt (for those who tolerate dairy), provide probiotics that support gut health and immune function.⁸⁰ Consume a serving of fermented foods daily. Rotate fermented foods since each offers unique probiotic microorganisms and health benefits.

Consuming clean water is an important way to assist

the body in healing from Lyme. Water moves nutrients throughout the body, and removes waste (including irritating, inflammatory substances), protects and cushions joints, maintains blood volume; a key part of blood pressure and heart function, and keeps the body at a normal temperature.⁸¹ Suggested daily intake: 13 cups (104oz)/day for men and 9 cups (72oz)/day for women⁸²

5. Rest and Self-Care Activities

Lyme disease is a complicated and potentially devastating illness. With symptoms varying dramatically from person to person, there is no single path to recovery. While healing from Lyme disease it's important to be patient and gentle with yourself. Typically, natural therapies are safe and effective but work at a slower pace than conventional medicine. Frustration and disappointment are normal experiences for Lyme disease sufferers. Ensuring you take good care of your physical and mental health are important steps in the recovery process.

Adequate Sleep: A good night's sleep enables your body to heal and better equips you to handle life's stressors. It may be frustrating to feel perpetually tired and lethargic while battling Lyme disease but fighting an illness takes a lot of energy; rest is an important part of recovery. Many immune processes that bolster your body's ability to fight off an illness or infection occur while you're asleep, as sleep and the circadian system exert a strong regulatory influence on immune functions.⁸³ Aim for 7 to 9 hours of sleep each night.⁸⁴

Herxheimer Reaction: The Herxheimer Reaction is a temporary worsening of symptoms that occurs when the Lyme spirochete, *B. burgdorferi*, is being killed off by antibiotic or antimicrobial treatments.⁸⁵ While the "Herx Reaction" may be a good indicator of healing, it can be extremely uncomfortable and distressing for those experiencing it. Symptoms can include fever, chills, severe pain, and skin rash.⁸⁶ It's important to maintain proper self-care if you experience a Herx reaction and keep in mind that it is temporary. If symptoms persist make sure to contact your healthcare provider to ensure you aren't having an allergic reaction.

Mindfulness: Mindfulness is an awareness of our thoughts, feelings, bodily sensations, and surrounding environment without judgment.⁸⁷ Mindfulness can help people living with chronic conditions, like Lyme disease, accept and respond to difficult feelings, including fear, loneliness, and sadness. Research shows that mindfulness-based programs like meditation, time in nature, and gratitude result in coping better with symptoms, improved overall well-being and quality of life, and enhanced health outcomes in people living with chronic health conditions.⁸⁸

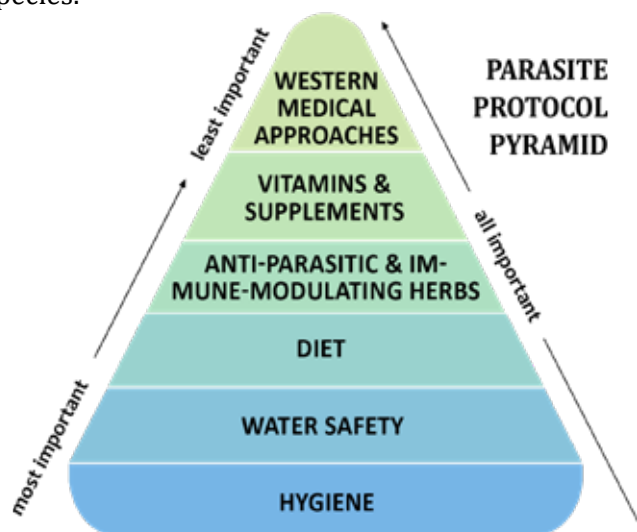
	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	NAC (N-acetylcysteine) -Take with juice away from food Japanese Knotweed & Chinese Skullcap Cryptolepis Wild Teasel tincture Multi-strain probiotic Vitamins C and D	Ashwagandha (take 20-30 min. before meals) Cat's Claw Reishi tincture Omega-3 FA capsule Zinc & Turmeric (w/ food) Sweet Wormwood (w/food)	NAC (N-acetylcysteine) (take w/ juice away from food) Japanese Knotweed & Chinese Skullcap Organic bone broth (1 cup) Wild Teasel tincture	Ashwagandha (take 20-30 min. before meal) Cryptolepis Turmeric Wild Teasel Reishi tincture	Japanese Knotweed Mindfulness practice Sleep tincture, if needed
MONDAY	NAC (N-acetylcysteine) -Take with juice away from food Japanese Knotweed & Chinese Skullcap Cryptolepis Wild Teasel tincture Multi-strain probiotic Vitamins C and D	Ashwagandha (before meals) Cat's Claw Reishi tincture Omega-3 FA capsule Zinc & Turmeric (w/ food) Sweet Wormwood (w/food)	NAC (N-acetylcysteine) (take w/ juice away from food) Japanese Knotweed & Chinese Skullcap Organic bone broth (1 cup) Wild Teasel tincture	Ashwagandha (before meal) Cryptolepis Turmeric Wild Teasel Reishi tincture	Japanese Knotweed Mindfulness practice Sleep tincture, if needed
TUESDAY	NAC (N-acetylcysteine) -Take with juice away from food Japanese Knotweed & Chinese Skullcap Cryptolepis Wild Teasel tincture Multi-strain probiotic Vitamins C and D	Ashwagandha (before meals) Cat's Claw Reishi tincture Omega-3 FA capsule Zinc & Turmeric (w/ food) Sweet Wormwood (w/food)	NAC (N-acetylcysteine) (take w/ juice away from food) Japanese Knotweed & Chinese Skullcap Organic bone broth (1 cup) Wild Teasel tincture	Ashwagandha (before meal) Cryptolepis Turmeric Wild Teasel Reishi tincture	Japanese Knotweed Mindfulness practice Sleep tincture, if needed
WEDNESDAY	NAC (N-acetylcysteine) -Take with juice away from food Japanese Knotweed & Chinese Skullcap Cryptolepis Wild Teasel tincture Multi-strain probiotic Vitamins C and D	Ashwagandha (before meals) Cat's Claw Reishi tincture Omega-3 FA capsule Zinc & Turmeric (w/ food) Sweet Wormwood (w/food)	NAC (N-acetylcysteine) (take w/ juice away from food) Japanese Knotweed & Chinese Skullcap Organic bone broth (1 cup) Wild Teasel tincture	Ashwagandha (before meal) Cryptolepis Turmeric Wild Teasel Reishi tincture	Japanese Knotweed Mindfulness practice Sleep tincture, if needed
THURSDAY	NAC (N-acetylcysteine) -Take with juice away from food Japanese Knotweed & Chinese Skullcap Cryptolepis Wild Teasel tincture Multi-strain probiotic Vitamins C and D	Ashwagandha (before meals) Cat's Claw Reishi tincture Omega-3 FA capsule Zinc & Turmeric (w/ food) Sweet Wormwood (w/food)	NAC (N-acetylcysteine) (take w/ juice away from food) Japanese Knotweed & Chinese Skullcap Organic bone broth (1 cup) Wild Teasel tincture	Ashwagandha (before meal) Cryptolepis Turmeric Wild Teasel Reishi tincture	Japanese Knotweed Mindfulness practice Sleep tincture, if needed
FRIDAY	NAC (N-acetylcysteine) -Take with juice away from food Japanese Knotweed & Chinese Skullcap Cryptolepis Wild Teasel tincture Multi-strain probiotic Vitamins C and D	Ashwagandha (before meals) Cat's Claw Reishi tincture Omega-3 FA capsule Zinc & Turmeric (w/ food) Sweet Wormwood (w/food)	NAC (N-acetylcysteine) (take w/ juice away from food) Japanese Knotweed & Chinese Skullcap Organic bone broth (1 cup) Wild Teasel tincture	Ashwagandha (before meal) Cryptolepis Turmeric Wild Teasel Reishi tincture	Japanese Knotweed Mindfulness practice Sleep tincture, if needed
SATURDAY	NAC (N-acetylcysteine) -Take with juice away from food Japanese Knotweed & Chinese Skullcap Cryptolepis Wild Teasel tincture Multi-strain probiotic Vitamins C and D	Ashwagandha (before meals) Cat's Claw Reishi tincture Omega-3 FA capsule Zinc & Turmeric (w/ food) Sweet Wormwood (w/food)	NAC (N-acetylcysteine) (take w/ juice away from food) Japanese Knotweed & Chinese Skullcap Organic bone broth (1 cup) Wild Teasel tincture	Ashwagandha (before meal) Cryptolepis Turmeric Wild Teasel Reishi tincture	Japanese Knotweed Mindfulness practice Sleep tincture, if needed

*Unless otherwise specified, all supplements and herbal preparations can be taken in either tincture or capsule form, depending on what's more convenient/available.

Ch. 21: Parasite Protocol

Parasites are organisms that live on or in a host, such as a human, and get their food from that host. There are three main types of parasites: protozoa, which are single-celled eukaryotic organisms; helminths, which include roundworms, tapeworms, and flukes; and ectoparasites, which live outside of the host, such as ticks, scabies, bed bugs, fleas, and lice. In this chapter, we'll discuss protozoan parasites and helminthic parasites. For more information on Lyme disease, which occurs when the bacterium *Borrelia burgdorferi* is transmitted to a human via tick bites, see our Lyme Protocol on page 133. For more information on other types of infections, see our Bacterial Protocol (see page 113), Anti-Fungal Protocol (see page 127), and Viral Protocol (see page 121).

PROTOZOAN PARASITES: Common protozoan parasites include *Giardia duodenalis*, *Entamoeba histolytica*, *Cryptosporidium* species, and *Plasmodium* species.



***Giardia duodenalis* (also Called *G. intestinalis* and *G. lamblia*):** Infection with *Giardia* occurs in the small intestine and causes giardiasis. Infections typically occur after exposure to water, food, objects, or surfaces contaminated with infectious feces. Symptoms include watery diarrhea, abdominal cramping and pain, fatigue, flatulence, diminished appetite, and foul-smelling stool.

***Entamoeba histolytica*:** *Entamoeba histolytica* causes amebiasis and is typically transmitted through oral contact with water, food, objects, or surfaces contaminated with infected feces. This protozoon infects the large intestine and causes loose stool, abdominal pain, and cramping. If the infection is severe, symptoms can also include fever and bloody stool.

***Cryptosporidium* spp.:** *Cryptosporidium* causes cryptosporidiosis. It is typically contracted by swallowing recreational water (e.g., water in lakes, streams, rivers, pools, hot tubs, water parks, etc.) that is contaminated with infected feces. Symptoms can include watery diarrhea, abdominal cramping, diminished appetite, weight loss, mild fever, and vomiting.

***Plasmodium* spp.:** The *Plasmodium* species causes

malaria. Malaria is contracted after a person is bitten by an infected mosquito. Malaria transmission typically occurs in South Asia, Africa, and some parts of Central and South America, the Caribbean, Southeast Asia, the Middle East, and Oceania. The most common symptoms of malaria include fever, chills, muscle aches, and fatigue. Symptoms can also include nausea, vomiting, diarrhea, anemia, and jaundice.

Trypanosoma cruzi*:** *Trypanosoma cruzi* is a protozoan parasite that causes Chagas disease (also known as trypanosomiasis). Symptoms can include fever, diminished appetite, body aches, headache, swelling at the site of the infection, eyelid swelling, and a rash. ***Trypanosoma brucei causes African sleeping sickness. Symptoms include fever, fatigue, enlarged lymph nodes, joint and muscle aches, and, in some cases, a rash.

***Toxoplasma gondii*:** *Toxoplasma gondii* causes toxoplasmosis. It is typically contracted by eating undercooked meat or being exposed to infected cat feces. A pregnant mother can pass it on to her unborn child. Symptoms can include swollen lymph nodes in the neck region, aching and painful muscles, headache, fever, malaise, and inflammation of the lungs, heart, and eye.

***Babesia* spp.:** *Babesia* species are protozoan parasites that infect red blood cells. People with babesiosis may report no symptoms at all, or they may have flu-like symptoms such as fever, chills, sweating, fatigue, body aches, nausea, or appetite changes. Infections with *Babesia* may resolve spontaneously or they may require treatment, depending on the severity of the disease.

HELMINTHIC PARASITES: Helminthic parasites are subdivided into three categories—roundworms (nematodes), tapeworms (cestodes), and flatworms or flukes (trematodes).

Roundworms: The most common roundworms that infect humans in the United States are pinworms. Other roundworms that infect humans include hookworms (*Ancylostoma duodenale*, *Necator americanus*), which you can get by walking barefoot on soil that contains the eggs; *Ascaris lumbricoides*, which is also transmitted via soil, and is most common in people living in or traveling to and from tropical or subtropical regions; and whipworms, which are spread from person to person through exposure to contaminated feces.

Tapeworms: Tapeworms (*Taenia saginatum*, *Taenia solium*, *Diphyllobothrium latum*, and *Hymenolepis nana*) are contracted by ingesting food or water that contains tapeworm eggs or larvae. These infections occur in the intestines, but if you ingest tapeworm eggs, they can leave your intestines and form cysts in other body tissues. This is known as an invasive infection.

Flatworms and Flukes: The most common flatworms that infect humans are the blood fluke (*Schistoma* spp.), the lung fluke (*Paragonimus westermani*), and the liver fluke (*Clonorchis sinensis*). Blood flukes are usually contracted through freshwater sources contaminated with the feces of someone who is infected with a blood fluke and has schistosomiasis. Lung flukes

are usually transmitted by eating raw or partially cooked crabs or crawfish that have been infected with the larvae. Liver flukes are usually contracted by eating watercress or other water plants that are contaminated with the immature larvae.

People with helminthic parasites can be asymptomatic if the infection is mild, or they can experience abdominal pain, diarrhea, blood loss, symptoms of low protein or malnutrition, rectal prolapse, and even delays in physical and cognitive growth if the infection is severe.

In this chapter, we'll discuss ways to address these common protozoan and helminthic parasitic infections.

1. Hygiene In order to reduce the likelihood of being infected with a parasite, it is important to practice principles of basic hygiene. These include:

Wash Hands Often with Soap and Water: To reduce the likelihood of contracting a parasite, wash your hands with soap and water before eating or handling food, after touching animals or cleaning animal feces or litter, after using the bathroom, after outdoor activities, and after shaking hands with others or touching surfaces that may be contaminated.

Clean Raw Vegetables Thoroughly and Cook Aquatic Plants: If your vegetables have been contaminated by animal or human feces (*e.g.*, if a food service worker handles your food without washing his or her hands), it increases your likelihood of contracting roundworms, tapeworms, and flatworms. To reduce this likelihood, clean your vegetables thoroughly. Aquatic plants like watercress should be cooked before eating to reduce the likelihood of developing a liver fluke infection.

Vacuum Carpets, Mop Floors, and Clean Surfaces in Your Home Often: Pinworm eggs can cling to surfaces and live there for up to 2 weeks. Regularly clean surfaces thoroughly and frequently to prevent spreading pinworms to others or reinfecting oneself.

Keep Nails Short and Don't Bite Them: Pinworms lead to anal itching. If an individual has scratched the anal region, pinworm eggs can get under the fingernails and live there for weeks. Keeping nails short and avoiding biting them reduces the spread of pinworms. It also reduces the likelihood of infected individuals being reinfecting.

Bathe and Change Underwear Daily: Pinworms lay their eggs on the skin near the anus at night. Showering in the morning helps to remove pinworm eggs from the anal region before they hatch. If you have pinworms, showering is preferable to bathing because it reduces the chance of reinfection. To prevent pinworm eggs from hatching and reinfecting you, change your underwear every morning. Washing your underwear in hot water and drying them on high heat also helps to kill the eggs.

Avoid Sharing Towels: Sharing towels or washcloths can lead to the transmission of parasites like pinworm eggs and immature roundworms from person to person.

140 Avoid Walking Barefoot on Dirt That May Be Contaminated with Feces: Hookworm eggs

release larvae when they hatch in soil. These larvae mature into a worm that can penetrate your skin when you step on them. If you walk barefoot on dirt that is contaminated with infected feces, hookworms may penetrate your skin, leading to an infection.

Keep Pets under the Care of a Veterinarian: To reduce the likelihood of contracting a parasitic infection, make sure your pet sees a veterinarian and that you treat your pet for worms. This reduces the likelihood of your pet spreading various types of worms.

Protect Yourself from Tick and Mosquito Bites: To protect yourself from tick bites, make sure you wear long sleeves and long pants when walking in the woods. For more information on Lyme disease, see our Lyme Disease Protocol. To reduce the likelihood of being bitten by a disease-carrying mosquito, use insect repellent. When traveling to areas where malaria is endemic, sleep with a mosquito net and look into anti-malarial options.

2. Water Safety In order to limit the spread of parasitic infections, practice principles of water safety. This includes drinking water and the water you are exposed to externally.

Purify Water from Natural Sources: Before drinking or cooking with water from a stream or other natural source, it is best to purify that water. Ideally, use a filter with an absolute pore size of at least 1 micron that has been rated for cyst removal, as that is needed to remove *Giardia*. If you do not have an appropriate water filter on hand, bring the water to a rolling boil for at least 1 minute. If you are in the mountains or in a location with a higher elevation, bring the water to a rolling boil for at least 3 minutes.

Avoid Getting Water in Your Mouth While Swimming: When swimming in swimming pools, water parks, hot tubs, lakes, rivers, or the ocean, avoid swallowing water or getting water in your mouth. This reduces your risk of contracting giardiasis and other parasites.

Avoid Ice and Untreated Water When Traveling: To limit the likelihood of contracting giardiasis and other infectious agents when traveling, avoid ice, raw unwashed vegetables, and any water that may be untreated. Opt for unopened, factory-sealed, bottled water or bring a water filter with an absolute pore size of at least 1 micron that has been rated for cyst removal with you. You may also boil your water (see above).

Avoid Swimming When Sick and for at Least 2 Weeks After Diarrhea Stops: In order to avoid spreading parasites like *Giardia*, do not swim in pools, hot tubs, or bodies of water while you have diarrhea or for at least 2 weeks after the diarrhea has stopped. You can continue to shed *Giardia* in your feces for weeks, even after your symptoms have resolved.

3. Diet Eating certain foods can reduce your risk of being infected by a parasite; other foods can increase that risk. Foods that reduce your risk include garlic, high-fiber foods, and foods rich in beta-carotene. Foods that can increase your risk of contracting a parasite

include raw meat or fish and highly processed foods.

Eat More Garlic: Garlic contains active constituents such as allicin that have potent effects against parasites. Allicin prevents parasites from replicating by partially or completely inhibiting DNA, RNA, and protein synthesis.¹

Optimize Your Fiber Intake: Experimental research suggests that diets low in fiber lead to more severe parasitic infections, whereas diets high in prebiotic fruit and vegetable fibers such as cellulose, pectin, and inulin tended to protect against parasitic infection.²

Eat Foods Rich in Beta-Carotene: Beta-carotene is converted into vitamin A in the body, and vitamin A helps your body resist parasitic worms.³ Foods high in beta-carotene include carrots, spinach, leafy greens, tomatoes, sweet potatoes, broccoli, cantaloupe, winter squash, and most yellow and orange fruits and vegetables.

Avoid Raw or Under-Cooked Meat or Seafood: Curing, drying, smoking, or microwaving meat is not enough to eliminate worms. Cooking meat to appropriate temperatures is the best way to consistently reduce your risk of contracting parasitic infections like trichinosis (trichinellosis), which can be contracted from bear, cougar, pig, or boar meat, and other parasitic roundworm, flatworm, or tapeworm infections.

Poultry should be cooked to at least 165°F; fresh meat chops, steaks, and roasts should be cooked to at least 145°F; and ground meats should be cooked to at least 160°F. Freezing pork that is under 6 inches thick for 20 days at 5°F can kill worms; however, some worm species that infect wild game meat cannot be killed by freezing.

Limit Highly Processed Foods: Highly processed foods such as refined flour, sugar, juices, and other simple carbohydrates disrupt the microbiome and have an inflammatory effect on the body.⁴ This may impair the body's ability to combat parasitic infections. If you need to use a sweetener, consider using raw honey since it has been found to have anti-parasitic activity against some parasitic pathogens.⁵

4. Anti-Parasitic and Immune-Modulating Herbs

There are several herbs that can help modulate the immune system and address parasitic infections. These include berberine-containing herbs (barberry, Oregon grape, Chinese goldthread, and goldenseal), wormwood, clove, yarrow, green tea, ginger, and cinnamon.

Berberine-Containing Herbs: Berberine-containing herbs, such as barberry (*Berberis vulgaris*), Oregon grape (*Berberis aquifolium*), Chinese goldthread (*Coptis chinensis*), and goldenseal (*Hydrastis canadensis*) exhibit anti-parasitic activity against a variety of parasites. Berberine has anti-parasitic activity against *Giardia duodenalis*, *Entamoeba histolytica*, *Trichomonas vaginalis*, and *Leishmania donovani*.^{6 7 8 9} People with an autoimmune disorder should consult their physician before taking goldenseal or use an alternate form of

berberine.

Suggested dose for barberry: 4 dropperfuls of a 1:4 tincture, twice daily.

Suggested dose for Oregon grape: 3 dropperfuls of a 1:4 tincture, twice daily.

Suggested dose for Chinese goldthread: 2 dropperfuls of a 1:5 tincture, twice daily.

Suggested dose for goldenseal: 2 dropperfuls of a 1:3 tincture, twice daily.

Wormwood (*Artemisia* spp.): In experimental models, treatment with wormwood significantly reduced *Giardia* counts, improved the health of the small intestine, and decreased pro-inflammatory chemicals.¹⁰ These effects are attributed to the main active constituent of wormwood, artemisinin. Artemisinin and its derivatives also exhibit anti-parasitic activity against *Plasmodium* spp., *Leishmania* spp., *Trypanosoma* spp., *Acanthamoeba castellanii*, *Naegleria fowleri*, *Babesia* spp., *Cryptosporidium parvum*, and *Toxoplasma gondii*.¹¹

While artemisinin is usually prescribed along with antimalarial drugs to treat malaria, research shows that dried *Artemisia annua* (sweet wormwood) leaves, when administered orally, reduce parasitic load more effectively than a comparable dose of purified artemisinin alone.¹²

Researchers noted that this is likely because the other anti-malarial compounds in the plant have a synergistic effect with artemisinin. Suggested dose is one 6-12 gram capsule, once daily with food for 7 days. Do not take for more than seven days. The dose can be repeated after three weeks. Do not use for more than 1-2 months.

Curly Mint (*Mentha crispa*), a Variety of Spearmint: *Mentha crispa* essential oil exhibits anti-parasitic activity against *Trypanosoma brucei*, a parasite that causes African sleeping sickness.¹³ When administered orally, *Mentha crispa* also shows anti-parasitic activity against *Trichomonas vaginalis*, which causes the STD trichomoniasis.

In a study comparing the effectiveness of *Mentha crispa* and the oral antibiotic secnidazole, there was no significant difference between treatment groups.¹⁴ *Mentha crispa* reduced vaginal discharge, malodorous vaginal secretion, pain with sexual intercourse, painful urination, pelvic pain, and burning and itching in the genital area in women with trichomoniasis, with significantly fewer side effects when compared to secnidazole. Suggested dose: a single dose of 24 mg *Mentha crispa*.¹⁵

Clove (*Syzygium aromaticum*): The eugenol in clove essential oil exhibits anti-parasitic activity against *Giardia* by preventing the parasite from adhering to the walls of the small intestine.¹⁶ Clove also has anti-parasitic activity against *Blastocystis* spp., leading researchers to conclude that clove essential oil may be a promising and safe option for the treatment of blastocystis.¹⁷

In addition to anti-parasitic activity against parasites that affect humans, extracts of clove exhibit anti-parasitic activity against many species of *Babesia* (*B. bovis*, *B. bigemina*, *B. divergens*, *B. caballi*, and *B. microti*) and various helminths.^{18 19} Suggested dose: 1 dropperful of a 1:3 tincture, 3 times daily.

Yarrow (*Achillea* spp.): Yarrow also contains eugenol. Yarrow exhibits anti-parasitic activity against *Trypanosoma cruzi*, which causes Chagas disease.²⁰ Sneezewort yarrow (*Achillea santolina*) was found to completely eradicate *Giardia duodenalis* in its trophozoite (motile/active) form.²¹ Suggested dose: 4 dropperfuls of a 1:3 yarrow tincture, twice daily.

Green Tea (*Camellia sinensis*): Green tea exhibits anti-parasitic activity against *Trypanosoma cruzi*, a parasite that causes Chagas disease.²² The catechins in green tea are anti-parasitic against two of the three developmental stages of the parasite. Green tea also has anti-parasitic activity against both developmental stages of *Acanthamoeba castellanii*, which causes inflammation of the cornea in humans.²³ An infusion of green tea leaves protects against complications of malaria, such as hemolysis.²⁴ Suggested dose: 450 mg green tea extract, twice daily.

Ginger (*Zingiber officinale*): Ginger exhibits anti-parasitic activity against blood flukes (*Schistosoma* spp.) and *Trichinella spiralis*, which causes trichinosis.^{25 26} Experimental studies show that ginger may be safe and effective for treating chronic toxoplasmosis.²⁷ Suggested dose: 1-2 teaspoons of chopped ginger per cup of water, three times daily; 0.5 to 1 dropperful of a 1:3 tincture, 3 times daily.

Ceylon Cinnamon (*Cinnamomum zeylanicum*): Cinnamon exhibits anti-parasitic activity against *Cryptosporidium parvum*, which causes cryptosporidiosis.²⁸ Cinnamon also exhibits anti-parasitic activity against *Giardia*²⁹ and *Ascaris suum*,³⁰ a roundworm that infects pigs and that can infect humans if they or their food have contact with pigs or pig manure. Suggested dose: 3 dropperfuls of a 1:4 tincture, twice daily.

5. Vitamins and Supplements There are vitamins and supplements that can be used to address parasitic infections. These include vitamin A, zinc, selenium, papaya and pumpkin seeds, and probiotics.

Papaya Seeds: Papaya seeds exhibit anti-helminthic and anti-amoebic activity against roundworms (*Ascaris lumbricoides*), threadworms (*Strongyloides stercoralis*), hookworms (*Necator americanus*), and *Entamoeba histolytica*.³¹ In a study comparing the effectiveness of papaya seeds with that of albendazole, children given papaya seeds experienced a more significant reduction in intestinal roundworms (*Ascaris lumbricoides*) than those given albendazole or placebo.³² The children given papaya seeds also experienced fewer fungal infections (*Tinea corporis*, ringworm) than the group who was given albendazole. Suggested dose: 10 grams of

ground papaya seeds daily, with meals in divided doses.

Pumpkin Seeds (*Cucurbita pepo*): Research suggests that the anti-helminthic activity of pumpkin seeds are due to these active constituents: cucurbitine, amino acids, fatty acids, berberine, and palmatine. In one experimental study, an alcoholic extract of pumpkin seeds led to a decrease in fecal egg counts and a significant reduction in intestinal worms.³³ Suggested dose: 2 dropperfuls of a 1:3 tincture, twice daily with meals.

Probiotics: Probiotics can kill or inhibit pathogens and can also stimulate the immune system so that it is better able to fight off pathogens.³⁴ Beneficial gut flora can stimulate the immune system and protect against *Toxoplasma gondii* infection.³⁵ Various strains of beneficial bacteria have also been found to be protective against *Babesia microti*, *Cryptosporidium parvum*, *Giardia lamblia*, *Trichinella spiralis*, and *Trypanosoma cruzi*.³⁶ Suggested dose: 100 billion CFU per day with meals, in divided doses.

Selenium: In an experimental study, selenium-deficient mice were not as able to efficiently expel helminthic parasites from their bodies. When fed a selenium-adequate diet, the mice efficiently expelled the worms.³⁷ The mice's impaired ability to effectively expel the parasites was due to negative changes in the immune system due to low selenium levels, suggesting that a diet with adequate selenium is important for expulsion of parasites. Suggested dose is: 55 mcg per day for most adults; 60 or 70 mcg for pregnant or lactating women, respectively. 1-2 Brazil nuts a day gives you adequate selenium³⁸.

Vitamin A: Vitamin A helps your body resist parasitic worms.³⁹ In a study conducted on children who were given albendazole to treat *Ascaris* infections, results showed that administering high-dose, supplemental vitamin A after medicinal deworming protects against reinfection with *Ascaris* in well-nourished children.⁴⁰ Do not take high-dose vitamin A if you are pregnant or may become pregnant because it can lead to birth defects. Suggested dose: one dose of 200,000 IU (60 mg retinol) vitamin A⁴¹.

Zinc: Experimental research demonstrates that parasites are more likely to thrive in a host with low zinc levels.⁴² Low levels of zinc lead to negative changes in immune function, specifically lower levels of immunoglobulin E (IgE), which is important for fighting parasites. Lower levels of IgE in zinc-deficient hosts allow parasites to survive longer. Consume adequate zinc, through diet, supplementation, or a combination of the two. Suggested dose: 11 mg per day for men and 8 mg per day for most women; 11 or 12 mg for pregnant or lactating women, respectively.

6. Western Medical Approaches There are several conventional medications used to address parasitic infections. These include antimalarial drugs, metronidazole and tinidazole, ivermectin, pyrantel pamoate, albendazole, and mebendazole.

Antimalarial Drugs: Antimalarial drugs are typically used for both prevention of malaria when traveling to a place where malaria is endemic and for treatment. When used for prevention, most drugs need to be continued for 1-4 weeks after returning home. Antimalarial drugs typically used for both prevention and treatment include chloroquine, mefloquine, atovaquone-proguanil, doxycycline, and, although it usually isn't preferred, quinine. Drugs that are usually only used for treatment include primaquine and artemether-lumefantrine.

Metronidazole and Tinidazole: Metronidazole and tinidazole are used to treat giardiasis, amebiasis, and trichomoniasis. These drugs are known to have multiple adverse effects, including nausea, bad taste in the mouth, skin issues, vaginal candida infection, dizziness, headache, peripheral neuropathy, seizures, and visual or auditory dysfunction. Other side effects may occur, so it's important to let your provider know or seek emergency medical care if you begin to experience concerning symptoms while on either of these drugs.

Ivermectin: Ivermectin is used off-label to treat pinworms and other parasitic worms. This drug paralyzes and kills the worms. Ivermectin is most effective when taken with a meal that contains fat. Side effects are infrequent, but may include fatigue, dizziness, abdominal pain, or rashes. Within the first two days of treatment, you may experience symptoms related to the death of the parasites, such as joint or muscle aches.

Pyrantel Pamoate: Pyrantel is available over-the-counter to treat pinworms. It is also used to treat roundworms and occasionally hookworms. Potential side effects include nausea, vomiting, appetite changes, and diarrhea. It can also cause dizziness, headache, or drowsiness.

Albendazole: Albendazole is typically used to treat various kinds of tapeworms. Providers also use it off-label to treat some roundworms, hookworms, and pinworms. It can be taken with or without food; if it is taken with food, more is absorbed systemically and if it is taken without food, the effects are more localized to the gastrointestinal tract. Side effects may include abdominal pain, nausea, vomiting, headache, dizziness, and reversible hair loss.

Mebendazole: Mebendazole is mostly used to treat pinworms. It is also used to treat whipworms, roundworms, and hookworms. It prevents worms from being able to absorb glucose. As a result, the worms run out of energy and die. People with bone marrow concerns or liver disease should consult with their primary care provider before taking mebendazole. Side effects may include abdominal pain, bloating, hair loss or thinning of the hair, appetite changes, flatulence, and weight loss. If you experience any of the following symptoms while taking this drug, contact your provider: changes in stool or urine color; hives or swelling, convulsions, unusual bleeding or bruising, yellowing of skin or eyes, or upper right abdominal pain.

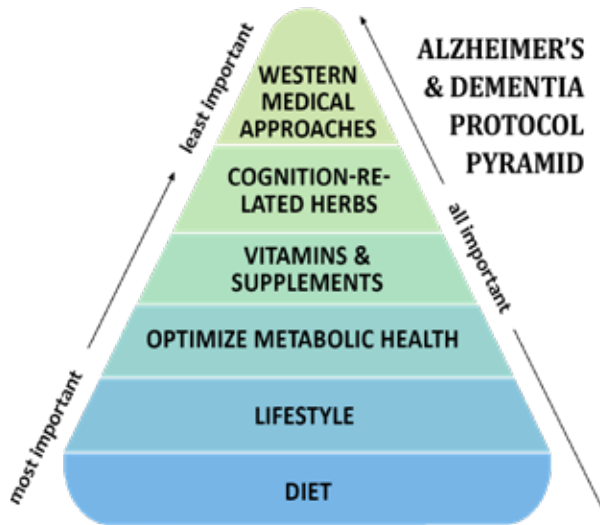
	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Dose 1: Anti-parasitic herb(s)* Wormwood extract w/ food** Clove tincture Probiotics Selenium: 2 Brazil nuts	Grocery shop for garlic, ginger, high fiber foods, & Beta-carotene-rich foods Clove tincture Green tea extract One-time dose of vitamin A or curly mint, if needed Garlic & ginger	Dose 2: Anti-parasitic herb(s)* Pumpkin or papaya seeds, if needed	Clove tincture Pumpkin or papaya seeds, if needed Physical activity (aim for 30-45 min.) Green tea extract Zinc	Unsweetened ginger tea (or tincture)
MONDAY	Dose 1: Anti-parasitic herb(s)* Wormwood extract** Clove tincture Probiotics Selenium: 2 Brazil nuts	Clove tincture Green tea extract Garlic & ginger	Dose 2: Anti-parasitic herb(s)* Pumpkin or papaya seeds	Clove tincture Pumpkin or papaya seeds Physical activity (30-45 min.) Green tea extract Zinc	Unsweetened ginger tea (or tincture)
TUESDAY	Dose 1: Anti-parasitic herb(s)* Wormwood extract** Clove tincture Probiotics Selenium: 2 Brazil nuts	Clove tincture Green tea extract Garlic & ginger	Dose 2: Anti-parasitic herb(s)* Pumpkin or papaya seeds	Clove tincture Pumpkin or papaya seeds Physical activity (30-45 min.) Green tea extract Zinc	Unsweetened ginger tea (or tincture)
WEDNESDAY	Dose 1: Anti-parasitic herb(s)* Wormwood extract** Clove tincture Probiotics Selenium: 2 Brazil nuts	Clove tincture Green tea extract Garlic & ginger	Dose 2: Anti-parasitic herb(s)* Pumpkin or papaya seeds	Clove tincture Pumpkin or papaya seeds Walk after dinner Green tea extract Zinc	Unsweetened ginger tea (or tincture)
THURSDAY	Dose 1: Anti-parasitic herb(s)* Wormwood extract** Clove tincture Probiotics Selenium: 2 Brazil nuts	Clove tincture Green tea extract Garlic & ginger	Dose 2: Anti-parasitic herb(s)* Pumpkin or papaya seeds	Clove tincture Pumpkin or papaya seeds Physical activity (30-45 min.) Green tea extract Zinc	Unsweetened ginger tea (or tincture)
FRIDAY	Dose 1: Anti-parasitic herb(s)* Wormwood extract** Clove tincture Probiotics Selenium: 2 Brazil nuts	Clove tincture Green tea extract Garlic & ginger	Dose 2: Anti-parasitic herb(s)* Pumpkin or papaya seeds	Clove tincture Pumpkin or papaya seeds Physical activity (30-45 min.) Green tea extract Zinc	Unsweetened ginger tea (or tincture)
SATURDAY	Dose 1: Anti-parasitic herb(s)* Wormwood extract** Clove tincture Probiotics Selenium: 2 Brazil nuts	Clove tincture Green tea extract Garlic & ginger	Dose 2: Anti-parasitic herb(s)* Pumpkin or papaya seeds	Clove tincture Pumpkin or papaya seeds Walk after dinner Green tea extract Zinc	Unsweetened ginger tea (or tincture)

*Anti-parasitic herbs: Yarrow, Ceylon cinnamon, Barberry, Oregon grape root, Chinese goldthread, Goldenseal (Berberine-containing herbs should not be taken for more than six weeks without a treatment break: See Protocol).

**Don't take wormwood for more than 7 days without a treatment break. See protocol for more information. Take w/ food.

Ch. 22: Alzheimer's and Dementia Protocol

In this chapter, we'll talk about mild cognitive impairment/decline and dementia. Mild cognitive impairment is an early stage of memory loss that may also include loss of other cognitive abilities, such as language and visual or spatial perception. Mild cognitive decline is different from dementia as individuals with mild cognitive decline maintain their ability to live independently and/or to carry out their activities of daily living independently.



Mild cognitive impairment is typically classified based on the thought processes that are affected.¹ For example, a person whose memory is primarily affected is said to have amnesic mild cognitive impairment, while a person whose other thinking skills are primarily affected (such as the ability to make sound decisions, visual processing, and/or the ability to judge the time or steps needed to complete a task) is said to have non-amnesic mild cognitive impairment.

A person could have mild cognitive impairment due to Alzheimer's disease; this is diagnosed using a combination of patient history, physical examination, and laboratory testing or brain imaging. Risk factors for mild cognitive impairment include a family history of Alzheimer's disease or another form of dementia, older age, and cardiovascular disease.

In contrast to mild cognitive impairment, dementia does affect a person's ability to complete their daily activities. The term "dementia" means memory loss and a decrease in language, problem-solving, and other thinking abilities that is severe enough to interfere with daily life. There are several types of dementia, with dementia due to Alzheimer's disease being the most common and vascular dementia being the second most common.

Dementia Due to Alzheimer's Disease: Dementia due to Alzheimer's disease is characterized by loss of mental function associated with brain changes, such as a decrease in brain volume and the formation of beta-amyloid plaques and neurofibrillary tangles in the brain. Alzheimer's disease is sub-categorized into the

following stages-early-stage Alzheimer's, middle-stage Alzheimer's, and late-stage Alzheimer's.²

In early-stage Alzheimer's, the person may retain some of their independence. They may still be able to drive, work, and take care of themselves. However, they may have a sense that they are having lapses in memory, such as forgetting everyday words or where they've put common objects.

In middle-stage Alzheimer's, the person may have a difficult time expressing thoughts or performing activities without help. They may also confuse their words, become frustrated or even angry because they sense their declining function, or act in ways they don't typically act.

In late-stage Alzheimer's, symptoms are severe. Their ability to converse is severely affected and may be completely lost, personality changes may arise, and the person tends to be largely unaware of their environment. The ability to walk, sit, and eventually swallow is compromised and risk for infection increases.

Vascular Dementia: Vascular dementia is a term for difficulty with reasoning, planning, judgment, memory, and other thought processes that is specifically due to disruptions in the brain's blood flow. In vascular dementia, impaired blood flow to the brain leads to brain damage, which affects mental functions. Most cases of vascular dementia develop after a stroke, although strokes don't always cause vascular dementia.

The most common symptoms of vascular dementia at its onset include slowing of thought processing; problems concentrating, understanding, and planning; and mood or behavior changes. As symptoms progress, mood and personality changes may become more severe; problems with urinary incontinence, balance, concentration, memory, and language may arise; and the person may feel disoriented or confused.

In this chapter, we'll discuss interventions that can help prevent or slow the progression of mild cognitive impairment and dementia.

1. Diet There are dietary habits associated with increased risk for dementia. These include diets that are high in saturated and trans-unsaturated fat, diets that contain high-mercury foods, and diets that are low in antioxidants.

Saturated and Trans-Unsaturated Fat: A meta-analysis involving 8,630 participants found that higher dietary intake of saturated fat was associated with a 39% increase in Alzheimer's disease risk and a 105% increase in risk for dementia of any kind.³ Trans-unsaturated fatty acids, which are usually obtained from baked goods and other processed foods in the form of partially hydrogenated vegetable oils, lead to negative changes in cholesterol level. These fats cause an increase in low-density lipoprotein (LDL) cholesterol and a decrease in high-density lipoprotein (HDL), both of which are associated with dementia risk.^{4 5} Opt for more foods that are lower in saturated and trans-unsaturated

fats, such as unprocessed or minimally processed plant-based foods.

High-Mercury Foods: Evidence suggests a correlation between mercury levels and the development of Alzheimer's disease. Some genetic factors that increase risk for Alzheimer's (such as the apolipoprotein E4 allele) increase the risk for mercury toxicity as well. Additionally, mercury inhibits DNA synthesis in the hippocampus; this has been associated with genetic mutations of presenilin 1 and 2, which are both linked with the development of Alzheimer's disease.⁶ Limiting your fish intake to low-mercury fish such as salmon, mackerel, anchovies, sardines, and herring may help reduce Alzheimer's disease risk.

Antioxidants: The formation of beta-amyloid plaques and neurofibrillary tangles in the development of Alzheimer's disease involves oxidative and inflammatory processes. Antioxidants, such as those found in foods that are rich in vitamins C and E may help reduce oxidative damage and inflammation and decrease neuronal cell loss as a result.^{7,8} Increasing your intake of high vitamin C and E foods, such as strawberries, cruciferous vegetables (e.g., broccoli, brussel sprouts, cabbage, cauliflower, kale, bok choy, etc.), kiwi and other citrus fruits, red bell peppers, sunflower seeds, and almonds may help reduce your risk of developing dementia.

2. Lifestyle

There are several lifestyle factors that can help limit the progression of cognitive decline and Alzheimer's disease. These include engaging in physical activity, getting adequate sleep, learning a second language, and doing cognitive training exercises.

Physical Activity: Research shows that physical activity, specifically aerobic exercise (e.g., swimming, cycling, using an elliptical trainer, walking, rowing, etc.), may improve cognitive decline and reduce dementia risk. Seniors who began an aerobic exercise program showed a reduction in age-related loss of gray matter in the brain.⁹ Seniors who engaged in aerobic exercise for one year had larger volumes of gray matter in the hippocampus and in other regions of the brain than seniors who did stretching and toning exercises.¹⁰

If you're not in the habit of exercising regularly, here's how researchers had the participants start: start by walking for 10 minutes per day for the first week, and then increase the time you spend walking by 5 minutes each week until you are walking for 40 minutes per day (e.g., 15 minutes per day in week 2, 20 minutes per day in week 3, etc.). Continue walking for 40 minutes each day, starting and ending each session with five minutes of stretching. (i.e. 5 minutes of stretching, 10 to 40 minutes of walking (depending on which week you are in), and then 5 more minutes of stretching. If possible, wear a heart-rate monitoring device and aim for 50–60% of the maximum heart rate reserve* for the first seven weeks and 60–75% of your maximum heart rate moving forward.

and 70 is associated with a 30% increase in dementia risk as compared to people with a sleep duration of 7 hours¹¹. Aim for at least 7 hours of sleep per night.

Learn a Second Language: Research suggests that being bilingual may delay the onset of Alzheimer's-related symptoms.¹² The evidence suggests that speaking two or more languages may contribute to the plasticity of the brain, which then contributes to cognitive reserve (think of this as helping the brain become more resilient) and protects against earlier onset of cognitive decline.

Cognitive Training: Research demonstrates that 60- to 75-minute cognitive training exercise sessions that focus on helping people build or maintain skills in the areas of memory, visual speed of processing, and reasoning lead to improvements in those skills. Those who participated in these cognitive training sessions were significantly less likely to experience decreases in quality of life up to 5 years after going through 10 sessions over 5 to 6 weeks.¹³

Older drivers who trained in the area of visual speed of processing were 40% less likely to have to stop driving up to three years after the sessions. Speed of processing and reasoning training led to a 50% decrease in at-fault motor vehicle collisions. Good examples of brain exercises include: learning an instrument, playing music, doing Sudoku and crossword puzzles, playing board games, puzzles, memory games, chess, and taking a class or starting a new hobby.

Meditation: A 2018 study showed that people with cognitive decline who meditated for just 12 minutes a day showed significant improvements in cognitive function, sleep, mood, and overall quality of life¹⁴. A 2021 study strongly suggests that long-term daily meditation may slow-down the neurodegenerative process in crucial areas of the brain of people with mild cognitive impairment or mild Alzheimer's disease. Long-term meditation practice led to beneficial changes in both cortical thickness and gray matter volume. Most of these changes were in the areas that are prominently at risk in neurodegenerative diseases – the brain areas related to executive control and memory¹⁵.

3. Optimize Metabolic Health

Poorly controlled blood sugar significantly increases your risk of developing Alzheimer's disease. Some researchers refer to Alzheimer's disease as “diabetes of the brain” or “type-III diabetes.”¹⁶ In order to minimize your risk of developing cognitive decline or Alzheimer's, it is critical to get your metabolic health under control. This includes optimizing your blood pressure, cholesterol, and blood sugar levels. It also includes maintaining a healthy weight through regular physical activity and a healthy diet. Herbs and supplements that can help you achieve and maintain optimal metabolic health include bitter melon, cordyceps, coenzyme Q10, fenugreek, *Gymnema sylvestre*, hawthorn, and many others. Please see the Heart Health Protocol on page 42 and the Diabetes Type-2 Protocol on page 96, if these are issues for you.

4. Vitamins and Supplements

There are several vitamins and supplements that help address concerns of cognitive decline and Alzheimer's disease. These include acetyl L-carnitine, citicoline, vitamin D, lithium orotate, omega-3 fatty acids, probiotics, and vitamin B12.

Acetyl L-Carnitine: Carnitine is an amino acid that is naturally found within our bodies. Research suggests that as people progress toward cognitive decline or Alzheimer's disease, our carnitine levels decline.¹⁷ Acetyl L-carnitine, a highly bioavailable form of carnitine, may improve brain health by enhancing mitochondrial function, supporting neurotransmitter activity, and improving cognition.^{18 19 20} Suggested dose: 2-3 grams per day.

Citicoline: Citicoline is a compound found within our bodies, typically within our cell membranes. Citicoline has neuroprotective properties. It can prevent cognitive decline after a stroke and help facilitate repair of the brain and nervous system in vascular (stroke-related) cognitive impairment, vascular dementia, and Alzheimer's disease.²¹ Suggested dose: 1000 to 1500 mg citicoline per day with meals, in divided doses.

Vitamin D: Research suggests that people with low vitamin D levels may be 53% more likely to develop dementia, and people who are severely deficient had 125% increased risk of developing dementia compared to people with normal vitamin D levels.²² Suggested dose: 5000-10,000 IU per day of vitamin D3 until 25-hydroxyvitamin D levels are at 40-60 ng/mL (first get your vitamin D levels tested) then decrease to a maintenance dose of 5000 IU per day.

Lithium Orotate: Lithium carbonate is usually prescribed in moderate doses as a mood stabilizer to address psychiatric concerns. At these dosages, kidney, thyroid, and parathyroid gland function must be monitored.²³ Lithium, however, is a naturally-occurring element. It is found in our food and water, and small amounts of lithium in drinking water correlates with improved mental health.²⁴ Small doses of lithium (e.g., 5 mg per day) are generally considered to be safe. Because of the improvements in mental well-being and cognition associated with small doses of lithium, researchers conducted a 15-month clinical trial. In this study, small doses (0.3 mg) of lithium slowed cognitive decline in people with Alzheimer's disease.²⁵ Research further suggests that lithium slows the progression of Alzheimer's-related changes in the brain and promotes brain cell regeneration in the hippocampus.²⁶ Lithium may interact with some antidepressants, non-steroidal anti-inflammatory drugs such as Celebrex, caffeine, theophylline, and blood pressure medications. Suggested dose: up to 5 mg of lithium (as lithium orotate) per day, with meals.

Omega-3 Fatty Acids: Docosahexaenoic acid (DHA) and eicosapentaenoic acid (EPA) are two types of omega-3 fatty acids that have beneficial effects on the brain. Specifically, the antioxidant and anti-inflammatory

properties of these fatty acids and their neurotrophic effects contribute to improved cognition in people with mild cognitive impairment and very mild Alzheimer's disease.^{27 28} Suggested dose: 2400 grams of omega-3 fatty acids (DHA and EPA) per day with meals, in divided doses.

Probiotics: A healthy gut microbiome is a critical part of achieving and maintaining normal brain function. In Alzheimer's disease, there is an increase in brain inflammation due to the presence of reactive oxygen species. There is also an increase in cell death, and a corresponding decrease in memory, cerebral glucose metabolism, and mental status. Research demonstrates that restoring the health of the gut microbiome through probiotics and prebiotics leads to decreases in brain inflammation, reactive oxygen species, beta-amyloid plaque formation, and anti-inflammatory chemical messengers.²⁹ Suggested dose: 100 billion CFU per day with meals, in divided doses.

Vitamin B12 and Folate: People who have been diagnosed with Alzheimer's disease tend to have lower levels of vitamin B12 than people who do not have Alzheimer's.³⁰ Elevated plasma homocysteine, which can be indicative of low levels of vitamin B12 or low folate, has been associated with the onset of dementia.³¹ Research suggests that B12 and folate supplementation helps slow the rate of brain atrophy in cases of mild cognitive impairment.³² Suggested dose: 1360 mcg dietary folate equivalents (DFE) and 0.5 mg vitamin B12 per day.

5. Cognition-Related Herbs

There are several cognition-related herbs that can be used to improve memory and delay the progression of cognitive decline. These include lemon balm, lion's mane mushroom, brahmi, Chinese skullcap, ginger, ginkgo, gotu kola, Indian gooseberry, reishi, and turmeric.

Lemon Balm (*Melissa officinalis*): Lemon balm is associated with improvements in memory, mood, and age-related cognitive performance, a reduction in stress and anxiety, and improved clarity and focus. Lemon balm is an antioxidant, helping to protect the brain and nerves. Lemon Balm also inhibits the brain's levels of acetylcholinesterase (AChE), an enzyme that helps break down acetylcholine (ACh), a critical neurotransmitter involved in cognition and memory. Reduced levels of acetylcholine have been associated with dementia, Parkinson's, and Alzheimer's disease. Suggested dose: 2 dropperfuls of lemon balm 1:3 tincture, 2 times daily.

Lion's Mane (*Hericium erinaceus*): The dominant components of lion's mane are hericenones and erinacines, which possess significant nerve repair ability. Lion's mane mushroom is neuroprotective and is known to help nervous system degeneration, decrease gut inflammation, boost brain function, lessen stress, anxiety, depression, and insomnia, and may increase bone density. It helps repair damaged nerve cells in the spinal cord and brain.³³ Research shows lion's mane mushroom stimulates Nerve Growth Factor (NGF), which is

crucial for the repair of the protective, fatty coating surrounding nerve fibers known as the myelin sheath³⁴; studies demonstrate great potential for myelination and regeneration of nerves. NGF proteins also help to maintain the neurons responsible for attention, learning, and memory. Lion's Mane helps reduce symptoms and brain cell degeneration³⁵ in multiple sclerosis, Alzheimer's, dementia, and Parkinson's in animal studies, as well as the progression of dementia.³⁶ Lion's mane boosts neuronal function³⁷ and significantly improves cognitive performance.³⁸ Lion's mane also speeds nervous system recovery; it is indicated for strokes due to the compound hericenone B, which shows strong anti-platelet activity, reduces nerve pain, and might be a novel compound for antithrombotic therapy. It is also an anti-oxidant and an anti-inflammatory. Suggested dose: 2 dropperfuls of a dual-extracted tincture, once to twice daily³⁹.

Bacopa/Brahmi (*Bacopa monnieri*): In healthy people, *Bacopa monnieri* significantly improves processing speed for visual information, learning rate, and memory consolidation, and also decreases anxiety.⁴⁰ Research also demonstrates that it improves cognition in people with Alzheimer's disease.⁴¹ Suggested dose: 300 mg of *Bacopa* extract, twice daily or 2 dropperfuls of a 1:2 tincture, twice daily.

Chinese Skullcap (*Scutellaria baicalensis*): Chinese skullcap has anti-inflammatory and antioxidative properties, and it slows the progression of neurodegenerative diseases.⁴² Chinese skullcap protects against beta-amyloid plaque formation and cerebral ischemia, which can increase your risk for vascular dementia and memory deficits. Its neuroprotective activity has been attributed to its main active constituents: baicalin, baicalein, wogonin, and scutebarbatine B. Suggested dose: 300-900 mg in capsule form or 2 dropperfuls of tincture, twice daily.

Ginger (*Zingiber officinale*): Preclinical research suggests that ginger may encourage brain cells to form new connections, leading to recognition of new objects and improvements in learning and memory.⁴³ Similar results have been noted in humans, with improvements in working memory and cognitive function noted after 2 months of middle-aged women taking the herb.⁴⁴ Suggested dose: 400-800 mg of ginger per day in capsule form; 2-3 teaspoons of chopped ginger per cup of water, three times daily; 3 to 4 dropperfuls of a 1:3 tincture, 3 times daily.

Ginkgo (*Ginkgo biloba*): Ginkgo has beneficial effects on cognition, neuropsychiatric symptoms, and completion of daily activities when taken by individuals with mild cognitive impairment and dementia.⁴⁵ Please note that ginkgo has an additive effect and should be taken for at least six weeks before reevaluation. Suggested dose: 240 mg ginkgo per day or 5 dropperfuls of a 1:5 tincture, twice daily.

Gotu Kola (*Centella asiatica*): Gotu kola helps reduce vascular cognitive impairment (memory problems after a stroke).⁴⁶ Suggested dose: 1000 mg of gotu

kola extract per day; 4 dropperfuls of a 1:5 tincture, twice daily.

Indian Gooseberry (*Phyllanthus emblica*): Indian gooseberry exhibits cognition-enhancing activity, which is likely due to its antioxidant, anti-inflammatory, and neuroprotective properties.⁴⁷ It may increase the levels of acetylcholine in the brain and enhance cognitive function as a result. It may also increase levels of superoxide dismutase, glutathione peroxidase, and other enzymes that have antioxidative effects. Suggested dose: 1-3 grams of powdered, dry fruit per day.

Reishi (*Ganoderma lucidum*): Reishi mushroom is a formidable adaptogenic anti-inflammatory that helps modulate inflammation and immune function. Reishi has neuroprotective qualities and is effective in calming the inflammation and immune response involved with Leaky Gut. It also helps to strengthen the body against the negative effects of stress—like inflammation, hormonal imbalances, high cortisol, fatigue, brain fog, and low energy. Research suggests that it could serve as a regenerative therapeutic agent for the treatment of cognitive decline associated with neurodegenerative diseases.⁴⁸ Reishi also suppresses certain inflammatory compounds, inhibits histamine release,⁴⁹ is an immunomodulator, and targets inflammatory cytokines.⁵⁰ Suggested dose: 2 dropperfuls of a dual-extracted reishi tincture, once to twice daily.

Turmeric (*Curcuma longa*): Research suggests that turmeric helps degrade beta-amyloid plaques, remove heavy metals from the body, and reduce inflammation and oxidative stress.^{51 52} Turmeric's ability to improve memory in patients with Alzheimer's disease has been attributed to these properties. Suggested dose: 2000 mg bioavailable turmeric per day, with fat-containing meals in divided doses or 4 dropperfuls of a 1:2 tincture with meals, twice daily.

6. Western Medical Approaches

There are a few conventional medications that are prescribed to address cognitive decline and Alzheimer's disease. These include galantamine, rivastigmine, and donepezil; aducanumab; memantine; and suvorexant.

Galantamine, Rivastigmine, and Donepezil: Donepezil (Brand name: Aricept®), rivastigmine (Brand name: Exelon®), and galantamine (Brand name: Razadyne®) are cholinesterase inhibitors, meaning they work by blocking the breakdown of a neurotransmitter called acetylcholine. These drugs are prescribed with a goal of improving memory, thinking, judgment, language and other thought processes in people with Alzheimer's dementia. While donepezil is approved by the Food and Drug Administration (FDA) to treat all stages of Alzheimer's, rivastigmine and galantamine are approved to treat mild-to-moderate Alzheimer's disease. Side effects may include nausea, vomiting, appetite changes, and an increase in bowel movement frequency.

Aducanumab: Aducanumab (Brand name: Aduhelm™) is a drug that is prescribed with a goal of delaying clinical decline in people with mild (early stage) cognitive

impairment due to Alzheimer's dementia. The drug is administered intravenously and a diagnostic test must be conducted before this medication can be prescribed.

The drug is believed to work by removing brain beta-amyloid plaques, which are suspected to contribute to the development of Alzheimer's disease. It's important to note that the approval of this drug by the Food and Drug Administration was controversial, because clinical trials originally did not find it to be effective at addressing Alzheimer's-related concerns.

Memantine: Memantine (Brand name: Namenda®) works by regulating a neurotransmitter called glutamate with a goal of improving memory, attention, language,

reasoning, and the ability to perform simple tasks. This drug is FDA-approved to treat moderate-to-severe Alzheimer's disease. Side effects of this drug may include confusion, headache, dizziness, and constipation.

Suvorexant: Suvorexant (Brand name: Belsomra®) is typically prescribed to treat insomnia in people who have dementia. This drug works by inhibiting a neurotransmitter called orexin, which is involved in regulating our patterns of sleeping and waking. It is FDA-approved for use in mild-to-moderate Alzheimer's disease. Potential side effects include headache, drowsiness (even into the following day), dizziness, abnormal dreams, increased risk for upper respiratory tract infections, cough, dry mouth, and diarrhea.

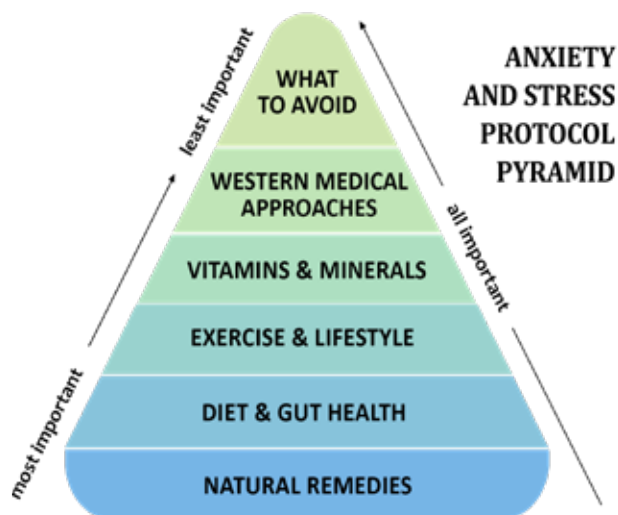
	Morning	Noon	Afternoon	Evening	Before Bed
S U N D A Y	Dose 1: Bacopa, Lemon Balm, Lion's mane, & Chinese Skullcap Tinctures Acetyl L-carnitine Vitamins B12, D, Folate & Probiotics	Weekly grocery shop for healthy foods Other cognition herbs of your choice* Citicoline, Omega-3s, Lithium orotate & Turmeric w/ lunch	Hydration check (Are you drinking enough water?) Reishi tincture Physical activity (work up to 30-45 min) and Cognitive brain exercises	Dose 2: Bacopa, Lion's mane, & Chinese Skullcap tinctures Other cognition herbs of your choice* Citicoline, Omega-3s, Lithium orotate & Turmeric w/ dinner	Meditation Lemon balm and ginger tea or tincture
M O N D A Y	Dose 1: Bacopa, Lemon Balm, Lion's mane, & Chinese Skullcap Tinctures Acetyl L-carnitine Vitamins B12, D, Folate & Probiotics	Other cognition herbs* Citicoline, Omega-3s, Lithium orotate & Turmeric w/ lunch Walk after lunch	Hydration check Reishi tincture Cognitive brain exercises	Dose 2: Bacopa, Lion's mane, & Chinese Skullcap Other cognition herbs * Walk after dinner Citicoline, Omega-3s, Lithium orotate & Turmeric w/ dinner	Meditation Lemon balm and ginger tea or tincture
T U E S D A Y	Dose 1: Anti-parasitic herb(s)* Wormwood extract** Clove tincture Probiotics Morning Walk	Other cognition herbs* Citicoline, Omega-3s, Lithium orotate & Turmeric w/ lunch Cognitive brain exercises	Hydration check Reishi tincture Physical activity (30-45 min.)	Dose 2: Bacopa, Lion's mane, & Chinese Skullcap Other cognition herbs Citicoline, Omega-3s, Lithium orotate & Turmeric w/ dinner	Meditation Lemon balm and ginger tea or tincture
W E D N E S D A Y	Dose 1: Bacopa, Lemon Balm, Lion's mane, & Chinese Skullcap Tinctures Acetyl L-carnitine Vitamins B12, D, Folate & Probiotics	Other cognition herbs* Citicoline, Omega-3s, Lithium orotate & Turmeric w/ lunch	Hydration check Reishi tincture Cognitive brain exercises	Dose 2: Bacopa, Lion's mane, & Chinese Skullcap Other cognition herbs * Walk after dinner Citicoline, Omega-3s, Lithium orotate & Turmeric w/ dinner	Meditation Lemon balm and ginger tea or tincture
T H U R S D A Y	Dose 1: Bacopa, Lemon Balm, Lion's mane, & Chinese Skullcap Tinctures Acetyl L-carnitine Vitamins B12, D, Folate & Probiotics	Other cognition herbs* Citicoline, Omega-3s, Lithium orotate & Turmeric w/ lunch Cognitive brain exercises	Hydration check Reishi tincture Physical activity (30-45 min.)	Dose 2: Bacopa, Lion's mane, & Chinese Skullcap Other cognition herbs Citicoline, Omega-3s, Lithium orotate & Turmeric w/ dinner	Meditation Lemon balm and ginger tea or tincture
F R I D A Y	Dose 1: Anti-parasitic herb(s)* Wormwood extract** Clove tincture Probiotics Morning Walk	Other cognition herbs* Citicoline, Omega-3s, Lithium orotate & Turmeric w/ lunch	Hydration check Reishi tincture Cognitive brain exercises	Dose 2: Bacopa, Lion's mane, & Chinese Skullcap Other cognition herbs * Walk after dinner Citicoline, Omega-3s, Lithium orotate & Turmeric w/ dinner	Meditation Lemon balm and ginger tea or tincture
S A T U R D A Y	Dose 1: Bacopa, Lemon Balm, Lion's mane, & Chinese Skullcap Tinctures Acetyl L-carnitine Vitamins B12, D, Folate & Probiotics	Other cognition herbs* Citicoline, Omega-3s, Lithium orotate & Turmeric w/ lunch Cognitive brain exercises	Hydration check Reishi tincture Physical activity (30-45 min.)	Dose 2: Bacopa, Lion's mane, & Chinese Skullcap Other cognition herbs Citicoline, Omega-3s, Lithium orotate & Turmeric w/ dinner	Meditation Lemon balm and ginger tea or tincture

*Options for other cognition-related herbs: Ginger, Gotu kola, Ginkgo, and Indian gooseberry.

** Examples include: learning a 2nd language or a new instrument, playing music, doing Sudoku and crossword puzzles, playing board games, puzzles, memory games, chess, and taking a class or starting a new hobby.

Ch. 23: Anxiety and Stress Protocol

Considered the most common group of mental health disorders, with women twice as likely to be diagnosed, anxiety disorders are a major global health problem.¹ Anxiety is defined as apprehension, tension, or uneasiness that stems from the anticipation of danger.² Those with anxiety often experience excessive fear and worry or avoidance of perceived threats. Anxiety is often accompanied by a variety of physiological experiences, such as lightheadedness, tingling in extremities, hyperventilation, and a sense of “unreality”.³



Anxiety can be a perfectly normal response to a dangerous or distressing situation, so it's important to determine if a person is experiencing anxiety that is appropriate to the situation or a mental health condition. There are many manifestations of anxiety disorders, including: chronic worrying, phobias, social anxiety, public speaking anxiety, shyness, OCD (Obsessive-Compulsive Disorder), PTSD (Post-Traumatic Stress Disorder), panic attacks, agoraphobia, and BDD (Body Dysmorphic Disorder).⁴

Research suggests anxiety disorders involve dysfunction in brain circuits that respond to danger.⁵ The exact cause is unknown, however, risk factors include distorted thought patterns, neurotransmitter dysregulation, genetic factors, trauma, nutritional status, and environmental factors.^{6, 7}

It's important to note that some illnesses like endocrine diseases (such as hyperthyroidism, hypoparathyroidism, Cushing's disease, carcinoid syndrome, and pheochromocytoma) can cause anxiety.⁸ In these scenarios, anxiety is not a result of a mental health disorder, but secondary to a medical condition that must be addressed.

Conventional medicine is effective for treating acute anxiety, however, it poses many limitations when used as a long-term treatment.⁹ As a result, alternative therapies have become popular treatment options for people struggling with anxiety disorders. Emerging research shows promising outcomes for a collaborative approach to treating anxiety disorders.¹⁰

1. Natural Remedies

Personally, I use my Anxiety and Stress tincture, a blend of ashwagandha, lemon balm, lion's mane mushroom, and reishi mushroom when I am feeling anxious or stressed. Many people with anxiety use it daily with excellent results. These four herbs plus lavender, holy basil, and CBD are my top choices for daytime use. Sleep herbs that also work for anxiety include hops, passionflower, chamomile, and valerian root. A sleep blend tincture of these herbs taken at night is a popular choice. See below for these and other herbs that may work for you as part of this holistic anxiety and stress protocol.

Calming adaptogens like reishi and ashwagandha help downregulate hormone levels if they are too high and upregulate hormone levels if they are too low. For example, if your cortisol levels have gotten high enough to invoke a “fight or flight” response, adaptogens work to lower these levels.

Ashwagandha (*Withania somnifera*): Ashwagandha is one of the most important herbs in traditional Ayurvedic medicine in India. It is sometimes called “Indian ginseng” because it acts primarily as an adaptogen to help mitigate the harmful effects of stress.¹¹ Like many adaptogens, it affects hormones and is commonly used for low libido and infertility. Unlike some adaptogens, it is calming rather than stimulating. Ashwagandha has a sedative effect and is often used for stressed individuals who are overexcited, nervous, and having difficulties with sleep.¹² It is generally considered safe, however, it may stimulate thyroid function so caution in hyperthyroidism.¹³ Suggested dose:¹⁴ 500mg daily or 2 dropperfuls of tincture 3 times daily.

Cannabidiol (CBD): A constituent of the plant *Cannabis sativa*, CBD has a promising role as an alternative therapy in the management of mental health and anxiety disorders.¹⁵ CBD is usually well tolerated with minimal adverse effects; the most common being fatigue and sedation.¹⁶ CBD is thought to interact with receptors found in the endocannabinoid system, a complex cell signaling network in the nervous system involved in feel-good neurotransmitters like serotonin.¹⁷ Serotonin plays an important role in mental health. Low levels of serotonin are correlated with anxiety.¹⁸ Unlike THC, the use of CBD doesn't result in alternated mental states, like feeling “high”. CBD is also associated with pain reduction.¹⁹ Though it's often well-tolerated, CBD can cause side effects, such as dry mouth, diarrhea, reduced appetite, drowsiness, and fatigue. CBD can also interact with other medications, such as blood thinners.²⁰ Suggested dose:²¹ 400mg daily (This dosage is not currently available in tincture form, but can be found in CBD gummies.).

German Chamomile (*Matricaria recutita*): A 2016 study found that long-term use of chamomile extract significantly reduced moderate to severe symptoms of generalized anxiety disorder.²² Additional research has shown that chamomile improves overall sleep quality.²³ Suggested dose: one cup of tea or a dropperful of tincture at night.

Holy Basil/Tulsi (*Ocimum sanctum*): A revered herb in Ayurvedic medicine, research suggests that Holy Basil is an effective treatment for psychological stress, such as nervous exhaustion, irregular mood disturbances, depression, anxiety, poor cognition, and attention deficit hyperactivity disorder (ADHD, ADD).²⁴ No studies have reported any significant adverse events with the use of Holy Basil.²⁵ Suggested dose:²⁶ 1000mg daily or 2.5 dropperfuls of tincture 3 times daily.

Hops (*Humulus lupulus*): A well-known component of beer, this plant has sedative properties that boost GABA levels. Hops significantly reduces anxiety, depression, and stress levels in those with mood disorders.²⁷ It is commonly used for those who have an over-stimulated nervous system, anxiety, restlessness, and tension headaches.²⁸ While it's considered safe, due to its sedative effects, it should not be combined with other sedatives such as benzodiazepines, barbiturates, alcohol, or sleep aids. Additionally, there have been some reports that hops may accentuate symptoms of depression, so caution should be used in those with depression.²⁹ If you would like the anti-anxiety effects of the herb without the strong sedative effects, use dried hops as fresh have a strong sedative quality. Suggested dose: 620mg per day, or 2 dropperfuls of tincture 3 times daily.

Kava Kava (*Piper mythesticum*): Kava Kava is best known for its relaxing qualities.³⁰ The active ingredients are a group of substances known as kava lactones.³¹ Research has shown its ability to mitigate over-excited states and mood disorders such as anxiety, insomnia, depression, and obsessive-compulsive disorder (OCD).³² Due to its sedative effects, it should not be combined with other sedatives such as benzodiazepines, barbiturates, alcohol, or sleep aids.³³ Some research has shown that long-term, heavy usage of the herb can cause a scaly skin rash called "kava dermatopathy" and possible liver damage.³⁴ Suggested dose:³⁵ 620mg daily or 2 dropperfuls of tincture per day.

Lavender (*Lavendula angustifolia*): For centuries, this plant has been used for its anxiolytic and calming properties. Several clinical trials have substantiated the efficacy of oral supplementation with concentrated lavender oil for symptoms of anxiety, depression, restlessness, agitation, and disturbed sleep.³⁶ Concentrated lavender oil is an effective and well-tolerated alternative to benzodiazepines for amelioration of generalized anxiety with no potential for drug abuse.³⁷ Adverse effects are very rare but could include upset stomach, lavender burps, or other GI effects.³⁸ Suggested dose: ³⁹160mg daily or 1 dropperful of tincture per day.

Lemon Balm (*Melissa officinalis*): A popular herb for anxiety, Lemon Balm is well known for its anxiolytic effects. Research has demonstrated the active compound rosmarinic acid improves mood and cognitive performance. It is known for reducing rumination and increasing levels of the neurotransmitter GABA in the brain.⁴⁰ It also has many brain benefits (see the Brain and Nerve protocol for more). Lemon Balm may interact with thyroid and HIV medications. If you are taking these ask

your doctor before taking this herb.⁴¹ Suggested dose:⁴² 1,600mg daily or 2 dropperfuls of tincture 3 times daily.

Lion's Mane (*Hericium erinaceus*): A well-known edible mushroom, Lion's Mane has numerous biological activities. Two active constituents, hericenones and erinacines are able to pass the blood-brain barrier and stimulate nerve growth factor (NGF) synthesis, which may reduce anxiety through its effects on the autonomic nervous system.⁴³ Research suggests lion's mane may have dual-function actions for improving sleep disruption while sustaining anxiolytic effects.⁴⁴ While generally safe, Lion's mane mushroom might slow blood clotting and reduce blood glucose levels. This might cause extra bleeding, and interfere with blood sugar control during and after surgery.⁴⁵ Suggested dose:⁴⁶ 3g daily or 4 dropperfuls of tincture per day.

Passion Flower (*Passiflora incarnata*): With a long history of use as a natural medicine for the treatment of insomnia and anxiety in Europe, and as a sedation tea in North America, Passion Flower is a popular herbal medicine.⁴⁷ Research suggests it helps reduce stress and can thus be helpful in the treatment of insomnia, anxiety, and depression.⁴⁸ It reduces nervous conditions following chronic disease or stress-induced restlessness, such as anxiety and insomnia.⁴⁹ While generally considered safe, due to its sedative effects, it should not be combined with other sedatives such as benzodiazepines, barbiturates, alcohol, or sleep aids.⁵⁰ Suggested dose:⁵¹ 800mg daily or 4 dropperfuls once per day.

Reishi (*Ganoderma lucidum*): Referred to as the "mushroom of immortality" due to its ability to promote longevity and calm the mind, Reishi has been a popular mushroom in Traditional Chinese Medicine for thousands of years. Traditionally, it's been used to reduce stress and anxiety.⁵² Research shows Reishi is a valuable medicinal fungus with many actions including anxiety and pain reduction.⁵³ While generally considered safe, taking Reishi mushroom along with medications that also slow blood clotting might increase the risk of bruising and bleeding.⁵⁴ Suggested dose:⁵⁵ 1-2 dropperfuls of a dual-extracted reishi tincture three times daily.

Rhodiola (*Rhodiola rosea*): Found in the arctic and at high altitudes, historically, it has been used in Europe, Asia, and Russia as an adaptogen, to help people cope with stress, and also to relieve fatigue.⁵⁶ Rhodiola also appears to improve anxiety and depression.⁵⁷ The antidepressant activity of Rhodiola is through its inhibition of monoamine oxidase (MAO), an enzyme involved in the modulation of various neurotransmitters involved in mood (e.g. serotonin, dopamine) and stress (epinephrine, norepinephrine).⁵⁸ In addition, Rhodiola may affect opioid receptors and has also historically been used to help with opioid addiction.⁵⁹ While Rhodiola is generally considered safe, due to its interaction with multiple neurotransmitters, it should not be combined with other mood-modifying medications without consulting a healthcare provider. Additionally, due to its potentially stimulating properties, it should

be taken in the morning to prevent insomnia.⁶⁰ It may be too stimulating for some people with autoimmune issues. Suggested dose:⁶¹ 400mg daily or 3 dropperfuls of tincture three times daily.

Skullcap (*Scutellaria lateriflora*): A native plant of North America, Skullcap has been used by Americans and Europeans as a nerve tonic for more than 200 years.⁶² Research has demonstrated its effectiveness in treating nervous conditions following chronic disease, stress-induced restlessness, anxiety, and insomnia.⁶³ Additionally, it has a positive effect on mood without a reduction in energy or cognition.⁶⁴ This herb is generally considered safe, however, there have been some case reports of liver toxicity when contaminated with Germander, a group of plants known to cause liver problems.⁶⁵ For this reason, it's important to ensure you obtain a pure source. Suggested dose:⁶⁶ 1050mg daily or 4 dropperfuls of tincture three times daily.

St. John's Wort (*Hypericum perforatum*): One of the most commonly used herbal medications in the United States, this plant has a long history of use for a variety of conditions associated with the nervous system.⁶⁷ The main active ingredient, hyperforin, has many effects on the nervous system including inhibiting the re-uptake of several neurotransmitters associated with mood (e.g. serotonin, dopamine, and GABA).⁶⁸ While mostly known for its antidepressant effects, St. John's Wort also reduces anxiety.⁶⁹ While generally considered safe St. John's Wort interferes with enzymes involved in drug metabolism, (CYP3A4, CYP2C9), which may decrease the effectiveness of certain drugs, including birth control and HIV drugs.⁷⁰ Additionally, this herb should not be used with other medications that alter neurotransmitter levels as it has the potential to induce serotonin syndrome.⁷¹ Please advise your doctor before taking this with any other medication. Suggested dose:⁷² 900mg or 4 dropperfuls of tincture three times daily.

Valerian (*Valeriana officinalis*): A popular nervine and sedative, valerian root has a calming effect on the nervous system by targeting several receptors and enzymes related to GABA, a main inhibitory neurotransmitter.⁷³ It is commonly prescribed for over-stimulated states and for patients feeling anxious.⁷⁴ Valerian is recommended acutely as a sleep aid, but for insomnia, best results are experienced with long-term use or in combination with herbs such as hops. Valerian has been used to help decrease patients' dependence on benzodiazepine drugs.⁷⁵ While generally considered safe, due to its sedative effects, it should not be combined with other sedatives such as benzodiazepines, barbiturates, alcohol, or sleep aids.⁷⁶ Suggested dose (2 hours before bed):⁷⁷ 600mg nightly or 2-4 dropperfuls of tincture.

2. Diet and Gut Health

Healthy eating is critical for managing anxiety. What we eat and how our digestive system functions play a major role in anxiety.

Blood Sugar Regulation: By maintaining healthy blood sugar levels, we send a physiological signal

to our bodies that we are safe. However, when our blood sugar levels are imbalanced, a physiological stress response is triggered by the body that can elicit or heighten feelings of anxiety. Foods that quickly spike our blood sugar (ex. simple carbohydrates) induce a rapid production of insulin. This process enables the sugar or "glucose" in our blood to be quickly absorbed into our cells; as a result we experience a rebound "dip" or "crash" in blood sugar levels. In an attempt to regulate these changing blood sugar levels, our body responds by releasing stress hormones like cortisol and adrenaline. These hormones initiate glycogen breakdown in the liver, a process that releases stored sugar into the blood to prevent the perceived threat of starvation.⁷⁸ The effects of these hormones can leave us feeling stressed, anxious, overwhelmed, and worried. By choosing foods that steadily release sugar into our bloodstream we prevent the spikes and dips that trigger stress hormone release. Nut butters, like almond butter, are great food choices as they are helpful in blood sugar regulation.⁷⁹

Simple vs. Complex Carbs: It's common knowledge that protein and vegetables are great foods to keep blood sugar levels steady, however, it's important to distinguish between simple and complex carbohydrates, as they have a major impact on our blood sugar. Simple carbs, like sweets and pasta, have a basic chemical structure that makes them easy for the body to digest. As a result, they enter the bloodstream quickly, causing spikes and dips in blood sugar.⁸⁰ Complex carbs, like sweet potatoes or quinoa, have longer more complex chemical structures, which makes them more difficult for the body to break down and digest. As a result, blood sugar levels rise much more slowly and are well regulated.

Microbiome Support: We all know the brain tells the rest of our body how to behave, but recent research reveals that the gut also tells our brain how to behave.⁸¹ The gut-brain axis is a term for the communication network between the gut and brain. Through the largest nerve in the body, the vagus nerve, and over 500 million neurons, your gut is in constant communication with the brain, impacting numerous processes in the body.⁸² Several neurotransmitters critical to mood are produced by the gut and by the trillions of microbes living there. It's estimated that gut bacteria manufacture about 95 percent of the body's supply of serotonin.⁸³ The most abundant bacteria in the gastrointestinal tract also produce large quantities of GABA.⁸⁴ It's no surprise that certain probiotics have been shown to improve symptoms of stress, anxiety, and depression.⁸⁵ Having a healthy microbiome can be supported by eating foods such as kimchi and sauerkraut that support the growth of good bacteria (probiotics) in the gut, and by taking probiotic supplements and herbs that help with gut health (please see our Leaky Gut Protocol at page 70 for more detail).

Eat Breakfast: Having even a small healthy snack in the morning may help stabilize blood sugar and prevent symptoms of anxiety. Research shows that skipping breakfast leads to increased concentrations of free cortisol in the body.⁸⁶ Higher levels of cortisol are present

in people who suffer from anxiety.⁸⁷ By taking steps to prevent excessive cortisol secretion you can reduce the likelihood of experiencing anxiety.

Avoid Caffeine: It has been well documented that caffeine can cause anxiety.⁸⁸ Excessive caffeine consumption can lead to anxiety disorders, increased hostility, and even psychotic symptoms.⁸⁹ While caffeine may not impact all individuals who suffer from anxiety, take a break from caffeine to see if it's impacting you. If you regularly drink caffeine, it is advised you slowly taper off over a period of a few weeks to prevent caffeine withdrawal symptoms.⁹⁰

Food Allergies and Intolerances: Food intolerances and sensitivities are extremely common. It's estimated that between 2–20% of people worldwide may suffer from food intolerance.⁹¹ Growing interest has emerged around the connection between food allergies and mental health. While food allergies and intolerances are not completely understood, a growing body of evidence has suggested allergies increase anxiety.⁹² An Elimination Diet (see page 300) is designed to avoid or remove certain foods that you may be sensitive or allergic to. While Elimination Diets require time and preparation, they are effective in the treatment of specific food allergies.⁹³

Avoid Intermittent Fasting (IF): While IF is extremely helpful for men and post-menopausal women in retraining their metabolism, intermittent fasting poses risks for women of reproductive age struggling with anxiety. By creating a state of ketosis and triggering starvation cascades, it may interfere with hormonal production (ovulation) and induce stress hormones which may worsen feelings of anxiety.⁹⁴

3. Exercise and Lifestyle **Exercise:** Research has repeatedly shown that exercise can be a powerful tool in treating anxiety.⁹⁵ Even a small amount of weekly exercise has a significant impact on anxiety.⁹⁶ By changing the chemistry in our brain, exercise increases the production of feel-good chemicals, resulting in a calming effect and reduction in feelings of anxiety.⁹⁷ Exercise improves mood in several ways. It diverts your attention from things you are anxious about,⁹⁸ physical activity decreases muscle tension, lowering the body's contribution to feeling anxious,⁹⁹ and it increases the heart rate thus increasing the availability of important anti-anxiety chemicals, including serotonin, gamma-aminobutyric acid (GABA), brain-derived neurotrophic factor (BDNF), endorphins, and endocannabinoids.¹⁰⁰ Additionally, exercise activates frontal regions of the brain responsible for executive function, which helps control the amygdala, our reaction system to real or imagined threats to our survival.¹⁰¹ Finally, exercising regularly enhances resilience to distressing emotions.¹⁰²

While research demonstrates that doing 30 minutes or more of exercise a day for three to five days a week may significantly improve anxiety symptoms, even smaller amounts of physical activity, as little as 10 to 15 minutes at a time, may lower anxiety.¹⁰³ An ideal amount of exercise is 3 times per week, for roughly 30-45 minutes

each session. By choosing activities you find enjoyable, like a brisk walk, a dance class, or playing sports, you increase the likelihood that you will remain committed to getting the exercise you need.

Breathing Exercises: Known as “deep breathing”, “lymphatic breathing”, “belly breathing”, and “diaphragmatic breathing”, breathing exercises are a wonderful way to reduce stress and improve overall health. Research shows that deep breathing or “diaphragmatic breathing” is especially helpful for reducing stress and symptoms of anxiety.¹⁰⁴ Deep breathing involves fully engaging the stomach, abdominal muscles, and diaphragm when inhaling and exhaling. It triggers body relaxation responses through the nervous system and benefits both physical and mental health.¹⁰⁵

Deep breathing also has a positive impact on the lymphatic system. A network of tissues, vessels, and organs that work together, the lymphatic system is a central component of the immune system. It is filled with sympathetic and parasympathetic nerves that are constantly sending signals throughout the body.¹⁰⁶ Research shows a relationship between the functioning of the lymphatic system and psychological stress.¹⁰⁷ Unlike the cardiovascular system, the lymph system is made of a series of one-way valves that require physical movement to function. When you engage in deep breathing exercises, the diaphragm engages one of these valves; sweeping lymph into the venous system where it can be processed. Deep breathing exercises thus enhance the function of the lymph system, which also positively impacts the nervous system and mood.

Mindfulness-Based Interventions (MBIs): Mindfulness is the practice of purposely bringing one's attention to the present moment without judgment. There are several interventions built around this concept. The most common include Mindfulness-Based Stress Reduction (MBSR) and Mindfulness-Based Cognitive Therapy (MBCT). MBIs reduce anxiety symptom severity in a broad range of individuals.¹⁰⁸ These therapies aim to change the way people think about situations, helping them focus on the present vs. the past or future. It also heightens awareness of negative thought processes to help people identify anxious thinking and reactions early to detach and remain centered and calm.¹⁰⁹

Time in Nature: While spending time in nature may seem like an obvious way to reduce stress, research has proven it's a potent anti-anxiety treatment. Research shows 5 specific effects of spending time in nature: improved immune system function (increase in natural killer cells/cancer prevention), reduction in hypertension/coronary artery disease, reduced allergies and respiratory disease, reduction in mood disorders and stress, mental relaxation, and increase in feelings of gratitude and selflessness.¹¹⁰

Cognitive Behavioural Therapy (CBT): CBT is a treatment approach that helps you recognize negative or unhelpful thought and behavior patterns. CBT uses a variety of techniques to address cognitive

distortions. These are thought patterns that distort a person's perception of reality causing anxious thoughts and feelings. For example, "magical thinking" is a common cognitive distortion in anxiety, where a person believes that the anxiety, although uncomfortable, protects him or her from some terrible catastrophe.¹¹¹ By working with a psychologist people with anxiety can learn to identify distorted thought patterns to learn strategies to stop anxious thoughts. CBT is an effective treatment for anxiety disorders, with outcomes equal to pharmaceutical treatment and more effective 6 months after study completion.¹¹²

Gratitude: Gratitude is defined as "the expression of appreciation for what one has"¹¹³. Gratitude is connected to having a less critical, less punishing, and more compassionate relationship with the self.¹¹⁴ Gratitude interventions result in increased positive affect, better life satisfaction, and fewer depression and anxiety symptoms.¹¹⁵ Gratitude also reduces "repetitive negative thinking", a common symptom of anxiety.¹¹⁶ Based on the available research the most convenient and effective way to practice gratitude is by using a gratitude app or keeping a daily gratitude journal.¹¹⁷

Epsom Salt Baths: Taking baths instead of showers improves both physical and emotional aspects of quality of life. Researchers suggest that bathing exerts hyperthermic action that activates the parasympathetic nervous system, a part of the body responsible for relaxation.¹¹⁸

Adding Epsom salts (a magnesium salt) is an excellent way to enhance the relaxation qualities of baths. Magnesium has shown benefits in the treatment of anxiety,^{119, 120} and prolonged soaking in Epsom salts increases blood magnesium concentrations.¹²¹ For a standard tub add 1-2 cups of Epsom salts to a warm bath.¹²² Adding lavender essential oil (15-20 drops) to a bath further enhances the relaxation effects.¹²³

Sleep: Please refer to the Sleep Protocol (see page 286).

4. Vitamins and Minerals

B-Complex Vitamin:

B vitamins are a group of eight different nutrients that work together to manage a variety of functions in the body and are one of the most commonly linked vitamins to reducing anxiety.¹²⁴ Research has repeatedly shown that a higher intake of B-vitamins is associated with a lower prevalence of depression, anxiety, and stress symptoms.¹²⁵ Suggested dose:¹²⁶ These can be found in a good B-complex vitamin. B1 (thiamine): 1.1 mg, B2 (riboflavin): 1.1 mg, B3 (niacin): 14 mg, B5 (pantothenic acid): 5 mg, B6 (pyridoxine): 1.3 mg, B7 (biotin): 30 mcg, B9 (folate): 400 mcg, B12 (cobalamin): 2.4 mcg.

Vitamin C: A water-soluble vitamin, best known as an antioxidant needed to form blood vessels, cartilage, muscle, and collagen in bones, vitamin C also plays a role in mental health as it possesses acute anxiolytic effects.¹²⁷ When combined with a B-complex vitamin, individuals experienced reduced stress, better mental health, and improved cognitive performance.¹²⁸ Adverse effects of vitamin C at doses over 1000 daily are rare but

could include loose stools, diarrhea, iron overload, and kidney stones.¹²⁹ Suggested dose:¹³⁰ 500mg daily.

Vitamin D: Emerging research shows that higher Vitamin D blood levels are associated with less depression and better moods. Vitamin D supplementation reduces anxiety symptoms by increasing serotonin concentrations and decreasing the levels of the inflammatory biomarkers associated with stress.¹³¹ Vitamin D-3 (cholecalciferol) is the most absorbable form of this fat-soluble vitamin. Suggested dose:¹³² 1000 IU daily.

Fish Oil: Fish oils contain high amounts of omega-3 essential fatty acids, specifically EPA (eicosapentaenoic acid) and DHA (docosahexaenoic acid). Fish oils support mood, lessen anxiety, and might help to ease stress-related symptoms.¹³³ It is not known exactly how fish oil supports mood, but researchers suggest it may have to do with reducing levels of prostaglandins, which in high levels are associated with anxiety and depression.^{134, 135}

Suggested dose¹³⁶ is 1,500mg of EPA and 500mg of DHA. Please keep ensure you buy high-quality fish oil that has been tested for heavy metals and keep it in the refrigerator to avoid rancidity.

GABA (Gamma-aminobutyric acid): An amino acid that modulates brain waves, GABA reduces excitability in the central nervous system, resulting in relaxation and reduced anxiety. It works by blocking brain signals (neurotransmissions) involved in stress.¹³⁷ GABA has shown similar effects to those observed with benzodiazepine (anti-anxiety) medications, but without any habituation and/or tolerance developing.¹³⁸ Suggested dose:¹³⁹ 300mg daily (if possible, 30-minutes before a meal).

Glycine: Glycine is helpful for individuals who worry a lot. It is an amino acid that your body uses to create proteins, needed for the growth and maintenance of tissue and for making important hormones and enzymes. Glycine antagonizes the release of norepinephrine, a neurotransmitter that is released when experiencing anxiety or panic. Glycine antagonizes the release of NE, thus mitigating anxiety and panic.¹⁴⁰ Suggested dose:¹⁴¹ 3 grams daily.

Inositol: Inositol is a type of sugar made by the body that influences both the body's insulin response and several hormones associated with mood and cognition.¹⁴² Supplementation with inositol can reduce general anxiety, panic, and OCD symptoms.¹⁴³ Suggested dose¹⁴⁴ is up to 12 grams daily. The easiest way to consume inositol is to mix it into a smoothie.

L-Theanine: An amino acid well known for being found in green tea, L-Theanine is associated with improvements in mood, cognition, and reduction of anxiety-like symptoms.¹⁴⁵ Supplementation assists in reducing acute stress and anxiety in people experiencing stressful situations.¹⁴⁶ Suggested dose:¹⁴⁷ 400mg daily (if possible, 30-minutes before a meal).

Magnesium: Magnesium is an essential mineral, playing a role in over 300 enzyme reactions in the human body.¹⁴⁸ Low levels of magnesium are highly associated with

stress levels.¹⁴⁹ This mineral plays an important role in regulating cortisol levels. It prevents excess cortisol production due to its ability to calm the nervous system. Magnesium also helps maintain healthy levels of the neurotransmitter GABA, which plays an important role in restorative sleep. In order for the body to have healthy GABA levels, it needs adequate magnesium. Magnesium can reduce anxiety, depression, stress, and insomnia.¹⁵⁰ Magnesium bis-glycinate is an easily absorbed form of magnesium. Suggested dose:¹⁵¹ 320mg nightly.

Niacinamide: A form of vitamin B3, Niacin has been shown in animals to work in the brain in ways similar to anti-anxiety medications. One study found that niacin helped people get through withdrawal from benzodiazepines (drugs used to treat anxiety).¹⁵² Suggested dose:¹⁵³ 500 mg daily with food.

Note: Niacin causes flushing, which is not serious and not indicative of an allergic reaction. The flush is not dangerous and appears to be beneficial in reducing muscle tension and anxiety, which aids in having a good night's sleep. The flush sensation usually begins on and around the forehead and then travels down the body to the belly button and sometimes to the toes. Non-flushing forms of niacin are available.¹⁵⁴

Probiotics: Probiotics may improve psychological symptoms, particularly those associated with anxiety.¹⁵⁵ Probiotics also aid by improving digestion, absorption, and many other mechanisms by increasing the number of beneficial bacteria in your small and large intestine. Suggested Dose:¹⁵⁶ 10-12 billion colony-forming units daily.

Taurine: Taurine is an amino acid that increases glycine and GABA to calm the brain and reduce anxiety. It also protects the brain by reducing the harmful effects of excess glutamate, a neurotransmitter thought to increase anxiety.¹⁵⁷ Research shows taurine could be beneficial for the control of anxiety in clinical situations.¹⁵⁸ Suggested dose:¹⁵⁹ 3 grams daily.

5-HTP (5-Hydroxytryptophan): 5-HTP is a chemical byproduct of the protein building block L-tryptophan. 5-HTP is used as an oral supplement to enhance levels of serotonin, a mood-enhancing neurotransmitter, in the brain and central nervous system.¹⁶⁰ Supplementing with 5-HTP can reduce depression, anxiety, and panic attacks.¹⁶¹ While 5-HTP is generally considered safe, there is the possibility that when taken with other medications it may cause serotonin levels to increase to dangerous levels, a condition called serotonin syndrome.¹⁶² Suggested dose:¹⁶³ 100mg daily with food.

5. Western Medical Approaches

Conventional medicine can be a helpful short-term approach for people suffering from anxiety disorders. Typically, western medical treatments rely on the use of pharmaceutical medications to stabilize a person's mood by altering levels of neurotransmitters in the brain that contribute to feelings of anxiety, worry, and depression. The most

common medications used are SSRIs (selective serotonin reuptake inhibitors), SNRIs (serotonin-norepinephrine reuptake inhibitors), and benzodiazepines. While these medications do a great job of minimizing emotional overwhelm initially, they may become ineffective over time. For example, symptoms return for up to 33% of people using antidepressants.¹⁶⁴ These medications often have significant side effects such as weight gain, loss of sexual function, and dependence; making them difficult to get off of.

Western Medicine usually attributes anxiety to the Monoamine Theory of Mood Disorders. This theory states that mood disorders are a result of chemical imbalances of neurotransmitters in the brain due to genetics. It ignores external determinants of mental health.¹⁶⁵ By ascribing to this model, western medicine often ignores many other important contributors to anxiety. These include nutrition, lifestyle factors, exercise, environment, maladaptive thought patterns, etc.

6. Things to Avoid

When treating anxiety, avoiding aggravating factors can be as important as seeking out supportive treatments. Some common triggers for anxiety include the following:

Stimulants: Through the release of certain chemicals (norepinephrine, dopamine, etc.), stimulants are substances that speed up activity in the central nervous system. While these effects can be great for improving alertness, energy, concentration, and memory they can dramatically worsen and even create anxiety. Common stimulants include medications like amphetamines, caffeine, and nicotine.

Emotional Disturbance: When it comes to anxiety, safety, community, and connection are often overlooked. If we are surrounded by people who don't treat us with kindness and respect, we are not getting our fundamental needs met. People who antagonize our sense of self can be damaging to our mood and overall well-being, leaving us feeling alone and isolated. As social creatures, to feel safe we must feel a sense of belonging and community. When we are in tense environments or feel ostracized we feel unsafe and inevitably anxious. Avoiding unsupportive environments, while seeking out those that offer support, is critical in reducing anxiety.

Blood Sugar Disruptors: Anything that causes abrupt drops in blood sugar levels should be avoided as they physiologically create feelings of anxiety. These include high sugar foods, alcohol, and extreme dieting. Too much sugar has been found to decrease a protein called brain-derived neurotrophic factor (BDNF), which is involved in the development of depression and anxiety.

Food Allergens: By avoiding foods that may cause immunological responses in our bodies we minimize our exposure to inflammation, which research has demonstrated is a cause of anxiety.¹⁶⁶ It's important to determine which foods may negatively impact your body.

Poor Sleeping Habits: Ensuring you get a healthy night's sleep is critical to preventing anxiety. Our

bodies need to recover both physically and mentally during sleep. If we are sleep deprived, we are far less equipped to handle daily stressors, let alone an anxiety disorder. Please refer to the sleep protocol to create healthy sleep habits.

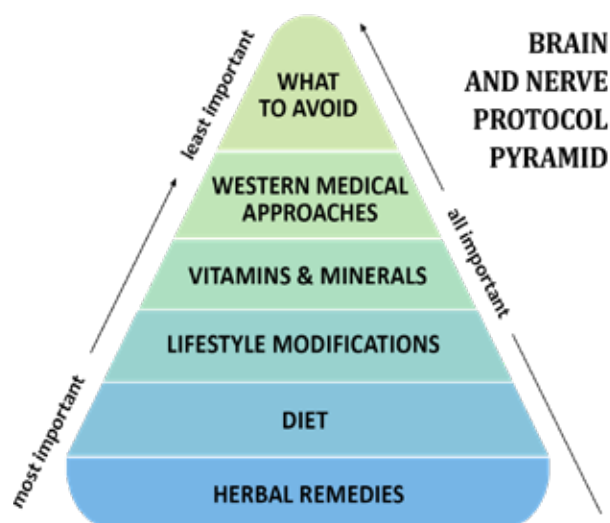
No longer dismissed as a “neurotransmitter deficiency”, anxiety is a serious and complex condition. It is a disorder that is impacted by many facets of a person’s life. While

conventional medicine may be an important part of an initial treatment plan, it’s important to note the side effects and possible dependency of anti-anxiety medications when used over a long period of time. Fortunately, natural therapies offer a variety of options to holistically address the many factors that contribute to anxiety. To optimize outcomes, treatments should be tailored to the individual’s unique needs and circumstances.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Morning gratitude practice Lion’s Mane & Reishi mushroom tinctures Low glycemic breakfast 5-HTP w/ food Vitamin B-complex, C, & D Niacinamide	GABA & L-Theanine (30 min. before your meal) Fish oil w/ lunch Ashwagandha & Holy Basil (Tulsi) Mindfulness &/or CBT	Inositol smoothie Glycine & Taurine Probiotic 30 min. of exercise	Lion’s Mane & Reishi mushroom dual-extracted tinctures Post-dinner walk outside	Lemon Balm tincture & CBD Deep lymphatic breathing Epsom salt bath Sleep blend tincture (w/ valerian, hops, passionflower, & chamomile) Magnesium glycinate
MONDAY	Morning gratitude practice Lion’s Mane & Reishi mushroom tinctures Low glycemic breakfast 5-HTP w/ food Vitamin B-complex, C, & D Niacinamide	GABA & L-Theanine (30 min. before your meal) Fish oil w/ lunch Ashwagandha & Holy Basil (Tulsi) Mindfulness &/or CBT	Inositol smoothie Glycine & Taurine Probiotic 30 min. of exercise	Lion’s Mane & Reishi mushroom tinctures Post-dinner walk outside	Lemon Balm tincture & CBD Deep lymphatic breathing Aromatherapy Sleep blend tincture Magnesium glycinate
TUESDAY	Morning gratitude practice Lion’s Mane & Reishi mushroom tinctures Low glycemic breakfast 5-HTP w/ food Vitamin B-complex, C, & D Niacinamide	GABA & L-Theanine (30 min. before your meal) Fish oil w/ lunch Ashwagandha & Holy Basil (Tulsi) Mindfulness &/or CBT	Inositol smoothie Glycine & Taurine Probiotic 30 min. of exercise	Lion’s Mane & Reishi mushroom tinctures Post-dinner walk outside	Lemon Balm tincture & CBD Deep lymphatic breathing Epsom salt bath Sleep blend tincture Magnesium glycinate
WEDNESDAY	Morning gratitude practice Lion’s Mane & Reishi mushroom tinctures Low glycemic breakfast 5-HTP w/ food Vitamin B-complex, C, & D Niacinamide	GABA & L-Theanine (30 min. before your meal) Fish oil w/ lunch Ashwagandha & Holy Basil (Tulsi) Mindfulness &/or CBT	Inositol smoothie Glycine & Taurine Probiotic 30 min. of exercise	Lion’s Mane & Reishi mushroom tinctures Post-dinner walk outside	Lemon Balm tincture & CBD Deep lymphatic breathing Aromatherapy Sleep blend tincture Magnesium glycinate
THURSDAY	Morning gratitude practice Lion’s Mane & Reishi mushroom tinctures Low glycemic breakfast 5-HTP w/ food Vitamin B-complex, C, & D Niacinamide	GABA & L-Theanine (30 min. before your meal) Fish oil w/ lunch Ashwagandha & Holy Basil (Tulsi) Mindfulness &/or CBT	Inositol smoothie Glycine & Taurine Probiotic 30 min. of exercise	Lion’s Mane & Reishi mushroom tinctures Post-dinner walk outside	Lemon Balm tincture & CBD Deep lymphatic breathing Aromatherapy Sleep blend tincture Magnesium glycinate
FRIDAY	Morning gratitude practice Lion’s Mane & Reishi mushroom tinctures Low glycemic breakfast 5-HTP w/ food Vitamin B-complex, C, & D Niacinamide	GABA & L-Theanine (30 min. before your meal) Fish oil w/ lunch Ashwagandha & Holy Basil (Tulsi) Mindfulness &/or CBT	Inositol smoothie Glycine & Taurine Probiotic 30 min. of exercise	Lion’s Mane & Reishi mushroom tinctures Post-dinner walk outside	Lemon Balm tincture & CBD Deep lymphatic breathing Aromatherapy Sleep blend tincture Magnesium glycinate
SATURDAY	Morning gratitude practice Lion’s Mane & Reishi mushroom tinctures Low glycemic breakfast 5-HTP w/ food Vitamin B-complex, C, & D Niacinamide	GABA & L-Theanine (30 min. before your meal) Fish oil w/ lunch Ashwagandha & Holy Basil (Tulsi) Mindfulness &/or CBT	Inositol smoothie Glycine & Taurine Probiotic 30 min. of exercise	Lion’s Mane & Reishi mushroom tinctures Post-dinner walk outside	Lemon Balm tincture & CBD Deep lymphatic breathing Aromatherapy Sleep blend tincture Magnesium glycinate

Ch. 24: Brain and Nerve Protocol - Including Stroke and Concussion

Considered the most complex thing in our universe, science has only scratched the surface of how the brain works.¹ Central to our experience as human beings, the brain regulates almost all of our physiological functions. It allows us to sense, feel, think, move, and interact with the world around us.² Unfortunately, one-sixth of the world's population suffers from neurological disorders.³ Given their potential to interfere with normal living, drastically reduce a person's quality of life, and cause disability and even death, cognitive health issues can be devastating. Accordingly, a rapidly growing field of study dedicated to brain health has emerged. It is focused on creating, preserving, protecting, and optimizing cognitive function. Increasing in popularity are nootropics, natural compounds that have a beneficial effect on brain function.⁴ Their effects range from boosting memory, motivation, creativity, and alertness, to improving general cognitive function. Nootropics may also reduce age-related declines in brain function.⁵ The ongoing research and advancements in this field provide an optimistic outlook for the future of brain health.



Please also visit the Alzheimer's and Dementia Protocol on page 145 and the Parkinson's Protocol on page 169.

1. Herbal Remedies

There are several herbs that address brain and nerve health. These include Lion's Mane, Cordyceps, Lemon Balm, Reishi, Ashwagandha, Bacopa, Ginkgo, Gotu Kola, Panax Ginseng, and Rhodiola.

Lion's Mane (*Hericium erinaceus*): Known for its anti-inflammatory and neuroprotective effects, Lion's Mane is said to have been used as a tea by Buddhist monks to heighten their ability to focus during meditation.⁶ Lion's Mane stimulates Nerve Growth Factor (NGF);⁷ NGF helps repair and boost the growth of new neurons in the brain. This process enables learning and memory.⁸ Lion's Mane may be beneficial for those suffering from Alzheimer's disease, dementia, and Parkinson's⁹ and it helps repair

damaged nerve cells in the spinal cord and brain.¹⁰ Lion's mane also speeds nervous system recovery; it is indicated for strokes due to the compound hericenone B, which shows strong anti-platelet activity, reduces nerve pain, and might be a novel compound for antithrombotic therapy.¹¹ Suggested dose is 2 dropperfuls of a dual-extracted tincture, twice daily. Possible interactions: May interact with blood thinning medications.

Cordyceps (*Cordyceps militaris* and *C. sinensis*): Known as the "Caterpillar Mushroom", this fungus grows on the larvae of insects like caterpillars. It has been used in Traditional Chinese Medicine for centuries to treat brain, respiratory, and kidney diseases.¹² Research shows supplementing with this mushroom enhances learning and memory.¹³ Other research shows a reduction in fatigue and improved exercise performance. It is believed its effects are due to increase blood flow, ATP synthesis, and reducing oxidative stress.¹⁴ Suggested dose is 2 dropperfuls of a dual-extracted tincture, once to twice daily. Possible interactions: May have a compounding effect on immune therapy.

Lemon Balm (*Melissa officinalis*): Lemon Balm is associated with improvements in memory, mood, and age-related cognitive performance, a reduction in stress and anxiety, and improved clarity and focus.¹⁵ Lemon balm is an antioxidant, helping to protect the brain and nerves.^{16, 17} Lemon Balm also inhibits the brain's levels of acetylcholinesterase (AChE), an enzyme that helps break down acetylcholine (ACh), a critical neurotransmitter involved in cognition and memory, reduced levels of acetylcholine have been associated with dementia, Parkinson's, and Alzheimer's disease.¹⁸ Suggested dose: 2 dropperfuls of a 1:3 tincture, 2 times daily.

Reishi (*Ganoderma lucidum*): Reishi mushroom is a formidable anti-inflammatory that helps modulate inflammation and immune function, and inhibits histamine release.¹⁹ It has neuroprotective qualities and is effective in calming the inflammation and immune response involved with Leaky Gut. It also helps to strengthen the body against the negative effects of stress-like inflammation, hormonal imbalances, high cortisol, fatigue, brain fog, and low energy. Research suggests that it could serve as a regenerative therapeutic agent for the treatment of cognitive decline associated with neurodegenerative diseases.²⁰ Suggested dose: 2 dropperfuls of a dual-extracted tincture, once to twice daily.

Ashwagandha (*Withania somnifera*): Considered a potent antioxidant and adaptogen, ashwagandha helps mitigate the harmful effects of stress.²¹ Ashwagandha helps regenerate axons and dendrites of brain nerve cells and helps reconstruct synapses, the junctions where nerve cells communicate with other cells.²² Supplementing with this herb enhances memory and has been helpful in the treatment of neurodegenerative diseases.²³ Suggested dose is 500mg daily or 3 dropperfuls of tincture daily.²⁴ Possible interactions: It is generally considered safe; however, it may stimulate thyroid function so caution with hyperthyroidism.²⁵

Bacopa (*Bacopa monnieri*): Well known in Ayurvedic medicine to boost memory, Bacopa helps

sharpen the mind and intellect.²⁶ It has a long history of use for memory loss, anxiety, poor cognition, and loss of concentration.²⁷ A meta-analysis of randomized controlled trials evaluating the cognitive effects of *Bacopa* determined that it has the potential to improve cognition, particularly speed of attention.²⁸ Additional research has demonstrated its ability to boost memory and cognition and improve mood.²⁹ Suggested dose is one 300 mg capsule or a daily tincture. It is best to take this herb upon waking and about 30-minutes away from food if possible.

Ginkgo (*Ginkgo biloba*): Ginkgo is one of the oldest living tree species in the world.³⁰ Numerous studies have been conducted on Ginkgo biloba leaf extract, particularly around its effect on memory loss, slow thinking and reasoning, and tinnitus.³¹ Ginkgo biloba has chronic memory-enhancing effects in both healthy subjects and patients with dementia.³² It is hypothesized that Ginkgo works by boosting cerebral blood flow, reducing oxidative stress by eliminating free radicals, and increasing nitric oxide. Suggested dose is 240 mg ginkgo per day; 3-5 dropperfuls tincture, twice daily. Possible interactions: anticoagulant, hypoglycemic, and anti-depressant medications³³.

Gotu Kola (*Centella asiatica*): Gotu kola is well known for its memory-enhancing properties, such as increased attention span and concentration.³⁴ Research shows it stimulates the growth of dendrites, a type of brain cell. It encourages dendrite branching, which boosts brain plasticity and communication between neurons. Supplementing with this herb results in the enhancement of mood, cognition, memory, and learning.³⁵ It has also been shown to be effective in improving cognitive function after stroke.³⁶ Suggested dose is 120mg daily or 2 dropperfuls of tincture.³⁷ Possible interactions: May interact with anti-platelet medications.

Korean Ginseng (*Panax ginseng*): The ginsenosides in ginseng are believed to be the active compound that gives this herb its remarkable properties. Studies have repeatedly demonstrated ginseng's ability to improve outcomes related to cognition and memory, specifically acute working memory and attention.³⁸ Suggested dose is 4 mL (dropperfuls) tincture daily. Possible interactions: May increase the effects of blood pressure medication.

Rhodiola (*Rhodiola rosea*): Found in the arctic and at high altitudes, Rhodiola is an adaptogen, used to reduce stress and fatigue, and increase energy, alertness, and stamina while boosting mental performance.³⁹ Rhodiola can increase attention to detail-oriented tasks by improving concentration over a prolonged period.⁴⁰ Researchers believe Rhodiola protects neurons from oxidative stress-induced cell death and may stimulate stem cell production.⁴¹ Suggested dose is 500mg daily or 3 dropperfuls of tincture three times daily.⁴² Possible interactions: Rhodiola may be stimulating so should be taken in the morning.

2. Diet Known as “brain food”, research shows that the unique nutritional composition of certain foods can support the health of your brain. These include:

158 Green, Leafy Vegetables: Leafy greens such as kale,

spinach, collards, and broccoli are rich in brain-healthy nutrients. Consumption of green leafy vegetables may help to slow the decline in cognitive abilities. Researchers believe this is due to the neuroprotective actions of lutein, folate, β -carotene, and phyloquinone. The addition of a daily serving of green leafy vegetables to one's diet may be a simple way to improve brain health.⁴³ The current Dietary Guidelines for Americans suggest consuming at least 2.5 cups of vegetables per day.⁴⁴

Fatty Fish: Fatty fish are abundant sources of omega-3 fatty acids, healthy unsaturated fats that have been linked to lower blood levels of beta-amyloid—the protein that forms damaging clumps in the brains of people with Alzheimer's disease.⁴⁵ Fish consumption also supports overall brain structure and improves cognitive ability in healthy people.⁴⁶ Make sure to choose varieties of fish that are less likely to be contaminated with toxins. These include SMASH fish: sardines, mackerel, anchovies, salmon, and herring.

Berries: Flavonoids, the natural plant pigments that give berries their bright colors, also help improve memory.⁴⁷ Flavonoids seem to work by protecting vulnerable brain cells and enhancing the function of existing brain cell structures.⁴⁸ A study found that women who consumed two or more servings of strawberries and blueberries each week delayed memory decline by up to two-and-a-half years.⁴⁹ The current Dietary Guidelines for Americans suggest consuming at least 1.5-2 cups of fruits per day.⁵⁰

Walnuts: Nuts are a delicious and convenient snack, and they also happen to be an excellent source of protein and healthy fats. Walnuts, in particular, might also improve memory. They are high in a type of omega-3 fatty acid called alpha-linolenic acid (ALA). Diets rich in ALA have been linked to lower blood pressure and cleaner arteries, factors that are brain protective. Research shows that higher walnut consumption is associated with improved cognitive test scores.⁵¹

Probiotic-Rich Foods: The gut-brain axis (GBA) consists of bidirectional communication between the central nervous system of the brain and the stomach and intestines.⁵² There is increasing evidence that the composition of the bacteria within the gastrointestinal tract can influence the brain and behavior, particularly with respect to cognitive function.⁵³ Based on these findings, a healthy gut microbiome may play an important role in cognitive function. By choosing probiotic-rich foods such as yogurt, kefir, sauerkraut, tempeh, kimchi, miso, kombucha, pickles, and more, we can support the health of our brain. Please see the Leaky Gut Protocol at page 70 for more information.

3. Lifestyle Modifications **Exercise:** Physical activity plays an essential role in maintaining a healthy brain. It provides unique benefits for the vascular and cellular systems that sustain a healthy brain. Exercise initiates diverse and powerful neuroprotective pathways that promote continued brain health into old age.⁵⁴ One way is through the production of BDNF. BDNF increases the connections between brain cells, helps the brain grow new cells,

and is shown to be effective in preventing Alzheimer's disease. BDNF levels are increased two to three-fold after acute exercise when compared to resting conditions.⁵⁵ As a general goal, aim for at least 30 minutes of moderate physical activity every day.⁵⁶

Meditation: Meditation is a free and easy way to enhance cognitive health. Studies show that meditation affects multiple pathways that could play a role in brain aging and mental fitness.⁵⁷ One kind of meditation is mindfulness meditation. It is the practice of cultivating a heightened state of awareness by bringing one's attention to the present moment, without judgment. In one study participants engaged in mindfulness meditation for 20 minutes every day. After only 8 weeks, they experienced growth in gray matter - the brain regions involved in learning and memory processes, emotion regulation, self-referential processing, and perspective.⁵⁸ Based on the research, the current recommendation is a minimum of 20 minutes of meditation per day.

Sleep: Many biological processes happen during sleep. The body repairs cells, restores energy and releases molecules like hormones and proteins.⁵⁹ The brain's glymphatic system is most active during sleep. It flushes out waste from the central nervous system and removes toxic byproducts from the brain, which build up throughout the day. This allows your brain to work well when you wake up.⁶⁰ As nerve cells communicate and reorganize, your brain also stores new information during sleep.

Research suggests that sleep contributes to memory function by converting short-term memories into long-term memories, as well as by erasing, or forgetting, unneeded information that might otherwise clutter the nervous system.⁶¹ Sleep has been proven to improve memory recall, regulate metabolism, and reduce mental fatigue.⁶² Evidence supports the notion that brain cell growth and recovery can be improved by managing sleep quality.⁶³ Additionally, sleep loss is associated with detrimental effects on brain cell recovery and growth.⁶⁴ Ensuring you get between 7 and 9 hours of sleep per night is critical in supporting brain health. See the Sleep Protocol at page 286, if sleep is an issue for you.

4. Vitamins and Minerals **Acetyl-L-Carnitine:**

Acetyl-L-Carnitine (ALC) is an amino acid produced in the body. Through stabilizing nerve cell membranes, supporting mitochondrial function, and maintaining healthy neurotransmitter activity it plays an important role in brain health.⁶⁵ ALC facilitates energy production in the body as well as antioxidant protection for cells.⁶⁶ In the absence of mitochondrial efficiency, individuals may experience common complaints related to aging, including loss of memory, learning, recall, and cognition.⁶⁷ Suggested dose is 2-3 grams daily. Possible interactions: May interact with serotonergic, blood clotting, seizure, and thyroid medications.⁶⁸

Fish Oil (Omega-3 Fatty Acids): Fish oils contain high amounts of omega-3 essential fatty acids, specifically

EPA (eicosapentaenoic acid) and DHA (docosahexaenoic acid). DHA is crucial for the healthy structure and function of the brain. In fact, almost half of the brain is composed of DHA. These fatty acids impact the brain's signaling systems, neurotransmitters, memory, learning, focus, and attention. Recent studies have concluded that omega-3s have tremendous benefits for cognitive function and that DHA specifically helps reduce and even repair cognitive decline.⁶⁹ Suggested dose is 1,500mg of EPA and 500mg of DHA.⁷⁰ Purchase high-quality fish oil that has been tested for heavy metals and keep it in the refrigerator; to avoid rancidity.

L-Theanine: An amino acid found in green tea, L-theanine is associated with several health benefits including improvements in mood, cognition, and reduction of anxiety-like symptoms.⁷¹ Research has shown supplementation can promote alpha brain waves, which lower stress, reduce anxiety, decrease depression, and improve creative thinking.⁷² Suggested dose: 400mg daily (if possible, 30 minutes before a meal)⁷³.

Magnesium: Magnesium is a cofactor in more than 300 enzymatic reactions in the body.⁷⁴ It also plays an important role in cognitive health. Increased levels of magnesium in the brain have been shown to promote multiple mechanisms of brain plasticity that can enhance learning and memory.⁷⁵ Brain plasticity is the biological, chemical, and physical capacity of the brain to reorganize its structure and function. It occurs as a result of learning, experience, and memory formation, or as a result of damage to the brain.⁷⁶ Magnesium levels are directly correlated to cognitive functions.⁷⁷ Magnesium glycinate is an easily absorbed form of magnesium. Suggested dose is 320mg nightly.⁷⁸ Possible interactions: May interfere with bisphosphonates and antibiotics.⁷⁹

Phosphatidylserine: Phosphatidylserine is a fatty substance made by the body that declines with age.⁸⁰ It helps cover and protect the cells in the brain and carries messages between them. Phosphatidylserine plays an important role in keeping your mind and memory sharp. It supports human cognitive functions, including the formation of short-term memory, the consolidation of long-term memory, the ability to create new memories, the ability to retrieve memories, the ability to learn and recall information, the ability to focus attention and concentrate, the ability to reason and solve problems, language skills, and the ability to communicate. It also supports locomotor functions, especially rapid reactions and reflexes.⁸¹ Suggested dose is 300mg daily. Possible interactions: May interfere with Anticholinergic and Cholinergic drugs.

Turmeric (*Curcuma longa*): Turmeric helps degrade beta-amyloid plaques, remove heavy metals from the body, and reduce inflammation and oxidative stress.^{82 83} Turmeric's ability to improve memory in patients with Alzheimer's is attributed to these properties. Suggested dose: 2000 mg bioavailable turmeric per day, with fat-containing meals in divided doses or 4 dropperfuls of a 1:2 tincture with meals, twice daily.

Vitamin B-complex: From their involvement in memory, cellular signaling, and neurotransmitter production, to mood and sleep, B-vitamins play a major role in brain health.⁸⁴ Research consistently shows that deficiencies in B-vitamins are associated with cognitive impairment.⁸⁵ In children, deficiencies in B-vitamins can cause delayed or impaired cognitive development or worsen symptoms related to learning and attention.⁸⁶ When considering their high concentration in the brain, it's no wonder B-vitamins are critical in cognitive functioning. Vitamin B9 is seen in the brain at a concentration four times higher than in the blood.⁸⁷ Concentrations of vitamins B7 and B5 are 50 times higher in the brain.⁸⁸ B-vitamin supplementation is associated with slowing of cognitive decline and dementia.⁸⁹ Suggested dose is 1 pill of B-complex vitamin daily with breakfast or lunch. Possible interactions: medications such as altretamine, cisplatin, certain antibiotics, certain anti-seizure drugs, and levodopa.⁹⁰

Vitamin D-3 (Cholecalciferol): Vitamin D is both a fat-soluble vitamin and a hormone made by our bodies. It plays an important role in brain maturation and growth.⁹¹ After exposure to sunlight, our body forms vitamin D naturally. Unfortunately, Vitamin D deficiency is extremely common. Up to 40% of U.S. adults have insufficient levels.⁹² Research shows a strong association between low vitamin D and a range of cognitive impairments.⁹³ Vitamin D receptors are widespread in brain tissue, and vitamin D's biologically active form has neuroprotective effects including the clearance of amyloid plaques, a hallmark of Alzheimer's Disease.⁹⁴ Suggested dose: Get your levels tested. If normal, then 800 IU daily.⁹⁵ Otherwise, 5000-10,000 IU per day until 25-hydroxyvitamin D levels are at 40-60 ng/mL then decrease to a maintenance dose of 2000-4000 IU per day. Possible interactions: medications containing calcitriol.

5. Western Medical Approaches

Typically, Western medicine utilizes pharmaceutical medications to enhance cognitive function. The medications most commonly prescribed are stimulants, such as Adderall, Modafinil, Ritalin, etc. Stimulants are a broad class of drugs that increase the activity of the central nervous system.⁹⁶ While these medications are highly effective at sustaining attention and alertness, they are often habit-forming and have a variety of side effects such as appetite suppression, anxiety, insomnia, cardiac involvement, and more. For medications for Alzheimer's, Dementia, MS, and Parkinson's, please visit those specific protocols.

6. Things to Avoid

EMFs: Electric and magnetic fields (EMFs) are invisible areas of energy that give off radiation.⁹⁷ They are associated with the use of electrical power and various forms of man-made lighting. Studies on EMFs have found they can induce changes in the central nervous system, including cell apoptosis, and changes in the function of the nerve myelin and ion channels.⁹⁸ Furthermore, studies show EMFs act as a stress source in living creatures.⁹⁹ Neurological cognitive disorders, such

as headache, tremor, dizziness, loss of memory, loss of concentration, and sleep disturbance due to EMFs have been reported by several studies.¹⁰⁰ While we live in a technological world and cannot escape EMFs completely, avoiding unnecessary exposure to them is a great way to protect brain health. Simple steps like moving your router or wifi booster at least 20 feet from your desk, disabling Bluetooth and wireless features on your phone, and using them only when needed can dramatically reduce your exposure to EMFs.

Organophosphates: Organophosphates are the most common pesticides used in agriculture.¹⁰¹ They are also the most common synthetic chemicals detected in the environment and in animal and human tissues.¹⁰² This is an increasing environmental concern due to their high toxicity. Adverse health effects associated with these chemicals have been documented for decades, and the nervous system is especially vulnerable to damage from organophosphates.¹⁰³ They cause multiple neurotoxic effects and have been linked to neurodegenerative diseases, dementia, attention deficit hyperactivity disorder, and Parkinson's disease.¹⁰⁴ Conventionally grown food has ten to 100 times more pesticides than organic food.¹⁰⁵ By choosing organically grown foods and washing our foods properly we can dramatically reduce our exposure to organophosphates.

Heavy Metals: Heavy metals such as lead, cadmium, mercury, and manganese have the potential to damage brain cells.¹⁰⁶ Known as "neurotoxins", heavy metal exposure is associated with risks of Alzheimer's disease, dementia, and cognitive decline.¹⁰⁷ Unfortunately, heavy metals are widespread in our environments. Exposure can come from contaminated foods, drinking water, inhalation of polluted air, and personal care products.¹⁰⁸ Mercury exposure induces changes in the central nervous system, resulting in a range of symptoms from irritability, fatigue, behavioral changes, tremors, headaches, hearing and cognitive loss, dysarthria, incoordination, hallucinations, and death.¹⁰⁹ Lead poisoning is especially dangerous for kids because it can cause irreversible brain damage and even death.¹¹⁰ Even low levels of heavy metals are associated with poor health outcomes.¹¹¹

You can reduce your heavy metal exposure by:

- Consuming organic food, as many pesticides contain heavy metals¹¹².
- Eating "SMASH" fish and avoiding large, predatory fish such as tuna, king mackerel, marlin, orange roughy, and swordfish.
- Drinking clean purified water; Lead pipes can leach lead into drinking water and well water is often contained with heavy metals from the soil.
- Choosing safe personal care products; Makeup, whitening toothpaste, sunscreen, eye drops, and nail polish commonly contain heavy metals.
- Breathing fresh air whenever possible; If you live in a city, ensure you spend plenty of time in green spaces or keep air purifying plants in your home.

Blood Sugar Dysregulation: Excess sugar is harmful, even to the brain. When there is too much sugar in the blood (hyperglycemia) a number of biological processes occur that can negatively impact the brain. Glycation occurs when excess sugar molecules get stuck to your cells, preventing them from performing properly. Over a long period of time, high-sugar diets can predispose individuals to chronic inflammation and devastating neurodegenerative disorders such as Alzheimer's disease.¹¹³ Even over a short period of time, excess sugar in the blood is associated with subtle brain injury and impaired attention and memory in young adults.¹¹⁴ By replacing high sugar foods and drinks with low sugar alternatives you can protect your body and brain.

Sedentary Lifestyle: Research shows that sedentary behavior is a risk factor for cognitive decline and is associated with lower cognitive function.¹¹⁵ Characterized by an inactive lifestyle, being sedentary means rarely

engaging in physical activity. Researchers believe that long periods of inactivity may lead to atrophy of the part of the brain responsible for memory (the medial temporal lobe).¹¹⁶ Being sedentary also contributes to poor glycemic control, which is a known risk factor for dementia. Even intermittent light intensity physical activity can help slow cognitive decline.¹¹⁷

Smoking: Smoking drastically increases the likelihood of vascular diseases, like strokes.¹¹⁸ Strokes can be devastating. If they don't result in death, a person may suffer from cognitive and physical impairment for the rest of their life. By avoiding smoking and secondhand smoke you dramatically reduce your risk for a number of serious health issues, including cognitive impairment.

As Aristotle said, "The energy of the mind is the essence of life." With the current advancements being made in cognitive health, there has never been a better time to enhance brain health.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps dual-extracted tinctures) Bacopa, Ginseng & Rhodiola Vit. D & B-complex	Shop for healthy foods Physical exercise Probiotic-rich lunch Turmeric, Omega-3s & Acetyl-L-Carnitine Ginkgo, Gotu Kola, & Lemon Balm tinctures	Snack: berries & walnuts Time outside/ nature walk	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps dual-extracted tinctures) Healthy Dinner Phosphatidylserine, L-Theanine & Turmeric	Magnesium Disconnect devices Mindfulness meditation Sleep herbs, if needed Ashwagandha & Lemon Balm tinctures
MONDAY	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Bacopa, Ginseng & Rhodiola Vit. D & B-complex	Shop for healthy foods Physical exercise Probiotic-rich lunch Turmeric, Omega-3s & Acetyl-L-Carnitine Ginkgo, Gotu Kola, & Lemon Balm tinctures	Berries & walnuts Time outside/ nature walk	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Healthy Dinner Phosphatidylserine, L-Theanine & Turmeric	Magnesium Disconnect devices Mindfulness meditation Sleep herbs, if needed Ashwagandha & Lemon Balm tinctures
TUESDAY	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Bacopa, Ginseng & Rhodiola Vit. D & B-complex	Physical exercise Probiotic-rich lunch Turmeric, Omega-3s & Acetyl-L-Carnitine Ginkgo, Gotu Kola, & Lemon Balm tinctures	Berries & walnuts	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Healthy Dinner Phosphatidylserine, L-Theanine & Turmeric Walk outside after dinner	Magnesium Disconnect devices Mindfulness meditation Sleep herbs, if needed Ashwagandha & Lemon Balm tinctures
WEDNESDAY	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Bacopa, Ginseng & Rhodiola Vit. D & B-complex	Physical exercise Probiotic-rich lunch Turmeric, Omega-3s & Acetyl-L-Carnitine Ginkgo, Gotu Kola, & Lemon Balm tinctures	Berries & walnuts Time outside/ nature walk	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Healthy Dinner Phosphatidylserine, L-Theanine & Turmeric	Magnesium Disconnect devices Mindfulness meditation Sleep herbs, if needed Ashwagandha & Lemon Balm tinctures
THURSDAY	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Bacopa, Ginseng & Rhodiola Vit. D & B-complex	Physical exercise Probiotic-rich lunch Turmeric, Omega-3s & Acetyl-L-Carnitine Ginkgo, Gotu Kola, & Lemon Balm tinctures	Berries & walnuts	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Healthy Dinner Phosphatidylserine, L-Theanine & Turmeric Walk outside after dinner	Magnesium Disconnect devices Mindfulness meditation Sleep herbs, if needed Ashwagandha & Lemon Balm tinctures
FRIDAY	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Bacopa, Ginseng & Rhodiola Vit. D & B-complex	Physical exercise Probiotic-rich lunch Turmeric, Omega-3s & Acetyl-L-Carnitine Ginkgo, Gotu Kola, & Lemon Balm tinctures	Berries & walnuts Time outside/ nature walk	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Healthy Dinner Phosphatidylserine, L-Theanine & Turmeric	Magnesium Disconnect devices Mindfulness meditation Sleep herbs, if needed Ashwagandha & Lemon Balm tinctures
SATURDAY	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Bacopa, Ginseng & Rhodiola Vit. D & B-complex	Physical exercise Probiotic-rich lunch Turmeric, Omega-3s & Acetyl-L-Carnitine Ginkgo, Gotu Kola, & Lemon Balm tinctures	Berries & walnuts	3 mushroom tinctures (Lion's Mane, Reishi, & Cordyceps tinctures) Healthy Dinner Phosphatidylserine, L-Theanine & Turmeric Walk outside after dinner	Magnesium Disconnect devices Mindfulness meditation Sleep herbs, if needed Ashwagandha & Lemon Balm tinctures

Ch. 25: Depression Protocol

According to the World Health Organization, depression is the leading cause of disability worldwide. Suicide is the fourth-highest cause of death for people between the ages of 15-29, and mental disorders are consistently in the top 10 for burden of disease worldwide¹. There are still misconceptions and areas of ignorance surrounding depression, partly due to the myriad ways depression presents itself, not only the variation in disorders but also the variation of the disorder in each individual presentation.



What is depression? Often this word is used when people simply feel sad, but situational depression and clinical depression differ. The existence of situational depression versus endogenous (biologically-based) depression has been a hotly debated topic among professionals. Some studies have failed to find any variation between the two types², but more-specific studies have found that situational depression has distinctive characteristics³. Field clinicians have long held that **situational depression** is a reactive response to adverse life events that are generally short-term and primarily due to the individual not being able to respond or cope effectively with a negative or traumatic event⁵. This is sometimes referred to as an adjustment disorder. Individuals can point to a specific incident that triggered the depression and, upon learning coping skills or seeking short-term medical treatment, may not suffer another episode.

Dysthymia (chronic mild depression with possible major episodes) is a persistent long-term depressive disorder. Adult diagnosis requires that symptoms present over two years. For children, it must last at least a year and may present as anger or irritability. While dysthymia may not be as severe as other forms of depression, its long-term nature can make it feel oppressive to sufferers and can wreak havoc on an individual's self-esteem.

Major Depressive Disorder is the most commonly recognized form of depression. This is recognized by feelings of guilt or worthlessness, low energy, poor concentration, appetite changes (having no appetite or a ravenous one), low or depressed mood,

decreased interest in previously enjoyable activities, slow body movement or markedly restless movement, sleep disturbances, or suicidal thoughts⁶. For diagnosis, an individual must have five of these symptoms, with at least one being depressed mood or lack of pleasure in activities with marked impairment in daily life⁷. This type of depression may have a direct trigger that individuals can point to or can simply be part of an individual's life. Clinical depression is believed to be due to an imbalance of neurotransmitters in the brain, commonly agreed to be tied to dopamine, serotonin, and norepinephrine. However, there is some evidence that depression may be connected to a difference in grey matter in the brain.^{8 9 10}

Bipolar Disorder, previously known as **manic depression**, cycles between markedly high or energetic moods and markedly low depressive moods. Between the cycles of depressive symptoms, an individual may experience mania characterized by less need for sleep, feelings of grandiosity, pressured speech or excessive need to talk, impaired judgment when it comes to risky or dangerous activities, and either an increase in goal-directed activity (such as creative work, school work, etc.) or what is known as psychomotor agitation, essentially an increase in activity for the sake of needing to keep moving. Mania can present in various levels of severity. If the primary symptoms of mania are present but do not markedly impair functioning in any area of life, it is classified as hypomania or Bipolar 2. Bipolar symptoms typically, but not always, first begin to appear in young adulthood, though there are instances of symptoms appearing in children; this is rare and should be considered carefully as many behaviors typical to childhood or other disorders of childhood can have similar looking symptoms.¹¹

Depression is also a large part of **schizoaffective disorder**. In schizoaffective disorder, aspects of schizophrenia and depression are present simultaneously. Depressive symptoms must be present for the entire time psychotic symptoms are present for diagnosis. Similar to other disorders associated with depressive symptoms, there is a suspected imbalance of the neurotransmitters dopamine, norepinephrine, and serotonin. However, structural differences in the brain associated with schizophrenia and schizoaffective disorders are also noted compared to brains of individuals without disordered symptoms.^{12 13}

Depression is a complex and varied illness with many available treatments that can present in many different ways. Some expressions require specific treatments, but there is a standard protocol of treatment options. It is essential to take care of yourself holistically if you have been diagnosed with any form of depression. If you have a close relative diagnosed with depression you are at increased risk of developing depressive symptoms and should pay close attention to how you treat your body and mind.

Certain lifestyle choices related to nutrition and supplementation, activity, and mental improvements enable optimal functioning of your mental health. This depression protocol demonstrates how to structure nutrition and activity and educates you about natural

remedies and mental exercises that can help keep depressive symptoms managed.

1. Diet and Nutrition

Diet and nutrition matter for both physical and mental health. Maintaining a lifelong commitment to fueling your body in a way that feeds your gut microbiome and your brain is key. You may have heard the phrase “you can’t outrun your fork.” The accepted premise is that you can’t exercise enough to undo the effects of a harmful diet. The nutrients we ingest help supply the brain, and your brain can’t function well with a poor diet.

Chemicals in the brain play a significant role in symptoms of depression. These are affected by the building blocks they are provided as well as interference such as inflammation. This foundation of the pyramid focuses on the Do’s and Don’ts of nutrition for prevention and to aid in the treatment of symptoms. Success lies in keeping everything in moderation and being gentle with self-expectations. Being hard on yourself makes it harder to maintain significant changes and good lifestyle choices.

Do - Eat the Colors of the Rainbow: Leafy greens, colorful fruits, and vegetables should create the basis for anyone’s nutritional plan; this is especially important for mental health. Leafy greens like kale, spinach, collards, swiss chard, mustard, and arugula have many health benefits for the brain and body. Leafy greens are low in calories but dense in nutrients vitamin A, vitamin C, antioxidants, fiber, folate, vitamin K, magnesium, calcium, iron, many B vitamins, and potassium. Vitamins B12, B9, and folate deficiencies have been correlated to symptoms similar to depression, such as fatigue, slowed thinking and memory decline, and irritability¹⁴. The first step in making sure your brain is functioning well is ensuring it has all the essential nutrients it needs.

Colorful fruits and vegetables provide a plethora of nutrients as well. Colorful fruits and vegetables provide vitamin A, C, sodium, potassium, fiber, folate, magnesium, and iron. Carotenoids are naturally occurring plant pigments that produce the red, orange, and yellow colors of fruits and vegetables. Carotenoids are antioxidative and anti-inflammatory, two characteristics that have been found to improve mental functioning.¹⁵

Do - Omega-3’s: Omega-3 fatty acids are vital for optimal brain function. These are essential for neurotransmission, gene expression, and the growth and development of neural tissue.¹⁶ Western diets provide a significant amount of Omega-6 fatty acids, but lacks Omega-3s, and this imbalance is a marked predictor of attention disorders and depressive symptoms¹⁷. Studies have shown between 200 mg and 2,200 mg daily is the range needed to help improve depression¹⁸. The amount varies by individual, so starting lower and gradually increasing until you find your balance is recommended. Omega-3s are found in fish and seafood and to a lesser extent, in grass-fed beef. Long-chain omega-3s are best obtained via diet; however, supplements may be added.

Do - Foods and Herbs that Feed Your Microbiome: If you struggle with depression, addressing the state of your gut microbiome is key. Known as the gut-brain axis,

scientists have shown that the gut has an extraordinary effect on our moods, clarity of thought, and overall brain health¹⁹. What you eat is related to what is in your gut, and what is in your gut is linked to your mental health.²⁰ ²¹ If you have a gut infection, have a gut inflammatory disease, or if your microbiome is out of balance, there is a propensity to experience depression, anxiety and/or negative thinking²². There is also an uptick in social anxiety.

A healthy microbiome thrives on a diet rich in resistant starches, prebiotics, probiotics, and insoluble fiber. Ways to feed your microbiome include yogurt, sauerkraut, kefir, kombucha, miso, apple cider vinegar, and kimchi. Eat these regularly. Adding in balanced gut herbs like lion’s mane mushroom, reishi, slippery elm, and marshmallow root plus prebiotics like turkey tail mushroom, flax seeds, and jerusalem artichoke also help maintain a healthy gut, as do daily probiotics.

Do - Herbs, Spices, and Medicinal Plants: There are numerous cooking herbs and spices that have anti-inflammatory and other benefits. Greens like parsley, basil, and cilantro have antioxidative and anti-inflammatory effects.²³ Cilantro/Coriander have shown promise in mediating blood sugar instability²⁴. Cinnamon is currently being studied for its anti-depressant properties. Cinnamon contains components such as phenol, cinnamaldehyde, turmerone, proanthocyanidin, and linalool, which are inflammatory with anti-depressant effects²⁵.

Don’t - High Amounts of Caffeine: There are some benefits to having one daily cup of coffee or green tea. However, when that cup of coffee turns into a pot of coffee or includes multiple sodas, it has a markedly negative effect²⁶, including increasing feelings of anxiety, which can increase feelings of depression²⁷. For children and adolescents, 2.5 mg/kg is the recommended maximum dose of caffeine daily²⁸; most adults best tolerate 200-400 mg daily though this sensitivity varies widely²⁹.

Don’t - Fast Foods/Fried Foods: The “western” diet, high in fast food and processed foods, is linked to depression. The reduced insulin response and increased inflammatory response that is tied to fried, fast, and processed foods may be responsible for the link between depression and other disorders that are linked to this less healthy style of eating.³⁰ Processed and fried foods also disrupt gut health, which is strongly correlated to physical and mental health.

Don’t - Excessive Sugar: Sugar is one of the most addictive substances we ingest. Studies show it has similar effects on the brain as drugs like cocaine. It has been posited that something similar to natural opioids are released when sugar is ingested, creating a cycle of addiction that leads to continued unhealthy eating.^{31 32} Additionally, processed sugar is a well-known inflammatory agent, and inflammation is linked with higher instances of depression.

While all of these together add up to optimal nutrition for brain and body health, it is important to note that improvement comes from the combined accrual of

benefits. No one part is likely to hold the magic key to mood improvement; rather it is the sum of taking care of your body holistically that makes the difference.

2. Activity Exercise was one of the earliest holistic recommendations regularly made to people with depression. In fact, many placed exercise as the foundation of mental and physical health over nutrition. However, recent studies show that an individual can have a great exercise routine, but if they eat poorly, they will still experience adverse health effects. If easier, focus on activity rather than exercise, along with diet.

Many people groan and grumble at the thought of exercise; it has a negative connotation and seems like work, especially to someone fighting depression. However, if you think of an activity that is typically somewhat enjoyable, like swimming, going for a walk outside, playing tennis, going dancing, and then simply think of it as increasing your activity level instead of exercising, it is a much easier task to undertake. If going to the gym and running on the treadmill or lifting weights feels daunting, try participating in a sport you enjoy with friends as both of these types of activities have positive effects on your mental health.^{33 34} Time spent outside in nature is also important.

Physical activity elicits positive changes in neuroplasticity, the endocrine system, inflammation, oxidative stress, self-esteem, and self-efficacy.³⁵ Adult neurogenesis, the growth of new neurons, has been negatively correlated with Major Depressive Disorder. It's been hypothesized that neurogenesis that occurs with physical activity accounts for much of the anti-depressant effect of activity³⁶.

Neuro-inflammation has been tied to depression; it is thus understandable then that anything that decreases inflammation potentially has an antidepressant effect. Regular activity is linked to decreases in inflammation. Additionally, maintaining a healthy habit of activity improves feelings of self-esteem and self-efficacy, both of which improve symptoms of depression.³⁷

3. Finding Purpose For many individuals, finding purpose in life is essential to feelings of joy and happiness. Altruism has long been considered an essential aspect of self-esteem and self-efficacy. Whether it is learning to focus outwardly on others' needs or the connection that having found purpose creates, feeling a sense of meaning in life is a significant buffer against depression and is correlated with improvement in depression^{38 39}.

Resilience is a critical factor in finding contentment. The ability to push forward despite adverse events is a significant protective factor in mental health, and finding purpose is tied to having this ability. Mindfulness, decentering from oneself, and finding a more significant role in the world is central to long-term feelings of contentment and well-being.⁴⁰ When looking to find your purpose, look to family, volunteering,⁴¹ or supporting a cause. Everyone's journey is different. It is vital that you find something that drives you, motivates you,

and gives you meaning.

4. Counseling There are several evidence-based therapies for individuals suffering from various types of depression. Recent studies emphasize the importance of interventions geared toward the individual client, reinforcing the importance of a therapist choosing methodologies based on their knowledge of the individual person and not taking a one size fits all approach⁴². The evidence-based therapies in this section are ones recommended in the guidelines for the treatment of depression published by the American Psychological Association.

Cognitive Behavioral Therapy (CBT): In CBT, individuals learn how their feelings and thoughts are intertwined, emphasizing how an individual perceives and thinks about events as integral to how they feel about the events and themselves in general. Through CBT, individuals learn to master their own minds in order to impact how they live.

Mindfulness and Gratitude Practices: Mindfulness and gratitude practices are often used to improve symptoms of depression. The goal is to learn to live in the present moment and intentionally focus on your surroundings while *non-judgmentally* noticing your reactions and working to respond in a more beneficial way⁴³. Mindfulness and daily gratitude have been found to be very beneficial for the treatment of anxiety and mood disorders^{44 45}. Mindfulness and cognitive therapy are combined in MBCT, and Mindfulness-based cognitive therapy is considered a best practice by the APA⁴⁶.

Interpersonal Counseling: This is an effective evidence-based treatment that focuses on interactions between a person and their social group and how changes have occurred to cause or exacerbate depressive symptoms. Therapist and patient work together to identify the most important interpersonal issues and address communication, supportive listening, and analysis of how the patient can improve interactions.^{47 48}

Psychoanalytic Therapy: Psychoanalytic therapy comes from the foundations of psychology and psychoanalysis; however, psychoanalytic therapy is much more active on the part of the therapist and focuses on repairing internal conflicts and encouraging positive defenses and self-esteem. It is more short-term than a full psychoanalysis process that focuses on reconstructing aspects of the person's defenses and personality.⁴⁹ Psychoanalytic therapy has been found to be effective in the treatment of depression in all ages of patients in both the short term (a few months) to the longer term (multiple years of treatment).^{50 51}

Supportive Therapy: Supportive therapy is what it sounds like; the therapist works to support the client. This may take the form of advising, comforting, active listening, and empathy. This may also include educating the client about their specific issue, educating their family and other systems about the best ways to help and support them, as well as teaching the client to learn to advocate for themselves. Supportive therapy may be

most appropriate when the client is distressed, unable to cope with life effectively, and in need of more direct intervention than more insight-oriented therapies offer.⁵²

⁵³Therapy or counseling is very effective at decreasing symptoms of depression. It works best in conjunction with lifestyle changes, nutrition, and healthy activity.

5. Natural Remedies

This section focuses on supplements, herbs, and natural treatments. These work best when coupled with good nutrition and healthy activity levels. The line between natural remedies and nutritive eating for depression is a bit blurry. Some of these supplements are found in food, but the amount needed to have a benefit may require supplementation beyond what you can get via diet. *Before adding supplements to your routine, talk to your healthcare practitioner if you are taking western medications as some may interact.*

Amino Acids: Amino acids are precursors for building neurotransmitters. There is an increasing amount of evidence that depression is often primarily a function of a problem in the metabolic processes of neurotransmitters in the brain. The same neurotransmitters that western medicines target with anti-depressants are the ones targeted by the supplements listed here⁵⁴.

5-Hydroxytryptophan (HTP): HTP has been recognized for its antidepressant effect since the early 1980s. 5-HTP increases brain serotonin levels as it is a serotonin precursor.⁵⁵ Best results have been found with the addition of carbidopa and tyrosine. L- Tryptophan also helps in the production of serotonin and has been found to double levels in the brain.⁵⁶ Preliminary studies show 5-HTP may work as well as SSRI antidepressant drugs (e.g. fluoxetine (Prozac) and sertraline (Zoloft)) for mild-to-moderate depression. Another study showed that the therapeutic effects of 5-HTP were the same as for the SSRI fluvoxamine (Luvox) with fewer side effects⁵⁷. Suggested dose: 200-300mg 1-2 times/day taken as a slow or extended release formula (5 HTP SR) or 50-250mg 3x/day, as 5-HTP does not last long in the body^{58 59}. Start low and see what works for you. Best taken with food.

DL-Phenylalanine: DL-Phenylalanine is believed to work as an enhancer when enhancing serotonin isn't enough. Individuals taking this supplement report higher mood and energy levels.⁶⁰ Individual recommended dosage varies quite a bit. Talk to your doctor as there are some people with the rare metabolic disorder phenylketonuria (PKU) who are unable to process phenylalanine. Suggested dose: 75mg to 200mg per day^{61 62}.

S-Adenosylmethionine (SAME): SAME is a naturally-occurring chemical that is believed to have antidepressant effects. SAME occurs in all living cells and is vitally important to cellular metabolism. Several clinical studies have found that supplementation with SAME has a beneficial effect on mood and may help relieve symptoms of depression.^{63 64} Suggested dose is 800-1600mg divided between two and four doses daily. It is recommended to start out taking 200 mg twice daily and then increase to 400 mg twice daily and more if needed. It is best taken

on an empty stomach for absorption.⁶⁵ This supplement should not be taken with an anti-depressant unless carefully monitored by your attending physician.

Gamma-Aminobutyric Acid GABA: This is an inhibitory amino that helps calm anxiety. When it inhibits excitatory action, it calms the brain. Imbalance in GABA and the glutamate systems have been tied to depression in multiple studies.^{66 67} Many studies show that appropriate levels of GABA are not found in depressed patients when CSF, plasma, and brain concentrations are studied versus non-depressed patients. Additionally, there is a potential negative impact that some antidepressants may have on the GABA system.⁶⁸ Other supplements also have an effect on GABA levels in the body, so start with a smaller dose and increase as needed. Suggested dose: 100-250 mg twice with a maximum of three times daily⁶⁹.

The following herbs can help with depression:

Lemon Balm (*Melissa officinalis*): Lemon balm has been recognized for many years in alternative medicine as having positive effects on mood⁷⁰. Recent studies link compounds in lemon balm with mediating cell death and its antioxidant properties as part of the process that improves mood disorders. Even after being subjected to stress, the effects of lemon balm helped mood improvement.^{71 72} It has anxiolytic (anti-anxiety) properties, helps with looped thinking, and encourages the production of GABA, a calming neurotransmitter that helps to soothe anxiety and stress. Lemon Balm interacts with some sedatives and thyroid medications; discuss with your physician if you are taking any western medications. Suggested dose: 2 dropperfuls of lemon balm tincture 2x/day or 150 mg supplement 20-30 minutes before bedtime^{73 74}.

Mushroom Supplements (Lion's Mane, Cordyceps, and Reishi): Mushroom supplements have neuroprotective effects attributed to their antioxidant, anti-neuroinflammatory, and cholinesterase inhibitory properties⁷⁵ and show great promise for depression as well^{76 77 78 79 80}. Some compounds in Lion's Mane significantly decrease chemicals tied to depression^{81 82}. They have the added benefit of helping with gut health and inflammation⁸³. See out Leaky Gut Protocol at page 70 for more. These mushrooms are best taken as a dual-extracted mushroom tincture. Suggested dose: 2 dropperfuls(ml) of Lion's Mane and Reishi Tincture twice a day, and Cordyceps tincture once a day.

St. John's Wort (*Hypericum perforatum*): It has long been used in places like Germany as an antidepressant and has significant evidence of its efficacy in the treatment of mild to moderate depression.⁸⁴ It is hypothesized that St. John's Wort works due to compounds that bind with neurotransmitters in the brain and decrease the reuptake process, which is similar to what pharmaceutical medications do.⁸⁵ St. John's Wort interacts with a lot of western medications including organ transplant anti-rejection and HIV anti-retroviral medications so talk to your doctor first before trying this supplement.⁸⁶ Suggested dose is 300 mg three times a day in

tincture or supplement form. Note it can take three to six weeks to be effective. Some studies have increased dose to 1200 mg total daily as needed.⁸⁷

Chamomile (*Matricaria chamomilla*): is a flower with chemical components that help with inflammation and have antioxidant properties. It has anxiolytic (anti-anxiety) properties and has been demonstrated as an effective antidepressant, especially when the depression has anxious characteristics⁸⁸. Chamomile is traditionally made into a tea and ingested by drinking after steeping for 10 minutes in a covered cup; however, tinctures and extracts are also effective. Suggested dose: tea or 200 mg powdered supplement once daily, which can be increased up to 800 mg if needed.⁸⁹ For more on anxiety, please see our Anxiety and Stress Protocol.

The following supplements can help with depression:

Vitamin D: Vitamin D supplementation favorably impacts depression ratings in major depression. Suggested dose: 3,000 IU daily.

B-Complex: Vitamins B1, B3, B6, B9 and B12 are essential for neuronal function and deficiencies have been linked to depression.⁹⁰

B1: Thiamine low levels of thiamine have been linked with the “brain fog” or confusion aspects of depression. Suggested dose: 1.1-1.2 mg daily⁹¹.

B9 (Folate): Do not confuse this with folic acid. Folate is the natural form of the popular synthetic supplement called folic acid. Folate absorbs into the body faster and more efficiently than its synthetic counterpart. It helps the body function and develop properly. It also helps the body produce genetic material like DNA. It has been shown to possibly help with depression.⁹² In one study “patients managed with l-methylfolate achieved statistically significant improvements in self-reported depression symptoms and functioning and greater satisfaction with their medication treatment.” Suggested dose: 50 mg daily.

B12: Vitamin B12 deficiencies and even those in the low end of standard range have been linked to depressive symptoms, memory concerns, and irritability. With supplementation, studies find these symptoms abate.⁹³ Suggested dose: 1 mg daily⁹⁴.

Omega 3's: Omega 3 fatty acids are vital for neurotransmission.⁹⁵ Suggested dose is 200 to 2,200 mg daily⁹⁶. Starting low and gradually increase as needed.

Zinc: This important mineral helps keep the immune system working and is often taken as part of a healthy vitamin regimen. Low zinc levels are associated with depression. Our bodies do not store excess zinc, so we need to get this mineral through diet or supplements. Ways to obtain B vitamins and Zinc were discussed in our nutrition section above. Suggested dose is 8 mg daily for women and 11 mg for men with 40 mg being the upper limit of tolerability. Too much zinc can interfere with other minerals such as copper and iron.⁹⁷

depression. Some studies have found that it is as effective as tricyclic antidepressant medications.⁹⁸ Suggested dose: 125 to 300 mg of magnesium glycinate with each meal and at bedtime.

Acupuncture and Reflexology: These stimulate the nervous and circulatory systems. It works via the autonomic nervous system and reducing muscle tension and anxiety, which in turn cause feelings of calm and general well-being.⁹⁹ Studies affirm the effects on the nervous system and the neurotransmitters in the brain as the methods of action for improving depression as well as other mental health disorders^{100 101}.

Massage Therapy: Humans are social animals and need physical and emotional connectedness. In our disconnected world, sometimes depression is related to a lack of contact. There are promising studies that show for those who lack physical touch, massage therapy relieves some symptoms of depression.¹⁰² Similar to reflexology, massage stimulates the pressure receptors/nervous system and encourages a vagal response and reduces cortisol levels. Research consistently shows improvement in depressive symptoms with regular massage therapy.^{103 104}

Yoga: The combination of focused breath control, paying attention to your body, and gentle physical activity especially helps depression symptoms that stem from anxiety.¹⁰⁵ The neuroplasticity of the brain plays a significant part in feelings of depression and yoga produces substantial antidepressant effects that correlate with the elevation of serum brain-derived neurotrophic factor (BDNF) levels, implying a neuroplastic mechanism of antidepressant action.¹⁰⁶

Light Therapy: Light therapy has primarily been used in the treatment of seasonal affective disorder (SAD) - depression caused by a lack of exposure to sunlight in winter. This has been accepted as the standard for many years.¹⁰⁷ Recent studies have begun to look into the effects of light therapy on non-seasonal forms of depression. Recommendations include bright light therapy at or above 5,000 lux for 30-60 minutes a day as an effective and efficient treatment for major depressive disorder that is not seasonal in nature.^{108 109} Time spent in green spaces/outside in nature is linked to elevated mental and physical health. Make sure to spend time outside.

6. Sleep Sleep is crucial to the function of your brain and hormones, both of which directly connect to mood. Since depression is often linked with wanting to sleep too much during the day or not being able to sleep enough at night, it is practical to follow sleep hygiene guidelines as part of your protocol for combating depression. Sleep herbs such as valerian, hops, and chamomile coupled with magnesium glycinate may be taken at night. Please see our Sleep Protocol Chapter for more detail.

7. Western Medical Approaches ¹¹⁰ Some may question our pyramid placement of western medicine, given that for many people medication is necessary for depressive

disorders. It is important to note that it is rarely recommended that an individual seek out medication alone. Recommendations for all forms of depression typically include medication plus counseling, exercise, or light therapy. Many medications have negative side effects, and it may take going through several medications before you find one that helps. Having the rest of your pyramid in place will make that search less challenging to manage should it be necessary for you.

Selective Serotonin Reuptake Inhibitors (SSRIs) are the most commonly prescribed medications for mild to moderate depression. They affect the neurotransmitter serotonin by decreasing the amount that is taken into neurons, leaving more serotonin available in the brain to transmit messages between neurons.

Citalopram (Celexa), Escitalopram (Lexapro), Fluoxetine (Prozac), Paroxetine (Paxil, Pexeva), and Sertraline (Zoloft) are examples of SSRI medications currently available.

While dosages vary, the general recommendation is to start on a low dose to ensure the best chance of the medication being tolerated with the least chance of side effects.

The biggest warning with these is that when individuals first start these medications, if they have previously been a suicide risk, then they may be at increased risk of suicidal ideations for the first few weeks of taking the medication. This risk is especially high for teenagers.

Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs): SNRIs work similarly to SSRIs, with the exception that it works on norepinephrine reuptake as well as serotonin. Examples of SNRIs are Desvenlafaxine (Pristiq), Duloxetine (Cymbalta), Levomilnacipran (Fetzima), and Venlafaxine (Effexor XR). SNRIs frequently have withdrawal effects and may have side effects that feel flu-like when first starting.

Tricyclic Antidepressants (TCAs): TCAs were some of the first antidepressants developed and have been largely replaced by newer medications due to their unpleasant side effects. These are still sometimes used if depression is resistant to other medications.

These include amitriptyline (Elavil), clomipramine (Anafranil), doxepin (Sinequan), imipramine (Tofranil), trimipramine (Surmontil), amoxapine (Amoxapine Tablets), desipramine (Norpramin), nortriptyline (Pamelor, Aventyl), and protriptyline (Vivactil).

Many of these carry a warning of a death potential with overdose due to the effect that they have on heart rate and other basic health functions. This category also carries a higher risk of abuse.

Tetracyclic Antidepressants: These are similar in profile to tricyclics in that they inhibit the reuptake of neurotransmitters associated with depression and have significant side effects. Tetracyclic antidepressants include Mirtazapine (Remeron) and Maprotiline (Ludiomil).

Dopamine and Norepinephrine Reuptake Blockers: These are newer medications that work on the neurotransmitters dopamine and norepinephrine. One of the key differences of this combination is that these typically do not have the side effects of weight gain or diminished sex drive as are common with other antidepressants.

Bupropion is known by the brand names Wellbutrin and Aplenzin. Due to the increased risk of restlessness and trouble sleeping doctors generally start patients on a low dose and work their way up if needed.

Monoamine Oxidase Inhibitors (MAOIs): These work to prevent the breakdown of all three neurotransmitters in the brain associated with depression. MAOIs are very effective medications in the treatment of depression but can be problematic due to the risk of dangerously high blood pressure if mixed with certain foods or other medications. These medications include Phenelzine (Nardil), Tranylcypromine (Parnate), Isocarboxazid (Marplan), and Selegiline (Zelapar, Eldepryl, Emsam).

5-HT_{1A/2/3} Receptor Antagonists: Some of these are also classified/known as atypical antidepressants. These are typically last-line treatment options as they have significantly more side effects and risks as they are intended to treat far more serious mental health disorders. Some are used for the treatment of schizophrenia, some treat hypertension, and some also treat bipolar disorder.

They all have some effectiveness in handling treatment-resistant depressive disorder but are only considered if everything else has failed to improve symptoms. These include: Clozapine, Trazodone, Alprenolol, Pindolol, Quetiapine, Asenapine, Paliperidone, Molindone, Doxepin, and Nefazodone. The use of these medications should be discussed at length with your doctor with a full discussion about side effects, contraindications, and risks versus benefits.

It is important to remember the reason that medications are at the lowest level of the pyramid is not that they aren't effective, but because, as a general rule, medications alone are not as effective as making other life changes. It is relatively easy to take a pill daily, but if you don't make the foundational lifestyle changes to accompany it then you will not see the full benefit of depression treatment.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Morning gratitude Vit. D, B complex 15-minute yoga routine Nutritious breakfast (including yogurt/kefir, flax/chia seeds, red/orange/yellow fruit, whole grain, cup of coffee or tea with cinnamon) Depression supplements	5-HTP Medicinal mushroom tinctures (Lion's Mane, Reishi, and Cordyceps) Nutritious lunch (large serving of leafy greens and beneficial herbs, blue/purple fruit, whole grain, seafood/healthy protein) Omega-3 supplement 5-min. mindfulness exercise	Weekly shopping/meal planning 20 min. walk outside Weekly goal planning: set 2-3 weekly goals Stop caffeine intake by 2 pm Probiotics	GABA support Lemon Balm tincture Zinc w/ dinner 15-min. exercise Reflective-gratitude journal Light nutritious dinner (leafy greens & beneficial herbs, colourful veggies, healthy protein) Lion's Mane & Reishi tinctures	5-HTP, Zinc, Magnesium GABA support Chamomile tea Balanced gut herbs/tincture 5-min. mindfulness Sleep routine Sleep herbs/sleep tincture, if needed
MONDAY	Morning gratitude Vit. D, B complex 15-minute yoga routine Nutritious breakfast Depression supplements	5-HTP Mushroom tinctures Nutritious lunch Omega-3 supplement 5-min. mindfulness exercise	20 min. walk outside Weekly therapy or support group meeting Stop caffeine by 2 pm Probiotics	GABA support Lemon Balm tincture Zinc w/ dinner 15-min. exercise Reflective-gratitude journal Light nutritious dinner Lion's Mane & Reishi tinctures	5-HTP, Zinc, Magnesium GABA support Chamomile tea Balanced gut herbs/tincture 5-min. mindfulness Sleep routine Sleep herbs/sleep tincture, if needed
TUESDAY	Morning gratitude Vit. D, B complex 15-minute yoga routine Nutritious breakfast Depression supplements	5-HTP Mushroom tinctures Nutritious lunch Omega-3 supplement 5-min. mindfulness exercise	20 min. walk outside Weekly volunteering Stop caffeine by 2 PM Probiotics	GABA support Lemon Balm tincture Zinc w/ dinner 15-min. exercise Reflective-gratitude journal Light nutritious dinner Lion's Mane & Reishi tinctures	5-HTP, Zn, Mg GABA support Chamomile tea Gut herbs/tincture 5-min. mindfulness Sleep routine/herbs
WEDNESDAY	Morning gratitude Vit. D, B complex 15-minute yoga routine Nutritious breakfast Depression supplements	5-HTP Mushroom tinctures Nutritious lunch Omega-3 supplement 5-min. mindfulness exercise	20 min. walk outside Check in: weekly goals Stop caffeine by 2 PM Probiotics	GABA support Lemon Balm tincture Zinc w/ dinner 15-min. exercise Reflective-gratitude journal Light nutritious dinner Lion's Mane & Reishi tinctures	5-HTP, Zn, Mg GABA support Chamomile tea Gut herbs/tincture 5-min. mindfulness Sleep routine/herbs
THURSDAY	Morning gratitude Vit. D, B complex 15-minute yoga routine Nutritious breakfast Depression supplements	5-HTP Mushroom tinctures Nutritious lunch Omega-3 supplement 5-min. mindfulness exercise	20 min. walk outside Weekly Creative Gathering or hobby Stop caffeine by 2 PM Probiotics	GABA support Lemon Balm tincture Zinc w/ dinner 15-min. exercise Reflective-gratitude journal Light nutritious dinner Lion's Mane & Reishi tinctures	5-HTP, Zn, Mg GABA support Chamomile tea Gut herbs/tincture 5-min. mindfulness Sleep routine/herbs
FRIDAY	Morning gratitude Vit. D, B complex 15-minute yoga routine Nutritious breakfast Depression supplements	5-HTP Mushroom tinctures Nutritious lunch Omega-3 supplement 5-min. mindfulness exercise	20 min. walk outside Weekly social gathering Stop caffeine by 2 PM Probiotics	GABA support Lemon Balm tincture Zinc w/ dinner 15-min. exercise Reflective-gratitude journal Light nutritious dinner Lion's Mane & Reishi tinctures	5-HTP, Zn, Mg GABA support Chamomile tea Gut herbs/tincture 5-min. mindfulness Sleep routine/herbs
SATURDAY	Morning gratitude Vit. D, B complex 15-minute yoga routine Nutritious breakfast Depression supplements	5-HTP Mushroom tinctures Nutritious lunch Omega-3 supplement 5-min. mindfulness exercise	Extended time outside Stop caffeine by 2 PM Probiotics	GABA support Lemon Balm tincture Zinc w/ dinner 15-min. exercise Reflective-gratitude journal Light nutritious dinner Lion's Mane & Reishi tinctures	5-HTP, Zn, Mg GABA support Chamomile tea Gut herbs/tincture 5-min. mindfulness Sleep routine/herbs

Ch. 26: Parkinson's Disease Protocol

Parkinson's disease is a brain disorder that leads to movement issues, such as tremors, stiffness, balance difficulties, and problems with coordination. As the disorder progresses, mood symptoms and difficulty with sleep may arise, and the ability to speak and walk may also be compromised.

Symptoms of Parkinson's disease may include:¹ tremors in the hands, arms, legs, jaw, or head; muscle stiffness, where muscle remains contracted for a long time; slow movement; impaired balance and coordination, sometimes leading to falls; parkinsonian gait, which includes a tendency to lean forward, take small and/or quick steps, and reduce swinging of the arms; depression and other emotional changes; difficulty swallowing, chewing, and speaking; urinary problems or constipation; and skin problems.



Although the causes of Parkinson's disease are not fully understood, we do have a fairly stable understanding of how the disorder develops and progresses. The condition usually develops gradually; however, the rate of progression varies from person to person.

When this disorder begins to develop, oxidative stress in a region of the brain called the nigrostriatal pathway (the region including and connecting the substantia nigra and the striatum) leads to the brain cells there being damaged. These brain cells are dopaminergic neurons, meaning that they produce dopamine. As the dopaminergic neurons are damaged, the body is tasked with the responsibility to clean out these damaged cells. This process is called autophagy. Over time and if the damage continues, autophagy can become impaired, leading to apoptosis, which is programmed cell death. The damaged dopaminergic neurons begin to die, and this leads to lower levels of dopamine in the brain.

The neurotransmitter dopamine is well known for its role in the pleasure response, but it also plays a critical role in movement. Dopamine levels in the brain need to be balanced with acetylcholine levels (another neurotransmitter) for normal movement to

occur. When the continual destruction of dopaminergic neurons leads to lower levels of dopamine, acetylcholine begins to dominate; this is what leads to many of the motor symptoms that we recognize as characteristic of Parkinson's disease, including the shaking, muscle rigidity, and issues with balance and coordination.

In order to address these hallmark symptoms of Parkinson's and help slow its progression, we need to combat oxidative stress with high levels of antioxidants, and address dysregulated mitochondrial dysfunction, autophagy, and apoptosis. In this chapter, you'll find several interventions found to be beneficial in the prevention and treatment of Parkinson's disease.

1. Diet There are several dietary interventions associated with decreased risk for Parkinson's disease. These include avoiding dairy products and consuming whole foods and organic, non-genetically modified soy.

Avoid Dairy: Research demonstrates that consuming milk and other dairy products may increase your risk for Parkinson's disease.^{2 3 4 5} The exact reason behind the correlation isn't clear, although preliminary research suggests that this may be due to dairy products leading to low serum uric acid levels.⁶ Serum urate and uric acid are believed to have a protective effect against the development of Parkinson's disease. Consider dairy alternatives, such as almond milk; organic, non-genetically modified soy milk; cashew milk; rice milk; coconut milk; or oat milk.

Consume Whole Foods: Eat whole foods with a focus on vegetables and fruits, as their high phytonutrient and antioxidant content has been associated with decreased functional decline and a slowing of the progression of Parkinson's disease.⁷ Fruits and vegetables contain vitamins A, B2 (riboflavin), C, and E, each of which tends to be lower in individuals with Parkinson's.⁸ As an example of neuroprotective phytochemicals found in fruits and vegetables, sulforaphane, a compound found in cruciferous vegetables like cabbage, broccoli, kale, bok choy, and cauliflower, has been found to exert a neuroprotective effect in neurodegenerative disorders like Parkinson's disease.⁹

Include Non-Genetically Modified Soy: Genistein, which is an isoflavone found in soybeans, has a protective effect on the brain and nervous system. Pre-clinical research suggests it may be especially beneficial in protecting menopausal women from Parkinson's disease.¹⁰ In another pre-clinical trial that was not limited to female models, genistein administration led to complete restoration of dopamine and its metabolites, dihydroxyphenylacetic acid (DOPAC) and homovanillic acid (HVA), in the striatum.¹¹

2. Protective Hobbies and Activities There are hobbies and activities that are associated with reduced risk for developing Parkinson's disease and with slowed progression of Parkinson's disease. These include dancing, therapeutic singing, and music therapy.

Dancing: Research suggests that dance is a beneficial adjunct therapy in the treatment of Parkinson's disease.¹² People who danced experienced a decrease in the amount of time it took for them to rise from a seated position and showed significant improvements in their movement and functional mobility.

Therapeutic Singing: In a therapeutic singing program for people with Parkinson's, group singing was associated with protection of the airway (e.g., trachea and lungs) from foreign materials for longer periods during swallowing.¹³ Research suggests that singing may be of benefit with regard to the speech of people with Parkinson's disease.¹⁴ Additionally, people with Parkinson's reported that choral singing helped them to better manage the social isolation, low mood, and communication difficulties that they faced as a result of their condition.¹⁵

Music Therapy: Music therapy has beneficial effects on motor symptoms, non-motor symptoms, and quality of life in people with Parkinson's disease.¹⁶ Music therapy may decrease freezing in patients with Parkinson's, leading to fewer falls and maintenance of long-term mobility; improve mood and depressive syndromes; and improve overall quality of life in people with Parkinson's disease.^{17 18 19}

3. Physical Activity

Physical activity is highly beneficial in the treatment of Parkinson's disease. Specifically, physical activity exerts a protective effect on the brain and nerves, which may be due to the fact that physical activity increases brain-derived neurotrophic factor (BDNF).^{20 21} BDNF helps the brain develop new connections, repairs failing brain cells, and protects healthy brain cells. This leads to improvements in behavior, mood, and cognition.²² These benefits may also be due to the fact that physical activity reduces cortisol levels, vascular risk, and neuroinflammation.²³

Furthermore, the hallmark of Parkinson's disease is a reduction in dopamine in the substantia nigra, which is a specific region of the brain. Lower levels of dopamine lead to the unwanted motor and mood symptoms that are typically seen with Parkinson's. Physical activity leads to the production of dihydroxyphenylacetic acid (DOPAC) and homovanillic acid (HVA), which are dopamine metabolites.²⁴

4. Vitamins and Supplements

More intense physical activity typically leads to higher levels of DOPAC and HVA, as well as higher levels of extracellular dopamine. In other words, physical activity increases the production of dopamine and dopamine-related compounds, leading to improvements in Parkinson's-related symptoms.

Participating in aerobic exercise like walking, cycling, or water aerobics, at least 3 days per week for 30 to 40 minutes per session may help delay decline in Parkinson's disease.

There are vitamins and supplements that are beneficial in addressing Parkinson's-related symptoms. These include vitamins B3, C, D, and E; and omega-3 fatty acids.

Vitamin B3 (Niacin): People with Parkinson's disease tend to have lower levels of vitamin B3, also known as niacin, than the general population. Niacin plays a major role in energy production, and researchers have suggested that some of the symptoms of Parkinson's, such as fatigue, sleep problems, and mood changes, may be due to low levels of niacin. Research suggests that high-niacin diets help reduce the risk of developing Parkinson's disease.^{25 26} Research also demonstrates that low-dose supplementation with niacin could slow disease progression and improve quality of life in individuals diagnosed with Parkinson's disease.²⁷ It's important to note that supplementing with niacin can lead to flushing, which can last for 30-45 minutes. Suggested dose: 250 mg slow-release niacin per day.

Vitamin C: Vitamin C plays an important role in the production of catecholamines like dopamine. In Parkinson's, the metabolism of dopamine leads to the accumulation of oxidative by-products, and this leads to the accumulation of abnormal proteins.²⁸ Vitamin C's antioxidant properties can be beneficial because vitamin C can be transported to the brain where its antioxidant activity can be expressed. Vitamin C can also help improve the effectiveness of the Parkinson's medication, levodopa.²⁹ Suggested dose: 200 mg ascorbic acid per day.

Vitamin D: Vitamin D plays an important role in muscle and bone health. Researchers have suggested that low vitamin D levels may play a critical role in the development of Parkinson's disease.³⁰ People with low levels of vitamin D tend to have more severe Parkinson's symptoms than people with normal vitamin D levels.^{31 32} Additionally, supplemental vitamin D significantly reduces the likelihood of clinical decline in people diagnosed with Parkinson's disease. Suggested dose is 5000-10,000 IU per day of vitamin D3 until 25-hydroxyvitamin D levels are at 40-60 ng/mL (get your vitamin D levels tested); then decrease to a maintenance dose of 5000 IU per day.

Vitamin E: Vitamin E is a potent antioxidant. High intake of vitamin E helps protect against the development of Parkinson's disease.³³ A pilot trial found that long-term treatment with vitamin E may help delay the onset of treatment with levodopa.³⁴ This is important because, while levodopa is the most effective conventional medical treatment for Parkinson's, it becomes increasingly less effective over time. This is why delaying the onset of levodopa treatment is important for people with Parkinson's. In this particular study, the time when levodopa treatment was initiated was delayed by 2.5 years in individuals receiving alpha-tocopherol (Vit E) and ascorbate (Vit C). Please note that people with bleeding or clotting disorders should not take high doses of supplemental vitamin E due to increased risk for bleeding. Suggested dose: 1000-1500 IU of natural vitamin E per day, with meals.

Omega-3 Fatty Acids: While there has not yet been any human trials conducted with the goal of discovering whether or not omega-3 fatty acids improve motor symptoms in Parkinson's disease, research does demonstrate that supplementation with omega-3 fatty acids reduces depression in individuals with Parkinson's disease.^{35 36} Omega-3 fatty acids have a beneficial effect on mood both in patients taking anti-depressants and in those who are not. Research conducted on animal models of Parkinson's disease show that the omega-3 fatty acid docosahexaenoic acid (DHA) helps protect the cells in the brain that produce dopamine from destruction.³⁷ DHA also helps reduce levodopa-related symptoms in primates by 40%.³⁸ Suggested dose: 1000-2000 mg of omega-3 fatty acids (e.g., 720 mg EPA and 480 mg DHA) per day, with meals.

5. Herbs There are several herbs that can be beneficial in addressing Parkinson's disease and symptoms. These include velvet bean, turmeric, green tea, ginseng, lemon balm, lion's mane, cordyceps, and reishi mushrooms, valerian, passionflower, St. John's wort, and ginkgo.

Velvet Bean (*Mucuna pruriens*): Velvet bean is a leguminous plant that is a natural source of L-DOPA. This is why researchers have been trying to evaluate the benefits associated with using this plant in the treatment of Parkinson's. In a 2017 study, people with Parkinson's who were given a single low dose of velvet bean (12.5 mg of velvet bean per kg of body weight) showed a similar response to people who were given a dose of levodopa and benserazine.³⁹ Both groups showed similar improvements in motor capabilities with the velvet bean group showing fewer involuntary movements and adverse effects than the levodopa plus benserazine group.

Those given a single higher dose of velvet bean (17.5 mg of velvet bean per kg of body weight) showed greater improvements in motor function at 90 and 180 minutes, fewer involuntary movements, and fewer adverse effects than the group given levodopa plus benserazine. In another study, people with Parkinson's were divided into three groups and were either given 15 grams of velvet bean, 30 grams of velvet bean, or 200 mg of levodopa plus 50 mg of carbidopa.⁴⁰ Everyone in the study rotated through all three treatment options on a weekly basis, but they did not know which treatment option they were being given each week.

Plasma L-dopa levels rose faster when participants were given 30 grams of velvet bean, and the treatment became active in about half the time that it took for participants to experience the effects of the levodopa-carbidopa. L-dopa levels were 110% higher when participants were given 30 grams of velvet bean compared to when they were given levodopa-carbidopa, and there were no significant differences in side effects or how well the study participants tolerated each of the treatments. Suggested dose: Additional research is needed to determine if single dosing or regular, repeated dosing of velvet bean is more beneficial in Parkinson's disease.

Turmeric (*Curcuma longa*): Curcumin, the main active constituent in turmeric, helps restore the health of degenerated neurons in the substantia nigra.^{41 42} In a pre-clinical trial, treatment with curcumin led to marked improvements in motor function as well as improvements at the cellular and biochemical levels. Curcumin also inhibits the enzyme monoamine oxidase B, which is responsible for the breakdown of dopamine in the brain.⁴³ The active constituents in turmeric exhibit anti-inflammatory activity that may help address the increased inflammation observed in Parkinson's disease; turmeric helps reduce levels of pro-inflammatory compounds like tumor necrosis factor- α , interleukin-1 β , and interleukin-6.⁴⁴ Suggested dose: 2 dropperfuls of a 1:4 tincture, 2 times daily.

Green Tea (*Camellia sinensis*): Green tea exhibits antioxidant, neuroprotective, and anti-inflammatory activity.⁴⁵ The main active component of green tea, epigallocatechin gallate (EGCG), prevents the aggregation of a protein found in the brain called α -synuclein. When α -synuclein aggregates, it leads to the destruction of neurons in the brain in the pattern recognized as characteristic of Parkinson's disease. EGCG binds to sites on the α -synuclein protein that are prone to aggregation and prevents aggregation. It also contributes to the disaggregation of α -synuclein, which leads to neuron preservation.^{46 47 48} Suggested dose: 450 mg green tea extract, twice daily.

Ginseng (*Panax ginseng*): Pre-clinical studies suggest that ginseng may exhibit neuroprotective activity in Parkinson's disease.⁴⁹ This benefit may be due to its antioxidant properties, as well as its ability to decrease apoptosis by reducing the levels of apoptotic proteins, such as Bax, Bcl-2, cytochrome c, and cleaved caspase-3.⁵⁰ Ginseng's antioxidant activity (which may help prevent neuronal damage) and its effects on apoptosis may make it more likely that the dopaminergic neurons of the substantia nigra, which are typically damaged and destroyed in Parkinson's, will be preserved. Suggested dose: 3-4 dropperfuls of a 1:5 tincture, 2 times daily.

Lemon Balm (*Melissa officinalis*): Lemon Balm is associated with improvements in memory, mood, and age-related cognitive performance; a reduction in stress and anxiety; and improved clarity and focus. It has potent antioxidant properties and helps protect brain cells and nerves.⁵¹ Pre-clinical trials demonstrate that lemon balm reduces oxidative damage in the brain and protects neurons in animal models of Parkinson's disease.⁵² Suggested dose: 2 dropperfuls of a 1:3 tincture, 2 times daily.

Lion's Mane (*Hericium erinaceus*): Lion's mane has neurotrophic activity, meaning that it helps promote the growth and maturation of nerve and brain cells.⁵³ Lion's mane does this by increasing levels of nerve growth factor (NGF). Pre-clinical trials suggest that, when administered orally for at least 25 days, lion's mane may significantly improve oxidative stress and reduce neuronal damage in the substantia nigra.⁵⁴ Suggested dose: 171

2 dropperfuls of a dual-extracted tincture, 2 times daily.

Cordyceps (*Cordyceps spp.*): Cordycepin, one of the main active constituents of cordyceps mushroom, exhibits neuroprotective, anti-inflammatory, antioxidant and anti-tumor activity. It helps improve the pathological features of Parkinson's disease. Specifically, cordycepin improves motor disorders and the loss of dopaminergic neurons, and it also reduces harmful effects on microglia, which are immune cells of the brain and spinal cord that are important to Parkinson's pathology.⁵⁵ Suggested dose: 2 dropperfuls of a dual-extracted tincture, 2 times daily.

Reishi (*Ganoderma lucidum*): Reishi also exerts a protective effect on neurons. Oxidative stress, mitochondrial dysfunction, impaired autophagy (the body's ability to repair damaged cells to generate healthy ones), and overactive apoptosis (programmed cell death, which typically occurs at a given rate normally but can become overactive leading to tissue destruction) have all been highlighted as contributors to Parkinson's disease. Research suggests that reishi medicinal mushroom protects dopaminergic neurons from oxidative stress, and regulates mitochondrial function, autophagy, and apoptosis.⁵⁶ Suggested dose: 2 dropperfuls of a dual-extracted tincture, 2 times daily.

Valerian (*Valeriana officinalis*): Valerian exhibits neuroprotective effects in Parkinson's disease by inhibiting apoptosis. Multiple studies have demonstrated that people who were exposed to the herbicide rotenone are at greater risk for Parkinson's disease. Treating neurons with rotenone leads to programmed cell death, similar to what occurs in Parkinson's; rotenone is used to induce a Parkinson's-like disorder in animal models for pre-clinical studies. Valerian extract brought about a significant increase in the viability of cells that were treated with rotenone; treatment with valerian extract significantly reduced the number of cells that went through programmed cell death.⁵⁷ Suggested dose: 6 dropperfuls of a 1:5 tincture, at bedtime (valerian is also a sleep aid).

Passionflower (*Passiflora incarnata*): Passionflower exhibits antioxidant and cognition-enhancing activity that may be beneficial in the treatment of Parkinson's disease. In one pre-clinical study, treatment with passionflower extract was associated with less body rigidity and fewer episodes of abnormal movements.⁵⁸ In the same study, the extract was found to have free radical-scavenging activity, which may have contributed to the improvements in cognition that were also observed. Suggested dose: 3 dropperfuls of a 1:5 tincture, 2 times daily.

St. John's Wort (*Hypericum perforatum*): St. John's wort's active constituents exhibit antioxidant and neuroprotective properties, and pre-clinical research demonstrates that this plant may help prevent degeneration of neurons in Parkinson's disease.⁵⁹ It

with the metabolism of many drugs, so if you are taking any conventional medication, speak with your provider about potential interactions before taking St. John's wort. Suggested dose: 4 dropperfuls of a 1:5 tincture, 2 times daily.

Ginkgo (*Ginkgo biloba*): Ginkgo extract inhibits the destruction of dopaminergic neurons.⁶⁰ Additional pre-clinical research demonstrates that the neuroprotective effects of ginkgo on the neurons of the nigrostriatal pathway (the area of the brain where dopaminergic neurons tend to be damaged and lost in Parkinson's disease) are likely, at least in part, due to the fact that ginkgo quenches free radicals, reducing the oxidative stress that leads to neuronal damage.⁶¹ Suggested dose: 4 dropperfuls of a 1:5 tincture, 2 times daily.

6. Western Medical Approaches

Conventional
m e d i c a l

approaches to addressing Parkinson's disease include carbidopa-levodopa, dopamine agonists, MAO-B inhibitors, amantadine, anti-cholinergics, and deep brain stimulation.

Carbidopa-Levodopa: Carbidopa-levodopa is a combination medication that is commonly prescribed to address Parkinson's disease and Parkinson's-like syndromes. Levodopa is a natural compound that travels through the body and enters the brain, where it is converted into dopamine. Carbidopa acts similarly. Combining carbidopa with levodopa makes it less likely that the carbidopa will be converted to dopamine before it reaches the brain. This reduces the likelihood of adverse effects, such as lightheadedness and nausea. This medication can become less effective with time, and it may lead to involuntary movement at higher doses.

Dopamine Agonists: Instead of being converted into dopamine within the body like carbidopa-levodopa, dopamine agonists are substances that are structurally similar to dopamine and can act like dopamine by stimulating dopamine receptors within the body. Examples include pramipexole (Brand name: Mirapex ER®), rotigotine (Brand name: Neupro®), and apomorphine (Brand name: Apokyn®). Although they aren't typically as effective as levodopa, these drugs tend to work for a longer period of time. Side effects may include hallucinations; sleepiness; and compulsive behaviors, such as hypersexuality, gambling, and overeating.

Monoamine Oxidase B (MAO-B) Inhibitors: Selegiline (Brand name: Zelapar®), rasagiline (Brand name: Azilect®), and safinamide (Brand name: Xadago®). These drugs block the monoamine oxidase B enzyme so that it is less able to break down dopamine that is present in the brain. Side effects may include nausea, headaches, or insomnia, and this drug may make hallucinations more likely if it is taken with carbidopa-levodopa.

Amantadine: Amantadine has been found to increase the release of dopamine in experimental models, but the exact mechanism of action is not clearly understood.

This drug is typically prescribed during mild, early-stage Parkinson's disease, or alongside carbidopa-levodopa in the later stages of Parkinson's to limit the involuntary movements that may occur with higher doses of carbidopa-levodopa. Side effects may include skin mottling, hallucinations, or ankle swelling.

Anti-Cholinergics: Anti-cholinergics, are usually prescribed alongside other drugs to treat symptoms of Parkinson's, such as tremors. Anti-cholinergic drugs include trihexyphenidyl (Brand name: Artane®), benztropine (Brand name: Cogentin®), orphenadrine (Brand names: Norflex®, Norgesic®, etc.), procyclidine (Brand name: Kemadrin®), and biperiden (Brand name: Akineton®). These drugs are prescribed in an attempt to restore the balance between the neurotransmitters acetylcholine and dopamine in the brain. Side effects

may include blurred vision, dry mouth, constipation, and difficulty urinating. In older adults, symptoms may also include confusion, hallucination, memory problems, and sedation.

Deep Brain Stimulation: Deep brain stimulation requires a surgical procedure where a surgeon implants electrodes in the brain to stimulate a specific part of the brain. This procedure is typically reserved for people with late-stage Parkinson's disease who are no longer improving with levodopa. Deep brain stimulation can lead to cognitive or memory problems, so it isn't recommended for people with dementia. It's also important to note that deep brain stimulation may offer benefit, but it does not prevent Parkinson's disease from progressing.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Dose 1: Green tea extract & Ginseng tincture Dose 1: 3 dual-extracted mushroom tinctures (Lion's mane, Cordyceps, & Reishi) Vitamins C, D, & E	Dose 2: Green tea extract & Ginseng Other Parkinson's-related herb(s) of your choice* Omega-3s & Niacin w/ food	Dose 2: 3 dual-extracted mushroom tinctures (Lion's mane, Cordyceps, & Reishi)	Physical activity (aim for 30-45 min. per day) Other Parkinson's-related herb(s) of your choice*	Valerian, passion-flower, and lemon balm tea or tincture before bed
MONDAY	Dose 1: Green tea extract & Ginseng tincture Dose 1: 3 dual-extracted mushroom tinctures (Lion's mane, Cordyceps, & Reishi) Vitamins C, D, & E	Dose 2: Green tea extract & Ginseng Other Parkinson's-related herb(s) of your choice* Omega-3s & Niacin w/ food	Dose 2: 3 dual-extracted mushroom tinctures (Lion's mane, Cordyceps, & Reishi)	Protective hobby of your choice** Other Parkinson's-related herb(s)*	Valerian, passion-flower, and lemon balm tea or tincture before bed
TUESDAY	Dose 1: Green tea & Ginseng Dose 1: 3 mushroom tinctures Vitamins C, D, & E	Dose 2: Green tea extract & Ginseng Other Parkinson's-related herb(s) * Omega-3s & Niacin w/ food	Dose 2: 3 dual-extracted mushroom tinctures (Lion's mane, Cordyceps, & Reishi)	Physical activity (aim for 30-45 min. per day) Other Parkinson's-related herb(s)*	Valerian, passion-flower, and lemon balm tea or tincture before bed
WEDNESDAY	Dose 1: Green tea & Ginseng Dose 1: 3 mushroom tinctures Vitamins C, D, & E	Dose 2: Green tea extract & Ginseng Other Parkinson's-related herb(s) * Omega-3s & Niacin w/ food	Dose 2: 3 dual-extracted mushroom tinctures (Lion's mane, Cordyceps, & Reishi)	Protective hobby of your choice** Other Parkinson's-related herb(s)*	Valerian, passion-flower, and lemon balm tea or tincture before bed
THURSDAY	Dose 1: Green tea & Ginseng Dose 1: 3 mushroom tinctures Vitamins C, D, & E	Dose 2: Green tea extract & Ginseng Other Parkinson's-related herb(s) * Omega-3s & Niacin w/ food	Dose 2: 3 dual-extracted mushroom tinctures (Lion's mane, Cordyceps, & Reishi)	Physical activity (aim for 30-45 min. per day) Other Parkinson's-related herb(s)*	Valerian, passion-flower, and lemon balm tea or tincture before bed
FRIDAY	Dose 1: Green tea & Ginseng Dose 1: 3 mushroom tinctures Vitamins C, D, & E	Dose 2: Green tea extract & Ginseng Other Parkinson's-related herb(s) * Omega-3s & Niacin w/ food	Dose 2: 3 dual-extracted mushroom tinctures (Lion's mane, Cordyceps, & Reishi)	Physical activity (aim for 30-45 min. per day) Other Parkinson's-related herb(s)*	Valerian, passion-flower, and lemon balm tea or tincture before bed
SATURDAY	Dose 1: Green tea & Ginseng Dose 1: 3 mushroom tinctures Vitamins C, D, & E	Dose 2: Green tea extract & Ginseng Other Parkinson's-related herb(s) * Omega-3s & Niacin w/ food	Dose 2: 3 dual-extracted mushroom tinctures (Lion's mane, Cordyceps, & Reishi)	Protective hobby** Other Parkinson's-related herb(s)*	Valerian, passion-flower, and lemon balm tea or tincture before bed

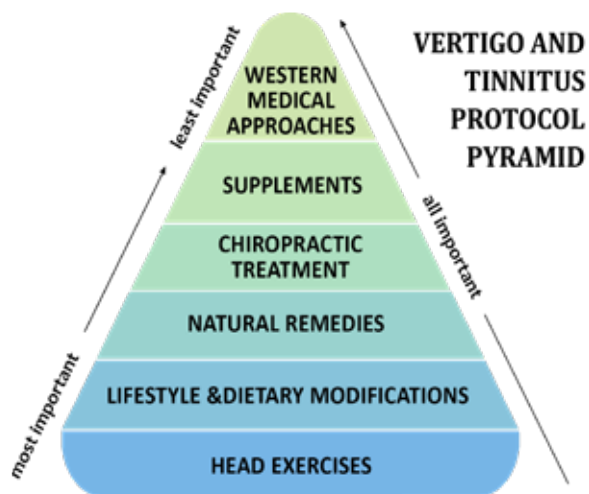
*Consider adding a one-time dose of velvet bean (See protocol for additional information). Talk with your provider first if you are currently taking carbidopa-levodopa.

*Options for other Parkinson's-related herbs: Turmeric, St. John's wort, Ginkgo, Lemon Balm (Please note that St. John's wort interacts with many drugs. Speak with your provider before taking it if you are currently taking medication).

**Protective hobbies: dancing, therapeutic singing, music therapy with familiar music.

Ch. 27: Vertigo and Tinnitus Protocol

The ears are one of the most sophisticated and sensitive areas of our body. Its two main functions are hearing and maintaining balance. The three main parts include: the outer ear, the middle ear, and the inner ear. Each serves to help transfer and transduce sound waves into nerve impulses so that our brain may interpret them.



We are able to maintain our balance due to mechanisms within the ear. They are designed to sense gravity and acceleration. When there is an issue with the inner ear, you usually notice it right away. Those with vertigo feel as if they are spinning, tilting, or off-balance when they are sitting still. It causes significant nausea and makes it difficult to function.

Vertigo can be caused by several different things. Benign paroxysmal positional vertigo (BPPV) is caused by tiny calcium particles called canaliths. These particles become dislodged from their proper location and begin to collect in one of the semicircular canals within the inner ear. This semicircular canal becomes sensitive to directional head changes, causing dizziness and disorientation. Sometimes BPPV can resolve on its own. However, interventions are often needed to correct the issue.

Sometimes a virus can cause an infection and inflammation in the inner ear. This affects nerves in the inner ear that help the body stay balanced. Vertigo can also be caused by Meniere's disease. Meniere's disease is when the buildup of fluid in the inner ear changes the pressure within the ear. This can trigger recurring episodes of vertigo as well as tinnitus.

Tinnitus is a ringing in the ears. While tinnitus is usually experienced as a ringing in the ears, it can also manifest as buzzing, humming, roaring, hissing, or clicking. The sounds are not coming from an external source, only the person with tinnitus can hear them.

Both tinnitus and vertigo can worsen as we age. Risk factors include:

amount of time. Those who work in loud environments like factory workers, musicians, soldiers, loggers, and construction workers are at a higher risk.

Aging: With age comes a decline in nerve fibers inside the ears. As these nerve fibers decline in function, the risk of tinnitus and hearing loss increases.

Cardiovascular and Blood Pressure Issues: Many inner ear problems like tinnitus and vertigo are linked to circulatory issues. It is important to make sure that the ears are receiving proper blood flow.

Gender: Men tend to have more issues with tinnitus than women.

Smoking and Alcohol: Smokers and those who drink alcohol regularly have an increased risk of ear problems.

1. Head Exercises There are three main head exercises for dealing with vertigo: the Brandt-Daroff Exercise, the Semont Maneuver, and the Epley Maneuver (I have personally used the Epley Maneuver at home to stop an acute case of vertigo – it worked beautifully and quickly). It is especially helpful to do these exercises during a vertigo episode. They can help shift the crystals within the ear to stop vertigo. These head exercises may also bring circulation back to the ear, which can help with tinnitus as well.

Epley Maneuver: The Epley Maneuver is one that can be performed without assistance while sitting on a bed. It is also called the particle repositioning or canalith repositioning procedure. Follow the steps below and do this at least twice daily, once in the morning and once in the evening:

- Sit on your bed with your feet and legs straight out in front of you. Place a pillow so that it will be under your shoulders (not your head) when you lie back.
- If your issue is in the right ear, turn your head 45 degrees to the right. If your issue is in the left ear, turn your head 45 degrees to the left. If you are unsure try both.
- Now quickly lie back, keeping your head turned. Keep your shoulders on the pillow with your head still turned and reclined. Your head should recline about 30 degrees below your shoulders.
- Wait 60 seconds.
- Without sitting back up, turn your head 90 degrees the opposite way. Your head should now be looking 45 degrees the other way.
- Wait 60 seconds.
- Turn your head and entire body another 90 degrees (i.e. roll to one side). If the problem is with your right ear, turn your body 90 degrees to the left. If the issue is your left ear, turn your body 90 degrees to the right. Your head should still be below your shoulders.
- Wait 60 seconds.
- If the problem is the right ear, sit up on your left. If the problem is the left ear, sit up on the right. If you are having trouble visualizing this there are how to examples online.

Brandt-Daroff Exercise: This is a relatively simple daily exercise. If you have recurring issues with vertigo, try this

exercise three times a day using the steps below:

- Sit in an upright position on the edge and toward the middle of your bed or couch.
- Slowly lie down on one side keeping your head looking up at a 45-degree angle and bringing your feet up onto the bed, keeping them bent.
- Remain lying down and looking up at an angle for thirty seconds before sitting back up into an upright position. Stay upright for 30 seconds.
- Repeat on the opposite side.
- Repeat the exercise 5 times.
- Lie on one side or the other until the vertigo disappears.

After an episode, do this daily for 1-2 weeks.

Semont Maneuver: The Semont Maneuver is another effective way to treat vertigo caused by BPPV (benign paroxysmal positional vertigo). It is often done with a physical therapist present, but you can do it at home. If you are practicing it at home, have someone else with you to make sure you don't become too dizzy. Follow the steps below:

- Sit in an upright position on the middle edge of your bed.
- Determine which side is affected.
- Have someone manually turn your head at a 45-degree angle away from the affected side.
- Have the person quickly lay you down on the bed on the affected side. Look up at the ceiling once you are lying down.
- Lie on your side until the dizziness subsides (if you have dizziness).
- Have someone move you up into a sitting position and then quickly lie you down on the opposite side (the unaffected side) with your head in the same position as it was on the first side. Once in position, your eyes will be looking down at the floor.
- Stay in this position until the vertigo passes.
- Have someone guide you back up into a sitting position.

Try to stay upright for several hours after this procedure. Have someone help you with this daily. If you are still experiencing issues after a week of doing this, BPPV may not be the cause of your vertigo. Try the Brandt-Daroff Exercise or the Epley Maneuver if this is the case – they work for BPPV as well as other causes of vertigo.

It is normal to feel dizzy during these exercises. Practicing them daily can help make a significant difference by getting your body used to the sensations and removing crystals in the ear. If you are concerned that you may lose your balance, make sure someone is around to assist you.

2. Lifestyle and Dietary Modifications

Lifestyle and dietary modifications can help reduce the amount of fluid in the ears that contributes to vertigo and tinnitus. Some foods can help reduce inflammation while others can make it worse. Below are recommended foods plus those to avoid for inner ear issues.

Sodium directly affects body fluid levels and the regulation

of body fluids. Limit your sodium intake to no more than 2,400 mg daily. If you want to address an inner ear issue, try to stay under 1,000 to 2,000 mg by being mindful of the foods you eat and by keeping track of your sodium intake.

Choose foods that are *naturally low in sodium*, such as fresh vegetables and fruits, fresh meat, poultry, and fish, and whole grains. Use herbs and spices to add flavor instead of salt.

Avoid the Following Foods: Avoid canned and processed foods, MSG, blackened and smoked foods, and fried foods. Don't salt to your food. Caffeine can trigger tinnitus. Use herbal tea instead. **Sugar** is problematic as it causes inflammation and fluctuations in body fluids. Significantly lowering your sugar intake can help reduce vertigo and tinnitus.

Stop Smoking/Ditch the Nicotine: Nicotine decreases blood supply to the inner ear and may also increase blood pressure.

Watch Migraine Triggers: Migraines can make you dizzy and worsen both vertigo and tinnitus. Pay attention to what may be triggering your migraines if you suffer from them. It may be certain foods, lighting, or stress.

Be Mindful of Medications: Some medications can trigger or worsen vertigo and tinnitus. Over the counter medications like antacids have a high amount of sodium. Pain relievers and fever reducers like aspirin and ibuprofen may also interfere with the body's fluid control mechanisms.

Change Your Lighting: Fluorescent lighting can trigger dizziness. If you are forced to work, shop, or dine in an environment with fluorescent lighting, try sitting with your back to it. Flickering candles can also worsen vertigo.

3. Natural Remedies Natural remedies can help improve blood circulation to the inner ear, restore balance, and support the health of the inner ear. They can also reduce side effects like nausea. Some herbs target the neurological system, helping with any issues that may be triggering vertigo and nausea. Try the remedies below for relief, and continue once daily for maintenance.

Cordyceps Tincture: Fluid accumulation in the inner ear can cause tinnitus and vertigo, as in the case of Ménière's disease.¹ Cordyceps mushroom is an effective treatment for tinnitus caused by inner ear fluid²; do note that it isn't helpful for those with long-term auditory nerve disorder. Suggested dose: 2 dropperfuls of a double-extracted tincture twice daily.

Lion's Mane and Reishi Mushroom Tinctures: Lion's mane and reishi mushrooms are powerful allies when it comes to brain health, increasing circulation, lowering inflammation, and as a neuroprotective³. Lion's mane has been shown to increase Nerve Growth Factor (NGF) in the brain⁴ and both mushrooms may help reduce the intensity of tinnitus. Suggested dose: Take two dropperfuls of each as a dual-extracted tincture twice daily.

Ginkgo Biloba: Research around ginkgo for tinnitus

is mixed.⁵ Try it to see if it works for you. Like reishi and lion's mane, it acts as a vasodilator, thus helping with circulation. It may help tinnitus by preventing free-radical damage to the cochlea or by increasing blood flow, improving the health of the inner ear⁶. Ginkgo extract was also shown to protect against noise-induced hearing loss⁷. Standardized Ginkgo leaf extracts have been used in clinical trials for tinnitus, cognitive issues, and cardiovascular disorders, at daily doses of 60 mg to 450 mg – best used as an extract/tincture.

Rosemary (*Rosmarinus officinalis*) Tincture and External Rosemary Massage Oil: Rosemary is often used to enhance circulation in the body. This makes it a promising treatment for ear issues that result from poor circulation. Suggested dose: Two dropperfuls tincture twice daily internally. For external use, combine ten to fifteen drops of rosemary essential oil with three ounces of olive oil. Suggested dose: Apply this to the outside of the ear, around the ear, and down the neck twice daily, massaging it into the skin thoroughly.

Butcher's Broom (*Ruscus aculeatus*): Butcher's broom is used for both tinnitus and vertigo caused by Meniere's disease. It can help restore circulation to the inner ear as well as lower inflammation. Suggested dose: Two dropperfuls of tincture up to three times daily (twice is usually sufficient) during an episode.

Ginger and Peppermint Aromatherapy or Tea (for Nausea): An unwelcome side effect of vertigo is nausea. A combination of ginger and peppermint helps calm an upset stomach and quell nausea. If you are experiencing vertigo, start by trying the head exercises in the first section of this protocol. Afterwards, take deep breaths of an aromatherapy inhaler that contains three drops of ginger and one drop of peppermint essential oils. Take slow, deep breaths in through your nose as you inhale. Then slowly exhale through your mouth. Do this for ten to fifteen minutes until the nausea subsides. If you don't have an aromatherapy inhaler, place the drops on a cotton ball and store it in a plastic bag to keep with you throughout the day. Ginger and peppermint may also be eaten or taken as a tea for nausea.

Cayenne: Cayenne helps to promote circulation in the body and alleviates inflammation and pain. For ear issues that stem from damage to the inner ear as a result of infection and inflammation, try adding cayenne to your diet. You can also make a fiery health drink by filling a jar with cayenne peppers and covering them in apple cider vinegar. Cover and let infuse for four weeks, then strain into a bottle and label. Take a half-ounce shot daily with food or after eating.

Gotu Kola: Gotu kola is known to treat insufficient blood flow as well as various nerve issues. It can be useful for both tinnitus and vertigo. Suggested dose: One dropperful of a tincture three times a day for relief up to 14 days. You may also take this as a powdered pill supplement. Do not take more than 500mg twice daily. Do not use for more than 6 weeks at a time (best to use 2-weeks at a time), and take a 2-week break before starting again.

4. Chiropractic Treatment

Chiropractic adjustments can help relieve symptoms of tinnitus and vertigo. Treatment focuses on the spine and joints and their relationship to the nervous system. The goal is to give the body the means to heal itself. Many people think that going to the chiropractor is just for those with a bad back, but chiropractic adjustments do much more for the body. Chiropractic care can help promote better drainage of inner ear fluids. Since tinnitus and vertigo can be caused by fluid trapped in the ear, by adjusting the neck and spine, fluids are released from the inner ear and inflammation is reduced.

Find a trained chiropractor in your area. Note that most chiropractors will need to adjust your body several times. After a few visits, you may notice that you have more energy, less fluid and drainage, and less inflammation. You may also notice a better range of motion.

5. Supplements

Some supplements have shown promise in helping to treat or prevent issues with both tinnitus and vertigo. Many people with Meniere's disease are lacking certain nutrients.

B Vitamins: Those who suffer from tinnitus are often deficient in B vitamins. Not only do they increase circulation to the ears, B vitamins can also help stabilize the nervous system. Find a quality B vitamin supplement that contains vitamins B12, B3, and B1, as these are the most important for tinnitus. Take as directed on the bottle.

Magnesium: Some studies have shown that when magnesium levels rise in ear fluid, instances of tinnitus stop.^{8 9} It is important to take magnesium daily as the ringing can come back when magnesium levels drop. Don't skip a dose. Suggested dose: 250-350 mg daily at night.

Lipo-Flavonoids: One of the most popular treatments for tinnitus, good scientific research is lacking. Suggested dose: six capsules daily, taken two at a time after each meal.

Turmeric/Curcumin: If inflammation in the ear canal is causing issues, a curcumin supplement may help. Curcumin is the active compound in turmeric. It is widely popular for lowering both inflammation and pain. It is especially helpful for those who suffer from ear issues that are caused by prior trauma to the ear canal (such as chronic ear infections). Follow the dosage directions on the bottle.

Calcium: A deficiency in calcium can directly affect your middle ear. Calcium is important for helping with muscle contractions and muscle tone. Low calcium can lead to hearing loss and inner ear issues. Suggested dose; 1,000 mg daily in two divided doses.

6. Western Medical Approaches

If you suspect you may have vertigo or tinnitus and see a physician, they may perform diagnostic tests to help determine the issue. A common test to start with is a hearing exam. You may be referred to an ear, nose, and throat specialist for the remainder of your treatment, or a physician who specializes in inner ear issues (audiologist).

After a hearing test, your doctor may require head scans to gain a better idea of what is taking place in the ear canal. They may also test for vitamin deficiencies or heart problems. For issues with vertigo, an eye and head movement test may be conducted. Sometimes, rotary chair testing is used to check balance. After the tests are complete, your doctor should have a better idea of the root cause of the hearing issue, such as problems with the blood vessels or an overabundance of ear wax. If earwax is the culprit, the doctor may remove it in the office.

Communicate clearly with your doctor about any noises you are hearing and all of your symptoms. Provide a list of all the medications you are taking as some medications can cause vertigo and tinnitus. Depending on the cause of your inner ear issues, you may be prescribed medication

like Piracetam and Mesoglycan. Mesoglycan is often prescribed for ear issues related to a blood vessel disorder. Piracetam is for neurological conditions and is marketed as a cognitive enhancer. For vertigo, medications that relieve dizziness and nausea may include antihistamines or anticholinergics. A common side effect of these is drowsiness. Sometimes, anti-anxiety medications like Valium and Xanax are prescribed. Keep in mind that these medications cause drowsiness and can be addictive.

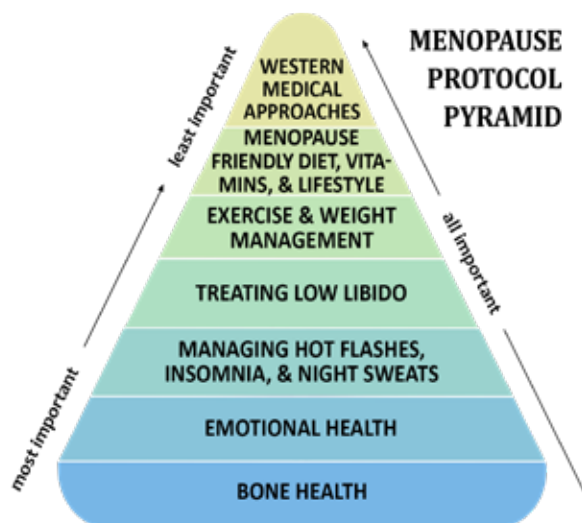
On rare occasions surgery may be required. Surgeries may include removal of inner ear organs (only for very serious cases) or injections of antibiotics into the ear to disable the balance function. These injections are given in the affected ear and proper balance function is expected to take over in the unaffected ear.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Epley maneuver Calcium, Lipo-flavonoids, B vitamins, & Curcumin (all after breakfast) Cordyceps, Lion's Mane, & Reishi dual-extracted tinctures Gotu Kola as/if needed Ginkgo extract	Weekly meal plan Weekly grocery shop Butcher's broom tincture Ginger & peppermint aromatherapy or tea	Lipo-flavonoids (after lunch) Gotu kola as needed Cordyceps tincture	Calcium, Lipo-flavonoids, (after dinner) Rosemary oil ear massage Butcher's broom tincture as needed	Brandt-Daroff exercise Magnesium Lion's mane & Reishi dual-extracted tinctures
MONDAY	Epley maneuver Calcium, Lipo-flavonoids, B vitamins, & Curcumin (all after breakfast) Cordyceps, Lion's Mane, & Reishi dual-extracted tinctures Gotu Kola as/if needed Ginkgo extract	Chiropractic treatment Butcher's broom tincture Ginger & peppermint aromatherapy or tea	Lipo-flavonoids (after lunch) Gotu kola as needed Cordyceps tincture	Calcium, Lipo-flavonoids, (after dinner) Rosemary oil ear massage Butcher's broom tincture as needed	Brandt-Daroff exercise Magnesium Lion's mane & Reishi dual-extracted tinctures
TUESDAY	Epley maneuver Calcium, Lipo-flavonoids, B vitamins, & Curcumin (all after breakfast) Cordyceps, Lion's Mane, & Reishi dual-extracted tinctures Gotu Kola as/if needed Ginkgo extract	Butcher's broom tincture Ginger & peppermint aromatherapy or tea	Lipo-flavonoids (after lunch) Gotu kola as needed Cordyceps tincture	Calcium, Lipo-flavonoids, (after dinner) Rosemary oil ear massage Butcher's broom tincture as needed	Brandt-Daroff exercise Magnesium Lion's mane & Reishi dual-extracted tinctures
WEDNESDAY	Epley maneuver Calcium, Lipo-flavonoids, B vitamins, & Curcumin (all after breakfast) Cordyceps, Lion's Mane, & Reishi dual-extracted tinctures Gotu Kola as/if needed Ginkgo extract	Butcher's broom tincture Ginger & peppermint aromatherapy or tea	Lipo-flavonoids (after lunch) Gotu kola as needed Cordyceps tincture	Calcium, Lipo-flavonoids, (after dinner) Rosemary oil ear massage Butcher's broom tincture as needed	Brandt-Daroff exercise Magnesium Lion's mane & Reishi dual-extracted tinctures
THURSDAY	Epley maneuver Calcium, Lipo-flavonoids, B vitamins, & Curcumin (all after breakfast) Cordyceps, Lion's Mane, & Reishi dual-extracted tinctures Gotu Kola as/if needed Ginkgo extract	Chiropractic treatment Butcher's broom tincture Ginger & peppermint aromatherapy or tea	Lipo-flavonoids (after lunch) Gotu kola as needed Cordyceps tincture	Calcium, Lipo-flavonoids, (after dinner) Rosemary oil ear massage Butcher's broom tincture as needed	Brandt-Daroff exercise Magnesium Lion's mane & Reishi dual-extracted tinctures
FRIDAY	Epley maneuver Calcium, Lipo-flavonoids, B vitamins, & Curcumin (all after breakfast) Cordyceps, Lion's Mane, & Reishi dual-extracted tinctures Gotu Kola as/if needed Ginkgo extract	Butcher's broom tincture Ginger & peppermint aromatherapy or tea	Lipo-flavonoids (after lunch) Gotu kola as needed Cordyceps tincture	Calcium, Lipo-flavonoids, (after dinner) Rosemary oil ear massage Butcher's broom tincture as needed	Brandt-Daroff exercise Magnesium Lion's mane & Reishi dual-extracted tinctures
SATURDAY	Epley maneuver Calcium, Lipo-flavonoids, B vitamins, & Curcumin (all after breakfast) Cordyceps, Lion's Mane, & Reishi dual-extracted tinctures Gotu Kola as/if needed Ginkgo extract	Butcher's broom tincture Ginger & peppermint aromatherapy or tea	Lipo-flavonoids (after lunch) Gotu kola as needed Cordyceps tincture	Calcium, Lipo-flavonoids, (after dinner) Rosemary oil ear massage Butcher's broom tincture as needed	Brandt-Daroff exercise Magnesium Lion's mane & Reishi dual-extracted tinctures

Ch. 28: Menopause Protocol

Menopause is a natural biological process that occurs when the menstrual cycle ends. Technically, you have reached menopause after you have gone one year without a menstrual cycle. While menopause can happen when a woman is in her forties, the average age for a woman to officially enter menopause in the United States is 51. The years leading up to menopause, when a woman may experience changes in her menstrual cycle, hot flashes, and other symptoms, is a transitional phase called *perimenopause*.

The duration of menopause depends on a variety of factors, such as whether a woman is a smoker, as well as a woman's race and ethnicity. On average, menopause is a process that lasts around seven years, but may be as long as 14 years for some women.



Every woman is affected by menopause in a different and unique way. While menopause is a normal part of aging for women, the symptoms can be quite unpleasant. Common menopause symptoms include hot flashes, insomnia, fatigue, mood swings, lowered libido, vaginal dryness, weight gain, irritability, and even depression. Because of the reduction of hormones like estrogen that help promote healthy bones, some women experience brittle bones and bone loss after menopause.

There are four stages of menopause: pre-menopause, perimenopause, menopause, and post-menopause. **Pre-menopause** begins at puberty when a woman has her first menstrual period. During puberty, a woman's body signals that it is ready to carry a child when her ovaries release its first egg. The lining of the uterus has built up in hopes of holding a fertilized egg, but when the egg is not fertilized, the lining of the uterus sheds, resulting in a menstrual period. This happens each month until perimenopause, when periods may become more unpredictable. Menstrual cycles vary for each woman, but it is not uncommon for women to experience symptoms such as uterine cramps, headaches and migraines, irritability, breast tenderness, and anxiety. Once puberty has started, a woman will have a menstrual cycle each month until perimenopause.

when a woman is in her 40s. During perimenopause many women experience more severe PMS (premenstrual syndrome) symptoms due to hormonal fluctuations and imbalances. These can cause irregular periods, hot flashes, night sweats, mood swings, decreased libido, and vaginal dryness.

Stage three is **menopause**. Eventually, a woman will have expended all of her eggs and will have her final menstrual cycle. This signals the start of menopause (although menopause doesn't technically start until twelve months after the last menstrual cycle). During menopause, you may still have perimenopausal symptoms like mood swings, weight gain, insomnia, hot flashes, night sweats, low libido, and irritability. The day after the twelfth month mark is a period of time known as **post-menopause**. This is the final stage. Post-menopausal women need to be extra vigilant about their health, as health complications can arise at this time due to hormonal changes. Health issues include bone loss and osteoporosis, heart disease, urinary tract infections, and incontinence. Due to increased vaginal dryness, it is not uncommon for post-menopausal women to experience pain with sexual activity (dyspareunia). If you have bleeding after you are post-menopausal (12 months after your last cycle) please see your OB/GYN.

This protocol is designed to help women in various stages of menopause with symptoms like insomnia, night sweats, hot flashes, mood swings, anxiety, bone loss, and low libido. While menopause is a natural life process, it doesn't have to consume your life. This protocol can help you manage the uncomfortable symptoms of menopause while also helping you live a healthy, full life post-menopause. Women who are struggling with menopause may seek help from their OB/GYN, especially if symptoms are interfering with their daily lives. Bloodwork can test for Follicle-stimulating hormone (FSH) levels, estrogen (estradiol) levels, and Thyroid-stimulating hormone (TSH) levels, plus rule out other potential hormonal issues. Vaginal estrogen may be prescribed to relieve vaginal dryness. Talk to your doctor about hormone replacement therapies and do your own research as well, as not all OB/GYNs are trained in menopause. Please see our Western Medical section of this protocol for more information.

1. Bone Health Estrogen plays an important role in bone health. During menopause, estrogen levels decline significantly. As a result, osteoporosis can develop and bones can become brittle and break more easily. There are several natural ways you can combat this. You can take herbs with phytoestrogens - compounds that act similarly to estrogen in the human body - to help build estrogen levels to support bone health. You can consume nutrient-dense herbs to give your body the calcium and nutrients it needs. You can take supplemental calcium, omega-3 fatty acids, Vitamins B, C, D, and K-2, magnesium, and trace minerals such as strontium, silicon, and boron. You can alter your diet and exercise patterns. Please see our Osteoporosis Protocol at page 234 for more detailed information.

Below are herbs that contain phytoestrogens, calcium,

and other nutrients that support bone health. Make sure to see a physician/ND before beginning a daily phytoestrogen routine to make sure you need additional estrogen. More information on these herbs and supplements can be found in the Osteoporosis Protocol.

Dandelion (*Taraxacum officinale*): Dandelion leaves contain high amounts of vitamins A, C, E, and K and also provide calcium, potassium, iron, folate, and magnesium. Add the fresh leaves to salads daily or gently sauté them. Dandelion has also been studied for reducing the risk of coronary artery disease¹, a disease that post-menopausal women are especially at risk for developing.

***Epimedium* spp. Plants² (Fairy Wing/Horny Goat Weed):** These have some of the same phytoestrogenic isoflavones as soy and inhibits osteoclast activity and slows down bone breakdown. When postmenopausal women took an *Epimedium* extract with their calcium supplement for 2 years, they had maintained bone density³, whereas the women their age who took calcium alone still lost density. Suggested dose: 200mg capsule or 2 dropperfuls tincture/extract per day.

Hops (*Humulus lupulus*): Hops flowers have potent phytoestrogenic activity.⁴ Hops may be used in combination with red clover if you have issues such as insomnia and anxiety, as hops is an excellent nervine and sedative. Read more about how to use hops for better sleep and less anxiety in the Sleep Protocol chapter.

Red Clover (*Trifolium pretense*): It contains phytoestrogens and may help combat bone loss, as estrogen plays a prominent role in bone health. A recent study⁵ showed that red clover influences estradiol concentration, exercise performance, and gut microbiota, all great news for menopausal women. I successfully use a black cohosh-red clover tincture daily to combat hot flashes.

Red Raspberry (*Rubus idaeus*): Red raspberry leaf can nourish the body with extra calcium, potassium, iron, manganese, and magnesium, and vitamins B, C, and E. It also supports a healthy uterus and has been used for centuries and studied for its effects on easing labor⁶ and delivery. Red raspberry leaf can help tone the uterus and may help prevent uterine fibroids that can worsen with menopause⁷ (see our Women's Sexual Health Protocol). Infuse two tsp. of chopped leaves into a cup of hot water and drink one to two cups daily for uterine and nutrient support.

Stinging Nettle (*Urtica dioica*): This is a nutritional powerhouse for bone-strengthening nutrients. It contains protein, fiber, calcium, and iron. It also helps with arthritis, inflammation, and allergies. Suggested dose is 2 dropperfuls stinging nettle tincture 1-2 times daily. Also add a cup of fresh (or ½-cup dried) stinging nettle to bone broth. Sip this nourishing broth throughout the day.

2. Emotional Health

Wildly fluctuating hormones during perimenopause and menopause can set the stage for mood swings, outbursts, and sudden bouts of crying. If you find that you are on edge, more emotional, and experiencing major mood shifts, you are not alone. This is a normal experience for

many menopausal women. The remedies in this section help to ease anxiety and stress, and bring emotional stability. For more on herbs for emotional health, please see our Anxiety and Stress Protocol.

Ashwagandha (*Withania somnifera*): Ashwagandha is an exceptional herb for balancing hormones. It improves mood by helping fortify your adrenals (via the HPA-axis) and also supports underactive (hypo) thyroid^{8,9}. This is helpful for women who are experiencing a slow metabolism post-menopause. A 2019 study¹⁰ found that adults who took 125-300 mg of ashwagandha extract twice daily for eight weeks experienced lowered stress, better sleep, and reduced blood cortisol levels. All three are important for balanced hormones, maintaining a healthy weight, and stabilizing moods. Do not take if you have hyperthyroid. Suggested dose: 125-300 mg (~2 dropperfuls) ashwagandha tincture twice daily.

Lavender (*Lavandula angustifolia*): Lavender can help calm nerves, reduce stress, and lower blood pressure. There are several ways to effectively use lavender ranging from inhalation of the essential oil to infusing in hot water to make tea. Because of its positive cardiovascular benefits, some doctors recommend lavender essential oil inhalation for cardiac rehabilitation¹¹. Since menopausal women are at a higher risk for cardiovascular issues, lavender is an especially useful herb. Try putting a few drops of lavender essential oil in an ultrasonic diffuser during the day or inhale the oil directly from the bottle as needed.

Lemon Balm (*Melissa officinalis*): Lemon balm is an effective nervine if you are struggling with anxiety. It has anxiolytic (anxiety reducing) effects, with rosmarinic acid as one of the active components shown to improve mood and cognitive performance. It is excellent for stopping looped thinking, and raising the levels of the neurotransmitter GABA in the brain. It significantly improves mean anxiety and depression scores¹² compared with a placebo. Suggested dose: Infuse two tsp. in a cup of hot water (add raw honey to taste) and drink one to two cups daily or using 2 dropperfuls of lemon balm tincture (my preference) daily/as needed.

Lion's Mane (*Hericium erinaceus*): It boosts cognitive and neurological function to help improve memory, focus, and Nerve Growth Factor. It reduces the toll stress and anxiety take on the body, while also lowering depression¹³, insomnia and inflammation. It is highly recommended for menopausal and postmenopausal women who struggle with brain fog and/or have a family history of Alzheimer's, dementia, and Parkinson's. It also helps with gut inflammation and is immunomodulatory, making it useful for those with an autoimmune disorder. Suggested dose: 2 dropperfuls of a dual-extracted tincture twice daily.

Motherwort (*Leonurus cardiaca*): Motherwort is another incredibly useful anxiolytic¹⁴ herb for calming anxiety and stress while also nourishing the heart and cardiovascular system. Motherwort is so effective for this that it was given the Latin name *Leonurus cardiaca* as a tribute to its cardiac-nourishing abilities. If you

find yourself getting easily frustrated due to anxiety and stress, motherwort can help calm the nerves and keep you grounded. Suggested dose: Take one dropperful as needed when you are feeling overwhelmed or infuse 1-2 tsp of motherwort aerial parts in a cup of hot water and drink as needed.

Reishi (*Ganoderma lucidum*): Reishi is an adaptogen that can strengthen the body against the negative effects of stress and anxiety, and has been associated with antidepressant activity¹⁵. It can also correct hormonal imbalances that result in mood swings and negatively affect emotional health. Reishi can help with brain fog, low energy, and even improve sleep, making it useful for menopausal women in a variety of ways. Medicinal mushrooms like reishi, lion's mane, turkey tail, and cordyceps may also help fight fatigue¹⁶ that leads to mood instability and they are all antiviral, anti-inflammatory, and immunomodulatory, making them a good choice for those who also have an autoimmune disorder.

Many menopausal women complain that they feel very fatigued and tired; these adaptogenic mushrooms help combat this issue. Sleep interruptions are another reason why some menopausal women feel tired. Learn how to address sleep issues in our Sleep Protocol. Suggested dose: Use reishi as a dual extracted tincture as it requires a double extraction to properly get all the medicinal compounds, as some compounds are water soluble and some are alcohol soluble. Take two dropperfuls up to three times daily.

St. John's Wort (*Hypericum perforatum*): It is helpful for those who suffer from mild to moderate depression and anxiety¹⁷. Suggested dose: Infuse 2 tsp fresh flowers into a cup of hot water to make tea. Drink 1-2 cups daily. As a tincture (extract) take two dropperfuls up to three times daily. If you are taking any Western meds be careful with interactions as St. John's wort interacts with many medications, including SSRIs.

Vitex/Chasteberry: Vitex specifically works to balance hormones that sometimes trigger mood imbalance. Vitex is a great option for perimenopausal women who wish to balance irregular periods, mood swings, and other symptoms of hormonal discord¹⁸. Studies show that vitex is a safe and efficacious alternative to be considered for the treatment of PMS/PMDD symptoms¹⁹, uterine fibroids, and PCOS²⁰ (for women with normal LH levels and high prolactin levels). Please see our Women's Sexual Health Protocol for more information. Suggested dose is 2 dropperfuls of vitex tincture up to twice daily. It may take a few months to see results.

Deep Breathing and Meditation: Along with the recommended remedies to combat stress and anxiety, try daily deep breathing and meditation to help reduce stress and help you cope with the hormonal fluctuations that trigger mood changes. Deep breathing is a simple yet effective way to calm the body and mind. Start your day by lying in bed and focusing on your breathing (and nothing else) for up to fifteen minutes. Take slow,

deep breaths in through your mouth and then slowly back out through your nose. Keep your mind focused on nothing but your breathing and feel free to use meditation strategies like repeating a mantra or a positive affirmation as you do this. End your day with this technique as well. Feel free to implement deep breathing as needed through the day. Studies show this helps improve mood, reduce stress levels, and lower heart rate and blood pressure.

3. Managing Hot Flashes, Insomnia, and Night Sweats

When a woman goes through perimenopause and menopause, she may experience hot flashes and night sweats that can significantly impact sleep. Fluctuating hormones are to blame for these uncomfortable phenomena. Hot flashes don't just happen at night, they can happen any time. In this section, you will learn what you can do to combat hot flashes, get a more restful night's sleep, and help eliminate night sweats.

HOT FLASH AND NIGHT SWEAT REMEDIES:

Black Cohosh Root (*Actaea racemosa*): Black cohosh is a phytoestrogenic plant that can help raise estrogen levels and suppress fluctuations that result in hot flashes and night sweats. Studies show black cohosh to be an effective remedy²¹ for hot flashes and more powerful than evening primrose oil. Suggested dose: 40-80 mg daily. Use as a tincture or supplement. I pair it with red clover as a blended tincture and take 2 dropperfuls as needed, especially before bed.

Dong Quai: This plant has been used with success to treat night sweats and hot flashes. It is sometimes called "female ginseng" due to all the health benefits it has for women. Women going through menopause and perimenopause may find that it they get a better night's sleep with this herb because it contains phytoestrogens that help raise low estrogen levels²² that contribute to fluctuations that cause hot flashes and night sweats. Additionally, it can improve metabolic action and strengthen the immune system. Suggested dose: Drink a cup of dong quai tea before bed.

Evening Primrose Oil (*Oenothera biennis*): This oil is made from the seeds of the evening primrose plant. These seeds are rich in gamma-linolenic acid, an essential fatty acid that is involved in the production of the hormone-like prostaglandins. As such, it can help counter hormonal fluctuations that cause hot flashes and night sweats. Additionally, evening primrose oil decreases postmenopausal psychological symptoms²³. Suggested dose: Start with 500 mg daily; you can increase up to 1000mg daily if needed.

Peppermint and Aloe Spray: For hot flashes that cause issues in the daytime, try combining the soothing effects of aloe with the invigorating and cooling effects of peppermint in a spray. This spray is especially soothing in the summer months when hot flashes can be even more problematic. To make this spray, combine one half ounce aloe vera juice with one half ounce of distilled water and shake well. Add three drops of peppermint essential oil and shake again. Put this in a one-ounce spray bottle and

shake well before each use. Thoroughly mist the skin as needed when you are feeling a hot flash coming on. Do not get this in mucus membranes or eyes.

REMEDIES FOR BETTER SLEEP: Chamomile (*Matricaria chamomilla*): Chamomile has long been a sleep-aid that helps to calm the body and mind, as well as reduce spasmodic activity that interrupts sleep. This gentle sedative can also soothe muscle aches and pain that disrupt sleep. Chamomile contains a compound called apigenin, an antioxidant that helps induce sleep²⁴. Suggested dose: Drink a cup of chamomile tea thirty minutes to one hour before bedtime or use a sleep tincture like my Sleep Blend, which contains chamomile plus hops, valerian root, passionflower, and magnesium glycinate.

Hops (*Humulus lupulus*): Hops have already been mentioned for their phytoestrogen content, but this flower also possesses the ability to help you get a full night's sleep by decreasing nocturnal activity in the circadian activity rhythm²⁵. Hops flowers have gentle sedative properties, and, like valerian, hops boost the production of GABA – a calming brain chemical that promotes sleep. Those taking hops report that it helps increase sleep time, while also reducing body temperature to help you enter a more relaxed state. This is great news if night sweats and hot flashes interfere with your sleep. Suggested dose: Drink a cup of hops tea at bedtime or take two dropperfuls of hops tincture or the Sleep Blend tincture 30 minutes before bed. In addition to the benefits hops has for sleep and hormone levels, flavonoids in hops have been found to help reduce weight gain, lower cholesterol, and reduce high blood sugar. Hops and valerian are often taken together for a more potent sleep remedy.

Lemon Balm (*Melissa officinalis*): This plant has already been mentioned for anxiety and mood balance, but it also helps with sleep since it calms the body and reduces anxiety. A review in the journal *Electron Physician*²⁶ found that several medicinal plants rich in phytoestrogens, including lemon balm, help calm menopausal hot flashes. In addition, lemon balm provides relief if your sleep issues are caused by hot flashes, hormone-induced anxiety and stress, or looped thinking. Suggested dose: Take two dropperfuls of lemon balm tincture as needed up to three times daily.

Passion Flower: Passionflower is an herbal sedative²⁷ that helps to soothe anxiety and calm the body and mind. Because of its nervine and sedative properties, passionflower helps to improve sleep quality. What's more, it has been shown to reduce hot flashes and mood swings. Suggested dose: Drink a cup of passionflower tea an hour before bed or take two dropperfuls of Sleep Blend tincture 30 minutes before bedtime.

Poppy (California): This pretty orange plant helps promote sleep and calm due to its nervine and sedative²⁸ effects. Suggested dose: Take 2-3 dropperfuls of tincture thirty minutes before bedtime.

Valerian Root: One of the most effective herbs for sleep, valerian root is a powerful Central Nervous System (CNS) depressant. It acts on the nerves to calm the body and

mind to help improve sleep quality and length. Valerian root can also help you fall asleep faster, decreases frequent waking at night, and calms restless sleep. It improves sleep by increasing the levels of the neurotransmitter gamma-aminobutyric acid (GABA). It also soothes stress and anxiety. One study conducted by the Foellinge Health Center showed that forty four percent of the participants reported perfect sleep and eighty nine percent reported improved sleep while taking valerian root²⁹, with no negative side effects reported. Suggested dose: Drink a cup of valerian root tea before bed or take two dropperfuls of valerian root or Sleep Blend tincture before bedtime.

Magnesium Glycinate: Magnesium glycinate is a sleep-inducing amino acid that is highly absorbable. It helps to improve the quality of sleep, promote healthy sleep patterns and REM cycles, and help the body produce melatonin. It can also relieve muscle tension to help you fall asleep and be more comfortable as you sleep. Magnesium glycinate has also been studied for its ability to reduce symptoms of depression³⁰. Suggested dose: 350 mg before bed.

Aromatherapy: Essential oils can be extremely helpful for soothing an overactive mind and calming the body. Some great sleep oils to try include cedarwood, vetiver, frankincense, chamomile, lavender, and clary sage. Combinations like three drops each of lavender and chamomile or vetiver and cedarwood are especially soothing and calming for sleep. One study showed that essential oils may help elderly dementia patients sleep better and correct sleep difficulties³¹, making aromatherapy a promising remedy to use alongside natural teas and tinctures for sleep.

Epsom Salts: Another form of magnesium for sleep is magnesium sulfate, or Epsom salts. If you have trouble falling asleep, try soaking in a warm bath with a cup or two of Epsom salts to relieve muscle tension. Consider a nightly bath before bed with Epsom salts if sore muscles or joints keep you from falling asleep.

Adaptogenic Mushrooms: Reishi and cordyceps mushrooms support the production of estrogen and help to counteract symptoms of menopause like low energy, daytime fatigue, brain fog, mood instability, insomnia, osteoporosis, and weight gain. These mushrooms can help strengthen bones while also helping you to feel better, including feeling more rested. Reishi may also improve hormonal imbalance, lower high cortisol levels, and encourage healthy immune function. Reishi can increase the production of estrogen and progesterone to counter the effects of diminishing hormones that result from menopause. Cordyceps also increases blood oxygenation to help with improved mental function and a healthier heart, muscles, brain, and body. Take reishi and cordyceps in a dual-extracted form daily for help with sleep, as well as a variety of other menopause-related issues. Both mushrooms have also been shown to help combat cancer³². As women age, their risk for cancer increases, so taking these as a preventative may help lower their risk. Suggested dose: 2 dropperfuls of a double extracted tincture 2-3 times daily.

4. Treating Low Libido

A decreased sex drive can be a stressful obstacle for those going through menopause. With the decrease in hormones comes an imbalance that results in a diminished desire to have sex. The libido changes that often come with menopause can affect relationships. Below are strategies to increase libido and to manage vaginal dryness naturally. Please see our Women's Sexual Health Protocol for even more information.

Coconut Oil: Vaginal dryness is common for women in menopause and post menopause. The drop in hormones means it is harder for the body to produce the lubrication needed to engage in sexual intercourse. Vaginal dryness can cause pain during sex, which can greatly affect sex drive. While there are plenty of lubricants you can use to make sex more enjoyable, there aren't many that are all-natural or toxin-free. Coconut oil is safe to use, is naturally antifungal, and can help prevent yeast infections and bacterial vaginosis.

Damiana: Damiana is a well-known aphrodisiac and studies back up its ability to boost libido³³. Damiana can increase sex drive, sexual desire, and overall sexual satisfaction for menopausal women. It may also help with vaginal dryness and lubrication. Suggested dose: Start with 200 mg daily in capsules or as a tincture.

Panax Ginseng: American ginseng is a popular herb for sexual dysfunction because it can boost mood, energy, and stamina. A recent study found that "Ginseng has significant effects in improving sexual function and quality of life and mitigating menopausal symptoms."³⁴ It may also help improve symptoms of hormonal imbalance for women going through menopause³⁵. This may in turn improve libido. Suggested dose: 2 dropperfuls of ginseng tincture up to twice daily or a 500mg supplement³⁶.

Maca: Maca root is known for being an effective fertility enhancer and may also increase sex drive³⁷. A 2018 study found that women taking 1.5 to 3.5 grams of maca every day for three months experienced a boost in libido. Other studies show that taking maca daily for at least six weeks can improve sexual desire. It must be taken in the correct dose to experience the benefits. Suggested dose: 3000 mg (3 g) daily to both improve mood and increase libido³⁸.

Saffron: Saffron has long been used as a libido and stamina enhancer. In addition to improving libido, studies show it can improve sexual dysfunction in both men and women³⁹. It may also improve mood. Some studies also point to saffron as a potential weight loss aid; helpful as women who are going through menopause gain weight more easily. Suggested dose: Take five dropperfuls of saffron tincture twice daily.

Shatavari: In addition to its antidepressant effects, shatavari may help improve libido for menopausal women. Shatavari has long been used to help with fertility, stress, and to prevent miscarriage. Studies back up the "folklore" associated with this herb's ability to enhance sex drive⁴⁰. Suggested dose: 500 mg capsules up to twice daily.

5. Exercise and Weight Management

Weight gain during menopause is very common, with the average woman gaining 1.5 pounds per year during their 50s and 60s⁴¹. Most of this weight tends to accumulate in the abdominal region. It is not entirely clear whether this weight gain is the result of hormones or aging. It is very likely a combination of both. Research shows that people lose three to eight percent of their muscle mass every decade after the age of 30⁴². After the age of 60, this loss accelerates. Women tend to gain more weight than men during their later years. Declining estrogen levels play a role for some women, as estrogen helps block the body's hunger signals. Women going through menopause have significantly higher total fat mass, with more abdominal and visceral fat than when they were not going through menopause.

Other factors play a role in weight gain during menopause. Women who exercise daily are better able to stave off menopausal weight gain as they age. It may get harder to motivate yourself to exercise daily as you age, but it is crucial to staying healthy, avoiding cardiovascular disease, heart disease, high blood pressure, and a plethora of other issues that can creep up as we age and gain weight.

Another factor that may play a role in weight gain during menopause is poor sleep. Women who don't get enough quality sleep due to hot flashes, night sweats, and anxiety are at a higher risk of gaining weight. This can turn into a vicious cycle as poor rest results in increased stress and anxiety. This raises cortisol levels and affects hormones to cause more weight gain. Not enough sleep at night may also make you too tired to exercise.

Whatever the cause of menopausal weight gain, the results range from diabetes to heart disease. It is very important to do what you can to mitigate these risks as you enter your perimenopausal years and beyond. Women are three times more likely to be obese or have metabolic syndrome after menopause than before it⁴³. In addition, sixty five percent of women age 40-65 are already obese, and a staggering 74 percent of women over 65 are obese. The key is starting a healthy lifestyle with daily exercise and healthy food choices in order to avoid putting yourself at risk for weight-related diseases in the future.

BENEFITS OF EXERCISE FOR MENOPAUSAL WOMEN:

When it comes to a healthy lifestyle, start with daily exercise and activity. Make it a point to do some sort of physical exercise each day for at least thirty minutes. Find an exercise that suits your needs and fitness level. As you become more fit, you may find that you need a more challenging type of exercise. Great exercises to start with are yoga and low-impact weight training. There are many levels of yoga, so you will be able to find different poses and exercises that suit your growing skill level. Yoga also helps improve balance and agility. Low-impact weight training involves using weights (in whatever size you can best handle to start off with) to strengthen your muscles. It not only helps build muscle; it may also combat bone loss⁴⁴.

Some other great ways to get the exercise you need is engaging in outdoor exercise. Being outdoors has been

shown to boost overall mood and lower depression⁴⁵. Try taking a daily walk for 30 minutes to one hour or even a daily jog. Set aside time to go on hikes and explore nature trails. Another outdoor choice is swimming. This helps gently build and strengthen muscles while also burning a lot of calories.

Exercise comes with many more benefits than weight loss and weight management. It is also affiliated with increased energy levels (to counter fatigue that comes with menopause) and better night's sleep (to combat insomnia that comes with menopause). Exercise is also associated with improved mood and lower stress and anxiety.

WATCH CALORIC INTAKE: In addition to daily exercise, watch the calories you consume each day. Women who are going through menopause burn around 200-220 fewer calories each day than women who are younger than 30. After age 60, the number of calories burned each day dwindles even more. While it may be useful to cut your caloric intake, it is not healthy to severely cut your caloric intake and go on a low-calorie diet. These diets can deprive us of the nutrients we need. A good example is to cut back to 1,800 calories a day if you were previously eating 2,000 calories a day before menopause (while maintaining a healthy weight). Eating an anti-inflammatory diet is an excellent way to eat well daily.

If you were already overweight, you may need to cut more calories than the recommended 200. See a nutritionist to learn how to lose or maintain your weight. In addition, follow the dietary advice in the next section.

REMEDIES FOR A SLUGGISH METABOLISM:

If you are eating healthy and exercising daily, but keep gaining weight, you may benefit from remedies that target a sluggish metabolism. Check out the remedies below to give your metabolism a boost:

Ashwagandha: Ashwagandha root extract can reduce psychological and physiological markers of stress, improve mental well-being, and reduce serum cortisol levels. It may also reduce food cravings and improve eating behaviors. A statistically significant reduction in body weight and body mass index (BMI)⁴⁶ was observed when patients took ashwagandha root extract (compared to placebo). It is a great remedy for menopausal women because it can help boost energy levels and mood while also lowering stress and anxiety. Suggested dose: 2 dropperfuls of ashwagandha up to twice daily.

Cayenne: Cayenne contains capsaicin. Capsaicin helps increase the number of calories burned each day and controls hunger to promote weight loss. Taking capsaicin in capsule form may help to increase the feeling of fullness⁴⁷ to keep you from snacking and overeating. Suggested dose: 100-150 mg of capsaicin daily in capsule form.

Fenugreek and Fennel Seeds: Fenugreek seeds are high in soluble fiber, help reduce blood glucose levels in those with type 2 diabetes⁴⁸, and fight body fat by slowing

digestion and the absorption of intestinal glucose and dietary fat⁴⁹, which promotes satiety⁵⁰. In one study⁵¹, participants reduced their dietary fat intake by 12% simply by supplementing with fenugreek. Fennel seeds also decrease hunger, aiding in weight loss⁵².

Ginger: Ginger helps decrease body weight and belly fat⁵³ by decreasing fat absorption and appetite while burning fat⁵⁴. Suggested dose: Take no more than four grams of ginger daily.

Lion's Mane and Reishi: Both Reishi and Lion's mane help with weight loss in animal studies^{55 56} by boosting fat metabolism. They also both help with blood sugar regulation^{57 58 59}. Obesity is often linked to inflammation and gut dysbiosis, and these medicinal mushrooms address these root causes⁶⁰. They also assist in the stress response, which is linked with weight gain. Suggested dose: 2 dropperfuls of dual-extracted reishi and lion's mane tinctures twice daily.

Please see our Weight Loss Protocol for more information, including herbal weight loss recipes.

6. Menopause Friendly Diet, Vitamins, and Lifestyle

MENOPAUSE FRIENDLY DIET:

Along with daily exercise and supplementation, a menopause-friendly diet can help ease symptoms, give you more energy, and boost mood. If you struggle as a result of diminished hormones, there are foods you can eat that are phytoestrogen-rich and may help boost estrogenic activity, especially when supplementing with daily herbal remedies like red clover. Phytoestrogen-rich foods include **chickpeas and soy**. Chickpeas are protein-rich and are the main ingredient used to make hummus. Eat a cup of baby carrots or snap peas with hummus dip for lunch or as a snack. Soy is another common source of phytoestrogens, but make sure that the products you buy contain only *organic soy*, as soy is often heavily processed and sprayed with numerous chemicals.

Flaxseed is a great fiber-rich addition because it contains a group of phytochemicals called lignins, which have estrogenic properties. There is evidence that flaxseed can protect the body from breast cancer and other hormonally-dependent cancers⁶¹. Grind fresh flaxseed into flax meal and sprinkle it on yogurt and use flax meal in your crust for your favorite quiche recipe or as a thickener in smoothies. Do not take extra supplements; take as food only.

With a depletion of hormones may also come a depletion of nutrients. Make sure you are eating plenty of **fresh fruits and vegetables**, including leafy greens such as spinach, kale, dandelion greens, and watercress. Add antioxidant-rich berries to meals for the free-radical eliminating benefits. Replace processed sugar snacks with berries.

Eat **lean meats** like chicken, turkey, and fish. Avoid red meat since the consumption of red meat is associated with heart and cardiovascular disease. Your body needs calcium when you are going through menopause and post-menopause, so make sure to incorporate more

calcium-rich foods like seeds, beans, lentils, almonds, leafy greens, and amaranth. Many people think they have to eat a lot of dairy to get calcium, but this is not necessarily true. While many dairy products contain calcium, many are also high in calories. If you wish to consume dairy for calcium, stick with plain Greek yogurt with probiotics.

Follow an **anti-inflammatory diet** that is rich in vegetables, lean protein, healthy fats, and fiber.

Other lifestyle tips for menopause include **avoiding endocrine disruptors**. Unfortunately, these compounds are in a lot of seemingly safe products. They have been associated with hormonal imbalance and even some cancers⁶². Take a close look at the everyday products you use and check for the presence of endocrine disruptors like flame-retardants, pesticides, herbicides, and artificial fragrances. Avoid the use of plastics, as many contain compounds like BPA that mimic estrogen⁶³. Beauty supplies like makeup, shampoo, conditioner, deodorant, and lotions often contain artificial fragrances, dyes, heavy metals, and other endocrine disruptors that are associated with hormonal imbalance. Read labels to look for parabens, dyes, fragrances, and sulfates. Choose brands that make organic and toxin-free products. Replace your Teflon-coated pots and pans with stainless steel or ceramic and consider replacing plastic cooking utensils with wood to reduce your exposure to potentially harmful chemicals when cooking.

VITAMINS⁶⁴: It's best to get your vitamins through diet. However, supplementation with the following vitamins may benefit menopausal and post-menopausal women.

Calcium with K-2: In addition to making sure you are eating plenty of calcium-rich foods, you can also take a calcium supplement. Make sure to purchase one that includes vitamin K2. Calcium is especially important for menopausal and post-menopausal women to counter the potential effects of a decline in estrogen that may result in bone loss, brittle bones, and osteoporosis. Vitamin K2 is a great addition because research indicates⁶⁵ it may have a positive impact on the health of postmenopausal women. It may help reduce the intensity of night sweats and may improve bone strength. If you are taking blood thinners, ask your doctor before beginning a daily vitamin K2/calcium regimen. Suggested dose: 500 mg twice a day, with food (Do not take 1000mg once a day as your body cannot absorb it.).

Vitamin D: Vitamin D is essential for calcium absorption and bone mineralization. It is positively associated with bone mineral density. Research⁶⁶ indicates that vitamin D may also improve muscle strength to help prevent falls, which are the biggest contributor to fracture incidence. One of the best ways our body naturally creates this vitamin is through exposure to sunlight (~ 20 min daily). Suggested dose is 2000 units daily. Take early in the day with your first Ca-K2 supplement.

B-Complex⁶⁷: For most people, getting enough vitamin B is not a problem, since it is prevalent in meat and dairy products. Those who are vegan may have issues

with getting enough B-vitamins. However, as we age, our bodies sometimes absorb micronutrients less efficiently, resulting in deficiencies. Research⁶⁸ indicates that women are not getting enough B9 (folate) and B2 (riboflavin), as well as other micronutrients they need to stay healthy. It may benefit menopausal and post-menopausal women to take a vitamin B complex to make sure they are getting sufficient amounts of B vitamins like B2, B6, B9, and B12. Vitamin B6 also plays a role in the regulation of the sleep hormone melatonin, which controls the sleep/wake cycle. Vitamin B12 plays a role in bone health by helping to keep homocysteine levels from rising and weakening bones. Suggested dose: Choose a vitamin B complex and take as directed on the bottle.

Vitamin E: Hot flashes affect around 75 percent of menopausal women. While the exact reason for the reduction in hot flashes as a result of taking vitamin E are still unknown, a placebo double blind-controlled trial⁶⁹ showed that women taking vitamin E reported a statistically significant reduction in the frequency and severity of hot flashes. Suggested dose: 400 IU vitamin E daily, if you suffer from hot flashes.

7. Western Medical Approaches

Women who are struggling with the effects of menopause may seek help from their OB/GYN, especially if symptoms interfere with their daily lives. Most women will undergo bloodwork to better determine hormone levels and a course of action. Bloodwork can give doctors an idea of Follicle-stimulating hormone (FSH) levels, estrogen (estradiol) levels, and Thyroid-stimulating hormone (TSH) levels.

FSH and estrogen levels often fluctuate during menopause, so FSH tests are not considered diagnostic. Symptoms are more often used for diagnoses due to this fluctuation in hormone levels so don't be alarmed if your doctor doesn't test for FSH. *TSH levels are helpful because symptoms of an underactive thyroid (e.g. hypothyroid and Hashimoto's Thyroiditis) are very similar to those of menopause.* Below are some common allopathic treatments for menopausal symptoms:

Vaginal Estrogen: Vaginal estrogen may be prescribed to relieve vaginal dryness. It is usually administered directly into the vagina. It comes in several forms, such as in a cream, a tablet, or a ring. It also helps with discomfort during intercourse. Side effects are usually mild, but can range from itching or stinging of the vagina or genitals to headache or nausea. Discharge and a mild odor have also been reported. Some women report feelings of pressure in the vagina or bleeding/spotting while taking vaginal estrogen. For most, it works very well.

Hormone Replacement Therapy (HRT): Sometimes called estrogen therapy, this is an effective treatment option for relieving symptoms like hot flashes. If you still have a uterus, you will take both estrogen and progesterin. An additional benefit is the potential prevention of bone loss. However, unopposed estrogen is not considered a safe long-term therapy, as it has been associated with an increased risk of cardiovascular issues and breast

and uterine cancer. For this reason, most doctors will recommend the shortest time frame (as well as the lowest dosage) possible to be effective but safe. Some HRT therapies now add in testosterone for women to help with libido. Do your research and talk with your doctor. Hormones make a huge difference for many menopausal women.

Antidepressants: If you struggle with mood swings and depression due to fluctuating hormones, some doctors will prescribe low-dose antidepressants. Often, these are selective serotonin reuptake inhibitors (SSRIs). These may help with mood and hot flashes. They are an option for women who cannot take estrogen for health reasons.

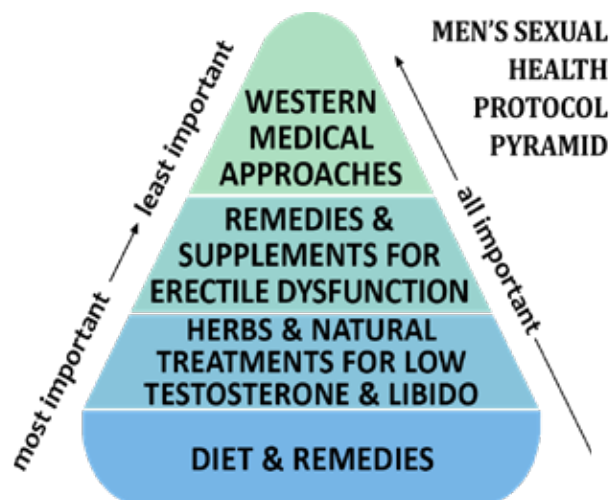
Many antidepressants come with a variety of side effects and can affect mental health so make sure to pay attention to your body and report any unusual symptoms or side effects to your doctor right away. Please see our Depression Protocol for natural alternatives.

Other Medications: Some doctors prescribe seizure medications to treat hot flashes. They include medications like Gralise, Horizant, and Neurontin. Some hypertension medications are also used to provide relief from hot flashes. These include Catapres and Kapvay. For women at risk for osteoporosis, medications to treat or prevent osteoporosis may be prescribed.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Healthy breakfast Vit. D, B-complex, Ca, & K-2 w/ breakfast Stinging nettle tincture Lion's Mane and Reishi mushroom tinctures Deep breathing and/or meditation	Weekly grocery shopping (for menopause-friendly foods) Healthy lunch Calcium w/ lunch Lemon balm tincture 30* min. exercise (yoga or low impact strength training)	Cordyceps and Reishi mushroom tinctures Healthy snack (e.g., cup of berries) Red clover tincture or infusion (one cup)	Healthy dinner Ashwagandha tincture Aromatherapy for calming and to prepare for sleep Lion's mane tincture Post-dinner walk	Deep breathing/meditation Black cohosh & red clover tincture or supplement (if prone to night sweats/hot flashes) Epsom salt bath Sleep blend tincture Magnesium glycinate
MONDAY	Healthy breakfast Vit. D, B-complex, Ca, & K-2 w/ breakfast Stinging nettle tincture Lion's Mane and Reishi mushroom tinctures Deep breathing and/or meditation	Healthy lunch Calcium w/ lunch Lemon balm tincture 30* min. exercise (yoga or low impact strength training)	Cordyceps and Reishi mushroom tinctures Healthy snack Red clover tincture or infusion (one cup)	Healthy dinner Ashwagandha tincture Aromatherapy Lion's mane tincture	Deep breathing/meditation Black cohosh & red clover tincture or supplement Epsom salt bath Sleep blend tincture Magnesium glycinate
TUESDAY	Healthy breakfast Vit. D, B-complex, Ca, & K-2 w/ breakfast Stinging nettle tincture Lion's Mane and Reishi mushroom tinctures Deep breathing and/or meditation	Healthy lunch Calcium w/ lunch Lemon balm tincture 30* min. exercise (yoga or low impact strength training)	Cordyceps and Reishi mushroom tinctures Healthy snack Red clover tincture or infusion (one cup)	Healthy dinner Ashwagandha tincture Aromatherapy Lion's mane tincture	Deep breathing/meditation Black cohosh & red clover tincture or supplement Epsom salt bath Sleep blend tincture Magnesium glycinate
WEDNESDAY	Healthy breakfast Vit. D, B-complex, Ca, & K-2 w/ breakfast Stinging nettle tincture Lion's Mane and Reishi mushroom tinctures Deep breathing and/or meditation	Healthy lunch Calcium w/ lunch Lemon balm tincture 30* min. exercise (yoga or low impact strength training)	Cordyceps and Reishi mushroom tinctures Healthy snack Red clover tincture or infusion (one cup)	Healthy dinner Ashwagandha tincture Aromatherapy Lion's mane tincture Post-dinner walk	Deep breathing/meditation Black cohosh & red clover tincture or supplement Epsom salt bath Sleep blend tincture Magnesium glycinate
THURSDAY	Healthy breakfast Vit. D, B-complex, Ca, & K-2 w/ breakfast Stinging nettle tincture Lion's Mane and Reishi mushroom tinctures Deep breathing and/or meditation	Healthy lunch Calcium w/ lunch Lemon balm tincture 30* min. exercise (yoga or low impact strength training)	Cordyceps and Reishi mushroom tinctures Healthy snack Red clover tincture or infusion (one cup)	Healthy dinner Ashwagandha tincture Aromatherapy Lion's mane tincture	Deep breathing/meditation Black cohosh & red clover tincture or supplement Epsom salt bath Sleep blend tincture Magnesium glycinate
FRIDAY	Healthy breakfast Vit. D, B-complex, Ca, & K-2 w/ breakfast Stinging nettle tincture Lion's Mane and Reishi mushroom tinctures Deep breathing and/or meditation	Healthy lunch Calcium w/ lunch Lemon balm tincture 30* min. exercise (yoga or low impact strength training)	Cordyceps and Reishi mushroom tinctures Healthy snack Red clover tincture or infusion (one cup)	Healthy dinner Ashwagandha tincture Aromatherapy Lion's mane tincture	Deep breathing/meditation Black cohosh & red clover tincture or supplement Epsom salt bath Sleep blend tincture Magnesium glycinate
SATURDAY	Healthy breakfast Vit. D, B-complex, Ca, & K-2 w/ breakfast Stinging nettle tincture Lion's Mane and Reishi mushroom tinctures Deep breathing and/or meditation	Healthy lunch Calcium w/ lunch Lemon balm tincture 30* min. exercise (yoga or low impact strength training)	Cordyceps and Reishi mushroom tinctures Healthy snack Red clover tincture or infusion (one cup)	Healthy dinner Ashwagandha tincture Aromatherapy Lion's mane tincture Post-dinner walk	Deep breathing/meditation Black cohosh & red clover tincture or supplement Epsom salt bath Sleep blend tincture Magnesium glycinate

Ch. 29: Men's Sexual Health Protocol

Men's sexual health can be defined as a state of well-being that allows a man to fully engage in and appreciate sexual activity. There are many factors that encompass true sexual health. These include a man's physical health, along with psychological, interpersonal, and social factors.



True sexual health means that a man can experience sexual desire (a healthy libido), the ability to attain and sustain an erection, and experience balanced hormones (the body producing the right amount of testosterone). It also includes the prevention and treatment of sexually transmitted diseases, male infertility (something that is becoming more prevalent for both men and women), and prostate health. Please see the Prostate Protocol for information on how to care for your prostate.

It is important to understand that when issues arise, they can be signs of something more serious. Pay attention to your body and any changes you notice, no matter how subtle you think they are. For example, erectile dysfunction (ED) - the inability to get or maintain an erection long enough for satisfying sexual activity - can be one of the first signs of heart disease. Cholesterol-clogged arteries can contribute to ED, so if you start having ED issues, see a health professional and make sure your cardiovascular system is healthy.

1. Diet and Remedies for Men's Sexual and Reproductive Health

One of the first things you can do to positively impact your sexual (and overall) health is to watch what you eat. Second to this is making sure you set time aside each day to be active and exercise (especially if you have a desk job, where you sit for prolonged periods). In this section, we discuss diet and exercise for men's sexual health, as well as what you can do to tackle other issues that may arise, such as infections and impaired fertility.

dietary choices exacerbate weight gain and physical health. There is a strong correlation between excessive weight gain and sexual health problems for men. Men who are overweight may suffer from issues with confidence, as well as lowered energy levels, hormonal issues, and impaired sexual function.

Weight gain also causes psychological changes that impact sexual performance. Men who have gained a significant amount of weight report lower energy levels, lower self-confidence, and even lowered testosterone levels. When testosterone levels become low, this impacts libido and erections. In addition, when men gain weight, this creates higher levels of Sex Hormone Binding Globulin (SHBG). This binds with testosterone and leaves less testosterone in the system to do its job, like helping create strong erections and improving libido.

Another negative impact of weight gain is the link between weight and high cholesterol. Diet plays a big role in this, because the foods we eat can contribute to clogged arteries. A diet full of trans fats, oils, and red meat can harm arteries. The extra weight that comes with an unhealthy diet can cause blood vessels in the genitals to narrow and this disrupts blood flow. If left untreated, this may lead to severe erectile dysfunction or impotence.

Losing just five percent of body weight can improve erectile function and increase libido for men¹. Losing weight and making a commitment to exercise every day takes effort and determination, but making these a priority in your life can have a significant toll on sexual health.

A good diet to promote weight loss and circulation includes plenty of fresh fruits and vegetables. Eat your fill of leafy greens with each meal and try to include a cup of antioxidant-rich berries like blueberries, raspberries, or blackberries. Red meat can be problematic when it comes to cardiovascular health; opt for lean meats like chicken and omega-3-rich fish.

Avoid foods that are fried, as these are high in trans fats. Steer clear of oils and excess carbohydrates that come from breads and other processed grains. Instead of a high-carbohydrate (from grain) diet, choose organic, seed and whole-grain comprised flatbreads that do not contain genetically modified ingredients.

Cut your processed sugar intake as much as possible. Sugar causes myriad health issues ranging from inflammation to obesity. If you need to sweeten a dish, try a natural sweetener like stevia (never use artificial sweeteners).

Along with a healthy diet, exercise can help boost low testosterone, as well as get more blood flowing to the pelvic area to help improve erections. Exercise also expedites weight loss. Exercise also reduce stress and promotes a happier mental state. Your mental health is a significant contributor to your sexual health.

The type of exercise isn't as important as doing it daily for at least twenty minutes. However, cardiovascular exercise like running, jogging, swimming, jumping jacks, jumping

rope, and power walking get the blood pumping and circulating throughout the body better than some other forms of exercise. This oxygenated blood circulating in the body will inevitably make its way to the pelvic region to help with a variety of sexual health concerns.

Infections, especially sexually transmitted infections, can occur in men who are sexually active. It is important to watch for symptoms so you can get treatment. Some common symptoms of a sexually transmitted infection in a man include:

- Burning or pain when urinating
- Pain in the lower abdomen
- An urge to urinate more frequently
- Pain with ejaculation
- Bumps, blisters, or sores on the genital area or penis
- Abnormal penile discharge, especially discharge that has a strange color or foul smell.

The symptoms above are signs of several common sexually transmitted infections. These include infections like chlamydia, herpes, genital warts, gonorrhea, and trichomoniasis. Infections like chlamydia, gonorrhea, and trichomoniasis are often treated with antibiotics. Viral infections like herpes, HIV, and genital warts may be treated with antiviral medications.

While it is best to see a doctor if you suspect an STD, you can also try the remedies below. Choose antiviral for a viral infection and antibiotic for a bacterial infection:

Echinacea: Echinacea is a great antiviral and antibacterial. It has been used successfully to treat a wide range of infections. Studies show direct virucidal activity². Suggested dose: 2 dropperfuls three times day at the first sign of illness. Do not use or use with care if you have an autoimmune issue as it may cause it to flare. Elderberry is also a proven antiviral. Please see our Flu, Viral, and Immune Protocol for more information.

Lemon Balm (*Melissa officinalis*): Lemon balm has surprising antiviral properties, with its most potent activity geared toward viruses that cause lesions, such as herpes³ (both genital herpes and fever blisters). Suggested dose: 2 dropperfuls of lemon balm tincture up to three times daily; also, apply it externally to lesions. If you want a potent external application, try purchasing lemon balm essential oil and adding five drops to a teaspoon of carrier oil (such as olive or coconut oil). Apply to the affected areas two to three times daily or until they clear up.

Berberine: Berberine is found in plants like goldenseal and Oregon grape root. It is a powerful antibiotic⁴ and works well for UTIs. It works by inhibiting DNA duplication, RNA transcription, and protein biosynthesis in bacterial cells by inhibiting enzymatic activities and modifying the surface structure of bacterial cell walls^{5 6 7}. It is also helpful for high cholesterol and diabetes. Suggested dose: 1,500 mg daily (divided into three 500mg doses).

Usnea: Usnea has powerful antibiotic, antifungal, antimicrobial, and antiviral properties⁸. It is one of the

most potent natural antibiotics⁹ available. It can be tinctured as a dual extraction for the treatment of various infections. Suggested dose: 2 dropperfuls three times daily for 10 days or until the infection is gone.

REPRODUCTIVE HEALTH AND FERTILITY:

There are many factors that can affect a man's ability to reproduce. These include low sperm quality, lack of sperm, stress, hormonal issues, genetic issues, genital tract blockages, anabolic steroid use, testicular injury, infections, and undetermined causes. In around fifty percent of male infertility cases, the cause cannot be determined. Ten to fifteen percent of men experience infertility due to a lack of sperm. The medical term for a lack of sperm is azoospermia. The medical term for a low sperm count is oligospermia. One of the most common causes for this is an enlarged vein in the testicle called a varicocele.

A healthy couple in their 20s or 30s trying to conceive should usually be able to do so after several months. If you go one full year with no success, it may be time to visit a fertility specialist. A man visiting a fertility doctor may be asked to provide a sperm sample, as well as bloodwork to check for hormonal issues. Below are remedies for male fertility. Choose what works best for you depending on your particular issue.

Tribulus: This plant contains steroidal saponins¹⁰ that may stimulate the synthesis of luteinizing hormone (LH), ultimately leading to higher testosterone levels in men. In vivo studies show that Tribulus can significantly impact sperm quality, motility, and viability to increase chances of conception. Tribulus also contains zinc, which has been proven to support sperm health. Its high antioxidant content can protect sperm from free radical damage. Suggested dose: 85-250 mg up to three times daily.

Panax ginseng: Panax ginseng is a powerful tool for men's sexual health. It can increase energy levels and stimulate smooth muscle relaxation of the penis via the release of nitric oxide¹¹, triggering erections. Ginseng is an adaptogen, and as such, can help the body better respond and deal with stress. Stress can be a major contributor to infertility. Ginseng also has cardiovascular benefits, which in turn can help ED. Suggested dose: 1,000 mg daily.

Maca: This root is often used by women for enhancing fertility, and studies show it may be helpful for male infertility as well. In animal studies, black maca increased sperm count as early as one day after administration¹². Suggested dose: Take in tincture form for best results. Start with 2 dropperfuls up to four times daily.

Cordyceps: Cordyceps are a valuable remedy for male fertility¹³. These powerful medicinal mushrooms improve both sperm production and motility¹⁴. Cordyceps are a great adaptogenic remedy as well, helping to relieve the effects of stress on the body. Since stress is a big cause of infertility, these are doubly beneficial. The best way to reap the benefits is to take a double extraction. This accesses both the water and alcohol soluble medicinal compounds. Suggested dose: 2 dropperfuls of a dual-extracted Cordyceps

tincture up to three times daily.

Ashwagandha (*Withania somnifera*): Ashwagandha is another adaptogen, helping with energy levels and the stress response. Ashwagandha has been used for centuries by those wishing to improve testosterone levels, increase sperm count¹⁵, enhance endurance, and control cortisol levels. Suggested dose: 2 dropperfuls up to three times daily.

2. Herbs and Natural Treatments for Low Testosterone and Libido

As men age, their chances of

being affected by low testosterone increase significantly. Low testosterone affects around forty percent of men over the age of 45. When testosterone levels are low, this can affect sexual health in significant ways, such as a lowered libido. There are several factors that contribute to low testosterone. These include a man's body mass index (BMI), diet, certain medications, illness, alcohol consumption, and age.

Common symptoms of low testosterone include: lowered libido, erectile dysfunction, depression, trouble focusing, memory issues, lethargy and fatigue, irritability, and loss of muscle tone and strength. In addition, some other changes that may occur in men with low testosterone include lowered blood hemoglobin and anemia, a decrease in body hair, osteoporosis, weight gain, breast development (gynecomastia), and infertility.

REMEDIES FOR LOW TESTOSTERONE: There are natural remedies that aim to increase testosterone levels naturally. Before beginning a regimen, see your doctor and have bloodwork conducted to make sure that low testosterone is in fact your issue.

Pine Pollen: There are certain plants that contain hormones that act very much like human hormones and can affect human hormonal production when taken regularly. Some plants contain estrogenic compounds (phytoestrogens) and some contain testosterone-like compounds (phytotestosterone). Pine pollen contains larger amounts of testosterone¹⁶. In the spring, the pine tree produces pollen. It can easily be harvested and tintured to create a natural testosterone supplement. Suggested dose: Tincture form, 1 dropperful up to four times daily.

Vitamin D: Vitamin D provides a number of benefits, including promoting bone health and supporting the immune system. It may also help increase testosterone levels. There is a link between men who have a vitamin D deficiency and low testosterone levels¹⁷. Taking a daily vitamin D supplement may help to increase testosterone for men who have a severe deficiency. Get your levels tested. Suggested dose: Take up to 4,000 IU daily and spend ~20 minutes in the sunlight daily, if possible, to get this vitamin naturally.

Zinc: Zinc is heavily involved in the production of testosterone. It also helps maintain sperm quality.

Taking a daily zinc supplement may help to improve testosterone levels for men who have a zinc

deficiency¹⁸. Suggested dose: 11 mg daily. You can also get this mineral from foods like poultry, seafood, nuts, beans, fortified cereals, and some dairy products. Do not exceed the daily recommended amount.

Saw Palmetto (*Serenoa repens*): Saw palmetto is popular for prostate health; it may also be beneficial for raising testosterone levels. It may also help boost libido and increase sperm production. A 2020 study¹⁹ on men with benign prostatic hyperplasia found that saw palmetto oil improved prostate symptoms and also increased levels of serum free testosterone. Suggested dose: 160-320 mg daily.

Reishi (*Ganoderma lucidum*): This adaptogenic mushroom is a great choice for a variety of men's health issues including hormonal imbalances. Its powerful antioxidant properties help protect and nourish many bodily systems, including the reproductive system. Reishi helps modulate the body's stress response²⁰, and studies show it has a positive impact on androgen deficiency in men²¹. Take reishi in a double extraction form for the full spectrum of medicinal compounds. Suggested dose: 2 dropperfuls of reishi dual-extracted tincture up to three times daily.

NATURAL REMEDIES TO ENHANCE LIBIDO:

Low libido and low testosterone seem to go hand-in-hand, but not always. A low sex drive can also be the result factors such as stress, medications, chronic illness, depression, low self-esteem, aging, insomnia, and alcohol and/or drug use. While it may be normal to lose interest in sexual activity every now and then, experiencing a low libido for an extended period of time may be an indicator of a more serious issue and cause for concern. If you have been experiencing a lowered libido for an extended amount of time, it may be prudent to see a doctor to help determine potential causes and rule out serious issues. There are several natural remedies that help with low libido. In addition to trying these remedies, combine these with proper diet, exercise, and treatment for other issues like low testosterone (see above).

American Ginseng (*Panax ginseng*): Panax ginseng has proven libido-enhancing effects²². It has been utilized for centuries to help increase energy, stamina, and libido. Its adaptogenic properties can help the body react to stress in a way that prevents damage and protects overall health. Suggested dose for libido: 1,000 mg daily and increase this slowly if necessary (not to exceed 4,500 mg).

Ashwagandha (*Withania somnifera*): This adaptogen has already been mentioned for its ability to increase sperm count and fertility, but it is also known for its effects on libido. In fact, the name translates to "like a horse" to insinuate the energy and stamina the remedy provides for a man's sex life. While it has been studied with positive results in women²³, it is also thought to improve male libido. Suggested dose: 2 dropperfuls tincture up to three times daily.

Rhodiola rosea: Another adaptogen, rhodiola has been used for centuries to help increase fertility, libido, energy, and stamina. If chronic fatigue²⁴ is holding you back from

enjoying a fulfilling sex life, rhodiola helps boost energy and athletic performance. Because it is an adaptogen, it also reduces the effects of physical stress on the body (which can further enhance libido). Rhodiola is known for its ability to regulate thyroid function²⁵ as well. Do not take this remedy if you have an autoimmune disorder; use reishi and cordyceps instead. Suggested dose: 200 mg twice daily.

3. Remedies and Supplements for ED

Erectile dysfunction (ED) is also called impotence. It occurs when the penis fails to become or stay erect during sexual activity. While most cases of ED aren't detrimental to a man's health, they can certainly be detrimental to a man's self-image and lead to relationship issues. ED may be an early warning sign of cardiovascular issues, as blood is unable to make its way to the penis due to blockages. See your doctor.

Diabetes, high cholesterol, high blood pressure, arteriosclerosis, and low testosterone levels are thought to be the most common causes of ED. In addition, psychological issues like depression and medications can be a trigger.

ED is the most common kind of sexual dysfunction in men. Statistics from the National Health Institute²⁶ show that an estimated 15 to 30 million men have experienced chronic erectile dysfunction in the United States. As men age, the numbers rise. Around five percent of men in their 40s are affected by ED, and 15-25 percent of men are affected by the time they turn 65. Both transient ED and issues with inadequate erections affect an estimated 50 percent of men ages 40-70.

HERBS FOR ED: It is important to see your doctor to determine the root cause of your ED. Once you know why you are experiencing ED, you can better treat it. Herbal options for ED include:

American Ginseng (*Panax ginseng*): This adaptogen has already been mentioned for its effects on libido, fertility, and stress; it also helps with erectile dysfunction²⁷. Suggested dose: Start with 1,000 mg daily and slowly work your way up if needed. Do not exceed 4,500 mg daily.

Garlic: For ED that is caused by cardiovascular issues, garlic improves blood flow and blood vessel function²⁸. It works well when combined with other natural remedies for ED, such as ginseng. It also improves nitric oxide levels, which are responsible for causing blood vessels to dilate for better blood flow. Suggested dose: Blend four cloves of garlic with raw honey and take daily, or take garlic capsules, between 17-24 grams daily.

Ginkgo biloba: Ginkgo is known for its effect on blood flow. It is thought to increase blood flow to the penis, as well as other areas of the body. It has been studied with positive results on improving blood flow in elderly adults²⁹. If you are on blood thinners or other medications for cardiovascular issues, check with your doctor before beginning any herbal regimen. Suggested dose: 240 mg daily.

Along with herbal remedies, implement these strategies: stop smoking ASAP, avoid alcohol, exercise regularly, minimize stress, get more sleep, and lose weight (if you are overweight).

SUPPLEMENTS FOR ED: In addition to herbs, there are supplements³⁰ that may improve men's sexual health.

L-Arginine: The amino acid L-arginine is a vasodilator. It may help open blood vessels to increase blood flow to the penis for men with ED. A 2017 study³¹ showed that men with ED tend to have lower levels of L-arginine. Do not take this supplement alongside other ED medications. Suggested dose: 6-30 grams daily, divided into three equal doses.

Propionyl-L-Carnitine: If diabetes is contributing to your ED, this supplement improves ED when taken along with sildenafil³² (as opposed to taking sildenafil alone). Sildenafil is the generic version of the popular ED pharmaceutical, Viagra. Suggested dose: 500-2,000 mg daily.

Dehydroepiandrosterone (DHEA): DHEA is responsible for helping to create sex hormones like testosterone and estrogen. Our natural levels decrease as we age, so this may contribute to a decrease in sexual function. Studies show that this supplement may help men with ED³³ by improving hormone function. Try taking around 50 milligrams daily.

4. Western Medical Approaches

In addition to natural options, there are many options in the allopathic world. There are several pharmaceutical options for men's health issues. There are also surgical options, depending on the problem. Below are some common western medical treatments for the health issues discussed in this protocol:

Medical Treatments for STDs: Most Sexually Transmitted Diseases and infections are treated with a round or two of antibiotics. When the nature of the disease is viral, antiviral medications may be prescribed. For some STDs, such as genital warts, there are surgical options and topical applications that help to remove the warts. For viral infections like HIV, strides have been made over the past few decades that have brought this disease from a death sentence to a manageable condition. However, those with HIV generally have to take pharmaceuticals as part of their daily life.

Medical Treatments for Infertility: When visiting a fertility doctor, they will test your sperm for motility and abundance and do bloodwork to check hormone levels. For some men, a surgical option exists to remove blockages in the testicle that affect sperm production. Not all infertility issues are treatable with pharmaceuticals and surgery. Today, Intrauterine Implantation (IUI) and In vitro Fertilization (IVF) are popular options. IUI involves the doctor taking your sperm and concentrating it to make it more potent. The sperm is inserted into the uterus with a device and drugs are given to

the woman to make her ovulate so conception is more likely. IUI is much more cost-effective than IVF, but not as successful. IVF involves taking the sperm and egg and causing fertilization in a lab setting. These fertilized eggs are then re-inserted into the uterus. This has a much higher success rate, but can cost \$30,000 or more depending on the doctor.

Medical Treatments for Low Testosterone and Libido:

A doctor will likely perform many tests to determine the cause of low testosterone levels. They may prescribe testosterone replacement treatment, which can be administered via a skin patch, topical gel, oral therapy, pellets, or injections. Men undergoing this therapy usually notice an improvement in mood and symptoms within six weeks, but usually do not notice an improvement in muscle mass for three to six months. This treatment may also improve libido.

Medical Treatments for Erectile Dysfunction (ED):

When visiting a doctor for ED concerns, they may run

tests such as bloodwork, urine tests, a physical exam, and an ultrasound. Some doctors may also perform a psychological exam or refer you to a specialist for this. Once more is known about the condition and serious cardiovascular concerns are ruled out, doctors often prescribe one of several pharmaceutical drugs for ED. These include Viagra (generic form is Sildenafil), Cialis (generic form is Tadalafil), Levitra or Staxyn (generic form is Vardenafil), and Stendra (generic form is Avanafil). These medications all help to increase nitric oxide levels to enhance blood flow and erections. These can produce some side effects, so your doctor will make sure you are not already taking nitrate drugs or supplements, have heart disease, or suffer from low blood pressure problems. Having these can increase your chances of serious side effects. If pharmaceutical options fail to work, other more invasive options exist such as the installation of a penile implant and the use of penis pumps that force blood to the area.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Healthy breakfast Zinc & Vit. D Reishi & Cordyceps dual extracted mushroom tinctures Panax ginseng Saw palmetto	Weekly grocery shop for healthy foods Healthy lunch Garlic supplement Reishi & Cordyceps mushroom tinctures	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Reishi & Cordyceps mushroom tinctures Pine pollen tincture (only if low T and libido)	Ashwagandha tincture Prep for bed (no screens, dim lights, etc)
MONDAY	Healthy breakfast Zinc & Vit. D Reishi & Cordyceps Panax ginseng Saw palmetto	Healthy lunch Garlic supplement Reishi & Cordyceps	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Reishi & Cordyceps Pine pollen tincture (only if low T and libido)	Ashwagandha tincture Prep for bed
TUESDAY	Healthy breakfast Zinc & Vit. D Reishi & Cordyceps Panax ginseng Saw palmetto	Healthy lunch Garlic supplement Reishi & Cordyceps	Exercise (at least 30 min.) Reishi Dual Extraction Garlic supplement Tribulus (If TTC)	Healthy dinner Reishi & Cordyceps Pine pollen tincture (only if low T and libido)	Ashwagandha tincture Prep for bed
WEDNESDAY	Healthy breakfast Zinc & Vit. D Reishi & Cordyceps Panax ginseng Saw palmetto	Healthy lunch Garlic supplement Reishi & Cordyceps	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Reishi & Cordyceps Pine pollen tincture (only if low T and libido)	Ashwagandha tincture Prep for bed
THURSDAY	Healthy breakfast Zinc & Vit. D Reishi & Cordyceps Panax ginseng Saw palmetto	Healthy lunch Garlic supplement Reishi & Cordyceps	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Reishi & Cordyceps Pine pollen tincture (only if low T and libido)	Ashwagandha tincture Prep for bed
FRIDAY	Healthy breakfast Zinc & Vit. D Reishi & Cordyceps Panax ginseng Saw palmetto	Healthy lunch Garlic supplement Reishi & Cordyceps	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Reishi & Cordyceps Pine pollen tincture (only if low T and libido)	Ashwagandha tincture Prep for bed
SATURDAY	Healthy breakfast Zinc & Vit. D Reishi & Cordyceps Panax ginseng Saw palmetto	Healthy lunch Garlic supplement Reishi & Cordyceps	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Reishi & Cordyceps Pine pollen tincture (only if low T and libido)	Ashwagandha tincture Prep for bed

Ch. 30: Prostate Protocol

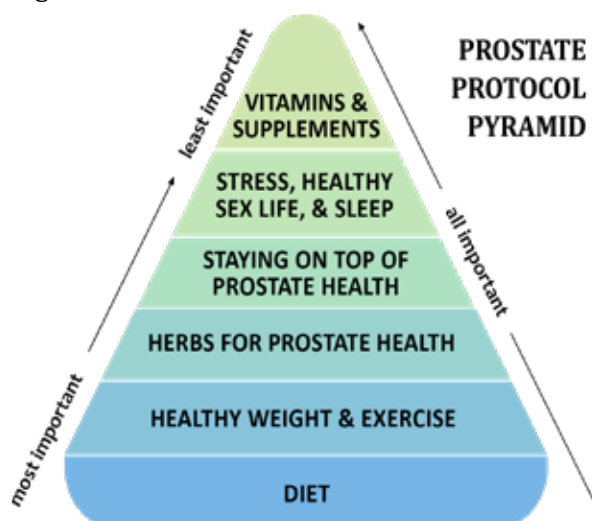
The prostate gland is a part of the male reproductive system located below the bladder just above the pelvic floor muscles, in front of the rectum. (Due to its close proximity, the prostate can be felt during a rectal examination). The urethra, which expels urine and sperm, runs through the center of this chestnut-sized gland. This gland performs several functions, the most important of which is the production of a type of fluid that makes up semen. To further ensure semen is able to exit forcefully (nature's way of helping the semen get as close as possible to the uterus to help fertilize an egg), the prostate has muscles that contract to help semen forcefully pass into the urethra and expel outwards during ejaculation. The fluid and sperm cells combine with fluid from other glands to form the semen that releases¹.

Anatomically, the tissue that makes up the prostate is divided into three zones:

The transition zone is on the outside of the gland. It is the smallest part of the gland and makes up about ten percent of the prostate. It surrounds the urethra between the bladder and upper third of the urethra.

The central zone surrounds the transition zone; it is about twenty five percent of the prostate gland. This zone is where the ducts (ejaculatory and seminal ducts) and seminal vesicles are located.

The peripheral zone is the largest and main part of the prostate. This zone makes up a little less than 70 percent of the gland.



These three zones surround the urethra much like layers of an onion. Because this gland surrounds the urethra, issues can arise, especially as men age. If the prostate gland grows (e.g., benign growth), the urethra can become compressed. This may make it difficult to pass urine. Most cases of benign prostatic hyperplasia or prostate enlargement occur in the transition zone that surrounds the urethra. Transversely, most cancers in the prostate occur in the outer peripheral zone.

Other Functions of the Prostate: Aside from making seminal fluid, other important functions of the prostate

include secretions that keep sperm cells healthy and functioning. Thus, the prostate is integral for fertility in men. It also produces a hormone-like substance to help sperm move better and make their way to the egg. Additionally, the prostate closes the urethra up to the bladder during ejaculation using a sphincter muscle that keeps semen from entering the bladder. The seminal ducts close during urination and the central zone muscles help close the prostate's ducts to keep urine from entering. This gland is also responsible for hormone metabolism. The male sex hormone testosterone is transformed to the biologically active form of testosterone, dihydrotestosterone (DHT), in the prostate.

If you notice any of the following symptoms, see your doctor right away:

- A frequent urge to urinate
- Getting up multiple times a night to urinate
- Pain or burning during urination
- Blood in urine or semen
- Painful ejaculation
- Dribbling of urine instead of a continual flow
- Pain or stiffness in the lower back, pelvis, hips, rectal area, or upper thighs.

The PSA Test: A PSA (Prostate-specific antigen) test is a test performed to measure the level of PSA in the blood. Results are reported as nanograms of PSA per milliliter of blood. Those with prostate cancer often have an elevated blood level of PSA, so it can be a helpful tool to detect potentially serious issues. In 1986, the FDA recommended using a PSA test in conjunction with a digital rectal exam (DRE) to help detect prostate cancer in men who are over the age of fifty. Prostate issues that can cause PSA levels to rise that do not involve cancer include prostatitis and benign prostatic hyperplasia.

Since other conditions can cause an elevated PSA, many medical organizations stopped routine PSA testing in 2008. PSA testing is often indicated for men who are at a higher risk of developing prostate cancer and are over 40-45 years old. Certain demographics, such as black men and those with a family history of prostate cancer, are at a higher risk and thus may be advised to test.

PSA levels of 4.0 nanograms per milliliter of blood used to be considered normal, but it has been shown that prostate cancer can still occur in those with lower PSA levels. Additionally, people with high levels may not have prostate cancer, as factors such as age, prostate gland size, infection, a recent biopsy, or inflammation can cause elevated levels. Ejaculation and vigorous exercise two days before testing, as well as certain medications, may interfere with PSA levels. A high PSA level is more strongly correlated with prostate cancer, but this is not always the case.

Digital Rectal Screening: This type of screening involves a physician using their gloved and lubricated finger to feel for anything out of the ordinary. Because the prostate can be felt through the rectum, physicians might be able to physically detect cancer and other issues. This type of screening can be helpful, but some groups, such as the US Preventative Services Task Force, do

not recommend this because they believe there is little evidence that it provides benefits.

Western Medications: There are a number of Western medicines used for different prostate issues. Speak to your urologist and naturopathic doctor for information, side effects, and how to integrate this protocol into your treatment plan if you choose this route.

How do you know if you are at risk and need to be tested? First, make sure to familiarize yourself with the symptoms listed above. Risk factors that make you more likely to have prostate issues include your family history, race, weight, and age. If you have more than one first-degree relative who had prostate cancer, including relatives in three generations on your father or mother's side of the family, you have an increased risk of developing prostate cancer. You are also at an increased risk if other family members have been diagnosed with ovarian, breast, or pancreatic cancer.

You are more 1.5 times more likely to get prostate cancer and more than twice as likely to die from it if you are African-American. African-American men are also more likely to get prostate cancer at a younger age, have more advanced disease when it is detected, and have more severe types of prostate cancer. Get tested and watch for symptoms.

The older a man is, the greater his odds of developing prostate cancer². Around six in ten cases of prostate cancer occur in men who are older than 65. Men with a body mass index over 30 are also at an increased risk of developing prostate cancer.

The three most common issues to affect the prostate include prostatitis, benign prostatic hyperplasia (BPH), and prostate cancer³.

Prostatitis: Prostatitis is a painful condition that causes inflammation of the prostate and the areas around the prostate. There are four types of prostatitis: chronic prostatitis or chronic pelvic pain syndrome, acute bacterial prostatitis, chronic bacterial prostatitis, and asymptomatic inflammatory prostatitis⁴. Asymptomatic inflammatory prostatitis doesn't have symptoms, but it is often diagnosed when testing for urinary tract or reproductive tract issues.

Those with chronic prostatitis or chronic pelvic pain syndrome still don't have answers as to the exact cause of the condition. It is thought that a microorganism, chemicals in the urine, an autoimmune response, previous urinary tract infections, and/or nerve damage to the pelvic area may be responsible for this inflammatory condition. Acute and chronic bacterial prostatitis are the result of a bacterial infection of the prostate. "Acute" means it comes on suddenly and doesn't last long if treated promptly and "chronic" means it develops slowly over time and can last a long time. This infection may occur when bacteria travel from the urethra into the tissue of the prostate.

Symptoms of prostatitis include pain in the pelvic region, pain between the scrotum and anus, pain in the central abdomen, penile pain, lower back pain, and scrotum pain. Other symptoms include pain during or after

urination in the urethra or penis. It can also cause urinary frequency, urgency, and a weak or interrupted urine stream.

With bacterial prostatitis, one may experience urinary urgency and frequency, fever, chills, burning during urination, pain in the genital area, groin, lower abdomen, or lower back, frequent urination during sleep, nausea and vomiting, body aches, urinary retention, trouble urinating, a urinary blockage, or a urinary tract infection.⁵

Benign Prostatic Hyperplasia (BPH)/Prostate Gland Enlargement: BPH often occurs when men get older. Since the prostate gland wraps around the urinary tract, it can lead to uncomfortable symptoms like blocked urinary flow, as well as bladder, kidney, or urinary tract issues. Other symptoms include a frequent or urgent need to urinate, increased frequency at night, difficulty starting urination, dribbling at the end of urination, a weak urine stream or a stream that stops and starts, and an inability to fully empty the bladder⁶.

Prostate Cancer: Prostate cancer is one of the most common types of cancer. It typically grows slowly and stays confined to the gland itself. While not as common, there are types of prostate cancer that are aggressive and spread quickly. The key to a good outcome is early detection when the cancer is still confined to the gland and hasn't spread. Early detection starts with paying close attention to your body and knowing the warning signs of prostate cancer, as well as understanding the risk factors. Some symptoms of prostate cancer include trouble urinating, a decreased force in the urine stream, blood in urine or semen, bone pain, losing weight without trying, and erectile dysfunction.

1. Diet Prostate cancer is one of the most common cancers for men. It is estimated that one in every eight men will get it⁷. Chances are higher if you have a family history. To keep your prostate healthy and reduce the risk of prostate cancer or other prostate diseases, a change in diet can help significantly. Eat foods that are considered "prostate-friendly" and avoid foods that can contribute to prostate problems. Make sure you are eating plenty of these foods:

Colorful Fruits and Vegetables: Nature has a way of telling us what to eat by the color of foods. Brightly colored fruits and vegetables have phytochemicals that benefit the body in many ways. For example, orange fruits and vegetables often have carotenoids (beta-carotene) that are converted in the body to vitamin A. Beta carotene also promotes proper cellular communication and supports the immune system. Blue and purple fruits and vegetables contain anthocyanins that protect the body from free radical damage (that can lead to cancers) and fight against heart disease and inflammation (beneficial for an inflamed or enlarged prostate). Red fruits and vegetables contain lycopene, an antioxidant that helps prevent cancer. Green vegetables and fruits contain high amounts of chlorophyll, a phytochemical that pumps the body with oxygen to promote wound healing, fight infection, and assist immune system health. Try to incorporate a serving of colorful fruits and vegetables with each meal.

Vitamin C-Rich Foods: As an antioxidant, vitamin C helps fight free radicals in the body so they don't lead to issues down the road. Vitamin C is needed for a healthy immune system to fight off disease. Examples of vitamin C-rich foods include citrus fruits, guava, kiwis, rose hips, cantaloupe, strawberries, broccoli, kale, red cabbage, and yellow peppers.

Zinc-Rich Foods: Incorporate zinc in your diet as zinc plays an important role in protecting the body against prostate cancer⁸. There is a close relationship between zinc deficiency and an enlarged prostate, putting zinc at the top of the list for a prostate-friendly diet aimed at prevention. Zinc is also a powerful tumor suppressor. As men age, zinc levels tend to decline in the prostate, which otherwise accumulates zinc. What's more, zinc deficiency and low testosterone seem to go hand-in-hand. Try eating more shellfish (if you're not allergic), legumes, nuts, seeds, whole-grains, and cage-free eggs for healthy sources of zinc.

Eat More Lycopene: Research shows that lycopene reduces the risk of developing prostate cancer⁹. This powerful antioxidant decreases cell damage, slows cancer cell production, and protects cells from damage¹⁰. One of the best ways to get lycopene is to eat more tomatoes. Lycopene is often tightly bound to the cell walls of the tomatoes, so try juicing them to make tomato juice. Other foods high in lycopene include rose hips, guava, papaya, watermelon, grapefruit, and red bell peppers.

Cruciferous Vegetables: Research shows a positive correlation between consumption of cruciferous vegetables and cancer prevention¹¹. Try eating a serving of cruciferous vegetables every meal. Examples to try include broccoli, arugula, cabbage, cauliflower, collard greens, radishes, rutabaga, and watercress.

Green Tea: Green tea consumption is associated with a reduced risk of prostate cancer¹². Antioxidant compounds in green tea¹³, specifically xanthine derivatives, epigallocatechin gallate (EGCG), and epicatechin, help reduce the risk of prostate cancer. If you love your morning coffee, try swapping it out with green tea. This is a great way to wean yourself off coffee (because caffeine is bad for the prostate-more on that below).

Natural Sources of Omega-3s: Researchers have found links between higher amounts of omega-3 fatty acids and a lower risk of prostate cancer¹⁴. The body doesn't make these so we have to depend on diet. Most western diets give us plenty of omega-6, but not enough omega-3. Great sources include fatty fish found in cold waters such as salmon, herring, mackerel, and trout. Make sure the fish you eat are wild-caught and not farm raised.

Antioxidants: Antioxidant-rich foods help prevent chronic diseases related to oxidative stress and have been well studied for their effects on the prostate. For example, pomegranates are an antioxidant-rich superfruit that have been studied for their ability to inhibit the proliferation of prostate cancer cells¹⁵. Try drinking a glass of pomegranate juice daily and add their seeds to salads. Other antioxidant-rich foods include strawberries, blackberries, blueberries, and raspberries. Add a cup of

fresh berries with one or two meals a day or snack on them when you have a sweet tooth between meals.

The following foods should be avoided:

Sugar: Too much sugar leads to obesity and high blood sugar, both of which are linked with prostate issues¹⁶. Elevated glucose levels put men at a much higher risk (around three times higher than average) for an enlarged prostate¹⁷. Don't swap your sugar to artificial sweeteners, as these are just as bad for your health as processed cane sugar. If you are craving something sweet, opt for berries instead.

Coffee: Coffee is a diuretic, which flush excess water. If you struggle with urination due to an enlarged prostate, coffee will increase your urge to urinate. One cup a day may not do much harm, but best to keep coffee to a minimum, especially if you are struggling with urinary issues due to an enlarged prostate.

Dairy Products: Dairy products may increase your risk of developing prostate cancer¹⁸, and can increase the risk of progression to fatal prostate cancer. Avoid milk, including skim and low-fat milks, full-fat cheeses, butters, cream cheese, ice cream, and full fat yogurts. There are many good non-dairy options such as almond milk, coconut milk, and cashew milk.

Processed Foods: Processed foods contain additives, preservatives, dyes, and other potentially hazardous substances that contribute to cancer and other diseases¹⁹. For your overall health, as well as your prostate health, avoid processed foods and opt for organic, whole foods that have not been denatured or processed.

Additional dietary tips include:

Cut the Carbs: Avoid white bread, pastas, and cereals because the body converts these carbohydrate-laden foods into glucose. Elevated glucose levels are associated with an enlarged prostate²⁰. Many breads and pastas also contain GMOs, gluten, and pesticides (unless you buy organic). If you want to enjoy occasional bread and pasta, opt for organic whole grain, sprouted, or gluten-free bread and lentil pasta instead.

Choose Protein Wisely: Instead of red meats and processed meats, opt for fish, skinless poultry, cage-free eggs, nuts, seeds, and beans.

Healthy Fats: Eat healthy fats like avocados and nuts instead of products with partially hydrogenated fats.

Everything in Moderation: Avoid large portion sizes. Use a smaller plate when you eat to train your brain into eating smaller portions. Eating in moderation helps prevent obesity, which is linked to prostate issues.

2. Healthy Weight and Exercise

Studies show a strong link between a high BMI (body mass index) and prostate issues.²¹ Exercise not only helps with weight; it helps modulate hormones responsible for prostate health²². Maintaining a healthy weight and exercising regularly are key to keeping your prostate healthy.

What is a healthy BMI? Adults should try to keep their BMI between 18 and 24.9. Anything over 25 is considered overweight. Anything over 30 is considered obese. The CDC has a BMI calculator²³. Type in your height and weight to get your BMI.

How does one maintain a healthy BMI? If your BMI shows you are overweight or obese, don't be discouraged. If you follow the dietary guidelines in the protocol, stick to healthy, whole foods, and exercise, you can lose weight and lower your BMI. Most people notice weight loss quickly when they cut out foods that contribute to obesity, such as sugar, processed foods, dairy, and bread products.

Watch your daily caloric intake. If you are gaining weight eating the recommended 2,200 to 3,200 calories a day, cut back until you notice weight loss. One pound of fat is around 3,500 calories worth of energy. This means that you need to burn 3,500 calories to lose one pound of fat. Taking in fewer calories than you expend is necessary for weight loss, but do not cut calories back so much that you experience dizziness, weakness, or malnutrition. Everyone is different when it comes to what they need to lose and maintain their weight. A healthy diet and being mindful of caloric intake must go hand-in-hand with daily exercise. See our Weight Loss Protocol for more tips.

Daily Exercise (Aerobic Exercise Is Optimal): Daily exercise is essential. The kind of exercise you choose isn't as important as exercising daily to burn calories, build strength, and improve blood circulation. If you don't currently exercise much, start slowly. According to an ongoing Harvard University-based health professional follow-up study, men who are physically active are less likely to experience Benign Prostatic Hyperplasia (BPH).

An Italian study²⁴ showed that men with prostatitis who walked briskly three times a week had fewer issues with prostatitis pain, less anxiety and depression, and an improved quality of life compared to men who did non-aerobic forms of exercise three times a week. Aerobic exercise must be a part of your exercise routine.

Examples include jogging, brisk walking, water aerobics, swimming, jumping rope, cycling, elliptical, kickboxing, and Zumba. You don't have to join a class to do these, but classes are available in many areas. Exercise daily for at least 20 minutes. Make sure to drink plenty of water. If you can't handle twenty straight minutes, set goals to exercise five to ten minutes before taking a breather and then starting over to complete another ten minutes. Eventually you will be able to complete 20 to 30 minutes daily.

When you get to a healthy weight, you will likely notice more energy, stamina, and even a more positive attitude since exercise is linked with decreased depression and stress. Stress is another factor that can worsen health and may put you at risk for prostate issues.

3. Herbs for Prostate Health There are science-backed herbs and mushrooms that have been shown to inhibit prostate cancer, protect the prostate, reduce inflammation in the prostate, and more. Below is a list of effective

remedies you can employ to protect your prostate, as well as help treat conditions like an enlarged prostate.

Reishi (*Ganoderma lucidum*) and Turkey Tail (*Coriolus/Trametes versicolor*) Mushrooms for Prostate Cancer: Reishi has been shown to suppress angiogenesis (the development of new blood vessels) in prostate cancer cells²⁵. It also inhibits cancer cell viability and induces apoptosis (cell death) in highly invasive prostate cancer cells. It may suppress the migration and invasion via inhibition of matrix metalloproteinase (MMP) expression. Reishi polysaccharides (GLP) have also been shown to inhibit other types of cancer²⁶ and act as a tumor suppressor. Turkey tail is effective in targeting prostate cancer stem/progenitor cells (CSCs), which are responsible for cancer initiation and disease progression. Conventional therapies target the more differentiated cancer cells and not the prostate CSCs. Research shows that PSP, an active component extracted from turkey tail mushrooms, inhibits prostate tumor formation^{27 28}. Both mushrooms also act as a cancer preventative. Suggested dose: three dropperfuls/ml of reishi dual-extracted tincture and dual-extracted turkey tail tincture 3x daily.

Reishi Mushroom for Enlarged Prostate (BPH): Reishi helps enlarged prostate/prostatic hyperplasia by inhibiting prostate growth.²⁹ It supports benign prostatic hyperplasia (BPH) by reducing 5 α -reductase, an enzyme responsible for hormonal changes and prostate enlargement. Of nineteen species of mushrooms studied, reishi was by far the most powerful in reducing the capacity of 5 α -reductase. Suggested dose: three dropperfuls/ml of reishi dual-extracted tincture twice daily.

Note: This valuable medicinal mushroom contains water-soluble polysaccharides as well as alcohol-soluble triterpenes and triterpenoids. All are crucial medicinal compounds. *Preparing the mushroom in tea or making a simple tincture isn't enough to get all the compounds needed because some are water-soluble and some are alcohol-soluble.* You need a double extraction. Look for a dual-extracted tincture made from mushroom fruiting bodies.

Chinese Skullcap (*Scutellaria baicalensis*): A popular plant for targeting viral infections, Chinese skullcap also shows promise in treating prostate cancer³⁰. Studies show that Chinese skullcap encourages apoptosis³¹ (programmed cell death) of skin, colon, prostate, and bladder cancers. In animal studies, prostate cancer growth was reduced by 55%, skin tumors were reduced by 66%, bladder tumors shrank by 30%, and it also shows promise in treating breast cancer³². Suggested dose is 2-5 grams daily. Take as a tincture, tea, or supplement.

Saw Palmetto (*Serenoa repens*) Berries: One of the most popular herbs for prostate health, saw palmetto berries are harvested from this subtropical palm for enlarged prostate treatment. The most notable improvement is relief from the constant urge to urinate, including at night³³. Saw palmetto appears to have the same effects as pharmaceuticals for enlarged prostate, but saw palmetto is tolerated³⁴ better and has fewer side effects. Fatty acids in saw palmetto extracts have been studied for their

ability to help with prostate tumor growth³⁵, and BPH symptoms. Your physician should test for prostate cancer before using saw palmetto for BPH. Standard dose: 320 milligrams daily (160mg twice/day). Liquid saw palmetto supplements (liquid capsules or extracts – not alcoholic tinctures) have the highest concentration of both fatty acids and phytosterols³⁶. Make sure the label states that it contains 85-95% fatty acids and sterols (a liposterolic extract) and that the contents are standardized³⁷.

Stinging Nettle (*Urtica dioica*): This diuretic has long been used to treat urinary tract issues; studies also show promise for treatment of prostate issues like BPH. In animal studies, a 20% methanolic extract (tincture) was effective³⁸ for treating an enlarged prostate, with a 51.4% inhibition of induced growth. Stinging nettle helps the bladder empty better, as well as improve urinary flow, which is another reason why it provides relief for those with an enlarged prostate who have issues with urination and emptying the bladder fully. Stinging nettle's anti-inflammatory qualities³⁹ help reduce inflammation that triggers prostate enlargement. Drink extra water when taking stinging nettle since it is a diuretic. Suggested dose: Drink 1-3 cups daily; avoid drinking it in the evening. An easier option is to take 2-3 dropperfuls of stinging nettle tincture every morning. It also helps with allergies.

Pygeum: This herbal remedy comes from the bark of an African plum tree and has been used for centuries to treat urinary issues and BPH. According to the Canadian Journal of Urology⁴⁰, it helps with bladder emptying and urine flow, two common issues for those with an enlarged prostate. Suggested dose is 100 mg daily for six to eight weeks; then take a break for two to three months before resuming.

Beta-Sitosterol: Beta-sitosterol can improve urinary symptoms and flow measures⁴¹ for those with urinary issues related to BPH. Beta-sitosterol is actually a combination of plants that contain cholesterol-like substances called sitosterols or phytosterols (plant-based fats). Saw palmetto is an example of one of the plants that contain these beneficial compounds.

Rye Grass Pollen: This extract is a blend of three types of grass. It may help those who suffer from the effects of BPH by improving nighttime symptoms⁴² like frequent urination. Those with an enlarged prostate often complain that certain symptoms like urinary frequency and urgency seem to get worse at night. Suggested dose: 80-120 mg daily.

4. Staying on Top of Prostate Health

There's a wise saying, "An ounce of prevention is worth a pound of cure." This could not be truer when it comes to prostate health. It is recommended that men age 50 and older see their doctor every year for a prostate exam. However, if you have a family history of prostate cancer or fall into a high-risk category, start getting yearly exams at age 40. If you are under the recommended age for yearly prostate exams, don't hesitate to schedule a visit with a urologist if you suspect you are having prostate issues. It never hurts to make sure everything is functioning as it should.

It is important to listen to your body and pay attention to changes. If you start having symptoms consistent with BPH or prostate cancer, see a medical professional right away and don't ignore it. Some common symptoms to be aware of include:

Issues with Urination: Urinary issues are one of the most common indicators that something is amiss with the prostate. This includes trouble emptying the bladder, urinary frequency, urinary urgency, and dribbling urine. Blood in the urine should never be ignored; see a doctor right away.

Pain and/or Stiffness: Watch for frequent pain or stiffness in the lower back, hips, pelvis, rectal, or upper thigh areas. These are common symptoms of prostate issues.

Pain During Urination and/or Ejaculation: Since the prostate surrounds the urethra and is involved in ejaculation, if something is wrong with your prostate you may experience pain during urination and/or ejaculation. Seek medical help to rule out anything serious.

Sexual Dysfunction: Because the prostate plays a key role in the male reproductive system, it is no surprise that prostate cancer can trigger sexual dysfunction. If you begin to experience problems getting or maintaining an erection, see a doctor right away. Don't ignore erectile dysfunction.

Regular checkups, especially if you are in the high-risk category, listening to your body, not hesitating to seek medical help if you suspect an issue, and an awareness of symptoms are the best ways to stay on top of your prostate health. Remember that if caught early, prostate cancer is treatable.

5. Stress, Healthy Sex Life, and Sleep

Chronic stress promotes the development of cancer⁴³. Stress weakens the body's immune system, making it harder to fight off unwelcome invaders, diseases, and free radicals that can lead to cancer. Chronic stress also activates the neuroendocrine system. This promotes the production of stress hormones that promote tumorigenesis and cancer development. Chronic stress also interferes with the body's inflammatory response, so those who suffer from an enlarged prostate may experience a worsening of symptoms. While some stress in life is inevitable, chronic stress is the most damaging. If you find that you are stressed more often than not, take steps to mitigate your stress response. One thing you can do is start taking adaptogens. Adaptogenic herbs and mushrooms are those that help the body better cope with stress and the effects of stress. Below are helpful adaptogens and nervine herbs to help your body cope with stress.

Ashwagandha (*Withania somnifera*): This adaptogenic plant helps balance hormones, boost stamina and energy, reduce depression, and mitigate the effects of stress⁴⁴. Ashwagandha also improves memory and immune system function, and lowers inflammation, which can lead to BPH. The roots of this plant are used in medicinal

preparations, in capsule or tincture form. Suggested dose: 250 mg daily supplement or a daily tincture 2x/day.

Lion's Mane (*Hericium erinaceus*): If you are looking for an herb that improves brain function, memory, focus, and stress, lion's mane is the perfect choice. Like other adaptogens, lion's mane counters the potentially dangerous effects of stress and stress hormones in the body. It is also known for its anti-inflammatory, neurotrophic⁴⁵, and immune modulating abilities. Suggested dose: two dropperfuls of tincture twice daily.

Reishi (*Ganoderma lucidum*): The benefits of reishi have already been discussed when it comes to prostate health and prostate cancer prevention. This adaptogenic mushroom is also an excellent choice for reducing stress, as an anti-inflammatory, as a neuroprotective⁴⁶, for protection from oxidative stress⁴⁷, and as an immunomodulator⁴⁸. Suggested dose: 2-3 dropperfuls of reishi double extracted tincture twice daily.

Lemon Balm (*Melissa officinalis*): Lemon balm is an effective nervine for stress and anxiety. Lemon balm can boost mood⁴⁹ and improve cognitive performance. Suggested dose: Drink one to three cups of lemon balm tea daily as needed to curb stress and anxiety. You can also take it as an easy-to-use tincture as needed up to 3x daily.

Lavender (*Lavandula angustifolia*): Chronic stress leads to more than oxidative damage and a weakened immune system. It can also lead to cardiovascular issues like high blood pressure. Lavender is known for its calming effects, including lowering high blood pressure⁵⁰. Keep lavender essential oil on hand and inhale during times of stress and anxiety. Even simple inhalation of the essential oil is enough to bring down the heart rate and reduce stress significantly.

A healthy sex life is good for the prostate. Frequent ejaculation lowers your risk of prostate cancer⁵¹. A 2016 study concluded that "men reporting higher compared to lower ejaculatory frequency in adulthood were less likely to be subsequently diagnosed with prostate cancer." Another study showed that men who reported ejaculating at least once a week were less likely to have moderate to severe BPH symptoms⁵² than men reporting no ejaculations. (Note that frequent ejaculation doesn't have to be achieved with a partner). Frequent ejaculation also helps reduce stress. Having a healthy relationship with your partner and a healthy sex life is great for helping cope with life's stresses.

Other stress relief tips include:

Deep Breathing and Meditation: Deep breathing helps lower blood pressure while reducing stress. It is also great for oxygenating the body and brain. Practice deep breathing daily. Find a quiet spot and focus on nothing but your breathing. Close your eyes to stay focused. Take deep, slow breaths in through your nose and then breathe slowly out through your mouth. Make sure to exhale all of the air in your lungs before inhaling again.

Get all the stagnant air out of your lungs for best results. Practice this during meditation for a deeper, more stress-relieving experience. Studies show a positive correlation between deep breathing and mood improvement as well⁵³. Do this in the morning when you wake up and in the evening before bedtime to help with chronic stress.

Do What You Love/Spend Time Outdoors: Find time to learn something you have always wanted to learn or practice a hobby you love. Try to make time to do what you love at least one day a week. Spending time in nature is another excellent way to lower stress levels. Please see the Anxiety and Stress Protocol for more tips.

Get the Sleep Your Body Needs: Studies show a link between sleep disorders and the development of prostate cancer⁵⁴. Lack of sleep also contributes to stress and anxiety. Establish a bedtime routine, avoid fluids in the evening (so you aren't up all-night urinating), and try to get eight hours of solid sleep each night. Please see the Sleep Protocol if you are having trouble with sleep.

6. Vitamins and Supplements **Vitamin D:**

Vitamin D is crucial to overall health, including prostate health. One study showed that taking up to 6,000 IU of vitamin D daily can help decrease prostate volume⁵⁵ for those suffering from BPH. Researchers are also studying the effects of vitamin D on prostate cancer and prostate cancer⁵⁶ prevention. One of the best ways to get this vitamin is to spend 20 minutes out in the sun each day. Our body converts this to vitamin D. It can also be incorporated into your diet by eating more mushrooms. Suggested supplement dose: 1000 IU daily (more if you live in a temperate zone and during the winter).

Omega-3s: Omega-3 fatty acids are mentioned in the diet section as the prostate needs these to function properly. Getting omega-3 fatty acids through diet is best, but you can also take a supplement if your doctor feels this would be beneficial. While there are many health benefits to taking an omega-3 fish oil supplement, having too much⁵⁷ may put you at a higher risk for prostate cancer, according to some studies. However, there isn't much evidence and further studies need to be conducted. Moderation is the key to benefiting from omega-3 fatty acid supplements. If your doctors think you need more of this in your diet, suggested dose is 1,000 mg daily or every other day.

Zinc: Studies show that zinc may help protect against prostate cancer⁵⁸. Zinc has been shown to have tumor-suppressing effects⁵⁹ for different cancers, including prostate cancers⁶⁰. The connection between zinc and the prostate is no surprise, as the prostate contains the highest concentrations of zinc of any soft tissue. Zinc levels tend to decline as people age, and this may be a reason why the older a man is, the more likely he is to have prostate issues. Suggested dose: Start with 10 mg daily. Do not take over 100mg/day, including dietary zinc⁶¹.

Selenium: Selenium has been studied for both prevention and treatment of prostate cancer⁶². Suggested dose: Start by taking 20-70 micrograms daily. Brazil nuts are a good

daily source of selenium – one brazil nut contains per 68–91 mcg per nut⁶³.

Lycopene: Lycopene reduces the risk of developing prostate cancer⁶⁴. This antioxidant suppresses cancer progression and proliferation and induces apoptosis of prostate cancer cells. Lycopene also modulates signaling pathways for both treatment and prevention⁶⁵. Suggested dose: 10-20 mg twice daily with meals.

Probiotics: A healthy microbiome can mean a healthy prostate. Recent studies⁶⁶ have shed light on the link between the microbiome and prostate cancer⁶⁷. The gut microbiome has an indirect but significant role in prostate cancer development and treatment. What does this mean for you? It means you need to make sure your

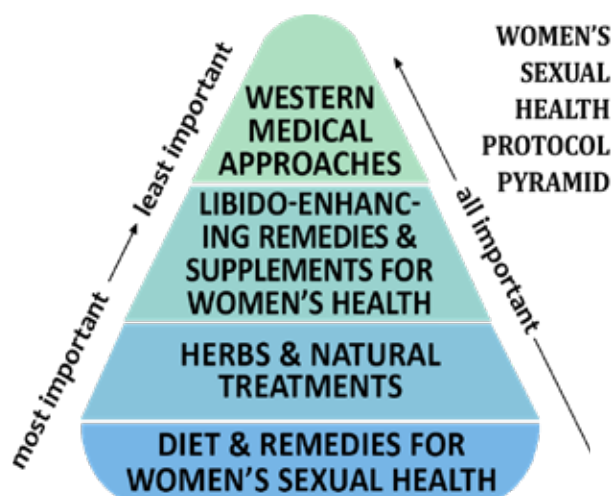
gut flora is balanced and healthy in order to further protect your prostate from harm. There are millions of bacteria (some good and some bad) living in our gut and if you follow a standard Western diet, the good bacteria may be lacking. Eating unhealthy, processed foods kills off the good bacteria, as does stress and other factors. When too many bad bacteria are allowed to thrive, this affects the body in a variety of ways, one of which is an increased risk of prostate cancer. You can make sure you are getting the good bacteria you need by taking a daily probiotic supplement and eating probiotic-rich foods. Make sure to take a supplement that offers a variety of different types of probiotics so you are getting multiple good bacteria in your microbiome. Suggested dose: one to ten billion CFUs daily.

	Morning	Noon	Afternoon	Evening	Before Bed
S U N D A Y	Deep breathing/meditation Reishi & Turkey tail dual-extracted mushroom tinctures Stinging nettle tincture Lycopene w/ breakfast Selenium (one Brazil nut), Vit. D, & Zinc Cup of green tea	Weekly grocery shopping for prostate-friendly foods Probiotic supplement Prostate-friendly lunch Saw palmetto 20 min. in the sun for Vit. D (nature walk) (+Vit. D supplement in winter) Lemon balm tea/tincture - if needed for stress add in daily to schedule below	Aerobic exercise (at least 20 min.) Make time for something you enjoy – a hobby or craft – for stress relief	Reishi & Lion's mane dual-extracted mushroom tinctures Prostate-friendly dinner Limit evening fluids Saw palmetto Chinese skullcap	Sleep blend tincture, if needed for sleep Deep breathing/meditation Lavender aromatherapy for stress relief and better sleep
M O N D A Y	Breathing/meditation Reishi & Turkey tail dual-extracted mushroom tinctures Stinging nettle tincture Lycopene w/ breakfast Selenium (one Brazil nut), Vit. D, & Zinc Cup of green tea	Probiotic supplement Prostate-friendly lunch Saw palmetto 20 min. walk in the sun for Vit. D & time in nature	Yearly prostate exam appointment Aerobic exercise (at least 20 min.) Hobby/craft time	Reishi & Lion's mane Prostate-friendly dinner Limit evening fluids Saw palmetto Chinese skullcap	Sleep blend tincture, if needed for sleep (use regularly as works best w/ daily use) Breathing/meditation Lavender aromatherapy
T U E S D A Y	Breathing/meditation Reishi & Turkey tail dual-extracted mushroom tinctures Stinging nettle tincture Lycopene w/ breakfast Selenium (one Brazil nut), Vit. D, & Zinc Cup of green tea	Probiotic supplement Prostate-friendly lunch Saw palmetto 20 min. walk in the sun for Vit. D & time in nature	Aerobic exercise (at least 20 min.) Hobby/craft time	Reishi & Lion's mane Prostate-friendly dinner Limit evening fluids Saw palmetto Chinese skullcap	Sleep blend tincture Breathing/meditation Lavender aromatherapy
W E D N E S D A Y	Breathing/meditation Reishi & Turkey tail dual-extracted mushroom tinctures Stinging nettle tincture Lycopene w/ breakfast Selenium (one Brazil nut), Vit. D, & Zinc Cup of green tea	Probiotic supplement Prostate-friendly lunch Saw palmetto 20 min. walk in the sun for Vit. D & time in nature	Aerobic exercise (at least 20 min.) Hobby/craft time	Reishi & Lion's mane Prostate-friendly dinner Limit evening fluids Saw palmetto Chinese skullcap	Sleep blend tincture Breathing/meditation Lavender aromatherapy
T H U R S D A Y	Breathing/meditation Reishi & Turkey tail dual-extracted mushroom tinctures Stinging nettle tincture Lycopene w/ breakfast Selenium (one Brazil nut), Vit. D, & Zinc Cup of green tea	Probiotic supplement Prostate-friendly lunch Saw palmetto 20 min. walk in the sun for Vit. D & time in nature	Aerobic exercise (at least 20 min.) Hobby/craft time	Reishi & Lion's mane Prostate-friendly dinner Limit evening fluids Saw palmetto Chinese skullcap	Sleep blend tincture Breathing/meditation Lavender aromatherapy
F R I D A Y	Breathing/meditation Reishi & Turkey tail dual-extracted mushroom tinctures Stinging nettle tincture Lycopene w/ breakfast Selenium (one Brazil nut), Vit. D, & Zinc Cup of green tea	Probiotic supplement Prostate-friendly lunch Saw palmetto 20 min. walk in the sun for Vit. D & time in nature	Aerobic exercise (at least 20 min.) Hobby/craft time	Reishi & Lion's mane Prostate-friendly dinner Limit evening fluids Saw palmetto Chinese skullcap	Sleep blend tincture Breathing/meditation Lavender aromatherapy
S A T U R D A Y	Breathing/meditation Reishi & Turkey tail dual-extracted mushroom tinctures Stinging nettle tincture Lycopene w/ breakfast Selenium (one Brazil nut), Vit. D, & Zinc Cup of green tea	Probiotic supplement Prostate-friendly lunch Saw palmetto 20 min. walk in the sun for Vit. D & time in nature	Aerobic exercise (at least 20 min.) Hobby/craft time	Reishi & Lion's mane Prostate-friendly dinner Limit evening fluids Saw palmetto Chinese skullcap	Sleep blend tincture Breathing/meditation Lavender aromatherapy

Ch. 31: Women's Sexual Health Protocol (PCOS, Endometriosis, Uterine Fibroids, Fertility, Birth Control, STDs)

True sexual health is a state of physical, emotional, mental, and social wellbeing in relation to sexuality. Sexual health isn't just about being physically healthy, but also having a positive and respectful approach to your own sexuality, relationships, and learning how to engage in safe sexual experiences. Nurturing your sexual health can help you live a happier, healthier life.

While sexual health encompasses a variety of aspects, a large part involves learning how to stay physically healthy so you can take control of your fertility, reproductive health, libido, and any reproductive issues you encounter.



The purpose of this protocol is to arm you with the information you need to take care of your sexual health in a multifaceted way.

Aspects of sexual health covered in this protocol include reproductive health, fertility, birth control options, libido, and issues such as polycystic ovary syndrome (PCOS), endometriosis, and uterine fibroids. In addition, you will learn how to manage menstrual issues, urinary tract infections, bacterial vaginosis, yeast infections, and sexually transmitted diseases (STDs). Since sexual health also involves mental and emotional aspects, you will learn how to boost your sexual confidence to feel better about yourself.

1. Diet and Remedies POLYCYSTIC OVARY SYNDROME (PCOS):

Polycystic ovary syndrome is a hormonal disorder common among women of reproductive age. Women with PCOS experience infrequent or prolonged menstrual cycles and/or elevated male hormone levels (androgens).

Ovaries of those with PCOS often develop multiple small collections of fluid (follicles) and can fail to

regularly release eggs. The specific cause is unknown. However, it is important to diagnose PCOS as soon as possible for early treatment and intervention. A common intervention is weight loss, which helps to reduce the risk of long-term complications like type 2 diabetes and heart disease.

• Symptoms

Often symptoms of PCOS develop around the time of the first menstrual period during puberty, but they can develop later in life. A lot of women experience symptoms later in life in response to substantial weight gain, which seems to directly affect PCOS. Symptoms of an underactive thyroid can mimic PCOS, as can endometriosis. Insulin resistance, inflammation, and excess androgen are symptoms of PCOS. Other signs and symptoms include:

Ovarian Issues: Women who suffer from PCOS may have enlarged ovaries that contain follicles that surround the eggs. The result is ovaries that fail to function as they should. This can affect fertility for women who wish to conceive. The follicles associated with PCOS can often be detected on an ultrasound. Some women report severe pain in one or both ovaries as cysts rupture occasionally or frequently.

Irregular Periods: One of the most common symptoms of PCOS is infrequent, irregular, or prolonged menstrual cycles. If you have fewer than nine periods each year or regularly go more than 35 days between periods, this can be a sign of PCOS. Abnormally heavy periods are also a symptom.

Excess Androgens: Women with PCOS often have elevated levels of male hormones that result in symptoms such as excess facial and body hair, severe acne, or male-pattern baldness.

Mood Swings/Depression: Some women experience issues with mood and depression as a result of interference with hormones that come with PCOS.

• Potential Causes

The exact cause of PCOS isn't known, but factors that seem to play a significant role include excess insulin, low-grade inflammation in the body, genetics, and excess androgen. Excess insulin may increase androgen production and interfere with ovulation. Women with PCOS have a type of low-grade inflammation that may stimulate polycystic ovaries to produce androgens. Over time, this may lead to heart and blood vessel issues. Research points to certain genes linked to this condition. Environmental factors cannot yet be ruled out.

• Diet for PCOS

A good diet for those with PCOS includes nourishing foods and cutting out inflammatory foods, as low-grade inflammation is a factor. Inflammatory foods and drinks to avoid include processed/refined sugars (PCOS is linked with insulin resistance), alcohol, fast food, and processed foods. Coffee is a diuretic and can make you lose more

water than you take in. As a stimulant, its effects are desirable for many people, but this stimulant effect can cause cortisol levels to increase. Those with PCOS are at a higher risk for adrenal fatigue, and a rise in cortisol effects hormones and inflammation. Get an IGG test to test for possible inflammatory trigger foods such as dairy, gluten, and eggs.

Replace inflammatory foods with anti-inflammatory foods and drinks. For example, if you are dependent on a sugary and dairy-rich coffee drink in the morning, try making the switch to green tea instead. Green tea has strong anti-inflammatory effects and may help improve the quality of life for people with inflammatory diseases¹.

If you are hooked on sugar, take steps to curb this inflammatory craving by swapping refined and processed sugars for a healthier alternative like organic stevia leaf or eat fresh, ripe fruit when you are craving refined sugar. Fresh fruit contains simple sugars that are much easier for the body to digest. Switching to a whole-foods diet helps reduce inflammation and decrease other health risks².

Fast foods and processed foods are full of additives, preservatives, and other substances that may interfere with hormones. If you eat fast food on a regular basis, try meal planning. Meal planning is an effective and healthy way to change your eating habits.

The key to managing PCOS is a low-fat, plant-based, whole foods diet. This diet can lead to a decrease in insulin resistance and thus dramatically reduce PCOS symptoms. Many women with PCOS are told to lose weight to reduce symptoms. Following a low-fat, plant-based, whole food diet will likely help you lose excess weight.

In addition to helping with insulin sensitivity and weight loss, following the recommended diet can help lower blood glucose and reduce the menstrual irregularities common with PCOS. The following foods are recommended for those with PCOS.

Fruits: Fruits are an excellent source of fiber and can decrease your saturated fat intake while reducing your fasting and post-meal blood glucose levels. Fruit provides carbohydrate energy, which helps fuel the brain, muscles, and liver. Fruits are a rich source of vitamins, minerals, and antioxidants as well. Try eating three to five servings of fruit with breakfast and lunch.

Whole Grains: Whole grains like quinoa, steel cut oats, amaranth, and buckwheat are a great source of whole carbohydrate energy and help you feel more satiated. It is recommended to eat one to two servings of these daily. Try sprinkling them in salads or tossing them in with various foods you eat.

Legumes: Legumes are full of nutrients. Legumes to incorporate into your diet include chickpeas, lentils, beans, and peas. Try eating one to two servings each day. While it is not necessary to cook these, they can be lightly sautéed, steamed, or baked to make them easier to consume. Hummus is made with chickpeas and can be a delicious and easy way to get your daily lentil serving.

Sprinkle some whole grains on the hummus and use one of the vegetables mentioned below to dip into it for a tasty and nutritious snack.

Vegetables (Non-Starchy): Choose vegetables that are not starchy, such as tomatoes, onions, bell peppers, asparagus, cauliflower, and broccoli. Eat three to five servings of non-starchy vegetables each day to get the nutrition you need.

Leafy Greens: The benefits of leafy greens in your diet cannot be understated. Greens like spinach, kale, watercress, Swiss chard, arugula, collard greens, and beet greens contain many vitamins and minerals, as well as fiber to help keep your blood glucose stable. Eat two to four cups of leafy greens each day. If you plan on eating more carbohydrates than usual, add additional leafy greens to your meal.

• Herbal Support for PCOS

Along with a PCOS-friendly diet, herbs can help balance hormones and keep symptoms at bay.

Reishi (*Ganoderma lucidum*): Reishi mushrooms provide health benefits for a wide variety of issues. This is in part due to their adaptogenic properties. Reishi mushroom is especially helpful for PCOS because it can support liver regeneration, detoxification, and hormonal balance. Most importantly for those with PCOS, it is anti-androgenic³. Take reishi in a double extraction form, as this form captures both the water and alcohol-soluble medicinal compounds in the mushroom. Suggested dose: 2-3 dropperfuls of a reishi dual-extracted tincture twice daily.

Vitex/Chaste Tree (*Vitex agnus-castus*): Vitex is known for its ability to bring hormonal balance. For women with normal LH levels and high prolactin levels, vitex tincture has beneficial effects in the treatment of PCOS⁴. Note: for women with high LH (luteinizing hormone) levels, vitex can worsen PCOS. Have your levels checked. Suggested dose is 2 dropperfuls of vitex tincture up to twice daily. It may take a few months to see results.

Yarrow (*Achillea millefolium*): Yarrow has been used for centuries as a women's herb to treat menstrual conditions, fibroids, ovarian cysts, endometriosis, urinary tract infections, cramping, and PCOS. A 2020 study on rats showed that yarrow extract improves the detrimental effects of PCOS on reproductive parameters including estrous cycles, body and ovary weights, number of ovarian cysts (causing them to reach normal levels), and sex hormones in a dose-dependent manner⁵. For those with PCOS with severely painful and heavy menstrual cycles, you may benefit from taking yarrow the week before and during your cycles. Yarrow suppresses bleeding and is also great for wound care thanks to its anti-inflammatory⁶ and antibacterial properties. Suggested dose: Take 2 dropperfuls of yarrow tincture twice daily the week before menstrual cycles, as well as during cycles to help with heavy flow.

Black Cohosh (*Actaea racemosa*): If you are suffering from PCOS and wish to conceive, black

cohoash has been studied for its ability (when used with Clomid) to increase a woman's chances of getting pregnant⁷. It may also help regulate cycles to make it easier to conceive. Suggested dose: Take 20-120 mg of black cohosh daily if you are trying to conceive.

False Unicorn Root (*Chamaelirium luteum*): Not to be confused with true unicorn root, *false* unicorn root has a history of use for women with difficulties getting pregnant, and who suffer from hormonal imbalance, menstrual issues, and miscarriages. It may be helpful for those with PCOS because it interacts with the body's estrogen receptor sites to increase estrogen levels⁸. In turn, this may help the ovaries produce mature eggs and ovulate as they should, helping women with PCOS conceive easier. In addition, the normalizing of hormones can help relieve symptoms of PCOS. Suggested dose: 1-2 grams up to three times daily.

Sweet Marjoram (*Origanum majorana*): Since many with PCOS struggle with insulin resistance, sweet marjoram may be helpful because it can reduce DHEA-S and fasting insulin levels. Sweet marjoram also significantly decreases ovary weight and estradiol levels for women with PCOS in addition to improving insulin sensitivity⁹. Suggested dose: 6-9 grams daily.

Berberine: Berberine is a compound found in several medicinal herbs. It may improve insulin sensitivity, support weight management and even improve cholesterol. It is popular for those with PCOS because of its ability to help with insulin issues¹⁰. Since more than fifty percent of women with PCOS are diagnosed with type two diabetes by the age of 40, managing insulin early is very important. Suggested dose: 1,500 mg daily (divided into three doses).

ENDOMETRIOSIS: Endometriosis is a painful disorder where the tissue that normally lines the uterus (the endometrium) grows outside the uterus as well. This tissue may be found growing on ovaries, fallopian tubes, and tissues lining the pelvis. In some rare cases, the tissue may even be found in areas beyond the pelvic organ regions.

This tissue that is not where it is supposed to be behaves like endometrial tissue. This means it thickens, breaks down, and bleeds when a woman has her monthly cycle. However, since this tissue has no way to exit the body, it becomes trapped and can result in irritation, scarring, adhesions, and severe pain.

• Symptoms

While it is not yet known what causes this debilitating condition, symptoms include pelvic pain, extremely painful periods with severe cramping, lower back and abdominal pain, pain with intercourse, pain with bowel movements or urination (especially during periods), excessive bleeding during menstrual periods and sometimes bleeding between periods, infertility, fatigue, diarrhea, constipation, and bloating or nausea (especially during periods).

women find relief through a change in diet and medicinal herbs.

• Endometriosis Diet

Certain foods can aggravate endometriosis.¹¹ Eliminate foods high in trans fats, red meat, gluten, processed sugars, high-estrogen foods like soy, and foods high in fermentable oligosaccharides, disaccharides, monosaccharides and polyols (FODMAPs).

Trans fats are often found in processed, fried, and fast foods. High-fat diets can make endometriosis significantly worse, so eliminate as much fat as possible¹². Stick to an organic, whole food diet of fresh fruits, vegetables, leafy greens, and seeds/nuts. Because of the effect on the bowels (especially for those with irritable bowel syndrome), a diet low in FODMAP foods can help prevent flares¹³. It is also a good idea to eliminate alcohol, dairy (this is a common inflammatory trigger for many), and caffeine. Common FODMAP-containing foods include garlic, onion, and fermented cabbage (sauerkraut). There is a link between gluten consumption and endometriosis, so limit your gluten intake and consider going gluten-free¹⁴. Gluten is a trigger for many inflammatory conditions.

Choose foods that are high in fiber, iron, and essential fatty acids. When shopping for fresh fruits and vegetables, look for organic as these won't contain potentially triggering chemicals like pesticides and herbicides. Eat more anti-inflammatory foods like fruits, vegetables, legumes, wild-caught salmon, walnuts, chia, and flaxseed. Incorporate more leafy greens like spinach, watercress, Swiss chard, and kale into your diet. Eat berries such as cherries, raspberries, and blueberries.

• Herbs for Endometriosis

In addition to following the dietary suggestions above, try the following herbs to help relieve symptoms of endometriosis such as heavy bleeding and inflammation that causes pain and irritation. When you are able to manage your endometriosis with a healthy diet and herbal remedies, you may reduce your risk of developing more serious complications from endometriosis like certain cancers¹⁵:

Evening Primrose (*Oenothera biennis*): Evening primrose seed oil may help reduce the severity of endometriosis because it may help lower estrogen. Endometriosis is linked with estrogen dominance, so the hypothesis is that it helps control this particular hormone to prevent severe symptoms¹⁶. If you are looking for something to bring more balanced estrogen levels, try taking evening primrose seed oil in capsules. Suggested dose: 6-8 grams daily.

Curcumin/Turmeric: Curcumin helps control inflammation¹⁷ that leads to pain and a worsening of symptoms. Research on endometriosis points toward inflammatory cytokines. Curcumin, the primary active medicinal compound found in turmeric, may help control these. Suggested dose: 900 mg daily.

Reishi (*Ganoderma lucidum*): Reishi mushroom is an anti-inflammatory medicinal mushroom that may help

endometriosis in several ways. First, it helps control inflammation and pain. Second, it helps to balance hormones. Third, it can help balance other issues in the body that are potentially triggering flares and pain¹⁸. Take in a double extraction formulation. Suggested dose: 2-3 dropperfuls of a reishi dual-extracted tincture twice daily.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): The high antioxidant content in turkey tail mushrooms can support a healthy body¹⁹ and its adaptogenic qualities can help bring much-needed balance. Like reishi, turkey tail mushrooms are anti-inflammatory and may help to relieve pain that is associated with inflamed tissues. They are also great for immune support. Suggested dose: 2-3 dropperfuls of a turkey tail dual-extracted tincture twice daily.

Cordyceps: Another medicinal and adaptogenic mushroom, cordyceps brings balance to the body while also reducing inflammation. They also balance immune function in addition to hormonal function²⁰. Suggested dose: 2-3 dropperfuls of a cordyceps dual-extracted tincture twice daily.

Yarrow (*Achillea millefolium*): Yarrow has a wide range of medicinal benefits; one that is especially helpful for endometriosis is its ability to reduce excessive bleeding, tone the uterus, and alleviate pelvic congestion. Many women with endometriosis experience especially heavy and painful menstrual cycles. Yarrow can help reduce this heavy bleeding and bring much-needed relief²¹. Suggested dose: Take 2 dropperfuls of yarrow tincture twice daily the week before menstrual cycles, as well as during cycles to help with heavy flow.

Berberine: Some research indicates that endometriosis may be caused by an overproduction of gram-negative bacteria and a bacterial toxin called lipopolysaccharide (LPS) in the pelvic region²². This is thought to potentially trigger inflammation in that region. Berberine is a compound found in plants such as goldenseal and Oregon grape root that may help neutralize LPS. Suggested dose: 500-1,500 mg daily.

Shepherd's Purse (*Capsella bursa-pastoris*): A friend of midwives, this common yard plant was commonly given to women after birth if they were bleeding too much. It may be helpful for the heavy bleeding associated with endometriosis²³. Suggested dose: Infuse 2 tsp. into a cup of hot water and drink up to twice daily during times of heavy menstrual flow.

UTERINE FIBROIDS: Uterine fibroids are non-cancerous tumors that grow in the uterine wall. They range in size from the size of a bean to the size of a large grapefruit. Sometimes, they are asymptomatic, but they may cause symptoms such as pain and menstrual changes.

• Diet for Uterine Fibroids

Follow the same dietary advice for uterine fibroids that is indicated above for endometriosis and PCOS. Eat more anti-inflammatory foods and avoid processed foods like fast food, refined sugar, caffeine, and alcohol.

• Herbal Support for Uterine Fibroids

Yarrow: Yarrow may be helpful for those with uterine fibroids because it can reduce heavy bleeding. It may also help with pain and inflammation in the uterine area. Suggested dose: Drink yarrow tea (2 tsp. yarrow/cup of hot water) twice daily or take 2 dropperfuls of yarrow tincture twice daily for pain and heavy bleeding.

Red Raspberry (*Rubus idaeus*) Leaf: This is a great remedy for toning the uterus and helping make menstrual periods lighter and less painful. Some women claim drinking red raspberry tea or infusions daily help relieve symptoms of fibroids or reduce the fibroids altogether. Suggested dose: Boil a quart of water and pour into a quart jar that has been filled with a cup or two of red raspberry leaves. Infuse overnight and then refrigerate. Drink up to three cups daily. You may also use a tincture (2 dropperfuls/day).

Vitex/Chaste Tree: This is an excellent herb for hormonal imbalances²⁴ that cause fibroids and other related uterine conditions. By restoring hormonal balance, you may experience less pain, inflammation, heavy bleeding, and other symptoms associated with fibroids, PCOS, and endometriosis. Suggested dose: Take 2 dropperfuls of vitex tincture up to twice daily. It may take a few months to see results.

Milk Thistle (*Silybum marianum*): Milk thistle seeds may help shrink uterine fibroids, even though it is more well-known for supporting liver health²⁵. However, some believe it may affect estrogen levels and worsen these conditions. While this is a concern, there is currently no solid evidence to point to milk thistle negatively affecting hormones. Milk thistle contains anti-inflammatory compounds that can help reduce pain and inflammation in the uterine area. Suggested dose: Take 12-15 grams daily or 200-400 mg of the active compound in the plant called silymarin.

Reishi (*Ganoderma lucidum*): Because it is an effective anti-inflammatory²⁶, reishi mushroom may help relieve pain and inflammation caused by fibroids. Additionally, reishi may help relieve hormonal imbalances that contribute to fibroids. Suggested dose: 2-3 dropperfuls of a reishi dual-extracted tincture twice daily.

URINARY TRACT INFECTIONS: Urinary Tract Infections (UTIs) are a common and troublesome occurrence for many women. There are several factors that increase the risk of a UTI. These include hygiene, the undergarments you wear, products you may be using, sexual intercourse, and not drinking enough water. Make sure that when you use the restroom, you wipe from front to back to keep bacteria away from the urethra. Wear breathable cotton underwear to prevent bacteria from growing. Avoid using soaps in the vaginal area, as these can irritate the urinary tract and affect pH balance. Warm water is plenty to keep the region clean. If you take a lot of baths, try taking showers instead, as baths can introduce bacteria to the urinary tract. Try urinating before and after sexual intercourse to cleanse the urinary

tract. Stay hydrated.

Another important tip for preventing UTIs is to avoid commercial pantyliners and pads and opt for organic cloth liners and pads instead. Commercial products are made using chemicals and bleach to make the cotton white. They rub against the urethra and can cause infections.

Symptoms of a UTI include burning during urination, the feeling like you need to urinate often and suddenly, strong smelling urine, or urine that is a different color than usual (such as brown, pink, red, or cloudy).

If left untreated, a urinary tract infection can migrate and turn into a bladder or kidney infection, which can be serious. These need quick and persistent treatment to avoid complications. In addition to following the advice above, there are natural remedies you can use to stave off an infection. However, if you try these remedies and symptoms persist, see a doctor immediately for treatment.

• **Herbal Remedies for UTIs**

Cranberry Juice: Cranberry juice²⁷ may help eliminate urinary tract and bladder infections by interfering with bacteria's ability to invade the lining of the bladder. Cranberry makes a great preventative if you are prone to bladder infections. Suggested dose: Take cranberry capsules twice daily or drink a cup of naturally sugar free cranberry juice daily.

Goldenrod (*Solidago* spp.): Goldenrod may help to eliminate the bacteria causing UTIs, as well as help flush it out more efficiently.²⁸ It is a natural diuretic, so it helps push water from the body to cleanse and purify. It is highly antioxidant and anti-inflammatory, helping bring relief from the symptoms of infection. Suggested dose: Infuse 2-3 tsp of dried goldenrod into a cup of hot water and drink up to three cups daily or take 2 dropperfuls tincture/day.

D-Mannose: This is a natural remedy made from simple sugars. It is thought to work by blocking UTI-causing bacteria from growing in the urinary tract²⁹. Suggested dose: Take 1,500 -2000 mg at the onset of a suspected UTI. Continue taking this dosage for up to three days.

Bearberry (*Uva ursi*): This berry is also called kinnikinnick. It has been used for centuries to treat UTIs and bladder inflammation like cystitis. It can be used as a treatment and a preventative. It helps to reduce bacteria in the urinary tract³⁰. Suggested dose: Take in tincture form at the onset of a suspected UTI. 2 dropperfuls up to four times daily until symptoms are gone. It also works well when combined with goldenseal or Oregon grape or another berberine-containing plant. Tincture equal parts together and take two dropperfuls up to three times daily.

Oregano: Oregano is a strong antibacterial and antimicrobial³¹ herb, so it is no wonder it works to get rid of bacteria in the urinary tract. Suggested dose: If you suspect a UTI, make a strong tea with oregano and drink several cups daily or add 2 dropperfuls of

oregano tincture to a big bottle of water and drink these two or three times a day. Alongside, drink as much water as you can to help flush things out.

YEAST INFECTION AND BACTERIAL

VAGINOSIS: Yeast infections are caused by yeast in the vaginal canal. This fungal infection can cause severe itching, rashes, white discharge, redness, and burning. The infection can appear inside or outside the vagina on the vulva, labia, and other areas. To prevent yeast infections, wear breathable underwear, avoid tight-fitting clothing, avoid douching, avoid hot tubs and bathing, change out of wet clothes instead of wearing them for extended periods, wipe front to back, manage diabetes, only use antibiotics when you have to, and maintain good hygiene when you are on your period. These practices will lower your chances of getting a yeast infection.

Bacterial vaginosis (BV) is similar to a yeast infection. It is caused by an overgrowth of bacteria in the vagina and often happens when something upsets the pH balance of the vagina. Symptoms include burning, increased discharge that smells like fish, burning with urination, and inflammation. The vagina operates in a delicate balance. It self-cleans and does not require douching or extra products to be clean. Women who use these increase their likelihood of upsetting the pH balance and having an infection. Follow the same protocols as you would to prevent a UTI or yeast infection. In addition, both partners should maintain good hygiene prior to sexual intercourse to prevent the introduction of bacteria during sex.

Below are some helpful remedies to treat both yeast infections and bacterial vaginosis.

Coconut Oil: Coconut oil is naturally antifungal³² and antimicrobial, and has a relatively neutral pH, so it makes a wonderful treatment for both BV and yeast infections. Wash your hands with soap and then take a small amount (the size of a nickel) and insert this into the vaginal canal. Lie down for as long as you can (do this before bed as well, so you can make sure it stays in there a while). Apply two to three times daily as needed. Do this at the onset of symptoms for maximum effectiveness. If your infection doesn't go away after a day or two, try the remedy below.

Tea Tree and Lavender Bolus: A bolus is a remedy molded into a ball to be inserted anally or vaginally. It is like an herbal suppository. Coconut oil is the perfect base for a bolus because it is solid at room temperature, but will melt quickly and gently when it is inserted. Tea tree and lavender essential oils added to this base make a potent antifungal³³ remedy. Melt a 1/2 cup of coconut oil in a double boiler and then infuse it with a tablespoon each of tea tree leaves and lavender buds (or 5 drops of the essential oil of each). Let this infuse for up to eight hours before straining. Pour it into a sterile jar to cool. When it has resolidified but is still malleable, take out a quarter-sized amount and roll it into a ball. Insert this into the vagina while lying down. Leave this in the vagina for as long as possible. Administer this bolus up to three times daily as needed for infections.

SEXUALLY TRANSMITTED DISEASES:

There are many types of sexually transmitted diseases (STDs) that affect both women and men. Unfortunately, women seem to be more at risk and suffer more severe complications from some STDs. It is important to get help as soon as you believe you have an issue to prevent possible complications like infertility. If you see a doctor, they may run tests like a blood test, fluid test, and a urine sample to better understand the problem. The following symptoms are consistent with common STDs: painful urination, excess vaginal discharge, lower abdominal pain, pain or itching in genital area, ulcers in the genital region, small red bumps or blisters in the genital region, anal itching, swollen lymph nodes, bleeding or pain with intercourse, and fatigue. Common STDs include gonorrhea, syphilis, chlamydia, herpes (HSV), human papilloma virus (HPV), hepatitis, and trichomoniasis. To prevent STDs use protection during sexual intercourse or sexual acts. If you have multiple partners, your risk is increased.

While it is best to see a medical professional if you suspect you have a sexually transmitted disease, there are herbal remedies that can help speed up treatment, as well as relieve symptoms and help the body heal.

• Herbs for Herpes and other STDs

Chinese Skullcap (*Scutellaria baicalensis*): The baicalin and wogonin compounds in Chinese skullcap exhibit strong antiviral activity against hepatitis B virus, hepatitis C virus, and herpes simplex virus 1 and 2.³⁴ Suggested dose: 3 dropperfuls of tincture, 3 times daily.

Lemon Balm (*Melissa officinalis*): Lemon balm helps treat cold sores, genital herpes, and shingles, both internally and topically.^{35 36 37} Lemon balm contains antiviral compounds³⁸ that work well to eradicate viral lesions on the lips and genitals as a result of herpes viruses. Suggested dose: Take 2 dropperfuls lemon balm tincture, three times daily. For external lesions, melt a tablespoon of coconut oil in a double boiler and add five to ten drops of lemon balm essential oil. Stir well and pour it in a jar to cool. Apply to viral lesions three to five times daily. You may also apply lemon balm tincture externally to stop cold sores in their tracks.

Licorice Root (*Glycyrrhiza glabra*): Licorice exhibits antiviral properties against several viruses, including hepatitis C, herpes simplex virus, and human immunodeficiency virus (HIV).³⁹ Note that individuals with hypertension, liver cirrhosis, congestive heart failure, or renal failure should not use licorice. Suggested dose: 1 dropperful of tincture, three times daily (Note: not DGL-licorice root as you need the triterpenoid glycyrrhetic acid for this use⁴⁰).

Myrrh: Myrrh extract exhibits antiviral activity against herpes simplex virus type 1, the type that most commonly causes cold sores.⁴¹ Suggested dose: 2 dropperfuls tincture twice daily.

Peppermint Oil: Exhibits antiviral activity against herpes

simplex viruses 1 and 2.⁴²

St. John's Wort (*Hypericum perforatum*): Most people know this plant for its ability to reduce depression, anxiety, and nerve pain. It also possesses strong anti-inflammatory and antiviral properties⁴³. Like lemon balm, it can help eradicate viral herpes lesions when applied topically. Make the lemon balm recipe above extra powerful by infusing St. John's wort flowers in coconut oil before adding the lemon balm essential oil. Suggested dose: Apply externally to lesions 3x/day.

Tea Tree: Tea tree essential oil is antimicrobial⁴⁴ and antifungal. It has been used to treat the STD's syphilis and trichomoniasis. It can be used in a sitz bath for STDs. Suggested dose: Add twenty drops of tea tree oil to a tablespoon of organic liquid castile soap. Blend this into a cup of Epsom salts and pour into warm bath water. Soak daily as needed for relief.

Aloe Vera: For topical infections and skin issues⁴⁵, aloe vera provides soothing relief while also reducing inflammation and pain. It is also antimicrobial and can help kill infections. Suggested dose: Apply aloe vera gel topically as needed for healing and relief.

This portion of the protocol will address natural ways to manage fertility and birth control. Western medical treatments for fertility and birth control are discussed in the last portion of the protocol under "Western Medical Treatments."

2. Herbs and Natural Treatments

FERTILITY: Infertility is becoming more and more prevalent in the United States, as well as around the world. Around nine percent of men and eleven percent of women of reproductive age in the United States have experienced some sort of fertility issue. After a year of having unprotected sex, twelve to fifteen percent of couples are unable to conceive. After two years, ten percent of couples still haven't had a live-born child. For couples younger than thirty and generally healthy, forty to sixty percent are able to conceive in the first three months of trying.

As we age, fertility declines for both men and women. Women are affected by age much more than men. Women in their 30s are half as fertile as they were in their early 20s and their chance of conception declines even more significantly after the age of 35. While male fertility declines with age, this progression is much more gradual.

There are many factors that contribute to infertility, and some have already been discussed above. Endometriosis and PCOS are two large contributors. Consider employing the strategies discussed in this protocol for treating these conditions to increase your chances of conceiving. Other problems that can lead to infertility include uterine fibroids (discussed above), autoimmune disorders, failure to ovulate, menstrual irregularity and problems with menstrual cycles, infections, failure of the egg to mature properly, implantation failure, and reproductive structural issues. These are just the

problems women face. Men face issues with low sperm counts (caused by a variety of factors).

If you are a generally healthy woman and have been trying to conceive for more than one year without success, consider seeing a fertility doctor to help diagnose what is interfering with conception. They will perform an exam, possibly an ultrasound to look at eggs, and conduct a sperm test to check your partner's sperm count. They may also do bloodwork to check hormone levels.

Once you have a better understanding of the source of the problem, you will be better equipped to treat it with success. The protocols below can help if you have an awareness of what is holding you back from conceiving. They are aimed toward treating hormone imbalances, regulating menstrual cycles, and getting familiar with your body and how it works.

• **Herbal Remedies for Fertility**

Herbs can be very helpful for correcting imbalances, relieving pelvic congestion, toning the uterus, and helping get your body where it needs to be in order to conceive. Below are some useful herbs for fertility in women:

Vitex/Chaste Tree: Of all the herbal remedies for fertility, vitex may be the best. This is because it is an adaptogenic plant that specifically targets hormones⁴⁶ to bring balance and stability. The beauty of vitex is that no matter the hormonal issue, it can help bring balance where it is needed (note the exception in the PCOS section). Every woman has different hormonal issues. Some may have higher estrogen or progesterone levels while others may have high prolactin levels. Vitex can help bring these levels to where they need to be, and as a result, restore fertility. It may take several months of taking vitex to see the effects, so be patient. Suggested dose: 2 dropperfuls 3x times daily in tincture form with breakfast, lunch, and dinner.

Rhodiola rosea and Reishi: Rhodiola and Reishi Mushroom help improve energy and reduce chronic fatigue. Rhodiola also boosts athletic performance⁴⁷, as does cordyceps mushroom. They are famous for enhancing physical strength and endurance, and allow athletes to minimize the effects of physical stress. Rhodiola and Reishi help regulate thyroid hormones so they function normally⁴⁸, which is part of the reason they help to increase energy. They may also increase fertility. *Avoid Rhodiola if you have an autoimmune disorder.* Suggested dose: 2 dropperfuls twice daily in tincture form of Reishi or Rhodiola.

Maca: Maca can help the body respond better to stress, and stress can affect fertility. Maca also supports a healthy libido and contains nutrients essential to a healthy reproductive system. It can be used by men and women⁴⁹ wanting to conceive and enhance fertility; it has shown promise in improving sperm count⁵⁰. Maca contains higher amounts of iodine, so do not take it if you have thyroid issues and are already on iodine supplements. Maca should be avoided if you have an androgen sensitive issue like PCOS. Suggested dose: 1,500 to 3,000 mg daily.

Motherwort (*Leonurus cardiaca*): Motherwort has been utilized for centuries for women who are stressed and need help with anxiety and cardiovascular support⁵¹. Motherwort is an effective remedy for menstrual issues like cramping and uterine spasms. It can lower high blood pressure, especially when induced by stress and anxiety. Since anxiety and stress can affect fertility, motherwort may be a useful remedy to support stress levels and healthy menstruation. Suggested dose: Take 1-2 dropperfuls of tincture as needed for painful periods, anxiety, and stress.

Red Raspberry Leaf: Many women find that drinking infusions of red raspberry leaf helps to tone the uterus⁵², reduce fibroids, and get the uterus ready for carrying a child. Red raspberry leaf is great for uterine support and despite misinformation that it stimulates the uterus, it actually tones it and helps keep uterine tissues healthy. It also helps with heavy or painful menstruation. Suggested dose: Boil a quart of water and pour into a quart jar that has been filled with a cup or two of red raspberry leaves. Infuse overnight and then refrigerate. Drink up to three cups daily. You may also use a tincture (2 dropperfuls/day).

Other Adaptogens (e.g., Reishi): Besides rhodiola, vitex, and maca, adaptogenic mushrooms like reishi, turkey tail, and cordyceps⁵³ may be helpful when trying to conceive because they help regulate hormone function in addition to immune function. This makes them promising for those with autoimmune diseases that may prevent them from conceiving. These adaptogenic mushrooms also help alleviate fatigue caused by hormonal imbalance⁵⁴ to give you the energy you need to increase stamina and vitality. Suggested dose: 2 dropperfuls of two dual-extracted mushroom tinctures twice daily (Reishi + one other).

Traditional Chinese Medicine (TCM) Herbs: Some women have scar tissue in their fallopian tubes that prevents the egg from being fertilized or being able to implant in the womb. There are two different herbs used in Traditional Chinese Medicine⁵⁵ that may help to remove this scar tissue and open up the fallopian tubes to aid in conception. These herbs are called XiXianCao and YiMuCao. They are best used together for this purpose. They come in powdered, granular form and can be dissolved easily into hot water to drink as tea. Suggested dose: Dissolve one tsp of each powder into a cup of tea twice daily for one to three months.

• **What to Avoid**

Below are foods to avoid if you are trying to conceive. Eliminating these foods from your diet may increase your chances of conceiving because they help to lower inflammation and acidity in the body.

Inflammatory Foods: Follow the anti-inflammatory diet described above for endometriosis and PCOS. Avoid inflammatory foods like fast food, processed foods, processed sugars, and alcohol. These foods may worsen inflammation in the body and make it harder to conceive. They may trigger autoimmune issues that affect fertility.

Acidic Foods: Acidic foods may affect the environment in the vagina, making it more acidic. An acidic vagina kills off sperm and makes it harder for to conceive. Try to eat alkaline foods that create an alkaline environment in the body. This pH neutral environment is much easier on sperm.

• Other Strategies

Yoni Steams: Yoni steams have long been used to help relieve pelvic congestion, promote healthy menstruation, and increase fertility. They are a relatively simple concept: Boil beneficial herbs like yarrow, rosemary, lavender, calendula, self-heal, and motherwort in a pot and then carefully set the pot under a special yoni chair that is open to allow the medicinal herb steam to reach the vagina and surrounding areas (at a safe distance). Many women wrap themselves in a blanket to trap in the steam as they sit for around 20 minutes. It is best to do this once or twice a week and never during menstruation.

Castor Oil Packs: Using castor oil packs may help promote circulation to the uterine area to increase fertility. They are often used when a woman has scarring in the fallopian tubes that prevents implantation or fertilization. They are easy to perform and require castor oil, a towel, and a heat pad. Warm a heat pad to a comfortable temperature and then massage a nickel-sized amount of castor oil into the abdomen where the reproductive organs are. Place a towel over this and then place the heat pad on top of the towel. Lie down with this for twenty minutes. Do this daily or as needed.

Ovulation Testing: There is no better way to get to know your unique cycle than performing regular ovulation tests. Every woman is different in terms of their menstrual cycles and ovulation times. Note that unlike pregnancy tests, which are best taken in the morning, most ovulation tests should be taken in the afternoon for better results. While many women typically ovulate 14 days before menstruation, this is not always the case. Testing regularly can also help you determine whether you are even ovulating at all.

Start taking ovulation tests around one week after your period and keep testing every day or every few days until you get a positive result. Chart these results (there are really helpful phone apps for this). After a few months of charting ovulation, you will have a good idea of when you ovulate. This is great if you are trying to conceive because you will know exactly when you are most fertile so you can have sex at the perfect time to increase your chances of getting pregnant.

Managing Stress: Do what you can to manage stress as you are trying to conceive. Stress works against the body and can be detrimental to those trying to conceive⁵⁶. It interferes with many bodily processes, and fertility is one. Some women find that weekly massage or acupuncture help to relieve stress to aid in conception. Take daily nature walks, get exercise, put time aside for self-care, and take stress-relieving remedies like motherwort, reishi, cordyceps, and turkey tail to help the body manage stress

better.

BIRTH CONTROL: Having a more intimate understanding of how your body works is a great tool for controlling your fertility. Many women are told to go on synthetic hormonal birth control rather than dare to understand when they are fertile and when they are not fertile. It may be easier to take a pill, but knowing how to chart your fertility has no adverse side effects (unlike hormonal birth control). Hormonal birth control has been handed out with little thought as to potential side effects like blood clots, stroke, major weight gain, heart attack, cancers, and there is even a strong link between birth control and autoimmune disease like inflammatory bowel disease and ulcerative colitis⁵⁷.

A woman is only fertile for around 24 hours every month (during ovulation). Sperm live in the vagina for three to five days before they die off. Every man's sperm can be different in terms of how fast it dies off. A good rule of thumb is to assume it will live for the maximum five days if you want to avoid getting pregnant. Avoid intercourse or use protection six days before and three to four days after ovulation. Chart your ovulation to know when you are ovulating. Once you know the pattern your body follows, you will know when to abstain from intercourse or use more protection if you are wanting to avoid pregnancy. This method of birth control is called Natural Family Planning (NFP). It involves no medications, and when followed correctly, works great.

When following the NFP method, there are several options you can use when looking for protection should you choose to have intercourse during your fertile week. While it is best to abstain during this time, some options for protection include condoms (there are latex-free lambskin condoms available for those allergic to latex), cervical caps or diaphragms, and sponges. These are not foolproof methods, so it may be a good idea to use spermicide as well. Some people are allergic to the chemicals in spermicide, so if you are looking for a natural spermicide to use in conjunction with other barrier methods, try diluting five drops of neem oil in a tablespoon of coconut oil and using this. Studies show neem is able to kill all sperm within 20 seconds when administered in a three-milligram dose⁵⁸.

Wild carrot seed is an herbal birth control option for those seeking methods to use in conjunction with NFP. Wild carrot (also called Queen Anne's lace or *Daucus carota*) has been used as a form of birth control for centuries. The seeds of the plants work best when they are harvested in the fall and tinctured. Women take this tincture daily during ovulation and the week after ovulation. Dosages range from ½ to two dropperfuls or seven drops three times daily for three days following intercourse. It is hypothesized that taking this makes the uterus "too slippery" for anything to implant, but this has not been confirmed fully. More research is needed to confirm its contraceptive effects but what little research exists⁵⁹ points to it as a promising natural form of birth control.

3. Libido-Enhancing Remedies and Supplements

Libido is a large factor holding a lot of women back from having a fulfilling sex life. A low libido can be the result of a wide range of factors, including stress, hormonal imbalance, certain medications, fatigue, relationship issues, menopause, and psychological issues. There are many natural libido-boosting remedies outlined in the Menopause Protocol, so see this for a more comprehensive list of herbs and remedies to help improve sex drive.

Try the following herbs for libido (See our Menopause Protocol as well):

Panax ginseng⁶⁰: American ginseng has been used for centuries to boost libido, provide more energy, protect the body from the effects of stress, and enhance stamina. While it has been long-used as a men's sexual health supplement, it can also be used by women who are looking to increase energy and sex drive. It may also help improve symptoms of hormonal imbalance for women going through menopause⁶¹. This may in turn improve libido. Suggested dose: 2 dropperfuls of ginseng tincture up to twice daily, or a 500mg supplement⁶².

Damiana: Damiana is a well-known aphrodisiac that has been utilized for centuries for its aphrodisiac properties, and studies back up its ability to boost libido⁶³. Damiana can increase sex drive, sexual desire, and overall sexual satisfaction for menopausal women. It may also help with vaginal dryness and lubrication to make sex more enjoyable for menopausal women. Suggested dose: Start with 200 mg daily in capsules or as a tincture.

Ginkgo⁶⁴: In a study on menopausal and post-menopausal women, a combination of *Ginkgo biloba* and *Muira puama* was shown to increase libido. Libido can be greatly affected during menopause due to hormone fluctuations, so the fact that these herbs had a positive effect on libido during this phase of life is promising for women looking for natural forms of libido-enhancement. This herbal combination is marketed under the name Herbal vX and should be taken as directed on the bottle.

A decreased sex drive isn't always the result of a physical issue. It can be a psychological or mental issue. One obstacle holding a lot of women back from enjoying a fuller sex life is a lack of self-esteem and self-confidence. Below are strategies you can employ:

Stress Reduction: Stress and anxiety affect libido in a negative way. It is important to manage stress so you can enjoy life and your partner. First, set time aside for self-care. Even a few hours a week set aside to do something you enjoy can work wonders when it comes to relieving stress. Take a vacation if you can. Spend more time outdoors, as this has been proven to lower stress levels and decrease depression⁶⁵. Focus on taking care of your body, such as getting regular massages, pedicures, manicures, hair appointments, or anything that makes you feel better about yourself. In addition, make sure to set at least 20 minutes aside every day for exercise.

you feel better about yourself mentally and physically. It is linked to lowered stress, anxiety, and depression⁶⁶.

In addition to these tips, try taking 2 dropperfuls of adaptogenic mushroom tinctures twice daily (choose 2 of the following: reishi, cordyceps, or turkey tail dual-extracted tinctures), plus nervine herbs to help lower anxiety, stress, and feelings of being overwhelmed:

Lemon Balm: Lemon balm is a wonderful nervine herb with the ability to help with anxiety and depression⁶⁷. It has long been used for its gentle calming effects. Suggested dose: 1-2 dropperfuls of lemon balm tincture as needed once to twice a day.

Motherwort: Motherwort has already been mentioned in this protocol for its positive effects on women's health. In addition, motherwort is an excellent herbal remedy for anxiety and stress⁶⁸ because it can promote feelings of calm and keep you from feeling overwhelmed. It can lower blood pressure and protect the cardiovascular system from the effects of stress as well. Suggested dose: Take one dropperful as needed when you are feeling overwhelmed.

Lavender: Just the scent of lavender lowers stress levels and high blood pressure⁶⁹. Lavender essential oil is an easy way to reap the benefits lavender has to offer. Carry a bottle with you and inhale it straight from the bottle as needed for stress and anxiety management.

Below you will find dietary supplements for sexual health:

Vitamin D: Not only can adequate vitamin D levels help improve your immune function, it may also counter symptoms of depression. While it has been studied for this effect in men, it may also be beneficial for women⁷⁰. Suggested dose: Spend twenty minutes outdoors in the sunlight each day if possible; as a supplement: 1,000 to 4,000 IU daily.

Vitamin B Complex: Taking a daily vitamin B complex may help improve your sex life. Vitamin B6 has been shown to help stabilize fluctuating hormones, especially for menopausal women. This can improve libido and overall sexual health. Vitamin B2 may help reduce vaginal dryness, and vitamin B3 may improve blood flow to the pelvic region to increase pleasure.

If you are trying to conceive, vitamin B complex is associated with fetal brain development and a reduction in birth defects⁷¹. It can even reduce nausea for women experiencing this during pregnancy. Suggested dose: Vitamin B complex supplements vary in dosage, so follow the recommended dosage for the supplement you are taking and do not take more than the recommended amount.

Vitamin E: This vitamin may help improve your sex life, but not in the way you might think. Break open the vitamin E capsule and massage the vitamin E oil on the vagina several times per week, even if you aren't having sex. This helps lubricate tissues and intensify sexual arousal and pleasure. This can be especially helpful for

menopausal and post-menopausal women who deal with vaginal dryness and lack of arousal⁷².

L-Arginine: L-arginine is an amino acid that helps the body make nitric oxide. This molecule helps relax and open blood vessels to help men achieve an erection. It may also be beneficial for women to help with blood flow to the pelvic region. Avoid this supplement if you have heart or cardiovascular issues. One study showed promising results with this supplement when combined with yohimbe and glutamate for menopausal women⁷³. Suggested dose: 2-6 grams daily.

4. Western Medical Approaches

Western treatments

are also employed to help with a variety of women's sexual health needs. Most of these treatments include hormonal replacement therapy and other pharmaceutical drugs. When visiting your doctor, they may run a series of tests to determine the issue.

Expect bloodwork, a urine analysis, and questions regarding your sexual health and sex life. Your doctor or OBGYN may also perform an ultrasound to get a better look at your reproductive organs and look for any abnormalities.

You should see your OBGYN yearly for pap tests and other reproductive health exams to make sure everything is healthy and keep an eye out for cancers, cysts, etc. When visiting your doctor for sexual health concerns, drugs may be prescribed depending on the issue or diagnosis. These include, but are not limited to, the drugs and procedures listed below:

Drugs for Low Libido: Your doctor may prescribe androgen or estrogen therapy depending on your bloodwork. They may also prescribe an antidepressant like Flibanserin or a drug like Bremelanotide for low sex drive.

Drugs and Procedures for Endometriosis: Some doctors prescribe hormonal birth control to help with symptoms of endometriosis, but this doesn't help all the time and is more of a temporary bandage than a way to cure the root of the issue. Some doctors prescribe pain medications and NSAIDs to help with the intense pain many women experience during menstruation when they have endometriosis.

Progestin therapy is also utilized and consists of using this hormone to combat the growth of endometrial tissue outside the uterus. For severe cases of endometriosis, laser surgery can help remove scar tissue causing issues in the areas surrounding the uterus. However, it can be temporary, as each menstrual cycle introduces more scar tissue back to the areas until menopause.

Drugs for PCOS: Doctors often prescribe birth control (a temporary bandage that doesn't address the root cause of the issue), progestin therapy, lifestyle changes such as losing weight, and androgen-blocking drugs. Androgen-blockers help reduce the effects of PCOS, such

as facial hair.

Drugs and Procedures for Fibroids: Some common drugs for fibroids include hormone therapy, GnRH agonists and antagonists, and drugs to reduce estrogen levels. Procedures that involve injections into the arteries that feed the fibroid can shrink them, but they often involve a hospital stay to manage the side-effects. Some surgical approaches also exist, including ablations and myomectomies to remove the fibroids.

Drugs and Procedures for Menstrual Issues: A doctor may perform a laparoscopy to help detect any underlying issues that cause painful and heavy periods. The most common drugs prescribed for painful or heavy periods include hormonal birth control and pain medications. In some cases, surgery to remove fibroids or manage endometriosis may be advised. In extreme cases, a doctor may suggest a hysterectomy.

Drugs and Procedures for Infertility: There is a long list of fertility drugs used to aid in conception. Common drugs include clomid and femara to stimulate ovulation. Alongside these drugs, procedures like intrauterine implantation (IUI) and in vitro fertilization are performed to increase the chances of conception. Your fertility doctor will help determine what steps you need to take to get started.

These procedures can be costly, with IUI being significantly cheaper than in vitro, which requires the egg and sperm to be extracted, made into an embryo, and then reimplanted into the uterus. IUI is less invasive and involves taking the sperm and making it more concentrated, then inserting this into the uterus during ovulation (which is induced pharmaceutically with clomid or other drugs).

Drugs and Procedures for Birth Control: Aside from condoms, barrier methods, spermicides, and natural contraceptive methods, other forms of birth control commonly used today include hormonal birth control such as implants, pills, injections, and male and female sterilization.

There is also IUD birth control, some hormonal and some not. It is important to research anything you put into your body, as everything comes with pros, cons, and side effects. Ask for the inserts that come with these products so you can read the side effects and make an informed decision regarding your sexual health.

Drugs for Infections: A wide range of antibiotics exist to combat various infections like urinary tract infections and certain sexually transmitted infections. Your doctor will choose an antibiotic they feel will work best at killing the bacteria responsible for the infection.

Drugs for Yeast Infections: There are several types of over-the-counter drugs used to treat yeast infections. These are commonly found in stores and are very easy to use. A pharmaceutical option to treat yeast infections is Diflucan. It is an oral pill that can be taken to clear up a yeast infection.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Healthy breakfast Reishi dual-extracted mushroom tincture Vitex Red raspberry leaf tea	Weekly grocery shopping for anti-inflammatory, GF, dairy-free, foods Berberine Turmeric/curcumin Daily exercise Healthy lunch	Yarrow tincture (as needed before and during menstruation for heavy bleeding) Red raspberry leaf tea Vitex	Healthy dinner Turmeric/curcumin Evening primrose oil Reishi dual-extracted mushroom tincture	Aromatherapy for stress and sleep: lavender Yarrow if needed for bleeding
MONDAY	Healthy breakfast Reishi tincture Vitex Red raspberry leaf tea	Berberine Turmeric/curcumin Daily exercise Healthy lunch	Yarrow tincture if needed Red raspberry leaf tea Vitex	Healthy dinner Turmeric/curcumin Evening primrose oil Reishi tincture	Aromatherapy for stress and sleep: lavender Yarrow if needed
TUESDAY	Healthy breakfast Reishi tincture Vitex Red raspberry leaf tea	Berberine Turmeric/curcumin Daily exercise Healthy lunch	Yarrow tincture if needed Red raspberry leaf tea Vitex	Healthy dinner Turmeric/curcumin Evening primrose oil Reishi tincture	Aromatherapy for stress and sleep: lavender Yarrow if needed
WEDNESDAY	Healthy breakfast Reishi tincture Vitex Red raspberry leaf tea	Berberine Turmeric/curcumin Daily exercise Healthy lunch	Yarrow tincture if needed Red raspberry leaf tea Vitex	Healthy dinner Turmeric/curcumin Evening primrose oil Reishi tincture	Aromatherapy for stress and sleep: lavender Yarrow if needed
THURSDAY	Healthy breakfast Reishi tincture Vitex Red raspberry leaf tea	Berberine Turmeric/curcumin Daily exercise Healthy lunch	Yarrow tincture if needed Red raspberry leaf tea Vitex	Healthy dinner Turmeric/curcumin Evening primrose oil Reishi tincture	Aromatherapy for stress and sleep: lavender Yarrow if needed
FRIDAY	Healthy breakfast Reishi tincture Vitex Red raspberry leaf tea	Berberine Turmeric/curcumin Daily exercise Healthy lunch	Yarrow tincture if needed Red raspberry leaf tea Vitex	Healthy dinner Turmeric/curcumin Evening primrose oil Reishi tincture	Aromatherapy for stress and sleep: lavender Yarrow if needed
SATURDAY	Healthy breakfast Reishi tincture Vitex Red raspberry leaf tea	Berberine Turmeric/curcumin Daily exercise Healthy lunch	Yarrow tincture if needed Red raspberry leaf tea Vitex	Healthy dinner Turmeric/curcumin Evening primrose oil Reishi tincture	Aromatherapy for stress and sleep: lavender Yarrow if needed

Note: Add in other herbs for your particular condition from the protocol, as needed.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Healthy breakfast Vitex Reishi dual-extracted mushroom tincture TCM herbal tea	Weekly grocery shopping for low-acid and anti-inflammatory foods Healthy lunch Red raspberry leaf tea or tincture Maca	Ovulation test Daily exercise (at least 20 min.) Reishi dual-extracted mushroom tincture Vitex	Healthy dinner Lemon balm tincture for stress and anxiety Yoni steam Vitex TCM herbal tea	Castor oil pack while lying in bed if ttc Motherwort tincture or tea
MONDAY	Healthy breakfast Vitex Reishi tincture TCM herbal tea	Healthy lunch Red raspberry leaf Maca	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Lemon balm tincture Vitex TCM herbal tea	Castor oil pack while lying in bed if ttc Motherwort tincture or tea
TUESDAY	Healthy breakfast Vitex Reishi tincture TCM herbal tea	Healthy lunch Red raspberry leaf Maca	Exercise (at least 30 min.) Reishi Dual Extraction Garlic supplement Tribulus (If TTC)	Healthy dinner Lemon balm Vitex TCM herbal tea	Castor oil pack while lying in bed if ttc Motherwort tincture or tea
WEDNESDAY	Healthy breakfast Vitex Reishi tincture TCM herbal tea	Healthy lunch Red raspberry leaf Maca	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Lemon balm Vitex TCM herbal tea	Castor oil pack while lying in bed if ttc Motherwort tincture or tea
THURSDAY	Healthy breakfast Vitex Reishi tincture TCM herbal tea	Healthy lunch Red raspberry leaf Maca	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Lemon balm Vitex TCM herbal tea	Castor oil pack while lying in bed if ttc Motherwort tincture or tea
FRIDAY	Healthy breakfast Vitex Reishi tincture TCM herbal tea	Healthy lunch Red raspberry leaf Maca	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Lemon balm Vitex TCM herbal tea	Castor oil pack while lying in bed if ttc Motherwort tincture or tea
SATURDAY	Healthy breakfast Vitex Reishi tincture TCM herbal tea	Healthy lunch Red raspberry leaf Maca	Exercise (at least 30 min.) Garlic supplement Tribulus (If TTC)	Healthy dinner Lemon balm Vitex TCM herbal tea	Castor oil pack while lying in bed if ttc Motherwort tincture or tea

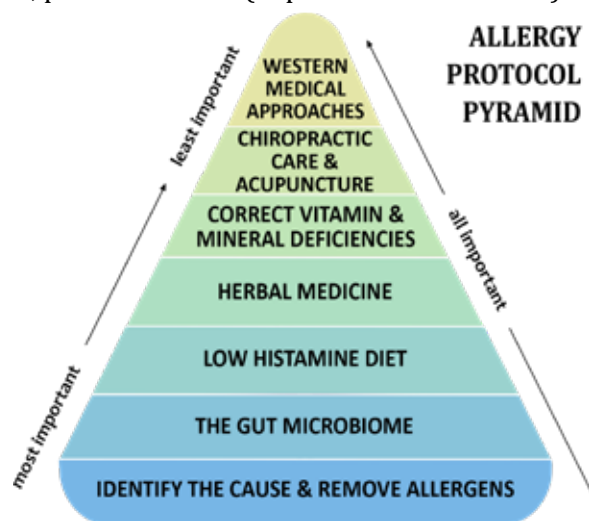
Note 1: Assume this chart represents the week of ovulation.

Note 2: If you are trying to conceive (ttc), this is the week to have intercourse.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Healthy breakfast Adaptogenic dual extracted mushroom tinctures (turkey tail, reishi, and/or cordyceps) Vitamin D supplement	Weekly grocery shopping for healthy, anti-inflammatory foods Healthy lunch Panax ginseng Vitamin B complex & L-arginine Self-care time	Daily exercise (>20 min.) Lemon balm tea or tincture Outdoor time in sun/nature (20 min. for Vit. D and stress reduction) Damiana	Healthy dinner Adaptogenic dual extracted mushroom tinctures (2)	Vitamin E capsule massage oil Motherwort
MONDAY	Healthy breakfast Adaptogenic tinctures (turkey tail, reishi, and/or cordyceps) Vitamin D supplement	Healthy lunch Panax ginseng Vitamin B complex & L-arginine Self-care time	Daily exercise (>20 min.) Lemon balm tea or tincture Outdoor time in sun/nature Damiana	Healthy dinner Mushroom tinctures	Vitamin E capsule massage oil Motherwort
TUESDAY	Healthy breakfast Adaptogenic tinctures (turkey tail, reishi, and/or cordyceps) Vitamin D supplement	Healthy lunch Panax ginseng Vitamin B complex & L-arginine Self-care time	Daily exercise (>20 min.) Lemon balm tea or tincture Outdoor time in sun/nature Damiana	Healthy dinner Mushroom tinctures	Vitamin E capsule massage oil Motherwort
WEDNESDAY	Healthy breakfast Adaptogenic tinctures (turkey tail, reishi, and/or cordyceps) Vitamin D supplement	Healthy lunch Panax ginseng Vitamin B complex & L-arginine Self-care time	Daily exercise (>20 min.) Lemon balm tea or tincture Outdoor time in sun/nature Damiana	Healthy dinner Mushroom tinctures	Vitamin E capsule massage oil Motherwort
THURSDAY	Healthy breakfast Adaptogenic tinctures (turkey tail, reishi, and/or cordyceps) Vitamin D supplement	Healthy lunch Panax ginseng Vitamin B complex & L-arginine Self-care time	Daily exercise (>20 min.) Lemon balm tea or tincture Outdoor time in sun/nature Damiana	Healthy dinner Mushroom tinctures	Vitamin E capsule massage oil Motherwort
FRIDAY	Healthy breakfast Adaptogenic tinctures (turkey tail, reishi, and/or cordyceps) Vitamin D supplement	Healthy lunch Panax ginseng Vitamin B complex & L-arginine Self-care time	Daily exercise (>20 min.) Lemon balm tea or tincture Outdoor time in sun/nature Damiana	Healthy dinner Mushroom tinctures	Vitamin E capsule massage oil Motherwort
SATURDAY	Healthy breakfast Adaptogenic tinctures (turkey tail, reishi, and/or cordyceps) Vitamin D supplement	Healthy lunch Panax ginseng Vitamin B complex & L-arginine Self-care time	Daily exercise (>20 min.) Lemon balm tea or tincture Outdoor time in sun/nature Damiana	Healthy dinner Mushroom tinctures	Vitamin E capsule massage oil Motherwort

Ch. 32: Allergy Protocol

The prevalence of allergies has dramatically increased in developed countries¹. Mild allergic responses include: hives (itchy red spots on the skin), general itchiness, nasal congestion (rhinitis), rashes, scratchy throat (due to post nasal drip), and watery and/or itchy eyes. More severe allergic reactions include: abdominal cramping, pain or tightness in the chest, diarrhea, difficulty swallowing, dizziness (vertigo), fear or anxiety, flushing of the face, nausea or vomiting, heart palpitations, swelling of the face, eyes, or tongue, weakness, wheezing, difficulty breathing, or even loss of consciousness. Typical allergens include: pet dander, insect venom (bites and stings), certain foods (including gluten, peanuts, dairy, and shellfish), certain medications (such as penicillin or aspirin), dyes, certain plants, pollen or molds (in particular black mold).



Allergies can occur as the more common immunoglobulin E (IgE) mediated allergies - when the body mistakes the identified allergen as a threat causing an immune cascade and triggering the production of histamine; or non-IgE cell-mediated reactions - as seen in food protein reactions and intolerances². Allergies can present in many different ways, including³:

- Anaphylaxis – a rapid onset life-threatening reaction
- Respiratory symptoms such as asthma, rhinitis, and sinusitis
- Skin presentations such as pruritis, eczema, dermatitis, and urticaria (hives, rash)
- Gastrointestinal symptoms such as abdominal pain, nausea, vomiting, & diarrhea
- Other symptoms such as swelling of eyes and/or lips, conjunctivitis, and lethargy.

Allergies are often inherited; in fact, a family history is the single most important factor that puts you at risk of developing allergies⁴. They occur when your immune system thinks that a substance, such as pet dander, is a foreign invader. When your body develops antigens against a substance, you become allergic. Antigens that provoke an allergic reaction are called allergens. Repeated exposure may increase the severity of the allergic response.

The immune system protects the body against antigens using cells such as lymphocytes. We have two kinds of lymphocytes (a type of white blood cell) —B cells and T cells. In a normal immune response, when the immune system registers a substance as a threat, B cells produce antibodies and T cells attack foreign substances directly. However, in an allergic response, our immune system misidentifies a harmless substance (an allergen) as a threat.

T cells are responsible for delayed allergic reactions, like contact dermatitis. Immediate allergic reactions are due to B cells. B cells help the body attack allergens by producing IgE antibodies (proteins that bind to and neutralize antigens), which attach to mast cells, the major allergy immune cell, which then release chemicals like histamines into the bloodstream. Histamines are an important signaling molecule but in an allergic reaction they cause inflammation and an allergic cascade, causing symptoms such as itching, sneezing, and watery eyes.

A more detailed explanation for allergic reactions with immediate effects due to B-cell stimulation follows here. B cells produce a class of allergen-specific antibodies called immunoglobulin E (IgE)-blocking antibodies, which mediate the most common type of allergic reaction⁵. Usually, IgE antibodies are only present in small amounts. People with allergies have an excess. Allergic reactions occur when a person develops these “allergic” IgE antibodies, each of which is specific for each type of allergen. The IgE antibodies bind tightly to granulated immune cells - mast cells or basophils - around blood vessels, and in the skin, connective tissue, airways, and GI tract. Mast cells and basophils have receptors that are specific for IgE antibodies⁶. They store messenger substances that trigger inflammatory reactions and symptoms. When the bound IgE recognizes an allergen, these triggers activate and rapidly break down (degranulate) and release histamines, cytokines (TNF-alpha), and proteases (chymase and tryptase), followed by the release of prostaglandins and leukotrienes⁷. Acute reactions are mostly due to mast cells and histamine release. *Thus mast cell stabilization and antihistamines are the basis for many allergy medicines. Leukotrienes act similar to histamines.* Cytokines, specifically interleukin-4, released by basophils, is thought to tell the body to make more IgE, leading to an even more intense reaction⁸.

For example, if you're allergic and exposed to a specific pollen, your body will produce a large amount of allergen-specific IgE antibodies. These IgE antibodies bind to mast cells, located in the skin, gut and airways (places where the body frequently comes into contact with allergens). When an allergen binds to the attached IgE antibodies, the mast cell releases its stored histamine plus other inflammatory messengers. Histamines play an important role in inflammation. The released messengers can cause itching, a runny nose, hives, a narrowing of the airways, wheezing, and swelling.

Antihistamines alone do not work for most allergies. Drugs like antihistamines treat the allergic symptoms, but not the allergy itself. Immunotherapy

helps the body produce less IgE and build long-term tolerance to allergens. In extreme cases of anaphylaxis, patients may require an adrenaline injection.

The most common way to diagnose an IgE-mediated allergy is through a blood test or a skin prick test, which looks for local inflammatory reactions after administration of the trigger allergen. For food allergies, one can begin with an Elimination Diet, as well as a food sensitivity test.

There are many contributing factors to this immune hypersensitivity, including gut microbiome dysbiosis, parasitic infections, family history of allergies, excess body weight, environmental toxins, and poor maternal and infant nutrition⁹.

1. Identify the Cause and Remove Allergens

While not all allergic responses are life threatening, an anaphylactic reaction is a clinical emergency and requires immediate medical attention.¹⁰ Anaphylaxis most commonly presents within an hour of exposure and is characterised by a multi-systemic reaction involving respiratory and circulatory distress. An anaphylactic reaction is most commonly caused by a food allergen but can also be triggered by bee and wasp stings, tick bites, and medications; for 20% of reactions the cause is unknown¹¹.

Allergies can occur as a result of genetic polymorphisms. These are important to identify for the correct 'root-cause' based treatment. A polymorphism of Methylene tetrahydrofolate reductase (MTHFR) occurs commonly in the general population and has been linked to an increased prevalence of allergic asthma and exaggerated immune response (atopy). The MTHFR gene is involved in the methylation cycle, which influences histamine expression and detoxification¹². While general treatment is usually approached in a similar way, understanding altered gene expressions gives insight into alternative or additional treatment options.

Other causes for hypersensitive immune reactions include changes to mast cell regulation such as mast cell activation syndrome (MCAS). MCAS is a hyperresponsive state of mast cells that presents as allergies and causes an increased release of histamine and inflammatory cytokines. MCAS is a genetic condition and can occur from birth or develop in child or adulthood¹³.

2. The Gut Microbiome

Research suggests that the increasing prevalence of allergic diseases has a strong link to changes in the gastrointestinal tract.¹⁴ With 70% of the immune system located in the gut, it's not surprising that alterations of the bacterial microbiome can lead to adverse immune responses such as allergies.

The gut microbiome consists of many different species that live symbiotically and directly influence your health. When these microflora become imbalanced, this leads to gut dysbiosis, which can cause an array of issues, including allergies.

A compromised microbiome reduces the production of short-chain fatty acids (like butyric acid) and lowers the acidity of the gut, allowing pathogenic and invasive microbes to colonize and influence inflammatory and immune responses¹⁵. The gut epithelial barrier is protected and regularly repaired by an optimally functioning microbiome. This prevents the release of antigens in response to allergens that come into contact with the digestive tract – through food or environmental exposure. When the gut barrier is compromised it becomes hypersensitive to certain external stimuli.

Small intestinal bacterial overgrowth (SIBO) is a condition that occurs when there is an overgrowth of bacteria in the small bowel. This condition occurs due to a malfunction of the migrating motor complex within the intestines that is responsible for clearing excess bacteria between meals. SIBO has been linked to allergies and also increases the risk of children developing respiratory allergic symptoms and atopy later in life¹⁶.

Ways to support gut barrier function are vast and situation dependent. Please see the Leaky Gut Protocol for more detailed information after reading the low-histamine diet section below. Ways to support a healthy microbiome and decrease your allergic and inflammatory response include:

- **Eat a diet high in fiber and low in refined sugar.**

In addition, a primarily vegetarian diet increases gut diversity and increases the population of protective bacteria¹⁷. Remove aggravating and symptom-causing factors and foods as there is an attached inflammatory response that disrupts the gut. Leafy green vegetables and starches encourage butyrate production in the gut, which is highly protective to the epithelial barrier.

- **Avoid gluten** as gluten-containing products can further autoimmune and inflammatory cascades and should be avoided or consumed in small amounts.

- **Avoid snacking and aim to have a 12-hour fasting period** overnight. The migrating motor complex in our intestines works to clear excess bacteria from our intestines during times of fasting. Working in 130-minute cycles it also helps prevent the development of SIBO¹⁸.

- **Probiotics** are an important aspect of gut healing as pathogenic bacteria and a dysbiotic environment heavily impact the gut microbiome. Certain strains of probiotics are used to correct dysbiosis and these strains are often found in reduced numbers in allergy-sufferers. *Lactobacillus* spp., and *Faecalibacterium prausnitzii* are involved in short-chain fatty acid production, which aids a healthy gut barrier and re-equilibrates the gut microbiome¹⁹. While probiotic rich foods are beneficial for gut health, these often come in the form of fermented food; these can exacerbate allergy symptoms early in the gut-healing process.

- **Prebiotic foods** act to feed the good bacteria in your gut, primarily the species of bacteria that produce short-chain fatty acids²⁰. These foods include Jerusalem

artichoke, flaxseeds, raw garlic, turkey tail mushroom, onions, apples, and oats.

- **Reduce stress** as stress has been shown to cause gut inflammation and reduce beneficial microbial diversity. Stressors raise the stress hormone cortisol an excessive amount or for a prolonged period of time²¹. These include a high-stress job, a stressful home life, poor or little sleep, high alcohol intake, and imbalanced blood sugar. Removing stressors and implementing stress-reducing techniques is an important part of long-term gut health. See our Anxiety and Stress Protocol.

- **Regular exercise** increases gut diversity and encourages the production of protective short chain fatty acids. Exercising increases blood flow and transit time, improving digestion and alleviating constipation²². Exercise 3-5 times per week when well rested and make sure to meet your body's caloric needs to protect the body from excessive cortisol stimulation during exercise.

- **Avoid excess antibiotics.** Certain medications are harmful to the gut, especially antibiotics. While they are often necessary, do not over-use and take with caution. Antibiotics kill both pathogenic and beneficial bacteria. This causes dysbiosis and your beneficial bacteria can remain altered for over 18 months²³.

- **Sleep 7-8 hours per night.** Sleep is important for our overall health as well as gut health. Our microbiome follows our circadian rhythms. Little sleep or poor sleep can impact our body's circadian rhythm and reduce our microbiome's functionality²⁴. Have your last meal at least an hour before bed, avoid blue lights and screens an hour before sleep, and expose your vision to sunlight for 15 minutes each morning. See our Sleep Protocol if sleep is an issue for you.

Herbal medicine can help combat dysbiosis and aid in the protection of the gut epithelial barrier. Herbs that help repair the intestinal tract and restore healthy gut function include lion's mane, reishi, and turkey tail mushrooms, marshmallow root, and slippery elm. All of these are found in my Balanced Gut Tincture. Suggested dose: Take 2 dropperfuls tincture 3x/day for a month. Then continue 1-2x/day for maintenance. See the Leaky Gut Protocol for more details on each herb. Suggested dose for Slippery Elm and Marshmallow Root as tea: 1-2 tsp. of ground bark/root in water 1-2 times daily²⁵.

Slippery Elm bark and **Marshmallow root** are demulcents, forming a protective coating within the gastrointestinal tract to prevent damage to the epithelial barrier and increase gut healing. **Lion's mane** helps tame gut inflammation and protects the gut from *H. pylori* and gastric ulcers²⁶. **Reishi** calms intestinal inflammation, modulates the immune response, and inhibits *Candida* yeast²⁷. **Turkey Tail** calms inflammation, contains prebiotics, encourages the growth of "good" gut bacteria, and controls *Candida* and small intestinal bacterial overgrowth (SIBO²⁸).

3. Low Histamine Diet

Histamine-rich foods are a trigger for allergies and asthma and contribute to increased airway obstruction and inflammation.²⁹ Histamines usually aren't an issue, as we have enzymes and encoded genes that manage histamine metabolism, but when there is a build-up or an inability to adequately clear them, they become problematic. Histamine intolerance primarily occurs in the gut and can exacerbate histamine-fueled immune conditions including allergies.

A low histamine diet is a short-term approach to reduce the body's exposure to high histamine foods to allow the healing of the gut and the clearance of any histamine build-up during times of immune hypersensitivity. It is important to note that this diet is followed concurrently with a gut-healing protocol as the main enzyme for histamine breakdown is made in the intestines and requires a healthy microbiome³⁰. High histamine foods may be reintroduced once healing has occurred. A low histamine diet helps symptoms and the process of deeper healing.

Foods high in histamine include fermented and aged foods, all alcohol, processed meats and fish (any meat that has been frozen, canned, cured, smoked, or salted), dried fish and fish sauce, aged/mature cheeses, milk, dried fruits, legumes (soy, beans, chickpeas, peanuts), processed foods, sugary drinks, and nightshade vegetables such as eggplant, tomato, white potatoes, and bell peppers³¹.

There are foods that may stimulate the release of endogenous histamine. These should be eaten in moderation but need not be entirely avoided³². These include nuts other than peanuts, citrus fruits, spinach, chocolate, avocados, and strawberries.

While the low histamine diet is effective in reducing the allergic symptoms, asthma, and the inflammatory response, any restrictive diet should ensure essential nutrients are not displaced during food avoidance.

Alongside the low histamine diet, there are foods that help fight inflammation, help the immune system stay regulated, and reduce allergy-symptoms:

Ginger: Ginger downgrades allergic immune responses and reduces airway inflammation. It reduces allergy symptoms such as itchy eyes and a blocked nose³³.

Local Honey and Bee Pollen: For those with seasonal allergies raw local honey allows minute exposure to local pollen. Over time this desensitizes the body to the allergen. Honey also works to reduce IgE-mediated reactions, which reduces the frequency and severity of allergies. Avoid honey, bee pollen, or propolis if you have a bee allergy³⁴.

Beef Liver: Liver and organ meats are packed full of nutrients, especially the immune essentials –Vitamin D and Zinc. Dosing up on these nutrients in allergy time supports the immune system and decreases hypersensitivity.

Pineapple/Bromelain: Pineapple in an enzymatic

powerhouse, mainly due to bromelain, which helps reduce hay fever symptoms. It is a potent anti-inflammatory and reduces the severity of allergy and asthma symptoms³⁵.

4. Herbal Medicine

Herbal medicine has been used for centuries to assist with the treatment of allergies through immune modulation, clearing of mucus from the lungs, reducing inflammation, and improving mucus membrane function and health. When the body is in a state of hypersensitivity, as seen with allergies, immune modulating herbs help reduce the production of antigens against the allergen, regulate the immune response, and reduce the immune-mediated inflammation associated with chronic hypersensitivity.

Cordyceps Mushroom (*Cordyceps militaris*): Cordyceps is a potent anti-inflammatory and immunomodulator. Cordyceps inhibits degranulation in activated mast cells and inhibits the synthesis of cytokines³⁶, it fights chronic inflammation, balances blood sugar, demonstrates great potential for acute lung injury, and supports cardiovascular and liver health. Suggested dose: 2 dropperfuls of a dual-extracted cordyceps tincture 2-3 times daily as needed.

Reishi (*Ganoderma lucidum*): Reishi is an immunomodulator, is anti-inflammatory, supports gut health, and is effective for treating chronic immune issues. Reishi has steroid-like compounds that suppress certain inflammatory compounds in the body (such as prostaglandins) and inhibits histamine release³⁷. One study found that reishi extracts compared favorably with the steroidal anti-inflammatory drug Prednisone®, without the serious side effects³⁸. Reishi is useful in reducing excess histamine.

As an adaptogen, it can help reduce fatigue and brain fog associated with seasonal allergies. Reishi also helps with liver function and blood sugar management. This is important, as it's been shown that it's possible to avoid the onset of seasonal allergies by keeping blood sugar levels balanced. Do note that reishi's steroidal triterpenoids are only found in an alcohol-extraction of reishi. Tea will not extract these compounds, as they are not water-soluble. A dual-extracted reishi tincture is recommended. Reishi works best with daily use. Suggested dose: 2 dropperfuls of dual-extracted tincture twice daily³⁹.

Stinging Nettle (*Urtica dioica*): Stinging nettle is a natural antihistamine. It reduces the inflammation that sparks hay fever symptoms, hinders the release of compounds that trigger allergic reactions, and reduces allergic histamine and inflammatory responses⁴⁰. It blocks histamine receptors, thus stopping the cascade allergic inflammatory reaction, inhibiting mast cell degranulation, and blocking proinflammatory pathways^{41 42 43 44}. Stinging nettle contains the following anti-inflammatory and anti-allergenic compounds: adenine, nicotinamide, synephrine and osthole. A double-blind study found that 57% of the participants reported that their seasonal allergies were significantly reduced with the use of nettle⁴⁵. It also reduces systemic inflammation, especially of the joints. Suggested

dose: 2 dropperfuls stinging nettle tincture 2-3 times per day as needed.

Herbs for allergy symptoms include:

Albizia (*Albizia lebbbeck*): An anti-allergic herb used to treat hay fever, hives and other allergic skin presentations, and asthma. It reduces sneezing and nasal-symptoms in allergies and blocks the production of histamine⁴⁶. Suggested dose: 2-3 dropperfuls tincture twice per day

Goldenseal (*Hydrastis canadensis*): A mucus membrane trophorestorative used for chronic allergy symptoms such as rhinitis and sinus headaches, hay fever, and lower respiratory conditions. It is used to modulate microbiome flora, which reduces the frequency and severity of allergic flare-ups. *Do not use if you have or may have an autoimmune condition.* Suggested dose: 1 dropperful tincture up to three times per day⁴⁷.

Mullein (*Verbascum thapsus*): An excellent herb for respiratory conditions from acute and chronic allergic responses. Mullein reduces mucus formation, calms coughs and wheezing, eases tightness of breathing, encourages the expulsion of phlegm, and improves asthma, allergy, and COPD symptoms. It assists with expelling excess mucus from the lungs while reducing the formation of more. It is also a potent expectorant^{48 49}. Suggested dose: 2 dropperfuls tincture 2-3 times daily as needed.

Chinese Skullcap (*Scutellaria baicalensis*): This herb is used to reduce IgE-mediated allergies such as asthma, atopic skin conditions, and food allergies. Its antihistamine qualities make it effective against a large variety of other allergy-associated symptoms such as urticaria (hives) and hay fever⁵⁰. Suggested dose: 2 dropperfuls 2-3 times per day.

Yarrow (*Achillea millefolium*): Yarrow helps lower blood pressure, relax blood vessels, and improve breathing. Yarrow is a bronchodilator and peripheral vasodilator, which reduces allergy symptoms such as asthma, hay fever, and elevated blood pressure⁵¹. Yarrow also has a natural drying property and soothes swelling in the nasal glands, providing relief for those suffering from allergies. Suggested dose: 1-2 dropperfuls yarrow tincture 2-3 times daily as needed.

Yerba Santa (*Eriodictyon californicum*): Yerba Santa is decongestant and expectorant, alleviating mucus and reducing inflammation⁵². It is my go-to to dry up mucus quickly. Suggested dose: 1 dropperful tincture up to three times per day, as needed for congestion.

5. Correct Vitamin and Mineral Deficiencies

An optimally functioning immune system requires certain levels of nutrients and minerals in the body. Correcting vitamin and mineral deficiencies should be carried out towards the end-stage of a gut healing protocol to ensure proper absorption and assimilation.

Vitamin D3 (*Cholecalciferol*): Vitamin D3 is the active form of vitamin D that we usually obtain from the sun.

Vitamin D directly impacts the immune system by reducing IgE antibody production, reducing immune hypersensitivity through the decrease of neutrophils and eosinophils, and improving mast cell differentiation, which reduces the attack of allergens. Populations with low levels of vitamin D have a higher chance of developing allergy-related illnesses such as asthma and food allergies⁵³. Suggested dose: Get your blood levels tested before supplementation. If deficient: 4000-5000IU per day.

For maintenance: 2000-3000IU per day.

Vitamin C: Vitamin C is an antioxidant widely used in maintaining immune health. Vitamin C is especially useful for allergies due to its role in the differentiation of immune cells, which reduces hypersensitivity to allergens⁵⁴. Suggested dose: 2000-3000mg twice daily, or until you reach bowel tolerance (High doses may cause diarrhea.).

Vitamin E: Vitamin E is found in large quantities in immune cells. Vitamin E is utilised in both cell-mediated and antibody immune reactions, such as those in allergies. It has been used successfully in reducing the severity of allergic asthma and deficiencies often manifest as increased immune-mediated inflammation⁵⁵. Suggested dose: 500-1000IU daily if needed.

Quercetin: Quercetin is an antioxidant that can prevent free radical damage in the body. It is also highly anti-inflammatory and can reduce allergy symptoms. It is a valuable supplement to have on hand during allergy season. It may also reduce histamine response, which is crucial for those with allergies seeking relief⁵⁶. Suggested dose: 500-1,000 milligrams daily.

Zinc: Zinc is essential for proper functioning of the immune system. Deficiencies can manifest as irregular immune reactions such as allergies and autoimmune conditions. Zinc is used in immune cell-signalling, cell maturation, and cytokine function. It is a key antioxidant, anti-inflammatory, and homeostatic controller of the immune system. Up to 20% of the general population are zinc deficient⁵⁷.

Suggested dose is 25 mg once daily. Start at a lower dose range and increase weekly until at the upper dose. Note the usual supplementation dose is 8mg for women and 11mg for men.

6. Chiropractic Care and Acupuncture

Chiropractic Care: Most people may not associate chiropractic care with a reduction in allergies, but many people report improvement with allergies after regular chiropractic

care⁵⁸. This is because chiropractic care opens up nerve pathways and reduces bodily inflammation, which in turn helps with allergies.

Acupuncture: Scientific studies around acupuncture and allergies are mixed, but many people report success. A 2014 study found acupuncture to be effective for allergic rhinitis, asthma, and allergic eczema, primarily due to cytokine profile regulation⁵⁹.

7. Western Medical Approaches

Although naturopathic philosophies focus on finding the root-cause and begin treating from there, there is still a place for western medicine for allergies.

Epinephrine: Allergies can occur as a life-threatening, rapid-onset, medical emergency and western medicine is essential in response to this. Epinephrine (EpiPen) is the first-line treatment for anaphylaxis. It is injected intramuscularly into the mid-outer thigh and should be administered as soon as possible.

Epinephrine works through its vasoconstrictive actions in the stabilisation of respiratory and cardiac changes⁶⁰. Those with anaphylaxis should have an allergy-emergency plan and carry an EpiPen with them at all times.

Antihistamines: For most non-anaphylaxis inducing allergies, except food intolerances, antihistamines are the go-to pharmaceutical. These are taken in doses dependant on the severity of the allergic response, often one dose will help mild allergies but up to four can be taken (depending on the brand)⁶¹. Common antihistamines are Cetirizine, Dimetane, Diphenhydramine, and Loratadine.

Steroids: Steroids, or corticosteroids, are used to treat and prevent allergy-related inflammation and symptoms such as rhinitis and conjunctivitis, and is used topically for skin allergies.

They are taken on a daily basis and can take up to a few weeks to start working. They are an effective way to treat allergies but can cause side effects such as diarrhea, hypertension, peptic ulcers when taken orally, and thinning of the skin when used topically^{62 63}.

Allergen Immunotherapy: Allergen immunotherapy is the nearest western 'cure' for allergies. During therapy a minute dose of the allergen is given via injection or under the tongue in a controlled environment. This desensitizes the body to the allergen without it becoming a medical emergency. Over time, the amount of the allergen is increased until the body no longer reacts to the allergen; the exposure period can take up to 5 years⁶⁴.

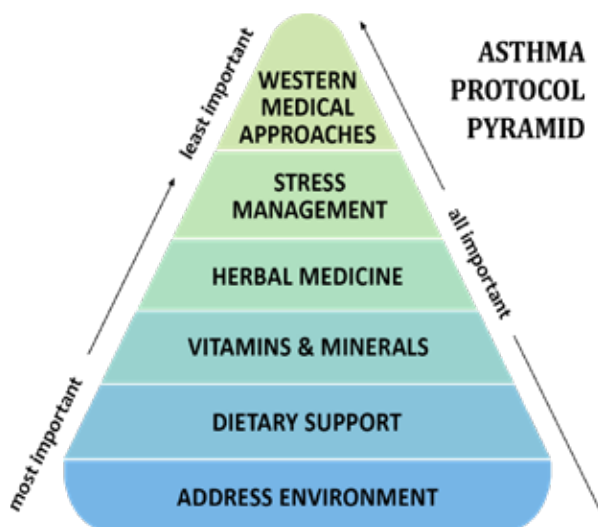
	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Savory vegetarian breakfast with fruit Vit. D, C, E, & Zinc supplements Local raw honey + Immune & Allergy tinctures: Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures 20-min. walk outside in sunshine	Weekly grocery shopping for low histamine foods Probiotics & Quercetin Healthy lunch (biggest meal of the day) Balanced/Leaky gut herbs	Plan and prepare meals for the week Healthy high-protein snack (chopped pineapple with full-fat yogurt)	Vegetarian dinner Immune & Allergy tinctures: Cordyceps, Reishi, Mullein, & Stinging Nettle	Hot shower 5 min. deep breathing Avoid screens and blue light at least one hour before bed Allow 8 hours for sleep (+Sleep herbs if needed)
MONDAY	Prebiotic rich breakfast (Oats & stewed apples with probiotic coconut yogurt) Vit. D, C, E, & Zinc supplements Ginger tea + Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures 20-min. walk outside in sunshine	Probiotics & Quercetin Healthy lunch (biggest meal of the day) Balanced/Leaky gut herbs	Engage in an active hobby for exercise and stress management	Vegetarian dinner Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures	Hot shower 5 min. deep breathing Avoid screens/blue light Allow 8 hours for sleep
TUESDAY	Savory vegetarian breakfast with fruit Vit. D, C, E, & Zinc supplements Local raw honey + Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures 20-min. walk outside in sunshine	Probiotics & Quercetin Healthy lunch Balanced/Leaky gut herbs	Healthy high protein snack	Vegetarian dinner Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures	Hot shower 5 min. deep breathing Avoid screens/blue light Allow 8 hours for sleep
WEDNESDAY	Prebiotic rich breakfast Vit. D, C, E, & Zinc supplements Ginger tea + Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures 20-min. walk outside in sunshine	Probiotics & Quercetin Healthy lunch Balanced/Leaky gut herbs	Engage in an active hobby for exercise and stress management	Vegetarian dinner Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures	Hot shower 5 min. deep breathing Avoid screens/blue light Allow 8 hours for sleep
THURSDAY	Savory vegetarian breakfast with fruit Vit. D, C, E, & Zinc supplements Local raw honey + Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures 20-min. walk outside in sunshine	Probiotics & Quercetin Healthy lunch Balanced/Leaky gut herbs	Healthy high protein snack	Vegetarian dinner Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures	Hot shower 5 min. deep breathing Avoid screens/blue light Allow 8 hours for sleep
FRIDAY	Prebiotic rich breakfast Vit. D, C, E, & Zinc supplements Ginger tea + Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures 20-min. walk outside in sunshine	Probiotics & Quercetin Healthy lunch Balanced/Leaky gut herbs	Healthy high protein snack	Vegetarian dinner Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures	Hot shower 5 min. deep breathing Avoid screens/blue light Allow 8 hours for sleep
SATURDAY	Savory vegetarian breakfast with fruit Vit. D, C, E, & Zinc supplements Local raw honey + Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures 20-min. walk outside in sunshine	Probiotics & Quercetin Healthy lunch Balanced/Leaky gut herbs	Engage in an active hobby for exercise and stress management	Vegetarian dinner Cordyceps, Reishi, Mullein, & Stinging Nettle tinctures	Hot shower 5 min. deep breathing Avoid screens/blue light Allow 8 hours for sleep

* Add in other herbs from protocol if needed for specific symptoms, etc.

Ch. 33: Asthma Protocol

Asthma is a disease that affects the respiratory system. While the combination of factors seems to be unique from person to person, environmental, allergic, and genetic factors can all lead to the development of asthma. Additionally, chronic psychosocial stress in childhood increases the likelihood of being diagnosed with asthma¹.

Inside the body, asthma interferes with three main processes of healthy breathing- airway hyperreactivity, airway constriction and inflammation, and increased mucus production. A high percentage of asthma sufferers have increased levels of immune system cells and antibodies² present in the respiratory tissues, leading to a cycle of inflammation and hypersensitivity. When triggered, this combination reduces available airspace in the bronchi and lungs for oxygen and carbon dioxide exchange to take place.



Asthma is characterized by recurrent symptoms of difficulty breathing, sensation of tightness in the chest, cough, and wheezing on expiration. Breathing can become ineffective, and oxygen levels drop. If left untreated, an asthma “attack” or episode when symptoms come on suddenly can be mild, but it can also be more extreme, resulting in emergency room visits to avoid respiratory failure and even death. The World Health Organization’s last estimate stated in 2019 that asthma affects 216 million people worldwide and was responsible for 455,000 deaths³.

Asthma attacks can be triggered or made worse by: exercise, cold air, dust and dust mites, animal dander and animal hair, mold, air pollution, exhaust, and smoke, pollen, respiratory infections, strong emotions (sadness, shock, laughter).

While the frequency and severity of asthma attacks differs from person to person, asthma is a chronic disease that has long lasting emotional and physical effects and is often associated with other inflammatory and immune related diseases like rhinitis, obesity, and cardiovascular disease⁴.

Diagnosis can be made by symptoms alone but can be accompanied by pulmonary function tests. Peak flow and spirometry tests are commonly performed to assess the volume of air in the lungs, the amount of air that can be forcibly exhaled, and how quickly. This helps aid diagnosis, treatment, and monitoring. Peak flow meter testing, for example, is quick, inexpensive, and available to many patients to utilize on their own to monitor lung function.

When diagnosing asthma, there are different categories depending on severity. How often symptoms occur, if symptoms interrupt sleep or daily activities, lung function test results, and how often medication is needed are all considered when classifying asthma.

1. Address Environment You’ll notice from the list of common asthma triggers below that many are particles and allergens found in the air, both indoors and outdoors. It is important that anyone with asthma become aware of their individual triggers and understand the importance of minimizing their environmental exposure.

Mold: A number of molds and fungi⁵ found outdoors and indoors are known to cause asthma symptoms. Even Candida, a yeast-like fungus that commonly lives on human skin and in the digestive tract, can irritate the immune system enough to cause trigger asthma attacks. Studies worldwide in different populations and climates have documented a connection between mold exposure and the development of asthma, and that mold exposure makes asthma worse. Regarding mold in the home environment, visible growing mold is certainly a concern, but mold does not need to be visible to cause problems⁶. Mold spores and particles released into the air can cause an immune response. There are many asthma studies concentrating on the presence of mold grown from dust⁷ found within the home or mold grown from sampling spores in the air⁸. This means that even homes that don’t appear moldy can cause problems for asthmatics.

Higher household mold burden is associated with poorer asthma control, more symptomatic days, and more unscheduled emergency care visits⁹. When indoor mold is removed from the home, asthma symptoms improve and people use less medication¹⁰.

Expert recommendations¹¹ for how to deal with indoor mold include:

- Keep bathrooms and other mold-prone areas clean.
- Use an air conditioner or dehumidifier that keeps home humidity levels below 50%.
- Locate and repair sources of moisture, like leaks or condensation that facilitate mold growth. Avoid air-drying clothes inside.
- Keep good indoor air ventilation by opening windows and/or using extractor fans.
- Isolate any sources of moisture like bathrooms and the kitchen when cooking.
- Remove existing mold via a mold removal specialist.

While outdoor mold sources are more difficult to control, it is recommended that people with asthma wear a mask or avoid going outside when mold counts are high and to keep your yard free of fallen leaves or rotten plant material.

Air Quality: Air pollution, both outdoors and indoors, is known to make asthma worse and is probably related to asthma development¹². Several large-scale scientific analyses show that communities exposed to higher levels of air pollution tend to have higher incidences of asthma¹³, and there's evidence that this affects children more¹⁴ than adolescents and adults. The air pollution effect in children is so strong that even a mother's exposure to air pollutants while pregnant and exposure within the newborn period¹⁵ can increase a child's risk of developing asthma.

Sources of air pollution vary from place to place and include¹⁶:

- Smoke: from wildfires, volcanoes, coal, and kerosene heating and cooking
- Traffic-related air pollution like car exhaust and ozone
- Volatile organic compounds (VOCs): found in building materials, carpets, paints, aerosol products, cleaning, cosmetics and degreasing products
- Carbon monoxide and carbon dioxide: from wood, fuel, and gas combustion, open fires, wick lamps, and tobacco products
- Lead and radon gas.

When we look at effects of prenatal tobacco smoke exposure specifically, the risk of developing asthma extends far into adulthood¹⁷, with diagnoses being made between ages 31-46. Tobacco smoke as an air pollutant has been studied extensively and the adverse health effects to the smoker and those exposed second-hand are well established. There is disagreement on whether or not being an active smoker can cause asthma, but it is clear that asthmatic smokers have more severe symptoms and require more medication¹⁸ than those who never smoked. They report that their quality of life is worse than non-smokers and they have more asthma-induced hospital visits¹⁹. The opposite effect has also been studied, showing that asthmatics who quit smoking²⁰ have better symptom control and use less medication.

Besides chemical air pollution, indoor and outdoor air can contain organic particles and allergens that commonly exacerbate asthma²¹, like dust, dust mites, cockroach particles, animal dander, pollens, and particles from grasses and weeds. Given the many possible airborne assaults to an asthmatic's health, experts advise taking the following measures in home and work environments to minimize exacerbations from poor indoor air quality²²:

- Reduce indoor humidity.
- Use an air purifier with a HEPA filter.
- Open windows, use extractor fans, and keep your environment well ventilated, but be aware of high pollen and high pollution days.

- Work to quit smoking or avoid smoking tobacco products indoors.
- Dust regularly with something that will trap dust, not disperse it.
- Remove carpets, curtains, and upholstery that can harbor particles, or vacuum and damp-clean often.
- Create pet-free areas. Wash hands or bathe after contact with animals.
- Avoid using scented candles and artificial fragrances.
- Use allergy resistant covers on pillows and mattresses.
- Choose furniture, flooring, and products that are low-VOC, low odor, and non-aerosol. Maintain adequate air flow when using chemical and paints indoors.
- Hire professional extermination services for cockroaches.

Additionally, the following measures are recommended to reduce and manage outdoor air pollution and allergen exposure²³:

- Check pollution forecasts and ratings in your area daily
- Limit time spent outdoors when air quality is poor. Avoid outdoor exercise when outdoor air quality is poor.
- Avoid high-traffic areas and traveling during rush hours. Go out early when you can before pollution levels rise and avoid exercising in high-traffic areas at all times.
- Use close fitting N95 facemasks if you must be outside when air quality is poor.
- Drive with windows closed and use car air filtration systems.

2. Dietary Support One of the known components of asthma is an immune system that is overly sensitive. In addition to maintaining awareness of external environmental asthma triggers, it is important for asthmatics to understand possible food triggers as well as foods that aggravate or support the immune system.

Assess/Avoid Food Allergies: There is a well-documented link between the development of asthma and food allergies, and the relationship holds true in both directions²⁴. Ingested food and food allergies can cause asthma, even life-threatening asthma, and having asthma is a risk factor for developing a life-threatening food allergy. People who suffer from both are more likely to have severe asthmatic episodes²⁵. While this relationship tends to be more common in children and adolescents, it can persist into adulthood²⁶. It is recommended that anyone with asthma be assessed for food allergies, particularly IgE mediated allergies, or immediate onset type.

Low-Histamine Diet: Histamine-rich foods are a trigger for asthma and allergies and contribute to increased airway obstruction and inflammation.²⁷ A low histamine diet is a short-term approach to reduce the body's exposure to high histamine foods to allow gut healing and the clearance of histamine build-up during times of immune

hypersensitivity. This diet is best followed concurrently with a gut-healing protocol (see the Leaky Gut Protocol) as the main enzyme for histamine breakdown is made in the intestines and requires a healthy microbiome.²⁸ High histamine foods may be reintroduced once healing has occurred. A low histamine diet helps symptoms and the process of deeper healing.

Foods high in histamine include fermented and aged foods, all alcohol, processed meats and fish (any meat or fish that has been frozen, canned, cured, smoked, dried, or salted), aged/mature cheeses, milk, dried fruits, legumes (soy, beans, chickpeas, peanuts), processed foods, sugary drinks, and nightshade vegetables such as eggplant, tomato, white potatoes, and bell peppers²⁹. There are also foods that may stimulate the release of endogenous histamine. These should be eaten in moderation but need not be entirely avoided³⁰. These include citrus fruits, spinach, chocolate, avocados, nuts, and strawberries.

Avoid Western Diet Traps: The Western style diet stems from increased availability of food and processed convenience foods, resulting in food choices that are high fat, high protein, high sugar, but also low fruit, low vegetable, and low fiber. Eating this way causes systemic inflammation, as this diet decreases antioxidant protection and decreases healthy, protective fats. Chronic low-grade inflammation is associated with the development of diabetes and obesity, cardiovascular diseases, liver disease, and neurodegenerative diseases³¹, and also affects the lungs and pulmonary system. Airway inflammation specifically, in addition to immune cell activation³², increases after high-fat, low-fiber meals. Immune response in the airway decreases when saturated fat in the diet is decreased³³. These findings suggest that moving away from a high fat, low-fiber diet helps improve asthma.

This is supported by research concentrating on other aspects of the Western Diet and asthma. In children, consuming fast food and sugary drinks³⁴ is associated with increased risk of asthma, whereas diets involving a high fruit and vegetable intake³⁵ results in decreased risk of wheezing and asthmatic episodes, and can reduce immune activity in the respiratory tract in as quickly as 7 days. Results are similar for adults, with asthma being better controlled with fewer episodes³⁶ when adhering to an intake of 5+ servings of vegetables and two servings of fruit daily.

It has been suggested that one reason for these findings is the levels of Vitamin A and carotenoids³⁷ present in fresh fruits and vegetables, offering an antioxidant effect that benefits those with asthma, but there are many interconnected reasons that someone with asthma should consider adding more fruits and vegetables to their diet and decrease their intake of sugary foods and saturated fats.

3. Vitamins and Minerals In addition to healthy dietary choices, nutritional supplementation can help ensure you are getting the nutrients needed to keep asthma attacks

at bay and support pulmonary health.

Vitamin D: Multiple studies looking at the prevalence of asthma have found its occurrence to be higher in people who are Vitamin D deficient, especially for people who have severe or uncontrolled asthma³⁸. Similarly, low levels of Vitamin D in asthmatics result in poor symptom control and increased medication use³⁹.

This could be due to any number of Vitamin D's effects in the body like improved immune function and decreased inflammation. Vitamin D supplementation is associated with fewer exacerbations and improved lung capacity and function⁴⁰. These effects are larger for people who had insufficient levels. Vitamin D supplementation can also help steroid-resistant asthmatics⁴¹ better respond to therapy.

It is difficult for us to make the amount of Vitamin D we need from sun exposure unless we spend a lot of time outdoors with most of our skin exposed. The amount of Vitamin D needed daily depends on your current level. Get a blood test. Vitamin D deficient adults can take from 5,000-10,000iu of Vitamin D per day over a period of months to reach normal levels. Vitamin D supplementation for asthmatics should be long-duration (over 12 months)⁴² to ensure adequate replenishment. Suggested dose: 2,000-5,000iu per day.

Magnesium: Although the exact role magnesium plays in asthma is not completely understood, it does function as an anti-inflammatory agent in addition to its role in inhibiting the effect of calcium to contract smooth muscle, which may contribute to bronchodilation. Cellular levels of magnesium⁴³ are lower in people who have asthma compared to those who do not.

In asthma treatment, magnesium may have several uses. In emergency settings, intravenous or inhaled routes can be used as a last resort for severe, persistent asthma that fails to respond to conventional treatment⁴⁴. Studies on oral magnesium supplementation for asthma show it may improve lung function. In children and adolescents with asthma who use bronchodilators, those also supplementing with 300mg oral magnesium had fewer episodes and used less medication⁴⁵. Similar improvements in lung function are seen in adults taking oral magnesium supplementation⁴⁶. Most studies use magnesium citrate although some use magnesium glycinate. Suggested dose: 300mg daily at night.

Omega-3 Fatty Acids: Omega-3 fatty acids have strong anti-inflammatory effects that help reduce lung and airway inflammation⁴⁷; they also reduce oxidative stress. Higher blood levels of Omega-3s are linked to better symptom control in asthmatics and decreased medication doses⁴⁸. Omega-3 supplements help address the underlying causes of asthma; people with moderately severe asthma⁴⁹ who took an omega-3 supplement saw greater improvements in lung function and decreases in inflammation compared to those taking a placebo. Asthmatics receiving supplementation with 50mg of omega-3s⁵⁰ had greater improvements in lung

function and wheezing than those who received olive oil.

There is some evidence that Omega-3 oil supplementation has no effect on asthma but it what might be most important is the combined effect of increasing Omega-3 intake while simultaneously decreasing Omega-6 fats that are common in the standard Western Diet to obtain a better anti-inflammatory ratio.⁵¹ Suggested dose: 2,000-3,000mg daily.

Nutrient Antioxidants: Reactive oxygen species, or free radicals, are unstable molecules that are a part of normal metabolism, but they can cause damage if left unbalanced by other resources in the body. Asthmatics have higher systemic levels⁵² of oxidative stress, and high oxidative stress is associated with increased asthmatic inflammation and disease severity⁵³. Adding antioxidants helps neutralize reactive free radicals.

Pineapple/Bromelain: Bromelain, found in pineapple cores, is a potent anti-inflammatory, reduces the severity of allergy and asthma symptoms⁵⁴, and helps break up and dissolve mucus that contributes to allergies, lung issues, and congestion⁵⁵. Suggested dose: 80-400 mg bromelain, up to three times daily.

Selenium: Selenium plays many roles in the body that pertain to asthma and its underlying processes; it helps support the immune system, controls inflammation, and helps reduce oxidative stress. Asthmatics have lower selenium levels⁵⁶ than the general population; some sources state that selenium deficiency increases the risk of asthma by 5 times⁵⁷. There is evidence that when dietary selenium⁵⁸ intake is increased, symptoms decrease and that selenium supplementation can lead to improved outcomes⁵⁹ and decreased medication⁶⁰ use. Suggested dose: 100mcg - 200mcg daily.

Zinc: Zinc is similar to selenium in that it is another mineral that has both antioxidant, immune, and anti-allergic roles. Zinc deficiency is also more common amongst asthmatics⁶¹ and zinc supplementation can help decrease symptom severity⁶². Suggested dose: 30mg per day (Talk to your doctor as this is higher than the RDA of 11 mg a day for men and 8 mg for women.).

Vitamin C: Vitamin C is a potent antioxidant. It is involved in the metabolism and recycling of inflammatory chemicals related to bronchoconstriction, like histamine⁶³, and there is strong evidence to support the use of Vitamin C in asthma treatment. Vitamin C supplementation in asthmatics (1,500mg per day)⁶⁴ decreases exercise-induced asthma symptoms and lung function tests. A study of adult smokers documented that increased dietary intake of Vitamin C led to decreased coughing and wheezing⁶⁵. After comparing asthmatic children supplemented with 250mg of Vitamin C to those who weren't, researchers concluded that Vitamin C supplementation might protect against respiratory irritants⁶⁶. This effect has been documented in adults with asthma who were exposed to respiratory irritants, with those given dietary sources of Vitamin

those who didn't supplement. Suggested dose: 1,000mg once or twice a day.

4. Herbal Medicine Herbal medicine should be considered for helping improve asthma symptoms and their underlying causes. There have been numerous studies on the efficacy of these extracts.

English Ivy (*Hedera helix*): English ivy leaves are useful to reduce mucus, swelling, and blockage in airways⁶⁷ and help improve lung function for people prone to bronchial swelling. It is approved for routine use in Germany⁶⁸ due to its antispasmodic, spasmolytic, antimicrobial, anti-inflammatory actions and is considered safe and effective. In children with grade-2 mild asthma, ivy leaf cough syrup (35mg extract twice a day) improved lung capacity and function⁶⁹. In adults and children with acute bronchitis⁷⁰, ivy extract syrup improved all symptoms including shortness of breath, coughing, sputum production, and sleep disturbance and it reduced acute cough symptoms in adults.⁷¹ It is a safe and effective cough treatment for children when taken as a syrup or lozenge for 14 days⁷². Suggested dose: 35mg once or twice a day, or 1-2 dropperfuls tincture (extract) up to twice daily.

Jujube (*Ziziphus jujuba*): Jujubes, or Chinese dates, have an over 3,000-year history of medicinal use⁷³. Jujube fruits contain high levels of cAMP⁷⁴, a molecule that relaxes bronchoconstriction, helps clear mucus, and decreases pro-inflammatory cell presence in the lungs. In vitro analysis shows that jujube both inhibits mucus production⁷⁵ and decreases inflammatory cell presence in lung tissue. Some studies have concentrated on its ability to decrease and regulate immune cell presence and allergic response⁷⁶ in lung tissue and decrease inflammation. Additionally, jujube fruit is thought to act as a bronchodilator through increasing nitric oxide⁷⁷, a naturally occurring compound known to help open the bronchi and blood vessels. Jujube fruits can be taken dried as a snack or infused in a tea. Suggested dose: 80 grams dried fruit or 300ml tea infusion three times a day.

Ginkgo (*Ginkgo biloba*): Animal studies⁷⁸ suggest ginkgo can help inhibit inflammation of the bronchial tissue and in-vitro analysis confirms that ginkgo extracts have an anti-inflammatory influence⁷⁹ on respiratory cells and improves symptoms and bronchial reactivity⁸⁰. In adults, ginkgo extract short-term use⁸¹ protects against exercise-induced asthma and bronchial reactivity, and in adults with steroid-dependent asthma, ginkgo extract tablets⁸² (40mg three times a day) improve several different lung function measurements. Suggested dose: 120mg or 2 ml tincture (~2 dropperfuls) daily.

Mullein (*Verbascum thapsus*): Mullein reduces the formation of mucus, acts as a potent expectorant, helps soothe irritated respiratory passages, eases tightness of breathing, encourages the expulsion of phlegm,⁸³ and calms coughs and wheezing⁸⁴. Mullein has anti-inflammatory and antioxidant properties⁸⁵. It is a valued herb for treating bronchitis, asthma, COPD,

and emphysema. Mullein can be taken in a variety of preparations, like water infusions (teas), syrups, and tinctures. Mullein can also be smoked. I have personally stopped asthma attacks in their tracks by having people inhale mullein smoke and it has been used in this way for centuries⁸⁶. Suggested dose: 1-2 teaspoons dried mullein per cup of water or syrup. Drink up to three times a day. As a tincture: 2 dropperfuls tincture 2-3 times daily as needed.

Cordyceps (*Cordyceps sinensis*) and Reishi (*Ganoderma lucidum*): Cordyceps and reishi have anti-inflammatory and immune modulating components that are helpful for asthma sufferers. In vitro and animal models⁸⁷ have shown components of cordyceps decrease airway hyper-responsiveness, mucus production, and immune cell presence in lung tissue. In adults with moderate to severe asthma, taking a cordyceps concentrate⁸⁸ capsule (1.2g three times a day) along with standard drug therapy resulted in more symptom free days, greater improvement on lung function tests, and greater decrease in inflammation markers. Results for reishi mushroom are similar. Reishi has steroid-like compounds that suppress certain inflammatory compounds in the body (such as prostaglandins) and inhibits histamine release⁸⁹. In vitro and animal models⁹⁰ show reishi extracts decrease immune and inflammatory markers in the lungs.

Medicinal mushrooms are best taken as a double-extracted tincture (extract) to access all of their medicinal components and for ease of administration. Suggested dose: 2 dropperfuls of a dual-extracted cordyceps tincture and a reishi tincture daily.

Stinging Nettle (*Urtica dioica*)⁹¹: Stinging nettle is a natural antihistamine. Stinging nettle reduces inflammation in asthma and allergic responses. Nettle has antioxidant and immune modulating properties⁹², and blocks numerous chemicals involved in an allergic cascade and is useful for allergic conditions like rhinitis⁹³. In animal models, it has been shown to decrease immune cell presence⁹⁴ and reactivity in lung tissue. Suggested dose: 150-300mg, or 2 dropperfuls stinging nettle tincture twice daily⁹⁵.

Yerba Santa (*Eriodictyon californicum*): Yerba Santa is a decongestant and expectorant, alleviating mucus and reducing inflammation⁹⁶. It is my go-to to dry up mucus quickly. It is for symptoms only; it doesn't treat the root cause. Suggested dose: 1 dropperful tincture up to three times per day, as needed for congestion.

5. Stress Management Several lifestyle factors have an impact on asthma and its development. Anxiety and panic disorders⁹⁷ are commonly seen in people with asthma and techniques known to strengthen coping mechanisms, self-regulation, and stress management have a beneficial effect on asthma symptoms⁹⁸. Please see the Anxiety and Stress Protocol for more details.

Yoga and Tai Chi: Yoga and tai chi are popular ways to enhance emotional and physical health. While there are

many different styles and disciplines, generally it is a combination of physical poses, breathing techniques, and meditation. Numerous research studies have investigated the practice of yoga and asthma.

Adolescents and teens taking regular asthma medication who practiced yoga for 45 minutes per day had more improvement in their lung function⁹⁹ tests and quality of life reports at their 3-month follow-up than their counterparts who took medication alone. Similar significant improvements in lung function and quality of life¹⁰⁰ were seen in adults who underwent an 8-week stress management-based yoga program involving yoga poses, meditation, breathing techniques, and group discussion.

A large-scale review analyzing the results from 15 different scientific studies¹⁰¹ on yoga interventions for asthma concluded that yoga can possibly improve quality of life and reduce symptoms in asthmatics. Yoga is accessible in community classes as well as online and through apps for practicing at home.

Meditation: Meditation is the practice of quieting and concentrating the mind, either through sitting in stillness, or completely engaging oneself in an activity thoroughly and mindfully. Meditation¹⁰² may lead to improvements in quality of life, decreased perceived stress, and it may help decrease medication use.

Acupuncture: Acupuncture may be administered in different ways, from traditional Chinese acupuncture with needles to electroacupuncture and laser acupuncture. All of these methods have been used to treat asthma symptoms. For children with asthma¹⁰³, acupuncture improves lung function tests, asthma-specific anxiety, general symptoms, and medication use. It may decrease symptoms and inflammation in adults and adolescents¹⁰⁴ as well when combined with conventional treatments.

Biofeedback: Biofeedback is a mind-body self-regulation technique that teaches people how to modify and adjust their physiology. Biofeedback practice requires the user to engage with specialty equipment that converts physical signals into audiovisual cues. The response from these cues assists the user in understanding their own physiology, accompanied by instruction from a trained biofeedback practitioner¹⁰⁵.

Adults with asthma can decrease their steroid drug dependence through the use of heart rate variability biofeedback¹⁰⁶. While best results are typically seen after multiple practice sessions, other biofeedback studies in adults have shown that even just a single 20-minute biofeedback¹⁰⁷ session improves lung function and helps protect against stress-induced decreases in lung function. Biofeedback seeks to give the user control over processes we typically assume cannot be consciously controlled. It should be considered as a non-invasive and natural option for assisting with breathing and asthma symptoms.

6. Western Medical Approaches Since asthma attacks can be fatal, pharmaceutical management is

usually recommended for asthma. The most common recommended asthma medication is an inhaled drug, usually a beta-agonist bronchodilator like albuterol or salbutamol, which helps open the lungs. This helps reverse the airway constriction associated with asthma attacks and assists with air passage through the lungs.

Bronchodilators are used as needed to address symptoms and are called “rescue” inhalers because they are essential to avoiding medical emergencies. Because this class of drugs acts on the central nervous system to achieve the desired effects, side effects can be rapid heartbeat, tremor, and headaches.

Since bronchodilators are only used to address active symptoms, and asthma severity is classified by how often

they are used, sometimes low dose inhaled daily steroid drugs are advised to help suppress airway inflammation¹⁰⁸ and hyperresponsiveness to reduce episodes. Side effects of daily steroid use are sore throat, oral fungal infection (thrush), and increased susceptibility to infections. There is also some evidence that long-term use can increase the risk of developing osteoporosis and cataracts¹⁰⁹.

Immunotherapy may be recommended for people whose symptoms are persistently triggered by allergens. In addition to these conventional recommendations, several natural and holistic treatment options and considerations that may improve asthma symptoms and help people avoid attacks.

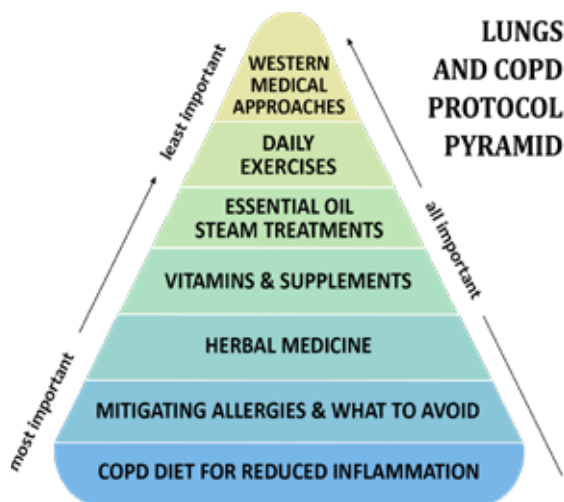
	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Dust + vacuum Bromelain (take on an empty stomach) Vitamins C & D Ivy + Ginkgo tinctures Meditation	Meal planning/Shop for asthma-friendly foods Clean/check air filters Omega-3s, Selenium, & Zinc w/ food Jujube snack and/or tea	Mullein & Stinging Nettle Tinctures Yerba Santa** (only if needed for symptoms) Bromelain (take on an empty stomach)	Cordyceps & Reishi dual-extracted mushroom tinctures*	Indoor humidity + mold check Mullein tincture or tea Magnesium
MONDAY	Bromelain Vitamins C & D Ivy + Ginkgo tinctures Meditation	Omega-3s, Selenium, & Zinc w/ food Jujube snack and/or tea Acupuncture	Mullein & Stinging Nettle Tinctures Yerba Santa** (if needed) Bromelain	Cordyceps & Reishi mushroom tinctures*	Mullein tincture or tea Yoga or tai chi Magnesium
TUESDAY	Bromelain Vitamins C & D Ivy + Ginkgo tinctures Meditation	Omega-3s, Selenium, & Zinc w/ food Jujube snack and/or tea	Mullein & Stinging Nettle Tinctures Yerba Santa** (if needed) Bromelain	Cordyceps & Reishi mushroom tinctures*	Mullein tincture or tea Magnesium
WEDNESDAY	Bromelain Vitamins C & D Ivy + Ginkgo tinctures Meditation	Omega-3s, Selenium, & Zinc w/ food Jujube snack and/or tea Acupuncture	Mullein & Stinging Nettle Tinctures Yerba Santa** (if needed) Bromelain	Cordyceps & Reishi mushroom tinctures*	Mullein tincture or tea Yoga or tai chi Magnesium
THURSDAY	Bromelain Vitamins C & D Ivy + Ginkgo tinctures Meditation	Omega-3s, Selenium, & Zinc w/ food Jujube snack and/or tea	Mullein & Stinging Nettle Tinctures Yerba Santa** (if needed) Bromelain	Cordyceps & Reishi mushroom tinctures*	Indoor humidity check Mullein tincture or tea Magnesium
FRIDAY	Bromelain Vitamins C & D Ivy + Ginkgo tinctures Meditation	Omega-3s, Selenium, & Zinc w/ food Jujube snack and/or tea	Mullein & Stinging Nettle Tinctures Yerba Santa** (if needed) Bromelain	Cordyceps & Reishi mushroom tinctures*	Mullein tincture or tea Yoga or tai chi Magnesium
SATURDAY	Bromelain Vitamins C & D Ivy + Ginkgo tinctures Meditation	Omega-3s, Selenium, & Zinc w/ food Jujube snack and/or tea	Mullein & Stinging Nettle Tinctures Yerba Santa** (if needed) Bromelain	Cordyceps & Reishi mushroom tinctures*	Mullein tincture or tea Magnesium

*If easier for you, you can take these anytime earlier in the day.

** Yerba Santa for symptom relief only. Take when needed.

Ch. 34: Lungs And COPD Protocol

Our lungs have the important task of allowing us to breathe. They help our body utilize oxygen and remove carbon dioxide from the blood. This process is called “gas exchange.” They are the biggest and most important organ of the respiratory system and act as a secondary organ for the cardiovascular system. We have two lungs, located in the chest cavity. Our right lung is comprised of three lobes (upper, middle, and lower) and the slightly smaller left lung has two lobes (upper and lower).



When we take a breath (inhalation), our chest expands and the diaphragm pushes against the lungs, which causes the lungs to expand and allows air to enter the cavity. When air enters the lungs through the trachea, it travels through the trachea into the alveoli (small, balloon-like structures responsible for gas exchange). These are surrounded by blood vessels that help deliver blood for the exchange of oxygen. Once this exchange has occurred, our body must remove the carbon dioxide via exhalation. When we exhale, our diaphragm relaxes and the lungs return to their normal position. Air is pushed out by the lungs and sent back through the trachea and out of the mouth.

Gas exchange happens via the alveoli, which fill with air when we inhale. Alveoli are surrounded by capillaries and when blood is pumped by the heart through the pulmonary vein, deoxygenated blood is sent to these capillaries to allow red blood cells to pick up oxygen. When the blood has this oxygen, it returns back to the heart and is pushed back into the body by arteries.

Our lungs are protected from injury by the rib cage. They have other defense mechanisms, such as cilia that move back and forth to remove germs and mucus from airways and white blood cells that destroy viruses and bacteria when they attempt to enter the body and cause harm. Sometimes these defense mechanisms fail and the lungs are harmed. One common lung disease is chronic obstructive pulmonary disease (COPD). **COPD** is a chronic, inflammatory progressive lung disease that causes obstructed airflow from the lungs. This exaggerated inflammation leads to oxidative stress, emphysema,

airway fibrosis, and mucus hypersecretion¹.

COPD Symptoms: Symptoms of COPD include difficulty breathing, coughing, mucus production, wheezing, shortness of breath (especially during physical activity), chest tightness, frequent respiratory infections, lack of energy, unintended weight loss (this usually happens in later stages), and swelling in the ankles and feet.

Causes and Risk Factors: Most cases of COPD are caused by long-term exposure to irritants like cigarette smoke. Genetics, pollutants, diet, obesity, and a sedentary lifestyle all contribute to COPD.

Other Risks: Those with COPD are at risk for developing lung cancer, heart disease, and other health conditions. Conditions that contribute to COPD include emphysema and chronic bronchitis. They usually occur together and vary in severity for each person. Chronic bronchitis is inflammation of the lining of the bronchial tubes that carry air to and from the alveoli of the lungs. This is characterized by a daily cough and mucus production. Emphysema involves the destruction of the alveoli due to long-term exposure to irritants like cigarette smoke and/or other gases and particulates.

While COPD varies in severity, it can be managed with a combination of holistic treatments.

1. COPD Diet for Reduced Inflammation

In order to treat COPD you must target the underlying cause: chronic inflammation². Diet plays a significant role. A standard Western diet, rich in refined foods, meat, sugar, and saturated fat has a negative effect on lung function and COPD risk. According to a 2019 study³, “foods and nutrients, especially those nutraceuticals endowed with antioxidant and anti-inflammatory properties and when consumed in combinations in the form of balanced dietary patterns, are associated with better pulmonary function, less lung function decline, and reduced risk of COPD.”

What to Eat and What to Avoid: Eat a diet that includes a “high consumption of fresh fruits, vegetables, wholegrains, plant oils and fish, low intake of alcohol, and avoidance of processed, refined, high-saturated fat foods, sweets, cured/red meats, and sugar-containing beverages”⁴. When preparing meals, choose fresh, whole foods that are not pre-packaged. Cook with olive oil instead of canola, vegetable, or corn oil. Be wary with foods that commonly trigger allergies or contain compounds that the body may react to. These include foods containing dairy and gluten, as these often trigger a worsening of allergies, sensitivities, and inflammation. If you are unsure what affects you, try an Elimination Diet.

This recent study also notes that the magnitude of effect of diet on lung function is comparable to that of chronic smoking⁵. This is an alarming revelation. For this reason, it is strongly advised that those with COPD or lung issues stick to an anti-inflammatory diet that follows the above advice. Learn more about this diet in the Anti-Inflammatory Diet section.

Other Issues: An unhealthy diet can trigger issues like increased intestinal permeability (leaky gut), an

unbalanced microbiome, and/or *H. pylori* bacteria in the stomach and intestines. These can worsen inflammation in the body and trigger a worsening of lung conditions like COPD. To learn how to control these, please see the Leaky Gut Protocol, the Candida Protocol and the Ulcer Protocol.

Weight: A healthy diet may help you lose weight. Being overweight exacerbates COPD as it impacts breathing and causes increased inflammation and pro-inflammatory cytokines in the body. Being overweight makes it harder to walk, complete daily activities, and exercise without getting out of breath. If you are overweight, start by lowering your caloric intake and attempt to exercise daily for at least 20 minutes to burn calories and build stamina. Read more on ways to exercise in the “Daily Exercises” section.

2. Mitigating Allergies and What to Avoid

If you are living with COPD and also suffer from allergies, this can severely impact lung health. Allergies occur when the body encounters foreign invaders (like pollen) and send histamines to fight off the invader. The result is additional inflammation in the sinuses and respiratory passages, as well as excess mucus. For someone who already suffers from lung issues, this additional mucus and inflammation compounds COPD. Do what you can to mitigate allergies and control inflammation. Please see the Allergy Protocol if this is an issue for you. A few basic tips are below.

Local Raw Honey: For those with seasonal allergies raw local honey allows minute exposure to local pollen. Over time this desensitizes the body to the allergen. Take a small amount (one to two teaspoons) of local raw honey daily⁶.

Stinging Nettle (*Urtica dioica*): Stinging nettle contains flavonoids that help suppress histamine production⁷. A double-blind study found that 57% of the participants reported that their seasonal allergies were significantly reduced with the use of nettle⁸. It also reduces systemic inflammation. Suggested dose: 2 dropperfuls stinging nettle tincture 2- 3 times per day as needed.

Avoid: In addition to controlling allergies, you should eliminate other triggers from your lifestyle to enhance lung and respiratory health and prevent future issues. First, stop smoking and vaping. If you are consistently exposed to secondhand smoke, take steps to avoid this. Do your best to stay active each day and try to avoid a sedentary lifestyle, which contributes to weight gain and a decline in overall health. Finally, do what you can to mitigate stress in your life, as chronic stress and anxiety can worsen lung and respiratory issues. See the Anxiety and Stress Protocol for helpful tips on how to support a stress-free lifestyle.

3. Herbal Medicine In addition to herbs and supplements listed in our Allergy Protocol, below are herbs and supplements that help nourish the lungs and respiratory system.

Mullein (*Verbascum thapsus*): An excellent

herb for respiratory conditions, mullein reduces mucus formation, calms coughs and wheezing, eases tightness of breathing, encourages the expulsion of phlegm, and improves asthma, allergy, and COPD symptoms. It assists with expelling excess mucus from the lungs while reducing the formation of more⁹. It is also a potent expectorant^{10 11}. Suggested dose: Take 2 dropperfuls tincture 2-3 times daily as needed for congestion and to open airways.

Elecampane (*Inula helenium*): Elecampane root helps support lung and respiratory health by removing stubborn phlegm from the lungs. Its antimicrobial and anti-inflammatory properties¹² make it useful for lung and bronchial infections. Suggested dose: 1-2 dropperfuls tincture up to three times daily.

Adaptogens are helpful for overall health, and are especially useful for those with autoimmune-related lung issues or inflammatory conditions as they are immunomodulatory, antioxidant, and anti-inflammatory^{13 14}. Adaptogens help protect the body from the effects of stress. Some excellent adaptogens include reishi, cordyceps, and Panax ginseng. Suggested dose: Take two dropperfuls of reishi and cordyceps dual extracted tinctures and/or Panax ginseng tincture twice daily.

Cordyceps (*Cordyceps militaris* and *C. sinensis*): Cordyceps is a potent anti-inflammatory and immunomodulator. Cordyceps fights chronic inflammation, inhibits degranulation in activated mast cells, inhibits the synthesis of cytokines¹⁵, balances blood sugar, demonstrates great potential for acute lung injury¹⁶, and supports cardiovascular and liver health. Both species of Cordyceps are clinically effective for chronic bronchitis^{17 18}. Suggested dose: 2 dropperfuls of a dual-extracted cordyceps tincture 2-3 times daily as needed.

Reishi (*Ganoderma lucidum*): Reishi has steroid-like compounds that suppress certain inflammatory compounds in the body¹⁹ (such as prostaglandins) and inhibits histamine release²⁰. Reishi extract compares favorably with the steroidal anti-inflammatory drug Prednisone®, without the serious side effects²¹. As an adaptogen, it can help reduce fatigue and brain fog. Reishi is an immunomodulator, anti-inflammatory, supports gut health, and is effective for chronic immune issues. Reishi also helps with liver function and blood sugar management. Do note that reishi's steroidal triterpenoids are only found in an alcohol-extraction of reishi. Tea will not extract these compounds, as they are not water-soluble. A dual-extracted reishi tincture is recommended. Reishi works best with daily use. Suggested dose: 2 dropperfuls of dual-extracted tincture twice daily²².

Panax ginseng: *Panax ginseng* and its active compounds, ginsenosides and their metabolites, interact with multiple body processes and inhibit both inflammation and oxidative stress. They inhibit kinase phosphorylation and pro-inflammatory mediators in COPD. They protect against oxidative stress by increasing anti-oxidative enzymes and reducing the production of oxidants²³.

These changes can be seen in pulmonary function tests²⁴. Suggested dose: 2 dropperfuls tincture before noon.

Anise Hyssop (*Agastache foeniculum*): During the time of the year when smoke and wildfires are burning, you can support lung health with anise hyssop²⁶. It is also antibacterial²⁷. Suggested dose: two dropperfuls of tincture 2-3 times daily.

English Ivy (*Hedera helix*): English ivy leaves are useful to reduce swelling and blockage in the airways²⁸, help improve lung function for people prone to bronchial swelling, and loosen mucus in the lungs and airways. Suggested dose: Take 1-2 dropperfuls tincture up to twice daily. It works well when blended with raw honey to soothe a cough or sore throat.

Indian Tobacco (*Lobelia inflata*): *Lobelia inflata*'s active compound is the alkaloid lobeline, known for stimulating breathing, supporting the cough reflex, and improving vascular tone²⁹. *Lobelia inflata* also helps relax airways and remove excess mucus. *Lobelia inflata* clears lungs better than other expectorants when a cough is productive³⁰. Take this plant carefully, as too much can induce vomiting. Suggested dose: 100-400 mg as needed.

Lungwort Lichen (*Lobaria pulmonaria*): This antibacterial lichen can soothe respiratory inflammation³¹ and clear the respiratory system of irritants. It is great for supporting the respiratory system when you are exposed to irritants like smoke and helps clear bronchial infections. Pair it with mullein for best results. Suggested dose: two dropperfuls of lungwort lichen tincture up to three times daily.

Red Sage (*Salvia miltiorrhiza*): Sage is a natural cough suppressant and antispasmodic. Sage can help quiet a spasmodic cough and relieve sore and irritated airways and mucus membranes. Sage is loaded with powerful polyphenols³² that help reduce inflammation and eliminate free radicals. Make tincture with sage leaves and add raw honey to make a homemade cough syrup. Suggested dose: Take 5-10 milliliters of sage tincture blended with a teaspoon of raw honey as needed for a cough.

Usnea: Usnea is known for its antimicrobial and antibiotic³³ properties, and is also useful for reducing inflammation, especially in the lungs and airways. Due to its ability to kill harmful bacteria while also reducing inflammation, usnea lichen can be a formidable opponent against bronchial and respiratory infections. Suggested dose: Take two dropperfuls of usnea dual extraction up to three times daily until the infection clears (~10 days).

Yarrow (*Achillea millefolium*): Yarrow helps lower blood pressure, relax blood vessels, and improve breathing. Yarrow is a bronchodilator and peripheral vasodilator, reducing allergy symptoms such as asthma, hay fever, and elevated blood pressure³⁴ and reducing inflammation³⁵ in the airways. Yarrow has a natural drying property and soothes swelling in the nasal glands. Yarrow is a strong antibacterial, so it can help kill bacteria that contribute

to infections. Suggested dose: Use as a tea or tincture. For tea: Infuse 1-2 tsp. dried yarrow leaves and/or flowers for 15 minutes. Drink 3 cups daily for infection. As a tincture, take two dropperfuls of yarrow tincture three times daily until the infection clears.

HERBS FOR UPPER RESPIRATORY ILLNESS:

For those who already have lung or respiratory issues like asthma and COPD, getting an upper respiratory virus can be a serious problem. Since these types of viruses attack the upper respiratory tract, additional congestion and restriction of airways can result. Watch for signs of a virus, such as fever, chills, congestion, body aches, and feeling "off." If you suspect you are coming down with a virus, start taking additional vitamin C to support your immune system. You should also consider taking one of the remedies below to prevent viral replication to help mitigate problems before they start. Herbs from the above section that are helpful with infections are Cordyceps, reishi, lungwort lichen, usnea, and yarrow. Please see the Bacterial Infection Protocol or the Flu, Viral, and Immune System Protocol, depending on the type of respiratory illness you have.

Elderberry: Elderberry helps fight and shortens the duration of viral illness^{36 37}. It is also a good source of quercetin, helping open airways and relieve congestion. It is a safe option for treating viral respiratory illness and does not overstimulate the immune system or create a 'cytokine storm.'^{38 39} It reduces inflammation in the airways and lungs. Suggested dose: At the first sign of illness, 15ml of elderberry syrup four times daily or 3 dropperfuls of elderberry tincture, 3 times daily.

Japanese Honeysuckle (*Lonicera japonica*): Japanese honeysuckle inhibits viral replication⁴⁰. It is great for viruses like influenza and respiratory viruses. Suggested dose: Take 1-2 dropperfuls Japanese honeysuckle tincture up to three times daily at the first sign of viral illness.

Sweet Wormwood (*Artemisia annua*): Sweet wormwood is specifically helpful for upper respiratory viruses. It works by inhibiting viral replication⁴¹. Take at the first sign of illness for best results. Suggested dose: Take 2 dropperfuls of sweet wormwood tincture three times daily for one week or until symptoms stop. **Note:** Do not confuse sweet wormwood (*Artemisia annua*) with wormwood (*Artemisia absinthium*).

4. Vitamins and Supplements

Vitamin C: Vitamin C is a necessary antioxidant for immune health⁴². It can be obtained through diet, but is poorly absorbed. For additional immune support, you can take vitamin C supplements. Suggested dose: Take 1,000 to 3,000 mg (1-3g) of vitamin C daily to bowel tolerance. Up your dose to 8g (8000mg) daily if you are starting to get sick with a virus or infection⁴³.

Vitamin D: Vitamin D is essential to many aspects of health, including lung health⁴⁴. Many people with COPD have low Vitamin D levels. Get your levels tested and supplement if needed, especially in the winter

months. Suggested dose: 1000-2000 IU daily.

Omega-3s: Omega-3 fatty acids have strong anti-inflammatory effects that help reduce lung and airway inflammation⁴⁵. People with COPD experience a narrowing of the airways, making it difficult to move air in and out of the lungs. Many with COPD find omega-3 supplementation helpful. Suggested dose: 250-500 mg daily.

NAC (N-acetylcysteine): NAC is used by the body to build antioxidants. NAC may help protect the organs (including the lungs⁴⁶) and improve detoxification pathways. It also has anti-inflammatory effects, helpful for those with inflammatory lung issues and allergies, and helps clear mucus. It does not stop COPD flares but it helps reduce the number of flare-ups. Suggested daily dose: 600-1,200 mg daily.

Bromelain: This compound, found in pineapple, is a potent anti-inflammatory, reduces the severity of allergy and asthma symptoms⁴⁷, and helps break up and dissolve mucus that contributes to allergies, lung issues, and congestion⁴⁸. Bromelain is also known for its ability to quell a stubborn cough. Note that that bromelain is primarily found in the core of the pineapple, which most people do not consume. Best used as a supplement. You can find supplements that contain both bromelain and quercetin so you can get the benefits of both together. **Quercetin** is a supplement used for allergies that also helps overall lung health. Suggested dose: 80-400 mg bromelain, up to three times daily.

5. Essential Oil Steam Treatments

Essential oils can be a powerful treatment for respiratory issues. These plant isolates are highly concentrated. As such, essential oils can be volatile and should be used with caution on the skin. Always dilute an essential oil in a carrier oil (like olive oil) before applying to skin. Adding a few drops of an essential oil to boiling water on the stove and inhaling this steam is a great way to support the respiratory system and clear up mucus.

Fill a small pot with water and bring to a boil. Add five to ten drops of one or two of the essential oils listed below and place your head over the steam (after turning off the burner). Place a towel over your head to trap in more steam as you inhale. Do this for ten to fifteen minutes daily or as needed for relief from congestion and blocked airways.

Best Essential Oils for Respiratory Health: Essential oils to use in a steam treatment include eucalyptus⁴⁹, peppermint, rosemary⁵⁰, frankincense, oregano⁵¹, thyme, tea tree⁵², cedarwood, pine, fir needle, ginger, orange, and lemon. These essential oils are known for their mucus-clearing properties and their ability to reduce inflammation in the respiratory tract. For children under 10, it is best to avoid oils that contain the constituent 1,8 cineole. These include peppermint, eucalyptus and rosemary. Use caution with oregano and thyme

around children, as they are considered “hot” oils and can burn the skin if used incorrectly.

Cool Ultrasonic Diffusion Option: If you don't want to do a hot steam treatment with essential oils, you can use a cool mist diffuser to disperse the oils into the air. There are many options for cool mist essential oil diffusers on the market. Fill it with the recommended amount of water and add five to eight drops of essential oil. Turn on the diffuser and sit nearby to inhale.

6. Daily Exercises There are two ways you can exercise to improve lung health: physical exercises and breathing exercises.

Daily activity/exercise is imperative to stay in physical shape. The more weight you have on your body, the harder your body has to work to do everyday activities, making it harder on your respiratory system. Studies show a direct correlation between daily exercise and positive respiratory health⁵³. Try exercising every day for at least 20 minutes and follow the dietary advice in the first portion of this protocol for weight management.

Daily breathing exercises help strengthen your lungs, maximize lung capacity, and help with symptoms of COPD and other respiratory issues. Think of your lungs like a door with a spring. This door opens and shuts on its own. However, for those with lung issues like asthma and COPD, this door can lose its ‘springiness’⁵⁴. The result is more stale air trapped in the lungs. Breathing exercises can help get rid of this stale air and leave more room for the diaphragm to contract to allow for fresh air to enter. Below are simple exercises to try daily:

Belly Breathing: This exercise starts with you taking a breath in through your nose. Be mindful of how your belly fills up with air as you do this. Place your hands gently on your stomach so you can feel the belly rise and fall. Breathe out through your mouth and relax the neck and shoulders as you work to train the diaphragm to fill and empty the lungs. Breathe like this mindfully for ten to fifteen minutes a day.

Breathing through Pursed Lips: This exercise involves training the body to reduce the number of breaths you take. It can also keep the airways open longer, which is great for those with respiratory issues. Pursed lip breathing allows more air to flow in and out of the lungs. To perform this breathing exercise, breathing in through your nose and breathe out through pursed lips, making sure that you breathe out twice as long through pursed lips. Try to perform this exercise for ten to fifteen minutes daily.

7. Western Medical Approaches Diagnosing COPD often requires a variety of tests. These may include a lung function test to measure the amount of air you can inhale and exhale and whether the lungs are delivering enough oxygen to the blood. This test involves blowing into a large tube connected to a machine that measures how much air the lungs can hold and how fast you can blow this air out

of the lungs. Tests like chest x-rays, CT scans, bloodwork, and arterial blood gas analysis may also be performed.

Western treatment for COPD may involve medications like bronchodilators and inhaled steroids. These are medications that relax airway muscles so you can breathe easier. These are also used by those with asthma. There are two types of bronchodilators: short-acting bronchodilators like albuterol and long-acting bronchodilators like arformoterol. Inhaled steroids can work to reduce airway inflammation and reduce exacerbations in airway obstruction. These include medications like fluticasone and budesonide. There are also combination inhalers like fluticasone and vilanterol that combine inhaled steroids with a bronchodilator. Other medications that may be prescribed include oral steroids, phosphodiesterase-4 inhibitors, theophylline, and antibiotics.

Lung therapies may be utilized for those with respiratory conditions like COPD. These include oxygen therapy, a therapy that delivers oxygen to the lungs via a portable unit. A doctor may allow someone to use this as needed or at all times, depending on the severity of the condition.

In some cases, surgery may be needed. Surgical procedures for COPD include lung volume reduction surgery, which removes damaged lung tissue from the upper lungs to create more space for the healthy tissue to expand. Another procedure is endoscopic lung volume reduction. This is much less invasive and involves the placement of a valve in the lung to allow the damaged lobe to shrink so the healthier areas have room to function and expand. A bullectomy is another surgical option. This involves a doctor removing bullae from the lungs to improve air flow. In serious cases, a lung transplant may be needed.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Healthy breakfast Vit. C, D, Omega-3s, NAC, & Bromelain w/ Quercetin after breakfast Dose1: Cordyceps, Reishi, & Ginseng tinctures	Grocery shopping for anti-inflammatory foods Healthy lunch Breathing exercises (15 min.) Mullein or Bronchial blend tincture**	Physical exercise (20 min.)	Healthy dinner Breathing exercises (15 min.) Dose 2: Cordyceps & Reishi dual-extracted Mushroom tinctures	Essential oil steam treatment Mullein and/or lungwort lichen tincture (Bronchial blend tincture)**
MONDAY	Healthy breakfast Vit. C, D, Omega-3s, NAC, & Bromelain w/ Quercetin Dose1: Cordyceps, Reishi, & Ginseng tinctures	Weekly chiropractic care Healthy lunch Breathing exercises (15 min.) Mullein or Bronchial blend tincture**	Physical exercise (20 min.)	Healthy dinner Breathing exercises Dose 2: Cordyceps & Reishi tinctures	Essential oil steam treatment Bronchial blend tincture (mullein and lungwort lichen)**
TUESDAY	Healthy breakfast Vit. C, D, Omega-3s, NAC, & Bromelain w/ Quercetin Dose1: Cordyceps, Reishi, & Ginseng tinctures	Healthy lunch Breathing exercises (15 min.) Mullein or Bronchial blend tincture**	Physical exercise (20 min.)	Healthy dinner Breathing exercises Dose 2: Cordyceps & Reishi tinctures	Essential oil steam treatment Bronchial blend tincture
WEDNESDAY	Healthy breakfast Vit. C, D, Omega-3s, NAC, & Bromelain w/ Quercetin Dose1: Cordyceps, Reishi, & Ginseng tinctures	Healthy lunch Breathing exercises (15 min.) Mullein or Bronchial blend tincture**	Physical exercise (20 min.)	Healthy dinner Breathing exercises Dose 2: Cordyceps & Reishi tinctures	Essential oil steam treatment Bronchial blend tincture
THURSDAY	Healthy breakfast Vit. C, D, Omega-3s, NAC, & Bromelain w/ Quercetin Dose1: Cordyceps, Reishi, & Ginseng tinctures	Healthy lunch Breathing exercises (15 min.) Mullein or Bronchial blend tincture**	Physical exercise (20 min.)	Healthy dinner Breathing exercises Dose 2: Cordyceps & Reishi tinctures	Essential oil steam treatment Bronchial blend tincture
FRIDAY	Healthy breakfast Vit. C, D, Omega-3s, NAC, & Bromelain w/ Quercetin Dose1: Cordyceps, Reishi, & Ginseng tinctures	Healthy lunch Breathing exercises (15 min.) Mullein or Bronchial blend tincture**	Physical exercise (20 min.)	Healthy dinner Breathing exercises Dose 2: Cordyceps & Reishi tinctures	Essential oil steam treatment Bronchial blend tincture
SATURDAY	Healthy breakfast Vit. C, D, Omega-3s, NAC, & Bromelain w/ Quercetin Dose1: Cordyceps, Reishi, & Ginseng tinctures	Healthy lunch Breathing exercises (15 min.) Mullein or Bronchial blend tincture**	Physical exercise (20 min.)	Healthy dinner Breathing exercises Dose 2: Cordyceps & Reishi tinctures	Essential oil steam treatment Bronchial blend tincture

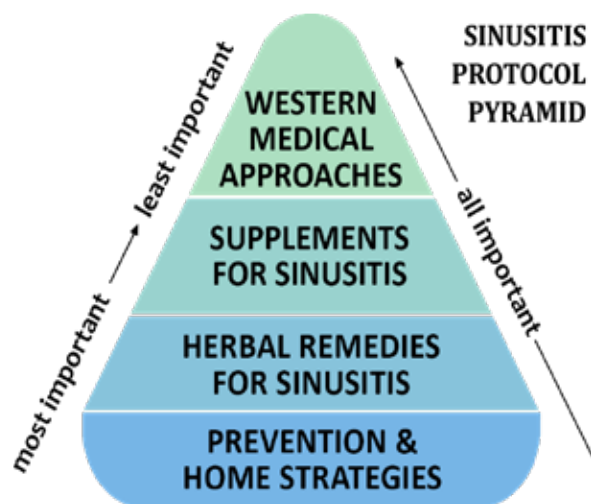
*Use remedies for allergies (stinging nettle) and respiratory illness as needed (see protocol).

** My Bronchial Blend contains both mullein and lungwort lichen. You may also use them separately.

Ch. 35: Sinusitis Protocol

Sinusitis is sometimes referred to as rhinosinusitis. It is characterized by inflammation of the mucous membranes that line the sinuses. When these tissues are inflamed, symptoms such as excess nasal mucus, post-nasal drip, facial pain and pressure, fever, headaches, a plugged nose, sore throat, loss of smell, bad breath, and a cough may develop. If the condition persists for four weeks, it is called acute sinusitis. Chronic sinusitis lasts over 12 weeks.

There are many causes of sinusitis. Some of the most common triggers are allergies or an infection (viral or bacterial). Those who suffer from chronic sinusitis often have another condition that triggers it, such as asthma, cystic fibrosis, or a weakened immune system. Occasionally, structural issues within the nose are to blame for sinusitis (these include sinus and nasal polyps). If you live in an area with poor air quality, this can also trigger sinusitis. While it may not always be possible to prevent sinusitis, you can lessen your chances of fighting this condition by avoiding smoking (and secondhand smoke) and washing your hands often.



If you suspect you have a sinus infection, start by assessing your symptoms. Do you have severe facial pain and pressure? Do you have chronic headaches? Has your fever lasted longer than 3-4 days? Is there blood in the mucus you are producing? If you go more than ten days without getting better, it may be time to seek medical treatment. However, most infections go away on their own within a week.

In the United States, around ten to 30 percent of the population is affected by sinusitis every year, and treating this problem costs around 11 billion dollars each year¹. The problem is that many people are taking antibiotics for sinusitis that is caused by a virus. Treating a viral infection with antibiotics is often ineffective and unnecessary. However, it can be difficult to determine whether or not it is viral. Most doctors will "watch and wait" to see if the condition goes away on its own before prescribing antibiotics, but unfortunately, not all doctors take this approach.

Instead of reaching for more antibiotics and/or steroids to treat recurring sinus infections, consider other options that are not as harmful and do not require prescription drugs. This keeps you from taking gut-damaging antibiotics unnecessarily. In this protocol, you will learn how to treat and prevent sinusitis using natural remedies, supplements, and strategies you can try at home for relief from inflamed sinus membranes and nasal congestion.

1. Prevention and Home Strategies

You may be able to cut down on the number of sinusitis episodes you experience by taking simple steps to prevent them from occurring. If you are prone to getting sinus infections, try to determine the triggers. Many times, this is allergies or some sort of irritant. Below are strategies to cut back on irritants in your home:

Use a HEPA Air Filter: To reduce the number of irritants in your home, change your air filter often. The Environmental Protection Agency and the American Lung Association recommend good air filtration for those who suffer from allergies and asthma. This isn't the solution all by itself, but it will help cut down on the number of irritants you are exposed to.

There are several types of air filters you can use; the most common is a mechanical filter. This forces air through a screen that traps irritants like pollen, dust mites, and pet dander. They may even capture irritants like tobacco smoke. It is highly recommended that you use a high-efficiency particulate air (HEPA) filter. These are able to capture around 90 percent of air particles, allowing you to breathe cleaner air in your house. Clean all air filters often, as well as air conditioner filters, duct filters, and any other filters you have in your home.

Cut Down on Dust: Dusting isn't a fun pastime, but it can help prevent triggering of allergies. If you are sensitive to dust, wear facial protection when you dust your home. To prevent more dust from getting in your home, keep the windows closed. Keep furniture to a minimum, as unnecessary and unused furniture gathers dust. If you are in the market for new flooring, avoid carpet, as it traps particles that trigger allergies. Use smooth flooring instead, such as hard wood, laminate, or cork flooring. You can cut down on the amount of dust mites in your home by using the hottest water possible when doing laundry. This helps to kill dust mites on your clothes.

The Pet Problem: We love our furry friends, but do not get a pet if you are sensitive to dust and dander in the home. If you already have a pet, let them play outside as much as possible or keep them outside. If this isn't an option, keep them out of rooms you frequent (like your bedroom). Keep them off of furniture, as they tend to collect pet dander and irritants. If you are in the market for a pet, consider one that is "hypo-allergenic." These pets do not shed much and don't contribute to dander in the home. While it is impossible to have a pet that is one hundred percent allergen-free, some pets produce very few allergens. Dog breeds in this category include miniature schnauzer, shih tzu, Maltese, and many breeds

of terrier.

No Smoking Zone: Most people with recurring sinus issues already know that smoking is a major irritant². If you aren't already, do not allow people to smoke in your home. Even being around those who are smokers can set some sensitive people off. Secondhand smoke is just as damaging, so make sure guests smoke outdoors or not at all. Consider limiting your use of a wood stove in the winter, as the smoke from some wood stoves can irritate sinus membranes.

Use a Humidifier at Night: Some people with sinus issues may benefit from running a humidifier in their room at night. A humidifier can put a little moisture in the air and provide relief from dried out sinuses. When sinus membranes get really dried out, they can become inflamed and irritated. Winter months seem to be the worst for dryness, especially if you use wood heat in your home. Try running a cool or warm mist humidifier next to your bed as you sleep to rehydrate the sinuses.

Mold in the Home: Many people are living in homes that are full of potentially dangerous and irritating mold and don't even realize it. If you have recurring sinus issues and cannot detect the culprit, look into whether or not your home has mold³. To start, check some common areas of the home that tend to grow mold. These include: areas below sinks, areas between furniture and exterior walls (especially in the basement or in a concrete block house), coil areas of HVAC handlers and ductwork, areas surrounding plumbing lines (the mold can be behind the drywall in these areas as well), the kitchen and bathroom (around and under tubs, toilets, etc.), wooden cutting boards, trash cans, behind the stove, windows and window sills.

If you aren't sure what to look for, start by looking for mold that is black, green, or gray. It will often have a smell that is musty and mildewy. If you find mold, it may be time to call in an expert to eliminate it successfully. To prevent mold from growing in your home, take the following steps:

- Keep your house dry and ventilated. When cooking or doing dishes, open a window or use a fan.
- Take out your trash often.
- Clean cabinets, counters, window sills, stoves, microwaves, and other areas often.
- Don't let dishes pile up in the sink.
- Keep your fridge clean.
- If you have a wet basement (or any wet area of the home), run a dehumidifier in that area to keep things as dry as possible.

Mold is one of the most aggravating substances for those with sinusitis; it is also dangerous. It can trigger severe allergies, asthma, or anaphylaxis. It can also trigger autoimmune flares, even in small amounts. Many homes have the potential for mold growth, so it is important to stay on top of issues like water leaks, stagnant water, and high levels of humidity in the home. By learning what to look for, as well as where to look, you can keep your home mold-free and support your overall health in the process.

Stay Hydrated: When you don't get enough water, your sinuses can get dried out (as well as the rest of your body). When this happens, you are at a higher risk for a sinus infection. Make sure you are drinking plenty of water each day to lubricate those sensitive sinus membranes. There is a correlation between headaches and dehydration, and some researchers believe it may be due to dried out sinus membranes. One study found that dehydration is found more commonly in people who suffer from recurring sinusitis. Women who are over the age of 60 seem to be at a higher risk for dehydration-induced sinusitis⁴.

HOME REMEDIES FOR SINUSITIS: If you start to notice a little more congestion than normal, slight pressure in the sinus area, or any other subtle signs of sinusitis beginning, you may be able to find relief by utilizing the remedies below. Catching the problem early is the best way to avoid serious complications. Pair these with herbal remedies for an even stronger and more targeted treatment.

Nasal Irrigation: Neti pots look like a small teapot. Using these can significantly help relieve congestion and treat/prevent sinusitis⁵. The idea is to pour the water from the spout into one nostril with your head tilted. This allows the water flow through the nasal passages and out the other nostril. Make sure to use over a sink to prevent messes. As you use the neti pot, keep your body bent over the sink at a 45-degree angle. Most often, a sterile saline solution is used for this procedure. It is not safe to use tap water or any other kind of water in a neti pot. Make sure to only purchase liquid meant for use in a neti pot. Keep your mouth open slightly as you use a neti pot and breathe through your mouth during the procedure. Do this once daily for prevention or as needed to lubricate and irrigate the sinuses.

Lymphatic Massage: Just below the ears on either side of the neck are lymph nodes. These tend to get congested and may protrude slightly from the neck when you have an infection in your body or a lymphatic blockage. Daily lymphatic massage can help unblock congestion in the glands and can open up pathways for lymph fluid to flow more easily through the body to prevent ear infections⁶.

Use an herb-infused oil or essential oils diluted in a carrier oil (see "essential oils" below for more information) to massage into the neck below the ears. Start just behind the ears and gently press with your pointer fingers, massaging downward; repeat tapping the lymph nodes gently as you go down the neck. Keep massaging downward to the clavicles and collarbones. Do this for ten to fifteen minutes each day, but especially when you are sick, have congestion, or experience drainage of any kind.

Spicy Foods: One of the quickest ways to open up congested sinuses and loosen mucus is by eating spicy foods. Let your nose flow and allow the mucus to leave your body before it causes an infection. Try blending a teaspoon of ground cayenne in a cup of hot water to make tea. Capsaicin in cayenne peppers can relieve inflammation and reduce pain, in addition to opening up the sinuses. Drink a cup daily when congested.

2. Herbal Remedies for Sinusitis

Warm Herbal Compress: A warm herbal compress can help with painful sinuses, pressure, and congestion. They are easy to make. Heat a cup or two of water on the stove and add a tablespoon of herbs like eucalyptus, peppermint, ginger, tea tree leaves, oregano, rosemary, or thyme. Allow to simmer uncovered for ten to fifteen minutes and then remove from heat. When it has cooled enough to safely place on the skin, dip a clean cloth into the liquid. Wring it out and place it on the face as long as needed to relieve pain and pressure.

Herbal Steam Inhalation: When herbs are boiled in water, they release small amounts of essential oils that emerge in the form of steam. This is how steam distillation of essential oils works. Your sinuses benefit from steam inhalation because the heat helps open up clogged airways while the medicinal compounds in the steam enter the airways to promote fast healing, kill germs, lower inflammation, and lessen pain⁷. Boil a pot of water on the stove and add a half cup of a sinus-friendly plant like eucalyptus, tea tree leaves, oregano, rosemary, thyme, goldenseal roots, garlic, and peppermint (or any other member of the mint family). Simmer fifteen minutes and remove from heat. Carefully place your head over the pot to inhale the steam, trapping in more steam by placing a towel over your head. Do this for twenty minutes daily.

Hang Eucalyptus in Shower: Steam can help release oils when you hang a plant in the shower under hot water. Tie a small bundle of eucalyptus in the shower. You'll be able to smell the invigorating and antimicrobial aroma of eucalyptus as it works to open and heal your sinuses⁸. Replace the bundle every few days or as needed.

Essential Oils: Essential oils are concentrated oils extracted from plants. They can be very useful for sinusitis⁹. When using essential oils, make sure to dilute them in a carrier oil (like olive oil). A good rule of thumb is one drop per teaspoon of carrier oil. Some useful essential oils for sinusitis include eucalyptus, rosemary, peppermint, ginger, and pine. After diluting these in a carrier oil, carefully massage them into the face (avoid contact with eyes and other mucus membranes). Try lymphatic massage (detailed above) with diluted essential oils to not only open up lymphatic pathways, but also speed healing and target infection.

Cordyceps: This adaptogenic mushroom has a long history of use for respiratory conditions. It is highly anti-inflammatory, and helps reduce the inflammation that comes with congested and irritated sinuses. Its antibacterial and antiviral properties help tackle a wide range of respiratory illnesses¹⁰. Suggested dose: 2-3 dropperfuls of cordyceps dual extracted tincture up to 3x daily.

Echinacea: If your sinusitis is caused by a viral infection, echinacea¹¹ can give your immune system the boost it needs to fight viruses that can lead to sinusitis. Do not use echinacea if you have an autoimmune disease (elderberry is a good replacement). Suggested dose: 2 dropperfuls up to 3x daily at the first sign of

sickness.

Elderberry: Elderberry is natural preventative to keep you from catching a virus like influenza. It is best used at the onset of a virus and should be taken as early as possible to lessen the duration and severity of illness¹². This antiviral and immune-supporting botanical has been well-researched and science backs up claims that this plant can help with viral illness¹³. Elderberry is also a source of quercetin, a beneficial flavonoid that can reduce swelling and histamine production to help with allergies. It works even better when combined with yarrow. Suggested dose: Take 1-3 dropperfuls of elderberry tincture/glycerite/syrup up to 4x daily at the onset of illness.

Garlic: This common kitchen staple is an antimicrobial powerhouse¹⁴. Garlic is excellent for targeting infections and one of the best ways to get this medicinal remedy into your sinuses is to inhale via steam. Chop four to five cloves of garlic and bring this to boil in a pot of water on the stove for ten to fifteen minutes. Remove the pot from heat and carefully place your head over the steam and inhale for up to twenty minutes, placing a towel over your head. You can also take garlic in capsule form. Suggested dose: 180 mg of allicin in capsule form daily.

Ginger: Not only is ginger powerfully anti-inflammatory¹⁵, it is also great for stimulating mucus secretion if you are congested. In addition to helping reduce swelling in the sinus membranes, ginger helps clean sinus cavities and reduce the pain and pressure you feel. Suggested dose: Infuse two teaspoons of peeled and chopped ginger root into a cup of hot water for ten to fifteen minutes and drink once or twice daily.

Goldenseal: Goldenseal has historically been a favorite for those seeking to prevent an upper respiratory tract infection. It is great for infections of the mucous membranes and can help kill bacteria¹⁶ responsible for these infections. In addition to using it in an herbal steam treatment (detailed above), you can also take goldenseal tincture. Recent findings suggest that tincturing the leaves and roots together make a more effective medicine. Suggested dose is 2 dropperfuls up to twice daily. Do not use goldenseal if you have an autoimmune disease or are pregnant.

Lungwort Lichen: This lichen is a great choice for respiratory complaints because it helps to clear mucus from the bronchial passages and reduces inflammation¹⁷. It is a natural antibiotic, so it helps to clear chest infections like bronchitis, whopping cough, pneumonia, and even tuberculosis. It may also be beneficial for those with asthma. Suggested dose: 2 dropperfuls of dual-extracted lungwort lichen tincture up to 3x daily. I pair lungwort lichen with mullein for a powerful bronchial blend tincture and take 3 dropperfuls 3x/day.

Mullein Leaf: For those with congestion, mucus, and stopped up airways, mullein can help the body expel mucus to allow healing¹⁸. Mullein also reduces the formation of mucus and can help conditions like bronchitis, asthma, and emphysema. This common North American plant can

be found just about anywhere on disturbed ground. The second-year leaves are best for respiratory complaints. Suggested dose: Take two dropperfuls of mullein tincture up to three times daily to open the airways and control congestion.

Mullein Flowers and Garlic Oil for Ears: Mullein flowers are used to treat ear infections because they help with pressure, pain, and inflammation inside the ear canal. Paired with garlic, they target and control the infection as well as the symptoms. Since ear problems often result when the sinuses are congested, have this ear oil on hand to manage ear infections before they become severe. To make your own garlic and mullein flower ear oil, chop two cloves of garlic and fill a sterile glass jar with the garlic and two (or more if you want) tablespoons of mullein flowers. Completely cover the flowers and chopped garlic with olive oil. Sit this jar in a pan of hot water (use the “warm” setting on the stove or place it on low heat). Let this infuse on low heat for six to eight hours and strain through cheesecloth. Use at room temperature. Suggested dose: Place 3-5 drops in the ears as needed for infection management. Place a cotton ball in the ear afterward and allow the oil to sit inside the ear as long as possible. Do not use if the eardrum is punctured.

Oregano: Another potent antibacterial¹⁹, oregano is great for targeting germs responsible for sinus infections. Try using fresh or dried oregano in a steam inhalation treatment in addition to taking it internally. You can take oregano internally via tinctures, teas, infusions, or oil capsules. It works much like garlic to clear infection from the inside out. Suggested dose: 500 mg daily in capsule form, or 2 dropperfuls of oregano tincture up to 3x daily.

Reishi: Reishi mushrooms are a very useful adaptogenic mushroom helpful for chronic bronchitis, allergies, and asthma. This anti-inflammatory mushroom also reduces histamine production to help control excess mucus and congestion. Like other adaptogens, reishi helps reduce fatigue and protects the body from stress that weakens the immune system²⁰. Suggested dose: 2 dropperfuls of reishi dual extracted tincture up to 3x daily.

Rosemary: This common culinary herb is expectorant, antispasmodic, antibacterial, anti-inflammatory²¹, and antiviral. It can help open up airways and rid the body of mucus. It has been used traditionally to help ease congestion and inflammation. Try infusing two teaspoons of rosemary leaves into a cup of hot water to make tea. Suggested dose: Drink one to two cups of this tea daily or as needed for the management of sinusitis.

Stinging Nettle: If allergies are contributing to your sinusitis, stinging nettle helps with a variety of allergy-related conditions ranging from hay fever to seasonal allergies²². Its diuretic properties help pull excess water from the body to flush out toxins, while its anti-inflammatory properties calm inflammation and swelling in the airways. Suggested dose: Take 2 dropperfuls of stinging nettle tincture, or a cup of stinging nettle tea 3x daily. Combine stinging nettle with other remedies (such as garlic, turmeric, reishi, cordyceps, yarrow, and

yerba santa) to more effectively target your sinus issues. Combining stinging nettle with local raw honey can increase its effectiveness, as local raw honey can also help prevent allergies.

Turmeric: Turmeric is a great anti-inflammatory that can help relieve congestion due to inflammation in the sinus cavities²³. In addition, it is a mild blood thinner, so it can help improve circulation and potentially relieve sinus headaches. Infuse two teaspoons of peeled and chopped turmeric root into a cup of hot water for ten to fifteen minutes to make a tea. Suggested dose: Drink one to two cups daily or as needed.

Usnea: This lichen possesses both antibiotic²⁴ and antiviral properties to help eliminate a variety of infections. This includes infections of the respiratory tract like bronchitis, pneumonia, sinus infections, and strep throat. Usnea is not a plant, but a lichen. As such, it tends to be more potent in a dual extracted tincture form. Suggested dose: Take 2 dropperfuls of usnea tincture up to 3x daily at the first sign of congestion or infection.

Yarrow: Yarrow can help lower blood pressure, relax blood vessels, and improve breathing. It has demonstrated bronchodilatory effects. It naturally dries and soothes swelling in the nasal glands to provide relief from allergies, asthma, and congestion. Yarrow reduces fevers by encouraging sweating and the opening of pores. A historical blood purifier, yarrow is also powerfully antibacterial and antifungal²⁵. Suggested dose: Take 2 dropperfuls of yarrow tincture up to 3x daily or one to three cups of yarrow tea daily.

Yerba Santa: This helpful herb can dry up mucus fast. It contains compounds that alleviate and loosen mucus in the chest and sinuses due to allergies, infection, colds, and other viruses. Yerba Santa is the perfect remedy for those seeking relief from excess mucus, asthma, allergies, sinusitis, and respiratory infections. It works so well at reducing inflammation, that it has been studied for reducing brain inflammation to treat Alzheimer's²⁶. A tincture made with the leaves is easy to utilize. In addition to the tincture, you can poultice the leaves and apply them to the chest to loosen congestion. Yerba Santa is also known for helping to lower fevers. Suggested dose: 2 dropperfuls of Yerba Santa tincture up to twice daily.

3. Supplements for Sinusitis

In addition to the effective home and herbal remedies for sinusitis (listed above), you can add a few supplements to your sinusitis treatment regimen to speed healing and/or prevent another infection. The supplements listed below may be able to provide additional support to the immune system, as well as heal structural issues of the sinuses:

Vitamin C: This antioxidant is essential to healthy immune function²⁷. Many of us may think we are getting enough vitamin C in our diet, but the body doesn't absorb it as easily as some other vitamins and minerals. Taking a vitamin C supplement is a great way to support healthy immune function to help your body more

effectively fight off illnesses that contribute to sinusitis. Suggested dose: Take 200-1000 mg of vitamin C daily. Start with the lowest recommended dose and work your way up if you suspect you are getting sick.

Vitamin D: This vitamin plays an important role in immune system function and health²⁸. If viral and bacterial infections keep contributing to your sinusitis issues, make sure you are getting enough vitamin D in your diet and spending enough time in the sunlight each day. Vitamin D supplements are available for additional support. Suggested dose: 1000 IU daily .

Probiotics: Since our gut health is directly tied to our immune system health²⁹, make sure you are supporting your microbiome. Eat plenty of prebiotic and probiotic foods and take a good probiotic supplement with a variety of different probiotics. Suggested dose: 10 billion CFU daily.

Serrapeptase: If nasal and sinus polyps are causing you problems, the enzyme serrapeptase may be able to help. It reduces inflammation by dissolving excess mucus³⁰ and other foreign objects blocking respiratory airways. It is even utilized by women looking to unblock scarring in fallopian tubes and those looking to dissolve blood clots in the body. It can support wound healing and wound cleaning, and makes a useful pre and post-surgery supplement to speed healing³¹. Suggested dose: 10 to 60 mg daily.

4. Western Medical Approaches

If you desire to seek allopathic medical treatment for sinusitis, there are many different approaches your doctor may take, depending on the severity and underlying causes of your sinusitis. A doctor will likely perform a few diagnostics before choosing a treatment. Diagnostics may include imaging tests to look at the sinuses more closely, looking into the sinuses with a fiberoptic light, an allergy test, or samples from your nasal and sinus discharge (to be cultured in a lab). All of these tests will help determine if the cause is bacterial, what antibiotics to use (if any), and whether or not surgery is needed. Below are some common western medical treatments for sinusitis:

Nasal Spray: To treat inflammation in the sinuses and nasal cavity, a doctor may prescribe nasal corticosteroids. These

are nasal sprays including fluticasone, triamcinolone, budesonide, mometasone, and beclomethasone. If these don't work as effectively as you like, your doctor may suggest saline nasal irrigations (with a neti pot device as described above).

Allergy Medications: Since allergies are the cause of sinusitis for many, a doctor may have you take allergy medications like Claritin or Zyrtec.

Aspirin Desensitization: For some people, a reaction to aspirin can cause sinusitis and/or nasal polyps. If this is the case for you, your doctor may suggest a treatment aimed at increasing your tolerance to aspirin. Under medical supervision, you would be gradually given larger doses of aspirin over a period of time.

Antibiotics: Antibiotics are commonly prescribed for sinusitis. Some commonly prescribed antibiotics include amoxicillin and doxycycline. Some sinusitis infections are viral and will not be cured with antibiotics, so it is important to make sure a bacterial infection is truly the underlying cause of your sinusitis.

Antifungal Medications: Sometimes a fungal infection is to blame for sinusitis. If this is the case, a doctor may prescribe an antifungal medication.

Oral Corticosteroids: Corticosteroids relieve inflammation and can help those with severe sinusitis, especially if they also have polyps. They are not a long-term solution and can cause serious side-effects.

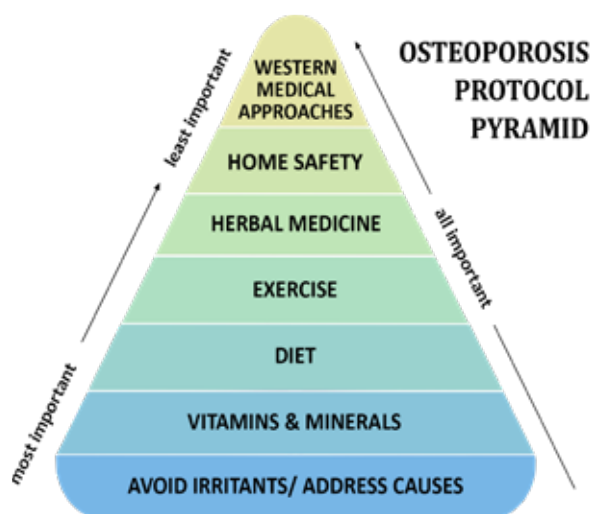
Injections: Injected corticosteroids may be used in lieu of oral corticosteroids for some people. Your doctor may suggest allergy shots to help desensitize you to the things you are allergic to. A doctor may also give an injection of dupilumab or omalizumab to treat nasal polyps. These injected medications may reduce the size of the polyps to lessen congestion.

Surgery: If you are resistant to treatment or medications, sinus surgery may be the next step. This is usually an endoscopic surgery that uses a thin, flexible tube with a light to explore the sinus passages. When a blockage (usually a polyp) is detected, the doctor may use instruments to remove it or shave it away. It is not uncommon for sinus surgery to involve enlarging a narrow sinus opening to help with drainage as well.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Garlic, Serrapeptase, and Vit. C & D with breakfast Stinging nettle tincture (if needed for allergies) Elderberry or Echinacea	Neti Pot Warm herbal compress Reishi and Cordyceps dual-extracted tinctures Oregano capsule Bronchial tincture	Hydration check (are you drinking enough water?) Yarrow & Yerba Santa tinctures Mullein/Garlic ear drops	Lymphatic massage Steam inhalation Reishi & Cordyceps dual-extracted tinctures Mullein tea or Bronchial tincture	Start cool or warm mist humidifier (allow to run all night as you sleep) Neti Pot Stinging nettle tincture (if needed for allergies) Elderberry or Echinacea Yarrow and Yerba Santa
MONDAY	Garlic, Serrapeptase, and Vit. C & D with breakfast Stinging nettle tincture Elderberry or Echinacea	Neti Pot Warm herbal compress Reishi and Cordyceps Oregano capsule Bronchial tincture	Hydration check Yarrow & Yerba Santa tinctures Mullein/Garlic ear drops	Lymphatic massage Steam inhalation Reishi & Cordyceps tinctures Mullein tea or Bronchial tincture	Humidifier Neti Pot Stinging nettle tincture (if needed for allergies) Elderberry or Echinacea Yarrow and Yerba Santa
TUESDAY	Garlic, Serrapeptase, and Vit. C & D with breakfast Stinging nettle tincture Elderberry or Echinacea	Neti Pot Warm herbal compress Reishi and Cordyceps Oregano capsule Bronchial tincture	Hydration check Yarrow & Yerba Santa Mullein/Garlic ear drops	Lymphatic massage Steam inhalation Reishi & Cordyceps Mullein tea or Bronchial tincture	Humidifier Neti Pot Stinging nettle tincture (if needed for allergies) Elderberry or Echinacea Yarrow and Yerba Santa
WEDNESDAY	Garlic, Serrapeptase, and Vit. C & D with breakfast Stinging nettle tincture Elderberry or Echinacea	Neti Pot Warm herbal compress Reishi and Cordyceps Oregano capsule Bronchial tincture	Hydration check Yarrow & Yerba Santa Mullein/Garlic ear drops	Lymphatic massage Steam inhalation Reishi & Cordyceps Mullein tea or Bronchial tincture	Humidifier Neti Pot Stinging nettle tincture (if needed for allergies) Elderberry or Echinacea Yarrow and Yerba Santa
THURSDAY	Garlic, Serrapeptase, and Vit. C & D with breakfast Stinging nettle tincture Elderberry or Echinacea	Neti Pot Warm herbal compress Reishi and Cordyceps Oregano capsule Bronchial tincture	Hydration check Yarrow & Yerba Santa Mullein/Garlic ear drops	Lymphatic massage Steam inhalation Reishi & Cordyceps Mullein tea or Bronchial tincture	Humidifier Neti Pot Stinging nettle tincture (if needed for allergies) Elderberry or Echinacea Yarrow and Yerba Santa
FRIDAY	Garlic, Serrapeptase, and Vit. C & D with breakfast Stinging nettle tincture Elderberry or Echinacea	Neti Pot Warm herbal compress Reishi and Cordyceps Oregano capsule Bronchial tincture	Hydration check Yarrow & Yerba Santa Mullein/Garlic ear drops	Lymphatic massage Steam inhalation Reishi & Cordyceps Mullein tea or Bronchial tincture	Humidifier Neti Pot Stinging nettle tincture (if needed for allergies) Elderberry or Echinacea Yarrow and Yerba Santa
SATURDAY	Garlic, Serrapeptase, and Vit. C & D with breakfast Stinging nettle tincture Elderberry or Echinacea	Neti Pot Warm herbal compress Reishi and Cordyceps Oregano capsule Bronchial tincture	Hydration check Yarrow & Yerba Santa Mullein/Garlic ear drops	Lymphatic massage Steam inhalation Reishi & Cordyceps Mullein tea or Bronchial tincture	Humidifier Neti Pot Stinging nettle tincture (if needed for allergies) Elderberry or Echinacea Yarrow and Yerba Santa

Ch. 36: Osteoporosis Protocol

Osteoporosis is a disease that weakens bones. Your bones are not permanently fixed; they undergo constant remodeling and maintenance, resulting in a new skeleton approximately every 10 years¹. Minerals such as calcium, phosphorus, and magnesium are deposited, taken out, and recycled regularly according to how much stress is placed on a bone by body weight and exercise, hormone levels, steroid use, diet, and family history. Healthy bone remodeling is a balance between cells that make bone, osteoblasts, and cells that degrade bone, osteoclasts. Osteoporosis is a condition involving loss of bone strength, density, and quality due to decreased amounts of mineral deposition and poor bone remodeling. The word itself means “porous bone” and it can be the result of excess osteoclast activity, inadequate or slow osteoblast activity, or both.



Loss of bone density and strength leads to increased risk of injury and fractures. In fact, this is often the only sign that osteoporosis is present. Worldwide, osteoporosis causes approximately 9 million fractures a year². It is otherwise asymptomatic. 20% of osteoporotic fractures are hip fractures, which incur high healthcare costs and severely limit mobility, plus commonly result in disability, decreased quality of life, and increased risk of death in the elderly. While the incidence varies from country to country, and depends on age, 20-75%^{3 4 5} of people over age 50 die within one year of a hip fracture, even after it has been surgically addressed and 8 years post-fracture, mortality is still twice as likely⁶ than if there were no fracture. It is a condition that is best prevented or treated early.

Osteoporosis is more common in women. Post-menopausal women have the highest risk as the decrease in estrogen during menopause also decreases the rate of calcium deposition in the bone. Men tend to have higher mortality rates after fractures than women. Universally recognized risk factors for developing osteoporosis are having a small or petite body frame, lack of exercise, Vitamin D deficiency, current smoking, excessive alcohol consumption, and family history. Celiac

disease and inflammatory bowel disease (Crohn's disease and ulcerative colitis) increase a person's risk of osteoporosis due to their decreased ability to digest and absorb calcium into the bloodstream. Other considerable risk factors include thyroid and parathyroid disease, history of eating disorder, gastric bypass surgery, or a prolonged period of immobilization. Medications such as steroids, estrogen and testosterone reducing drugs, diuretics, antidepressants, and antacids increase risk. What many of these risk factors have in common is that they decrease the amount of minerals available for bone remodeling, resulting in weaker bones.

Bone density and bone loss is measured with a special type of x-ray called a DEXA (Dual Energy X-ray Absorptiometry). Measurements are taken in the spinal vertebrae and at the hip and assigned a score relating to the density of the bone in those places. The results of the DEXA scan place the patient in one of three main categories of bone density:

- Normal: T score above -1.
- Osteopenia: T score between -1 and -2.5.
- Osteoporosis: T score of -2.5 or less.

Due to the increased risk of osteoporosis in post-menopausal women, a screening DEXA scan to measure bone density is often recommended for any woman over the age of 65 and follow-up scans may be done if the levels are abnormal or if the woman has other risk factors for the development of osteoporosis. Men over 65 with risk factors should also receive a screening DEXA.

1. Avoid Irritants Since many risk factors for the development of osteoporosis are known and several are modifiable it makes good practical sense to protect the bones by avoiding known assaults to bone health. To best maintain healthy rates of bone construction and remodeling, the body needs adequate calcium. Factors that contribute to available calcium levels include:

- Incoming calcium in the diet
- The ability for the intestines to digest and absorb calcium
- The amount of calcium being deposited or taken from the bone
- The amount of calcium being eliminated by the kidneys through urination.

Any threats to optimal calcium levels must be addressed. Additionally, diseases or lifestyle factors that change hormonal instructions to bone synthesis should also be controlled. Signals that tell the body to breakdown bone or that result in poor-quality bone should be reduced. Signals that tell the body to make and strengthen bone should be amplified.

Reduce Alcohol Consumption and Smoking^{7 8 9}: Addressing these two behaviors greatly improves bone health and osteoporosis risk. Besides promoting inflammation and depleting the body of essential vitamins, alcohol is directly toxic to bone metabolism,

with increased intake causing lower bone mass and strength^{10 11} and increased risk of fractures¹².

Cigarette smoking is a well-known risk factor for developing osteoporosis. The heavier the smoking habit, the worse the bone mass density and fracture risk. Recent research proposes this might be due to toxic levels of cadmium¹³ absorbed when smoking. The increase in fracture risk among long-time smokers is so high that it is not proportional to the changes seen in bone density; thus smoking must impair the quality of bone being constructed over time. Making a plan to quit smoking and reduce alcohol consumption to below two units¹⁴ per day should be among the first steps in treating and preventing osteoporosis.

Maintain Healthy Sodium Intake: Eating a diet high sodium causes the kidneys to get rid of calcium¹⁵ through the urine. Over time, this effect impacts the bones and studies show that bone density is lower¹⁶ in people who have high sodium diets (over 4,000mg per day. There is some evidence that heavily restricting sodium may also lead to decreased bone density, so the goal is simply to avoid excess sodium, not to tightly restrict it.

A healthy target sodium intake is around 1,500-2,000 mg per day. However, it is estimated that many people eat 3,000mg of sodium or more daily. Keep an eye on how many high sodium foods you eat- like canned soups, frozen convenience dinners, and cured or salted meats. Avoiding very high sodium foods and maintaining a healthy level of sodium in your diet can help preserve your calcium stores and protect you against osteoporosis.

Reduce Refined Sugars: Ingesting refined sugar, especially in excess, can cause a variety of health issues due to the hormonal and inflammatory changes in the body sugar causes. Excess sugar intake results in both reduced calcium absorption in the intestines and increased kidney elimination of calcium through the urine- both undesirable for bone health.

Additionally, a high sugar diet and having high circulating blood sugar are associated with Vitamin D deficiency and slower rates of tooth and bone growth¹⁷. Type 2 diabetes, a disease involving poor blood sugar regulation and high blood sugar, is connected to an increased risk of developing osteoporosis and osteoporotic fractures.

Maintain a Healthy Protein Intake: Research shows that a healthy dietary intake of protein is essential to healthy bone building and maintenance. Our bodies require sufficient levels of protein to make collagen- the connective tissue matrix, or scaffolding, that supports our bones and skin. It is estimated that our bone contents are 30% collagen, with calcium and other minerals built into this matrix. For people with protein intake that is too low, collagen synthesis and the growth factors and hormones needed for bone building are impaired, resulting in decreased bone building and bone mass. Additionally, when dietary protein is too high, it causes the kidneys to eliminate calcium through the urine, which again increases risk for poor bone health and strength.

Increased risk of fractures is seen amongst people with both very low¹⁸ and high protein^{19 20} intake, so similar to sodium, the goal is to keep normal healthy levels.

Reduce Intake of Coffee: There are mixed results in studies evaluating the relationship between coffee intake and osteoporosis, with many concluding that higher intake of coffee results in lower bone densities²¹. Sometimes this results in increased occurrence of fracture and sometimes it does not, with some researchers stating the risk is only minor. It is well established that increasing coffee intake results in increased kidney elimination of calcium²², a result to be avoided for anyone with osteoporosis. "Excess" is defined as more than²³ 2 cups per day²⁴.

Control and Monitor Other Diseases: Certain diseases are known to be risk factors for developing osteoporosis or can make bone loss worse, often through the loss of nutrients needed for healthy bone building. Underlying diseases such as hyperthyroidism, inflammatory bowel diseases, metabolic syndrome, diabetes, and kidney disease should be appropriately treated and monitored in order to improve calcium absorption from the gut, reduce calcium loss from the kidneys, and reduce the rate of bone breakdown. If you are taking prescription medications for another illness, evaluate your medication list and substitute wherever possible for drugs that carry less risk of nutrient loss and osteoporosis.

2. Vitamins and Minerals

We can help promote healthy bone building by making sure the body has an abundance of materials and resources in circulation. You often need more than the recommended daily intake amount of a vitamin or mineral in order to create positive changes and improvements in a condition instead of simply maintaining homeostasis. The following nutrients are key for bone building, and in some cases also influence the levels of bone building signals or hormones in the body.

Calcium: 99% of calcium in the body is found in the bones and calcium makes up about 40% of a bone's total weight. It is required for construction, formation, and maintenance of bone. While supplementation with calcium alone has been shown to positively affect bone density²⁵, it has best results when taken with Vitamin D to optimize its use and absorption. Supplementation with calcium with Vitamin D reduces²⁶ overall fractures and hip fractures specifically. Good dietary sources of calcium are dairy products (like milk, cheese, and yogurt), soybeans and tofu, sardines, salmon, almonds, and leafy green vegetables. Suggested dose: Take 1,000-2,000mg total per day taken in 2 doses. Doses higher than 2,000mg per day are not recommended²⁷ and can cause side effects like kidney stones and increased risk for cardiovascular events.

Vitamin D: Vitamin D is essential to bone health. Vitamin D increases absorption of calcium and phosphorus in the digestive tract and supports calcium and phosphorus deposition in bone for bone construction. Low levels of Vitamin D are common in people²⁸ with osteoporosis, osteopenia, and osteomalacia - another condition

resulting in weak bones. Optimizing Vitamin D improves bone mass density²⁹ and is key to treating osteoporosis.

Suggested dose: It is difficult for us to make the amount of Vitamin D we need from sun exposure unless we spend a lot of time outdoors with the majority of our skin exposed. The amount of Vitamin D needed daily varies from person to person, depending on their current level. This is easily measured by a blood test. Vitamin D deficient adults can take 5,000-10,000iu of Vitamin D per day over a period of months to reach normal levels. If you are uncertain of your Vitamin D level, a daily dose of 2,000-5,000iu per day is recommended.

Magnesium: It is estimated that 50-60% of all magnesium in the body is found in the bone. It contributes to bone health in a few different ways³⁰: it is needed for activation of Vitamin D, which then aids calcium recruitment into the bone, it comprises part of the bone mineral structure by binding to calcium, and it influences parathyroid hormone, which gives instructions to the bone on whether to keep calcium or get rid of it. Low levels of magnesium are connected to osteopenia³¹ and osteoporosis^{32 33}. While most of these studies analyzed post-menopausal women, other studies evaluating men³⁴ or both men and women³⁵ also correlate low blood magnesium with increased fracture risk.

Increasing your intake of magnesium-rich foods like green leafy vegetables, legumes, nuts, seeds, and whole grains can help assure you do not become magnesium deficient. Studies indicate that magnesium supplementation has positive effects on many different measures of bone health and metabolism and that supplementation increases bone mineral density more than a magnesium-rich diet alone³⁶ or no treatment³⁷. Suggested dose: Magnesium supplementation works best in combination with other minerals like calcium and should be present in about half or one-third³⁸ the amount as supplemental calcium, so if supplementing with 1,000mg calcium take 300-500mg of magnesium. The most commonly studied³⁹ forms of magnesium for osteoporosis are magnesium citrate, carbonate or oxide.

Vitamin K (K2): Vitamin K refers to a collection of three active compounds in the body - K1, K2, and K3. Vitamin K plays a role in bone metabolism and strength through direct regulation of bone building and remodeling cells, osteoblasts, and osteoclasts. It is also necessary for building bone matrix and calcium binding. Low circulating Vitamin K levels are associated with osteoporosis⁴⁰ and increased fractures^{41 42}. Supplementation with Vitamin K decreases fracture risk and improves bone mass density. It may enhance the use and efficacy of calcium and Vitamin D when taken together⁴³. Vitamin K is most abundant in foods like green leafy vegetables, broccoli, cauliflower, fish, eggs, and kiwi.

Supplementation with vitamin K2 is thought to be best for bone health. Supplemental doses of 45mg Vitamin K2 daily have been a routine treatment⁴⁴ for osteoporosis in Japan since the 1960s, but recent research shows 5mg a day is sufficient⁴⁵. Because of Vitamin K's role

in the body of assisting blood clotting, typically Vitamin K supplements are not to be taken by patients on blood thinning medication. Suggested dose: 5mg daily.

Vitamin C: Vitamin C is important to bone health because it helps promote healthy collagen formation, and collagen forms the core matrix in bone on which minerals are deposited. Higher levels of Vitamin C in the diet result in stronger bone density^{46 47}, reduced risk of osteoporosis by as much as 33%⁴⁸, and reduced incidences of hip fracture. Vitamin C also seems to be helpful for those already experiencing bone loss. Increasing Vitamin C levels seems to decrease some of the bone loss seen amongst smokers and people with low levels of physical activity⁴⁹, and 1,000mg of Vitamin C per day results in fewer signs of bone degradation⁵⁰ in the elderly when taken with 500mg of Vitamin E. Good dietary sources of Vitamin C are citrus fruits, bell peppers, berries, broccoli, and brussel sprouts. Suggested dose: 1,000mg once or twice a day.

Trace Minerals: Trace minerals are sometimes the forgotten key to bone building and bone strength. Although calcium, magnesium, and phosphorus are the most abundant minerals in the bone, each of the following trace minerals play an important role in bone health and are helpful additions for osteoporosis:

- **Strontium** is a trace mineral with a similar chemical structure to calcium that is deposited into healthy bone. Strontium improves bone strength, promotes bone building, and suppresses breakdown⁵¹ by influencing osteoblasts and osteoclasts. In some countries, strontium ranelate is available as a medication, and strontium is available over the counter in the form of strontium citrate. Strontium citrate suggested dose: Take 450mg-600mg per day⁵². Because of its chemical similarity, strontium is best taken away from calcium to maximize absorption.
- **Silicon** is similar to Vitamin C in that it aids in collagen formation and increases bone strength. Higher amounts of dietary silicon result in better bone mass densities later in life for men and women⁵³. When supplemented, silicon may increase bone density in people with osteoporosis^{54 55}, and may protect against later bone loss. It is commonly added to trace mineral and bone health supplements and has been shown to be safe up to 700mg per day⁵⁶, although more regular doses range from 6-25mg per day.
- **Boron** is known as an antioxidant and anti-inflammatory and has been researched for its wound healing and cancer treatment benefits. Boron has also been shown to stimulate bone growth and increase bone strength⁵⁷. It is thought to improve calcium absorption from the gut and assist with calcium and magnesium deposition on bone through enhancing Vitamin D activation⁵⁸. Suggested dose: 3mg per day⁵⁹.

Omega-3 Fatty Acids: Omega-3 oils contain healthy anti-inflammatory fatty acids that support several body systems, most notably the cardiovascular and neurological systems. It also promotes healthy bone metabolism⁶⁰ and remodeling. Elderly people with higher levels of Omega-3

fatty acids in their blood⁶¹ tend to have higher bone mass density than those with lower levels. Other studies suggest that dietary Omega-3 both⁶² increase bone building and decrease rates of bone breakdown and that increasing Omega-3 oil intake through supplementation may help preserve⁶³ bone density in the hip and spine.

It is easy to increase your daily intake of these beneficial fats by including more of the following in your diet: cold water fish (like salmon, tuna, sardines), flaxseeds and flax oil, chia seeds, walnuts, hemp seeds and hemp seed oil. Omega-3 oils in supplemental form are commonly sourced from fish and are sometimes referred to as 'fish oil.' The most common vegetarian Omega-3 supplement is from flax oil. Suggested dose: 1,000mg – 3,000mg daily.

B-Vitamins⁶⁴: There are eight different B-vitamins, each with numerous responsibilities in the body. The majority of studies about bone health focus on the relationship between low folic acid and increased fracture risk and decreased bone density, some showing how folate supplementation helps improve bone mass. There is also evidence that higher B12 and B6⁶⁵ intake are protective against fractures and that higher levels of dietary B3, niacin, are correlated with better bone density at the hip.

B-complex vitamins are naturally abundant in foods like beans, legumes, eggs, and green leafy vegetables so they should comprise a healthy part of your diet. A high-quality B-complex vitamin may assist in bone health maintenance. One per day should suffice.

3. Diet While our previous section on avoiding irritants focuses mainly on which aspects of the diet should be removed or monitored, there are dietary choices that benefit osteoporosis risk. Healthy dietary choices represent the first line of defense for providing nutrients essential to bone health, and should include anti-inflammatory oils and bioflavonoids.

A whole-foods based Mediterranean Diet is very supportive of bone health, with countries surrounding the Mediterranean⁶⁶ having significantly lower rates of osteoporosis and osteoporotic fractures⁶⁷ than their neighboring countries in Europe. The focus on fresh fruits and vegetables, whole grains, and beans, over proteins, fats, and sweets provides high levels of several nutrients such as Vitamin C, Vitamin K, calcium, magnesium, and B-vitamins. It has also been suggested that the increased vegetable intake may make the blood slightly more alkaline, which decreases⁶⁸ the release of calcium from the bones. The reliance on olive oil as a primary fat source also offers some protection⁶⁹ for the bones.

Soybeans and soy-based products have long been known for their ability to weakly mimic estrogen in the human body, which is why it is sometimes referred to as a phytoestrogen, or plant-based estrogen. This phytoestrogenic property allows soy to influence osteoblasts and osteoclasts⁷⁰ similarly to estrogen with positive effects on bone metabolism⁷¹, especially for post-menopausal women. Osteoporotic fractures are lower in countries⁷² where soy consumption is high compared

to populations with a more Western diet. Adding organic soybeans and organic soy-based products like tofu, tempeh, and miso are good ways to naturally add supportive phytoestrogens to your diet. In supplement form, extracts of soy bioflavonoids are used, called isoflavones. Suggested dose of soy isoflavones⁷³: 100-150mg per day.

4. Exercise An inactive lifestyle is a known risk factor for the development of osteoporosis. Increasing physical activity improves bone mineral density for people with osteoporosis. This effect has been seen across different types of exercise⁷⁴. Weight bearing exercise, aerobic dance therapy, and resistance⁷⁵ exercises all help bone mineral density in post-menopausal women.

To further improve outcomes, target exercise to the area of concern. Lower extremity⁷⁶ resistance training is best for increasing density at the head of the femur and hip while a combination of different exercises is best for increasing density of the spine. Activities aimed at increasing muscle strength and balance (like yoga) may increase proprioceptive awareness and decrease falls⁷⁷. For maximum benefit, studies suggest a frequency of three times a week, at a minimum, for one hour.

5. Herbal Medicine Like soy, there are other plants that have phytoestrogenic properties that have been investigated for their bone health benefits. Instead of being eaten as food, these botanicals are usually consumed as medicinal extracts/tinctures.

Dandelion Leaves (*Taraxacum officinale*): These contain high amounts of vitamins A, C, E, and K and provide calcium, potassium, iron, folate, and magnesium. Harvest from an area you know hasn't been sprayed with harmful chemicals. Add fresh leaves to salads or gently sauté. Steaming is also a good way to consume them, but too much heat damages some of the more sensitive phytochemicals in plants and may reduce the amount of nutrients.

***Epimedium*⁷⁸ spp. Plants (Fairy Wings or Horny Goat Weed):** These have been used in Chinese medicine as antiosteoporotic agents for a long time. It has some of the same phytoestrogenic isoflavones as soy and it is thought to be beneficial for osteoporosis because it saves calcium from being eliminated in the urine and promotes bone formation. They help inhibit osteoclast activity and slow bone breakdown. When postmenopausal women took an *Epimedium* extract with their calcium supplement for 2 years, they had maintained bone density⁷⁹, whereas the women their age who took calcium alone still lost density. Suggested dose: 200mg capsule or 2 dropperfuls tincture/extract per day.

Licorice Root (*Glycyrrhiza glabra*): Licorice root extracts are used for many bone diseases^{80 81} such as osteoporosis, osteopenia, fractures, osteomalacia, and periodontal disease. It acts⁸² as a phytoestrogen and enhances bone formation^{83 84} and reduces breakdown in animal studies. Similar results have been seen in human cells

and investigations into licorice as a natural treatment for osteoporosis are ongoing⁸⁵. Suggested dose: Take 500mg-1000mg capsule or 1 dropperful three times a day (note: not DGL-licorice root). Use with care.

Red Clover (*Trifolium pratense*): Red clover exhibits positive effects on bone density⁸⁶ through phytoestrogenic stimulation. There is new evidence supporting the use of red clover alongside supplementation with Vitamin D and calcium - the addition of red clover resulted in more beneficial changes⁸⁷ in bone density than the nutrients alone. A recent study⁸⁸ showed that red clover influences estradiol concentration, exercise performance, and gut microbiota. Suggested dose: one 700 mg capsule per day or 3 dropperfuls tincture twice a day.

Stinging Nettle (*Urtica dioica*): may be covered in stinging hairs, but the sting is completely eliminated if the plant is dried or cooked. It is a rich source of nutrition and contains protein, crude fiber, calcium, and iron. Studies have shown stinging nettle to be nutritionally superior⁸⁹ to other plants (commonly used in flours) and it contains significantly more bone-strengthening nutrients. It also helps with arthritis, inflammation, and allergies. Best taken as a twice-daily tincture. Also add a cup of fresh (or a half cup of dried) stinging nettle to a nourishing bone broth. Sip this nourishing broth throughout the day.

6. Home Safety Due to the increased risk of fracture associated with osteoporosis and the disabling effects of fractures in locations like the hip, trip and fall risks within the home must be addressed. It is recommended that the home environment is equipped with:

- Adequate, bright lighting and nightlights, especially around stairs, entrance ways and bathrooms
- Floor surfaces clear of loose cords, throw rugs
- Walking paths clear of furniture
- Railings, non-skid surfaces, secured rugs on stairs
- Grab bars and non-slip strips or rubber mat in showers/bathtubs
- Grippy socks and full-foot slippers (nothing slippery, floppy, or flimsy).

7. Western Medical Approaches Considering how common osteoporosis is, there aren't very many options amongst conventional treatments. It is common for doctors to recommend a good calcium supplement and Vitamin D. Besides that, there are two kinds of approaches used: stopping or slowing bone loss and stimulating bone growth.

Anti-Resorptive Drugs: Anti-Resorptive Drugs are known to slow or inhibit bone remodeling are anti-resorptive drugs.

Bisphosphonates are a category of antiresorptive therapy and the preferred treatment for osteoporosis. Most of these medications are taken orally, although some

can be administered intravenously. The majority work by limiting the lifespan osteoclasts or limiting their ability to bind to bone to break it down. The most common side effects of oral bisphosphonates are abnormal taste in the mouth, difficulty swallowing, esophageal ulcers and acid regurgitation, nausea, abdominal pain and gastritis, passing gas, and stool changes like constipation or diarrhea. Severe side effects are increased fractures and a degenerative disease of the jaw called osteonecrosis. If side effects become intolerable, administration is switched to IV.

Another kind of anti-resorptive drug is an **antibody treatment** that binds to osteoclasts to stop their growth and development. This treatment is only given through injection and has many of the same side effects as bisphosphonates, and is reserved for patients with kidney disease.

There are two kinds of treatment aimed at optimizing and mimicking natural hormonal stimulation of the bone that are also considered to be anti-resorptive:

1. Estrogens and Hormone Replacement Therapy can be given in capsule form, skin patches, or subdermal tabs. While replacement of hormones in postmenopausal women is not considered first line therapy for osteoporosis, they help slow bone breakdown by mimicking or supplementing the body's low estrogen.
2. Selective Estrogen Receptor Modulators or SERMs do exactly as their name states- they stimulate estrogen receptors without actually being estrogen. SERMs are sometimes given instead of hormone replacement therapy to reduce stimulation of the breast and uterine tissue. SERMs are only used in post-menopausal women.

Calcitonin is a hormone that tells the body to put circulating calcium into the bones, and it is sometimes given artificially as second choice osteoporosis treatment. It can be given orally, injected, or intranasally but is generally only used when first choice treatments are ineffective or poorly tolerated. Patients may become allergic and calcitonin has side effects such as low blood calcium, intranasal injury, and increased incidence of prostate cancer.

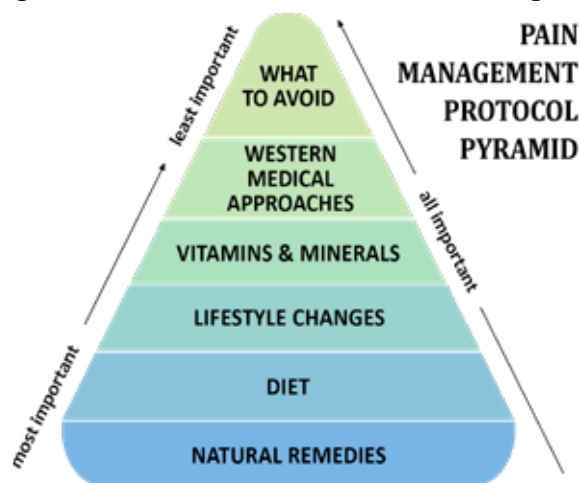
Anabolic Drugs: Anabolic drugs are a newer approach to osteoporosis treatment that is typically for those with very low bone density due to their high cost. They mimic one of our own hormones, parathyroid hormone (PTH). They are given by daily or monthly injection. Administering it in this way makes it act differently than PTH and it stimulates construction of bone- a method that is absent or very weak in other therapies. Because these therapies are relatively new, there have not been studies on long-term use, but short-term side effects include nausea, headache, and dizziness. There is a concern, albeit small, over a few documented incidences of bone cancer believed to be caused by this class of drug.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	B-complex vitamin Vitamin C	Home safety inspection Calcium-Magnesium Epimedium & Red clover tinctures Vitamin D & Omega-3 oils with food	Mediterranean Diet meal planning Weekly shopping	Vitamin K-2 & trace minerals Check protein intake- consider organic soy protein	Restorative yoga & balance exercises Calcium-Magnesium Red clover tincture
MONDAY	B-complex vitamin Vitamin C	Calcium-Magnesium Epimedium & Red clover tinctures Vitamin D & Omega-3 oils with food	30-min. resistance training	Vitamin K-2 & trace minerals	Calcium-Magnesium Red clover tincture
TUESDAY	B-complex vitamin Vitamin C	Calcium-Magnesium Epimedium & Red clover tinctures Vitamin D & Omega-3 oils with food		Vitamin K-2 & trace minerals Check protein intake- consider organic soy protein	Restorative yoga & balance exercises Calcium-Magnesium Red clover tincture
WEDNESDAY	B-complex vitamin Vitamin C	Calcium-Magnesium Epimedium & Red clover tinctures Vitamin D & Omega-3 oils with food	30-min. resistance training	Vitamin K-2 & trace minerals	Calcium-Magnesium Red clover tincture
THURSDAY	B-complex vitamin Vitamin C	Calcium-Magnesium Epimedium & Red clover tinctures Vitamin D & Omega-3 oils with food		Vitamin K-2 & trace minerals Check protein intake- consider organic soy protein	Restorative yoga & balance exercises Calcium-Magnesium Red clover tincture
FRIDAY	B-complex vitamin Vitamin C	Calcium-Magnesium Epimedium & Red clover tinctures Vitamin D & Omega-3 oils with food	30-min. resistance training	Vitamin K-2 & trace minerals	Calcium-Magnesium Red clover tincture
SATURDAY	B-complex vitamin Vitamin C	Calcium-Magnesium Epimedium & Red clover tinctures Vitamin D & Omega-3 oils with food	Outdoor walk	Vitamin K-2 & trace minerals	Restorative yoga & balance exercises Calcium-Magnesium Red clover tincture

Ch. 37: Pain Management Protocol

About 20 percent of American adults live with chronic pain most days or every day.¹ It is one of the top reasons people see their doctor, with low back pain being the most common type of chronic pain reported.² Pain can be acute, lasting a short period of time or chronic, coming and going for more than three months.³ Causes of pain include tissue damage, illness, musculoskeletal problems, injury, surgery, and cancer. Sometimes there is no apparent cause.

Pain is associated with significant emotional distress and functional disability. Up to 85% of patients with chronic pain are affected by severe depression.⁴ Accidental overdose, mainly from opioid pain medication, is the leading cause of death of Americans under the age of 50.⁵



People feel pain when specific neurons called nociceptors detect tissue damage and transmit this information along the spinal cord to the brain. Once the message arrives, it causes you to feel the unpleasant sensation of pain. The brain's interpretation of these signals and the efficiency of the communication pathways dictate how a person experiences pain.⁶ Pain can persist for long periods of time if signals become dysregulated and keep firing in the nervous system.⁷

Pain management is a major public health issue with an enormous social and economic impact. Opioid medications, commonly prescribed for moderate to severe pain, are highly addictive. Opioid pain relievers are used by 10% of the American population, and 80% of heroin users said the first opioid they ever used was a prescription opioid.⁸ Given the significance of the opioid crisis, there is an urgent need to find safe, non-addictive approaches to pain relief.

1. Natural Remedies

Below are listed internal herbs that help with pain management:

Boswellia/Indian Frankincense (*Boswellia serrata*):

Indian Frankincense is an herbal extract taken from the *Boswellia serrata* tree. Boswellia has been used

for centuries to treat chronic inflammatory illnesses (like arthritis) and pain; it has anti-inflammatory and analgesic (pain relieving) effects,⁹ considerably improves joint function,¹⁰ and significantly increases individuals' pain threshold and pain tolerance.¹¹ It is believed that Boswellia works by inhibiting leukotrienes, inflammatory chemicals in the body involved in pain.¹² Frankincense also works topically for pain. Suggested dose: Take 350mg daily or 2-3 dropperfuls of tincture, daily. Possible interactions: Do not use internally if pregnant.¹³

Cannabidiol (CBD): Cannabidiol, a compound derived from *Cannabis* plants, is a popular research topic in pain management. The legalization of CBD in most U.S. states has ignited interest in its potential for pain management.¹⁴ CBD has several advantages for pain management. Most significantly, it does not seem to be addictive, and it appears to be relatively safe in humans, with few negative side effects.¹⁵ CBD can improve nerve pain unresponsive to standard treatments,¹⁶ and is an effective treatment for peripheral neuropathic pain, rheumatoid arthritis, and cancer pain.¹⁷ CBD exerts its pain-relieving effects through its various interactions and modulation of the endocannabinoid, inflammatory, and pain-sensing systems (nociceptive).¹⁸ Suggested dose: Take 20 to 40 mg daily. Possible interactions: Research shows even high doses of up to 1,500 mg a day are well tolerated. However, if taken with other medications high doses may cause liver damage. Side effects may include diarrhea, appetite changes, and fatigue.¹⁹

Curcumin/Turmeric Root (*Curcuma longa*):

Turmeric contains curcumin. Research has repeatedly demonstrated curcumin's antioxidant, anti-inflammatory, and immunomodulatory properties²⁰. It has the potential to help treat a wide variety of inflammatory diseases including cancer, diabetes, cardiovascular diseases, arthritis, Alzheimer's disease, psoriasis, etc., through modulation of numerous molecular targets²¹. It also impacts pain. A 2021 study demonstrated improvements in both pain and improved physical movement for osteoarthritis²². It is an effective adjunct for neurologic pain relief²³ and neuropathic pain.²⁴ Researchers believe curcumin works by suppressing the NF-κB pathway²⁵. This pathway is implicated in a variety of pain conditions such as rheumatoid arthritis, migraines, and nerve injury²⁶. Choose a supplement formulated with black pepper (piperine) to enhance bioavailability, and take curcumin with fat to enhance absorption. Suggested dose: 750 mg following breakfast and 750 mg following dinner.

Feverfew (*Tanacetum parthenium*): Feverfew has a long history of use in traditional Greek medicine for migraines and toothaches.²⁷ While research has demonstrated its anti-inflammatory and anti-migraine properties, newer research has focused on its use for treating general pain. Feverfew extract behaves as a potent pain reliever in acute, inflammatory, articular, and neuropathic pain.²⁸ Suggested dose: Take 100 mg daily. Possible interactions: Blood-thinning medications such as warfarin (Coumadin), clopidogrel (Plavix), or aspirin and medications broken down by the liver.²⁹

Reishi (*Ganoderma lucidum*): Reishi mushroom is a formidable adaptogenic anti-inflammatory, targets inflammatory cytokines, helps reduce pain in rheumatoid arthritis, and helps modulate the immune system^{30 31 32}. Reishi is a good choice for inflammatory, arthritic, and autoimmune disorders that cause pain^{34 35}. Suggested dose: 2 dropperfuls of a dual-extracted reishi tincture, twice daily.

St. John's Wort (*Hypericum perforatum*): One of the most commonly used herbal medications in the United States, St. John's wort (SJW) is used internally and externally for nerve pain and depression.³⁶ The main active ingredient, hyperforin, has many effects on the nervous system including inhibiting the re-uptake of several neurotransmitters.³⁷ St. John's wort can reduce neuropathic pain.³⁸ Neuropathic pain is caused by damage or injury to the nerves that transfer information between the brain and spinal cord from the skin, muscles, and other parts of the body. It is typically treated with opioid drugs.³⁹ SJW can be used alone or in combination with opioid medications. Utilizing SJW may reduce dependence on opioid medications and prevent addiction.⁴⁰ Suggested dose: Take 900mg daily or 3 dropperfuls of tincture three times daily.⁴¹ It may also be used topically.

Possible interactions: SJW interferes with enzymes involved in drug metabolism, (CYP3A4, CYP2C9), which may decrease the effectiveness of certain drugs, including birth control and HIV drugs.⁴² SJW shouldn't be used with other medications that alter neurotransmitter levels (*e.g.*, certain antidepressants) as it has the potential to induce serotonin syndrome.⁴³ Ask your doctor before taking St. John's wort with any western medication.

White Willow (*Salix alba*): Willow bark has been used for thousands of years; people chew on the bark to reduce fever and inflammation.⁴⁴ Willow contains salicylic acid, which was used as a precursor to make aspirin (acetylsalicylic acid). Acetylsalicylic acid was patented by the Bayer Pharmaceutical Company in the 1800s and is the most common household drug.⁴⁵ Salicylic acid inhibits cyclooxygenase enzymes (COX) involved in making prostaglandins that stimulate inflammation and fever.⁴⁶ People commonly use willow bark for back pain, osteoarthritis, fever, flu, muscle pain, and many other conditions. In contrast to synthetic aspirin, willow bark does not damage the gastrointestinal mucosa.⁴⁷ Willow bark is especially effective for back pain.⁴⁸ Suggested dose: Take 120 mg daily. Possible interactions: Blood-thinning medications such as warfarin (Coumadin), Aspirin. Contraindicated in pregnancy, lactation, and for children as it may cause Reye's syndrome.

Below are listed external analgesics and anti-inflammatories:

Capsaicin/Cayenne (*Capsicum annuum*): Capsaicin, a constituent of cayenne peppers, is an effective external pain treatment. Capsaicin binds to "hot" receptors in the skin to produce a sensation of heat, which in turn relieves the perception of pain. Research shows its effectiveness

in treating neuropathic and soft-tissue pain.^{49 50} Due to its high concentration, for topical use capsaicin should always be diluted in a cream or salve.

Inhalation of Lavender (*Lavandula angustifolia*): Lavender is well known for its calming, anti-inflammatory, and antioxidant effects.⁵¹ Inhalation of lavender oil may relieve pain. In a study investigating migraine headaches, inhaling lavender essential oil resulted in significant pain relief.⁵²

Peppermint (*Mentha piperita*) Oil: Topical use of peppermint oil is effective in treating pain.⁵³ In contrast to capsaicin, peppermint oil binds to "cold" receptors in the skin that produces a sensation of cooling, relieving minor pain and irritation. Researchers believe that menthol, a constituent of peppermint, gives it a cooling effect. Research has shown peppermint oil to be effective in treating a number of conditions including neuralgia, headaches, and arthritis.^{54 55} Due to its high concentration, peppermint oil should always be diluted before topical use.

I make a Joint and Movement salve that curbs pain and inflammation. It is an outstanding remedy for arthritis, muscle and joint pain, carpal tunnel, general aches and pain, nerve pain, and tennis elbow. Apply 3x/day. It contains: organic olive oil infused with Calendula, Arnica, St. John's Wort, Cottonwood Buds (Balm of Gilead), and Cayenne, plus organic beeswax, organic Helichrysum essential oil, organic Frankincense essential oil, and vitamin E. Specific herbal uses are below:

Arnica: Anti-inflammatory, stimulates blood flow, and analgesic. Excellent for pain, strains, sprains, bruising, carpal tunnel, and swelling.

Calendula: Reduces pain by soothing inflammation. Supports cell rejuvenation.

Cayenne: Rich in capsaicin to help relieve arthritis, joint, muscle, tendon, and diabetic neuropathy pain. Works best with regular use (see above section).

Cottonwood Buds (Balm of Gilead): Anti-inflammatory, antibacterial, antifungal, and analgesic. Rich in salicylic acids for pain relief and to reduce swelling.

Helichrysum and Frankincense (*Boswellia*): Provides effective pain relief through strong anti-inflammatory action and may fight cartilage loss. See above for internal use.

St. John's Wort: Consistent external use reduces inflammation and pain, including nerve pain. See internal uses above.

2. Diet Anti-Inflammatory Diet: Nutrition plays a major role in pain management. Persistent low-grade systemic inflammation is associated with chronic pain, while healthy eating patterns are associated with less systemic inflammation and reduced pain.⁵⁶ Carefully selecting foods that are anti-inflammatory while avoiding foods that are pro-inflammatory can effectively reduce pain and improve quality of life.⁵⁷

Incorporating anti-inflammatory foods such as fruits, vegetables, fatty fish, antioxidants, and fiber leads to a reduction in pain severity.⁵⁸ Visit the Anti-Inflammatory Diet chapter for full details and dietary suggestions.

Microbiome Support: Gut microbiota may regulate pain. The gut microbiome is a crucial modulator of visceral pain, vague pain that occurs in the abdomen, chest, intestines, or pelvis. Recent evidence suggests that gut microbiota may also play a critical role in many other types of chronic pain, including inflammatory pain, headache, neuropathic pain, and opioid tolerance.⁵⁹ This is likely due to the gut-brain axis, a term for the communication network between the gut and brain. Through the largest nerve in the body, the vagus nerve, and over 500 million neurons, your gut is in constant communication with the brain, impacting numerous body processes.⁶⁰ Researchers believe that signaling molecules from the gut microbiota regulate the peripheral and central nervous system, which in turn mediate the development and perception of chronic pain.⁶¹ Having a healthy microbiome can be supported through eating probiotic foods that support the growth of good bacteria in the gut and by taking supplements. See the Leaky Gut Protocol for details.

Dark Chocolate: Dark chocolate has a positive effect on pain. Women who ate between 40–120 grams of dark chocolate daily during their menstrual period experienced reduced pain.⁶² Chocolate is made from cacao beans, which exert anti-inflammatory and antioxidant actions that may have a positive effect on pain.⁶³ Cacao releases enkephalin, an endorphin-like peptide that acts on opiate receptors in the brain, reducing pain.⁶⁴

Ginger Root Tea (*Zingiber officinale*): Ginger possesses anti-inflammatory properties and is a natural pain reliever.⁶⁵ Researchers believe ginger modulates pain through the inhibition of prostaglandins and inflammatory cytokines. Ginger can be added to food, consumed as tea, and/or taken as a supplement.

3. Lifestyle Changes **Acupuncture:** Acupuncture is a treatment that has been used in Traditional Chinese Medicine for thousands of years. A trained acupuncturist inserts very thin steel needles into the skin to stimulate specific points in the body.⁶⁶ Research repeatedly shows that acupuncture is an effective treatment in pain management. Acupuncture enhances the release of the body's natural opiates, such as dynorphin, endorphin, and enkephalin, and release corticosteroids, relieving pain and enhancing the healing process.⁶⁷ A common treatment plan for a single complaint would typically involve one or two treatments a week. The number of treatments will depend on the condition being treated and its severity. In general, it's common to receive 6 to 8 treatments. Treatments are typically 30-60 minutes.⁶⁸ It's important to work with a qualified acupuncturist.

Chiropractic Care: Around 7.7 million Americans each year find back pain relief through chiropractic care. This profession focuses on the spine and joints and their relationship to the body's nervous

system. A chiropractor manually adjusts your body and may also integrate massage and electrical stimulation. Chiropractic adjustments help improve your range and quality of motion and help reduce inflammation and pain. A chiropractor may also incorporate exercise and a rehabilitation plan. Your initial assessment may include an X-ray to help them target the root cause of your pain.

Epsom Salt Bath (Magnesium Sulfate): Taking baths improves physical and emotional health. Bathing activates the parasympathetic nervous system, a part of the body responsible for relaxation.⁶⁹ After taking saltwater baths, patients report significant reductions in pain.⁷⁰ Hot baths are especially effective for low back pain.⁷¹ Epsom salts made from magnesium sulfate enhance muscle relaxation, and soaking in Epsom salts increases blood magnesium concentrations.⁷² Add 1-2 cups of Epsom salts to a warm bath.⁷³

Light Exercise: Light or gentle exercise can be an effective way to reduce chronic pain, and over time help those with chronic pain engage in more essential and enjoyable activities of daily living.⁷⁴ Exercise changes how the brain responds to pain by normalizing the pain signal process and promoting the release of analgesics, such as natural pain relievers and serotonin that turn off pain signals.⁷⁵ Gentle exercises such as water aerobics, gentle yoga, walking, Tai Chi, and Qigong are effective interventions used to prevent and treat pain.⁷⁶

Mindfulness Meditation: Research repeatedly shows that mindfulness-based meditation significantly reduces pain.^{77 78} Mindfulness meditation involves cultivating a heightened state of awareness of the present moment without judgment. It often involves body scanning and paying attention to parts of the body and bodily sensations in a gradual sequence from feet to head. For pain relief, experts advise practicing it every day for 45 minutes, even if it seems boring or doesn't seem to be initially helping.⁷⁹

Sleep: The link between sleep and pain is well documented. When you are asleep, the body repairs cells, restores energy, and releases important hormones and proteins.⁸⁰ Growth hormone is released during sleep, which helps with healing, muscle repair, and growth of new tissue.⁸¹ Improvements in sleep quality and duration reduce pain in people with chronic pain,⁸² while sleep deprivation dramatically increases pain sensitivity.⁸³ Poor sleep not only affects general health but has a direct impact on inflammation, pain response, and emotional well-being. Improving sleep in people living with chronic pain has the potential to dramatically increase their quality of life.⁸⁴ Often people with chronic pain have an increased need for sleep, so ensure you have at least 7 to 9 hours of sleep per night. Please see the Sleep Protocol for more detail.

4. Vitamins and Minerals **Vitamin D (Cholecalciferol):** New research suggests Vitamin D may play a role in the signaling pathways involved in chronic pain.⁸⁵ Several studies have shown low Vitamin levels are associated with a higher incidence of acute and chronic pain.⁸⁶ Additionally, inadequate levels of vitamin D are associated with higher use of narcotic

medication by patients with chronic pain.⁸⁷ While it isn't exactly clear what role this vitamin plays, scientists believe it may influence immune cells and pain sensitization.⁸⁸ Vitamin D deficiency is extremely common; up to 40% of U.S. adults are considered to have insufficient levels.⁸⁹ Maintaining optimal vitamin D levels may prevent and reduce pain. Get your levels tested. Suggested dose: 2000 IU daily⁹⁰.

Vitamin B-12 (Methylcobalamin): Vitamin B-12 is a key player in the function, development, and maintenance of the nervous system. Methylcobalamin, the activated form of vitamin B12, can alleviate pain in diabetic neuropathy, low back pain, and neuralgia.⁹¹ This is likely due to its supportive role in the function of sensory and motor peripheral nerves, which perceive and respond to pain.⁹² Maintaining healthy B-12 levels is an important step in reducing chronic pain, especially nerve pain. Suggested dose: 50 mcg daily.

Fish Oil/Omega 3-s: Fish oils contain high amounts of omega-3 essential fatty acids, specifically EPA (eicosapentaenoic acid) and DHA (docosahexaenoic acid). These fatty acids modulate inflammation and the immune system. Research compared ibuprofen and omega-3 EFA fish oil for pain management and found they had an equivalent effect.⁹³ Fish oil appears to be a safer alternative to NSAIDs for the treatment of nonsurgical neck or back pain. People taking omega-3 fatty acids were more likely to be able to decrease the amount of pain medication they were taking.⁹⁴ Suggested dose is 1,500mg of EPA and 500mg of DHA.⁹⁵ Please keep ensure you buy high-quality fish oil that has been tested for heavy metals and keep it in the refrigerator; otherwise, it could go rancid quickly.

Magnesium: Magnesium is a cofactor in more than 300 enzymatic reactions in the body.⁹⁶ Magnesium intake is protective against chronic pain⁹⁷ and emerging evidence supports the use of magnesium in chronic pain management. The use of magnesium for the treatment of chronic pain has been studied in patients with migraine headaches, complex regional pain syndrome, neuropathic pain, chronic low back pain, and fibromyalgia.⁹⁸ Magnesium exerts its analgesic effects by blocking the N-methyl-d-aspartate (NMDA) receptors located on the postsynaptic spinal neurons in the dorsal horn of the spinal cord.⁹⁹ Magnesium bis-glycinate is an easily absorbed form of magnesium. Suggested dose: 320 mg nightly¹⁰⁰.

Probiotics: Probiotics improve outcomes in digestive, emotional, and pain-related disorders.¹⁰¹ By increasing the number of beneficial bacteria in your small and large intestine, probiotics aid digestion and absorption of nutrients, ultimately reducing inflammation.¹⁰² Suggested dose: 10-12 billion colony-forming units daily.

Vitamin C: Vitamin C is vital to a multitude of metabolic pathways, including those affecting tissue regeneration, stress response, and pain perception.¹⁰³ Vitamin C appears to be a safe and effective adjunctive therapy for acute and chronic pain and may ease opioid withdrawal

and drug cravings.¹⁰⁴ Administered orally in high doses, vitamin C has been observed to relieve pain and reduce opioid use in cancer patients.¹⁰⁵ Vitamin C may support the production of endorphins, the body's natural pain reliever.¹⁰⁶ Suggested dose: 500-1,000mg once daily with food.

5. Western Medical Approaches The following pharmaceutical drugs are used to treat acute and chronic pain:¹⁰⁷

Acetaminophen: Used for mild to moderate pain, and temporary reduction of fever. Side effects are rare; however, overuse can result in liver damage and death. An estimated 458 deaths in the United States are due to acute liver failure from acetaminophen each year.¹⁰⁸

Nonsteroidal Anti-Inflammatory Drugs (NSAIDs): Used for mild-to-moderate pain, pain associated with inflammation, and temporary reduction of fever. Side effects include kidney damage and peptic ulcers, and they may also increase your risk for heart attack, stroke, or high blood pressure. It is estimated that between 5,000–16,500 deaths from gastrointestinal bleeding in the United States are a direct consequence of NSAID overuse.¹⁰⁹

Antidepressant Medications: Used as first line treatment for neuropathic pain to enhance mood and pain perception. Side effects include anxiety, restlessness, insomnia, dry mouth, weight gain, nausea, diarrhea, sweating, headaches, dizziness, decreased libido, tremor, and sexual difficulties.¹¹⁰

Antiepileptic Medications: Used for neuropathic pain. The most common antiepileptics used for pain treatment are gabapentin and pregabalin. Side effects include dizziness, drowsiness, and mental slowing; other side effects like weight gain, metabolic acidosis, nephrolithiasis, angle closure glaucoma, skin rash, hepatotoxicity, colitis, and movement and behavioral disorders.¹¹¹

Local Anesthetics: Used for peripheral neuropathic pain. Side effects include pain at the injection site, tingling sensation, minor bruising, bleeding, or soreness. In rare cases, there can be permanent paralysis of the area being treated.¹¹²

Opioid Agents: Used for mild to severe pain. These are the most widely used drugs in treating severe pain. Side effects include sedation, dizziness, nausea, vomiting, constipation, physical dependence, tolerance, respiratory depression, physical dependence, and addiction.¹¹³ In 2020, over 16,000 people in the United States died from overdosing on prescription opioids.¹¹⁴

6. What to Avoid **Inflammatory Foods:** Specific foods are known to exert strong effects on inflammatory pathways within the body. Research has repeatedly demonstrated that pro-inflammatory foods are associated with chronic pain, especially low back pain in adults.¹¹⁵ ¹¹⁶ Diets that promote inflammation are high in refined starches, sugar, saturated and trans-fats.¹¹⁷ These foods are

often highly processed.¹¹⁸ Processed foods usually have 5 or more ingredients, come from a package, contain a combination of sugar, starch, or unhealthy oils, are deep-fried, or are made with additives and preservatives. See the Anti-Inflammatory Diet section.

Food Sensitivities: Research shows food sensitivities create inflammation and pain in the body¹¹⁹ and are linked to migraine headaches.¹²⁰ To rule out food sensitivities, get an IgG food sensitivity test and/or try an elimination diet to reduce exposure to foods that commonly cause inflammation. Refer to the Elimination Diet chapter.

High-Stress Levels: The negative effects of stress on health outcomes on physiology and behavior, including pain, are well documented.¹²¹ Stress is strongly correlated with a poor prognosis in people with a wide array of pain conditions.¹²² Researchers believe that a prolonged or exaggerated stress response may perpetuate cortisol dysfunction, widespread inflammation, and pain.¹²³

This can create a cycle where stress and pain perpetuate one another. To prevent the detrimental effects of stress, eliminate or reduce unnecessary sources of stress in your life. This may mean creating healthy boundaries with people, avoiding stressful situations, and finding ways to cultivate a sense of calm in your life. See the Anxiety and Stress Protocol for details.

Smoking: Smoking is a major risk factor for pain. Smokers are nearly three times as likely to get lower back pain.¹²⁴ While the exact mechanism is unknown, smoking can lead to the development of chronic pain, likely to inflammatory processes, and pain can motivate further use of nicotine

or tobacco creating a maladaptive cycle.¹²⁵

By quitting or avoiding smoking you dramatically reduce your risk for chronic pain and a host of other health issues. Plus, quitting smoking adds roughly 10 years to life expectancy.¹²⁶

A Sedentary Lifestyle: Chronic pain is partly attributed to a sedentary and inactive lifestyle. There is a strong correlation between sedentary behavior and low back pain.¹²⁷ This association is partially attributed to prolonged periods of sitting. Even a little bit of movement helps reduce pain; small improvements in the amount of daily movement are better than none at all.¹²⁸ To offset the effects of prolonged sitting researchers recommend getting up and moving around, even briefly, every 30 minutes.¹²⁹

Pain is an enormous global health problem, affecting 1 in 5 people.¹³⁰ Pain interrupts daily living and dramatically reduces a person's quality of life. While the cause of pain is often unclear, finding safe effective treatment strategies is often elusive. Perhaps the most devastating outcome of this health crisis has been the emergence of the opioid epidemic.

From 1999-2020 there have been roughly 932,000 fatal overdoses from opioids.¹³¹ Despite this terrible situation, there is hope for the millions of people struggling with pain. Research has consistently demonstrated safe and natural treatments for pain relief. From medicinal herbs to nutritional supplements, an anti-inflammatory diet, and lifestyle changes, people can reduce their pain and enhance their well-being.

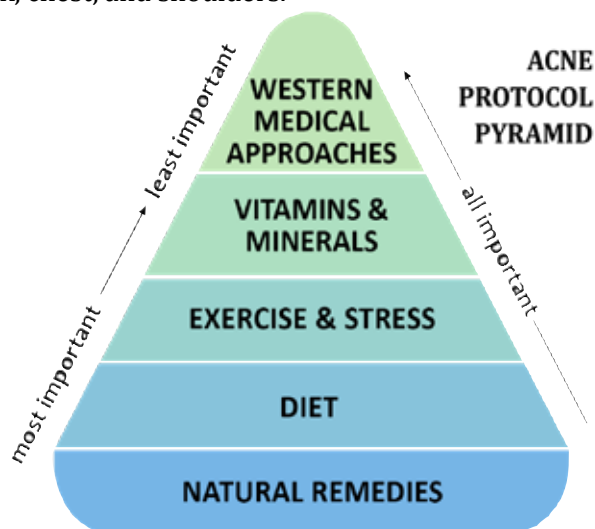
	Morning	Noon	Afternoon	Evening	Before Bed
S U N D A Y	Boswellia & Reishi* St. John's Wort Vit. D, C & B-12 Probiotics Curcumin after eating	Grocery shop for anti-inflammatory foods Capsaicin salve and/or Peppermint oil** Anti-inflammatory lunch Omega 3-s White willow	Feverfew & Reishi* Mindfulness meditation Light exercise	Anti-inflammatory dinner Curcumin after eating Dark chocolate (small amount) CBD	Ginger tea Epsom salt bath Lavender inhalation Capsaicin salve and/or Peppermint oil** Magnesium
M O N D A Y	Boswellia & Reishi* St. John's Wort Vit. D, C & B-12 Probiotics Curcumin after eating	Capsaicin salve and/or Peppermint oil** Anti-inflammatory lunch Omega 3-s White willow	Acupuncture Feverfew & Reishi Mindfulness meditation Light exercise	Anti-inflammatory dinner Curcumin after eating Dark chocolate (small amount) CBD	Ginger tea Epsom salt bath Lavender inhalation Capsaicin salve and/or Peppermint oil** Magnesium
T U E S D A Y	Boswellia & Reishi* St. John's Wort Vit. D, C & B-12 Probiotics Curcumin after eating	Capsaicin salve and/or Peppermint oil** Anti-inflammatory lunch Omega 3-s White willow	Feverfew & Reishi* Mindfulness meditation Light exercise	Anti-inflammatory dinner Curcumin after eating Dark chocolate (small amount) CBD	Ginger tea Epsom salt bath Lavender inhalation Capsaicin salve and/or Peppermint oil** Magnesium
W E D N E S D A Y	Boswellia & Reishi* St. John's Wort Vit. D, C & B-12 Probiotics Curcumin after eating	Capsaicin salve and/or Peppermint oil** Anti-inflammatory lunch Omega 3-s White willow	Feverfew & Reishi* Mindfulness meditation Light exercise	Anti-inflammatory dinner Curcumin after eating Dark chocolate (small amount) CBD	Ginger tea Epsom salt bath Lavender inhalation Capsaicin salve and/or Peppermint oil** Magnesium
T H U R S D A Y	Boswellia & Reishi* St. John's Wort Vit. D, C & B-12 Probiotics Curcumin after eating	Capsaicin salve and/or Peppermint oil** Anti-inflammatory lunch Omega 3-s White willow	Acupuncture Feverfew & Reishi Mindfulness meditation Light exercise	Anti-inflammatory dinner Curcumin after eating Dark chocolate (small amount) CBD	Ginger tea Epsom salt bath Lavender inhalation Capsaicin salve and/or Peppermint oil** Magnesium
F R I D A Y	Boswellia & Reishi* St. John's Wort Vit. D, C & B-12 Probiotics Curcumin after eating	Capsaicin salve and/or Peppermint oil** Anti-inflammatory lunch Omega 3-s White willow	Feverfew & Reishi* Mindfulness meditation Light exercise	Anti-inflammatory dinner Curcumin after eating Dark chocolate (small amount) CBD	Ginger tea Epsom salt bath Lavender inhalation Capsaicin salve and/or Peppermint oil** Magnesium
S A T U R D A Y	Boswellia & Reishi* St. John's Wort Vit. D, C & B-12 Probiotics Curcumin after eating	Capsaicin salve and/or Peppermint oil** Anti-inflammatory lunch Omega 3-s White willow	Feverfew & Reishi* Mindfulness meditation Light exercise	Anti-inflammatory dinner Curcumin after eating Dark chocolate (small amount) CBD	Ginger tea Epsom salt bath Lavender inhalation Capsaicin salve and/or Peppermint oil** Magnesium

*Make sure to use a dual-extracted Reishi tincture.

**Or use Joint & Movement salve 2-3x/day.

Ch. 38: Acne Protocol

Acne is the most common skin disease worldwide.¹ It is caused by an overproduction of sebum due to testosterone, excess production of skin cells, inflammation, and skin colonization of the gram-positive bacteria, *Cutibacterium acnes* (formerly known as *Propionibacterium acnes*). These factors lead to oil and dead skin cells clogging the pores, resulting in outbreaks or lesions (often called pimples or zits). Outbreaks usually occur on the face, back, chest, and shoulders.



While acne is typically associated with teenagers, recent data has revealed a growing incidence of adult female acne. No definitive explanation for this increase has been established, though stress appears to be a major influence². Acne usually subsides by age thirty-five but may persist through one's forties and fifties. PCOS and acne may appear together in women, signaling a hormonal imbalance. However, only a small percentage of women with acne will demonstrate true hyperandrogenism secondary to pathologic conditions such as PCOS. Acne flares in women may be related to increased testosterone levels induced by stress, menses, or pregnancy³.

Treating acne requires time as skin cells must go through a full cycle of shedding before clear skin can emerge. Human skin cells turn over every 2-4 weeks⁴, which is typically the minimum amount of time required to see skin changes. If a treatment works for you, you should notice some improvement between 4 weeks and 3 months.

Acne is associated with residual scarring and psychological issues, such as poor self-image, depression, and anxiety, which can negatively impact quality of life.⁵ Emotionally, acne can cause embarrassment and social isolation. Physically, acne can be a symptom of a more serious hormonal imbalance or food allergy. In advanced cases, it is wise to seek medical attention, hopefully with a holistic practitioner.

1. Natural Remedies Below you will find multiple natural oral treatments that help with acne:

Herbal Adaptogenic Tincture for Acne: This tincture helps decrease inflammation, kills acne-

causing bacteria, and assists with stress reduction and hormonal balance. If you have an autoimmune condition, talk to a naturopathic doctor before taking the tincture below, as some ingredients are immunostimulating, or replace Rhodiola and Echinacea with a double-extracted Reishi Mushroom tincture.

You will need: 2 ounces Rhodiola, 3 ounces Echinacea leaves/root, 3 ounces Holy Basil (Tulsi) leaves, 2 ounces Dandelion root, and 2 ounces Goji berries. Rhodiola is antioxidant, immunomodulator, and metabolic stabilizer. It regulates the nervous system and is an adaptogen and blood sugar stabilizer. Echinacea is an antibiotic that helps decrease pro-inflammatory cytokines⁶ ⁷. Dandelion purifies the blood and cleanses the liver. Tulsi is adaptogenic, anti-microbial, and antibiotic⁸. Goji berries are antioxidant and encourage a healthy microbiome. Suggested dose: Take one dropperful of this tincture three times daily.

How to make the tincture: Add your dried herbs to a large glass jar, filling it halfway. Then add 40% (80-proof) vodka, leaving a space of ½" at the top of the jar. Seal and label with date and ingredients. Let this mixture infuse for 6-8 weeks, shaking daily. When ready, strain using cheesecloth. Pour into a tincture dropper bottle for ease of use.

Chaste Tree (*Vitex agnus-castus*): Chaste Tree fruits are widely used to treat premenstrual symptoms and researchers have found a reduction in acne in those taking this herb⁹. It is hypothesized that Vitex works by decreasing levels of the hormone prolactin. This helps rebalance other hormones that may cause acne. Suggested dose: Take 160 mg of dried extract once daily, either in the morning away from food, or at bedtime.

Black Cohosh (*Actaea racemosa*): Black Cohosh has phytoestrogens; plant compounds that mimic estrogen in the body. By binding to human estrogen receptors in the body, testosterone, a major contributor to acne, is reduced. Suggested dose¹⁰: 80 mg tablets of a standardized extract taken once a day.

Borage Oil (*Borago officinalis*): Research has shown Borage oil's ability to regulate hormones and reduce acne¹¹. It's thought that the high gamma-Linoleic acid content is converted by the body into prostaglandins, reducing inflammation and acne lesions. Suggested dose: 1000-1300 mg daily.

DIM (Diindolylmethane): DIM is a natural substance generated when the body breaks down indole-3-carbinol, a compound found in cruciferous vegetables such as broccoli and kale. Available in supplement form, DIM offers a potential means of controlling acne due to its antibiofilm and antibiotic properties¹². Suggested dose: 200 mg of DIM in supplement form per day.

Reishi (*Ganoderma lucidum*): Reishi significantly reduces 5-alpha reductase levels, preventing the conversion of testosterone into DHT, which binds oil gland androgen receptors, stimulating excessive sebum (oil) production and bacterial growth, which leads to

acne. Reishi is also an adaptogen, helping with the stress response that also leads to acne breakouts. Suggested dose¹³: 2 dropperfuls of a dual-extracted tincture once to twice daily.

Below you will find multiple natural topical treatments that help with acne:

Cleanser: Wash the face gently with a mild cleanser, such as Cerave (pH of 5.5) or Cetaphil, (pH of 6.5 - both available at pharmacies or online). Acne-causing bacteria thrive in basic environments, so it's important to ensure your skin is kept at a healthy pH. The recommended pH to treat acne and maintain the skin's acid mantle is 4.5-6.5. The acid mantle is a fine, slightly acidic film on the surface of human skin acting as a barrier to bacteria, viruses, and other potential contaminants that might penetrate the skin.¹⁴ It also maintains the skin's flora; maintaining a healthy and balanced skin microbiome.

Toner: Toner further cleanses your skin and prepares it for topical treatments. Use natural floral water mixed with 1/6 Witch Hazel (*Hamamelis virginiana*). Natural floral waters consist of distilled waters that contain floral extracts like rose (*Rosa damascena*) or the antibacterial yarrow (*Achillea millefolium*). Research has demonstrated their ability to calm inflamed skin¹⁵ and protect the it from damage associated with inflammation.¹⁶ Witch Hazel has astringent properties that assist in closing pores, which prevents clogging and debris infiltration.¹⁷ All these products can be purchased at local health food stores.

Raw Honey/Orange Peel Face Mask: Combine 1 teaspoon of raw honey with ½ teaspoon of grated orange zest. Gently smooth this paste over the affected area. Let it sit for 20 minutes. Gently wash it off and dry.

Natural Yogurt: Putting raw yogurt on the face has been shown to reduce acne lesions. This is likely due to its high *Lactobacillus* content, which inhibits growth of acne-causing bacteria.¹⁸ Apply yogurt to the face, leave on for 20 minutes and wash off.

Propolis Skin Cream: Apply propolis skin cream daily, there are many commercially available. Do not touch or try to pop or scrape your acne sores.

Tea Tree (*Melaleuca alternifolia*): Tea tree oil contains approximately 100 different compounds. Tea tree oil reduces skin inflammation by inhibiting histamine release and has broad-spectrum antibiotic activity against both Gram-negative and Gram-positive bacteria¹⁹. Gels containing at least 5 percent tea tree oil may be as effective as lotions containing 5 percent benzoyl peroxide, although tea tree oil may work more slowly. Possible side effects include minor itching, burning, redness, and dryness. Tea tree oil should be used only topically²⁰. Dilute the essential oil or use a topical gel with tea tree.

Yarrow (*Achillea millefolium*): Yarrow is a well-known anti-bacterial and anti-inflammatory herb often used for wound healing. Topically it reduces acne-causing bacteria²¹. Yarrow may be purchased as a salve, gel, or moisturizer and applied topically.

2. Diet Recent research suggests that diet plays a central role for acne and skin health.²²

Anti-Inflammatory Diet: Inflammation is known to create and exacerbate acne. It's important to avoid pro-inflammatory foods in the diet, which can trigger cytokines that create acne. Avoid the following: sugar, dairy, gluten, processed and fried foods, red meat and pork, heavily processed soy products, alcohol, omega-6 fatty acids (vegetable oil, corn oil, sunflower oil, cottonseed oil). See our Anti-Inflammatory Diet section for more information.

Low Glycemic Diet: Research has demonstrated a strong link between acne and a high glycemic load (HGL) diet.²³ The glycemic index has to do with a food's potential to increase blood glucose and insulin levels. Research reports the absence of acne in populations where people do not consume high HGL diets. Several studies have shown low glycemic diets improve acne²⁴. When blood sugar spikes, it causes a cascade involving insulin and inflammation. It's important to keep your blood sugar stable.

Lean Towards a High Fiber Diet: Eat fruits and vegetables, vegetable protein (nuts, seeds & legumes), and supplement with essential fatty acids (fish oil, flax).

Avoid Dairy: Research shows a strong link between acne and the consumption of dairy products.²⁵ High amounts of dairy are mucus-producing and also high in sugar. Oat milk, coconut milk, or almond milk are great alternatives. Higher quality cheeses are all right sometimes but not habitually.

Microbiome Support: A growing body of evidence suggests that acne sufferers benefit from microbiome support. A gut-brain-skin hypothesis suggests that alternations in the microbial flora of the GI tract lead to local and systemic inflammation. Several studies have demonstrated the ability of probiotics like *Lactobacillus* to reduce skin inflammation and improve stress-induced skin inflammation that cause acne²⁶. Incorporating probiotic-rich foods into your diet is an excellent way to optimize your digestion and reduce inflammation. Sources of dietary probiotics include kombucha, kimchi, sauerkraut, and miso broth. A probiotic supplement is also recommended. As a dessert alternative, cook apples, pears, strawberries, blackberries, raspberries, and blueberries with ⅓ teaspoon cinnamon. These berries combined with cinnamon stabilize blood sugar levels. You may add stevia to this if you like.

Pumpkin Seed Oil: The oil found in pumpkin seeds is a powerful testosterone blocker. Pumpkin seed oil blocks the action of 5-alpha reductase in the body through the action of beta-sitosterol, a healthy plant fat. Blocking this steroid hormone means that testosterone can't metabolize into its potent acne-causing form called Dihydrotestosterone, or DHT. Pumpkin seed oil is also high in zinc. It makes a great oil for salad dressing or the seeds flavor food in a trail mix, or as nut butter.

Hydration: Adequate clean water intake is imperative for healthy digestion as a lubricant and detoxifier. It reduces your cravings and helps stabilize blood sugar. Drinking water is primarily thought to prevent acne

by promoting proper skin hydration. Dry skin can trigger excess oil production, which can contribute to acne²⁷. Suggested dose: 12 cups (3 liters) to 15 cups (4 liters) of fluids a day²⁸.

Spearmint Tea: Spearmint tea is helpful for hormonal acne. Studies performed on women show testosterone levels were significantly reduced over 30 days after drinking spearmint tea twice a day. Testosterone is a driving force in some women's acne, especially adult acne that typically appears along the jawline and which flares during menstruation and stress²⁹. Suggested dose: Drink one to two cups of spearmint tea during or after dinner.

Clear Skin Smoothie Recipe: You will need: 1 cup kale stripped from the stem and washed, 1/2 cup berries (blueberries or strawberries), 1 cup unsweetened nut milk, 2 tbsp. pumpkin seeds, 2 tbsp. flaxseed, 1 tbsp. bee pollen, optional: 1/2 banana for flavor. Add all ingredients into a blender, blend, and enjoy!

Indole-3-carbinol is present at relatively high levels in cruciferous vegetables such as kale. Indole-3-carbinol offers a potential means of controlling acne due to its antibiofilm and antibiotic properties³⁰. Strawberries contain slow-release carbohydrates that do not cause blood sugar spikes. The glycemic index in strawberries is below 55, which makes it a low-sugar food. Additionally, they are rich in antioxidants and phytochemicals that support the skin³¹. Nut milks, like almond milk, are a great alternative to dairy, which has been linked to acne.³² Almond milk also has anti-inflammatory and anti-oxidant properties.³³

Pumpkin Seeds: Pumpkin seeds contain oil that is a powerful testosterone blocker, which is a known cause of acne. Additionally, pumpkin seeds are high in zinc, an important mineral in the reduction of acne³⁴.

Flaxseed: Flaxseeds are high in omega-3s, which help reduce inflammatory acne. Additionally, flaxseed has been shown to regulate sex hormones that exacerbate acne³⁵.

Bee Pollen: Bee pollen contains several vitamins, minerals, antibiotics, and antioxidants that support overall health. It also fights inflammatory acne through its potent anti-inflammatory activity³⁶.

Food Sensitivities: Food sensitivities are commonly linked to acne. To rule out food sensitivities, you may want to pursue IgG food sensitivity testing. Also, an elimination diet will reduce exposure to foods that commonly cause acne. These include sugar, dairy, eggs, alcohol, gluten, and yeast. See our Elimination Diet for more info.

Meal planning is essential. Create a healthy seven-day meal plan and a shopping list. Shop weekly and prepare your vegetables weekly on one prep day. This strategy saves time, prevents unhealthy snacking patterns, and prevents low blood sugar from dictating food choices. It can even save money. Reuse your weekly plans and rotate them with others so you don't get stuck in a food rut. Keep a food diary to see what triggers your acne.

functional medical physician specializing in acne can help you evaluate possible allergies and hormonal imbalances.

3. Exercise and Stress

Do at least thirty minutes of cardiovascular exercise daily, one hour is ideal (5,000 to 8,000 steps per day). Best done outside if possible. Add in Yoga 2-3x/week as well. Plan activities outdoors as much as you can. Breathe fresh air. Get sunlight.

Stress: Stress has repeatedly been shown to increase acne, especially in adult women. Research shows that increase in severity of stress is strongly correlated with an increase in acne severity³⁷. During times of stress, your body releases a number of hormones including cortisol, corticotrophin-releasing hormone, adrenaline, and androgens. These hormones can create insulin dysregulation, imbalanced ratios of male to female sex hormones, inflammation, and excessive sebum production, all of which contribute to or cause acne.

Try reducing stress through Deep Lymphatic Breathing. Gradually work your way from 5 minutes of deep breathing 3x daily to 15 minutes 3x daily. While you breathe focus on positive thoughts like a daily inspirational quote, prayer, mantra, or a soothing visual such as a candle flame or ocean waves.

Use an Imperative/Important Daily Task List to help you prioritize realistically and let go of what is not imperative. Mindfully focus on your priorities, one step at a time.

Simplify/Delegate or Eliminate tasks from your daily list that interfere with your mental and physical health and learn to say no to new tasks.

It may be important to see a medical practitioner to test your hormones. Hormonal imbalance is often indicated in more serious acne cases.

Consistent and steady sleep patterns are critical for hormonal regulation. Sound sleep and healthy eating patterns go hand in hand. Please cross-reference the Sleep Protocol. You may also wish to refer to the Anxiety Protocol or the Depression Protocol if those issues pertain to you.

4. Vitamins and Minerals

Vitamin A: Vitamin A is a fat-soluble nutrient that promotes skin development and fights inflammation. Multiple studies have linked low vitamin A levels to the presence of acne³⁸. Suggested dose³⁹: 900 mcg for men and 700 mcg for women per day.

Vitamin B3 (Niacin): Niacin is a water-soluble vitamin known for its anti-inflammatory properties and regulation in oil production. It's an important vitamin for skin health. Research shows that supplementing with niacin reduces acne⁴⁰. Suggested dose⁴¹: 250 mg per day before eating.

Vitamin D: Vitamin D is a fat-soluble vitamin that helps the body absorb calcium and is essential to bone health. It also plays an important role in the body's ability to fight off germs, like acne-causing bacteria. Research shows that people with acne have lower levels of vitamin D than those without acne and that a Vitamin D deficiency

may be a factor in the development of acne⁴². Suggested dose⁴³: 800 IU/day.

Zinc: Zinc is an essential mineral that plays a key role in multiple aspects of your health. Studies show that zinc decreases oil production in the skin and protects against bacterial infection and inflammation⁴⁴. Suggested dose⁴⁵: 15–30 mg of elemental zinc with food to prevent nausea.

5. Western Medical Approaches

The goals of treatment

are to help heal lesions, stop new lesions from forming, and prevent scarring. Typically, doctors will prescribe medications, medicated topical solutions, or laser light therapy for acne. The topical antibiotic clindamycin is often used successfully for acne. Surgical chemical peels remove the outer layer of the skin. Surgical procedures can remove some scars from acne lesions⁴⁶.

The following are typical antibiotics treatments for acne:

- Mild acne: External Clindamycin - this works well for

many with minimal side effects.

- Moderate acne: Doxycycline
- Severe acne: Isotretinoin

Other treatments include hormonal therapy to reduce free testosterone, medicated topical solutions such as benzoyl peroxide and salicylic acid, antibiotics such as tetracycline, and laser light therapy.

Treatment failure is high, with approximately 80% of women failing multiple courses of systemic antibiotic medications and approximately 30% to 40% failing after a course of isotretinoin⁴⁷. Additionally, antibiotic acne treatments contribute to the global problem of antibiotic resistance⁴⁸.

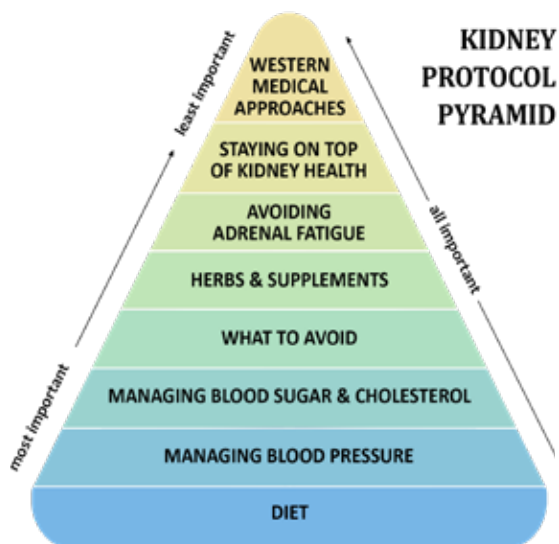
Some functional medical doctors have started to take a more holistic approach and connect the root causes of acne. A holistically trained physician will help you evaluate possible allergies and hormonal imbalances. In general, holistic approaches will focus more on a diet, lifestyle changes, supplementation, and hormonal support.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Herbal acne tincture Cleanser & Yarrow toner Propolis skin cream Vit. A, B3, & D DIM Reishi tincture	Weekly food plan Weekly food prep ½ - 1 h. cardio (outside if possible) Clear skin smoothie Vitex (Chaste Berry) Black Cohosh	Borage oil Herbal acne tincture Lymphatic breathing	Herbal acne tincture Yoga Spearmint tea Zinc w/ food Reishi tincture	Cleanser & Yarrow toner Honey/Orange/Yogurt face mask Deep lymphatic breathing Topical Tea Tree
MONDAY	Herbal acne tincture Cleanser & Yarrow toner Propolis skin cream Vit. A, B3, & D DIM Reishi tincture	½ - 1 h. cardio (outside if possible) Clear skin smoothie Vitex (Chaste Berry) Black Cohosh	Borage oil Herbal acne tincture Lymphatic breathing	Herbal acne tincture Yoga or meditation Spearmint tea Zinc w/ food Reishi tincture	Cleanser & Yarrow toner Honey/Orange/Yogurt face mask Deep lymphatic breathing Topical Tea Tree
TUESDAY	Herbal acne tincture Cleanser & Yarrow toner Propolis skin cream Vit. A, B3, & D DIM Reishi tincture	½ - 1 h. cardio (outside if possible) Clear skin smoothie Vitex (Chaste Berry) Black Cohosh	Borage oil Herbal acne tincture Lymphatic breathing	Herbal acne tincture Yoga Spearmint tea Zinc w/ food Reishi tincture	Cleanser & Yarrow toner Honey/Orange/Yogurt face mask Deep lymphatic breathing Topical Tea Tree
WEDNESDAY	Herbal acne tincture Cleanser & Yarrow toner Propolis skin cream Vit. A, B3, & D DIM Reishi tincture	½ - 1 h. cardio (outside if possible) Clear skin smoothie Vitex (Chaste Berry) Black Cohosh	Borage oil Herbal acne tincture Lymphatic breathing	Herbal acne tincture Yoga or meditation Spearmint tea Zinc w/ food Reishi tincture	Cleanser & Yarrow toner Honey/Orange/Yogurt face mask Deep lymphatic breathing Topical Tea Tree
THURSDAY	Herbal acne tincture Cleanser & Yarrow toner Propolis skin cream Vit. A, B3, & D DIM Reishi tincture	½ - 1 h. cardio (outside if possible) Clear skin smoothie Vitex (Chaste Berry) Black Cohosh	Borage oil Herbal acne tincture Lymphatic breathing	Herbal acne tincture Yoga Spearmint tea Zinc w/ food Reishi tincture	Cleanser & Yarrow toner Honey/Orange/Yogurt face mask Deep lymphatic breathing Topical Tea Tree
FRIDAY	Herbal acne tincture Cleanser & Yarrow toner Propolis skin cream Vit. A, B3, & D DIM Reishi tincture	½ - 1 h. cardio (outside if possible) Clear skin smoothie Vitex (Chaste Berry) Black Cohosh	Borage oil Herbal acne tincture Lymphatic breathing	Herbal acne tincture Yoga or meditation Spearmint tea Zinc w/ food Reishi tincture	Cleanser & Yarrow toner Honey/Orange/Yogurt face mask Deep lymphatic breathing Topical Tea Tree
SATURDAY	Herbal acne tincture Cleanser & Yarrow toner Propolis skin cream Vit. A, B3, & D DIM Reishi tincture	½ - 1 h. cardio (outside if possible) Clear skin smoothie Vitex (Chaste Berry) Black Cohosh	Borage oil Herbal acne tincture Lymphatic breathing	Herbal acne tincture Yoga Spearmint tea Zinc w/ food Reishi tincture	Cleanser & Yarrow toner Honey/Orange/Yogurt face mask Deep lymphatic breathing Topical Tea Tree

Ch. 39: Kidney Protocol

Our two kidneys are located below the ribs and behind the belly on the left and right side of the spine, in a region called the retroperitoneal space.¹ These red/brown organs reach about 4 ½ inches (11cm) in length. They control the volume of various bodily fluids like blood and urine. They also control osmolality, pH balance, and electrolyte concentrations. One of the most important functions of the kidneys is to filter toxins.

The kidneys receive blood from the renal arteries, filtering about 150 quarts of blood daily using nephrons; this blood then exits the kidneys via the renal veins. The kidneys have ureters that carry excreted urine (~ 1-2 quarts/day) to the bladder for removal. The kidneys allow for reabsorption of some substances and the excretion of others. Some reabsorbed substances include water, sodium, amino acids, and glucose. Excreted substances include ammonium, potassium, uric acid, and hydrogen. The kidneys also convert vitamin D to its active form, and synthesize the hormones erythropoietin, and renin.



Because the kidneys are responsible for the filtration of toxins from the body, many people in today's world suffer from kidney-related issues due to overexposure to toxins in the food they eat, products they use, and environmental toxins. Chronic kidney disease is one of the leading public health problems in the world. Kidneys are one of the most in-demand organs with many waiting on transplant lists. Globally, it is estimated that 13.4 percent of the population suffers from chronic kidney disease and around five to seven million people need renal replacement therapy.

Your adrenal glands are located on top of each kidney. They are part of the endocrine system and are responsible for the production of hormones that help regulate metabolism, electrolytes (sodium and potassium), blood pressure, the immune system, and the body's response to stress. They interact with the kidneys as your adrenal glands make the hormone aldosterone, which signals the kidneys to absorb sodium and release potassium, thus helping regulate blood pH and blood pressure. Please see the Adrenal Fatigue Protocol Chapter for more information.

Symptoms of kidney disease include:

Trouble Sleeping: When the kidneys cannot filter properly, the toxins that are normally excreted stay in the blood. This can affect sleep. There is also a link between chronic kidney disease and sleep apnea.

Fatigue, Low Energy, and Trouble Focusing: When toxins build up in the blood, some people experience fatigue and have trouble concentrating. Anemia is linked with kidney disease and can also cause fatigue.

Frequent Urination, Especially at Night: Kidney disease can cause damage to the filters and increase the urge to urinate. This also leads to frequent urinary tract infections and an enlarged prostate (for men).

Dry, Itchy Skin: Dry and itchy skin may be the result of a loss of minerals and bone disease that accompanies kidney disease. When the kidneys aren't able to keep the right balance of nutrients and minerals, this manifests from the inside out.

Blood in Urine: This is one of the most common signs of a kidney issue; see a health professional right away. Healthy kidneys keep blood cells in the body when filtering waste from the blood to make urine, but when the filters are damaged, blood cells can leak into the urine. Blood in the urine may also be a sign of an infection, kidney stones, or tumors.

Swollen Ankles and Feet: The body may retain more sodium when the kidneys are damaged, leading to swelling in the feet and ankles. Swelling in the lower regions of the body may also be a sign of heart or liver damage.

Foamy Urine: If you notice an excessive number of bubbles in your urine, this may indicate a kidney issue. An excessive amount would present like having to flush multiple times for the bubbles to go away. This foam may look like the white bubbly foam you see when scrambling eggs. This is because this protein (called albumin) is the same protein found in eggs.

Decreased Appetite: When the body has an issue, one of the first symptoms is a decreased appetite. The body wants to conserve its energy to healing the issue, not digesting food. The buildup of toxins in the body (caused by poor kidney function) can cause a dramatic decrease in appetite.

Puffiness Around the Eyes: If you notice persistent puffiness around the eyes, this could indicate an issue with the kidneys.

Muscle Cramps: Because the kidneys help control the balance of electrolytes in the body, an imbalance due to kidney disease can result in low calcium levels and poorly controlled phosphorus. This causes muscles to cramp, often for no reason.

Feeling Cold: Sometimes kidney disease can cause anemia, which is known to make one feel cold, even in warm weather.

Shortness of Breath: When the kidneys cannot function properly, this may cause extra fluid to build up in the

lungs. In addition, anemia can cause a shortage of oxygen-carrying red blood cells. Both of these kidney-related issues can leave one feeling short of breath.

Metallic Taste in Food: When toxins build up in the blood, this can make food taste different. You may even stop liking meat, as this can taste especially metallic. This toxic build up may also cause bad breath.

Ammonia Breath: In addition to bad breath, those with kidney issues may have breath that smells like ammonia due to the buildup of uremia in the body.

Nausea: It isn't uncommon to frequently feel nauseated or even vomit regularly when you have kidney disease due to a buildup of toxins in the body. This can lead to weight loss and lethargy.

Common causes of kidney disease are²:

Diabetes: High blood sugar is the #1 cause and high blood pressure is the #2 cause of kidney damage. People with diabetes often have both issues, and one third of people with diabetes develop kidney disease³. Please see our chapters on Diabetes for more information.

High Blood Pressure: Hypertension can narrow and restrict blood flow to the kidneys, leading to extra fluid in the body, raising blood pressure even more. Please see our Heart Health Protocol for more information.

Reduced Blood Flow to Kidneys: This is often due to hypertension, blood clots, cancer, cysts, or injury.

Kidney Injury: An injury to the kidneys can cause one (or both in some cases) to work improperly. For example, car accidents, sports injuries, and horseback riding incidents can cause blows to the kidneys that result in kidney trauma.

Heart Disease: Heart disease may result in inadequate blood flow to the kidneys.

Infections: Frequent infections of the urinary tract, bladder, or kidneys can damage the kidneys over time.

Liver Disease: The liver also filters toxins. When it isn't functioning correctly, the kidneys may have to work extra hard.

Too Many NSAIDs: Pain relievers/fever reducers like aspirin, naproxen, and ibuprofen are hard on the kidneys. They have to filter these drugs, so if you take them on a daily basis damage can occur.

Dehydration: The kidneys need plenty of water to help filter toxins. Many people do not drink enough water and chronic dehydration is a major risk factor for kidney disease.

Blood Clots: Because these can inhibit blood flow to the kidneys, they can cause kidney damage.

Unhealthy Lifestyle: When you put toxins in your body, the kidneys have to work harder to filter them out. This includes toxins like drugs and alcohol and dietary toxins like processed foods, sugar, and foods tainted with pesticides or herbicides.

Kidney Stones: These painful mineral deposits can lodge in the kidneys and disrupt kidney function.

Taking care of your kidneys means doing what you can to prevent and/or treat the problems that lead to kidney disease. Address symptoms as soon as they manifest to prevent further damage. This protocol addresses various treatments for kidney disease by tackling the most common root causes.

1. Diet The food you consume can help heal or harm your kidneys. In this section, we discuss foods that help protect and support your kidneys.

Eat Antioxidant-Rich Foods: When grocery shopping, choose antioxidant-rich, organic berries like cranberries, blueberries, pomegranate, cherries, raspberries, strawberries, goji berries, and blackberries. Multiple studies show antioxidants play a significant role in kidney health⁴. Not only do they protect the kidneys, they also mitigate renal oxidative damage. Avocados contain compounds that help the body absorb antioxidants better. When eating meals, pair avocados with berries for maximum kidney support. Avocados also contain antioxidants. Note: because avocados are also a source of phosphorus, they are not recommended for those with kidney failure, as the kidneys may not be able to filter out the excess phosphorus. For those who do not have kidney failure or advanced kidney disease, an avocado a day is fine.

Vitamin C is an antioxidant necessary for immune system health, as well as the growth and repair of bodily tissues. The body doesn't absorb vitamin C very well. Get the majority of your vitamin C through the foods you eat, as this is the best way to ensure it absorbs into the body effectively. Vitamin C-rich foods include citrus fruits, rose hips, bell peppers, broccoli, brussel sprouts, kiwi, cabbage, cauliflower, tomatoes, pineapple, melons, and sweet potatoes. Heat significantly reduces the amount of vitamin C, so eating raw fruits and/or lightly steamed vegetables is the best way to avoid destroying this nutrient in the foods you eat.

Beware of Oxalates: Organic fruits and vegetables contain a plethora of essential vitamins, minerals, and other nutrients. However, for some people, the effects of eating certain foods raw can be surprisingly negative. This is because some healthy foods contain oxalates⁵, a naturally occurring compound in plants that binds to calcium. When this happens, kidney stones can form. It generally takes a diet high in oxalate-rich foods to cause kidney stones, but some people are more sensitive than others. If you struggle with kidney stones or kidney issues, heat significantly reduces oxalates. Don't overcook your vegetables, but steaming them is a great way to negate some of the oxalates in your food. In addition to steaming your foods, drink plenty of water to help the body flush them out. Some foods high in oxalates include spinach, kale, beets, almonds, potatoes, raspberries, cranberries, navy beans, and dates. There is no need to avoid these foods because of oxalates—simply blanch or steam them before consuming. Dehydrating foods (e.g., cranberries) also reduces oxalates.

Electrolytes: For kidney health, it is important to eat foods rich in electrolytes⁶. Electrolytes are minerals that carry an electric charge, sparking cell function throughout the body. Most know them for their ability to support proper hydration. Another important function of electrolytes is stimulating muscle contractions, among the most important, keeping the heart beating.

The kidneys hold a central role in maintaining normal electrolyte levels; those with kidney disease may have abnormal electrolyte levels. Electrolytes also play a role in toxin buildup as they help move nutrients into the cells – and waste out. Eat foods that naturally contain electrolytes such as avocados, kale, broccoli, beans, almonds, peanuts, strawberries, watermelon, oranges, bananas, tomatoes, avocados, yogurt, fish, turkey, chicken, raisins, and olives. Some of these (like spinach and kale) may contain oxalates; cook them to negate this.

Resveratrol: Resveratrol is a compound that helps eliminate uric acid and may improve kidney inflammation⁷. Uric acid can be dangerous for those with kidney issues and it contributes to chronic kidney disease. Resveratrol helps boost kidney health and may also protect the kidneys from further damage. It is found in several healthy foods, such as blueberries (many berries contain resveratrol), grapes, black cherries, and nuts. Some of the highest concentrations of resveratrol are found in the invasive Japanese knotweed plant. You can help the environment and heal your kidneys by eating this plant. If you wildcraft Japanese knotweed, harvest in areas you know haven't been sprayed with pesticides or herbicides. Harvest young shoots in the spring. These stems have a tart and pleasant taste, similar to rhubarb. Japanese knotweed is also an excellent source of minerals, including zinc, manganese, Vitamin C, and Vitamin A. Since it also contains phosphorus and potassium, limit your intake if you have active kidney disease or kidney failure.

Foods to Help Eliminate Heavy Metals: If your kidneys are not functioning properly, your body may have higher amounts of toxins as a result. Some toxins, like heavy metals, can build up and cause significant health issues⁸. There are foods you can eat to help eliminate heavy metals. Even if your kidneys are functioning properly, you can have a buildup of heavy metals from environmental exposure. Choose foods like seaweed, spirulina, chlorella, kelp, dulse, dandelion root, and cilantro for heavy metal detoxification.

Foods for Heart and Blood Sugar Health: High blood pressure, heart disease, blood clots, and diabetes put you at a much higher risk for kidney disease⁹. Because these all directly impact the kidneys, it is vital to manage circulatory and blood sugar issues before they cause damage. Daily exercise and a healthy diet help circulatory function and blood sugar management. Please see our Heart Health and Diabetes Protocols for more information.

Foods to Avoid: For diabetics, and everyone else, high-sugar foods are toxic. If you are craving something

Even natural sugars from raw honey or maple syrup aren't ideal for those with diabetes, but stevia is okay when used in moderation. Avoid artery-clogging foods and oils, as well as processed foods. Choose lean meats (no red meat) and no fried foods of any kind. When possible, choose organic and non-GMO produce. Avoid alcoholic drinks. Get carbohydrates from healthy sources and not foods like white rice, breads, and pasta. Canned foods often have very high sodium, so avoid these or choose low sodium canned foods. Dairy is okay in moderation, but it can be inflammatory and high in fat. Choose dairy products that are low in sugar like Greek yogurt. In place of milk, consider drinking almond milk.

2. Managing Blood Pressure The connection between blood pressure and kidney health is well documented. Our kidneys filter waste and extra fluid from the blood. If the blood vessels become damaged, the nephrons that filter the blood do not receive the oxygen and nutrients they need to function. Over time the kidneys lose their ability to filter blood effectively, regulate fluids, and keep adequate levels of acids and salts in the body. In addition, the kidneys may fail to regulate blood pressure, putting you at an even greater risk for stroke or heart damage (especially if you already have high blood pressure).

The nephrons in our kidneys have a thick network of blood vessels with a high volume of blood flow. If blood pressure is allowed to stay high and unchecked, over time this can cause arteries around the kidneys to narrow and weaken. It may also result in the hardening of the arteries in this region, leaving them unable to do their job delivering much-needed blood to the kidneys, leading to kidney failure.

If you have high blood pressure, it is imperative that you take care of it as soon as possible to prevent damage to the kidneys and heart. Take your blood pressure often so you can keep track of your progress. Please see our Heart Health Protocol for more information on managing hypertension.

The following herbs are recommended for high blood pressure:

Garlic: Garlic is a powerful ally for those seeking to lower blood pressure and support heart health. Garlic contains allicin, and has been shown to benefit the cardiovascular system by preventing clots, having strong antioxidant effects, improving circulation, reducing homocysteine levels, and helping with vasodilation and relaxing smooth muscles and blood vessels¹⁰. Note that garlic helps decrease blood pressure in people with increased systolic pressure, but not in patients without increased systolic pressure¹¹. Suggested dose: 600-1,200 mg garlic extract daily with meals.

Hawthorn (*Crataegus monogyna*): The berries on this tree are famous for their heart-nourishing effects¹². Hawthorn berries support a healthy cardiovascular system, lower blood pressure, and strengthen the heart (especially a heart damaged by high blood pressure). Both clinical and pharmacological studies have shown that hawthorn lowers blood pressure via its potent

antioxidants: flavonoids and oligomeric procyanidins¹³. Suggested dose: 2 dropperfuls hawthorn tincture up to three times daily.

Holy Basil/Tulsi: Extract of *O. basilicum* is antihypertensive. It causes a fall in systolic, diastolic, and mean BP in a dose-dependent manner. This cardiovascular effect has been attributed to eugenol, which exerts its effect by blocking calcium channels¹⁴. Suggested dose: The median effective supplement dose is 30 mg/kg or take as a twice daily tincture¹⁵.

Lemon Balm (*Melissa officinalis*): A powerful nervine, lemon balm can help lower blood pressure due to stress and anxiety. If you struggle with anxiety and stress, this can take a toll on cardiovascular health over time. Lemon balm calms the nerves and in turn stabilizes blood pressure. It may also help control blood glucose levels for those with blood sugar issues. Suggested dose: 2 dropperfuls lemon balm tincture up to three times daily.

Lion's Mane (*Hericium erinaceus*): Lion's mane is a powerful adaptogenic mushroom linked to neurological health. Lion's mane can also act as an ACE inhibitor with blood pressure-lowering effects. It also stabilizes blood sugar, making it especially helpful for those wanting to nourish their kidneys. Suggested dose: 2 dropperfuls of lion's mane dual extraction twice daily.

Motherwort (*Leonurus cardiaca*): Motherwort nourishes the heart and circulatory system¹⁶. Not only can it help lower blood pressure, but it may also prevent and treat heart damage. It helps strengthen the heart muscle, arteries, and blood vessels. Additionally, motherwort has been shown to help relieve stress, a major contributor to high blood pressure. Suggested dose: Take 2 dropperfuls tincture twice daily, or as needed for anxiety and stress.

Reishi (*Ganoderma lucidum*): This adaptogenic mushroom is called the "mushroom of longevity" and helps the body better respond to stress. When we are continually stressed, our body tends to react in a negative way by producing hormones that can actually be harmful to our health. Reishi restores balance to the body in a variety of ways, including stabilizing blood pressure. Reishi may also help stabilize or lower blood sugar levels. Suggested dose: 2 dropperfuls dual-extracted reishi tincture twice daily.

Other ways to support healthy blood pressure are:

Daily Meditation and Deep Breathing: Practicing meditation with deep breathing twice daily for twenty minutes can help keep your blood pressure in check¹⁷. Those who practice relaxation techniques have higher measurable levels of nitric oxide, a molecule the body makes to help relax and widen blood vessels to help keep blood pressure under control.

To successfully practice meditation with deep breathing, try finding a quiet place to lie down. Close your eyes and focus on your breathing. It helps to utter a word or phrase over and over to keep you focused when your mind begins to wander (as it usually does with most people). Take slow, deep breaths in through your nose and then exhale

slowly through your mouth with your lips slightly open. Make sure to fully exhale each breath to get the stale air out of the lungs and make room for fresh air to come in. Try to do this when you wake in the morning and before you go to bed in the evening, and whenever you feel the need to focus and destress.

Yoga: Yoga helps lower blood pressure and calms the body and mind¹⁸. Try twenty to thirty minutes of yoga daily.

Weight Loss: One of the most common causes of high blood pressure is obesity¹⁹. Obese people have more fatty tissue, increasing their vascular resistance thus causing the heart to work harder in order to pump blood throughout the body. Over time, damage can be done to the kidneys as well as other organs. If you need to lose weight, start by cutting sugar and white breads/carbohydrates. Reach for healthy fruits and vegetables and strictly limit the number of processed foods and oils you consume. Choose lean meats like chicken and turkey or healthy fish like salmon in lieu of red meat.

Manage Stress ASAP: Stress is a big contributor to high blood pressure, yet many people don't take it seriously²⁰. Daily anxiety and stress take a heavy toll on the body and high blood pressure is just one way this manifests. Take time for self-care, try meditation and a daily gratitude practice, spend time in nature, and consider regular massages or acupuncture. Support your body's response to stress by taking adaptogenic remedies like Reishi and Lion's Mane. Please also visit our Anxiety and Stress Protocol in this book.

3. Managing Blood Sugar and Cholesterol

People with diabetes are at an increased risk of developing kidney disease²¹. If blood sugar gets too high, the kidneys start to spill sugar into the urine. Over time, this causes damage to the blood vessels in the kidney, as well as your filters. More water and salt are retained, as well as toxins and waste materials. These build in in the blood and cause a wide range of issues including damaged kidneys and even death if left untreated. The average person will likely not experience any issues with high blood sugar affecting kidneys, but diabetics are at a higher risk.

Cholesterol clogs arteries that supply the kidneys with blood. When the kidneys are unable to get the blood they need to function, they can be damaged. Learning how to control both high blood sugar and cholesterol are important in order to keep your kidneys healthy.

Natural remedies for blood sugar management are found below:

B Vitamins: B vitamins (especially vitamin B6) help support healthy blood sugar levels²². B vitamins help with glucose metabolism and serve as cofactors in cellular reactions utilizing glucose. They are being studied to determine the benefits of helping with blood sugar levels, and the research looks promising. Suggested dose: Dosage will vary depending on the brand you purchase; follow instructions on the supplement bottle.

Berberine: Berberine is a compound found in

some plants that increases insulin sensitivity, promotes insulin production, regulates metabolism, increases the breakdown of glucose, reduces glucose production in the liver, increases nitric oxide production to widen arteries, and slows carbohydrate absorption from the gut²³. While you can get this compound by taking certain herbal extracts like goldenseal or Oregon grape root, you can get more of it by taking a berberine supplement. Suggested dose: 500-1,500 mg daily.

Bilberry: Bilberries are used to lower blood sugar levels in people with type 2 diabetes. The berries help prevent the breakdown and absorption of carbs in your gut, similar to some blood sugar-lowering medication. Suggested dose: 160-480 mg of high concentration bilberry powder/extract daily.

Bitter Melon: Bitter melon contains compounds that act like insulin in the body²⁴. Studies show a definite reduction in blood sugar levels after taking this plant. Suggested dose: 50-100 milliliters of fresh bitter melon juice or five grams of the powdered fruit three times daily.

Cinnamon: Cinnamon reduces blood sugar levels by mimicking insulin (the hormone the body uses to allow cells to convert sugar into energy)²⁵. It emulates insulin by attaching to insulin receptors on the surface of cells and creating a channel by which sugar can enter the cells. This results in less sugar in the blood. If you want to take cinnamon in supplement form, keep in mind that it also has a blood-thinning effect, so do not take it with other blood thinning herbs (like clove) or blood-thinning medications. Suggested dose: two to four grams of cinnamon daily, divided into two or three doses.

Cordyceps: These mushrooms contain a compound that acts like insulin to help lower blood glucose levels and treat diabetes. They have been shown to improve kidney function, as well as inflammation and immune system weakness. Suggested dose: 2 dropperfuls dual-extracted Cordyceps tincture twice daily.

Fenugreek Seeds: Fenugreek has been studied for its effects on blood sugar levels and may help lower blood glucose while leaving serum insulin levels unaffected. In addition to being helpful for blood sugar regulation, fenugreek may also improve blood pressure – a win-win for kidney health²⁶. Suggested dose: eight to ten grams of fenugreek seed powder daily, or two dropperfuls of fenugreek seed tincture up to three times daily.

Ginseng: American ginseng supports insulin by enhancing insulin secretion, binding and sensitivity. It also promotes the transport of glucose to increase the movement of sugar from the bloodstream into cells. Third, it reduces glucose absorption in the digestive tract²⁷. Suggested dose: two dropperfuls ginseng root tincture up to three times daily.

Holy Basil/Tulsi: This adaptogenic herb stabilizes blood sugar (glucose) levels, improves lipid profiles, helps with insulin resistance, prevents weight gain, and protects the liver and kidneys from high blood glucose²⁸. Because tulsi (holy basil) helps the body respond better to stress, it is also associated with an improvement in

blood pressure. Suggested dose: two dropperfuls of holy basil/tulsi tincture up to three times daily.

Lemon Balm: In addition to having blood pressure-lowering effects, this nervine herb may also help reduce blood sugar levels and oxidative stress to keep the kidneys thriving. Suggested dose: two dropperfuls lemon balm tincture up to three times daily.

Reishi: Reishi mushroom helps both high blood pressure and high blood sugar. It may also help lower high cholesterol, and it protects against insulin resistance^{29 30}. It is a powerful tool to have on hand for kidney health, and helps with stress due to its adaptogenic properties. Suggested dose: 2 dropperfuls dual-extracted reishi tincture twice daily.

Turkey Tail: Turkey tail mushrooms are a powerhouse for kidney health. They are extremely high in antioxidants and have been proven effective at reducing blood glucose and managing blood sugar levels for those with diabetes. They also help lower cholesterol and blood pressure. Suggested dose: 2 dropperfuls dual-extracted turkey tail tincture twice daily.

Natural remedies for high cholesterol are found below:

In addition to previously discussed herbs, such as **cinnamon, fenugreek, hawthorn, reishi, and turkey tail mushrooms**, the below remedies are also known to help lower cholesterol.

Ginger: Anti-inflammatory ginger helps lower “bad” cholesterol levels to reduce the risk of heart disease³¹. Suggested dose: Take no more than four grams of ginger root daily.

Guggul: For those with a family history of high cholesterol, guggul, made from the gum resin of the guggul tree, may help keep cholesterol and triglyceride levels within the normal range³². It contains guggulsterone, which affects cholesterol production by the liver. Note that studies are still mixed for this supplement. It seems to work for people who follow certain diets (like a classic Indian diet) but not for those who follow a Western diet. Suggested dose: 500 mg up to three times daily.

Niacin: Niacin is a B vitamin that has been shown to increase high-density lipoprotein (HDL) cholesterol (the “good” cholesterol) while also removing low-density lipoprotein (LDL) (the “bad” cholesterol)³³. However, if you are already taking cholesterol medication, niacin supplementation may not make much of a difference. Suggested dose: 14-16 mg daily.

Soluble Fiber: Soluble fiber has been shown to lower LDL cholesterol by binding with cholesterol so that it can exit the body, while also reducing the risk of coronary heart disease³⁴. For best results, it is recommended that you eat anywhere from five to ten grams of soluble fiber a day. Psyllium powder, flax seeds, and chia seeds (or a combo of these) are good supplement options. Suggested dose: 14 grams of soluble fiber for every 1,000 calories consumed each day.

Turmeric: Turmeric may affect cholesterol and

triglyceride levels³⁵. Turmeric appears to lower LDL and may also prevent it from being oxidized. Suggested dose: two dropperfuls of turmeric tincture up to three times daily or 500-2,000 mg turmeric capsules daily.

To further support healthy cholesterol levels, eat a healthy diet. Eat foods rich in soluble fiber and avoid high-fat foods like fried foods and sweets. Eat plenty of vegetables and fruit. Watch your dairy intake and consider substituting almond or coconut milk for cow milk.

4. What to Avoid Just as there are steps you can take to prevent kidney issues and protect kidney health, there are also things to avoid. The things you choose not to do are as important as the things you choose to do when focusing on restoring and supporting kidney health. Below I discuss toxic foods, beverages, drugs, and habits that interfere with kidney function.

Soda: Most nephrologists will tell you that one of the most toxic beverages you can consume is dark soda. Soda is bad because of the high amount of sugar and empty calories, but what makes it especially bad for the kidneys is that dark sodas have additives that contain phosphorus. Phosphorus is added to dark sodas during processing to enhance flavor, shelf-life, and to prevent discoloration. The problem with phosphorus is that the body absorbs more than we need. Synthetic phosphorus in dark sodas absorbs into the intestinal tract very easily. Damaged kidneys cannot remove phosphorus well. Excess phosphorus pulls calcium from the bones and it can also combine with calcium, leading to deposits.

Caffeine and Alcohol: Two other offenders are coffee and alcoholic beverages. Caffeine and alcohol dehydrate you. You need to stay as hydrated as possible to keep your kidneys functioning properly. While some teas are great for kidney health, like antioxidant-rich green tea, avoid caffeinated teas and teas with sugar. If you want to drink tea for kidney health, try goldenrod or cleavers with no sugar or with a small amount of stevia. The absolute best beverage you can drink for kidney health is water.

High Sodium: Canned foods often contain high amounts of sodium. For those with kidney disease, the kidneys may have a hard time filtering out excess sodium. Other high sodium foods include processed meats, pickles, olives, relish, pretzels, chips, and crackers. Avoid packaged, instant, and pre-made meals, as these tend to be high in sodium.

Foods that are high in **phosphorus and potassium** are best to avoid, or at least eat in moderation. If you don't have kidney disease, you can consume these, but for those with kidney disease, extra potassium and phosphorus are not filtered well and can cause serious issues. The following foods are high in these minerals: avocados (these are very healthy but are not recommended for those with severe kidney disease), whole wheat breads, brown rice, potatoes and sweet potatoes, apricots, Swiss chard, beet greens, spinach, bananas, prunes, raisins, dates, oranges (and orange juice), and dairy. Many of the foods listed here are very good for you. Keep in mind that

while they are not good for those with kidney disease, they are fine (in moderation) for those who do not have kidney disease.

Sleep: Not getting enough sleep (you should be getting around eight hours of sleep every night) is bad for the kidneys. Kidney function is regulated by the sleep-wake cycle. Avoid blue light a few hours before bedtime; diffuse sleep-friendly essential oils (like lavender or chamomile), and try sleep herbs like valerian and hops in tincture-form before bed. Also, see our Sleep Protocol.

Dehydration makes it hard for the kidneys to clear excess sodium and other toxins from the body. You may be able to avoid kidney stones by drinking enough water. Fluid intake needs to be monitored and sometimes restricted by those in active kidney failure, but for everyone else, drinking at least half your weight in ounces of water each day is a good goal.

Smoking: Those who smoke regularly often have more protein in their urine – a sign of kidney issues. This toxic habit is linked to heart and lung diseases, cancer, and more. The link between smoking and kidney health is yet another reason to avoid lighting up.

A sedentary lifestyle is linked to kidney disease. Researchers are still not entirely sure why, but what is known is that more physical activity is associated with better blood pressure and glucose metabolism, which are great for kidney health. Stay active. Try to exercise at least twenty minutes a day. If you work long stretches at a desk, take breaks and stretch/walk around every hour.

NSAIDs: If you're in the habit of taking NSAIDs on a regular basis, be aware that these are strongly linked to kidney damage, especially those containing aspirin and naproxen. If you need something for aches and pains, try using white willow bark instead. It has the benefits of aspirin, but not the side effects (when used in moderation).

5. Herbs and Supplements to Promote Healthy Kidney Function

There are herbs and supplements you can utilize to not only support kidney health, but also help heal issues that can lead to kidney damage, such as kidney stones and frequent urinary/bladder/kidney infections.

Remedies for infections and kidney support are found below:

Cleavers (*Galium aparine*): In early spring, sticky plants called cleavers (also known as Bedstraw, Goosegrass, and Galium) commonly appear. Cleavers activate the lymphatic system, pump toxins from the body, and cleanse the bloodstream³⁶. They work to help flush the kidneys of unwanted toxins and waste, and are great for flushing out infections as well. I like to add these to my smoothies or juice them in spring.

Cordyceps Mushrooms: These adaptogenic mushrooms have been studied to help reverse chronic kidney disease (CKD)³⁷. They show "potential promise to decrease serum creatinine, increase creatine clearance,

reduce proteinuria and alleviate CKD-associated complications, such as increased haemoglobin and serum albumin³⁸. Suggested dose: 2 dropperfuls dual-extracted cordyceps tincture twice daily.

Corn Silk: Corn silk performs two important functions for the kidneys³⁹. Its diuretic properties help flush out waste and it is helpful for urinary tract infections as it lubricates the urinary tract and bladder to relieve pain and spasms. This part of corn is so often overlooked and thrown away; consider saving it the next time you shuck corn. Suggested dose: 2 dropperfuls tincture up to three times daily.

Goldenrod (*Solidago* spp.): Not only is goldenrod an effective diuretic to help flush waste from the kidneys and urinary tract, it also has disinfectant properties⁴⁰. It can be a great tool for helping flush out an infection, while also killing the bacteria responsible. Suggested dose: Drink one to two cups of goldenrod tea daily (2 tsp. infused in a cup of hot water for fifteen minutes).

Lungwort Lichen (*Lobaria pulmonaria*): This lichen can help flush out excess fluid and waste, making it useful for kidney health and detoxification. Like usnea, it has antimicrobial properties that can help heal infections. Suggested dose: 2 dropperfuls 1-2 times daily.

Stinging Nettle (*Urtica dioica*): Stinging Nettle supports kidney health in a variety of ways. It is nutrient-dense and especially high in vitamin C, which has been shown to help flush out waste from the body. Additionally, stinging nettle is anti-inflammatory and contains flavonoids that help suppress histamine production, making it useful for conditions that affect the kidney, bladder, and urinary tract (like Interstitial Cystitis, or IC). Suggested dose: 2 dropperfuls tincture twice daily or three cups of stinging nettle tea daily.

Usnea: Usnea has potent antimicrobial and antibiotic properties that can help heal infections of the urinary tract, bladder, and kidneys. Suggested dose: Take two dropperfuls of usnea dual extraction three times daily for 10 days if you have an active infection.

Remedies for kidney stones are found below:

Chanca piedra: Chanca piedra not only dissolves existing stones, it prevents the formation of new stones. It helps protect the liver and reduce oxidative stress and inflammation to help fight infections. Suggested dose: Take five to ten drops of chanca piedra tincture up to three times daily.

Gravel Root: This indigenous remedy for kidney stones comes from a plant called “Joe Pye Weed”. The root of the plant is used to tone the kidneys and urinary tract⁴¹. It contains a compound called euparin that destroys potentially harmful microorganisms to prevent infection. Suggested dose: one dropperful tincture up to four times daily.

Pipsissewa: If you’re looking for remedy to break up kidney stones, this evergreen plant’s name translates to “to break into small pieces” in Cree. I know many people who have used this with great success. Not

only does this plant contain compounds to dissolve stones, it also contains hydroquinone to help disinfect the urinary tract and help heal infections. Suggested dose: 1-2 dropperfuls tincture up to three times daily.

Remedies for detoxification are found below:

Bilberry (*Vaccinium myrtillus*): Bilberry may help improve the function of the kidneys and liver by removing harmful toxins and heavy metals. Compounds in bilberry may chelate arsenic, lead, cadmium, and iron from the blood. Suggested dose: 160-480 mg of high concentration bilberry powder/extract daily.

Cat’s Claw (*Uncaria tomentosa*): This herb is great for detoxification because it can remove heavy metals, toxins, drugs, and other potentially harmful waste from the body. It is known to support kidney health, pancreatic health, and digestive health. Careful if you have an autoimmune disorder. Suggested dose: one dropperful tincture up to three times daily.

Dandelion Root (*Taraxacum officinale*): Dandelion root is a great detoxification herb along with milk thistle seeds. It is diuretic, so it can flush waste from the body. Make sure to harvest this common plant from clean areas that haven’t been sprayed. Suggested dose: one cup of dandelion root tea up to three times daily (Infuse two tsp. of the chopped root into a cup of hot water.).

Lemon Water: A simple yet effective remedy, lemon water or lemon juice (sans sugar) can help keep kidneys functioning properly. Lemon water can aid in detoxification and may even prevent the formation of kidney stones. Use caution if you have Interstitial Cystitis (IC), as acidic foods and beverages can inflame the bladder and worsen symptoms. Suggested dose: Have one to three cups of lemon water daily. Squeeze the juice of ½ a lemon into an 8oz glass of water.

To avoid infections of the kidneys, urinary tract, and bladder, start by making sure you are staying hydrated⁴². Avoid dehydrating beverages like coffee and alcohol. Foods and beverages high in sugar are triggers for infection. Maintain good hygiene and wear breathable cotton underwear. Urinate and wash before and after sexual intercourse to keep infections at bay.

Kidney stones are caused by excess calcium, oxalates, and phosphorus in the urine. These are influenced by the foods and drinks you consume. Some foods contain high amounts of oxalates. To negate these effects, steam or lightly cook leafy greens and other high-oxalate vegetables. Watch your milk and dairy intake. See the list of phosphorus-containing foods in the “toxic food” section.

In addition to herbs for kidney health, there are supplements that help kidney function. Make sure to check with a healthcare professional before beginning a supplement regimen.

Alpha Lipoic Acid: Also called ALA, Alpha lipoic acid is a kind of antioxidant that may address cardiovascular disease. Since cardiovascular disease is a complication often seen with kidney disease, it can be beneficial for

kidney health in this way. In one study⁴³, those undergoing hemodialysis found that ALA supplementation increased the activity of certain antioxidant enzymes that protect against oxidative stress and oxidative disorders like cardiovascular disease. Suggested dose: 400-600 mg once daily with a meal.

NAC: NAC stands for N-acetylcysteine. This is an amino acid and precursor to glutathione, another antioxidant in the body that targets oxygen free radicals. In one study⁴⁴, patients receiving NAC supplementation while undergoing hemodialysis were found to benefit from NAC, as it helped reduce oxidative stress and improved the outcomes of uremic anemia. Suggested dose: 600-1,800 mg daily.

Resveratrol: This polyphenol is found in foods like grapes, peanuts, and berries. It is a powerful antioxidant. It can protect the heart and cardiovascular system, as well as the kidneys. In studies on animals, resveratrol was shown to protect against kidney issues like drug-induced injury, diabetic neuropathy, and hyperuricemic injury⁴⁵. It may target inflammation in the body, as well as oxidative stress associated with chronic kidney disease progression. Suggested dose: 500-1,500 mg daily.

6. Avoiding Adrenal Fatigue

If you want healthy kidneys, you need healthy adrenals. The adrenal glands are located on top of each kidney and are often overlooked when it comes to kidney health. These glands are imperative for the production of hormones that regulate metabolism, immune function, blood pressure, and the stress response. One of the most common ways people harm the adrenals is with stress.

Constant stress causes these glands to overproduce hormones responsible for the body's stress response, like cortisol. While cortisol has an essential function, overproduction can result in health issues like weight gain. When these glands are constantly producing, they can become weak and unable to perform. This is called adrenal fatigue or HPA axis dysfunction. Symptoms include brain fog, hair loss, chronic fatigue, sleep issues, irritability and mood swings, weight gain, and depression.

It is important to take care of your adrenals if you want to take care of your kidneys. To do this, keep stress in check. Gentle exercise like yoga can relieve stress, as can meditation, deep breathing, and adaptogenic herbal supplements to help your body better respond to stress. Some herbs to try for stress management include reishi, cordyceps, lion's mane mushrooms, and ashwagandha to help lower cortisol, reduce depression and anxiety, and protect the adrenals from the effects of stress. Please see our Adrenal Fatigue/HPA Axis Protocol for more information.

7. Staying on Top of Kidney Health

Staying healthy and catching signs early gives you the chance to manage your kidney health naturally.

Take your blood pressure often, documenting any changes, especially if it begins creeping up above the normal threshold. Utilize treatments for high blood pressure and

see a doctor if you are unable to keep it down naturally. Monitor blood sugar often, especially if you have a family history of diabetes. Consider the blood sugar remedies mentioned in this book and see your doctor annually for a physical and to have bloodwork/labs conducted to keep an eye on cholesterol and triglycerides.

Stay hydrated and try to drink three to four pints of water a day. Avoid consuming foods and drinks that are counterproductive to kidney health. Watch your oxalate intake to avoid kidney stones. Never hesitate to see a physician or schedule an appointment with a nephrologist if you suspect any issues. They can perform tests to better determine exactly what is wrong.

8. Western Medical Approaches

Upon seeking western medical treatment for kidney disfunction, your doctor will first perform tests to better understand the condition. Expect blood tests to look at the level of waste products in your blood, urine tests, imaging tests, and in some cases, you may also have a sample of kidney tissue removed for testing. A kidney biopsy is usually done under anesthesia with a long, thin needle that takes a small sample of the kidney.

Once doctors have a better understanding of your particular issue, they may go in one of several directions with treatment depending on your problem. Many times, treatment options may include tackling an underlying condition that is putting stress on the kidneys, such as diabetes mellitus or a problem with high blood pressure.

Pharmaceuticals: If blood pressure problems are behind your kidney issue, you may be prescribed a medication to lower blood pressure. Angiotensin is a popular drug for this purpose. Sometimes, high blood pressure drugs can decrease kidney function and change electrolyte levels, so your doctor may perform frequent blood tests to make sure your condition isn't changing too much. Another drug commonly prescribed is a diuretic (water pill). This can help push more water from the body to help the kidneys filter better. Other drugs commonly prescribed for kidney issues include medications to relieve swelling, medications to treat anemia, medications to lower cholesterol (many with chronic kidney disease also have bad cholesterol levels), and supplements like vitamin D and calcium to protect bones. In addition, a doctor may prescribe a phosphate binder to help protect blood vessels from the effects of calcium deposits.

Diet: A doctor may recommend a low salt diet to help improve kidney function. They may also recommend a low protein diet that aims to minimize waste in the blood. The more waste-producing foods you eat, the harder your kidneys have to work to filter it.

Dialysis: If you are in the later stages of kidney disease, a doctor may recommend dialysis. This is a process that artificially removes waste and extra fluid from the blood when your kidneys cannot. A machine filters all the waste and fluids from the blood when a tube is inserted into the abdomen. This tube fills the abdominal cavity with a dialysis solution that absorbs excess fluids. After

some time, the solution drains from the body, carrying the waste products with it. Some people have to go once a week and some have to go more than once a week, depending on their situation.

Kidney Transplant: A kidney transplant is for those

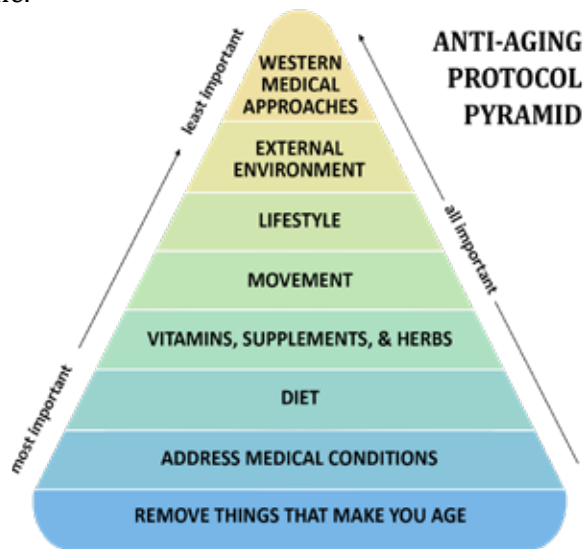
in later stages of kidney disease. It involves surgically placing a healthy donor kidney into the body. If you have a kidney transplant, you will need to take medications for the rest of your life to keep your body from rejecting the organ.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Meditation/deep breathing Kidney-friendly breakfast (e.g., berries w/ Greek yogurt. Monitor blood sugar afterwards, if needed.) NAC & Resveratrol Cordyceps, Reishi & Stinging Nettle tinctures Heart health herbs (Hawthorn, Fenugreek, Tulsi, Bilberry)	Weekly grocery shop for kidney-friendly foods Hydration check Kidney-friendly lunch (e.g., high-fiber veggies stir-fried w/ lean meat. Monitor blood sugar afterwards, if needed.) Dandelion root tea Alpha-lipoic acid w/lunch	Walk in nature Corn silk Lemon water Healthy snack (e.g., pomegranate with walnuts) Heart health herbs Cordyceps, Reishi & Stinging Nettle tinctures	Monitor BP Goldenrod tea Hydration check (Goal is 3-4 pints) Kidney-friendly dinner (e.g., baked chicken with non-starchy veggies. Monitor blood sugar after, if needed.) Prepare for bed (no blue light, etc) Lemon balm tea/tincture after dinner	Meditation/deep breathing Bedtime routine for better sleep (bath, reading, dim lights, etc.) Sleep herbs if needed Lion's Mane tincture
MONDAY	Meditation/deep breathing Kidney-friendly breakfast NAC & Resveratrol Cordyceps, Reishi & Stinging Nettle tinctures Heart health herbs	Hydration check Kidney-friendly lunch Dandelion root tea Alpha-lipoic acid w/lunch	Yoga Corn silk Lemon water Healthy snack Heart health herbs Cordyceps, Reishi & Stinging Nettle	Monitor BP Goldenrod tea Hydration check Kidney-friendly dinner (monitor blood sugar afterward if needed) Prepare for bed Lemon balm after dinner	Meditation/deep breathing Bedtime routine for better sleep Sleep herbs if needed Lion's Mane tincture
TUESDAY	Meditation/deep breathing Kidney-friendly breakfast NAC & Resveratrol Cordyceps, Reishi & Stinging Nettle tinctures Heart health herbs	Hydration check Kidney-friendly lunch Dandelion root tea Alpha-lipoic acid w/lunch	Yoga Corn silk Lemon water Healthy snack Heart health herbs Cordyceps, Reishi & Stinging Nettle	Monitor BP Goldenrod tea Hydration check Kidney-friendly dinner Prepare for bed Lemon balm after dinner	Meditation/deep breathing Bedtime routine for better sleep Sleep herbs if needed Lion's Mane tincture
WEDNESDAY	Meditation/deep breathing Kidney-friendly breakfast NAC & Resveratrol Cordyceps, Reishi & Stinging Nettle tinctures Heart health herbs	Hydration check Kidney-friendly lunch Dandelion root tea Alpha-lipoic acid w/lunch	Yoga Corn silk Lemon water Healthy snack Heart health herbs Cordyceps, Reishi & Stinging Nettle	Monitor BP Goldenrod tea Hydration check Kidney-friendly dinner Prepare for bed Lemon balm after dinner	Meditation/deep breathing Bedtime routine for better sleep Sleep herbs if needed Lion's Mane tincture
THURSDAY	Meditation/deep breathing Kidney-friendly breakfast NAC & Resveratrol Cordyceps, Reishi & Stinging Nettle tinctures Heart health herbs	Hydration check Kidney-friendly lunch Dandelion root tea Alpha-lipoic acid w/lunch	Yoga Corn silk Lemon water Healthy snack Heart health herbs Cordyceps, Reishi & Stinging Nettle	Monitor BP Goldenrod tea Hydration check Kidney-friendly dinner Prepare for bed Lemon balm after dinner	Meditation/deep breathing Bedtime routine for better sleep Sleep herbs if needed Lion's Mane tincture
FRIDAY	Meditation/deep breathing Kidney-friendly breakfast NAC & Resveratrol Cordyceps, Reishi & Stinging Nettle tinctures Heart health herbs	Hydration check Kidney-friendly lunch Dandelion root tea Alpha-lipoic acid w/lunch	Yoga Corn silk Lemon water Healthy snack Heart health herbs Cordyceps, Reishi & Stinging Nettle	Monitor BP Goldenrod tea Hydration check Kidney-friendly dinner Prepare for bed Lemon balm after dinner	Meditation/deep breathing Bedtime routine for better sleep Sleep herbs if needed Lion's Mane tincture
SATURDAY	Meditation/deep breathing Kidney-friendly breakfast NAC & Resveratrol Cordyceps, Reishi & Stinging Nettle tinctures Heart health herbs	Hydration check Kidney-friendly lunch Dandelion root tea Alpha-lipoic acid w/lunch	Yoga Corn silk Lemon water Healthy snack Heart health herbs Cordyceps, Reishi & Stinging Nettle	Monitor BP Goldenrod tea Hydration check Kidney-friendly dinner Prepare for bed Lemon balm after dinner	Meditation/deep breathing Bedtime routine for better sleep Sleep herbs if needed Lion's Mane tincture

* Add in Cholesterol, Blood Pressure, and/or Blood Sugar Herbs, if needed.

Ch. 40: Anti-Aging Protocol

We're all familiar with the signs of aging. Our skin changes, showing more fine lines and wrinkles. Our muscles aren't as strong, our endurance seems to dwindle, and we become more forgetful. This gradual decrease in physical and mental function is an inevitable part of our life cycle. But just because aging is inevitable, doesn't mean it needs to be severe or painful. There are many ways to support our bodies as we age and to live longer, better. It is estimated that there will be 3.7 million centenarians across the globe by 2050¹. Focusing on nutrition and lifestyle can improve the quality of our additional years of life.



What exactly is aging and what is happening internally to drive these changes? Why do some of us age faster or slower than others? Are there ways to positively impact or even slow the processes of aging? The good news is there are.

Cells are the smallest unit of life and humans are composed of over 37 trillion cells. These cells are responsible for carrying out every biological process in your body, from digesting nutrients to thinking. Every cell in your body contains DNA (your "blueprint"), directions that dictate how the cell is to carry out its function (division, duplication, functionality, death, etc.). When our cells struggle to replicate or function, disease and aging occur. Both cellular genetics (DNA) and epigenetics (the environment that DNA works within) are important for aging.

Epigenetics: Longevity is thought to be a combination of genetics (our internal "blueprint") and lifestyle (our internal and external environmental conditions). Genetics were originally thought to be out of our control, but we now know that environment and behavior cause changes in the way our genes are expressed; this is epigenetics. It does not change our DNA sequences, but how our bodies read and execute our DNA sequences. Genetic changes are not reversible, but epigenetic changes are. Having our genes work in less than ideal environments over long periods of time attributes to the aging process, often

accelerating cellular damage and dysfunction.

There are additional theories as to why and how we age. These include: Cellular Senescence (sometimes known as "limited cell replication" theory), DNA Damage (sometimes referred to as "free radical/oxidative stress" theory), and Telomere Shortening (sometimes referred to as the "Molecular Clock" theory).

Cellular Senescence: Even with ample space and nutrition, cells can lose their ability to divide and grow, known as cellular senescence.² This impacts aging in 2 ways:

1. When these non-functioning cells accumulate, they degrade the strength and functional capacity of tissues and organ systems. When systems work less efficiently, they become vulnerable to disease states and breakdown.
2. When cellular senescence occurs in stem cells, we lose our ability to regenerate and repair, leading to loss of tissue strength and functional capacity.

DNA damage and telomere shortening can both trigger the process of cellular senescence. I'll discuss these next plus how we can support our bodies to help mitigate these issues.

DNA Damage: We've discussed how a gene's environment can impact its function (epigenetics). Now let's discuss genes themselves. When we experience damage to our DNA blueprint our cells don't function as efficiently, adding to the aging process. Whether DNA damage occurs due to external forces (like radiation, tobacco smoke, and genotoxic drugs) or internal forces (like oxidative stress as a result of normal metabolic processes), DNA damage is implicated in mutagenesis (disease states), carcinogenesis (cancer), and aging³. Mitochondria, where energy is produced within a cell, creates normal byproducts of metabolism. These byproducts, in excess, are believed to cause DNA damage due to free radical formation. Free radicals are molecules that contain an unpaired electron, making them highly reactive to other molecules in the body. They cause chain reactions of damage known as oxidative stress. When oxidation happens faster than the body can address it, DNA damage occurs; DNA damage significantly diminishes health and accelerates aging. Oxidative stress can be combated with antioxidant rich foods and avoidance of risk factors like smoking, too much sun exposure, and environmental pollutants. Antioxidants have an abundance of extra electrons, neutralizing free radicals and stopping damage. Antioxidants are naturally produced in the body (e.g. glutathione) and are found in foods we eat (e.g. Vitamin C). We'll discuss antioxidants in more detail later in this protocol.

Outside of aging, signs of increased oxidative stress (oxidation faster than our bodies can address) include the following (all of which add to the acceleration of aging): fatigue, brain fog, muscle soreness, joint pain, and memory loss⁴.

Telomere Shortening: Telomeres are the time

markers on the ends of DNA (think of the sealed end of a shoelace). Telomeres shorten with each chromosome replication. Eventually they get so short that chromosome replication is impossible. This leads to cellular senescence (discussed above). Studies associate shorter telomere lengths with several diseases related to aging including cancer, Alzheimer's disease, and coronary artery disease^{5 6 7}.

Chronological age is commonly used as a measurement for predicting health. However, functional status, or how well one can accomplish the desired tasks of daily living, is more indicative of health than chronological age. Because these biological changes are neither linear nor consistent, we cannot look to chronological age alone to predict health. The pace at which we age is not predetermined or predictable. Biological, environmental, psychosocial, and economic factors all impact how and to what degree we age.

1. Remove Things that Make You Age

Because aging is highly linked to environment and oxidative stress, first remove anything adding to and/or accelerating aging.

Sun (UVA/UVB Rays): Sun leads to premature aging and skin damage (lines, wrinkles, discoloration).

Limit Radiation Exposure: Radiation has been linked to health issues such as atherosclerosis, "altered immune-inflammatory serum protein levels", and cataracts⁸.

Tobacco Smoke: Tobacco smoke causes "premature skin aging, squamous cell carcinoma, melanoma, oral cancer, acne, psoriasis, and hair loss"⁹. Stopping smoking significantly diminishes signs of early aging as well as many pathologies and disease states (both internally and externally)¹⁰.

High Fat, High Carbohydrate, Low Antioxidant Diet: Diets especially high in fat and carbohydrates and low in nutrients are associated with oxidative stress. Oxidative stress further expedites the aging process¹¹.

Alcohol Consumption: Alcohol directly accelerates aging by damaging DNA in telomeres. Keeping telomeres healthy helps with longevity¹².

2. Address Medical Conditions

When medical conditions are left unchecked, these disease states can significantly impact and degrade otherwise healthy tissues, adding to biological damage and the aging process. Hormonal imbalances, diabetes, and autoimmune diseases can all add to the aging process when mismanaged. Age-appropriate screenings, annual physicals, and regular lab work can help catch and address disease states that can accelerate aging.

Hormones, the Endocrine System, and Aging: The effects of aging are inherently connected to our endocrine system. Hormones are the chemical messengers of the body, responsible for communicating important needs (such as hunger, sleep, and body temperature).

When hormones aren't functioning correctly these functions are affected, causing changes that can accelerate aging. Endocrine production and receptor sensitivity are both vulnerable to decline as we age. Aging impacts most endocrine organs, including the thyroid, adrenal glands, pituitary gland, ovaries, and testes. Aging alters hormone production and sensitivity, and endocrine changes can accelerate aging.

For example, changes in estrogen can cause atrophic skin changes and age acceleration. Estrogen insufficiency slows our defenses against oxidative stress, leaving us vulnerable to thinner skin due to less collagen and decreased elasticity. These changes can be seen in the increase of fine lines and wrinkles, increased skin dryness, reduced blood flow, slower wound healing, and hair loss.

To protect against the severity of these changes, get regular testing to check for hormonal disturbances. If a medical condition arises, addressing it as soon as possible will help alleviate accelerated aging variables¹³.

Diabetes and Aging: Insulin mediates essential biological functions such as metabolism, growth, proliferation, and differentiation. Impaired insulin action in conditions such as insulin resistance and diabetes has been linked to premature aging, sarcopenia (reduced muscle mass), body composition changes, nerve damage, and mental acuity changes. The brain runs on glucose so an impaired ability to effectively utilize glucose will inevitably affect the nervous system, causing diabetic neuropathy and cognitive impairment. Improving insulin production and receptor sensitivity is essential for longevity. Correcting pathologies related to insulin resistance is a key component to preventing accelerated aging and to protect against the deterioration of essential metabolic pathways^{14 15}.

Autoimmune Diseases and Aging: When our immune system can't tell the difference between healthy cells and foreign cells, the body mistakenly attacks healthy cells, degrading health and accelerating the aging process, especially if left unchecked. When the immune system is in a constant state of hyperactivity, signaling and regulatory molecules produce high levels of cytokines - inflammation producing substances. The immune system response is thus exacerbated even further, worsening autoimmune disease states. Autoimmune diseases can affect nearly any tissue and organ in your body. This degrades health over time, negatively affecting longevity. Our ability to prevent and address infection is also an essential marker of health and wellness¹⁶.

3. Diet Nutrition significantly impacts longevity. By eating a whole-foods, low-processed diet and adding substances that support the body's biological processes (especially those associated with aging like oxidative stress, metabolism, and hormone balance), we can slow the aging process. See our Dietary Section in the book for good diets to follow.

There are 3 major classes of phytochemicals that are responsible for giving plants their vibrant colors:

Red, Orange, Yellow: Carotenoids produce red, yellow, and orange hues in foods such as carrots, sweet potatoes,

tomatoes, watermelon, and corn. Examples include beta-carotene, lycopene, and lutein. It is estimated there are over 600 types of carotenoids in nature, about 50 within the human diet, and about 20 present in human tissue and blood.

The human body converts beta-carotene into vitamin A (retinol). Vitamin A is a significant component in healthy skin and mucous membranes and seen in many anti-aging face/skin serums because of its ability to improve skin appearance and texture. Vitamin A also supports a healthy immune system and eyesight. We are unable to produce carotenoids; they must be consumed via diet. Carotenoids lower the prevalence of age-related diseases as they reduce oxidative stress and the presence of free radicals. Between 2-20mg of carotenoids daily enhances health and prevents ocular pathologies. The tolerable upper limit of Vitamin A is 3000 mcg (2,500 mcg in those pregnant or breastfeeding as too much Vitamin A has been linked to birth defects). Vitamin A is a fat-soluble vitamin and excess is stored in the liver. Too much Vitamin A can cause liver damage, so make sure any supplementation is below the UL^{17 18 19}.

Green: Chlorophyll produces the green color in many vegetables and herbs such as spinach, kale, and parsley. Vitamins A, C, E, and K are abundant in chlorophyll and all contribute significantly to the radiant, youthful appearance of skin. Dark spots, fine lines, and wrinkles, which are frequently caused by sun damage, can be visibly diminished when eating adequate levels of chlorophyll.

Chronological aging and photo-aging are two types of skin aging. Topical chlorophyll may have anti-aging properties for aging due to sun exposure. Internal chlorophyll can help treat and prevent wrinkles by enhancing the body's production of collagen and getting rid of pollutants that harm the skin. Studies show a substantial improvement in wrinkles and skin tone after consuming between 100-300 mg of chlorophyll each day, in at least three different dosages or meals. As long as your diet contains moderate to high amounts of green vegetables, you can easily reach these goals. Just 1 cup of spinach has approximately 24 mg of chlorophyll.²⁰

Blue, Purple, Red: Anthocyanins are responsible for the rich blues, purples, and reds in fruits and vegetables like blueberries and beets. Anthocyanins are highly antioxidant and protect the skin by scavenging free radicals that destroy connective tissue, protecting the skin from damage, fine lines, and wrinkles. Anthocyanins also protect against internal disease states associated with aging such as CVD (cardiovascular disease-namely hypertension and atherosclerosis²¹), certain cancers, neurodegenerative disorders, and aging-associated bone loss (osteopenia and osteoporosis).²² Anthocyanins may also increase cell proliferation and decrease apoptosis (cell death) that lead to the development and progression of certain cancers, namely human oral, breast, colon, and prostate. These studies examined a variety of berry extracts at concentrations ranging from 25 to 200 µg/mL with promising results.²³

Anthocyanins may help prevent neurons from degenerating. Their high anti-oxidative qualities and

their capacity to cross the blood brain barrier has been linked to their potential positive effects on CNS structures, helping prevent dementia and Alzheimer's. Study dosages range between 50-250 mg per day.

Healthy Fatty Acids: Long-chain omega-3 polyunsaturated fatty acids (PUFAs) may prevent and/or lessen inflammation, hyperlipidemia, platelet aggregation, and hypertension. These effects are mediated by a variety of processes, including regulation of gene expression, eicosanoid synthesis, and cell membrane composition and function. Omega-3 fatty acids are increasingly recognized as safe and beneficial in the treatment and prevention of a number of age-related problems²⁴.

People who significantly increase the proportion of omega-3s to other fatty acids in their diet have immune system cells with longer telomeres. Omega-3 supplements may be a unique nutritional intervention with the potential to reduce the risk for a variety of aging-related diseases, including coronary heart disease, Type-2 diabetes, arthritis, and Alzheimer's disease. Dosages ranging from 1.25-2.5 grams of Omega 3 fatty acids (both EPA and DHA) are associated with a variety of health benefits²⁵.

Fatty acids are building blocks to essential biological substances that wane as we age, such as sex hormones and joint lubrication. Fatty acids also help build collagen and fibrin to keep skin, hair, and nails youthful. Omega-3 fatty acids are widely acknowledged as important nutrients to stop degenerative problems linked to aging. Good food sources include mackerel, salmon, cod liver oil, flax seeds, and chia seeds²⁶.

Fiber: Fiber is present in plant-based foods like beans, fruit, grains, nuts, and vegetables. Since it isn't digested and absorbed, it isn't technically a nutrient. However, that is what makes it so useful. Fiber is divided into two groups: soluble and insoluble. Soft and soluble fiber dissolves in water to create a gel-like material. Your stool becomes more substantial and simpler to pass. Sources include fleshy fruit, beans, oats, and sweet potatoes. Vegetables, fruits, and whole grains all contain insoluble fiber, which encourages the transport of food and waste through the digestive tract. Fiber has a potent impact on longevity and health. It lowers cholesterol, helps prevent diabetes, regulates weight, reduces inflammation, lowers the chance of developing arthritis, feeds your gut microbiome, and helps with telomere length.

Intermittent Fasting: A growing body of research shows that having an eating window of six hours and fasting for 18 hours causes a metabolic switch from glucose-based to ketone-based energy, increasing stress tolerance, lengthening lifespan, and lowering the prevalence of diseases like cancer and obesity. Read about Intermittent Fasting on page 303.

4. Vitamins, Supplements, and Herbs

The most effective anti-aging regimens employ both targeted nutrition and topical agents to address both internal and external signs of aging. To effectively combat

wrinkles, cognitive decline, and poor physical health, a combination of quality nutrition and supplementation is often more effective than nutrition alone. The following vitamins and minerals are recommended:

Vitamin A (Retinol): This super vitamin protects and delays signs of aging internally and externally. Retinol is the main component of many anti-aging supplements and serums. Tretinoin, a derivative of vitamin A, increases collagen synthesis and lessens wrinkles by inhibiting the enzymes that degrade collagen in the dermal layer of the skin and encouraging the production of new collagen in their place. Vitamin A supports healthy heart and lung function, higher immunity, and improved vision. Suggested dose²⁷: 700 mcg for women and 900 mcg for men.

Vitamin C: Vitamin C is an antioxidant. It is important for protein metabolism, the manufacture of L-carnitine, collagen, and certain neurotransmitters, iron absorption, immune system health, wound healing, and the preservation of cartilage, bones, and teeth. Collagen is a crucial component of connective tissue, which is essential in wound healing and skin health. Suggested dose: The RDA for women is 75 mg and 90 mg for men, although doses as high as 1,000-2,000 mg can be well tolerated. Doses higher than this can sometimes cause GI upset and osmotic diarrhea²⁸.

Vitamin D: Vitamin D is a fat-soluble vitamin primarily produced through sun exposure. The body uses vitamin D for the reduction of inflammation, the modulation of processes like cell growth, neuromuscular and immune function, and glucose metabolism. Vitamin D maintains adequate serum calcium and phosphate concentrations and encourages calcium absorption in the gut, supporting healthy bone mineralization and helping prevent osteoporosis. Osteoblasts and osteoclasts need Vitamin D to repair and build new bone. Vitamin D is linked to increased longevity and a potential decline in age-related disorders and illnesses^{29 30}. Vitamin D also helps shield the skin from UV light, one of the main causes of fine lines and wrinkles. Suggested dose³¹: 2000 IU per day w/ Vitamin K-2.

Vitamin E: Vitamin E serves as one of the skin's initial lines of defense. Vitamin E is a potent antioxidant that combats free radicals. In addition to preventing wrinkles and fine lines, vitamin E increases the production of collagen, accelerating cell renewal and promoting the formation of new skin. Taking vitamin E supplements orally significantly improves defense against collagen deterioration. Vitamin E also potentially protects against cognitive decline and reduces the risk of heart disease. Suggested dose^{32 33 34}: 15 mg daily.

Zinc: Zinc is essential for protein synthesis, immune system function, and wound healing. Zinc deficiencies have been shown to accelerate aging while sufficient zinc intake has been linked to lowering the incidence of age-related diseases and infections. Zinc is essential for preventing hair loss, as zinc is essential for the

growth and repair of hair tissue. Zinc helps antioxidants prevent new free radicals from damaging the skin, essentially preventing wrinkle formation at the molecular level. Zinc strengthens cell membranes, bringing new youthful cells to the surface. Suggested dose^{35 36 37}: 8 mg for women and 11 mg for men.

The following supplements are known to help slow down aging:

Berberine and Metformin: Berberine and Metformin act similarly in the body, despite their chemical differences. Berberine is a bitter tasting, yellow-colored natural alkaloid that limits cellular senescence and protects DNA³⁸. Several alkaloids exhibit antibacterial, antiviral, antiproliferation, and antimetastatic effects on various types of cancers, including berberine^{39 40}. Berberine also helps regulate blood sugar, reduce swelling and inflammation, and is antimicrobial. Berberine can help lower cholesterol and triglyceride levels (500mg twice a day). Metformin, like berberine, is first-line treatment for type 2 diabetes mellitus (T2DM). Metformin activates AMP-activated protein kinase (AMPK) and lowers the risk of cancer, as does berberine⁴¹. Metformin dramatically lowers levels of free fatty acids, triglycerides, and soluble vascular cell adhesion molecule-1 (sVCAM-1) in the body, which reduces the risk of CVD. Metformin is a prescription medication; berberine is over the counter. Suggested dose: Berberine: 1000-2000mg/day divided doses with meals (e.g., 3 doses of 500mg)⁴²; Metformin: 250-750mg twice daily with meals or 800mg in the evening (Dosing suggestions vary in studies.).

Collagen: Collagen is at the top of the list of anti-aging supplements. As we age, our bodies create less collagen, which causes wrinkles and drooping of the skin. Collagen synthesis is thought to slow down as early as our early twenties. Supplemental collagen increases skin suppleness and minimizes wrinkles. A study of 72 women over 12 weeks showed that regularly taking a supplement containing 2.5 grams of collagen significantly improved the appearance of skin elasticity and softness. Adding collagen to your daily regimen can support healthy skin, joints, hair, skin, and nails, helping to slow the aging process in these tissues^{43 44 45}.

Nicotinamide Mononucleotide (NMN) and NAD+ (Tru Niagen): Nicotinamide could potentially slow the aging process by increasing the enzyme NAD+ (nicotinamide adenine dinucleotide). NAD+ works with proteins to preserve health via DNA repair and cellular energy. The effects of exercise and food restriction are mediated in part by sirtuin deacetylases (SIRT1-7), which are activated by NAD+ precursors⁴⁶. Our capacity to exercise and benefit from it decreases as we age, in large part due to a decline in endothelium NAD+. Vascular deterioration is one of the main causes of aging. Exercise promotes neovascularization, delaying the effects of aging. Nicotinamide mononucleotide (NMN), a precursor to NAD+, promotes SIRT1-dependent improvements in capillary density, which enhances blood flow and extends

endurance. These findings have implications for raising human performance, enhancing blood flow to organs and tissues, and restoring mobility. A more recent strategy is to take NAD⁺ precursors such as nicotinamide riboside (NR) or nicotinamide mononucleotide (NMN) to restore NAD⁺ levels. Dosages of 300mg per day have shown promising results for nicotinamide riboside (NR) and nicotinamide mononucleotide (NMN)'s antiaging effects^{47 48}. Suggested dose: Tru Niagen Brand 300mg every morning and/or 1g NMN in the morning.

Resveratrol: Resveratrol is a natural phenolic compound. Polyphenols are bioactive substances that have a wide range of health-promoting benefits through a variety of pathways, including antioxidation, anti-inflammation, immunomodulation, and gut microbiota modulation. Numerous foods, including grapes, red wine, peanuts, and blueberries, contain resveratrol. Resveratrol functions as an effective and safe substance for the prevention and treatment of aging and age-related disorders by reducing oxidative stress, reducing inflammation, increasing mitochondrial function, and controlling apoptosis (cell death). In animal models, resveratrol has been shown to increase lifespan primarily via promoting autophagy, lowering oxidative stress, and enhancing neuroprotection for neurodegenerative conditions. The use of resveratrol enhances behavioral performance, boosts the generation of new neurons, and encourages hippocampus neurogenesis.

Age-associated ovarian aging and infertility are accelerated by the loss of oocytes and follicles as well as the deterioration of oocyte quality. Treatment with resveratrol dramatically raises the ratios of fertilization, reduces oxidative stress, improves apoptosis, and decreases sperm binding loss while enhancing oocyte quality. Female mice demonstrated a greater follicular pool than age-matched controls after receiving resveratrol for 12 months. Resveratrol enhanced cortical thickness, bone volume, and bone trabecular number while decreasing bone trabecular spacing. Another study found that resveratrol increased bone formation by promoting osteoblast activity and bone growth. Overall, resveratrol supports multiple organ systems to stave off the signs of aging^{49 50 51}. Suggested dose: 1000mg (1g) daily in the morning.

Spermidine (Extracted from Wheat Germ): Spermidine may lower cardiovascular and cancer-related mortality in humans. High dietary spermidine intake is associated with lower blood pressure, a decreased incidence of cardiovascular disease, and reduced mortality (it predicted a lower incidence of death even after accounting for potential confounding variables like age, body mass index, use of alcohol or aspirin, diabetes, metabolic syndrome, physical activity, sex, socioeconomic status, and dietary quality). This finding supports spermidine's ability to cause biochemical changes that are comparable to those caused by caloric restriction. Similar to the benefits of intermittent fasting, spermidine can aid in delaying the aging process in humans^{52 53}. Suggested

dose: 1mg daily in the morning.

Turmeric: Turmeric contains curcumin, which fights free radical damage and minimizes inflammation both internally (preventing disease/cellular damage) and externally (protecting the skin from harmful UVA/UVB rays). Curcumin has been linked to benefits in cognition and memory, and is currently being researched as a potential treatment for Alzheimer's disease. Curcumin has a favorable effect on a variety of aging-related factors, including oxidative stress, cell damage, and telomere length decrease. Curcumin is protective against lipid peroxide-induced DNA damage and new studies suggest that curcumin promotes the DNA repair pathway in bone marrow. It also has anti-cancer activity. Turmeric is high in antioxidants, offering protection against free radical damage. Because of this, turmeric is also used topically to address skin damage and prevent damage associated to excess sun exposure.^{54 55 56 57 58} Suggested dose: 500-2000mg daily (Take curcumin with piperine for best absorption.).

Adaptogens are herbal medicines that help the body deal with physical and emotional stress. Adaptogens support healthy stress management, decrease damaging stress hormone output, and support adrenal (HPA-axis) function. The following adaptogens have research around their use for longevity and aging well:

Reishi (*Ganoderma lucidum*): Reishi mushroom has long been revered as the mushroom of immortality. Extractions from reishi assist with anti-aging, primarily through immunomodulation and inhibiting oxidation and neurodegeneration. They help with diabetes, are anti-cancer, help with cellular senescence, protect the liver, modulate the immune cascade, inhibit systemic inflammation, scavenge free radicals, help with stress, fatigue, hormonal balance, and brain fog, and are used to treat autoimmune disorders and slow the progress of neurological disorders.^{59 60} Suggested dose: Take 2 dropperfuls of a dual-extracted reishi tincture twice daily. Note that most medicinal mushrooms are better in a dual-extracted tincture form to access both the water-soluble and alcohol-soluble medicinal compounds. A tea is not enough.

Ashwagandha (*Withania somnifera*): Ashwagandha root extract boosts telomerase activity; thus ashwagandha can delay aging by protecting telomeres, ensuring genes can continue to replicate longer. An increase in telomerase activity of ~45% was seen at a dosage of 10 to 50 g of ashwagandha root extract. Ashwagandha also acts as a neuroprotective protecting the brain and nervous system from damage from stress and potentially prolonging life expectancy. Ashwagandha shows significant promise in degenerative and adult onset health diseases, with significant potential in supporting anti-aging pathways^{61 62 63}. Suggested dose: 2 dropperfuls tincture 1-2 times daily.

Cordyceps: Cordyceps increases antioxidants in elderly mice, helping with memory and sexual

function. It's this antioxidant concentration that may account for their anti-aging effects. Antioxidants scavenge free radicals, which otherwise promote illness and aging. Mice and fruit flies administered cordyceps lived significantly longer than those given a placebo. It is unknown whether cordyceps has the same anti-aging properties in humans, but animal trials have been quite promising. By preventing mitochondrial swelling and boosting antioxidant activity, cordyceps fruiting bodies defend mitochondria and as a result, may have significant medicinal benefits for age prevention and mitochondrial protection, effectively slowing down the aging process⁶⁴. Suggested dose: 2 dropperfuls of a dual-extracted cordyceps tincture 1-2 times daily.

Lion's Mane (*Hericium erinaceus*): Lion's Mane has promising antiaging benefits both neurologically and for the skin. Key components in lion's mane are effective in inhibiting the effects of cellular senescence on human dermal fibroblasts (HDFs), the cells responsible for skin repair. The polysaccharides in lion's mane boost the activity of antioxidant enzymes and raise collagen levels in the skin of elderly rats. A byproduct of aging in both humans and animals is lipofuscin.

Lipofuscin is made up of lipid-containing lysosomal breakdown byproducts and is found in increasing amounts in the liver, kidney, heart muscle, retina, adrenals, nerve cells, and ganglion cells as we age; it's known as the "wear-and-tear" pigment. It is continuously building up as cells deteriorate, causing cells to atrophy (waste away). Lion's mane polysaccharides dramatically decrease amounts of lipofuscin, suggesting a strong anti-aging effect on these organ systems. Lion's mane has also reduced markers of aging in human cell cultures.

As a result, there is support that lion's mane may create neuroprotective metabolites that could be used as effective treatments and preventative measures for age-related neurodegenerative illnesses. Lion's mane encourages the production of nerve growth factor (NFG) and myelin,⁶⁵ the insulation around nerve fibers. Both myelin and NFG are crucial to brain function and health. Lion's mane is a good option if you're looking for brain-boosting effects. It improves cognition and concentration, relieves anxiety⁶⁶, and decreases gut inflammation^{67 68}. Suggested dose: 2 dropperfuls of lion's mane tincture 1-2 times daily.

Panax Ginseng: This root's active ingredients are ginsenosides. Ginsenosides exhibit anti-aging activities in the skin that result from an increase of type I collagen production and the suppression of UV-induced cell damage. Numerous homeostatic illnesses, metabolic disorders, cardiovascular diseases, neurological diseases, and cancer are predisposed by the aging process.

Panax ginseng helps with age-related disorders as it assists with inflammation, oxidative stress, mitochondrial dysfunction, apoptosis, neurodegenerative and metabolic disorders, cardiovascular illnesses, and cancer.

that cause inflammation, and inflammatory cell activity can be inhibited by ginsenosides.

In addition, ginsenosides' anti-apoptotic, antioxidant, and anti-inflammatory properties play a role in neuroprotection. Ginseng extract benefits vascular endothelial function, lipid homeostasis, and the regulation of cholesterol and triglyceride levels. It elevates mood, lessens anxiety, and eases fatigue⁶⁹. Suggested dose: 2 dropperfuls of a dual-extracted tincture once daily before noon.

The two main contributing aspects to skin aging are: photo-aging brought on by exposure to UV rays and chronic aging brought on by the passage of time. As we age, the skin barrier breaks down and develops characteristics like dryness, wrinkles, and spots. Below are 3 powerful topical herbs to directly address the skin's signs of aging. I use them as an infused oil or topical salve; I make an herbal face oil with these 3 herbs that I use daily.

Calendula: Calendula is an effective collagen-boosting plant that also slows down collagen deterioration. It is anti-inflammatory and antimicrobial. Collagen synthesis and skin renewal are encouraged by calendula extract. Calendula stimulates the growth of epithelial cells, which helps to speed up cell turnover⁷⁰.

Calendula softens skin, promotes cell renewal, lessens the visibility of scars, and treats photo-damage, wrinkles, and dry skin. Calendula's vibrant orange hue is a result of carotenoids, chemicals linked to vitamin A. These fat-soluble compounds include lycopene, zeaxanthin, leutein, and beta-carotene. Carotenoids can improve immunity, reduce inflammation, increase the proliferation of epithelial cells to speed up cell replacement, and help prevent UV-induced skin damage.

Quercetin, flavonoids, terpenoids, and polyphenols are additional active ingredients in calendula; many of them are antioxidants and anti-inflammatory. These aid in preventing oxidative skin damage and skin stress from the sun, weather, pollution, and other environmental factors. Calendula also enhances skin tightness, firmness, elasticity, and moisture. The herb's flavonoids aid in skin brightness and age spot reduction. Calendula promotes faster wound healing, cures and prevents acne, and may help with psoriasis.

Cottonwood Buds (Balm of Gilead): Cottonwood buds have a strong antioxidant profile, are anti-inflammatory, and scavenge free radicals⁷¹, all of which help with aging skin. They are also anti-bacterial, anti-microbial, and anti-fungal and help with pain due to their salicylic acid content. Due to their role in cell regeneration and skin renewal, these buds are renowned for their anti-aging benefits⁷².

Rosehips: Rosehip oil preserves the integrity of skin cell walls by supplying necessary fatty acids, allowing the skin to retain water and stay hydrated. Vitamins A and C, which assist with skin cell renewal and turnover, are abundant in rosehips. Additionally, these vitamins aid in collagen

formation, which firms skin and increases elasticity.

Rosehips have a lot of vitamin E, which works in conjunction with vitamins A and C to prevent photo-aging and sun damage. Rosehips are antioxidant and anti-inflammatory. When applied topically, the fats in rosehip oil protect against age spots and reduce the appearance of acne scarring and stretch marks. Rosehip extract increases moisture levels, increases skin elasticity, promotes skin brightness, reduces the appearance of wrinkles, and decreases the depth of crow's feet. Rosehips provide a strong defense against UV rays and the ensuing skin damage⁷³.

5. Movement

Lifelong exercise helps people stay healthier for longer, delays the onset of chronic disorders and diseases, prevents cognitive decline, lowers the risk of falling, reduces stress and anxiety, and lengthens lives. Normal aging is associated with changes in body composition and physical resilience and strength are lowered with age. On average, there is a decline in lean body mass of 2 to 3% per decade from age 30 to 70.

Regular physical activity, including strengthening and flexibility exercises, contributes to the maintenance of muscles. Exercise can't reverse aging but there's clear evidence that exercise can activate the machinery necessary for DNA repair.

Builds Muscle Strength and Helps Coordination:

Sarcopenia is a condition in which people lose muscular mass and strength as they age. Resistance exercise is one of the best strategies to slow that decrease. It keeps your muscles strong and powerful, and makes daily tasks like cooking, cleaning, and climbing stairs easier. Additionally, it lessens your vulnerability to illness, enhances your mood and cognitive health, and supports extended periods of independence.

Improves Bone Density: The body breaks down old bone and replaces it with new bone tissue to maintain strong bones. However, after the age of 30, bone mass stops increasing. You gradually start making less bone than you are losing in your 40s and 50s. Exercise helps prevent osteoporosis, a condition that weakens bone and raises the chance of breakage as you age, by helping you build bone density.

Lengthens Telomeres: Age-related shortening of telomeres contributes to cell senescence, which renders cells incapable of dividing. Telomere length is associated with some chronic diseases, particularly heart disease, stroke, and high blood pressure. In comparison to people who are sedentary, research shows that higher levels of physical activity are associated with longer telomere lengths.

Improves Cognition: According to the National Institute on Aging, being able to switch between tasks rapidly, organize an activity, and reject extraneous information are all indications of excellent cognitive function. One of the most effective ways to maintain cognitive function and lower the risk of age-related cognitive decline is through

physical activity.

While it is still unclear if exercise can help prevent dementia, studies have shown a correlation between increased physical activity and a lower incidence of dementia, including Alzheimer's disease. Moving your body regularly is one of the best things you can do to stave off aging. Exercising 30 minutes per day, 5 days per week with a combination of aerobic and resistance exercises provides the most benefit.⁷⁴

6. Lifestyle

Sleep: Sleep is where we do most repairing, processing, and rejuvenating. Without quality sleep, our body's ability to heal and regenerate new cells and tissues significantly declines, as does our mental acuity. Read more about how to get quality sleep in the Sleep Protocol.

Stress: When we become stressed, our nervous system releases catecholamines in response. These substances (like adrenaline and the stress hormone cortisol), when released in excessive amounts over time, can damage otherwise healthy cells, accelerating the aging process.

Long term, chronic, stress causes us to age more quickly both internally and externally and makes us vulnerable to disease. See the Anxiety & Stress Protocol.

Meditation and Hobbies: Our parasympathetic (or rest and digest) nervous system is where repair and healing take place.

To shift from fight or flight (our sympathetic nervous system) to the rest and digest nervous system, we need to learn to intentionally relax; sleep alone won't make up for long stints of high stress levels. Breath-work, meditation, and creativity all help us move into this nervous system setting, increasing our health, wellness and vitality, preventing accelerated aging in the process.

These relaxing modalities also improve mental and emotional health and overall life fulfillment. Read more in the Anxiety & Stress Protocol.

7. External Environment

What we put on and around our bodies effects our biochemistry. It is important to create an external environment that supports healthy aging and prevents the acceleration of age-related disease states and skin damage. Here is a list of things to avoid and/or add to support longevity and stave off unwanted disease states associated with aging:

Hormone Disruptors: Phthalates, parabens, and BPA are examples of common endocrine disruptors found in common body products (creams, lotions, soaps) and home products (plastics, non-stick coatings, cleaning supplies). These can all add to premature aging and health problems.

They interfere with chemical messengers (hormones) and can decrease the efficiency of these pathways, exacerbating the aging process. Read ingredient labels and remove anything containing Bisphenol A (BPA), Perfluoroalkyl and Polyfluoroalkyl Substances

(PFAS), Phthalates, and Triclosan.⁷⁵

Sun: There is a fine line between preventing sun damage while getting enough Vitamin D. The primary way we make Vitamin D is from sun exposure. About 30 minutes in the sun can produce enough Vitamin D for an entire week. The RDA for Vitamin D is 800 IU per day; getting the recommended 15-30 minutes of midday sun can produce between 5,000-20,000 IU. Wear a (healthy) sunscreen when needed to prevent sun damage.

Topical Serums, Peels, and Creams: The most popular anti-aging skin serum/cream ingredients are Vitamin A, Vitamin C, and peptides (small protein building blocks for skin). These nutrients specifically target skin cells, addressing current damage and protecting skin from future damage.

Hazardous Environmental Substances and Products: Mold, poor air quality, air fresheners, perfumes, dryer sheets, etc. all add to our oxidative environmental stress load, add to the toxic load of the liver, and accelerate the aging process.

8. Western Medical Approaches

Hormone Replacement (Pellets, Injections, Creams to Give Aging Women and

Men the Hormone Profile of a Much Younger Person): Hormone replacement is one of the pillars of anti-aging. Women currently spend a third of their lives in a sex hormone deficient state.

Age-related testosterone decrease also affects men, yet andropause is still frequently misdiagnosed and poorly managed. The relevance of dehydroepiandrosterone (DHEA) in the generation of steroid hormones rises with age as a result of the decline in hormone production from gonads in both sexes.

DHEA declines with age in both sexes, as do estrogen and progesterone during menopause in women. Additionally,

there are age-related declines in growth hormone and insulin growth factor-1 levels.

When hormones decrease, it causes a cascade of biological changes associated with aging. We experience skin changes, lean body mass loss, erectile dysfunction, decreased sexual desire, and loss in bone mineral density.

Hormone replacement therapy may help ease the severity of these hormonal declines. In order to more precisely define treatment modalities, large-scale studies are still required along with long-term safety research.⁷⁶

Injectables (Muscle Relaxants Like Botox and Dermal Fillers Like Juvederm): Botox, or botulinum toxin, is most frequently injected into the face to smooth out wrinkles and fine lines.

It eases the tension in the muscles that cause our skin to furrow and wrinkle, preventing them from contracting as forcefully and giving us a more youthful appearance.

Although its primary function is cosmetic, it can also be used to treat hyperhidrosis (excessive sweating) as well as jaw clenching and migraines. Dermal fillers are designed to give volume to wrinkles and folds. Hyaluronic acid, a naturally occurring substance in our skin that is lost with aging, is injected into the skin.⁷⁷

Pharmaceuticals / Medications: Compounds such as Dasatinib (used with quercetin⁷⁸) and Navitoclax are a part of a newer class of pharmaceuticals used to enhance and perhaps slow down aging. In addition, drugs like metformin and NAD+ precursors are promising anti-aging candidates because they appear to support improved organ function, increased physical resistance, disease resistance, and prolonged life expectancy, particularly in terms of cardiovascular and cancer mortality.

More long-term clinical trials are needed to demonstrate effectiveness and safety of anti-aging prescription medication.⁷⁹

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	10 min. stretching or tai chi + 10 min. meditation Healthy breakfast Collagen + protein powder Drink 1 liter water Panax Ginseng & Reishi tincture Resveratrol, NAD+, NMN, & Spermidine supplements	Grocery Shop: focus on anti-aging foods (produce of ALL colors, fiber-rich foods, antioxidants, fish) Healthy lunch w/ fiber Probiotics, Omega-3s, vit. A, C, D, & Zinc	20 min. sun exposure 30 min. movement (dance, yoga, weights, walk, etc.) Drink 1 liter water Berberine	Healthy high fiber dinner 3-mushroom tinctures** Drink 1 L water Berberine Turmeric	Sleep hygiene (no screens 1-2 hrs. before bed; write down thoughts to clear mind, sleep herbs if needed) Turmeric milk latte*** 10 min. light stretching Ashwagandha tincture Sleep 8 hrs.
MONDAY	20 min.: stretching + meditation Healthy breakfast Collagen + protein powder Drink 1 liter water Panax Ginseng & Reishi tincture Resveratrol, NAD+, NMN, & Spermidine supplements	Healthy lunch w/ fiber Probiotics, Omega-3s, Vit. A, C, D, & Zinc Happy hobby 1 hour	20 min. sun exposure 30 min. movement Drink 1 liter water Berberine	FAST; water only 3-mushroom tinctures** Drink 1 L water	FAST; water only Sleep hygiene Gratitude journal Ashwagandha tincture Sleep 8 hours
TUESDAY	FAST 20 min.: stretching + meditation Healthy breakfast Collagen + protein powder Drink 1 liter water Panax Ginseng & Reishi tincture Resveratrol, NAD+, NMN, & Spermidine supplements	Healthy lunch w/ fiber Probiotics, Omega-3s, Vit. A, C, D, & Zinc Call someone you love for a chat or see a friend	20 min. sun exposure 30 min. movement Drink 1 liter water Berberine	Healthy dinner 3-mushroom tinctures** Drink 1 L water Berberine Turmeric	Sleep hygiene Turmeric milk latte*** 10 min. light stretching Ashwagandha tincture Sleep 8 hrs.
WEDNESDAY	20 min.: stretching + meditation Healthy breakfast Collagen + protein powder Drink 1 liter water Panax Ginseng & Reishi tincture Resveratrol, NAD+, NMN, & Spermidine supplements	Healthy lunch w/ fiber Probiotics, Omega-3s, Vit. A, C, D, & Zinc	20 min. sun exposure 30 min. movement Drink 1 liter water Berberine	Healthy dinner 3-mushroom tinctures** Drink 1 L water Berberine Turmeric	Sleep hygiene Turmeric milk latte*** 10 min. light stretching Ashwagandha tincture Sleep 8 hrs.
THURSDAY	20 min.: stretching + meditation Healthy breakfast Collagen + protein powder Drink 1 liter water Panax Ginseng & Reishi tincture Resveratrol, NAD+, NMN, & Spermidine supplements	Healthy lunch w/ fiber Probiotics, Omega-3s, Vit. A, C, D, & Zinc Happy hobby 1 hour	20 min. sun exposure 30 min. movement Drink 1 liter water Berberine	Healthy dinner 3-mushroom tinctures** Drink 1 L water Berberine Turmeric	Sleep hygiene Turmeric milk latte*** 10 min. light stretching Ashwagandha tincture Sleep 8 hrs.
FRIDAY	20 min.: stretching + meditation Healthy breakfast Collagen + protein powder Drink 1 liter water Panax Ginseng & Reishi tincture Resveratrol, NAD+, NMN, & Spermidine supplements	Healthy lunch w/ fiber Probiotics, Omega-3s, Vit. A, C, D, & Zinc Call someone you love for a chat or see a friend	20 min. sun exposure 30 min. movement Drink 1 liter water Berberine	Healthy dinner 3-mushroom tinctures** Drink 1 L water Berberine Turmeric	FAST; water only Sleep hygiene Gratitude journal Ashwagandha tincture Sleep 8 hours
SATURDAY	FAST 20 min.: stretching + meditation Healthy breakfast Collagen + protein powder Drink 1 liter water Panax Ginseng & Reishi tincture Resveratrol, NAD+, NMN, & Spermidine supplements	Healthy lunch w/ fiber Probiotics, Omega-3s, Vit. A, C, D, & Zinc	20 min. sun exposure 30 min. movement Drink 1 liter water Berberine	Healthy dinner 3-mushroom tinctures** Drink 1 L water Berberine Turmeric	Sleep hygiene Turmeric milk latte*** 10 min. light stretching Ashwagandha tincture Sleep 8 hrs.

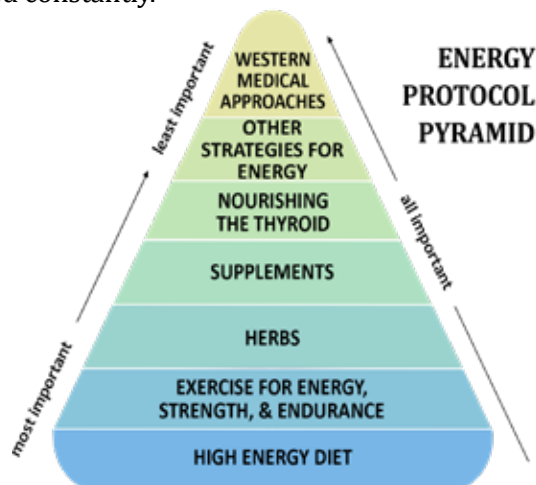
*In the mornings: Apply Vitamin C Serum w/ retinol and SPF (products w/o hormone disruptors) and herbal oil (see protocol).

** Take 3 dual-extracted mushroom tinctures: Reishi, Cordyceps, and Lion's Mane.

*** Turmeric, warm dairy free milk, dash of black pepper, drizzle of raw honey.

Ch. 41: Energy Protocol

Do you feel like you don't have the energy you used to have? Is it harder to get motivated? If you feel like your loss of energy has become a chronic problem, this protocol focuses on ways to help you regain the energy you need to live your best life, while also helping you determine the cause of your energy depletion and/or brain fog. Symptoms of low energy levels and fatigue are: very dry skin, allergies, overweight or underweight, depression, fertility issues, food cravings, brain fog, and feeling fatigued constantly.



If you find yourself compensating for a lack of energy by treating the symptoms, start looking into the root cause. Common causes of a chronic lack of energy include: underlying infection, thyroid issues, systemic inflammation, autoimmune disease, stress and anxiety, poor diet, lack of self-care, lack of exercise, lack of quality sleep, menopause or a hormonal imbalance, anemia or low iron, and/or a vitamin deficiency (especially vitamin B).

It is important to identify the root cause. Think about your life and habits. How long have you been suffering from a lack of energy? Can you correlate this with anything else going on in your life? How is your diet? Keep in mind that your energy issue can be the result of more than one factor, such as a combination of stress and autoimmune disease or a lack of sleep coupled with poor diet.

The effects of low energy levels can creep into many areas of your life and can cause you to neglect putting forth effort because you don't have the vitality to tackle everyday tasks. The good news is that whatever the cause of your energy depletion, there are medicinal herbs and lifestyle changes you can make to regain your strength.

1. High Energy Diet Food is your body's main source of fuel and without the proper nutrition your energy levels can suffer significantly. Eating an anti-inflammatory whole foods diet is a great start. Please see our Anti-Inflammatory Diet for more.

Watch Your Portions: Eating large portions is bad for digestion, your waistline, and it can stretch your stomach. Instead of three big meals, try eating small,

frequent meals instead. These give you a steady source of energy throughout the day. Try five to six small meals a day with plenty of fruit, vegetables, and leafy greens.

Drink Plenty of Water: Water is the main component of blood. It helps bring nutrients to the cells and carries away waste. When you are dehydrated, one of the first symptoms is fatigue. This is because the body isn't getting the water it needs to energize and hydrate your cells. Stay hydrated.

Limit Alcohol and Other Depressants: Alcohol is a depressant; it slows down the nervous system, our breathing, and our heart rate. All of which can cause a lack of energy. Many drugs – legal and illegal – are also depressants.

Don't Starve to Lose Weight: Avoid the latest fad or crash diet. Starving yourself is a sure way to deplete nutrients needed for energy. Eat frequent, healthy, small meals.

Ditch the Processed Foods: Processed foods contain preservatives, additives, food dyes, and trans fats. These ingredients are inflammatory, can trigger autoimmune flares, upset the digestive system, and are associated with weight gain. If it comes in a box or package, think twice.

Eat More Fruits and Vegetables: Fruit has sugar, but fruit sugar is much easier for the body to process. Fruit contains vitamins (especially vitamin C) and simple carbohydrates. Simple carbohydrates can give us the energy our body needs to function effectively. Avoid complex carbohydrates that come from processed foods like breads and pasta. Eat seasonal organic fruits, including antioxidant-rich berries. Vegetables and leafy greens are full of vitamins and minerals that help give you energy. Not only are they nutrient-dense, they contain beneficial prebiotics that help support a healthy microbiome. Start your meal by eating plenty of fruit, vegetables, and leafy greens and you may find that you don't crave sweets after a meal. If you are still hungry, try a lean source of protein (see below).

Eat Lean Protein: Lean sources of protein are good sources of energy. Eat small portions (six ounces) of baked, grilled, boiled, or broiled (not fried) chicken, turkey, or fish once or twice a day with a meal. Avoid processed or fried meats. Organic grass-fed red meat is okay once or twice a week. If possible, eat wild caught fish instead of farm raised. Some great fish to eat include salmon, mackerel, cod, and herring. If you are vegan, you can obtain protein by eating more nuts like walnuts, almonds, or pecans.

Sugar and Caffeine (Unhealthy Stimulants): While both sugar and caffeine give you a kick of energy, both have negative consequences. You are very likely to have a "sugar crash" after your temporary bout of energy. Sugar is associated with weight gain and systemic inflammation, both of which deplete energy levels. Instead of sugar, try a healthier substitute like stevia. Caffeine in excess of a morning cup of coffee can cause an increased heart rate and can worsen anxiety. If you have to have a cup of coffee every few hours to function effectively, rethink this strategy.

Feed Your Gut: Low energy, inflammation, autoimmune

issues, anxiety, depression and more often stem from your microbiome. Many people have increased intestinal permeability, also known as “leaky gut”. Make sure to eat plenty of prebiotic and probiotic foods plus take a probiotic supplement. Prebiotics feed the “good” gut bacteria (probiotics). **Prebiotic foods** that can help keep your gut bacteria healthy and balanced include¹: leeks, onions, garlic, asparagus, apples, bananas, seaweed, chia and flax seeds, Jerusalem artichokes, mushrooms, oats, dandelion leaves, burdock root, jicama root, chicory root (great coffee substitute), lentils, split peas, beans, chickpeas, spinach, and blueberries. Foods that contain probiotics have usually been fermented to naturally produce these bacteria. Fermented **probiotic foods** to incorporate into your daily diet include: kombucha, kimchi, miso soup, fresh (not canned) sauerkraut, plain full fat yogurt with live or active cultures, kefir, tempeh, and natto. For more information, please see our Leaky Gut Protocol.

Lose Weight if You Are Overweight: Obesity is a big contributor to depleted energy levels. When people are overweight, they have more weight to carry around, which makes simple tasks harder. Extra weight puts a stressful load on the joints and also makes the heart work harder. See a nutritionist or health coach to learn how much weight you need to lose and how many calories you need to eat per day to lose weight. Also see our Weight Loss Protocol. Begin eating more fruit, vegetables, leafy greens, and lean meats to meet your caloric needs.

Bone Broth: Bone broth is full of valuable nutrients, collagen, and bone-strengthening minerals. It is nourishing, especially when made with healthy vegetables and greens like stinging nettle and dandelion leaves. It contains compounds that may ease inflammation, keep your gut healthy, and increase energy levels. Drink 1-2 cups daily. It's a great drink first thing in the morning.

“Energy” Foods: Examples of nutrient-dense foods that provide natural energy that sticks with you throughout the day are: edamame, avocados, wild greens, apples, fish. Edamame is a superfood which contains 27 grams of protein per cup. It is an excellent protein source for those who don't eat or limit meat consumption. It is also high in iron, and can help counter the energy-depleting effects of anemia. Avocados are packed with fiber, antioxidants, and B vitamins. They also contain fatty acids to help with nutrient absorption and sustain energy levels. Leafy greens (spinach, kale, chard) are full of vitamins C and K, lutein, folic acid, magnesium, iron, and fiber. Apples are rich in vitamins, fiber, and simple carbohydrates. They are a low-glycemic index fruit, so they will not cause a sugar rush and crash. Their antioxidant content may contribute to the slowing of carb digestion, which can prolong their energizing effects. Fish like salmon, tuna, and sardines contain protein, B vitamins, and omega 3 fatty acids that can reduce the inflammation responsible for fatigue.

2. Exercise

In addition to diet, daily physical activity sends oxygen and nutrients to the body's cells, which in turn help the heart and lungs work more efficiently to boost energy levels. Spend at least 30 minutes every day doing some sort of exercise. It may not

be easy at first, especially if you are fatigued, but it will get easier over time if you commit to making it a daily habit.

It isn't advisable to jump into daily exercise by pushing yourself to finish long, intense workouts. Start out with exercise routines that won't burn you out or push your body further than it can currently go. Below are exercises to do daily for energy and stamina, from beginner to intermediate levels:

The exercises below are perfect for those just starting out. Feel free to switch it up each day and try something new. Make sure to always stretch before exercising!

Power Walking: Whether it is on a treadmill or outdoors, take a 30-minute power walk. The time passes quickly when you are listening to music or enjoying beautiful outdoor scenery. Walking not only protects the joints, it also increases oxygen flow throughout the body.

Beginners' Yoga: Yoga is great for improving strength, balance, and flexibility, in addition to helping energy levels. Yoga has an abundance of moves ranging from easy to difficult (and everything in between). Try some simple yoga poses and hold them for ten to thirty seconds. Good poses to start with include the warrior pose, planks, hero, high lunge, and triangle.

Low Impact Cardio: Cardio may sound intense, but you can still complete a cardiovascular exercise if you are a beginner. Low impact cardio exercises like jumping jacks, burpees, and high-knees (start by doing ten and work your way up over time to 30 or more if you can).

Low Impact Strength Training: Grab small dumbbells (whatever weight is comfortable for you) for low-impact strength training exercises. Examples include bicep curls, staggered deadlifts (with a small weight), calf raises, and glute bridges.

Once you are ready to take on a new challenge, try some intermediate exercises. It is important to work your way to intermediate, as your body will become stronger and more able to accomplish the tasks you set out to complete. Gently challenging your body builds endurance and energy over time. Try one of the following exercises 30 minutes daily:

Bike Riding: Outdoor biking is a great workout and helps burn a lot of calories. If you live in an area with lots of hills and mountains, it may not be very easy. In this case, you can use an indoor bike that inclines to provide a challenge.

Jogging/Running: Gentle jogging/running can work wonders for building endurance, stamina, and energy levels. You can exercise outdoors or indoors on a treadmill.

Intermediate Yoga: Once you have mastered beginner yoga poses, try some intermediate poses. Good poses to try include bridge pose, bow pose, dolphin pose, crane pose, and eagle pose.

Intermediate Cardio: After you have mastered low impact cardio exercises, you may be ready for intermediate cardio. Examples include 100-meter sprints, running stairs, and jumping rope.

Exercising outdoors is one of the best ways to exercise. Besides the abundance of fresh air, you get to spend time in nature, which is effective against stress and anxiety². This is good for those wishing to increase energy levels, as stress and anxiety contribute to energy depletion. Below are exercises to try outdoors:

Swimming: Swimming is a great way to build endurance. It causes the body to use muscle groups that other exercises cannot touch. It also boosts cardiorespiratory health. It may also help decrease anxiety and depression while boosting the mood.

Outdoor Walking/Jogging/Running: The best way to enjoy walking, jogging, or running is outdoors in nature.

Hiking: If you like exploring nature, hiking is a great option. Hiking can help lower blood pressure, improve mental health, and stabilize cholesterol levels. It is also an excellent way to burn calories.

Whatever exercises you choose to complete each day, you will reap the many health benefits that come with daily exercise, including improved cardiovascular health, increased endurance, increased strength, and increased energy.

3. Herbs

Diet and exercise are your first line of defense against fatigue and drained energy. However, there are herbs and supplements you can take to give your body the boost it needs to get through each day with energy to spare. Below are excellent options for increasing energy naturally:

Ashwagandha (*Withania somnifera*): Ashwagandha is an adaptogenic root that has been used for centuries to bring energy, reduce fatigue, and increase stamina³. Ashwagandha can be helpful for those with energy issues caused by hypothyroidism. It is also beneficial for those who suffer from stress and anxiety; its adaptogenic properties help protect the body from the effects of stress. Suggested dose: two dropperfuls of ashwagandha tincture up to three times daily.

Lion's Mane (*Hericum erinaceus*): This anti-inflammatory immunomodulatory adaptogenic mushroom is known to increase energy levels, relieve fatigue, and help with gut inflammation. Lion's mane is high in antioxidants that help reduce lactic acid buildup in the blood, increase blood oxygen levels, and lower muscle fatigue. It also increases glycogen levels in muscles to provide a readily available energy source for the body. Like other adaptogens, lion's mane can protect the body from the damaging effects of stress. This beneficial effect can work wonders for those with low energy due to stress, anxiety, and depression. Studies⁴ report that lion's mane contains anti-fatigue, antibiotic, anticarcinogenic, antidiabetic, neuroprotective, cardioprotective, and hepatoprotective properties. Suggested dose: two dropperfuls of dual-extracted lion's mane tincture two to three times daily.

Reishi (*Ganoderma lucidum*): Another powerful

adaptogenic immunomodulatory mushroom, reishi can significantly reduce fatigue, increase mental health, and improve overall well-being and quality of life⁵. It is great for those with adrenal fatigue, as it can regulate hormones and bring bodily systems back into balance. Reishi can protect the body from the effects of stress, making it useful for those who have low-energy due to stress, anxiety, and depression. It is also anti-inflammatory. Suggested dose: two dropperfuls of reishi dual-extracted tincture two to three times daily.

Cordyceps: Cordyceps is a very beneficial adaptogenic anti-inflammatory mushroom for fatigue because it improves athletic performance and circulation while also boosting energy levels. It can restore health after exhaustion and is well-known in Asia for enhancing endurance and strength⁶. Its antioxidant, antimicrobial, antiviral, and immunomodulatory properties make it an especially valuable remedy for a variety of health issues. Like the other immunomodulatory mushrooms (reishi, lion's mane, and turkey tail) cordyceps can help modulate the immune system for those with autoimmune disorders. Suggested dose: two dropperfuls of cordyceps dual-extracted tincture two to three times daily.

Panax Ginseng: Ginseng is a plant adaptogen, so it can help protect the body from the effects of stress while balancing other bodily systems. American (panax) ginseng is well-known for its ability to provide energy, strength, stamina, and endurance⁷. Suggested dose: one to three dropperfuls of ginseng tincture up to twice daily.

Rhodiola: This plant can help improve energy, reduce chronic fatigue, and boost athletic performance⁸. It is famous for its ability to enhance physical strength and endurance to allow athletes to minimize the effects of physical stress. It may also help regulate thyroid hormones, which is one reason this plant can help increase energy⁹. Rhodiola may also help improve sexual health, libido, stamina, and increase fertility. Do not use rhodiola if you have an autoimmune disease. Suggested dose: two dropperfuls of rhodiola tincture up to two times daily.

Cat's Claw (*Uncaria tomentosa*): For energy reserves impacted by a high viral load¹⁰, cat's claw may help fight the virus so you can feel better faster and have more energy. It is rich in quinovic acid glycosides that treat viruses like herpes, Epstein-Barr, hepatitis, HPV, HIV, dengue fever, and more. Its anti-inflammatory properties help to reduce energy-draining inflammation, while its antibacterial properties attack invaders to help you heal. It is also effective for chronic fatigue syndrome. Use with care if you have an autoimmune disorder. Suggested dose: one dropperful of cat's claw tincture up to three times daily.

Moringa: This versatile herb is rich in antioxidants and active compounds that boost thyroid health to fight fatigue and encourage high energy levels. Moringa also contains significant amounts of tryptophan. This helps to stabilize nerve and brain function by increasing the production of serotonin, a neurotransmitter that calms mood swings, insomnia, and depression¹¹. Moringa can help to combat

fatigue as well. Suggested dose: one dropperful up to three times daily.

Green Tea: Instead of coffee, try drinking a cup of antioxidant green tea in the mornings for energy. It is also rich in inflammation-fighting polyphenols. Green tea may also boost the metabolic rate to help you burn fat and lose weight. Suggested dose: one to two teaspoons of dried and powdered green tea leaves infused in a cup of hot water daily.

Peppermint: Peppermint has long been known to provide an energizing and stimulating boost in just about any form. Research shows that the scent of peppermint essential oil is effective for increasing energy and improving memory and alertness¹². Suggested dose: one to two teaspoons of dried peppermint leaves infused in a cup of water daily and/or inhalation of peppermint essential oil as needed.

Essential Oils: In addition to peppermint, there are other essential oils that can boost energy and mood. These include lemon, orange, grapefruit, and other citrus essential oils, as well as eucalyptus and rosemary. Essential oils are concentrated plant isolates, and as such, they don't require invasive uses to experience a positive effect. Simple inhalation of most essential oils is all that is needed to feel refreshed and revitalized. Suggested dose: Place two drops on a cotton ball and place the cotton ball inside a plastic baggie. Carry with you throughout the day and remove as needed to inhale.

4. Supplements

Like herbs, supplements can help support the body and provide nutrients to address low energy and fatigue.

B Vitamins: Certain autoimmune diseases and health conditions can deplete B vitamins and leave you feeling drained. A deficiency in B vitamins such as B12 may result in shakiness, fatigue, low blood pressure, muscle stiffness, and mood disturbances. If you are not getting enough B vitamins, you can take a B vitamin complex to provide you with the different B vitamins your body needs for enhanced energy¹³. Suggested dose: Dosages will vary depending on the supplement you purchase so make sure to follow the directions on the bottle when using a B complex supplement.

Iron: If anemia or low iron levels have you feeling depleted, an iron supplement can help build iron levels back up¹⁴. A naturally derived iron supplement made with beet root powder is a great option. Only take iron supplements if you are indeed low on iron or anemic. Get a blood test to be sure. Suggested dose: Dosage varies depending on your iron levels, so check with your doctor before beginning an iron regimen.

Magnesium: Magnesium is sometimes called the “sleep mineral” but it can help fight fatigue as well. It creates more energy in the body by activating ATP at the cellular level. ATP is the main source of energy in our cells, and it binds to magnesium to be biologically active¹⁵. Getting enough magnesium can help you get a better night's sleep as well, which will in turn help you feel more energetic

during the day. Suggested dose: 310-420 mg daily.

CoQ10: Coenzyme Q10 is a compound that the body produces naturally. You can also get it in small amounts from certain foods such as fish, organ meat, and whole grains. As we age, we produce less and less of this compound. It is a powerful antioxidant that helps to reduce oxidative damage of the body's cells. This can improve heart health¹⁶ and boost energy levels. It may also help improve exercise performance and muscle recovery. Suggested dose: 90-200 mg of COQ10 daily.

5. Nourishing the Thyroid

The thyroid is often the cause of fatigue and low energy because it is responsible for releasing thyroid stimulating hormone (TSH) to regulate the amount of thyroxine (T4) and triiodothyronine (T3) hormones produced by the gland. T4 is the main hormone responsible for transporting T3 to organs in the body. T3 also works to help with the delivery of oxygen and energy to cells, tissues, and other areas of the body. Sometimes the thyroid has trouble converting T4 to T3, and as a result metabolism slows and energy levels drop significantly.

In addition to fatigue and low energy, other symptoms of a thyroid disorder include: constipation, depression, weight gain or loss, slowed heart rate, dry skin or hair, infertility, muscle and joint aches, feeling cold, and reduced sweating.

If you feel like you have had low energy over a period of time and cannot figure out why, have your thyroid hormones checked to rule out a thyroid issue. Do not take herbal remedies or supplements for thyroid issues without first having an understanding of your particular issue. If your tests show that you have a thyroid disorder, there are many treatment options. See the Thyroid Protocol for more information on how to successfully treat thyroid disorders. Note that strategies and herbs differ depending on whether you have hyper- or hypothyroidism.

6. Other Strategies For Energy

Sleep: Getting enough sleep is a vital part of making sure you have the energy it takes to conquer the day. Many people do not get the sleep they need for their body to function properly and this can significantly affect energy levels. You need 7-8 hours of uninterrupted sleep each night. If you are unable to sleep, see our Sleep Protocol.

Control Stress: Chronic stress and anxiety will wreak havoc on your body and over time this can affect energy levels. Taking adaptogens like the ones mentioned in the herbal remedies section can help protect your body from the damaging effects of stress. Also, see our Anxiety and Stress Protocol.

Smoking: Smoking affects many aspects of health, including energy levels. Smoking affects circulation in the body, which in turn affects oxygenation. When the body's tissues are unable to get the oxygen they need, fatigue can develop. If you are a smoker, take steps to quit and don't expose yourself to secondhand smoke.

Take a Break: Sometimes you just need to get away and take a short break. This can restore energy levels and leave you feeling rejuvenated. Make sure to use your vacation time and get away when you can. Spend more time in nature, gardening, or just doing what you love. Find a place you enjoy spending time to escape to as needed. Even a short break can lift your spirits, boost your mood, and leave you feeling more invigorated.

Watch Your Mindset: Your attitude plays a large role in how you perceive the world around you. A bad attitude can affect your mental and physical health in a negative way.

If you constantly focus on being tired and direct your thoughts to your low energy levels, your energy levels can be impacted. Speak positive affirmations, give daily gratitudes, and think positive, energetic thoughts. Focus on what you want and what you can do to improve.

Listen to Upbeat Music: Listening to upbeat music can energize and invigorate the mind and body. Get together some of your favorite upbeat tunes and make a playlist. Put your earbuds in and disconnect for a while. You can listen to music while you work out for a helpful boost to let you push through or you can listen to positive music when you drive to work to get your day started on a positive note.

Keep a Journal: If a cluttered mind has you feeling weighed down, it may help to get your feelings out in a journal. This can also make room for you to focus on other important tasks, be present, and give you more energy. Writing is also a great way to stimulate creative thinking and give you the motivation to live each day with a positive attitude.

Stay Hydrated: This has already been mentioned, but it is worth mentioning again because if you aren't getting enough water, dehydration can set in. Dehydration can cause severe fatigue and less oxygen in your bloodstream. Eight eight-ounce glasses of water daily is a good place to start.

Deep Breathing: Taking deep breaths (inhaling slowly and deeply through your nose and then slowly exhaling through the mouth for ten to fifteen minutes) is a great way to increase energy and focus.

Go somewhere quiet and close your eyes to practice uninterrupted deep breathing. Try focusing on your breathing and not letting your mind wander. Do this a few times a day to reap its oxygenating and stress-relieving benefits.

Get Hormones Balanced: See your doctor to have bloodwork and diagnostics to determine if a hormonal imbalance is to blame for your low energy levels. Women often suffer from hormonal imbalances when they are

going through perimenopause and menopause.

For more information on how to manage hormonal imbalance, see the Women's and Men's Sexual Health Protocols, as well as the Menopause Protocol.

7. Western Medical Approaches

Treatment for fatigue

and low energy vary greatly depending on the cause of the problem. If you visit your doctor for low energy issues, they will likely run a series of diagnostics to better determine the root cause of the issue.

The cause could range from thyroid issues to lack of sleep or depression. Expect bloodwork, as this helps determine if the issue is being caused by a thyroid disorder. Other tests may include a mental health assessment or sleep assessment. Below are some common medications and treatments for fatigue:

Low Energy Caused by Anxiety or Depression: Chronic fatigue and depression go hand-in-hand. A doctor may screen you for depression and prescribe antidepressants. In addition to antidepressants, your doctor may prescribe anti-anxiety medications like Xanax.

Low Energy Caused by Thyroid Disorder: If a blood test shows that your thyroid is to blame for your energy issues, you may be prescribed an iodine supplement or drugs like levothyroxine, Synthroid, or Levoxyl.

Low Energy Caused by Lack of Sleep: If sleep deprivation is behind your low energy, your doctor may have you change your bedtime routine to promote better sleep or test you for sleep apnea. If sleep apnea is the cause of your sleep issues, you may be advised to use a sleep apnea machine. Lastly, a doctor may prescribe sedative drugs to help you sleep.

Low Energy Caused by Chronic Pain: If chronic pain is slowing you down, your doctor may prescribe NSAIDs like ibuprofen, naproxen, or prescription drugs like pregabalin, duloxetine, amitriptyline, or gabapentin.

However, it is important to get to the root of your pain rather than take pills to treat the symptom. Try to find the root cause of your pain and treat this rather than rely on these drugs, which can affect the organs negatively over time.

Orthostatic Intolerance: Sometimes those with chronic fatigue syndrome or low energy feel faint or nauseated when they get up too fast or sit upright. If this is the case, your doctor may prescribe something to regulate blood pressure or heart rhythms.

Low Energy Caused by Hormonal Imbalance: Depending on the imbalance, a doctor may prescribe hormone therapy to compensate for the lack of a certain hormone.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Deep breathing 10 min. Cup of green tea with small, healthy breakfast B vitamins, Probiotics, and CoQ10 Ginseng tincture	Small, healthy lunch w/ pre- and probiotics Adaptogenic mushroom dual extracted tinctures* Exercise 30 min. Inhale peppermint essential oil	Small, healthy snack (have another snack in a few hours if you want) Ginseng tincture Deep breathing 5-10 min. Outdoor hike/walk	Small, healthy dinner Hydration check (Have you drank enough water today?) Adaptogenic mushroom dual extracted tinctures*	Deep breathing 10 min. Bedtime routine (no blue light, wind down, sleep herbs if needed, etc.) Ashwagandha tincture Magnesium
MONDAY	Deep breathing 10 min. Cup of green tea with small, healthy breakfast B vitamins, Probiotics, and CoQ10 Ginseng tincture	Small, healthy lunch w/ pre- and probiotics Mushroom tinctures (2)* Exercise 30 min. Peppermint essential oil	Small, healthy snack Ginseng tincture Deep breathing 5-10 min. Outdoor hike/walk	Small, healthy dinner Hydration check Mushroom tinctures (2)*	Deep breathing 10 min. Bedtime routine Ashwagandha tincture Magnesium
TUESDAY	Deep breathing 10 min. Cup of green tea with small, healthy breakfast B vitamins, Probiotics, and CoQ10 Ginseng tincture	Small, healthy lunch w/ pre- and probiotics Mushroom tinctures (2)* Exercise 30 min. Peppermint essential oil	Self-care/break time (at least an hour) Small, healthy snack Ginseng tincture Deep breathing 5-10 min. Outdoor hike/walk	Small, healthy dinner Hydration check Mushroom tinctures (2)*	Deep breathing 10 min. Bedtime routine Ashwagandha tincture Magnesium
WEDNESDAY	Deep breathing 10 min. Cup of green tea with small, healthy breakfast B vitamins, Probiotics, and CoQ10 Ginseng tincture	Small, healthy lunch w/ pre- and probiotics Mushroom tinctures (2)* Exercise 30 min. Peppermint essential oil	Small, healthy snack Ginseng tincture Deep breathing 5-10 min. Outdoor hike/walk	Small, healthy dinner Hydration check Mushroom tinctures (2)*	Epsom salt bath Bedtime routine Ashwagandha tincture Magnesium
THURSDAY	Deep breathing 10 min. Cup of green tea with small, healthy breakfast B vitamins, Probiotics, and CoQ10 Ginseng tincture	Small, healthy lunch w/ pre- and probiotics Mushroom tinctures (2)* Exercise 30 min. Peppermint essential oil	Self-care/break time (at least an hour) Small, healthy snack Ginseng tincture Deep breathing 5-10 min. Outdoor hike/walk	Small, healthy dinner Hydration check Mushroom tinctures (2)*	Deep breathing 10 min. Bedtime routine Ashwagandha tincture Magnesium
FRIDAY	Deep breathing 10 min. Cup of green tea with small, healthy breakfast B vitamins, Probiotics, and CoQ10 Ginseng tincture	Small, healthy lunch w/ pre- and probiotics Mushroom tinctures (2)* Exercise 30 min. Peppermint essential oil	Small, healthy snack Ginseng tincture Deep breathing 5-10 min. Outdoor hike/walk	Small, healthy dinner Hydration check Mushroom tinctures (2)*	Epsom salt bath Bedtime routine Ashwagandha tincture Magnesium
SATURDAY	Deep breathing 10 min. Cup of green tea with small, healthy breakfast B vitamins, Probiotics, and CoQ10 Ginseng tincture	Small, healthy lunch w/ pre- and probiotics Mushroom tinctures (2)* Exercise 30 min. Peppermint essential oil	Outdoor hike/walk Small, healthy snack Ginseng tincture Deep breathing 5-10 min.	Small, healthy dinner Hydration check Mushroom tinctures (2)*	Deep breathing 10 min. Bedtime routine Ashwagandha tincture Magnesium

*Choose two of the following: Lion's Mane, Reishi, Cordyceps.

Ch. 42: Hair Loss Protocol

Hair loss is also referred to as baldness or alopecia. Severity ranges from mild to complete hair loss. There are a range of factors that can contribute to hair loss. These include stress, genetics, hormone imbalance, bad hair habits, some medications, aging, and illness. The average person loses up to 100 hairs each day. This may seem like a lot, but for most people, this hair is replaced, making up for the loss. There are circumstances that can cause more hair to fall out or for hair to not grow back as it's supposed to. Determining what is causing your hair loss is essential to figuring out how to reverse it.



The earlier you act, the better your chances of stopping hair loss. Warning signs include:

Thinning on Top of the Head: For men, the top of the head may not look as full as it once did. Watch for a receding hairline. While a receding hairline affects more men than women, women can also be affected.

Hair Is Looser than Normal: Run your fingers through your hair gently. Do you notice more hair coming out than usual? This could be a sign that your hair is in distress. Those who have gone through a traumatic event or shock sometimes notice this phenomenon. If sudden trauma is the cause, it will likely grow back over time.

Bald Patches: Some people have circular patches of missing hair on their head. The spot may be irritated or itchy. This may be a sign of a dermatological issue.

Scaly Patches that Spread on the Scalp: This needs treatment right away, as it may be ringworm – a fungal infection. The good news is that it is easily treatable with applications of tea tree oil diluted in coconut oil. See our Fungal Protocol for more information.

If your parent had a receding hairline, baldness, or hair loss, you are at a higher risk. The genetics of hair loss are somewhat complex, involving at least 287 independent genetic signals. The research that elicited this fairly recent discovery also demonstrated that hair loss is heritable through common genetic variants on autosomes as well as on the X chromosome.¹ This means that,

although many people have believed that hair loss is inherited solely through the maternal line, hair loss can be inherited through either parent's genes. That said, people with the "AR gene" for baldness (located on the X-chromosome) are twice as likely to have male-pattern baldness², though more than 80% of men with male androgenetic alopecia (male pattern baldness) have a father with the same condition³.

Certain medications can put you at a higher risk for hair loss. These include drugs for cancer, arthritis, depression, heart problems, gout, and high blood pressure. Research potential side effects and discuss these with your doctor before beginning a drug regimen.

Women with hormonal issues like polycystic ovarian syndrome (PCOS) are at a higher risk for hair loss due to higher androgen levels. Other conditions that put women at a higher risk include menopause, thyroid disorders, stress, toxicity, pregnancy (and other conditions that cause fluctuating hormones), and birth control pills. A major risk factor for men is a sensitivity to the androgenic hormone dihydrotestosterone (DHT). If a man's hair follicles are sensitive to DHT, then the hormone can bind to vulnerable follicles, causing them to shrink. The hair then becomes thinner and shorter, as the shrinking follicle makes it more difficult for the hair to grow normally. This eventually leads to hair loss. Other risk factors for men include aging, rapid weight loss, stress, poor nutrition, and medical conditions like diabetes.

Both men and women can suffer from alopecia areata. This is a type of autoimmune disorder in which the immune system attacks hair follicles and can cause patches of hair to fall out. It can affect facial hair, eyebrows, and eyelashes, in addition to hair on the head. Having any autoimmune disease can put you at greater risk for developing additional autoimmune concerns (including alopecia), so be sure to address your autoimmunity.

Another risk factor that is fairly new is illness like COVID-19. This virus can deplete nutrients from the body. Many people, both women and men, notice hair loss in the months following viral infection.

Early intervention and treatment can help restore hair damage and loss. This protocol will address the major causes of hair loss, provide you with effective treatments you can utilize to rejuvenate and stimulate your hair follicles, and help you develop a more targeted treatment that addresses your unique issues and circumstances.

1. Diet and Nutrients We are what we eat and diet plays a significant role in the health of our hair. The foods we eat can either reduce or contribute to inflammation, so choosing healthy foods is extremely important (especially for those with an autoimmune hair loss issue). For healthy, thick hair, steer clear of trans fats, refined sugar, processed foods, alcohol, and large amounts of caffeine. These substances are big offenders when it comes to inflammation.

Eat foods that are rich in minerals like zinc, as well as antioxidants, omega-3 fatty acids, healthy fats, collagen,

and lean protein. Good examples of foods to eat more of include pumpkin seeds, wild caught fish, bone broth, berries, and flax, chia, or hemp seeds. When choosing foods, look for organic foods, as these won't have the chemicals that can worsen endocrine issues.

If you're having a hard time eliminating coffee, try green tea. It gives you energy and is a rich source of antioxidants. It may be especially helpful for men; it has been studied for its ability to block the conversion of testosterone into DHT⁴, an androgenic hormone that contributes to male hair loss.

If you have hair loss caused by an autoimmune disorder, try a Mediterranean or Anti-Inflammatory diet. These diets include anti-inflammatory foods that may alter the expression of inflammatory genes to affect chronic disease.⁵

Another way diet influences hair health is via the microbiome. An imbalanced gut microbiome can play a significant role in the development of alopecia areata.⁶ Gut bacteria can degrade the intestinal epithelial barrier and lead to inflammation and autoimmunity. Diet influences gut microbiota.

Support your gut health by avoiding inflammatory foods, processed foods, and only taking antibiotics when absolutely necessary. Eat foods that nourish and support the microbiome, such as prebiotic-rich vegetables, fruits, and leafy greens. In addition, eat probiotic-rich foods like sauerkraut, kimchi, and kombucha. See our Leaky Gut Protocol for more.

High-fat diets have a detrimental effect on hair follicle stem cells, accelerating hair thinning.⁷ Diets high in saturated fat also lead to inflammation and negative changes in the gut microbiome.^{8 9} Higher levels of inflammation can accelerate hair loss in many cases. To optimize your fat consumption and reduce your risk of hair loss, limit your consumption of saturated fats (sources include animal-based foods like beef, pork, poultry, full-fat dairy products, and eggs, as well as coconut and palm oils) and consume a healthy amount of unsaturated fats (sources include nuts, seeds, and plant-based oils).

Bone broth is especially good for hair health because it is a rich source of protein collagen. Collagen is a potent natural remedy for those seeking to thicken hair, reduce hair loss, and accelerate the speed of hair growth. As an added bonus, it helps support healthy skin and nails. The antioxidants in collagen can also help nourish hair by creating healthy hair follicles. Bone broth is also a source of glycine, which is anti-inflammatory.¹⁰ This is especially helpful for those with hair issues as a result of autoimmune inflammation.

In addition to diet, another way to get the nutrients your hair needs is through supplementation. There are several beneficial supplements you can take daily to help with hair health and growth.

Vitamin B6: In a study involving post-pubertal women under the age of 30 who had symptoms of diffuse alopecia,

vitamin B6 helped improve the condition or quality of hair and also reduced hair loss.¹¹ Topical applications of vitamin B6, especially when used in combination with topical zinc sulfate, led to improvements in hormonal (androgenic) hair loss.¹² Suggested dose: 1.3 mg daily for individuals 19-50 years old; for men over 50 - 1.7 mg daily; for women over 50 - 1.5 mg daily.

Folate: While the evidence is mixed on whether or not folate deficiency is a significant contributor to hair loss, research suggests hair loss may be related to abnormalities in the metabolism of folate (vitamin B9). Specifically, researchers found that a C677T mutation in the methylene-tetrahydrofolate reductase (MTHFR) gene may increase risk for alopecia areata.¹³ People who have this mutation have a difficult time converting synthetic folic acid to the form of folate that our bodies can use, and this may negatively impact hair growth and thickness. As a result, people with this MTHFR mutation should limit their consumption of or completely avoid synthetic folic acid and increase their intake of natural folate-rich foods. Suggested dose: 400 mcg Dietary Folate Equivalents (DFE) daily for most adults; for pregnant women - 600 mcg DFE daily; for breastfeeding women - 500 mcg DFE daily.

Biotin: Biotin is a water-soluble B vitamin. Your body needs this nutrient to convert into keratin. Keratin is a protein that makes up hair, skin, and nails. A 2012 study showed that taking an oral biotin supplement helped to increase overall volume and thickness for older women.¹⁴ You can obtain the biotin you need through a healthy and balanced diet, but if you suspect you are deficient, this could explain hair loss issues. Suggested dose: 30 mcg daily.

Collagen: Collagen can be obtained via diet; it can also be taken via supplementation. As we age, our body becomes less efficient at producing collagen.¹⁵ The result of a reduction in collagen is fine lines, wrinkles, and thinning hair. A collagen supplement can be especially helpful for those with age-related hair loss or thinning hair. It can fight damage to hair follicles and may also provide the body with amino acids that can be used to build healthy hair.¹⁶ Suggested dose: 2.5 to 15 grams of collagen daily.

Zinc: There is a link between zinc deficiency and hair loss.¹⁷ Zinc plays a role in hair tissue growth and repair and works to keep the oil glands around the hair follicles healthy. While it is a vital nutrient for hair health, getting too much zinc may have the opposite-of-intended effect. Make sure you need zinc before beginning a supplementation regimen. A diet rich in foods like lentils, pumpkin seeds, oysters, and spinach may provide enough zinc for hair health. If you wish to supplement, the suggested dose is 8 mg for women and 11 mg for men.

Multivitamin: Make sure you are getting enough vitamins and minerals via daily supplementation of a multivitamin. Choose one that contains vitamins D, A, C, and E.¹⁸ There is an association between the incidence and severity of alopecia areata and vitamin deficiency. It may be possible to treat this condition with the right diet,

vitamins, and herbs. The antioxidants in these vitamins support healthy hair, skin, and nails. Daily dosage for a multivitamin varies depending on brand. Follow the directions on the bottle.

Fish Oil: The omega-3s in fish oil help strengthen hair, promote growth¹⁹, and nourish hair follicles. It also helps thicken hair that is thin and damaged.²⁰ It also supports cardiovascular health and lowers inflammation. Suggested dose: 1,000-1,200 mg daily.

2. Hormonal Hair Loss Remedies

One of the most common causes of hair loss is hormonal imbalance. This can be the result of issues like polycystic ovarian syndrome (PCOS), menopause, thyroid issues, or too much DHT (in men). Many women experience postpartum hair loss, as hormones fluctuate in the months following birth. Usually, postpartum hair loss corrects itself when hormones begin to balance after a few months to one year. If you are concerned about this type of hair loss, try the remedies in the “Nourishing Your Hair with Natural Recipes” section of this protocol. For those struggling with hair loss as a result of hormonal conditions, try the remedies below to balance hormones and tackle the root cause of your hair loss.

MALE HAIR LOSS AND DHT: DHT stands for dihydrotestosterone; it is an androgen that aids in the development of male sex characteristics like body hair. However, too much of this hormone can cause hair to fall out faster and earlier than normal. Around ten percent of testosterone in men is converted to DHT with the help of an enzyme. When it is freely flowing in the bloodstream, DHT can link to receptors on hair follicles in the scalp. This causes them to shrink and become unable to support a healthy and full head of hair. As men age, more DHT can get into the bloodstream. Below are some remedies to naturally block it and help reverse hair loss:

Saw Palmetto (*Serenoa repens*): Saw palmetto is a man’s best friend for many reasons. It has been shown to help shrink an enlarged prostate and even prevent prostate cancer.²¹ ²² It can also block DHT. One study showed that saw palmetto supplementation helped lower DHT by 32 percent.²³ Suggested dose is 160-320 mg daily or two dropperfuls of saw palmetto tincture twice daily. In addition to internal use, saw palmetto berries can be infused into a carrier oil (like jojoba or coconut oil) and massaged into the scalp. Doing this in conjunction with taking it internally is optimal.

Pumpkin Seed Oil: Pumpkin seed oil has been studied for its effects on hair growth in men with positive results. One study showed 40 percent more hair growth in men taking 400 mg of pumpkin seed oil over the course of 24 weeks with no adverse reactions.²⁴ Suggested dose: 400 mg daily.

Pygeum: Pygeum can block DHT that causes shrinking of follicles, as well as thinning and brittle hair. It is also anti-inflammatory and antioxidant. Like saw palmetto, it has been studied for its positive effects on an enlarged prostate.²⁵ Some report that it works best

when taken alongside saw palmetto. Suggested dose: two dropperfuls of pygeum tincture twice daily.

FEMALE HAIR LOSS REMEDIES: There are many issues that can affect female hormones and cause hair loss. Below are some helpful remedies to target the different causes of hormone-related female hair loss. Make sure to check out more remedies and information specific to your hormonal condition in the Menopause Protocol, Thyroid Protocol, And PCOS Protocol.

Lemon Balm (*Melissa officinalis*) for Overactive Thyroid: For those with an overactive thyroid (hyperthyroidism), lemon balm may inhibit stimulation of thyroid hormone production by both TSH and Graves’ antibodies.²⁶ It may also reduce excessive thyroid activity and inhibit TSH, which helps reduce the excessive secretion of thyroid hormone. Lemon balm is also a powerful nervine that can help calm anxiety and stress. Suggested dose: two dropperfuls of lemon balm tincture two to three times daily.

Stinging Nettle (*Urtica dioica*) for Thyroid Health: The antihistamine and anti-inflammatory effects²⁷ of stinging nettle work with both an underactive and overactive thyroid to calm the inflammation that triggers both issues. It can help reverse iodine deficiency for those with an underactive thyroid, and help the thyroid to function more optimally. Suggested dose: two to three cups of stinging nettle tea, or 2 dropperfuls of stinging nettle tincture daily.

Cordyceps for Thyroid Health: These adaptogenic mushrooms are a great remedy for a variety of thyroid conditions because they improve adrenal function, help balance the thyroid, increase helper T cells and cytotoxic T cells, and lower levels of thyroid antibodies for those with Hashimoto’s and Graves’ Disease. They are rich in the mineral selenium,²⁸ which has been shown to reduce thyroid antibody levels.²⁹ Suggested dose: 2 dropperfuls twice a day of a dual-extracted tincture.

Bugleweed for Overactive Thyroid: Bugleweed may help improve symptoms of mild hyperthyroidism and Graves’ disease. It acts against the antibodies that cause Graves’ and may inhibit T4 output, while also decreasing T3 levels.³⁰ Suggested dose: two dropperfuls tincture twice daily.

Black Cohosh (*Actaea racemosa*) for Hormonal Balance: For women with menopause-induced hormonal imbalance, black cohosh may be able to reduce the symptoms of menopause³¹ and restore hormonal balance naturally as a phytoestrogenic plant. Suggested dose: 40-80 mg daily or one to two dropperfuls of black cohosh tincture up to three times daily.

Vitex (Chasteberry) for Hormonal Balance: Perhaps one of the best hormonal balance remedies, vitex helps balance hormones for a variety of conditions. It is an adaptogen that seems to specifically target hormonal balance. This means that it can adjust the hormones to where they need to be, regardless of the condition or the hormone that is causing issues. It is great for those

with endometriosis, PCOS, uterine fibroids, and more.³² Suggested dose: two dropperfuls of vitex tincture twice daily.

Evening Primrose (*Oenothera biennis*) - for Both Women and Men - and Rosemary (*Rosmarinus officinalis*): Evening primrose seed oil can help attack the hormonal causes of male pattern baldness and androgenic alopecia in women. It helps balance hormones and prevent further hair loss.³³ It may also help hair grow back. It can be applied externally to the scalp and taken internally. Doing both is recommended. Suggested dose is one to three grams of evening primrose oil daily, along with daily head massage of evening primrose oil (blended with diluted rosemary essential oil). Rosemary essential oil was found to be as effective as the conventional treatment minoxidil (Rogaine) in preventing hair loss.³⁴ Recipe in Section 5 below.

3. Stress, Sleep, and Exercise Another major reason for hair loss is stress, anxiety, and trauma. Our mental health affects our physical health in a substantial way. Chronic stress is especially harmful to the body, as extra stress hormones take a toll on our health. There are many strategies you can employ to deal with anxiety in order to keep it from affecting your overall health. See the Anxiety and Stress Protocol for more detailed information.

Adaptogens: Some of the most effective remedies for stress are adaptogens because they protect the body from the effects of stress. If you suffer from chronic stress and anxiety, consider adaptogenic mushrooms like reishi, cordyceps, and turkey tail to stay healthy, both mentally and physically.

Reishi (*Ganoderma lucidum*): Known as the “mushroom of immortality,” this immunomodulatory mushroom has a long history of use against the negative effects of stress. It can help curb inflammation, hormonal imbalance, high cortisol (the “stress hormone”), and fatigue.³⁵ It can also help you get more rest (which is vital to staying healthy), relieve adrenal fatigue, and encourage a healthy immune system. Suggested dose: two dropperfuls of reishi double extracted tincture twice daily.

Cordyceps: These mushrooms are used for respiratory issues, viruses, and liver damage. They also boost energy levels and offset the effects of stress.³⁶ In addition, they help immune function, protect the body from cancer, regulate blood sugar levels, and relieve inflammation. Like many other mushroom remedies, it is best to take these in a dual/double extraction form. Suggested dose: two dropperfuls of cordyceps dual extracted tincture twice daily.

Turkey Tail (*Trametes versicolor* or *Coriolus versicolor*): Another adaptogenic mushroom, turkey tail mushrooms are used to treat inflammation, autoimmune disorders, chronic inflammatory diseases, cardiovascular concerns, high cholesterol, high blood pressure, and immune system issues like viral and bacterial infections.³⁷ Take in a dual extraction form for best results. Suggested dose: two dropperfuls of turkey tail dual extracted

tincture twice daily.

Get Quality Sleep: In addition to nurturing your body with adaptogens, it is important to get the sleep your body needs to stay healthy. A lack of sleep can take a serious toll on your overall health.³⁸ For more details on the importance of sleep and how to get the sleep you need (naturally), see the Sleep Protocol. The remedies below can help you fall asleep faster and stay asleep longer:

Valerian (*Valeriana officinalis*): One of the most potent remedies for sleep, valerian root is a central nervous system (CNS) depressant that acts gently on the nerves to induce sleep.³⁹ It can also calm anxiety and stress. Suggested dose: Have one cup of valerian root tea (made with two teaspoons of dried and chopped valerian roots) one hour prior to bedtime or 2 dropperfuls of tincture 30 minutes before bed.

Passionflower: Passionflower is a gentle sedative and nervine that can help quiet a busy mind and body.⁴⁰ Suggested dose: two dropperfuls tincture thirty minutes before bedtime.

Lemon Balm: Lemon balm is an effective nervine that helps calm the body and mind to help you get a more restful night's sleep. In addition, it is great for stress and anxiety.⁴¹ Suggested dose: Take two dropperfuls tincture as needed for anxiety or two-three dropperfuls thirty minutes before bedtime. Do not use if you have hypothyroidism.

Exercise: Exercise is critical for maintaining hair thickness and growth, because it leads to the release of endorphins, which improve our mood and help us better handle stress.⁴² Exercise also modulates testosterone levels,⁴³ which could be beneficial in hormone-related hair loss. Furthermore, regular moderate- to high-intensity exercise protects the body from oxidative damage by engaging the body's natural antioxidant mechanisms.⁴⁴ Physical activity can also have significant benefits on our perception of stress and how our bodies react to stressors. Physical activity can lead to changes in the body's hypothalamic-pituitary-adrenal (HPA) axis, which controls our response to chronic stress, and these changes can help modulate our reactivity to stress and anxiety.⁴⁵ ⁴⁶ People who suffer from anxiety tend to develop a fear for the physiological effects of anxiety, such as increased heart rate, and experiencing these symptoms typically leads to greater anxiety. Engaging in regular physical activity (specifically short-term, aerobic exercise) can help increase anxiety-sensitive individuals' tolerance for these symptoms.^{47 48 49 50} This helps reinforce the belief that, while these symptoms may be uncomfortable, they are not life-threatening, and it frequently leads to individuals handling anxiety in a healthier way, both psychologically and physiologically. Aim to get 30 minutes or more of moderate- to high-intensity physical activity, 5 times per week.

4. Eliminating Bad Hair Habits and Being Proactive

If you suspect your hair is starting to thin or fall out, take a closer look at bad hair habits. These habits are hard on hair, but they're especially hard on hair that

is already falling out.

Ditch the Heat: Heat damages hair, making it more brittle and more likely to fall out. These tools include straightening irons, curling irons, hair dryers, hot curlers, and anything else that exposes the hair to heat. It is also advisable to stop taking really hot showers if that is something you tend to do. Wash your hair with warm, not hot, water.

Stop Daily Washing: Many people think they need to wash their hair every day for it to be healthy, but this is not the case. Daily washing can stress hair out, especially if you use commercial shampoos and hair products and follow that with heat styling. See how long you can go without washing your hair. If your hair tends to get oily and greasy when you go too long without washing it, try using the dry shampoo recipe below under the “nourishing your hair with natural recipes” section of this protocol.

Avoid Chemicals: Many shampoos, conditioners, and other hair treatments have hair-harming chemicals like parabens, sulfates, polyethylene glycol, benzene, fragrance (phthalates), dye, alcohol, sodium hydroxide, formaldehyde, cocamidopropyl betaine, dimethicone, triclosan, and diethanolamine. These chemicals are the top offenders for causing hair damage and many are linked to cancer and endocrine disruption. Look for natural hair products without these ingredients. You can also make your own nourishing shampoo by following the recipe below under the “nourishing your hair with natural recipes” section of this protocol.

Pat to Dry: Instead of rubbing your towel on your head, gently pat your head to dry hair. Rubbing can cause more to fall out, especially if you are already having issues. In the same vein, sleeping with a wet head of hair can be damaging to hair. Let it fully dry (naturally without heat) before going to bed.

Wear Down, not Up: Wearing your hair up all the time puts stress on the follicles and roots. Give your hair a break as much as you can by wearing it down.

Let it Breathe: If you have lost a lot of hair and wear wigs, make sure you only wear them when you need to, as they can stifle oxygen on the scalp and prevent more hair from coming in. Only wear hats when you need to as well, as wearing hats all the time can harm the hair on your scalp and cause more to fall out or thin.

Salon Treatments: Salon treatments like Brazilian blowouts, perms, and coloring are all very hard on hair. Many salon treatments utilize chemicals that are linked to cancer, endocrine disruption, and major hair damage. Coloring your hair can be especially hard on the health and integrity of hair. Instead of coloring your hair with chemical hair dyes, consider using henna or non-permanent dyes.

Be Proactive: Being proactive is one of the best ways to ensure your hair is healthy for years to come. If you have a family history of premature hair loss, or if

you suffer from a hormonal imbalance that can cause hair loss, the time to start protecting your hair is now.

5. Nourishing Your Hair with Natural Recipes

There are plenty of commercial hair products that claim to promote growth and increased volume, but these products are often full of hair-damaging chemicals. It isn't hard to make your own nourishing hair products.

Natural Shampoo: Some plants contain compounds called saponins that naturally froth. These plants have been used for centuries in natural body and hair washes because they help to gently clean the body and hair. Some common saponin-containing plants include soapwort (bouncing betty), yucca root, soapbark tree, English ivy, horse chestnut, soap nuts, clematis, and California soaproot. There is likely a plant in your area that you can use to make your own natural shampoo. When a saponin-containing plant is combined with nourishing plants like lavender, stinging nettle, calendula, rosemary, and horsetail, you create a nutrient-dense shampoo and body wash that doesn't harm your hair. If you don't have soapwort, substitute it with one of the saponin-containing plants above.

Homemade Natural Shampoo Recipe: Ingredients: 4 cups of distilled water, 4 tablespoons of dried and chopped soapwort (all parts of the plant-double this if you are using fresh plant material), 2-3 teaspoons each of lavender, rosemary, stinging nettle, calendula, horsetail, and any other nourishing herb you wish to add. Directions: Bring water to a boil on the stove. Add the soapwort and simmer for 20 minutes. Add the additional herbs and simmer for another 5-10 minutes. Remove from heat and cool before straining. Bottle your shampoo and store in a cool, dark place (the refrigerator is best between uses). It should keep in the refrigerator up to 10 days. When showering, shake the bottle to help the saponins begin frothing and apply a liberal amount to the hair and body. Leave on your hair as long as you want to get the most out of all the beneficial nutrients in the shampoo. Rinse.

Simple Dry Shampoo: If you want to give your hair a break from daily washing, but feel you cannot because you have oily hair, try a natural dry shampoo. This helps soak up some of the oil on the scalp and hair shaft so your hair feels less weighed down and greasy. If you have blonde hair, simply apply a small amount of organic arrowroot flour to the scalp and hair as needed. Gently brush your hair after you apply it to get any residue out. If you have dark hair, try using organic cacao powder in place of arrowroot flour.

Rosemary: Rosemary essential oil is one of the best oils to help increase scalp circulation and promote hair health. It works well for those with androgenic alopecia as well.⁵¹ A 2015 study found rosemary essential oil as effective as the conventional treatment minoxidil (Rogaine) in preventing hair loss.⁵² Blend with jojoba and castor oil to create a powerful hair treatment that promotes hair health and growth. *Nourishing Rosemary Hair Mask*

Recipe: In a bowl, combine one tablespoon jojoba oil, one half tablespoon castor oil, and seven drops of rosemary essential oil. Apply a liberal amount of this to the scalp and then work your way down the shaft of the hair. Leave in for twenty minutes then rinse out with warm water and a natural, chemical-free shampoo. Repeat up to twice a week or as needed.

Other Beneficial Oils for Hair: In addition to rosemary, there are several other essential oils you can use to keep your hair healthy. These include sage, spikenard, lavender, cedarwood, geranium, and frankincense essential oils. Sage essential oil can encourage healthy hair and works well when combined with rosemary. Spikenard essential oil has been shown to promote hair growth and slow the graying process.⁵³ When using essential oils, it is important to dilute them in a carrier oil (like jojoba oil) before applying them to the hair or skin. Try blending five drops of an essential oil with one tablespoon of carrier oil to make a nourishing scalp and hair treatment.

Horsetail (*Equisetum arvense*) and Aloe Rinse: Horsetail contains silica,⁵⁴ a substance that can increase hair elasticity and resistance while also strengthening and lubricating dry and damaged hair shafts. It helps to decrease breakage and protect fragile hair. It may also promote the thickening of hair.

Aloe vera infuses hair with moisture and nutrients, while also promoting stronger and thicker hair. When these are combined, you get a powerful natural treatment for hair health and restoration.

Horsetail and Aloe Rinse Recipe: To make a rinse with horsetail and *Aloe vera*, bring two cups of distilled water to boil on the stove. Add one half cup of chopped horsetail. Simmer for twenty minutes.

Remove the pot from heat and let cool. Strain out the liquid and pour it in a glass mason jar. Add one cup of *Aloe vera* juice to the liquid and place the lid on the jar. Shake vigorously to blend. Apply a liberal amount to the hair after shampooing and leave it in for ten minutes before rinsing out. Store in the refrigerator between uses; it should last up to ten days.

6. Western Medical Approaches

When seeking medical help for hair loss, a dermatologist will usually run diagnostic testing. This is because hair loss is often an outward manifestation of something else going on in the body.

A doctor may call for blood work, a scalp biopsy, light microscopy (close examination of hairs trimmed at the bases), or a pull test (this is where the doctor gently pulls out a few hairs to see how easily they come out. This helps determine the stage of hair loss). Depending on

results, your doctor may prescribe medication, perform topical treatments, and in some cases, undergo transplant surgery or laser therapy.

Medications for Hair Loss: A doctor may prescribe one of several drugs, including minoxidil (Rogaine), finasteride (Propecia), Carospir, Aldactone, or dutasteride (Avodart) taken orally. These medications may help slow the rate of loss.

Like many pharmaceuticals, they come with their own set of side effects, such as scalp irritation, unwanted hair growth on the adjacent skin of the face or hands, diminished sex drive and sexual function, and an increased risk of prostate cancer. Make sure you research each medication thoroughly and are aware of side effects before beginning a medication regimen.

Laser Therapy: This is a somewhat new hair loss therapy. It consists of using lasers to treat hereditary hair loss in both men and women. There aren't many studies to back up claims that it improves hair density, so time will tell whether or not this treatment is viable. It is less invasive than some medications and transplant surgery.

Hair Transplant Surgery: Hair transplant surgery involves a cosmetic surgeon or dermatologist moving hair from one part of the scalp to an area of the scalp where the hair is thinning. This procedure is similar to a skin graft. During hair transplant surgery, the surgeon numbs the area he or she will be working with, removes a 6- to 10-inch strip of hair (usually from the back of the head), divides that strip into 500-2000 individual grafts, and then transplants the small grafts to the area of the scalp where the hair is thinning.

This process can take four to eight hours and may require repeat treatments if the area where the hair is thinning is large. Two to three weeks after hair transplant surgery, the grafted hair falls out, but it then begins to grow again within a few months.

Platelet-Rich Plasma Therapy (PRP): Platelet-rich plasma therapy involves a practitioner drawing your blood, processing it, and then injecting it into the area of the scalp where hair is thinning. The premise behind PRP is that the platelet-rich plasma injections stimulate and maintain increased hair thickness and growth in areas of the scalp that were previously thinning.

This method of addressing hair loss requires maintenance treatments every 4-6 months. If you are on blood thinners, are a heavy smoker, or have a history of alcohol- or drug-dependence, PRP may not be the best option for you. The ideal candidate for PRP is fairly healthy and has not been diagnosed with any chronic, systemic disorders.

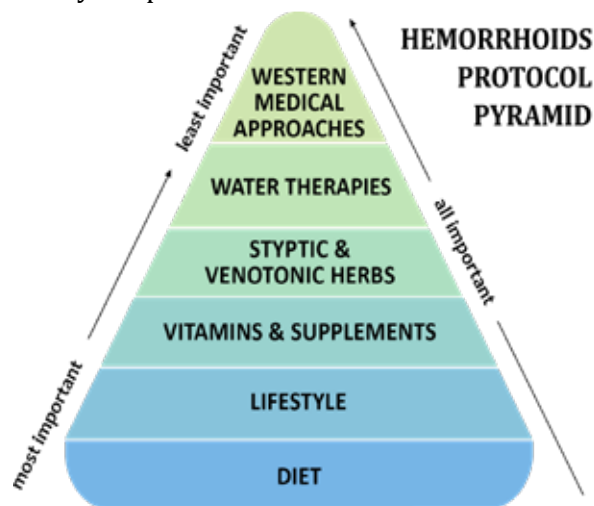
	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Multivitamin, biotin, and collagen (after breakfast) Rosemary hair mask Wash hair with natural shampoo & use horsetail and aloe rinse	Grocery shop for anti-inflammatory foods Cordyceps dual extracted mushroom tincture Stinging nettle tincture One cup bone broth	*Choose and take remedies for hormonal hair loss, if needed, and adaptogenic tinctures (e.g., Reishi and Turkey Tail) Pumpkin seed oil Outdoor hike/walk	Cordyceps dual extracted mushroom tincture One cup bone broth Vit. B6, folate, & fish oil	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Sleep blend tincture Bedtime routine (dim lights, avoid screens, Epsom salt bath, etc)
MONDAY	Multivitamin, biotin, and collagen (after breakfast) Natural dry shampoo (if needed) Morning exercise	Cordyceps tincture Stinging nettle tincture One cup bone broth	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Pumpkin seed oil	Cordyceps tincture One cup bone broth Vit. B6, folate, & fish oil	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Sleep blend tincture Bedtime routine
TUESDAY	Multivitamin, biotin, and collagen (after breakfast) Natural dry shampoo (if needed) Morning exercise	Cordyceps tincture Stinging nettle tincture One cup bone broth	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Pumpkin seed oil	Cordyceps tincture One cup bone broth Vit. B6, folate, & fish oil	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Sleep blend tincture Bedtime routine
WEDNESDAY	Multivitamin, biotin, and collagen (after breakfast) Rosemary hair mask Wash hair with natural shampoo & use horsetail and aloe rinse Morning exercise	Cordyceps tincture Stinging nettle tincture One cup bone broth	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Pumpkin seed oil	Cordyceps tincture One cup bone broth Vit. B6, folate, & fish oil	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Sleep blend tincture Bedtime routine
THURSDAY	Multivitamin, biotin, and collagen (after breakfast) Natural dry shampoo (if needed) Morning exercise	Cordyceps tincture Stinging nettle tincture One cup bone broth	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Pumpkin seed oil	Cordyceps tincture One cup bone broth Vit. B6, folate, & fish oil	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Sleep blend tincture Bedtime routine
FRIDAY	Multivitamin, biotin, and collagen (after breakfast) Natural dry shampoo (if needed) Morning exercise	Cordyceps tincture Stinging nettle tincture One cup bone broth	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Pumpkin seed oil	Cordyceps tincture One cup bone broth Vit. B6, folate, & fish oil	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Sleep blend tincture Bedtime routine
SATURDAY	Multivitamin, biotin, and collagen (after breakfast) Rosemary hair mask Wash hair with natural shampoo & use horsetail and aloe rinse	Cordyceps tincture Stinging nettle tincture One cup bone broth Outdoor hike/walk	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Pumpkin seed oil	Cordyceps tincture One cup bone broth Vit. B6, folate, & fish oil	*Remedies for hormonal hair loss & adaptogenic mushroom tinctures (Reishi & Turkey Tail) Sleep blend tincture Bedtime routine

*Hormone-related hair loss: Choose what you need (may be more than one): Saw palmetto (male hormonal hair loss), Pygeum (male hormonal hair loss), Bugleweed (overactive thyroid), Lemon balm (overactive thyroid), Stinging nettle (thyroid modulation), Black cohosh (reproductive hormone balance), Vitex (reproductive hormone balance). See Thyroid Protocol and Men & Women's Sexual Health Protocols for more information.

*Adaptogens for stress-related hair loss: Reishi and Turkey tail dual-extracted mushroom tinctures – you can take these in the morning if easier for your routine. Recommended twice a day.

Ch. 43: Hemorrhoids Protocol

The area surrounding the anus and rectum contains many blood vessels. When the integrity of the veins in that region is compromised, the veins swell in a manner similar to varicose veins, causing itching, discomfort, and pain. This is hemorrhoids. Hemorrhoids have several potential causes and contributing factors. These include straining during bowel movements, prolonged sitting on the toilet, constipation, multiple bowel movements per day, not consuming enough fiber, and trauma to the anal region. Symptoms of hemorrhoids may include pain with defecation, discomfort when sitting or during bowel movements, and itching, bleeding, and swollen veins that may or may not protrude out of the anus.



Hemorrhoids are categorized as internal or external. Internal hemorrhoids develop within the rectum, above what is known as the dentate line. You cannot usually see internal hemorrhoids from outside the body, but you can feel them. Internal hemorrhoids tend to be less painful than external hemorrhoids. External hemorrhoids can usually be seen by looking at the anus. They tend to be more painful and sensitive, and they are more likely to resolve spontaneously than internal hemorrhoids. Whether your hemorrhoids are internal or external may affect the conventional treatment options available to you. For example, some procedures, like external hemorrhoid thrombectomy, can only be performed on external hemorrhoids, while hemorrhoid stapling is only done on internal hemorrhoids.

In addition to being categorized as internal or external, hemorrhoids are classified by grades. Hemorrhoid grading is important because some treatments are effective at addressing only specific grades of hemorrhoids.

Grade I: Hemorrhoids do not protrude from the anus, but they may bleed.

Grade II: Hemorrhoids protrude from the anus during defecation, but they spontaneously reduce (re-enter the rectum) afterward.

Grade III: Hemorrhoids protrude from the rectum. They do not spontaneously reduce; they can be manually

reduced by pushing the protruding hemorrhoids back into the rectum.

Grade IV: Hemorrhoids are permanently prolapsed out of or protruding from the rectum; they do not spontaneously re-enter the rectum and they cannot be manually reduced.

Hemorrhoids can improve, and their grading can change. In this chapter, I'll discuss several interventions beneficial in addressing symptoms of hemorrhoidal disease, plus treatment options that can lead to an objective decrease in hemorrhoid grading.

1. Diet Diet plays an essential role in hemorrhoid prevention and treatment. Two critical dietary factors for hemorrhoid prevention and helping hemorrhoids heal are avoiding highly processed foods and consuming high-fiber foods.

Avoid Highly Processed Foods: Constipation is a significant contributing factor to hemorrhoids. Constipation often leads to straining during bowel movements. Straining increases blood flow to the lower rectum and anus, causing swelling of the veins in that region. These swollen veins (hemorrhoids) cause the itching, redness, and pain associated with hemorrhoidal disease. Highly processed foods are usually lacking in fiber and contribute to constipation. Replace highly processed foods with whole foods. See our Dietary Chapters for whole food dietary recommendations.

Consume High-Fiber Foods: Consuming high-fiber foods, like fruits, vegetables, and whole grains, helps reduce constipation, straining, and hemorrhoid formation. Fiber increases the weight of stool; reduces colon transit time, which is the amount of time stool spends in the colon; and helps the body pass stool more easily. High-fiber diets (at least 25 to 35 grams of fiber daily) reduce bleeding and pain in patients with hemorrhoids^{1 2} and taking a fiber supplement such as psyllium (*Plantago ovata*) leads to a reduction in hemorrhoidal bleeding on contact.³ The effects of taking a fiber supplement may not be felt immediately; in this particular study, benefits were noted after 30 days of treatment and continued to improve with time.

2. Lifestyle Lifestyle interventions play an important role in hemorrhoid prevention and treatment. TONE is a mnemonic used to remind patients of the treatment goals for hemorrhoidal disease. The T in TONE stands for three minutes for defecation, the O stands for once-per-day bowel movement frequency, the N stands for no straining, and the E stands for enough fiber. Research demonstrates that adhering to TONE and taking an adequate fiber supplement leads to a significant reduction in the progression of hemorrhoids and bleeding, and prevents surgery in most patients with advanced hemorrhoidal disease.⁴

Avoiding Prolonged Sitting on the Toilet: The center is cut out in toilet seats, so your anus and rectum are lower than the rest of your backside when you sit on them. Because your anus and rectum are lower

when sitting on a toilet, the blood pools in this region and pressure increases in the rectal and anal veins. This increase in pressure aggravates these veins and increases the risk for hemorrhoids or causes existing hemorrhoids to worsen. This is why prolonged sitting on the toilet or completing activities like reading while defecating should be avoided. If you struggle with hemorrhoids, you should spend no more than 3 minutes seated on the toilet at a time.

Avoiding Constipation: Constipation increases the pressure in the veins that collect blood from the rectum and anus; this leads to a worsening of hemorrhoidal disease. In addition to a high-fiber diet and potentially supplementing with fiber, ensuring proper hydration and adequate physical activity are important keys to limiting constipation so that it does not lead to hemorrhoid formation or worsening of existing hemorrhoids.

Avoiding Straining: Straining increases pressure in the anal and rectal vasculature, and can worsen hemorrhoids as a result. If you feel the need to defecate, but you go to the toilet and are unable to do so, do not remain seated on the toilet or attempt to strain to force the feces out; instead, come back later and try again if you are able to do so.

Ensuring Adequate Intake of Dietary Fiber: Fiber helps soften stool and reduces constipation. As previously mentioned, consuming high-fiber diets and supplementing with fiber can help reduce hemorrhoidal bleeding and eradicate the need for surgery in the majority of patients with advanced hemorrhoids.

3. Vitamins and Supplements

There are several vitamins and supplements shown to be beneficial in addressing hemorrhoids. These include flavonoids, such as diosmin, hesperidin, rutin, and borage oil; vitamin C; psyllium; and topical MSM, tea tree oil, and hyaluronic acid.

FLAVONOIDS: Flavonoids are compounds naturally found in fruits and vegetables; they have antioxidant properties and they help reduce inflammation and modulate the immune response. In the context of hemorrhoids, flavonoids reduce capillary permeability by increasing the vascular walls' resistance. By tonifying the walls of veins, they significantly reduce bleeding, pain, itching, and the likelihood of hemorrhoid recurrence.⁵

Suggested dose: 1350 mg of diosmin plus 150 mg of hesperidin, twice daily for four days followed by 900 mg of diosmin and 100 mg of hesperidin, twice daily for three days.

Suggested dose: 4 grams of O-(beta-hydroxyethyl)-rutosides per day.

Suggested dose: 1000 mg borage oil per day, with meals.

Diosmin: Diosmin is perhaps the most well-studied flavonoid. It stimulates blood flow and restores the function of swollen veins. Diosmin has been found to be safe for most people, specifically when it is used short-term for up to three months. For hemorrhoids, diosmin use leads to fewer episodes of bleeding,

discomfort, pain, and discharge.⁶ Diosmin also led to improvements in inflammation levels, congestion, edema, and prolapse.

Hesperidin and Rutin: Diosmin is frequently administered alongside other flavonoids.⁷ In a study that combined diosmin, hesperidin, and troxerutin (a flavonoid derived from rutin), researchers found that the combination led to a significant decrease in pain, bleeding, edema, and thrombosis after twelve days of treatment.⁸ People taking flavonoids significantly reduced their acetaminophen use after six days of treatment. In another study involving O-(beta-hydroxyethyl)-rutosides, which are flavonoids derived from rutin, these flavonoids were found to be effective at reducing hemorrhoidal pain and bleeding without significant unwanted effects.⁹

Borage Oil: In a study where participants were given rutin and borage oil (administered orally) combined with a topically administered ointment consisting of flavonoids and styptic and venotonic herbs, hemorrhoids stopped bleeding an entire week sooner when participants were administered this oral and topical treatment versus a placebo.¹⁰

OTHER VITAMINS AND SUPPLEMENTS:

Vitamin C: In a 2021 study involving patients with hemorrhoids who were not following a special diet and had not received any previous treatment, patients took vitamin C combined with flavonoids and venotonic and styptic herbs for a seven-day period. After taking the supplement, edema, prolapse, itching, thrombosis, burning, pain, feeling of needing to defecate even when bowels are empty, and bleeding decreased significantly and 89.8% of patients experienced a reduction in their hemorrhoid grade.¹¹ No adverse events were noted during supplement use. Suggested dose: at least 160 mg vitamin C per day.

Psyllium (*Plantago ovata*): Psyllium is used as a bulking agent because of its high fiber content. It is the main ingredient in products like Metamucil and Laxomucil. In a study involving people with advanced hemorrhoids (grades III and IV) who only took psyllium and incorporated healthier lifestyle habits (*i.e.*, TONE mnemonic) into their daily routines without taking any additional supplements, prolapse improved in over half of the participants, and the number of people who experienced bleeding decreased from 71.8% to 29.4%.¹² In another study, people who took psyllium after hemorrhoid removal surgery (hemorrhoidectomy) had significantly shorter hospital stays post-operation and their pain was significantly more tolerable.¹³ Suggested dose: 5-6 teaspoons of psyllium husk in at least 600 ml of water.

Topical MSM, Tea Tree Oil, and Hyaluronic Acid: In one study, a product called Proctoial Gel Rectal, which contains a proprietary combination of hyaluronic acid, tea tree oil, and methyl-sulfonyl-methane was used by people with grades I through III hemorrhoids. After 14 days of treatment, participants who used the gel reported significant reductions in anal pain, pain during defecation, visible bleeding, itching, irritation, and inflammation.¹⁴

4. Styptic and Venotonic Herbs

Styptic herbs help stop bleeding and venotonic herbs help support the integrity of veins. There are several styptic and venotonic herbs beneficial for hemorrhoids. These include witch hazel, horse chestnut, butcher's broom, gotu kola, white oak, common myrtle, bilberry, yarrow, and plantain. These herbs can be applied topically and/or used orally.

Witch Hazel (*Hamamelis virginiana*): Witch hazel was traditionally used internally to address bleeding; it is now more commonly used topically. In a study where participants applied witch hazel topically in the form of a rectal ointment that also included other herbs and flavonoids, participants who used the ointment experienced less hemorrhoidal pain, itching, and edema.¹⁵ Furthermore, a scientific review of available treatment options concluded that oral supplementation with horse chestnut, butcher's broom, gotu kola, and witch hazel may prevent the time-consuming, expensive, and painful complications of hemorrhoids.¹⁶ Suggested dose: 5 to 10 grams of herb to 1 cup of water to create a decoction for topical use (e.g., as a compress).

Butcher's Broom (*Ruscus aculeatus*): Butcher's broom contains ruscogenins, which have anti-inflammatory and vasoconstricting properties; they help strengthen and improve the integrity of veins.¹⁷ These properties make it effective in the treatment of both hemorrhoids and chronic venous insufficiency.^{18 19} Suggested dose: 2 dropperfuls of a 1:3 tincture, twice daily.

Gotu Kola (*Centella asiatica*): Treatment with gotu kola led to a reduction in the amount of time it took for hemorrhoids to stop bleeding, as well as a reduction in anal irritation, compared to participants who underwent routine conservative and surgical treatment.²⁰ Gotu kola can be taken internally or applied topically (e.g., as an ointment). Suggested dose: 4 dropperfuls of a 1:5 tincture, twice daily.

Horse Chestnut (*Aesculus hippocastanum*): Horse chestnut bark exhibits potent anti-inflammatory properties.²¹ It also has potent venotonic properties and has a long history of use to address chronic venous insufficiency and other vascular disorders.²² In a double-blind, placebo-controlled trial involving people with hemorrhoids, use of a horse chestnut product led to participants reporting improvements in hemorrhoidal bleeding, swelling, and pain after one week of use.²³ After 2 weeks, medical evaluators examining participants noted improvements. Horse chestnut can be used internally as capsules, a tea, or a tincture, and it can also be used topically, for example as a gel. Suggested dose: Take 4 dropperfuls of a 1:5 tincture, twice daily. Possible interactions: Horse chestnut should not be taken by anyone who is pregnant, breastfeeding, or under the age of four, or anyone with acute kidney inflammation or gastric ulcers.

Bilberry (*Vaccinium myrtillus*): In a study involving people with grades II and III hemorrhoids who took a combination product that included bilberry, other herbs,

vitamin C, and flavonoids, participants experienced a significant reduction in edema, prolapse, itching, thrombosis, burning, pain, feeling of needing to defecate even when bowels are empty, and bleeding; 89.8% of patients experienced a reduction in their hemorrhoid grade.²⁴ Researchers reported the product was both safe and effective. Suggested dose: 160 mg of bilberry per day or 5 dropperfuls of a 1:5 tincture, twice daily.

Common Myrtle (*Myrtus communis*): The leaves of common myrtle contain tannins, flavonoids, and healing volatile oils.²⁵ Decoctions made from its leaves are taken orally and applied topically to treat hemorrhoids; recent research has focused on the topical application of the essential oil for the treatment of hemorrhoids. Common myrtle essential oil ointment was as effective as conventional treatment options including hydrocortisone and lidocaine in addressing grades I and II hemorrhoids.²⁶ Both ointments led to significant reductions in bleeding, permanent pain, pain during defecation, anal itching and irritation, and tenesmus, which is the feeling that you need to defecate even though your bowels are already empty. Furthermore, common myrtle ointment was effective at reducing symptoms even in people whose hemorrhoids did not improve with the conventional ointment. Suggested dose: Oral use: 2 dropperfuls of a 1:3 tincture, twice daily.

Narrowleaf and Broadleaf Plantain (*Plantago lanceolata* & *P. major*): Plantain has a long history of use for gastrointestinal disorders.²⁷ Like many other herbs we've discussed, plantain has astringent and styptic properties. The two main types of plantain—narrowleaf plantain and broadleaf plantain—have very similar medicinal effects. Suggested dose: Oral use: 5 dropperfuls of a 1:5 tincture, twice daily.

White Oak (*Quercus alba*): Research on white oak in the context of hemorrhoidal disease is limited. However, white oak contains a high concentration of tannins, which contribute to its potent astringent, venotonic, and styptic properties.²⁸ These properties are helpful in addressing distended veins such as typically occur with hemorrhoidal disease. White oak can be taken internally (by mouth) or used topically in the form of an enema, a suppository, a compress, or in a bath.²⁹ Suggested dose: Oral use: 3 dropperfuls of a 1:5 tincture, twice daily.

Yarrow (*Achillea millefolium*): Yarrow is frequently used to address hemorrhoidal disease because of its astringent, styptic, and anti-inflammatory properties as well as its favorable safety profile.^{30 31} While yarrow can be taken internally, its localized styptic properties also make it an excellent herb to use topically. Suggested dose: Oral use: 5 dropperfuls of a 1:3 tincture, twice daily. Possible interactions: People taking lithium and certain blood thinners should avoid using yarrow internally as it may interact with their medication.

5. Water Therapies Water therapies can be beneficial in addressing hemorrhoids. The water therapy that is backed by the most research is a sitz bath. A sitz bath consists

of removing your clothing and undergarments from the lower half of the body and sitting in warm water for approximately ten minutes. In a study that tested the effectiveness of sitz baths at, 40°C (104°F), 45°C (113°F), and 50°C (122°F), higher temperatures led to more significant pain reduction³² and pain relief lasted longer when higher temperatures were used. The pain-reducing effects of sitz baths are due to the relaxation of the internal anal sphincter as well as the reduction of pressure in the rectal neck.

How to Perform a Sitz Bath: Fill a basin with about 3 inches of warm water, ensuring that the water is at a tolerable temperature and that you will not be burned. You may place the basin in your tub in case water spills out. Sit in the water for about ten minutes. Sitz baths help reduce the swelling and pain associated with hemorrhoids. You may repeat this procedure several times per day, as needed.

6. Western Medical Approaches

There are several conventional interventions used to address hemorrhoids. These include topical hydrocortisone, external hemorrhoid thrombectomy, rubber band ligation, sclerotherapy, infrared photocoagulation, hemorrhoid stapling, and hemorrhoidectomy.

Topical Hydrocortisone: Hydrocortisone is a steroid that is applied topically, usually as a suppository that is inserted into the rectum or as a cream that is applied to the affected area. In some cases, hydrocortisone is administered in the form of an enema or a foam that is used inside the rectum. Hydrocortisone works by activating compounds in the skin that help reduce the swelling, redness, and itching associated with hemorrhoids.

Because hydrocortisone can be acquired over the counter, it's important to note that if symptoms do not improve within 7 days of using this medication, you should see your provider. Common side effects of topical hydrocortisone include stinging, burning, irritation, dryness, or redness at the site of application. Side effects may also include headaches, dizziness, muscle weakness, and changes in menstruation or sleep patterns.

External Hemorrhoid Thrombectomy: At times, blood clots can form within hemorrhoids. When a clot forms within an external hemorrhoid, it causes severe pain. An external hemorrhoid thrombectomy is usually considered at this point.

This procedure involves administering localized anesthesia to numb the pain, and then making a small cut in the clot to drain it. This procedure is most effective if it is performed within three days of the hemorrhoid appearing. The pain is usually reduced fairly quickly after this procedure; note the risk of recurrence is high.

Rubber Band Ligation: Rubber band ligation is usually performed on internal hemorrhoids. Because the sensory

nerve fibers in internal hemorrhoids differ from those of external hemorrhoids, ligating internal hemorrhoids is usually less painful than it would be to ligate an external hemorrhoid.

This process involves using a device to immobilize the hemorrhoid and then apply a rubber band to its base. This effectively cuts off the blood supply to the hemorrhoid, and the tissue dies and falls off within a few days. Some people are able to go back to their normal activities right away after this procedure, while others may require 2-3 days of bed rest.

Sclerotherapy: In sclerotherapy, a sclerotic agent (such as 5% phenol in almond or vegetable oil, quinine, ethanolamine, or hypertonic saline) is injected into the hemorrhoid. This induces inflammation, which leads to fibrosis of the hemorrhoidal tissue.

With the blood supply to the hemorrhoid now cut off, the tissue dies and falls off within four to six weeks. People typically need to avoid strenuous exercise the day of this procedure; however, they are usually able to return to work the following day.

Infrared Photocoagulation: Infrared photocoagulation is helpful for hemorrhoids of grades I through IV. This procedure involves using a probe to introduce heat via infrared rays into the hemorrhoid.

This causes coagulation of the hemorrhoid's blood supply, which leads to the hemorrhoidal tissue becoming necrotic and dying within seven to ten days. This procedure typically requires individuals to take a few days off work to rest and recover after the procedure. Avoid heavy lifting and straining with bowel movements during recovery.

Hemorrhoid Stapling: In cases where an internal hemorrhoid protrudes out of the anus, hemorrhoid stapling or hemorrhoidectomy is usually recommended. In hemorrhoid stapling, also known as stapled hemorrhoidopexy, the prolapsing part of the hemorrhoid is moved upward and then stapled to the rectal mucosa.

This procedure is only conducted on internal hemorrhoids that do not show signs of thrombosis. This procedure is less painful than undergoing a hemorrhoidectomy; however, the risk of recurrence is greater with hemorrhoid stapling.

Hemorrhoidectomy: In cases where the hemorrhoid prolapses out of the anus and is visible without the person bearing down, hemorrhoidectomy is considered. This procedure involves a surgeon surgically removing the hemorrhoids.

This procedure can be more painful than many of the other available treatment options, and the potential for complications is greater. For example, if the surgeon needs to do an extensive resection, then widespread fibrosis and strictures may result. If the dissection is not performed carefully, fecal incontinence may result.

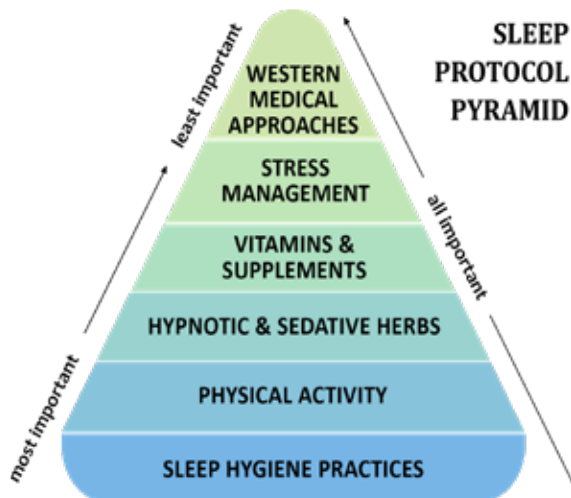
	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Diosmin, hesperidin, and rutosides/rutin Psyllium (with water) Vitamin C	Weekly grocery shop for high fiber & whole foods Dose 1: Take 3-4 styptic/venotonic herbal tinctures of your choice (see options below)*	Diosmin, hesperidin, and borage oil (w/ food) Psyllium (with water)	Physical activity (aim for 30-45 min.) Dose 2: Take 3-4 styptic/venotonic herbal tinctures of your choice (see options below)*	Sitz bath Topical witch hazel**
MONDAY	Diosmin, hesperidin, and rutosides/rutin Psyllium (with water) Vitamin C	Dose 1: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Diosmin, hesperidin, and borage oil (w/ food) Psyllium (with water)	Walk after dinner Dose 2: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Sitz bath Topical witch hazel**
TUESDAY	Diosmin, hesperidin, and rutosides/rutin Psyllium (with water) Vitamin C	Dose 1: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Diosmin, hesperidin, and borage oil (w/ food) Psyllium (with water)	Physical activity (30-45 min.) Dose 2: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Sitz bath Topical witch hazel**
WEDNESDAY	Diosmin, hesperidin, and rutosides/rutin Psyllium (with water) Vitamin C	Dose 1: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Diosmin, hesperidin, and borage oil (w/ food) Psyllium (with water)	Walk after dinner Dose 2: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Sitz bath Topical witch hazel**
THURSDAY	Diosmin, hesperidin, and rutosides/rutin Psyllium (with water) Vitamin C	Dose 1: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Diosmin, hesperidin, and borage oil (w/ food) Psyllium (with water)	Physical activity (30-45 min.) Dose 2: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Sitz bath Topical witch hazel**
FRIDAY	Diosmin, hesperidin, and rutosides/rutin Psyllium (with water) Vitamin C	Dose 1: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Diosmin, hesperidin, and borage oil (w/ food) Psyllium (with water)	Walk after dinner Dose 2: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Sitz bath Topical witch hazel**
SATURDAY	Diosmin, hesperidin, and rutosides/rutin Psyllium (with water) Vitamin C	Dose 1: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Diosmin, hesperidin, and borage oil (w/ food) Psyllium (with water)	Physical activity (30-45 min.) Dose 2: Take 3-4 styptic/venotonic herbal tinctures of your choice*	Sitz bath Topical witch hazel**

*Styptic/venotonic herbs (use as tinctures): Butcher's broom, gotu kola, horse chestnut, bilberry, common myrtle, plantain, white oak, yarrow.

**Add in topical MSM, diluted tea tree oil, and hyaluronic acid blend if needed.

Ch. 44: Sleep Protocol

Sleep is essential to our physical and mental health. It helps reset our brain and body, and some experts have suggested that if we eat well and exercise, but we don't get enough sleep, we may be undermining our efforts for optimal health. Sleep is a recurring period of time in which we enter into an altered level of consciousness. During this time, much of our sensory activity like smelling, hearing, and seeing, is inhibited; our muscle activity is significantly reduced or almost fully inhibited (depending on the stage of sleep we are in); and our ability to interact with our surroundings is severely limited.



Sleep enables our bodies to repair damaged cells and tissues, ward off diseases, and consolidate memories. When we sleep, our brains' glymphatic system removes metabolic waste from the central nervous system so that our brains and nerves can function optimally the following day. Sub-par amounts of sleep and low-quality sleep lead to an inflammatory and oxidative state in the body; this state leads to an increase in chronic disease risk.¹ Specifically, inadequate sleep increases our risk for cardiometabolic concerns, such as coronary artery disease, high blood pressure, obesity, metabolic syndrome, and stroke.

Regarding how much sleep is enough, research suggests that we need 7-9 hours of sleep per night if we are between the ages of 18 and 64 years old. Children need more sleep and adults over the age of 65 may require slightly less sleep. Sleep is divided into four stages-stage I non-rapid eye movement (NREM) sleep, stage II NREM sleep, stage III NREM sleep, and rapid eye movement (REM) sleep.

The first NREM stage is considered light sleep. This is where we transition from being fully awake to beginning to sleep. In the second NREM stage, our sleep deepens, our heart rate and breathing slow down, and our muscles relax. In the third NREM stage, our heart rate and breathing continue to slow. This stage is important in order for you to feel refreshed and well rested when you wake up.

In the REM stage, our respiratory rate, heart rate, and blood pressure begin to increase, we dream, and our arms and legs become temporarily paralyzed

(we believe this is to keep us from acting out our dreams during sleep). The REM stage plays a crucial role in learning and the consolidation of memories.²

The four stages of sleep repeat throughout the night, with the REM stage becoming longer and longer with each cycle. Each sleep cycle lasts about 90 to 120 minutes, with non-REM sleep making up about 75% to 80% of each cycle.

People who struggle with insomnia can be divided into three main categories-those who have difficulty initiating or falling asleep, those who have difficulty maintaining or staying asleep, and those who have difficulty with both sleep initiation and sleep maintenance. The available treatment options can target issues with sleep initiation, issues with sleep maintenance, or both.

In this chapter, I discuss several interventions found to be beneficial in addressing insomnia and promoting optimal quantity and high-quality sleep.

1. Sleep Hygiene Practices

Adopting practices that promote

sufficient, high-quality sleep is fundamental to improving sleep. This includes implementing a consistent nightly routine, ensuring your room is comfortable, and sleeping and eating on a regular schedule.

Implement a Consistent Nightly Routine: Establishing a bedtime routine is one of the best ways to ensure you get a better night's sleep. A bedtime routine can look different for each individual, but may consist of winding down one to two hours before bed by turning off screens (this includes your phone), turning down the lights, taking a warm bath, having a cup of herbal tea or taking an herbal sleep tincture, reading a book, and using aromatherapy and/or meditation. Our bodies function best when we are on a schedule. Carefully choosing the elements of your bedtime routine and consistently doing them around the same time each night helps the brain send signals to the rest of the body that it is nearing time to go to sleep.

One of the most important things you can do to help your brain send signals that it is time for bed is to turn off screens at least one hour before you plan on going to sleep. Blue light from screens is notorious for sending the wrong signals to the brain and body. These signals essentially tell the brain and body to stay awake and alert. If you absolutely need to use your devices at night, download an app for your laptop or set your phone to warm up the color temperature of your display as the sun begins to set and as the night progresses. By removing some of the blue light from your devices, you make it less likely that their light will disrupt your melatonin production and your sleep cycle. If possible, set your phone to go into "sleep mode" at a certain time each night so it isn't constantly going off and waking you up. You can set it to only ring for certain people if you are concerned you may miss an emergency. It may be tempting to do work in the evenings on your laptop or computer while lying in bed, but resist the temptation to do this. Keep your office separate from your bedroom.

A warm bath helps with peace and tranquility before bed. Consider doing this nightly if you have trouble falling asleep. In addition, research shows that being exposed to different colors and intensities of sunlight is important for regulating the sleep-wake cycle and the circadian rhythm as a whole.^{3 4} For example, watching the sunset sends signals to the brain that bedtime is approaching. To support optimal sleep, take a few moments to watch the sunset and practice deep breathing as you do this for added relaxation. As soon as possible after you wake up in the morning, expose yourself to sunlight, as this helps synchronize your sleep-wake patterns with the external environment.

Make Your Room as Comfortable as Possible: If your room is too hot or too cold, you may find yourself waking up throughout the night and not getting quality rest. Make sure your thermostat is set to a comfortable temperature at night. Use a fan or blankets to ensure that you are comfortable and will be able to remain asleep for the duration of the night.

If there is too much light or sound in your room, this can affect your ability to remain asleep at night. If you have light-emitting devices on in your room, turn them off or cover them up. Consider purchasing blackout curtains to prevent light outside from lightening your room or try a sleep mask. Avoid sleeping with the television on. Even if the TV doesn't wake you up, the sound and light will negatively affect sleep quality. If you cannot sleep in silence, try a sound machine. Sound machines play a continuous loop of white noise. In a clinical trial, participants who slept with a sound machine were able to fall asleep faster than they did when they were exposed to normal noise around them.⁵ If you live near a road or are otherwise exposed to noise when sleeping and a sound machine does not work for you, consider sleeping with ear plugs.

Sleep and Eat on a Regular Schedule: The times that you go to bed may not always be within your control. For example, an unexpected event may occur that requires you to delay your bedtime, or maybe you work a rotating schedule so you don't always get home at the same time each night. However, as much as it is within your control, it's important for you to have a regular time when you begin your bedtime routine. Our bodies operate on a 24-hour (circadian) cycle, and we thrive when we operate within the context of routine. Research demonstrates that having regular times for going to sleep and waking up (even on the weekends) is important for overall well-being. People with irregular sleep patterns are more likely to weigh more, have higher blood sugar and blood pressure, are more likely to report stress and depression, and they have a higher projected risk of having a stroke or a heart attack within ten years compared to people with regular sleep and wake patterns.⁶ Try to avoid napping during the day, as this can negatively affect your sleep-wake cycles; however, if you absolutely must nap, choose a time and aim to regularly nap at that time everyday.

Eating times also affect sleep. Meal timing regulates the circadian rhythm. In one study, delaying a meal by 5.5

hours led to significant changes in hormone production and other elements of the circadian rhythm.⁷ Eating meals at consistent times is critical for the synchronization of our internal clocks. To help maintain optimal circadian function, aim to eat your breakfast, lunch, and dinner around the same times each day. Avoid eating within 2-3 hours of your bedtime, and do not consume any caffeine after 2 p.m.

2. Physical Activity Your physical activity level and the type of physical activity you engage in can significantly impact the quality and quantity of sleep that you get. For example, in a review that included four randomized, controlled clinical trials, low-moderate levels of aerobic exercise noticeably improved sleep quality.⁸ However, lower levels of physical activity (i.e., yoga) did not have a significant effect on insomnia sleep index scoring or sleep quality.

In another study, both conventional exercise and tai chi led to improved sleep, and the beneficial effects of that exercise program persisted for 24 months.⁹ Another study revealed that moderate-intensity resistance exercise and stretching led to improvements in sleep for patients with chronic insomnia.¹⁰

Finally, research demonstrates that moderate-intensity aerobic exercise coupled with sleep hygiene education is extremely beneficial to sleep. Sedentary individuals with chronic insomnia who engaged in physical activity experienced improvements in the amount of time it took to fall asleep, the length of time they stayed asleep, daytime dysfunction, sleep quality, and sleep efficiency.¹¹ Participants also experienced reduced depressive symptoms and daytime sleepiness, as well as improvements in vitality as compared to their baseline scores.

3. Hypnotic and Sedative Herbs **Aerobic exercise,** like swimming, power-walking, and cycling, may be the most beneficial to individuals with sleep issues. Before beginning a new exercise program, speak with your provider about your plans, start with low-intensity exercise, and gradually increase the intensity of your physical activity. Note that that exercising late in the evening may be overly stimulating and may negatively impact your ability to fall asleep. Avoid vigorous physical activity for at least one hour before your planned bedtime.

There are several hypnotic (promotes sleep at the appropriate time, *e.g.*, at night) and sedative (promotes sleep immediately, at the time you take it) herbs that can support people who need help falling or staying asleep. These include valerian, hops, California poppy, chamomile, passionflower, American skullcap, lemon balm, reishi, and wild jujube. I make a sleep blend tincture from valerian root, hops, passionflower, chamomile, and magnesium glycinate. I find tinctures easy to take, fast-acting, convenient for travel, and without the extra water you get from tea, which may wake you up at night.

Valerian (*Valeriana officinalis*): Valerian root is perhaps the most potent of all the sleep aid herbs.

It improves sleep by increasing the levels of the calming neurotransmitter gamma-aminobutyric acid (GABA). It also decreases frequent waking, calms restless sleep, shortens the time it takes to fall asleep, and improves overall sleep quality and length. A double-blind study of valerian root found that, “forty-four percent [of the participants] reported perfect sleep and 89% reported improved sleep from the preparation” without any side effects.¹² Valerian is most effective when used regularly. Suggested dose: Have one cup of valerian root tea (made with two teaspoons of dried and chopped valerian roots) one hour prior to bedtime or 2 dropperfuls of valerian tincture, 30 minutes before bedtime.

Hops (*Humulus lupulus*): The hypnotic and sedative properties of hops help relieve insomnia and anxiety. Hops boost the production of GABA, a calming brain neurotransmitter that promotes sleep. Hops increases sleep time and works well in combination with valerian root. Research suggests that, when used together, hops and valerian may improve the length of time it takes to fall asleep and the quality of sleep.¹³ It also reduces body temperature, which helps you enter into a relaxed state for sleep. (An additional note: a flavonoid in hops has been found to help reduce weight gain, lower elevated cholesterol and reduce high blood sugar)¹⁴. Suggested dose: 4 dropperfuls of a 1:5 tincture, 30 minutes before bedtime.

California Poppy (*Eschscholzia californica*): California poppy is a gentle nervine with sedative and hypnotic properties. Supplementing with California poppy and valerian leads to increased sleep duration, enhanced sleep efficiency, and reductions in the number of awakenings.¹⁵ Study participants also reported decreased anxiety, and the supplement was well-tolerated. Infuse one to two teaspoons of California poppy into one cup of hot water for ten to fifteen minutes. Drink one cup one hour before bed. If you usually like tea with sugar, skip the sugar with hypnotic and sedative teas. Sugar is a stimulant and can be counterproductive to helping you fall asleep. Suggested dose: 1-2 teaspoons of dried herb in 1 cup of water or 4 dropperfuls of a 1:3 tincture, 30 minutes before bedtime.

Chamomile (*Matricaria chamomilla*): Chamomile calms the nervous system and is a slight sedative. It contains apigenin, an antioxidant that can induce sleep. It also soothes muscle aches and pains that may disrupt sound sleep. Research demonstrates that chamomile significantly improves sleep quality.¹⁶ Suggested dose: 400 mg chamomile, twice daily or 4 dropperfuls of a 1:3 tincture, 2 times daily.

Passionflower (*Passiflora incarnata*): Passionflower is a hypnotic, moderate sedative, and anxiolytic (soothes anxiety) herb. It helps calm the body and mind, improves sleep quality, helps with ADHD symptoms, and reduces the effects of menopause — including depression and hot flashes. In a double-blind, placebo-controlled trial, participants who drank one cup of passionflower tea per night over a 7-day period reported improved sleep quality.¹⁷ A study that compared the effectiveness of the pharmaceutical sedative-hypnotic drug zolpidem

with a polyherbal hypnotic-sedative product that included valerian, hops, and passionflower demonstrated that, when taken together at bedtime for two weeks, the effects of valerian, hops, and passionflower on sleep duration, time it took to fall asleep, number of awakenings per night, and insomnia severity were comparable to the effects of zolpidem.¹⁸ No significant difference was observed between the two treatment groups. Suggested dose: 2 dropperfuls of a 1:5 tincture, 30 minutes before bedtime.

American Skullcap (*Scutellaria lateriflora*): Skullcap is a nervine with hypnotic and moderate sedative properties. It has anxiolytic properties and can help with muscle spasms.¹⁹ Skullcap works by binding to GABA receptors in order to reduce anxiety and promote restful sleep. Skullcap may increase total sleep time.²⁰ Interestingly, skullcap exhibits a synergistic effect when taken with romaine lettuce leaf extracts, meaning that their effect on sleep when taken in combination was superior to the sum of the effects that were observed when participants took romaine lettuce leaf extract or skullcap alone. Suggested dose: 4 dropperfuls of a 1:5 tincture, 30 minutes before bedtime.

Lemon Balm (*Melissa officinalis*): Lemon balm is a nervine with mild sedative, anti-spasmodic (helpful in addressing muscle spasms), and anti-depressant properties. In a clinical trial, 85% of participants who took a lemon balm extract reported full remission of their insomnia.²¹ By the end of this trial, 95% of participants responded to the treatment, 85% achieved full remission of their anxiety, and 70% achieved full remission of both their anxiety and their insomnia. It also helps stop looped thinking that may keep you up at night. Another study involving participants with anxiety, depression, and insomnia showed that lemon balm, when used in conjunction with another herb (*Nepeta menthoides* Boiss & Buhse), led to improvements in anxiety, depression, and sleep.²² Suggested dose: Take two dropperfuls of a 1:3 tincture, twice daily. Do not use if you have hypothyroidism.

Reishi (*Ganoderma lucidum*): Although it does not have direct sedative properties, reishi mushroom can be very helpful for insomnia. Reishi is a powerful adaptogen. Adaptogens help the body adapt to stress while improving fatigue and providing stamina and restoration. When taken regularly, reishi can help alleviate insomnia and other sleep disorders. Research suggests that reishi may promote sleep through its positive effects on the gut microbiome.²³ In one pre-clinical trial, reishi supplementation led to an increase in sleep duration and a reduction in the amount of time it took to fall asleep. These improvements were attributed to the fact that reishi increased levels of 5-hydroxytryptophan in the sleep-regulating serotonergic synapse pathway of the hypothalamus. Researchers believe these effects were due to reishi's effect on the gut microbiome. Suggested dose: 2 dropperfuls of a dual-extracted tincture, 1-2 times daily.

Wild/Chinese Jujube (*Ziziphus jujuba* var. *spinosa*): Wild jujube is a fruit commonly used to address anxiety

and insomnia in traditional Chinese medicine. Wild jujube supplementation leads to significant improvements in sleep quality, sleep duration, sleep efficiency, and the amount of time it takes to fall asleep.²⁴ Suggested dose: 2,000 mg 1 hour before bedtime.

4. Vitamins and Supplements

There are several vitamins and supplements that are beneficial in regulating sleep and improving insomnia. These include magnesium glycinate, melatonin, myoinositol, vitamin C, B complex, gamma-aminobutyric acid, L-theanine, 5-hydroxytryptophan, and exogenous cannabinoids like cannabidiol and tetrahydrocannabinol.

Magnesium Glycinate: Magnesium glycinate consists of magnesium bound to glycine, which is an amino acid that helps the body achieve deep sleep more quickly. Magnesium glycinate is one of the most absorbable forms of magnesium, and glycine improves sleep quality; promotes natural, healthy sleep patterns; and supports healthy REM cycles. Magnesium glycinate helps your body produce melatonin, the sleep hormone, and it also helps relieve muscle tension. Magnesium supplementation may help improve sleep patterns by reducing a person's likelihood of falling asleep during the day, and it may also help improve sleep efficiency, sleep time, sleep onset latency, early morning waking, and melatonin and cortisol levels.^{25–26} A 2021 study demonstrated magnesium improves the time it takes to fall asleep.²⁷ Suggested dose: 350–500 mg magnesium per day.

Melatonin: Melatonin is a hormone that tells your body when to sleep and when to wake up. Some research suggests that melatonin supplements can ease sleep issues like jet lag and trouble falling or staying asleep. Melatonin is safe for healthy adults if taken for a few weeks or months. In a randomized, controlled trial involving ICU patients, research demonstrated that oral melatonin (10 mg) for up to 10 consecutive nights led to improved sleep quality.²⁸ The College of Family Physicians of Canada states that melatonin can be a useful treatment for children with delayed sleep phase syndrome (alongside a thorough insomnia work-up and appropriate follow-up) and that melatonin use should be considered for children with undiagnosed insomnia and optimal sleep hygiene habits.²⁹ Side effects of melatonin can include headache, dizziness, and nausea. You can develop tolerance to melatonin and we don't currently have reliable data on the effects of supplementing with this hormone in the long-term,³⁰ so it's best to use lower doses of melatonin on a short-term basis to reset your circadian rhythm and then discontinue use. Suggested dose: 1–3 mg, approximately 2 hours before going to bed.

Myoinositol: Myoinositol is a pseudovitamin that works closely with the B vitamins. It is believed to play a critical role in various sleep processes. Research suggests that having low levels of myoinositol in the frontal cortex may be associated with increased risk for depression and sleep disturbances.³¹ Research shows that supplementing with myoinositol may improve sleep quality and sleep

duration.³² Myoinositol supplementation is generally recognized as safe, even at doses well above the typically recommended dose for addressing sleep concerns. Suggested dose: 500–2000 mg per day.

Vitamin C: People who don't get enough sleep are at greater risk for metabolic syndrome, and research suggests that vitamin C is crucial to this link between sleep duration and metabolic syndrome.³³ People with higher levels of vitamin C report better sleep than individuals with lower levels. Both inadequate intake and low serum levels of this antioxidant vitamin have been associated with short sleep duration, difficulty falling and staying asleep, daytime sleepiness, and non-restorative sleep.^{34–35} Research conducted on cancer patients who were receiving intravenous and oral vitamin C revealed that their insomnia improved significantly within one month of vitamin C therapy.³⁶ Suggested dose: 1–3 grams of buffered vitamin C per day, in divided doses.

B complex: B complex vitamins have been associated with sleep. For example, vitamin B6 plays an important role in enhancing dream recall.³⁷ It also helps reduce nighttime waking.³⁸ A study that evaluated the effects of supplementation with B complex vitamins along with magnesium and melatonin in participants who had mild to moderate insomnia found that this combination led to improvement in insomnia-related symptoms, regardless of the cause of the insomnia.³⁹ Suggested dose: 10 mg vitamin B6, 16 µg vitamin B12, and 600µg folate per day.

Gamma-Aminobutyric Acid (GABA) and L-Theanine: GABA is a brain neurotransmitter that boosts relaxation and sleep. Pharmaceutical sedative-hypnotic drugs, which are commonly prescribed to treat insomnia, work by binding GABA to its receptors in order to enhance its effects. Treatment with GABA and L-theanine, another sleep-promoting amino acid, led to reductions in the amount of time it took to fall asleep and increase in sleep duration.⁴⁰ Researchers noted that GABA and L-theanine had a synergistic effect, meaning that their effect on sleep when used together was superior to the sum of the effects that were observed when participants took GABA or L-theanine alone. Suggested dose: 100–200 mg GABA daily; 200 mg L-theanine daily.

5-Hydroxytryptophan (5-HTP): 5-hydroxytryptophan, more commonly referred to as 5-HTP, is a compound that your body derives from foods. It plays an important role in the production of melatonin. Supplementing with tryptophan, particularly in combination with vitamin B6 and melatonin, led to improvements in nighttime awakenings in a study conducted on children with headaches and concomitant sleep disorders.⁴¹ Suggested dose: Take 100–300 mg of 5-HTP before bedtime. If you get a headache or experience nausea afterward, stop taking it.

Cannabidiol (CBD) and Tetrahydrocannabinol (THC): Cannabidiol (CBD) and Tetrahydrocannabinol (THC) are compounds found in cannabis plants. Note that CBD doesn't get you high; THC is the main

psychoactive compound in marijuana. Current research suggests these work best if lack of sleep is due to anxiety, PTSD, and/or pain. They help relax the body, calm the mind, and lower inflammation. While CBD, and especially CBN for sleep, is popular and widely available, THC products often require an application process or card (depending on where you live). Most use an Indica strain of THC for sleep. Exogenous cannabinoids like CBD and THC are best for occasional help as your body can develop a tolerance, and it may affect quality of sleep over time. Suggested dose: Start off with a dose of 5 mg and increase if needed to 25–175 mg.⁴² See what works best for you. Many dose it based on weight: 1-6mg CBD/10 pounds of body weight (ex. if you are 150 lbs., you would take between 15-90 mg).

5. Stress Management

High stress levels contribute to insomnia, which is why appropriately managing stress is a critical component of ensuring quality sleep. Methods of managing stress include prioritizing daily exposure to green space; using breathing exercises; taking adaptogenic herbs to help your body moderate your levels of cortisol, which is your major chronic stress hormone; meditation; aromatherapy; working with a cognitive behavioral therapist to process and work through past and current stressful experiences; and journaling about your day or your emotions. I discuss these and other stress management techniques extensively in the Anxiety and Stress Protocol.

Research demonstrates that aromatherapy using lavender essential oil combined with healthy sleep hygiene practices help improve overall sleep pattern, sleep quality and quantity, sleep efficiency, and mood.⁴³ ⁴⁴ Pre-clinical research suggests that inhalation of the blended Anshen essential oil may increase concentrations of 5-hydroxytryptophan and GABA in the brain, has a sedative and hypnotic effect, and may reduce the length of time it takes to fall asleep and increase the length of sleep duration.⁴⁵

Other great essential oils to help the mind and body relax and wind down include vetiver, patchouli, frankincense, bergamot, valerian, ylang ylang, valerian root, lavender, chamomile, and clary sage. Combine these in any way that works best for you. Use around three drops of each oil in a diffuser when combining, instead of five to eight drops for a single oil.

6. Western Medical Approaches

There are several conventional approaches used to address sleep-related concerns. These include cognitive behavioral therapy and pharmaceutical drugs like eszopiclone, zaleplon, zolpidem, and ramelteon.

Cognitive Behavioral Therapy-Insomnia (CBT-I): Cognitive behavioral therapy-insomnia (CBT-I) practices are typically the first intervention that is tried when a person reports difficulty sleeping. It tends to be more effective than sleep medication in the long run. ²⁹⁰ The goal of CBT-I is to help you control or replace

negative thoughts or behaviors that keep you from getting quality sleep. CBT-I addresses harmful beliefs, such as worries and negative thoughts that keep you awake at night, and helps you adopt positive sleeping habits that promote restful sleep. This focus on thoughts (cognitive) and behaviors (behavioral) is characteristic of all forms of CBT, not just CBT-I.

Pharmaceutical Medication to Address Insomnia:

The drugs most commonly prescribed to address insomnia can be categorized as pharmaceutical sedative-hypnotics and melatonin receptor agonists. There have been concerns raised over the past several years about unwanted effects associated with the use of pharmaceutical sedative-hypnotics. In April 2019, the Food and Drug Administration (FDA) placed black box warnings, which are their strongest level of warnings, on the pharmaceutical sedative-hypnotics. The FDA cited reports that, while taking these drugs, people have engaged in life-threatening sleep behaviors, such as getting out of bed and driving cars, preparing and eating food, having sex, making phone calls, sleep walking, and engaging in other activities without being fully awake and without being able to recall the details of these experiences upon waking. These drugs also have a high potential for addiction and abuse. It's important to weigh the risks and benefits associated with taking medications before making a treatment decision. Melatonin receptor agonists work by stimulating melatonin receptors to help initiate sleep.

Eszopiclone (Brand name: Lunesta®) is a pharmaceutical sedative-hypnotic. It is typically prescribed both short- and long-term to assist with the initiation and maintenance of sleep. It works by stimulating the brain's GABA receptors in order to slow brain activity and allow a person to fall asleep. It is usually taken immediately before bed as it starts working within an hour of you taking it. If you drink grapefruit juice, it's important to talk with your doctor about the potential interactions that exist between grapefruit juice and eszopiclone. Common potential side effects include difficulty waking up, grogginess throughout the following day, dry mouth, a bitter or metallic taste in the mouth, headache, and dizziness.

Zaleplon (Brand name: Sonata®) is a pharmaceutical sedative-hypnotic. It binds GABA receptors in the central nervous system to help initiate sleep, but because it has a relatively short half-life, it is not as effective at maintaining sleep.⁴⁶ It works fairly quickly, so it should be taken within an hour of or as close to your intended bedtime as possible. Common potential side effects associated with zaleplon include drowsiness; dizziness; lightheadedness; poor coordination; numbness, burning, or tingling in the hands or feet; headache; appetite changes; and vision problems.

Zolpidem (Brand names: Ambien®, Edluar®, Intermezzo®, and Zolpimist®) is a sedative-hypnotic. It works by activating GABA and binding it to the GABA receptors in order to block the brain activity that is

associated with insomnia. It is most commonly prescribed to help with initiating sleep. It's effects on sleep duration or maintenance appear to be minimal.⁴⁷ Like the other pharmaceutical sedative-hypnotics, zolpidem begins to work fairly quickly and should be taken immediately before bed. Common potential side effects associated with zolpidem use include nausea, dizziness, diarrhea, headache, feelings of agitation, abnormal thoughts (i.e., hallucinations), behavioral changes, next-day fatigue or grogginess, and depression.

Remelteon (Brand name: Rozerem®) is in a completely different drug class. It is in the drug category of melatonin receptor agonists, and it helps with sleep initiation. It is typically prescribed to people who have a hard time falling asleep who don't a difficult time remaining asleep. Ramelteon works by binding to melatonin receptors in order to regulate the sleep-wake cycle. It is typically taken within 30 minutes of bedtime. Common potential side effects include dizziness, tiredness, daytime drowsiness, nausea, and worsening sleep problems.

	Morning	Noon	Afternoon	Evening	Before Bed
S U N D A Y	Dose #1: Reishi tincture Vit. C & B complex		Dose #2: Reishi tincture plus Dose #1: Lemon Balm tincture Physical activity (aim for 30-45 min.)	Chamomile tea & Lemon Balm tincture after dinner	Bedtime routine Valerian, hops, & passionflower tinctures* Optional: add California poppy, skullcap, jujube, and/ or lemon balm + GABA & L-theanine or 5-HTP Magnesium and Myoinositol
M O N D A Y	Dose #1: Reishi tincture Vit. C & B complex		Dose #2: Reishi tincture plus Dose #1: Lemon Balm tincture Physical activity (aim for 30-45 min.)	Chamomile tea & Lemon Balm tincture after dinner	Bedtime routine Valerian, hops, & passionflower tinctures* Optional: add California poppy, skullcap, jujube, and/ or lemon balm + GABA & L-theanine or 5-HTP Magnesium and Myoinositol
T U E S D A Y	Dose #1: Reishi tincture Vit. C & B complex		Dose #2: Reishi tincture plus Dose #1: Lemon Balm tincture Physical activity (aim for 30-45 min.)	Chamomile tea & Lemon Balm tincture after dinner	Bedtime routine Valerian, hops, & passionflower tinctures* Optional: add California poppy, skullcap, jujube, and/ or lemon balm + GABA & L-theanine or 5-HTP Magnesium and Myoinositol
W E D N E S D A Y	Dose #1: Reishi tincture Vit. C & B complex		Dose #2: Reishi tincture plus Dose #1: Lemon Balm tincture Physical activity (aim for 30-45 min.)	Chamomile tea & Lemon Balm tincture after dinner	Bedtime routine Valerian, hops, & passionflower tinctures* Optional: add California poppy, skullcap, jujube, and/ or lemon balm + GABA & L-theanine or 5-HTP Magnesium and Myoinositol
T H U R S D A Y	Dose #1: Reishi tincture Vit. C & B complex		Dose #2: Reishi tincture plus Dose #1: Lemon Balm tincture Physical activity (aim for 30-45 min.)	Chamomile tea & Lemon Balm tincture after dinner	Bedtime routine Valerian, hops, & passionflower tinctures* Optional: add California poppy, skullcap, jujube, and/ or lemon balm + GABA & L-theanine or 5-HTP Magnesium and Myoinositol
F R I D A Y	Dose #1: Reishi tincture Vit. C & B complex		Dose #2: Reishi tincture plus Dose #1: Lemon Balm tincture Physical activity (aim for 30-45 min.)	Chamomile tea & Lemon Balm tincture after dinner	Bedtime routine Valerian, hops, & passionflower tinctures* Optional: add California poppy, skullcap, jujube, and/ or lemon balm + GABA & L-theanine or 5-HTP Magnesium and Myoinositol
S A T U R D A Y	Dose #1: Reishi tincture Vit. C & B complex		Dose #2: Reishi tincture plus Dose #1: Lemon Balm tincture Physical activity (aim for 30-45 min.)	Chamomile tea & Lemon Balm tincture after dinner	Bedtime routine Valerian, hops, & passionflower tinctures* Optional: add California poppy, skullcap, jujube, and/ or lemon balm + GABA & L-theanine or 5-HTP Magnesium and Myoinositol

*My sleep blend tincture has valerian, hops, passionflower, and chamomile, and magnesium glycinate.

Ch. 45: Weight Loss Protocol

For some people, losing weight comes easily, while for others it's a lifelong struggle filled with failed diets and the dreadful yo-yo effect where you lose weight only to put it back on...and then some. This Weight Loss Protocol is an integrated approach. Many programs are based only on diet and physical exercise. You'll learn how your body utilizes the calories you take in and find ways to improve your metabolism.

Note that undiagnosed conditions can make it difficult to achieve your weight loss goals. It's important to establish if you have an underlying condition, such as an autoimmune disease, candida overgrowth, heavy metal, chemical, or environmental toxicity, thyroid disease, hormonal imbalance, or hypothalamic-pituitary-adrenal (HPA axis) dysfunction (otherwise known as "adrenal fatigue") that may be responsible for your extra weight.

1. Calorie Deficit

No matter what diet you choose, maintaining a calorie deficit is crucial if you want to lose weight. The first way to achieve this is by reducing the number of calories you consume. The second is through expending more calories by increasing your physical activity.



First determine the number of calories you should eat each day to maintain a caloric deficit. It's easy to do using an online calculator: <https://www.calculator.net/calorie-calculator>. Enter your age, height, and current weight plus the number of times you exercise per week and the intensity of your workouts. This will give you a rough estimate of how many calories you need daily. To lose weight, your daily intake needs to be under that benchmark. How much weight you can expect to lose? If you are in a 500-calorie-a-day deficit you'll lose about one pound a week. If you plan to lose 25 pounds, it would take you 25 consistent weeks to do so. This is a rough estimate as not all food sources are created equal.

2. Macronutrients

Think of your body being like a car that can use two types of fuel: gas and diesel. If you eat a carb-rich diet, your body will always burn only one type of fuel. If you restrict carbs, like with a ketogenic diet, you will switch to the second and more efficient fuel type.

Keto helps you lose weight by putting your body into ketosis. By drastically cutting your carbohydrate intake, your liver will burn fat to produce ketones for energy.

Since the invention of agriculture, the number of carbs in our diet has skyrocketed. Carbs are found in bread, sweets, crackers, and pasta-things that didn't exist until relatively recently. Our ancestors did eat starchy tubers and roots but these are high in dietary fiber and nutrients. On keto, your body burns fat (broken down into ketones) instead of carbs (glucose) for energy. You can read more about this in our Ketogenic Diet chapter found in this book.

Below are guidelines for whole foods you should be eating daily to lose weight:

- **Fresh vegetables** - steamed or lightly sauteed (two or more 1-cup servings daily); these are best lightly cooked so that they are easier on your gut.
- **Fresh fruits** - any kind (but not before going to bed); try to avoid eating a lot of bananas. If you are on a strict keto diet, watch your carb intake from fruit.
- **Dark, leafy greens** - very important: 1 or more cups steamed or sauteed daily.
- **Oily fish** - like salmon, mackerel, or sardines (3 servings weekly).
- **Protein**: chicken or turkey (4 servings weekly).
- **Whole grains** - brown rice, millet, oats and oat bran, quinoa, and buckwheat (1 serving daily, unless you are on a keto diet)
- **Flax meal and/or chia seeds** - 1 tbsp. each daily); these contain fiber, protein, essential fatty acids, and other key vitamins.
- **Olive oil** - almond, apricot, avocado, coconut, and grapeseed oil are also okay.
- **Nuts/Seeds** - soaked raw almonds, sunflower seeds, or pumpkin seeds (1 tbsp. daily); nuts are best soaked because they are easier to digest.
- **Eggs and almonds** - these contain glutamine, which has been shown to help digestion; include these foods in your diet as needed and/or supplement with glutamine.
- **Probiotic foods** - these often-fermented foods improve digestion, helping you lose unwanted pounds; at every meal, eat ½ cup of any of the following: Kombucha, unsweetened yogurt with live cultures, sauerkraut (not canned), miso broth, kimchi.
- **Water** - drink half your body weight in ounces of water daily; adequate water is imperative for healthy digestion as a lubricant and detoxifier.

When the body has the correct nutritional balance, it can perform at its optimal level. The correct fuel can reduce hunger as getting the proper nutrients may cause the

body to stop seeking other foods to fill the void.

3. Metabolism, Digestion, and Herbal Support

Increasing your metabolism can have a huge impact on how

much weight you lose:

- It makes it much easier to stay in a caloric deficit. When you naturally burn more calories, you can eat more food and still be in a caloric deficit.
- You need to increase metabolism to overcome weight loss plateaus.

Most of us trying to lose weight will eventually reach a plateau. The body, fighting the caloric deficit, will lower your metabolism to restore balance. So even if you eat 500 fewer calories each day, after a while you won't lose any more fat because your body has reduced its daily energy requirements. I've found that some of the best ways to increase your metabolism is through natural remedies. They also help improve your digestion and shorten your elimination cycle.

Massage for Weight Loss: The digestive system is muscular. By toning these muscles, you can help optimize your elimination cycle, which can help complement weight loss. Laying in a supine position with your abdomen exposed, put five drops of essential oil in a nickel-sized amount of lotion or carrier oil: olive oil, almond oil, avocado oil, grapeseed oil. Then place your palms side by side and lay them on the right side of your abdominal wall, closer to your right hip than your navel. Always move in a clockwise direction with abdominal massage because this is the direction of your elimination cycle. By doing so, you enhance its efficiency. Move your hands slowly in circles clockwise thirty times. You may feel warmth develop. This is normal and healthy. Manual massage restores heat and circulation. Take note of areas that seem congested or "bunched up." Go a little more slowly over those areas. It may be undigested matter that has been slowing down your elimination cycle and releasing toxins into your system over time. It is recommended that you do this massage two to three times each week. Besides the aromatherapy massage, use two herbal support teas or tinctures (see next section).

Herbal Support: There are a number of herbs helpful with weight loss. To help your metabolism, try using an adaptogenic tincture to encourage an optimal "fat-burning state" and a fat-burning tincture or tea. Take these three times daily before each meal. Drink plenty of water.

Fenugreek and Fennel seeds - Fenugreek seeds help with a variety of health issues — including metabolic syndrome and weight loss. Its high soluble fiber content¹ helps reduce blood glucose levels in those with type 2 diabetes. Additionally, fenugreek fights body fat by slowing digestion and the absorption of intestinal glucose and dietary fat², which promotes satiety³ and less caloric intake. In one study⁴, participants reduced their dietary fat intake by 12% simply by supplementing with fenugreek. Fennel seeds also decrease hunger, aiding in weight loss.

Lion's Mane and Reishi Mushrooms - Both Reishi and Lion's

mane have been shown to help with weight loss in animal studies^{5 6} by boosting fat metabolism. In addition, they both help with the regulation of blood sugar⁷. Obesity is often linked to inflammation and gut dysbiosis, and these medicinal mushrooms address these root causes⁸. Finally, they assist in the stress response, which is linked with weight gain. Suggested dose: 2 dropperfuls of dual-extracted reishi and lion's mane tinctures twice daily.

Rhodiola⁹ - An excellent herbal remedy for burning belly fat, *Rhodiola rosea* contains active compounds that can help to prevent diet-induced obesity by encouraging the production of an enzyme called hormone-sensitive lipase. This compound has the ability to break down fat tissue in the belly area. What's more, rhodiola lowers the effects of stress and suppresses the fat-storing hormone cortisol¹⁰, further assisting with weight loss. *Rhodiola is contraindicated for those with autoimmune disorders*. Alternative adaptogens are reishi dual-extracted tincture and American (panax) ginseng.

Herbal Tea for Weight Loss: You will need: 3 Tbsp. fenugreek seeds, 2 Tbsp. chickweed, 2 Tbsp. cleavers, 1 ½ inch cinnamon stick. You may also add fennel seeds. Boil a cinnamon stick in 32oz water for five minutes. Turn the burner down to medium, add the rest of the ingredients and cover. Simmer for three minutes. Turn the burner off and let it steep covered for 15 minutes. Strain into clean glass jars. Take 2 cups daily with your intermittent fasting meal.

Metabolism Tea: You will need: 1 cup warm water, 1-2 Tbsp. raw honey, 2 Tbsp. fresh lemon juice, 2 Tbsp. raw apple cider vinegar, 2 Tbsp. fresh ginger (grated), a dash of cinnamon, a dash of cayenne pepper. Combine the warm water and raw honey. Add in the fresh lemon juice and apple cider vinegar. Grate the fresh ginger and add into a mason jar. Add a dash of fresh cinnamon and cayenne pepper. Combine with a whisk or cover and shake.

Adaptogenic Tincture: Combine the following in a two-quart glass jar: 3 oz. of dried amla fruit /Indian Gooseberry (Eases indigestion and flatulence, high vitamin C, and increases digestion), 3 oz. of lycium goji berries (Promotes healthy bowel flora to improve digestion), 2 oz. of American panax ginseng (Enhances digestion and absorption), 2 oz. of licorice root (Digestive tonic that also cleanses the liver, making it more efficient at burning fat), 2 oz. of Rhodiola (To break down belly fat and suppress the fat-storing hormone cortisol; *only use Rhodiola if you don't have an autoimmune issue, otherwise skip this ingredient*). Cover the herbs with 60- to 80- proof vodka, leaving ½-inch to accommodate the herbs as they expand. Label and store in a cool, dark place for 6-8 weeks. Then strain into a clean glass vessel or dropper bottles. This recipe is safe to take long term, up to three months. Take 2-3 dropperfuls of this tincture two to three times daily before meals, same as the teas. It will help stabilize your metabolism, prevent cravings, and provide extra stamina.

Fat-Burning Tincture: Ingredients and equipment: 2 Tbsp. sliced ginger, 2 Tbsp. sliced ginseng root,

1 Tbsp. fenugreek seeds, 1 Tbsp. cumin seeds, 1 Tbsp. oregano, 1 Tbsp. fennel seeds, ¼ cup chopped alfalfa, 1 tsp. of black pepper, 1 tsp. of chili flakes or paprika, 1 tsp. of cinnamon, 8 oz (200–250 ml) of high proof alcohol (I use 40% vodka or brandy), a glass jar, tea strainer or coffee filter paper.

Combine all of the ingredients in a glass jar covering the herbs with the alcohol. Leave ½-inch of free space at the top. Shake. Infuse for at least four weeks and shake daily. When the tincture is ready, strain it with a tea strainer. If you used ground spices, strain using coffee filter paper. Label your tincture and store in a cool dark place or cupboard. There are a few ways that you can take this tincture. It has a warm, spicy taste you can use to enhance your cooking. You can also add a few drops to a cup of coffee or tea. Or apply a few drops directly under your tongue and allow it to sit for a minute before swallowing. Best taken before each meal.

4. Meal Timing/Fasting Structure the way you eat by using intermittent fasting. Intermittent fasting is not a diet but an eating protocol, meaning you are not restricted in what you eat but when you eat.

For weight loss, I recommend using the 16-8 or 18-6 rule. That means you eat all your meals in either an 8-hour or a 6-hour window. The rest of the time, you only consume water, unsweetened tea, or unsweetened coffee, refraining from anything with calories. This will address one of the root causes of why people struggle to lose weight: insulin resistance. Intermittent fasting helps your body burn more fat after the 12-hour fasting mark. There are people who fast for longer periods, but I do not recommend it for weight loss alone. Please see our dietary section for more information on IF.

16-8 Fasting Protocol Example: This plan allows for two meals per day. Choose the two times per day that you tend to be the hungriest before 6:00 p.m. Your two times might be 10:00 a.m. and 4:00 p.m. or noon and 5:00 p.m. If you're a very early riser it might be 8:00 a.m. and 3:00 p.m. Choose a start time when you are usually hungry. These are the times when your personal metabolic rates are the highest.

Chew your food at least twenty times before swallowing, especially foods that are difficult to digest. Try to avoid snacking. Focus on drinking water instead. Keep your blood sugar levels stable by eating fresh, healthy foods.

Each meal can only last 55 minutes maximum. After 55 minutes, the insulin cycle begins again. You want to prevent the insulin cycle from gearing up again between meals. This is what allows the plan to work because then the body uses the excess fuel that is in storage.

On this plan, especially if you drink plenty of water during your fasting period, you will not feel as hungry. Nevertheless, try not to fill your stomach more than ¾ full at each meal. Overeating at a meal often means that some portions of the food are not digested fully because the system is overloaded. Do not eat before bedtime.

5. Good Sleep Sound sleep and healthy eating patterns go hand in hand, so it's important to regulate sleep function. Wind down and go to bed at the same time each night with a soothing routine. This is crucial. Please see the Sleep Protocol if you are having trouble in this area. You may also wish to refer to the Anxiety and Stress Protocol or the Depression Protocol.

6. Strength Training and HIIT One of the most widely accepted weight loss myths is that cardio is king when it comes to losing pounds. But as you've already seen in the weight loss pyramid, cardio comes in last. Instead, try strength training combined with high intensity interval training (HIIT)¹¹ as this will burn the most calories with the lowest time investment. Note that HIIT is not safe for beginners – best to work your way up to it, if you are just starting out.

Weight training helps speed up your metabolism more than regular cardio, as muscles burn more calories than fat. The more muscle mass you develop, the more calories you burn throughout the day to maintain that muscle mass and the easier it becomes to stay in a caloric deficit.

The best types of exercises involve compound movements: deadlifts, squats, bench presses, pull-ups, shoulder presses, renegade rows, resistance band seated rows.

Start with a moderate weight and work your way up. Aim for strength training three to four times a week. Keep your workout to a maximum of 45 minutes per session to avoid cortisol spikes and to stay in the desired anabolic state (muscle growth phase).

High-intensity interval training (HIIT) has been shown to be more effective for weight loss than moderate exercise such as brisk, continuous walking or biking for long distances. Performing short bursts of intense exercise — such as sprinting, burpees, or push-ups — for 20 seconds then resting for 10 seconds, burn more sugar stored in the muscles. This forces the body to rapidly re-fuel by drawing glucose from other areas of the body, which requires insulin. Over time, the body becomes more sensitive to smaller levels of insulin, helping to combat insulin resistance. HIIT also depletes oxygen in the muscles, forcing the body to burn fat for energy. Ultimately this causes the body to burn calories for hours after exercise due to the oxygen debt. Another advantage is that HIIT requires little time investment — about 10 minutes per session with twenty rounds of intense exercise/rest several times per week.

7. Supplements, Vitamins, and Minerals **Caffeine:** Caffeine can boost your metabolism by up to 16% for a couple of hours. Drink a cup in the morning to help you during the last few hours of intermittent fasting, when the body's natural fat-burning abilities are at their peak levels. It also helps your body burn more fat as fuel and increases ketosis.

Chromium Polynicotinate: Chromium is a trace mineral

that enhances insulin, the most important hormone for turning food into energy. Chromium helps regulate cholesterol, metabolism, and blood sugar^{12 13}, Chromium polynicotinate is composed of chromium and niacin (B3). A common problem people face when trying to lose weight is insulin resistance. A cause of insulin resistance is the lack of chromium in our modern diets. Note that most supplements are Chromium picolinate, which has mixed reviews regarding weight loss. Suggested dose: 200 mcg/day if needed.

Psyllium: Psyllium is a good source of prebiotic fiber, which helps your elimination process and helps you feel fuller. Suggested dose: Take one teaspoon (4 grams) of psyllium powder daily before bedtime. Mix with water.

Magnesium: Magnesium helps control insulin and glucose blood levels. Supplementation assists with weight loss for those who are magnesium deficient, obese, and/or have issues with insulin resistance¹⁴. Suggested dose: 350 mg daily.

Vitamin B-12: Vitamin B-12 is required to help convert fat and protein into energy. Suggested dose: Vitamin B Complex - follow dosage on the bottle as supplements vary.

Vitamin C: Vitamin C deficiency can slow down fat loss. Suggested dose: 3,000 mg daily.

Vitamin D: Getting enough Vitamin D can keep your hormone levels in check and may help enhance weight loss and decrease body fat. Suggested dose: 3,000 IU Vit D drops daily.

L-Glutamine: L-Glutamine is a major metabolic fuel, helping to maintain intestinal cells. Suggested dose: Take 500 mg two times a day on an empty stomach with water (never milk). Best taken with Vitamins B and C to enhance absorption.

Chlorophyll: Take 1 teaspoon of liquid chlorophyll in water each day for detoxification.

8. Cardio Daily cardio is important for health. Do at least thirty minutes of cardiovascular exercise daily; one hour is ideal (5,000 to 10,000 steps per day).

If your health allows it, I recommend incorporating high-intensity interval training (HIIT) into your workout routine. This type of intense exercise makes your body burn fat hours after exercising. I do not recommend it for elderly people or people with heart problems. See above for more about HIIT.

9. Other Strategies **Breathing Practices:** Stress and weight gain go hand-in-hand. Not only do chronically stressed people eat more food more often, they are less efficient at burning calories. There are two breathing exercises below. One helps reduce stress and controls food cravings, and the other helps curb stress and tone your digestive tract.

Deep Lymphatic Breathing reduces stress. Gradually increase your sessions from five minutes of deep breathing to 15 minutes three times a day. While you breathe, focus on positive thoughts such as a daily inspirational quote, prayer, mantra, or soothing visual like a candle flame or ocean waves.

Kapalabhati Breathing/Breath of Fire is a yogic breathing exercise that helps stimulate and tone the digestive tract, energizes the body, and decreases stress¹⁵. Do not practice this if you have a mental illness, are pregnant, have heart disease, or are recovering from surgery or stroke.

For this breathing, your inhale is passive (allow your lungs to fill naturally) and your exhale is active. Inhale, then forcefully exhale, drawing the naval up and inward toward your spine. Repeat in rapid succession, at a rate of about 1 second per exhalation. Begin with 20 repetitions, slowly building to 100¹⁶.

10. Western Medical Approaches The four weight loss drugs approved by the FDA generally work by reducing cravings and appetite so you'll eat less. Carefully read about their side effects if you choose this route.

Bupropion-naltrexone (Contrave): Side effects include severe headaches, dry mouth, dizziness, skin rashes, constipation, tiredness, stomach pains, anxiety, loose stool, modified flora, and nausea.

Liraglutide (Saxenda): Side effects include nausea, vomiting, stomach pains, diarrhea, headaches, dizziness, nausea, heartburn, fatigue, and irritability.

Orlistat (Xenical): Side effects include heartburn, bowel problems, gas, orange and oily stools, vomiting, stomach pain, and constipation.

Phentermine-Topiramate (Qsymia): Side effects include insomnia, dry mouth, constipation, fatigue, depression, numbness, brain fog, and tingling in hands and feet.

	Morning	Noon	Afternoon	Evening	Before Bed
SUNDAY	Herbal tea* Reishi & Lion's Mane tinctures L-Glutamine, Vit. B, C, D Exercise (strength training & HIIT)	Weekly food shop Weekly meal plan Herbal tea* Fat-burning tincture	Breathing exercises American (Panax) ginseng or Rhodiola** Reishi & Lion's Mane dual-ex- tracted mushroom tinctures	Herbal tea* Fat-burning tincture L-Glutamine, Vit. C, Chlorophyll	Psyllium Magnesium Sleep herbs if needed (see Sleep Protocol)
MONDAY	Herbal tea* Reishi & Lion's Mane tinctures L-Glutamine, Vit. B, C, D Exercise (cardio)	Herbal tea* Fat-burning tincture	Breathing exercises Ginseng or Rhodiola** Reishi & Lion's Mane	Herbal tea* Fat-burning tincture L-Glutamine, Vit. C, Chlorophyll	Psyllium Magnesium Good sleep
TUESDAY	Herbal tea* Reishi & Lion's Mane tinctures L-Glutamine, Vit. B, C, D Exercise (strength training & HIIT)	Herbal tea* Fat-burning tincture	Breathing exercises Ginseng or Rhodiola** Reishi & Lion's Mane	Herbal tea* Fat-burning tincture L-Glutamine, Vit. C, Chlorophyll	Psyllium Magnesium Good sleep
WEDNESDAY	Herbal tea* Reishi & Lion's Mane tinctures L-Glutamine, Vit. B, C, D Exercise (cardio)	Herbal tea* Fat-burning tincture	Breathing exercises Ginseng or Rhodiola** Reishi & Lion's Mane	Herbal tea* Fat-burning tincture L-Glutamine, Vit. C, Chlorophyll	Psyllium Magnesium Good sleep
THURSDAY	Herbal tea* Reishi & Lion's Mane tinctures L-Glutamine, Vit. B, C, D Exercise (strength training & HIIT)	Herbal tea* Fat-burning tincture	Breathing exercises Ginseng or Rhodiola** Reishi & Lion's Mane	Herbal tea* Fat-burning tincture L-Glutamine, Vit. C, Chlorophyll	Psyllium Magnesium Good sleep
FRIDAY	Herbal tea* Reishi & Lion's Mane tinctures L-Glutamine, Vit. B, C, D Exercise (cardio)	Herbal tea* Fat-burning tincture	Breathing exercises Ginseng or Rhodiola** Reishi & Lion's Mane	Herbal tea* Fat-burning tincture L-Glutamine, Vit. C, Chlorophyll	Psyllium Magnesium Good sleep
SATURDAY	Herbal tea* Reishi & Lion's Mane tinctures L-Glutamine, Vit. B, C, D Exercise (strength training & HIIT)	Herbal tea* Fat-burning tincture	Breathing exercises Ginseng or Rhodiola** Reishi & Lion's Mane Outdoor walk	Herbal tea* Fat-burning tincture L-Glutamine, Vit. C, Chlorophyll	Psyllium Magnesium Good sleep

* Weight loss and/or metabolism tea.

** Or adaptogenic tincture from protocol.

Ch. 46: Anti-Inflammatory Diet

Inflammation is your body's natural and necessary response to repair injury and fight illness. Your body increases the production of cytokines and white blood cells to fight off infection, which leads to inflammation. Acute inflammation includes symptoms such as swelling, heat, and redness. Chronic inflammation, which can last from weeks to years, can lead to a plethora of diseases, including autoimmune disorders.

Inflammation can occur in any part of the body. Many people experience inflammation in their joints (arthritis), inflammation in the digestive tract (that leads to diseases like IBS and Ulcerative Colitis), and inflammation that affects other parts of the body and even the organs (which can happen with autoimmune diseases like Lupus). Autoimmune disease is a major cause of inflammation, as it is the result of the immune system erroneously attacking bodily tissues like they are foreign invaders, when they are not.

Other factors that lead to inflammation include a sedentary lifestyle, stress, poor eating habits, and leaky gut/increased intestinal permeability. Leaky gut is a major contributor to inflammation and autoimmune disease because it is the result of toxins leaking out into the body from the intestines during digestion¹. Learn how to manage leaky gut in the Leaky Gut, Gastritis, and IBS Protocol. There are many things you can do to decrease systemic inflammation. One of the most important is to consume anti-inflammatory foods and to avoid inflammatory foods. Many people who suffer from inflammation and inflammatory diseases find that a diet change does wonders for their issues.

The consumption of an anti-inflammatory diet is associated with a decreased risk of dementia, rheumatoid arthritis and other autoimmune disorders, heart disease, cancer, diabetes, and nonalcoholic fatty liver disease^{2 3 4 5 6}. Additionally, an anti-inflammatory diet is linked to a longer life span⁷. Those with autoimmune disorders and leaky gut syndrome have a decrease in symptoms when following an anti-inflammatory diet, because inflammation is linked to autoimmune response. It may be helpful to explore an Elimination Diet in order to identify your individual triggers (e.g., dairy). For many, gluten is a trigger. As a result, the best anti-inflammatory diet is also gluten-free.

Research shows that people who follow vegetarian diets and the Mediterranean Diet also experience anti-inflammatory effects^{8 9}. Both of these diets are composed of nutrient-dense whole foods and rely on foods high in antioxidants.

Anti-Inflammatory Foods: Foods that are considered anti-inflammatory include foods that are high in antioxidants. These antioxidants neutralize free radicals in the body that cause mutations leading to cell damage. Antioxidants can decrease inflammatory markers, such as TNF alpha, CRP, and IL-6. Regularly consuming the

following foods helps fight chronic inflammation. Aim to consume some of these foods at every meal: vegetables (kale, brussel sprouts, bell peppers, broccoli, cabbage, wild greens, etc.), fruits (berries, grapes, cherries), healthy fats (avocado, nuts, salmon, sardines, olives, olive oil, avocado oil), spices (turmeric, cinnamon, oregano, etc.), green tea and dark chocolate (occasionally).

Bell peppers are loaded with antioxidants. Specifically, bell peppers contain quercetin, which has been shown to decrease the risk of developing diabetes. Dark chocolate is packed with flavonols, which help fight inflammation and protect your heart. Cherries contain anthocyanins and catechins, which decrease the inflammatory marker CRP. Green tea consumption is associated with a decreased risk of developing obesity, diabetes, heart disease, and Alzheimer's. Turmeric is a strong anti-inflammatory often used for arthritis and diabetes.

Healthy fats are important to consume to fight inflammation. When choosing oils, choose high quality oils that have a high smoke point. Low quality oils with low smoke points are heated during processing, which makes them turn rancid. This rancidity causes oxidation of the oil, which can lead to free radicals in the body. Consume plenty of healthy fats including anti-inflammatory omega-3 fatty acids.

Fermented Foods - Fermented foods are highly recommended to lower inflammation. These include foods like sauerkraut (not canned), kimchi, beet kvass, and unsweetened yogurt. Yogurt should only be consumed by those who can tolerate dairy products. These foods provide the microbiome with essential probiotics that support gut health and immune function. Fermented foods also help with leaky gut/intestinal permeability. Rotate the fermented foods you consume to benefit from the unique probiotic microorganisms each food offers.

Bone Broth - Organic bone broth is a highly valuable addition to an anti-inflammatory diet. This superfood contains beneficial nutrients like collagen, glycine, calcium, glucosamine, and chondroitin. The glycine¹⁰ in bone broth has both anti-inflammatory and antioxidant benefits¹¹. Bone broth may help with joint pain and inflammation because it is a source of glucosamine and chondroitin¹². For arthritic diseases like RA and osteoarthritis, this can be highly beneficial. Bone broth is an asset for those seeking to support bone health because it contains nutrients like calcium. Bone broth is also full of prebiotic nutrition that can support a healthy microbiome. Try drinking a cup of bone broth daily to lower inflammation and support overall health.

To make your own bone broth, you will need: 2-3 pounds of organic bones (animal bones such as beef, chicken, turkey, lamb, salmon, etc.), 1 onion, 4 cloves of garlic, 1 gallon of water, 2 tbsp. apple cider vinegar, salt and pepper to taste, herbs of your choice (some great herbs include oregano, rosemary, thyme, and sage). Other optional (and very beneficial) ingredients you can add include bay leaves, chopped celery, carrots, ginger, parsley, citrus rinds, and stinging nettle.

Place the bones and vegetables into a large pot or slow cooker. Pour enough water into the pot to cover everything. Add the vinegar and bring to a boil over high heat (for a slow cooker start on high and then change to low). After the pot comes to a boil, reduce heat immediately and simmer.

Add salt and pepper, as well as herbs and spices of your choice. Allow to simmer on low heat for as long as possible (minimum of 8-12 hours). Some people simmer their bone broth for 24 hours. The longer you simmer, the more nutrient-dense your broth will be. Allow the broth to cool and strain out the solids. Bottle the broth and freeze what you won't use right away.

Stay Hydrated - Along with a diet rich in the foods mentioned above, make sure to drink plenty of water every day. Dehydration can worsen inflammatory conditions, resulting in more pain and inflammation in the body. It can trigger or worsen inflammatory autoimmune conditions.

Drinking enough water is also important for those with inflammatory bowel diseases, as you need a sufficient amount of water to aid in proper digestion. Try drinking your body weight in ounces of water each day if you can. At the very least, drink half your body weight in ounces daily.

Foods to Limit: Inflammation is promoted by the consumption of junk foods and trigger foods, such as gluten and dairy. While not inflammatory for all people, dairy and gluten are triggering for some. Perhaps the most inflammatory food you can consume is anything containing processed sugar.

Aim to limit or avoid the following foods: desserts (cakes, pastries, candy, cookies, ice cream)¹³, processed meats (hot dogs, deli meat, sausages, etc.)¹⁴, processed snack foods (crackers, chips, pretzels, etc.), fried foods (French fries, onion rings, etc.), sugar and sugary beverages (juice, soda, etc.)¹⁵, refined carbohydrates (white pasta, white rice, white bread, etc.), industrial seed oils (canola, corn, cottonseed, soybean, safflower oils), gluten, alcohol, trans fats¹⁶, dairy (for some but not all people) and eggs (for some, but not all; rarer than dairy).

Use the Elimination Diet and/or take an **IgG test** to find your trigger foods. IgG testing is a type of food antibody assessment. IgG stands for Immunoglobulin G, a type of antibody. This blood test measures antibodies to more

than 87 commonly consumed foods.

This test can show exactly which foods you may be allergic or sensitive to so you know which specific foods to eliminate from your diet to help resolve inflammation. Some IgG tests can be ordered and taken from home, but the best way to get an accurate IgG test is to see a doctor, naturopath, or health professional to draw blood. Do note that you must be eating the foods prior to testing for them to show up on the test.

Lectins - Peppers and other nightshades rarely trigger allergies though they may lead to nightshade intolerance, though there are not any large-scale studies currently supporting this. But, for some, lectins do seem to be a trigger. Lectins are sticky proteins plants use to defend themselves and can be hard to digest¹⁷. Foods that contain lectins include plants in the nightshade family (like tomatoes and potatoes), and legumes like peanuts and peas, grains, and beans¹⁸. If concerned, you can reduce their lectin content by soaking grains, nuts, seeds, and legumes and by cooking nightshades, grains, and legumes¹⁹.

Anti-Inflammatory Lifestyle: Outside of diet, there are lifestyle factors that influence inflammation. Sleep is very important for decreased risk of inflammation²⁰. See more information in the Sleep Protocol for help in this area.

Additionally, while too much exercise can be proinflammatory, regular moderate exercise reduces inflammation²¹. Combine an anti-inflammatory diet with regular exercise and plenty of sleep to best reduce inflammation in the body.

Stress can cause imbalances in the body in more ways than most people realize. At a molecular and cellular level, these stress-induced imbalances occur in cytokine pathways, mitochondrial networks, redox systems, and other signaling pathways to ultimately contribute to systemic inflammation²².

Chronic stress is detrimental to overall health, but it is especially troublesome for those who suffer from autoimmune diseases and systemic inflammation. Curbing the stress response in the body is essential to managing and lowering its inflammatory response. Thankfully, there are steps you can take to mitigate and control your body's response to stress. See the Anxiety and Stress protocol for more information on how to mitigate stress in your daily life.

Day of the week	Breakfast	Lunch	Snack	Dinner
SUNDAY	Blueberry & cherry gluten-free oatmeal	Arugula salad with tuna, dried cranberries, walnuts, tomatoes, & balsamic dressing One cup of organic bone broth	Air popped popcorn with salt and nutritional yeast	Burrito bowl: brown rice, sautéed bell peppers, black beans, avocado
MONDAY	Avocado sprouted gluten-free toast with 2 hard-boiled eggs	Tomato slices, cottage cheese, sardines, gluten free crackers One cup of organic bone broth	Apple with almond butter	Honey Dijon wild salmon with brown rice, cherry tomatoes, & green beans
TUESDAY	Blueberry & cherry gluten-free oatmeal	Spinach salad with olives, grilled chicken, & bell pepper with organic olive oil and lemon juice One cup of organic bone broth	Almonds & dried figs	Chickpea or lentil pasta with red sauce & steamed broccoli
WEDNESDAY	Avocado sprouted gluten-free toast with 2 hard-boiled eggs	Cauliflower red lentil turmeric curry over brown rice One cup of organic bone broth	Frozen cherry & banana "nice" cream (blended frozen fruit – may add coconut or almond milk)	Sheet-pan roasted broccoli, kale, & tofu over brown rice with teriyaki sauce
THURSDAY	Blueberry & cherry gluten-free oatmeal	Kale salad with apple, dates, walnuts, & balsamic dressing One cup of organic bone broth	Green tea & grapes	Vegetable lentil soup
FRIDAY	Avocado sprouted gluten-free toast with 2 hard-boiled eggs	Vegetable lentil soup in organic bone broth	Dark chocolate & cashews	Napa cabbage salad with chicken, mandarin oranges, & walnuts
SATURDAY	Blueberry & cherry gluten-free oatmeal	Gluten-free bread with fried eggs, tomato salsa, & fruit salad One cup of organic bone broth	Apple chips & walnuts	Vegetable lentil soup

Ch. 47: Elimination Diet

The elimination diet is an eating plan that identifies food intolerances or sensitivities. These occur when a person experiences adverse reactions to commonly tolerated foods¹. Food intolerances are estimated to affect up to 20% of the population². The elimination diet involves temporarily removing certain foods from the diet and later reintroducing them, one at a time, to identify which foods are creating health issues. An elimination diet usually lasts for 5–6 weeks³. Once triggering foods have been identified, they can be removed from the diet to prevent future adverse effects⁴.

Typically, an elimination diet begins with a 28-day period of removing all potentially triggering or “culprit” foods. These include processed foods, eggs, gluten, dairy, caffeine, sugar, aspartame, soy, corn, peanuts, tree nuts, citrus fruits, nightshade vegetables, certain legumes, processed meats, shellfish, hydrogenated fats, flavor enhancers, artificial food coloring, nitrate and nitrites, and preservatives.⁵

Food intolerances and sensitivities are often confused with food allergies, they are not the same. Food allergies are related to IgE antibodies, whereas food intolerances and sensitivities are related to IgG antibodies. IgE antibodies are responsible for causing severe allergic reactions, like those seen in peanut or bee allergies. These immune responses occur suddenly, creating a major immune response. They can be serious and sometimes fatal. In contrast, food intolerances or sensitivities are related to IgG antibodies, which have a delayed response. Their effects are much milder and typically contribute to chronic, as opposed to acute, health issues⁶.

Symptoms of food intolerances and sensitivities are numerous and vary widely from person to person. They include gastrointestinal issues, aches and pains, migraines, flu-like symptoms, and skin conditions⁷. The concept of food intolerance is somewhat controversial in conventional medicine as the specific mechanisms involved in IgG food reactions are unclear. Unfortunately, clinical tests to identify food intolerances are sometimes unreliable and often expensive⁸. Elimination diets can be a safe and inexpensive alternative to identifying such intolerances or sensitivities. You may of course pair it with the results of an IgG test.

While the elimination diet is challenging and requires a tremendous amount of willpower, research shows it has a number of health benefits. It reduces symptoms of irritable bowel syndrome⁹, it improves symptoms of eosinophilic esophagitis¹⁰, it reduces symptoms in skin conditions like atopic dermatitis and eczema¹¹, it improves symptoms of ADHD¹², reduces the frequency of migraines¹³, and it improves symptoms associated with rheumatoid arthritis and other autoimmune disorders¹⁴. While the diet doesn't involve any caloric restriction, due to the elimination of many unhealthy foods, people often lose weight during the process.

It's important to note that this diet is contraindicated in anyone suffering from an eating disorder, such as anorexia, body dysmorphic disorder, or orthorexia. Food restriction may provoke or aggravate these serious, life-threatening disorders.

The elimination diet consists of two phases: the elimination phase and the reintroduction phase. In order to effectively identify symptoms, clinicians usually recommend keeping a food journal throughout the diet, documenting everything you eat and the symptoms you experience afterward¹⁵.

Phase 1 - Elimination: The first phase is typically 28 days long. It is the most demanding part of the diet, as it requires considerable planning and discipline. During this phase, you must remove all “culprit” foods from your diet, without any exceptions. This means carefully prepping meals, reading food labels, and ensuring prepared foods don't contain hidden triggers, like additives or preservatives. This makes eating out especially difficult. Perhaps the most challenging aspect is that if you do consume a culprit food, the diet must be restarted.¹⁶ This phase of your diet should consist mostly of whole and unprocessed foods. The following foods are suggested: fruits (most fruits, excluding citrus fruits), vegetables (most vegetables, excluding nightshades), gluten-free grains (quinoa and buckwheat), certain meat and fish (organic beef, turkey, lamb, wild game, and cold-water fish like salmon), dairy substitutes (unsweetened coconut and almond milk -without preservatives or additives), fats (cold-pressed olive oil, flaxseed oil, and coconut oil), beverages (water and herbal teas), spices, condiments, and others (black pepper, fresh herbs and spices - excluding cayenne pepper and paprika), and apple cider vinegar¹⁷.

During the elimination phase, it is important to eat foods that provide the same nutrients as those you are avoiding in order to prevent nutritional deficiencies. After 28 days you can initiate the second phase of the diet. However, if you are still experiencing symptoms of food sensitivities, it is recommended you remain on phase one of the diet for an additional two weeks until your symptoms have disappeared. If symptoms don't disappear contact your healthcare professional.

Phase 2 - Reintroduction: The next phase is the reintroduction or “challenge” phase. It involves slowly and systemically reintroducing eliminated foods back into your diet over a period of time. This enables you to monitor any adverse food reactions you may experience. A food diary or journal is especially helpful during this phase.

Each food should be reintroduced to the diet one at a time, eating it a few times a day over a 2-day period (often 1 day is enough), then waiting 24 hours before introducing the next food. It is important to carefully monitor symptoms that may emerge. When introducing food back into the diet, it is recommended you consume the food in its purest form. For example, when re-introducing eggs, try

to eat scrambled or hard-boiled eggs as opposed to egg salad. Additional ingredients in a food may obscure your ability to determine where symptoms are coming from. The food should be consumed on several occasions over the suggested time period. By repeatedly re-introducing a specific food over a 1-2 day period, you increase the likelihood of identifying food intolerances.¹⁸ Once you have finished one food or food group you can move on to the next, and so on.

Common symptoms that may indicate a food intolerance include headache, itching, bloating, fatigue, diarrhea, constipation, joint pain, indigestion, or worsening of

chronic health conditions.¹⁹ If a food does not cause symptoms during the reintroduction phase, it can be added back into the diet once the elimination diet is complete.

After an elimination diet is complete, you will develop a better understanding of which foods support or harm your health. Based on your symptoms, your health care provider can help you create an individualized meal plan to better assist your well-being. While this diet is most commonly used to help people suffering from gastrointestinal issues, it is a helpful tool in treating many health conditions²⁰.

Day of the week	Breakfast	Lunch	Snack	Dinner
SUNDAY	Berry-coconut smoothie	Chickpea & veggie salad	Pear with almond butter Herbal tea	Grilled salmon with cauliflower "rice"
MONDAY	Chia seed blueberry pudding	Vegan lettuce wraps	Garlic hummus with cucumber slices Herbal tea	Grilled lamb chops with thyme & baked sweet potato
TUESDAY	Ginger-beet juice	Cucumber & avocado salad	Fruit cup (blueberries, sliced bananas, melon, kiwi) Herbal tea	Chickpea & quinoa bowl
WEDNESDAY	Lemon-blueberry energy balls	Squash soup	Nut mix (cashews, almonds, pumpkin seeds) Herbal tea	Braised beef with fennel
THURSDAY	Banana buckwheat pancakes	Sweet potato skins with guacamole	Beet chips (thinly sliced beets, cold-pressed olive oil, salt) Herbal tea	Vegan cabbage soup
FRIDAY	Vegan smoothie bowl	Roasted vegetable & cauliflower rice bowl	Apple with pumpkin seed butter Herbal tea	Thai salad with grilled steak
SATURDAY	Banana pistachio chia pudding	Cucumber vinegar salad	Baked banana chips (bananas, coconut oil, ground cinnamon, salt) Herbal tea	Slow-cooker moroccan-spiced turkey stew

ELIMINATION DIET - WEEKLY MEAL PLAN CHART*

***Berry-Coconut Smoothie:** unsweetened coconut milk beverage, frozen mixed berries, frozen sliced banana, unsweetened shredded coconut, 3 ice cubes.

Chickpea & Veggie Salad: mixed salad greens, chopped cucumbers, rinsed and drained chickpeas, avocado, apple cider vinegar, cold-pressed olive oil, kosher salt, ground pepper.

Grilled Salmon with Cauliflower "Rice": cold-pressed olive oil, ground cumin, dried marjoram, salt, wild salmon, fresh riced cauliflower, parsley leaves, diced

cucumber, sliced scallions.

Chia Seed Blueberry Pudding: unsweetened coconut milk, chia seeds, ground cinnamon, blueberries.

Vegan Lettuce Wraps: bibb lettuce, cooked quinoa, chickpeas, garlic, kosher salt, cold-pressed olive oil, ground cumin, ground cinnamon, diced carrot, fresh parsley, fresh mint.

Grilled Lamb Chops with Thyme and Baked Sweet Potato: grilled lamb, cold-pressed olive oil, fresh rosemary, fresh thyme, baked sweet potato,

coconut oil.

Ginger-Beet Juice: juiced beet, juiced carrot, juiced apple, fresh ginger, ice cubes.

Cucumber & Avocado Salad: shallots, extra-virgin olive oil, fresh mint, fresh basil, salt, thinly sliced English cucumber, chopped ripe avocado.

Chickpea & Quinoa Bowl: quinoa, chickpeas, cucumber slices, diced avocado, hummus, fresh parsley, cold-pressed olive oil, salt, ground pepper.

Lemon-Blueberry Energy Balls: ground walnuts, pitted dates, dried blueberries, chia seeds, almond butter.

Squash Soup: squash, cold-pressed olive oil, thinly sliced shallots, smashed garlic, ground cumin, vegetable broth, salt, fresh basil, dry-roasted chopped pistachios.

Braised Beef with Fennel: Braised organic beef, minced garlic, bay leaf, cold-pressed olive oil, fennel.

Banana Buckwheat Pancakes: buckwheat flour, mashed ripe bananas, coconut oil, ground cinnamon, ground flax seeds, apple cider vinegar, baking soda, salt.

Sweet Potato Skins with Guacamole: sweet potatoes, cold-pressed olive oil, salt, ripe avocado, minced garlic, minced red onion, fresh chopped cilantro.

Vegan Cabbage Soup: cold-pressed olive oil, chopped carrots, sliced fennel, chopped onion, minced garlic, ground coriander, salt, vegetable broth, oregano, green cabbage, cold-pressed olive oil.

Vegan Smoothie Bowl: chia seeds, sliced kiwi, frozen blueberries, unsweetened coconut milk, sliced almonds, unsweetened toasted coconut flakes.

Roasted Vegetable and Cauliflower Rice Bowl: roasted vegetables, diced avocado, sliced scallions, cold-pressed olive oil, chopped fresh cilantro, cauliflower rice.

Thai Salad with Grilled Steak: sliced cucumber, coconut oil, red cabbage, shredded carrot, organic sirloin, fresh cilantro, sliced radishes, chopped green onions.

Banana Pistachio Chia Pudding: unsweetened coconut milk, chia seeds, ground cinnamon, ground cardamom, sliced banana, chopped unsalted roasted pistachios.

Cucumber Vinegar Salad: cucumbers, salt, thinly sliced red onion, apple cider vinegar, cold-pressed olive oil, chopped fresh dill, ground pepper.

Slow-Cooker Moroccan-Spiced Turkey Stew: ground cumin, ground cinnamon, ground coriander, boneless turkey, olive oil, yellow onions, minced fresh ginger, turkey stock, chopped peeled sweet potato (from 1 sweet potato, chickpeas, salt, fresh cilantro sprigs).

Ch. 48: Intermittent Fasting

Intermittent fasting (IF) is gaining popularity due to its proven health benefits. Intermittent fasting refers to any diet where there is a voluntary fasting period, followed by a period of eating freely. With intermittent fasting, you are not told what foods you should eat but rather when you should eat.

The most popular kind of intermittent fasting is daily time-restricted eating. This means that there is a daily fast, typically between 16 and 18 hours (note that you are asleep during ~8 of these hours), during which no foods or drinks with energy are consumed, followed by a period of 6 to 8 hours of eating during the day¹. For example, you could start by eating lunch at noon and finish with dinner at 6:00pm. You then fast until lunch the following day.

Other popular intermittent fasting options include modified fasting regimens, which include full days in which your total calorie intake is significantly decreased to between 500 and 1000 calories. The 5:2 method of intermittent fasting refers to picking two days a week as fasting days¹ and consuming only 500-600 calories during two, non-consecutive days of the week, and eating normally for the remaining five days.

Intermittent fasting somewhat mimics how our hunter-gatherer ancestors would have eaten. Our ancestors likely didn't have access to food at regular intervals; the changes we've seen as a culture in terms of caloric intake and the timing of food consumption is likely leading to obesity and type 2 diabetes². This metabolic switch from utilizing readily available glucose for energy to using stored energy during intermittent fasting is likely built into our biology.

The Many Health Benefits of IF: Intermittent fasting can have a profound impact on your health. It has an especially high rate of success in helping people lose weight and regulate blood sugar, insulin levels, blood pressure, and heart rate, making it beneficial for those with cardiovascular disease². It was previously believed that you had to fully exhaust your glycogen reserves, which could take up to 48 hours of fasting, before seeing benefits. However, recent studies show that time-restricted fasting, where you eat within a 6-8 hour window, resets the immune system; improves resting heart rate, blood pressure, and insulin resistance; and improves cholesterol, triglyceride, and glucose levels. Intermittent fasting also reduces markers of oxidative stress and systemic inflammation associated with atherosclerosis^{3 4 5}. Cholesterol, resting heart rate, and markers of atherosclerosis see improvement within two to four weeks of beginning an intermittent fasting diet².

When you practice intermittent fasting, your human growth hormone (HGH) levels significantly increase, which leads to fat loss, lean muscle gain,¹ and improved insulin sensitivity.¹ Intermittent fasting also enhances our immune response and may help us live longer⁶ as IF initiates cellular repair processes.³

Intermittent fasting may have the same effects on immune function as a 3-day complete fast. Fasting for 3 days reduces white blood cell counts, which trigger the formation of new white blood cells by the immune system⁷. These cells are an important part of fighting infectious disease.

Intermittent fasting may be a possible “priming tool for host defense” against serious and potentially life-threatening infectious respiratory infections⁸: “Calorie restriction in the form of intermittent fasting (IF) in several clinical settings...promote(s) several health benefits, including priming of the immune response. This dietary restriction also activates autophagy, a cell surveillance system that boosts up immunity.”⁹

Intermittent fasting shows huge promise for neurodegenerative disorders, such as Parkinson's and Alzheimer's. Decreased caloric intake improves verbal memory, executive function, and cognition, likely due to the improved ability to suppress inflammation in the brain and increasing the stress resistance of neurons⁴. This eating pattern can delay the onset and progression of some neurodegenerative disorders; it is an exciting new area of research¹⁰.

Another benefit of IF is increased stress resistance². In our current society, our body systems are constantly dealing with stressors. We face emotional stress, dietary stress, and environmental toxins. Intermittent fasting allows our cells to adapt more quickly to stressors and react faster to repair DNA, suppress inflammation, and increase antioxidant defenses². This means our bodies are better able to fight disease and stay healthy. Almost every chronic disease includes inflammation, and intermittent fasting may help fight symptoms and health issues from diseases like cancer, diabetes, and autoimmune disorders.

How it Works and How to Support IF: The secret behind intermittent fasting is accessing ketosis. When our bodies are in a state of low caloric intake for 8 to 12 hours, they begin to break down fat for energy rather than relying on food alone¹¹. The process of breaking down fat creates molecules called ketones. These ketones regulate metabolic reactions, stress resistance, cognitive functions, and the expression of proteins and molecules in our bodies.² These changes allow you to reap many benefits from your fasting state.

Starting an intermittent fasting regimen may be difficult at first. Many people complain of hunger and fatigue as their bodies adjust to a new way of operating. Fasting can sometimes lead to electrolyte imbalance or muscle loss; which is why it is key to include good amounts of protein in your diet, as well as supplementing electrolytes, especially at the beginning of an intermittent fasting journey³. For menstruating women, note that longer fasting, such as 5:2 fasting or modified fasting regimens, may disrupt hormonal messengers that trigger different parts of your monthly cycle; and that longer fasts are not typically recommended¹². With that being said, time restricted eating cycles can be successful and very useful for menstruating women to help regulate

cycles, hormones, and weight⁵.

Overall, intermittent fasting is recommended for individuals who are overweight, obese, struggling from type 2 diabetes, those with chronic inflammatory issues, autoimmune disorders, neurodegenerative diseases, and cardiac problems. It is safest to begin with daily time restricted eating, and then, if needed, expand into a more intense 5:2 fasting state once your body has time to

adjust to being in a state of ketosis regularly.

Remember that this style of eating requires you eat nutrient dense foods, supplement with electrolytes, and hydrate regularly. Focus on anti-inflammatory whole foods such as fruits, vegetables, protein, nuts, and seeds, versus processed foods. Below is an example of a daily time restricted eating meal plan utilizing a 16-hour fasting period.

Time of day	6 - 10 am	10 am	1 pm	3:30 pm	5:30 pm	6:00 pm
SUNDAY	Water or tea of choice (0 cal.)	Green smoothie 2 eggs over easy ½ cup GF* oats with flaxseed, berries, & walnuts	Organic chicken breast veggie wrap Banana with nut butter	Hummus Carrots & peppers 1-2 servings GF** nut-based crackers	4-6 oz. baked lemon salmon ½ cup quinoa with olive oil & salt Green beans, 1 cup cooked	Water or tea
MONDAY	Water or tea	Green smoothie Chia seed pudding topped with berries of choice &/or nut butter	Mediterranean chicken bowl	Apple with nut butter	Ground turkey & black bean burger patty wrapped in lettuce Kale salad	Water or tea
TUESDAY	Water or tea	Green smoothie Chia seed pudding topped with berries of choice &/or nut butter	Wild salmon & veggie wrap Coconut yogurt & grain-free granola with bananas	Hummus Carrots & peppers 1-2 servings GF nut-based crackers	Pulled BBQ pork stuffed sweet potato (4-6 oz. organic meat) 1 cup steamed veggie of choice	Water or tea
WEDNESDAY	Water or tea	Fruit & GF * nut bar ½ cup fresh berries Green smoothie	Large cobb salad	Apple with nut butter	Chicken breast (4-6 oz.) 1 serving chickpea-based pasta with with marinara sauce 1 cup steamed veggie of choice	Water or tea
THURSDAY	Water or tea	½ cup fresh fruit salad 3-egg veggie omelet 1 piece GF* whole grain toast with nut butter	Mexican chicken or tofu rice bowl	Hummus Carrots & peppers 1-2 servings GF nut-based crackers	4-6 oz. baked lemon salmon ½ cup quinoa with olive oil & salt Green beans, 1 cup cooked	Water or tea
FRIDAY	Water or tea	Green smoothie Banana pancakes, topped with nut butter	Wild salmon & veggie wrap (or favorite lunch from the week)	Rice cake with nut butter, fruit, & dark cocoa chips	Stir fry	Water or tea
SATURDAY	Water or tea	Green smoothie 2 eggs over easy ½ cup GF* oats with flaxseed, berries, & walnuts	Organic chicken breast veggie wrap Banana with nut butter	Hummus Carrots & peppers 1-2 servings GF nut-based crackers	- GF lasagna (rice noodles, ground beef, marinara sauce, cottage cheese with spices, spinach, mushrooms)	Water or tea

***Green Smoothie:** spinach, mango, pineapple, coconut water, strawberries, protein powder, if desired.

Organic Chicken Breast and Veggie Wrap: 4 oz. chicken, peppers, spinach, onion, tomatoes.

3 Egg Veggie Omelet: spinach, peppers, onion, tomatoes, cheese or goat cheese.

Mediterranean Chicken Bowl: 4- 6 oz. chicken, quinoa, tomatoes, olives, red onion, spinach, taziki or hummus.

Chia Seed Pudding: chia seeds, almond milk, honey.

Large Cobb Salad: mixed greens, hard-boiled egg, 3-4 oz chicken breast, tomatoes, onion, olive oil-based dressing.

Mexican Chicken or Tofu Rice Bowl: 4- 6 oz. chicken, brown rice, salsa, grilled veggies, avocado.

Banana Pancakes: 1 mashed banana, ½ cup GF oats, 2 eggs, cinnamon; cook on griddle, top with real maple syrup.

Stir Fry: 4-6 oz. ground turkey; grilled veggies, brown rice, teriyaki sauce to top.

****GF=** gluten free

GF lasagna: ice noodles, ground beef, marinara sauce, cottage cheese with spices, spinach, mushrooms, layered and baked.

Ch. 49: Ketogenic Diet

The Ketogenic or “Keto” Diet has become an extremely popular strategy for weight loss and a variety of health conditions. Originally used as a therapeutic diet for epilepsy decades ago, many other health benefits have since been discovered.¹

The ketogenic diet refers to a high-fat, low-carb approach to eating. It dramatically limits your carbohydrate intake while increasing your fat and protein intake. When most of your calories come from fat, your body is forced to use different energy pathways. Instead of using carbohydrates for energy, the body burns its own fat stores as a source of fuel. During this process, called ketosis, your liver produces compounds called ketones, a byproduct of fatty acid breakdown.² If you follow the diet strictly, your body will reach ketosis in about 4 days. You will likely even see signs of weight loss in the first week.³

The diet consists of eating a high amount of fat, a moderate amount of protein, and a very low amount of carbohydrates. The general goal is to swap out carbohydrates for fat. The diet consists of approximately 55% to 60% fat, 30% to 35% protein, and 5% to 10% carbohydrates.⁴ The Dietary Guidelines for Americans roughly suggest a 2000-calorie-per-day diet.⁵ Based on this recommendation, while on the ketogenic diet, carbohydrate consumption would amount to 20 to 50 grams per day.⁶ Considering a medium-sized banana has about 27 grams of carbohydrates, this diet can be quite restrictive.⁷

Health Benefits and Risks of a Ketogenic Diet: Research has demonstrated several health benefits from the ketogenic diet. It results in significant weight loss and a reduction of harmful abdominal fat (known to cause serious metabolic problems)^{8 9}. This diet lowers blood sugar and insulin levels¹⁰ and reduces blood pressure and the risk of developing Type 2 Diabetes.¹¹ This diet can be beneficial for women, especially those struggling with Polycystic Ovarian Syndrome (PCOS).¹² It is beneficial in treating epilepsy¹³ and evolving research suggests this diet may also help treat symptoms of Alzheimer’s and Parkinson’s disease.¹⁴

Despite its many advantages, most research focuses on short-term benefits. There is conflicting information

regarding its long-term health benefits beyond epilepsy.¹⁵ New research shows that diets both high and low in carbohydrate consumption are associated with increased mortality.¹⁶ Based on these findings a ketogenic diet may not be a healthy long-term diet. One major concern is the increased consumption of saturated fats. Saturated fats are found in animal-based foods like beef, pork, poultry, full-fat dairy products, and eggs and tropical oils like coconut, and palm.¹⁷ Saturated fats are a known risk factor for heart disease and other health complications. Another concern is the development of nutrient deficiencies associated with reduced consumption of nutrient-rich fruits, vegetables, and grains.¹⁸ Additionally, a healthy gut microbiome is supported by fiber. Due to the limited grain consumption in a ketogenic diet, the ketogenic diet provides significantly less fiber than other diets. A lack of fiber may reduce the concentration of good bacteria in your gut.¹⁹

The ketogenic diet is contraindicated in people with pancreatitis, kidney problems, liver problems, and disorders of fat metabolism, as the ketogenic may overload the organs involved in processing protein and fat.²⁰

How to Support the Diet: Foods recommended on the ketogenic diet include: animal proteins (eggs, beef, pork, poultry, and seafood), high-fat, unsweetened dairy and dairy alternatives (cheese, plain yogurt, cottage cheese, cream, half-and-half, and certain unsweetened plant milk), fats and oils (butter, ghee, avocado, olive oil, and coconut oils), non-starchy vegetables (leafy greens, summer squash, peppers, avocados, and olives), fruits that have a low carbohydrate content (berries, avocados, watermelon, cantaloupe, lemons, tomatoes, peaches, and starfruit), nuts and seeds (almonds, peanuts, cashews, pumpkin seeds, etc.), dark chocolate is allowed, unsweetened beverages (coffee, tea, and sparkling water).

Vegetables not allowed on the ketogenic diet include potatoes and sweet potatoes, certain winter squashes, such as acorn squash and butternut squash, corn, and beets.

Adopting the ketogenic diet consists of strictly consuming the recommended foods and ensuring proportions are roughly 60% fat, 35% protein, and no more than 10% carbohydrates.

Day of the week	Breakfast	Lunch	Snack	Dinner
SUNDAY	Keto Protein Bar Coffee/Tea	Salmon Salad Zucchini Cups	Keto Chocolate Chip Cookies	Keto Turkey Sausage Balls
MONDAY	Keto Grain-Free Oatmeal Coffee/Tea	Keto Tuscan Soup	Pepperoni Slices with Cheese	Keto Spaghetti Vegetable Soup
TUESDAY	Keto Turkey Sausage Omelette Coffee/Tea	Keto Caesar Salad Wraps	Hard Boiled Eggs	Keto Burgers with Mushroom Sauce
WEDNESDAY	Keto Pecan Muffins Coffee/Tea	Crust-Free Keto Quiche	Keto Trail Mix	Keto Pesto Pasta with Zesty Salmon
THURSDAY	Savory Breakfast Pancakes Coffee/Tea	Keto Chicken Fajita Salad	Chia Seed Pudding	Keto Chicken Fajitas
FRIDAY	Keto Hash Browns Coffee/Tea	Keto Cauliflower "Potato" Salad	Kale Chips	Garlic Butter Steak
SATURDAY	Keto Protein Bar Coffee/Tea	Grain-Free Pasta Salad	Caprese Salad	Keto Thai Ginger Chicken Noodles

***Keto Protein Bar:** Nut butter of choice (almond or cashew are most popular), melted coconut oil, melted vanilla protein, vanilla stevia, pink salt, cacao nibs, and cinnamon.

Keto Chocolate Chip Cookies: Almond butter, monkfruit sweetener, organic egg, vanilla extract, baking soda, pink salt, vanilla-flavored liquid stevia, sugar-free chocolate chips, and chopped raw walnuts.

Salmon Salad Zucchini Cups: Medium zucchini, wild-caught salmon, avocado oil mayonnaise, shredded cheese, ground pepper, and sea salt.

Keto Turkey Sausage Balls: Ground turkey, organic egg, blanched almond flour, diced onions, onion powder, dried rosemary, paprika, dried oregano, red pepper flakes, black pepper, and sea salt.

Keto Grain-Free Oatmeal: Hemp hearts, organic coconut milk, chia seeds, vanilla extract, liquid stevia or monk fruit sweetener, sea salt, berries, and almonds.

Pepperoni Slices with Cheese: Sliced pepperoni and cheddar (or dairy free) cheese.

large onion, minced garlic, dried oregano, sea salt and pepper, sun-dried tomatoes, chicken broth, chopped kale, heavy cream or non-dairy heavy cream, grated Parmesan cheese or dairy-free Parmesan cheese.

Keto Spaghetti Vegetable Soup: Baked spaghetti squash baked, chopped onion, chopped kale, chopped red and green bell pepper, extra virgin olive oil, vegetable broth, fresh rosemary, black pepper, and sea salt.

Keto Turkey Sausage Omelette: Organic eggs, sea salt, pepper, sliced turkey sausage, dried thyme, fresh chopped parsley, coconut oil, and sliced goat cheese.

Hard Boiled Eggs: Organic eggs

Keto Caesar Salad Wraps: Turkey bacon, shredded chicken, crisp lettuce, cucumber, feta cheese, grass-fed sour cream, dijon mustard, white wine or apple cider vinegar, minced garlic, extra virgin olive oil anchovy filets.

Keto Burgers with Mushroom Sauce: Grass-fed beef burger patties, grass-fed butter, extra virgin olive oil, sliced mushrooms, organic heavy cream or dairy-free coconut heavy cream, Worcestershire sauce, dijon

mustard, sea salt and pepper, and fresh chopped parsley.

Keto Pecan Muffins: Almond flour, coconut flour, liquid stevia, sea salt, baking powder, organic eggs, melted butter, keto maple syrup, chopped pecans, and vanilla extract.

Keto Trail Mix: Pumpkin seeds, almonds, cashews, peanuts, cacao nibs.

Crust-Free Keto Quiche: Organic eggs, cream cheese or dairy-free cream cheese, feta cheese or dairy-free cheese, sea salt or pink salt, chopped fresh spinach, and diced red peppers.

Keto Pesto Pasta with Zesty Salmon: Zucchini (spiralized into noodles), wild Alaskan salmon, virgin coconut oil, fresh lemon juice, salmon seasoning, raw spinach, fresh basil, pine nuts, Parmesan cheese, pepper, and sea salt.

Savory Breakfast Pancakes: blended cream cheese, organic eggs, cinnamon (for pancake), cream cheese, chopped fresh fennel, smoked trout or salmon, fresh lemon juice (for cream cheese spread), coconut oil, butter (for cooking).

Chia Seed Pudding: Chia seeds, unsweetened organic coconut milk, cinnamon.

Keto Chicken Fajita Salad: Sliced chicken breasts, thinly sliced red, yellow & green bell peppers, minced garlic clove, finely chopped yellow or red onion, olive oil, Mexican spice blend, keto ranch dressing, mixed green, and sliced avocado.

Keto Chicken Fajitas: Sliced chicken breasts, cumin, garlic powder, extra-virgin olive oil, coconut aminos, sliced red and green bell pepper, sliced yellow onion, cauliflower rice, pepper, and sea salt.

Keto Hash Browns: Stemmed and roughly chopped cauliflower, egg, crumbled cheddar cheese, fresh chives, sea salt or pink salt, black pepper, cayenne pepper, and garlic powder.

Kale Chips: Kale, olive oil, nutritional yeast, and salt.

Keto Cauliflower “Potato” Salad: Steamed cauliflower, turkey or grass-fed beef bacon, chopped celery stalks, finely chopped red onion, chopped fresh chives, avocado oil mayonnaise, dijon mustard, apple cider vinegar, salt, and pepper.

Garlic Butter Steak: Boneless New York or rib-eye steak, olive oil, minced garlic, finely chopped fresh rosemary, finely chopped thyme, butter, pepper, and sea salt.

Keto Protein Bar: Nut butter of choice (almond or cashew are most popular), melted coconut oil, melted vanilla protein, vanilla stevia, pink salt, cacao nibs, and cinnamon.

Caprese Salad: Sliced fresh mozzarella cheese, chopped fresh basil, olive oil, Worcestershire sauce, salt, and pepper.

Grain-Free Pasta Salad: Yellow squash (peeled and spiraled into noodles), sliced green onion, quartered and sliced cucumber, kalamata olives, extra virgin olive oil, apple cider vinegar, garlic powder, dried oregano, thyme, dried basil, chopped parsley, and sea salt.

Keto Thai Ginger Chicken Noodles: Zucchini (spiralized into noodles), boneless chicken breast, coconut oil, sea salt, sliced bok choy, almond butter, coconut aminos, lime juice, water, minced garlic cloves, minced ginger, olive oil, coconut vinegar, and liquid stevia.

Ch. 50: Mediterranean Diet

The Mediterranean Diet is well known for its numerous health benefits. Worldwide interest in the diet emerged in the 1950s when it was noted that heart disease was relatively rare in Mediterranean countries¹. Researchers observed that these people were exceptionally healthy compared to the rest of the world². Inspired by these findings, a mass amount of research was conducted on the lifestyle habits of people living in the region. The Mediterranean diet reflects the traditional eating patterns of Greece, Italy, Spain, and other countries that border the Mediterranean Sea³.

While there are a few versions of the Mediterranean diet, it typically consists of plant-based foods, such as whole grains, vegetables, legumes, fruits, nuts, seeds, herbs, and spices⁴. Seafood, specifically fish, is the main source of animal protein⁵. Olive oil is consumed in high quantities and is the main source of fat⁶. There is moderate red wine consumption with meals⁷. Dairy and poultry are also consumed in moderation⁸. The diet doesn't exclude many foods but emphasizes minimal red meat and sugar/dessert consumption. The diet isn't restrictive as it doesn't require eliminating certain foods, calorie counting, or portion measurements⁹. Instead, it encourages freely consuming fresh, nutrient-dense, unprocessed foods native to the region. This makes it a popular choice for people who want to enhance their

health and longevity without making too many sacrifices.

Health Benefits of the Mediterranean Diet: The Mediterranean diet has been linked to a number of health benefits like reduced mortality risk and lower incidence of cardiovascular disease¹⁰. It reduces the risk of premature death from all causes in both men and women¹¹, reduces obesity and Type 2 Diabetes¹², reduces strokes by 40% or more in high-risk patients¹³, lowers high blood pressure¹⁴, improves gut health¹⁵, reduces risk of dementia and cognitive decline¹⁶, and lower risks for common cancers, such as colorectal, breast, and prostate cancer¹⁷. It also improves mental health conditions, such as anxiety and depression¹⁸.

How to Start: Although there are no strict rules for following the Mediterranean diet, processed foods, added sugar, and refined grains should be restricted¹⁹. Generally speaking, the diet encourages freely consuming fruits, vegetables, whole grains, legumes, nuts, seeds, seafood, and heart-healthy fats. General guidelines you can follow to incorporate the principles of the diet into your daily routine include; building meals around vegetables, beans, and whole grains, eating fish at least twice a week, using olive oil instead of butter in preparing food, and serving fresh fruit for dessert²⁰. For many, eating a Mediterranean diet is more of a lifestyle choice than a diet. It can be as simple as making a habit of choosing nutrient-dense foods while avoiding processed and sweetened food.

Day of the week	Breakfast	Lunch	Snack	Dinner
SUNDAY	Muesli with Fruit Fresh juice Coffee/Tea (Unsweetened)	White Bean Soup & Greek Salad	Spicy Lentil Soup	Sautéed Scallops with Vegetables Mixed Berries for Dessert
MONDAY	Vegetable & Egg Frittata Fresh juice Coffee/Tea (Unsweetened)	Salmon Salad	Fruit Cup (watermelon blueberries, banana)	Mediterranean Chickpea & Farro Salad Mixed Berries for Dessert
TUESDAY	Greek Yogurt Parfait Fresh juice Coffee/Tea (Unsweetened)	Avocado Toast	Hummus with Celery and Cucumber	Grilled Salmon Filet with Vegetables Mixed Berries for Dessert
WEDNESDAY	Poached Eggs with Sweet Potato Hash Fresh juice Coffee/Tea (Unsweetened)	Grilled Veggie Sandwich	Blueberry Coconut Energy Bites	Greek Stuffed Eggplant Glass of Red Wine Mixed Berries for Dessert
THURSDAY	Shakshuka Fresh juice Coffee/Tea (Unsweetened)	Tuna Salad	Herb Roasted Chickpeas	Garlic Baked Cod with Roasted Vegetables Mixed Berries for Dessert
FRIDAY	Oatmeal with Fruit Fresh juice Coffee/Tea (Unsweetened)	Lentil Soup & Crackers	Green Smoothie (Blueberries, Spinach, Greek yogurt)	Grilled Chicken Breast with Vegetables Mixed Berries for Dessert

S A T U R D A Y	Goat Cheese Omelet Fresh juice Coffee/Tea (Unsweetened)	Greek Chicken Quinoa Bowl	Fruit & Nut Butter (Banana, Almond Butter)	Mediterranean Pasta Glass of Red Wine Mixed Berries for Dessert
--------------------------------------	---	---------------------------	---	---

*** Muesli with Fruit:** Muesli, blueberries, bananas, almond milk.

Spicy Lentil Soup: Lentils, garlic, olive oil, lemon juice, vegetable broth, spinach, oregano, salt, pepper.

White Bean Soup and Greek Salad: White beans, vegetable broth, onion, garlic, olive oil, parsley, thyme, oregano, lemon juice, salt, pepper.

Sautéed Scallops with Vegetables: Scallops, shallot, garlic, olive oil, lemon juice, red bell pepper, cauliflower, brown rice, cumin, paprika, oregano, salt, pepper.

Vegetable and Egg Frittata: Eggs, zucchini, tomatoes, broccoli, white onion, olive oil, green onion, feta cheese, parsley, thyme, salt, pepper.

Salmon Salad: Salmon fillet, lemon juice, olive oil, lettuce, red onions, feta cheese, quinoa, tomatoes, salt, pepper.

Mediterranean Chickpea and Farro Salad: Farro, red onion, bell peppers, sun-dried tomatoes, parsley, chickpeas, feta cheese, avocado, lemon juice, olive oil, salt, pepper.

Greek Yogurt Parfait: Greek yogurt, walnuts, berries, almonds, chia seeds.

Avocado Toast: Whole-gran toast, olive oil, avocado, tomatoes, red pepper flakes, lemon juice, salt, pepper.

Grilled Salmon Filet with Vegetables: Salmon, lemon juice, garlic, olive oil, brown rice, carrots, broccoli, salt, pepper.

Poached Eggs with Sweet Potato Hash: Sweet potato, eggs, olive oil, garlic, red bell pepper, kale, cumin, garlic powder, paprika, green onion, salt, pepper.

Blueberry Coconut Energy Bites: Oats, chia seeds, flaxseed, almond butter, honey, cinnamon, vanilla extract.

Grilled Veggie Sandwich: Whole-grain bread, olive oil,

feta cheese, kale, red onion, tomato, black ripe olives, salt, pepper.

Greek Stuffed Eggplant: Eggplant, olive oil, red onion, tomato sauce, cumin, garlic, tomatoes, red bell pepper, parsley, kalamata olives, red wine vinegar, feta cheese, dill, oregano, salt, pepper.

Shakshuka: Poached eggs, tomato sauce, olive oil, peppers, onion, garlic, cumin, paprika, cayenne pepper, salt, pepper.

Herb Roasted Chickpeas: Chickpeas, olive oil, garlic, parsley, thyme, oregano, salt, pepper.

Tuna Salad: Lettuce, tuna, cucumber, olive oil, green onions, fresh dill, balsamic vinegar, lemon juice, salt, pepper.

Garlic Baked Cod with Roasted Vegetables: Cod, lemon juice, olive oil, garlic potatoes, asparagus, salt, pepper.

Oatmeal with Fruit: Steel cut oats, fresh fruit, sliced almonds, cinnamon, honey.

Lentil Soup and Crackers: Lentils, garlic, onion, carrot, celery, tomatoes, onion, olive oil, paprika, garlic, lemon juice, parsley, salt, pepper, whole-grain crackers.

Grilled Chicken Breast with Vegetables: Chicken, thyme, olive oil, garlic, lemon juice, spinach, red potatoes, onions, salt, pepper.

Goat Cheese Omelet: Eggs, olive oil, goat cheese, tomatoes, green pepper, onion, salt, pepper.

Greek Chicken Quinoa Bowl: Chicken, quinoa, onions, garlic, lemon juice, olive oil, feta cheese, olive cucumber, red onions, avocado, salt, pepper.

Mediterranean Pasta: Whole wheat or GF pasta, tomato sauce, garlic, onion, olive oil, steamed broccoli, parmesan cheese.

Ch. 51: The Wahls Protocol Diet

The Wahls protocol has become a popular diet for people suffering from auto-immune diseases, especially Multiple Sclerosis (MS). It was developed by Dr. Terry Wahls, a physician who has MS. After becoming wheelchair-bound, she researched the role of food and nutrition in MS. She developed the Wahls protocol, a dietary approach aimed to rejuvenate cells and slow the process of MS.¹ She was able to rid herself of most of her debilitating symptoms from MS using this diet.²

The Wahls diet combines principles from the Paleo (Paleolithic) and Elimination diets. It advocates for eating nutrient-dense foods-high in vitamins, minerals, antioxidants, and essential fatty acids.³ The diet consists of eating plenty of vegetables, meat, fish, and fresh fruits, while strictly avoiding all processed foods. It also eliminates any foods that might activate the immune system through intolerances or sensitivities. The diet specifically focuses on nutrients that reduce inflammation and support brain and cell health, such as carotenoids, vitamin K, sulfur, retinol, and fermented foods.⁴

There are three levels to the Wahls diet. The levels are meant to build on each other, making the diet more manageable. According to Dr. Wahls, the higher the level of the diet that you follow, the greater benefit you'll see.⁵

Health Benefits: The Wahls protocol has a large and enthusiastic following. Many people report health benefits from being on the diet. It is especially popular in the MS community. Currently, there is limited research on the Wahls Protocol, although a major study is presently underway.

One small study found that people with MS who switched to a Wahls-style diet for a year were much less tired. However, they also received a number of other therapies including, exercise, stretching, meditation, massage, and electrical stimulation therapy, making it difficult to draw specific conclusions about the diet.⁶ Other research has shown elimination diets improve symptoms and inflammation in patients with autoimmune diseases, like

irritable bowel disease (IBD)⁷ and rheumatoid arthritis (RA).⁸ Another study demonstrated that a paleo-elimination diet was effective as part of a multi-disciplinary approach to treating the autoimmune disease Hashimoto's thyroiditis.⁹ Despite the clear link between dietary patterns and the progression of autoimmune diseases, there is a lack of research on the topic.

How to Start: The Wahls protocol consists of eating lots of meat and fish, with an emphasis on organ meat like liver and kidneys. It encourages large servings of vegetables, especially dark leafy vegetables like kale and sulfur-containing vegetables like cabbage. Initially, brightly colored fruit, like berries, are also consumed in high quantities. Fat from both animal and plant sources is an important part of the diet, with an emphasis on omega-3 fatty acids. It also includes seaweed, nutritional yeast, and fermented foods.¹⁰ The diet prohibits eating processed foods, dairy, sugar, sweeteners (including natural sweeteners like honey), grains (including wheat, rice, and oatmeal), legumes (beans and lentils), and many nightshade vegetables.¹¹

Level One: Eliminate all gluten and dairy, and focus on consuming 9 cups of fruits and vegetables each day.

Level Two: Eliminate all grains, and limit potatoes and beans to two servings a week. You'll incorporate seaweed and fermented foods.

Level Three: This level is a ketogenic diet with intermittent fasting. It eliminates potatoes, beans, and legumes, and limits fruit to one serving, preferably berries. It incorporates coconut milk and limits meals to twice daily, with a long fast.¹²

The Wahl's protocol has become a popular diet for people suffering from MS. There are countless anecdotal reports of its health benefits and symptom relief. This is encouraging considering the limited treatment options for people suffering from MS.¹³ This protocol has laid the foundation for more research to be conducted on the important relationship between nutrition and autoimmune disease.

To learn more, visit her website.

Day of the week	Breakfast	Lunch	Snack	Dinner
SUNDAY	Cocoa Smoothie	Onion Parsley Sumac Salad	Healthy Chips	Sauteed Scallops with Cauliflower
MONDAY	Bacon Kimchi Skillet	Grilled Salmon with Asparagus	Fruit Cup (Watermelon, Blueberries, Banana, Orange, Kiwi, Honeydew melon)	Chicken Liver Fried "Rice"
TUESDAY	Fresh Fruit Salad	Rainbow Chard with Bone Broth and Bacon	Wahls Fudge	Grilled Salmon Filet with Vegetables

W E D N E S D A Y	Chicken Zucchini Poppers	Turkey Tacos	Blueberry Coconut Energy Bites	Sweet Potato Salad with Bacon
T H U R S D A Y	Matcha (Green) Tea Smoothie	Winter Squash And Sweet Potato "Risotto"	Red Cabbage Coleslaw	Garlic Baked Cod with Roasted Vegetables
F R I D A Y	Aronia Smoothie	Spicy Beef Collard Wraps	Trail Mix (Sunflower & Pumpkin Seeds, Cashews, Almonds, Raisins)	Pesto Roasted Chicken
S A T U R D A Y	Turkey Apple Breakfast Hash	Salmon & Kimchi Salad	Chia Seed Pudding	Slow Cooker Spaghetti Squash

***Cocoa Smoothie:** Hemp milk, water (or ice), unsweetened organic cocoa, cinnamon, banana, algae, dried kelp.

Healthy Chips: Baked beet or zucchini chips, olive oil, sea salt, pepper.

Onion Parsley Sumac Salad: Spanish onion, mint leaves, ground sumac berries, fresh parsley, cumin, lemon juice, olive oil, sea salt, pepper.

Sauteed Scallops with Cauliflower: Scallops, shallot, garlic, olive oil, lemon juice, cauliflower, cumin, paprika, oregano, sea salt, pepper.

Bacon Kimchi Skillet: Bacon, ghee, cauliflower, carrots, onions, kale, kimchi, sea salt, pepper.

Grilled Salmon with Asparagus: Salmon fillet, ghee, lemon juice, olive oil, garlic, fresh dill, asparagus, salt, pepper.

Chicken Liver Fried "Rice": Coconut oil, carrot, scallops, chicken livers, coconut aminos, cauliflower, sea salt, pepper.

Fresh Fruit Salad: Blueberries, strawberries, bananas, orange, kiwi, chia seeds, coconut flakes.

Wahls Fudge: Coconut oil, avocado, raisins, dried unsweetened coconut, unsweetened cocoa powder.

Rainbow Chard with Bone Broth and Bacon: Olive Oil, organic bacon, onion, rainbow chard, raisins, bone broth, sea salt, pepper.

Grilled Salmon Filet with Vegetables: Salmon, lemon juice, garlic, olive oil, carrots, broccoli, sea salt, pepper.

Chicken Zucchini Poppers: Zucchini, coconut oil, ground chicken, ground pork, garlic powder, onion powder, fresh parsley, pepper, sea salt.

Blueberry Coconut Energy Bites: Chia seeds, flaxseed, almond butter, cinnamon, vanilla extract.

Turkey Tacos: Ghee, ground turkey, onion, garlic cloves, cilantro, hot sauce, kale, avocado, lime juice, sea salt,

pepper.

Sweet Potato Salad with Bacon: Sweet potato, bacon, celery, fresno chili, mustard, garlic, white wine vinegar, olive oil, green onions, sea salt, pepper.

Matcha (Green) Tea Smoothie: Hemp milk, boiling water, matcha green tea powder, ground cardamom, cinnamon, banana, algae, dried kelp.

Red Cabbage Coleslaw: Red Cabbage, fennel, grated carrots, red onion, jicama, apple cider vinegar, olive oil, horseradish.

Winter Squash and Sweet Potato "Risotto": Ghee, coconut oil, leek, butternut squash, sweet potato, fresh thyme, fresh oregano, chicken broth, sea salt, pepper.

Garlic Baked Cod with Roasted Vegetables: Cod, lemon juice, olive oil, garlic potatoes, asparagus sea salt, pepper.

Aronia Smoothie: Aronia berries, hemp milk, cinnamon, ground cardamom.

Spicy Beef Collard Wraps: Onions, cilantro, green peppers, jalapenos, celery, cauliflower, organic beef, chili powder, hot pepper flakes, lime juice, sea salt, pepper.

Pesto Roasted Chicken: Garlic, fresh basil leaves, toasted pumpkin seeds, olive oil, whole chicken, lemon, lemon juice, sea salt, pepper.

Turkey Apple Breakfast Hash: Ground turkey, coconut oil, thyme, cinnamon, onion, zucchini, shredded carrots, butternut squash (or sweet potato), apple, spinach, powdered ginger, garlic powder, turmeric, sea salt, pepper.

Chia Seed Pudding: Chia seeds, hemp milk, cinnamon, bananas, strawberries, blueberries.

Salmon and Kimchi Salad: Salmon, ghee, fresh dill, lettuce, onions, kale, broccoli, avocado, olive oil, lemon juice, kimchi.

Slow Cooker Spaghetti Squash: Spaghetti squash, ghee, nutritional yeast, sea salt, pepper.

Index

5-HT1A/2/3 receptor antagonists: 167
 5-HTP: 155, 156, 165, 168, 289, 291
 5-hydroxytryptophan: 153, 163, 285, 286, 339, 340, 343
A
 Acetaminophen: 78, 83, 243, 282
 Acetyl-L-Carnitine: 159, 161
 Achillea millefolium: 86, 115, 124, 199, 201, 214, 225, 247, 283
 Acid reflux: 60, 63, 84
 Acid Reflux Protocol: 60, 63
 Acne: 9, 119, 246, 247, 248, 249, 264, 265
 Actaea racemosa: 180, 199, 246, 276
 Activated charcoal: 56, 59
 Acyclovir: 125
 Adrenal Fatigue Protocol: 90, 95, 250
 Aducanumab: 148
 Aesculus hippocastanum: 283
 Agastache foeniculum: 225
 Agrimonia eupatoria: 50, 72
 ALA: 98, 256
 Albendazole: 142, 143
 Albizia: 214
 Allergen immunotherapy: 215
 Allergy: 211, 212, 213, 214, 215, 216, 220, 224, 225, 226, 231, 232, 246
 Allergy Protocol: 211, 216, 224
 Aloe vera: 32, 66, 69, 72, 75, 85, 89, 129, 132, 180, 203, 279
 Alpha lipoic acid: 98, 99, 102, 256, 258
 Althaea officinalis: 10, 16, 30, 37, 50, 62, 66, 71, 79, 86, 87, 106
 Alzheimer's: 145, 146, 147, 148, 149, 157, 158, 159, 160, 161, 240, 261, 263, 265, 305
 Alzheimer's Protocol: 145, 149, 157
 Amantadine: 172
 American ginseng: 16, 101, 182, 188, 189, 206, 254
 American skullcap: 287, 288
 Amino acids: 22, 76, 96, 107, 109, 136, 137, 142, 165, 275
 Amoxicillin: 119, 232
 Anabolic drugs: 238
 Antacids: 62, 63, 74, 88
 Anti-Aging Protocol: 259, 267
 Antibiotics: 58, 68, 88, 92, 116, 118, 119, 187, 189, 207, 227, 232, 249
 Anti-cholinergics: 172, 173
 Anti-Inflammatory Diet: 21, 27, 29, 35, 36, 60, 64, 65, 70, 104, 137, 183, 184, 204, 223, 241, 242, 244, 247, 268, 275, 297, 298, 299
 Anxiety: 9, 10, 16, 17, 19, 23, 25, 26, 27, 32, 37, 44, 52, 73, 75, 85, 86, 90, 108,

111, 135, 174, 148, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 164, 165, 166, 171, 177, 179, 180, 181, 183, 194, 196, 203, 204, 206, 209, 253, 257, 264, 265, 270, 276, 277, 288, 308
 Anxiety and Stress Protocol: 10, 18, 33, 40, 43, 52, 73, 85, 91, 98, 150, 156, 166, 179, 196, 213, 221, 224, 244, 253, 271, 277, 290, 294, 298
 Anxiety Protocol: 150, 156
 Apple cider vinegar: 22, 36, 49, 61, 97, 117, 129, 132, 163, 293
 Arctium lappa: 31
 Arnica: 39, 41, 241
 Artemisia: 135, 141, 225
 Ashwagandha: 11, 16, 19, 31, 52, 67, 69, 73, 85, 91, 95, 100, 102, 106, 111, 135, 138, 150, 156, 157, 161, 179, 183, 185, 188, 190, 195, 257, 263, 267, 270, 273
 Asthma: 214, 215, 217, 218, 219, 220, 221, 222, 224, 225, 226, 227, 230, 231
 Asthma Protocol: 217, 222
 Astragalus: 114, 120, 122, 126
 Autoimmune Protocol: 7, 13, 16, 23, 32
 Avocado-soybean unsaponifiables: 39, 41
 Azithromycin: 116, 119
B
 Bacopa: 106, 148, 149, 157, 158, 161
 Bacterial Infection Protocol: 113, 120, 225
 Bacterial vaginosis: 51, 182, 198, 202
 Baloxavir: 125
 Bearberry: 202
 Bedstraw: 31, 51, 79, 255
 Bentonite clay: 47, 56, 59
 Berberine: 16, 30, 50, 53, 75, 84, 86, 100, 102, 116, 141, 142, 187, 200, 201, 202, 208, 253, 254, 262, 267
 Beta-sitosterol: 195, 247
 Bilberry: 81, 100, 102, 254, 256, 258, 283, 285
 Biologics: 19, 33, 40, 68
 Biotin: 32, 154, 275, 280
 Bismuth: 87, 88
 Bisphosphonates: 238
 Bitter melon: 86, 100, 102, 146, 254
 Bitter orange: 129, 130, 132
 Black cohosh: 180, 185, 199, 200, 246, 249, 276, 280
 Black walnut: 30, 50, 56, 57, 59, 106, 130, 131, 132
 Bladderwrack: 107
 Bone broth: 8, 15, 22, 27, 35, 61, 63, 71, 74, 75, 107, 117, 120, 121, 126, 136, 137, 138, 179, 238, 269, 275, 280, 297, 298, 299, 310, 311
 Boneset: 124, 126
 Boron: 178, 236

Brahmi: 147, 148
 Brain: 7, 16, 18, 23, 24, 25, 26, 91, 92, 109, 120, 135, 136, 146, 147, 148, 149, 151, 155, 157, 158, 159, 160, 161, 163, 165, 166, 170, 171, 173, 175, 179, 181, 196, 196, 199, 231, 263, 264, 270, 286, 290, 303, 310
 Brain and Nerve Protocol: 151, 157, 161
 Broadleaf plantain: 16, 283
 Bromelain: 213, 214, 220, 222, 226, 227
 Bugleweed: 107, 108, 112, 276, 280
 Bupropion-naltrexone: 295
 Burdock: 8, 31, 34, 36, 70, 105, 269
 Butcher's broom: 176, 177, 283, 285
C
 Calcium: 24, 27, 43, 163, 176, 177, 178, 179, 183, 184, 185, 234, 235, 237, 238, 239, 257, 297
 Camellia sinensis: 142, 171
 Candida: 9, 10, 16, 22, 30, 37, 46, 47, 48, 49, 50, 51, 52, 53, 62, 72, 75, 79, 80, 87, 91, 106, 127, 128, 129, 130, 131, 132, 213, 224
 Candida Protocol: 46, 47, 53, 72, 80, 91, 127, 224
 Cannabidiol: 150, 240, 289
 Caprylic acid: 50, 53, 129, 130, 132
 Capsaicin: 39, 183, 229, 241, 245
 Capsella bursa-pastoris: 201
 Capsicum annum: 241
 Carbidopa: 165, 171, 172
 Cat's claw: 38, 41, 79, 135, 138, 256, 270
 Cayenne: 39, 41, 176, 183, 229, 241, 293, 307, 309
 CBD: 150, 156, 240, 245, 289, 290
 Centella asiatica: 148, 158, 283
 Cephalexin: 119
 Ceylon cinnamon: 142, 144
 Chamaelirium luteum: 200
 Chamomile: 11, 13, 18, 32, 50, 52, 62, 72, 73, 75, 86, 95, 110, 150, 156, 166, 168, 181, 255, 287, 288, 290, 291
 Chanca piedra: 256
 Chasteberry: 180, 276
 Chickweed: 17, 32, 49, 79, 293
 Chinese skullcap: 8, 10, 23, 24, 27, 39, 41, 108, 112, 122, 126, 135, 138, 147, 148, 149, 194, 197, 203, 214
 Chlorophyll: 192, 261, 295, 296
 Cholecalciferol: 136, 154, 160, 214, 242
 Chromium: 99, 102, 294, 295
 Cilantro: 106, 163, 252, 302, 311
 Cinnamomum zeylanicum: 142
 Cinnamon: 100, 102, 141, 142,

144, 163, 168, 247, 254, 293, 294, 297, 301, 302, 304, 306, 307, 309, 311
 Ciprofloxacin: 68, 120
 Citicoline: 147, 149
 Clavulanic acid: 119
 Cleavers: 255, 293
 Clindamycin: 119, 249
 Clotrimazole: 129, 131
 Clove: 49, 56, 57, 59, 134, 141, 142, 144, 149, 254
 Coconut oil: 8, 15, 32, 47, 48, 49, 51, 53, 129, 130, 132, 182, 187, 202, 203, 205, 274, 276, 300, 301, 302, 306, 307, 311
 Coenzyme Q10/CoQ10: 11, 13, 17, 19, 24, 27, 43, 45, 146, 271, 273
 Collagen powder: 22, 74
 Common myrtle: 283, 285
 Contrave: 295
 COPD: 214, 220, 223, 224, 226, 227
 COPD Protocol: 223, 227
 Copper: 108, 124, 166
 Cordyceps spp.: 9, 23, 37, 42, 79, 81, 92, 100, 107, 172
 Coriolus versicolor: 9, 10, 16, 23, 30, 37, 49, 51, 62, 67, 72, 79, 81, 87, 101, 106, 123, 201, 277
 Corn silk: 116, 256, 258
 Corticosteroids: 13, 18, 33, 38, 40, 68, 94, 215, 232, 242
 Cottonwood: 17, 19, 39, 41, 241, 264
 Crataegus monogyna: 42, 81, 252
 Crohn's Disease: 7, 26, 64, 65, 66, 67
 Cucurbita pepo: 142
 Cumin: 9, 11, 16, 24, 31, 38, 50, 56, 57, 115, 131, 171, 176, 177, 200, 208, 240, 245, 263, 294, 301, 302, 307, 309, 311
 Curcuma longa: 109, 115, 136, 148, 159, 171, 240
 Curcumin: 9, 11, 16, 24, 31, 38, 50, 115, 131, 171, 176, 177, 200, 208, 240, 245, 263
 Cymbopogon citratus: 49
D
 Damiana: 182, 206, 210
 Dandelion: 8, 15, 16, 31, 34, 36, 48, 51, 53, 66, 70, 71, 75, 77, 79, 82, 86, 105, 179, 237, 246, 252, 256, 258, 269
 DASH diet: 43, 45
 Deep brain stimulation: 172, 173
 Deglycyrrhized licorice: 62, 66, 69, 86
 Dehydroepiandrosterone: 189, 266
 Dementia: 9, 10, 23, 24, 145, 146, 147, 148, 149, 157, 158, 160, 179, 181, 261, 265, 297, 308
 Dementia Protocol: 145, 149, 157
 Depression: 8, 11, 16, 17, 18,

21, 22, 23, 24, 25, 31, 37, 147, 151, 154, 155, 159, 162, 163, 164, 165, 166, 167, 168, 171, 179, 180, 185, 194, 195, 203, 206, 241, 248, 257, 270, 288, 294
 Depression Protocol: 18, 162, 168, 185, 248, 294
 DGL: 62, 63
 DHEA: 189, 200
 Diabetes: 10, 43, 81, 96, 97, 98, 99, 100, 101, 102, 146, 187, 252, 254, 262, 263, 293, 297, 304, 305, 308
 Diabetes Type 2 Protocol: 81, 96, 102, 146, 252
 Dialysis: 257
 Digestive enzymes: 61, 62, 63, 73, 75, 87, 89, 109
 Diindolylmethane: 246
 Diosmin: 282, 285
 Dipsacus fullonum: 134, 135
 D-mannose: 116, 202
 DMARDs: 40
 Donepezil: 148
 Dopamine: 167, 172
 Doxycycline: 93, 119, 133, 134, 143, 232, 249

E
 Eczema: 17, 28, 29, 30, 31, 32, 33, 34, 215, 300
 Eczema Protocol: 28, 34
 Elderberry: 9, 10, 16, 23, 50, 51, 122, 123, 126, 187, 225, 230, 233
 Elecampane: 116, 123, 126, 224
 Elimination diet: 12, 15, 36, 153, 212, 223, 244, 248, 297, 298, 300, 310
 Endometriosis: 198, 199, 200, 201, 204, 207, 208, 277
 Energy Protocol: 268, 273
 English ivy: 220, 225, 278
 Epimedium: 179, 237, 239
 Epinephrine: 167, 215
 Equisetum arvense: 279
 Eriodictyon californicum: 123, 214, 221
 Eszopiclone: 290
 Eucalyptus pauciflora: 129, 130
 Eupatorium perfoliatum: 124
 Evening primrose: 39, 107, 180, 200, 208, 277

F
 Fenugreek: 42, 81, 100, 102, 146, 183, 254, 258, 293
 Fermented foods: 109, 137, 292, 297, 310
 Fertility: 182, 187, 188, 189, 198, 203, 204, 205, 207, 209, 270
 Feverfew: 240, 245
 Fiber: 242, 254, 255, 261, 269, 282, 295, 305
 Fish oil: 11, 21, 32, 43, 45, 93, 95, 99, 136, 154, 156, 159, 243, 247, 276, 280
 Flu: 9, 23, 92, 121, 123, 124,

126, 241, 225
 Flu, Viral, and Immune System Protocol: 121, 126, 225
 Folate: 61, 62, 147, 149, 154, 158, 163, 166, 179, 237, 275, 280, 289
 Folic acid: 24, 27, 62, 269
 Food poisoning: 54, 55, 56, 57, 58, 59
 Food Poisoning Protocol: 54, 59
 Frankincense: 38, 39, 41, 52, 181, 226, 240, 241, 279, 290
 Fungal Infection Protocol: 127, 132

G
 GABA: 151, 152, 153, 154, 155, 156, 165, 168, 179, 181, 288, 289, 290, 291
 Galantamine: 148
 Gamma-aminobutyric acid: 153, 154, 165, 181, 288, 289
 Gamma-linolenic acid: 39, 41, 180
 Ganoderma lucidum: 9, 10, 16, 23, 30, 37, 49, 51, 62, 67, 72, 78, 79, 81, 87, 91, 101, 106, 107, 115, 123, 136, 148, 151, 157, 172, 180, 188, 194, 196, 199, 200, 201, 214, 221, 224, 241, 246, 253, 263, 270, 277, 288
 Gastritis: 10, 70, 71, 72, 73, 74, 75, 297
 GERD: 60, 61, 62, 63
 German Chamomile: 150
 Ginger: 31, 34, 38, 41, 50, 56, 57, 59, 61, 62, 72, 75, 86, 89, 107, 111, 117, 141, 142, 144, 147, 148, 149, 176, 177, 183, 213, 216, 226, 230, 242, 245, 254, 293, 297, 301, 302, 306, 307, 311
 Gingko Biloba: 42, 158
 GLA: 39
 Glucosamine: 32, 36, 37, 297
 Glutathione: 12, 13, 77, 136, 148, 257, 259
 Glycine: 136, 154, 155, 156, 275, 289, 297
 Glycyrrhiza glabra: 38, 86, 92, 100, 107, 123, 203, 237
 Goldenseal: 30, 50, 72, 84, 86, 89, 106, 115, 116, 120, 122, 123, 126, 141, 144, 187, 201, 202, 214, 230, 254
 Goosegrass: 255
 Gotu kola: 147, 148, 149, 157, 158, 161, 176, 177, 283, 285
 Graves' disease: 103, 104, 107, 108, 110, 111, 112, 276
 Graves' Disease Protocol: 103, 112
 Green tea: 15, 44, 77, 84, 91, 141, 142, 144, 154, 159, 171, 173, 193, 197, 199, 255, 271, 273, 275, 297, 299, 311
 Grifola frondosa: 42
 Gromwell: 108
 Guggul: 107, 111, 254
 Gymnema sylvestre: 100, 146

H
 H2 Blockers: 62, 63
 Hair loss: 17, 107, 262, 274, 275, 276, 277, 278, 279, 280
 Hair Loss Protocol: 17, 274, 280
 Hair transplant surgery: 279
 Hamamelis virginiana: 50, 117, 247, 283
 Hashimoto's thyroiditis: 103, 108, 110, 111, 310
 Hashimoto's Thyroiditis Protocol: 103, 111
 Hawthorn: 10, 42, 45, 81, 146, 252, 253, 254, 258
 Heart Health Protocol: 42, 45, 81, 146, 251, 252
 Hedera helix: 220, 225
 Helianthus tuberosus: 70
 Helichrysum: 39, 41, 241
 Hemorrhoidectomy: 282, 284
 Hemorrhoids: 50, 281, 282, 283, 284, 285
 Hemorrhoids Protocol: 281, 285
 Hericium erinaceus: 9, 10, 16, 23, 30, 37, 49, 62, 67, 71, 79, 86, 92, 106, 116, 123, 147, 151, 157, 171, 179, 196, 253, 264, 270
 Hesperidin: 282, 285
 Hibiscus: 42, 45
 Holy Basil: 42, 52, 81, 101, 150, 151, 156, 246, 253, 254
 Honey: 48, 84, 92, 117, 129, 132, 141, 189, 213, 216, 224, 225, 231, 247, 249, 267, 299
 Horse chestnut: 278, 283, 285
 Horsetail: 278, 279, 280
 HPA axis: 16, 90, 95, 98, 100, 135, 257, 263, 277
 Humulus lupulus: 116, 151, 179, 288
 Hyaluronic acid: 282, 285
 Hydrastis canadensis: 72, 86, 115, 123, 141, 214
 Hydrocortisone: 94, 283, 284
 Hypericum perforatum: 31, 39, 152, 165, 172, 180, 203, 241

I
 IBD: 64, 65, 66, 67, 68, 69, 86, 310
 Iberogast: 62, 63
 IBS: 10, 66, 70, 71, 73, 74, 75, 297
 IBS Protocol: 70, 75, 297
 Immune System Protocol: 121, 126, 225
 Immunosuppressants: 19
 Indian gooseberry: 147, 148, 149, 293
 Inflammatory bowel disease: 64, 65, 66, 67, 68, 69, 70, 73, 86, 109, 235, 298
 Inflammatory Bowel Disease Protocol: 64, 69
 Intermittent fasting: 7, 21, 35, 70, 121, 126, 261, 263, 293, 294, 303, 304, 310
 Inula helenium: 116, 123, 224
 Iron: 12, 52, 78, 109, 111, 163,

179, 200, 237, 238, 256, 262, 269, 271
 Ivermectin: 142, 143

J
 Japanese honeysuckle: 10, 23, 24, 50, 53, 122, 123, 126, 225
 Jerusalem artichokes: 8, 15, 66, 70, 105, 269
 Juglans nigra: 50, 57, 131
 Jujube: 220, 222, 287, 288, 289, 291

K
 Kava kava: 151
 Ketogenic Diet: 7, 21, 292, 305, 310
 Kidney Protocol: 250, 258
 Kinnikinnick: 116, 202

L
 L-Arginine: 44, 45, 189, 207, 210
 Lavandula angustifolia: 17, 32, 117, 129, 179, 196, 241
 Lavender: 11, 13, 17, 18, 19, 32, 47, 49, 51, 52, 73, 85, 95, 110, 117, 129, 132, 134, 150, 151, 154, 179, 181, 196, 197, 202, 205, 206, 208, 241, 245, 255, 278, 279, 290
 LDN: 19, 26
 Leaky gut: 8, 9, 10, 14, 15, 16, 22, 23, 28, 29, 30, 34, 36, 37, 41, 49, 50, 62, 70, 72, 73, 74, 75, 79, 82, 86, 87, 97, 105, 106, 107, 109, 148, 152, 157, 158, 165, 212, 213, 216, 219, 224, 242, 269, 275, 297
 Leaky Gut Protocol: 8, 15, 36, 79, 97, 105, 152, 158, 165, 212, 213, 219, 224, 242, 269, 275
 Lemon balm: 11, 52, 73, 85, 86, 89, 100, 102, 108, 112, 122, 123, 126, 147, 149, 150, 151, 156, 157, 161, 165, 168, 171, 173, 179, 181, 185, 187, 196, 197, 203, 206, 209, 210, 253, 254, 258, 276, 277, 280, 287, 288, 291
 Lemongrass: 49
 Leonurus cardiaca: 108, 179, 204, 253
 Levofloxacin: 116, 119
 L-Glutamine: 12, 13, 22, 74, 75, 87, 89, 109, 111, 295, 296
 Libido: 150, 178, 182, 185, 186, 188, 189, 190, 198, 204, 206, 207, 210, 270
 Licorice root: 38, 41, 66, 86, 92, 100, 107, 123, 203, 237
 Lion's mane: 8, 9, 10, 11, 13, 16, 19, 22, 23, 27, 30, 34, 37, 41, 49, 50, 52, 53, 62, 63, 67, 69, 71, 72, 73, 75, 79, 84, 85, 86, 87, 91, 92, 95, 106, 108, 110, 111, 112, 116, 120, 122, 123, 126, 147, 148, 149, 150, 151, 156, 157, 161, 163, 165, 168, 171, 173, 175, 176, 177, 179, 180, 183, 185, 196, 197, 213, 253, 257, 258, 264, 267, 270, 273, 293, 296

Lipo-flavonoids: 176, 177
 Liraglutide: 295
 Lithium orotate: 147, 149
 Lithospermum officianalis: 108
 Lithospermum ruderales: 108
 Lobaria pulmonaria: 225, 256
 Lobelia inflata: 225
 Lonicera japonica: 50, 225
 L-Theanine: 44, 45, 154, 156, 159, 161, 289, 291
 Lungs: 79, 119, 120, 124, 170, 214, 217, 220, 221, 222, 223, 224, 225, 226, 227, 269
 Lungs and COPD Protocol: 223, 227
 Lungwort: 116, 124, 225, 227, 230, 256
 Lupus: 14, 15, 16, 17, 18, 19, 122
 Lupus Protocol: 14, 19
 Lycopene: 192, 193, 197, 264
 Lycopodium europeus: 107
 Lycopodium virginicus: 107
 Lyme Disease Protocol: 133, 138, 140

M

Maca: 182, 187, 204, 209
 Magnesium: 43, 44, 45, 50, 52, 56, 63, 93, 95, 98, 99, 102, 107, 108, 110, 111, 112, 124, 130, 154, 155, 156, 159, 161, 163, 166, 168, 176, 177, 178, 179, 181, 185, 219, 222, 234, 236, 237, 239, 242, 243, 245, 269, 271, 273, 287, 289, 291, 295, 296
 Magnesium glycinate: 44, 156, 159, 166, 181, 185, 219, 287, 289, 291
 Maitake: 42, 45, 108
 Manganese: 109, 160, 179, 252
 MAO-B: 172
 MAOIs: 167
 Marshmallow: 10, 16, 19, 22, 30, 37, 50, 62, 63, 66, 69, 71, 72, 75, 79, 86, 87, 89, 106, 116, 163, 213
 Matricaria chamomilla: 50, 72, 166, 181, 288
 Matricaria recutita: 150
 Mebendazole: 142, 143
 Mediterranean Diet: 21, 35, 43, 97, 104, 237, 239, 297, 308
 Melaleuca alternifolia: 49, 125, 129, 247
 Melatonin: 25, 27, 61, 63, 181, 184, 286, 289, 290, 291
 Melissa officinalis: 86, 100, 108, 123, 147, 151, 157, 165, 171, 179, 181, 187, 196, 203, 253, 276, 288
 Memantine: 148, 149
 Menopause: 46, 178, 179, 180, 181, 182, 183, 184, 185, 206, 207, 234, 266, 268, 272, 274, 276, 288
 Menopause Protocol: 178, 206, 272, 276

Men's Sexual Health: 186, 187, 189, 206, 272

Men's Sexual Health Protocol: 186
 Mentha crisper: 141
 Mentha piperita: 50, 56, 60, 125, 241
 Metformin: 97, 100, 101, 262, 266
 Methyl sulfonylmethane: 39
 Metronidazole: 68, 120, 142, 143
 Miconazole: 131
 Milk thistle: 48, 51, 53, 56, 59, 75, 79, 82, 86, 101, 201, 256
 Mistletoe: 43, 45
 Molybdenum: 48, 53
 Moringa: 270
 Motherwort: 11, 73, 85, 108, 110, 112, 179, 180, 204, 205, 206, 209, 210, 253
 MSM: 32, 39, 41, 282, 285
 Mucuna pruriens: 171
 Mullein: 93, 116, 124, 126, 214, 216, 220, 221, 222, 224, 225, 227, 230, 231, 233
 Multiple Sclerosis Protocol: 20, 27
 Multivitamin: 79, 82, 109, 112, 275, 276, 280
 Myoinositol: 289, 291
 Myrrh: 122, 123, 126, 203
 Myrtus communis: 283

N

NAC: 25, 27, 124, 126, 136, 138, 226, 227, 257, 258
 N-Acetylcysteine: 25, 136, 138, 226, 257
 NAD+: 262, 263, 266, 267
 Nerve: 7, 9, 17, 20, 21, 23, 24, 26, 31, 38, 39, 86, 108, 133, 135, 147, 148, 151, 152, 153, 157, 159, 160, 161, 170, 171, 174, 175, 176, 179, 180, 181, 192, 203, 215, 240, 241, 242, 243, 253, 260, 264, 270, 277, 284, 286
 Nicotinamide: 17, 50, 53, 124, 126, 214, 262, 263
 Nicotinamide mononucleotide: 262, 263
 Nicotinamide riboside: 17, 124, 126, 263
 NMN: 262, 263, 267
 NSAIDs: 18, 40, 78, 83, 85, 87, 207, 243, 251, 255, 272

O

Ocimum sanctum: 42, 151
 Ocimum tenuiflorum: 42
 Oenothera biennis: 107, 180, 200, 277
 Olea europaea: 43
 Olive Leaf: 43, 45, 115, 120
 Omega-3 fatty acids: 24, 32, 34, 35, 39, 66, 71, 96, 97, 98, 99, 102, 104, 136, 147, 158, 159, 163, 170, 171, 178, 193, 196, 219, 226, 236, 243, 261, 274, 297, 310
 Opioid Agents: 243
 Oregano: 10, 49, 56, 57, 83, 92, 115, 120, 121, 134, 202, 226, 230, 231, 233, 294, 297, 302,

306, 307, 309, 311
 Origanum majorana: 200
 Origanum vulgare: 57, 115
 Orlistat: 295
 Oseltamivir: 125
 Osteoporosis: 35, 178, 179, 181, 184, 185, 188, 222, 234, 235, 236, 237, 238, 239, 261, 262, 265
 Osteoporosis Protocol: 178, 179, 234, 239

P

Pain management: 240, 241, 242, 243, 245
 Pain Management Protocol: 240, 245
 Panax ginseng: 16, 101, 107, 157, 158, 171, 182, 187, 188, 189, 190, 210, 224, 264, 267, 270, 293, 296
 Papaya seeds: 142, 144
 Parasite Protocol: 139, 144
 Parkinson's: 11, 148, 157, 160, 169, 170, 171, 172, 173, 303, 305
 Parkinson's Disease Protocol: 169, 173
 Passion flower: 151, 181
 Pau d'arco: 49, 130, 132
 Pennyroyal: 56, 57, 79
 Peppermint: 50, 56, 57, 59, 60, 61, 92, 118, 125, 176, 177, 180, 203, 226, 230, 241, 245, 271, 273
 Peramivir: 125
 Phellinus linteus: 38
 Phosphatidylserine: 159, 161
 Phyllanthus emblica: 148
 Pineapple: 213, 216, 220, 226, 251, 304
 Pine pollen: 188, 190
 Piper myristicum: 151
 Pipsissewa: 256
 Plantago lanceolata: 283
 Plantago major: 10, 16, 30, 50, 62, 67, 79
 Plantago ovata: 281, 282
 Plantain: 10, 16, 19, 30, 50, 62, 63, 67, 69, 79, 86, 87, 283, 285
 Platelet-rich plasma therapy: 279
 Polycystic ovary syndrome: 198
 Poppy: 52, 110, 181, 287, 288, 291
 Populus trichocarpa: 17, 39
 PPIs: 63, 88
 Prebiotic: 8, 9, 15, 23, 36, 37, 62, 67, 70, 71, 84, 105, 141, 212, 216, 232, 269, 275, 295, 297
 Probiotic: 7, 8, 15, 22, 23, 30, 34, 36, 37, 48, 56, 62, 70, 71, 72, 73, 74, 75, 84, 86, 87, 88, 92, 94, 95, 105, 109, 112, 121, 126, 136, 137, 138, 152, 156, 158, 161, 197, 212, 216, 232, 242, 247, 269, 275, 292, 297
 Prostate: 186, 188, 191, 192, 193, 194, 195, 196, 197, 238, 250, 261, 276, 279, 308
 Protectants: 88

Proton pump inhibitors: 62, 63, 88
 PRP: 279
 Psoriasis: 7, 9, 28, 29, 30, 31, 32, 33, 34, 90, 240, 260, 264
 Psyllium: 22, 97, 99, 102, 254, 281, 282, 285, 295, 296
 Pumpkin seeds: 99, 106, 142, 247, 248, 275, 292, 301, 305, 307, 311
 Pygeum: 195, 276, 280

Q

Qsymia: 295
 Quercetin: 9, 39, 41, 44, 45, 83, 87, 89, 117, 215, 216, 225, 226, 227, 230, 264, 297
 Quercus alba: 283

R

Ramelteon: 290, 291
 Red Sage: 225
 Reishi: 8, 9, 10, 11, 13, 16, 19, 22, 23, 27, 30, 31, 34, 37, 41, 49, 50, 51, 52, 53, 62, 63, 67, 69, 71, 72, 73, 75, 78, 79, 81, 82, 85, 87, 91, 95, 101, 102, 106, 107, 108, 110, 111, 112, 115, 120, 122, 123, 126, 136, 138, 147, 148, 149, 150, 151, 156, 157, 161, 163, 165, 168, 171, 172, 173, 175, 176, 177, 180, 181, 183, 185, 188, 189, 190, 194, 196, 197, 199, 200, 201, 204, 205, 206, 208, 209, 210, 213, 214, 216, 221, 222, 224, 225, 227, 231, 233, 241, 245, 246, 247, 249, 253, 254, 257, 258, 263, 267, 270, 273, 277, 280, 287, 288, 291, 293, 296
 Resveratrol: 10, 12, 13, 24, 77, 135, 252, 257, 258, 263, 267
 Retinol: 142, 261, 262, 267, 310
 Rheumatoid arthritis (RA): 7, 11, 31, 35, 36, 37, 38, 39, 40, 41, 122, 240, 241, 297, 300, 310
 Rheumatoid Arthritis Protocol: 35, 41
 Rivastigmine: 148
 Rosehips: 51, 93, 99, 264, 265
 Rubber band ligation: 284
 Ruscus aculeatus: 176, 283
 Rutin: 282, 285

S

Sarsaparilla: 9, 30, 34
 Saw palmetto: 188, 190, 194, 197, 276, 280
 Saxenda: 295
 Sclerotherapy: 284
 Scutellaria lateriflora: 152, 288
 Seaweed: 8, 70, 105, 107, 252, 269, 310
 Selenium: 93, 95, 99, 104, 107, 108, 111, 112, 142, 144, 196, 197, 220, 222, 276
 Serenoa repens: 188, 194, 276
 Serrapeptase: 232, 233
 Sexual health: 179, 180, 182, 186, 187, 189, 190, 198, 206, 207, 270, 272

Shatavari: 182
 Silicon: 178, 236
 Silybum marianum: 48, 56, 79, 101, 201
 Sinusitis: 228, 229, 230, 231, 232, 233
 Sinusitis Protocol: 228, 233
 Sleep: 11, 13, 18, 19, 25, 27, 33, 34, 40, 41, 52, 53, 81, 82, 85, 95, 110, 111, 112, 118, 120, 122, 126, 137, 138, 146, 154, 156, 159, 161, 166, 168, 179, 181, 185, 196, 197, 208, 213, 216, 242, 255, 258, 265, 267, 271, 273, 277, 280, 286, 287, 288, 289, 290, 291, 294, 296, 298
 Sleep Protocol: 13, 18, 33, 40, 52, 81, 85, 95, 110, 118, 122, 126, 154, 159, 166, 179, 196, 213, 242, 255, 265, 271, 286, 291, 296, 298
 Slippery elm: 10, 16, 19, 22, 30, 37, 50, 62, 63, 66, 69, 71, 72, 75, 79, 86, 87, 89, 106, 163, 213
 Smilax glabra: 9, 30
 SNRIs: 155, 167
 Soluble fiber: 163, 183, 254, 255, 261, 293
 Spermidine: 263, 267
 SSRIs: 155, 167, 185
 Stellaria media: 17, 32, 49, 79
 Steroids: 94, 215, 227, 232, 242
 Stinging nettle: 8, 15, 31, 34, 38, 41, 71, 107, 108, 111, 112, 116, 121, 179, 185, 195, 197, 214, 316, 221, 222, 224, 227, 231, 233, 238, 256, 258, 269, 276, 278, 280, 297
 St. john's wort: 31, 34, 39, 41, 152, 165, 171, 172, 173, 180, 203, 241, 245
 Stoneseed: 108, 112
 Stress Protocol: 10, 18, 33, 40, 43, 52, 73, 85, 91, 98, 150, 156, 166, 179, 196, 213, 221, 224, 244, 253, 265, 271, 277, 290, 294, 298
 Strontium: 178, 236
 STW-5: 62
 Sulfamethoxazole: 119
 Surgery: 177, 189, 207, 227, 232, 234, 279, 281, 282, 295
 Suvorexant: 148, 149
 Swedish bitters: 48
 Sweet marjoram: 200
 Sweet wormwood: 135, 138, 141, 225
 Syzygium aromaticum: 141
T
 Tanacetum parthenium: 240
 Taraxacum officinale: 31, 48, 51, 79, 179, 237, 256
 Taurine: 155, 156
 TCAs: 167
 Tea tree: 32, 49, 51, 125, 129, 132, 202, 203, 226, 230, 247, 249, 274, 282, 285
 Terbinafine: 131

Tetracyclic antidepressants: 167
 Tetrahydrocannabinol: 289
 THC: 289, 290
 Thunder god vine: 38, 41
 Thyme: 10, 49, 56, 57, 59, 92, 115, 120, 121, 226, 230, 297, 301, 306, 307, 309, 311
 Thymus vulgaris: 57, 115
 Thyroid: 12, 52, 90, 91, 93, 94, 95, 103, 104, 105, 106, 107, 108, 109, 110, 111, 135, 147, 150, 151, 157, 159, 165, 178, 179, 184, 189, 198, 204, 234, 260, 268, 270, 271, 274, 276, 280, 292
 Thyroid Protocol: 93, 94, 103, 271, 276, 280
 Tinnitus: 158, 174, 175, 176, 177
 Tinnitus Protocol: 174, 177
 Trametes versicolor: 9, 10, 16, 23, 30, 37, 49, 51, 62, 67, 72, 79, 81, 87, 101, 106, 123, 194, 201, 277
 Tribulus: 187, 190, 209
 Trigonella foenum-graecum: 42
 Trimethoprim: 119
 Tripterygium wilfordii: 38
 Tulsi: 10, 42, 81, 101, 102, 151, 156, 246, 253, 254, 258
 Turkey tail: 8, 9, 10, 13, 16, 19, 22, 23, 27, 30, 34, 37, 41, 49, 50, 53, 62, 63, 67, 69, 71, 72, 75, 79, 81, 87, 101, 102, 106, 108, 111, 112, 123, 126, 163, 180, 194, 197, 201, 204, 205, 206, 210, 213, 254, 270, 277, 280
 Turmeric: 9, 11, 13, 16, 19, 24, 27, 31, 34, 38, 41, 50, 53, 74, 75, 109, 111, 112, 115, 120, 130, 131, 132, 136, 138, 147, 148, 149, 159, 161, 171, 173, 176, 200, 208, 231, 240, 254, 255, 263, 267, 297, 299, 311
U
 UC: 64, 65, 66, 67, 68, 234, 297
 Ulcer: 10, 16, 17, 22, 30, 37, 40, 64, 66, 70, 72, 83, 84, 85, 86, 87, 88, 89, 106, 113, 116, 117, 213, 215, 224, 238, 243, 283
 Ulcer Protocol: 16, 83, 224
 Ulmus rubra: 10, 16, 30, 37, 50, 62, 66, 71, 79, 86, 87, 106
 Uncaria tomentosa: 38, 79, 135, 256, 270
 Undecylenic acid: 129, 130, 132
 Ursolic acid: 9, 10, 23, 50, 51
 Urtica dioica: 31, 38, 107, 108, 179, 195, 214, 221, 224, 238, 256, 276
 Usnea: 10, 23, 49, 93, 115, 116, 117, 120, 122, 123, 126, 129, 130, 131, 132, 187, 225, 231, 256
 Usnea barbata: 115, 116, 123, 129, 130, 131
 Uterine fibroids: 179, 180, 198, 201, 203, 277

Uva ursi: 116, 202
V
 Vaccinium myrtillus: 81, 100, 256, 283
 Velvet bean: 171, 173
 Verbascum thapsus: 93, 124, 214, 224
 Vertigo: 20, 133, 174, 175, 176, 177, 211
 Vertigo and Tinnitus Protocol: 174, 177
 Viral Protocol: 139
 Viscum alba: 43
 Vitamin A: 33, 62, 124, 141, 142, 144, 163, 192, 219, 248, 252, 261, 262, 264, 266
 Vitamin B-3: 50
 Vitamin B-12: 11, 17, 24, 32, 243, 295
 Vitamin B complex: 93, 95, 99, 154, 156, 160, 161, 166, 184, 185, 206, 210, 237, 239, 295
 Vitamin C: 24, 27, 44, 45, 48, 51, 53, 62, 80, 93, 99, 117, 118, 120, 124, 136, 137, 146, 154, 163, 170, 193, 215, 220, 225, 231, 232, 236, 237, 239, 243, 251, 252, 256, 259, 262, 266, 267, 268, 282, 283, 285, 289, 293, 295
 Vitamin D: 11, 16, 24, 32, 34, 39, 42, 66, 69, 73, 80, 85, 93, 95, 99, 104, 108, 118, 124, 136, 147, 154, 160, 166, 170, 184, 188, 196, 206, 210, 213, 214, 215, 219, 225, 232, 235, 236, 238, 239, 242, 243, 248, 250, 257, 262, 266, 295
 Vitamin D-3: 154, 160
 Vitamin E: 39, 62, 80, 118, 170, 184, 206, 210, 215, 236, 241, 262, 265
 Vitamin K2: 11, 16, 24, 184, 236
 Vitex: 180, 199, 201, 204, 208, 209, 246, 249, 276, 277, 280
 Vitex agnus-castus: 199, 246
W
 Wahls Protocol Diet: 7, 21, 29, 310
 Weight Loss: 8, 33, 80, 96, 97, 98, 100, 182, 183, 186, 194, 198, 199, 269, 292, 293, 294, 295, 296, 305
 Weight Loss Protocol: 33, 183, 194, 269
 White oak: 283, 285
 White willow: 241, 245, 255
 Witch hazel: 50, 117, 247, 283, 285
 Withania somnifera: 16, 31, 67, 91, 100, 106, 107, 135, 150, 157, 179, 188, 195, 263, 270
 Women's Sexual Health Protocol: 179, 180, 182, 198, 280
 Wormwood: 30, 79, 106, 135, 138, 141, 144, 149, 225
X
 Xenical: 295

Y
 Yarrow: 32, 86, 115, 116, 117, 120, 124, 126, 141, 142, 144, 199, 201, 205, 208, 214, 225, 230, 231, 233, 247, 249, 283, 285
 Yerba santa: 123, 126, 214, 221, 222, 231, 233
Z
 Zaleplon: 290
 Zanamivir: 125
 Zinc: 24, 87, 89, 99, 102, 107, 108, 111, 112, 124, 126, 130, 136, 138, 142, 144, 166, 168, 187, 188, 190, 193, 196, 197, 213, 215, 216, 220, 222, 247, 248, 249, 252, 262, 267, 274, 275
 Zingiber officinale: 31, 38, 50, 57, 62, 107, 117, 142, 148, 242
 Ziziphus jujuba: 220
 Ziziphus jujuba var. spinosa: 288
 Zolpidem: 288, 290, 291

References

Autoimmune Protocol

- 1 <https://pathology.jhu.edu/autoimmune/> Johns Hopkins Institute of Pathology
- 2 <https://journal.restorativemedicine.org/index.php/journal/article/view/9/15> Zampieron, Eugene & Kamhi, Ellen. (2012). Natural Support for Autoimmune and Inflammatory Disease. *Journal of Restorative Medicine*. 1. 38-47. 10.14200/jrm.2012.1.1003
- 3 <https://journal.restorativemedicine.org/index.php/journal/article/view/9/15> Bishehsari, F., Magno, E., Swanson, G., Desai, V., Voigt, R. M., Forsyth, C. B., & Keshavarzian, A. (2017). Alcohol and Gut-Derived Inflammation. *Alcohol research: current reviews*, 38(2), 163-171
- 4 <https://journal.restorativemedicine.org/index.php/journal/article/view/9/15>: Zampieron, Eugene & Kamhi, Ellen. (2012). Natural Support for Autoimmune and Inflammatory Disease. *Journal of Restorative Medicine*. 1. 38-47. 10.14200/jrm.2012.1.1003
- 5 Bhardwaj, N., Katyal, P., & Sharma, A. K. (2014). Suppression of inflammatory and allergic responses by pharmacologically potent fungus *Ganoderma lucidum*. *Recent patents on inflammation & allergy drug discovery*, 8(2), 104-117. <https://doi.org/10.2174/1872213x08666140619110657>
- 6 Bhardwaj, Neha et al. "Suppression of inflammatory and allergic responses by pharmacologically potent fungus *Ganoderma lucidum*." *Recent patents on inflammation & allergy drug discovery* vol. 8,2 (2014): 104-17. doi:10.2174/1872213x08666140619110657
- 7 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, *Herichium erinaceus* (Higher Basidiomycetes) from Malaysia. *International journal of medicinal mushrooms*, 15(6), 539-554. <https://doi.org/10.1615/intjmed-mushrv15.i6.30>
- 8 Deshmukh, Sunil & Sridhar, K. & MK, Gupta. (2021). *Herichium erinaceus* -A Rich Source of Diverse Bioactive Metabolites. *Fungal Biotech*. 1. 10-38. 10.5943/FunBio-1/2/2. https://www.researchgate.net/publication/355143838_Herichium_erinaceus_-_A_Rich_Source_of_Diverse_Bioactive_Metabolites
- 9 Mori, K., Obara, Y., Moriya, T., Inatomi, S., & Nakahata, N. (2011). Effects of *Herichium erinaceus* on amyloid β (25-35) peptide-induced learning and memory deficits in mice. *Biomedical research* (Tokyo, Japan), 32(1), 67-72. <https://doi.org/10.2220/biomedres.32.67>
- 10 <https://journal.restorativemedicine.org/index.php/journal/article/view/9/15>
- 11 Xu J, Li X, Zhang P, Li ZL, Wang Y. Antiinflammatory constituents from the roots of *Smilax bockii* warb. *Arch Pharm Res*. 2005; 28(4):395-9/
- 12 Spelman K, Burns J, Nichols D, Winters N, Ottersberg S, Tenborg M. Modulation of cytokine expression by traditional medicines: A review of herbal immunomodulators. *Alternative Medicine Review*. 2006; 11(2):128-50
- 13 Golan R. *Optimal Wellness: Where Mainstream and Alternative Medicine Meet*. 1995. Ballantine. New York
- 14 Sartor RB. Importance of Intestinal Mucosal Immunity and Luminal Bacterial-Cell Wall Polymers in the Etiology of Inflammatory Joint Diseases. *Baillieres Clinical Rheuma*. 1989; 3(2):223-45
- 15 Garaeva ZS, Safina NA, Tyurin YA, Kuklin VT, Zinkevich OD. Intestinal dysbiosis as the cause of systemic endotoxemia in psoriatic patients. *Vestn Dermatol Venerol*. 2007(1):23-7
- 16 <https://journal.restorativemedicine.org/index.php/journal/article/view/9/15> Bishehsari, F., Magno, E., Swanson, G., Desai, V., Voigt, R. M., Forsyth, C. B., & Keshavarzian, A. (2017). Alcohol and Gut-Derived Inflammation. *Alcohol research: current reviews*, 38(2), 163-171
- 17 https://www.ema.europa.eu/en/documents/herbal-report/final-assessment-report-sambucus-nigra-l-fructus_en.pdf
- 18 He, J., & Giusti, M. M. (2010). Anthocyanins: natural colorants with health-promoting properties. *Annual review of food science and technology*, 1, 163-187. <https://doi.org/10.1146/annurev.food.080708.100754>
- 19 Zakay-Rones, Z., Thom, E., Wollan, T., & Wadstein, J. (2004). Randomized study of the efficacy and safety of oral elderberry extract in the treatment of influenza A and B virus infections. *The Journal of international medical research*, 32(2), 132-140. <https://doi.org/10.1177/147323000403200205>
- 20 Tiralongo, E., Wee, S. S., & Lea, R. A. (2016). Elderberry Supplementation Reduces Cold Duration and Symptoms in Air-Travellers: A Randomized, Double-Blind Placebo-Controlled Clinical Trial. *Nutrients*, 8(4), 182. <https://doi.org/10.3390/nu8040182>
- 21 Wieland, L. S., Piechotta, V., Feinberg, T., Ludeman, E., Hutton, B., Kanji, S., Seely, D., & Garritty, C. (2021). Elderberry for prevention and treatment of viral respiratory illnesses: a systematic review. *BMC complementary medicine and therapies*, 21(1), 112. <https://doi.org/10.1186/s12906-021-03283-5>
- 22 Wieland, L. S., Piechotta, V., Feinberg, T., Ludeman, E., Hutton, B., Kanji, S., Seely, D., & Garritty, C. (2021). Elderberry for prevention and treatment of viral respiratory illnesses: a systematic review. *BMC complementary medicine and therapies*, 21(1), 112. <https://doi.org/10.1186/s12906-021-03283-5>
- 23 Kronbichler, A., Effenberger, M., Eisenhut, M., Lee, K. H., & Shin, J. I. (2020). Seven recommendations to rescue the patients and reduce the mortality from

- COVID-19 infection: An immunological point of view. *Autoimmunity reviews*, 19(7), 102570. <https://doi.org/10.1016/j.autrev.2020.102570>
- 24 <https://pubmed.ncbi.nlm.nih.gov/25491337/> Gleńsk, M., Gliński, J. A., Włodarczyk, M., & Stefanowicz, P. (2014). Determination of ursolic and oleanolic acid in *Sambuci fructus*. *Chemistry & biodiversity*, 11(12), 1939-1944. <https://doi.org/10.1002/cbdv.201400118>
- 25 <https://neurosciencenews.com/ursolic-acid-multiple-sclerosis-16089/>
- 26 Zhang, Y., Li, X., Ciric, B., Curtis, M. T., Chen, W. J., Rostami, A., & Zhang, G. X. (2020). A dual effect of ursolic acid to the treatment of multiple sclerosis through both immunomodulation and direct remyelination. *Proceedings of the National Academy of Sciences of the United States of America*, 117(16), 9082-9093. <https://doi.org/10.1073/pnas.2000208117>
- 27 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7260130/>
- 28 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6025220/>
- 29 <https://www.healthcmi.com/Acupuncture-Continuing-Education-News/2039-covid-19-coronavirus-herbs-found-effective>
- 30 <https://pubmed.ncbi.nlm.nih.gov/30296293/>
- 31 https://www.researchgate.net/publication/272020690_Polygonum_cuspidatum_and_Its_Active_Components_Inhibit_Replication_of_the_Influenza_Virus_through_Toll-Like_Receptor_9-Induced_Interferon_Beta_Expression
- 32 <https://pubmed.ncbi.nlm.nih.gov/28283884/>
- 33 Wang, Mingxing et al. "Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion's Mane Medicinal Mushroom, *Herichium erinaceus* (Higher Basidiomycetes)." *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055-60. doi:10.1615/intjmedmushrooms.v17.i11.50. <https://pubmed.ncbi.nlm.nih.gov/26853960/>
- 34 <https://pubmed.ncbi.nlm.nih.gov/>

Lupus Protocol

- 1 <https://www.niddk.nih.gov/health-information/kidney-disease/lupus-nephritis>
- 2 Pons-Estel GJ, Alarcón GS, Scofield L, Reinlib L, Cooper GS. Understanding the epidemiology and progression of systemic lupus erythematosus. *Seminars in Arthritis and Rheumatism*. 2010;39(4):257-268
- 3 Cross-Reaction between Gliadin and Different Food and Tissue Antigens; Vojdani, A. & Tarash, I., *Food and Nutrition Sciences*, 2013, 4, 20-32 <http://dx.doi.org/10.4236/fns.2013.41005> Published Online January 2013 (<http://www.scirp.org/journal/fns>)
- 4 Wahls, T. L., Chenard, C. A., & Snetselaar, L. G. (2019). Review of two popular eating plans within the multiple sclerosis community: low saturated fat and modified paleolithic. *Nutrients*, 11(2), 352
- 5 Mu Q, Kirby J, Reilly CM, Luo XM. Leaky Gut ss a Danger Signal for Autoimmune Diseases. *Front Immunol*. 2017 May 23;8:598. doi: 10.3389/fimmu.2017.00598. PMID: 28588585; PMCID: PMC5440529
- 6 Schneeman B. O. (2002). Gastrointestinal physiology and functions. *The British journal of nutrition*, 88 Suppl 2, S159-S163. <https://doi.org/10.1079/BJN2002681>
- 7 Lin Z. B. (2005). Cellular and molecular mechanisms of immuno-modulation by *Ganoderma lucidum*. *Journal of pharmacological sciences*, 99(2), 144-153. <https://doi.org/10.1254/jphs.crj05008x>
- 8 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, *Herichium erinaceus* (Higher Basidiomycetes) from Malaysia. *International journal of medicinal mushrooms*, 15(6), 539-554. <https://doi.org/10.1615/intjmed-mushrv15.i6.30>
- 9 Pallav, K., Dowd, S. E., Villafuerte, J., Yang, X., Kabbani, T., Hansen, J., Dennis, M., Leffler, D. A., Newburg, D. S., & Kelly, C. P. (2014). Effects of polysaccharopeptide from *Trametes versicolor* and amoxicillin on the gut microbiome of healthy volunteers: a randomized clinical trial. *Gut microbes*, 5(4), 458-467. <https://doi.org/10.4161/gmic.29558>
- 10 Lopresti, A. L., Smith, S. J., Malvi, H., & Kodgule, R. (2019). An investigation into the stress-relieving and pharmacological actions of an ashwagandha (*Withania somnifera*) extract: A randomized, double-blind, placebo-controlled study. *Medicine*, 98(37), e17186. <https://doi.org/10.1097/MD.00000000000017186>
- 11 Arring, N. M., Millstine, D., Marks, L. A., & Nail, L. M. (2018). Ginseng as a Treatment for Fatigue: A Systematic Review. *Journal of alternative and complementary medicine* (New York, N.Y.), 24(7), 624-633. <https://doi.org/10.1089/acm.2017.0361>
- 12 Alexandra Detters, Janina Zippel, Nils Hellenbrand, Dirk Pappai, Cathleen Possemeyer, Andreas Hensel, Aqueous extracts and polysaccharides from Marshmallow roots (*Althea officinalis* L.): Cellular internalisation and stimulation of cell physiology of human epithelial cells in vitro, *Journal of Ethnopharmacology*, Volume 127, Issue 1, 2010, Pages 62-69, ISSN 0378-8741, <https://doi.org/10.1016/j.jep.2009.09.050>.
- 13 Antioxidant effects of herbal therapies used by patients with inflammatory bowel disease: an in vitro study L. Langmead,C. Dawson,C. Hawkins,N. Banna,S. Loo,D. S. RamptonFirst published: 18 January 2002 <https://doi.org/10.1046/j.1365-2036.2002.01157.x>
- 14 Hussan F, Mansor AS, Hassan SN, Tengku Nor Effendy Kamaruddin TN, Budin SB, Othman F. Anti-Inflammatory Property of Plantago major Leaf Extract Reduces the Inflammatory Reaction in Experimental Acetaminophen-Induced Liver Injury. *Evid Based Complement Alternat Med*. 2015;2015:347861. doi: 10.1155/2015/347861. Epub 2015 Aug 2. PMID: 26300946; PMCID: PMC4537734.
- 15 Awaad, A. (2012, February 3). Natural products in treatment of ulcerative colitis and peptic ulcer. *Science Direct*. Retrieved July 25, 2022, from <https://www.science->

- direct.com and Zhang L, Wu X, Yang R, Chen F, Liao Y, Zhu Z, Wu Z, Sun X, Wang L. Effects of Berberine on the Gastrointestinal Microbiota. *Front Cell Infect Microbiol*. 2021 Feb 19;10:588517. doi: 10.3389/fcimb.2020.588517. PMID: 33680978; PMCID: PMC7933196.
- 16 Jurenka J. S. (2009). Anti-inflammatory properties of curcumin, a major constituent of *Curcuma longa*: a review of preclinical and clinical research. *Alternative medicine review: a journal of clinical therapeutics*, 14(2), 141–153
- 17 Murdaca, G., Tonacci, A., Negrini, S., Greco, M., Borro, M., Puppo, F., & Gangemi, S. (2019). Emerging role of vitamin D in autoimmune diseases: An update on evidence and therapeutic implications. *Autoimmunity reviews*, 18(9), 102350. <https://doi.org/10.1016/j.autrev.2019.102350>
- 18 Simopoulos A. P. (2002). Omega-3 fatty acids in inflammation and autoimmune diseases. *Journal of the American College of Nutrition*, 21(6), 495–505. <https://doi.org/10.1080/07315724.2002.10719248>
- 19 Cojocaru M, Cojocaru IM, Silosi I, Vrabie CD. Gastrointestinal manifestations in systemic autoimmune diseases. *Maedica (Bucur)*. 2011 Jan;6(1):45-51. PMID: 21977190; PMCID: PMC3150032
- 20 Djouder N. (2015). Boosting NAD (+) for the prevention and treatment of liver cancer. *Molecular & cellular oncology*, 2(4), e1001199. <https://doi.org/10.1080/23723556.2014.1001199>
- 21 Boosting NAD+ blunts TLR4-induced type I IFN in control and systemic lupus erythematosus monocytes Jing Wu, Mariana J. Kaplan, Michael N. Sack Published January 13, 2022. Citation Information: *J Clin Invest*. 2022;132(5):e139828. <https://doi.org/10.1172/JCI139828>
- 22 López-Pedrerá C, Villalba JM, Patiño-Trives AM, Luque-Tévar M, Barbarroja N, Aguirre MÁ, Escudero-Contreras A, Pérez-Sánchez C. Therapeutic Potential and Immunomodulatory Role of Coenzyme Q10 and Its Analogues in Systemic Autoimmune Diseases. *Antioxidants (Basel)*. 2021 Apr 13;10(4):600. doi: 10.3390/antiox10040600. PMID: 33924642; PMCID: PMC8069673
- 23 López-Pedrerá C, Villalba JM, Patiño-Trives AM, Luque-Tévar M, Barbarroja N, Aguirre MÁ, Escudero-Contreras A, Pérez-Sánchez C. Therapeutic Potential and Immunomodulatory Role of Coenzyme Q10 and its Analogues in Systemic Autoimmune Diseases. *Antioxidants (Basel)*. 2021 Apr 13;10(4):600. doi: 10.3390/antiox10040600. PMID: 33924642; PMCID: PMC8069673
- 24 Eghdampour, F., Jahdie, F., Kheyrikhah, M., Taghizadeh, M., Naghizadeh, S., & Hagani, H. (2013). The Impact of Aloe vera and Calendula on Perineal Healing after Episiotomy in Primiparous Women: A Randomized Clinical Trial. *Journal of caring sciences*, 2(4), 279–286. <https://doi.org/10.5681/jcs.2013.033>. <https://pubmed.ncbi.nlm.nih.gov/25276736/>
- 25 Amiel, E., Ofir, R., Dudai, N., Soloway, E., Rabinsky, T., & Rachmilevitch, S. (2012). β -Caryophyllene, a Compound Isolated from the Biblical Balm of Gilead (*Commiphora gileadensis*), Is a Selective Apoptosis Inducer for Tumor Cell Lines. Evidence-based complementary and alternative medicine: eCAM, 2012, 872394. <https://doi.org/10.1155/2012/872394> <https://pubmed.ncbi.nlm.nih.gov/22567036/>
- 26 Sniezek DP, Siddiqui IJ. Acupuncture for Treating Anxiety and Depression in Women: A Clinical Systematic Review. *Med Acupunct*. 2013 Jun;25(3):164-172. doi: 10.1089/acu.2012.0900. PMID: 24761171; PMCID: PMC3689180
- 27 Roy RA, Boucher JP, Comtois AS. Inflammatory response following a short-term course of chiropractic treatment in subjects with and without chronic low back pain. *J Chiropr Med*. 2010 Sep;9(3):107-14. doi: 10.1016/j.jcm.2010.06.002. PMID: 22027032; PMCID: PMC3188345
- 28 Wu, M. L., Yu, K. H., & Tsai, J. C. (2017). The Effectiveness of Exercise in Adults with Systemic Lupus Erythematosus: A Systematic Review and Meta-Analysis to Guide Evidence-Based Practice. *Worldviews on evidence-based nursing*, 14(4), 306–315. <https://doi.org/10.1111/wvn.12221>
- ## Multiple Sclerosis Protocol
- 1 Bjornevik K, Cortese M, Healy BC, et al. Longitudinal analysis reveals high prevalence of Epstein-Barr virus associated with multiple sclerosis. *Science*. 2022;375(6578):296-301. doi:10.1126/science.abj8222
- 2 Handel AE, Williamson AJ, Disanto G, Handunnetthi L, Giovannoni G, Ramagopalan SV. An updated meta-analysis of risk of multiple sclerosis following infectious mononucleosis. *PLoS One*. 2010;5(9):e12496. Published 2010 Sep 1. doi:10.1371/journal.pone.0012496
- 3 Hounen G, Trier NH, Frederiksen JL. Epstein-Barr Virus and Multiple Sclerosis. *Front Immunol*. 2020;11:587078. Published 2020 Dec 17. doi:10.3389/fimmu.2020.587078
- 4 Gough SM, Casella A, Ortega KJ, Hackam AS. Neuroprotection by the Ketogenic Diet: Evidence and Controversies. *Front Nutr*. 2021;8:782657. Published 2021 Nov 23. doi:10.3389/fnut.2021.782657
- 5 Bock M, Karber M, Kuhn H. Ketogenic diets attenuate cyclooxygenase and lipoxigenase gene expression in multiple sclerosis. *EBioMedicine*. 2018;36:293-303. doi:10.1016/j.ebiom.2018.08.057
- 6 Brenton JN, Banwell B, Bergqvist AGC, et al. Pilot study of a ketogenic diet in relapsing-remitting MS. *Neurol Neuroimmunol Neuroinflamm*. 2019;6(4):e565. Published 2019 Apr 12. doi:10.1212/NXI.0000000000000565
- 7 Fitzgerald KC, Vizthum D, Henry-Barron B, et al. Effect of intermittent vs. daily calorie restriction on changes in weight and patient-reported outcomes in people with multiple sclerosis. *Mult Scler Relat Disord*. 2018;23:33-39. doi:10.1016/j.msard.2018.05.002
- 8 Cignarella F, Cantoni C, Ghezzi L, et al. Intermittent Fasting Confers Protection in CNS Autoimmunity by Altering the Gut Microbiota. *Cell Metab*. 2018;27(6):1222-1235.e6. doi:10.1016/j.cmet.2018.05.006
- 9 Irish AK, Erickson CM, Wahls TL, Snetselaar LG, Darling WG. Randomized control trial evaluation of a modified Paleolithic dietary intervention in the treatment of relapsing-remitting multiple sclerosis: a pilot study. *Degener Neurol Neuromuscul Dis*. 2017;7:1-18. Published 2017 Jan 4. doi:10.2147/DNND.S116949
- 10 Lee JE, Bisht B, Hall MJ, et al. A Multimodal, Nonpharmacologic Intervention Improves Mood and Cognitive Function in People with Multiple Sclerosis. *J Am Coll Nutr*. 2017;36(3):150-168. doi:10.1080/07315724.2016.1255160
- 11 <https://www.health.harvard.edu/staying-healthy/the-truth-about-fats-bad-and-good>
- 12 <https://www.health.harvard.edu/staying-healthy/the-truth-about-fats-bad-and-good> 2. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8303934/> El Soury, M., Fornasari, B. E., Carta, G., Zen, F., Haastert-Talini, K., & Ronchi, G. (2021). The Role of Dietary Nutrients in Peripheral Nerve Regeneration. *International journal of molecular sciences*, 22(14), 7417. <https://doi.org/10.3390/ijms22147417>
- 13 Rodrigo, L., et al. “Randomised clinical trial comparing the efficacy of a gluten-free diet versus a regular diet in a series of relapsing-remitting multiple sclerosis patients.” *Int. J. Neurol. Neurother* 1.1 (2014): 1-6
- 14 Monte SM, Kril JJ. Human alcohol-related neuropathology. *Acta Neuropathol*. 2014;127(1):71-90. doi:10.1007/s00401-013-1233-3
- 15 Pakpoor J, Goldacre R, Disanto G, Giovannoni G, Goldacre MJ. Alcohol Misuse Disorders and Multiple Sclerosis Risk. *JAMA Neurol*. 2014;71(9):1188–1189. doi:10.1001/jamaneurol.2014.1795
- 16 Arneth B. Multiple Sclerosis and Smoking. *Am J Med*. 2020;133(7):783-788. doi:10.1016/j.amjmed.2020.03.008
- 17 Rosso M, Chitnis T. Association Between Cigarette Smoking and Multiple Sclerosis: A Review. *JAMA Neurol*. 2020;77(2):245-253. doi:10.1001/jamaneurol.2019.4271
- 18 Nouri, Mehrnaz et al. “Intestinal barrier dysfunction develops at the onset of experimental autoimmune encephalomyelitis, and can be induced by adoptive transfer of auto-reactive T cells.” *PloS one* vol. 9,9 e106335. 3 Sep. 2014, doi:10.1371/journal.pone.0106335. <https://pubmed.ncbi.nlm.nih.gov/25184418/>
- 19 Takewaki D, Yamamura T. Gut microbiome research in multiple sclerosis. *Neurosci Res*. 2021;168:28-31. doi:10.1016/j.neures.2021.05.001
- 20 Boziki MK, Kesidou E, Theotokis P, et al. Microbiome in Multiple Sclerosis; Where Are We, What We Know and Do Not Know. *Brain Sci*. 2020;10(4):234. Published 2020 Apr 14. doi:10.3390/brainsci10040234
- 21 Tankou SK, Regev K, Healy BC, et al. A probiotic modulates the microbiome and immunity in multiple sclerosis. *Ann Neurol*. 2018;83(6):1147-1161. doi:10.1002/ana.25244
- 22 Moine L, Díaz de Barboza G, Pérez A, Benedetto M, Tolosa de Talamoni N. Glutamine protects intestinal calcium absorption against oxidative stress and apoptosis. *Comp Biochem Physiol A Mol Integr Physiol*. 2017;212:64-71. doi:10.1016/j.cbpa.2017.07.006
- 23 Wang B, Wu G, Zhou Z, et al. Glutamine and intestinal barrier function. *Amino Acids*. 2015;47(10):2143-2154. doi:10.1007/s00726-014-1773-4
- 24 Achamrah N, Déchelotte P, Coëffier M. Glutamine and the regulation of intestinal permeability: from bench to bedside. *Curr Opin Clin Nutr Metab Care*. 2017;20(1):86-91. doi:10.1097/MCO.0000000000000339
- 25 Swann OG, Kilpatrick M, Breslin M, Oddy WH. Dietary fiber and its associations with depression and inflammation. *Nutr Rev*. 2020;78(5):394-411. doi:10.1093/nutrit/nuz072
- 26 Chen Q, Chen O, Martins IM, et al. Collagen peptides ameliorate intestinal epithelial barrier dysfunction in immunostimulatory Caco-2 cell monolayers via enhancing tight junctions. *Food Funct*. 2017;8(3):1144-1151. doi:10.1039/c6fo01347c
- 27 Frasca G, Cardile V, Puglia C, Bonina C, Bonina F. Gelatin tannate reduces the proinflammatory effects of lipopolysaccharide in human intestinal epithelial cells. *Clin Exp Gastroenterol*. 2012;5:61-67. doi:10.2147/CEG.S28792
- 28 Wang, Mingxing et al. “Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion’s Mane Medicinal Mushroom, *Herichium erinaceus* (Higher Basidiomycetes).” *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055-60. doi:10.1615/intjmedmushrooms.v17.i11.50. <https://pubmed.ncbi.nlm.nih.gov/26853960/>
- 29 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion’s mane medicinal mushroom, *Herichium erinaceus* (Higher Basidiomycetes) from Malaysia. *International journal of medicinal mushrooms*, 15(6), 539–554. <https://doi.org/10.1615/intjmed-mushr.v15.i6.30> <https://pubmed.ncbi.nlm.nih.gov/24266378/>
- 30 Kuo, Hsing-Chun et al. “*Herichium erinaceus* mycelium and its isolated erinacine A protection from MPTP-induced neurotoxicity through the ER stress, triggering an apoptosis cascade.” *Journal of translational medicine* vol. 14 78. 18 Mar. 2016
- 31 Mori, K., Obara, Y., Moriya, T., Inatomi, S., & Nakahata, N. (2011). Effects of *Herichium erinaceus* on amyloid β (25-35) peptide-induced learning and memory deficits in mice. *Biomedical research (Tokyo, Japan)*, 32(1), 67–72. <https://doi.org/10.2220/biomedres.32.67> <https://pubmed.ncbi.nlm.nih.gov/21383512/>
- 32 Brandalise, Federico et al. “Dietary Supplementation of *Herichium erinaceus* Increases Mossy Fiber-CA3 Hippocampal Neurotransmission and Recognition Memory in Wild-Type Mice.” *Evidence-based complementary and alternative medicine: eCAM* vol. 2017 (2017): 3864340. doi:10.1155/2017/3864340. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5237458/>
- 33 Mori, Koichiro et al. “Improving effects of the mushroom Yamabushitake (*Herichium erinaceus*) on mild cognitive impairment: a double-blind placebo-controlled clinical trial.” *Phytotherapy research: PTR* vol. 23,3 (2009): 367-72. doi:10.1002/ptr.2634. <https://pubmed.ncbi.nlm.nih.gov/18844328/>
- 34 Bhardwaj, Neha et al. “Suppression of inflammatory and allergic re-

- sponses by pharmacologically potent fungus *Ganoderma lucidum*." Recent patents on inflammation & allergy drug discovery vol. 8,2 (2014): 104-17. doi:10.2174/1872213x08666140619110657
- 35 Huang, Shichao et al. "Polysaccharides from *Ganoderma lucidum* Promote Cognitive Function and Neural Progenitor Proliferation in Mouse Model of Alzheimer's Disease." *Stem cell reports* vol. 8,1 (2017): 84-94. doi:10.1016/j.stemcr.2016.12.007. <https://pubmed.ncbi.nlm.nih.gov/28076758/>
- 36 <https://www.sciencedirect.com/science/article/pii/S1319562X1830192X> Young Ock Kim, Hak Jae Kim, Gasem Mohammad Abu-Taweel, Junsang Oh, Gi-Ho Sung, Neuro-protective and therapeutic effect of Cordyceps militaris on ischemia-induced neuronal death and cognitive impairments, *Saudi Journal of Biological Sciences*, Volume 26, Issue 7, 2019, Pages 1352-1357, ISSN 1319-562X, <https://doi.org/10.1016/j.sjbs.2018.08.011>
- 37 Li, X. T., Li, H. C., Li, C. B., Dou, D. Q., & Gao, M. B. (2010). Protective effects on mitochondria and anti-aging activity of polysaccharides from cultivated fruiting bodies of *Cordyceps militaris*. *The American journal of Chinese medicine*, 38(6), 1093-1106. <https://doi.org/10.1142/S0192415X10008494> <https://pubmed.ncbi.nlm.nih.gov/21061463/>
- 38 <https://pubmed.ncbi.nlm.nih.gov/18803231/> Ji, D. B., Ye, J., Li, C. L., Wang, Y. H., Zhao, J., & Cai, S. Q. (2009). Antiaging effect of *Cordyceps sinensis* extract. *Phytotherapy research: PTR*, 23(1), 116-122. <https://doi.org/10.1002/ptr.2576>
- 39 <https://pubmed.ncbi.nlm.nih.gov/20724804/> Guo, P., Kai, Q., Gao, J., Lian, Z. Q., Wu, C. M., Wu, C. A., & Zhu, H. B. (2010). Cordycepin prevents hyperlipidemia in hamsters fed a high-fat diet via activation of AMP-activated protein kinase. *Journal of pharmacological sciences*, 113(4), 395-403. <https://doi.org/10.1254/jphs.10041fp>
- 40 Liao H, Ye J, Gao L, Liu Y. The main bioactive compounds of *Scutellaria baicalensis* Georgi. for alleviation of inflammatory cytokines: A comprehensive review. *Biomed Pharmacother*. 2021;133:110917. doi:10.1016/j.biopha.2020.110917
- 41 <https://www.ncbi.nlm.nih.gov/labs/pmc/articles/PMC6025220/>
- 42 Sowndhararajan, K., Deepa, P., Kim, M., Park, S. J., & Kim, S. (2018). Neuroprotective and Cognitive Enhancement Potentials of Baicalin: A Review. *Brain sciences*, 8(6), 104. <https://doi.org/10.3390/brainsci8060104>
- 43 Heo HJ, Kim DO, Choi SJ, Shin DH, Lee CY. Potent Inhibitory effect of flavonoids in *Scutellaria baicalensis* on amyloid beta protein-induced neurotoxicity. *J Agric Food Chem*. 2004;52(13):4128-4132. doi:10.1021/jf049953x
- 44 He, J., & Giusti, M. M. (2010). Anthocyanins: natural colorants with health-promoting properties. *Annual review of food science and technology*, 1, 163-187. <https://doi.org/10.1146/annurev.food.080708.100754> <https://pubmed.ncbi.nlm.nih.gov/22129334/>
- 45 Wieland, L. S., Piechotta, V., Feinberg, T., Ludeman, E., Hutton, B., Kanji, S., Seely, D., & Garritty, C. (2021). Elderberry for prevention and treatment of viral respiratory illnesses: a systematic review. *BMC complementary medicine and therapies*, 21(1), 112. <https://doi.org/10.1186/s12906-021-03283-5> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8026097/>
- 46 Wieland, L. S., Piechotta, V., Feinberg, T., Ludeman, E., Hutton, B., Kanji, S., Seely, D., & Garritty, C. (2021). Elderberry for prevention and treatment of viral respiratory illnesses: a systematic review. *BMC complementary medicine and therapies*, 21(1), 112. <https://doi.org/10.1186/s12906-021-03283-5> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8026097/>
- 47 <https://pubmed.ncbi.nlm.nih.gov/25491337/> Gleńsk, M., Gliński, J. A., Włodarczyk, M., & Stefanowicz, P. (2014). Determination of ursolic and oleanolic acid in *Sambuci fructus*. *Chemistry & biodiversity*, 11(12), 1939-1944. <https://doi.org/10.1002/cbdv.201400118>
- 48 Zhang, Y., Li, X., Ciric, B., Curtis, M. T., Chen, W. J., Rostami, A., & Zhang, G. X. (2020). A dual effect of ursolic acid to the treatment of multiple sclerosis through both immunomodulation and direct remyelination. *Proceedings of the National Academy of Sciences of the United States of America*, 117(16), 9082-9093. <https://doi.org/10.1073/pnas.2000208117> <https://pubmed.ncbi.nlm.nih.gov/32253301/>
- 49 <https://www.ncbi.nlm.nih.gov/labs/pmc/articles/PMC7260130/>
- 50 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6025220/>
- 51 <https://www.healthcni.com/Acupuncture-Continuing-Education-News/2039-covid-19-coronavirus-herbs-found-effective>
- 52 <https://pubmed.ncbi.nlm.nih.gov/30296293/>
- 53 <https://www.researchgate.net/publication/272020690>
- 54 <https://pubmed.ncbi.nlm.nih.gov/28283884>
- 55 <https://www.ncbi.nlm.nih.gov/>
- 56 Soilu-Hänninen M, Aivo J, Lindström BM, et al. A randomised, double blind, placebo controlled trial with vitamin D3 as an add on treatment to interferon β -1b in patients with multiple sclerosis. *J Neurol Neurosurg Psychiatry*. 2012;83(5):565-571. doi:10.1136/jnnp-2011-301876
- 57 Rotstein DL, Healy BC, Malik MT, et al. Effect of vitamin D on MS activity by disease-modifying therapy class. *Neurol Neuroimmunol Neuroinflamm*. 2015;2(6):e167. Published 2015 Oct 29. doi:10.1212/NXI.0000000000000167
- 58 Darwish H, Haddad R, Osman S, et al. Effect of Vitamin D Replacement on Cognition in Multiple Sclerosis Patients. *Sci Rep*. 2017;7:45926. Published 2017 Apr 4. doi:10.1038/srep45926
- 59 Gallai V, Sarchielli P, Trequattrini A, et al. Cytokine secretion and eicosanoid production in the peripheral blood mononuclear cells of MS patients undergoing dietary supplementation with n-3 polyunsaturated fatty acids. *J Neuroimmunol*. 1995;56(2):143-153. doi:10.1016/0165-5728(94)00140-j
- 60 Miller A, Korem M, Almog R, Galboiz Y. Vitamin B12, demyelination, remyelination and repair in multiple sclerosis. *J Neurol Sci*. 2005;233(1-2):93-97. doi:10.1016/j.jns.2005.03.009
- 61 Reynolds EH, Linnell JC, Faludy JE. Multiple sclerosis associated with vitamin B12 deficiency. *Arch Neurol*. 1991;48(8):808-811. doi:10.1001/archneur.1991.00530200044017
- 62 Khosravi-Largani M, Pourvali-Talatappeh P, Roustae AM, et al. A review on potential roles of vitamins in incidence, progression, and improvement of multiple sclerosis. *eNeurologicalSci*. 2018;10:37-44. Published 2018 Jan 28. doi:10.1016/j.ensci.2018.01.007
- 63 Reynolds EH. Multiple sclerosis and vitamin B12 metabolism. *J Neuroimmunol*. 1992;40(2-3):225-230. doi:10.1016/0165-5728(92)90137-a
- 64 Nozari E, Ghavamzadeh S, Razazian N. The Effect of Vitamin B12 and Folic Acid Supplementation on Serum Homocysteine, Anemia Status and Quality of Life of Patients with Multiple Sclerosis. *Clin Nutr Res*. 2019;8(1):36-45. Published 2019 Jan 25. doi:10.7762/cnr.2019.8.1.36
- 65 Jin, Cy., Lee, Jd., Park, C. et al. Curcumin attenuates the release of pro-inflammatory cytokines in lipopolysaccharide-stimulated BV2 microglia. *Acta Pharmacol Sin* 28, 1645-1651 (2007). <https://doi.org/10.1111/j.1745-7254.2007.00651.x>
- 66 Tegenge MA, Rajbhandari L, Shrestha S, Mithal A, Hosmane S, Venkatesan A. Curcumin protects axons from degeneration in the setting of local neuroinflammation. *Exp Neurol*. 2014;253:102-110. doi:10.1016/j.expneurol.2013.12.016
- 67 Kumar Ranjan Ravi, Singh Lovekesh, Thakur Amandeep, Singh Shamsher and Kumar Bhupinder*, Role of Vitamins in Neurodegenerative Diseases: A Review, *CNS & Neurological Disorders - Drug Targets* 2022; 21(9) . <https://dx.doi.org/10.2174/187152732006621119122150>
- 68 Guo YE, Suo N, Cui X, Yuan Q, Xie X. Vitamin C promotes oligodendrocytes generation and remyelination. *Glia*. 2018;66(7):1302-1316. doi:10.1002/glia.23306
- 69 Bredholt M, Frederiksen JL. Zinc in Multiple Sclerosis: A Systematic Review and Meta-Analysis. *ASN Neuro*. 2016;8(3):1759091416651511. Published 2016 Jun 9. doi:10.1177/1759091416651511
- 70 Ghaiaad, H.R., Nooh, M.M., El-Sawalhi, M.M. et al. Resveratrol Promotes Remyelination in Cuprizone Model of Multiple Sclerosis: Biochemical and Histological Study. *Mol Neurobiol* 54, 3219-3229 (2017). <https://doi.org/10.1007/s12035-016-9891-5>
- 71 Shindler KS, Ventura E, Dutt M, Elliott P, Fitzgerald DC, Rostami A. Oral resveratrol reduces neuronal damage in a model of multiple sclerosis. *J Neuroophthalmol*. 2010;30(4):328-339. doi:10.1097/WNO.0b013e3181f7f833
- 72 <https://pubmed.ncbi.nlm.nih.gov/>
- 73 Sanoobar M, Eghtesadi S, Azimi A, et al. Coenzyme Q10 supplementation ameliorates inflammatory markers in patients with multiple sclerosis: a double blind, placebo, controlled randomized clinical trial. *Nutr Neurosci*. 2015;18(4):169-176. doi:10.1179/1476830513X00000000106
- 74 Schwalfenberger GK. N-Acetylcysteine: A Review of Clinical Usefulness (an Old Drug with New Tricks). *J Nutr Metab*. 2021;2021:9949453. Published 2021 Jun 9. doi:10.1155/2021/9949453
- 75 Shahrampour S, Heholt J, Wang A, et al. N-acetyl cysteine administration affects cerebral blood flow as measured by arterial spin labeling MRI in patients with multiple sclerosis. *Heliyon*. 2021;7(7):e07615. Published 2021 Jul 16. doi:10.1016/j.heliyon.2021.e07615
- 76 Krysko KM, Bischof A, Nourbakhsh B, et al. A pilot study of oxidative pathways in MS fatigue: randomized trial of N-acetyl cysteine. *Ann Clin Transl Neurol*. 2021;8(4):811-824. doi:10.1002/acn3.51325
- 77 Lovera J, Reza T. Stress in multiple sclerosis: review of new developments and future directions. *Curr Neurol Neurosci Rep*. 2013;13(11):398. doi:10.1007/s11910-013-0398-4
- 78 Tavee J, Rensel M, Planchon SM, Butler RS, Stone L. Effects of meditation on pain and quality of life in multiple sclerosis and peripheral neuropathy: a pilot study. *Int J MS Care*. 2011;13(4):163-168. doi:10.7224/1537-2073-13.4.163
- 79 Levin AB, Hadjkiss EJ, Weiland TJ, et al. Can meditation influence quality of life, depression, and disease outcome in multiple sclerosis? Findings from a large international web-based study. *Behav Neurol*. 2014;2014:916519. doi:10.1155/2014/916519
- 80 Cavalera C, Rovaris M, Mendozzi L, et al. Online meditation training for people with multiple sclerosis: A randomized controlled trial. *Mult Scler*. 2019;25(4):610-617. doi:10.1177/1352458518761187
- 81 Oz HS, Oz F. A psychoeducation program for stress management and psychosocial problems in multiple sclerosis. *Niger J Clin Pract*. 2020;23(11):1598-1606. doi:10.4103/njcp.njcp_462_19
- 82 Agnostoni M, Babili I, Chrousos G, Artemiadis A, Darviri C. A novel cognitive-behavioral stress management method for multiple sclerosis. A brief report of an observational study. *Neurol Res*. 2019;41(3):223-226. doi:10.1080/01616412.2018.1548745
- 83 White, M.P., Alcock, I., Grellier, J. et al. Spending at least 120 minutes a week in nature is associated with good health and wellbeing. *Sci Rep* 9, 7730 (2019). <https://doi.org/10.1038/s41598-019-44097-3>
- 84 Caminero A, Bartolomé M. Sleep disturbances in multiple sclerosis. *J Neurol Sci*. 2011;309(1-2):86-91. doi:10.1016/j.jns.2011.07.015
- 85 Veauthier C. Sleep disorders in multiple sclerosis. *Review. Curr Neurol Neurosci Rep*. 2015;15(5):21. doi:10.1007/s11910-015-0546-0
- 86 Kołtuniuk A, Kazimierska-Zajac M, Pogódek D, Chojdak-Lukasiewicz J. Sleep Disturbances, Degree of Disability and the Quality of Life in Multiple Sclerosis Patients. *Int J Environ Res Public Health*. 2022;19(6):3271. Published 2022 Mar 10. doi:10.3390/ijerph19063271
- 87 Skarlis C, Agnostoni M. The role of melatonin in Multiple Sclerosis. *Neurol Sci*.

2020;41(4):769-781. doi:10.1007/s10072-019-04137-2

88 Hsu WY, Anderson A, Rowles W, et al. Effects of melatonin on sleep disturbances in multiple sclerosis: A randomized, controlled pilot study. *Mult Scler J Exp Transl Clin*. 2021;7(4):20552173211048756. Published 2021 Nov 5. doi:10.1177/20552173211048756

89 Lorenz RA, Auerbach S, Nisbet P, et al. Improving Sleep among Adults with Multiple Sclerosis using Mindfulness plus Sleep Education. *West J Nurs Res*. 2021;43(3):273-283. doi:10.1177/0193945920947409

90 Heo JY, Kim K, Fava M, et al. Effects of smartphone use with and without blue light at night in healthy adults: A randomized, double-blind, cross-over, placebo-controlled comparison. *J Psychiatr Res*. 2017;87:61-70. doi:10.1016/j.jpsychires.2016.12.010

91 Interface of Multiple Sclerosis, Depression, Vascular Disease, and Mortality - A Population-Based Matched Cohort Study; Raffaele Palladino, Jeremy Chataway, Azeem Majeed, Ruth Ann Marrie; *Neurology* Sep 2021, 97 (13) e1322-e1333; DOI: 10.1212/WNL.0000000000012610 <https://n.neurology.org/content/97/13/e1322>

92 Mincu, R. I., Magda, L. S., Florescu, M., Velcea, A., Mihaila, S., Mihalcea, D., Popescu, B. O., Chiru, A., Tiu, C., Cinteza, M., & Vinereanu, D. (2015). Cardiovascular Dysfunction in Multiple Sclerosis. *Maedica*, 10(4), 364–370. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5394435/>

93 Magyari, M., & Sorensen, P. S. (2020). Comorbidity in Multiple Sclerosis. *Frontiers in neurology*, 11, 851. <https://doi.org/10.3389/fneur.2020.00851>

94 Aalbers G, McNally RJ, Heeren A, de Wit S, Fried EI. Social media and depression symptoms: A network perspective. *J Exp Psychol Gen*. 2019;148(8):1454-1462. doi:10.1037/xge0000528

95 de Hoog N, Verboon P. Is the news making us unhappy? The influence of daily news exposure on emotional states. *Br J Psychol*. 2020;111(2):157-173. doi:10.1111/bjop.12389

96 Sandroff BM, Johnson CL, Motl RW. Exercise training effects on memory and hippocampal viscoelasticity in multiple sclerosis: a novel application of magnetic resonance elastography. *Neuroradiology*. 2017;59(1):61-67. doi:10.1007/s00234-016-1767-x

97 Halabchi F, Alizadeh Z, Sahraian MA, Abolhasani M. Exercise prescription for patients with multiple sclerosis; potential benefits and practical recommendations. *BMC Neurol*. 2017;17(1):185. Published 2017 Sep 16. doi:10.1186/s12883-017-0960-9

98 Prakash RS, Snook EM, Motl RW, Kramer AF. Aerobic fitness is associated with gray matter volume and white matter integrity in multiple sclerosis. *Brain Res*. 2010;1341:41-51. doi:10.1016/j.brainres.2009.06.063

99 Cramer H, Lauche R, Azizi H, Dobos G, Langhorst J. Yoga for multiple sclerosis: a systematic review and meta-analysis. *PLoS One*. 2014;9(11):e112414. Published 2014 Nov 12. doi:10.1371/journal.pone.0112414

100 Rogers KA, MacDonald M. Therapeutic Yoga: Symptom Management for Multiple Sclerosis. *J Altern Complement Med*. 2015;21(11):655-659. doi:10.1089/acm.2015.0015

101 Kjølshede T, Vissing K, Dalgas U. Multiple sclerosis and progressive resistance training: a systematic review. *Mult Scler*. 2012;18(9):1215-1228. doi:10.1177/1352458512437418

102 Patil S, Sen S, Bral M, et al. The Role of Acupuncture in Pain Management. *Curr Pain Headache Rep*. 2016;20(4):22. doi:10.1007/s11916-016-0552-1

103 Smith CA, Hay PP. Acupuncture for depression. *Cochrane Database Syst Rev*. 2005;(2):CD004046. Published 2005 Apr 18. doi:10.1002/14651858.CD004046.pub2

104 Bussell J. The effect of acupuncture on working memory and anxiety. *J Acupunct Meridian Stud*. 2013;6(5):241-246. doi:10.1016/j.jams.2012.12.006

105 Khodaie F, Abbasi N, Kazemi Motlagh AH, Zhao B, Naser Moghadasi A. Acupuncture for multiple sclerosis: A literature review. *Mult Scler Relat Disord*. 2022;60:103715. doi:10.1016/j.msard.2022.103715

106 Criado MB, Santos MJ, Machado J, Gonçalves AM, Greten HJ. Effects of Acupuncture on Gait of Patients with Multiple Sclerosis. *J Altern Complement Med*. 2017;23(11):852-857. doi:10.1089/acm.2016.0355

107 Donnellan CP, Shanley J. Comparison of the effect of two types of acupuncture on quality of life in secondary progressive multiple sclerosis: a preliminary single-blind randomized controlled trial. *Clin Rehabil*. 2008;22(3):195-205. doi:10.1177/0269215507082738

108 Ding Y, Shi X. *Zhongguo Zhen Jiu*. 2013;33(9):793-795

109 Wang C, Chen Z, Wang L, et al. *Zhongguo Zhen Jiu*. 2017;37(6):576-580. doi:10.13703/j.0255-2930.2017.06.002

110 Kim PS, Fishman MA. Low-Dose Naltrexone for Chronic Pain: Update and Systemic Review. *Curr Pain Headache Rep*. 2020;24(10):64. Published 2020 Aug 26. doi:10.1007/s11916-020-00898-0

111 Toljan K, Vrooman B. Low-Dose Naltrexone (LDN)-Review of Therapeutic Utilization. *Med Sci (Basel)*. 2018;6(4):82. Published 2018 Sep 21. doi:10.3390/medsci6040082

112 Patten DK, Schultz BG, Berlau DJ. The Safety and Efficacy of Low-Dose Naltrexone in the Management of Chronic Pain and Inflammation in Multiple Sclerosis, Fibromyalgia, Crohn's Disease, and Other Chronic Pain Disorders. *Pharmacotherapy*. 2018;38(3):382-389. doi:10.1002/phar.2086

113 Trofimovitch D, Baumrucker SJ. Pharmacology Update: Low-Dose Naltrexone as a Possible Nonopioid Modality for Some Chronic, Nonmalignant Pain Syndromes. *Am J Hosp Palliat Care*. 2019;36(10):907-912. doi:10.1177/1049909119838974

114 Gironi M, Martinelli-Boneschi F, Sacerdote P, et al. A pilot trial of low-dose naltrexone in primary progressive multiple sclerosis. *Mult Scler*. 2008;14(8):1076-1083. doi:10.1177/1352458508095828

115 Younger J, Noor N, McCue R, Mackey S. Low-dose naltrexone for the treatment of fibromyalgia: findings of a small, randomized, double-blind, placebo-controlled, counterbalanced, crossover trial assessing daily pain levels. *Arthritis Rheum*. 2013;65(2):529-538. doi:10.1002/art.37734

116 Winkelman, A., Loebermann, M., Reisinger, E. et al. Disease-modifying therapies and infectious risks in multiple sclerosis. *Nat Rev Neurol* 12, 217–233 (2016). <https://doi.org/10.1038/nrneurol.2016.21>

Psoriasis and Eczema Protocol

1 <https://www.mayoclinic.org/>

2 Roesner, L. M., & Werfel, T. (2019). Autoimmunity (or Not) in Atopic Dermatitis. *Frontiers in immunology*, 10, 2128. <https://doi.org/10.3389/fimmu.2019.02128> <https://pubmed.ncbi.nlm.nih.gov/31552053>

3 Roesner, L. M., & Werfel, T. (2019). Autoimmunity (or Not) in Atopic Dermatitis. *Frontiers in immunology*, 10, 2128. <https://doi.org/10.3389/fimmu.2019.02128> <https://pubmed.ncbi.nlm.nih.gov/31552053/>

4 <https://www.manhattanmedicalresearch.com>

5 Papagrigroraki, A., Maurelli, M., Del Giglio, M., Gisondi, P., & Girolomoni, G. (2017). Advanced Glycation End Products in the Pathogenesis of Psoriasis. *International journal of molecular sciences*, 18(11), 2471. <https://doi.org/10.3390/ijms18112471> <https://pubmed.ncbi.nlm.nih.gov/29156622/>

6 Zivkovic, A. M., Telis, N., German, J. B., & Hammock, B. D. (2011). Dietary omega-3 fatty acids aid in the modulation of inflammation and metabolic health. *California agriculture*, 65(3), 106–111. <https://doi.org/10.3733/ca.v065n03p106> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4030645/>

7 Wong, J. Y., Abdulla, M. A., Raman, J., Phan, C. W., Kuppusamy, U. R., Golbabapour, S., & Sabaratnam, V. (2013). Gastroprotective Effects of Lion's Mane Mushroom *Hericium erinaceus* (Bull.-Fr.) Pers. (Aphyllophoromycetidae) Extract against Ethanol-Induced Ulcer in Rats. Evidence-based complementary and alternative medicine: eCAM, 2013, 492976. <https://doi.org/10.1155/2013/492976>

8 Wachtel-Galor S, Yuen J, Buswell JA, et al. *Ganoderma lucidum* (Lingzhi or Reishi): A Medicinal Mushroom. In: Benzie IFF, Wachtel-Galor S, editors. *Herbal Medicine: Biomolecular and Clinical Aspects*. 2nd edition. Boca Raton (FL): CRC Press/Taylor & Francis; 2011. Chapter 9. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK92757/>

9 Peterson, C. T., Sharma, V., Uchitel, S., Denniston, K., Chopra, D., Mills, P. J., & Peterson, S. N. (2018). Prebiotic Potential of Herbal Medicines Used in Digestive Health and Disease. *Journal of alternative and complementary medicine (New York, N.Y.)*, 24(7), 656–665. <https://doi.org/10.1089/acm.2017.0422>

10 Fink, C., Schmidt, M., & Kraft, K. (2018). *Complementary medicine research*, 25(5), 299–305. <https://doi.org/10.1159/000489560>

11 Najafian, Y., Hamed, S. S., Farshchi, M. K., & Feyzabadi, Z. (2018). *Plantago major* in Traditional Persian Medicine and modern phytotherapy: a narrative review. *Electronic physician*, 10(2), 6390–6399. <https://doi.org/10.19082/6390>

12 Benson, K. F., Stamets, P., Davis, R., Nally, R., Taylor, A., Slater, S., & Jensen, G. S. (2019). The mycelium of the *Trametes versicolor* (Turkey tail) mushroom and its fermented substrate each show potent and complementary immune activating properties in vitro. *BMC complementary and alternative medicine*, 19(1), 342. <https://doi.org/10.1186/s12906-019-2681-7>

13 Fahey, J. W., Stephenson, K. K., & Wallace, A. J. (2015). Dietary amelioration of *Helicobacter* infection. *Nutrition research (New York, N.Y.)*, 35(6), 461–473. <https://doi.org/10.1016/j.nutres.2015.03.001>

14 Mravčáková, D., Komáromyová, M., Babják, M., Urda Dolinská, M., Königová, A., Petrič, D., Čobanová, K., Šlusarczyk, S., Cieslak, A., Várady, M., & Váradyová, Z. (2020). Anthelmintic Activity of Wormwood (*Artemisia absinthium* L.) and Mallow (*Malva sylvestris* L.) against *Haemonchus contortus* in Sheep. *Animals: an open access journal from MDPI*, 10(2), 219. <https://doi.org/10.3390/ani10020219>

15 Di, T. T., Ruan, Z. T., Zhao, J. X., Wang, Y., Liu, X., Wang, Y., & Li, P. (2016). Astilbin inhibits Th17 cell differentiation and ameliorates imiquimod-induced psoriasis-like skin lesions in BALB/c mice via Jak3/Stat3 signaling pathway. *International immunopharmacology*, 32, 32–38. <https://doi.org/10.1016/j.intimp.2015.12.035>

16 Guggenheim, A. G., Wright, K. M., & Zwickey, H. L. (2014). Immune Modulation From Five Major Mushrooms: Application to Integrative Oncology. *Integrative medicine (Encinitas, Calif.)*, 13(1), 32–44

17 Ziauddin, M., Phansalkar, N., Patki, P., Diwanay, S., & Patwardhan, B. (1996). Studies on the immunomodulatory effects of Ashwagandha. *Journal of ethnopharmacology*, 50(2), 69–76. [https://doi.org/10.1016/0378-8741\(95\)01318-0](https://doi.org/10.1016/0378-8741(95)01318-0)

18 Park, C. M., Cho, C. W., & Song, Y. S. (2014). TOP 1 and 2, polysaccharides from *Taraxacum officinale*, inhibit NFκB-mediated inflammation and accelerate Nrf2-induced antioxidative potential through the modulation of PI3K-Akt signaling pathway in RAW 264.7 cells. *Food and chemical toxicology: an international journal published for the British Industrial Biological Research Association*, 66, 56–64. <https://doi.org/10.1016/j.fct.2014.01.019>

19 Roschek, B., Jr, Fink, R. C., McMichael, M., & Alberte, R. S. (2009). Nettle extract (*Urtica dioica*) affects key receptors and enzymes associated with allergic rhinitis. *Phytotherapy research: PTR*, 23(7), 920–926. <https://doi.org/10.1002/ptr.2763>

20 Chan, YS., Cheng, LN., Wu, JH. et al. A review of the pharmacological effects of *Arctium lappa* (burdock). *Inflammopharmacol* 19, 245–254 (2011). <https://doi.org/10.1007/s10787-010-0062-4>

21 Iliina T, Kashpur N, Granica S, Bazylko A, Shinkovenko I, Kovalyova A, Goryacha O, Koshovy O. Phytochemical Profiles and In Vitro Immunomodulatory Activity of Ethanolic Extracts from *Galium aparine* L. *Plants*. 2019; 8(12):541. <https://doi.org/10.3390/plants8120541>

- 22 Jurenka J. S. (2009). Anti-inflammatory properties of curcumin, a major constituent of *Curcuma longa*: a review of preclinical and clinical research. *Alternative medicine review: a journal of clinical therapeutic*, 14(2), 141–153
- 23 Mashhadi, N. S., Ghiasvand, R., Askari, G., Hariri, M., Darvishi, L., & Mofid, M. R. (2013). Anti-oxidative and anti-inflammatory effects of ginger in health and physical activity: review of current evidence. *International journal of preventive medicine*, 4(Suppl 1), S36–S42
- 24 Olajide O. A. (2009). Inhibitory effects of St. John's Wort on inflammation: ignored potential of a popular herb. *Journal of dietary supplements*, 6(1), 28–32. <https://doi.org/10.1080/19390210802687247>
- 25 Pommier, P. (n.d.). Phase III Randomized Trial of *Calendula Officinalis* Compared With Trolamine for the Prevention of Acute Dermatitis During Irradiation for Breast Cancer | *Journal of Clinical Oncology*. Phase III Randomized Trial of *Calendula Officinalis* Compared With Trolamine for the Prevention of Acute Dermatitis During Irradiation for Breast Cancer | *Journal of Clinical Oncology*; ascopubs.org. Retrieved May 9, 2022, from <https://ascopubs.org/doi/10.1200/JCO.2004.07.063>
- 26 Chandra, S., & Rawat, D. S. (2015). Medicinal plants of the family Caryophyllaceae: a review of ethno-medicinal uses and pharmacological properties. *Integrative medicine research*, 4(3), 123–131. <https://doi.org/10.1016/j.imr.2015.06.004>
- 27 Reuter, J., Jocher, A., Stump, J., Grossjohann, B., Franke, G., & Schempp, C. M. (2008). Investigation of the anti-inflammatory potential of Aloe vera gel (97.5%) in the ultra-violet erythema test. *Skin pharmacology and physiology*, 21(2), 106–110. <https://doi.org/10.1159/000114871>
- 28 Cerio, R., Dohil, M., Jeanine, D., Magina, S., Mahé, E., & Stratigos, A. J. (2010). Mechanism of action and clinical benefits of colloidal oatmeal for dermatologic practice. *Journal of drugs in dermatology: JDD*, 9(9), 1116–1120
- 29 Miceli A, Schmieder GJ. Palmo-plantar Psoriasis. [Updated 2021 Aug 22]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK448142/>
- 30 Barrea, L., Savanelli, M. C., Di Somma, C., Napolitano, M., Megna, M., Colao, A., & Savastano, S. (2017). Vitamin D and its role in psoriasis: An overview of the dermatologist and nutritionist. *Reviews in endocrine & metabolic disorders*, 18(2), 195–205. <https://doi.org/10.1007/s11154-017-9411-6>
- 31 Lin, X., Meng, X., & Song, Z. (2019). Homocysteine and psoriasis. *Bioscience reports*, 39(11), BSR20190867. <https://doi.org/10.1042/BSR20190867>
- 32 Upala, S., Yong, W. C., Theparee, T., & Sanguaneko, A. (2017). Effect of omega-3 fatty acids on disease severity in patients with psoriasis: A systematic review. *International journal of rheumatic diseases*, 20(4), 442–450. <https://doi.org/10.1111/1756-185X.13051>
- 33 Zhu, X., Sang, L., Wu, D., Rong, J., & Jiang, L. (2018). Effectiveness and safety of glucosamine and chondroitin for the treatment of osteoarthritis: a meta-analysis of randomized controlled trials. *Journal of orthopaedic surgery and research*, 13(1), 170. <https://doi.org/10.1186/s13018-018-0871-5>
- 34 Butawan, M., Benjamin, R. L., & Bloomer, R. J. (2017). Methylsulfonylmethane: Applications and Safety of a Novel Dietary Supplement. *Nutrients*, 9(3), 290. <https://doi.org/10.3390/nu9030290>
- 35 Zhang, P., & Wu, M. X. (2018). A clinical review of phototherapy for psoriasis. *Lasers in medical science*, 33(1), 173–180. <https://doi.org/10.1007/s10103-017-2360-1>
- 36 Jensen, P., & Skov, L. (2016). Psoriasis and Obesity. *Dermatology (Basel, Switzerland)*, 232(6), 633–639. <https://doi.org/10.1159/000455840>
- 37 Psoriasis.org. 2022. Treatment and Care. [online] Available at: <<https://www.psoriasis.org/treatment-and-care/>> [Accessed 9 May 2022]

Rheumatoid Arthritis Protocol

- 1 Doğan SC, Hizmetli S, Hayta E, Kaptanoğlu E, Erselcan T, Güler E. Sarcopenia in women with rheumatoid arthritis. *Eur J Rheumatol*. 2015 Jun;2(2):57–61. doi: 10.5152/eur-jrheum.2015.0038. Epub 2015 Mar 31. PMID: 27708927; PMCID: PMC5047263
- 2 Chen YJ, Chang YT, Wang CB, Wu CY. The risk of cancer in patients with rheumatoid arthritis: a nationwide cohort study in Taiwan. *Arthritis Rheum*. 2011;63:352–358
- 3 Guo Q, Wang Y, Xu D, Nossent J, Pavlos NJ, Xu J. Rheumatoid arthritis: pathological mechanisms and modern pharmacologic therapies. *Bone Res*. 2018 Apr 27;6:15. doi: 10.1038/s41413-018-0016-9. PMID: 29736302; PMCID: PMC5920070. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5920070/>
- 4 Dube, S. R., Fairweather, D., Pearson, W. S., Felitti, V. J., Anda, R. F., & Croft, J. B. (2009). Cumulative childhood stress and autoimmune diseases in adults. *Psychosomatic medicine*, 71(2), 243–250. <https://doi.org/10.1097/PSY.0b013e3181907888>
- 5 Vadell AKE, Bärebring L, Hulander E, Gertsson I, Lindqvist HM, Winkvist A. Anti-inflammatory Diet In Rheumatoid Arthritis (ADIRA)-a randomized, controlled crossover trial indicating effects on disease activity. *Am J Clin Nutr*. 2020 Jun 1;111(6):1203–1213. doi: 10.1093/ajcn/nqaa019. PMID: 32055820; PMCID: PMC7266686
- 6 Müller, H., de Toledo, F. W., & Resch, K. L. (2001). Fasting followed by vegetarian diet in patients with rheumatoid arthritis: a systematic review. *Scandinavian journal of rheumatology*, 30(1), 1–10. <https://doi.org/10.1080/030097401750065256>
- 7 Kjeldsen-Kragh, J., Haugen, M., Borchgrevink, C. F., Laerum, E., Eek, M., Mowinkel, P., Hovi, K., & Førre, O. (1991). Controlled trial of fasting and one-year vegetarian diet in rheumatoid arthritis. *Lancet (London, England)*, 338(8772), 899–902. [https://doi.org/10.1016/0140-6736\(91\)91770-u](https://doi.org/10.1016/0140-6736(91)91770-u)
- 8 Dourado, E., Ferro, M., Sousa Guerreiro, C., & Fonseca, J. E. (2020). Diet as a Modulator of Intestinal Microbiota in Rheumatoid Arthritis. *Nutrients*, 12(11), 3504. <https://doi.org/10.3390/nu12113504>

- Fully Convincing Evidence?. *Journal of clinical medicine*, 8(11), 1837. <https://doi.org/10.3390/jcm8111837>
- 10 Fujinami, R. S., von Herrath, M. G., Christen, U., & Whitton, J. L. (2006). Molecular mimicry, bystander activation, or viral persistence: infections and autoimmune disease. *Clinical microbiology reviews*, 19(1), 80–94. <https://doi.org/10.1128/CMR.19.1.80-94.2006>
- 11 Taneja V. (2014). Arthritis susceptibility and the gut microbiome. *FEBS letters*, 588(22), 4244–4249. <https://doi.org/10.1016/j.febslet.2014.05.034>
- 12 Vaahtovu, J., Munukka, E., Korkeamäki, M., Luukkainen, R., & Toivanen, P. (2008). Fecal microbiota in early rheumatoid arthritis. *The Journal of rheumatology*, 35(8), 1500–1505.
- 13 Gérard, H. C., Wang, Z., Wang, G. F., El-Gabalawy, H., Goldbach-Mansky, R., Li, Y., Majeed, W., Zhang, H., Ngai, N., Hudson, A. P., & Schumacher, H. R. (2001). Chromosomal DNA from a variety of bacterial species is present in synovial tissue from patients with various forms of arthritis. *Arthritis and rheumatism*, 44(7), 1689–1697. [https://doi.org/10.1002/1529-0131\(200107\)44:7<1689::AID-ART293>3.0.CO;2-K](https://doi.org/10.1002/1529-0131(200107)44:7<1689::AID-ART293>3.0.CO;2-K)
- 14 Azuma, K., Osaki, T., Wakuda, T., Tsuka, T., Imagawa, T., Okamoto, Y., & Minami, S. (2012). Suppressive effects of N-acetyl-D-glucosamine on rheumatoid arthritis mouse models. *Inflammation*, 35(4), 1462–1465. <https://doi.org/10.1007/s10753-012-9459-0>
- 15 Richter, J., Capková, K., Hříbalová, V., Vannucci, L., Danyi, I., Malý, M., & Fišerová, A. (2014). Collagen-induced arthritis: severity and immune response attenuation using multivalent N-acetyl glucosamine. *Clinical and experimental immunology*, 177(1), 121–133. <https://doi.org/10.1111/cei.12313>
- 16 Shmagel, A., Demmer, R., Knights, D., Butler, M., Langsetmo, L., Lane, N. E., & Ensrud, K. (2019). The Effects of Glucosamine and Chondroitin Sulfate on Gut Microbial Composition: A Systematic Review of Evidence from Animal and Human Studies. *Nutrients*, 11(2), 294. <https://doi.org/10.3390/nu11020294>
- 17 Arafa, N. M., Hamuda, H. M., Melek, S. T., & Darwish, S. K. (2013). The effectiveness of Echinacea extract or composite glucosamine, chondroitin and methyl sulfonyl methane supplements on acute and chronic rheumatoid arthritis rat model. *Toxicology and industrial health*, 29(2), 187–201. <https://doi.org/10.1177/0748233711428643>
- 18 Wang, Mingxing et al. "Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion's Mane Medicinal Mushroom, *Herichium erinaceus* (Higher Basidiomycetes)." *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055-60. doi:10.1615/intjmedmushrooms.v17.i11.50. <https://pubmed.ncbi.nlm.nih.gov/26853960/>
- 19 Siqueira, U. S., Orsini Valente, L. G., de Mello, M. T., Szejnfeld, V. L., & Pinheiro, M. M. (2017). Effectiveness of Aquatic Exercises in Women With Rheumatoid Arthritis: A Randomized, Controlled, 16-Week Intervention-The HydRA Trial. *American journal of physical medicine & rehabilitation*, 96(3), 167–175. <https://doi.org/10.1097/PHM.0000000000000564>
- 20 Baxter, S. V., Hale, L. A., Stebbings, S., Gray, A. R., Smith, C. M., & Trehan, G. J. (2016). Walking is a Feasible Physical Activity for People with Rheumatoid Arthritis: A Feasibility Randomized Controlled Trial. *Musculoskeletal care*, 14(1), 47–56. <https://doi.org/10.1002/msc.1112>
- 21 Cooney, J. K., Law, R. J., Matschke, V., Lemmey, A. B., Moore, J. P., Ahmad, Y., Jones, J. G., Maddison, P., & Thom, J. M. (2011). Benefits of exercise in rheumatoid arthritis. *Journal of aging research*, 2011, 681640. <https://doi.org/10.4061/2011/681640>
- 22 Fentem P. H. (1994). ABC of sports medicine. Benefits of exercise in health and disease. *BMJ (Clinical research ed.)*, 308(6939), 1291–1295. <https://doi.org/10.1136/bmj.308.6939.1291>
- 23 Han, A., Robinson, V., Judd, M., Taixiang, W., Wells, G., & Tugwell, P. (2004). Tai chi for treating rheumatoid arthritis. *The Cochrane database of systematic reviews*, (3), CD004849. <https://doi.org/10.1002/14651858.CD004849>
- 24 Mori, K., Ouchi, K., & Hirasawa, N. (2015). The Anti-Inflammatory Effects of Lion's Mane Culinary-Medicinal Mushroom, *Herichium erinaceus* (Higher Basidiomycetes) in a Coculture System of 3T3-L1 Adipocytes and RAW264 Macrophages. *International journal of medicinal mushrooms*, 17(7), 609–618. <https://doi.org/10.1615/intjmed-mushrooms.v17.i7.10>
- 25 Sheena, N., Ajith, T. A., & Janardhanan, K. K. (2003). Anti-inflammatory and Antinociceptive Activities of *Ganoderma lucidum* Occurring in South India. *Pharmaceutical Biology*, 41:4, 301–304. DOI: 10.1076/phbi.41.4.301.15677
- 26 Li, E. K., Tam, L. S., Wong, C. K., Li, W. C., Lam, C. W., Wachtel-Galor, S., Benzie, I. F., Bao, Y. X., Leung, P. C., & Tomlinson, B. (2007). Safety and efficacy of *Ganoderma lucidum* (lingzhi) and San Miao San supplementation in patients with rheumatoid arthritis: a double-blind, randomized, placebo-controlled pilot trial. *Arthritis and rheumatism*, 57(7), 1143–1150. <https://doi.org/10.1002/art.22994>
- 27 Bhardwaj, N., Katyal, P., & Sharma, A. K. (2014). Suppression of inflammatory and allergic responses by pharmacologically potent fungus *Ganoderma lucidum*. *Recent patents on inflammation & allergy drug discovery*, 8(2), 104–117. <https://doi.org/10.2174/1872213x08666140619110657>
- 28 Saleh, M. H., Rashedi, I., & Keating, A. (2017). Immunomodulatory Properties of *Coriolus versicolor*: The Role of Polysaccharopeptide. *Frontiers in immunology*, 8, 1087. <https://doi.org/10.3389/fimmu.2017.01087>
- 29 Yu, Z. T., Liu, B., Mukherjee, P., & Newburg, D. S. (2013). *Trametes versicolor* extract modifies human fecal microbiota composition in vitro. *Plant foods for human nutrition (Dordrecht, Netherlands)*, 68(2), 107–112. <https://doi.org/10.1007/s11300-013-0342-4>
- 30 Noh, E. M., Kim, J. S., Hur, H., Park, B. H., Song, E. K., Han, M. K., Kwon, K. B., Yoo, W. H., Shim, I. K., Lee, S. J., Youn, H. J., & Lee, Y. R. (2009). Cordycepin inhibits IL-1β-induced MMP-1 and MMP-3 expression in rheumatoid arthritis synovial fibroblasts. *Rheumatology (Oxford, England)*, 48(1), 45–48. <https://doi.org/10.1093/rheumatology/ken417>
- 31 Guo, P., Kai, Q., Gao, J., Lian, Z. Q., Wu, C. M., Wu, C. A., & Zhu, H. B. (2010). Cordycepin

- prevents hyperlipidemia in hamsters fed a high-fat diet via activation of AMP-activated protein kinase. *Journal of pharmacological sciences*, 113(4), 395–403. <https://doi.org/10.1254/jphs.10041fp>
- 32 Chandra, L., Alexander, H., Traoré, D., Lucas, E. A., Clarke, S. L., Smith, B. J., Lightfoot, S. A., & Kuvibidila, S. (2011). White button and shiitake mushrooms reduce the incidence and severity of collagen-induced arthritis in dilute brown non-agouti mice. *The Journal of nutrition*, 141(1), 131–136. <https://doi.org/10.3945/jn.110.127134>
- 33 Kim, G. Y., Kim, S. H., Hwang, S. Y., Kim, H. Y., Park, Y. M., Park, S. K., Lee, M. K., Lee, S. H., Lee, T. H., & Lee, J. D. (2003). Oral administration of proteoglycan isolated from *Phellinus linteus* in the prevention and treatment of collagen-induced arthritis in mice. *Biological & pharmaceutical bulletin*, 26(6), 823–831. <https://doi.org/10.1248/bpb.26.823>
- 34 Semalty, M., Adhikari, L., Semwal, D., Chauhan, A., Mishra, A., Kotiyal, R., & Semalty, A. (2017). A comprehensive review on phytochemistry and pharmacological effects of stinging nettle (*Urtica dioica*). *Current Traditional Medicine*, 3(3), 156–167
- 35 Abd-Nikfarjam, B., Abbasi, M., Memarzadeh, M., Farzam, S. A., Jamshidian, A., & Dolati-Somarin, A. (2022). Therapeutic efficacy of *Urtica dioica* and evening primrose in patients with rheumatoid arthritis: A randomized double-blind, placebo-controlled clinical trial. *Journal of Herbal Medicine*, 32, 100556. <https://www.sciencedirect.com/science/article/abs/pii/S2210803322000240>
- 36 Johnson TA, Sohn J, Inman WD, Bjeldanes LF, Rayburn K. Lipophilic stinging nettle extracts possess potent anti-inflammatory activity, are not cytotoxic and may be superior to traditional tinctures for treating inflammatory disorders.
- 37 Kessenbrock, K., Fröhlich, L., Sixt, M., Lämmermann, T., Pfister, H., Bateman, A., Belaouaj, A., Ring, J., Öllert, M., Fässler, R., & Jenne, D. E. (2008). Proteinase 3 and neutrophil elastase enhance inflammation in mice by inactivating antiinflammatory programulin. *The Journal of clinical investigation*, 118(7), 2438–2447. <https://doi.org/10.1172/JCI34694>
- 38 Jacquet, A., Girodet, P. O., Pariente, A., Forest, K., Mallet, L., & Moore, N. (2009). Phytalgic, a food supplement, vs placebo in patients with osteoarthritis of the knee or hip: a randomised double-blind placebo-controlled clinical trial. *Arthritis research & therapy*, 11(6), R192. <https://doi.org/10.1186/ar2891> <https://pubmed.ncbi.nlm.nih.gov/20015358/>
- 39 Goldbach-Mansky, R., Wilson, M., Fleischmann, R., Olsen, N., Silverfield, J., Kempf, P., Kivitz, A., Sherrer, Y., Pucino, F., Csako, G., Costello, R., Pham, T. H., Snyder, C., van der Heijde, D., Tao, X., Wesley, R., & Lipsky, P. E. (2009). Comparison of Tripterygium wilfordii Hook F versus sulfasalazine in the treatment of rheumatoid arthritis: a randomized trial. *Annals of internal medicine*, 151(4), 229–W51. <https://doi.org/10.7326/0003-4819-151-4-200908180-00005>
- 40 Huang, Q. C., Wang, M. J., Chen, X. M., Yu, W. L., Chu, Y. L., He, X. H., & Huang, R. Y. (2016). Can active components of licorice, glycyrrhizin and glycyrrhetic acid, lick rheumatoid arthritis? *Oncotarget*, 7(2), 1193–1202. <https://doi.org/10.18632/oncotarget.6200>
- 41 Feng, Y., Mei, L., Wang, M., Huang, Q., & Huang, R. (2021). Anti-inflammatory and Pro-apoptotic Effects of 18beta-Glycyrrhetic Acid In Vitro and In Vivo Models of Rheumatoid Arthritis. *Frontiers in pharmacology*, 12, 681525. <https://doi.org/10.3389/fphar.2021.681525>
- 42 Sharma, M. L., Bani, S., & Singh, G. B. (1989). Anti-arthritis activity of boswellic acids in bovine serum albumin (BSA)-induced arthritis. *International journal of immunopharmacology*, 11(6), 647–652. [https://doi.org/10.1016/0192-0561\(89\)90150-1](https://doi.org/10.1016/0192-0561(89)90150-1)
- 43 Sharma, M. L., Bani, S., & Singh, G. B. (1989). Anti-arthritis activity of boswellic acids in bovine serum albumin (BSA)-induced arthritis. *International journal of immunopharmacology*, 11(6), 647–652. [https://doi.org/10.1016/0192-0561\(89\)90150-1](https://doi.org/10.1016/0192-0561(89)90150-1)
- 44 Siddiqui M. Z. (2011). Boswellia serrata, a potential antiinflammatory agent: an overview. *Indian journal of pharmaceutical sciences*, 73(3), 255–261. <https://doi.org/10.4103/0250-474X.93507>
- 45 Sotebeer, M., & White, N. D. (2022). Potential Benefits of Curcumin in Rheumatoid Arthritis. *American Journal of Lifestyle Medicine*, 15598276221116063
- 46 Daily, J. W., Yang, M., & Park, S. (2016). Efficacy of turmeric extracts and curcumin for alleviating the symptoms of joint arthritis: a systematic review and meta-analysis of randomized clinical trials. *Journal of medicinal food*, 19(8), 717–729. <https://www.liebertpub.com/doi/full/10.1089/jmf.2016.3705>
- 47 Al-Nahain, A., Jahan, R., & Rahmatullah, M. (2014). Zingiber officinale: A Potential Plant against Rheumatoid Arthritis. *Arthritis*, 2014, 159089. <https://doi.org/10.1155/2014/159089>
- 48 Funk, J. L., Frye, J. B., Oyarzo, J. N., & Timmermann, B. N. (2009). Comparative effects of two gingerol-containing Zingiber officinale extracts on experimental rheumatoid arthritis. *Journal of natural products*, 72(3), 403–407. <https://doi.org/10.1021/np8006183>
- 49 Srivastava, K. C., & Mustafa, T. (1992). Ginger (Zingiber officinale) in rheumatism and musculoskeletal disorders. *Medical hypotheses*, 39(4), 342–348. [https://doi.org/10.1016/0306-9877\(92\)90059-1](https://doi.org/10.1016/0306-9877(92)90059-1)
- 50 Al-Nahain A, Jahan R, Rahmatullah M. Zingiber officinale: A Potential Plant against Rheumatoid Arthritis. *Arthritis*. 2014;2014:159089. doi: 10.1155/2014/159089. Epub 2014 May 27. PMID: 24982806; PMCID: PMC4058601
- 51 Mur, E., Hartig, F., Eibl, G., & Schirmer, M. (2002). Randomized double blind trial of an extract from the pentacyclic alkaloid-chelomotype of *Uncaria tomentosa* for the treatment of rheumatoid arthritis. *The Journal of rheumatology*, 29(4), 678–681.
- 52 Kim, E. H., Shim, B., Kang, S., Jeong, G., Lee, J. S., Yu, Y. B., & Chun, M. (2009). Anti-inflammatory effects of *Scutellaria baicalensis* extract via suppression of immune modulators and MAP kinase signaling molecules. *Journal of ethnopharmacology*, 126(2), 320–331. <https://doi.org/10.1016/j.jep.2009.08.027>
- 53 Yang, X., Yang, J., & Zou, H. (2013). Baicalin inhibits IL-17-mediated joint inflammation in murine adjuvant-induced arthritis. *Clinical & developmental immunology*, 2013, 268065. <https://doi.org/10.1155/2013/268065>
- 54 Kostoglou-Athanassiou, I., Athanassiou, P., Lyraki, A., Raftakis, I., & Antoniadis, C. (2012). Vitamin D and rheumatoid arthritis. *Therapeutic advances in endocrinology and metabolism*, 3(6), 181–187. <https://doi.org/10.1177/2042018812471070>
- 55 Javadi, F., Ahmadvadeh, A., Eghtesadi, S., Aryaeian, N., Zabihyeganeh, M., Rahimi Foroushani, A., & Jazayeri, S. (2017). The Effect of Quercetin on Inflammatory Factors and Clinical Symptoms in Women with Rheumatoid Arthritis: A Double-Blind, Randomized Controlled Trial. *Journal of the American College of Nutrition*, 36(1), 9–15. <https://doi.org/10.1080/07315724.2016.1140093>
- 56 Kostoglou-Athanassiou, I., Athanassiou, L., & Athanassiou, P. (2020). The Effect of Omega-3 Fatty Acids on Rheumatoid Arthritis. *Mediterranean journal of rheumatology*, 31(2), 190–194. <https://doi.org/10.31138/mjr.31.2.190>
- 57 Rajaei, E., Mowla, K., Ghorbani, A., Bahadoram, S., Bahadoram, M., & Dargahi-Malamir, M. (2015). The Effect of Omega-3 Fatty Acids in Patients With Active Rheumatoid Arthritis Receiving DMARDs Therapy: Double-Blind Randomized Controlled Trial. *Global journal of health science*, 8(7), 18–25. <https://doi.org/10.5539/gjhs.v8n7p18>
- 58 Leventhal, L. J., Boyce, E. G., & Zurier, R. B. (1993). Treatment of rheumatoid arthritis with gammalinolenic acid. *Annals of internal medicine*, 119(9), 867–873. <https://doi.org/10.7326/0003-4819-119-9-199311010-00001>
- 59 Salehi, B., Rescigno, A., Dettori, T., Calina, D., Docea, A. O., Singh, L., Cebeci, F., Özçelik, B., Bhia, M., Dowlati Beirami, A., Sharifi-Rad, J., Sharopov, F., Cho, W. C., & Martins, N. (2020). Avocado-Soybean Unsaponifiables: A Panoply of Potentialities to Be Exploited. *Biomolecules*, 10(1), 130. <https://doi.org/10.3390/biom10010130>
- 60 Cinelli, M., Guiducci, S., Del Rosso, A., Pignone, A., Del Rosso, M., Fibbi, G., Serrati, S., Gabrielli, A., Giacomelli, R., Piccardi, N., & Maticci Cerinic, M. (2006). Piascledine modulates the production of VEGF and TIMP-1 and reduces the invasiveness of rheumatoid arthritis synoviocytes. *Scandinavian journal of rheumatology*, 35(5), 346–350. <https://doi.org/10.1080/03009740600709865>
- 61 Butawan, M., Benjamin, R. L., & Bloomer, R. J. (2017). Methylsulfonylmethane: Applications and Safety of a Novel Dietary Supplement. *Nutrients*, 9(3), 290. <https://doi.org/10.3390/nu9030290>
- 62 Dube, S. R., Fairweather, D., Pearson, W. S., Felitti, V. J., Anda, R. F., & Croft, J. B. (2009). Cumulative childhood stress and autoimmune diseases in adults. *Psychosomatic medicine*, 71(2), 243–250. <https://doi.org/10.1097/PSY.0b013e3181907888>
- 63 Hassett, A. L., & Clauw, D. J. (2010). The role of stress in rheumatic diseases. *Arthritis research & therapy*, 12(3), 123. <https://doi.org/10.1186/ar3024>
- 64 Grabovac, I., Haider, S., Berner, C., Lamprecht, T., Fenzl, K. H., Erlacher, L., Quittan, M., & Dorner, T. E. (2018). Sleep Quality in Patients with Rheumatoid Arthritis and Associations with Pain, Disability, Disease Duration, and Activity. *Journal of clinical medicine*, 7(10), 336. <https://doi.org/10.3390/jcm7100336>

Heart Health Protocol

- Centers for Disease Control and Prevention. (2022, July 15). *Heart disease facts*. Centers for Disease Control and Prevention. Retrieved July 18, 2022, from <https://www.cdc.gov/heartdisease/facts.htm>
- Semeco, A. (2017, October 12). *9 benefits of coenzyme Q10 (coq10)*. Healthline. Retrieved July 28, 2022, from https://www.healthline.com/nutrition/coenzyme-q10#TOC_TITLE_HDR_1
- World Health Organization. (n.d.). *Cardiovascular diseases: Avoiding heart attacks and strokes*. World Health Organization. Retrieved August 3, 2022, from <https://www.who.int/news-room/questions-and-answers/item/cardiovascular-diseases-avoiding-heart-attacks-and-strokes>
- Guo, P., Kai, Q., Gao, J., Lian, Z. Q., Wu, C. M., Wu, C. A., & Zhu, H. B. (2010). Cordycepin prevents hyperlipidemia in hamsters fed a high-fat diet via activation of AMP-activated protein kinase. *Journal of pharmacological sciences*, 113(4), 395–403. <https://doi.org/10.1254/jphs.10041fp> <https://pubmed.ncbi.nlm.nih.gov/20724804/>
- Heshmat-Gahdarjani, K., Mashayekhlasi, N., Amerizadeh, A., Teimouri Jervevani, Z., & Sadeghi, M. (2020). Effect of fenugreek consumption on serum lipid profile: A systematic review and meta-analysis. *Phytotherapy Research*, 34(9), 2230–2245. <https://doi.org/10.1002/ptr.6690>
- Fenugreek uses, benefits & dosage - drugs.com herbal database*. Drugs.com. (n.d.). Retrieved July 25, 2022, from <https://www.drugs.com/npp/fenugreek.html>
- Wu, Y.-Z., Li, S.-Q., Zu, X.-G., Du, J., & Wang, F.-F. (2008). Ginkgo biloba extract improves coronary artery circulation in patients with coronary artery disease: Contribution of plasma nitric oxide and endothelin-1. *Phytotherapy Research*, 22(6), 734–739. <https://doi.org/10.1002/ptr.2335>
- Isah, T. (2015). Rethinking ginkgo biloba L.: Medicinal uses and conservation. *Pharmacognosy Reviews*, 9(18), 140. <https://doi.org/10.4103/0973-7847.162137>
- Diamond, B. J., & Bailey, M. R. (2013). Ginkgo biloba. *Psychiatric Clinics of North America*, 36(1), 73–83. <https://doi.org/10.1016/j.jpsc.2012.12.006>
- Hawthorn*. Mount Sinai Health System. (n.d.). Retrieved July 20, 2022, from <https://www.mountsinai.org/health-library/herb/hawthorn>
- Hawthorn - Tassell, M. C., Kingston, R., Gilroy, D., Lehane, M., & Furey, A. (2010). Hawthorn (*Crataegus* spp.) in the treatment of cardiovascular disease. *Pharmacognosy reviews*, 4(7), 32–41. <https://doi.org/10.4103/0973-7847.65324> <https://www.ncbi.nlm.nih.gov/labs/pmc/articles/PMC3249900/>
- Hawthorn*. Mount Sinai Health System. (n.d.). Retrieved July 20, 2022, from <https://www.mountsinai.org/health-library/herb/hawthorn>
- Da-Costa-Rocha, I., Bonnlaender, B., Sievers, H., Pischel, I., & Heinrich, M. (2014). *Hibiscus Sabdariffa* L. – a phytochemical and pharmacological review.

- 14 Najafpour Boushehri, S., Karimbeiki, R., Ghasempour, S., Ghalishourani, S. S., Pourmasoumi, M., Hadi, A., Mbabazi, M., pour, Z. K., Assarroudi, M., Mahmoodi, M., Khosravi, A., Mansour-Ghanaei, F., & Joukar, F. (2020). The efficacy of sour tea (*hibiscus sabdariffa* L.) on selected Cardiovascular disease risk factors: A systematic review and meta-analysis of randomized clinical trials. *Phytotherapy Research*, 34(2), 329–339. <https://doi.org/10.1002/ptr.6541>
- 15 Cohen MM. Tulsi - Ocimum sanctum: A herb for all reasons. *J Ayurveda Integr Med*. 2014 Oct-Dec;5(4):251-9. doi: 10.4103/0975-9476.146554. PMID: 25624701; PMCID: PMC4296439
- 16 Azhar I, Aftab K, Usmanghani K. Naturally occurring calcium channel blockers. *Hamdard Medicus*. 1995;38:5–16
- 17 Tabassum N, Ahmad F. Role of natural herbs in the treatment of hypertension. *Pharmacogn Rev*. 2011 Jan;5(9):30-40. doi: 10.4103/0973-7847.79097. PMID: 22096316; PMCID: PMC3210006
- 18 Wu, J.-Y., Siu, K.-C., & Geng, P. (2021). Bioactive ingredients and medicinal values of grifola frondosa (maitake). *Foods*, 10(1), 95. <https://doi.org/10.3390/foods10010095>
- 19 ADACHI, K. Y. O. K. O., NANBA, H. I. R. O. A. K. I., OTSUKA, M. A. S. A. M. I. C. H. I., & KURODA, H. I. S. A. T. O. R. A. (1988). Blood pressure-lowering activity present in the fruit body of grifola frondosa (Maitake). *i. Chemical and Pharmaceutical Bulletin*, 36(3), 1000–1006. <https://doi.org/10.1248/cpb.36.1000>
- 20 Indication-specific dosing for grifola frondosa (maitake), frequency-based adverse effects, comprehensive interactions, contraindications, pregnancy & lactation schedules, and cost information. (2019, September 28). Retrieved July 18, 2022, from <https://reference.medscape.com/drug/grifola-frondosa-maitake-344489>
- 21 Suveren, E., Baxter, G. F., Iskit, A. B., & Turker, A. U. (2017). Cardioprotective effects of Viscum album L. subsp. album (European Mistletoe) leaf extracts in myocardial ischemia and reperfusion. *Journal of Ethnopharmacology*, 209, 203–209. <https://doi.org/10.1016/j.jep.2017.07.010>
- 22 Mistletoe uses, benefits & dosage - drugs.com herbal database. Drugs.com. (n.d.). Retrieved July 20, 2022, from <https://www.drugs.com/npp/mistletoe.html>
- 23 Olive leaf. Memorial Sloan Kettering Cancer Center. (n.d.). Retrieved July 18, 2022, from <https://www.mskcc.org/cancer-care/integrative-medicine/herbs/olive-leaf>
- 24 Brennan, D. (2020, November 6). Olive leaf extract: *Health benefits, uses, dosage, and more*. WebMD. Retrieved July 20, 2022, from <https://www.webmd.com/diet/health-benefits-olive-leaf-extract#1>
- 25 Brennan, D. (2020, November 6). Olive leaf extract: *Health benefits, uses, dosage, and more*. WebMD. Retrieved July 20, 2022, from <https://www.webmd.com/diet/health-benefits-olive-leaf-extract#1>
- 26 Martínez-González, M. A., Gea, A., & Ruiz-Canela, M. (2019). The Mediterranean diet and Cardiovascular Health. *Circulation Research*, 124(5), 779–798. <https://doi.org/10.1161/circresaha.118.313348>
- 27 Martínez-González, M. A., Salas-Salvadó, J., Estruch, R., Corella, D., Fitó, M., & Ros, E. (2015). Benefits of the Mediterranean diet: Insights from the predimed study. *Progress in Cardiovascular Diseases*, 58(1), 50–60. <https://doi.org/10.1016/j.pcad.2015.04.003>
- 28 Mayo Foundation for Medical Education and Research. (2021, June 25). *How to make the Dash Diet Work For you*. Mayo Clinic. Retrieved July 25, 2022, from <https://www.mayoclinic.org/healthy-lifestyle/nutrition-and-healthy-eating/in-depth/dash-diet/art-20048456>
- 29 Kahn, J. K. (2016). The Whole Heart Solution: Halt heart disease now with the best alternative and Traditional Medicine. The Reader's Digest Association, Inc.
- 30 Just how healthy are pomegranates? www.heart.org. (2022, January 3). Retrieved August 7, 2022, from <https://www.heart.org/en/news/2021/09/09/just-how-healthy-are-pomegranates>
- 31 Asgary, S., Javanmard, S. H., & Zarfeshany, A. (2014). Potent health effects of pomegranate. *Advanced Biomedical Research*, 3(1), 100. <https://doi.org/10.4103/2277-9175.129371>
- 32 Sahebkar, A., Ferri, C., Giorgini, P., Bo, S., Nachtigal, P., & Grassi, D. (2017). Effects of pomegranate juice on blood pressure: A systematic review and meta-analysis of randomized controlled trials. *Pharmacological Research*, 115, 149–161. <https://doi.org/10.1016/j.phrs.2016.11.018>
- 33 Tian, D., & Meng, J. (2019). Exercise for prevention and relief of cardiovascular disease: Prognoses, mechanisms, and approaches. *Oxidative Medicine and Cellular Longevity*, 2019, 1–11. <https://doi.org/10.1155/2019/3756750>
- 34 Tian, D., & Meng, J. (2019). Exercise for prevention and relief of cardiovascular disease: Prognoses, mechanisms, and approaches. *Oxidative Medicine and Cellular Longevity*, 2019, 1–11. <https://doi.org/10.1155/2019/3756750>
- 35 Yeates, K., Lohfeld, L., Sleeth, J., Morales, F., Rajkotia, Y., & Ogedegbe, O. (2015). A global perspective on cardiovascular disease in vulnerable populations. *Canadian Journal of Cardiology*, 31(9), 1081–1093. <https://doi.org/10.1016/j.cjca.2015.06.035>
- 36 Murtagh, E. M., Murphy, M. H., & Boone-Heinonen, J. (2010). Walking: The first steps in cardiovascular disease prevention. *Current Opinion in Cardiology*, 25(5), 490–496. <https://doi.org/10.1097/hco.0b013e32833ce972>
- 37 Steptoe, A., & Kivimäki, M. (2012). Stress and cardiovascular disease. *Nature Reviews Cardiology*, 9(6), 360–370. <https://doi.org/10.1038/nrcardio.2012.45>
- 38 Dar, T., Radfar, A., Abohashem, S., Pittman, R. K., Tawakol, A., & Osborne, M. T. (2019). Psychosocial stress and cardiovascular disease. *Current Treatment Options in Cardiovascular Medicine*, 21(5). <https://doi.org/10.1007/s11936-019-0724-5>
- 39 Semeco, A. (2017, October 12). *9 benefits of coenzyme Q10 (coq10)*. Healthline. Retrieved July 28, 2022, from https://www.healthline.com/nutrition/coenzyme-q10#TOC_TITLE_HDR_1
- 40 Vaduganathan, M., Butler, J., Pitt, B., & Gheorghiad, M. (2015). Contemporary drug development in heart failure. *Circulation: Heart Failure*, 8(4), 826–831. <https://doi.org/10.1161/circheartfailure.115.002271>
- 41 Deichmann, R., Lavie, C., & Andrews, S. (2010). Coenzyme q10 and statin-induced mitochondrial dysfunction. *The Ochsner Journal*, 10(1), 16–21.
- 42 Mortensen, S. A., Rosenfeldt, F., Kumar, A., Dolliner, P., Filipiak, K. J., Pella, D., Alehagen, U., Steurer, G., & Littarru, G. P. (2014). The effect of coenzyme Q 10 on morbidity and mortality in chronic heart failure. *JACC: Heart Failure*, 2(6), 641–649. <https://doi.org/10.1016/j.jchf.2014.06.008>
- 43 Morisco, C., Trimarco, B., & Condorelli, M. (1993). Effect of coenzyme Q10 therapy in patients with congestive heart failure: A long-term Multicenter Randomized Study. *The Clinical Investigator*, 71(S8). <https://doi.org/10.1007/bf00226854>
- 44 DiNicolantonio, J. J., Bhutani, J., McCarty, M. F., & O'Keefe, J. H. (2015). Coenzyme Q10 for the treatment of heart failure: A review of the literature. *Open Heart*, 2(1). <https://doi.org/10.1136/openhrt-2015-000326>
- 45 U.S. National Library of Medicine. (n.d.). *Omega-3 fats - good for your heart: Medlineplus medical encyclopedia*. MedlinePlus. Retrieved August 7, 2022, from <https://medlineplus.gov/ency/patientinstructions/000767.htm#:~:text=Omega>
- 46 Kar S, Webl R. Fish oil supplementation & coronary artery disease: does it help? *Mo Med*. 2012 Mar-Apr;109(2):142-5. PMID: 22675796; PMCID: PMC6181727.
- 47 Swanson, D., Block, R., & Mousa, S. A. (2012). Omega-3 fatty acids EPA and DHA: Health benefits throughout life. *Advances in Nutrition*, 3(1), 1–7. <https://doi.org/10.3945/an.111.000893>
- 48 Mayo Foundation for Medical Education and Research. (2021, February 4). *L-arginine*. Mayo Clinic. Retrieved July 28, 2022, from <https://www.mayoclinic.org/drugs-supplements-l-arginine/art-20364681>
- 49 Tousoulis, D., Antoniadis, C., Tentolouris, C., Goumas, G., Stefanadis, C., & Toutouzias, P. (2002). L-arginine in cardiovascular disease: Dream or reality? *Vascular Medicine*, 7(3), 203–211. <https://doi.org/10.1191/1358863x02vm434ra>
- 50 Griffin, R. M. (n.d.). *Arginine (L-arginine) supplements: Uses, side effects, and more*. WebMD. Retrieved August 5, 2022, from <https://www.webmd.com/diet/supplement-guide-l-arginine>
- 51 Lopes Sakamoto, F., Metzker Pereira Ribeiro, R., Amador Bueno, A., & Oliveira Santos, H. (2019). Psychotropic effects of L-Theanine and its clinical properties: From the management of anxiety and stress to a potential use in schizophrenia. *Pharmacological Research*, 147, 104395. <https://doi.org/10.1016/j.phrs.2019.104395>
- 52 Adhikary, R., & Mandal, V. (2017). L-Theanine: A potential multifaceted natural bio-active amide as health supplement. *Asian Pacific Journal of Tropical Biomedicine*, 7(9), 842–848. <https://doi.org/10.1016/j.apjtb.2017.08.005>
- 53 Williams, J. L., Everett, J. M., D'Cunha, N. M., Sergi, D., Georgousopoulou, E. N., Keegan, R. J., McKune, A. J., Mellor, D. D., Anstice, N., & Naumovski, N. (2019). The effects of green tea amino acid L-theanine consumption on the ability to manage stress and anxiety levels: A systematic review. *Plant Foods for Human Nutrition*, 75(1), 12–23. <https://doi.org/10.1007/s11130-019-00771-5>
- 54 U.S. Department of Health and Human Services. (n.d.). *Office of dietary supplements - magnesium*. NIH Office of Dietary Supplements. Retrieved June 22, 2022, from <https://ods.od.nih.gov/factsheets/Magnesium-HealthProfessional/>
- 55 Tangvoraphonkchai, K., & Davenport, A. (2018). Magnesium and cardiovascular disease. *Advances in Chronic Kidney Disease*, 25(3), 251–260. <https://doi.org/10.1053/j.ackd.2018.02.010>
- 56 Houston, M. (2011). The role of magnesium in hypertension and cardiovascular disease. *The Journal of Clinical Hypertension*, 13(11), 843–847. <https://doi.org/10.1111/j.1751-7176.2011.00538.x>
- 57 Dibaba, D. T., Xun, P., Song, Y., Rosanoff, A., Shechter, M., & He, K. (2017). The effect of magnesium supplementation on blood pressure in individuals with insulin resistance, prediabetes, or Noncommunicable Chronic Diseases: A meta-analysis of randomized controlled trials. *The American Journal of Clinical Nutrition*. <https://doi.org/10.3945/ajcn.117.155291>
- 58 Serban, M. C., Sahebkar, A., Zanchetti, A., Mikhailidis, D. P., Howard, G., Antal, D., Andrica, F., Ahmed, A., Aronow, W. S., Muntner, P., Lip, G. Y., Graham, I., Wong, N., Rysz, J., & Banach, M. (2016). Effects of quercetin on blood pressure: A systematic review and meta-analysis of Randomized Controlled Trials. *Journal of the American Heart Association*, 5(7). <https://doi.org/10.1161/jaha.115.002713>
- 59 Patel, R. V., Mistry, B. M., Shinde, S. K., Syed, R., Singh, V., & Shin, H.-S. (2018). Therapeutic potential of quercetin as a cardiovascular agent. *European Journal of Medicinal Chemistry*, 155, 889–904. <https://doi.org/10.1016/j.ejmech.2018.06.053>
- 60 Ostadmohammadi, V., Milajerdi, A., Ayati, E., Kolahdooz, F., & Asemi, Z. (2019). Effects of quercetin supplementation on glycemic control among patients with metabolic syndrome and related disorders: A systematic review and meta-analysis of Randomized Controlled Trials. *Phytotherapy Research*, 33(5), 1330–1340. <https://doi.org/10.1002/ptr.6334>
- 61 Raman, R. (2020, July 1). *What is quercetin? benefits, foods, dosage, and side effects*. Healthline. Retrieved July 30, 2022, from <https://www.healthline.com/nutrition/quercetin#sources-dosage>
- 62 Moser, M., & Chun, O. (2016). Vitamin C and heart health: A review based on findings from Epidemiologic Studies. *International Journal of Molecular Sciences*, 17(8), 1328. <https://doi.org/10.3390/ijms17081328>
- 63 Oliveira, I. J., de Souza, V. V., Motta, V., & Da-Silva, S. L. (2014). Effects of oral vitamin C supplementation on anxiety in students: A double-blind, randomized, placebo-controlled trial. *Pakistan Journal of Biological Sciences*, 18(1), 11–18. <https://doi.org/10.1161/circheartfailure.115.002271>

- 64 Centers for Disease Control and Prevention. (2022, July 12). *Facts about hypertension*. Centers for Disease Control and Prevention. Retrieved July 18, 2022, from <https://www.cdc.gov/bloodpressure/facts.htm>
- 65 *Department of Health*. Physical Inactivity and Cardiovascular Disease. (n.d.). Retrieved August 8, 2022, from <https://www.health.ny.gov/diseases/chronic/cvd.htm>
- 66 Health, B. (2018, August 31). *How sitting too much can lead to heart disease*. Beaumont Health. Retrieved August 8, 2022, from <https://www.beaumont.org/health-wellness/blogs/how-sitting-too-much-can-lead-to-heart-disease>
- 67 Murtagh, E. M., Murphy, M. H., & Boone-Heinonen, J. (2010). Walking: The first steps in cardiovascular disease prevention. *Current Opinion in Cardiology*, 25(5), 490–496. <https://doi.org/10.1097/hco.0b013e32833ce972>
- 68 *Department of Health*. Physical Inactivity and Cardiovascular Disease. (n.d.). Retrieved August 8, 2022, from <https://www.health.ny.gov/diseases/chronic/cvd.htm>
- 69 Temple, N. (2018). Fat, sugar, whole grains and heart disease: 50 years of confusion. *Nutrients*, 10(1), 39. <https://doi.org/10.3390/nu10010039>
- 70 Livesey, G., & Livesey, H. (2019). Coronary heart disease and dietary carbohydrate, glycemic index, and glycemic load: Dose-response meta-analyses of prospective cohort studies. *Mayo Clinic Proceedings: Innovations, Quality & Outcomes*, 3(1), 52–69. <https://doi.org/10.1016/j.mayocpiqo.2018.12.007>
- 71 NHS. (n.d.). NHS choices. Retrieved August 3, 2022, from <https://www.nhs.uk/live-well/eat-well/how-to-eat-a-balanced-diet/what-are-processed-foods/>
- 72 Juul, F., Vaidean, G., & Parekh, N. (2021). Ultra-processed foods and cardiovascular diseases: Potential mechanisms of action. *Advances in Nutrition*, 12(5), 1673–1680. <https://doi.org/10.1093/advances/nmab049>
- 73 Steptoe, A., & Kivimäki, M. (2012). Stress and cardiovascular disease. *Nature Reviews Cardiology*, 9(6), 360–370. <https://doi.org/10.1038/nrcardio.2012.45>
- 74 Michos, E. D. (2021, August 8). *Sitting disease: How a sedentary lifestyle affects heart health*. Johns Hopkins Medicine. Retrieved August 8, 2022, from <https://www.hopkins-medicine.org/health/wellness-and-prevention/sitting-disease-how-a-sedentary-lifestyle-affects-heart-health>
- 75 *Smoking and heart disease*. Smoking and heart disease - Better Health Channel. (n.d.). Retrieved August 3, 2022, from <https://www.betterhealth.vic.gov.au/health/healthyliving/smoking-and-heart-disease>
- 76 U.S. Department of Health and Human Services. (n.d.). *Smoking and your heart - how smoking affects the heart and blood vessels*. National Heart Lung and Blood Institute. Retrieved August 3, 2022, from <https://www.nhlbi.nih.gov/health/heart/smoking>
- 77 Taylor, D. H., Hasselblad, V., Henley, S. J., Thun, M. J., & Sloan, F. A. (2002). Benefits of smoking cessation for longevity. *American Journal of Public Health*, 92(6), 990–996. <https://doi.org/10.2105/ajph.92.6.990>

Candida Protocol

- 1 <https://hms.harvard.edu/news/combating-candida-albicans>
- 2 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3163673/>
- 3 <https://www.cdc.gov/fungal/diseases/candidiasis/index.html>
- 4 <https://www.cdc.gov/fungal/diseases/candidiasis/invasive/index.html>
- 5 <https://www.cdc.gov/fungal/diseases/candidiasis/genital/index.html>
- 6 <https://www.cdc.gov/fungal/diseases/candidiasis/thrush/index.html>
- 7 <https://www.ei-resource.org/articles/candida-and-gut-dysbiosis-articles/the-candida-aldehyde-detox-pathway-and-the-molybdenum-connection/>
- 8 <https://pubmed.ncbi.nlm.nih.gov/34661898/>
- 9 <https://pubmed.ncbi.nlm.nih.gov/31315227/>
- 10 Kunyeyt, L., Kurrey, N. K., Anu-Appaiah, K. A., & Rao, R. P. (2019). Probiotic Yeasts Inhibit Virulence of Non-albicans Candida Species. *mBio*, 10(5), e02307-19. <https://doi.org/10.1128/mBio.02307-19>
- 11 Wu, Y., Hu, S., Wu, C., Gu, F., & Yang, Y. (2022). Probiotics: Potential Novel Therapeutics Against Fungal Infections. *Frontiers in cellular and infection microbiology*, 11, 793419. <https://doi.org/10.3389/fcimb.2021.793419>
- 12 <https://pubmed.ncbi.nlm.nih.gov/30166063/>
- 13 Donohoe, D. R., Garge, N., Zhang, X., Sun, W., O'Connell, T. M., Bunger, M. K., & Bultman, S. J. (2011). The microbiome and butyrate regulate energy metabolism and autophagy in the mammalian colon. *Cell metabolism*, 13(5), 517–526. <https://doi.org/10.1016/j.cmet.2011.02.018> <https://pubmed.ncbi.nlm.nih.gov/21531334/>
- 14 <https://pubmed.ncbi.nlm.nih.gov/31315227/>
- 15 Talattof, Z., Azad, A., Zahed, M., & Shahradian, N. (2018). Antifungal Activity of Xylitol against Candida albicans: An in vitro Study. *The journal of contemporary dental practice*, 19(2), 125–129. <https://doi.org/10.5005/jp-journals-10024-2225>
- 16 Pizzo, G., Giuliana, G., Milici, M. E., & Giangreco, R. (2000). Effect of dietary carbohydrates on the in vitro epithelial adhesion of Candida albicans, Candida tropicalis, and Candida krusei. *The new microbiologica*, 23(1), 63–71
- 17 Yagnik, D., Serafin, V., & J Shah, A. (2018). Antimicrobial activity of apple cider vinegar against Escherichia coli, Staphylococcus aureus and Candida albicans; downregulating cytokine and microbial protein expression. *Scientific reports*, 8(1), 1732. <https://doi.org/10.1038/s41598-017-18618-x>
- 18 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3768360/>
- 19 Ogbolu, D. O., Oni, A. A., Daini, O. A., & Oloko, A. P. (2007). In vitro antimicrobial properties of coconut oil on Candida species in Ibadan, Nigeria. *Journal of medicinal food*, 10(2), 384–387. <https://doi.org/10.1089/jmf.2006.1209>

- 20 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5748643/>
- 21 Behmanesh, F., Pasha, H., Sefidgar, A. A., Taghizadeh, M., Moghadamnia, A. A., Adib Rad, H., & Shirkhani, L. (2015). Antifungal Effect of Lavender Essential Oil (Lavandula angustifolia) and Clotrimazole on Candida albicans: An In Vitro Study. *Scientifica*, 2015, 261397. <https://doi.org/10.1155/2015/261397>
- 22 Nenoff, P., Haustein, U. F., & Brandt, W. (1996). Antifungal activity of the essential oil of Melaleuca alternifolia (tea tree oil) against pathogenic fungi in vitro. *Skin pharmacology: the official journal of the Skin Pharmacology Society*, 9(6), 388–394. <https://doi.org/10.1159/000211450>
- 23 Bharti, S., Nayaka, S., & Kumar, R. (2022). EVALUATION OF SOME TRADITIONAL THERAPEUTIC PROPERTIES OF USNEA LONGISSIMA (ASCOMYCOTA, LICHENIZED FUNGI): ANTIMICROBIAL, ANTIQUORUM AND ANTIOXIDANT. *Journal of Microbiology, Biotechnology and Food Sciences*, 11(4), e3163. <https://doi.org/10.55251/jmbfs.3163> <https://office2.jmbfs.org/index.php/JMBFS/article/view/3163>
- 24 Jaramillo-Ordoñez, Cristhina Estefanía. (2020). In vitro antifungal and antibacterial activity of the ethanolic extract of Usnea laevis against Candida albicans, Staphylococcus aureus and Pseudomonas aeruginosa. *Revista Medica Herediana*, 31 (3), 169-174. <https://dx.doi.org/10.2453/rmh.v31i3.3806> http://www.scielo.org.pe/scielo.php?pid=S1018-130X2020000300169&script=sci_arttext
- 25 Popovici V, Bucur L, Calcan SI, Cuculea EI, Costache T, Rambu D, Schröder V, Gîrd CE, Gherghel D, Vochita G, Caraiane A, Badea V. Elemental Analysis and In Vitro Evaluation of Antibacterial and Antifungal Activities of Usnea barbata (L.) Weber ex F.H. Wigg from Călimani Mountains, Romania. *Plants*. 2022; 11(1):32. <https://doi.org/10.3390/plants11010032> <https://www.mdpi.com/2223-7747/11/1/32/hhtm>
- 26 <https://pubmed.ncbi.nlm.nih.gov/18553017/>
- 27 Yamashita, M., Sawano, J., Umeda, R., Tatsumi, A., Kumeda, Y., & Iida, A. (2021). Structure-Activity Relationship Studies of Antimicrobial Naphthoquinones Derived from Constituents of Tabebuia avellanaeae. *Chemical & pharmaceutical bulletin*, 69(7), 661–673. <https://doi.org/10.1248/cpb.c21-00208>
- 28 Gómez Castellanos, J. R., Prieto, J. M., & Heinrich, M. (2009). Red Lapacho (Tabebuia impetiginosa)-a global ethnopharmacological commodity?. *Journal of ethnopharmacology*, 121(1), 1–13. <https://doi.org/10.1016/j.jep.2008.10.004>
- 29 Bhardwaj A, Gupta P, Kumar N, Mishra J, Kumar A, Rakhee R, Misra K. Lingzhi or Reishi Medicinal Mushroom, Ganoderma lucidum (Agaricomycetes), Inhibits Candida Biofilms: A Metabolomic Approach. *Int J Med Mushrooms*. 2017;19(8):685-696. doi: 10.1615/IntJMedMushrooms.2017021225. PMID: 29199568. <https://pubmed.ncbi.nlm.nih.gov/29199568/>
- 30 BHAT, Z. A., WANI, A. H., WAR, J. M., & BHAT, M. Y. (2021). Major bioactive properties of ganoderma polysaccharides: A review. *Asian Journal of Pharmaceutical and Clinical Research*, 11-24. <https://www.researchgate.net>
- 31 <https://www.ncbi.nlm.nih.gov/books/NBK92757/>
- 32 Knežević A, Stajić M, Sofrenić I, Stanojković T, Milovanović I, Tešević V, et al. (2018) Antioxidative, antifungal, cytotoxic and antineurodegenerative activity of selected Trametes species from Serbia. *PLoS ONE* 13(8): e0203064
- 33 SK, D., Sridhar, K. R., & Gupta, M. K. Hericium erinaceus-A Rich Source of Diverse Bioactive Metabolites. https://www.fungalbiotec.org/pdf/FGBT_1_2_2.pdf
- 34 Song X, Gaascht F, Schmidt-Dannert C, Salomon CE. 2020 – Discovery of antifungal and biofilm preventative compounds from mycelial cultures of a unique North American Hericium sp. fungus. *Molecules* 25 (4), p 963. doi: 10.3390/molecules25040963
- 35 Xie, Y., Liu, X., & Zhou, P. (2020). In vitro Antifungal Effects of Berberine Against Candida spp. In Planktonic and Biofilm Conditions. *Drug design, development and therapy*, 14, 87–101. <https://doi.org/10.2147/DDDT.S230857> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6957002>
- 36 Chen, W.-C., Liou, S.-S., Tzeng, T.-F., Lee, S. L., and Liu, I. M. (2012). Wound repair and anti-inflammatory potential of Lonicera japonica in excision wound-induced rats. *BMC Complement. Altern. Med.* 12:226. doi: 10.1186/1472-6882-12-226
- 37 Xing, X., Liao, Z., Tan, F., Zhu, Z., Jiang, Y., & Cao, Y. (2019). Effect of Nicotinamide Against Candida albicans. *Frontiers in microbiology*, 10, 595. <https://doi.org/10.3389/fmicb.2019.00595>
- 38 Omura, Y., O'Young, B., Jones, M., Pallos, A., Duvvi, H., & Shimotsuura, Y. (2011). Caprylic acid in the effective treatment of intractable medical problems of frequent urination, incontinence, chronic upper respiratory infection, root canal tooth infection, ALS, etc., caused by asbestos & mixed infections of Candida albicans, Helicobacter pylori & cytomegalovirus with or without other microorganisms & mercury. *Acupuncture & electro-therapeutics research*, 36(1-2), 19–64. <https://doi.org/10.3727/036012911803860886>
- 39 Bae, Y. S., & Rhee, M. S. (2019). Short-Term Antifungal Treatments of Caprylic Acid with Carvacrol or Thymol Induce Synergistic 6-Log Reduction of Pathogenic Candida albicans by Cell Membrane Disruption and Efflux Pump Inhibition. *Cellular physiology and biochemistry: international journal of experimental cellular physiology, biochemistry, and pharmacology*, 53(2), 285–300. <https://doi.org/10.33594/000000139>
- 40 Ho, K. V., Lei, Z., Sumner, L. W., Coggeshall, M. V., Hsieh, H. Y., Stewart, G. C., & Lin, C. H. (2018). Identifying Antibacterial Compounds in Black Walnuts (Juglans nigra) Using a Metabolomics Approach. *Metabolites*, 8(4), 58. <https://doi.org/10.3390/metabo8040058>
- 41 Yun, J., Lee, H., Ko, H. J., Woo, E. R., & Lee, D. G. (2015). Fungicidal effect of isoquercitrin via inducing membrane disturbance. *Biochimica et biophysica acta*, 1848(2), 695–701. <https://doi.org/10.1016/j.bbame.2014.11.019>
- 42 Muruges, J., Annigeri, R. G., Mangala, G. K., Mythily, P. H., & Chandrakala, J. (2019). Evaluation of the antifungal efficacy of different concentrations of Curcuma longa on Candida albicans: An in vitro study. *Journal of oral and maxillofacial pathology*: JOMFP, 23(2), 305. https://doi.org/10.4103/jomfp.JOMFP.200_18

- 43 <https://pubmed.ncbi.nlm.nih.gov/33827515/>
- 44 Wermig-Morgan, Julia. (2020). Elderberry is anti-bacterial, anti-viral and modulates the immune system: anti-bacterial, anti-viral and immunomodulatory non-clinical (in-vitro) effects of elderberry fruit and flowers (*Sambucus nigra*): a systematic review
- 45 Pereira, V. V., Pereira, N. R., Pereira, R., Duarte, L. P., Takahashi, J. A., & Silva, R. R. (2022). Synthesis and Antimicrobial Activity of Ursolic Acid Ester Derivatives. *Chemistry & biodiversity*, 19(1), e202100566. <https://doi.org/10.1002/cbdv.202100566>
- 46 Gleńsk, M., Czapirńska, E., Woźniak, M., Ceremuga, I., Włodarczyk, M., Terlecki, G., Ziółkowski, P., & Seweryn, E. (2017). Triterpenoid Acids as Important Antiproliferative Constituents of European Elderberry Fruits. *Nutrition and cancer*, 69(4), 643–651. <https://doi.org/10.1080/01635581.2017.1295085>
- 47 <https://www.ncbi.nlm.nih.gov/books/NBK92757/>
- 48 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5485961/>
- 49 <https://pubmed.ncbi.nlm.nih.gov/17651080/>
- 50 <https://pubmed.ncbi.nlm.nih.gov/25012787/>
- 51 <https://pubmed.ncbi.nlm.nih.gov/32824103>
- 52 <https://www.msn.com/en-us/health/medical/chemicals-in-shampoo-and-makeup-are-linked-to-early-death-study-finds/ar-AApqwT2>
- 53 <https://abcnews.go.com/Health/wireStory/study-half-us-cosmetics-toxic-chemicals-78298811>
- 54 <https://www.hsph.harvard.edu/news/features/fluoride-childrens-health-grand-jean-choi/>
- 55 <https://pubmed.ncbi.nlm.nih.gov/28913736/>
- 56 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7068600/>
- 57 <https://scholarworks.unt.edu/cgi/viewcontent.cgi?article=1079&context=utpp>
- 58 <https://pubmed.ncbi.nlm.nih.gov/27995346/>
- 59 <https://www.washingtontimes.com/news/2019/aug/20/exercising-out-doors-will-reduce-more-stress-indoor/>
- 60 de Oliveira Santos, G. C., Vasconcelos, C. C., Lopes, A., de Sousa Cartágenes, M., Filho, A., do Nascimento, F., Ramos, R. M., Pires, E., de Andrade, M. S., Rocha, F., & de Andrade Monteiro, C. (2018). Candida Infections and Therapeutic Strategies: Mechanisms of Action for Traditional and Alternative Agents. *Frontiers in microbiology*, 9, 1351. <https://doi.org/10.3389/fmicb.2018.01351>

Food Poisoning Protocol

- 1 Metreveli, M., Bulia, S., Shalamberidze, I., Tevzadze, L., Tsanova, S., Goenaga, J. C., Stingl, K., & Imnadze, P. (2022). Campylobacteriosis, Shigellosis and Salmonellosis in Hospitalized Children with Acute Inflammatory Diarrhea in Georgia. *Pathogens (Basel, Switzerland)*, 11(2), 232. <https://doi.org/10.3390/pathogens11020232>
- 2 Vanhauwaert, E., Matthys, C., Verdonck, L., & De Preter, V. (2015). Low-residue and low-fiber diets in gastrointestinal disease management. *Advances in nutrition (Bethesda, Md.)*, 6(6), 820–827. <https://doi.org/10.3945/an.115.009688>
- 3 Weir, S. B. S. & Akhondi, H. (2021). Bland Diet. In *Statpearls [Internet]*. Treasure Island (FL): StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK538142/>
- 4 United States Department of Agriculture. (2016, December 2). *Food Safety and Inspection Service*. Cleanliness Helps Prevent Foodborne Illness | Food Safety and Inspection Service. Retrieved July 7, 2022, from <https://www.fsis.usda.gov/food-safety/safe-food-handling-and-preparation/food-safety-basics/cleanliness-helps-prevent>
- 5 United States Food Safety. (2022, June 23). *Cook to a safe minimum internal temperature*. FoodSafety.gov. Retrieved July 7, 2022, from <https://www.foodsafety.gov/food-safety-charts/safe-minimum-internal-temperatures>
- 6 United States Department of Agriculture. (2020, July 31). *Leftovers and Food Safety*. USDA Food Safety and Inspection Service. Retrieved July 7, 2022, from https://www.fsis.usda.gov/food-safety/safe-food-handling-and-preparation/food-safety-basics/leftovers-and-food-safety#_Cool
- 7 Shaheen, N. A., Alqahtani, A. A., Assiri, H., Alkhodair, R., & Hussein, M. A. (2018). Public knowledge of dehydration and fluid intake practices: variation by participants' characteristics. *BMC public health*, 18(1), 1346. <https://doi.org/10.1186/s12889-018-6252-5>
- 8 <https://illinoiscancericare.com/news/electrolytedrinkrecipe/>
- 9 Chyka, P. A., Seger, D., Krenzelok, E. P., Vale, J. A., American Academy of Clinical Toxicology, & European Association of Poisons Centres and Clinical Toxicologists (2005). Position paper: Single-dose activated charcoal. *Clinical toxicology (Philadelphia, Pa.)*, 43(2), 61–87. <https://doi.org/10.1081/clt-200051867>
- 10 Zellner, T., Prasa, D., Färber, E., Hoffmann-Walbeck, P., Genser, D., & Eyer, F. (2019). The Use of Activated Charcoal to Treat Intoxications. *Deutsches Arzteblatt international*, 116(18), 311–317. <https://doi.org/10.3238/arztebl.2019.0311>
- 11 Zellner, T., Prasa, D., Färber, E., Hoffmann-Walbeck, P., Genser, D., & Eyer, F. (2019). The Use of Activated Charcoal to Treat Intoxications. *Deutsches Arzteblatt international*, 116(18), 311–317. <https://doi.org/10.3238/arztebl.2019.0311>
- 12 Moosavi M. (2017). Bentonite Clay as a Natural Remedy: A Brief Review. *Iranian journal of public health*, 46(9), 1176–1183
- 13 Damrau F. (1961). The value of bentonite for diarrhea. *The Medical annals of the District of Columbia*, 30, 326–328
- 14 U.S. Food and Drug Administration. (2016, March 23). *FDA warns consumers not to use "Best bentonite clay"*. FDA.gov. Retrieved July 7, 2022, from <https://www.fda.gov/drugs/drug-safety-and-availability/fda-warns-consumers-not-use-best-bentonite-clay>
- 15 Billoo, A. G., Memon, M. A., Khaskheli, S. A., Murtaza, G., Iqbal, K., Saeed Shekhan, M., & Siddiqi, A. Q. (2006). Role of a probiotic (*Saccharomyces boulardii*)

- in management and prevention of diarrhoea. *World journal of gastroenterology*, 12(28), 4557–4560. <https://doi.org/10.3748/wjg.v12.i28.4557>
- 16 Isolauri E. (2003). Probiotics for infectious diarrhoea. *Gut*, 52(3), 436–437. <https://doi.org/10.1136/gut.52.3.436>
- 17 Lacombe, G., & St-Onge, M. (2016). Towards evidence-based emergency medicine: best BETs from the Manchester Royal Infirmary. BET 1: Silibinin in suspected amatoxin-containing mushroom poisoning. *Emergency medicine journal: EMJ*, 33(1), 76–77. <https://doi.org/10.1136/emmermed-2015-205558.1>
- 18 Mengs, U., Pohl, R. T., & Mitchell, T. (2012). Legalon® SIL: the antidote of choice in patients with acute hepatotoxicity from amatoxin poisoning. *Current pharmaceutical biotechnology*, 13(10), 1964–1970. <https://doi.org/10.2174/138920112802273353>
- 19 Mohr, C., Jensen, C., Padden, N., Besel, J. M., & Brant, J. M. (2021). Peppermint Essential Oil for Nausea and Vomiting in Hospitalized Patients: Incorporating Holistic Patient Decision Making Into the Research Design. *Journal of holistic nursing: official journal of the American Holistic Nurses' Association*, 39(2), 126–134. <https://doi.org/10.1177/0898010120961579>
- 20 Maghami, M., Afazel, M. R., Azizi-Fini, I., & Maghami, M. (2020). The effect of aromatherapy with peppermint essential oil on nausea and vomiting after cardiac surgery: A randomized clinical trial. *Complementary therapies in clinical practice*, 40, 101199. <https://doi.org/10.1016/j.ctcp.2020.101199>
- 21 Jafarimanesh, H., Akbari, M., Hoseinian, R., Zarei, M., & Harorani, M. (2020). The Effect of Peppermint (*Mentha piperita*) Extract on the Severity of Nausea, Vomiting and Anorexia in Patients with Breast Cancer Undergoing Chemotherapy: A Randomized Controlled Trial. *Integrative cancer therapies*, 19, 1534735420967084. <https://doi.org/10.1177/1534735420967084> Format:
- 22 McKay, D. L., & Blumberg, J. B. (2006). A review of the bioactivity and potential health benefits of peppermint tea (*Mentha piperita* L.). *Phytotherapy research: PTR*, 20(8), 619–633. <https://doi.org/10.1002/ptr.1936>
- 23 Lete, I., & Allué, J. (2016). The Effectiveness of Ginger in the Prevention of Nausea and Vomiting during Pregnancy and Chemotherapy. *Integrative medicine insights*, 11, 11–17. <https://doi.org/10.4137/IMI.S36273>
- 24 Mostafa, A. A., Al-Askar, A. A., Almaary, K. S., Dawoud, T. M., Sholkamy, E. N., & Bakri, M. M. (2018). Antimicrobial activity of some plant extracts against bacterial strains causing food poisoning diseases. *Saudi journal of biological sciences*, 25(2), 361–366. <https://doi.org/10.1016/j.sjbs.2017.02.004>
- 25 Amarowicz, R., Dykes, G. A., & Pegg, R. B. (2008). Antibacterial activity of tannin constituents from *Phaseolus vulgaris*, *Fagopyrum esculentum*, *Corylus avellana* and *Juglans nigra*. *Fitoterapia*, 79(3), 217–219. <https://doi.org/10.1016/j.fitote.2007.11.019>
- 26 Cheraghali, F., Shojaee-Aliabadi, S., Hosseini, S. M., Mirmoghaddae, L., Mortazavian, A. M., Ghanati, K., Abedi, A. S., & Moslemi, M. (2018). Characterization of Microcapsule Containing Walnut (*Juglans regia* L.) Green Husk Extract as Preventive Antioxidant and Antimicrobial Agent. *International journal of preventive medicine*, 9, 101. https://doi.org/10.4103/ijpvm.IJPVM_308_18
- 27 Su, P. W., Yang, C. H., Yang, J. F., Su, P. Y., & Chuang, L. Y. (2015). Antibacterial Activities and Antibacterial Mechanism of *Polygonum cuspidatum* Extracts against Nosocomial Drug-Resistant Pathogens. *Molecules (Basel, Switzerland)*, 20(6), 11119–11130. <https://doi.org/10.3390/molecules200611119>
- 28 Heimesaat, M. M., Mousavi, S., Weschka, D., & Bereswill, S. (2021). Garlic Essential Oil as Promising Option for the Treatment of Acute Campylobacteriosis-Results from a Preclinical Placebo-Controlled Intervention Study. *Microorganisms*, 9(6), 1140. <https://doi.org/10.3390/microorganisms9061140>
- 29 Lu, X., Rasco, B. A., Jabal, J. M., Aston, D. E., Lin, M., & Konkel, M. E. (2011). Investigating antibacterial effects of garlic (*Allium sativum*) concentrate and garlic-derived organosulfur compounds on *Campylobacter jejuni* by using Fourier transform infrared spectroscopy, Raman spectroscopy, and electron microscopy. *Applied and environmental microbiology*, 77(15), 5257–5269. <https://doi.org/10.1128/AEM.02845-10>
- 30 Chowdhury, A. K., Ahsan, M., Islam, S. N., & Ahmed, Z. U. (1991). Efficacy of aqueous extract of garlic & allicin in experimental shigellosis in rabbits. *The Indian journal of medical research*, 93, 33–36
- 31 Tavakoli, H. R., Mashak, Z., Moradi, B., & Sodagari, H. R. (2015). Antimicrobial Activities of the Combined Use of Cuminum Cuminum L. Essential Oil, Nisin and Storage Temperature Against *Salmonella typhimurium* and *Staphylococcus aureus* In Vitro. *Jundishapur journal of microbiology*, 8(4), e24838. [https://doi.org/10.5812/jjm.8\(4\)2015.24838](https://doi.org/10.5812/jjm.8(4)2015.24838)
- 32 Mostafa, A. A., Al-Askar, A. A., Almaary, K. S., Dawoud, T. M., Sholkamy, E. N., & Bakri, M. M. (2018). Antimicrobial activity of some plant extracts against bacterial strains causing food poisoning diseases. *Saudi journal of biological sciences*, 25(2), 361–366. <https://doi.org/10.1016/j.sjbs.2017.02.004>
- 33 Silva, N., Alves, S., Gonçalves, A., Amaral, J. S., & Poeta, P. (2013). Antimicrobial activity of essential oils from Mediterranean aromatic plants against several foodborne and spoilage bacteria. *Food science and technology international = Ciencia y tecnologia de los alimentos internacional*, 19(6), 503–510. <https://doi.org/10.1177/1082013212442198>
- 34 Holtz, L. R., Neill, M. A., & Tarr, P. I. (2009). Acute bloody diarrhea: a medical emergency for patients of all ages. *Gastroenterology*, 136(6), 1887–1898. <https://doi.org/10.1053/j.gastro.2009.02.059>

Gerd and Acid Reflux Protocol

- 1 Mittal, R.K. & Goyal, R.K. (2006). Sphincter mechanisms at the lower end of the esophagus. *GI Motility Online*, www.GIMotilityonline.com; doi:10.1038/gimo14, 2006
- 2 Jarosz, M., & Taraszewska, A. (2014). Risk factors for gastroesophageal reflux disease: the role of diet. *Przegląd gastroenterologiczny*, 9(5), 297–301. <https://doi.org/10.5114/pg.2014.46166>

- 3 Morozov, S., Isakov, V., & Konovalova, M. (2018). Fiber-enriched diet helps to control symptoms and improves esophageal motility in patients with non-erosive gastroesophageal reflux disease. *World journal of gastroenterology*, 24(21), 2291–2299. <https://doi.org/10.3748/wjg.v24.i21.2291>
- 4 Jarosz, M., & Taraszewska, A. (2014). Risk factors for gastroesophageal reflux disease: the role of diet. *Przegląd gastroenterologiczny*, 9(5), 297–301. <https://doi.org/10.5114/pg.2014.46166>
- 5 Song, J. H., Chung, S. J., Lee, J. H., Kim, Y. H., Chang, D. K., Son, H. J., Kim, J. J., Rhee, J. C., & Rhee, P. L. (2011). Relationship between gastroesophageal reflux symptoms and dietary factors in Korea. *Journal of neurogastroenterology and motility*, 17(1), 54–60. <https://doi.org/10.5056/jnm.2011.17.1.54>
- 6 Iwakiri, K., Kobayashi, M., Kotoyori, M., Yamada, H., Sugiura, T., & Nakagawa, Y. (1996). Relationship between postprandial esophageal acid exposure and meal volume and fat content. *Digestive diseases and sciences*, 41(5), 926–930. <https://doi.org/10.1007/BF02091532>
- 7 Holloway, R. H., Lyrenas, E., Ireland, A., & Dent, J. (1997). Effect of intraduodenal fat on lower oesophageal sphincter function and gastro-oesophageal reflux. *Gut*, 40(4), 449–453. <https://doi.org/10.1136/gut.40.4.449>
- 8 Śmiechowska M. & Cugowska M. (2011). Rola żywności i żywienia w chorobie reflowej [Polish] *Bromat Chem Toksykol.* 3, 298–304
- 9 Rodriguez-Stanley, S., Collings, K. L., Robinson, M., Owen, W., & Miner, P. B., Jr (2000). The effects of capsaicin on reflux, gastric emptying and dyspepsia. *Alimentary pharmacology & therapeutics*, 14(1), 129–134. <https://doi.org/10.1046/j.1365-2036.2000.00682.x>
- 10 Jarosz, M., & Taraszewska, A. (2014). Risk factors for gastroesophageal reflux disease: the role of diet. *Przegląd gastroenterologiczny*, 9(5), 297–301. <https://doi.org/10.5114/pg.2014.46166>
- 11 Morales-González, J. A., Madrigal-Bujaidar, E., Sánchez-Gutiérrez, M., Izquierdo-Vega, J. A., Valadez-Vega, M., Álvarez-González, I., Morales-González, Á., & Madrigal-Santillán, E. (2019). Garlic (*Allium sativum* L.): A Brief Review of Its Antigenotoxic Effects. *Foods (Basel, Switzerland)*, 8(8), 343. <https://doi.org/10.3390/foods8080343>
- 12 Allen, M. L., Mellow, M. H., Robinson, M. G., & Orr, W. C. (1990). The effect of raw onions on acid reflux and reflux symptoms. *The American journal of gastroenterology*, 85(4), 377–380
- 13 Dibley, L. B., Norton, C., & Jones, R. (2010). Don't eat tomatoes: patient's self-reported experiences of causes of symptoms in gastro-oesophageal reflux disease. *Family practice*, 27(4), 410–417. <https://doi.org/10.1093/fampra/cm020>
- 14 Surdea-Blaga, T., Negrutiu, D. E., Palage, M., & Dumitrascu, D. L. (2019). Food and Gastroesophageal Reflux Disease. *Current medical chemistry*, 26(19), 3497–3511. <https://doi.org/10.2174/0929867324666170515123807>
- 15 Song, J. H., Chung, S. J., Lee, J. H., Kim, Y. H., Chang, D. K., Son, H. J., Kim, J. J., Rhee, J. C., & Rhee, P. L. (2011). Relationship between gastroesophageal reflux symptoms and dietary factors in Korea. *Journal of neurogastroenterology and motility*, 17(1), 54–60. <https://doi.org/10.5056/jnm.2011.17.1.54>
- 16 Johnson, T., Gerson, L., Hershcovici, T., Stave, C., & Fass, R. (2010). Systematic review: the effects of carbonated beverages on gastro-oesophageal reflux disease. *Alimentary pharmacology & therapeutics*, 31(6), 607–614. <https://doi.org/10.1111/j.1365-2036.2010.04232.x>
- 17 Jarosz, M., & Taraszewska, A. (2014). Risk factors for gastroesophageal reflux disease: the role of diet. *Przegląd gastroenterologiczny*, 9(5), 297–301. <https://doi.org/10.5114/pg.2014.46166>
- 18 Morozov S, Isakov V, Konovalova M. Fiber-enriched diet helps to control symptoms and improves esophageal motility in patients with non-erosive gastroesophageal reflux disease. *World J Gastroenterol*. 2018;24(21):2291-2299. doi:10.3748/wjg.v24.i21.229. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5989243/>
- 19 Karim, S., Jafri, W., Faryal, A., Majid, S., Salih, M., Jafri, F., Hamid, S., Shah, H. A., Nawaz, Z., & Tariq, U. (2011). Regular post dinner walk; can be a useful lifestyle modification for gastroesophageal reflux. *JPM. The Journal of the Pakistan Medical Association*, 61(6), 526–530.
- 20 Taraszewska A. (2021). Risk factors for gastroesophageal reflux disease symptoms related to lifestyle and diet. *Roczniki Panstwowego Zakladu Higieny*, 72(1), 21–28. <https://doi.org/10.32394/rpzh.2021.0145>
- 21 Mendes-Filho, A. M., Moraes-Filho, J. P., Nasi, A., Eisig, J. N., Rodrigues, T. N., Barbutti, R. C., Campos, J. M., & Chinzon, D. (2014). Influence of exercise testing in gastroesophageal reflux in patients with gastroesophageal reflux disease. *Arquivos brasileiros de cirurgia digestiva: ABCD = Brazilian archives of digestive surgery*, 27(1), 3–8. <https://doi.org/10.1590/s0102-67202014000100002>
- 22 Karim, S., Jafri, W., Faryal, A., Majid, S., Salih, M., Jafri, F., Hamid, S., Shah, H. A., Nawaz, Z., & Tariq, U. (2011). Regular post dinner walk; can be a useful lifestyle modification for gastroesophageal reflux. *JPM. The Journal of the Pakistan Medical Association*, 61(6), 526–530.
- 23 Fujiwara, Y., Machida, A., Watanabe, Y., Shiba, M., Tominaga, K., Watanabe, T., Oshitani, N., Higuchi, K., & Arakawa, T. (2005). Association between dinner-to-bed time and gastro-esophageal reflux disease. *The American journal of gastroenterology*, 100(12), 2633–2636. <https://doi.org/10.1111/j.1572-0241.2005.00354.x>
- 24 Kaltenbach, T., Crockett, S., & Gerson, L. B. (2006). Are lifestyle measures effective in patients with gastroesophageal reflux disease? An evidence-based approach. *Archives of internal medicine*, 166(9), 965–971. <https://doi.org/10.1001/archinte.166.9.965>
- 25 El-Serag H. (2008). The association between obesity and GERD: a review of the epidemiological evidence. *Digestive diseases and sciences*, 53(9), 2307–2312. <https://doi.org/10.1007/s10620-008-0413-9>
- 26 Mitchell, D. R., Derakhshan, M. H., Wirz, A. A., Ballantyne, S. A., & McColl, K. (2017). Abdominal Compression by Waist Belt Aggravates Gastroesophageal Reflux, Primarily by Impairing Esophageal Clearance. *Gastroenterology*, 152(8), 1881–1888. <https://doi.org/10.1053/j.gastro.2017.02.036>
- 27 Song, J. H., Chung, S. J., Lee, J. H., Kim, Y. H., Chang, D. K., Son, H. J., Kim, J. J., Rhee, J. C., & Rhee, P. L. (2011). Relationship between gastroesophageal reflux symptoms and dietary factors in Korea. *Journal of neurogastroenterology and motility*, 17(1), 54–60. <https://doi.org/10.5056/jnm.2011.17.1.54>
- 28 Pan, J., Cen, L., Chen, W., Yu, C., Li, Y., & Shen, Z. (2019). Alcohol Consumption and the Risk of Gastroesophageal Reflux Disease: A Systematic Review and Meta-analysis. *Alcohol and alcoholism (Oxford, Oxfordshire)*, 54(1), 62–69. <https://doi.org/10.1093/alcalc/agy063>
- 29 Kahrilas, P. J., & Gupta, R. R. (1990). Mechanisms of acid reflux associated with cigarette smoking. *Gut*, 31(1), 4–10. <https://doi.org/10.1136/gut.31.1.4>
- 30 Ness-Jensen, E., Lindam, A., Lagergren, J., & Hveem, K. (2014). Tobacco smoking cessation and improved gastroesophageal reflux: a prospective population-based cohort study: the HUNT study. *The American journal of gastroenterology*, 109(2), 171–177. <https://doi.org/10.1038/ajg.2013.414>
- 31 Mendes-Filho, A. M., Moraes-Filho, J. P., Nasi, A., Eisig, J. N., Rodrigues, T. N., Barbutti, R. C., Campos, J. M., & Chinzon, D. (2014). Influence of exercise testing in gastroesophageal reflux in patients with gastroesophageal reflux disease. *Arquivos brasileiros de cirurgia digestiva: ABCD = Brazilian archives of digestive surgery*, 27(1), 3–8. <https://doi.org/10.1590/s0102-67202014000100002>
- 32 Kandil, T. S., Mousa, A. A., El-Gendy, A. A., & Abbas, A. M. (2010). The potential therapeutic effect of melatonin in Gastro-Esophageal Reflux Disease. *BMC gastroenterology*, 10, 7. <https://doi.org/10.1186/1471-230X-10-7>
- 33 Pereira R. (2006). Regression of gastroesophageal reflux disease symptoms using dietary supplementation with melatonin, vitamins and aminoacids: comparison with omeprazole. *Journal of pineal research*, 41(3), 195–200. <https://doi.org/10.1111/j.1600-079X.2006.00359.x>
- 34 Werbach M. R. (2008). Melatonin for the treatment of gastroesophageal reflux disease. *Alternative therapies in health and medicine*, 14(4), 54–58.
- 35 Lukić, M., Segec, A., Segec, I., Pinotić, L., Pinotić, K., Atalić, B., Solić, K., & Vcev, A. (2012). The impact of the vitamins A, C and E in the prevention of gastroesophageal reflux disease, Barrett's oesophagus and oesophageal adenocarcinoma. *Collegium antropologicum*, 36(3), 867–872.
- 36 Jurna I. (1998). Analgetische und analgesiepotenzierende Wirkung von B-Vitaminen [Analgesic and analgesia-potentiating action of B vitamins]. *Schmerz (Berlin, Germany)*, 12(2), 136–141. <https://doi.org/10.1007/s004829800054>
- 37 Gualano, M., Stramentinoli, G., & Berti, F. (1983). Anti-inflammatory activity of S-adenosyl-L-methionine: interference with the eicosanoid system. *Pharmacological research communications*, 15(7), 683–696. [https://doi.org/10.1016/s0031-6989\(83\)80040-x](https://doi.org/10.1016/s0031-6989(83)80040-x)
- 38 Pereira R. (2006). Regression of gastroesophageal reflux disease symptoms using dietary supplementation with melatonin, vitamins and aminoacids: comparison with omeprazole. *Journal of pineal research*, 41(3), 195–200. <https://doi.org/10.1111/j.1600-079X.2006.00359.x>
- 39 Cheng, J., & Ouwehand, A. C. (2020). Gastroesophageal Reflux Disease and Probiotics: A Systematic Review. *Nutrients*, 12(1), 132. <https://doi.org/10.3390/nu12010132>
- 40 Brown, R., Sam, C. H., Green, T., & Wood, S. (2015). Effect of GutsyGum (tm), A Novel Gum, on Subjective Ratings of Gastro Esophageal Reflux Following A Refluxogenic Meal. *Journal of dietary supplements*, 12(2), 138–145. <https://doi.org/10.3109/19390211.2014.950783>
- 41 Muss, C., Mosgoeller, W., & Endler, T. (2013). Papaya preparation (Caricol®) in digestive disorders. *Neuro endocrinology letters*, 34(1), 38–46.
- 42 Ottillinger, B., Storr, M., Malferteiner, P., & Allescher, H. D. (2013). STW 5 (Ibero-gast®) - a safe and effective standard in the treatment of functional gastrointestinal disorders. *Wiener medizinische Wochenschrift (1946)*, 163(3-4), 65–72. <https://doi.org/10.1007/s10354-012-0169-x>
- 43 Yeh, A. M., & Golianu, B. (2014). Integrative Treatment of Reflux and Functional Dyspepsia in Children. *Children (Basel, Switzerland)*, 1(2), 119–133. <https://doi.org/10.3390/children1020119>
- 44 Yeh, A. M., & Golianu, B. (2014). Integrative Treatment of Reflux and Functional Dyspepsia in Children. *Children (Basel, Switzerland)*, 1(2), 119–133. <https://doi.org/10.3390/children1020119>
- 45 Setright, R. (2017). Prevention of Symptoms of Gastric Irritation (GERD) Using Two Herbal Formulas: An observational study. *Journal of the Australian Traditional-Medicine Society*, 23(2), 68+. <https://link.gale.com/apps/doc/A544602451/AONE?u=anon~cd-12c9a0&sid=google&id=1fac696f>
- 46 Yeh, A. M., & Golianu, B. (2014). Integrative Treatment of Reflux and Functional Dyspepsia in Children. *Children (Basel, Switzerland)*, 1(2), 119–133. <https://doi.org/10.3390/children1020119>
- 47 Joseph, S., Sabulal, B., George, V., Antony, K. R., & Janardhanan, K. K. (2011). Antitumor and anti-inflammatory activities of polysaccharides isolated from *Ganoderma lucidum*. *Acta pharmaceutica (Zagreb, Croatia)*, 61(3), 335–342. <https://doi.org/10.2478/v10007-011-0030-6>
- 48 I. Wong, J. Y., Abdulla, M. A., Raman, J., Phan, C. W., Kuppusamy, U. R., Golbabapour, S., & Sabaratnam, V. (2013). Gastroprotective Effects of Lion's Mane Mushroom *Hericium erinaceus* (Bull.:Fr.) Pers. (Aphyllophoromycetidae) Extract against Ethanol-Induced Ulcer in Rats. *Evidence-based complementary and alternative medicine: eCAM*, 2013, 492976. <https://doi.org/10.1155/2013/492976>
- 49 S. Hussan, F., Mansoor, A. S., Hassan, S. N., Tengku Nor Effendy Kamaruddin, T. N., Budin, S. B., & Othman, F. (2015). Anti-Inflammatory Property of Plantago major Leaf Extract Reduces the Inflammatory Reaction in Experimental

Acetaminophen-Induced Liver Injury. *Evidence-based complementary and alternative medicine: eCAM*, 2015, 347861. <https://doi.org/10.1155/2015/347861>

50 Deters, A., Zippel, J., Hellenbrand, N., Pappai, D., Possemeyer, C., & Hensel, A. (2010). Aqueous extracts and polysaccharides from Marshmallow roots (*Althea officinalis* L.): cellular internalisation and stimulation of cell physiology of human epithelial cells in vitro. *Journal of ethnopharmacology*, 127(1), 62–69. <https://doi.org/10.1016/j.jep.2009.09.050>

51 Setright, R. (2017). Prevention of Symptoms of Gastric Irritation (GERD) Using Two Herbal Formulas: An observational study. *Journal of the Australian Traditional-Medicine Society*, 23(2), 68+. <https://link.gale.com/apps/doc/A544602451/AONE?u=anon~cd-12c9a0&sid=googleScholar&xid=1fac696f>

52 Nehra, A. K., Alexander, J. A., Loftus, R. N., G., & Nehra, V. (2018). Proton Pump Inhibitors: Review of Emerging Concerns. *Mayo Clinic proceedings*, 93(2), 240–246. <https://doi.org/10.1016/j.mayocp.2017.10.022>

Inflammatory Bowel Disease (IBD) Protocol

1 Hou, J. K., Abraham, B., & El-Serag, H. (2011). Dietary intake and risk of developing inflammatory bowel disease: a systematic review of the literature. *The American journal of gastroenterology*, 106(4), 563–573. <https://doi.org/10.1038/ajg.2011.44>

2 David, L. A., Maurice, C. F., Carmody, R. N., Gootenberg, D. B., Button, J. E., Wolfe, B. E., Ling, A. V., Devlin, A. S., Varma, Y., Fischbach, M. A., Biddinger, S. B., Dutton, R. J., & Turnbaugh, P. J. (2014). Diet rapidly and reproducibly alters the human gut microbiome. *Nature*, 505(7484), 559–563. <https://doi.org/10.1038/nature12820>

3 Dalal, S. R., & Chang, E. B. (2014). The microbial basis of inflammatory bowel diseases. *The Journal of clinical investigation*, 124(10), 4190–4196. <https://doi.org/10.1172/JCI72330>

4 Haskey, N., & Gibson, D. L. (2017). An Examination of Diet for the Maintenance of Remission in Inflammatory Bowel Disease. *Nutrients*, 9(3), 259. <https://doi.org/10.3390/nu9030259>

5 Sandefur, K., Kahleova, H., Desmond, A. N., Elfrink, E., & Barnard, N. D. (2019). Crohn's Disease Remission with a Plant-Based Diet: A Case Report. *Nutrients*, 11(6), 1385. <https://doi.org/10.3390/nu11061385>

6 Brown, A. C., Rampertab, S. D., & Mullin, G. E. (2011). Existing dietary guidelines for Crohn's disease and ulcerative colitis. *Expert review of gastroenterology & hepatology*, 5(3), 411–425. <https://doi.org/10.1586/egh.11.29>

7 Wang, X., Guo, J., Liu, Y., Yu, H., & Qin, X. (2019). Sucralose Increased Susceptibility to Colitis in Rats. *Inflammatory bowel diseases*, 25(2), e3–e4. <https://doi.org/10.1093/ibd/izy196>

8 Rodriguez-Palacios, A., Harding, A., Menghini, P., Himmelman, C., Retuerto, M., Nickerson, K. P., Lam, M., Croniger, C. M., McLean, M. H., Durum, S. K., Pizarro, T. T., Ghannoum, M. A., Ilic, S., McDonald, C., & Cominelli, F. (2018). The Artificial Sweetener Splenda Promotes Gut Proteobacteria, Dysbiosis, and Myeloperoxidase Reactivity in Crohn's Disease-Like Ileitis. *Inflammatory bowel diseases*, 24(5), 1005–1020. <https://doi.org/10.1093/ibd/izy060>

9 Li, X., Liu, Y., Wang, Y., Li, X., Liu, X., Guo, M., Tan, Y., Qin, X., Wang, X., & Jiang, M. (2020). Sucralose Promotes Colitis-Associated Colorectal Cancer Risk in a Murine Model Along With Changes in Microbiota. *Frontiers in oncology*, 10, 710. <https://doi.org/10.3389/fonc.2020.00710>

10 Li, X., Liu, Y., Wang, Y., Li, X., Liu, X., Guo, M., Tan, Y., Qin, X., Wang, X., & Jiang, M. (2020). Sucralose Promotes Colitis-Associated Colorectal Cancer Risk in a Murine Model Along With Changes in Microbiota. *Frontiers in oncology*, 10, 710. <https://doi.org/10.3389/fonc.2020.00710>

11 Kawano, R., Okamura, T., Hashimoto, Y., Majima, S., Senmaru, T., Ushigome, E., Asano, M., Yamazaki, M., Takakuwa, H., Sasano, R., Nakanishi, N., Hamaguchi, M., & Fukui, M. (2021). Erythritol Ameliorates Small Intestinal Inflammation Induced by High-Fat Diets and Improves Glucose Tolerance. *International journal of molecular sciences*, 22(11), 5558. <https://doi.org/10.3390/ijms22115558>

12 Zhang, J., Chen, X., Yang, R., Ma, Q., Qi, W., Sanidad, K. Z., Park, Y., Kim, D., Decker, E. A., & Zhang, G. (2019). Thermally Processed Oil Exaggerates Colonic Inflammation and Colitis-Associated Colon Tumorigenesis in Mice. *Cancer prevention research (Philadelphia, Pa.)*, 12(11), 741–750. <https://doi.org/10.1158/1940-6207.CAPR-19-0226>

13 Karban, A., & Eliakim, R. (2007). Effect of smoking on inflammatory bowel disease: Is it disease or organ specific?. *World journal of gastroenterology*, 13(15), 2150–2152. <https://doi.org/10.3748/wjg.v13.i15.2150>

14 Fletcher, J., Cooper, S. C., Ghosh, S., & Hewison, M. (2019). The Role of Vitamin D in Inflammatory Bowel Disease: Mechanism to Management. *Nutrients*, 11(5), 1019. <https://doi.org/10.3390/nu11051019>

15 Picardo, S., Altuwaijri, M., Devlin, S. M., & Seow, C. H. (2020). Complementary and alternative medications in the management of inflammatory bowel disease. *Therapeutic advances in gastroenterology*, 13, 1756284820927550. <https://doi.org/10.1177/1756284820927550>

16 Mozaffari, H., Daneshzad, E., Larijani, B., Bellissimo, N., & Azadbakht, L. (2020). Dietary intake of fish, n-3 polyunsaturated fatty acids, and risk of inflammatory bowel disease: a systematic review and meta-analysis of observational studies. *European journal of nutrition*, 59(1), 1–17. <https://doi.org/10.1007/s00394-019-01901-0>

17 Tamboli, C. P., Neut, C., Desreumaux, P., & Colombel, J. F. (2004). Dysbiosis in inflammatory bowel disease. *Gut*, 53(1), 1–4. <https://doi.org/10.1136/gut.53.1.1>

18 Parian, A., & Limketkai, B. N. (2016). Dietary Supplement Therapies for Inflammatory Bowel Disease: Crohn's Disease and Ulcerative Colitis. *Current pharmaceutical design*, 22(2), 180–188. <https://doi.org/10.2174/1381612822666151112145033>

19 Tang, X., Li, X., Wang, Y., Zhang, Z., Deng, A., Wang, W., Zhang, H., Qin, H., & Wu, L. (2020). Butyric Acid Increases the Therapeutic Effect of EHL7 on Ulcerative

Colitis by Inhibiting JAK2/STAT3/SOCS1 Signaling Pathway. *Frontiers in pharmacology*, 10, 1553. <https://doi.org/10.3389/fphar.2019.01553>

20 Scheppach W. (1994). Effects of short chain fatty acids on gut morphology and function. *Gut*, 35(1 Suppl), S35–S38. https://doi.org/10.1136/gut.35.1_suppl.s35

21 Banasiewicz, T., Domagalska, D., Borycka-Kiciak, K., & Rydzewska, G. (2020). Determination of butyric acid dosage based on clinical and experimental studies - a literature review. *Przegląd gastroenterologiczny*, 15(2), 119–125. <https://doi.org/10.5114/pg.2020.95556>

22 Bangodi, N., et al. (2021). Evaluation of Ulmus Fulva Extract in Experimentally Induced Inflammatory Bowel Disease in Rats. *IOSR Journal of Pharmacy and Biological Sciences (IOSR-JPBS)*, 16(4), 1–7.

23 Langmead, L., Dawson, C., Hawkins, C., Banna, N., Loo, S., & Rampton, D. S. (2002). Antioxidant effects of herbal therapies used by patients with inflammatory bowel disease: an in vitro study. *Alimentary pharmacology & therapeutics*, 16(2), 197–205. <https://doi.org/10.1046/j.1365-2036.2002.01157.x>

24 Czige, S., Bittner Fialová, S., Tóth, J., Mučaji, P., Nagy, M., & On Behalf Of The Oemonom (2022). Treatment of Gastrointestinal Disorders-Plants and Potential Mechanisms of Action of Their Constituents. *Molecules (Basel, Switzerland)*, 27(9), 2881. <https://doi.org/10.3390/molecules27092881>

25 Bahrami, G., Malekshahi, H., Miraghaee, S., Madani, H., Babaei, A., Mohammadi, B., & Hatami, R. (2020). Protective and Therapeutic Effects of Aloe Vera Gel on Ulcerative Colitis Induced by Acetic Acid in Rats. *Clinical nutrition research*, 9(3), 223–234. <https://doi.org/10.7762/cnr.2020.9.3.223>

26 Langmead, L., Feakins, R. M., Goldthorpe, S., Holt, H., Tsironi, E., De Silva, A., Jewell, D. P., & Rampton, D. S. (2004). Randomized, double-blind, placebo-controlled trial of oral aloe vera gel for active ulcerative colitis. *Alimentary pharmacology & therapeutics*, 19(7), 739–747. <https://doi.org/10.1111/j.1365-2036.2004.01902.x>

27 Liu, D., Gao, L., Zhang, J., Huo, X., Ni, H., & Cao, L. (2017). Anti-inflammatory and Anti-oxidant Effects of Licorice Flavonoids on Ulcerative Colitis in Mouse Model. *Chinese Herbal Medicines*, 9(4), 358–368.

28 Qin, M., Geng, Y., Lu, Z., Xu, H., Shi, J. S., Xu, X., & Xu, Z. H. (2016). Anti-Inflammatory Effects of Ethanol Extract of Lion's Mane Medicinal Mushroom, *Hericium erinaceus* (Agaricomycetes), in Mice with Ulcerative Colitis. *International journal of medicinal mushrooms*, 18(3), 227–234. <https://doi.org/10.1615/IntJMedMushrooms.v18.i3.50>

29 Wang, D., Zhang, Y., Yang, S., Zhao, D., & Wang, M. (2019). A polysaccharide from cultured mycelium of *Herichium erinaceus* attenuates colitis by counteracting oxidative stress and improving mitochondrial function. *International journal of biological macromolecules*, 125, 572–579. <https://doi.org/10.1016/j.ijbiomac.2018.12.092>

30 Ren, Y., Geng, Y., Du, Y., Li, W., Lu, Z. M., Xu, H. Y., Xu, G. H., Shi, J. S., & Xu, Z. H. (2018). Polysaccharide of *Herichium erinaceus* attenuates colitis in C57BL/6 mice via regulation of oxidative stress, inflammation-related signaling pathways and modulating the composition of the gut microbiota. *The journal of nutritional biochemistry*, 57, 67–76. <https://doi.org/10.1016/j.jnutbio.2018.03.005>

31 Pallav, K., Dowd, S. E., Villafuerte, J., Yang, X., Kabbani, T., Hansen, J., Dennis, M., Leffler, D. A., Newburg, D. S., & Kelly, C. P. (2014). Effects of polysaccharopeptide from *Trametes versicolor* and amoxicillin on the gut microbiome of healthy volunteers: a randomized clinical trial. *Gut microbes*, 5(4), 458–467. <https://doi.org/10.4161/gmic.29558>

32 Lim B. O. (2011). *Coriolus versicolor* suppresses inflammatory bowel disease by inhibiting the expression of STAT1 and STAT6 associated with IFN-γ and IL-4 expression. *Phytotherapy research: PTR*, 25(8), 1257–1261. <https://doi.org/10.1002/ptr.3378>

33 Saleh, M. H., Rashedi, I., & Keating, A. (2017). Immunomodulatory Properties of *Coriolus versicolor*: The Role of Polysaccharopeptide. *Frontiers in immunology*, 8, 1087. <https://doi.org/10.3389/fimmu.2017.01087>

34 Jin, M., Zhou, W., Jin, C., Jiang, Z., Diao, S., Jin, Z., & Li, G. (2019). Anti-inflammatory activities of the chemical constituents isolated from *Trametes versicolor*. *Natural product research*, 33(16), 2422–2425. <https://doi.org/10.1080/14786419.2018.1446011>

35 Xie, J., Liu, Y., Chen, B., Zhang, G., Ou, S., Luo, J., & Peng, X. (2019). *Ganoderma lucidum* polysaccharide improves rat DSS-induced colitis by altering cecal microbiota and gene expression of colonic epithelial cells. *Food & nutrition research*, 63, 10.29219/fnr.v63.1559. <https://doi.org/10.29219/fnr.v63.1559>

36 Liu, C., Dunkin, D., Lai, J., Song, Y., Ceballos, C., Benkov, K., & Li, X. M. (2015). Anti-inflammatory Effects of *Ganoderma lucidum* Triterpenoid in Human Crohn's Disease Associated with Downregulation of NF-κB Signaling. *Inflammatory bowel diseases*, 21(8), 1918–1925. <https://doi.org/10.1097/MIB.0000000000000439>

37 Mawdsley, J. E., & Rampton, D. S. (2005). Psychological stress in IBD: new insights into pathogenic and therapeutic implications. *Gut*, 54(10), 1481–1491. <https://doi.org/10.1136/gut.2005.064261>

38 Sajadieh, A., Nielsen, O. W., Rasmussen, V., Hein, H. O., Abedini, S., & Hansen, J. F. (2004). Increased heart rate and reduced heart-rate variability are associated with subclinical inflammation in middle-aged and elderly subjects with no apparent heart disease. *European heart journal*, 25(5), 363–370. <https://doi.org/10.1016/j.ehj.2003.12.003>

39 Danner, M., Kasl, S. V., Abramson, J. L., & Vaccarino, V. (2003). Association between depression and elevated C-reactive protein. *Psychosomatic medicine*, 65(3), 347–356. <https://doi.org/10.1097/01.psy.0000041542.29808.01>

40 Chandrasekhar, K., Kapoor, J., & Anishetty, S. (2012). A prospective, randomized double-blind, placebo-controlled study of safety and efficacy of a high-concentration full-spectrum extract of ashwagandha root in reducing stress and anxiety in adults. *Indian journal of psychological medicine*, 34(3), 255–262. <https://doi.org/10.4103/0253-7176.106022>

41 Lopresti, A. L., Smith, S. J., Malvi, H., & Kodgule, R. (2019). An investigation into the stress-relieving and pharmacological actions of an ashwagandha (*Withania somnifera*)

era) extract: A randomized, double-blind, placebo-controlled study. *Medicine*, 98(37), e17186. <https://doi.org/10.1097/MD.00000000000017186>

- 42 Khan, M. A., Subramanayan, M., Arora, V. K., Banerjee, B. D., & Ahmed, R. S. (2015). Effect of Withania somnifera (Ashwagandha) root extract on amelioration of oxidative stress and autoantibodies production in collagen-induced arthritic rats. *Journal of complementary & integrative medicine*, 12(2), 117–125. <https://doi.org/10.1515/jcim-2014-0075>
 - 43 Triantafyllidis, J. K., Triantafyllidis, A., Vagianos, C., & Papalois, A. (2016). Favorable results from the use of herbal and plant products in inflammatory bowel disease: evidence from experimental animal studies. *Annals of gastroenterology*, 29(3), 268–281. <https://doi.org/10.20524/aog.2016.0059>
 - 44 Chong, P. S., Fung, M. L., Wong, K. H., & Lim, L. W. (2019). Therapeutic Potential of *Hericium erinaceus* for Depressive Disorder. *International journal of molecular sciences*, 21(1), 163. <https://doi.org/10.3390/ijms21010163>
 - 45 Liao, L. Y., He, Y. F., Li, L., Meng, H., Dong, Y. M., Yi, F., & Xiao, P. G. (2018). A preliminary review of studies on adaptogens: comparison of their bioactivity in TCM with that of ginseng-like herbs used worldwide. *Chinese medicine*, 13, 57. <https://doi.org/10.1186/s13020-018-0214-9>
 - 46 Shashidhar, M. G., Giridhar, P., Udaya Sankar, K., & Manohar, B. (2013). Bioactive principles from *Cordyceps sinensis*: A potent food supplement - A review. *Journal of functional foods*, 5(3), 1013–1030. <https://doi.org/10.1016/j.jff.2013.04.018>
 - 47 Han, E. S., Oh, J. Y., & Park, H. J. (2011). Cordyceps militaris extract suppresses dextran sodium sulfate-induced acute colitis in mice and production of inflammatory mediators from macrophages and mast cells. *Journal of ethnopharmacology*, 134(3), 703–710. <https://doi.org/10.1016/j.jep.2011.01.022>
 - 48 Roe, J. J., Thompson, C. W., Aspinall, P. A., Brewer, M. J., Duff, E. I., Miller, D., Mitchell, R., & Clow, A. (2013). Green space and stress: evidence from cortisol measures in deprived urban communities. *International journal of environmental research and public health*, 10(9), 4086–4103. <https://doi.org/10.3390/ijerph10094086>
 - 49 Andersen, L., Corazon, S., & Stigsdottir, U. (2021). Nature Exposure and Its Effects on Immune System Functioning: A Systematic Review. *International journal of environmental research and public health*, 18(4), 1416. <https://doi.org/10.3390/ijerph18041416>
- ## Leaky Gut, Gastritis, and IBS Protocol
- 1 Cristofori, F., Dargenio, V. N., Dargenio, C., Miniello, V. L., Barone, M., & Francavilla, R. (2021). Anti-Inflammatory and Immunomodulatory Effects of Probiotics in Gut Inflammation: A Door to the Body. *Frontiers in immunology*, 12, 578386. <https://doi.org/10.3389/fimmu.2021.578386> <https://pubmed.ncbi.nlm.nih.gov/33717063/>
 - 2 <https://pubmed.ncbi.nlm.nih.gov/29742432/>
 - 3 Cignarella F, Cantoni C, Ghezzi L, Salter A, Dorsett Y, Chen L, Phillips D, Weinstock G.M., Fontana L., Cross A.H., et al. Intermittent fasting confers protection in CNS autoimmunity by altering the gut microbiota. *Cell Metab*. 2018;27:1222–1235. doi: 10.1016/j.cmet.2018.05.006.
 - 4 Rangan P, Choi I, Wei M., Navarrete G., Guen E., Brandhorst S., Enyati N., Pasia G., Maesincee D., Ocon V., et al. Fasting-mimicking diet modulates microbiota and promotes intestinal regeneration to reduce inflammatory bowel disease pathology. *Cell Rep*. 2019;26:2704–2719. doi: 10.1016/j.celrep.2019.02.019.
 - 5 Catterson J.H., Khericha M., Dyson M.C., Vincent A.J., Callard R., Haveron S.M., Rajasingam A., Ahmad M., Partridge L. Short-term, intermittent fasting induces long-lasting gut health and TOR-independent lifespan extension. *Curr. Biol*. 2018;28:1714–1724. doi: 10.1016/j.cub.2018.04.015
 - 6 Hills RD Jr, Pontefract BA, Mishcon HR, Black CA, Sutton SC, Theberge CR. Gut Microbiome: Profound Implications for Diet and Disease. *Nutrients*. 2019 Jul 16;11(7):1613. doi: 10.3390/nu11071613. PMID: 31315227; PMCID: PMC6682904. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6682904/>
 - 7 Jayachandran M, Xiao J, Xu B. A Critical Review on Health Promoting Benefits of Edible Mushrooms through Gut Microbiota. *International Journal of Molecular Sciences*. 2017; 18(9):1934. <https://doi.org/10.3390/ijms18091934>
 - 8 Rebello CJ, O'Neil CE, Greenway FL. Dietary fiber and satiety: the effects of oats on satiety. *Nutr Rev*. 2016 Feb;74(2):131–47. doi: 10.1093/nutrit/nuv063. Epub 2016 Jan 2. PMID: 26724486; PMCID: PMC4757923.
 - 9 Wang, Mingxing et al. "Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion's Mane Medicinal Mushroom, *Herichium erinaceus* (Higher Basidiomycetes)." *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055–60. doi:10.1615/intjmedmushrooms.v17.i11.50. <https://pubmed.ncbi.nlm.nih.gov/26853960/>
 - 10 Wang, Mingxing et al. "Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion's Mane Medicinal Mushroom, *Herichium erinaceus* (Higher Basidiomycetes)." *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055–60. doi:10.1615/intjmedmushrooms.v17.i11.50. <https://pubmed.ncbi.nlm.nih.gov/26853960/>
 - 11 Czige, S., Bittner Fialová, S., Tóth, J., Mučaji, P., Nagy, M., & On Behalf of The Oeonom (2022). Treatment of Gastrointestinal Disorders-Plants and Potential Mechanisms of Action of Their Constituents. *Molecules* (Basel, Switzerland), 27(9), 2881. <https://doi.org/10.3390/molecules27092881>
 - 12 Hawrelak JA, Myers SP. Effects of two natural medicine formulations on irritable bowel syndrome symptoms: A pilot study. *J Altern Complement Med* 2010;16:1065–1071
 - 13 Ke F, Yadav PK, Ju LZ. Herbal medicine in the treatment of ulcerative colitis. *Saudi J Gastroenterol* 2012;18:3–10
 - 14 Available Chemical Constituents and Activities of Ganoderma Lucidum (Lingzhi or Red Reishi) Utilizing in Disease Treatment: A mini review. Vo et al. *J Res Clin Med*.

2021;9: 32. doi: 10.34172/jrcm.2021.032

- 15 Pallav, K., Dowd, S. E., Villafuerte, J., Yang, X., Kabbani, T., Hansen, J., Dennis, M., Leffler, D. A., Newburg, D. S., & Kelly, C. P. (2014). Effects of polysaccharopeptide from *Trametes versicolor* and amoxicillin on the gut microbiome of healthy volunteers: a randomized clinical trial. *Gut microbes*, 5(4), 458–467. <https://doi.org/10.4161/gmic.29558> <https://pubmed.ncbi.nlm.nih.gov/25006989/>
 - 16 Wang, Mingxing et al. "Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion's Mane Medicinal Mushroom, *Herichium erinaceus* (Higher Basidiomycetes)." *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055–60. doi:10.1615/intjmedmushrooms.v17.i11.50. <https://pubmed.ncbi.nlm.nih.gov/26853960/>
 - 17 Fletcher, J., Cooper, S. C., Ghosh, S., & Hewison, M. (2019). The Role of Vitamin D in Inflammatory Bowel Disease: Mechanism to Management. *Nutrients*, 11(5), 1019. <https://doi.org/10.3390/nu11051019>
 - 18 Rastgou S, Ebrahimi-Daryani N, Agah S, Karimi S, Taher M, Rashidkhani B, Hejazi E, Mohseni F, Ahmadzadeh M, Sadeghi A, Hekmatdoost A. Glutamine Supplementation Enhances the Effects of a Low FODMAP Diet in Irritable Bowel Syndrome Management. *Front Nutr*. 2021 Dec 16;8:746703. doi: 10.3389/fnut.2021.746703. PMID: 34977110; PMCID: PMC8716871. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8716871>
 - 19 Song W, Chen Q, Wang Y, Han Y, Zhang H, Li B. Identification and Structure-Activity Relationship of Intestinal Epithelial Barrier Function Protective Collagen Peptides from Alaska Pollock Skin. *Mar Drugs*. 2019 Jul 31;17(8):450. doi: 10.3390/md17080450. PMID: 31370332; PMCID: PMC6723256
 - 20 Abrahams, M., O'Grady, R., & Prawitt, J. (2022). Effect of a Daily Collagen Peptide Supplement on Digestive Symptoms in Healthy Women: 2-Phase Mixed Methods Study. *JMIR formative research*, 6(5), e36339. <https://doi.org/10.2196/36339>
- ## Liver Health Protocol
- 1 <https://www.cdc.gov/hepatitis/theliver.htm>
 - 2 <https://www.hopkinsmedicine.org/health/conditions-and-diseases/liver-anatomy-and-functions>
 - 3 <https://www.hopkinsmedicine.org/health/conditions-and-diseases/liver-anatomy-and-functions>
 - 4 <https://www.mayoclinic.org/diseases-conditions/liver-problems/symptoms-causes/syc-20374502>
 - 5 Wang YP, Cheng ML, Zhang BF, Mu M, Wu J. Effects of blueberry on hepatic fibrosis and transcription factor Nrf2 in rats. *World J Gastroenterol*. 2010 Jun 7;16(21):2657–63. doi: 10.3748/wjg.v16.i21.2657. PMID: 20518088; PMCID: PMC2880779
 - 6 Bingül I, Başaran-Küçükgergin C, Aydın AF, Soluk-Tekkeşin M, Olgaç V, Doğru-Abbasoğlu S, Uysal M. Blueberry treatment attenuated cirrhotic and preneoplastic lesions and oxidative stress in the liver of diethylnitrosamine-treated rats. *Int J Immunopathol Pharmacol*. 2016 Sep;29(3):426–37. doi: 10.1177/0394632015621319. Epub 2015 Dec 18. PMID: 26684621; PMCID: PMC5806765
 - 7 Yi, Weiguang & Akoh, Casimir & Fischer, Joan & Krewer, Gerard. (2006). Effects of phenolic compounds in blueberries and muscadine grapes on HepG2 cell viability and apoptosis. *Food Research International - FOOD RES INT*. 39. 628–638. 10.1016/j.foodres.2006.01.001
 - 8 Chen YJ, Wallig MA, Jeffery EH. Dietary Broccoli Lessens Development of Fatty Liver and Liver Cancer in Mice Given Diethylnitrosamine and Fed a Western or Control Diet. *J Nutr*. 2016 Mar;146(3):542–50. doi: 10.3945/jn.115.228148. Epub 2016 Feb 10. PMID: 26865612; PMCID: PMC4763488
 - 9 Chen BB, Han Y, Pan X, Yan J, Liu W, Li Y, Lin X, Xu S, Peng XE. Association between nut intake and non-alcoholic fatty liver disease risk: a retrospective case-control study in a sample of Chinese Han adults. *BMJ Open*. 2019 Sep 4;9(9):e028961. doi: 10.1136/bmjopen-2019-028961. PMID: 31488477; PMCID: PMC6731785
 - 10 Lu W, Li S, Li J, Wang J, Zhang R, Zhou Y, Yin Q, Zheng Y, Wang F, Xia Y, Chen K, Liu T, Lu J, Zhou Y, Guo C. Effects of Omega-3 Fatty Acid in Nonalcoholic Fatty Liver Disease: A Meta-Analysis. *Gastroenterol Res Pract*. 2016;2016:1459790. doi: 10.1155/2016/1459790. Epub 2016 Aug 29. PMID: 27651787; PMCID: PMC5019889
 - 11 Spooner MH, Jump DB. Omega-3 fatty acids and nonalcoholic fatty liver disease in adults and children: where do we stand? *Curr Opin Clin Nutr Metab Care*. 2019 Mar;22(2):103–110. doi: 10.1097/MCO.0000000000000539. PMID: 30601174; PMCID: PMC6355343
 - 12 Zamani, H., de Joode, M., Hossein, I. J., Henckens, N., Guggeis, M. A., Berends, J. E., de Kok, T., & van Breda, S. (2021). The benefits and risks of beetroot juice consumption: a systematic review. *Critical reviews in food science and nutrition*, 61(5), 788–804. <https://doi.org/10.1080/10408398.2020.1746629> <https://pubmed.ncbi.nlm.nih.gov/32292042/>
 - 13 Chen, L., Zhu, Y., Hu, Z., Wu, S., & Jin, C. (2021). Beetroot as a functional food with huge health benefits: Antioxidant, antitumor, physical function, and chronic metabolomics activity. *Food science & nutrition*, 9(11), 6406–6420. <https://doi.org/10.1002/fsn3.2577> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8565237/>
 - 14 Clifford, T., Howatson, G., West, D. J., & Stevenson, E. J. (2015). The potential benefits of red beetroot supplementation in health and disease. *Nutrients*, 7(4), 2801–2822. <https://doi.org/10.3390/nu7042801> <https://pubmed.ncbi.nlm.nih.gov/pmc/articles/PMC4425174>
 - 15 Khoshbaten M, Aliasgarzadeh A, Masnadi K, Farhang S, Tarzamani MK, Babaei H, Kiani J, Zaare M, Najafipour F. Grape seed extract to improve liver function in patients with nonalcoholic fatty liver change. *Saudi J Gastroenterol*. 2010 Jul-Sep;16(3):194–7. doi: 10.4103/1319-3767.65197. PMID: 20616415; PMCID: PMC3003214
 - 16 Pezeshki A, Safi S, Feizi A, Askari G, Karami F. The Effect of Green Tea Extract Supplementation on Liver Enzymes in Patients with Nonalcoholic Fatty Liver Disease. *Int J Prev Med*. 2016 Feb 1;7:28. doi: 10.4103/2008-7802.173051.

PMID: 26955458; PMCID: PMC4763469

17 Yin X, Yang J, Li T, Song L, Han T, Yang M, Liao H, He J, Zhong X. The effect of green tea intake on risk of liver disease: a meta analysis. *Int J Clin Exp Med*. 2015 Jun 15;8(6):8339-46. PMID: 26309486; PMCID: PMC4538013

18 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4529263>

19 Palumbo, B., Efthimiou, Y., Stamatopoulos, J., Oguogho, A., Budinsky, A., Palumbo, R., & Sinzinger, H. (2003). Prickly pear induces upregulation of liver LDL binding in familial heterozygous hypercholesterolemia. *Nuclear medicine review. Central & Eastern Europe*, 6(1), 35–39

20 <https://www.ncbi.nlm.nih.gov/labs/pmc/articles/PMC5174149/>

21 <https://www.webmd.com/hepatitis/toxic-liver-disease>

22 <https://www.allcareservices.ca/health-effects-of-cleaning-with-aerosol-sprays/>

23 <https://pubmed.ncbi.nlm.nih.gov/28913736>

24 <https://www.theguardian.com/fashion/2021/jun/15/pfas-makeup-forever-chemicals>

25 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7068600/>

26 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5651828>

27 <https://www.ethicalconsumer.org/health-beauty/toxic-beauty>

28 <https://www.sciencedirect.com/science/article/abs/pii/S00489697163134>

29 Soares, A. A., de Sá-Nakanishi, A. B., Bracht, A., da Costa, S. M., Koehnlein, E. A., de Souza, C. G., & Peralta, R. M. (2013). Hepatoprotective effects of mushrooms. *Molecules* (Basel, Switzerland), 18(7), 7609–7630. <https://doi.org/10.3390/molecules18077609>

30 Wasser S.P. Reishi or Ling Zhi (*Ganoderma lucidum*) *Enc. Diet.Suppl.* 2005;2005:603–

31 Soares, A. A., de Sá-Nakanishi, A. B., Bracht, A., da Costa, S. M., Koehnlein, E. A., de Souza, C. G., & Peralta, R. M. (2013). Hepatoprotective effects of mushrooms. *Molecules* (Basel, Switzerland), 18(7), 7609–7630. <https://doi.org/10.3390/molecules18077609>

32 Lin, Z. B., Wang, M. Y., Liu, Q., & Che, Q. M. (2002). Effects of total triterpenoids extract from *Ganoderma lucidum* (Curt.: Fr.) P. Karst. (Reishi Mushroom) on experimental liver injury models induced by carbon tetrachloride or D-galactosamine in mice. *International Journal of Medicinal Mushrooms*, 4(4).

33 <https://www.researchgate.net/publication/281815032>

34 <https://pubmed.ncbi.nlm.nih.gov/26141646/>

35 Mahboubi M, Mahboubi M. Hepatoprotection by dandelion (*Taraxacum officinale*) and mechanisms. *Asian Pac J Trop Biomed* 2020;10:1-10

36 Hamza, A.A., Mohamed, M.G., Lashin, F.M. et al. Dandelion prevents liver fibrosis, inflammatory response, and oxidative stress in rats. *JoBAZ* 81, 43 (2020). <https://doi.org/10.1186/s41936-020-00177-9>

37 Abenavoli, L., Capasso, R., Milic, N., & Capasso, F. (2010). Milk thistle in liver diseases: past, present, future. *Phytotherapy research: PTR*, 24(10), 1423–1432. <https://doi.org/10.1002/ptr.320>

38 Hackett, E. S., Twedt, D. C., & Gustafson, D. L. (2013). Milk thistle and its derivative compounds: a review of opportunities for treatment of liver disease. *Journal of veterinary internal medicine*, 27(1), 10–16. <https://doi.org/10.1111/jvim.12002>

39 Arturo Alejandro Dreifuss, Amanda Leite Bastos-Pereira, Thiago Vinicius Ávila, Bruna da Silva Soley, Armando J. Rivero, José Luis Aguilar, Alexandra Acco, Antitumoral and antioxidant effects of a hydroalcoholic extract of cat's claw (*Uncaria tomentosa*) (Willd. Ex Roem. & Schult) in an in vivo carcinosarcoma model, *Journal of Ethnopharmacology*, Volume 130, Issue 1, 2010, Pages 127-133, ISSN 0378-8741, <https://doi.org/10.1016/j.jep.2010.04.029>

40 Elgawish, R. A., Abdelrazek, H., Ismail, S., Loutfy, N. M., & Soliman, M. (2019). Hepatoprotective activity of *Uncaria tomentosa* extract against sub-chronic exposure to fipronil in male rats. *Environmental science and pollution research international*, 26(1), 199–207. <https://doi.org/10.1007/s11356-018-3615-5>

41 Ma L, Song J, Shi Y, Wang C, Chen B, Xie D, Jia X. Anti-Hepatitis B Virus Activity of Chickweed [*Stellaria media* (L.) Vill.] Extracts in HepG2.2.15 Cells. *Molecules*. 2012; 17(7):8633-8646. <https://doi.org/10.3390/molecules17078633>

42 <https://ods.od.nih.gov/factsheets/vitamin-d-health-professional/>

43 Iruzubieta P, Terán Á, Crespo J, Fábrega E. Vitamin D deficiency in chronic liver disease. *World J Hepatol*. 2014 Dec 27;6(12):901-15. doi: 10.4254/wjh.v6.i12.901. PMID: 25544877; PMCID: PMC4269909

44 Stokes, C. S., Volmer, D. A., Grünhage, F., & Lammert, F. (2013). Vitamin D in chronic liver disease. *Liver international: official journal of the International Association for the Study of the Liver*, 33(3), 338–352. <https://doi.org/10.1111/liv.12106>

45 <https://www.livestrong.com/article/359556-risks-of-too-much-vitamin-b12/>

46 Yan X, Gao R, Hu Y, Jin J. Pernicious anemia associated with cryptogenic cirrhosis: Two case reports and a literature review. *Medicine (Baltimore)*. 2018 Sep;97(39):e12547. doi: 10.1097/MD.00000000000012547. PMID: 30278550; PMCID: PMC6181571

47 El Hadi H, Vettor R, Rossato M. Vitamin E as a Treatment for Nonalcoholic Fatty Liver Disease: Reality or Myth? *Antioxidants* (Basel). 2018 Jan 16;7(1):12. doi: 10.3390/antiox7010012. PMID: 29337849; PMCID: PMC5789322

48 He, Z., Li, X., Yang, H., Wu, P., Wang, S., Cao, D., Guo, X., Xu, Z., Gao, J., Zhang, W., & Luo, X. (2021). Effects of Oral Vitamin C Supplementation on Liver Health and Associated Parameters in Patients With Non-Alcoholic Fatty Liver Disease: A Randomized Clinical Trial. *Frontiers in nutrition*, 8, 745609. <https://doi.org/10.3389/fnut.2021.745609>

49 El Hadi H, Vettor R, Rossato M. Vitamin E as a Treatment for Nonalcoholic Fatty Liver Disease: Reality or Myth? *Antioxidants* (Basel). 2018 Jan 16;7(1):12. doi: 10.3390/antiox7010012. PMID: 29337849; PMCID: PMC5789322

50 <https://ods.od.nih.gov/factsheets/VitaminC-HealthProfessional/>

51 <https://my.clevelandclinic.org/health/diseases/15831-fatty-liver-disease>

52 <https://www.health.harvard.edu/blog/fatty-liver-disease-what-it-is-and-what-to-do-about-it-2019011015746>

53 <https://medlineplus.gov/cholesterol.html>

54 <https://thefhfoundation.org/liver-cholesterol>

55 Röder PV, Wu B, Liu Y, Han W. Pancreatic regulation of glucose homeostasis. *Exp Mol Med*. 2016 Mar 11;48(3):e219. doi: 10.1038/emm.2016.6. PMID: 26964835; PMCID: PMC4892884

56 McDougall, G. J., Kulkarni, N. N., & Stewart, D. (2008). Current developments on the inhibitory effects of berry polyphenols on digestive enzymes. *BioFactors* (Oxford, England), 34(1), 73–80. <https://doi.org/10.1002/biof.5520340108>

57 Li, Y., Talbot, C.L., Chandravanshi, B. et al. Cordyceps inhibits ceramide biosynthesis and improves insulin resistance and hepatic steatosis. *Sci Rep* 12, 7273 (2022). <https://doi.org/10.1038/s41598-022-11219-3>

58 Cheng, Y. W., Chen, Y. I., Tzeng, C. Y., Chen, H. C., Tsai, C. C., Lee, Y. C., Lin, J. G., Lai, Y. K., & Chang, S. L. (2012). Extracts of *Cordyceps militaris* lower blood glucose via the stimulation of cholinergic activation and insulin secretion in normal rats. *Phytotherapy research: PTR*, 26(8), 1173–1177. <https://doi.org/10.1002/ptr.3709>

59 Ma H.-T., Hsieh J.-F., Chen S.-T. Anti-diabetic effects of *Ganoderma lucidum*. *Phytochemistry*. 2015;114:109–113. doi: 10.1016/j.phytochem.2015.02.017

60 Xiao C, Wu Q, Zhang J, Xie Y, Cai W, Tan J. Antidiabetic activity of *Ganoderma lucidum* polysaccharides F31 down-regulated hepatic glucose regulatory enzymes in diabetic mice. *J Ethnopharmacol*. 2017;196:47-57

61 Xiao C, Wu Q, Zhang J, Xie Y, Cai W, Tan J. Antidiabetic activity of *Ganoderma lucidum* polysaccharides F31 down-regulated hepatic glucose regulatory enzymes in diabetic mice. *J Ethnopharmacol*. 2017;196:47-57

62 Xiao C, Wu Q, Zhang J, Xie Y, Cai W, Tan J. Antidiabetic activity of *Ganoderma lucidum* polysaccharides F31 down-regulated hepatic glucose regulatory enzymes in diabetic mice. *J Ethnopharmacol*. 2017;196:47-57

63 <https://pubmed.ncbi.nlm.nih.gov/25790910/>

64 <https://pubmed.ncbi.nlm.nih.gov/19839001/>

65 Tassell, M. C., Kingston, R., Gilroy, D., Lehane, M., & Furey, A. (2010). Hawthorn (*Crataegus* spp.) in the treatment of cardiovascular disease. *Pharmacognosy reviews*, 4(7), 32–41. <https://doi.org/10.4103/0973-7847.65324>

66 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5376420/>

67 <https://www.ncbi.nlm.nih.gov/labs/pmc/articles/PMC3210006>

68 <https://pubmed.ncbi.nlm.nih.gov/24199995/>

69 <https://www.mayoclinic.org/tests-procedures/liver-function-tests/about/pac-20394595>

70 <https://www.drugs.com/condition/liver-disease.html>

Ulcer Protocol

1 <https://www.ncbi.nlm.nih.gov/books/NBK537128/>

2 <https://www.healthdirect.gov.au/duodenal-ulcer>

3 <https://www.mayoclinic.org/diseases-conditions/peptic-ulcer/diagnosis-treatment/drc-20354229>

4 <https://www.niddk.nih.gov/health-information/digestive-diseases/peptic-ulcers-stomach-ulcers/symptoms-causes>

5 <https://www.niddk.nih.gov/health-information/digestive-diseases/peptic-ulcers-stomach-ulcers/symptoms-causes>

6 <https://pubmed.ncbi.nlm.nih.gov/2072799/>

7 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4458761/>

8 <https://pubmed.ncbi.nlm.nih.gov/33050668/>

9 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3758027/>

10 https://www.researchgate.net/publication/264934441_Effect_of_garlic_and_cabbage_on_healing_of_gastric_ulcer_in_experimental_rats

11 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4906699/>

12 <https://www.advresearch.org/>

13 <https://www.ncbi.nlm.nih.gov/books/NBK564659/>

14 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4958626/>

15 <https://pubmed.ncbi.nlm.nih.gov/25111233/>

16 <https://pubmed.ncbi.nlm.nih.gov/27995346/>

17 <https://pubmed.ncbi.nlm.nih.gov/22933142/>

18 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5103537/>

19 Zaghlool SS, Abo-Seif AA, Rabeh MA, Abdelmohsen UR, Messiha BAS. Gastro-Protective and Anti-Oxidant Potential of *Althaea officinalis* and *Solanum nigrum* on Pyloric Ligation/Indomethacin-Induced Ulceration in Rats. *Antioxidants* (Basel). 2019 Oct 25;8(11):512. doi: 10.3390/antiox8110512. PMID: 31731465; PMCID: PMC6912529

20 Zaghlool SS, Shehata BA, Abo-Seif AA, Abd El-Latif HA. Protective effects of ginger and marshmallow extracts on indomethacin-induced peptic ulcer in rats. *J Nat Sci Biol*

Med. 2015 Jul-Dec;6(2):421-8. doi: 10.4103/0976-9668.160026. PMID: 26283843; PMCID: PMC4518423

21 Adom, M. B., Taher, M., Mutalabisin, M. F., Amri, M. S., Kudos, M. B. A., Sulaiman, M. W. A. W., ... & Susanti, D. (2017). Chemical constituents and medical benefits of *Plantago major*: Biomedicine & pharmacotherapy, 96, 348-360

22 Setright, R. (2017). Prevention of symptoms of gastric irritation (GERD) using two herbal formulas: An observational study. *Journal of the Australian Traditional-Medicine Society*, 23(2), 68–71. <https://search.informit.org/doi/10.3316/informit.950298610899394>

23 Yadav, A. K., Pushpesh, K. M., Jain, P. K., Chandana, V. R., Tiwari, S., & Singh, V. (2016). Investigation of *Calendula officinalis* whole plant as a gastroprotective and antioxidant in peptic ulcer. *Br J Med Health Res*, 3(7), 67-76

24 Dizaye, K., & Ali, R. H. (2012). Gastroprotective effects of *calendula officinalis* extract. Impact of a health educational program on knowledge of female nursing staff and students regarding breast Cancer prevention and early detection in Erbil city, 88

25 Zaghlool SS, Shehata BA, Abo-Seif AA, Abd El-Latif HA. Protective effects of ginger and marshmallow extracts on indomethacin-induced peptic ulcer in rats. *J Nat Sci Biol Med*. 2015 Jul-Dec;6(2):421-8. doi: 10.4103/0976-9668.160026. PMID: 26283843; PMCID: PMC4518423

26 <https://pubmed.ncbi.nlm.nih.gov/23612703/>

27 Świąder, K., Startek, K., & Wijaya, C. H. (2019). The therapeutic properties of Lemon balm (*Melissa officinalis* L.): Reviewing novel findings and medical indications. *J. Appl. Bot. Food Qual*, 92, 327-335

28 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6836258/>

29 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3615700/>

30 <https://pubmed.ncbi.nlm.nih.gov/19879118/>

31 Ardalani, H., Hadipana, A., & Sahebkar, A. (2020). Medicinal Plants in the Treatment of Peptic Ulcer Disease: A Review. Mini reviews in medicinal chemistry, 20(8), 662–702. <https://doi.org/10.2174/1389557520666191227151939>

32 Wang, Mingxing et al. "Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion's Mane Medicinal Mushroom, *Hericium erinaceus* (Higher Basidiomycetes)." *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055-60. doi:10.1615/intjmedmushrooms.v17.i11.50. <https://pubmed.ncbi.nlm.nih.gov/26853960/>

33 <https://news.harvard.edu/gazette/story/2009/05/glutamine-supplements-show-promise-in-treating-stomach-ulcers/>

34 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4906699/>

35 <https://pubmed.ncbi.nlm.nih.gov/8309988/>

36 <https://pubmed.ncbi.nlm.nih.gov/8726273/>

37 <https://pubmed.ncbi.nlm.nih.gov/3047853/>

38 <https://pubmed.ncbi.nlm.nih.gov/27656478/>

39 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6249666>

40 <https://newsnetwork.mayoclinic.org/discussion/mayo-clinic-q-and-a-treating-your-peptic-ulcer/>

Adrenal Fatigue Protocol

1 Adrenal Insufficiency. In: De Groot LJ, Chrousos G, Dungan K, Feingold KR, Grossman A, Hershman JM, Koch C, Korbonits M, McLachlan R, New M, Purnell J, Rebar R, Singer F, Vinik A, editors. *Endotext* [Internet]. South Dartmouth (MA): MDText.com, Inc.; 2000-2017 Oct 14.

2 Stephens, M. A., & Wand, G. (2012). Stress and the HPA axis: role of glucocorticoids in alcohol dependence. *Alcohol research: current reviews*, 34(4), 468–483

3 Jameson, D. (2016) Persistent Burnout Theory of Chronic Fatigue Syndrome. *Neuroscience and Medicine*, 7, 66-73. doi: 10.4236/nm.2016.72008

4 Kamani Liyanarachchi, Richard Ross, Miguel Debono, Human Studies on Hypothalmo-Pituitary-Adrenal (HPA) Axis, Best Practice & Research Clinical Endocrinology & Metabolism, Volume 31, Issue 5, 2017, Pages 459-473, ISSN 1521-690X, <https://doi.org/10.1016/j.beem.2017.10.011>

5 Neary, N., & Nieman, L. (2010). Adrenal insufficiency: etiology, diagnosis and treatment. Current opinion in endocrinology, diabetes, and obesity, 17(3), 217–223. <https://doi.org/10.1097/MED.0b013e328338f608>

6 Bentham Science Publishers. (2015, November 11). Early Life Stress in Depressive Patients: Role of Glucocorticoid. <https://www.ingentaconnect.com/content/ben/cpd/2015/00000021/00000011/art00003>

7 Diagnosis and Treatment of Primary Adrenal Insufficiency: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*. 2016 Feb;101(2):364–89

8 Burfeind, K. G., Yadav, V., & Marks, D. L. (2016). Hypothalamic Dysfunction and Multiple Sclerosis: Implications for Fatigue and Weight Dysregulation. *Current neurology and neuroscience reports*, 16(11), 98. <https://doi.org/10.1007/s11910-016-0700-3>

9 Liao, L. Y., He, Y. F., Li, L., Meng, H., Dong, Y. M., Yi, F., & Xiao, P. G. (2018). A preliminary review of studies on adaptogens: comparison of their bioactivity in TCM with that of ginseng-like herbs used worldwide. *Chinese medicine*, 13, 57. <https://doi.org/10.1186/s13020-018-0214-9>

10 Head, K. A., & Kelly, G. S. (2009). Nutrients and botanicals for treatment of stress: adrenal fatigue, neurotransmitter imbalance, anxiety, and restless sleep. *Alternative medicine review: a journal of clinical therapeutic*, 14(2), 114–140

11 Kalani, A., Bahtiyar, G., & Sacerdote, A. (2012). Ashwagandha root in the treatment of non-classical adrenal hyperplasia. *BMJ case reports*, 2012, bcr2012006989. <https://doi.org/10.1136/bcr-2012-006989>

12 Wachtel-Galor, S., Yuen, J., Buswell, J. A., & Benzie, I. (2011). *Ganoderma lucidum* (Lingzhi or Reishi): A Medicinal Mushroom. In I. Benzie (Eds.) et. al, *Herbal Medicine: Biomolecular and Clinical Aspects*. (2nd ed.). CRC Press/Taylor & Francis

13 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, *Hericium erinaceus* (Higher Basidiomycetes) from Malaysia. *International journal of medicinal mushrooms*, 15(6), 539–554. <https://doi.org/10.1615/intjmedmushr.v15.i6.30>

14 Ji, D. B., Ye, J., Li, C. L., Wang, Y. H., Zhao, J., & Cai, S. Q. (2009). Antiaging effect of *Cordyceps sinensis* extract. *Phytotherapy research: PTR*, 23(1), 116–122. <https://doi.org/10.1002/ptr.2576>

15 Irfan, M., Kwak, Y. S., Han, C. K., Hyun, S. H., & Rhee, M. H. (2020). Adaptogenic effects of *Panax ginseng* on modulation of cardiovascular functions. *Journal of ginseng research*, 44(4), 538–543. <https://doi.org/10.1016/j.jgr.2020.03.001>

16 Al-Dujaili, E. A., Kenyon, C. J., Nicol, M. R., & Mason, J. I. (2011). Liquorice and glycyrrhetic acid increase DHEA and deoxycorticosterone levels in vivo and in vitro by inhibiting adrenal SULT2A1 activity. *Molecular and cellular endocrinology*, 336(1-2), 102–109. <https://doi.org/10.1016/j.mce.2010.12.011>

17 Li, G., Ma, X., Deng, L., Zhao, X., Wei, Y., Gao, Z., Jia, J., Xu, J., & Sun, C. (2015). Fresh Garlic Extract Enhances the Antimicrobial Activities of Antibiotics on Resistant Strains in Vitro. *Jundishapur journal of microbiology*, 8(5), e14814. <https://doi.org/10.5812/jjm.14814>

18 Mandal, M. D., & Mandal, S. (2011). Honey: its medicinal property and antibacterial activity. *Asian Pacific journal of tropical biomedicine*, 1(2), 154–160. [https://doi.org/10.1016/S2221-1691\(11\)60016-6](https://doi.org/10.1016/S2221-1691(11)60016-6)

19 Sienkiewicz, M., Wasiela, M., & Glowacka, A. (2012). Aktywność przeciwbakteryjna olejku oreganowego (*Origanum heracleoticum* L.) wobec szczepów klinicznych *Escherichia coli* i *Pseudomonas aeruginosa* [The antibacterial activity of oregano essential oil (*Origanum heracleoticum* L.) against clinical strains of *Escherichia coli* and *Pseudomonas aeruginosa*]. *Medycyna doświadczalna i mikrobiologia*, 64(4), 297–307

20 Sepahvand, A., Studzińska-Sroka, E., Ramak, P., & Karimian, V. (2021). *Unsea* sp.: Antimicrobial potential, bioactive compounds, ethnopharmacological uses and other pharmacological properties; a review article. *Journal of ethnopharmacology*, 268, 113656. <https://doi.org/10.1016/j.jep.2020.113656>

21 Turker, A. U., & Gurel, E. (2005). Common mullein (*Verbascum thapsus* L.): recent advances in research. *Phytotherapy research: PTR*, 19(9), 733–739. <https://doi.org/10.1002/ptr.1653>

22 Young, L. M., Pipingas, A., White, D. J., Gauci, S., & Scholey, A. (2019). A Systematic Review and Meta-Analysis of B Vitamin Supplementation on Depressive Symptoms, Anxiety, and Stress: Effects on Healthy and 'At-Risk' Individuals. *Nutrients*, 11(9), 2232. <https://doi.org/10.3390/nu11092232>

23 Voitsekhovskaia, I., Orlikov, G. A., Voskresenskaia, N., Umnova, L. M., Ivanov, I. V., Gredzena, P., Karpov, I., Ianovskaia, I. a., Voitsekhovskii, V. V., & Maulinysh, E. (2013). *Terapevticheskii arkhiv*, 85(10), 76–78

24 J Delarue, O Matzinger, C Binnert, P Schneiter, R Chioléro, L Tappy, Fish oil prevents the adrenal activation elicited by mental stress in healthy men, *Diabetes & Metabolism*, Volume 29, Issue 3, 2003, Pages 289-295, ISSN 1262-3636, [https://doi.org/10.1016/S1262-3636\(07\)70039-3](https://doi.org/10.1016/S1262-3636(07)70039-3)

25 Muscogiuri, G., Altieri, B., Penna-Martinez, M., & Badenhop, K. (2015). Focus on vitamin D and the adrenal gland. *Hormone and metabolic research = Hormon- und Stoffwechselforschung = Hormones et métabolisme*, 47(4), 239–246. <https://doi.org/10.1055/s-0034-1396893>

26 Das, D., Sen, C., & Goswami, A. (2016). Effect of Vitamin C on adrenal suppression by etomidate induction in patients undergoing cardiac surgery: A randomized controlled trial. *Annals of cardiac anaesthesia*, 19(3), 410–417. <https://doi.org/10.4103/0971-9784.185522>

27 Boyle, N. B., Lawton, C., & Dye, L. (2017). The Effects of Magnesium Supplementation on Subjective Anxiety and Stress-A Systematic Review. *Nutrients*, 9(5), 429. <https://doi.org/10.3390/nu9050429>

28 Cater R. E., 2nd (1995). Chronic intestinal candidiasis as a possible etiological factor in the chronic fatigue syndrome. *Medical hypotheses*, 44(6), 507–515. [https://doi.org/10.1016/0306-9877\(95\)90515-4](https://doi.org/10.1016/0306-9877(95)90515-4)

29 Endocrine Testing Protocols: Hypothalamic Pituitary Adrenal Axis. In: De Groot LJ, Chrousos G, Dungan K, Feingold KR, Grossman A, Hershman JM, Koch C, Korbonits M, McLachlan R, New M, Purnell J, Rebar R, Singer F, Vinik A, editors. *Endotext* [Internet]. South Dartmouth (MA): MDText.com, Inc.; 2000- 2017 May 17

Diabetes Type 2 Protocol

1 Sarris, J., & Wardle, J. (2014). *Clinical naturopathy* (2nd ed.). Australia: Elsevier

2 Jiang, S., Young, J. L., Wang, K., Qian, Y., & Cai, L. (2020). Diabetic induced alterations in hepatic glucose and lipid metabolism: The role of type 1 and type 2 diabetes mellitus (Review). *Molecular medicine reports*, 22(2), 603–611. <https://doi.org/10.3892/mmr.2020.11175>

3 Sarris, J., & Wardle, J. (2014). *Clinical naturopathy* (2nd ed.). Australia: Elsevier

4 Goldenberg, J. Z., Day, A., Brinkworth, G. D., Sato, J., Yamada, S., Jönsson, T., Beardsley, J., Johnson, J. A., Thabane, L., & Johnston, B. C. (2021). Efficacy and safety of low and very low carbohydrate diets for type 2 diabetes remission: Systematic review and meta-analysis of published and unpublished Randomized Trial Data. *BMJ*, m4743. <https://doi.org/10.1136/bmj.m4743>

5 Malaeb, S., Bakker, C., Chow, L. S., & Bantle, A. E. (2019). High-Protein Diets for Treatment of Type 2 Diabetes Mellitus: A Systematic Review. *Advances in nutrition* (Bethesda, Md.), 10(4), 621–633. <https://doi.org/10.1093/advances/>

- 6 Sarris, J., & Wardle, J. (2014). Clinical naturopathy (2nd ed.). Australia: Elsevier
- 7 Estruch, R., Ros, E., Salas-Salvadó, J., Covas, M. I., Corella, D., Arós, F., Gómez-Gracia, E., Ruiz-Gutiérrez, V., Fiol, M., Lapetra, J., Lamuela-Raventós, R. M., Serra-Majem, L., Pintó, X., Basora, J., Muñoz, M. A., Sorlí, J. V., Martínez, J. A., Fitó, M., Gea, A., Hernán, M. A., ... PRED-IMED Study Investigators (2018). Primary Prevention of Cardiovascular Disease with a Mediterranean Diet Supplemented with Extra-Virgin Olive Oil or Nuts. *The New England journal of medicine*, 378(25), e34. <https://doi.org/10.1056/NEJMoa1800389>
- 8 Mani, U. V., Mani, L., Biswas, M., & Kumar, S. N. (2011). An open-label study on the effect of flax seed powder (*Linum usitatissimum*) supplementation in the management of diabetes mellitus. *Journal of Dietary Supplements*, 8(3), 257–265. <https://doi.org/10.3109/19390211.2011.593615>
- 9 Johnston, C. S., Kim, C. M., & Buller, A. J. (2004). Vinegar improves insulin sensitivity to a high-carbohydrate meal in subjects with insulin resistance or type 2 diabetes. *Diabetes Care*, 27(1), 281–282. doi:10.2337/diacare.27.1.281
- 10 Helland, A., Bratlie, M., Hagen, I., Mjøs, S., Sørnes, S., Ingvar Halstensen, A., Gudbrandsen, O. (2017). High intake of fatty fish, but not of lean fish, improved postprandial glucose regulation and increased the n-3 PUFA content in the leucocyte membrane in healthy overweight adults: A randomised trial. *British Journal of Nutrition*, 117(10), 1368–1378. doi:10.1017/S0007114517001234
- 11 https://lavierebelle.org/IMG/pdf/a_review_on_diabetes_and_okra_abelmoschus_esculentus.pdf Dubey, P., & Mishra, S. (2017). A review on: Diabetes and okra (*Abelmoschus esculentus*). *J. Med. Plants Stud*, 5(3), 23–26
- 12 <https://www.tandfonline.com/doi/abs/10.1080/14786419.2019.1637874> Daliu, P., Annunziata, G., Tenore, G. C., & Santini, A. (2020). Abscisic acid identification in Okra, *Abelmoschus esculentus* L. (Moench): Perspective nutraceutical use for the treatment of diabetes. *Natural product research*, 34(1), 3–9
- 13 Sikalidis, A. K., & Maykish, A. (2020). The Gut Microbiome and Type 2 Diabetes Mellitus: Discussing a Complex Relationship. *Biomedicines*, 8(1), 8. <https://doi.org/10.3390/biomedicines8010008>
- 14 Qin, J., Li, Y., Cai, Z. et al. A metagenome-wide association study of gut microbiota in type 2 diabetes. *Nature* 490, 55–60 (2012). <https://doi.org/10.1038/nature11450>
- 15 Lee, H., & Ko, G. (2014). Effect of metformin on metabolic improvement and gut microbiota. *Applied and environmental microbiology*, 80(19), 5935–5943. <https://doi.org/10.1128/AEM.01357-14>
- 16 Chen, Z., Radjabzadeh, D., Chen, L., Kurilshikov, A., Kavousi, M., & Ahmadizar, F. et al. (2021). Association of Insulin Resistance and Type 2 Diabetes with Gut Microbial Diversity. *JAMA Network Open*, 4(7), e2118811. doi: 10.1001/jamanetworkopen.2021.18811
- 17 Koutnikova, H., Genser, B., Monteiro-Sepulveda, M., Faurie, J. M., Rizkalla, S., Schrezenmeir, J., & Clément, K. (2019). Impact of bacterial probiotics on obesity, diabetes and non-alcoholic fatty liver disease related variables: a systematic review and meta-analysis of randomised controlled trials. *BMJ open*, 9(3), e017995. <https://doi.org/10.1136/bmjopen-2017-017995>
- 18 Colberg, S., Sigal, R., Yardley, J., Riddell, M., Dunstan, D., & Dempsey, P. et al. (2016). Physical Activity/Exercise and Diabetes: A Position Statement of the American Diabetes Association. *Diabetes Care*, 39(11), 2065–2079. doi: 10.2337/dc16-1728
- 19 Mishra, A., Podder, V., Modgil, S., Khosla, R., Anand, A., Nagarathna, R., Malhotra, R., & Nagendra, H. R. (2020). Higher Perceived Stress and Poor Glycemic Changes in Pre-diabetics and Diabetics Among Indian Population. *Journal of medicine and life*, 13(2), 132–137. <https://doi.org/10.25122/jml-2019-0055>
- 20 Ahola, A. J., Mutter, S., Forsblom, C., Harjutsalo, V., & Groop, P. H. (2019). Meal timing, meal frequency, and breakfast skipping in adult individuals with type 1 diabetes - associations with glycaemic control. *Scientific reports*, 9(1), 20063
- 21 Diane, A., Mahmoud, N., Bensmail, I., Khattab, N., Abunada, H. A., & Dehbi, M. (2020). Alpha lipoic acid attenuates ER stress and improves glucose uptake through DNAJB3 cochaperone. *Scientific reports*, 10(1), 1–15
- 22 Rochette, L., Ghibu, S., Muresan, A., & Vergely, C. (2015). Alpha-lipoic acid: molecular mechanisms and therapeutic potential in diabetes. *Canadian journal of physiology and pharmacology*, 93(12), 1021–1027
- 23 Jacob, S., Ruus, P., Hermann, R., Tritschler, H. J., Maerker, E., Renn, W., ... & Rett, K. (1999). Oral administration of RAC- α -lipoic acid modulates insulin sensitivity in patients with type-2 diabetes mellitus: a placebo-controlled pilot trial. *Free Radical Biology and Medicine*, 27(3–4), 309–314
- 24 Laquale, Kathleen M. (2006). B-complex vitamins' role in energy release. In *Movement Arts, Health Promotion and Leisure Studies* Faculty Publications. Paper25. http://vc.bridgew.edu/mahpls_fac/25
- 25 Pácal, L., Kuricová, K., & Kaňková, K. (2014). Evidence for altered thiamine metabolism in diabetes: Is there a potential to oppose gluco- and lipotoxicity by rational supplementation?. *World journal of diabetes*, 5(3), 288–295. <https://doi.org/10.4239/wjdv5.i3.288>
- 26 Braun, L., & Cohen, M. (2005). Herbs & natural supplements: an evidence-based guide (2nd ed.). Sydney: Elsevier Mosby
- 27 Anderson RA: Chromium, glucose intolerance and diabetes. *J Am Col Nutr* 17:548–555, 1998 Mertz W: Interaction of chromium with insulin: a progress report. *Nutr Rev* 56:174–177, 1998
- 28 Hechtman, L. (2012). Advanced clinical naturopathic medicine. NSW, Australia: Elsevier
- 29 Lepretti M, Martucciello S, Burgos Aceves MA, Putti R, Lionetti L. Omega-3 Fatty Acids and Insulin Resistance: Focus on the Regulation of Mitochondria and Endoplasmic Reticulum Stress. *Nutrients*. 2018 Mar 14;10(3):350. doi: 10.3390/nu10030350. PMID: 29538286; PMCID: PMC5872768. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5872768/>
- 30 Pitocco, D., Martini, F., Zaccardi, F., & Ghirlanda, G. (2012). Antioxidants, Healthy Diet, and Diabetes. *Nutritional And Therapeutic Interventions for Diabetes and Metabolic Syndrome*, 299–313. doi: 10.1016/b978-0-12-385083-6.00024-3
- 31 Braun, L., & Cohen, M. (2005). Herbs & natural supplements: an evidence-based guide (2nd ed.). Sydney: Elsevier Mosby
- 32 Ranasinghe, P., Pigera, S., Galappatthy, P., Katulanda, P., & Constantine, G. R. (2015). Zinc and diabetes mellitus: understanding molecular mechanisms and clinical implications. *Daru: journal of Faculty of Pharmacy, Tehran University of Medical Sciences*, 23(1), 44. <https://doi.org/10.1186/s40199-015-0127-4>
- 33 Braun, L., & Cohen, M. (2005). Herbs & natural supplements: an evidence-based guide (2nd ed.). Sydney: Elsevier Mosby
- 34 Udayakumar, R., Kasthurirengan, S., Mariashibu, T. S., Rajesh, M., Anbazhagan, V. R., Kim, S. C., Ganapathi, A., & Choi, C. W. (2009). Hypoglycaemic and hypolipidaemic effects of *Withania somnifera* root and leaf extracts on alloxan-induced diabetic rats. *International journal of molecular sciences*, 10(5), 2367–2382. <https://doi.org/10.3390/ijms10052367>
- 35 Hechtman, L. (2012). Advanced clinical naturopathic medicine. NSW, Australia: Elsevier
- 36 Braun, L., & Cohen, M. (2005). Herbs & natural supplements: an evidence-based guide (2nd ed.). Sydney: Elsevier Mosby
- 37 Dong, H., Wang, N., Zhao, L., & Lu, F. (2012). Berberine in the treatment of type 2 diabetes mellitus: a systemic review and meta-analysis. *Evidence-based complementary and alternative medicine*, 2012
- 38 Yin, J., Ye, J., & Jia, W. (2012). Effects and mechanisms of berberine in diabetes treatment. *Acta Pharmaceutica Sinica B*, 2(4), 327–334. doi: 10.1016/j.apsb.2012.06.003
- 39 Dong, H., Wang, N., Zhao, L., & Lu, F. (2012). Berberine in the treatment of type 2 diabetes mellitus: a systemic review and meta-analysis. *Evidence-based complementary and alternative medicine: eCAM*, 2012, 591654. <https://doi.org/10.1155/2012/591654>
- 40 Braun, L., & Cohen, M. (2005). Herbs & natural supplements: an evidence-based guide (2nd ed.). Sydney: Elsevier Mosby
- 41 Joseph, B., & Jini, D. (2013). Antidiabetic effects of *Momordica charantia* (bitter melon) and its medicinal potency. *Asian Pacific Journal of Tropical Disease*, 3(2), 93–102. [https://doi.org/10.1016/S2222-1808\(13\)60052-3](https://doi.org/10.1016/S2222-1808(13)60052-3)
- 42 Braun, L., & Cohen, M. (2005). Herbs & natural supplements: an evidence-based guide (2nd ed.). Sydney: Elsevier Mosby
- 43 Braun, L., & Cohen, M. (2005). Herbs & natural supplements: an evidence-based guide (2nd ed.). Sydney: Elsevier Mosby
- 44 <https://www.sciencedirect.com/topics/medicine-and-dentistry/insulin-sensitivity>
- 45 Anderson, R. A., Zhan, Z., Luo, R., Guo, X., Guo, Q., Zhou, J., ... & Stoecker, B. J. (2016). Cinnamon extract lowers glucose, insulin and cholesterol in people with elevated serum glucose. *Journal of traditional and complementary medicine*, 6(4), 332–336
- 46 <https://www.sciencedirect.com/topics/medicine-and-dentistry/low-density-lipo-protein-cholesterol>
- 47 Shang, C., Lin, H., Fang, X., Wang, Y., Jiang, Z., Qu, Y., ... & Cui, X. (2021). Beneficial effects of cinnamon and its extracts in the management of cardiovascular diseases and diabetes. *Food & Function*
- 48 Braun, L., & Cohen, M. (2005). Herbs & natural supplements: an evidence-based guide (2nd ed.). Sydney: Elsevier Mosby
- 49 Yu SH, Chen SY, Li WS, Dubey NK, Chen WH, Chuu JJ, Leu SJ, Deng WP. Hypoglycemic Activity through a Novel Combination of Fruiting Body and Mycelia of *Cordyceps militaris* in High-Fat Diet-Induced Type 2 Diabetes Mellitus Mice. *J Diabetes Res*. 2015;2015:723190. doi: 10.1155/2015/723190. Epub 2015 Jul 16. Erratum in: *J Diabetes Res*. 2017;2017:7947401. PMID: 26258146; PMCID: PMC4519550
- 50 Lo, H. C., Tu, S. T., Lin, K. C., & Lin, S. C. (2004). The anti-hyperglycemic activity of the fruiting body of *Cordyceps* in diabetic rats induced by nicotinamide and streptozotocin. *Life sciences*, 74(23), 2897–2908. <https://doi.org/10.1016/j.lfs.2003.11.003>
- 51 Liu, C., Song, J., Teng, M., Zheng, X., Li, X., Tian, Y., Pan, M., Li, Y., Lee, R. J., & Wang, D. (2016). Antidiabetic and Antinephritic Activities of Aqueous Extract of *Cordyceps militaris* Fruit Body in Diet-Streptozotocin-Induced Diabetic Sprague Dawley Rats. *Oxidative medicine and cellular longevity*, 2016, 9685257. <https://doi.org/10.1155/2016/9685257>
- 52 Yu SH, Chen SY, Li WS, Dubey NK, Chen WH, Chuu JJ, Leu SJ, Deng WP. Hypoglycemic Activity through a Novel Combination of Fruiting Body and Mycelia of *Cordyceps militaris* in High-Fat Diet-Induced Type 2 Diabetes Mellitus Mice. *J Diabetes Res*. 2015;2015:723190. doi: 10.1155/2015/723190. Epub 2015 Jul 16. Erratum in: *J Diabetes Res*. 2017;2017:7947401. PMID: 26258146; PMCID: PMC4519550
- 53 Guo, P., Kai, Q., Gao, J., Lian, Z. Q., Wu, C. M., Wu, C. A., & Zhu, H. B. (2010). Cordycepin prevents hyperlipidemia in hamsters fed a high-fat diet via activation of AMP-activated protein kinase. *Journal of pharmacological sciences*, 113(4), 395–403. <https://doi.org/10.1254/jphs.10041fp> <https://pubmed.ncbi.nlm.nih.gov/20724804/>
- 54 Zhang, H. W., Lin, Z. X., Tung, Y. S., Kwan, T. H., Mok, C. K., Leung, C., & Chan, L. S. (2014). Cordyceps sinensis (a traditional Chinese medicine) for treating chronic kidney disease. *The Cochrane database of systematic reviews*, (12), CD008353. <https://doi.org/10.1002/14651858.CD008353.pub2>
- 55 Shin, S., Lee, S., Kwon, J., Moon, S., Lee, S., Lee, C. K., Cho, K., Ha, N. J., & Kim, K. (2009). Cordycepin Suppresses Expression of Diabetes Regulating Genes by Inhibition of Lipopolysaccharide-induced Inflammation in Macrophages. *Immune network*, 9(3), 98–105. <https://doi.org/10.4110/in.2009.9.3.98>
- 56 Tiwari P, Mishra B. N., Sangwan N. S. (2014). Phytochemical and pharmacological

properties of *Gymnema sylvestre*: An important medicinal plant. *Biomed. Res. Int.* 2014, 1–18. 10.1155/2014/830285

57 Khan F, Sarker MMR, Ming LC, Mohamed IN, Zhao C, Sheikh BY, Tsong HF, Rashid MA. Comprehensive Review on Phytochemicals, Pharmacological and Clinical Potentials of *Gymnema sylvestre*. *Front Pharmacol.* 2019 Oct 29;10:1223. doi: 10.3389/fphar.2019.01223. PMID: 31736747; PMCID: PMC6830388

58 Sherwani SI, Khan HA, Ekhzaimy A, Masood A, Saktharkar MK. Significance of HbA1c Test in Diagnosis and Prognosis of Diabetic Patients. *Biomark Insights.* 2016 Jul 3;11:95-104. doi: 10.4137/BMIS.38440. PMID: 27398023; PMCID: PMC4933534

59 Turner S, Diako C, Kruger R, Wong M, Wood W, Rutherford-Markwick K, Ali A. Consuming *Gymnema sylvestre* Reduces the Desire for High-Sugar Sweet Foods. *Nutrients.* 2020 Apr 10;12(4):1046. doi: 10.3390/nu12041046. PMID: 32290122; PMCID: PMC7230589

60 Asadi, A., Shidfar, F., Safari, M., Hosseini, A. F., Fallah Huseini, H., Heidari, I., & Rajab, A. (2019). Efficacy of *Melissa officinalis* L. (lemon balm) extract on glycemic control and cardiovascular risk factors in individuals with type 2 diabetes: A randomized, double-blind, clinical trial. *Phytotherapy research*, 33(3), 651-659

61 Khodsooz, S., Moshtaghian, J., & Eivani, M. (2016). Antihyperglycemic and antihyperlipidemic effects of hydroalcoholic extract of *Melissa officinalis* (Lemon Balm) in alloxan-induced diabetic rats. *Physiology and pharmacology*, 20(1), 24-30

62 Akhondali Z, Dianat M, Radan M. Negative Chronotropic and Antidysrhythmic Effects of Hydroalcoholic Extract of Lemon Balm (*Melissa officinalis* L.) on CaCl₂-Induced Arrhythmias in Rats. *Electron Physician.* 2015 Mar 1;7(1):971-6. doi: 10.14661/2015.971-976. PMID: 26052407; PMCID: PMC4455297

63 Hechtman, L. (2012). *Advanced clinical naturopathic medicine*. NSW, Australia: Elsevier

64 [https://pubmed.ncbi.nlm.nih.gov/Wińska, K., Mączka, W., Gabryelska, K., & Grabarczyk, M. \(2019\). Mushrooms of the Genus Ganoderma Used to Treat Diabetes and Insulin Resistance. *Molecules \(Basel, Switzerland\)*, 24\(22\), 4075. https://doi.org/10.3390/molecules24224075](https://pubmed.ncbi.nlm.nih.gov/Wińska, K., Mączka, W., Gabryelska, K., & Grabarczyk, M. (2019). Mushrooms of the Genus Ganoderma Used to Treat Diabetes and Insulin Resistance. Molecules (Basel, Switzerland), 24(22), 4075. https://doi.org/10.3390/molecules24224075)

65 Kazazis CE, Evangelopoulos AA, Kollas A, Vallianou NG. The therapeutic potential of milk thistle in diabetes. *Rev Diabet Stud.* 2014 Summer;11(2):167-74. doi: 10.1900/RDS.2014.11.167. Epub 2014 Aug 10. PMID: 25396404; PMCID: PMC4310066

66 Ebrahimipour-Koujan, S., Gargari, B. P., Mobasser, M., Valizadeh, H., & Asghari-Jafarabadi, M. (2018). Lower glycemic indices and lipid profile among type 2 diabetes mellitus patients who received novel dose of *Silybum marianum* (L.) Gaertn. (silymarin) extract supplement: A Triple-blinded randomized controlled clinical trial. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 44, 39–44. https://doi.org/10.1016/j.phymed.2018.03.050

67 Yuan, H. D., Kim, J. T., Kim, S. H., & Chung, S. H. (2012). Ginseng and diabetes: the evidences from in vitro, animal and human studies. *Journal of ginseng research*, 36(1), 27–39. https://doi.org/10.5142/jgr.2012.36.1.27

68 Braun, L., & Cohen, M. (2005). *Herbs & natural supplements: an evidence-based guide* (2nd ed.). Sydney: Elsevier Mosby

69 Ma H.-T., Hsieh J.-F., Chen S.-T. Anti-diabetic effects of *Ganoderma lucidum*. *Phytochemistry.* 2015;114:109–113. doi: 10.1016/j.phytochem.2015.02.017

70 Xiao C, Wu Q, Zhang J, Xie Y, Cai W, Tan J. Antidiabetic activity of *Ganoderma lucidum* polysaccharides F31 down-regulated hepatic glucose regulatory enzymes in diabetic mice. *J Ethnopharmacol.* 2017;196:47-57

71 <https://pubmed.ncbi.nlm.nih.gov/25790910/>

72 <https://www.ncbi.nlm.nih.gov>

73 Cohen MM. *Tulsi - Ocimum sanctum*: A herb for all reasons. *J Ayurveda Integr Med.* 2014 Oct-Dec;5(4):251-9. doi: 10.4103/0975-9476.146554. PMID: 25624701; PMCID: PMC4296439

74 Jamshidi N, Cohen MM. The Clinical Efficacy and Safety of Tulsi in Humans: A Systematic Review of the Literature. *Evid Based Complement Alternat Med.* 2017;2017:9217567. doi: 10.1155/2017/9217567. Epub 2017 Mar 16. PMID: 28400848; PMCID: PMC5376420

75 Sherpa, S. (2022). *Turkey Tail Mushroom: Health Benefits, Side Effects, And Usage Guide*. Benefits

76 Lo, H. C., Hsu, T. H., & Lee, C. H. (2020). Extracellular polysaccharopeptides from fermented Turkey Tail medicinal mushroom, *Trametes versicolor* (Agaricomycetes), mitigate oxidative stress, hyperglycemia, and hyperlipidemia in rats with type 2 diabetes mellitus. *International Journal of Medicinal Mushrooms*, 22(5).

77 Wang, Z., Wang, J., & Chan, P. (2013). Treating type 2 diabetes mellitus with traditional chinese and Indian medicinal herbs. Evidence-based complementary and alternative medicine: eCAM, 2013, 343594. https://doi.org/10.1155/2013/343594

78 Sarris, J., & Wardle, J. (2014). *Clinical naturopathy* (2nd ed.). Australia: Elsevier

Thyroid Protocol

1 <https://www.ncbi.nlm.nih.gov/books/NBK279388/>

2 <https://pubmed.ncbi.nlm.nih.gov/24744751/>

3 <https://www.niddk.nih.gov/health-information/endocrine-diseases/hypothyroidism>

4 <https://www.niddk.nih.gov/health-information/endocrine-diseases/hyperthyroidism>

5 <https://www.ncbi.nlm.nih.gov/books/NBK459466/>

6 <https://www.mayoclinic.org/diseases-conditions/hashimotos-disease/symptoms-causes/syc-20351855>

7 <https://my.clevelandclinic.org/health/diseases/17665-hashimotos-disease>

8 <https://www.thyroid.org/thyroid-nodules/>

9 <https://www.thyroid.org/thyroiditis/>

10 <https://www.cdc.gov/cancer/thyroid/index.htm>

11 Ruggeri, R. M., Giovino, S., Barbalace, M. C., Cristani, M., Alibrandi, A., Vicchio, T. M., ... & Cannavò, S. (2021). Influence of dietary habits on oxidative stress markers in Hashimoto's thyroiditis. *Thyroid*, 31(1), 96-105

12 Szczuko, M., Syrenicz, A., Szymkowiak, K., Przybylska, A., Szczuko, U., Półlocki, J., & Kulpa, D. (2022). Doubtful Justification of the Gluten-Free Diet in the Course of Hashimoto's Disease. *Nutrients*, 14(9), 1727

13 Krysiak, R., Szkróbka, W., & Okopień, B. (2019). The effect of gluten-free diet on thyroid autoimmunity in drug-naïve women with Hashimoto's thyroiditis: a pilot study. *Experimental and Clinical Endocrinology & Diabetes*, 127(07), 417-422

14 Ihnatowicz, P., Drywień, M., Wątor, P., & Wojsiat, J. (2020). The importance of nutritional factors and dietary management of Hashimoto's thyroiditis. *Annals of agricultural and environmental medicine*, 27(2)

15 <https://pubmed.ncbi.nlm.nih.gov/20351569/>

16 <https://pubmed.ncbi.nlm.nih.gov/30294759/>

17 <https://pubmed.ncbi.nlm.nih.gov/31257166/>

18 <https://academic.oup.com/jcem/article/92/11/4180/2598186>

19 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4056127/>

20 <https://www.sciencedirect.com/science/article/abs/pii/S0303720717300758?via%3Dihub>

21 Knezevic, J., Starchl, C., Tmava Berisha, A., & Amrein, K. (2020). Thyroid-Gut-Axis: How Does the Microbiota Influence Thyroid Function?. *Nutrients*, 12(6), 1769. https://doi.org/10.3390/nu12061769

22 Wang, Mingxing et al. "Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion's Mane Medicinal Mushroom, *Hericium erinaceus* (Higher Basidiomycetes)." *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055-60. doi:10.1615/intjmedmushrooms.v17.i11.50. https://pubmed.ncbi.nlm.nih.gov/26853960/.

23 <https://pubmed.ncbi.nlm.nih.gov/17240130/>

24 Wang, Mingxing et al. "Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion's Mane Medicinal Mushroom, *Hericium erinaceus* (Higher Basidiomycetes)." *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055-60. doi:10.1615/intjmedmushrooms.v17.i11.50. https://pubmed.ncbi.nlm.nih.gov/26853960/.

25 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7783124/>

26 <https://pubmed.ncbi.nlm.nih.gov/25591468/>

27 <https://www.liebertpub.com/doi/abs/10.1089/acm.2017.0183>

28 <https://pubmed.ncbi.nlm.nih.gov/28829155/>

29 <https://pubmed.ncbi.nlm.nih.gov/11535365/>

30 He, T., Zhao, R., Lu, Y., Li, W., Hou, X., Sun, Y., Dong, M., & Chen, L. (2016). Dual-Directional Immunomodulatory Effects of Corbrin Capsule on Autoimmune Thyroid Diseases. Evidence-based complementary and alternative medicine: eCAM, 2016, 1360386. https://doi.org/10.1155/2016/1360386

31 He, T., Zhao, R., Lu, Y., Li, W., Hou, X., Sun, Y., Dong, M., & Chen, L. (2016). Dual-Directional Immunomodulatory Effects of Corbrin Capsule on Autoimmune Thyroid Diseases. Evidence-based complementary and alternative medicine: eCAM, 2016, 1360386. https://doi.org/10.1155/2016/1360386

32 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6287405/>

33 <https://pubmed.ncbi.nlm.nih.gov/30136914/>

34 <https://www.ncbi.nlm.nih.gov/pubmedhealth/PMH0046740/>

35 <https://www.mskcc.org/cancer-care/integrative-medicine/herbs/reishi-mushroom>

36 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4049553/>

37 Arbaizar, B., & Llorca, J. (2011). Hipertiroidismo inducido por la ingestión de *Fucus vesiculosus* en un paciente en tratamiento concomitante con litio [*Fucus vesiculosus* induced hyperthyroidism in a patient undergoing concomitant treatment with lithium]. *Actas españolas de psiquiatria*, 39(6), 401–403

38 Laurberg P. (1984). Forskolin stimulation of thyroid secretion of T₄ and T₃. *FEBS letters*, 170(2), 273–276. https://doi.org/10.1016/0014-5793(84)81327-7

39 Sharma, A. K., Basu, I., & Singh, S. (2018). Efficacy and Safety of Ashwagandha Root Extract in Subclinical Hypothyroid Patients: A Double-Blind, Randomized Placebo-Controlled Trial. *Journal of alternative and complementary medicine (New York, N.Y.)*, 24(3), 243–248. https://doi.org/10.1089/acm.2017.0183

40 Sharma, A. K., Basu, I., & Singh, S. (2018). Efficacy and Safety of Ashwagandha Root Extract in Subclinical Hypothyroid Patients: A Double-Blind, Randomized Placebo-Controlled Trial. *Journal of alternative and complementary medicine (New York, N.Y.)*, 24(3), 243–248. https://doi.org/10.1089/acm.2017.0183

41 Eiling R, Wieland V, Niestroj M. Improvement of symptoms in mild hyperthyroidism with an extract of *Lycopus europaeus* (Thyreogutt® mono). *Wien Med Wochenschr.* 2013;163(3-4):95-101. doi:10.1007/s10354-012-0167-z

42 Kaplan, D., & Dosiou, C. (2021). Two Cases of Graves' Hyperthyroidism Treated With Homeopathic Remedies Containing Herbal Extracts from *Lycopus* spp. and *Melissa officinalis*. *Journal of the Endocrine Society*, 5(Suppl 1), A971. https://doi.org/10.1210/jendso/bvab048.1984

43 Auf'Molk, M., et al. "Extracts and Auto-Oxidized Constituents of Certain Plants Inhibit the Receptor-Binding and the Biological Activity of Graves' Igs*"

- 44 Winterhoff, H., et al. "Endocrine Effects of Lycopodium Europaeus L. Following Oral Application." *Arzneim. Forsch.*, vol. I, no. 44, 1944
- 45 Kaplan, D., & Dosiou, C. (2021). Two Cases of Graves' Hyperthyroidism Treated with Homeopathic Remedies Containing Herbal Extracts from Lycopodium spp. and Melissa officinalis. *Journal of the Endocrine Society*, 5(Suppl 1), A971. <https://doi.org/10.1210/jendo/bvab048.1984>
- 46 Beer, A. M., Wiebelitz, K. R., & Schmidt-Gayk, H. (2008). Lycopodium europaeus (Gypsowort): effects on the thyroidal parameters and symptoms associated with thyroid function. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 15(1-2), 16–22
- 47 Kaplan, D., & Dosiou, C. (2021). Two Cases of Graves' Hyperthyroidism Treated with Homeopathic Remedies Containing Herbal Extracts from Lycopodium spp. and Melissa officinalis. *Journal of the Endocrine Society*, 5(Suppl 1), A971. <https://doi.org/10.1210/jendo/bvab048.1984>
- 48 He, T., Zhao, R., Lu, Y., Li, W., Hou, X., Sun, Y., Dong, M., & Chen, L. (2016). Dual-Directional Immunomodulatory Effects of Corbrin Capsule on Autoimmune Thyroid Diseases. Evidence-based complementary and alternative medicine: eCAM, 2016, 1360386. <https://doi.org/10.1155/2016/1360386>
- 49 Kaplan, D., & Dosiou, C. (2021). Two Cases of Graves' Hyperthyroidism Treated with Homeopathic Remedies Containing Herbal Extracts from Lycopodium spp. and Melissa officinalis. *Journal of the Endocrine Society*, 5(Suppl 1), A971. <https://doi.org/10.1210/jendo/bvab048.1984>
- 50 Kaplan, D., & Dosiou, C. (2021). Two Cases of Graves' Hyperthyroidism Treated with Homeopathic Remedies Containing Herbal Extracts from Lycopodium spp. and Melissa officinalis. *Journal of the Endocrine Society*, 5(Suppl 1), A971. <https://doi.org/10.1210/jendo/bvab048.1984>
- 51 Shikov, A. N., Pozharitskaya, O. N., Makarov, V. G., Demchenko, D. V., & Shikh, E. V. (2011). Effect of Leonurus cardiaca oil extract in patients with arterial hypertension accompanied by anxiety and sleep disorders. *Phytotherapy research: PTR*, 25(4), 540–543. <https://doi.org/10.1002/ptr.3292>
- 52 Fierascu, R. C., Fierascu, I., Ortan, A., Fierascu, I. C., Anuta, V., Velescu, B. S., Pituru, S. M., & Dinu-Pirvu, C. E. (2019). Leonurus cardiaca L. as a Source of Bioactive Compounds: An Update of the European Medicines Agency Assessment Report (2010). *BioMed research international*, 2019, 4303215. <https://doi.org/10.1155/2019/4303215>
- 53 Kim, M., & Lee, B. C. (2019). Therapeutic Effect of Scutellaria baicalensis on L-Thyroxine-Induced Hyperthyroidism Rats. Evidence-based complementary and alternative medicine: eCAM, 2019, 3239649. <https://doi.org/10.1155/2019/3239649>
- 54 Winterhoff, H., Sourgens, H., & Kemper, F. H. (1983). Antihormonal effects of plant extracts. Pharmacodynamic effects of lithospermum officinale on the thyroid gland of rats; comparison with the effects of iodide. *Hormone and metabolic research = Hormon- und Stoffwechselforschung = Hormones et métabolisme*, 15(10), 503–507. <https://doi.org/10.1055/s-2007-1018769>
- 55 Breneman, W. R., & Zeller, F. J. (1975). Lithospermum inhibition of anterior pituitary hormones. *Biochemical and biophysical research communications*, 65(3), 1047–1053. [https://doi.org/10.1016/s0006-291x\(75\)80491-8](https://doi.org/10.1016/s0006-291x(75)80491-8)
- 56 He, T., Zhao, R., Lu, Y., Li, W., Hou, X., Sun, Y., Dong, M., & Chen, L. (2016). Dual-Directional Immunomodulatory Effects of Corbrin Capsule on Autoimmune Thyroid Diseases. Evidence-based complementary and alternative medicine: eCAM, 2016, 1360386. <https://doi.org/10.1155/2016/1360386>
- 57 Jin, M., Zhou, W., Jin, C., Jiang, Z., Diao, S., Jin, Z., & Li, G. (2019). Anti-inflammatory activities of the chemical constituents isolated from Trametes versicolor. *Natural product research*, 33(16), 2422–2425. <https://doi.org/10.1080/14786419.2018.1446011>
- 58 Chen, J. L., Chen, Y. C., Yang, S. H., Ko, Y. F., & Chen, S. Y. (2009). Immunological alterations in lupus-prone autoimmune (NZB/NZW) F1 mice by mycelia Chinese medicinal fungus Cordyceps sinensis-induced redistributions of peripheral mononuclear T lymphocytes. *Clinical and experimental medicine*, 9(4), 277–284. <https://doi.org/10.1007/s10238-009-0043-3>
- 59 Friedman M. (2015). Chemistry, Nutrition, and Health-Promoting Properties of Hericium erinaceus (Lion's Mane) Mushroom Fruiting Bodies and Mycelia and Their Bioactive Compounds. *Journal of agricultural and food chemistry*, 63(32), 7108–7123. <https://doi.org/10.1021/acs.jafc.5b02914>
- 60 Cao, J., Su, Y., Chen, Z., Ma, C., & Xiong, W. (2022). The risk factors for Graves' ophthalmopathy. Graefes's archive for clinical and experimental ophthalmology = Albrecht von Graefes Archiv für klinische und experimentelle Ophthalmologie, 260(4), 1043–1054. <https://doi.org/10.1007/s00417-021-05456-x>
- 61 Winther, K. H., Rayman, M. P., Bonnema, S. J., & Hegedüs, L. (2020). Selenium in thyroid disorders - essential knowledge for clinicians. *Nature reviews. Endocrinology*, 16(3), 165–176. <https://doi.org/10.1038/s41574-019-0311-6>
- 62 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5618598/>
- 63 Chao, G., Zhu, Y., & Fang, L. (2020). Correlation Between Hashimoto's Thyroiditis-Related Thyroid Hormone Levels and 25-Hydroxyvitamin D. *Frontiers in endocrinology*, 11, 4. <https://doi.org/10.3389/fendo.2020.00004>
- 64 Ahn, H. Y., Chung, Y. J., & Cho, B. Y. (2017). Serum 25-hydroxyvitamin D might be an independent prognostic factor for Graves' disease recurrence. *Medicine*, 96(31), e7700. <https://doi.org/10.1097/MD.00000000000007700>
- 65 Taheriniya, S., Arab, A., Hadi, A., Fadel, A., & Askari, G. (2021). Vitamin D and thyroid disorders: a systematic review and Meta-analysis of observational studies. *BMC endocrine disorders*, 21(1), 171. <https://doi.org/10.1186/s12902-021-00831-5>
- 66 <https://pubmed.ncbi.nlm.nih.gov/19693445/>

- Stress and Therapeutic Potential of Antioxidants in Graves' Ophthalmopathy. *Biomedicines*, 9(12), 1871. <https://doi.org/10.3390/biomedicines9121871>
- 68 <https://pubmed.ncbi.nlm.nih.gov/16500878/>
- 69 Jabczyk, M., Nowak, J., Hudzik, B., & Zubelewicz-Szkodzińska, B. (2021). Curcumin in Metabolic Health and Disease. *Nutrients*, 13(12), 4440. <https://doi.org/10.3390/nu13124440>
- 70 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3569681/>
- 71 Ferrari, S. M., Fallahi, P., Antonelli, A., & Benvenega, S. (2017). Environmental Issues in Thyroid Diseases. *Frontiers in endocrinology*, 8, 50. <https://doi.org/10.3389/fendo.2017.00050>
- 72 Benvenega, S., Elia, G., Ragusa, F., Paparo, S. R., Sturniolo, M. M., Ferrari, S. M., Antonelli, A., & Fallahi, P. (2020). Endocrine disruptors and thyroid autoimmunity. Best practice & research. *Clinical endocrinology & metabolism*, 34(1), 101377. <https://doi.org/10.1016/j.beem.2020.101377>
- 73 <https://pubmed.ncbi.nlm.nih.gov/12762077/>
- 74 <https://pubs.acs.org/doi/abs/10.1021/es1041942>
- 75 <https://link.springer.com/article/10.2478/s11536-009-0092-8>
- 76 <https://www.mayoclinic.org/adrenal-glands/img-20008674>
- 77 <https://thyroidresearchjournal.biomedcentral.com/articles/10.1186/1756-6614-5-13>
- 78 Fukao, A., Takamatsu, J., Murakami, Y., Sakane, S., Miyauchi, A., Kuma, K., Hayashi, S., & Hanafusa, T. (2003). The relationship of psychological factors to the prognosis of hyperthyroidism in antithyroid drug-treated patients with Graves' disease. *Clinical endocrinology*, 58(5), 550–555. <https://doi.org/10.1046/j.1365-2265.2003.01625.x>
- 79 <https://pubmed.ncbi.nlm.nih.gov/15650357/>
- 80 <https://pubmed.ncbi.nlm.nih.gov/15650357/>
- 81 Lindahl O, Lindwall L. Double blind study of a valerian preparation. *Pharmacol Biochem Behav.* 1989 Apr;32(4):1065-6. doi: 10.1016/0091-3057(89)90082-8. PMID: 2678162. <https://pubmed.ncbi.nlm.nih.gov/2678162/>
- 82 <https://stanfordmedicine25.stanford.edu/the25/thyroid.html>

Bacterial Infection Protocol

- 1 Shomali, N., Mahmoudi, J., Mahmoodpoor, A., Zamiri, R. E., Akbari, M., Xu, H., & Shotorbani, S. S. (2021). Harmful effects of high amounts of glucose on the immune system: An updated review. *Biotechnology and applied biochemistry*, 68(2), 404–410. <https://doi.org/10.1002/bab.1938>
- 2 Yu, Z., Wang, Y., Lu, J., Bond, P. L., & Guo, J. (2021). Nonnutritive sweeteners can promote the dissemination of antibiotic resistance through conjugative gene transfer. *ISME J*, 15:2117–2130. <https://doi.org/10.1038/s41396-021-00909-x>
- 3 Dahl-Jørgensen, K., Joner, G., & Hanssen, K. F. (1991). Relationship between cows' milk consumption and incidence of IDDM in childhood. *Diabetes care*, 14(11), 1081–1083. <https://doi.org/10.2337/diacare.14.11.1081>
- 4 Gerstein H. C. (1994). Cow's milk exposure and type I diabetes mellitus. A critical overview of the clinical literature. *Diabetes care*, 17(1), 13–19. <https://doi.org/10.2337/diacare.17.1.13>
- 5 Malosse, D., Perron, H., Sasso, A., & Seigneurin, J. M. (1992). Correlation between milk and dairy product consumption and multiple sclerosis prevalence: a worldwide study. *Neuroepidemiology*, 11(4-6), 304–312. <https://doi.org/10.1159/000110946>
- 6 Agranoff, B. W., & Goldberg, D. (1974). Diet and the geographical distribution of multiple sclerosis. *Lancet (London, England)*, 2(7888), 1061–1066. [https://doi.org/10.1016/s0140-6736\(74\)92163-1](https://doi.org/10.1016/s0140-6736(74)92163-1)
- 7 Frosh, A., Cruz, C., Wellsted, D., & Stephens, J. (2019). Effect of a dairy diet on nasopharyngeal mucus secretion. *The Laryngoscope*, 129(1), 13–17. <https://doi.org/10.1002/lary.27287>
- 8 Phillips, B. L., & Bland, R. D. (1972). Otitis media, milk allergy, and folk medicine. *Pediatrics*, 50(2), 346.
- 9 Sarkar, D., Jung, M. K., & Wang, H. J. (2015). Alcohol and the Immune System. *Alcohol Research: Current Reviews*, 37(2), 153–155.
- 10 Sarkar, D., Jung, M. K., & Wang, H. J. (2015). Alcohol and the Immune System. *Alcohol Research: Current Reviews*, 37(2), 153–155.
- 11 Qiu, F., Liang, C. L., Liu, H., Zeng, Y. Q., Hou, S., Huang, S., Lai, X., & Dai, Z. (2017). Impacts of cigarette smoking on immune responsiveness: Up and down or upside down?. *Oncotarget*, 8(1), 268–284. <https://doi.org/10.18632/oncotarget.13613>
- 12 Myles I. A. (2014). Fast food fever: reviewing the impacts of the Western diet on immunity. *Nutrition journal*, 13, 61. <https://doi.org/10.1186/1475-2891-13-61>
- 13 Samuel, A. O., Huang, B. T., Chen, Y., Guo, F. X., Yang, D. D., & Jin, J. Q. (2021). Antioxidant and antibacterial insights into the leaves, leaf tea and medicinal roots from Astragalus membranaceus (Fisch.) Bge. *Scientific reports*, 11(1), 19625. <https://doi.org/10.1038/s41598-021-97109-6>
- 14 Sullivan, A. M., Laba, J. G., Moore, J. A., & Lee, T. D. (2008). Echinacea-induced macrophage activation. *Immunopharmacology and immunotoxicology*, 30(3), 553–574. <https://doi.org/10.1080/08923970802135534>
- 15 Moghtaderi, M., Mirzaie, A., Zabet, N., Moammeri, A., Mansoori-Kermani, A., Akbarzadeh, I., Eshrati Yeganeh, F., Chitgarzadeh, A., Bagheri Kashtali, A., & Ren, Q. (2021). Enhanced Antibacterial Activity of Echinacea angustifolia Extract against Multidrug-Resistant Klebsiella pneumoniae through Niosome Encapsulation. *Nanomaterials (Basel, Switzerland)*, 11(6), 1573. <https://doi.org/10.3390/nano11061573>
- 16 Hudson J. B. (2012). Applications of the phytochemistry Echinacea purpurea (Purple Coneflower) in infectious diseases. *Journal of biomedicine & biotechnology*, 2012,

- 17 Cech, N. B., Junio, H. A., Ackermann, L. W., Kavanaugh, J. S., & Horswill, A. R. (2012). Quorum quenching and antimicrobial activity of goldenseal (*Hydrastis canadensis*) against methicillin-resistant *Staphylococcus aureus* (MRSA). *Planta medica*, 78(14), 1556–1561. <https://doi.org/10.1055/s-0032-1315042>
- 18 Omar S. H. (2010). Oleuropein in olive and its pharmacological effects. *Scientia pharmaceutica*, 78(2), 133–154. <https://doi.org/10.3797/scipharm.0912-18>
- 19 Sudjana, A. N., D'Orazio, C., Ryan, V., Rasool, N., Ng, J., Islam, N., Riley, T. V., & Hammer, K. A. (2009). Antimicrobial activity of commercial *Olea europaea* (olive) leaf extract. *International journal of antimicrobial agents*, 33(5), 461–463. <https://doi.org/10.1016/j.ijantimicag.2008.10.026>
- 20 Kosakowska, O., Węglarz, Z., Pióro-Jabucka, E., Przybył, J. L., Krasniewska, K., Gniewosz, M., & Bączek, K. (2021). Antioxidant and Antibacterial Activity of Essential Oils and Hydroethanolic Extracts of Greek Oregano (*O. vulgare* L. subsp. *hirtum* (Link) letswardt) and Common Oregano (*O. vulgare* L. subsp. *vulgare*). *Molecules (Basel, Switzerland)*, 26(4), 988. <https://doi.org/10.3390/molecules26040988>
- 21 Antih, J., Houdkova, M., Urbanova, K., & Kokoska, L. (2021). Antibacterial Activity of *Thymus vulgaris* L. Essential Oil Vapours and Their GC/MS Analysis Using Solid-Phase Microextraction and Syringe Headspace Sampling Techniques. *Molecules (Basel, Switzerland)*, 26(21), 6553. <https://doi.org/10.3390/molecules26216553>
- 22 Micucci, M., Protti, M., Aldini, R., Frosini, M., Corazza, L., Marzetti, C., Mattioli, L. B., Tocci, G., Chiarini, A., Mercolini, L., & Budriesi, R. (2020). *Thymus vulgaris* L. Essential Oil Solid Formulation: Chemical Profile and Spasmolytic and Antimicrobial Effects. *Biomolecules*, 10(6), 860. <https://doi.org/10.3390/biom10060860>
- 23 Patil, S. M., Ramu, R., Shirahatti, P. S., Shivamallu, C., & Amachawadi, R. G. (2021). A systematic review on ethnopharmacology, phytochemistry and pharmacological aspects of *Thymus vulgaris* Linn. *Heliyon*, 7(5), e07054. <https://doi.org/10.1016/j.heliyon.2021.e07054>
- 24 Patil, S. M., Ramu, R., Shirahatti, P. S., Shivamallu, C., & Amachawadi, R. G. (2021). A systematic review on ethnopharmacology, phytochemistry and pharmacological aspects of *Thymus vulgaris* Linn. *Heliyon*, 7(5), e07054. <https://doi.org/10.1016/j.heliyon.2021.e07054>
- 25 Niamsa, N., & Sittiwet, C. (2009). Antimicrobial activity of *Curcuma longa* aqueous extract. *Journal of Pharmacology and Toxicology*, 4(4):173–177.
- 26 Ungphai boon, S., Supavita, T., Singchangchai, P., Sungkarak, S., Rattanasuwan, P., & Itharat, A. (2005). Study on antioxidant and antimicrobial activities of turmeric clear liquid soap for wound treatment of HIV patients. *Songklanakarin Journal of Science and Technology*, 27(2):269–278
- 27 Popovici, V., Bucur, L., Calcan, S. I., Cuculea, E. I., Costache, T., Rambu, D., Schröder, V., Gird, C. E., Gherghel, D., Vochita, G., Caraiene, A., & Badea, V. (2021). Elemental Analysis and In Vitro Evaluation of Antibacterial and Antifungal Activities of *Usnea barbata* (L.) Weber ex F.H. Wigg from Călimani Mountains, Romania. *Plants (Basel, Switzerland)*, 11(1), 32. <https://doi.org/10.3390/plants11010032>
- 28 El-Kalamouni, C., Venskutonis, P. R., Zebib, B., Merah, O., Raynaud, C., & Talou, T. (2017). Antioxidant and Antimicrobial Activities of the Essential Oil of *Achillea millefolium* L. Grown in France. *Medicines (Basel, Switzerland)*, 4(2), 30. <https://doi.org/10.3390/medicines4020030>
- 29 Cör, D., Knez, Ž., & Knez Hrnič, M. (2018). Antitumour, Antimicrobial, Antioxidant and Antiacetylcholinesterase Effect of *Ganoderma lucidum* Terpenoids and Polysaccharides: A Review. *Molecules (Basel, Switzerland)*, 23(3), 649. <https://doi.org/10.3390/molecules23030649>
- 30 Li, D. G., & Ren, Z. X. (2017). *Cordyceps sinensis* promotes immune regulation and enhances bacteriostatic activity of PA-824 via IL-10 in *Mycobacterium tuberculosis* disease. *Brazilian journal of medical and biological research = Revista brasileira de pesquisas medicas e biologicas*, 50(9), e6188. <https://doi.org/10.1590/1414-431X20176188>
- 31 Shang, X., Tan, Q., Liu, R., Yu, K., Li, P., & Zhao, G. P. (2013). In vitro anti-*Helicobacter pylori* effects of medicinal mushroom extracts, with special emphasis on the Lion's Mane mushroom, *Hericium erinaceus* (higher Basidiomycetes). *International journal of medicinal mushrooms*, 15(2), 165–174. <https://doi.org/10.1615/intjmedmushr.v15.i2.50>
- 32 Pilzk, Z., Mycol, A.J., Timbreza, L.P., Reyes, J.L., Flores, C.H., Perez, R., Stockel, M.A., & Santiago, K.A. (2017). Antibacterial activities of the lichen *Ramalina* and *Usnea* collected from Mt. Banoi, Batangas and Dahilayan, Bukidnon, against multi-drug resistant (MDR) bacteria.
- 33 Sleha, R., Radochova, V., Malis, J., Mikyska, A., Houska, M., Krofta, K., Bogdanova, K., Janovska, S., Pejchal, J., Kolar, M., Cermak, P., & Bostik, P. (2021). Strong Antimicrobial and Healing Effects of Beta-Acids from Hops in Methicillin-Resistant *Staphylococcus aureus*-Infected External Wounds In Vivo. *Antibiotics (Basel, Switzerland)*, 10(6), 708. <https://doi.org/10.3390/antibiotics10060708>
- 34 Budán, F., Nan, M., Kocsis, B., & Laczkó-Zöld, E. (2021). Antimicrobial activity and potential secondary signal transduction mechanisms of *elecampane* (*Inula helenium* L.) root extract. *Plant cell biotechnology and molecular biology*, 22(7-8), 86-92.
- 35 Doyle, T. (2016). *Mahonia aquifolia*, commonly known as Oregon Grape, is an effective antibacterial, including treatment of Methicillin-resistant *Staphylococcus aureus* (MRSA). *American College of Healthcare*.
- 36 Zuo, G. Y., Li, Y., Han, J., Wang, G. C., Zhang, Y. L., & Bian, Z. Q. (2012). Antibacterial and synergy of berberines with antibacterial agents against clinical multi-drug resistant isolates of methicillin-resistant *Staphylococcus aureus* (MRSA). *Molecules (Basel, Switzerland)*, 17(9), 10322–10330. <https://doi.org/10.3390/molecules170910322>
- 37 Gatea-Kaabi, S.A., Abdulrazaq, R.A., Rasool, K.H., & Khassaf, S.A. (2020). Western herbal remedies for Urinary Tract infections. *Archive of Urological Research*, 4(1):49-60.
- 38 Fazly Bazzaz, B.S., Darvishi Fork, S., Ahmadi, R., & Khameneh, B. (2021). Deep insights into urinary tract infections and effective natural remedies. *Afr J Urol* 27(6). <https://doi.org/10.1186/s12301-020-00111-z>
- 39 Das S. (2020). Natural therapeutics for urinary tract infections-a review. *Future journal of pharmaceutical sciences*, 6(1), 64. <https://doi.org/10.1186/s43094-020-00086-2>
- 40 Tsao, S. M., Liu, W. H., & Yin, M. C. (2007). Two diallyl sulphides derived from garlic inhibit methicillin-resistant *Staphylococcus aureus* infection in diabetic mice. *Journal of medical microbiology*, 56(Pt 6), 803–808. <https://doi.org/10.1099/jmm.0.46998-0>
- 41 Nidadavolu, P., Amor, W., Tran, P. L., Dertien, J., Colmer-Hamood, J. A., & Hamood, A. N. (2012). Garlic ointment inhibits biofilm formation by bacterial pathogens from burn wounds. *Journal of medical microbiology*, 61(Pt 5), 662–671. <https://doi.org/10.1099/jmm.0.038638-0>
- 42 Cutler, R. R., & Wilson, P. (2004). Antibacterial activity of a new, stable, aqueous extract of allicin against methicillin-resistant *Staphylococcus aureus*. *British journal of biomedical science*, 61(2), 71–74. <https://doi.org/10.1080/09674845.2004.11732646>
- 43 Li, G., Ma, X., Deng, L., Zhao, X., Wei, Y., Gao, Z., Jia, J., Xu, J., & Sun, C. (2015). Fresh Garlic Extract Enhances the Antimicrobial Activities of Antibiotics on Resistant Strains in Vitro. *Jundishapur journal of microbiology*, 8(5), e14814. <https://doi.org/10.5812/jjm.14814>
- 44 Kim J. H. (1997). Anti-bacterial action of onion (*Allium cepa* L.) extracts against oral pathogenic bacteria. *The Journal of Nihon University School of Dentistry*, 39(3), 136–141. <https://doi.org/10.2334/josnuds1959.39.136>
- 45 Sharma, K., Mahato, N., & Lee, Y. R. (2018). Systematic study on active compounds as antibacterial and antibiofilm agent in aging onions. *Journal of food and drug analysis*, 26(2), 518–528. <https://doi.org/10.1016/j.jfda.2017.06.009>
- 46 Hannan, A., Humayun, T., Hussain, M. B., Yasir, M., & Sikandar, S. (2010). In vitro antibacterial activity of onion (*Allium cepa*) against clinical isolates of *Vibrio cholerae*. *Journal of Ayub Medical College, Abbottabad: JAMC*, 22(2), 160–163.
- 47 Mahomoodally, F., Ramcharun, S., & Zengin, G. (2018). Onion and Garlic Extracts Potentiate the Efficacy of Conventional Antibiotics against Standard and Clinical Bacterial Isolates. *Current topics in medicinal chemistry*, 18(9), 787–796. <https://doi.org/10.2174/1568026618666180604083313>
- 48 Mandal, M. D., & Mandal, S. (2011). Honey: its medicinal property and antibacterial activity. *Asian Pacific journal of tropical biomedicine*, 1(2), 154–160. [https://doi.org/10.1016/S2221-1691\(11\)60016-6](https://doi.org/10.1016/S2221-1691(11)60016-6)
- 49 Tonks, A. J., Dudley, E., Porter, N. G., Parton, J., Brazier, J., Smith, E. L., & Tonks, A. (2007). A 5.8-kDa component of manuka honey stimulates immune cells via TLR4. *Journal of leukocyte biology*, 82(5), 1147–1155. <https://doi.org/10.1189/jlb.1106683>
- 50 Lusby, P. E., Coombes, A. L., & Wilkinson, J. M. (2005). Bactericidal activity of different honeys against pathogenic bacteria. *Archives of medical research*, 36(5), 464–467. <https://doi.org/10.1016/j.arcmed.2005.03.038>
- 51 Yagnik, D., Serafin, V., & J Shah, A. (2018). Antimicrobial activity of apple cider vinegar against *Escherichia coli*, *Staphylococcus aureus* and *Candida albicans*; downregulating cytokine and microbial protein expression. *Scientific reports*, 8(1), 1732. <https://doi.org/10.1038/s41598-017-18618-x>
- 52 Yagnik, D., Ward, M. & Shah, A.J. (2021). Antibacterial apple cider vinegar eradicates methicillin resistant *Staphylococcus aureus* and resistant *Escherichia coli*. *Sci Rep* 11(1854). <https://doi.org/10.1038/s41598-020-78407-x>
- 53 Manjunatha, J. R., Bettadaiah, B. K., Negi, P. S., & Srinivas, P. (2013). Synthesis of quinoline derivatives of tetrahydrocurcumin and zingerone and evaluation of their antioxidant and antibacterial attributes. *Food chemistry*, 136(2), 650–658. <https://doi.org/10.1016/j.foodchem.2012.08.052>
- 54 Mahady, G. B., Pendland, S. L., Yun, G. S., Lu, Z. Z., & Stoia, A. (2003). Ginger (*Zingiber officinale* Roscoe) and the gingerols inhibit the growth of *Cag A+* strains of *Helicobacter pylori*. *Anticancer research*, 23(5A), 3699–3702.
- 55 Azu N, Onyeagba R. (2007). Antimicrobial Properties Of Extracts Of *Allium cepa* (Onions) and *Zingiber officinale* (Ginger) On *Escherichia coli*, *Salmonella typhi* and *Bacillus subtilis*. *The Internet Journal of Tropical Medicine*. 2007(3):1–10
- 56 Hiserodt, R.D., Franzblau, S.G., Rosen, R.T. (1998). Isolation of 6-, 8-, and 10-Gingerol from Ginger Rhizome by HPLC and Preliminary Evaluation of Inhibition of *Mycobacterium avium* and *Mycobacterium tuberculosis*. *Agric Food Chem*. 1998(46):2504–2508.
- 57 Chambial, S., Dwivedi, S., Shukla, K. K., John, P. J., & Sharma, P. (2013). Vitamin C in disease prevention and cure: an overview. *Indian journal of clinical biochemistry: IJCB*, 28(4), 314–328. <https://doi.org/10.1007/s12291-013-0375-3>
- 58 Mar-Solis, L. M., Soto-Domínguez, A., Rodríguez-Tovar, L. E., Rodríguez-Rocha, H., García-García, A., Aguirre-Arzola, V. E., Zamora-Ávila, D. E., Garza-Arredondo, A. J., & Castillo-Velázquez, U. (2021). Analysis of the Anti-Inflammatory Capacity of Bone Broth in a Murine Model of Ulcerative Colitis. *Medicina (Kaunas, Lithuania)*, 57(11), 1138. <https://doi.org/10.3390/medicina57111138>
- 59 Satomura, K., Kitamura, T., Kawamura, T., Shimbo, T., Watanabe, M., Kamei, M., Takano, Y., Tamakoshi, A., & Great Cold Investigators-I (2005). Prevention of upper respiratory tract infections by gargling: a randomized trial. *American journal of preventive medicine*, 29(4), 302–307. <https://doi.org/10.1016/j.amepre.2005.06.013>
- 60 Zobell, C. E., Anderson, D. Q., & Smith, W. W. (1937). The Bacteriostatic and Bactericidal Action of Great Salt Lake Water. *Journal of bacteriology*, 33(3), 253–262. <https://doi.org/10.1128/jb.33.3.253-262.1937>
- 61 Jantsch, J., Schatz, V., Friedrich, D., Schröder, A., Kopp, C., Siegert, I., Maronna, A., Wendelborn, D., Linz, P., Binger, K. J., Gebhardt, M., Heinig, M., Neubert, P., Fischer, F., Teufel, S., David, J. P., Neufert, C., Cavallaro, A., Rakova, N., Küper, C., ... Titze, J. (2015). Cutaneous Na⁺ storage strengthens the antimicrobial barrier function of the skin and boosts macrophage-driven host defense. *Cell metabolism*, 21(3), 493–501. <https://doi.org/10.1016/j.cmet.2015.02.003>

- 62 Emamian, M. H., Hassani, A. M., & Fateh, M. (2013). Respiratory Tract Infections and its Preventive Measures among Hajj Pilgrims, 2010: A Nested Case Control Study. *International journal of preventive medicine*, 4(9), 1030–1035.
- 63 Rabago, D., & Zgierska, A. (2009). Saline nasal irrigation for upper respiratory conditions. *American family physician*, 80(10), 1117–1119.
- 64 Swamy, M.K., Akhtar, M.S., & Sinniah, U.R. (2016). Antimicrobial Properties of Plant Essential Oils against Human Pathogens and Their Mode of Action: An Updated Review. Evidence-based complementary and alternative medicine: eCAM, 2016, 3012462. <https://doi.org/10.1155/2016/3012462>
- 65 Swamy, M.K., Akhtar, M.S., & Sinniah, U.R. (2016). Antimicrobial Properties of Plant Essential Oils against Human Pathogens and Their Mode of Action: An Updated Review. Evidence-based complementary and alternative medicine: eCAM, 2016, 3012462. <https://doi.org/10.1155/2016/3012462>
- 66 Imai, H., Osawa, K., Yasuda, H., Hamashima, H., Arai, T., & Sasatsu, M. (2001). Inhibition by the essential oils of peppermint and spearmint of the growth of pathogenic bacteria. *Microbios*, 106 Suppl 1, 31–39.
- 67 Hemilä H. (2017). Vitamin C and Infections. *Nutrients*, 9(4), 339. <https://doi.org/10.3390/nu9040339>
- 68 Gunville, C. F., Mourani, P. M., & Ginde, A. A. (2013). The role of vitamin D in prevention and treatment of infection. *Inflammation & allergy drug targets*, 12(4), 239–245. <https://doi.org/10.2174/18715281113129990046>
- 69 Lewis, E. D., Meydani, S. N., & Wu, D. (2019). Regulatory role of vitamin E in the immune system and inflammation. *IUBMB life*, 71(4), 487–494. <https://doi.org/10.1002/iub.1976>
- 70 Yan, F., & Polk, D. B. (2011). Probiotics and immune health. *Current opinion in gastroenterology*, 27(6), 496–501. <https://doi.org/10.1097/MOG.0b013e32834baa4d>
- 71 Li, X., Wang, Q., Hu, X., & Liu, W. (2022). Current Status of Probiotics as Supplements in the Prevention and Treatment of Infectious Diseases. *Frontiers in cellular and infection microbiology*, 12, 789063. <https://doi.org/10.3389/fcimb.2022.789063>
- 72 Besedovsky, L., Lange, T., & Born, J. (2012). Sleep and immune function. *Pflügers Archiv: European journal of physiology*, 463(1), 121–137. <https://doi.org/10.1007/s00424-011-1044-0>
- 73 Everson C. A. (1993). Sustained sleep deprivation impairs host defense. *The American journal of physiology*, 265(5 Pt 2), R1148–R1154. <https://doi.org/10.1152/ajpregu.1993.265.5.R1148>
- 74 Besedovsky, L., Lange, T., & Born, J. (2012). Sleep and immune function. *Pflügers Archiv: European journal of physiology*, 463(1), 121–137. <https://doi.org/10.1007/s00424-011-1044-0>
- 75 Akhavan, B.J., Khanna, N.R., & Vijhani, P. Amoxicillin. [Updated 2021 Aug 17]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK482250/>
- 76 Bailey, A., Walker, A., Hadley, A., & James, D. G. (1970). Cephalixin--a new oral antibiotic. *Postgraduate medical journal*, 46(533), 157–158. <https://doi.org/10.1136/pgmj.46.533.157>
- 77 Herman TF, Hashmi MF. Cephalixin. [Updated 2022 Feb 16]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK549780/>

Flu, Viral, and Immune System Protocol

- 1 Yu, S., Zhang, G., & Jin, L. H. (2018). A high-sugar diet affects cellular and humoral immune responses in Drosophila. *Experimental cell research*, 368(2), 215–224. <https://doi.org/10.1016/j.yexcr.2018.04.032>
- 2 Alkhatib A. (2020). Antiviral Functional Foods and Exercise Lifestyle Prevention of Coronavirus. *Nutrients*, 12(9), 2633. <https://doi.org/10.3390/nu12092633>
- 3 Rouf, R., Uddin, S. J., Sarker, D. K., Islam, M. T., Ali, E. S., Shilpi, J. A., Nahar, L., Tiralongo, E., & Sarker, S. D. (2020). Antiviral potential of garlic (*Allium sativum*) and its organo-sulfur compounds: A systematic update of pre-clinical and clinical data. *Trends in food science & technology*, 104, 219–234. <https://doi.org/10.1016/j.tifs.2020.08.006>
- 4 Cheng, C. W., Adams, G. B., Perin, L., Wei, M., Zhou, X., Lam, B. S., Da Sacco, S., Mirisola, M., Quinn, D. I., Dorff, T. B., Kopchick, J. J., & Longo, V. D. (2014). Prolonged fasting reduces IGF-1/PKA to promote hematopoietic-stem-cell-based regeneration and reverse immunosuppression. *Cell stem cell*, 14(6), 810–823. <https://doi.org/10.1016/j.stem.2014.04.014>
- 5 de Cabo, R., & Mattson, M. P. (2019). Effects of Intermittent Fasting on Health, Aging, and Disease. *The New England journal of medicine*, 381(26), 2541–2551. <https://doi.org/10.1056/NEJMr1905136>
- 6 Hannan, M. A., Rahman, M. A., Rahman, M. S., Sohag, A., Dash, R., Hossain, K. S., Farjana, M., & Uddin, M. J. (2020). Intermittent fasting, a possible priming tool for host defense against SARS-CoV-2 infection: Crosstalk among calorie restriction, autophagy and immune response. *Immunology letters*, 226, 38–45. <https://doi.org/10.1016/j.imlet.2020.07.001>
- 7 Jakubowicz, D., Landau, Z., Tsameret, S., Wainstein, J., Raz, I., Ahren, B., Chapnik, N., Barnea, M., Ganz, T., Menaged, M., Mor, N., Bar-Dayana, Y., & Froy, O. (2019). Reduction in Glycated Hemoglobin and Daily Insulin Dose Alongside Circadian Clock Upregulation in Patients With Type 2 Diabetes Consuming a Three-Meal Diet: A Randomized Clinical Trial. *Diabetes care*, 42(12), 2171–2180. <https://doi.org/10.2337/dc19-1142>
- 8 Besedovsky, L., Lange, T., & Born, J. (2012). Sleep and immune function. *Pflügers Archiv : European journal of physiology*, 463(1), 121–137. <https://doi.org/10.1007/s00424-011-1044-0>
- 9 Vgontzas, A. N., Zoumakis, E., Bixler, E. O., Lin, H. M., Follett, H., Kales, A., & Chrousos, G. P. (2004). Adverse effects of modest sleep restriction on sleepiness,

- performance, and inflammatory cytokines. *The Journal of clinical endocrinology and metabolism*, 89(5), 2119–2126. <https://doi.org/10.1210/jc.2003-031562>
- 10 Amatriain-Fernández, S., Gronwald, T., Murillo-Rodríguez, E., Imperatori, C., Solano, A. F., Latini, A., & Budde, H. (2020). Physical Exercise Potentials Against Viral Diseases Like COVID-19 in the Elderly. *Frontiers in medicine*, 7, 379. <https://doi.org/10.3389/fmed.2020.00379>
- 11 Garber, C. E., Blissmer, B., Deschenes, M. R., Franklin, B. A., Lamonte, M. J., Lee, I. M., Nieman, D. C., Swain, D. P., & American College of Sports Medicine (2011). American College of Sports Medicine position stand. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: guidance for prescribing exercise. *Medicine and science in sports and exercise*, 43(7), 1334–1359. <https://doi.org/10.1249/MSS.0b013e318213feb>
- 12 Fortes, M. B., Diment, B. C., Di Felice, U., & Walsh, N. P. (2012). Dehydration decreases saliva antimicrobial proteins important for mucosal immunity. *Applied physiology, nutrition, and metabolism = Physiologie appliquee, nutrition et metabolisme*, 37(5), 850–859. <https://doi.org/10.1139/h2012-054>
- 13 Adhikari, B., Marasini, B. P., Rayamajhee, B., Bhattarai, B. R., Lamichhane, G., Khadayat, K., Adhikari, A., Khanal, S., & Parajuli, N. (2021). Potential roles of medicinal plants for the treatment of viral diseases focusing on COVID-19: A review. *Phytotherapy research: PTR*, 35(3), 1298–1312. <https://doi.org/10.1002/ptr.6893>
- 14 Zheng, Y., Ren, W., Zhang, L., Zhang, Y., Liu, D., & Liu, Y. (2020). A Review of the Pharmacological Action of Astragalus Polysaccharide. *Frontiers in pharmacology*, 11, 349. <https://doi.org/10.3389/fphar.2020.00349>
- 15 Tronina, T., Mrozowska, M., Bartmańska, A., Popłowski, J., Sordon, S., & Huszcza, E. (2021). Simple and Rapid Method for Wogonin Preparation and Its Biotransformation. *International journal of molecular sciences*, 22(16), 8973. <https://doi.org/10.3390/ijms22168973>
- 16 Li, R. F., Zhou, X. B., Zhou, H. X., Yang, Z. F., Jiang, H. M., Wu, X., Li, W. J., Qiu, J. J., Mi, J. N., Chen, M., Zhong, N. S., Zhu, G. Y., & Jiang, Z. H. (2021). Novel Fatty Acid in Cordyceps Suppresses Influenza A (H1N1) Virus-Induced Proinflammatory Response Through Regulating Innate Signaling Pathways. *ACS omega*, 6(2), 1505–1515. <https://doi.org/10.1021/acsomega.0c05264>
- 17 Verma AK. Cordycepin: a bioactive metabolite of Cordyceps militaris and polyadenylation inhibitor with therapeutic potential against COVID-19. *J Biomol Struct Dyn*. 2022 May;40(8):3745–3752. doi: 10.1080/07391102.2020.1850352. Epub 2020 Nov 23. PMID: 33225826; PMCID: PMC7754931
- 18 Verma A. K. (2022). Cordycepin: a bioactive metabolite of Cordyceps militaris and polyadenylation inhibitor with therapeutic potential against COVID-19. *Journal of biomolecular structure & dynamics*, 40(8), 3745–3752. <https://doi.org/10.1080/07391102.2020.1850352>
- 19 Ueda, Y., Mori, K., Satoh, S., Dansako, H., Ikeda, M., & Kato, N. (2014). Anti-HCV activity of the Chinese medicinal fungus Cordyceps militaris. *Biochemical and biophysical research communications*, 447(2), 341–345. <https://doi.org/10.1016/j.bbrc.2014.03.150>
- 20 Chun S, Gopal J, Muthu M. Antioxidant Activity of Mushroom Extracts/Polysaccharides—Their Antiviral Properties and Plausible AntiCOVID-19 Properties. *Antioxidants*. 2021; 10(12):1899. <https://doi.org/10.3390/antiox10121899>
- 21 Nahas, R., & Balla, A. (2011). Complementary and alternative medicine for prevention and treatment of the common cold. *Canadian family physician Medecin de famille canadien*, 57(1), 31–36.
- 22 Nahas, R., & Balla, A. (2011). Complementary and alternative medicine for prevention and treatment of the common cold. *Canadian family physician Medecin de famille canadien*, 57(1), 31–36.
- 23 Zakay-Rones, Z., Thom, E., Wollan, T., & Wadstein, J. (2004). Randomized study of the efficacy and safety of oral elderberry extract in the treatment of influenza A and B virus infections. *The Journal of international medical research*, 32(2), 132–140. <https://doi.org/10.1177/147323000403200205>
- 24 Wieland, L. S., Piechotta, V., Feinberg, T., Ludeman, E., Hutton, B., Kanji, S., Seely, D., & Garrity, C. (2021). Elderberry for prevention and treatment of viral respiratory illness: a systematic review. *BMC complementary medicine and therapies*, 21(1), 112. <https://doi.org/10.1186/s12906-021-03283-5>
- 25 Kronbichler, A., Effenberger, M., Eisenhut, M., Lee, K. H., & Shin, J. I. (2020). Seven recommendations to rescue the patients and reduce the mortality from COVID-19 infection: An immunological point of view. *Autoimmunity reviews*, 19(7), 102570. <https://doi.org/10.1016/j.autrev.2020.102570>
- 26 Cecil, C. E., Davis, J. M., Cech, N. B., & Laster, S. M. (2011). Inhibition of H1N1 influenza A virus growth and induction of inflammatory mediators by the isquinoline alkaloid berberine and extracts of goldenseal (*Hydrastis canadensis*). *International immunopharmacology*, 11(11), 1706–1714. <https://doi.org/10.1016/j.intimp.2011.06.002>
- 27 Shang, X., Pan, H., Li, M., Miao, X., & Ding, H. (2011). *Lonicera japonica* Thunb.: ethnopharmacology, phytochemistry and pharmacology of an important traditional Chinese medicine. *Journal of ethnopharmacology*, 138(1), 1–21. <https://doi.org/10.1016/j.jep.2011.08.016>
- 28 Yeh, Y. C., Doan, L. H., Huang, Z. Y., Chu, L. W., Shi, T. H., Lee, Y. R., Wu, C. T., Lin, C. H., Chang, S. T., Liu, H. K., Chuang, T. H., Ping, Y. H., Liu, H. S., & Huang, C. F. (2022). Honey-suckle (*Lonicera japonica*) and Huangqi (*Astragalus membranaceus*) Suppress SARS-CoV-2 Entry and COVID-19 Related Cytokine Storm in Vitro. *Frontiers in pharmacology*, 12, 765553. <https://doi.org/10.3389/fphar.2021.765553>
- 29 Astani, A., Navid, M. H., & Schnitzler, P. (2014). Attachment and penetration of acyclovir-resistant herpes simplex virus are inhibited by *Melissa officinalis* extract. *Phytotherapy research: PTR*, 28(10), 1547–1552. <https://doi.org/10.1002/ptr.5166>
- 30 Vanti, G., Ntallis, S. G., Panagiotidis, C. A., Dourdouni, V., Patsoura, C., Bergonzi, M. C., Liazari, D., & Bilia, A. R. (2020). Glycosomes of *Melissa officinalis* L. Essential Oil for

Effective Anti-HSV Type 1. *Molecules (Basel, Switzerland)*, 25(14), 3111. <https://doi.org/10.3390/molecules25143111>

31 Mazzanti, G., Battinelli, L., Pompeo, C., Serrilli, A. M., Rossi, R., Sauzullo, I., Mengoni, F., & Vullo, V. (2008). Inhibitory activity of *Melissa officinalis* L. extract on Herpes simplex virus type 2 replication. *Natural product research*, 22(16), 1433–1440. <https://doi.org/10.1080/14786410802075939>

32 Mehrbod, P., Safari, H., Mollai, Z., Fotouhi, F., Mirfakhraei, Y., Entezari, H., Goodarzi, S., & Tofighi, Z. (2021). Potential antiviral effects of some native Iranian medicinal plants extracts and fractions against influenza A virus. *BMC complementary medicine and therapies*, 21(1), 246. <https://doi.org/10.1186/s12906-021-03423-x>

33 Wang, L., Yang, R., Yuan, B., Liu, Y., & Liu, C. (2015). The antiviral and antimicrobial activities of licorice, a widely-used Chinese herb. *Acta pharmaceutica Sinica*, 36(5), 310–315. <https://doi.org/10.1016/j.japsb.2015.05.005>

34 Adhikari, B., Marasini, B. P., Rayamajhee, B., Bhattarai, B. R., Lamichhane, G., Khadayat, K., Adhikari, A., Khanal, S., & Parajuli, N. (2021). Potential roles of medicinal plants for the treatment of viral diseases focusing on COVID-19: A review. *Phytotherapy research: PTR*, 35(3), 1298–1312. <https://doi.org/10.1002/ptr.6893>

35 Diling, C., Chaoqun, Z., Jian, Y., Jian, L., Jiyan, S., Yizhen, X., & Guoxiao, L. (2017). Immunomodulatory Activities of a Fungal Protein Extracted from *Hericium erinaceus* through Regulating the Gut Microbiota. *Frontiers in immunology*, 8, 666. <https://doi.org/10.3389/fimmu.2017.00666>

36 Madia, V. N., De Angelis, M., De Vita, D., Messori, A., De Leo, A., Ialongo, D., Tudino, V., Saccoliti, F., De Chiara, G., Garzoli, S., Scipione, L., Palamara, A. T., Di Santo, R., Nencioni, L., & Costi, R. (2021). Investigation of *Commiphora myrrha* (Nees) Engl. Oil and Its Main Components for Antiviral Activity. *Pharmaceuticals (Basel, Switzerland)*, 14(3), 243. <https://doi.org/10.3390/ph14030243>

37 Brochet, A., Guilbot, A., Haddioui, L., & Roques, C. (2017). Antibacterial, antifungal, and antiviral effects of three essential oil blends. *MicrobiologyOpen*, 6(4), e00459. <https://doi.org/10.1002/mbo3.459>

38 Ahmad, M. F., Ahmad, F. A., Khan, M. I., Alsayegh, A. A., Wahab, S., Alam, M. I., & Ahmed, F. (2021). *Ganoderma lucidum*: A potential source to surmount viral infections through β -glucans immunomodulatory and triterpenoids antiviral properties. *International journal of biological macromolecules*, 187, 769–779. <https://doi.org/10.1016/j.ijbiomac.2021.06.122>

39 Zhi-Bin Lin. "Cellular and Molecular Mechanisms of Immuno-modulation by *Ganoderma lucidum*." *J Pharmacol Sci* 99, 144-153. 2005

40 Elmekaw, S.; Meselhy, M.R.; Nakamura, N.; Tezuka, Y.; Hattori, M.; Kakiuchi, N.; Shimotohno, K.; Kawahata, T.; Otake, T. Anti-HIV-1 and anti-HIV-1-protease substances from *Ganoderma lucidum*. *Phytochemistry* 1998, 49, 1651–1657

41 Min, B. S., Nakamura, N., Miyashiro, H., Bae, K. W. & Hattori, M. Triterpenes from the spores of *Ganoderma lucidum* and their inhibitory activity against HIV-1 protease. *Chem Pharm Bull* 46, 1607–1612 (1998)

42 Zhu, Q. C. et al. Inhibition of neuraminidase by *Ganoderma* triterpenoids and implications for neuraminidase inhibitor design. *Scientific reports* 5, 13194 (2015)

43 Zheng, D. S. & Chen, L. S. Triterpenoids from *Ganoderma lucidum* inhibit the activation of EBV antigens as telomerase inhibitors. *Exp Ther Med* 14, 3273–3278 (2017).

44 Bharadwaj, S., Lee, K.E., Dwivedi, V.D. et al. Discovery of *Ganoderma lucidum* triterpenoids as potential inhibitors against Dengue virus NS2B-NS3 protease. *Sci Rep* 9, 19059 (2019). <https://doi.org/10.1038/s41598-019-55723-5>

45 Ng, T. B., Wang, H., & Wan, D. C. C. (2006). Polysaccharopeptide from the Turkey Tail Fungus *Trametes versicolor* (L.: Fr.) Pilát Inhibits Human Immunodeficiency Virus Type 1 Reverse Transcriptase and Protease. *International Journal of Medicinal Mushrooms*, 8(1).

46 Ingólfssdóttir K. (2002). Usnic acid. *Phytochemistry*, 61(7), 729–736. [https://doi.org/10.1016/S0031-9422\(02\)00383-7](https://doi.org/10.1016/S0031-9422(02)00383-7)

47 RxList. (2021, June 11). *Yerba Santa: Health benefits, side effects, uses, Dose & precautions*. RxList. Retrieved July 20, 2022, from https://www.rxlist.com/yerba_santa/supplements.htm

48 Turker, A. U., & Camper, N. D. (2002). Biological activity of common mullein, a medicinal plant. *Journal of ethnopharmacology*, 82(2-3), 117–125. [https://doi.org/10.1016/S0378-8741\(02\)00186-1](https://doi.org/10.1016/S0378-8741(02)00186-1)

49 Blanco-Salas, J., Hortigón-Vinagre, M. P., Morales-Jadán, D., & Ruiz-Téllez, T. (2021). Searching for Scientific Explanations for the Uses of Spanish Folk Medicine: A Review on the Case of Mullein (*Verbascum*, *Scrophulariaceae*). *Biology*, 10(7), 618. <https://doi.org/10.3390/biology10070618>

50 Turker, A. U., & Camper, N. D. (2002). Biological activity of common mullein, a medicinal plant. *Journal of ethnopharmacology*, 82(2-3), 117–125. [https://doi.org/10.1016/S0378-8741\(02\)00186-1](https://doi.org/10.1016/S0378-8741(02)00186-1)

51 Süleyman, H., Odabasoglu, F., Aslan, A., Cakir, A., Karagoz, Y., Gocer, F., Halici, M., & Bayir, Y. (2003). Anti-inflammatory and antiulcerogenic effects of the aqueous extract of *Lobaria pulmonaria* (L.) Hoffm. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 10(6-7), 552–557. <https://doi.org/10.1078/094471103322331539>

52 Augspole, I., Duma, M., Cinkmanis, I., Ozola, B. (2018). Herbal teas as a rich source of phenolic compounds. *Analytical Chemistry*. 29(4), <https://doi.org/10.6001/chemija.v29i4.3841>

53 Jaruporn Rakmai, J. Rakmai, Benjamas Cheirsilp, B. Cheirsilp, Ana Torrado-Agrasar, A. Torrado-Agrasar, Jesús Simal-Gándara, J. Simal-Gándara, & Juan Carlos Mejuto, J. Carlos Mejuto. (2017). Encapsulation of yarrow essential oil in hydroxypropyl-beta-cyclodextrin: physicochemical characterization and evaluation of bio-efficacies. *CyTA: journal of food*, 15, 409-417. doi: 10.1080/19476337.2017.1286523

54 Sales, A.J., Malekzadeh, P., Ebrahimzadeh, M., Kondlaji, K.B., Kaleybar, V.P., Dizaji, A.S. (2015). Evaluation of the anti-bacterial effects of ethanolic extract of yarrow (*Achillea wilhelmsii*) on antibiotic-resistant strains of *Staphylococcus aureus*. *Journal of Science and Today's World*. 4(6), 189-192. DOI: 10.13140/RG.2.2.10488.49921

55 Nahas, R., & Balla, A. (2011). Complementary and alternative medicine for prevention and treatment of the common cold. *Canadian family physician Medecin de famille canadien*, 57(1), 31–36.

56 Gorton, H. C., & Jarvis, K. (1999). The effectiveness of vitamin C in preventing and relieving the symptoms of virus-induced respiratory infections. *Journal of manipulative and physiological therapeutics*, 22(8), 530–533. [https://doi.org/10.1016/S0161-4754\(99\)70005-9](https://doi.org/10.1016/S0161-4754(99)70005-9)

57 Grant, W. B., Lahore, H., McDonnell, S. L., Baggerly, C. A., French, C. B., Aliano, J. L., & Bhattoa, H. P. (2020). Evidence that Vitamin D Supplementation Could Reduce Risk of Influenza and COVID-19 Infections and Deaths. *Nutrients*, 12(4), 988. <https://doi.org/10.3390/nu12040988>

58 Stephensen, C. B., & Lietz, G. (2021). Vitamin A in resistance to and recovery from infection: relevance to SARS-CoV2. *The British journal of nutrition*, 126(11), 1663–1672. <https://doi.org/10.1017/S0007114521000246>

59 Mossad, S. B., Macknin, M. L., Medendorp, S. V., & Mason, P. (1996). Zinc gluconate lozenges for treating the common cold. A randomized, double-blind, placebo-controlled study. *Annals of internal medicine*, 125(2), 81–88. <https://doi.org/10.7326/0003-4819-125-2-199607150-00001>

60 Prasad, A. S., Fitzgerald, J. T., Bao, B., Beck, F. W., & Chandrasekar, P. H. (2000). Duration of symptoms and plasma cytokine levels in patients with the common cold treated with zinc acetate. A randomized, double-blind, placebo-controlled trial. *Annals of internal medicine*, 133(4), 245–252. <https://doi.org/10.7326/0003-4819-133-4-200008150-00006>

61 Schloss, J., Leach, M., Brown, D., Hannan, N., Kendall-Reed, P., & Steel, A. (2020). The effects of N-acetyl cysteine on acute viral respiratory infections in humans: A rapid review. *Advances in integrative medicine*, 7(4), 232–239. <https://doi.org/10.1016/j.aimed.2020.07.006>

62 Heer, C. D., Sanderson, D. J., Voth, L. S., Alhammad, Y., Schmidt, M. S., Trammell, S., Perlman, S., Cohen, M. S., Fehr, A. R., & Brenner, C. (2020). Coronavirus infection and PARP expression dysregulate the NAD metabolome: An actionable component of innate immunity. *The Journal of biological chemistry*, 295(52), 17986–17996. <https://doi.org/10.1074/jbc.RA120.015138>

63 Buijze, G. A., Siersevelt, I. N., van der Heijden, B. C., Dijkgraaf, M. G., & Frings-Dresen, M. H. (2016). The Effect of Cold Showering on Health and Work: A Randomized Controlled Trial. *PloS one*, 11(9), e0161749. <https://doi.org/10.1371/journal.pone.0161749>

64 Kox, M., van Eijk, L. T., Zwaag, J., van den Wildenberg, J., Sweep, F. C., van der Hoeven, J. G., & Pickkers, P. (2014). Voluntary activation of the sympathetic nervous system and attenuation of the innate immune response in humans. *Proceedings of the National Academy of Sciences of the United States of America*, 111(20), 7379–7384. <https://doi.org/10.1073/pnas.1322174111>

65 Ophir, D., & Elad, Y. (1987). Effects of steam inhalation on nasal patency and nasal symptoms in patients with the common cold. *American journal of otolaryngology*, 8(3), 149–153. [https://doi.org/10.1016/S0196-0709\(87\)80037-6](https://doi.org/10.1016/S0196-0709(87)80037-6)

66 Garozzo, A., Timpanaro, R., Stivala, A., Bisignano, G., & Castro, A. (2011). Activity of *Melaleuca alternifolia* (tea tree) oil on Influenza virus A/PR/8: study on the mechanism of action. *Antiviral research*, 89(1), 83–88. <https://doi.org/10.1016/j.antiviral.2010.11.010>

67 Mieres-Castro, D., Ahmar, S., Shabbir, R., & Mora-Poblete, F. (2021). Antiviral Activities of Eucalyptus Essential Oils: Their Effectiveness as Therapeutic Targets against Human Viruses. *Pharmaceuticals (Basel, Switzerland)*, 14(12), 1210. <https://doi.org/10.3390/ph14121210>

68 Asif, M., Saleem, M., Saadullah, M., Yaseen, H. S., & Al Zarzour, R. (2020). COVID-19 and therapy with essential oils having antiviral, anti-inflammatory, and immunomodulatory properties. *Inflammopharmacology*, 28(5), 1153–1161. <https://doi.org/10.1007/s10787-020-00744-0>

69 Asif, M., Saleem, M., Saadullah, M., Yaseen, H. S., & Al Zarzour, R. (2020). COVID-19 and therapy with essential oils having antiviral, anti-inflammatory, and immunomodulatory properties. *Inflammopharmacology*, 28(5), 1153–1161. <https://doi.org/10.1007/s10787-020-00744-0>

70 Schuhmacher, A., Reichling, J., & Schnitzler, P. (2003). Virucidal effect of peppermint oil on the enveloped viruses herpes simplex virus type 1 and type 2 in vitro. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 10(6-7), 504–510. <https://doi.org/10.1078/094471103322331467>

71 Ramalingam, S., Graham, C., Dove, J., Morrice, L., & Sheikh, A. (2019). A pilot, open labelled, randomised controlled trial of hypertonic saline nasal irrigation and gargling for the common cold. *Scientific reports*, 9(1), 1015. <https://doi.org/10.1038/s41598-018-37703-3>

Fungal Infection Protocol

1 Gupta, A. K., Nicol, K., & Batra, R. (2004). Role of antifungal agents in the treatment of seborrheic dermatitis. *American journal of clinical dermatology*, 5(6), 417–422. <https://doi.org/10.2165/00128071-200405060-00006>

2 Kuruvella T, Pandey S. Tinea Barbae. (2022). Tinea barbae. In *Statpearls [Internet]*. Treasure Island (FL): StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK563204/>

3 Elewski B. E. (1998). Onychomycosis: pathogenesis, diagnosis, and management. *Clinical microbiology reviews*, 11(3), 415–429. <https://doi.org/10.1128/CMR.11.3.415>

- 4 Yu, S., Zhang, G., & Jin, L. H. (2018). A high-sugar diet affects cellular and humoral immune responses in *Drosophila*. *Experimental cell research*, 368(2), 215–224. <https://doi.org/10.1016/j.yexcr.2018.04.032>
- 5 Sanchez, A., Reeser, J. L., Lau, H. S., Yahiku, P. Y., Willard, R. E., McMillan, P. J., Cho, S. Y., Magie, A. R., & Register, U. D. (1973). Role of sugars in human neutrophilic phagocytosis. *The American journal of clinical nutrition*, 26(11), 1180–1184. <https://doi.org/10.1093/ajcn/26.11.1180>
- 6 Kiselar, J. G., Wang, X., Dubyak, G. R., El Sanadi, C., Ghosh, S. K., Lundberg, K., & Williams, W. M. (2015). Modification of β -Defensin-2 by Dicarbonyls Methylglyoxal and Glyoxal Inhibits Antibacterial and Chemotactic Function In Vitro. *PLoS one*, 10(8), e0130533. <https://doi.org/10.1371/journal.pone.0130533>
- 7 Rahiman, F., & Pool, E. J. (2014). The in vitro effects of artificial and natural sweeteners on the immune system using whole blood culture assays. *Journal of immunoassay & immunochemistry*, 35(1), 26–36. <https://doi.org/10.1080/15321819.2013.784197>
- 8 Vanegas, S. M., Meydani, M., Barnett, J. B., Goldin, B., Kane, A., Rasmussen, H., Brown, C., Vangay, P., Knights, D., Jonnalagadda, S., Koecher, K., Karl, J. P., Thomas, M., Dolnikowski, G., Li, L., Saltzman, E., Wu, D., & Meydani, S. N. (2017). Substituting whole grains for refined grains in a 6-wk randomized trial has a modest effect on gut microbiota and immune and inflammatory markers of healthy adults. *The American journal of clinical nutrition*, 105(3), 635–650. <https://doi.org/10.3945/ajcn.116.146928>
- 9 Bodemer, A. (2018, July 23). *Seborrheic Dermatitis*. Whole Health Library. Retrieved July 15, 2022, from <https://www.va.gov/WHOLEHEALTHLIBRARY/tools/seborrheic-dermatitis.asp>
- 10 Talattof, Z., Azad, A., Zahed, M., & Shahradian, N. (2018). Antifungal Activity of Xylitol against *Candida albicans*: An in vitro Study. *The journal of contemporary dental practice*, 19(2), 125–129. <https://doi.org/10.5005/jp-journals-10024-2225>
- 11 Pizzo, G., Giuliana, G., Milici, M. E., & Giangreco, R. (2000). Effect of dietary carbohydrates on the in vitro epithelial adhesion of *Candida albicans*, *Candida tropicalis*, and *Candida krusei*. *The new microbiologica*, 23(1), 63–71.
- 12 Yagnik, D., Serafin, V., & J Shah, A. (2018). Antimicrobial activity of apple cider vinegar against *Escherichia coli*, *Staphylococcus aureus* and *Candida albicans*: downregulating cytokine and microbial protein expression. *Scientific reports*, 8(1), 1732. <https://doi.org/10.1038/s41598-017-18618-x>
- 13 Mota, A. C., de Castro, R. D., de Araújo Oliveira, J., & de Oliveira Lima, E. (2015). Antifungal Activity of Apple Cider Vinegar on *Candida* Species Involved in Denture Stomatitis. *Journal of prosthodontics: official journal of the American College of Prosthodontists*, 24(4), 296–302. <https://doi.org/10.1111/jopr.12207>
- 14 Ogbolu, D. O., Oni, A. A., Daini, O. A., & Oloko, A. P. (2007). In vitro antimicrobial properties of coconut oil on *Candida* species in Ibadan, Nigeria. *Journal of medicinal food*, 10(2), 384–387. <https://doi.org/10.1089/jmf.2006.1209>
- 15 Khalid Thebo, N., Ahmed Simair, A., Sughra Mangrio, G., Ansari, K. A., Ali Bhutto, A., Lu, C., & Ali Sheikh, W. (2016). Antifungal Potential and Antioxidant Efficacy in the Shell Extract of *Cocos nucifera* (L.) (Arecaceae) against Pathogenic Dermal Mycosis. *Medicines (Basel, Switzerland)*, 3(2), 12. <https://doi.org/10.3390/medicines3020012>
- 16 Saniasiaya, J., Salim, R., Mohamad, I., & Harun, A. (2017). Antifungal Effect of Malaysian Aloe vera Leaf Extract on Selected Fungal Species of Pathogenic Otomycosis Species in In Vitro Culture Medium. *Oman medical journal*, 32(1), 41–46. <https://doi.org/10.5001/omj.2017.08>
- 17 Nabila, V. K., & Putra, I. B. (2020). The effect of Aloe vera ethanol extract on the growth inhibition of *Candida albicans*. *Medicinski glasnik : official publication of the Medical Association of Zenica-Doboj Canton, Bosnia and Herzegovina*, 17(2), 485–489. <https://doi.org/10.17392/1098-20>
- 18 Saniasiaya, J., Salim, R., Mohamad, I., & Harun, A. (2017). Antifungal Effect of Malaysian Aloe vera Leaf Extract on Selected Fungal Species of Pathogenic Otomycosis Species in In Vitro Culture Medium. *Oman medical journal*, 32(1), 41–46. <https://doi.org/10.5001/omj.2017.08>
- 19 Nenoff, P., Hausteine, U. F., & Brandt, W. (1996). Antifungal activity of the essential oil of *Melaleuca alternifolia* (tea tree oil) against pathogenic fungi in vitro. *Skin pharmacology: the official journal of the Skin Pharmacology Society*, 9(6), 388–394. <https://doi.org/10.1159/000211450>
- 20 Behmanesh, F., Pasha, H., Sefidgar, A. A., Taghizadeh, M., Moghadamnia, A. A., Adib Rad, H., & Shirkhani, L. (2015). Antifungal Effect of Lavender Essential Oil (*Lavandula angustifolia*) and Clotrimazole on *Candida albicans*: An In Vitro Study. *Scientifica*, 2015, 261397. <https://doi.org/10.1155/2015/261397>
- 21 Cassella, S., Cassella, J. P., & Smith, I. (2002). Synergistic antifungal activity of tea tree (*Melaleuca alternifolia*) and lavender (*Lavandula angustifolia*) essential oils against dermatophyte infection. *International Journal of Aromatherapy*, 12(1), 2–15. <https://doi.org/10.1054/ijar.2001.0127>
- 22 Burlando, B., & Cornara, L. (2013). Honey in dermatology and skin care: a review. *Journal of cosmetic dermatology*, 12(4), 306–313. <https://doi.org/10.1111/jocd.12058>
- 23 Irish, J., Carter, D. A., Shokohi, T., & Blair, S. E. (2006). Honey has an antifungal effect against *Candida* species. *Medical mycology*, 44(3), 289–291. <https://doi.org/10.1080/13693780500417037>
- 24 Moussa, A., Noureddine, D., Saad, A., Abdelmelek, M., & Abdelkader, B. (2012). Antifungal activity of four honeys of different types from Algeria against pathogenic yeast: *Candida albicans* and *Rhodotorula* sp. *Asian Pacific journal of tropical biomedicine*, 2(7), 554–557. [https://doi.org/10.1016/S2221-1691\(12\)60096-3](https://doi.org/10.1016/S2221-1691(12)60096-3)
- 25 Anand, S., Deighton, M., Livanos, G., Pang, E., & Mantri, N. (2019). Agastache honey has superior antifungal activity in comparison with important commercial honeys. *Scientific reports*, 9(1), 18197. <https://doi.org/10.1038/s41598-019-54679-w>
- 26 Shahi, S. K., Shukla, A. C., Bajaj, A. K., Banerjee, U., Rimek, D., Midgely, G., & Dikshit, A. (2000). Broad spectrum herbal therapy against superficial fungal infections. *Skin pharmacology and applied skin physiology*, 13(1), 60–64. <https://doi.org/10.1159/000029909>
- 27 Omura, Y., O'Young, B., Jones, M., Pallos, A., Duvvi, H., & Shimotsuura, Y. (2011). Caprylic acid in the effective treatment of intractable medical problems of frequent urination, incontinence, chronic upper respiratory infection, root canal tooth infection, ALS, etc., caused by asbestos & mixed infections of *Candida albicans*, *Helicobacter pylori* & cytomegalovirus with or without other microorganisms & mercury. *Acupuncture & electro-therapeutics research*, 36(1-2), 19–64. <https://doi.org/10.3727/036012911803860886>
- 28 Bae, Y. S., & Rhee, M. S. (2019). Short-Term Antifungal Treatments of Caprylic Acid with Carvacrol or Thymol Induce Synergistic 6-Log Reduction of Pathogenic *Candida albicans* by Cell Membrane Disruption and Efflux Pump Inhibition. *Cellular physiology and biochemistry: international journal of experimental cellular physiology, biochemistry, and pharmacology*, 53(2), 285–300. <https://doi.org/10.33594/000000139>
- 29 Smith, E. B., Powell, R. P., Graham, J. L., & Ulrich, J. A. (1977). Topical undecylenic acid in tinea pedis: a new look. *International journal of dermatology*, 16(1), 52–56. <https://doi.org/10.1111/j.1365-4362.1977.tb00741.x>
- 30 Chretien, J. H., Esswein, J. G., Sharpe, L. M., Kiely, J. J., & Lyddon, F. E. (1980). Efficacy of undecylenic acid-zinc undecylenate powder in culture positive tinea pedis. *International journal of dermatology*, 19(1), 51–54. <https://doi.org/10.1111/j.1365-4362.1980.tb01997.x>
- 31 Ramadan, W., Mourad, B., Ibrahim, S., & Sonbol, F. (1996). Oil of bitter orange: new topical antifungal agent. *International journal of dermatology*, 35(6), 448–449. <https://doi.org/10.1111/j.1365-4362.1996.tb03032.x>
- 32 Kunyeit, L., Kurrey, N. K., Anu-Appaiah, K. A., & Rao, R. P. (2019). Probiotic Yeasts Inhibit Virulence of Non-albicans *Candida* Species. *mBio*, 10(5), e02307-19. <https://doi.org/10.1128/mBio.02307-19>
- 33 Vilela, S. F., Barbosa, J. O., Rossoni, R. D., Santos, J. D., Prata, M. C., Anbinder, A. L., Jorge, A. O., & Junqueira, J. C. (2015). *Lactobacillus acidophilus* ATCC 4356 inhibits biofilm formation by *C. albicans* and attenuates the experimental candidiasis in *Galleria mellonella*. *Virulence*, 6(1), 29–39. <https://doi.org/10.4161/21505594.2014.981486>
- 34 Matsubara, V. H., Wang, Y., Bandara, H., Mayer, M., & Samaranyake, L. P. (2016). Probiotic lactobacilli inhibit early stages of *Candida albicans* biofilm development by reducing their growth, cell adhesion, and filamentation. *Applied microbiology and biotechnology*, 100(14), 6415–6426. <https://doi.org/10.1007/s00253-016-7527-3>
- 35 Kunyeit, L., Kurrey, N. K., Anu-Appaiah, K. A., & Rao, R. P. (2019). Probiotic Yeasts Inhibit Virulence of Non-albicans *Candida* Species. *mBio*, 10(5), e02307-19. <https://doi.org/10.1128/mBio.02307-19>
- 36 Wu, Y., Hu, S., Wu, C., Gu, F., & Yang, Y. (2022). Probiotics: Potential Novel Therapeutics Against Fungal Infections. *Frontiers in cellular and infection microbiology*, 11, 793419. <https://doi.org/10.3389/fcimb.2021.793419>
- 37 Singhal, D., Baker, L., Wormald, P. J., & Tan, L. (2011). *Aspergillus fumigatus* biofilm on primary human sinonasal epithelial culture. *American journal of rhinology & allergy*, 25(4), 219–225. <https://doi.org/10.2500/ajra.2011.25.3622>
- 38 Kean, R., & Ramage, G. (2019). Combined Antifungal Resistance and Biofilm Tolerance: The Global Threat of *Candida auris*. *mSphere*, 4(4), e00458-19. <https://doi.org/10.1128/mSphere.00458-19>
- 39 Matsubara, V. H., Wang, Y., Bandara, H., Mayer, M., & Samaranyake, L. P. (2016). Probiotic lactobacilli inhibit early stages of *Candida albicans* biofilm development by reducing their growth, cell adhesion, and filamentation. *Applied microbiology and biotechnology*, 100(14), 6415–6426. <https://doi.org/10.1007/s00253-016-7527-3>
- 40 Yamashita, M., Sawano, J., Umeda, R., Tatsumi, A., Kumeda, Y., & Iida, A. (2021). Structure-Activity Relationship Studies of Antimicrobial Naphthoquinones Derived from Constituents of *Tabebuia avellanae*. *Chemical & pharmaceutical bulletin*, 69(7), 661–673. <https://doi.org/10.1248/cpb.c21-00208>
- 41 Gómez Castellanos, J. R., Prieto, J. M., & Heinrich, M. (2009). Red Lapacho (*Tabebuia impetiginosa*) - a global ethnopharmacological commodity?. *Journal of ethnopharmacology*, 121(1), 1–13. <https://doi.org/10.1016/j.jep.2008.10.004>
- 42 Pärnu, M., Mojt, C. A., Pärnu, A. E., Mircea, C., Stoeber, L., Roşca-Casian, O., & Ţigu, A. B. (2019). *Allium sativum* Extract Chemical Composition, Antioxidant Activity and Antifungal Effect against *Mycobacterium guilliermondii* and *Rhodotorula mucilaginosa* Causing Onychomycosis. *Molecules (Basel, Switzerland)*, 24(21), 3958. <https://doi.org/10.3390/molecules24213958>
- 43 Ranković, B., Kosanić, M., Stanojković, T., Vasiljević, P., & Manojlović, N. (2012). Biological activities of *Toninia candida* and *Usnea barbata* together with their norstictic acid and usnic acid constituents. *International journal of molecular sciences*, 13(11), 14707–14722. <https://doi.org/10.3390/ijms131114707>
- 44 Madamombe, I. T., & Afolayan, A. J. (2008). Evaluation of Antimicrobial Activity of Extracts from South African *Usnea barbata*. *Pharmaceutical Biology*, 41:3, 199–202, DOI: 10.1076/phbi.41.3.199.15089
- 45 Ho, K. V., Lei, Z., Sumner, L. W., Coggeshall, M. V., Hsieh, H. Y., Stewart, G. C., & Lin, C. H. (2018). Identifying Antibacterial Compounds in Black Walnuts (*Juglans nigra*) Using a Metabolomics Approach. *Metabolites*, 8(4), 58. <https://doi.org/10.3390/metabo8040058>
- 46 Yun, J., Lee, H., Ko, H. J., Woo, E. R., & Lee, D. G. (2015). Fungicidal effect of isoquercitrin via inducing membrane disturbance. *Biochimica et biophysica acta*, 1848(2), 695–701. <https://doi.org/10.1016/j.bbame.2014.11.019>
- 47 Muruges, J., Annigeri, R. G., Mangala, G. K., Mythily, P. H., & Chandrakala, J. (2019). Evaluation of the antifungal efficacy of different concentrations of *Curcuma longa* on *Candida albicans*: An in vitro study. *Journal of oral and maxillofacial pathology: JOMFP*, 23(2), 305. https://doi.org/10.4103/jomfp.JOMFP_200_18
- 48 Moghadamtousi, S. Z., Kadir, H. A., Hassandaravish, P., Tajik, H., Abubakar, S., & Zandi,

- K. (2014). A review on antibacterial, antiviral, and antifungal activity of curcumin. *BioMed research international*, 2014, 186864. <https://doi.org/10.1155/2014/186864>
 - 49 National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 4189, Miconazole. Retrieved July 14, 2022 from <https://pubchem.ncbi.nlm.nih.gov/compound/Miconazole>.
 - 50 Khatter, N.J. & Khan, M.A.B. (2022). Clotrimazole. In Statpearls [Internet]. Treasure Island (FL): StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK560643/>
 - 51 Maxfield, L., Preuss, C.V., & Bermudez, R. (2022). Terbinafine. In Statpearls [Internet]. Treasure Island (FL): StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK545218/>
 - 52 Maxfield, L., Preuss, C.V., & Bermudez, R. (2022). Terbinafine. In Statpearls [Internet]. Treasure Island (FL): StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK545218/>
 - 53 Kalińska-Bienias, A., Kowalewski, C., & Woźniak, K. (2013). Terbinafine-induced subacute cutaneous lupus erythematosus in two patients with systemic lupus erythematosus successfully treated with topical corticosteroids. *Postępy dermatologii i alergologii*, 30(4), 261–264. <https://doi.org/10.5114/pdia.2013.37038>
- ## Lyme Disease Protocol
- 1 Centers for Disease Control and Prevention. (2020, May 15). *Vector-borne diseases in the United States, 2004-2018*. Centers for Disease Control and Prevention. Retrieved March 18, 2022, from <https://www.cdc.gov/nczid/dvbd/vital-signs/index.html>
 - 2 Morrison, G. (2002). *The invisible kingdom*. Titan.
 - 3 Guerra, M. (2002). Predicting the risk of lyme disease: Habitat suitability for Ixodes scapularis in the north central United States. *Emerging Infectious Diseases*, 8(3), 289–297. <https://doi.org/10.3201/eid0803.010166>
 - 4 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.
 - 5 Lyme disease. (2020, October 01). Retrieved March 24, 2022, from [https://www.cdc.gov/ticks/tickbornediseases/lyme.html#:~:text=The%20erythema%20migrans%20\(EM\)%20rash,red%20ring%20with%20central%20clearing](https://www.cdc.gov/ticks/tickbornediseases/lyme.html#:~:text=The%20erythema%20migrans%20(EM)%20rash,red%20ring%20with%20central%20clearing).
 - 6 Hersh, E. (2017, August 29). Erythema migrans: Identification, treatment, and more. Retrieved March 18, 2022, from <https://www.healthline.com/health/erythema-migrans#:~:text=Erythema%20chronicum%20migrans%20is%20a,appear%20as%20a%20solid%20circle>.
 - 7 Heart block. (n.d.). Retrieved March 23, 2022, from <https://www.hopkinsmedicine.org/health/conditions-and-diseases/heart-block#:~:text=Heart%20block%20occurs%20when%20the,block%20can%20be%20life%20threatening>.
 - 8 New-onset Bell palsy and Lyme disease. (2017). Canadian family physician Medecin de famille canadien, 63(12), 941.
 - 9 Steere, A. C. (1994). Pathogenesis, diagnosis and treatment of lyme arthritis. *Lyme Borreliosis*, 1-5. doi:10.1007/978-1-4615-2415-1_1
 - 10 Anastasia Climán, R. (2021, May 15). What you need to know about the stages of lyme disease. Retrieved March 23, 2022, from [https://www.verywellhealth.com/lyme-disease-stages-5176671#:~:text=Early%20Disseminated%20Lyme%20Disease%20\(Stage%202\)&text=Inflammation%20of%20the%20nervous%20system,in%20the%20arms%20or%20legs](https://www.verywellhealth.com/lyme-disease-stages-5176671#:~:text=Early%20Disseminated%20Lyme%20Disease%20(Stage%202)&text=Inflammation%20of%20the%20nervous%20system,in%20the%20arms%20or%20legs)
 - 11 Tick bite prophylaxis. (2019, May 30). Retrieved March 22, 2022, from <https://www.cdc.gov/ticks/tickbornediseases/tick-bite-prophylaxis.html>
 - 12 Feng, J., Leone, J., Schweig, S., & Zhang, Y. (2020). Evaluation of natural and botanical medicines for activity against growing and non-growing forms of B. Burgdorferi. *Frontiers in Medicine*, 7. doi:10.3389/fmed.2020.00006
 - 13 Feng, J., Leone, J., Schweig, S., & Zhang, Y. (2020). Evaluation of natural and botanical medicines for activity against growing and non-growing forms of B. Burgdorferi. *Frontiers in Medicine*, 7. doi:10.3389/fmed.2020.00006
 - 14 Anastasia Climán, R. (2021, May 15). What you need to know about the stages of lyme disease. Retrieved March 21, 2022, from [https://www.verywellhealth.com/lyme-disease-stages-5176671#:~:text=Early%20Disseminated%20Lyme%20Disease%20\(Stage%202\)&text=Inflammation%20of%20the%20nervous%20system,in%20the%20arms%20or%20legs](https://www.verywellhealth.com/lyme-disease-stages-5176671#:~:text=Early%20Disseminated%20Lyme%20Disease%20(Stage%202)&text=Inflammation%20of%20the%20nervous%20system,in%20the%20arms%20or%20legs)
 - 15 <https://www.arizonaadvancedmedicine.com/articles/2013/june/lyme-disease/>
 - 16 Grapefruit seeds extract - Brorson, O., & Brorson, S. H. (2007). Grapefruit seed extract is a powerful in vitro agent against motile and cystic forms of Borrelia burgdorferi sensu lato. *Infection*, 35(3), 206–208. <https://doi.org/10.1007/s15010-007-6105-0> <https://pubmed.ncbi.nlm.nih.gov/17565468/>
 - 17 Centers for Disease Control and Prevention. Post-treatment Lyme disease syndrome.
 - 18 Campos M. Lyme disease: resolving the “Lyme wars.” Harvard Health Publishing.
 - 19 Karvonen, K., Nykky, J., Marjomäki, V., & Gilbert, L. (2021). Distinctive Evasion Mechanisms to Allow Persistence of Borrelia burgdorferi in Different Human Cell Lines. *Frontiers in microbiology*, 12, 711291. <https://doi.org/10.3389/fmicb.2021.711291> <https://pubmed.ncbi.nlm.nih.gov/34712208/>
 - 20 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.
 - 21 Hyde, J., Plaza, J. A., & Kaffenberger, J. (2021). Interstitial granuloma annulare triggered by Lyme disease. *Dermatology online journal*, 27(5), 10.5070/D327553618. <https://doi.org/10.5070/D327553618>
 - 22 Lyme Disease Prevention & Risk Factors. (n.d.). Retrieved March 21, 2022, from <https://www.webmd.com/arthritis/understanding-lyme-disease-prevention#:~:text=How%20Can%20I%20Prevent%20Lyme,of%20bare%20skin%2C%20and%20clothes>.
 - 23 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.
 - 24 Nelson, C. A., Hayes, C. M., Markowitz, M. A., Flynn, J. J., Graham, A. C., Delorey, M. J., Mead, P. S., & Dolan, M. C. (2016). The heat is on: Killing blacklegged ticks in residential washers and dryers to prevent tickborne diseases. *Ticks and tick-borne diseases*, 7(5), 958–963. <https://doi.org/10.1016/j.ttbdis.2016.04.016>
 - 25 Identifying tick bites. (2022, February 25). Retrieved March 22, 2022, from <https://igenex.com/tick-talk/identifying-tick-bites/>
 - 26 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.
 - 27 Transmission. (2020, January 29). Retrieved March 24, 2022, from <https://www.cdc.gov/lyme/transmission/index.html>
 - 28 Lyme Disease Prevention & Risk Factors. (n.d.). Retrieved March 21, 2022, from <https://www.webmd.com/arthritis/understanding-lyme-disease-prevention#:~:text=How%20Can%20I%20Prevent%20Lyme,of%20bare%20skin%2C%20and%20clothes>.
 - 29 Osimitz, T., Murphy, J., Fell, L., & Page, B. (2010). Adverse events associated with the use of insect repellents containing N,N-diethyl-M-toluamide (DEET). *Regulatory Toxicology and Pharmacology*, 56(1), 93–99. doi:10.1016/j.yrtph.2009.09.004
 - 30 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.
 - 31 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.
 - 32 Right to know hazardous substance fact sheet. (n.d.). Retrieved March 23, 2022, from <https://nj.gov/health/eoh/rtkweb/documents/fs/3422.pdf>
 - 33 Saar-Reismaa, P., Bragina, O., Kuhtinskaja, M., Reile, I., Laanet, P. R., Kulp, M., & Vaher, M. (2022). Extraction and Fractionation of Bioactives from Dipsacus fullonum L. Leaves and Evaluation of Their Anti-Borrelia Activity. *Pharmaceuticals* (Basel, Switzerland), 15(1), 87. <https://doi.org/10.3390/ph15010087> <https://pubmed.ncbi.nlm.nih.gov/35056144/>
 - 34 Oszmiański, J., Wojdyło, A., Juszczak, P., & Nowicka, P. (2020). Roots and Leaf Extracts of Dipsacus fullonum L. and Their Biological Activities. *Plants* (Basel, Switzerland), 9(1), 78. <https://doi.org/10.3390/plants9010078> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7020454/>
 - 35 Saar-Reismaa, P., Bragina, O., Kuhtinskaja, M., Reile, I., Laanet, P. R., Kulp, M., & Vaher, M. (2022). Extraction and Fractionation of Bioactives from Dipsacus fullonum L. Leaves and Evaluation of Their Anti-Borrelia Activity. *Pharmaceuticals* (Basel, Switzerland), 15(1), 87. <https://doi.org/10.3390/ph15010087> <https://pubmed.ncbi.nlm.nih.gov/35056144/>
 - 36 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.
 - 37 Chaturvedi, R., & Chaturvedi, S. (2011). A review on different therapeutic uses and antimicrobial potential of garlic. *South Asian Journal of Experimental Biology*, 1(5), 20–25. doi:10.38150/sajeb.1(5). p20-25
 - 38 Garlic (allicin) - herbs database. (2018, September 05). Retrieved March 22, 2022, from <https://rawlsmd.com/herbs/garlic-allicin>
 - 39 Feng, J., Leone, J., Schweig, S., & Zhang, Y. (2020). Evaluation of natural and botanical medicines for activity against growing and non-growing forms of B. Burgdorferi. *Frontiers in Medicine*, 7. doi:10.3389/fmed.2020.00006
 - 40 Galiniak, S., Aebisher, D., & Bartusik-Aebisher, D. (2019). Health benefits of Resveratrol Administration. *Acta Biochimica Polonica*. doi:10.18388/abp.2018_2749
 - 41 Ma, X., Leone, J., Schweig, S., & Zhang, Y. (2021). Botanical medicines with activity against Stationary phase bartonella henselae. *Infectious Microbes and Diseases*, 3(3), 158–167. doi:10.1097/im9.0000000000000069
 - 42 Feng, J., Leone, J., Schweig, S., & Zhang, Y. (2020). Evaluation of natural and botanical medicines for activity against growing and non-growing forms of B. Burgdorferi. *Frontiers in Medicine*, 7. doi:10.3389/fmed.2020.00006
 - 43 Feng, J., Leone, J., Schweig, S., & Zhang, Y. (2020). Evaluation of natural and botanical medicines for activity against growing and non-growing forms of B. Burgdorferi. *Frontiers in Medicine*, 7. doi:10.3389/fmed.2020.00006
 - 44 Ansah, C., & Mensah, K. B. (2013). A review of the anticancer potential of the anti-malarial herbal cryptolepis sanguinolenta and its major alkaloid cryptolepine. *Ghana medical journal*, 47(3), 137–147.
 - 45 Ma, X., Leone, J., Schweig, S., & Zhang, Y. (2021). Botanical medicines with activity against Stationary phase bartonella henselae. *Infectious Microbes and Diseases*, 3(3), 158–167. doi:10.1097/im9.0000000000000069
 - 46 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.
 - 47 Zhao, T., Tang, H., Xie, L., Zheng, Y., Ma, Z., Sun, Q., & Li, X. (2019). Scutellaria baicalensis georgi. (Lamiaceae): A review of its traditional uses, botany, phytochemistry, pharmacology and toxicology. *Journal of Pharmacy and Pharmacology*, 71(9), 1353–1369. doi:10.1111/jphp.13129
 - 48 Liao, H., Ye, J., Gao, L., & Liu, Y. (2021). The main bioactive compounds of scutellaria baicalensis Georgi. for alleviation of inflammatory cytokines: A comprehensive review. *Biomedicine & Pharmacotherapy*, 133, 110917. doi:10.1016/j.biopha.2020.110917
 - 49 Su, X., & Miller, L. H. (2015). The discovery of artemisinin and the nobel prize in physiology or medicine. *Science China Life Sciences*, 58(11), 1175–1179. doi:10.1007/s11427-015-4948-7
 - 50 Eckstein-Ludwig, U., Webb, R. J., Van Goethem, I. D., East, J. M., Lee, A. G., Kimura, M., . . . Krishna, S. (2003). Artemisinins target the Serca of Plasmodium

falciparum. *Nature*, 424(6951), 957-961. doi:10.1038/nature01813

51 Feng, J., Leone, J., Schweig, S., & Zhang, Y. (2020). Evaluation of natural and botanical medicines for activity against growing and non-growing forms of *B. Burgdorferi*. *Frontiers in Medicine*, 7. doi:10.3389/fmed.2020.00006

52 Singh, N., Bhalla, M., De Jager, P., & Gilca, M. (2011). An overview on ashwagandha: A rasayana (rejuvenator) of ayurveda. *African Journal of Traditional, Complementary and Alternative Medicines*, 8(5S). doi:10.4314/ajtcam.v8i5s.9

53 Croypley, M., Banks, A. P., & Boyle, J. (2015). The effects of rhodiola ROSEAL extract on anxiety, stress, cognition and other mood symptoms. *Phytotherapy Research*, 29(12), 1934-1939. doi:10.1002/ptr.5486

54 Saar-Reismaa, P., Bragina, O., Kuhtinskaja, M., Reile, I., Laanet, P. R., Kulp, M., & Vaher, M. (2022). Extraction and Fractionation of Bioactives from *Dipsacus fullonum* L. Leaves and Evaluation of Their Anti-Borrelia Activity. *Pharmaceuticals* (Basel, Switzerland), 15(1), 87. https://doi.org/10.3390/ph15010087 https://pubmed.ncbi.nlm.nih.gov/35056144/

55 Oszmianński, J., Wojdyło, A., Juszczak, P., & Nowicka, P. (2020). Roots and Leaf Extracts of *Dipsacus fullonum* L. and Their Biological Activities. *Plants* (Basel, Switzerland), 9(1), 78. https://doi.org/10.3390/plants9010078 https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7020454/

56 Saar-Reismaa, P., Bragina, O., Kuhtinskaja, M., Reile, I., Laanet, P. R., Kulp, M., & Vaher, M. (2022). Extraction and Fractionation of Bioactives from *Dipsacus fullonum* L. Leaves and Evaluation of Their Anti-Borrelia Activity. *Pharmaceuticals* (Basel, Switzerland), 15(1), 87. https://doi.org/10.3390/ph15010087 https://pubmed.ncbi.nlm.nih.gov/35056144/

57 Grant, W. B., & Holick, M. F. (2005). Benefits and requirements of vitamin D for optimal health: a review. *Alternative medicine review: a journal of clinical therapeutic*, 10(2), 94-111.

58 T Donta, S. (2012). Issues in the diagnosis and treatment of lyme disease. *The Open Neurology Journal*, 6(1), 140-145. doi:10.2174/1874205x01206010140

59 Dleland. (2021, December 08). How good nutrition can help you recover from lyme disease. Retrieved April 3, 2022, from https://www.lymedisease.org/how-good-nutrition-can-help-you-recover-from-lyme-disease/

60 Van Gorkom, G., Klein Wolterink, R., Van Elssen, C., Wieten, L., Germeaard, W., & Bos, G. (2018). Influence of vitamin C on lymphocytes: An overview. *Antioxidants*, 7(3), 41. doi:10.3390/antiox7030041

61 Muller, K. E. (2012). Damage of collagen and elastic fibres by borrelia burgdorferi – known and new clinical and histopathological aspects. *The Open Neurology Journal*, 6(1), 179-186. doi:10.2174/1874205x01206010179

62 Prasad, A. S. (2008). Zinc in human health: Effect of zinc on immune cells. *Molecular Medicine*, 14(5-6), 353-357. doi:10.2119/2008-00033.prasad

63 Kruse-Jarres J. D. (1989). The significance of zinc for humoral and cellular immunity. *Journal of trace elements and electrolytes in health and disease*, 3(1), 1-8.

64 Gao, X., Lampraki, E., Al-Khalidi, S., Qureshi, M. A., Desai, R., & Wilson, J. B. (2017). N-acetylcysteine (NAC) ameliorates Epstein-Barr virus latent membrane protein 1 induced chronic inflammation. *PLOS ONE*, 12(12). doi:10.1371/journal.pone.0189167

65 Zhao Z, et al. NF-κB is a key modulator in the signaling pathway of Borrelia burgdorferi BmpA-induced inflammatory chemokines in murine microglia BV2 cells. *Mol Med Rep*. 2018; 17(4): 4953-4958

66 Gutierrez-Hoffmann, M. G., O'Meally, R. N., Cole, R. N., Tiniakou, E., Darrah, E., & Soloski, M. J. (2020). Borrelia burgdorferi-induced changes in the class II self-immunopeptidome displayed on HLA-DR molecules expressed by dendritic cells. *Frontiers in Medicine*, 7. doi:10.3389/fmed.2020.00568

67 Inflammation: What is it, causes, symptoms & treatment. (n.d.). Retrieved April 2, 2022, from https://my.clevelandclinic.org/health/symptoms/21660-inflammation

68 Furman, D., Campisi, J., Verdin, E., Carrera-Bastos, P., Targ, S., Franceschi, C., ... Slavich, G. M. (2019). Chronic inflammation in the etiology of disease across the life span. *Nature Medicine*, 25(12), 1822-1832. doi:10.1038/s41591-019-0675-0

69 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.

70 Cör, D., Knez, Ž., & Knez Hrncič, M. (2018). Antitumour, antimicrobial, antioxidant and antiacetylcholinesterase effect of ganoderma lucidum terpenoids and Polysaccharides: A Review. *Molecules*, 23(3), 649. doi:10.3390/molecules23030649

71 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.

72 Gutiérrez, S., Svahn, S. L., & Johansson, M. E. (2019). Effects of omega-3 fatty acids on immune cells. *International Journal of Molecular Sciences*, 20(20), 5028. doi:10.3390/ijms20205028

73 Dyall, S. C. (2015). Long-chain omega-3 fatty acids and the brain: A review of the independent and shared effects of EPA, DPA and dha. *Frontiers in Aging Neuroscience*, 7. doi:10.3389/fnagi.2015.00052

74 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.

75 Thipe, V. C., Mentor, S., Lima, C. S., Freitas, L. F., Fonseca, A. C., Nogueira, K. M., ... Lugão, A. B. (2022). The role of probiotics in maintaining immune homeostasis. *Probiotics in the Prevention and Management of Human Diseases*, 41-58. doi:10.1016/b978-0-12-823733-5.00024-6

76 Dleland. (2021, December 08). How good nutrition can help you recover from lyme disease. Retrieved April 3, 2022, from https://www.lymedisease.org/how-good-nutrition-can-help-you-recover-from-lyme-disease/

77 Gutiérrez, S., Svahn, S. L., & Johansson, M. E. (2019). Effects of omega-3

fatty acids on immune cells. *International Journal of Molecular Sciences*, 20(20), 5028. doi:10.3390/ijms20205028

78 Anti-inflammatory diet – made easy: Arthritis NSW. (2021, December 20). Retrieved April 2, 2022, from https://www.arthritisnsw.org.au/anti-inflammatory-diet-made-easy/

79 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.

80 Rawls, W. (2017). Unlocking lyme: Myths, truths, and practical solutions for chronic lyme disease. Cary, NC: FirstDoNoHarm Publishing.

81 N. (2021, December 22). Drink water for less pain -- 4 ways drinking water helps. Retrieved April 2, 2022, from https://peerwell.co/blog/drink-water-for-less-pain-4-ways-drinking-water-helps/

82 Water. (2021, July 06). Retrieved April 2, 2022, from https://www.hsph.harvard.edu/nutritionsource/water/#:~:text=General%20recommendations,exposed%20to%20very%20warm%20climates.

83 Besedovsky, L., Lange, T., & Born, J. (2011). Sleep and immune function. *Pflügers Archiv - European Journal of Physiology*, 463(1), 121-137. doi:10.1007/s00424-011-1044-0

84 Liu, Y., Wheaton, A. G., Chapman, D. P., & Croft, J. B. (2013). Sleep duration and chronic diseases among US adults age 45 years and older: Evidence from the 2010 Behavioral Risk Factor Surveillance System. *Sleep*, 36(10), 1421-1427. doi:10.5665/sleep.3028

85 Butler, T. (2016). The Jarisch–herxheimer reaction after antibiotic treatment of spirochetal infections: A review of recent cases and our understanding of pathogenesis. *The American Journal of Tropical Medicine and Hygiene*, 96(1), 46-52. doi:10.4269/ajtmh.16-0434

86 Nykytyuk, S., Boyarchuk, O., Klymnyuk, S., & Levenets, S. (2020). The jarisch-herxheimer reaction associated with doxycycline in a patient with lyme arthritis. *Reumatologia/Rheumatology*, 58(5), 335-338. doi:10.5114/reum.2020.99143

87 Keng, S., Smoski, M. J., & Robins, C. J. (2011). Effects of mindfulness on psychological health: A review of empirical studies. *Clinical Psychology Review*, 31(6), 1041-1056. doi:10.1016/j.cpr.2011.04.006

88 Merkes, M. (2010). Mindfulness-based stress reduction for people with chronic diseases. *Australian Journal of Primary Health*, 16(3), 200. doi:10.1071/py09063

Parasite Protocol

1 Lepczyńska, M., Białkowska, J., Dzika, E., Piskorz-Ogórek, K., & Korycińska, J. (2017). Blastocystis: how do specific diets and human gut microbiota affect its development and pathogenicity?. *European journal of clinical microbiology & infectious diseases : official publication of the European Society of Clinical Microbiology*, 36(9), 1531-1540. https://doi.org/10.1007/s10096-017-2965-0

2 Oliveira, B., Bresciani, K., & Widmer, G. (2019). Deprivation of dietary fiber enhances susceptibility of mice to cryptosporidiosis. *PLoS neglected tropical diseases*, 13(9), e0007411. https://doi.org/10.1371/journal.pntd.0007411

3 Shea-Donohue, T., Qin, B., & Smith, A. (2017). Parasites, nutrition, immune responses and biology of metabolic tissues. *Parasite immunology*, 39(5), 10.1111/pim.12422. https://doi.org/10.1111/pim.12422

4 Myles I. A. (2014). Fast food fever: reviewing the impacts of the Western diet on immunity. *Nutrition journal*, 13, 61. https://doi.org/10.1186/1475-2891-13-61

5 Mohammed, S.E., Kabbashi, A., Koko, W., Ansari, M., & Al-Ghamdi, A. (2017). In vitro activity of some natural honeys against Entamoeba histolytica and Giardia lamblia trophozoites. *Saudi Journal of Biological Sciences*. 26. 10.1016/j.sjbs.2017.06.004.

6 Kaneda, Y., Torii, M., Tanaka, T., & Aikawa, M. (1991). In vitro effects of berberine sulphate on the growth and structure of Entamoeba histolytica, Giardia lamblia and Trichomonas vaginalis. *Annals of tropical medicine and parasitology*, 85(4), 417-425. https://doi.org/10.1080/00034983.1991.11812586

7 Ghosh, A. K., Bhattacharyya, F. K., & Ghosh, D. K. (1985). Leishmania donovani: amastigote inhibition and mode of action of berberine. *Experimental parasitology*, 60(3), 404-413. https://doi.org/10.1016/0014-4894(85)90047-5

8 Gupte S. (1975). Use of berberine in treatment of giardiasis. American journal of diseases of children (1960), 129(7), 866. https://doi.org/10.1001/archpedi.1975.02120440082020

9 Subbaiah, T. V., & Amin, A. H. (1967). Effect of berberine sulphate on Entamoeba histolytica. *Nature*, 215(5100), 527-528. https://doi.org/10.1038/215527a0

10 Abd-Elhamid, T. H., Abdel-Rahman, I., Mahmoud, A. R., Almaillem, K. S., Almatroudi, A., Fouad, S. S., Abdella, O. H., Elshabrawy, H. A., & El-Kady, A. M. (2021). A Complementary Herbal Product for Controlling Giardiasis. *Antibiotics* (Basel, Switzerland), 10(5), 477. https://doi.org/10.3390/antibiotics10050477

11 Loo, C. S., Lam, N. S., Yu, D., Su, X. Z., & Lu, F. (2017). Artemisinin and its derivatives in treating protozoan infections beyond malaria. *Pharmacological research*, 117, 192-217. https://doi.org/10.1016/j.phrs.2016.11.012

12 Elfawal, M. A., Towler, M. J., Reich, N. G., Golenbock, D., Weathers, P. J., & Rich, S. M. (2012). Dried whole plant Artemisia annua as an antimalarial therapy. *PLoS one*, 7(12), e52746. https://doi.org/10.1371/journal.pone.0052746

13 de Sousa, D. P., Lima, T. C., & Steverding, D. (2016). Evaluation of Antiparasitic Activity of Mentha crissa Essential Oil, Its Major Constituent Rotundifolone and Analogues against Trypanosoma brucei. *Planta medica*, 82(15), 1346-1350. https://doi.org/10.1055/s-0042-107082

14 Moraes, M. E., Cunha, G. H., Bezerra, M. M., Fachine, F. V., Pontes, A. V., Andrade, W. S., Frota Bezerra, F. A., Moraes, M. O., & Cavalcanti, P. P. (2012). Efficacy of the Mentha crissa in the treatment of women with Trichomonas vaginalis infection. *Archives of gynecology and obstetrics*, 286(1), 125-130. https://doi.org/10.1007/s00404-012-2251-4

15 Moraes, M. E., Cunha, G. H., Bezerra, M. M., Fachine, F. V., Pontes, A. V., Andrade, W. S.,

- Fruta Bezerra, F. A., Moraes, M. O., & Cavalcanti, P. P. (2012). Efficacy of the *Mentha crisper* in the treatment of women with *Trichomonas vaginalis* infection. *Archives of gynecology and obstetrics*, 286(1), 125–130. <https://doi.org/10.1007/s00404-012-2251-4>
- 16 Machado, M., Dinis, A. M., Salgueiro, L., Custódio, J. B., Cavaleiro, C., & Sousa, M. C. (2011). Anti-Giardia activity of Syzygium aromaticum essential oil and eugenol: effects on growth, viability, adherence and ultrastructure. *Experimental parasitology*, 127(4), 732–739. <https://doi.org/10.1016/j.exppara.2011.01.011>
- 17 Ezz Eldin, H. (2019). Potent lethal effect of Syzygium aromaticum essential oil on *Blastocystis* spp.: An in vitro study. *Parasitologists United Journal*, 12(1), 61–67. doi: 10.21608/puj.2019.10650.1035
- 18 Batiha, G. E., Beshbishy, A. M., Tayebwa, D. S., Shaheen, H. M., Yokoyama, N., & Igarashi, I. (2019). Inhibitory effects of Syzygium aromaticum and Camellia sinensis methanolic extracts on the growth of Babesia and Theileria parasites. *Ticks and tick-borne diseases*, 10(5), 949–958. <https://doi.org/10.1016/j.ttbdis.2019.04.016>
- 19 Charitha, V. & J, Dr Adeppa & Malakondaiha, P. (2017). In vitro anthelmintic activity of Syzygium aromaticum and Melia dubia against Haemonchus contortus of sheep. *Indian Journal of Animal Sciences*. 87. 968–970.
- 20 Santoro, G. F., Cardoso, M. G., Guimarães, L. G., Mendonça, L. Z., & Soares, M. J. (2007). Trypanosoma cruzi: activity of essential oils from Achillea millefolium L., Syzygium aromaticum L. and Ocimum basilicum L. on epimastigotes and trypomastigotes. *Experimental parasitology*, 116(3), 283–290. <https://doi.org/10.1016/j.exppara.2007.01.018>
- 21 Alnomasy, S., Al-Awsi, G., Raziani, Y., Albalawi, A. E., Alanazi, A. D., Niazi, M., & Mahmoudvand, H. (2021). Systematic review on medicinal plants used for the treatment of Giardia infection. *Saudi journal of biological sciences*, 28(9), 5391–5402. <https://doi.org/10.1016/j.sjbs.2021.05.069>
- 22 Paveto, C., Güida, M. C., Esteve, M. I., Martino, V., Coussio, J., Flawiá, M. M., & Torres, H. N. (2004). Anti-Trypanosoma cruzi activity of green tea (Camellia sinensis) catechins. *Antimicrobial agents and chemotherapy*, 48(1), 69–74. <https://doi.org/10.1128/AAC.48.1.69-74.2004>
- 23 Fakae, L. B., Stevenson, C. W., Zhu, X. Q., & Elsheikha, H. M. (2020). In vitro activity of Camellia sinensis (green tea) against trophozoites and cysts of Acanthamoeba castellanii. *International journal for parasitology. Drugs and drug resistance*, 13, 59–72. <https://doi.org/10.1016/j.ijpddr.2020.05.001>
- 24 Audomkasok, S., Singpha, W., Chachiy, S., & Somsak, V. (2014). Antihelmolytic Activities of Green Tea, Safflower, and Mulberry Extracts during Plasmodium berghei Infection in Mice. *Journal of pathogens*, 2014, 203154. <https://doi.org/10.1155/2014/203154>
- 25 Sanderson, L., Bartlett, A., & Whitfield, P. J. (2002). In vitro and in vivo studies on the bioactivity of a ginger (Zingiber officinale) extract towards adult schistosomes and their egg production. *Journal of helminthology*, 76(3), 241–247. <https://doi.org/10.1079/JOH2002116>
- 26 El-Melegy, M. A., El-Saify, G. H., Hassab-El-Nabi, S. E. (2006). Evaluation of therapeutic effect of ginger compared to flubendazole on experimental trichinellosis in mice. *Egypt J Med Sci*. 27(2):25–48.
- 27 El-Kady, A. M., Al-Megrin, W., Abdel-Rahman, I., Sayed, E., Alshehri, E. A., Wakid, M. H., Baakdah, F. M., Mohamed, K., Elshazly, H., Alobaid, H. M., Qahl, S. H., Elshabrawy, H. A., & Younis, S. S. (2022). Ginger Is a Potential Therapeutic for Chronic Toxoplasmosis. *Pathogens (Basel, Switzerland)*, 11(7), 798. <https://doi.org/10.3390/pathogens11070798>
- 28 Abu El Ezz, N. M., Khalil, F. A., Shaapan, R. M. (2011). Therapeutic effect of onion (Allium cepa) and Cinnamon (C. zeylanicum) oils on cryptosporidiosis in experimentally infected mice. *Global Veterinaria*. 2011;7(2):179–183.
- 29 Mahmoud, A., Attia, R., Said, S., & Ibrahim, Z. (2014). Ginger and cinnamon: can this household remedy treat giardiasis? Parasitological and histopathological studies. *Iranian journal of parasitology*, 9(4), 530–540.
- 30 Williams, A. R., Ramsay, A., Hansen, T. V., Ropiak, H. M., Mejer, H., Nejsum, P., Mueller-Harvey, I., & Thamsborg, S. M. (2015). Anthelmintic activity of trans-cinnamaldehyde and A- and B-type proanthocyanidins derived from cinnamon (Cinnamomum verum). *Scientific reports*, 5, 14791. <https://doi.org/10.1038/srep14791>
- 31 Moraes, D., Levenhagen, M. A., Costa-Cruz, J. M., Costa, A. P., Netto, & Rodrigues, R. M. (2017). In vitro efficacy of latex and purified papain from Carica papaya against Strongyloides venezuelensis eggs and larvae. *Revista do Instituto de Medicina Tropical de Sao Paulo*, 59, e7. <https://doi.org/10.1590/S1678-9946201759007>
- 32 Kugo, M., Keter, L., Maiyo, A., Kinyua, J., Ndemwa, P., Maina, G., Otieno, P., & Songok, E. M. (2018). Fortification of Carica papaya fruit seeds to school meal snacks may aid Africa mass deworming programs: a preliminary survey. *BMC complementary and alternative medicine*, 18(1), 327. <https://doi.org/10.1186/s12906-018-2379-2>
- 33 Grzybek, M., Kukula-Koch, W., Strachecka, A., Jaworska, A., Phiri, A. M., Paleolog, J., & Tomczuk, K. (2016). Evaluation of Anthelmintic Activity and Composition of Pumpkin (Cucurbita pepo L.) Seed Extracts-In Vitro and in Vivo Studies. *International journal of molecular sciences*, 17(9), 1456. <https://doi.org/10.3390/ijms17091456>
- 34 Travers, M. A., Florent, I., Kohl, L., & Grellier, P. (2011). Probiotics for the control of parasites: an overview. *Journal of parasitology research*, 2011, 610769. <https://doi.org/10.1155/2011/610769>
- 35 Benson, A., Pifer, R., Behrendt, C. L., Hooper, L. V., & Yarovinsky, F. (2009). Gut commensal bacteria direct a protective immune response against Toxoplasma gondii. *Cell host & microbe*, 6(2), 187–196. <https://doi.org/10.1016/j.chom.2009.06.005>
- 36 Travers, M. A., Florent, I., Kohl, L., & Grellier, P. (2011). Probiotics for the control of parasites: an overview. *Journal of parasitology research*, 2011, 610769. <https://doi.org/10.1155/2011/610769>
- 37 Smith, A. D., Cheung, L., Beshah, E., Shea-Donohue, T., & Urban, J. F., Jr (2013). Selenium status alters the immune response and expulsion of adult Heligmosomoides bakeri worms in mice. *Infection and immunity*, 81(7), 2546–2553. <https://doi.org/10.1128/IAI.01047-12>
- 38 Christine D Thomson, Alexandra Chisholm, Sarah K McLachlan, Jennifer M Campbell, Brazil nuts: an effective way to improve selenium status, The American Journal of Clinical Nutrition, Volume 87, Issue 2, February 2008, Pages 379–384, <https://doi.org/10.1093/ajcn/87.2.379>
- 39 Shea-Donohue, T., Qin, B., & Smith, A. (2017). Parasites, nutrition, immune responses and biology of metabolic tissues. *Parasite immunology*, 39(5), 10.1111/pim.12422. <https://doi.org/10.1111/pim.12422>
- 40 Payne, L. G., Koski, K. G., Ortega-Barria, E., & Scott, M. E. (2007). Benefit of vitamin A supplementation on ascariis reinfection is less evident in stunted children. *The Journal of nutrition*, 137(6), 1455–1459. <https://doi.org/10.1093/jn/137.6.1455>
- 41 Payne, L. G., Koski, K. G., Ortega-Barria, E., & Scott, M. E. (2007). Benefit of vitamin A supplementation on ascariis reinfection is less evident in stunted children. *The Journal of nutrition*, 137(6), 1455–1459. <https://doi.org/10.1093/jn/137.6.1455>
- 42 Scott, M. E., & Koski, K. G. (2000). Zinc deficiency impairs immune responses against parasitic nematode infections at intestinal and systemic sites. *The Journal of nutrition*, 130(5S Suppl), 1412S–20S. <https://doi.org/10.1093/jn/130.5.1412S>

Alzheimer's and Dementia Protocol

- 1 Alzheimer's Association. (n.d.). *Mild cognitive impairment (MCI)*. Alzheimer's Disease and Dementia. Retrieved August 19, 2022, from https://www.alz.org/alzheimers-dementia/what-is-dementia/related_conditions/mild-cognitive-impairment
- 2 Alzheimer's Association. (n.d.). *Stages of Alzheimer's*. Alzheimer's Disease and Dementia. Retrieved August 19, 2022, from <https://www.alz.org/alzheimers-dementia/stages>
- 3 Ruan, Y., Tang, J., Guo, X., Li, K., & Li, D. (2018). Dietary Fat Intake and Risk of Alzheimer's Disease and Dementia: A Meta-Analysis of Cohort Studies. *Current Alzheimer research*, 15(9), 869–876. <https://doi.org/10.2174/1567205015666180427142350>
- 4 Mensink, R. P., & Katan, M. B. (1990). Effect of dietary trans fatty acids on high-density and low-density lipoprotein cholesterol levels in healthy subjects. *The New England journal of medicine*, 323(7), 439–445. <https://doi.org/10.1056/NEJM199008163230703>
- 5 Morris M. C. (2004). Diet and Alzheimer's disease: what the evidence shows. *Med-GenMed: Medscape general medicine*, 6(1), 48.
- 6 Sibilrud, R., Mutter, J., Moore, E., Naumann, J., & Walach, H. (2019). A Hypothesis and Evidence That Mercury May be an Etiological Factor in Alzheimer's Disease. *International journal of environmental research and public health*, 16(24), 5152. <https://doi.org/10.3390/ijerph16245152>
- 7 Joseph, J. A., Shukitt-Hale, B., Denisova, N. A., Prior, R. L., Cao, G., Martin, A., Taglialetta, G., & Bickford, P. C. (1998). Long-term dietary strawberry, spinach, or vitamin E supplementation retards the onset of age-related neuronal signal-transduction and cognitive behavioral deficits. *The Journal of neuroscience: the official journal of the Society for Neuroscience*, 18(19), 8047–8055. <https://doi.org/10.1523/JNEUROSCI.18-19-08047.1998>
- 8 Guerrero, A. L., Dorado-Martínez, C., Rodríguez, A., Pedroza-Ríos, K., Borronio-Pérez, G., & Rivas-Arancibia, S. (1999). Effects of vitamin E on ozone-induced memory deficits and lipid peroxidation in rats. *Neuroreport*, 10(8), 1689–1692. <https://doi.org/10.1097/00001756-199906030-00012>
- 9 Ahlsgog, J. E., Geda, Y. E., Graff-Radford, N. R., & Petersen, R. C. (2011). Physical exercise as a preventive or disease-modifying treatment of dementia and brain aging. *Mayo Clinic proceedings*, 86(9), 876–884. <https://doi.org/10.4065/mcp.2011.0252>
- 10 Erickson, K. I., Voss, M. W., Prakash, R. S., Basak, C., Szabo, A., Chaddock, L., Kim, J. S., Heo, S., Alves, H., White, S. M., Wojcicki, T. R., Mailey, E., Vieira, V. J., Martin, S. A., Pence, B. D., Woods, J. A., McAuley, E., & Kramer, A. F. (2011). Exercise training increases size of hippocampus and improves memory. *Proceedings of the National Academy of Sciences of the United States of America*, 108(7), 3017–3022. <https://doi.org/10.1073/pnas.1015950108> * Calculate using the Karvonen method, which takes your resting heart rate (heart rate reserve is the difference between your resting and your maximum heart rate) into account
- 11 Sabia, S., Fayosse, A., Dumurgier, J., van Hees, V. T., Paquet, C., Sommerlad, A., Kivimäki, M., Dugravot, A., & Singh-Manoux, A. (2021). Association of sleep duration in middle and old age with incidence of dementia. *Nature communications*, 12(1), 2289. <https://doi.org/10.1038/s41467-021-22354-2>
- 12 Anderson, J., Hawrylewicz, K., & Grundy, J. G. (2020). Does bilingualism protect against dementia? A meta-analysis. *Psychonomic bulletin & review*, 27(5), 952–965. <https://doi.org/10.3758/s13423-020-01736-5>
- 13 Tennstedt, S. L., & Unverzagt, F. W. (2013). The ACTIVE study: study overview and major findings. *Journal of aging and health*, 25(8 Suppl), 3S–20S. <https://doi.org/10.1177/0898264313518133>
- 14 Innes, K. E., Selfe, T. K., Brundage, K., Montgomery, C., Wen, S., Kandati, S., Bowles, H., Khalsa, D. S., & Huysmans, Z. (2018). Effects of Meditation and Music-Listening on Blood Biomarkers of Cellular Aging and Alzheimer's Disease in Adults with Subjective Cognitive Decline: An Exploratory Randomized Clinical Trial. *Journal of Alzheimer's disease: JAD*, 66(3), 947–970. <https://doi.org/10.3233/JAD-180164> <https://pubmed.ncbi.nlm.nih.gov/30320574/>
- 15 Dwivedi M, Dubey N, Pansari AJ, Bapi RS, Das M, Guha M, Banerjee R, Pramanick G, Basu J, Ghosh A. Effects of Meditation on Structural Changes of the Brain in Patients with Mild Cognitive Impairment or Alzheimer's Disease Dementia. *Front Hum Neurosci*. 2021 Nov 12;15:728993. doi: 10.3389/fnhum.2021.728993. PMID: 34867239; PMCID: PMC8633496. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8633496/>
- 16 Nguyen, T. T., Ta, Q., Nguyen, T., Nguyen, T., & Giau, V. V. (2020). Type 3 Diabetes and Its Role Implications in Alzheimer's Disease. *International journal of molecular sciences*, 21(9), 3165. <https://doi.org/10.3390/ijms21093165>
- 17 Lodeiro, M., Ibáñez, C., Cifuentes, A., Simó, C., & Cedazo-Minguez, Á. (2014). Decreased cerebrospinal fluid levels of L-carnitine in non-apolipoprotein E4

carriers at early stages of Alzheimer's disease. *Journal of Alzheimer's disease: JAD*, 41(1), 223–232. <https://doi.org/10.3233/JAD-132063>

18 Pennisi, M., Lanza, G., Cantone, M., D'Amico, E., Fisicaro, F., Puglisi, V., Vinciguerra, L., Bella, R., Vicari, E., & Malaguarnera, G. (2020). Acetyl-L-Carnitine in Dementia and Other Cognitive Disorders: A Critical Update. *Nutrients*, 12(5), 1389. <https://doi.org/10.3390/nu12051389>

19 Malaguarnera, M., Cammalleri, L., Gargante, M. P., Vacante, M., Colonna, V., & Motta, M. (2007). L-Carnitine treatment reduces severity of physical and mental fatigue and increases cognitive functions in centenarians: a randomized and controlled clinical trial. *The American journal of clinical nutrition*, 86(6), 1738–1744. <https://doi.org/10.1093/ajcn/86.5.1738>

20 Malaguarnera, M., Gargante, M. P., Cristaldi, E., Colonna, V., Messano, M., Koverech, A., Neri, S., Vacante, M., Cammalleri, L., & Motta, M. (2008). Acetyl L-carnitine (ALC) treatment in elderly patients with fatigue. *Archives of gerontology and geriatrics*, 46(2), 181–190. <https://doi.org/10.1016/j.archger.2007.03.012>

21 Alvarez-Sabín, J., & Román, G. C. (2011). Citicoline in vascular cognitive impairment and vascular dementia after stroke. *Stroke*, 42(1 Suppl), S40–S43. <https://doi.org/10.1161/STROKEAHA.110.606509>

22 Littlejohns, T. J., Henley, W. E., Lang, I. A., Annweiler, C., Beauchet, O., Chaves, P. H., Fried, L., Kestenbaum, B. R., Kuller, L. H., Langa, K. M., Lopez, O. L., Kos, K., Soni, M., & Llewellyn, D. J. (2014). Vitamin D and the risk of dementia and Alzheimer disease. *Neurology*, 83(10), 920–928. <https://doi.org/10.1212/WNL.0000000000000755>

23 Shine, B., McKnight, R. F., Leaver, L., & Geddes, J. R. (2015). Long-term effects of lithium on renal, thyroid, and parathyroid function: a retrospective analysis of laboratory data. *Lancet (London, England)*, 386(9992), 461–468. [https://doi.org/10.1016/S0140-6736\(14\)61842-0](https://doi.org/10.1016/S0140-6736(14)61842-0)

24 Memon, A., Rogers, I., Fitzsimmons, S., Carter, B., Strawbridge, R., Hidalgo-Mazzei, D., & Young, A. H. (2020). Association between naturally occurring lithium in drinking water and suicide rates: systematic review and meta-analysis of ecological studies. *The British journal of psychiatry: the journal of mental science*, 217(6), 667–678. <https://doi.org/10.1192/bjp.2020.128>

25 Nunes, M. A., Viel, T. A., & Buck, H. S. (2013). Microdose lithium treatment stabilized cognitive impairment in patients with Alzheimer's disease. *Current Alzheimer research*, 10(1), 104–107. <https://doi.org/10.2174/1567205011310010014>

26 Haussmann, R., Noppes, F., Brandt, M. D., Bauer, M., & Donix, M. (2021). Lithium: A therapeutic option in Alzheimer's disease and its prodromal stages?. *Neuroscience letters*, 760, 136044. <https://doi.org/10.1016/j.neulet.2021.136044>

27 Thomas, J., Thomas, C. J., Radcliffe, J., & Itsiopoulos, C. (2015). Omega-3 Fatty Acids in Early Prevention of Inflammatory Neurodegenerative Disease: A Focus on Alzheimer's Disease. *BioMed research international*, 2015, 172801. <https://doi.org/10.1155/2015/172801>

28 Ajith T. A. (2018). A Recent Update on the Effects of Omega-3 Fatty Acids in Alzheimer's Disease. *Current clinical pharmacology*, 13(4), 252–260. <https://doi.org/10.2174/1574884713666180807145648>

29 Arora, K., Green, M., & Prakash, S. (2020). The Microbiome and Alzheimer's Disease: Potential and Limitations of Prebiotic, Synbiotic, and Probiotic Formulations. *Frontiers in bioengineering and biotechnology*, 8, 537847. <https://doi.org/10.3389/fbioe.2020.537847>

30 Refsum, H., & Smith, A. D. (2003). Low vitamin B-12 status in confirmed Alzheimer's disease as revealed by serum holotranscobalamin. *Journal of neurology, neurosurgery, and psychiatry*, 74(7), 959–961. <https://doi.org/10.1136/jnnp.74.7.959>

31 Health Quality Ontario (2013). Vitamin B12 and cognitive function: an evidence-based analysis. *Ontario health technology assessment series*, 13(23), 1–45.

32 Health Quality Ontario (2013). Vitamin B12 and cognitive function: an evidence-based analysis. *Ontario health technology assessment series*, 13(23), 1–45.

33 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, *Herichium erinaceus* (Higher Basidiomycetes) from Malaysia. *International journal of medicinal mushrooms*, 15(6), 539–554. <https://doi.org/10.1615/intjmed-mushrv15.i6.30>

34 Deshmukh, S., Sridhar, K., & Gupta, M.K. (2021). *Herichium erinaceus* -A Rich Source of Diverse Bioactive Metabolites. *Fungal Biotech.* 1. 10-38. 10.5943/FunBiotech/1/2/2. <https://www.researchgate.net/publication/355143838>

35 Kuo, H. C., Lu, C. C., Shen, C. H., Tung, S. Y., Hsieh, M. C., Lee, K. C., Lee, L. Y., Chen, C. C., Teng, C. C., Huang, W. S., Chen, T. C., & Lee, K. F. (2016). *Herichium erinaceus* mycelium and its isolated erinacine A protection from MPTP-induced neurotoxicity through the ER stress, triggering an apoptosis cascade. *Journal of translational medicine*, 14, 78. <https://doi.org/10.1186/s12967-016-0831-y> (Retraction published J *Transl Med*. 2021 Feb 15;19(1):67)

36 Mori, K., Obara, Y., Moriya, T., Inatomi, S., & Nakahata, N. (2011). Effects of *Herichium erinaceus* on amyloid β (25-35) peptide-induced learning and memory deficits in mice. *Biomedical research (Tokyo, Japan)*, 32(1), 67–72. <https://doi.org/10.2220/biomedres.32.67>

37 Brandalise, F., Cesaroni, V., Gregori, A., Repetti, M., Romano, C., Orrù, G., Botta, L., Girometta, C., Guglielminetti, M. L., Savino, E., & Rossi, P. (2017). Dietary Supplementation of *Herichium erinaceus* Increases Mossy Fiber-CA3 Hippocampal Neurotransmission and Recognition Memory in Wild-Type Mice. Evidence-based complementary and alternative medicine: eCAM, 2017, 3864340. <https://doi.org/10.1155/2017/3864340>

38 Mori, K., Obara, Y., Moriya, T., Inatomi, S., & Nakahata, N. (2011). Effects of *Herichium erinaceus* on amyloid β (25-35) peptide-induced learning and memory deficits in mice. *Biomedical research (Tokyo, Japan)*, 32(1), 67–72. <https://doi.org/10.2220/biomedres.32.67>

39 Mori, K., Kikuchi, H., Obara, Y., Iwashita, M., Azumi, Y., Kinugasa, S., Inatomi, S., Oshima, Y., & Nakahata, N. (2010). Inhibitory effect of hericenone B from *Herichium erinaceus* on collagen-induced platelet aggregation. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 17(14), 1082–1085. <https://doi.org/10.1016/j.phymed.2010.05.004>

40 Stough, C., Lloyd, J., Clarke, J., Downey, L. A., Hutchison, C. W., Rodgers, T., & Nathan, P. J. (2001). The chronic effects of an extract of *Bacopa monniera* (Brahmi) on cognitive function in healthy human subjects. *Psychopharmacology*, 156(4), 481–484. <https://doi.org/10.1007/s002130100815>

41 Abdul Manap, A. S., Vijayabalan, S., Madhavan, P., Chia, Y. Y., Arya, A., Wong, E. H., Rizwan, F., Bindal, U., & Koshy, S. (2019). *Bacopa monnieri*, a Neuroprotective Lead in Alzheimer Disease: A Review on Its Properties, Mechanisms of Action, and Preclinical and Clinical Studies. *Drug target insights*, 13, 1177392819866412. <https://doi.org/10.1177/1177392819866412>

42 EghbaliFeriz, S., Taleghani, A., & Tayarani-Najaran, Z. (2018). Central nervous system diseases and Scutellaria: a review of current mechanism studies. *Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie*, 102, 185–195. <https://doi.org/10.1016/j.biopha.2018.03.021>

43 Saenghong, N., Wattanathorn, J., Muchimapura, S., Tongun, T., Piyavhatkul, N., Ban-chonglikitkul, C., & Kajsongkram, T. (2012). Zingiber officinale Improves Cognitive Function of the Middle-Aged Healthy Women. *Evidence-based complementary and alternative medicine: eCAM*, 2012, 383062. <https://doi.org/10.1155/2012/383062>

44 Saenghong, N., Wattanathorn, J., Muchimapura, S., Tongun, T., Piyavhatkul, N., Ban-chonglikitkul, C., & Kajsongkram, T. (2012). Zingiber officinale Improves Cognitive Function of the Middle-Aged Healthy Women. *Evidence-based complementary and alternative medicine: eCAM*, 2012, 383062. <https://doi.org/10.1155/2012/383062>

45 Zhang, H. F., Huang, L. B., Zhong, Y. B., Zhou, Q. H., Wang, H. L., Zheng, G. Q., & Lin, Y. (2016). An Overview of Systematic Reviews of Ginkgo biloba Extracts for Mild Cognitive Impairment and Dementia. *Frontiers in aging neuroscience*, 8, 276. <https://doi.org/10.3389/fnagi.2016.00276>

46 Farhana, K. M., Malueka, R. G., Wibowo, S., & Gofir, A. (2016). Effectiveness of Gotu Kola Extract 750 mg and 1000 mg Compared with Folic Acid 3 mg in Improving Vascular Cognitive Impairment after Stroke. *Evidence-based complementary and alternative medicine: eCAM*, 2016, 2795915. <https://doi.org/10.1155/2016/2795915>

47 Uddin, M. S., Mamun, A. A., Hossain, M. S., Akter, F., Iqbal, M. A., & Asaduzzaman, M. (2016). Exploring the Effect of *Phyllanthus emblica* L. on Cognitive Performance, Brain Antioxidant Markers and Acetylcholinesterase Activity in Rats: Promising Natural Gift for the Mitigation of Alzheimer's Disease. *Annals of neurosciences*, 23(4), 218–229. <https://doi.org/10.1159/000449482>

48 Huang, S., Mao, J., Ding, K., Zhou, Y., Zeng, X., Yang, W., Wang, P., Zhao, C., Yao, J., Xia, P., & Pei, G. (2017). Polysaccharides from *Ganoderma lucidum* Promote Cognitive Function and Neural Progenitor Proliferation in Mouse Model of Alzheimer's Disease. *Stem cell reports*, 8(1), 84–94. <https://doi.org/10.1016/j.stemcr.2016.12.007>

49 Bhardwaj, N., Katyal, P., & Sharma, A. K. (2014). Suppression of inflammatory and allergic responses by pharmacologically potent fungus *Ganoderma lucidum*. *Recent patents on inflammation & allergy drug discovery*, 8(2), 104–117. <https://doi.org/10.2174/1872213x08666140619110657>

50 Bhardwaj, N., Katyal, P., & Sharma, A. K. (2014). Suppression of inflammatory and allergic responses by pharmacologically potent fungus *Ganoderma lucidum*. *Recent patents on inflammation & allergy drug discovery*, 8(2), 104–117. <https://doi.org/10.2174/1872213x08666140619110657>

51 Mishra, S., & Palanivelu, K. (2008). The effect of curcumin (turmeric) on Alzheimer's disease: An overview. *Annals of Indian Academy of Neurology*, 11(1), 13–19. <https://doi.org/10.4103/0972-2327.40220>

52 Sarker, M. R., & Franks, S. F. (2018). Efficacy of curcumin for age-associated cognitive decline: a narrative review of preclinical and clinical studies. *GeroScience*, 40(2), 73–95. <https://doi.org/10.1007/s11357-018-0017-z>

Anxiety and Stress Protocol

1 McLean, C. P., Asnaani, A., Litz, B. T., & Hofmann, S. G. (2011). Gender differences in anxiety disorders: Prevalence, course of illness, comorbidity and burden of illness. *Journal of Psychiatric Research*, 45(8), 1027–1035. <https://doi.org/10.1016/j.jpsy-chires.2011.03.006>

2 Griffin JB JR. Anxiety. In: Walker HK, Hall WD, Hurst JW, editors. *Clinical Methods: The History, Physical, and Laboratory Examinations*. 3rd ed. Boston: Butterworths; 1990. Chapter 202. PMID: 21250157

3 Penninx, B. W. J. H., Pine, D. S., Holmes, E. A., & Reif, A. (2021). Anxiety disorders. *The Lancet*, 397(10277), 914–927. [https://doi.org/10.1016/s0140-6736\(21\)00359-7](https://doi.org/10.1016/s0140-6736(21)00359-7)

4 Burns, D. D. (2007). When panic attacks: The new, drug-free anxiety therapy that can change your life. Morgan Road Books

5 Shin, L. M., & Liberzon, I. (2009). The neurocircuitry of fear, stress, and anxiety disorders. *Neuropsychopharmacology*, 35(1), 169–191. <https://doi.org/10.1038/npp.2009.83>

6 Sathyanarayana Rao, T. S., Asha, M. R., Ramesh, B. N., & Jagannatha Rao, K. S. (2008). Understanding nutrition, depression and mental illnesses. *Indian Journal of Psychiatry*, 50(2), 77. <https://doi.org/10.4103/0019-5545.42391>

7 Liberzon, I., Duval, E., & Javanbakht, A. (2015). Neural circuits in anxiety and stress disorders: A focused review. *Therapeutics and Clinical Risk Management*, 115. <https://doi.org/10.2147/tcrm.s48528>

8 Rogers, M. P., Warshaw, M. G., Goisman, R. M., Goldenberg, I., Rodriguez-Villa, F., Mallya, G., Freeman, S. A., & Keller, M. B. (1999). Comparing primary and secondary generalized anxiety disorder in a long-term naturalistic study of anxiety disorders. *Depression and Anxiety*, 10(1), 1–7. [https://doi.org/10.1002/\(sici\)1520-6394\(1999\)10:1<1:aid-](https://doi.org/10.1002/(sici)1520-6394(1999)10:1<1:aid-)

- 9 Bystritsky, A., Hovav, S., Sherbourne, C., Stein, M. B., Rose, R. D., Campbell-Sills, L., Golinelli, D., Sullivan, G., Craske, M. G., & Roy-Byrne, P. P. (2012). Use of complementary and alternative medicine in a large sample of anxiety patients. *Psychosomatics*, 53(3), 266–272. <https://doi.org/10.1016/j.psych.2011.11.009>
- 10 Kemp, C. G., Johnson, L. C. M., Sagar, R., Poongothai, S., Tandon, N., Anjana, R. M., Aravind, S., Sridhar, G. R., Patel, S. A., Emmert-Fees, K., Rao, D., Narayan, K. M. V., Mohan, V., Ali, M. K., & Chwastiak, L. A. (2022). Effect of a collaborative care model on anxiety symptoms among patients with depression and diabetes in India: The Independent Randomized Clinical Trial. *General Hospital Psychiatry*, 74, 39–45. <https://doi.org/10.1016/j.genhosppsych.2021.11.003>
- 11 Liao, L.-ying, He, Y.-fan, Li, L., Meng, H., Dong, Y.-mao, Yi, F., & Xiao, P.-gen. (2018). A preliminary review of studies on adaptogens: Comparison of their bioactivity in TCM with that of ginseng-like herbs used worldwide. *Chinese Medicine*, 13(1). <https://doi.org/10.1186/s13020-018-0214-9>
- 12 Singh, N., Bhalla, M., De Jager, P., & Gilca, M. (2011). An overview on ashwagandha: A rasayana (rejuvenator) of ayurveda. *African Journal of Traditional, Complementary and Alternative Medicines*, 8(55). <https://doi.org/10.4314/ajtcam.v8i5.5.9>
- 13 Kamal, H. I., Patel, K., Brdak, A., Heffernan, J., & Ahmad, N. (2022). Ashwagandha as a unique cause of thyrotoxicosis presenting with supraventricular tachycardia. *Cureus*. <https://doi.org/10.7759/cureus.23494>
- 14 Petre, A. (2018, September 1). *Ashwagandha dosage: How much should you take per day?* Healthline. Retrieved June 17, 2022, from <https://www.healthline.com/nutrition/ashwagandha-dosage>
- 15 Skelley, J. W., Deas, C. M., Curren, Z., & Ennis, J. (2020). Use of cannabidiol in anxiety and anxiety-related disorders. *Journal of the American Pharmacists Association*, 60(1), 253–261. <https://doi.org/10.1016/j.japh.2019.11.008>
- 16 Blessing, E. M., Steenkamp, M. M., Manzanares, J., & Marmar, C. R. (2015). Cannabidiol as a potential treatment for anxiety disorders. *Neurotherapeutics*, 12(4), 825–836. <https://doi.org/10.1007/s13311-015-0387-1>
- 17 Shannon, S. (2019). Cannabidiol in anxiety and sleep: A large case series. *The Permanente Journal*, 23. <https://doi.org/10.7812/tpp/18-041>
- 18 Crippa, J. A., Derenusson, G. N., Ferrari, T. B., Wichert-Ana, L., Duran, F. L. S., Martin-Santos, R., Simões, M. V., Bhattacharyya, S., Fusar-Poli, P., Atakan, Z., Filho, A. S., Freitas-Ferrari, M. C., McGuire, P. K., Zuardi, A. W., Busatto, G. F., & Hallak, J. E. (2010). Neural basis of anxiolytic effects of cannabidiol (CBD) in Generalized Social Anxiety Disorder: A preliminary report. *Journal of Psychopharmacology*, 25(1), 121–130. <https://doi.org/10.1177/0269881110379283>
- 19 Johnson, J. R., Burnell-Nugent, M., Lossignol, D., Ganae-Motan, E. D., Potts, R., & Fallon, M. T. (2010). Multicenter, double-blind, randomized, placebo-controlled, parallel-group study of the efficacy, safety, and tolerability of THC: CBD extract and THC extract in patients with intractable cancer-related pain. *Journal of Pain and Symptom Management*, 39(2), 167–179. <https://doi.org/10.1016/j.jpainsymman.2009.06.008>
- 20 Brent A. Bauer, M. D. (2020, December 18). *CBD: Safe and effective?* Mayo Clinic. Retrieved June 17, 2022, from <https://www.mayoclinic.org/healthy-lifestyle/consumer-health/expert-answers/is-cbd-safe-and-effective/faq-20446700#:~:text=CBD%20use%20also%20carries%20some,taking%2C%20such%20as%20blood%20thinners.>
- 21 Linares, I. M., Zuardi, A. W., Pereira, L. C., Queiroz, R. H., Mechoulam, R., Guimarães, F. S., & Crippa, J. A. (2019). Cannabidiol presents an inverted U-shaped dose-response curve in a simulated public speaking test. *Brazilian Journal of Psychiatry*, 41(1), 9–14. <https://doi.org/10.1590/1516-4446-2017-0015>
- 22 Mao, J. J., Xie, S. X., Keefe, J. R., Soeller, I., Li, Q. S., & Amsterdam, J. D. (2016). Long-term chamomile (*Matricaria Chamomilla* L.) treatment for generalized anxiety disorder: A randomized clinical trial. *Phytomedicine*, 23(14), 1735–1742. <https://doi.org/10.1016/j.phymed.2016.10.012>
- 23 Chang, S.-M., & Chen, C.-H. (2015). Effects of an intervention with drinking chamomile tea on sleep quality and depression in sleep disturbed postnatal women: A randomized controlled trial. *Journal of Advanced Nursing*, 72(2), 306–315. <https://doi.org/10.1111/jan.12836>
- 24 Cohen, M. M. (2014). Tulsi - *Ocimum sanctum*: A herb for all reasons. *Journal of Ayurveda and Integrative Medicine*, 5(4), 251. <https://doi.org/10.4103/0975-9476.146554>
- 25 Jamshidi, N., & Cohen, M. M. (2017). The clinical efficacy and safety of tulsi in humans: A systematic review of the literature. *Evidence-Based Complementary and Alternative Medicine*, 2017, 1–13. <https://doi.org/10.1155/2017/9217567>
- 26 RxList. (2021, June 11). *Holy basil: Health benefits, side effects, uses, Dose & precautions*. RxList. Retrieved June 17, 2022, from https://www.rxlist.com/holy_basil/supplements.htm
- 27 Awad, R., Levac, D., Cybulska, P., Merali, Z., Trudeau, V. L., & Arnason, J. T. (2007). Effects of traditionally used anxiolytic botanicals on enzymes of the γ-aminobutyric acid (GABA) system: this article is one of a selection of papers published in this special issue (part 1 of 2) on the safety and efficacy of natural health products. *Canadian Journal of Physiology and Pharmacology*, 85(9), 933–942. <https://doi.org/10.1139/y07-083>
- 28 Franco, L., Sánchez, C., Bravo, R., Rodríguez, A., Barriga, C., & Juárez, J. (2012). The sedative effects of hops (*humulus lupulus*), a component of beer, on the activity/rest rhythm. *Acta Physiologica Hungarica*, 99(2), 133–139. <https://doi.org/10.1556/aphysiol.99.2012.2.6>
- 29 Carballido, E., & Editorial. (2022, January 3). *Hop contraindications*. Botanical online. Retrieved June 17, 2022, from <https://www.botanical-online.com/en/medicinal-plants/hop-contraindications>
- 30 *Kava Kava*. Mount Sinai Health System. (n.d.). Retrieved June 27, 2022, from <https://www.mountsinai.org/health-library/herb/kava-kava>
- 31 Piper methysticum (kava kava). *Altern Med Rev*. 1998 Dec;3(6):458-60. PMID: 9855570.
- 32 Feltenstein, M. W., Lambdin, L. C., Ganzera, M., Ranjith, H., Dharmaratne, W., Nanayakara, N. P., Khan, I. A., & Sufka, K. J. (2003). Anxiolytic properties of Piper methysticum extract samples and fractions in the chick social-separation-stress procedure. *Phytotherapy Research*, 17(3), 210–216. <https://doi.org/10.1002/ptr.1107>
- 33 Singh, Y. N. (n.d.). An introduction to kava piper methysticum. *Kava*, 7–15. <https://doi.org/10.1201/9781420023374-3>
- 34 Norton, S. A., & Ruze, P. (1994). Kava Dermopathy. *Journal of the American Academy of Dermatology*, 31(1), 89–97. [https://doi.org/10.1016/s0190-9622\(94\)70142-3](https://doi.org/10.1016/s0190-9622(94)70142-3)
- 35 *Kava uses, benefits & dosage - drugs.com herbal database*. Drugs.com. (n.d.). Retrieved June 17, 2022, from <https://www.drugs.com/npp/kava.html>
- 36 Yap, W. S., Dolzhenko, A. V., Jalal, Z., Hadi, M. A., & Khan, T. M. (2019). Efficacy and safety of lavender essential oil (Silexan) capsules among patients suffering from anxiety disorders: A network meta-analysis. *Scientific Reports*, 9(1). <https://doi.org/10.1038/s41598-019-54529-9>
- 37 Woelk, H., & Schläpke, S. (2010). A multi-center, double-blind, randomised study of the lavender oil preparation Silexan in comparison to lorazepam for generalized anxiety disorder. *Phytomedicine*, 17(2), 94–99. <https://doi.org/10.1016/j.phymed.2009.10.006>
- 38 Koulivand, P. H., Khaleghi Ghadiri, M., & Gorji, A. (2013). Lavender and the nervous system. *Evidence-Based Complementary and Alternative Medicine*, 2013, 1–10. <https://doi.org/10.1155/2013/681304>
- 39 Woelk, H., & Schläpke, S. (2010). A multi-center, double-blind, randomised study of the lavender oil preparation Silexan in comparison to lorazepam for generalized anxiety disorder. *Phytomedicine*, 17(2), 94–99. <https://doi.org/10.1016/j.phymed.2009.10.006>
- 40 *Valerian root and lemon balm extracts - nursingcenter.com*. (n.d.). Retrieved June 15, 2022, from https://www.nursingcenter.com/wkhlrp/Handlers/articleContent.pdf?key=pdf_00004650-201511000-00009
- 41 *Lemon Balm*. Mount Sinai Health System. (n.d.). Retrieved June 17, 2022, from <https://www.mountsinai.org/health-library/herb/lemon-balm>
- 42 *Lemon Balm uses, benefits & dosage - drugs.com herbal database*. Drugs.com. (n.d.). Retrieved June 17, 2022, from <https://www.drugs.com/npp/lemon-balm.html>
- 43 Nagano, M., Shimizu, K., Kondo, R., Hayashi, C., Sato, D., Kitagawa, K., & Ohnuki, K. (2010). Reduction of depression and anxiety by 4 weeks Hericium Erinaceus Intake. *Biomedical Research*, 31(4), 231–237. <https://doi.org/10.2220/biomedres.31.231>
- 44 Li, T.-J., Lee, T.-Y., Lo, Y., Lee, L.-Y., Li, I.-C., Chen, C.-C., & Chang, F.-C. (2021). Hericium erinaceus mycelium ameliorate anxiety induced by continuous sleep disturbance in vivo. *BMC Complementary Medicine and Therapies*, 21(1). <https://doi.org/10.1186/s12906-021-03463-3>
- 45 WebMD. (n.d.). *Lion's Mane Mushroom: Overview, uses, side effects, precautions, interactions, dosing and reviews*. WebMD. Retrieved June 17, 2022, from <https://www.webmd.com/vitamins/ai/ingredientmono-1536/lions-mane-mushroom>
- 46 *Lion's mane mushroom uses, benefits & dosage - drugs.com herbal database*. Drugs.com. (n.d.). Retrieved June 16, 2022, from <https://www.drugs.com/npp/lion-s-mane-mushroom.html>
- 47 Kim, M., Lim, H.-S., Lee, H.-H., & Kim, T.-H. (2017). Role identification of *passiflora incarnata* linnaeus: A mini review. *Journal of Menopausal Medicine*, 23(3), 156. <https://doi.org/10.6118/jmm.2017.23.3.156>
- 48 Janda, K., Wojtkowska, K., Jakubczyk, K., Antoniewicz, J., & Skonieczna-Żydecka, K. (2020). *Passiflora incarnata* in neuropsychiatric disorders—a systematic review. *Nutrients*, 12(12), 3894. <https://doi.org/10.3390/nu12123894>
- 49 Guerrero, F. A., & Medina, G. M. (2017). Effect of a medicinal plant (*passiflora incarnata* L.) on sleep. *Sleep Science*, 10(3), 96–100. <https://doi.org/10.5935/1984-0063.20170018>
- 50 McMillen, M. (n.d.). *Passionflower: Uses and risks*. WebMD. Retrieved June 17, 2022, from <https://www.webmd.com/vitamins-and-supplements/passionflower-uses-and-risks#:~:text=Passionflower%20may%20interact%20with%20certain,to%20lessen%20anxiety%20and%20insomnia.>
- 51 RxList. (2021, June 11). *Passion flower: Health benefits, side effects, uses, Dose & precautions*. RxList. Retrieved June 17, 2022, from https://www.rxlist.com/passion_flow-er/supplements.htm
- 52 Singh, R., Dhingra, G. S., & Shri, R. (2016). Evaluation of antianxiety potential of four ganoderma (agaricomycetes) species from India in mice. *International Journal of Medicinal Mushrooms*, 18(11), 991–998. <https://doi.org/10.1615/intjmedmushrooms.v18i11.40>
- 53 Hossen, S. M., Islam, M. J., Hossain, M. R., Barua, A., Uddin, M. G., & Emon, N. U. (2021). CNS anti-depressant, anxiolytic and analgesic effects of ganoderma applanatum (mushroom) along with ligand-receptor binding screening provide new insights: Multi-disciplinary approaches. *Biochemistry and Biophysics Reports*, 27, 101062. <https://doi.org/10.1016/j.bbrep.2021.101062>
- 54 WebMD. (n.d.). *Reishi Mushroom: Overview, uses, side effects, precautions, interactions, dosing and reviews*. WebMD. Retrieved June 17, 2022, from <https://www.webmd.com/vitamins/ai/ingredientmono-905/reishi-mushroom#:~:text=Medications%20that%20slow%20blood%20clotting,risk%20of%20bruising%20and%20bleeding.>
- 55 *Reishi Mushroom uses, benefits & dosage - drugs.com herbal database*. Drugs.com. (n.d.). Retrieved June 17, 2022, from <https://www.drugs.com/npp/reishi-mushroom.html>
- 56 Ishaque, S., Shamseer, L., Bukutu, C., & Vohra, S. (2012). *Rhodiola rosea* for physical and mental fatigue: A systematic review. *BMC Complementary and Alternative Medicine*, 12(1). <https://doi.org/10.1186/1472-6882-12-70>
- 57 Mao, J. J., Xie, S. X., Zee, J., Soeller, I., Li, Q. S., Rockwell, K., & Amsterdam, J. D.

- (2015). Rhodiola rosea versus sertraline for major depressive disorder: A randomized placebo-controlled trial. *Phytotherapy Research*, 22(3), 394–399. <https://doi.org/10.1016/j.phymed.2015.01.010>
- 58 Williams, T. D., Langley, H. N., Roberson, C. C., Rogers, R. R., & Ballmann, C. G. (2021). Effects of short-term golden root extract (rhodiola rosea) supplementation on resistance exercise performance. *International Journal of Environmental Research and Public Health*, 18(13), 6953. <https://doi.org/10.3390/ijerph18136953>
- 59 Mattioli, L., Titomanlio, F., & Perfumi, M. (2012). Effects of a rhodiola rosea L. Extract on the acquisition, expression, extinction, and reinstatement of morphine-induced conditioned place preference in mice. *Psychopharmacology*, 221(2), 183–193. <https://doi.org/10.1007/s00213-012-2686-0>
- 60 *Rhodiola rosea*. Mental Health America. (n.d.). Retrieved June 17, 2022, from <https://www.mhanational.org/rhodiola-rosea>
- 61 Edwards, D., Heufelder, A., & Zimmermann, A. (2012). Therapeutic effects and safety of rhodiola rosea extract WS® 1375 in subjects with life-stress symptoms - results of an open-label study. *Phytotherapy Research*, 26(8), 1220–1225. <https://doi.org/10.1002/ptr.3712>
- 62 Scutellaria lateriflora L. (2013). *Natural Compounds*, 59–59. https://doi.org/10.1007/978-1-4614-0535-1_185
- 63 Brock, C., Whitehouse, J., Tewfik, I., & Towell, T. (2013). American skullcap (*scutellaria lateriflora*): A randomised, double-blind placebo-controlled crossover study of its effects on mood in healthy volunteers. *Phytotherapy Research*, 28(5), 692–698. <https://doi.org/10.1002/ptr.5044>
- 64 Brock, C., Whitehouse, J., Tewfik, I., & Towell, T. (2013). American skullcap (*scutellaria lateriflora*): A randomised, double-blind placebo-controlled crossover study of its effects on mood in healthy volunteers. *Phytotherapy Research*, 28(5), 692–698. <https://doi.org/10.1002/ptr.5044>
- 65 *Skullcap*. Mount Sinai Health System. (n.d.). Retrieved June 27, 2022, from <https://www.mountsinai.org/health-library/herb/skullcap>
- 66 *SCULLCAP uses, benefits & dosage - drugs.com herbal database*. Drugs.com. (n.d.). Retrieved June 20, 2022, from <https://www.drugs.com/npp/skullcap.html>
- 67 H., P. B. N. (n.d.). *St. John's wort*. National Center for Biotechnology Information. Retrieved June 17, 2022, from <https://pubmed.ncbi.nlm.nih.gov/32491397/>
- 68 Butterweck, V. (2003). Mechanism of action of St John's wort in depression. *CNS Drugs*, 17(8), 539–562. <https://doi.org/10.2165/00023210-200317080-00001>
- 69 Davidson, J. R., & Connor, K. M. (2001). St. John's wort in generalized anxiety disorder: Three case reports. *Journal of Clinical Psychopharmacology*, 21(6), 635–636. <https://doi.org/10.1097/00004714-200112000-00026>
- 70 Markowitz, J. S. (2003). Effect of St John's wort on drug metabolism by induction of cytochrome P450 3A4 enzyme. *JAMA*, 290(11), 1500. <https://doi.org/10.1001/jama.290.11.1500>
- 71 Dannawi, M. (2002). Possible serotonin syndrome after combination of buspirone and St John's wort. *Journal of Psychopharmacology*, 16(4), 401–401. <https://doi.org/10.1177/026988110201600420>
- 72 Nicolussi, S., Drewe, J., Butterweck, V., & Meyer zu Schwabedissen, H. E. (2020). Clinical relevance of St. John's wort drug interactions revisited. *British Journal of Pharmacology*, 177(6), 1212–1226. <https://doi.org/10.1111/bph.14936>
- 73 Trauner, G., Khom, S., Baburin, I., Benedek, B., Hering, S., & Kopp, B. (2008). Modulation of GABA_A receptors by valerian extracts is related to the content of valerenic acid. *Planta Medica*, 74(1), 19–24. <https://doi.org/10.1055/s-2007-993761>
- 74 Andreatini, R., Sartori, V. A., Seabra, M. L., & Leite, J. R. (2002). Effect of valepotriates (valerian extract) in generalized anxiety disorder: A randomized placebo-controlled pilot study. *Phytotherapy Research*, 16(7), 650–654. <https://doi.org/10.1002/ptr.1027>
- 75 Poyares, D. R., Guilleminault, C., Ohayon, M. M., & Tufik, S. (2002). Can valerian improve the sleep of insomniacs after benzodiazepine withdrawal? *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 26(3), 539–545. [https://doi.org/10.1016/s0278-5846\(01\)00305-0](https://doi.org/10.1016/s0278-5846(01)00305-0)
- 76 Brent A. Bauer, M. D. (2018, February 15). *Valerian for insomnia? maybe*. Mayo Clinic. Retrieved June 17, 2022, from <https://www.mayoclinic.org/diseases-conditions/insomnia/expert-answers/valerian/faq-20057875#:~:text=Drug%20interactions,%20are%20possible,dietary%20supplements%2C%20such%20as%20St.>
- 77 Cafasso, J. (2019, March 8). *How much valerian root can I take safely?* Healthline. Retrieved June 17, 2022, from <https://www.healthline.com/health/food-nutrition/valerian-root>
- 78 Vora, E. (2022). *The Anatomy of Anxiety: Understanding and overcoming the body's fear response*. Orion Spring.
- 79 Hou, Y.-Y., Ojo, O., Wang, L.-L., Wang, Q., Jiang, Q., Shao, X.-Y., & Wang, X.-H. (2018). A randomized controlled trial to compare the effect of peanuts and almonds on the cardio-metabolic and inflammatory parameters in patients with type 2 diabetes mellitus. *Nutrients*, 10(11), 1565. <https://doi.org/10.3390/nu10111565>
- 80 Ferretti, F., & Mariani, M. (2017). Simple vs. complex carbohydrate dietary patterns and the global overweight and obesity pandemic. *International Journal of Environmental Research and Public Health*, 14(10), 1174. <https://doi.org/10.3390/ijerph14101174>
- 81 Clarke, G., Stilling, R. M., Kennedy, P. J., Stanton, C., Cryan, J. F., & Dinan, T. G. (2014). Minireview: Gut microbiota: The neglected endocrine organ. *Molecular Endocrinology*, 28(8), 1221–1238. <https://doi.org/10.1210/me.2014-1108>
- 82 Mayer, E. A. (2011). Gut feelings: The emerging biology of Gut–Brain Communication. *Nature Reviews Neuroscience*, 12(8), 453–466. <https://doi.org/10.1038/nrn3071>
- 83 Carpenter, D. S. (2012, September). *That gut feeling*. Monitor on Psychology. Retrieved June 20, 2022, from <https://www.apa.org/monitor/2012/09/gut-feeling#:~:text=For%20example%2C%20gut%20bacteria%20manufac->
- ture,both%20mood%20and%20GI%20activity.
- 84 *The human microbiota in health and disease - molmed.nl*. (n.d.). Retrieved June 21, 2022, from <https://www.molmed.nl/uploads/abstracts/1639/John%20Hays%20Microbiota%20Cursus%20Presentatie%2022.08.2018.3%20USE.pdf>
- 85 Akkasheh, G., Kashani-Poor, Z., Tajabadi-Ebrahimi, M., Jafari, P., Akbari, H., Taghizadeh, M., Memarzadeh, M. R., Asemi, Z., & Esmaillzadeh, A. (2016). Clinical and metabolic response to probiotic administration in patients with major depressive disorder: A randomized, double-blind, placebo-controlled trial. *Nutrition*, 32(3), 315–320. <https://doi.org/10.1016/j.nut.2015.09.003>
- 86 Willcox, B. J., & Willcox, D. C. (2013). Caloric restriction, caloric restriction mimetics, and healthy aging in Okinawa. *Current Opinion in Clinical Nutrition and Metabolic Care*, 1. <https://doi.org/10.1097/mco.0000000000000019>
- 87 Vreeburg, S. A., Zitman, F. G., van Pelt, J., DeRijk, R. H., Verhagen, J. C., van Dyck, R., Hoogendijk, W. J., Smit, J. H., & Penninx, B. W. (2010). Salivary cortisol levels in persons with and without different anxiety disorders. *Psychosomatic Medicine*, 72(4), 340–347. <https://doi.org/10.1097/psy.0b013e3181d2f0c8>
- 88 Kalant, H. (1962). Book review: Alcohol and caffeine. A study of their psychological effects. *Canadian Psychiatric Association Journal*, 7(5), 273–274. <https://doi.org/10.1177/070674376200700511>
- 89 Wang, H. R., Woo, Y. S., & Bahk, W.-M. (2015). Caffeine-induced psychiatric manifestations. *International Clinical Psychopharmacology*, 30(4), 179–182. <https://doi.org/10.1097/yci.0000000000000076>
- 90 Juliano, L. M., & Griffiths, R. R. (2004). A critical review of caffeine withdrawal: Empirical validation of symptoms and signs, incidence, severity, and associated features. *Psychopharmacology*, 176(1), 1–29. <https://doi.org/10.1007/s00213-004-2000-x>
- 91 Nelson, M., & Ogden, J. (2008). An exploration of food intolerance in the primary care setting: The general practitioner's experience. *Social Science & Medicine*, 67(6), 1038–1045. <https://doi.org/10.1016/j.socscimed.2008.05.025>
- 92 Polloni, L., & Muraro, A. (2020). Anxiety and food allergy: A review of the last two decades. *Clinical & Experimental Allergy*, 50(4), 420–441. <https://doi.org/10.1111/cea.13548>
- 93 Arvola, T., & Holmberg-marttila, D. (1999). Benefits and risks of elimination diets. *Annals of Medicine*, 31(4), 293–298. <https://doi.org/10.3109/07853899908995893>
- 94 Harvie, M., & Howell, A. (2017). Potential benefits and harms of intermittent energy restriction and intermittent fasting amongst obese, overweight and normal weight subjects—a narrative review of human and animal evidence. *Behavioral Sciences*, 7(4), 4. <https://doi.org/10.3390/bs7010004>
- 95 John J. Ratey, M. D. (2019, October 24). *Can exercise help treat anxiety?* Harvard Health. Retrieved June 21, 2022, from <https://www.health.harvard.edu/blog/can-exercise-help-treat-anxiety-2019102418096>
- 96 Herbert, C., Meixner, F., Wiebking, C., & Gilg, V. (2020). Regular physical activity, short-term exercise, mental health, and well-being among university students: The results of an online and a laboratory study. *Frontiers in Psychology*, 11. <https://doi.org/10.3389/fpsyg.2020.00509>
- 97 Anderson, E., & Shivakumar, G. (2013). Effects of exercise and physical activity on anxiety. *Frontiers in Psychiatry*, 4. <https://doi.org/10.3389/fpsyg.2013.00027>
- 98 Lago, T. R., Hsiung, A., Leitner, B. P., Duckworth, C. J., Balderston, N. L., Chen, K. Y., Grillon, C., & Ernst, M. (2018). Exercise modulates the interaction between cognition and anxiety in humans. *Cognition and Emotion*, 33(4), 863–870. <https://doi.org/10.1080/02699931.2018.1500445>
- 99 Carlson, C. R., Collins, F. L., Nitz, A. J., Sturgis, E. T., & Rogers, J. L. (1990). Muscle stretching as an alternative relaxation training procedure. *Journal of Behavior Therapy and Experimental Psychiatry*, 21(1), 29–38. [https://doi.org/10.1016/0005-7916\(90\)90046-n](https://doi.org/10.1016/0005-7916(90)90046-n)
- 100 Brellenthin, A. G., Crombie, K. M., Hillard, C. J., & Koltyn, K. F. (2017). Endo-cannabinoid and mood responses to exercise in adults with varying activity levels. *Medicine & Science in Sports & Exercise*, 49(8), 1688–1696. <https://doi.org/10.1249/mss.0000000000001276>
- 101 Schoenfeld, T. J., Rada, P., Pieruzzini, P. R., Hsueh, B., & Gould, E. (2013). Physical exercise prevents stress-induced activation of granule neurons and enhances local inhibitory mechanisms in the dentate gyrus. *Journal of Neuroscience*, 33(18), 7770–7777. <https://doi.org/10.1523/jneurosci.5352-12.2013>
- 102 Audiffren, M., & André, N. (2019). The exercise–cognition relationship: A virtuous circle. *Journal of Sport and Health Science*, 8(4), 339–347. <https://doi.org/10.1016/j.jshs.2019.03.001>
- 103 Mayo Foundation for Medical Education and Research. (2017, September 27). *Depression and anxiety: Exercise eases symptoms*. Mayo Clinic. Retrieved June 21, 2022, from <https://www.mayoclinic.org/diseases-conditions/depression/in-depth/depression-and-exercise/art-20046495#:~:text=How%20much%20is%20enough%3F,time%20%E2%80%94%20may%20make%20a%20difference.>
- 104 Hopper, S. I., Murray, S. L., Ferrara, L. R., & Singleton, J. K. (2019). Effectiveness of diaphragmatic breathing for reducing physiological and psychological stress in adults. *JBI Database of Systematic Reviews and Implementation Reports*, 17(9), 1855–1876. <https://doi.org/10.11124/jbisrir-2017-003848>
- 105 Ma, X., Yue, Z.-Q., Gong, Z.-Q., Zhang, H., Duan, N.-Y., Shi, Y.-T., Wei, G.-X., & Li, Y.-F. (2017). The effect of diaphragmatic breathing on attention, negative affect and stress in healthy adults. *Frontiers in Psychology*, 8. <https://doi.org/10.3389/fpsyg.2017.00874>
- 106 Lemole, G. M., McLanahan, S. A., McKee, D. L., Spiker, T., Chen, N., & Hyman, M. (2021). *Lymph & longevity: The untapped secret to health*. Headline.
- 107 Segerstrom, S. C., & Miller, G. E. (2004). Psychological stress and the human immune system: A meta-analytic study of 30 years of inquiry. *Psychological Bulletin*,

- 130(4), 601–630. <https://doi.org/10.1037/0033-2909.130.4.601>
- 108 Hofmann, S. G., & Gómez, A. F. (2017). Mindfulness-based interventions for anxiety and depression. *Psychiatric Clinics of North America*, 40(4), 739–749. <https://doi.org/10.1016/j.psc.2017.08.008>
- 109 Serpa, J. G., Taylor, S. L., & Tillisch, K. (2014). Mindfulness-based stress reduction (MBSR) reduces anxiety, depression, and suicidal ideation in veterans. *Medical Care*, 52(Supplement 5). <https://doi.org/10.1097/mlr.0000000000000202>
- 110 Hansen, M. M., Jones, R., & Tocchini, K. (2017). Shinrin-yoku (forest bathing) and nature therapy: A state-of-the-art review. *International Journal of Environmental Research and Public Health*, 14(8), 851. <https://doi.org/10.3390/ijerph14080851>
- 111 Burns, D. D. (2007). When panic attacks: The new, drug-free anxiety therapy that can change your life. Morgan Road Books.
- 112 Borza, L. (2017). Cognitive-behavioral therapy for generalized anxiety. *Dialogues in Clinical Neuroscience*, 19(2), 203–208. <https://doi.org/10.31887/dcn.2017.19.2/lborza>
- 113 Sussex Publishers. (n.d.). *Gratitude*. Psychology Today. Retrieved June 21, 2022, from <https://www.psychologytoday.com/ca/basics/gratitude>
- 114 Petrocchi, N., & Couyoumdjian, A. (2015). The impact of gratitude on depression and anxiety: The mediating role of criticizing, attacking, and reassuring the self. *Self and Identity*, 15(2), 191–205. <https://doi.org/10.1080/15298868.2015.1095794>
- 115 Cunha, L. F., Pellanda, L. C., & Reppold, C. T. (2019). Positive psychology and gratitude interventions: A randomized clinical trial. *Frontiers in Psychology*, 10. <https://doi.org/10.3389/fpsyg.2019.00584>
- 116 Heckendorff, H., Lehr, D., Ebert, D. D., & Freund, H. (2019). Efficacy of an internet and app-based gratitude intervention in reducing repetitive negative thinking and mechanisms of change in the intervention's effect on anxiety and depression: Results from a randomized controlled trial. *Behaviour Research and Therapy*, 119, 103415. <https://doi.org/10.1016/j.brat.2019.103415>
- 117 O'Connell, B. H., O'Shea, D., & Gallagher, S. (2017). Feeling thanks and saying thanks: A randomized controlled trial examining if and how socially oriented gratitude journals work. *Journal of Clinical Psychology*, 73(10), 1280–1300. <https://doi.org/10.1002/jclp.22469>
- 118 Goto, Y., Hayasaka, S., Kurihara, S., & Nakamura, Y. (2018). Physical and mental effects of bathing: A randomized intervention study. *Evidence-Based Complementary and Alternative Medicine*, 2018, 1–5. <https://doi.org/10.1155/2018/9521086>
- 119 Ismail, A. A. A., & Ismail, N. A. (2016). Magnesium: A mineral essential for health yet generally underestimated or even ignored. *Journal of Nutrition & Food Sciences*, 6(4). <https://doi.org/10.4172/2155-9600.1000523>
- 120 Boyle, N., Lawton, C., & Dye, L. (2017). The effects of magnesium supplementation on subjective anxiety and stress—a systematic review. *Nutrients*, 9(5), 429. <https://doi.org/10.3390/nu9050429>
- 121 Gröber, U., Werner, T., Vormann, J., & Kisters, K. (2017). Myth or reality—transdermal magnesium? *Nutrients*, 9(8), 813. <https://doi.org/10.3390/nu9080813>
- 122 Rath, L. (n.d.). *Epsom salts bath: How to take one, what it does*. WebMD. Retrieved June 21, 2022, from <https://www.webmd.com/a-to-z-guides/epsom-salt-bath#:~:text=For%20a%20standard%20size%20tub,for%20at%20least%2012%20minutes>
- 123 Field, T., Field, T., Cullen, C., Largent, S., Diego, M., Schanberg, S., & Kuhn, C. (2008). Lavender bath oil reduces stress and crying and enhances sleep in very young infants. *Early Human Development*, 84(6), 399–401. <https://doi.org/10.1016/j.earlhumdev.2007.10.008>
- 124 Kennedy, D. (2016). B vitamins and the brain: Mechanisms, dose and efficacy—a review. *Nutrients*, 8(2), 68. <https://doi.org/10.3390/nu8020068>
- 125 Mahdavi, B., Hosseinzadeh, M., Salehi-Abargouei, A., Mirzaei, M., & Vafa, M. (2021). Dietary intake of B vitamins and their association with depression, anxiety, and stress symptoms: A cross-sectional, population-based survey. *Journal of Affective Disorders*, 288, 92–98. <https://doi.org/10.1016/j.jad.2021.03.055>
- 126 Kubala, J. (2022, April 4). *B-complex vitamins*. Healthline. Retrieved June 23, 2022, from <https://www.healthline.com/nutrition/vitamin-b-complex>
- 127 Moritz, B., Schwarzbald, M. L., Guarnieri, R., Diaz, A. P., S Rodrigues, A. L., & Dafre, A. L. (2017). Effects of ascorbic acid on anxiety state and affect in a non-clinical sample. *Acta neurobiologiae experimentalis*, 77(4), 362–372.
- 128 Kennedy, D. O., Veasey, R., Watson, A., Dodd, F., Jones, E., Maggini, S., & Haskell, C. F. (2010). Effects of high-dose B vitamin complex with vitamin C and minerals on subjective mood and performance in healthy males. *Psychopharmacology*, 211(1), 55–68. <https://doi.org/10.1007/s00213-010-1870-3>
- 129 Elliott, B. (2019, December 14). *Does too much vitamin C cause side effects?* Healthline. Retrieved June 28, 2022, from <https://www.healthline.com/nutrition/side-effects-of-too-much-vitamin-c#vitamin-c>
- 130 Oliveira, I. J., de Souza, V. V., Motta, V., & Da-Silva, S. L. (2014). Effects of oral vitamin C supplementation on anxiety in students: A double-blind, randomized, placebo-controlled trial. *Pakistan Journal of Biological Sciences*, 18(1), 11–18. <https://doi.org/10.3923/pjbs.2015.11.18>
- 131 Eid, A., Khoja, S., AlGhamdi, S., Alsufiani, H., Alzeben, F., Alhejaili, N., Tayeb, H. O., & Tarazi, F. I. (2019). Vitamin D supplementation ameliorates severity of generalized anxiety disorder (GAD). *Metabolic Brain Disease*, 34(6), 1781–1786. <https://doi.org/10.1007/s11011-019-00486-1>
- 132 Bjarnadottir, A. (2021, February 18). *How much vitamin D should you take for optimal health?* Healthline. Retrieved June 22, 2022, from <https://www.healthline.com/nutrition/how-much-vitamin-d-to-take>
- 133 Hegarty, B., & Parker, G. (2013). Fish oil as a management component for mood disorders – an evolving signal. *Current Opinion in Psychiatry*, 26(1), 33–40. <https://doi.org/10.1097/ycp.0b013e32835ab4a7>
- 134 Ohishi, K., Ueno, R., Nishino, S., Sakai, T., & Hayaishi, O. (1988). Increased level of salivary prostaglandins in patients with major depression. *Biological Psychiatry*, 23(4), 326–334. [https://doi.org/10.1016/0006-3223\(88\)90283-1](https://doi.org/10.1016/0006-3223(88)90283-1)
- 135 Gądek-Michalska, A., Tadeusz, J., Rachwalska, P., & Bugajski, J. (2013). Cytokines, prostaglandins and nitric oxide in the regulation of stress-response systems. *Pharmacological Reports*, 65(6), 1655–1662. [https://doi.org/10.1016/s1734-1140\(13\)71527-5](https://doi.org/10.1016/s1734-1140(13)71527-5)
- 136 Swanson, D., Block, R., & Mousa, S. A. (2012). Omega-3 fatty acids EPA and DHA: Health benefits throughout life. *Advances in Nutrition*, 3(1), 1–7. <https://doi.org/10.3945/an.111.000893>
- 137 Gauthier, L., & Nuss, P. (2015). Anxiety disorders and GABA neurotransmission: A disturbance of modulation. *Neuropsychiatric Disease and Treatment*, 165. <https://doi.org/10.2147/ndt.s58841>
- 138 Hepsomali, P., Groeger, J. A., Nishihira, J., & Scholey, A. (2020). Effects of oral gamma-aminobutyric acid (GABA) administration on stress and sleep in humans: A systematic review. *Frontiers in Neuroscience*, 14. <https://doi.org/10.3389/fnins.2020.00923>
- 139 Byun, J.-I., Shin, Y. Y., Chung, S.-E., & Shin, W. C. (2018). Safety and efficacy of gamma-aminobutyric acid from fermented rice germ in patients with insomnia symptoms: A randomized, double-blind trial. *Journal of Clinical Neurology*, 14(3), 291. <https://doi.org/10.3988/jcn.2018.14.3.291>
- 140 Schmidt, R. W., & Thompson, M. L. (2016). Glycinergic signaling in the human nervous system: An overview of therapeutic drug targets and clinical effects. *Mental Health Clinician*, 6(6), 266–276. <https://doi.org/10.9740/mhc.2016.11.266>
- 141 WebMD. (n.d.). *Glycine: Overview, uses, side effects, precautions, interactions, dosing and reviews*. WebMD. Retrieved June 23, 2022, from <https://www.webmd.com/vitamins/ai/ingredientmono-1072/glycine>
- 142 Cherry, K. (2021, December 11). *Can inositol relieve panic and anxiety?* Verywell Mind. Retrieved June 23, 2022, from <https://www.verywellmind.com/inositol-what-should-i-know-about-it-89466>
- 143 Palatnik, A., Frolov, K., Fux, M., & Benjamin, J. (2001). Double-blind, controlled, crossover trial of inositol versus fluvoxamine for the treatment of panic disorder. *Journal of Clinical Psychopharmacology*, 21(3), 335–339. <https://doi.org/10.1097/00004714-200106000-00014>
- 144 Levine, J. (1997). Controlled trials of inositol in psychiatry. *European Neuropsychopharmacology*, 7(2), 147–155. [https://doi.org/10.1016/s0924-977x\(97\)00409-4](https://doi.org/10.1016/s0924-977x(97)00409-4)
- 145 Lopes Sakamoto, F., Metzker Pereira Ribeiro, R., Amador Bueno, A., & Oliveira Santos, H. (2019). Psychotropic effects of L-Theanine and its clinical properties: From the management of anxiety and stress to a potential use in schizophrenia. *Pharmacological Research*, 147, 104395. <https://doi.org/10.1016/j.phrs.2019.104395>
- 146 Everett, J. M., Gunathilake, D., Dufficy, L., Roach, P., Thomas, J., Upton, D., & Naumovski, N. (2016). Theanine consumption, stress and anxiety in human clinical trials: A systematic review. *Journal of Nutrition & Intermediary Metabolism*, 4, 41–42. <https://doi.org/10.1016/j.jnim.2015.12.308>
- 147 Williams, J. K., Everett, J. M., D'Cunha, N. M., Sergi, D., Georgousopoulou, E. N., Keegan, R. J., McKune, A. J., Mellor, D. D., Anstine, N., & Naumovski, N. (2019). The effects of green tea amino acid L-theanine consumption on the ability to manage stress and anxiety levels: A systematic review. *Plant Foods for Human Nutrition*, 75(1), 12–23. <https://doi.org/10.1007/s11130-019-00771-5>
- 148 U.S. Department of Health and Human Services. (n.d.). *Office of dietary supplements - magnesium*. NIH Office of Dietary Supplements. Retrieved June 22, 2022, from <https://ods.od.nih.gov/factsheets/Magnesium-HealthProfessional/>
- 149 Kirkland, A., Sarlo, G., & Holton, K. (2018). The role of magnesium in neurological disorders. *Nutrients*, 10(6), 730. <https://doi.org/10.3390/nu10060730>
- 150 Boyle, N., Lawton, C., & Dye, L. (2017). The effects of magnesium supplementation on subjective anxiety and stress—a systematic review. *Nutrients*, 9(5), 429. <https://doi.org/10.3390/nu9050429>
- 151 Pickering, G., Mazur, A., Trousselard, M., Bienkowski, P., Yaltseva, N., Amessou, M., Noah, L., & Pouteau, E. (2020). Magnesium status and stress: The Vicious Circle Concept revisited. *Nutrients*, 12(12), 3672. <https://doi.org/10.3390/nu12123672>
- 152 Möhler, H., Polc, R., Cumin, R., Pieri, L., & Kettler, R. (1979). Nicotinamide is a brain constituent with benzodiazepine-like actions. *Nature*, 278(5704), 563–565. <https://doi.org/10.1038/278563a0>
- 153 *Niacin - statpearls - NCBI bookshelf*. (n.d.). Retrieved June 23, 2022, from <https://www.ncbi.nlm.nih.gov/books/NBK541036/>
- 154 Prousky, J. E. (2005). Supplemental Niacinamide Mitigates Anxiety Symptoms. *Journal of Orthomolecular Medicine*, 20(3), 167–178.
- 155 Vitellio, P., Chira, A., De Angelis, M., Dumitrascu, D. L., & Portincasa, P. (2020). Probiotics in psychosocial stress and anxiety. A systematic review. *Journal of Gastrointestinal and Liver Diseases*, 29(1), 77–83. <https://doi.org/10.15403/jgl-352>
- 156 *Niacin - statpearls - NCBI bookshelf*. (n.d.). Retrieved June 23, 2022, from <https://www.ncbi.nlm.nih.gov/books/NBK541036/>
- 157 Jakaria, M., Azam, S., Haque, M. E., Jo, S.-H., Uddin, M. S., Kim, I.-S., & Choi, D.-K. (2019). Taurine and its analogs in neurological disorders: Focus on therapeutic potential and molecular mechanisms. *Redox Biology*, 24, 101223. <https://doi.org/10.1016/j.redox.2019.101223>
- 158 Zhang, C. G., & Kim, S.-J. (2007). Taurine induces anti-anxiety by activating strychnine-sensitive glycine receptor in vivo. *Annals of Nutrition and Metabolism*, 51(4), 379–386. <https://doi.org/10.1159/000107687>
- 159 Walle, G. V. D. (2022, February 1). *What is taurine? benefits, side effects, and more*. Healthline. Retrieved June 23, 2022, from <https://www.healthline.com/nutrition/what-is-aurine#side-effects-precautions>
- 160 WebMD. (n.d.). *5-HTP: Overview, uses, side effects, precautions, interactions*,

dosing and reviews. WebMD. Retrieved June 23, 2022, from <https://www.webmd.com/vitamins/ai/ingredientmono-794/5-htp>

161 Maffei, M. E. (2020). 5-hydroxytryptophan (5-HTP): Natural occurrence, analysis, biosynthesis, biotechnology, physiology and toxicology. *International Journal of Molecular Sciences*, 22(1), 181. <https://doi.org/10.3390/ijms22010181>

162 5-hydroxytryptophan (5-HTP). Mount Sinai Health System. (n.d.). Retrieved June 27, 2022, from <https://www.mountsinai.org/health-library/supplement/5-hydroxytryptophan-5-htp#:~:text=Taking%205%2DHTP%20with%20dextromethorphan,a%20condition%20called%20serotonin%20syndrome>

163 Gijssman, H. J., van Gerven, J. M., de Kam, M. L., Schoemaker, R. C., Pieters, M. S., Weemaes, M., de Rijk, R., van der Post, J., & Cohen, A. F. (2002). Placebo-controlled comparison of three dose-regimens of 5-hydroxytryptophan challenge test in healthy volunteers. *Journal of Clinical Psychopharmacology*, 22(2), 183–189. <https://doi.org/10.1097/00004714-200204000-00012>

164 Nestadt, P. (2021, October 13). *Why aren't my antidepressants working? Why Aren't My Antidepressants Working?* | Johns Hopkins Medicine. Retrieved June 24, 2022, from <https://www.hopkinsmedicine.org/health/wellness-and-prevention/why-arent-my-antidepressants-working#:~:text=If%20you%20feel%20like%20your,antidepressants%20E%80%94%20it's%20called%20breakthrough%20depression>

165 Vora, E. (2022). *The Anatomy of Anxiety: Understanding and overcoming the body's fear response*. Orion Spring.

166 Salim, S., Chugh, G., & Asghar, M. (2012). Inflammation in anxiety. *Advances in Protein Chemistry and Structural Biology Volume 88*, 1–25. <https://doi.org/10.1016/b978-0-12-398314-5.00001-5>

Brain and Nerve Protocol

1 Ackerman S. Discovering the Brain. Washington (DC): National Academies Press (US); 1992. Foreword. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK234155/>

2 Wang, Y., Pan, Y., & Li, H. (2020). What is Brain Health and why is it important? *BMJ*, m3683. <https://doi.org/10.1136/bmj.m3683>

3 World Health Organization. (n.d.). *Neurological disorders affect millions globally: Who report*. World Health Organization. Retrieved August 23, 2022, from <https://www.who.int/news/item/27-02-2007-neurological-disorders-affect-millions-globally-who-report>

4 Froestl, W., Muhs, A., & Pfeifer, A. (2012). Cognitive enhancers (nootropics). part 1: Drugs interacting with receptors. *Journal of Alzheimer's Disease*, 32(4), 793–887. <https://doi.org/10.3233/jad-2012-121186>

5 Onaolapo, A. Y., Obelawo, A. Y., & Onaolapo, O. J. (2019). Brain Ageing, cognition and Diet: A review of the emerging roles of food-based Nootropics in mitigating age-related memory decline. *Current Aging Science*, 12(1), 2–14. <https://doi.org/10.2174/1874609812666190311160754>

6 Chugh, R. M., Mittal, P., MP, N., Arora, T., Bhattacharya, T., Chopra, H., Cavalu, S., & Gautam, R. K. (2022). Fungal mushrooms: A natural compound with therapeutic applications. *Frontiers in Pharmacology*, 13. <https://doi.org/10.3389/fphar.2022.925387>

7 Lai, P.-L., Naidu, M., Sabaratnam, V., Wong, K.-H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, *hericium erinaceus* (higher basidiomycetes) from Malaysia. *International Journal of Medicinal Mushrooms*, 15(6), 539–554. <https://doi.org/10.1615/intjmed-mushrv15.i6.30>

8 Aguiar, S., & Borowski, T. (2013). Neuropharmacological review of the nootropic herb *bacopa monnieri*. *Rejuvenation Research*, 16(4), 313–326. <https://doi.org/10.1089/rej.2013.1431>

9 Mori, K., Inatomi, S., Ouchi, K., Azumi, Y., & Tsuchida, T. (2009). Improving effects of the mushroom yamabushitake (*hericium erinaceus*) on mild cognitive impairment: A double-blind placebo-controlled clinical trial. *Phytotherapy Research*, 23(3), 367–372. <https://doi.org/10.1002/ptr.2634>

10 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, *Hericium erinaceus* (Higher Basidiomycetes) from Malaysia. *International journal of medicinal mushrooms*, 15(6), 539–554. <https://doi.org/10.1615/intjmed-mushrv15.i6.30>

11 Mori, K., Kikuchi, H., Obara, Y., Iwashita, M., Azumi, Y., Kinugasa, S., Inatomi, S., Oshima, Y., & Nakahata, N. (2010). Inhibitory effect of hericenone B from *Hericium erinaceus* on collagen-induced platelet aggregation. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 17(14), 1082–1085. <https://doi.org/10.1016/j.phymed.2010.05.004>

12 Song, J., Wang, Y., Teng, M., Cai, G., Xu, H., Guo, H., Liu, Y., Wang, D., & Teng, L. (2015). Studies on the antifatigue activities of *cordyceps militaris* fruit body extract in mouse model. *Evidence-Based Complementary and Alternative Medicine*, 2015, 1–15. <https://doi.org/10.1155/2015/174616>

13 Yuan, G., An, L., Sun, Y., Xu, G., & Du, P. (2018). Improvement of learning and memory induced by *cordyceps* polypeptide treatment and the underlying mechanism. *Evidence-Based Complementary and Alternative Medicine*, 2018, 1–10. <https://doi.org/10.1155/2018/9419264>

14 Chen, S., Li, Z., Krochmal, R., Abrizado, M., Kim, W., & Cooper, C. B. (2010). Effect of CS-4® (*cordyceps sinensis*) on exercise performance in healthy older subjects: A double-blind, placebo-controlled trial. *The Journal of Alternative and Complementary Medicine*, 16(5), 585–590. <https://doi.org/10.1089/acm.2009.0226>

15 Doukani, K., Selles, A. S. M., & Bouhenni, H. (2021). *Melissa officinalis* (lemon balm). In *Naturally Occurring Chemicals Against Alzheimer's Disease* (pp. 225–241). Academic Press

16 Kamdem, J. P., Adeniran, A., Boligon, A. A., Klimaczewski, C. V., Elekofehinti, O. O., Hassan, W., ... & Athayde, M. L. (2013). Antioxidant activity, genotoxicity

and cytotoxicity evaluation of lemon balm (*Melissa officinalis* L.) ethanolic extract: Its potential role in neuroprotection. *Industrial Crops and Products*, 51, 26–34.

17 Eivani, M., & Khosronezhad, N. (2020). *Melissa officinalis*: a memory enhancer remedy. *Physiology and Pharmacology*, 24(3), 159–164

18 Eivani, M., & Khosronezhad, N. (2020). *Melissa officinalis*: a memory enhancer remedy. *Physiology and Pharmacology*, 24(3), 159–164

19 Bhardwaj, N., Katyal, P., & Sharma, A. K. (2014). Suppression of inflammatory and allergic responses by pharmacologically potent fungus *Ganoderma lucidum*. *Recent patents on inflammation & allergy drug discovery*, 8(2), 104–117. <https://doi.org/10.2174/1872213x08666140619110657>

20 Huang, S., Mao, J., Ding, K., Zhou, Y., Zeng, X., Yang, W., Wang, P., Zhao, C., Yao, J., Xia, P., & Pei, G. (2017). Polysaccharides from *Ganoderma lucidum* Promote Cognitive Function and Neural Progenitor Proliferation in Mouse Model of Alzheimer's Disease. *Stem cell reports*, 8(1), 84–94. <https://doi.org/10.1016/j.stemcr.2016.12.007>

21 Liao, L.-ying, He, Y.-fan, Li, L., Meng, H., Dong, Y.-mao, Yi, F., & Xiao, P.-gen. (2018). A preliminary review of studies on adaptogens: Comparison of their bioactivity in TCM with that of ginseng-like herbs used worldwide. *Chinese Medicine*, 13(1). <https://doi.org/10.1186/s13020-018-0214-9>

22 Kuboyama, T., Tohda, C., & Komatsu, K. (2005). Neuritic regeneration and synaptic reconstruction induced by withanolide A. *British Journal of Pharmacology*, 144(7), 961–971. <https://doi.org/10.1038/sj.bjp.0706122>

23 TOHDA, C. (2008). Overcoming several neurodegenerative diseases by traditional medicines. *YAKUGAKU ZASSHI*, 128(8), 1159–1167. <https://doi.org/10.1248/yakushi.128.1159>

24 Petre, A. (2018, September 1). *Ashwagandha dosage: How much should you take per day?* Healthline. Retrieved June 17, 2022, from <https://www.healthline.com/nutrition/ashwagandha-dosage>

25 Kamal, H. I., Patel, K., Brdak, A., Heffernan, J., & Ahmad, N. (2022). Ashwagandha as a unique cause of thyrotoxicosis presenting with supraventricular tachycardia. *Cureus*. <https://doi.org/10.7759/cureus.23494>

26 Aguiar, S., & Borowski, T. (2013). Neuropharmacological review of the nootropic herb *bacopa monnieri*. *Rejuvenation Research*, 16(4), 313–326. <https://doi.org/10.1089/rej.2013.1431>

27 Nemetchek, M. D., Stierle, A. A., Stierle, D. B., & Lurie, D. I. (2017). The Ayurvedic Plant *Bacopa Monnieri* inhibits inflammatory pathways in the brain. *Journal of Ethnopharmacology*, 197, 92–100. <https://doi.org/10.1016/j.jep.2016.07.073>

28 Kongkaew, C., Dilokthornsakul, P., Thanarangsarit, P., Limpeanchob, N., & Norman Scholfield, C. (2014). Meta-analysis of randomized controlled trials on cognitive effects of *Bacopa Monnieri* extract. *Journal of Ethnopharmacology*, 151(1), 528–535. <https://doi.org/10.1016/j.jep.2013.11.008>

29 Brimson, J. M., Brimson, S., Prasanth, M. I., Thitilertdech, P., Malar, D. S., & Tencom-nao, T. (2021). The effectiveness of *bacopa monnieri* (Linn.) wettst. as a nootropic, neuroprotective, or antidepressant supplement: Analysis of the available clinical data. *Scientific Reports*, 11(1). <https://doi.org/10.1038/s41598-020-80045-2>

30 Roger Cohn • May 1, Roger Cohn, Roger Cohn, • Hurdle, J., Hurdle, J., Marinelli, J., & Marinelli, J. (n.d.). *The life story of the oldest tree on Earth*. Yale E360. Retrieved August 4, 2022, from https://e360.yale.edu/features/peter_crane_history_of_ginkgo_earths_oldest_tree

31 Nowak, A., Kojder, K., Zielonka-Brzezicka, J., Wróbel, J., Bosiacki, M., Fabiańska, M., Wróbel, M., Sotek-Pastuszka, J., & Klimowicz, A. (2021). The use of ginkgo biloba L. as a neuroprotective agent in the alzheimer's disease. *Frontiers in Pharmacology*, 12. <https://doi.org/10.3389/fphar.2021.775034>

32 Nathan, P. J., Ricketts, E., Wesnes, K., Mrazek, L., Greville, W., & Stough, C. (2002). The acute nootropic effects of Ginkgo biloba in healthy older human subjects: A preliminary investigation. *Human Psychopharmacology: Clinical and Experimental*, 17(1), 45–49. <https://doi.org/10.1002/hup.353>

33 Diamond, B. J., & Bailey, M. R. (2013). Ginkgo biloba. *Psychiatric Clinics of North America*, 36(1), 73–83. <https://doi.org/10.1016/j.psc.2012.12.006>

34 Brinkhaus, B., Lindner, M., Schuppan, D., & Hahn, E. G. (2000). Chemical, pharmacological and clinical profile of the East Asian Medical Plant *Centella Asiatica*. *Phytomedicine*, 7(5), 427–448. [https://doi.org/10.1016/s0944-7113\(00\)80065-3](https://doi.org/10.1016/s0944-7113(00)80065-3)

35 Puttarak, P., Dilokthornsakul, P., Saokaew, S., Dhippayom, T., Kongkaew, C., Srumsiri, R., Chuthaputti, A., & Chaiyakunapruk, N. (2017). Effects of *Centella Asiatica* (L.) urb. on cognitive function and mood related outcomes: A systematic review and meta-analysis. *Scientific Reports*, 7(1). <https://doi.org/10.1038/s41598-017-09823-9>

36 Farhana, K. M., Malueka, R. G., Wibowo, S., & Gofir, A. (2016). Effectiveness of gotu kola extract 750 mg and 1000 mg compared with folic acid 3 mg in improving vascular cognitive impairment after stroke. *Evidence-Based Complementary and Alternative Medicine*, 2016, 1–6. <https://doi.org/10.1155/2016/2795915>

37 *Gotu Kola uses, benefits & dosage - drugs.com herbal database*. Drugs.com. (n.d.). Retrieved August 19, 2022, from <https://www.drugs.com/npp/gotu-kola.html>

38 Park, K.-C., Jin, H., Zheng, R., Kim, S., Lee, S.-E., Kim, B.-H., & Yim, S.-V. (2019). Cognition enhancing effect of *Panax ginseng* in Korean volunteers with mild cognitive impairment: A randomized, double-blind, placebo-controlled clinical trial. *Translational and Clinical Pharmacology*, 27(3), 92. <https://doi.org/10.12793/tcp.2019.27.3.92>

39 Ishaque, S., Shamseer, L., Bukutu, C., & Vohra, S. (2012). *Rhodiola rosea* for physical and mental fatigue: A systematic review. *BMC Complementary and Alternative Medicine*, 12(1). <https://doi.org/10.1186/1472-6882-12-70>

40 Koop, T., Dienel, A., Heldmann, M., & Münte, T. F. (2020). Effects of the *rhodiola rosea* extract on mental resource allocation and attention: An event-related potential dual task study. *Phytotherapy Research*, 34(12), 3287–3297. <https://doi.org/10.1002/ptr.6778>

- 41 Ismail, Z. M., Morcos, M. A., Mohammad, M. D., & Aboulkhair, A. G. (2014). Enhancement of neural stem cells after induction of depression in Male albino rats (a histological & immunohistochemical study). *International Journal of Stem Cells*, 7(2), 70–78. <https://doi.org/10.15283/ijsc.2014.7.2.70>
- 42 Edwards, D., Heufelder, A., & Zimmermann, A. (2012). Therapeutic effects and safety of rhodiola rosea extract WS® 1375 in subjects with life-stress symptoms - results of an open-label study. *Phytotherapy Research*, 26(8), 1220–1225. <https://doi.org/10.1002/ptr.3712>
- 43 Morris, M. C., Wang, Y., Barnes, L. L., Bennett, D. A., Dawson-Hughes, B., & Booth, S. L. (2017). Nutrients and bioactives in green leafy vegetables and cognitive decline. *Neurology*, 90(3). <https://doi.org/10.1212/wnl.0000000000004815>
- 44 Home: Dietary guidelines for Americans. Home | Dietary Guidelines for Americans. (n.d.). Retrieved August 22, 2022, from <https://www.dietaryguidelines.gov/>
- 45 Foods linked to better brainpower. Harvard Health. (2021, March 6). Retrieved August 22, 2022, from <https://www.health.harvard.edu/healthbeat/foods-linked-to-better-brainpower>
- 46 Kokubun, K., Nemoto, K., & Yamakawa, Y. (2020). Fish intake may affect brain structure and improve cognitive ability in healthy people. *Frontiers in Aging Neuroscience*, 12. <https://doi.org/10.3389/fnagi.2020.00076>
- 47 Foods linked to better brainpower. Harvard Health. (2021, March 6). Retrieved August 22, 2022, from <https://www.health.harvard.edu/healthbeat/foods-linked-to-better-brainpower>
- 48 Spencer, J. P. (2010). The impact of fruit flavonoids on memory and cognition. *British Journal of Nutrition*, 104(S3). <https://doi.org/10.1017/s0007114510003934>
- 49 Devore, E. E., Kang, J. H., Breteler, M. M., & Grodstein, F. (2012). Dietary intakes of berries and flavonoids in relation to cognitive decline. *Annals of Neurology*, 72(1), 135–143. <https://doi.org/10.1002/ana.23594>
- 50 Home: Dietary guidelines for Americans. Home | Dietary Guidelines for Americans. (n.d.). Retrieved August 22, 2022, from <https://www.dietaryguidelines.gov/>
- 51 Arab, L., & Ang, A. (2014). A cross sectional study of the association between walnut consumption and cognitive function among adult us populations represented in nhanes. *The Journal of Nutrition, Health & Aging*, 19(3), 284–290. <https://doi.org/10.1007/s12603-014-0569-2>
- 52 Mayer, E. A., Nance, K., & Chen, S. (2022). The gut-brain axis. *Annual Review of Medicine*, 73(1), 439–453. <https://doi.org/10.1146/annurev-med-042320-014032>
- 53 Gareau, M. G. (2016). Cognitive function and the microbiome. *International Review of Neurobiology*, 227–246. <https://doi.org/10.1016/bs.irn.2016.08.001>
- 54 Vecchio, L. M., Meng, Y., Khima, K., Lipsman, N., Hamani, C., & Aubert, I. (2018). The neuroprotective effects of exercise: Maintaining a healthy brain throughout aging. *Brain Plasticity*, 4(1), 17–52. <https://doi.org/10.3233/bpl-180069>
- 55 De la Rosa, A., Solana, E., Corpas, R., Bartrés-Faz, D., Pallàs, M., Vina, J., Sanfeliu, C., & Gomez-Cabrera, M. C. (2019). Long-term exercise training improves memory in middle-aged men and modulates peripheral levels of BDNF and Cathepsin B. *Scientific Reports*, 9(1). <https://doi.org/10.1038/s41598-019-40040-8>
- 56 Edward R. Laskowski, M. D. (2021, September 22). *How much exercise do you really need?* Mayo Clinic. Retrieved August 22, 2022, from <https://www.mayoclinic.org/healthy-lifestyle/fitness/expert-answers/exercise/faq-20057916#:~:text=As%20a%20general%20goal%2C%20aim,your%20risk%20of%20metabolic%20problems.>
- 57 Xiong, G. L., & Doraiswamy, P. M. (2009). Does meditation enhance cognition and brain plasticity? *Annals of the New York Academy of Sciences*, 1172(1), 63–69. <https://doi.org/10.1196/annals.1393.002>
- 58 Hölzel, B. K., Carmody, J., Vangel, M., Congleton, C., Yerramsetti, S. M., Gard, T., & Lazar, S. W. (2011). Mindfulness practice leads to increases in regional brain gray matter density. *Psychiatry Research: Neuroimaging*, 191(1), 36–43. <https://doi.org/10.1016/j.psychres.2010.08.006>
- 59 Nunez, K. (2020, July 20). *Why do we sleep?* Healthline. Retrieved August 20, 2022, from <https://www.healthline.com/health/why-do-we-sleep#:~:text=Many%20biological%20processes%20happen%20during,molecules%20like%20hormones%20and%20proteins.>
- 60 Nunez, K. (2020, July 20). *Why do we sleep?* Healthline. Retrieved August 20, 2022, from <https://www.healthline.com/health/why-do-we-sleep#brain-function>
- 61 Poe, G. R. (2017). Sleep is for forgetting. *The Journal of Neuroscience*, 37(3), 464–473. <https://doi.org/10.1523/jneurosci.0820-16.2017>
- 62 Eugene AR, Masiak J. The Neuroprotective Aspects of Sleep. *MEDtube Sci*. 2015 Mar;3(1):35-40. PMID: 26594659; PMCID: PMC4651462.
- 63 Gorgoni, M., D'Atti, A., Lauri, G., Rossini, P. M., Ferlazzo, F., & De Gennaro, L. (2013). Is sleep essential for neural plasticity in humans, and how does it affect motor and cognitive recovery? *Neural Plasticity*, 2013, 1–13. <https://doi.org/10.1155/2013/103949>
- 64 U.S. Department of Health and Human Services. (2016, March 31). *How sleep clears the brain*. National Institutes of Health. Retrieved August 15, 2022, from <https://www.nih.gov/news-events/nih-research-matters/how-sleep-clears-brain#:~:text=A%20mouse%20study%20suggests%20that,sleep%20in%20health%20and%20disease.>
- 65 Pennisi, M., Lanza, G., Cantone, M., D'Amico, E., Fiscaro, F., Puglisi, V., Vinciguerra, L., Bella, R., Vicari, E., & Malaguarnera, G. (2020). Acetyl-L-carnitine in dementia and other cognitive disorders: A critical update. *Nutrients*, 12(5), 1389. <https://doi.org/10.3390/nu12051389>
- 66 Virmani, M. A., & Cirulli, M. (2022). The role of L-carnitine in mitochondria, prevention of metabolic inflexibility and disease initiation. *International Journal of Molecular Sciences*, 23(5), 2717. <https://doi.org/10.3390/ijms23052717>
- 67 Rump, T. J., Muneer, P. M. A., Szlachetka, A. M., Lamb, A., Haorei, C., Alikunju, S., Xiong, H., Keblesh, J., Liu, J., Zimmerman, M. C., Jones, J., Donohue, T. M., Persidsky, Y., & Haorah, J. (2010). Acetyl-L-carnitine protects neuronal function from alcohol-induced oxidative damage in the brain. *Free Radical Biology and Medicine*, 49(10), 1494–1504. <https://doi.org/10.1016/j.freeradbiomed.2010.08.011>
- 68 WebMD. (n.d.). *Acetyl-L-carnitine: Overview, uses, side effects, precautions, interactions, dosing and reviews*. WebMD. Retrieved August 22, 2022, from <https://www.webmd.com/vitamins/ai/ingredientmono-834/acetyl-l-carnitine>
- 69 Chu, C.-S., Hung, C.-F., Ponnusamy, V. K., Chen, K.-C., & Chen, N.-C. (2022). Higher serum DHA and slower cognitive decline in patients with alzheimer's disease: Two-year follow-up. *Nutrients*, 14(6), 1159. <https://doi.org/10.3390/nu14061159>
- 70 Swanson, D., Block, R., & Mousa, S. A. (2012). Omega-3 fatty acids EPA and DHA: Health benefits throughout life. *Advances in Nutrition*, 3(1), 1–7. <https://doi.org/10.3945/an.111.000893>
- 71 Lopes Sakamoto, F., Metzker Pereira Ribeiro, R., Amador Bueno, A., & Oliveira Santos, H. (2019). Psychotropic effects of L-Theanine and its clinical properties: From the management of anxiety and stress to a potential use in schizophrenia. *Pharmacological Research*, 147, 104395. <https://doi.org/10.1016/j.phrs.2019.104395>
- 72 Everett, J. M., Gunathilake, D., Dufficy, L., Roach, P., Thomas, J., Upton, D., & Naumovski, N. (2016). Theanine consumption, stress and anxiety in human clinical trials: A systematic review. *Journal of Nutrition & Intermediary Metabolism*, 4, 41–42. <https://doi.org/10.1016/j.jnim.2015.12.308>
- 73 Williams, J. L., Everett, J. M., D'Cunha, N. M., Sergi, D., Georgousopoulou, E. N., Keegan, R. J., McKune, A. J., Mellor, D. D., Anstice, N., & Naumovski, N. (2019). The effects of green tea amino acid L-theanine consumption on the ability to manage stress and anxiety levels: A systematic review. *Plant Foods for Human Nutrition*, 75(1), 12–23. <https://doi.org/10.1007/s11130-019-00771-5>
- 74 Schwalfenberg, G. K., & Genuis, S. J. (2017). The importance of magnesium in clinical healthcare. *Scientifica*, 2017, 1–14. <https://doi.org/10.1155/2017/4179326>
- 75 Mateos-Aparicio, P., & Rodríguez-Moreno, A. (2019). The impact of studying brain plasticity. *Frontiers in Cellular Neuroscience*, 13. <https://doi.org/10.3389/fn-cel.2019.00066>
- 76 Gamma, E. (2021, March 24). *What is brain plasticity?* Brain Plasticity (Neuroplasticity) | Simply Psychology. Retrieved August 22, 2022, from <https://www.simplypsychology.org/brain-plasticity.html>
- 77 Kirkland, A., Sarlo, G., & Holton, K. (2018). The role of magnesium in neurological disorders. *Nutrients*, 10(6), 730. <https://doi.org/10.3390/nu10060730>
- 78 Pickering, G., Mazur, A., Troussellard, M., Bienkowski, P., Yaltsewa, N., Amessou, M., Noah, L., & Pouteau, E. (2020). Magnesium status and stress: The Vicious Circle Concept revisited. *Nutrients*, 12(12), 3672. <https://doi.org/10.3390/nu12123672>
- 79 MediLexicon International. (n.d.). *Magnesium glycinate: Uses, benefits, and side effects*. Medical News Today. Retrieved August 22, 2022, from <https://www.medicalnewstoday.com/articles/315372#risks-and-complications>
- 80 Kim, B.-K., & Park, S.-K. (2020). Phosphatidylserine modulates response to oxidative stress through hormesis and increases lifespan via DAF-16 in *Caenorhabditis elegans*. *Biogerontology*, 21(2), 231–244. <https://doi.org/10.1007/s10522-020-09856-0>
- 81 Glade, M. J., & Smith, K. (2015). Phosphatidylserine and the human brain. *Nutrition*, 31(6), 781–786. <https://doi.org/10.1016/j.nut.2014.10.014>
- 82 Mishra, S., & Palanivelu, K. (2008). The effect of curcumin (turmeric) on Alzheimer's disease: An overview. *Annals of Indian Academy of Neurology*, 11(1), 13–19. <https://doi.org/10.4103/0972-2327.40220>
- 83 Sarker, M. R., & Franks, S. F. (2018). Efficacy of curcumin for age-associated cognitive decline: a narrative review of preclinical and clinical studies. *GeroScience*, 40(2), 73–95. <https://doi.org/10.1007/s11357-018-0017-z>
- 84 Chamberlain, M. (2022, April 23). *The little known reason B vitamins are essential for cognitive function**. mindbodygreen. Retrieved August 23, 2022, from <https://www.mindbodygreen.com/articles/b-vitamins-methylation-and-brain-health>
- 85 Moore, E., Mander, A., Ames, D., Carne, R., Sanders, K., & Watters, D. (2012). Cognitive impairment and vitamin B12: A Review. *International Psychogeriatrics*, 24(4), 541–556. <https://doi.org/10.1017/s1041610211002511>
- 86 Venkatramanan, S., Armata, I. E., Strupp, B. J., & Finkelstein, J. L. (2016). Vitamin B-12 and cognition in children. *Advances in Nutrition*, 7(5), 879–888. <https://doi.org/10.3945/an.115.012021>
- 87 Spector, R. (2014). Vitamin Transport Diseases of brain: Focus on folates, thiamine and riboflavin. *Brain Disorders & Therapy*, 03(02). <https://doi.org/10.4172/2168-975x.1000120>
- 88 Uchida, Y., Ito, K., Ohtsuki, S., Kubo, Y., Suzuki, T., & Terasaki, T. (2015). Major involvement of Na⁺-dependent multivitamin transporter (SLC5A6/SMVT) in uptake of biotin and pantothenic acid by human brain capillary endothelial cells. *Journal of Neurochemistry*, 134(1), 97–112. <https://doi.org/10.1111/jnc.13092>
- 89 Wang, Z., Zhu, W., Xing, Y., Jia, J., & Tang, Y. (2021). B vitamins and prevention of cognitive decline and incident dementia: A systematic review and meta-analysis. *Nutrition Reviews*, 80(4), 931–949. <https://doi.org/10.1093/nutrit/nuab057>
- 90 WebMD. (n.d.). *Vitamins B complex oral: Uses, side effects, interactions, pictures, warnings & dosing*. WebMD. Retrieved August 23, 2022, from <https://www.webmd.com/drugs/2/drug-3387/vitamins-b-complex-oral/details#:~:text=Some%20products%20that%20may%20interact%2C%20other%20vitamin%2Fnutritional%20supplements.>
- 91 Anjum, I., Jaffery, S. S., Fayyaz, M., Samoo, Z., & Anjum, S. (2018). *The role of Vitamin D in brain health: A mini literature review*. Cureus. <https://doi.org/10.7759/cureus.2960>
- 92 Wartenberg, L. (2022, March 10). *Vitamin D deficiency: Symptoms, treatments, and causes*. Healthline. Retrieved August 22, 2022, from <https://www.healthline.com/nutrition/vitamin-d-deficiency-symptoms>
- 93 Sultan, S., Taimuri, U., Basnan, S. A., Ai-Orabi, W. K., Awadallah, A.,

Almowald, F., & Hazazi, A. (2020). Low vitamin D and its association with cognitive impairment and dementia. *Journal of Aging Research*, 2020, 1–10. <https://doi.org/10.1155/2020/6097820>

94 Soni M, Kos K, Lang IA, Jones K, Melzer D, Llewellyn DJ. Vitamin D and cognitive function. *Scand J Clin Lab Invest Suppl*. 2012;243:79-82. doi: 10.3109/00365513.2012.681969. PMID: 22536767.

95 Bjarnadottir, A. (2021, February 18). *How much vitamin D should you take for optimal health?* Healthline. Retrieved June 22, 2022, from <https://www.healthline.com/nutrition/how-much-vitamin-d-to-take>

96 Farzam K, Faizy RM, Saadabadi A. Stimulants. 2022 Jun 16. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. PMID: 30969718.

97 Miah, T., & Kamat, D. (2017). Current understanding of the health effects of electromagnetic fields. *Pediatric Annals*, 46(4). <https://doi.org/10.3928/19382359-20170316-01>

98 U.S. Department of Health and Human Services. (n.d.). *Electric & Magnetic Fields*. National Institute of Environmental Health Sciences. Retrieved August 15, 2022, from <https://www.niehs.nih.gov/health/topics/agents/emf/index.cfm>

99 Kivrak, E. G., Yurt, K. K., Kaplan, A. A., Alkan, I., & Altun, G. (2017). Effects of electromagnetic fields exposure on the antioxidant defense system. *Journal of Microscopy and Ultrastructure*, 5(4), 167. <https://doi.org/10.1016/j.jm.2017.07.003>

100 Kim, J. H., Lee, J.-K., Kim, H.-G., Kim, K.-B., & Kim, H. R. (2019). Possible effects of radiofrequency electromagnetic field exposure on central nerve system. *Biomolecules & Therapeutics*, 27(3), 265–275. <https://doi.org/10.4062/biomolther.2018.152>

101 Suratman, S., Edwards, J. W., & Babina, K. (2015). Organophosphate pesticides exposure among farmworkers: Pathways and risk of adverse health effects. *Reviews on Environmental Health*, 30(1), 65–79. <https://doi.org/10.1515/revveh-2014-0072>

102 Naughton, S. X., & Terry, A. V. (2018). Neurotoxicity in acute and repeated organophosphate exposure. *Toxicology*, 408, 101–112. <https://doi.org/10.1016/j.tox.2018.08.011>

103 Tsai, Y.-H., & Lein, P. J. (2021). Mechanisms of organophosphate neurotoxicity. *Current Opinion in Toxicology*, 26, 49–60. <https://doi.org/10.1016/j.cotox.2021.04.002>

104 Jokanović, M. (2018). Neurotoxic effects of organophosphorus pesticides and possible association with Neurodegenerative Diseases in man: A Review. *Toxicology*, 410, 125–131. <https://doi.org/10.1016/j.tox.2018.09.009>

105 Denis Lairon. Nutritional quality and safety of organic food. A review. *Agronomy for Sustainable Development*, Springer Verlag/EDP Sciences/INRA, 2010, 30 (1), pp.33-41. [ff10.1051/agro/2009019ff.fhal-00886513f](https://doi.org/10.1051/agro/2009019ff.fhal-00886513f)

106 Liu J, Lewis G. Environmental toxicity and poor cognitive outcomes in children and adults. *J Environ Health*. 2014 Jan-Feb;76(6):130-8. PMID: 24645424; PMCID: PMC4247328.

107 Bakulski, K. M., Seo, Y. A., Hickman, R. C., Brandt, D., Vadari, H. S., Hu, H., & Park, S. K. (2020). Heavy metals exposure and alzheimer's disease and related Dementias. *Journal of Alzheimer's Disease*, 76(4), 1215–1242. <https://doi.org/10.3233/jad-200282>

108 Tchounwou, P. B., Yedjou, C. G., Patlolla, A. K., & Sutton, D. J. (2012). Heavy metal toxicity and the environment. *Experientia Supplementum*, 133–164. https://doi.org/10.1007/978-3-7643-8340-4_6

109 Fernandes Azevedo, B., Barros Furieri, L., Peçanha, F. M., Wiggers, G. A., Frizzera Vassallo, P., Ronacher Simões, M., Fiorini, R., Rossi de Batista, P., Fiorelli, M., Rossoni, L., Stefanon, I., Alonso, M. J., Salas, M., & Valentim Vassallo, D. (2012). Toxic effects of mercury on the cardiovascular and Central Nervous Systems. *Journal of Biomedicine and Biotechnology*, 2012, 1–11. <https://doi.org/10.1155/2012/949048>

110 Wani, A. L., Ara, A., & Usmani, J. A. (2015). Lead toxicity: A Review. *Interdisciplinary Toxicology*, 8(2), 55–64. <https://doi.org/10.1515/intox-2015-0009>

111 Bakulski, K. M., Seo, Y. A., Hickman, R. C., Brandt, D., Vadari, H. S., Hu, H., & Park, S. K. (2020). Heavy metals exposure and alzheimer's disease and related Dementias. *Journal of Alzheimer's Disease*, 76(4), 1215–1242. <https://doi.org/10.3233/jad-200282>

112 Gimeno-García, E., Andreu, V., & Boluda, R. (1996). Heavy metals incidence in the application of inorganic fertilizers and pesticides to rice farming soils. *Environmental Pollution*, 92(1), 19–25. [https://doi.org/10.1016/0269-7491\(95\)00090-9](https://doi.org/10.1016/0269-7491(95)00090-9)

113 Moreira, P. I. (2013). High-sugar diets, type 2 diabetes and Alzheimer's disease. *Current Opinion in Clinical Nutrition and Metabolic Care*, 16(4), 440–445. <https://doi.org/10.1097/mco.0b013e328361c7d1>

114 Weinstein, G., Maillard, P., Himali, J. J., Beiser, A. S., Au, R., Wolf, P. A., Seshadri, S., & DeCarli, C. (2015). Glucose indices are associated with cognitive and structural brain measures in Young Adults. *Neurology*, 84(23), 2329–2337. <https://doi.org/10.1212/wnl.0000000000001655>

115 Wheeler, M. J., Dempsey, P. C., Grace, M. S., Ellis, K. A., Gardiner, P. A., Green, D. J., & Dunstan, D. W. (2017). Sedentary behavior as a risk factor for cognitive decline? A focus on the influence of glycemic control in Brain Health. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 3(3), 291–300. <https://doi.org/10.1016/j.trci.2017.04.001>

116 Siddharth, P., Burggren, A. C., Eyre, H. A., Small, G. W., & Merrill, D. A. (2018). Sedentary behavior associated with reduced medial temporal lobe thickness in middle-aged and older adults. *PLOS ONE*, 13(4). <https://doi.org/10.1371/journal.pone.0195549>

117 Wheeler, M. J., Dempsey, P. C., Grace, M. S., Ellis, K. A., Gardiner, P. A., Green, D. J., & Dunstan, D. W. (2017). Sedentary behavior as a risk factor for cognitive decline? A focus on the influence of glycemic control in Brain Health. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 3(3), 291–300. <https://doi.org/10.1016/j.trci.2017.04.001>

118 Lewis, C. R., Talboom, J. S., De Both, M. D., Schmidt, A. M., Naymik, M. A., Häberg, A. K., Rundek, T., Levin, B. E., Hoscheidt, S., Bolla, Y., Brinton, R. D., Hay,

M., Barnes, C. A., Glisky, E., Ryan, L., & Huentelman, M. J. (2021). Smoking is associated with impaired verbal learning and memory performance in women more than men. *Scientific Reports*, 11(1). <https://doi.org/10.1038/s41598-021-88923-z>

Depression Protocol

1 <https://www.who.int/news-room/fact-sheets/detail/depression>

2 Garvey, M. J., Tollefson, G. D., Mungas, D., & Hoffmann, N. (1984). Is the distinction between situational and nonsituational primary depression valid? *Comprehensive Psychiatry*, 25(3), 372–375. [https://doi.org/10.1016/0010-440x\(84\)90070-1](https://doi.org/10.1016/0010-440x(84)90070-1)

3 Joffe, R. T., Levitt, A. J., Bagby, M., & Regan, J. J. (1993). Clinical Features of Situational and Nonsituational Major Depression. *Psychopathology*, 26(3–4), 138–144. <https://doi.org/10.1159/000284813>

4 Coryell, W., Winokur, G., Maser, J. D., Akiskal, H. S., Keller, M. B., & Endicott, J. (1994a). Recurrently situational (reactive) depression: A study of course, phenomenology and familial psychopathology. *Journal of Affective Disorders*, 31(3), 203–210. [https://doi.org/10.1016/0165-0327\(94\)90030-2](https://doi.org/10.1016/0165-0327(94)90030-2)

5 Blatt, S. J. (2004). Experiences of depression: Theoretical, clinical, and research perspectives. *Experiences of Depression: Theoretical, Clinical, and Research Perspectives*. <https://doi.org/10.1037/10749-000>

6 Bains N, Abdijadid S. Major Depressive Disorder. [Updated 2022 Apr 14]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK559078/>

7 APA Council of Representatives. (2019, February). APA Clinical Practice Guideline for the Treatment of Depression Across Three Age Cohorts. American Psychological Association. Retrieved May 2022, from <https://www.apa.org/depression-guideline/guideline.pdf>

8 Belujon, P., & Grace, A. A. (2017). Dopamine System Dysregulation in Major Depressive Disorders. *International Journal of Neuropsychopharmacology*, 20(12), 1036–1046. <https://doi.org/10.1093/ijnp/pyx056>

9 Pan, J. X., Xia, J. J., Deng, F. L., Liang, W. W., Wu, J., Yin, B. M., Dong, M. X., Chen, J. J., Ye, F., Wang, H. Y., Zheng, P., & Xie, P. (2018). Diagnosis of major depressive disorder based on changes in multiple plasma neurotransmitters: a targeted metabolomics study. *Translational Psychiatry*, 8(1). <https://doi.org/10.1038/s41398-018-0183-x>

10 Zhang, F. F., Peng, W., Sweeney, J. A., Jia, Z. Y., & Gong, Q. Y. (2018). Brain structure alterations in depression: Psychoradiological evidence. *CNS Neuroscience & Therapeutics*, 24(11), 994–1003. <https://doi.org/10.1111/cns.12835>

11 American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). <https://doi-org.ezproxy.frederick.edu/10.1176/appi.books.9780890425596>

12 Wy, T. J. P., & Saadabadi, A. (2021). *Schizoaffective Disorder*. StatPearls Publishing.

13 Malaspina, D., Owen, M. J., Heckers, S., Tandon, R., Bustillo, J., Schultz, S., Barch, D. M., Gaebel, W., Gur, R. E., Tsuang, M., van Os, J., & Carpenter, W. (2013). Schizoaffective Disorder in the DSM-5. *Schizophrenia Research*, 150(1), 21–25. <https://doi.org/10.1016/j.schres.2013.04.026>

14 Lachance MD, L., & Ramsey MD, D. (2015). Food, Mood, and Brain Health: Implications for the Modern Clinician. *Missouri Medicine*, 112(2), 111–115.

15 Kang, J. H., Ascherio, A., & Grodstein, F. (2005). Fruit and vegetable consumption and cognitive decline in aging women. *Annals of Neurology: Official Journal of the American Neurological Association and the Child Neurology Society*, 57(5), 713–720

16 Sinclair, A. J., Begg, D., Mathai, M., & Weisinger, R. S. (2007). Omega 3 fatty acids and the brain: review of studies in depression. *Asia Pacific Journal of Clinical Nutrition*, 16(S1), 391–397.

17 Simopoulos, A. P. (2011). Evolutionary aspects of diet: the omega-6/omega-3 ratio and the brain. *Molecular neurobiology*, 44(2), 203–215.

18 Sublette, M. E., Ellis, S. P., Geant, A. L., & Mann, J. J. (2011). Meta-analysis of the effects of eicosapentaenoic acid (EPA) in clinical trials in depression. *The Journal of clinical psychiatry*, 72(12), 1577–1584. <https://doi.org/10.4088/JCP.10m06634>

19 Gros, D. F., Antony, M. M., McCabe, R. E., & Swinson, R. P. (2009). Frequency and severity of the symptoms of irritable bowel syndrome across the anxiety disorders and depression. *Journal of anxiety disorders*, 23(2), 290–296. <https://doi.org/10.1016/j.janxdis.2008.08.004>

20 Chen, Y., Xu, J., & Chen, Y. (2021). Regulation of Neurotransmitters by the Gut Microbiota and Effects on Cognition in Neurological Disorders. *Nutrients*, 13(6), 2099. <https://doi.org/10.3390/nu13062099>

21 Kennedy, P. J., Murphy, A. B., Cryan, J. F., Ross, P. R., Dinan, T. G., & Stanton, C. (2016). Microbiome in brain function and mental health. *Trends in Food Science & Technology*, 57, 289–301. <https://doi.org/10.1016/j.tifs.2016.05.001>

22 Dinan, T. G., & Cryan, J. F. (2017). The Microbiome-Gut-Brain Axis in Health and Disease. *Gastroenterology clinics of North America*, 46(1), 77–89. <https://doi.org/10.1016/j.gtc.2016.09.007>

23 Sharma, M., Gupta, A., & Prasad, R. (2017). A review on herbs, spices and functional food used in diseases. *International Journal of Research & Review*, 4(1), 103–108.

24 Singletary, K. (2016). Coriander: Overview of Potential Health Benefits. *Nutrition Today*, 51(3), 151–161. <https://doi.org/10.1097/nt.0000000000000159>

25 Amin, C. A., Afiahana Andatia, Fathia Daffa Putri, Siti Nurhayati Utami, & M. Yusuf Wijahaska. (2021). The Potential of Cinnamon as Anti-Depressant. *Eureka Herba Indonesia*, 2(2), 104–108. <https://doi.org/10.37275/ehi.v2i2.19>

26 Grosso, G., Micek, A., Castellano, S., Pajak, A., & Galvano, F. (2015). Coffee, tea, caffeine and risk of depression: A systematic review and dose-response meta-analysis of observational studies. *Molecular Nutrition & Food Research*, 60(1), 223–234. <https://doi.org/10.1002/mnfr.201500620>

- 27 Hall, S., Desbrow, B., Anoopkumar-Dukie, S., Davey, A. K., Arora, D., McDermott, C., Schubert, M. M., Perkins, A. V., Kiehl, M. J., & Grant, G. D. (2015). A review of the bioactivity of coffee, caffeine and key coffee constituents on inflammatory responses linked to depression. *Food Research International*, 76, 626–636. <https://doi.org/10.1016/j.foodres.2015.07.027>
- 28 Jin, M. J., Yoon, C. H., Ko, H. J., Kim, H. M., Kim, A. S., Moon, H. N., & Jung, S. P. (2016). The Relationship of Caffeine Intake with Depression, Anxiety, Stress, and Sleep in Korean Adolescents. *Korean Journal of Family Medicine*, 37(2), 111. <https://doi.org/10.4082/kjfm.2016.37.2.111>
- 29 Wang, L., Shen, X., Wu, Y., & Zhang, D. (2015). Coffee and caffeine consumption and depression: A meta-analysis of observational studies. *Australian & New Zealand Journal of Psychiatry*, 50(3), 228–242. <https://doi.org/10.1177/0004867415603131>
- 30 Sanchez-Villegas, A., & Martínez-González, M. A. (2013). Diet, a new target to prevent depression? *BMC Medicine*, 11(1). <https://doi.org/10.1186/1741-7015-11-3>
- 31 Ahmed, S. H., Guillem, K., & Vandaele, Y. (2013). Sugar addiction. Current Opinion in Clinical Nutrition and Metabolic Care, 16(4), 434–439. <https://doi.org/10.1097/mco.0b013e328361c8b8>
- 32 DiNicolantonio, J. J., O'Keefe, J. H., & Wilson, W. L. (2017). Sugar addiction: is it real? A narrative review. *British Journal of Sports Medicine*, 52(14), 910–913. <https://doi.org/10.1136/bjsports-2017-097971>
- 33 Singh, N. A., Clements, K. M., & Singh, M. A. F. (2001). The Efficacy of Exercise as a Long-term Antidepressant in Elderly Subjects: A Randomized, Controlled Trial. *The Journals of Gerontology Series A: Biological Sciences and Medical Sciences*, 56(8), M497–M504. <https://doi.org/10.1093/gerona/56.8.m497>
- 34 Legrand, F., & Heuze, J. P. (2007). Antidepressant Effects Associated with Different Exercise Conditions in Participants with Depression: A Pilot Study. *Journal of Sport and Exercise Psychology*, 29(3), 348–364. <https://doi.org/10.1123/jsep.29.3.348>
- 35 Kandola, A., Ashdown-Franks, G., Hendrikse, J., Sabiston, C. M., & Stubbs, B. (2019). Physical activity and depression: Towards understanding the antidepressant mechanisms of physical activity. *Neuroscience & Biobehavioral Reviews*, 107, 525–539. <https://doi.org/10.1016/j.neubiorev.2019.09.040>
- 36 Ernst, C., Olsen, A. K., Pinel, J. P. J., Lam, R. W., & Christie, B. R. (2006). Antidepressant effects of exercise: Evidence for an adult-neurogenesis hypothesis? *Journal of Psychiatry Neuroscience*, 31(2), 84–92.
- 37 Ekkekakis, P., & Murri, M. B. (2017). REMOVED: Exercise as antidepressant treatment: Time for the transition from trials to clinic? *General Hospital Psychiatry*, 49, 1. <https://doi.org/10.1016/j.genhosppsych.2017.04.008>
- 38 Psarra, E., & Kleftharas, G. (2013). Adaptation to Physical Disabilities: The Role of Meaning in Life and Depression. *The European Journal of Counselling Psychology*, 2(1), 79–99. <https://doi.org/10.5964/ejocp.v2i1.7>
- 39 Heisel, M. J., & Flett, G. L. (2004). Purpose in Life, Satisfaction with Life, and Suicide Ideation in a Clinical Sample. *Journal of Psychopathology and Behavioral Assessment*, 26(2), 127–135. <https://doi.org/10.1023/b:joba.0000013660.22413.e0>
- 40 Pearson, M. R., Brown, D. B., Bravo, A. J., & Witkiewitz, K. (2014). Staying in the Moment and Finding Purpose: The Associations of Trait Mindfulness, Decentering, and Purpose in Life with Depressive Symptoms, Anxiety Symptoms, and Alcohol-Related Problems. *Mindfulness*, 6(3), 645–653. <https://doi.org/10.1007/s12671-014-0300-8>
- 41 Salt, E., Crofford, L. J., & Segerstrom, S. (2017). The Mediating and Moderating Effect of Volunteering on Pain and Depression, Life Purpose, Well-Being, and Physical Activity. *Pain Management Nursing*, 18(4), 243–249. <https://doi.org/10.1016/j.pmn.2017.04.004>
- 42 Delgadillo, J., & Gonzalez Salas Duhne, P. (2020). Targeted prescription of cognitive-behavioral therapy versus person-centered counseling for depression using a machine learning approach. *Journal of Consulting and Clinical Psychology*, 88(1), 14–24. <https://doi.org/10.1037/ccp0000476>
- 43 Williams, J. M. G. (2008). Mindfulness, Depression and Modes of Mind. *Cognitive Therapy and Research*, 32(6), 721–733. <https://doi.org/10.1007/s10608-008-9204-z>
- 44 Hofmann, S. G., Sawyer, A. T., Witt, A. A., & Oh, D. (2010). The effect of mindfulness-based therapy on anxiety and depression: A meta-analytic review. *Journal of Consulting and Clinical Psychology*, 78(2), 169–183. <https://doi.org/10.1037/a0018555>
- 45 Teasdale, J. D., Segal, Z. V., Williams, J. M. G., Ridgeway, V. A., Soulsby, J. M., & Lau, M. A. (2000). Prevention of relapse/recurrence in major depression by mindfulness-based cognitive therapy. *Journal of Consulting and Clinical Psychology*, 68(4), 615–623. <https://doi.org/10.1037/0022-006x.68.4.615>
- 46 APA Council of Representatives. (2019, February). APA Clinical Practice Guideline for the Treatment of Depression Across Three Age Cohorts. American Psychological Association. Retrieved May 2022, from <https://www.apa.org/depression-guideline/guideline.pdf>
- 47 Psychology Today Staff. (n.d.). *Interpersonal Psychotherapy*. Psychology Today. Retrieved May 2022, from <https://www.psychologytoday.com/us/therapy-types/interpersonal-psychotherapy>
- 48 Cuijpers, P., Geraedts, A. S., van Oppen, P., Andersson, G., Markowitz, J. C., & van Straten, A. (2011). Interpersonal Psychotherapy for Depression: A Meta-Analysis. *American Journal of Psychiatry*, 168(6), 581–592. <https://doi.org/10.1176/appi.ajp.2010.10101411>
- 49 Marcus, E. R. (2002). Psychoanalytic Psychotherapy and Psychoanalysis, Overview. *Encyclopedia of Psychotherapy*, 423–430. <https://doi.org/10.1016/b0-12-343010-0/00171-9>
- 50 Taylor, D., Carlyle, J. A., McPherson, S., Rost, F., Thomas, R., & Fonagy, P. (2012). Tavistock Adult Depression Study (TADS): a randomised controlled trial of psychoanalytic psychotherapy for treatment-resistant/treatment-refractory forms of depression. *BMC Psychiatry*, 12(1). <https://doi.org/10.1186/1471-244x-12-60>
- 51 Weitkamp, K., Daniels, J. K., Hofmann, H., Timmermann, H., Romer, G., & Wiegand-Grefe, S. (2014). Psychoanalytic psychotherapy for children and adolescents with severe depressive psychopathology: Preliminary results of an effectiveness trial. *Psychotherapy*, 51(1), 138–147. <https://doi.org/10.1037/a0034178>
- 52 de Jonghe, F., de Maat, S., Van, R., Hendriksen, M., Kool, S., van Aalst, G., & Dekker, J. (2013). Short-Term Psychoanalytic Supportive Psychotherapy for Depressed Patients. *Psychoanalytic Inquiry*, 33(6), 614–625. <https://doi.org/10.1080/07351690.2013.835184>
- 53 Neuman M.D., F. (2013, June 2). *Supportive Psychotherapy*. Psychology Today. Retrieved May 2022, from <https://www.psychologytoday.com/us/blog/fighting-fear/201306/supportive-psychotherapy>
- 54 Abdallah, C. G., Jiang, L., de Feyter, H. M., Fasula, M., Krystal, J. H., Rothman, D. L., Mason, G. F., & Sanacora, G. (2014). Glutamate Metabolism in Major Depressive Disorder. *American Journal of Psychiatry*, 171(12), 1320–1327. <https://doi.org/10.1176/appi.ajp.2014.14010067>
- 55 van Praag, H., & de Haan, S. (1980). Depression vulnerability and 5-hydroxytryptophan prophylaxis. *Psychiatry Research*, 3(1), 75–83. [https://doi.org/10.1016/0165-1781\(80\)90049-9](https://doi.org/10.1016/0165-1781(80)90049-9)
- 56 Meyers, S. (2000). use of neurotransmitter precursors for treatment of depression. *Alternative Medicine Review*, 5(1), 64–71.
- 57 Jangid, P., Malik, P., Singh, P., Sharma, M., & Gulia, A. K. (2013). Comparative study of efficacy of l-5-hydroxytryptophan and fluoxetine in patients presenting with first depressive episode. *Asian journal of psychiatry*, 6(1), 29–34. <https://doi.org/10.1016/j.ajp.2012.05.011>
- 58 Iovieno N, Dalton ED, Fava M, Mischoulon D. Second-tier natural antidepressants: review and critique. *J Affect Disord*. 2011;130(3):343-357.20579741
- 59 Turner EH, Loftis JM, Blackwell AD. Serotonin a la carte: supplementation with the serotonin precursor 5-hydroxytryptophan. *Pharmacol Ther*. 2006;109(3):325-338.16023217
- 60 Kravitz, H. M., Sabelli, H. C., & Fawcett, J. (1984). Dietary supplements of phenylalanine and other amino acid precursors of brain neuroamines in the treatment of depressive disorders. *The Journal of the American Osteopathic Association*, 84(9), 119–123. <https://doi.org/10.1515/jom-1984-840928>
- 61 Beckmann, H., Strauss, M. A., & Ludolph, E. (1977). DL-phenylalanine in depressed patients: an open study. *Journal of neural transmission*, 41(2-3), 123–134. <https://doi.org/10.1007/BF01670277>
- 62 Fischer, E., Heller, B., Nachon, M., & Spatz, H. (1975). Therapy of depression by phenylalanine. Preliminary note. *Arzneimittel-Forschung*, 25(1), 132
- 63 Williams, A. L., Girard, C., Jui, D., Alyse, S., & Katz, D. L. (2009). S-Adenosylmethionine (SAME) as Treatment for Depression: A Systematic Review. *Clinical and Investigative Medicine*, 28(3), 132–139.
- 64 Cuomo, A., Beccarini Crescenzi, B., Bolognesi, S., Goracci, A., Koukouna, D., Rossi, R., & Fagioli, A. (2020). S-Adenosylmethionine (SAME) in major depressive disorder (MDD): a clinician-oriented systematic review. *Annals of General Psychiatry*, 19(1). <https://doi.org/10.1186/s12991-020-00298-z>
- 65 Sharma, A., Gerberg, P., Bottiglieri, T., Massoumi, L., Carpenter, L. L., Lavretsky, H., Muskin, P. R., Brown, R. P., Mischoulon, D., & as Work Group of the American Psychiatric Association Council on Research (2017). S-Adenosylmethionine (SAME) for Neuropsychiatric Disorders: A Clinician-Oriented Review of Research. *The Journal of clinical psychiatry*, 78(6), e656–e667. <https://doi.org/10.4088/JCP.16r11113>
- 66 Abdallah, C. G., Jiang, L., de Feyter, H. M., Fasula, M., Krystal, J. H., Rothman, D. L., Mason, G. F., & Sanacora, G. (2014). Glutamate Metabolism in Major Depressive Disorder. *American Journal of Psychiatry*, 171(12), 1320–1327. <https://doi.org/10.1176/appi.ajp.2014.14010067>
- 67 Musazzi, L., Treccani, G., & Popoli, M. (2012). Glutamate hypothesis of depression and its consequences for antidepressant treatments. *Expert review of neurotherapeutics*, 12(10), 1169–1172.
- 68 Sanacora, G., & Saricicek, A. (2007). GABAergic Contributions to the Pathophysiology of Depression and the Mechanism of Antidepressant Action. *CNS & Neurological Disorders - Drug Targets*, 6(2), 127–140. <https://doi.org/10.2174/187152707780363294>
- 69 Amen M.D., D., & Routh, L. C. (2004). Healing Anxiety and Depression: Based on Cutting-Edge Brain-Imaging Science (Reprint ed.). Berkley.
- 70 Ghazizadeh, J., Sadigh-Eteghad, S., Marx, W., Fakhari, A., Hamedeyazdan, S., Torbati, M., Taheri-Tarighi, S., Araj-khodaei, M., & Mirghafourvand, M. (2021). The effects of lemon balm (*Melissa officinalis* L.) on depression and anxiety in clinical trials: A systematic review and meta-analysis. *Phytotherapy Research*, 35(12), 6690–6705. <https://doi.org/10.1002/ptr.7252>
- 71 Ghazizadeh, J., Hamedeyazdan, S., Torbati, M., Farajdokht, F., Fakhari, A., Mahmoudi, J., Araj-khodaei, M., & Sadigh-Eteghad, S. (2020). *Melissa officinalis* L. hydro-alcoholic extract inhibits anxiety and depression through prevention of central oxidative stress and apoptosis. *Experimental Physiology*, 105(4), 707–720. <https://doi.org/10.1113/ep088254>
- 72 Lin, S. H., Chou, M. L., Chen, W. C., Lai, Y. S., Lu, K. H., Hao, C. W., & Sheen, L. Y. (2015). A medicinal herb, *Melissa officinalis* L. ameliorates depressive-like behavior of rats in the forced swimming test via regulating the serotonergic neurotransmitter. *Journal of Ethnopharmacology*, 175, 266–272. <https://doi.org/10.1016/j.jep.2015.09.018>
- 73 Cases, J., Ibarra, A., Feuillere, N., Roller, M., & Sukkar, S. G. (2011). Pilot trial of *Melissa officinalis* L. leaf extract in the treatment of volunteers suffering from mild-to-moderate anxiety disorders and sleep disturbances. *Mediterranean journal of nutrition and metabolism*, 4(3), 211–218
- 74 Kennedy, D. O., Scholey, A. B., Tildesley, N. T. J., Perry, E. K., & Wesnes, K. A.

(2002). Modulation of mood and cognitive performance following acute administration of *Melissa officinalis* (lemon balm). *Pharmacology Biochemistry and Behavior*, 72(4), 953-964.

75 Lew, S. Y., Teoh, S. L., Lim, S. H., Lim, L. W., & Wong, K. H. (2020). Discovering the Potentials of Medicinal Mushrooms in Combating Depression – A Review. *Mini-Reviews in Medicinal Chemistry*, 20(15), 1518–1531. <https://doi.org/10.2174/1389557520666200526125534>

76 Mi, Xue, Gui-Rong Zeng, Jie-Qing Liu, Zhou-Song Luo, Ling Zhang, Xiao-Man Dai, Wen-Ting Fang, Jing Zhang, and Xiao-Chun Chen. 2022. "Ganoderma Lucidum Triterpenoids Improve Maternal Separation-Induced Anxiety- and Depression-like Behaviors in Mice by Mitigating Inflammation in the Periphery and Brain" *Nutrients* 14, no. 11: 2268. <https://doi.org/10.3390/nu14112268>

77 Zhang, X., Wang, M., Qiao, Y., Shan, Z., Yang, M., Li, G., Xiao, Y., Wei, L., Bi, H., & Gao, T. (2022). Exploring the mechanisms of action of *Cordyceps sinensis* for the treatment of depression using network pharmacology and molecular docking. *Annals of translational medicine*, 10(6), 282. <https://doi.org/10.21037/atm-22-762>

78 Lin, Y. E., Chen, Y. C., Lu, K. H., Huang, Y. J., Panyod, S., Liu, W. T., Yang, S. H., Lu, Y. S., Chen, M. H., & Sheen, L. Y. (2021). Antidepressant-like effects of water extract of *Cordyceps militaris* (Linn.) Link by modulation of ROCK2/PTEN/Akt signaling in an unpredictable chronic mild stress-induced animal model. *Journal of ethnopharmacology*, 276, 114194. <https://doi.org/10.1016/j.jep.2021.114194>

79 Ashraf, S. A., Elkhalfi, A., Siddiqui, A. J., Patel, M., Awadelkareem, A. M., Snoussi, M., Ashraf, M. S., Adnan, M., & Hadi, S. (2020). Cordycepin for Health and Wellbeing: A Potent Bioactive Metabolite of an Entomopathogenic *Cordyceps* Medicinal Fungus and Its Nutraceutical and Therapeutic Potential. *Molecules* (Basel, Switzerland), 25(12), 2735. <https://doi.org/10.3390/molecules25122735>

80 Zaidi, Dr.Kamal. (2017). The health benefits of *Cordyceps militaris* - A review. *Kavaka*. 48. 27-32

81 Chong, P. S., Fung, M. L., Wong, K. H., & Lim, L. W. (2019). Therapeutic Potential of *Hericium erinaceus* for Depressive Disorder. *International Journal of Molecular Sciences*, 21(1), 163. <https://doi.org/10.3390/ijms21010163>

82 Nagano, M., Shimizu, K., Kondo, R., Hayashi, C., Sato, D., Kitagawa, K., & Ohnuki, K. (2010). Reduction of depression and anxiety by 4 weeks *Hericium erinaceus* intake. *Biomedical research* (Tokyo, Japan), 31(4), 231–237. <https://doi.org/10.2220/biomedres.31.231>

83 Yilin Ren, Yan Geng, Yan Du, Wang Li, Zhen-Ming Lu, Hong-Yu Xu, Guo-Hua Xu, Jin-Song Shi, Zheng-Hong Xu. Polysaccharide of *Hericium erinaceus* attenuates colitis in C57BL/6 mice via regulation of oxidative stress, inflammation-related signaling pathways and modulating the composition of the gut microbiota. *The Journal of Nutritional Biochemistry*, Vol 57, 2018, Pages 67-76, <https://doi.org/10.1016/j.jnutbio.2018.03.005>

84 Cauffield, J. S., & Forbes, H. J. (1999). dietary supplements used in the treatment of depression, anxiety, and sleep disorders. *Lippincott's Primary Care Practice*, 3(3), 290–304. https://europepmc.org/article/med/10711131?utm_medium=27156&client=bot&client=bot&client=bot

85 Gaster, B., & Holroyd, J. (2000). St John's Wort for Depression. *Archives of Internal Medicine*, 160(2), 152. <https://doi.org/10.1001/archinte.160.2.152>

86 St. John's Wort. (2020, October). NCCIH. Retrieved May 2022, from <https://www.nccih.nih.gov/health/st-johns-wort>

87 Kasper S. Phytopharmaceutical treatment of anxiety, depression, and dementia in the elderly: evidence from randomized, controlled clinical trials. *Wien Med Wochenschr*. 2015;165(11-12):217-228.26092515

88 Amsterdam, J. D., Li, Q. S., Xie, S. X., & Mao, J. J. (2020). Putative Antidepressant Effect of Chamomile (*Matricaria chamomilla* L.) Oral Extract in Subjects with Comorbid Generalized Anxiety Disorder and Depression. *The Journal of Alternative and Complementary Medicine*, 26(9), 815–821. <https://doi.org/10.1089/acm.2019.0252>

89 Amsterdam, J. D., Shults, J., Soeller, L., Mao, J. J., Rockwell, K., & Newberg, A. B. (2012). Chamomile (*Matricaria recutita*) may provide antidepressant activity in anxious, depressed humans: an exploratory study. *Alternative therapies in health and medicine*, 18(5), 44–49.

90 Mikkelsen, K., Stojanovska, L., & Apostolopoulos, V. (2016). The Effects of Vitamin B in Depression. *Current medicinal chemistry*, 23(38), 4317–4337. <https://doi.org/10.2174/0929867323666160920110810>

91 <https://ods.od.nih.gov/factsheets/Thiamin-HealthProfessional/>

92 Tiemeier, H., van Tuijl, H. R., Hofman, A., Meijer, J., Kiliaan, A. J., & Breteler, M. M. (2002). Vitamin B12, folate, and homocysteine in depression: the Rotterdam Study. *The American journal of psychiatry*, 159(12), 2099–2101. <https://doi.org/10.1176/appi.ajp.159.12.2099>

93 Syed, E. U., Wasay, M., & Awan, S. (2013). Vitamin B12 supplementation in treating major depressive disorder: a randomized controlled trial. *The open neurology journal*, 7, 44–48. <https://doi.org/10.2174/1874205X01307010044>

94 Wong C. W. (2015). Vitamin B12 deficiency in the elderly: is it worth screening? *Hong Kong medical journal* = *Xianggang yi xue za zhi*, 21(2), 155–164. <https://doi.org/10.12809/hkmj144383>

95 Sinclair, A. J., Begg, D., Mathai, M., & Weisinger, R. S. (2007). Omega 3 fatty acids and the brain: review of studies in depression. *Asia Pacific Journal of Clinical Nutrition*, 16(S1), 391–397.

96 Sublette, M. E., Ellis, S. P., Geant, A. L., & Mann, J. J. (2011). Meta-analysis of the effects of eicosapentaenoic acid (EPA) in clinical trials in depression. *The Journal of clinical psychiatry*, 72(12), 1577–1584. <https://doi.org/10.4088/JCP.10m06634>

97 Office of Dietary Supplements - Zinc. (2021, December). National Institute of Health. Retrieved May 2022, from <https://ods.od.nih.gov/factsheets/Zinc-HealthProfessional/>

98 Eby, G. A., & Eby, K. L. (2010). Magnesium for treatment-resistant depression: A review and hypothesis. *Medical Hypotheses*, 74(4), 649–660. <https://doi.org/10.1016/j.mehy.2009.10.051>

99 Wang, W. L., Hung, H. Y., Chen, Y. R., Chen, K. H., Yang, S. N., Chu, C. M., & Chan, Y. Y. (2020). The Effect of Foot Reflexology Intervention on Depression, Anxiety, and Sleep Quality in Adults: A Meta-Analysis and Meta-regression of Randomized Controlled Trials. *Evidence-Based Complementary and Alternative Medicine*, 2020, 1–21. <https://doi.org/10.1155/2020/2654353>

100 Tu, C. H., MacDonald, I., & Chen, Y. H. (2019). The Effects of Acupuncture on Glutamate Neurotransmission in Depression, Anxiety, Schizophrenia, and Alzheimer's Disease: A Review of the Literature. *Frontiers in Psychiatry*, 10. <https://doi.org/10.3389/fpsy.2019.00014>

101 Mahdavi-pour, F., Rahemi, Z., Sadat, Z., & Ajorpaz, N. M. (2019). The effects of foot reflexology on depression during menopause: A randomized controlled clinical trial. *Complementary Therapies in Medicine*, 47, 102195. <https://doi.org/10.1016/j.ctim.2019.102195>

102 Hou, W. H., Chiang, P. T., Hsu, T. Y., Chiu, S. Y., & Yen, Y. C. (2010). Treatment Effects of Massage Therapy in Depressed People. *The Journal of Clinical Psychiatry*, 71(07), 894–901. <https://doi.org/10.4088/jcp.09r05009blu>

103 Field, T. (2016). Massage therapy research review. *Complementary Therapies in Clinical Practice*, 24, 19–31. <https://doi.org/10.1016/j.ctcp.2016.04.005>

104 Field, T. M. (1998). Massage therapy effects. *American Psychologist*, 53(12), 1270–1281. <https://doi.org/10.1037/0003-066x.53.12.1270>

105 Javnbakht, M., Hejazi Kenari, R., & Ghasemi, M. (2009). Effects of yoga on depression and anxiety of women. *Complementary Therapies in Clinical Practice*, 15(2), 102–104. <https://doi.org/10.1016/j.ctcp.2009.01.003>

106 Thirthalli, J., Naveen, G., Rao, M., Varambally, S., Christopher, R., & Gangadhar, B. (2013). Positive therapeutic and neurotropic effects of yoga in depression: A comparative study. *Indian Journal of Psychiatry*, 55(7), 400. <https://doi.org/10.4103/0019-5545.116313>

107 Even, C., Schröder, C. M., Friedman, S., & Rouillon, F. (2008). Efficacy of light therapy in nonseasonal depression: A systematic review. *Journal of Affective Disorders*, 108(1–2), 11–23. <https://doi.org/10.1016/j.jad.2007.09.008>

108 Tao, L., Jiang, R., Zhang, K., Qian, Z., Chen, P., Lv, Y., & Yao, Y. (2020). Light therapy in non-seasonal depression: An update meta-analysis. *Psychiatry Research*, 291, 113247. <https://doi.org/10.1016/j.psychres.2020.113247>

109 Penders, T. M., Stanciu, C. N., Schoemann, A. M., Ninan, P. T., Bloch, R., & Saeed, S. A. (2016). Bright Light Therapy as Augmentation of Pharmacotherapy for Treatment of Depression. *The Primary Care Companion For CNS Disorders*. <https://doi.org/10.4088/pcc.15r01906>

110 Information about medications taken from Stahl, S. M. (2014). *Prescriber's Guide: Stahl's Essential Psychopharmacology* (5th ed.). Cambridge University Press

Parkinson's Disease Protocol

1 U.S. Department of Health and Human Services. (2022, April 4). *Parkinson's disease: Causes, symptoms, and treatments*. National Institute on Aging. Retrieved August 23, 2022, from <https://www.nia.nih.gov/health/parkinsons-disease>

2 Chen, H., Zhang, S. M., Hernán, M. A., Willett, W. C., & Ascherio, A. (2002). Diet and Parkinson's disease: a potential role of dairy products in men. *Annals of neurology*, 52(6), 793–801. <https://doi.org/10.1002/ana.10381>

3 Park, M., Ross, G. W., Petrovitch, H., White, L. R., Masaki, K. H., Nelson, J. S., Tanner, C. M., Curb, J. D., Blanchette, P. L., & Abbott, R. D. (2005). Consumption of milk and calcium in midlife and the future risk of Parkinson disease. *Neurology*, 64(6), 1047–1051. <https://doi.org/10.1212/01.WNL.0000154532.98495.BF>

4 Kyzozis, A., Ghika, A., Stathopoulos, P., Vassilopoulos, D., Trichopoulos, D., & Trichopolou, A. (2013). Dietary and lifestyle variables in relation to incidence of Parkinson's disease in Greece. *European journal of epidemiology*, 28(1), 67–77. <https://doi.org/10.1007/s10654-012-9760-0>

5 Sääksjärvi, K., Knekt, P., Lundqvist, A., Männistö, S., Heliövaara, M., Rissanen, H., & Järvinen, R. (2013). A cohort study on diet and the risk of Parkinson's disease: the role of food groups and diet quality. *The British journal of nutrition*, 109(2), 329–337. <https://doi.org/10.1017/S0007114512000955>

6 Choi, H. K., Liu, S., & Curhan, G. (2005). Intake of purine-rich foods, protein, and dairy products and relationship to serum levels of uric acid: the Third National Health and Nutrition Examination Survey. *Arthritis and rheumatism*, 52(1), 283–289. <https://doi.org/10.1002/art.20761>

7 Liu R. H. (2003). Health benefits of fruit and vegetables are from additive and synergistic combinations of phytochemicals. *The American journal of clinical nutrition*, 78(3 Suppl), 517S–520S. <https://doi.org/10.1093/ajcn/78.3.517S>

8 Seidl, S. E., Santiago, J. A., Bilyk, H., & Potashkin, J. A. (2014). The emerging role of nutrition in Parkinson's disease. *Frontiers in aging neuroscience*, 6, 36. <https://doi.org/10.3389/fnagi.2014.00036>

9 Tarozzi, A., Angeloni, C., Malaguti, M., Morroni, F., Hrelia, S., & Hrelia, P. (2013). Sulforaphane as a potential protective phytochemical against neurodegenerative diseases. *Oxidative medicine and cellular longevity*, 2013, 415078. <https://doi.org/10.1155/2013/415078>

10 Kyuhou S. (2018). Preventive effects of ginsenoside on motor dysfunction following 6-hydroxydopamine injection in ovariectomized rats. *Neuroscience letters*, 448(1), 10–14. <https://doi.org/10.1016/j.neulet.2008.10.045>

11 Liu, L. X., Chen, W. F., Xie, J. X., & Wong, M. S. (2008). Neuroprotective effects of ginsenoside on dopaminergic neurons in the mice model of Parkinson's disease. *Neuroscience research*, 60(2), 156–161. <https://doi.org/10.1016/j.neures.2007.10.005>

- 12 Dos Santos Delabary, M., Komerowski, I. G., Monteiro, E. P., Costa, R. R., & Haas, A. N. (2018). Effects of dance practice on functional mobility, motor symptoms and quality of life in people with Parkinson's disease: a systematic review with meta-analysis. *Aging clinical and experimental research*, 30(7), 727–735. <https://doi.org/10.1007/s40520-017-0836-2>
- 13 Stegemöller, E. L., Hurt, T. R., O'Connor, M. C., Camp, R. D., Green, C. W., Pattee, J. C., & Williams, E. K. (2018). Experiences of Persons With Parkinson's Disease Engaged in Group Therapeutic Singing. *Journal of music therapy*, 54(4), 405–431. <https://doi.org/10.1093/jmt/thx012>
- 14 Barnish, J., Atkinson, R. A., Barran, S. M., & Barnish, M. S. (2016). Potential Benefit of Singing for People with Parkinson's Disease: A Systematic Review. *Journal of Parkinson's disease*, 6(3), 473–484. <https://doi.org/10.3233/JPD-160837>
- 15 Fogg-Rogers, L., Buetow, S., Talmage, A., McCann, C. M., Leão, S. H., Tippet, L., Leung, J., McPherson, K. M., & Purdy, S. C. (2016). Choral singing therapy following stroke or Parkinson's disease: an exploration of participants' experiences. *Disability and rehabilitation*, 38(10), 952–962. <https://doi.org/10.3109/09638288.2015.1068875>
- 16 García-Casares, N., Martín-Colom, J. E., & García-Arnés, J. A. (2018). Music Therapy in Parkinson's Disease. *Journal of the American Medical Directors Association*, 19(12), 1054–1062. <https://doi.org/10.1016/j.jamda.2018.09.025>
- 17 Machado Sotomayor, M. J., Arufe-Giráldez, V., Ruiz-Rico, G., & Navarro-Patón, R. (2021). Music Therapy and Parkinson's Disease: A Systematic Review from 2015-2020. *International journal of environmental research and public health*, 18(21), 11618. <https://doi.org/10.3390/ijerph182111618>
- 18 Morris, I. B., Vasudevan, E., Schedel, M., Weymouth, D., Loomis, J., Pinkhasov, T., & Muratori, L. M. (2019). Music to One's Ears: Familiarity and Music Engagement in People with Parkinson's Disease. *Frontiers in neuroscience*, 13, 661. <https://doi.org/10.3389/fnins.2019.00661>
- 19 Pohl, P., Wressle, E., Lundin, F., Enthoven, P., & Dizdar, N. (2020). Group-based music intervention in Parkinson's disease - findings from a mixed-methods study. *Clinical rehabilitation*, 34(4), 533–544. <https://doi.org/10.1177/0269215520907669>
- 20 Vecchio, L. M., Meng, Y., Xhima, K., Lipsman, N., Hamani, C., & Aubert, I. (2018). The Neuroprotective Effects of Exercise: Maintaining a Healthy Brain Throughout Aging. *Brain plasticity (Amsterdam, Netherlands)*, 4(1), 17–52. <https://doi.org/10.3233/BPL-180069>
- 21 Nag, N., & Jelinek, G. A. (2019). A Narrative Review of Lifestyle Factors Associated with Parkinson's Disease Risk and Progression. *Neuro-degenerative diseases*, 19(2), 51–59. <https://doi.org/10.1159/000502292>
- 22 Oliveira de Carvalho, A., Filho, A., Murillo-Rodriguez, E., Rocha, N. B., Carta, M. G., & Machado, S. (2018). Physical Exercise for Parkinson's Disease: Clinical and Experimental Evidence. *Clinical practice and epidemiology in mental health: CP & EMH*, 14, 89–98. <https://doi.org/10.2174/1745017901814010089>
- 23 McKenzie, J. A., Spielman, L. J., Pointer, C. B., Lowry, J. R., Bajwa, E., Lee, C. W., & Klegeris, A. (2017). Neuroinflammation as a Common Mechanism Associated with the Modifiable Risk Factors for Alzheimer's and Parkinson's Diseases. *Current aging science*, 10(3), 158–176. <https://doi.org/10.2174/1874609810666170315113244>
- 24 Oliveira de Carvalho, A., Filho, A., Murillo-Rodriguez, E., Rocha, N. B., Carta, M. G., & Machado, S. (2018). Physical Exercise for Parkinson's Disease: Clinical and Experimental Evidence. *Clinical practice and epidemiology in mental health: CP & EMH*, 14, 89–98. <https://doi.org/10.2174/1745017901814010089>
- 25 Fall, P. A., Fredrikson, M., Axelson, O., & Granérus, A. K. (1999). Nutritional and occupational factors influencing the risk of Parkinson's disease: a case-control study in southeastern Sweden. *Movement disorders: official journal of the Movement Disorder Society*, 14(1), 28–37. [https://doi.org/10.1002/1531-8257\(199901\)14:1<28::aid-mds1007>3.0.co;2-o](https://doi.org/10.1002/1531-8257(199901)14:1<28::aid-mds1007>3.0.co;2-o)
- 26 Hellenbrand, W., Boeing, H., Robra, B. P., Seidler, A., Vieregge, P., Nischan, P., Joerg, J., Oertel, W. H., Schneider, E., & Ulm, G. (1996). Diet and Parkinson's disease. II: A possible role for the past intake of specific nutrients. Results from a self-administered food-frequency questionnaire in a case-control study. *Neurology*, 47(3), 644–650. <https://doi.org/10.1212/wnl.47.3.644>
- 27 Chong, R., Wakade, C., Seamon, M., Giri, B., Morgan, J., & Purohit, S. (2021). Niacin Enhancement for Parkinson's Disease: An Effectiveness Trial. *Frontiers in aging neuroscience*, 13, 667032. <https://doi.org/10.3389/fnagi.2021.667032>
- 28 Zhao, X., Zhang, M., Li, C., Jiang, X., Su, Y., & Zhang, Y. (2019). Benefits of Vitamins in the Treatment of Parkinson's Disease. *Oxidative medicine and cellular longevity*, 2019, 9426867. <https://doi.org/10.1155/2019/9426867>
- 29 Nagayama, H., Hamamoto, M., Ueda, M., Nito, C., Yamaguchi, H., & Katayama, Y. (2004). The effect of ascorbic acid on the pharmacokinetics of levodopa in elderly patients with Parkinson disease. *Clinical neuropharmacology*, 27(6), 270–273. <https://doi.org/10.1097/01.wnf.0000150865.21759.bc>
- 30 Newmark, H. L., & Newmark, J. (2007). Vitamin D and Parkinson's disease—a hypothesis. *Movement disorders: official journal of the Movement Disorder Society*, 22(4), 461–468. <https://doi.org/10.1002/mds.21317>
- 31 Liu, Y., & Zhang, B. S. (2014). Serum 25-hydroxyvitamin D predicts severity in Parkinson's disease patients. *Neurological sciences: official journal of the Italian Neurological Society and of the Italian Society of Clinical Neurophysiology*, 35(1), 67–71. <https://doi.org/10.1007/s10072-013-1539-x>
- 32 Sleeman, I., Aspray, T., Lawson, R., Coleman, S., Duncan, G., Khoo, T. K., Schoenmakers, I., Rochester, L., Burn, D., & Yarnall, A. (2017). The Role of Vitamin D in Disease Progression in Early Parkinson's Disease. *Journal of Parkinson's disease*, 7(4), 669–675. <https://doi.org/10.3233/JPD-171122>
- 33 de Rijk, M. C., Breteler, M. M., den Breeijen, J. H., Launer, J. J., Grobbee, D. E., van der Meché, F. G., & Hofman, A. (1997). Dietary antioxidants and Parkinson disease. The Rotterdam Study. *Archives of neurology*, 54(6), 762–765. <https://doi.org/10.1001/archneur.1997.00550180070015>
- 34 Fahn S. (1992). A pilot trial of high-dose alpha-tocopherol and ascorbate in early Parkinson's disease. *Annals of neurology*, 32 Suppl, S128–S132. <https://doi.org/10.1002/ana.410320722>
- 35 da Silva, T. M., Munhoz, R. P., Alvarez, C., Naliwaiko, K., Kiss, A., Andreatini, R., & Ferraz, A. C. (2008). Depression in Parkinson's disease: a double-blind, randomized, placebo-controlled pilot study of omega-3 fatty-acid supplementation. *Journal of affective disorders*, 111(2–3), 351–359. <https://doi.org/10.1016/j.jad.2008.03.008>
- 36 Seidl, S. E., Santiago, J. A., Bilyk, H., & Potashkin, J. A. (2014). The emerging role of nutrition in Parkinson's disease. *Frontiers in aging neuroscience*, 6, 36. <https://doi.org/10.3389/fnagi.2014.00036>
- 37 Ozsoy, O., Seval-Celik, Y., Hacıoglu, G., Yargicoglu, P., Demir, R., Agar, A., & Aslan, M. (2011). The influence and the mechanism of docosahexaenoic acid on a mouse model of Parkinson's disease. *Neurochemistry international*, 59(5), 664–670. <https://doi.org/10.1016/j.neuint.2011.06.012>
- 38 Samadi, P., Grégoire, L., Rouillard, C., Bédard, P. J., Di Paolo, T., & Lévesque, D. (2006). Docosahexaenoic acid reduces levodopa-induced dyskinesias in 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine monkeys. *Annals of neurology*, 59(2), 282–288. <https://doi.org/10.1002/ana.20738>
- 39 Cilia, R., Laguna, J., Cassani, E., Cereda, E., Pozzi, N. G., Isaia, I. U., Contin, M., Barichella, M., & Pezzoli, G. (2017). Mucuna pruriens in Parkinson disease: A double-blind, randomized, controlled, crossover study. *Neurology*, 89(5), 432–438. <https://doi.org/10.1212/WNL.0000000000004175>
- 40 Katzenschlager, R., Evans, A., Manson, A., Patsalos, P. N., Ratnaraj, N., Watt, H., Timmermann, L., Van der Giessen, R., & Lees, A. J. (2004). Mucuna pruriens in Parkinson's disease: a double blind clinical and pharmacological study. *Journal of neurology, neurosurgery, and psychiatry*, 75(12), 1672–1677. <https://doi.org/10.1136/jnnp.2003.028761>
- 41 Song, S., Nie, Q., Li, Z., & Du, G. (2016). Curcumin improves neurofunctions of 6-OHDA-induced parkinsonian rats. *Pathology, research and practice*, 212(4), 247–251. <https://doi.org/10.1016/j.prp.2015.11.012>
- 42 Khatri, D. K., & Juvekar, A. R. (2016). Kinetics of Inhibition of Monoamine Oxidase Using Curcumin and Ellagic Acid. *Pharmacognosy magazine*, 12(Suppl 2), S116–S120. <https://doi.org/10.4103/0973-1296.182168>
- 43 Khatri, D. K., & Juvekar, A. R. (2016). Kinetics of Inhibition of Monoamine Oxidase Using Curcumin and Ellagic Acid. *Pharmacognosy magazine*, 12(Suppl 2), S116–S120. <https://doi.org/10.4103/0973-1296.182168>
- 44 Nebriisi, E. E. (2021). Neuroprotective Activities of Curcumin in Parkinson's Disease: A Review of the Literature. *International journal of molecular sciences*, 22(20), 11248. <https://doi.org/10.3390/ijms222011248>
- 45 Malar, D. S., Prasanth, M. I., Brimson, J. M., Sharika, R., Sivamaruthi, B. S., Chaiyasut, C., & Tencomnao, T. (2020). Neuroprotective Properties of Green Tea (Camellia sinensis) in Parkinson's Disease: A Review. *Molecules (Basel, Switzerland)*, 25(17), 3926. <https://doi.org/10.3390/molecules25173926>
- 46 Xu, Y., Zhang, Y., Quan, Z., Wong, W., Guo, J., Zhang, R., Yang, Q., Dai, R., McGeer, P. L., & Qing, H. (2016). Epigallocatechin gallate (EGCG) Inhibits Alpha-Synuclein Aggregation: A Potential Agent for Parkinson's Disease. *Neurochemical research*, 41(10), 2788–2796. <https://doi.org/10.1007/s11064-016-1995-9>
- 47 Zhao, J., Liang, Q., Sun, Q., Chen, C., Xu, L., Ding, Y., & Zhou P. (2017). (-)-Epigallocatechin-3-gallate (EGCG) inhibits fibrillation, disaggregates amyloid fibrils of α -synuclein, and protects PC12 cells against α -synuclein-induced toxicity. *RSC Adv*. 7:32508–32517. doi: 10.1039/C7RA03752J.
- 48 Bieschke, J., Russ, J., Friedrich, R. P., Ehrnhoefer, D. E., Wobst, H., Neugebauer, K., & Wanker, E. E. (2010). EGCG remodels mature alpha-synuclein and amyloid-beta fibrils and reduces cellular toxicity. *Proceedings of the National Academy of Sciences of the United States of America*, 107(17), 7710–7715. <https://doi.org/10.1073/pnas.0910723107>
- 49 Kim, K. H., Lee, D., Lee, H. L., Kim, C. E., Jung, K., & Kang, K. S. (2018). Beneficial effects of Panax ginseng for the treatment and prevention of neurodegenerative diseases: past findings and future directions. *Journal of ginseng research*, 42(3), 239–247. <https://doi.org/10.1016/j.jgr.2017.03.011>
- 50 Hu, S., Han, R., Mak, S., & Han, Y. (2011). Protection against 1-methyl-4-phenylpyridinium ion (MPP+)-induced apoptosis by water extract of ginseng (Panax ginseng C.A. Meyer) in SH-SY5Y cells. *Journal of ethnopharmacology*, 135(1), 34–42. <https://doi.org/10.1016/j.jep.2011.02.017>
- 51 Miraj, S., Rafieian-Kopaei, & Kiani, S. (2017). Melissa officinalis L: A Review Study With an Antioxidant Prospective. *Journal of evidence-based complementary & alternative medicine*, 22(3), 385–394. <https://doi.org/10.1177/2156587216663433>
- 52 Martins, E. N., Pessano, N. T., Leal, L., Roos, D. H., Folmer, V., Puntel, G. O., Rocha, J. B., Aschner, M., Ávila, D. S., & Puntel, R. L. (2012). Protective effect of Melissa officinalis aqueous extract against Mn-induced oxidative stress in chronically exposed mice. *Brain research bulletin*, 87(1), 74–79. <https://doi.org/10.1016/j.brainresbull.2011.10.003>
- 53 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, Hericium erinaceus (Higher Basidiomycetes) from Malaysia. *International journal of medicinal mushrooms*, 15(6), 539–554. <https://doi.org/10.1615/intjmed-mushrv15.i6.30>
- 54 Wong, K. H., Naidu, M., David, P., Abdullah, M. A., Abdullah, N., Kuppusamy, U. R., & Sabaratnam, V. (2011). Peripheral Nerve Regeneration Following Crush Injury to Rat Peroneal Nerve by Aqueous Extract of Medicinal Mushroom Hericium erinaceus (Bull.: Fr) Pers. (Aphyllphoromycetidae). *Evidence-based complementary and alternative medicine: eCAM*, 2011, 580752. <https://doi.org/10.1093/ecam/neq662>
- 55 Cheng, C., & Zhu, X. (2019). Cordycepin mitigates MPTP-induced Parkinson's disease through inhibiting TLR/NF- κ B signaling pathway. *Life sciences*, 223, 120–127. <https://doi.org/10.1016/j.lfs.2019.02.037>

56 Ren, Z. L., Wang, C. D., Wang, T., Ding, H., Zhou, M., Yang, N., Liu, Y. Y., & Chan, P. (2019). Ganoderma lucidum extract ameliorates MPTP-induced parkinsonism and protects dopaminergic neurons from oxidative stress via regulating mitochondrial function, autophagy, and apoptosis. *Acta pharmacologica Sinica*, 40(4), 441–450. <https://doi.org/10.1038/s41401-018-0077-8>

57 de Oliveria, D. M., Barreto, G., De Andrade, D. V., Saraceno, E., Aon-Bertolino, L., Capani, F., Dos Santos El Bachá, R., & Giraldez, L. D. (2009). Cytoprotective effect of Valeriana officinalis extract on an in vitro experimental model of Parkinson disease. *Neurochemical research*, 34(2), 215–220. <https://doi.org/10.1007/s11064-008-9749-y>

58 Ingale, S. P., & Kasture, S. B. (2017). Protective Effect of Standardized Extract of Passiflora incarnata Flower in Parkinson's and Alzheimer's Disease. *Ancient science of life*, 36(4), 200–206. https://doi.org/10.4103/asl.ASL_231_16

59 Universidad Complutense de Madrid. (2009, May 13). Medicinal Plant, St John's Wort, May Reduce Neuronal Degeneration Caused By Parkinson's Disease. *ScienceDaily*. Retrieved August 21, 2022 from www.sciencedaily.com/releases/2009/05/090511181252.htm

60 Ramassamy, C., Clostre, F., Christen, Y., & Costentin, J. (1990). Prevention by a Ginkgo biloba extract (GBE 761) of the dopaminergic neurotoxicity of MPTP. *The Journal of pharmacy and pharmacology*, 42(11), 785–789. <https://doi.org/10.1111/j.2042-7158.1990.tb07021.x>

61 Rojas, P., Serrano-García, N., Mares-Sámano, J. J., Medina-Campos, O. N., Pedraza-Chaverri, J., & Ogren, S. O. (2008). EGB761 protects against nigrostriatal dopaminergic neurotoxicity in 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine-induced Parkinsonism in mice: role of oxidative stress. *The European journal of neuroscience*, 28(1), 41–50. <https://doi.org/10.1111/j.1460-9568.2008.06314.x>

Vertigo and Tinnitus Protocol

1 <https://www.nidcd.nih.gov/health/menieres-disease>

2 Zhuang JM, Chen HL (1985) Treatment of tinnitus with Cordyceps infusion: a report of 23 cases. *Fujian Med J*; <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6926930/>

3 Shaito A, Thuan DTB, Phu HT, Nguyen THD, Hasan H, Halabi S, Abdelhady S, Nasrallah GK, Eid AH, Pintus G. Herbal Medicine for Cardiovascular Diseases: Efficacy, Mechanisms, and Safety. *Front Pharmacol*. 2020 Apr 7;11:422. doi: 10.3389/fphar.2020.00422. PMID: 32317975; PMCID: PMC7155419. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7155419/>

4 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, *Hericium erinaceus* (Higher Basidiomycetes) from Malaysia. *International journal of medicinal mushrooms*, 15(6), 539–554. <https://doi.org/10.1615/intjmed-mushrv15.i6.30>

5 Sereda M, Xia J, Scutt P, Hilton MP, El Refaie A, Hoare DJ. Ginkgo biloba for tinnitus. *Cochrane Database Syst Rev*. 2019 Dec 23;2019(12):CD013514. doi: 10.1002/14651858.CD013514. PMCID: PMC6926930

6 Smith GS, Romanelli-Gobbi M, Gray-Karagrigroriou E, Artz GJ. Complementary and integrative treatments: tinnitus. *Otolaryngologic Clinics of North America* 2013;46:389-408

7 Dogan R, Sjostrand AP, Yenigun A, Karatas E, Kocyigit A, Ozturan O. Influence of Ginkgo Biloba extract (EGB 761) on expression of IL-1 Beta, IL-6, TNF-alfa, HSP-70, HSF-1 and COX-2 after noise exposure in the rat cochlea. *Auris Nasus Larynx* 2018;45(4):680-5

8 Tagoe, T., Deeping, D., & Hamann, M. (2017). Saturation of long-term potentiation in the dorsal cochlear nucleus and its pharmacological reversal in an experimental model of tinnitus. *Experimental neurology*, 292, 1–10. <https://doi.org/10.1016/j.expneurol.2017.02.011> <https://pubmed.ncbi.nlm.nih.gov/28214516/>

9 Cevette, M. J., Barrs, D. M., Patel, A., Conroy, K. P., Sydlowski, S., Noble, B. N., Nelson, G. A., & Stepanek, J. (2011). Phase 2 study examining magnesium-dependent tinnitus. *The international tinnitus journal*, 16(2), 168–173. <https://pubmed.ncbi.nlm.nih.gov/22249877/>

Menopause Protocol

1 Choi, U. K., Lee, O. H., Yim, J. H., Cho, C. W., Rhee, Y. K., Lim, S. I., & Kim, Y. C. (2010). Hypolipidemic and antioxidant effects of dandelion (*Taraxacum officinale*) root and leaf on cholesterol-fed rabbits. *International journal of molecular sciences*, 11(1), 67–78. <https://doi.org/10.3390/ijms11010067>

2 Jolly JJ, Chin KY, Alias E, Chua KH, Soelaiman IN. Protective Effects of Selected Botanical Agents on Bone. *Int J Environ Res Public Health*. 2018;15(5):963. Published 2018 May 11. doi:10.3390/ijerph15050963

3 Zhang G, Qin L, Shi Y. Epimedium-derived phytoestrogen flavonoids exert beneficial effect on preventing bone loss in late postmenopausal women: a 24-month randomized, double-blind and placebo-controlled trial. *J Bone Miner Res*. 2007;22(7):1072-1079. doi:10.1359/jbmr.070405

4 Milligan, S. R., Kalita, J. C., Heyerick, A., Rong, H., De Cooman, L., & De Keukeleire, D. (1999). Identification of a potent phytoestrogen in hops (*Humulus lupulus* L.) and beer. *The Journal of clinical endocrinology and metabolism*, 84(6), 2249–2252. <https://doi.org/10.1210/jcem.84.6.5887>

5 Chen, Y. M., Wang, I. L., Zhu, X. Y., Chiu, W. C., & Chiu, Y. S. (2021). Red Clover Isoflavones Influence Estradiol Concentration, Exercise Performance, and Gut Microbiota in Female Mice. *Frontiers in nutrition*, 8, 623698. <https://doi.org/10.3389/fnut.2021.623698>

6 Parsons, M., Simpson, M., & Ponton, T. (1999). Raspberry leaf and its effect on labour: safety and efficacy. *Australian College of Midwives Incorporated journal*, 12(3), 20–25. [https://doi.org/10.1016/s1031-170x\(99\)80008-7](https://doi.org/10.1016/s1031-170x(99)80008-7)

7 Adhikari, B. M., Bajracharya, A., & Shrestha, A. K. (2015). Comparison of nutritional properties of Stinging nettle (*Urtica dioica*) flour with wheat and barley

flours. *Food science & nutrition*, 4(1), 119–124. <https://doi.org/10.1002/fsn3.259>

8 Modi MB, Donga SB, Dei L. Clinical evaluation of Ashokarishta, Ashwagandha Churna and Praval Pishti in the management of menopausal syndrome. *Ayu*. 2012 Oct;33(4):511-6. doi: 10.4103/0974-8520.110529. PMID: 23723668; PMCID: PMC3665193

9 Lopresti AL, Smith SJ, Malvi H, Kodgule R. An investigation into the stress-relieving and pharmacological actions of an ashwagandha (*Withania somnifera*) extract: A randomized, double-blind, placebo-controlled study. *Medicine (Baltimore)*. 2019 Sep;98(37):e17186. doi: 10.1097/MD.00000000000017186. PMID: 31517876; PMCID: PMC6750292

10 Salve, J., Pate, S., Debnath, K., & Langade, D. (2019). Adaptogenic and Anxiolytic Effects of Ashwagandha Root Extract in Healthy Adults: A Double-blind, Randomized, Placebo-controlled Clinical Study. *Cureus*, 11(12), e6466. <https://doi.org/10.7759/cureus.6466>

11 Sebastian, L. A., & Kear, T. (2021). The Effect of Lavender Aromatherapy on Heart Rate, Blood Pressure, and Perceived Stress Among Cardiac Rehabilitation Patients: A Pilot Study. *Holistic nursing practice*, 10.1097/HNP.0000000000000430. Advance online publication. <https://doi.org/10.1097/HNP.0000000000000430>

12 Ghazizadeh, J., Sadigh-Eteghad, S., Marx, W., Fakhari, A., Hamedeyazdan, S., Torbati, M., Taheri-Tarighi, S., Araj-Khodaei, M., & Mirghafourvand, M. (2021). The effects of lemon balm (*Melissa officinalis* L.) on depression and anxiety in clinical trials: A systematic review and meta-analysis. *Phytotherapy research: PTR*, 35(12), 6690–6705. <https://doi.org/10.1002/ptr.7252>

13 Chong, P. S., Fung, M. L., Wong, K. H., & Lim, L. W. (2019). Therapeutic Potential of *Hericium erinaceus* for Depressive Disorder. *International journal of molecular sciences*, 21(1), 163. <https://doi.org/10.3390/ijms21010163>

14 Koshovyi, O., Raal, A., Kireyev, I., Tryshchuk, N., Ilina, T., Romanenko, Y., Kovalenko, S. M., & Bunyatyan, N. (2021). Phytochemical and Psychotropic Research of Motherwort (*Leonurus cardiaca* L.) Modified Dry Extracts. *Plants (Basel, Switzerland)*, 10(2), 230. <https://doi.org/10.3390/plants10020230>

15 Matsuzaki, H., Shimizu, Y., Iwata, N., Kamiuchi, S., Suzuki, F., Iizuka, H., Hibino, Y., & Okazaki, M. (2013). Antidepressant-like effects of a water-soluble extract from the culture medium of *Ganoderma lucidum* mycelia in rats. *BMC complementary and alternative medicine*, 13, 370. <https://doi.org/10.1186/1472-6882-13-370>

16 Geng, P., Siu, K. C., Wang, Z., & Wu, J. Y. (2017). Antifatigue Functions and Mechanisms of Edible and Medicinal Mushrooms. *BioMed research international*, 2017, 9648496. <https://doi.org/10.1155/2017/9648496>

17 Pilkington, K., Boshnakova, A., & Richardson, J. (2006). St John's wort for depression: time for a different perspective?. *Complementary therapies in medicine*, 14(4), 268–281. <https://doi.org/10.1016/j.ctim.2006.01.003>

18 van Die, M. D., Burger, H. G., Teede, H. J., & Bone, K. M. (2013). Vitex agnus-castus extracts for female reproductive disorders: a systematic review of clinical trials. *Planta medica*, 79(7), 562–575. <https://doi.org/10.1055/s-0032-1327831>

19 Cerqueira, R. O., Frey, B. N., Leclerc, E., Brietzke, E. (2017). Vitex agnus castus for premenstrual syndrome and premenstrual dysphoric disorder: a systematic review. *Archives of women's mental health*, 20(6), 713–719. <https://doi.org/10.1007/s00737-017-0791-0>

20 Feyzollahi, Z., Mohseni Kouchesfehiani, H., Jalali, H., Eslimi-Esfahani, D., & Sheikh Hosseini, A. (2021). Effect of Vitex agnus-castus ethanolic extract on hypothalamic KISS-1 gene expression in a rat model of polycystic ovary syndrome. *Avicenna journal of phytomedicine*, 11(3), 292–301

21 Mehrpooya, M., Rabiee, S., Larki-Harchegani, A., Fallahian, A. M., Moradi, A., Ataei, S., & Javad, M. T. (2018). A comparative study on the effect of "black cohosh" and "evening primrose oil" on menopausal hot flashes. *Journal of education and health promotion*, 7, 36. https://doi.org/10.4103/jehp.jehp_81_17

22 Amato, P., Christophe, S., & Mellon, P. L. (2002). Estrogenic activity of herbs commonly used as remedies for menopausal symptoms. *Menopause (New York, N.Y.)*, 9(2), 145–150. <https://doi.org/10.1097/00042192-200203000-00010>

23 Safdari, F., Motaghi Dastenaee, B., Kheiri, S., & Karimiankakolaki, Z. (2021). Effect of Evening Primrose Oil on Postmenopausal Psychological Symptoms: A Triple-Blind Randomized Clinical Trial. *Journal of menopausal medicine*, 27(2), 58–65. <https://doi.org/10.6118/jmm.21010>

24 Srivastava, J. K., Shankar, E., & Gupta, S. (2010). Chamomile: A herbal medicine of the past with bright future. *Molecular medicine reports*, 3(6), 895–901. <https://doi.org/10.3892/mmr.2010.377>

25 Franco, L., Sánchez, C., Bravo, R., Rodriguez, A., Barriga, C., & Juárez, J. C. (2012). The sedative effects of hops (*Humulus lupulus*), a component of beer, on the activity/rest rhythm. *Acta physiologica Hungarica*, 99(2), 133–139. <https://doi.org/10.1556/APhysiol.99.2012.2.6>

26 Kargozar, R., Azizi, H., & Salari, R. (2017). A review of effective herbal medicines in controlling menopausal symptoms. *Electronic physician*, 9(11), 5826–5833. <https://doi.org/10.19082/5826>

27 Guerrero, F. A., & Medina, G. M. (2017). Effect of a medicinal plant (*Passiflora incarnata* L.) on sleep. *Sleep science (Sao Paulo, Brazil)*, 10(3), 96–100. <https://doi.org/10.5935/1984-0063.20170018>

28 Fedurco, M., Gregorová, J., Šebřlová, K., Kantorová, J., Peš, O., Baur, R., Sigel, E., & Táborská, E. (2015). Modulatory Effects of *Eschscholzia californica* Alkaloids on Recombinant GABAA Receptors. *Biochemistry research international*, 2015, 617620. <https://doi.org/10.1155/2015/617620>

29 Lindahl, O., & Lindwall, L. (1989). Double blind study of a valerian preparation. *Pharmacology, biochemistry, and behavior*, 32(4), 1065–1066. [https://doi.org/10.1016/0091-3057\(89\)90082-8](https://doi.org/10.1016/0091-3057(89)90082-8)

30 Eby, G. A., & Eby, K. L. (2006). Rapid recovery from major depression using mag-

- nesium treatment. Medical hypotheses, 67(2), 362–370. <https://doi.org/10.1016/j.mehy.2006.01.047>
- 31 Takeda, A., Watanuki, E., & Koyama, S. (2017). Effects of Inhalation Aromatherapy on Symptoms of Sleep Disturbance in the Elderly with Dementia. Evidence-based complementary and alternative medicine: eCAM, 2017, 1902807. <https://doi.org/10.1155/2017/1902807>
- 32 Patel, S., & Goyal, A. (2012). Recent developments in mushrooms as anti-cancer therapeutics: a review. 3 Biotech, 2(1), 1–15. <https://doi.org/10.1007/s13205-011-0036-2>
- 33 Zhao, J., Dasmahapatra, A. K., Khan, S. I., & Khan, I. A. (2008). Anti-aromatase activity of the constituents from damiana (*Turnera diffusa*). Journal of ethnopharmacology, 120(3), 387–393. <https://doi.org/10.1016/j.jep.2008.09.016>
- 34 Ghorbani, Z., Mirghafourvand, M., Charandabi, S. M., & Javadzadeh, Y. (2019). The effect of ginseng on sexual dysfunction in menopausal women: A double-blind, randomized, controlled trial. Complementary therapies in medicine, 45, 57–64. <https://doi.org/10.1016/j.ctim.2019.05.015>
- 35 Lee, H. W., Choi, J., Lee, Y., Kil, K. J., & Lee, M. S. (2016). Ginseng for managing menopausal woman's health: A systematic review of double-blind, randomized, placebo-controlled trials. Medicine, 95(38), e4914. <https://doi.org/10.1097/MD.0000000000004914>
- 36 Ghorbani, Z., Mirghafourvand, M., Charandabi, S. M., & Javadzadeh, Y. (2019). The effect of ginseng on sexual dysfunction in menopausal women: A double-blind, randomized, controlled trial. Complementary therapies in medicine, 45, 57–64. <https://doi.org/10.1016/j.ctim.2019.05.015>
- 37 Cruz, D. A. C. (2018, February 12). The action of herbal medicine on the libido: aspects of nutritional intervention in increasing sexual desire. SpringerLink. Retrieved May 18, 2022, from <https://link.springer.com/article/10.1186>
- 38 rding, C. M., Fisher, L., Papakostas, G., Farabaugh, A., Sonawalla, S., Fava, M., & Mischoulon, D. (2008). A double-blind, randomized, pilot dose-finding study of maca root (*L. meyenii*) for the management of SSRI-induced sexual dysfunction. CNS neuroscience & therapeutics, 14(3), 182–191. <https://doi.org/10.1111/j.1755-5949.2008.00052.x>
- 39 Ranjbar, H., & Ashrafzadeh, A. (2019). Effects of saffron (*Crocus sativus*) on sexual dysfunction among men and women: A systematic review and meta-analysis. Avicenna journal of phytomedicine, 9(5), 419–427
- 40 Alok, S., Jain, S. K., Verma, A., Kumar, M., Mahor, A., & Sabharwal, M. (2013). Plant profile, phytochemistry and pharmacology of *Asparagus racemosus* (Shatavari): A review. Asian Pacific Journal of Tropical Disease, 3(3), 242–251. [https://doi.org/10.1016/S2222-1808\(13\)60049-3](https://doi.org/10.1016/S2222-1808(13)60049-3)
- 41 Kapoor E, Collazo-Clavell ML, Faubion SS. Weight gain in women at midlife: a concise review of the pathophysiology and strategies for management. Mayo Clinic Proceedings. 2017;92(10):1552–1558. doi:10.1016/j.mayocp.2017.08.004
- 42 Abdunour J, Doucet E, Brochu M, et al. The effect of the menopausal transition on body composition and cardiometabolic risk factors: a Montreal-Ottawa New Emerging Team group study. Menopause. 2012;19(7):760-767. doi:10.1097/gme.0b013e318240f6f3
- 43 Kozakowski J, Gietka-Czernel M, Leszczyńska D, Majos A. Obesity in menopause—our negligence or an unfortunate inevitability? Prz Menopauzalny. 2017;16(2):61-65. doi:10.5114/pm.2017.68594
- 44 Santos, L., Elliott-Sale, K. J., & Sale, C. (2017). Exercise and bone health across the lifespan. Biogerontology, 18(6), 931–946. <https://doi.org/10.1007/s10522-017-9732-6>
- 45 Beyer, K. M., Szabo, A. M., & Nattinger, A. B. (2016). Time Spent Outdoors, Depressive Symptoms, and Variation by Race and Ethnicity. American journal of preventive medicine, 51(3), 281–290. <https://doi.org/10.1016/j.amepre.2016.05.004>
- 46 Choudhary, D., Bhattacharyya, S., & Joshi, K. (2017). Body Weight Management in Adults Under Chronic Stress Through Treatment with Ashwagandha Root Extract: A Double-Blind, Randomized, Placebo-Controlled Trial. Journal of evidence-based complementary & alternative medicine, 22(1), 96–106. <https://doi.org/10.1177/2156587216641830>
- 47 Westerterp-Plantenga, M. S., Smeets, A., & Lejeune, M. P. (2005). Sensory and gastrointestinal satiety effects of capsaicin on food intake. International journal of obesity (2005), 29(6), 682–688. <https://doi.org/10.1038/sj.ijo.0802862>
- 48 Ranade, Manjiri, and Nikhil Mudgalkar. "A simple dietary addition of fenugreek seed leads to the reduction in blood glucose levels: A parallel group, randomized single-blind trial." Ayu vol. 38,1-2 (2017): 24-27. doi:10.4103/ayu.AYU_209_15 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5954247/>
- 49 Mathern, Jocelyn R et al. "Effect of fenugreek fiber on satiety, blood glucose and insulin response and energy intake in obese subjects." Phytotherapy research: PTR vol. 23,11 (2009): 1543-8. doi:10.1002/ptr.2795 [https://pubmed.ncbi.nlm.nih.gov/19353539/#:~:text=Fenugreek%20fiber%20\(8%20g\)%20significantly,related%20to%20postprandial%20blood%20glucose](https://pubmed.ncbi.nlm.nih.gov/19353539/#:~:text=Fenugreek%20fiber%20(8%20g)%20significantly,related%20to%20postprandial%20blood%20glucose)
- 50 Bae, JiYoung et al. "Fennel (*Foeniculum vulgare*) and Fenugreek (*Trigonella foenum-graecum*) Tea Drinking Suppresses Subjective Short-term Appetite in Overweight Women." Clinical nutrition research vol. 4,3 (2015): 168-74. doi:10.7762/cnr.2015.4.3.168 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4525133/>
- 51 Chevassus, Hugues et al. "A fenugreek seed extract selectively reduces spontaneous fat consumption in healthy volunteers." European journal of clinical pharmacology vol. 65,12 (2009): 1175-8. doi:10.1007/s00228-009-0733-5 <https://pubmed.ncbi.nlm.nih.gov/19809809/>
- 52 Bae J, Kim J, Choue R, Lim H. Fennel (*Foeniculum vulgare*) and Fenugreek (*Trigonella foenum-graecum*) Tea Drinking Suppresses Subjective Short-term Appetite in Overweight Women. Clin Nutr Res. 2015 Jul;4(3):168-74. doi: 10.7762/cnr.2015.4.3.168. Epub 2015 Jul 31. PMID: 2625133; PMCID: PMC4525133
- 53 Maharlouei, N., Tabrizi, R., Lankarani, K. B., Rezaianzadeh, A., Akbari, M., Kolahdooz, F., Rahimi, M., Keneshlou, F., & Asemi, Z. (2019). The effects of ginger intake on weight loss and metabolic profiles among overweight and obese subjects: A systematic review and meta-analysis of randomized controlled trials. Critical reviews in food science and nutrition, 59(11), 1753–1766. <https://doi.org/10.1080/10408398.2018.1427044>
- 54 Ebrahimzadeh Attari, V., Malek Mahdavi, A., Javadi, Z., Mahluji, S., Zununi Vahed, S., & Ostadrahimi, A. (2018). A systematic review of the anti-obesity and weight lowering effect of ginger (*Zingiber officinale* Roscoe) and its mechanisms of action. Phytotherapy research: PTR, 32(4), 577–585. <https://doi.org/10.1002/ptr.5986>
- 55 Hiwatashi K, Kosaka Y, Suzuki N, Hata K, Mukaiyama T, Sakamoto K, Shirakawa H, Komai M. Yamabushitake mushroom (*Herichium erinaceus*) improved lipid metabolism in mice fed a high-fat diet. Biosci Biotechnol Biochem. 2010;74(7):1447-51
- 56 hen Diling, Guo Yinrui, Qi Longkai, Tang Xiaocui, Liu Yadi, Feng Jiaxin, Zhu Xiangxiang, Zeng Miao, Shuai Ou, Wang Dongdong, Xie Yizhen, Burton B. Yang, Wu Qingping. Metabolic regulation of *Ganoderma lucidum* extracts in high sugar and fat diet-induced obese mice by regulating the gut-brain axis. Journal of Functional Foods. 2020;65:103639
- 57 Gao, Yihuai & Lan, Jin & Dai, Xihu & Ye, Jingxian & Zhou, Shufeng. A Phase I/II Study of Ling Zhi Mushroom *Ganoderma lucidum* (W.Curt.:Fr.) Lloyd (Aphyllophoromycetidae) Extract in Patients with Type II Diabetes Mellitus. International Journal of Medicinal Mushrooms. 2004;6. 33-40
- 58 Liang B, Guo Z, Xie F, Zhao A. Antihyperglycemic and antihyperlipidemic activities of aqueous extract of *Herichium erinaceus* in experimental diabetic rats. BMC Complement Altern Med. 2013;13:253
- 59 He X, Wang X, Fang J, Chang Y, Ning N, Guo H, Huang L, Huang X, Zhao Z. Structures, biological activities, and industrial applications of the polysaccharides from *Herichium erinaceus* (Lion's Mane) mushroom: A review. Int J Biol Macromol. 2017 Apr;97:228-237
- 60 Chang CJ, Lin CS, Lu CC, Martel J, Ko YF, Ojcius DM, Tseng SF, Wu TR, Chen YY, Young JD, Lai HC. *Ganoderma lucidum* reduces obesity in mice by modulating the composition of the gut microbiota. Nat Commun. 2015 Jun 23;6:7489. doi: 10.1038/ncomms8489. Erratum in: Nat Commun. 2017 Jul 11;8:16130. PMID: 26102296; PMCID: PMC4557287
- 61 Phipps, W. R., Martini, M. C., Lampe, J. W., Slavin, J. L., & Kurzer, M. S. (1993). Effect of flax seed ingestion on the menstrual cycle. The Journal of clinical endocrinology and metabolism, 77(5), 1215–1219. <https://doi.org/10.1210/jcem.77.5.8077314>
- 62 Harvey, P. W., & Darbre, P. (2004). Endocrine disruptors and human health: could oestrogenic chemicals in body care cosmetics adversely affect breast cancer incidence in women? Journal of applied toxicology: JAT, 24(3), 167–176. <https://doi.org/10.1002/jat.978>
- 63 Yang CZ, Yaniger SI, Jordan VC, Klein DJ, Bittner GD. Most plastic products release estrogenic chemicals: a potential health problem that can be solved. Environ Health Perspect. 2011 Jul;119(7):989-96. doi: 10.1289/ehp.1003220. Epub 2011 Mar 2. PMID: 21367689; PMCID: PMC3222987
- 64 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6372850/> Milart, P., Woźniakowska, E., & Wrona, W. (2018). Selected vitamins and quality of life in menopausal women. Przegląd menopauzalny = Menopause review, 17(4), 175–179. <https://doi.org/10.5114/pm.2018.81742>
- 65 Iwamoto, J., Takeda, T., & Sato, Y. (2006). Role of vitamin K2 in the treatment of postmenopausal osteoporosis. Current drug safety, 1(1), 87–97. <https://doi.org/10.2174/157488606775252629>
- 66 Laird, E., Ward, M., McSorley, E., Strain, J. J., & Wallace, J. (2010). Vitamin D and bone health: potential mechanisms. Nutrients, 2(7), 693–724. <https://doi.org/10.3390/nu2070693>
- 67 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6372850/> Milart, P., Woźniakowska, E., & Wrona, W. (2018). Selected vitamins and quality of life in menopausal women. Przegląd menopauzalny = Menopause review, 17(4), 175–179. <https://doi.org/10.5114/pm.2018.81742>
- 68 Derbyshire E. (2018). Micronutrient Intakes of British Adults Across Mid-Life: A Secondary Analysis of the UK National Diet and Nutrition Survey. Frontiers in nutrition, 5, 55. <https://doi.org/10.3389/fnut.2018.00055>
- 69 Ziaei, S., Kazemnejad, A., & Zareai, M. (2007). The effect of vitamin E on hot flashes in menopausal women. Gynecologic and obstetric investigation, 64(4), 204–207. <https://doi.org/10.1159/000106491>

Men's Sexual Health Protocol

- Washington University School of Medicine. (2016, February 22). In obese patients, 5 percent weight loss has significant health benefits: Initial weight loss lowers risk for diabetes, cardiovascular disease. ScienceDaily. Retrieved June 1, 2022 from www.sciencedaily.com/releases/2016/02/160222134137.htm
- Hudson, J., & Vimalanathan, S. (2011). Echinacea—A Source of Potent Antivirals for Respiratory Virus Infections. Pharmaceuticals, 4(7), 1019–1031. <https://doi.org/10.3390/ph4071019>
- Mazzanti, G., Battinelli, L., Pompeo, C., Serrilli, A. M., Rossi, R., Sauzullo, I., Mengoni, F., & Vullo, V. (2008). Inhibitory activity of *Melissa officinalis* L. extract on Herpes simplex virus type 2 replication. Natural product research, 22(16), 1433–1440. <https://doi.org/10.1080/14786410802075939>
- Chu, M., Ding, R., Chu, Z. Y., Zhang, M. B., Liu, X. Y., Xie, S. H., Zhai, Y. J., & Wang, Y. D. (2014). Role of berberine in anti-bacterial as a high-affinity LPS antagonist binding to TLR4/MD-2 receptor. BMC complementary and alternative medicine, 14, 89. <https://doi.org/10.1186/1472-6882-14-89>
- Gokgoz N.B., Avci F.G., Yoneten K.K., Alaybeyoglu B., Ozkirimli E., Sayar N.A., Kazan D., Sariyar Akbulut B. Response of *Escherichia coli* to prolonged berberine exposure. Microb. Drug Resist. 2017;23:531–544. doi: 10.1089/mdr.2016.0063
- Karaosmanoglu K., Sayar N.A., Kurnaz I.A., Akbulut B.S. Assessment of

Berberine as a multi-target antimicrobial: A multi-omics study for drug discovery and repositioning. *Omics J. Integr. Biol.* 2014;18:42–53. doi: 10.1089/omi.2013.0100

7 Petronio Petronio G, Cutuli MA, Magnifico I, Venditti N, Pietrangelo L, Vergalito F, Pane A, Scapagnini G, Di Marco R. In Vitro and In Vivo Biological Activity of Berberine Chloride against Uropathogenic *E. coli* Strains Using *Galleria mellonella* as a Host Model. *Molecules*. 2020 Oct 29;25(21):5010. doi: 10.3390/molecules25215010. PMID: 33137930; PMCID: PMC7662377

8 Ingólfsdóttir K. (2002). Usnic acid. *Phytochemistry*, 61(7), 729–736. [https://doi.org/10.1016/S0031-9422\(02\)00383-7](https://doi.org/10.1016/S0031-9422(02)00383-7)

9 Dobrescu, D., Tănăsescu, M., Mezdrea, A., Ivan, C., Ordosch, E., Neagoe, F., Rizeanu, A., Trifu, L., & Enescu, V. (1993). Contributions to the complex study of some lichens-*Usnea* genus. *Pharmacological studies on Usnea barbata and Usnea hirta* species. *Romanian journal of physiology: physiological sciences*, 30(1-2), 101–107

10 Khaleghi, S., Bakhtiari, M., Asadmobini, A., & Esmaeili, F. (2017). *Tribulus terrestris* Extract Improves Human Sperm Parameters In Vitro. *Journal of evidence-based complementary & alternative medicine*, 22(3), 407–412. <https://doi.org/10.1177/2156587216668110>

11 Cruz, D. A. C. (2018b, February 12). The action of herbal medicine on the libido: aspects of nutritional intervention in increasing sexual desire. *SpringerLink*. Retrieved June 1, 2022, from <https://link.springer.com>

12 Gonzales, G. F., Nieto, J., Rubio, J., & Gasco, M. (2006). Effect of Black maca (*Lepidium meyenii*) on one spermatogenic cycle in rats. *Andrologia*, 38(5), 166–172. <https://doi.org/10.1111/j.1439-0272.2006.00733.x>

13 Chen, Y. (2017, January 1). Functional study of *Cordyceps sinensis* and cordycepin in male reproduction: A review. *Science Direct*. Retrieved June 2, 2022, from <https://www.sciencedirect.com>

14 Chang, Y. (2008). Effect of *Cordyceps Militaris* Supplementation on Sperm Production, Sperm Motility and Hormones in Sprague-Dawley Rats. *World Scientific*. Retrieved February 6, 2022, from <https://www.worldscientific.com>

15 Ambiyi, V. R., Langade, D., Dongre, S., Aptikar, P., Kulkarni, M., & Dongre, A. (2013). Clinical Evaluation of the Spermatogenic Activity of the Root Extract of *Ashwagandha* (*Withania somnifera*) in Oligospermic Males: A Pilot Study. *Evidence-based complementary and alternative medicine: eCAM*, 2013, 571420. <https://doi.org/10.1155/2013/571420>

16 Saden-Krehula, M., Tajić, M., & Kolbah, D. (1971). Testosterone, epitestosterone and androstenedione in the pollen of Scotch pine *P. silvestris* L. *Experientia*, 27(1), 108–109. <https://doi.org/10.1007/BF02137770>

17 Pilz, S., Frisch, S., Koertke, H., Kuhn, J., Dreier, J., Obermayer-Pietsch, B., Wehr, E., & Zittermann, A. (2011). Effect of vitamin D supplementation on testosterone levels in men. *Hormone and metabolic research = Hormon- und Stoffwechselsforschung = Hormones et métabolisme*, 43(3), 223–225. <https://doi.org/10.1055/s-0030-1269854>

18 Prasad, A. S., Mantzoros, C. S., Beck, F. W., Hess, J. W., & Brewer, G. J. (1996). Zinc status and serum testosterone levels of healthy adults. *Nutrition* (Burbank, Los Angeles County, Calif.), 12(5), 344–348. [https://doi.org/10.1016/S0899-9007\(96\)80058-X](https://doi.org/10.1016/S0899-9007(96)80058-X)

19 Sudeep, H. V., Thomas, J. V., & Shyamprasad, K. (2020). A double blind, placebo-controlled randomized comparative study on the efficacy of phytosterol-enriched and conventional saw palmetto oil in mitigating benign prostate hyperplasia and androgen deficiency. *BMC urology*, 20(1), 86. <https://doi.org/10.1186/s12894-020-00648-9>

20 Liu, G. T. (1999b). Recent Advances in Research of Pharmacology and Clinical Applications of *Ganoderma* P.Karst. Species (*Aphyllphoromycetidae*) in China - *International Journal of Medicinal Mushrooms*, Volume 1, 1999, Issue 1 - Begell House Digital Library. Begell House Digital Library. Retrieved 2022, from <https://www.dl.begellhouse.com>

21 Peng, Z. (2001). Effect of Sporoderm-broken Spores of *Ganoderma lucidum* in the Treatment of Partial Androgen Deficiency of the Aging Male | *Journal of Guangzhou University of Traditional Chinese Medicine*; (6)2001. | WPRIM. Biblioteca. Retrieved 2022, from <https://pesquisa.bvsalud.org>

22 Kotta, S., Ansari, S. H., & Ali, J. (2013). Exploring scientifically proven herbal aphrodisiacs. *Pharmacognosy reviews*, 7(13), 1–10. <https://doi.org/10.4103/0973-7847.112832>

23 Dongre, S., Langade, D., & Bhattacharyya, S. (2015). Efficacy and Safety of *Ashwagandha* (*Withania somnifera*) Root Extract in Improving Sexual Function in Women: A Pilot Study. *BioMed research international*, 2015, 284154. <https://doi.org/10.1155/2015/284154>

24 Lekomtseva, Y., Zhukova, I., & Wacker, A. (2017). *Rhodiola rosea* in Subjects with Prolonged or Chronic Fatigue Symptoms: Results of an Open-Label Clinical Trial. *Complementary medicine research*, 24(1), 46–52. <https://doi.org/10.1159/000457918>

25 Bawa, A. S. (2006, November 20). Just a moment. . . Institute of Food Technologists (IFT). Retrieved May 16, 2022, from <https://ift.onlinelibrary.wiley.com/doi/abs/10.1111/j.1541-4337.2005.tb00073.x>

26 Definition & Facts for Erectile Dysfunction. (2021, July 29). National Institute of Diabetes and Digestive and Kidney Diseases. <https://www.niddk.nih.gov/health-information/urologic-diseases/erectile-dysfunction/definition-facts>

27 Dietary supplements for erectile dysfunction: A natural treatment for ED? (2021, April 10). Mayo Clinic. Retrieved January 6, 2022, from <https://www.mayoclinic.org/diseases-conditions/erectile-dysfunction/in-depth/erectile-dysfunction-herbs/art-20044394>

28 Mahdavi-Roshan, M., Mirmiran, P., Arjmand, M., & Nasrollahzadeh, J. (2017). Effects of garlic on brachial endothelial function and capacity of plasma to mediate cholesterol efflux in patients with coronary artery disease. *Anatolian journal of cardiology*, 18(2), 116–121. <https://doi.org/10.14744/AnatolJCardiol.2017.7669>

29 Wu, Y., Li, S., Cui, W., Zu, X., Du, J., & Wang, F. (2008). Ginkgo biloba extract improves coronary blood flow in healthy elderly adults: role of endotheli-

um-dependent vasodilation. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 15(3), 164–169. <https://doi.org/10.1016/j.phymed.2007.12.002>

30 Dietary supplements for erectile dysfunction: A natural treatment for ED? (2021b, April 10). Mayo Clinic. <https://www.mayoclinic.org/diseases-conditions/erectile-dysfunction/in-depth/erectile-dysfunction-herbs/art-20044394>

31 Barassi, A. (2017, February 8). Levels of l-arginine and l-citrulline in patients with erectile dysfunction of different etiology. *Wiley Online Library*. Retrieved June 2, 2022, from <https://onlinelibrary.wiley.com/doi/full/10.1111/andr.12293>

32 Gentile, V., Vicini, P., Prigiotti, G., Koverech, A., & Di Silverio, F. (2004). Preliminary observations on the use of propionyl-L-carnitine in combination with sildenafil in patients with erectile dysfunction and diabetes. *Current medical research and opinion*, 20(9), 1377–1384. <https://doi.org/10.1185/030079904X2394>

33 El-Sakka A. I. (2018). Dehydroepiandrosterone and Erectile Function: A Review. *The world journal of men's health*, 36(3), 183–191. <https://doi.org/10.5534/wjmh.180005>

Prostate Protocol

1 <https://www.ncbi.nlm.nih.gov/books/NBK279291>

2 Leitzmann, M. F., & Rohrmann, S. (2012). Risk factors for the onset of prostatic cancer: age, location, and behavioral correlates. *Clinical epidemiology*, 4, 1–11. <https://doi.org/10.2147/CLEP.S16747>

3 Harvard Health. (n.d.). Retrieved April 12, 2022, from <https://www.health.harvard.edu/topics/prostate-health>

4 <https://www.niddk.nih.gov/health-information/urologic-diseases/prostate-problems/prostatitis-inflammation-prostate>

5 <https://bphnews.com/prostatitis-symptoms/>

6 <https://www.mayoclinic.org/diseases-conditions/benign-prostatic-hyperplasia/symptoms-causes/syc-20370087>

7 Key statistics for prostate cancer: Prostate cancer facts. American Cancer Society. (n.d.). Retrieved April 12, 2022, from <https://www.cancer.org/cancer/prostate-cancer/about/key-statistics.html>

8 Ho, E., & Song, Y. (2009). Zinc and prostatic cancer. *Current opinion in clinical nutrition and metabolic care*, 12(6), 640–645. <https://doi.org/10.1097/MCO.0b013e328333106ee>

9 Chen, P., Zhang, W., Wang, X., Zhao, K., Negi, D. S., Zhuo, L., Qi, M., Wang, X., & Zhang, X. (2015). Lycopene and Risk of Prostate Cancer: A Systematic Review and Meta-Analysis. *Medicine*, 94(33), e1260. <https://doi.org/10.1097/MD.0000000000001260>

10 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7464847/> Imran, M., Ghorat, F., Ul-Haq, I., Ur-Rehman, H., Aslam, F., Heydari, M., Shariati, M. A., Okuskhanova, E., Yessimbekov, Z., Thiruvengadam, M., Hashempur, M. H., & Rebezov, M. (2020). Lycopene as a Natural Antioxidant Used to Prevent Human Health Disorders. *Antioxidants* (Basel, Switzerland), 9(8), 706. <https://doi.org/10.3390/antiox9080706>

11 Murillo, G., & Mehta, R. G. (2001). Cruciferous vegetables and cancer prevention. *Nutrition and cancer*, 41(1–2), 17–28. <https://doi.org/10.1080/01635581.2001.9680607>

12 Guo, Y., Zhi, F., Chen, P., Zhao, K., Xiang, H., Mao, Q., Wang, X., & Zhang, X. (2017). Green tea and the risk of prostate cancer: A systematic review and meta-analysis. *Medicine*, 96(13), e6426. <https://doi.org/10.1097/MD.0000000000006426>

13 Miyata, Y., Shida, Y., Hakariya, T., & Sakai, H. (2019). Anti-Cancer Effects of Green Tea Polyphenols Against Prostate Cancer. *Molecules* (Basel, Switzerland), 24(1), 193. <https://doi.org/10.3390/molecules24010193>

14 Aucuin, M., Cooley, K., Knee, C., Fritz, H., Balneaves, L. G., Breau, R., Fergusson, D., Skidmore, B., Wong, R., & Seely, D. (2017). Fish-Derived Omega-3 Fatty Acids and Prostate Cancer: A Systematic Review. *Integrative cancer therapies*, 16(1), 32–62. <https://doi.org/10.1177/1534735416656052>

15 Wang, L., & Martins-Green, M. (2014). Pomegranate and its components as alternative treatment for prostate cancer. *International journal of molecular sciences*, 15(9), 14949–14966. <https://doi.org/10.3390/ijms150914949>

16 Parikesit, D., Mochtar, C. A., Umbas, R., & Hamid, A. R. (2016). The impact of obesity towards prostate diseases. *Prostate international*, 4(1), 1–6. <https://doi.org/10.1016/j.pnrl.2015.08.001>

17 Kim, W. T., Yun, S. J., Choi, Y. D., Kim, G. Y., Moon, S. K., Choi, Y. H., Kim, I. Y., & Kim, W. J. (2011). Prostate size correlates with fasting blood glucose in non-diabetic benign prostatic hyperplasia patients with normal testosterone levels. *Journal of Korean medical science*, 26(9), 1214–1218. <https://doi.org/10.3346/jkms.2011.26.9.1214>

18 Yan Song, Jorge E. Chavarro, Yin Cao, Weiliang Qiu, Lorelei Mucci, Howard D. Sesso, Meir J. Stampfer, Edward Giovannucci, Michael Pollak, Simin Liu, Jing Ma, Whole Milk Intake Is Associated with Prostate Cancer-Specific Mortality among U.S. Male Physicians, *The Journal of Nutrition*, Volume 143, Issue 2, February 2013, Pages 189–196, <https://doi.org/10.3945/jn.112.168484>

19 Trudeau, K., Rousseau, M. C., & Parent, M. É. (2020). Extent of Food Processing and Risk of Prostate Cancer: The PRoTEuS Study in Montreal, Canada. *Nutrients*, 12(3), 637. <https://doi.org/10.3390/nu12030637>

20 Kim, W. T., Yun, S. J., Choi, Y. D., Kim, G. Y., Moon, S. K., Choi, Y. H., Kim, I. Y., & Kim, W. J. (2011). Prostate size correlates with fasting blood glucose in non-diabetic benign prostatic hyperplasia patients with normal testosterone levels. *Journal of Korean medical science*, 26(9), 1214–1218. <https://doi.org/10.3346/jkms.2011.26.9.1214>

21 Davies, N. M., Gaunt, T. R., Lewis, S. J., Holly, J., Donovan, J. L., Hamdy, F. C., Kemp, J. P., Eeles, R., Easton, D., Kote-Jarai, Z., Al Olama, A. A., Benlloch, S., Muir, K., Giles, G. G., Wiklund, F., Gronberg, H., Haiman, C. A., Schleutker, J., Nordestgaard, B. G., Travis, R. C., ... Martin, R. M. (2015). The effects of height and BMI on prostate cancer incidence and mortality: a Mendelian randomization study in 20,848 cases and 20,214 controls from the PRACTICAL consortium. *Cancer causes & control: CCC*, 26(11), 1603–1616. <https://doi.org/10.1007/s10551-015-2611-1>

- 22 Campos, C., Sotomayor, P., Jerez, D., González, J., Schmidt, C. B., Schmidt, K., Banzer, W., & Godoy, A. S. (2018). Exercise and prostate cancer: From basic science to clinical applications. *The Prostate*, 78(9), 639–645. <https://doi.org/10.1002/pros.23502>
- 23 Centers for Disease Control and Prevention. (2022, January 21). Adult BMI calculator. Centers for Disease Control and Prevention. Retrieved April 12, 2022, from https://www.cdc.gov/healthyweight/assessing/bmi/adult_bmi/english_bmi_calculator/bmi_calculator.html
- 24 3 ways exercise helps the prostate (yes, the prostate). Harvard Health. (2014, May 15). Retrieved May 30, 2022, from <https://www.health.harvard.edu/mens-health/3-ways-exercise-helps-the-prostate-yes-the-prostate>
- 25 Stanley, G., Harvey, K., Slivova, V., Jiang, J., & Sliva, D. (2005). Ganoderma lucidum suppresses angiogenesis through the inhibition of secretion of VEGF and TGF-beta1 from prostate cancer cells. *Biochemical and biophysical research communications*, 330(1), 46–52. <https://doi.org/10.1016/j.bbrc.2005.02.116>
- 26 Zhao, X., Zhou, D., Liu, Y., Li, C., Zhao, X., Li, Y., & Li, W. (2018). Ganoderma lucidum polysaccharide inhibits prostate cancer cell migration via the protein arginine methyltransferase 6 signaling pathway. *Molecular Medicine Reports*, 17, 147–157. <https://doi.org/10.3892/mmr.2017.7904>
- 27 Luk S-U, Lee TK-W, Liu J, Lee DT-W, Chiu Y-T, Ma S, et al. (2011) Chemopreventive Effect of PSP Through Targeting of Prostate Cancer Stem Cell-Like Population. *PLoS ONE* 6(5): e19804. <https://doi.org/10.1371/journal.pone.0019804>
- 28 Blagodatski, A., Yatsunskaya, M., Mikhailova, V., Tiasto, V., Kagansky, A., & Katanaev, V. L. (2018). Medicinal mushrooms as an attractive new source of natural compounds for future cancer therapy. *Oncotarget*, 9(49), 29259–29274. <https://doi.org/10.18632/oncotarget.25660> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6044372/>
- 29 Nahata, A. and Dixit, V.K. (2012). Ganoderma lucidum is an inhibitor of testosterone-induced prostatic hyperplasia in rats. *Andrologia*, 44: 160–174. <https://doi.org/10.1111/j.1439-0272.2010.1155.x>
- 30 Zhang, D. Y., Wu, J., Ye, F., Xue, L., Jiang, S., Yi, J., Zhang, W., Wei, H., Sung, M., Wang, W., & Li, X. (2003). Inhibition of cancer cell proliferation and prostaglandin E2 synthesis by *Scutellaria baicalensis*. *Cancer research*, 63(14), 4037–4043
- 31 Ikemoto, S., Sugimura, K., Yoshida, N., Yasumoto, R., Wada, S., Yamamoto, K., & Kishimoto, T. (2000). Antitumor effects of *Scutellariae radix* and its components baicalin, baicalin, and wogonin on bladder cancer cell lines. *Urology*, 55(6), 951–955. [https://doi.org/10.1016/s0090-4295\(00\)00467-2](https://doi.org/10.1016/s0090-4295(00)00467-2)
- 32 Ma, X., Yan, W., Dai, Z., Gao, X., Ma, Y., Xu, Q., Jiang, J., & Zhang, S. (2016). Baicalin suppresses metastasis of breast cancer cells by inhibiting EMT via downregulation of SATB1 and Wnt/ β -catenin pathway. *Drug design, development and therapy*, 10, 1419–1441. <https://doi.org/10.2147/DDDT.S102541>
- 33 Boyle, P., Robertson, C., Lowe, F., & Roehrborn, C. (n.d.). Meta-analysis of clinical trials of Permixon in the treatment of symptomatic benign prostatic hyperplasia. *Gold Journal*. Retrieved April 13, 2022, from [https://www.goldjournal.net/article/S0090-4295\(99\)00593-2/fulltext](https://www.goldjournal.net/article/S0090-4295(99)00593-2/fulltext)
- 34 Gordon, A. E., & Shaughnessy, A. F. (2003). Saw palmetto for prostate disorders. *American family physician*, 67(6), 1281–1283
- 35 Penugonda, K., & Lindshield, B. L. (2013). Fatty acid and phytosterol content of commercial saw palmetto supplements. *Nutrients*, 5(9), 3617–3633. <https://doi.org/10.3390/nu5093617>
- 36 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3798925/>
- Penugonda, K., & Lindshield, B. L. (2013). Fatty acid and phytosterol content of commercial saw palmetto supplements. *Nutrients*, 5(9), 3617–3633. <https://doi.org/10.3390/nu5093617>
- 37 <https://www.mountsinai.org/health-library/herb/saw-palmetto>
- 38 Lichius, J. J., & Muth, C. (1997). The inhibiting effects of *Urtica dioica* root extracts on experimentally induced prostatic hyperplasia in the mouse. *Planta medica*, 63(4), 307–310. <https://doi.org/10.1055/s-2006-957688>
- 39 Ghorbanibirgani, A., Khalili, A., & Zamani, L. (2013). The efficacy of stinging nettle (*urtica dioica*) in patients with benign prostatic hyperplasia: a randomized double-blind study in 100 patients. *Iranian Red Crescent medical journal*, 15(1), 9–10. <https://doi.org/10.5812/ircmj.2386>
- 40 Keehn, A., & Lowe, F. (n.d.). Complementary and alternative medications for benign prostatic ... - canjurol.com. *Canadian Journal of Urology*. Retrieved April 13, 2022, from https://www.canjurol.com/html/free-articles/JUV22I5S1F_08_DrLowe.pdf
- 41 Wilt, T., Ishani, A., MacDonald, R., Stark, G., Mulrow, C., & Lau, J. (2000). Beta-sitosterols for benign prostatic hyperplasia. *The Cochrane database of systematic reviews*, 1999(2), CD001043. <https://doi.org/10.1002/14651858.CD001043>
- 42 MacDonald, R., Ishani, A., Rutks, I., & Wilt, T. J. (2000). A systematic review of Cernilton for the treatment of benign prostatic hyperplasia. *BJU international*, 85(7), 836–841. <https://doi.org/10.1046/j.1464-410x.2000.00365.x>
- 43 Dai, S., Mo, Y., Wang, Y., Xiang, B., Liao, Q., Zhou, M., Li, X., Li, Y., Xiong, W., Li, G., Guo, C., & Zeng, Z. (2020). Chronic Stress Promotes Cancer Development. *Frontiers in oncology*, 10, 1492. <https://doi.org/10.3389/fonc.2020.01492>
- 44 Salve, J., Pate, S., Debnath, K., & Langade, D. (2019). Adaptogenic and Anxiolytic Effects of Ashwagandha Root Extract in Healthy Adults: A Double-blind, Randomized, Placebo-controlled Clinical Study. *Cureus*, 11(12), e6466. <https://doi.org/10.7759/cureus.6466>
- 45 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, *Hericium erinaceus* (Higher Basidiomycetes) from Malaysia. *International journal of medicinal mushrooms*, 15(6), 539–554. <https://doi.org/10.1615/intjmed-mushr.v15i6.30>
- 46 Kuypers K. (2021). Self-Medication with Ganoderma lucidum (“Reishi”) to Combat Parkinson’s Disease Symptoms: A Single Case Study. *Journal of medicinal food*, 24(7), 766–773. <https://doi.org/10.1089/jmf.2020.0137>
- 47 Rossi, P., Buonocore, D., Altobelli, E., Brandalise, F., Cesaroni, V., Iozzi, D., Savino, E., & Marzatico, F. (2014). Improving Training Condition Assessment in Endurance Cyclists: Effects of Ganoderma lucidum and Ophiocordyceps sinensis Dietary Supplementation. Evidence-based complementary and alternative medicine: eCAM, 2014, 979613. <https://doi.org/10.1155/2014/979613>
- 48 Rossi, P., Buonocore, D., Altobelli, E., Brandalise, F., Cesaroni, V., Iozzi, D., Savino, E., & Marzatico, F. (2014). Improving Training Condition Assessment in Endurance Cyclists: Effects of Ganoderma lucidum and Ophiocordyceps sinensis Dietary Supplementation. Evidence-based complementary and alternative medicine: eCAM, 2014, 979613. <https://doi.org/10.1155/2014/979613>
- 49 Scholey, A., Gibbs, A., Neale, C., Perry, N., Ossoukhova, A., Bilog, V., Kras, M., Scholz, C., Sass, M., & Buchwald-Werner, S. (2014). Anti-stress effects of lemon balm-containing foods. *Nutrients*, 6(11), 4805–4821. <https://doi.org/10.3390/nu6114805>
- 50 Sayorwan, W., Siripornpanich, V., Piriyaunaporn, T., Hongratanaworakit, T., Kotchabhakdi, N., & Ruangrunsi, N. (2012). The effects of lavender oil inhalation on emotional states, autonomic nervous system, and brain electrical activity. *Journal of the Medical Association of Thailand = Chotmaihet thangphaet*, 95(4), 598–606
- 51 Rider, J. R., Wilson, K. M., Sinnott, J. A., Kelly, R. S., Mucci, L. A., & Giovannucci, E. L. (2016). Ejaculation Frequency and Risk of Prostate Cancer: Updated Results with an Additional Decade of Follow-up. *European urology*, 70(6), 974–982. <https://doi.org/10.1016/j.eururo.2016.03.027>
- 52 Jacobsen, S. J., Jacobson, D. J., Rohe, D. E., Girman, C. J., Roberts, R. O., & Lieber, M. M. (2003). Frequency of sexual activity and prostatic health: fact or fairy tale
- 53 Percivalle, V., Blandini, M., Fecarotta, P., Buscemi, A., Di Corrado, D., Bertolo, L., Fichera, F., & Coco, M. (2017). The role of deep breathing on stress. *Neurological sciences: official journal of the Italian Neurological Society and of the Italian Society of Clinical Neurophysiology*, 38(3), 451–458. <https://doi.org/10.1007/s10072-016-2790-8>
- 54 Porcacchia, A. S., Câmara, D., Andersen, M. L., & Tufik, S. (2022). Sleep disorders and prostate cancer prognosis: biology, epidemiology, and association with cancer development risk. *European journal of cancer prevention: the official journal of the European Cancer Prevention Organisation (ECP)*, 31(2), 178–189. <https://doi.org/10.1097/CEJ.0000000000000685>
- 55 Espinosa, G., Esposito, R., Kazzazi, A., & Djavan, B. (2013). Vitamin D and benign prostatic hyperplasia -- a review. *The Canadian journal of urology*, 20(4), 6820–6825
- 56 Trump, D. L., & Aragon-Ching, J. B. (2018). Vitamin D in prostate cancer. *Asian journal of andrology*, 20(3), 244–252. https://doi.org/10.4103/aja.aja_14_18
- 57 Alexander W. (2013). Prostate cancer risk and omega-3 Fatty Acid intake from fish oil: a closer look at media messages versus research findings. P & T: a peer-reviewed journal for formulary management, 38(9), 561–564
- 58 Ho, E., & Song, Y. (2009). Zinc and prostatic cancer: Current opinion in clinical nutrition and metabolic care, 12(6), 640–645. <https://doi.org/10.1097/MCO.0b013e32833106ee>
- 59 Franklin, R. B., & Costello, L. C. (2007). Zinc as an anti-tumor agent in prostate cancer and in other cancers. *Archives of biochemistry and biophysics*, 463(2), 211–217. <https://doi.org/10.1016/j.abb.2007.02.033>
- 60 Sauer, A. K., Vela, H., Vela, G., Stark, P., Barrera-Juarez, E., & Grabrucker, A. M. (2020). Zinc Deficiency in Men Over 50 and Its Implications in Prostate Disorders. *Frontiers in oncology*, 10, 1293. <https://doi.org/10.3389/fonc.2020.01293> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7424038/>
- 61 Sauer, A. K., Vela, H., Vela, G., Stark, P., Barrera-Juarez, E., & Grabrucker, A. M. (2020). Zinc Deficiency in Men Over 50 and Its Implications in Prostate Disorders. *Frontiers in oncology*, 10, 1293. <https://doi.org/10.3389/fonc.2020.01293> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7424038>
- 62 Sayehmiri, K., Azami, M., Mohammadi, Y., Soleymani, A., & Tardeh, Z. (2018). The association between Selenium and Prostate Cancer: A Systematic Review and Meta-Analysis. *Asian Pacific journal of cancer prevention: APJCP*, 19(6), 1431–1437. <https://doi.org/10.22034/APJCP.2018.19.6.1431>
- 63 <https://ods.od.nih.gov/factsheets/Selenium-Consumer/>
- 64 Chen, P., Zhang, W., Wang, X., Zhao, K., Negi, D. S., Zhuo, L., Qi, M., Wang, X., & Zhang, X. (2015). Lycopene and Risk of Prostate Cancer: A Systematic Review and Meta-Analysis. *Medicine*, 94(33), e1260. <https://doi.org/10.1097/MD.0000000000001260>
- 65 Mirahmadi, M., Azimi-Hashemi, S., Saburi, E., Kamali, H., Pishbin, M., & Hadizadeh, F. (2020). Potential inhibitory effect of lycopene on prostate cancer. *Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie*, 129, 110459. <https://doi.org/10.1016/j.biopha.2020.110459>
- 66 Mirahmadi, M., Azimi-Hashemi, S., Saburi, E., Kamali, H., Pishbin, M., & Hadizadeh, F. (2020). Potential inhibitory effect of lycopene on prostate cancer. *Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie*, 129, 110459. <https://doi.org/10.1016/j.biopha.2020.110459>
- 67 Wheeler, K. M., & Liss, M. A. (2019). The Microbiome and Prostate Cancer Risk. *Current urology reports*, 20(10), 66. <https://doi.org/10.1007/s11934-019-0922-4>

Women’s Sexual Health Protocol

- 1 Ohishi, T., Goto, S., Monira, P., Isemura, M., & Nakamura, Y. (2016). Anti-inflammatory Action of Green Tea. Anti-inflammatory & anti-allergy agents in medicinal chemistry, 15(2), 74–90. <https://doi.org/10.2174/1871523015666160915154443>
- 2 Giugliano, D., Ceriello, A., & Esposito, K. (2006). The effects of diet on

- inflammation: emphasis on the metabolic syndrome. *Journal of the American College of Cardiology*, 48(4), 677–685. <https://doi.org/10.1016/j.jacc.2006.03.052>
- 3 Grant, P., & Ramasamy, S. (2012). An update on plant derived anti-androgens. *International journal of endocrinology and metabolism*, 10(2), 497–502. <https://doi.org/10.5812/ijem.3644>
- 4 Feyzollahi, Z., Mohseni Kouchesfehiani, H., Jalali, H., Eslimi-Esfahani, D., & Sheikh Hosseini, A. (2021). Effect of Vitex agnus-castus ethanolic extract on hypothalamic KISS-1 gene expression in a rat model of polycystic ovary syndrome. *Avicenna journal of phytomedicine*, 11(3), 292–301
- 5 Hadijafari, M., Khani, S., Abouhamzeh, B., Abdollahi, M., & Mirghazanfari, S. M. (2020). Yarrow (*Achillea millefolium* L.) Extract Produces Beneficial effects on Reproductive Parameters in Estradiol Valerate-Induced Polycystic Ovarian Syndrome in Rats. *Future Natural Products*, 6(1), 1–13
- 6 Tadić, V., Arsić, I., Zvezdanović, J., Zugić, A., Cvetković, D., & Pavkov, S. (2017). The estimation of the traditionally used yarrow (*Achillea millefolium* L. Asteraceae) oil extracts with anti-inflammatory potential in topical application. *Journal of ethnopharmacology*, 199, 138–148. <https://doi.org/10.1016/j.jep.2017.02.002>
- 7 Shahin, A. Y., & Mohammed, S. A. (2014). Adding the phytoestrogen Cimicifugae Racemosae to clomiphene induction cycles with timed intercourse in polycystic ovary syndrome improves cycle outcomes and pregnancy rates - a randomized trial. *Gynecological endocrinology: the official journal of the International Society of Gynecological Endocrinology*, 30(7), 505–510. <https://doi.org/10.3109/09513590.2014.895983>
- 8 Structure and Bioactivity of Steroidal Saponins Isolated from the Roots of *Chamaelirium luteum* (False Unicorn) Victoria L. Challinor, Julia M. U. Stuthe, Peter G. Parsons, Lynette K. Lambert, Reginald P. Lehmann, William Kitching, and James J. De Voss *Journal of Natural Products* 2012 75 (8), 1469–1479 DOI: 10.1021/np300393y
- 9 Rababa'h, A. M., Matani, B. R., & Ababneh, M. A. (2020). The ameliorative effects of marjoram in dehydroepiandrosterone induced polycystic ovary syndrome in rats. *Life sciences*, 261, 118353. <https://doi.org/10.1016/j.lfs.2020.118353>
- 10 Wang, H., Zhu, C., Ying, Y., Luo, L., Huang, D., & Luo, Z. (2017). Metformin and berberine, two versatile drugs in treatment of common metabolic diseases. *Oncotarget*, 9(11), 10135–10146. <https://doi.org/10.18632/oncotarget.20807>
- 11 Jurkiewicz-Przondziona, J., Lemm, M., Kwiatkowska-Pamuła, A., Ziółko, E., & Wójtowicz, M. K. (2017). Influence of diet on the risk of developing endometriosis. *Ginekologia polska*, 88(2), 96–102. <https://doi.org/10.5603/GPa.2017.0017>
- 12 Stacey A. Missmer, Jorge E. Chavarro, Susan Malspeis, Elizabeth R. Bertone-Johnson, Mark D. Hornstein, Donna Spiegelman, Robert L. Barbieri, Walter C. Willett, Susan E. Hankinson, A prospective study of dietary fat consumption and endometriosis risk, *Human Reproduction*, Volume 25, Issue 6, 1 June 2010, Pages 1528–1535, <https://doi.org/10.1093/humrep/deq044>. <https://academic.oup.com/humrep/article/25/6/1528/2915756>
- 13 Moore, J. S., Gibson, P. R., Perry, R. E., & Burgell, R. E. (2017). Endometriosis in patients with irritable bowel syndrome: Specific symptomatic and demographic profile, and response to the low FODMAP diet. *The Australian & New Zealand journal of obstetrics & gynaecology*, 57(2), 201–205. <https://doi.org/10.1111/ajo.12594>. <https://pubmed.ncbi.nlm.nih.gov/28303579/>
- 14 Marziali, M., Venza, M., Lazzaro, S., Lazzaro, A., Micossi, C., & Stolfi, V. M. (2012). Gluten-free diet: a new strategy for management of painful endometriosis related symptoms? *Minerva chirurgica*, 67(6), 499–504. <https://pubmed.ncbi.nlm.nih.gov/23334113/>
- 15 Keita, M., Bessette, P., Pelmus, M., Ainmelk, Y., & Aris, A. (2010). Expression of interleukin-1 (IL-1) ligands system in the most common endometriosis-associated ovarian cancer subtypes. *Journal of ovarian research*, 3, 3. <https://doi.org/10.1186/1757-2215-3-3>. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2832771/>
- 16 Mahboubi M. (2019). Evening Primrose (*Oenothera biennis*) Oil in Management of Female Ailments. *Journal of menopausal medicine*, 25(2), 74–82. <https://doi.org/10.6118/jmm.18190>
- 17 Peng, Y., Ao, M., Dong, B., Jiang, Y., Yu, L., Chen, Z., Hu, C., & Xu, R. (2021). Anti-Inflammatory Effects of Curcumin in the Inflammatory Diseases: Status, Limitations and Countermeasures. *Drug design, development and therapy*, 15, 4503–4525. <https://doi.org/10.2147/DDDT.S327378>
- 18 Suarez-Arroyo, I. J., Rosario-Acevedo, R., Aguilar-Perez, A., Clemente, P. L., Cubano, L. A., Serrano, J., Schneider, R. J., & Martínez-Montemayor, M. M. (2013). Anti-tumor effects of *Ganoderma lucidum* (reishi) in inflammatory breast cancer in vivo and in vitro models. *PloS one*, 8(2), e57431. <https://doi.org/10.1371/journal.pone.0057431>
- 19 Torkelson, C. J., Sweet, E., Martzen, M. R., Sasagawa, M., Wenner, C. A., Gay, J., Putiri, A., & Standish, L. J. (2012). Phase 1 Clinical Trial of *Trametes versicolor* in Women with Breast Cancer. *ISRN oncology*, 2012, 251632. <https://doi.org/10.5402/2012/251632>
- 20 Tuli, H. S., Sandhu, S. S., & Sharma, A. K. (2014). Pharmacological and therapeutic potential of *Cordyceps* with special reference to *Cordycepin*. *3 Biotech*, 4(1), 1–12. <https://doi.org/10.1007/s13205-013-0121-9>
- 21 Livdands-Forret, A. B., Harvey, P. J., & Larkin-Thier, S. M. (2007). Menorrhagia: a synopsis of management focusing on herbal and nutritional supplements, and chiropractic. *The Journal of the Canadian Chiropractic Association*, 51(4), 235–246
- 22 Gu, Y., & Zhou, Z. (2021). Berberine inhibits the proliferation, invasion and migration of endometrial stromal cells by downregulating miR-429. *Molecular medicine reports*, 23(6), 416. <https://doi.org/10.3892/mmr.2021.12055>
- 23 Naafe M, Kariman N, Keshavarz Z, Khademi N, Mojab F, Mohammadbeigi A. The effect of hydroalcoholic extracts of *Capsella bursa-pastoris* on heavy menstrual bleeding: a randomized clinical trial. *J Altern Complement Med*. July 2018;24(7):694-700. doi: 10.1089/acm.2017.0267
- 24 van Die, M. D., Burger, H. G., Teede, H. J., & Bone, K. M. (2013). Vitex agnus-castus extracts for female reproductive disorders: a systematic review of clinical trials. *Planta medica*, 79(7), 562–575. <https://doi.org/10.1055/s-0032-1327831>
- 25 Mulrow C, Lawrence V, Jacobs B, et al. Milk Thistle: Effects on Liver Disease and Cirrhosis and Clinical Adverse Effects: Summary. 2000. In: AHRQ Evidence Report Summaries. Rockville (MD): Agency for Healthcare Research and Quality (US); 1998-2005. 21. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK11896/>
- 26 Wachtel-Galor S, Yuen J, Buswell JA, et al. *Ganoderma lucidum* (Lingzhi or Reishi): A Medicinal Mushroom. In: Benzie IFF, Wachtel-Galor S, editors. *Herbal Medicine: Biomolecular and Clinical Aspects*. 2nd edition. Boca Raton (FL): CRC Press/Taylor & Francis; 2011. Chapter 9. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK92757/>
- 27 Gbinigie, O. A., Spencer, E. A., Heneghan, C. J., Lee, J. J., & Butler, C. C. (2020). Cranberry Extract for Symptoms of Acute, Uncomplicated Urinary Tract Infection: A Systematic Review. *Antibiotics* (Basel, Switzerland), 10(1), 12. <https://doi.org/10.3390/antibiotics10010012>
- 28 Melzig M. F. (2004). Echtes Goldrutenkraut--ein Klassiker in der urologischen Phytotherapie [Goldenrod--a classical exponent in the urological phytotherapy]. *Wiener medizinische Wochenschrift* (1946), 154(21-22), 523–527. <https://doi.org/10.1007/s10354-004-0118-4>
- 29 Domenici, L., Monti, M., Bracchi, C., Giorgini, M., Colagiovanni, V., Muzii, L., & Benedetti Panici, P. (2016). D-mannose: a promising support for acute urinary tract infections in women. A pilot study. *European review for medical and pharmacological sciences*, 20(13), 2920–2925
- 30 Trill, J., Simpson, C., Webley, F., Radford, M., Stanton, L., Maishman, T., Galanopoulou, A., Flower, A., Eyles, C., Willcox, M., Hay, A., Griffiths, G., Little, P., Lewith, G., & Moore, M. (2017). Uva-ursi extract and ibuprofen as alternative treatments of adult female urinary tract infection (ATAFUTI): study protocol for a randomised controlled trial. *Trials*, 18(1), 421. <https://doi.org/10.1186/s13063-017-2145-7>
- 31 Sienkiewicz, M., Wasiela, M., & Glowacka, A. (2012). Aktywność przeciwbakteryjna olejku oreganowego (*Origanum heracleoticum* L.) wobec szczepów klinicznych *Escherichia coli* i *Pseudomonas aeruginosa* [The antibacterial activity of oregano essential oil (*Origanum heracleoticum* L.) against clinical strains of *Escherichia coli* and *Pseudomonas aeruginosa*]. *Medycyna doświadczalna i mikrobiologia*, 64(4), 297–307
- 32 Ogbolu, D. O., Oni, A. A., Daini, O. A., & Oloko, A. P. (2007). In vitro antimicrobial properties of coconut oil on *Candida* species in Ibadan, Nigeria. *Journal of medicinal food*, 10(2), 384–387. <https://doi.org/10.1089/jmf.2006.1209>
- 33 Yu, D., Wang, J., Shao, X., Xu, F., & Wang, H. (2015). Antifungal modes of action of tea tree oil and its two characteristic components against *Botrytis cinerea*. *Journal of applied microbiology*, 119(5), 1253–1262. <https://doi.org/10.1111/jam.12939>
- 34 Tronina, T., Mrozowska, M., Bartmańska, A., Popłoński, J., Sordon, S., & Huszcza, E. (2021). Simple and Rapid Method for Wogonin Preparation and Its Biotransformation. *International journal of molecular sciences*, 22(16), 8973. <https://doi.org/10.3390/ijms22168973>
- 35 Astani, A., Navid, M. H., & Schnitzler, P. (2014). Attachment and penetration of acyclovir-resistant herpes simplex virus are inhibited by *Melissa officinalis* extract. *Phytotherapy research: PTR*, 28(10), 1547–1552. <https://doi.org/10.1002/ptr.5166>
- 36 Vanti, G., Ntallis, S. G., Panagiotidis, C. A., Dourdouni, V., Patsoura, C., Bergonzi, M. C., Lazari, D., & Bilia, A. R. (2020). Glycerosome of *Melissa officinalis* L. Essential Oil for Effective Anti-HSV Type 1. *Molecules* (Basel, Switzerland), 25(14), 3111. <https://doi.org/10.3390/molecules25143111>
- 37 Mazzanti, G., Battinelli, L., Pompeo, C., Serrilli, A. M., Rossi, R., Sauzullo, I., Mengoni, F., & Vullo, V. (2008). Inhibitory activity of *Melissa officinalis* L. extract on Herpes simplex virus type 2 replication. *Natural product research*, 22(16), 1433–1440. <https://doi.org/10.1080/14786410802075939>
- 38 Mazzanti, G., Battinelli, L., Pompeo, C., Serrilli, A. M., Rossi, R., Sauzullo, I., Mengoni, F., & Vullo, V. (2008). Inhibitory activity of *Melissa officinalis* L. extract on Herpes simplex virus type 2 replication. *Natural product research*, 22(16), 1433–1440. <https://doi.org/10.1080/14786410802075939>
- 39 Wang, L., Yang, R., Yuan, B., Liu, Y., & Liu, C. (2015). The antiviral and antimicrobial activities of licorice, a widely-used Chinese herb. *Acta pharmaceutica Sinica*. B, 5(4), 310–315. <https://doi.org/10.1016/j.apsb.2015.05.005>
- 40 Huan C, Xu Y, Zhang W, Guo T, Pan H, Gao S. Research Progress on the Antiviral Activity of Glycyrrhizin and its Derivatives in Liquorice. *Front Pharmacol*. 2021 Jul 6;12:680674. doi: 10.3389/fphar.2021.680674. PMID: 34295250; PMCID: PMC8290359
- 41 Brochot, A., Guilbot, A., Haddioui, L., & Roques, C. (2017). Antibacterial, antifungal, and antiviral effects of three essential oil blends. *MicrobiologyOpen*, 6(4), e00459. <https://doi.org/10.1002/mbo3.459>
- 42 Schuhmacher, A., Reichling, J., & Schnitzler, P. (2003). Virucidal effect of peppermint oil on the enveloped viruses herpes simplex virus type 1 and type 2 in vitro. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 10(6-7), 504–510. <https://doi.org/10.1078/094471103322331467>
- 43 Chen, H., Muhammad, I., Zhang, Y., Ren, Y., Zhang, R., Huang, X., Diao, L., Liu, H., Li, X., Sun, X., Abbas, G., & Li, G. (2019). Antiviral Activity Against Infectious Bronchitis Virus and Bioactive Components of *Hypericum perforatum* L. *Frontiers in pharmacology*, 10, 1272. <https://doi.org/10.3389/fphar.2019.01272>
- 44 Carson, C. F., Hammer, K. A., & Riley, T. V. (2006). *Melaleuca alternifolia* (Tea Tree) oil: a review of antimicrobial and other medicinal properties. *Clinical microbiology reviews*, 19(1), 50–62. <https://doi.org/10.1128/CMR.19.1.50-62.2006>
- 45 Hekmatpou, D., Mehrabi, F., Rahzani, K., & Aminiyan, A. (2019). The Effect of Aloe Vera Clinical Trials on Prevention and Healing of Skin Wound: A Systematic Review. *Iranian journal of medical sciences*, 44(1), 1–9
- 46 van Die, M. D., Burger, H. G., Teede, H. J., & Bone, K. M. (2013). Vitex agnus-castus extracts for female reproductive disorders: a systematic review of clinical trials. *Planta*

medica, 79(7), 562–575. <https://doi.org/10.1055/s-0032-1327831>

- 47 Lekomtseva, Y., Zhukova, I., & Wacker, A. (2017). Rhodiola rosea in Subjects with Prolonged or Chronic Fatigue Symptoms: Results of an Open-Label Clinical Trial. *Complementary medicine research*, 24(1), 46–52. <https://doi.org/10.1159/000457918>
- 48 Bawa, A. S. (2006, November 20). Just a moment. . . Institute of Food Technologists (IFT). Retrieved May 16, 2022, from <https://ift.onlinelibrary.wiley.com/doi/abs/10.1111/j.1541-4337.2005.tb00073.x>
- 49 Cruz, D. A. C. (2018, February 12). The action of herbal medicine on the libido: aspects of nutritional intervention in increasing sexual desire. SpringerLink. Retrieved May 18, 2022, from <https://link.springer.com>
- 50 Gonzales, G. F., Nieto, J., Rubio, J., & Gasco, M. (2006). Effect of Black maca (*Lepidium meyenii*) on one spermatogenic cycle in rats. *Andrologia*, 38(5), 166–172. <https://doi.org/10.1111/j.1439-0272.2006.00733.x>
- 51 Fierascu, R. C., Fierascu, I., Ortan, A., Fierascu, I. C., Anuta, V., Velescu, B. S., Pituru, S. M., & Dinu-Pirvu, C. E. (2019). *Leonurus cardiaca* L. as a Source of Bioactive Compounds: An Update of the European Medicines Agency Assessment Report (2010). *BioMed research international*, 2019, 4303215. <https://doi.org/10.1155/2019/4303215>
- 52 Jing Zheng, Pistilli, M. J., Holloway, A. C., & Crankshaw, D. J. (2010). The effects of commercial preparations of red raspberry leaf on the contractility of the rat's uterus in vitro. *Reproductive sciences* (Thousand Oaks, Calif.), 17(5), 494–501. <https://doi.org/10.1177/1933719109359703>
- 53 Liu, G. T. (1999). Recent Advances in Research of Pharmacology and Clinical Applications of *Ganoderma P.Karst. Species* (Aphyllorhizomycetideae) in China - *International Journal of Medicinal Mushrooms*, Volume 1, 1999, Issue 1 - Begell House Digital Library. *International Journal of Medicinal Mushrooms*. Retrieved May 18, 2022, from <https://www.dl.begellhouse.com>
- 54 Geng, P., Siu, K. C., Wang, Z., & Wu, J. Y. (2017). Antifatigue Functions and Mechanisms of Edible and Medicinal Mushrooms. *BioMed research international*, 2017, 9648496. <https://doi.org/10.1155/2017/9648496>
- 55 Liu, C., Qiu, H., Huang, R., Chai, H., Yuan, G., & Shan, S. (2021). Therapeutic Mechanism and Clinical Observation of Traditional Chinese Medicine Combined with Interventional Recanalization for Tubal Infertility. Evidence-based complementary and alternative medicine: eCAM, 2021, 2842250. <https://doi.org/10.1155/2021/2842250>
- 56 Hjøllund, N. H., Jensen, T. K., Bonde, J. P., Henriksen, T. B., Andersson, A. M., Kolstad, H. A., Ernst, E., Giwercman, A. J., Skakkebaek, N. E., & Olsen, J. (2000). Stress og frugtbarhed. En forløbsundersøgelse blandt par, som forsøger at blive gravide for første gang [Stress and fertility. A follow-up study among couples planning the first pregnancy]. *Ugeskrift for læger*, 162(38), 5081–5086
- 57 Khalili, H., Neovius, M., Ekholm, A., Ludvigsson, J. F., Askling, J., Chan, A. T., & Olen, O. (2016). Oral Contraceptive Use and Risk of Ulcerative Colitis Progression: A Nationwide Study. *The American journal of gastroenterology*, 111(11), 1614–1620. <https://doi.org/10.1038/ajg.2016.464>
- 58 Khillare, B., & Shrivastav, T. G. (2003). Spermicidal activity of *Azadirachta indica* (neem) leaf extract. *Contraception*, 68(3), 225–229. [https://doi.org/10.1016/s0010-7824\(03\)00165-3](https://doi.org/10.1016/s0010-7824(03)00165-3)
- 59 Jansen, G. C. (2014, January 1). ResearchGate - Temporarily Unavailable. ResearchGate. Retrieved May 18, 2022, from <https://www.researchgate.net>
- 60 Kotta, S., Ansari, S. H., & Ali, J. (2013). Exploring scientifically proven herbal aphrodisiacs. *Pharmacognosy reviews*, 7(13), 1–10. <https://doi.org/10.4103/0973-7847.112832>
- 61 Lee, H. W., Choi, J., Lee, Y., Kil, K. J., & Lee, M. S. (2016). Ginseng for managing menopausal woman's health: A systematic review of double-blind, randomized, placebo-controlled trials. *Medicine*, 95(38), e4914. <https://doi.org/10.1097/MD.0000000000004914>
- 62 Ghorbani, Z., Mirghafourvand, M., Charandabi, S. M., & Javadzadeh, Y. (2019). The effect of ginseng on sexual dysfunction in menopausal women: A double-blind, randomized, controlled trial. *Complementary therapies in medicine*, 45, 57–64. <https://doi.org/10.1016/j.ctim.2019.05.015>
- 63 Zhao, J., Dasmahapatra, A. K., Khan, S. I., & Khan, I. A. (2008). Anti-aromatase activity of the constituents from *damiana* (*Turnera diffusa*). *Journal of ethnopharmacology*, 120(3), 387–393. <https://doi.org/10.1016/j.jep.2008.09.016>
- 64 Wayne, J. (2000, September 1). Effects of Herbal vX on libido and sexual activity in premenopausal and postmenopausal women. SpringerLink. Retrieved May 18, 2022, from <https://link.springer.com>
- 65 Beyer, K. M., Szabo, A., & Nattinger, A. B. (2016). Time Spent Outdoors, Depressive Symptoms, and Variation by Race and Ethnicity. *American journal of preventive medicine*, 51(3), 281–290. <https://doi.org/10.1016/j.amepre.2016.05.004>
- 66 Stults-Kolehmainen, M. A., & Sinha, R. (2014). The effects of stress on physical activity and exercise. *Sports medicine* (Auckland, N.Z.), 44(1), 81–121. <https://doi.org/10.1007/s40279-013-0090-5>
- 67 Ghazizadeh, J., Sadigh-Eteghad, S., Marx, W., Fakhari, A., Hamedeyazdan, S., Torbati, M., Taheri-Tarighi, S., Araj-Khodaei, M., & Mirghafourvand, M. (2021). The effects of lemon balm (*Melissa officinalis* L.) on depression and anxiety in clinical trials: A systematic review and meta-analysis. *Phytotherapy research*: PTR, 35(12), 6690–6705. <https://doi.org/10.1002/ptr.7252>
- 68 Koshoviy, O., Raal, A., Kireyev, I., Tryshchuk, N., Ilina, T., Romanenko, Y., Kovalenko, S. M., & Bunyatyan, N. (2021). Phytochemical and Psychotropic Research of Motherwort (*Leonurus cardiaca* L.) Modified Dry Extracts. *Plants* (Basel, Switzerland), 10(2), 230. <https://doi.org/10.3390/plants10020230>
- 69 Sebastian, L. A., & Kear, T. (2021). The Effect of Lavender Aromatherapy on Heart Rate, Blood Pressure, and Perceived Stress Among Cardiac Rehabilitation Patients: A Pilot Study. *Holistic nursing practice*, 10.1097/HNP.0000000000000430. Advance online

publication. <https://doi.org/10.1097/HNP.0000000000000430>

- 70 Krysiak, R., Szwajkosz, A., & Okopień, B. (2018). The effect of low vitamin D status on sexual functioning and depressive symptoms in apparently healthy men: a pilot study. *International journal of impotence research*, 30(5), 224–229. <https://doi.org/10.1038/s41443-018-0041-7>
- 71 Finkelstein, J. L., Layden, A. J., & Stover, P. J. (2015). Vitamin B-12 and Perinatal Health. *Advances in nutrition* (Bethesda, Md.), 6(5), 552–563. <https://doi.org/10.3945/an.115.008201>
- 72 Parnam Emamverdikhani, A., Golmakani, N., Tabassi, S. A., Hassanzadeh, M., Sharifi, N., & Shakeri, M. T. (2016). A survey of the therapeutic effects of Vitamin E suppositories on vaginal atrophy in postmenopausal women. *Iranian journal of nursing and midwifery research*, 21(5), 475–481. <https://doi.org/10.4103/1735-9066.193393>
- 73 Meston, C. M., & Worcel, M. (2002). The effects of yohimbine plus L-arginine glutamate on sexual arousal in postmenopausal women with sexual arousal disorder. *Archives of sexual behavior*, 31(4), 323–332. <https://doi.org/10.1023/a:1016220225392>

Allergy Protocol

- 1 Muche-Borowski, C., Kopp, M., Reese, I., Sitter, H., Werfel, T., & Schäfer, T. (2009). Allergy prevention. *Deutsches Arzteblatt international*, 106(39), 625–631. <https://doi.org/10.3238/arztebl.2009.0625>
- 2 Muraro, A., Werfel, T., Hoffmann-Sommergruber, K., Roberts, G., Beyer, K., Bindslev-Jensen, C., Cardona, V., Dubois, A., duToit, G., Eigenmann, P., Fernandez Rivas, M., Halken, S., Hickstein, L., Høst, A., Knol, E., Lack, G., Marchisotto, M. J., Niggemann, B., Nwaru, B. I., Papadopoulos, N. G., ... EAACI Food Allergy and Anaphylaxis Guidelines Group (2014). EAACI food allergy and anaphylaxis guidelines: diagnosis and management of food allergy. *Allergy*, 69(8), 1008–1025. <https://doi.org/10.1111/all.12429>
- 3 Muraro, A., Werfel, T., Hoffmann-Sommergruber, K., Roberts, G., Beyer, K., Bindslev-Jensen, C., Cardona, V., Dubois, A., duToit, G., Eigenmann, P., Fernandez Rivas, M., Halken, S., Hickstein, L., Høst, A., Knol, E., Lack, G., Marchisotto, M. J., Niggemann, B., Nwaru, B. I., Papadopoulos, N. G., ... EAACI Food Allergy and Anaphylaxis Guidelines Group (2014). EAACI food allergy and anaphylaxis guidelines: diagnosis and management of food allergy. *Allergy*, 69(8), 1008–1025. <https://doi.org/10.1111/all.12429>
- 4 American Academy of Allergy, Asthma and Immunology: <https://www.aaaai.org/tools-for-the-public/conditions-library/allergies/allergic-reactions>
- 5 The Immune System in Health and Disease. 5th edition. Janeway CA Jr; Travers P, Walport M, et al. New York: Garland Science; 2001
- 6 Stone KD, Prussin C, Metcalfe DD. IgE, mast cells, basophils, and eosinophils. *J Allergy Clin Immunol*. 2010 Feb;125(2 Suppl 2): S73-80. doi: 10.1016/j.jaci.2009.11.017. PMID: 20176269; PMCID: PMC2847274
- 7 Galli SJ, Tsai M. IgE and mast cells in allergic disease. *Nat Med*. 2012 May 4;18(5):693-704. doi: 10.1038/nm.2755. PMID: 22561833; PMCID: PMC3597223
- 8 Thangam, E. B., Jemima, E. A., Singh, H., Baig, M. S., Khan, M., Mathias, C. B., Church, M. K., & Saluja, R. (2018). The Role of Histamine and Histamine Receptors in Mast Cell-Mediated Allergy and Inflammation: The Hunt for New Therapeutic Targets. *Frontiers in immunology*, 9, 1873. <https://doi.org/10.3389/fimmu.2018.01873>
- 9 Muche-Borowski, C., Kopp, M., Reese, I., Sitter, H., Werfel, T., & Schäfer, T. (2009). Allergy prevention. *Deutsches Arzteblatt international*, 106(39), 625–631. <https://doi.org/10.3238/arztebl.2009.0625>
- 10 Muraro, A., Roberts, G., Worm, M., Bilò, M. B., Brockow, K., Fernández Rivas, M., Santos, A. F., Zolkipli, Z. Q., Bellou, A., Beyer, K., Bindslev-Jensen, C., Cardona, V., Clark, A. T., Demoly, P., Dubois, A. E., DunnGalvin, A., Eigenmann, P., Halken, S., Harada, L., Lack, G., ... EAACI Food Allergy and Anaphylaxis Guidelines Group (2014). Anaphylaxis: guidelines from the European Academy of Allergy and Clinical Immunology. *Allergy*, 69(8), 1026–1045. <https://doi.org/10.1111/all.12437>
- 11 Muraro, A., Roberts, G., Worm, M., Bilò, M. B., Brockow, K., Fernández Rivas, M., Santos, A. F., Zolkipli, Z. Q., Bellou, A., Beyer, K., Bindslev-Jensen, C., Cardona, V., Clark, A. T., Demoly, P., Dubois, A. E., DunnGalvin, A., Eigenmann, P., Halken, S., Harada, L., Lack, G., ... EAACI Food Allergy and Anaphylaxis Guidelines Group (2014). Anaphylaxis: guidelines from the European Academy of Allergy and Clinical Immunology. *Allergy*, 69(8), 1026–1045. <https://doi.org/10.1111/all.12437>
- 12 Eyring, K. R., Pedersen, B. S., Maclean, K. N., Stabler, S. P., Yang, I. V., & Schwartz, D. A. (2018). Methylene-tetrahydrofolate reductase contributes to allergic airway disease. *PloS one*, 13(1), e0190916. <https://doi.org/10.1371/journal.pone.0190916>
- 13 Akin, C., Valent, P., & Metcalfe, D. D. (2010). Mast cell activation syndrome: Proposed diagnostic criteria. *The Journal of allergy and clinical immunology*, 126(6), 1099–1104. <https://doi.org/10.1016/j.jaci.2010.08.035>
- 14 Pascal, M., Perez-Gordo, M., Caballero, T., Escribese, M., Lopez Longo, M., & Luengo, O. et al. (2018). Microbiome and Allergic Diseases. *Frontiers In Immunology*, 9. doi: 10.3389/fimmu.2018.01584
- 15 Fujimura, K. E., & Lynch, S. V. (2015). Microbiota in allergy and asthma and the emerging relationship with the gut microbiome. *Cell host & microbe*, 17(5), 592–602. <https://doi.org/10.1016/j.chom.2015.04.007>
- 16 Peña-Vélez, R., Toro-Monjaraz, E., Avelar-Rodríguez, D., Ignorosa-Arellano, K., Zárate-Mondragón, F., Cervantes-Bustamante, R., Montijo-Barrios, E., Cadena-León, J., & Ramírez-Mayans, J. (2019). Small intestinal bacterial overgrowth: could it be associated with chronic abdominal pain in children with allergic diseases?. *Revista española de enfermedades digestivas: organo oficial de la Sociedad Española de Patología Digestiva*, 111(12), 927–930. <https://doi.org/10.17235/reed.2019.6321/2019>
- 17 Gomaa E. Z. (2020). Human gut microbiota/microbiome in health and diseases: a review. *Antonie van Leeuwenhoek*, 113(12), 2019–2040. <https://doi.org/10.1007/s10482-020-01474-7>
- 18 Deloese, E., Janssen, P., Depoortere, I., & Tack, J. (2012). The migrating motor

complex: control mechanisms and its role in health and disease. *Nature reviews. Gastroenterology & hepatology*, 9(5), 271–285. <https://doi.org/10.1038/nrgastro.2012.57>

19 Gomaa E. Z. (2020). Human gut microbiota/microbiome in health and diseases: a review. *Antonie van Leeuwenhoek*, 113(12), 2019–2040. <https://doi.org/10.1007/s10482-020-01474-7>

20 Quigley E. (2019). Prebiotics and Probiotics in Digestive Health. *Clinical gastroenterology and hepatology: the official clinical practice journal of the American Gastroenterological Association*, 17(2), 333–344. <https://doi.org/10.1016/j.cgh.2018.09.028>

21 Molina-Torres, G., Rodríguez-Arrastia, M., Roman, P., Sanchez-Labrac, N., & Cardona, D. (2019). Stress and the gut microbiota-brain axis. *Behavioural pharmacology*, 30(2 and 3-Spec Issue), 187–200. <https://doi.org/10.1097/FBP.0000000000000478>

22 Gomaa E. Z. (2020). Human gut microbiota/microbiome in health and diseases: a review. *Antonie van Leeuwenhoek*, 113(12), 2019–2040. <https://doi.org/10.1007/s10482-020-01474-7>

23 Gomaa E. Z. (2020). Human gut microbiota/microbiome in health and diseases: a review. *Antonie van Leeuwenhoek*, 113(12), 2019–2040. <https://doi.org/10.1007/s10482-020-01474-7>

24 Matenchuk, B. A., Mandhane, P. J., & Kozyskyj, A. L. (2020). Sleep, circadian rhythm, and gut microbiota. *Sleep medicine reviews*, 53, 101340. <https://doi.org/10.1016/j.smrv.2020.101340>

25 Bone, K. (2009). *The Ultimate Herbal Compendium: A desktop guide for herbal prescribers*

26 Wang, Mingxing et al. "Anti-Gastric Ulcer Activity of Polysaccharide Fraction Isolated from Mycelium Culture of Lion's Mane Medicinal Mushroom, *Hericium erinaceus* (Higher Basidiomycetes)." *International journal of medicinal mushrooms* vol. 17,11 (2015): 1055-60. doi:10.1615/intjmedmushrooms.v17i11.50https://pubmed.ncbi.nlm.nih.gov/26853960/)

27 Available Chemical Constituents and Activities of *Ganoderma lucidum* (Lingzhi or Red Reishi) Utilizing in Disease Treatment: A mini review. Vo et al. *J Res Clin Med*. 2021;9: 32. doi: 10.34172/jrcm.2021.032

28 Pallav, K., Dowd, S. E., Villafuerte, J., Yang, X., Kabbani, T., Hansen, J., Dennis, M., Leffler, D. A., Newburg, D. S., & Kelly, C. P. (2014). Effects of polysaccharopeptide from *Trametes versicolor* and amoxicillin on the gut microbiome of healthy volunteers: a randomized clinical trial. *Gut microbes*, 5(4), 458–467. <https://doi.org/10.4161/gmic.29558> <https://pubmed.ncbi.nlm.nih.gov/25006989/>

29 Vassilopoulou, E., Konstantinou, G. N., Dimitriou, A., Manios, Y., Koumbi, L., & Papadopoulos, N. G. (2020). The Impact of Food Histamine Intake on Asthma Activity: A Pilot Study. *Nutrients*, 12(11), 3402. <https://doi.org/10.3390/nu12113402>

30 Son, J. H., Chung, B. Y., Kim, H. O., & Park, C. W. (2018). A Histamine-Free Diet Is Helpful for Treatment of Adult Patients with Chronic Spontaneous Urticaria. *Annals of dermatology*, 30(2), 164–172. <https://doi.org/10.5021/ad.2018.30.2.164>

31 Vassilopoulou, Emilia & Konstantinou, George & Dimitriou, Anastasia & Manios, Yannis & Koumbi, Lemonica & Papadopoulos, Nikolaos. (2020). The Impact of Food Histamine Intake on Asthma Activity: A Pilot Study. *Nutrients*. 12. 3402. 10.3390/nu12113402

32 Sánchez-Pérez, S., Comas-Basté, O., Veciana-Nogués, M. T., Latorre-Moratalla, M. L., & Vidal-Carou, M. C. (2021). Low-Histamine Diets: Is the Exclusion of Foods Justified by Their Histamine Content?. *Nutrients*, 13(5), 1395. <https://doi.org/10.3390/nu13051395>

33 Kawamoto, Y., Ueno, Y., Nakahashi, E., Obayashi, M., Sugihara, K., Qiao, S., Iida, M., Kumasaka, M. Y., Yajima, I., Goto, Y., Ohgami, N., Kato, M., & Takeda, K. (2016). Prevention of allergic rhinitis by ginger and the molecular basis of immunosuppression by 6-gingerol through T cell inactivation. *The Journal of Nutritional Biochemistry*, 27, 112–122. <https://doi.org/10.1016/j.jnutbio.2015.08.025>

34 Asha'ari, Z. A., Ahmad, M. Z., Jihan, W. S., Che, C. M., & Leman, I. (2013). Ingestion of honey improves the symptoms of allergic rhinitis: evidence from a randomized placebo-controlled trial in the East coast of Peninsular Malaysia. *Annals of Saudi medicine*, 33(5), 469–475. <https://doi.org/10.5144/0256-4947.2013.469>

35 Secor ER Jr, Szczepanek SM, Castater CA, Adami AJ, Matson AP, Rafti ET, Guernsey L, Natarajan P, McNamara JT, Schramm CM, Thrall RS, Silbart LK. Bromelain Inhibits Allergic Sensitization and Murine Asthma via Modulation of Dendritic Cells. *Evid Based Complement Alternat Med*. 2013;2013:702196. doi: 10.1155/2013/702196. Epub 2013 Dec 5. PMID: 24381635; PMCID: PMC3870104. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3870104/>

36 Wu TF, Chan YY, Shi WY, Jhong MT. Uncovering the Molecular Mechanism of Anti-Allergic Activity of Silkworm Pupa-Grown *Cordyceps militaris* Fruit Body. *Am J Chin Med*. 2017;45(3):497-513. doi: 10.1142/S0192415X17500306. Epub 2017 Apr 2. PMID: 28367714

37 Bhardwaj N, Katyal P, Sharma AK. Suppression of inflammatory and allergic responses by pharmacologically potent fungus *Ganoderma lucidum*. *Recent Pat Inflamm Allergy Drug Discov*. 2014;8(2):104-17. doi: 10.2174/1872213x08666140619110657. PMID: 24948193

38 Stavinocha, W.B., 1990. Study of the anti-inflammatory activity of *Ganoderma lucidum*. Presented at the Third Academic Joint Conference, pp: 201-208

39 Bone, K. (2009). *The Ultimate Herbal Compendium: A desktop guide for herbal prescribers*

40 Roschek, B., Jr, Fink, R. C., McMichael, M., & Alberte, R. S. (2009). Nettle extract (*Urtica dioica*) affects key receptors and enzymes associated with allergic rhinitis. *Phytotherapy research*: PTR, 23(7), 920–926. <https://doi.org/10.1002/ptr.2763> <https://pubmed.ncbi.nlm.nih.gov/19140159/>

41 Upton R. Stinging Nettles leaf (*Urtica dioica* L.): Extraordinary vegetable medicine. *Journal of Herbal Medicine*. 2013;3(1):9-38. doi:10.1016/j.hermed.2012.11.001

42 Mittman P. Randomized, Double-Blind Study of Freeze-Dried *Urtica dioica* in the Treatment of Allergic Rhinitis. *Planta Medica*. 1990;56(01):44-47. doi:10.1055/s-2006-960881

43 Roschek B, Fink R, McMichael M, Alberte R. Nettle extract (*Urtica dioica*) affects key receptors and enzymes associated with allergic rhinitis. *Phytotherapy Research*

44 Ayers S, Roschek Jr B, Williams J, Alberte R. Pharmacokinetic analysis of anti-allergy and anti-inflammatory bioactives in a nettle (*Urtica dioica*) extract. *Online Journal of Pharmacology and Pharmacokinetics*. 2008;5:6-21

45 Mittman P. (1990). Randomized, double-blind study of freeze-dried *Urtica dioica* in the treatment of allergic rhinitis. *Planta medica*, 56(1), 44–47. <https://doi.org/10.1055/s-2006-960881>

46 Nurul, I. M., Mizuguchi, H., Shahriar, M., Venkatesh, P., Maeyama, K., Mukherjee, P. K., Hattori, M., Choudhuri, M. S., Takeda, N., & Fukui, H. (2011). *Albizia lebbbeck* suppresses histamine signaling by the inhibition of histamine H1 receptor and histidine decarboxylase gene transcriptions. *International immunopharmacology*, 11(11), 1766–1772. <https://doi.org/10.1016/j.intimp.2011.07.003>

47 Bone, K. (2009). *The Ultimate Herbal Compendium: A desktop guide for herbal prescribers*

48 Klotter, J. (2006). Coughs, cough medicines, & upper respiratory infections. Townsend Letter: The Examiner of Alternative Medicine, (281), 38+. <https://link.gale.com/apps/doc/A155218063/AONE?u=anon~42a579ea&sid=googleScholar&xid=9f-dada15>

49 Turker, A. U., & Camper, N. D. (2002). Biological activity of common mullein, a medicinal plant. *Journal of ethnopharmacology*, 82(2-3), 117-125

50 Shin, H. S., Bae, M. J., Choi, D. W., & Shon, D. H. (2014). Skullcap (*Scutellaria baicalensis*) extract and its active compound, wogonin, inhibit ovalbumin-induced Th2-mediated response. *Molecules (Basel, Switzerland)*, 19(2), 2536–2545. <https://doi.org/10.3390/molecules19022536>

51 Khan, A. U., & Gilani, A. H. (2011). Blood pressure lowering, cardiovascular inhibitory and bronchodilatory actions of *Achillea millefolium*. *Phytotherapy research: PTR*, 25(4), 577–583. <https://doi.org/10.1002/ptr.3303> <https://pubmed.ncbi.nlm.nih.gov/20857434/>

52 Ziment, I. (2002). Herbal antitussives. *Pulmonary Pharmacology & Therapeutics*, 15(3), 327-333

53 Murdaca, G., Gerosa, A., Paladin, F., Petrocchi, L., Banchero, S., & Gangemi, S. (2021). Vitamin D and Microbiota: Is There a Link with Allergies?. *International journal of molecular sciences*, 22(8), 4288. <https://doi.org/10.3390/ijms22084288>

54 Doseděl, M., Jirkovský, E., Macáková, K., Krčmová, L. K., Javorská, L., Pourová, J., Mercoliní, L., Remião, F., Nováková, L., Mladěnka, P., & on Behalf of The Oeonomom (2021). Vitamin C-Sources, Physiological Role, Kinetics, Deficiency, Use, Toxicity, and Determination. *Nutrients*, 13(2), 615. <https://doi.org/10.3390/nu13020615>

55 Lewis, E. D., Meydani, S. N., & Wu, D. (2019). Regulatory role of vitamin E in the immune system and inflammation. *IUBMB life*, 71(4), 487–494. <https://doi.org/10.1002/iub.1976>

56 Mlecek, J., Jurikova, T., Skrovankova, S., & Sochor, J. (2016). Quercetin and Its Anti-Allergic Immune Response. *Molecules (Basel, Switzerland)*, 21(5), 623. <https://doi.org/10.3390/molecules21050623>

57 Wessels, I., Maywald, M., & Rink, L. (2017). Zinc as a Gatekeeper of Immune Function. *Nutrients*, 9(12), 1286. <https://doi.org/10.3390/nu9121286>

58 Balon, J. W., & Mior, S. A. (2004). Chiropractic care in asthma and allergy. *Annals of allergy, asthma & immunology: official publication of the American College of Allergy, Asthma, & Immunology*, 93(2 Suppl 1), S55–S60. [https://doi.org/10.1016/s1081-1206\(10\)61487-1](https://doi.org/10.1016/s1081-1206(10)61487-1)

59 Hauswald B, Yarin YM. Acupuncture in allergic rhinitis: A Mini-Review. *Allergo J Int*. 2014;23(4):115-119. doi: 10.1007/s40629-014-0015-3. Epub 2014 Jun 21. PMID: 26120523; PMCID: PMC4479426

60 Sicherer, S. H., Simons, F., & SECTION ON ALLERGY AND IMMUNOLOGY (2017). Epinephrine for First-aid Management of Anaphylaxis. *Pediatrics*, 139(3), e20164006. <https://doi.org/10.1542/peds.2016-4006>

61 Simon D. (2019). Recent Advances in Clinical Allergy and Immunology 2019. *International archives of allergy and immunology*, 180(4), 291–305. <https://doi.org/10.1159/000504364>

62 Head, K., Chong, L. Y., Hopkins, C., Philpott, C., Burton, M. J., & Schilder, A. G. (2016). Short-course oral steroids alone for chronic rhinosinusitis. *The Cochrane database of systematic reviews*, 4(4), CD011991. <https://doi.org/10.1002/14651858.CD011991.pub2>

63 Lax, S. J., Harvey, J., Axon, E., Howells, L., Santer, M., Ridd, M. J., Lawton, S., Langan, S., Roberts, A., Ahmed, A., Muller, I., Ming, L. C., Panda, S., Chernyshov, P., Carter, B., Williams, H. C., Thomas, K. S., & Chalmers, J. R. (2022). Strategies for using topical corticosteroids in children and adults with eczema. *The Cochrane database of systematic reviews*, 3(3), CD013356. <https://doi.org/10.1002/14651858.CD013356.pub2>

64 <https://www.allergy.org.au/patients/allergy-treatment/immunotherapy>

Asthma Protocol

1 Barnthouse M, Jones BL. The Impact of Environmental Chronic and Toxic Stress on Asthma. *Clin Rev Allergy Immunol*. 2019;57(3):427-438. doi:10.1007/s12016-019-08736-x Rosa MJ, Lee AG, Wright RJ. Evidence establishing a link between prenatal and early-life stress and asthma development. *Curr Opin Allergy Clin Immunol*. 2018;18(2):148-158. doi:10.1097/ACI.0000000000000421

2 Wang N, Li J, Huang X, Chen W, Chen Y. Herbal Medicine *Cordyceps sinensis* Improves Health-Related Quality of Life in Moderate-to-Severe Asthma. *Evid Based Complement Alternat Med*. 2016; 2016:6134593. doi:10.1155/2016/6134593

- 3 WHO: <https://www.who.int/news-room/fact-sheets/detail/asthma>
- 4 Gibson P. G., Reddel H., McDonald V. M., et al. Effectiveness and response predictors of omalizumab in a severe allergic asthma population with a high prevalence of comorbidities: the Australian Xolair Registry. *Internal Medicine Journal*. 2016;46(9):1054–1062. doi: 10.1111/imj.13166.
- 5 Akiyama K. Nihon Ishinkin Gakkai Zasshi. 2000;41(3):149–155. doi:10.3314/jjmm.41.149
- 6 Zhang Z, Reponen T, Hershey GK. Fungal Exposure and Asthma: IgE and Non-IgE-Mediated Mechanisms. *Curr Allergy Asthma Rep*. 2016;16(12):86. doi:10.1007/s11882-016-0667-9 Quansah R, Jaakkola MS, Hugg TT, Heikkinen SA, Jaakkola JJ. Residential dampness and molds and the risk of developing asthma: a systematic review and meta-analysis [published correction appears in *PLoS One*. 2014;9(3):e93454]. *PLoS One*. 2012;7(11):e47526. doi:10.1371/journal.pone.0047526
- 7 Dannemiller KC, Gent JF, Leaderer BP, Pecchia J. Influence of housing characteristics on bacterial and fungal communities in homes of asthmatic children. *Indoor Air*. 2016;26(2):179–192. doi:10.1111/ina.12205
- 8 Maheswaran D, Zeng Y, Chan-Yeung M, Scott J, Osornio-Vargas A, Becker AB, et al. (2014) Exposure to Beta-(1,3)-D-Glucan in House Dust at Age 7–10 Is Associated with Airway Hyperresponsiveness and Atopic Asthma by Age 11–14. *PLoS ONE* 9(6): e98878. <https://doi.org/10.1371/journal.pone.0098878> Gent JF, Kezik JM, Hill ME, Tsai E, Li DW, Leaderer BP. Household mold and dust allergens: exposure, sensitization and childhood asthma morbidity. *Environ Res*. 2012;118:86–93. doi:10.1016/j.envres.2012.07.005
- 9 Gent JF, Kezik JM, Hill ME, Tsai E, Li DW, Leaderer BP. Household mold and dust allergens: exposure, sensitization and childhood asthma morbidity. *Environ Res*. 2012;118:86–93. doi:10.1016/j.envres.2012.07.005 Pongracic JA, O'Connor GT, Muilenberg ML, et al. Differential effects of outdoor versus indoor fungal spores on asthma morbidity in inner-city children. *J Allergy Clin Immunol*. 2010;125(3):593–599. doi:10.1016/j.jaci.2009.10.036
- 10 Pongracic JA, O'Connor GT, Muilenberg ML, et al. Differential effects of outdoor versus indoor fungal spores on asthma morbidity in inner-city children. *J Allergy Clin Immunol*. 2010;125(3):593–599. doi:10.1016/j.jaci.2009.10.036 Segura-Medina P, Vargas MH, Aguilar-Romero JM, Arreola-Ramirez JL, Miguel-Reyes JL, Salas-Hernández J. Mold burden in house dust and its relationship with asthma control. *Respir Med*. 2019;150:74–80. doi:10.1016/j.rmed.2019.02.014
- 11 Burr ML, Matthews IP, Arthur RA, et al. Effects on patients with asthma of eradicating visible indoor mould: a randomised controlled trial. *Thorax*. 2007;62(9):767–772. doi:10.1136/thx.2006.070847
- 12 <https://www.asthma.org.uk/> and American Lung Association www.lung.org/blog/mold-and-asthma
- 13 Guarnieri M, Balmes JR. Outdoor air pollution and asthma. *Lancet*. 2014;383(9928):1581–1592. doi:10.1016/S0140-6736(14)60617-6
- 14 Tiotiu AI, Novakova P, Nedeva D, et al. Impact of Air Pollution on Asthma Outcomes. *Int J Environ Res Public Health*. 2020;17(17):6212. Published 2020 Aug 27. doi:10.3390/ijerph17176212
- 15 Khreis H, Kelly C, Tate J, Parslow R, Lucas K, Nieuwenhuijsen M. Exposure to traffic-related air pollution and risk of development of childhood asthma: A systematic review and meta-analysis. *Environ Int*. 2017;100:1–31. doi:10.1016/j.envint.2016.11.012
- 16 Jung CR, Chen WT, Tang YH, Hwang BF. Fine particulate matter exposure during pregnancy and infancy and incident asthma. *J Allergy Clin Immunol*. 2019;143(6):2254–2262.e5. doi:10.1016/j.jaci.2019.03.024 Silvestri M, Franchi S, Pistorio A, Petecchia L, Rusconi F. Smoke exposure, wheezing, and asthma development: a systematic review and meta-analysis in unselected birth cohorts. *Pediatr Pulmonol*. 2015;50(4):353–362. doi:10.1002/ppul.23037
- 17 World Health Organization:<https://www.who.int/teams/environment-climate-change-and-health/air-quality-and-health/health-impacts/types-of-pollutants> Tiotiu AI, Novakova P, Nedeva D, et al. Impact of Air Pollution on Asthma Outcomes. *Int J Environ Res Public Health*. 2020;17(17):6212. Published 2020 Aug 27. doi:10.3390/ijerph17176212
- 18 Toppila-Salmi S, Luukkainen AT, Xu B, et al. Maternal smoking during pregnancy affects adult onset of asthma in offspring: a follow up from birth to age 46 years. *Eur Respir J*. 2020;55(6):1901857. Published 2020 Jun 11. doi:10.1183/13993003.01857-2019
- 19 Thomson NC, Chaudhuri R, Livingston E. Asthma and cigarette smoking. *Eur Respir J*. 2004;24(5):822–833. doi:10.1183/09031936.04.00039004
- 20 Sippel JM, Pedula KL, Vollmer WM, Buist AS, Osborne ML. Associations of smoking with hospital-based care and quality of life in patients with obstructive airway disease. *Chest*. 1999;115(3):691–696. doi:10.1378/chest.115.3.691
- 21 Tønnesen P, Pisinger C., Hvidberg S., Wenneke P, Bremann L., Westin A., Thomsen C., Nilsson F. Effects of smoking cessation and reduction in asthmatics. *Nicotine Tob. Res*. 2005;7:139–148. doi: 10.1080/14622200412331328411.
- 22 American Lung Association <https://www.lung.org/blog/how-the-air-in-there-asthma.ca/3-ways-to-improve-your-indoor-air-quality-manage-asthma/>
- 23 American Lung Association
- 24 Tiotiu AI, Novakova P, Nedeva D, et al. Impact of Air Pollution on Asthma Outcomes. *Int J Environ Res Public Health*. 2020;17(17):6212. Published 2020 Aug 27. doi:10.3390/ijerph17176212
- 25 American Academy of Allergy, Asthma, & Immunology <https://www.aaaai.org/Tools-for-the-Public/Conditions-Library/Asthma/Your-Questions-Answered-on-Air-Pollution-and-Asthma> Asthma and Lung UK <https://www.asthma.org.uk/advice/triggers/pollution/> American Lung Association: <https://www.lung.org/clean-air/outdoors/10-tips-to-protect-yourself>
- 26 Roberts G, Lack G. Food allergy and asthma—what is the link?. *Paediatr Respir Rev*. 2003;4(3):205–212. doi:10.1016/s1526-0542(03)00058-7
- 27 Foong RX, du Toit G, Fox AT. Mini Review - Asthma and Food Allergy. *Curr Pediatr Rev*. 2018;14(3):164–170. doi:10.2174/1573396314666180507121136
- 28 Bedolla-Pulido TR, Bedolla-Barajas M, Uribe-Cota B, González-Mendoza T, Morales-Romero J, Mariscal-Castro J. Alergia a alimentos en adultos con enfermedades respiratorias alérgicas: prevalencia y manifestaciones clínicas [Allergy to food in adults with allergic respiratory diseases: prevalence and clinical manifestations]. *Rev Alerg Mex*. 2019;66(1):1–8. doi:10.29262/ram.v66i1.400
- 29 Vassilopoulou E, Konstantinou G, N., Dimitriou A., Manios, Y., Koumbi, L., & Papadopoulos, N. G. (2020). The Impact of Food Histamine Intake on Asthma Activity: A Pilot Study. *Nutrients*, 12(11), 3402. <https://doi.org/10.3390/nu12113402>
- 30 Son, J. H., Chung, B. Y., Kim, H. O., & Park, C. W. (2018). A Histamine-Free Diet Is Helpful for Treatment of Adult Patients with Chronic Spontaneous Urticaria. *Annals of dermatology*, 30(2), 164–172. <https://doi.org/10.5021/ad.2018.30.2.164>
- 31 Vassilopoulou, Emilia & Konstantinou, George & Dimitriou, Anastasia & Manios, Yannis & Koumbi, Lemonica & Papadopoulos, Nikolaos. (2020). The Impact of Food Histamine Intake on Asthma Activity: A Pilot Study. *Nutrients*. 12. 3402. 10.3390/nu12113402.
- 32 Sánchez-Pérez, S., Comas-Basté, O., Veciana-Nogués, M. T., Latorre-Moratalla, M. L., & Vidal-Carou, M. C. (2021). Low-Histamine Diets: Is the Exclusion of Foods Justified by Their Histamine Content?. *Nutrients*, 13(5), 1395. <https://doi.org/10.3390/nu13051395>
- 33 Christ A, Lauterbach M, Latz E. Western Diet and the Immune System: An Inflammatory Connection. *Immunity*. 2019;51(5):794–811. doi:10.1016/j.immuni.2019.09.020
- 34 Guilleminault L, Williams EJ, Scott HA, Berthon BS, Jensen M, Wood LG. Diet and Asthma: Is It Time to Adapt Our Message?. *Nutrients*. 2017;9(11):1227. Published 2017 Nov 8. doi:10.3390/nu9111227
- 35 Scott HA, Gibson PG, Garg ML, et al. Dietary restriction and exercise improve airway inflammation and clinical outcomes in overweight and obese asthma: a randomized trial. *Clin Exp Allergy*. 2013;43(1):36–49. doi:10.1111/cea.12004
- 36 DeChristopher LR, Uribarri J, Tucker KL. Intakes of apple juice, fruit drinks and soda are associated with prevalent asthma in US children aged 2–9 years. *Public Health Nutr*. 2016;19(1):123–130. doi:10.1017/S1368980015000865 Castro-Rodriguez JA, Ramirez-Hernandez M, Padilla O, Pacheco-Gonzalez RM, Pérez-Fernández V, Garcia-Marcos L. Effect of foods and Mediterranean diet during pregnancy and first years of life on wheezing, rhinitis and dermatitis in preschoolers. *Allergol Immunopathol (Madr)*. 2016;44(5):400–409. doi:10.1016/j.aller.2015.12.002
- 37 Ensiyeh Seyedrezazadeh, Masoud Pour Moghaddam, Khalil Ansarin, Mohammad Reza Vafa, Sangita Sharma, Fariba Kolahdooz, Fruit and vegetable intake and risk of wheezing and asthma: a systematic review and meta-analysis, *Nutrition Reviews*, Volume 72, Issue 7, 1 July 2014, Pages 411–428, <https://doi.org/10.1111/nure.12121> Hosseini B, Berthon BS, Wark P, Wood LG. Effects of Fruit and Vegetable Consumption on Risk of Asthma, Wheezing and Immune Responses: A Systematic Review and Meta-Analysis. *Nutrients*. 2017;9(4):341. Published 2017 Mar 29. doi:10.3390/nu9040341
- 38 Patel B.D., Welch A.A., Bingham S.A., Luben R.N., Day N.E., Khaw K.T., Lomas D.A., Wareham N.J. Dietary antioxidants and asthma in adults. *Thorax*. 2006;61:388–393. doi: 10.1136/thx.2004.024935. Wood L.G., Garg M.L., Smart J.M., Scott H.A., Barker D., Gibson P.G. Manipulating antioxidant intake in asthma: A randomized controlled trial. *Am. J. Clin. Nutr*. 2012;96:534–543. doi: 10.3945/ajcn.111.032623.
- 39 Whyand, T, Hurst, J.R., Beckles, M. et al. Pollution and respiratory disease: can diet or supplements help? A review. *Respir Res* 19, 79 (2018). <https://doi.org/10.1186/s12931-018-0785-0>
- 40 Korn S, Hübner M, Jung M, Blettner M, Buhl R. Severe and uncontrolled adult asthma is associated with vitamin D insufficiency and deficiency. *Respir Res*. 2013;14(1):25. Published 2013 Feb 22. doi:10.1186/1465-9921-14-25 Litonjua AA. Childhood asthma may be a consequence of vitamin D deficiency. *Curr Opin Allergy Clin Immunol*. 2009;9(3):202–207. doi:10.1097/ACI.0b013e32832b36cd
- 41 Kerley CP, Elnazir B, Faul J, Cormican L. Vitamin D as an adjunctive therapy in asthma. Part 2: A review of human studies. *Pulm Pharmacol Ther*. 2015;32:75–92. doi:10.1016/j.pupt.2015.02.010
- 42 Wang M, Liu M, Wang C, et al. Association between vitamin D status and asthma control: A meta-analysis of randomized trials. *Respir Med*. 2019;150:85–94. doi:10.1016/j.rmed.2019.02.016
- 43 Yan YX, Li YN. *Zhongguo Dang Dai Er Ke Za Zhi*. 2019;21(7):724–729. doi:10.7499/j.issn.1008-8830.2019.07.020
- 44 Canguen O, El Ansari W, Yassin A. Vitamin D Supplementation As a Potential therapeutic Mediator in Asthma: Does Dose Really Matter? a Critical Review of the Literature. *Aging Male*. 2020;23(4):300–307. doi:10.1080/13685538.2018.1506433
- 45 Fantidis P, Ruiz Cacho J, Marín M, Madero Jarabo R, Solera J, Herrero E. Intracellular (polymorphonuclear) magnesium content in patients with bronchial asthma between attacks. *J R Soc Med*. 1995 Aug;88(8):441–5. PMID: 7562826; PMCID: PMC1295297.
- 46 Abuabat F, AlAlwan A, Masuadi E, Murad MH, Jahdali HA, Ferwana MS. The role of oral magnesium supplements for the management of stable bronchial asthma: a systematic review and meta-analysis. *NPJ Prim Care Respir Med*. 2019;29(1):4. Published 2019 Feb 18. doi:10.1038/s41533-019-0116-z
- 47 Gontijo-Amaral, C., Ribeiro, M., Gontijo, L. et al. Oral magnesium supplementation in asthmatic children: a double-blind randomized placebo-controlled trial. *Eur J Clin Nutr* 61, 54–60 (2007). <https://doi.org/10.1038/sj.ejcn.1602475>
- 48 Kazaks, Alexandra G., et al. “Effect of oral magnesium supplementation on measures of airway resistance and subjective assessment of asthma control and quality of life in men and women with mild to moderate asthma: a randomized placebo-controlled trial.” *Journal of Asthma* 47.1 (2010): 83–92.

- 47 Yu H, Su X, Lei T, Zhang C, Zhang M, Wang Y, Zhu L, Liu J. Effect of Omega-3 Fatty Acids on Chronic Obstructive Pulmonary Disease: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *Int J Chron Obstruct Pulmon Dis*. 2021 Sep 23;16:2677-2686. doi: 10.2147/COPD.S331154. PMID: 34588776; PMCID: PMC8476109.
- 48 Stoodley I, Garg M, Scott H, Macdonald-Wicks L, Berthon B, Wood L. Higher Omega-3 Index Is Associated with Better Asthma Control and Lower Medication Dose: A Cross-Sectional Study. *Nutrients*. 2019;12(1):74. Published 2019 Dec 27. doi:10.3390/nu12010074
- 49 Okamoto M, Mitsunobu F, Ashida K, et al. Effects of dietary supplementation with n-3 fatty acids compared with n-6 fatty acids on bronchial asthma. *Intern Med*. 2000;39(2):107-111. doi:10.2169/internalmedicine.39.10
- 50 Emelyanov A, Fedoseev G, Krasnoschekova O, Abulimity A, Trendelewa T, Barnes PJ. Treatment of asthma with lipid extract of New Zealand green-lipped mussel: a randomised clinical trial. *Eur Respir J*. 2002;20(3):596-600. doi:10.1183/09031936.02.02632001
- 51 de Luis DA, León R, Izaola O. Influencia de la ingesta de los ácidos grasos Omega 3 en el asma [Influence of omega 3 fatty acid dietary intake on asthma]. *Rev Clin Esp*. 2007;207(1):24-25. doi:10.1157/13098496
- 52 Bazan-Socha S, Wójcik K, Olchawa M, et al. Increased Oxidative Stress in Asthma-Relation to Inflammatory Blood and Lung Biomarkers and Airway Remodeling Indices. *Biomedicines*. 2022;10(7):1499. Published 2022 Jun 24. doi:10.3390/biomedicines10071499
- 53 Sahiner UM, Birben E, Erzurum S, Sackesen C, Kalayci Ö. Oxidative stress in asthma: Part of the puzzle. *Pediatr Allergy Immunol*. 2018;29(8):789-800. doi:10.1111/pai.12965
- 54 Zajac D. Mineral Micronutrients in Asthma. *Nutrients*. 2021;13(11):4001. Published 2021 Nov 10. doi:10.3390/nu13114001
- 54 Secor ER Jr, Szczepanek SM, Castater CA, Adami AJ, Matson AP, Rafti ET, Guernsey L, Natarajan P, McNamara JT, Schramm CM, Thrall RS, Silbart LK. Bromelain Inhibits Allergic Sensitization and Murine Asthma via Modulation of Dendritic Cells. *Evid Based Complement Alternat Med*. 2013;2013:702196. doi: 10.1155/2013/702196. Epub 2013 Dec 5. PMID: 24381635; PMCID: PMC3870104. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3870104/>
- 55 Secor, E. R., Jr; Shah, S. J., Guernsey, L. A., Schramm, C. M., & Thrall, R. S. (2012). Bromelain limits airway inflammation in an ovalbumin-induced murine model of established asthma. *Alternative therapies in health and medicine*, 18(5), 9–17.
- 56 Chen M, Sun Y, Wu Y. Lower circulating zinc and selenium levels are associated with an increased risk of asthma: evidence from a meta-analysis. *Public Health Nutr*. 2020;23(9):1555-1562. doi:10.1017/S1368980019003021
- 57 Greene LS. Asthma and oxidant stress: nutritional, environmental, and genetic risk factors. *J Am Coll Nutr*. 1995;14(4):317-324. doi: 10.1080/07315724.1995.10718516
- 58 Hu G, Cassano PA. Antioxidant nutrients and pulmonary function: the Third National Health and Nutrition Examination Survey (NHANES III). *Am J Epidemiol*. 2000;151(10):975-981. doi:10.1093/oxfordjournals.aje.a010141
- 59 Hu G, Cassano PA. Antioxidant nutrients and pulmonary function: the Third National Health and Nutrition Examination Survey (NHANES III). *Am J Epidemiol*. 2000;151(10):975-981. doi:10.1093/oxfordjournals.aje.a010141
- 60 F. Gazdík, J. Kadrabová, K. Gazdík. Decreased consumption of corticosteroids after selenium supplementation in corticoid-dependent asthmatics *Bratisl Med J*, 103 (2002), pp. 22-25
- 61 Devirgiliis C, Zalewski PD, Perozzi G, Murgia C. Zinc fluxes and zinc transporter genes in chronic diseases. *Mutat Res*. 2007;622(1-2):84-93. doi:10.1016/j.mrfmm.2007.01.013
- 62 Soutar A, Seaton A, Brown K. Bronchial reactivity and dietary antioxidants. *Thorax*. 1997;52(2):166-170. doi:10.1136/thx.52.2.166
- 62 Rerksupphaphol S, Rerksupphaphol L. Zinc Supplementation in Children with Asthma Exacerbation. *Pediatr Rep*. 2016;8(4):6685. Published 2016 Dec 9. doi:10.4081/pr.2016.6685
- 63 Olin JT, Wechsler M E. Asthma: pathogenesis and novel drugs for treatment *BMJ* 2014; 349 :g5517 doi:10.1136/bmj.g5517
- 64 Tecklenburg SL, Mickleborough TD, Fly AD, Bai Y, Stager JM. Ascorbic acid supplementation attenuates exercise-induced bronchoconstriction in patients with asthma. *Respir Med*. 2007;101(8):1770-1778. doi:10.1016/j.rmed.2007.02.014
- 65 Omenaas E, Fluge O, Buist AS, Vollmer WM, Gulsvik A. Dietary vitamin C intake is inversely related to cough and wheeze in young smokers. *Respir Med* 2003;97:134–142
- 66 Romieu I, Sienra-Monge JJ, Ramírez-Aguilar M, et al. Antioxidant supplementation and lung functions among children with asthma exposed to high levels of air pollutants. *Am J Respir Crit Care Med*. 2002;166(5):703-709. doi:10.1164/rccm.2112074
- 67 Barnes LA, Leach M, Anheyer D, Brown D, Carè J, Lauche R, Medina DN, Pinder TA, Bugarcic A, Steel A. The effects of Hedera helix on viral respiratory infections in humans: A rapid review. *Adv Integr Med*. 2020 Dec;7(4):222-226. doi: 10.1016/j.aimed.2020.07.012. Epub 2020 Aug 13. PMID: 32837900; PMCID: PMC7424313.
- 68 Baharara, H., Moghadam, A.T., Sahebkar, A., Emami, S.A., Tayebi, T., Mohammadpour, A.H. (2021). The Effects of Ivy (Hedera helix) on Respiratory Problems and Cough in Humans: A Review. In: Sahebkar, A., Sathyapalan, T. (eds) *Natural Products and Human Diseases*. *Advances in Experimental Medicine and Biology()*, vol 1328. Springer, Cham. https://doi.org/10.1007/978-3-030-73234-9_23
- 69 Zeil S, Schwanebeck U, Vogelberg C. Tolerance and effect of an add-on treatment with a cough medicine containing ivy leaves dry extract on lung function in children with bronchial asthma. *Phytomedicine*. 2014;21(10):1216-1220. doi:10.1016/j.phymed.2014.05.006
- 70 Bücking S. Assessment of the Efficacy and Safety of Ivy Leaf (Hedera helix) Cough Syrup Compared with Acetylcysteine in Adults and Children with Acute Bronchitis. *Evid Based Complement Alternat Med*. 2020;2020:1910656. Published 2020 May 4. doi:10.1155/2020/1910656
- 71 Schaefer A, Kehr MS, Giannetti BM, Bulitta M, Staiger C. A randomized, controlled, double-blind, multi-center trial to evaluate the efficacy and safety of a liquid containing ivy leaves dry extract (EA 575®) vs. placebo in the treatment of adults with acute cough. *Pharmazie*. 2016;71(9):504-509.
- 72 Schmidt M, Thomsen M, Schmidt U. Suitability of ivy extract for the treatment of paediatric cough. *Phytother Res*. 2012;26(12):1942-1947. doi:10.1002/ptr.4671
- 73 Chen J, Liu X, Li Z, et al. A Review of Dietary Ziziphus jujuba Fruit (Jujube): Developing Health Food Supplements for Brain Protection. *Evid Based Complement Alternat Med*. 2017;2017:3019568. doi:10.1155/2017/3019568
- 74 Jiaqi Ma, Xuemin Chen, Lei Zhang, Liyan Ma, Jiangui Li, Jingming Li, Jiachen Zang, The stability and absorption of naturally occurring cAMP by its weak interactions with jujube polysaccharides were greatly improved. *Food Hydrocolloids*, Vol. 133, 2022. ISSN 0268-005X, <https://doi.org/10.1016/j.foodhyd.2022.107957>.
- 75 Ninave PB, Patil SD. Antiasthmatic potential of Zizyphus jujuba Mill and Jujuboside B. - Possible role in the treatment of asthma. *Respir Physiol Neurobiol*. 2019;260:28-36. doi:10.1016/j.resp.2018.12.001
- 76 Kim, Y., Oh, J. and Kim, J. (2020). Anti-inflammatory Effect of Hydrolyzed Jujube Ethanolic Extract. *The FASEB Journal*, 34: 1-1. <https://doi.org/10.1096/fasebj.2020.34.s1.07063>
- 77 Suresh R. Naik, Sushant Bhagat, Priyank D. Shah, Abhishek A. Tare, Deepa Ingawale, Raju R. Wadekar, Evaluation of anti-allergic and anti-anaphylactic activity of ethanolic extract of Zizyphus jujuba fruits in rodents, *Revista Brasileira de Farmacognosia*, Volume 23, Issue 5, 2013. <https://doi.org/10.1590/S0102-695X2013000500014>.
- 77 Jujuboside B Reduces Vascular Tension by Increasing Ca2+ Influx and Activating Endothelial Nitric Oxide Synthase Zhao Y, Zhang X, Li J, Bian Y, Sheng M, et al. (2016) Jujuboside B Reduces Vascular Tension by Increasing Ca2+ Influx and Activating Endothelial Nitric Oxide Synthase. *PLOS ONE* 11(2): e0149386. <https://doi.org/10.1371/journal.pone.0149386>
- 78 Tao Z, Jin W, Ao M, Zhai S, Xu H, Yu L. Evaluation of the anti-inflammatory properties of the active constituents in Ginkgo biloba for the treatment of pulmonary diseases. *Food Funct*. 2019;10(4):2209-2220. doi:10.1039/c8fo02506a
- 79 Tang YJ, Xu YJ, Zhang ZX. *Zhongguo Zhong Xi Yi Jie He Za Zhi*. 2006;26 Suppl:47-50.
- 79 Mahmoud F, Abul H, Onadeko B, Khadadad M, Haines D, Morgan G. In vitro effects of Ginkgolide B on lymphocyte activation in atopie asthma: comparison with cyclosporin A. *Jpn J Pharmacol*. 2000;83(3):241-245. doi:10.1254/jip.83.241
- 80 Li, Mh., Yang, By, Yu, H. et al. Clinical observation of the therapeutic effect of Ginkgo leaf concentrated oral liquor on bronchial asthma. *CJIM* 3, 264–267 (1997). <https://doi.org/10.1007/BF02934827>
- 81 Tang Y, Xu Y, Xiong S, et al. The effect of Ginkgo Biloba extract on the expression of PKCα in the inflammatory cells and the level of IL-5 in induced sputum of asthmatic patients. *J Huazhong Univ Sci Technolog Med Sci*. 2007;27(4):375-380. doi:10.1007/s11596-007-0407-4
- 81 Wilkens, J., Wilkens, H., Uffmann, J., Bovers, J., Fabel, H. and Frolich, J. (1990), Effects of a PAF-antagonist (BN 52063) on bronchoconstriction and platelet activation during exercise induced asthma. *British Journal of Clinical Pharmacology*, 29: 85-91. <https://doi.org/10.1111/j.1365-2125.1990.tb03606.x>
- 82 Jiahui ZHENG; Guoshi WAN; Yuanxiu ZHENG; Fangfang HUANG; Zhumei WANG. Effects of ginkgo biloba tablets plus combined with routine therapy on HPA axis in patients with steroid-dependent asthma *Chinese Journal of Immunology*; (12): 97-100, 2016.
- 83 Arzu Ucar Turker, N.D Camper, Biological activity of common mullein, a medicinal plant, *Journal of Ethnopharmacology*, Volume 82, Issues 2–3, 2002, Pages 117-125, ISSN 0378-8741, [https://doi.org/10.1016/S0378-8741\(02\)00186-1](https://doi.org/10.1016/S0378-8741(02)00186-1).
- 84 Blanco-Salas J, Hortigón-Vinagre MP, Morales-Jadán D, Ruiz-Téllez T. Searching for Scientific Explanations for the Uses of Spanish Folk Medicine: A Review on the Case of Mullein (Verbascum, Scrophulariaceae). *Biology*. 2021; 10(7):618. <https://doi.org/10.3390/biology10070618>
- 85 Saeidi, Karamatollah, and Zahra Lorigooini. “Determination of mucilage content of mullein (Verbascum songaricum) populations.” *Journal of Pharmaceutical Sciences and Research* 9.12 (2017): 2641-2643.
- 86 Turker, A. U., & Gurel, E. (2005). Common mullein (Verbascum thapsus L.): recent advances in research. *Phytotherapy Research: An International Journal Devoted to Pharmacological and Toxicological Evaluation of Natural Product Derivatives*, 19(9), 733-739
- 87 Yang X, Li Y, He Y, et al. Cordycepin alleviates airway hyperreactivity in a murine model of asthma by attenuating the inflammatory process. *International Immunopharmacology*. 2015;26(2):401–408. doi: 10.1016/j.intimp.2015.04.017.
- 88 Kim H. G., Shrestha B., Lim S. Y., et al. Cordycepin inhibits lipopolysaccharide-induced inflammation by the suppression of NF-κB through Akt and p38 inhibition in RAW 264.7 macrophage cells. *European Journal of Pharmacology*. 2006;545(2-3):192–199. doi: 10.1016/j.ejphar.2006.06.047.
- 88 Wang N, Li J, Huang X, Chen W, Chen Y. Herbal Medicine Cordyceps sinensis Improves Health-Related Quality of Life in Moderate-to-Severe Asthma. *Evid Based Complement Alternat Med*. 2016;2016:6134593. doi:10.1155/2016/6134593
- 89 Bhardwaj N, Katyal P, Sharma AK. Suppression of inflammatory and allergic responses by pharmacologically potent fungus *Ganoderma lucidum*. *Recent Pat Inflamm Allergy Drug Discov*. 2014;8(2):104-17. doi: 10.2174/1872213x08666140619110657. PMID: 24948193
- 90 Li Y, Li M, Wang R, Wang B, Athari SS, Wang J. *Ganoderma* modulates allergic asthma pathologic features via anti-inflammatory effects. *Respir Physiol Neurobiol*. 2022;299:103843. doi:10.1016/j.resp.2022.103843
- 91 Chen ML, Lin BF. Effects of triterpenoid-rich extracts of *Ganoderma tsugae* on airway hyperreactivity and Th2 responses in vivo. *Int Arch Allergy Immunol*. 2007;143(1):21-30. doi:10.1159/000098222
- 91 Roschek B Jr, Fink RC, McMichael M, Albert RS. Nettle extract (Urtica dioica)

- affects key receptors and enzymes associated with allergic rhinitis. *Phytother Res*. 2009;23:920-6. CrossRef 27. Mittman P. Randomized, double-blind study of freeze-dried *Urtica dioica* in the treatment of allergic rhinitis. *Planta Med*. 1990;56:44-7. Zem-mouri, H., Sekiou, O., Ammar, S., El Feki, A., Bouaziz, M., Messarah, M., & Boumendjel, A. (2017). *Urtica dioica* attenuates ovalbumin-induced inflammation and lipid peroxidation of lung tissues in rat asthma model. *Pharmaceutical biology*, 55(1), 1561-1568. <https://www.tandfonline.com/doi/full/10.1080/13880209.2017.1310905>
- 92 Said, A. Ait Haj, et al. "Highlights on nutritional and therapeutic value of stinging nettle (*Urtica dioica*).²" *International Journal of Pharmacy and Pharmaceutical Sciences* 7.10 (2015): 8-14. APA
- 93 Bakhshae M, Mohammad Pour AH, Esmaeili M, et al. Efficacy of Supportive Therapy of Allergic Rhinitis by Stinging Nettle (*Urtica dioica*) root extract: a Randomized, Double-Blind, Placebo- Controlled, Clinical Trial. *Iran J Pharm Res*. 2017;16(Suppl):112-118
- 94 M. Irani, R. Choopani, S. Esmaeili, T. Dargahi, S.M. Athari, S.S. Athari. Effect of nettle seed on immune response in a murine model of allergic asthma. *Revue Française d'Allergologie*, Volume 60, Issue 5, 2020, Pages 417-422, ISSN 1877-0320, <https://doi.org/10.1016/j.rval.2020.03.007>.
- 95 Bone, K. (2009). *The Ultimate Herbal Compendium: A desktop guide for herbal prescribers*
- 96 Ziment, I. (2002). Herbal antitussives. *Pulmonary Pharmacology & Therapeutics*, 15(3), 327-333.
- 97 Shavitt RG, Gentil V, Mandetta R. The association of panic/agoraphobia and asthma. Contributing factors and clinical implications. *Gen Hosp Psychiatry*. 1992;14(6):420-423. doi:10.1016/0163-8343(92)90010-8
- 98 Lehrer, P.M. Emotionally Triggered Asthma: A Review of Research Literature and Some Hypotheses for Self-Regulation Therapies. *Appl Psychophysiol Biofeedback* 23, 13-41 (1998). <https://doi.org/10.1023/A:1022170028979>
- 99 Yadav P, Jain PK, Sharma BS, Sharma M. Yoga Therapy as an Adjuvant in Management of Asthma. *Indian J Pediatr*. 2021;88(11):1127-1134. doi:10.1007/s12098-021-03675-y
- 100 Vempati, R., Bijlani, R.L. & Deepak, K.K. The efficacy of a comprehensive lifestyle modification programme based on yoga in the management of bronchial asthma: a randomized controlled trial. *BMC Pulm Med* 9, 37 (2009). <https://doi.org/10.1186/1471-2466-9-37>
- 101 Yang ZY, Zhong HB, Mao C, et al. Yoga for asthma. *Sao Paulo Med J*. 2016;134(4):368. doi:10.1590/1516-3180.20161344T2
- 102 Paudyal P, Jones C, Grindley C, Dawood R, Smith H. Meditation for asthma: Systematic review and meta-analysis. *J Asthma*. 2018;55(7):771-778. doi:10.1080/02770903.2017.1365887
- 103 Liu CF, Chien LW. Efficacy of acupuncture in children with asthma: a systematic review. *Ital J Pediatr*. 2015;41:48. Published 2015 Jul 7. doi:10.1186/s13052-015-0155-1
- 104 Jiang C, Jiang L, Qin Q. Conventional Treatments plus Acupuncture for Asthma in Adults and Adolescent: A Systematic Review and Meta-Analysis. *Evid Based Complement Alternat Med*. 2019;2019:9580670. Published 2019 Jan 17. doi:10.1155/2019/9580670
- 105 Frank DL, Khorshid L, Kiffer JF, Moravec CS, McKee MG. Biofeedback in medicine: who, when, why and how?. *Ment Health Fam Med*. 2010;7(2):85-91.
- 106 Paul M, Lehrer, Evgeny Vaschillo, Bronya Vaschillo, Shou-En Lu, Anthony Scardella, Mahmood Siddique, Robert H. Habib. Biofeedback Treatment for Asthma. *Chest*. Vol 126, Issue 2, 2004, 352-361, ISSN 0012-3692, <https://doi.org/10.1378/chest.126.2.352>.
- 107 Taghizadeh N, Eslaminejad A, Raoofy MR. Protective effect of heart rate variability biofeedback on stress-induced lung function impairment in asthma. *Respir Physiol Neurobiol*. 2019;262:49-56. doi:10.1016/j.rresp.2019.01.011
- 108 Allam, V.S.R.R., Paudel, K.R., Gupta, G. et al. Nutraceuicals and mitochondrial oxidative stress: bridging the gap in the management of bronchial asthma. *Environ Sci Pollut Res* (2022). <https://doi.org/10.1007/s11356-022-21454-w>
- 109 Patel R, Naqvi SA, Griffiths C, Bloom CI. Systemic adverse effects from inhaled corticosteroid use in asthma: a systematic review. *BMJ Open Respir Res*. 2020;7(1):e000756. doi:10.1136/bmjresp-2020-000756
- ## Lungs And COPD Protocol
- 1 Shergis, J. L., Di, Y. M., Zhang, A. L., Vlahos, R., Helliwell, R., Ye, J. M., & Xue, C. C. (2014). Therapeutic potential of Panax ginseng and ginsenosides in the treatment of chronic obstructive pulmonary disease. *Complementary Therapies in Medicine*, 22(5), 944-953
- 2 Shergis, J. L., Di, Y. M., Zhang, A. L., Vlahos, R., Helliwell, R., Ye, J. M., & Xue, C. C. (2014). Therapeutic potential of Panax ginseng and ginsenosides in the treatment of chronic obstructive pulmonary disease. *Complementary Therapies in Medicine*, 22(5), 944-953
- 3 Scoditti E, Massaro M, Garbarino S, Toraldo DM. Role of Diet in Chronic Obstructive Pulmonary Disease Prevention and Treatment. *Nutrients*. 2019 Jun 16;11(6):1357. doi: 10.3390/nu11061357. PMID: 31208151; PMCID: PMC6627281
- 4 Scoditti E, Massaro M, Garbarino S, Toraldo DM. Role of Diet in Chronic Obstructive Pulmonary Disease Prevention and Treatment. *Nutrients*. 2019 Jun 16;11(6):1357. doi: 10.3390/nu11061357. PMID: 31208151; PMCID: PMC6627281
- 5 Scoditti E, Massaro M, Garbarino S, Toraldo DM. Role of Diet in Chronic Obstructive Pulmonary Disease Prevention and Treatment. *Nutrients*. 2019 Jun 16;11(6):1357. doi: 10.3390/nu11061357. PMID: 31208151; PMCID: PMC6627281
- 6 Asha'ari ZA, Ahmad MZ, Jihan WS, Che CM, Leman I. Ingestion of honey improves the symptoms of allergic rhinitis: evidence from a randomized placebo-controlled trial in the East coast of Peninsular Malaysia. *Ann Saudi Med*. 2013 Sep-Oct;33(5):469-75. doi: 10.5144/0256-4947.2013.469. PMID: 24188941; PMCID: PMC6074882
- 7 Bakhshae M, Mohammad Pour, A. H., Esmaeili, M., Jabbari Azad, F., Alipour Talesh, G., Salehi, M., & Noorollahian Mohajer, M. (2017). Efficacy of Supportive Therapy of Allergic Rhinitis by Stinging Nettle (*Urtica dioica*) root extract: a Randomized, Double-Blind, Placebo- Controlled, Clinical Trial. *Iranian journal of pharmaceutical research: IJPR*, 16(Suppl), 112-118
- 8 Mittman P. (1990). Randomized, double-blind study of freeze-dried *Urtica dioica* in the treatment of allergic rhinitis. *Planta medica*, 56(1), 44-47. <https://doi.org/10.1055/s-2006-960881>
- 9 Bone, K. (2009). *The Ultimate Herbal Compendium: A desktop guide for herbal prescribers* Klotter, J. (2006). Coughs, cough medicines, & upper respiratory infections. *Townsend Letter: The Examiner of Alternative Medicine*, (281), 38+. <https://link.gale.com/apps/doc/A155218063/AONE?u=anon~42a579ea&sid=googleScholar&xid=9f-dada15>
- 10 Klotter, J. (2006). Coughs, cough medicines, & upper respiratory infections. *Townsend Letter: The Examiner of Alternative Medicine*, (281), 38+. <https://link.gale.com/apps/doc/A155218063/AONE?u=anon~42a579ea&sid=googleScholar&xid=9f-dada15>
- 11 Turker, A. U., & Camper, N. D. (2002). Biological activity of common mullein, a medicinal plant. *Journal of ethnopharmacology*, 82(2-3), 117-125
- 12 Gierlikowska, B., Gierlikowski, W., Bekier, K., Skaliczka-Woźniak, K., Czerwińska, M. E., & Kiss, A. K. (2020). Inula helenium and Grindelia squarrosa as a source of compounds with anti-inflammatory activity in human neutrophils and cultured human respiratory epithelium. *Journal of ethnopharmacology*, 249, 112311. <https://doi.org/10.1016/j.jep.2019.112311>
- 13 Benson, K. F., Stamets, P., Davis, R., Nally, R., Taylor, A., Slater, S., & Jensen, G. S. (2019). The mycelium of the Trametes versicolor (Turkey tail) mushroom and its fermented substrate each show potent and complementary immune activating properties in vitro. *BMC complementary and alternative medicine*, 19(1), 342. <https://doi.org/10.1186/s12906-019-2681-7>
- 14 Lin Z. B. (2005). Cellular and molecular mechanisms of immuno-modulation by *Ganoderma lucidum*. *Journal of pharmacological sciences*, 99(2), 144-153. <https://doi.org/10.1254/jphs.crj05008x>
- 15 Wu TF, Chan YY, Shi WY, Jhong MT. Uncovering the Molecular Mechanism of Anti-Allergic Activity of Silkworm Pupa-Grown *Cordyceps militaris* Fruit Body. *Am J Chin Med*. 2017;45(3):497-513. doi: 10.1142/S0192415X17500306. Epub 2017 Apr 2. PMID: 28367714
- 16 Yang, L., Jiao, X., Wu, J., Zhao, J., Liu, T., Xu, J., ... & Dong, L. (2018). *Cordyceps sinensis* inhibits airway remodeling in rats with chronic obstructive pulmonary disease. *Experimental and Therapeutic Medicine*, 15(3), 2731-2738
- 17 Yue, G. G. L., Bik-San Lau, C., Fung, K. P., Leung, P. C., & Ko, W. H. (2008). Effects of *Cordyceps sinensis*, *Cordyceps militaris* and their isolated compounds on ion transport in Calu-3 human airway epithelial cells. *Journal of Ethnopharmacology*, 117(1), 92-101
- 18 Fung, C. K., & Ko, W. H. (2012). *Cordyceps* extracts and the major ingredient, cordycepin: possible cellular mechanisms of their therapeutic effects on respiratory disease. In *Respiratory Diseases*. IntechOpen
- 19 Lin, C. H., Hsiao, Y. M., Ou, C. C., Lin, Y. W., Chiu, Y. L., Lue, K. H., ... & Ko, J. L. (2010). GMI, a *Ganoderma* immunomodulatory protein, down-regulates tumor necrosis factor α -induced expression of matrix metalloproteinase 9 via NF- κ B pathway in human alveolar epithelial A549 cells. *Journal of agricultural and food chemistry*, 58(22), 12014-12021
- 20 Bhardwaj N, Katyal P, Sharma AK. Suppression of inflammatory and allergic responses by pharmacologically potent fungus *Ganoderma lucidum*. *Recent Pat Inflamm Allergy Drug Discov*. 2014;8(2):104-17. doi: 10.2174/1872213x08666140619110657. PMID: 24948193
- 21 Stavinoha, W.B., 1990. Study of the anti-inflammatory activity of *Ganoderma lucidum*. Presented at the Third Academic Joint Conference, pp: 201-208
- 22 Bone, K. (2009). *The Ultimate Herbal Compendium: A desktop guide for herbal prescribers*
- 23 Shergis, J. L., Di, Y. M., Zhang, A. L., Vlahos, R., Helliwell, R., Ye, J. M., & Xue, C. C. (2014). Therapeutic potential of Panax ginseng and ginsenosides in the treatment of chronic obstructive pulmonary disease. *Complementary Therapies in Medicine*, 22(5), 944-953
- 24 Gross, D., Shenkman, Z., Bleiberg, B., Dayan, M., Gittelson, M., & Efrat, R. (2002). Ginseng improves pulmonary functions and exercise capacity in patients with COPD. *Monaldi archives for chest disease*, 57(5/6), 242-246
- 25 uo R, Pittler MH, Ernst E. Herbal medicines for the treatment of COPD: a systematic review. 2006. In: Database of Abstracts of Reviews of Effects (DARE): Quality-assessed Reviews [Internet]. York (UK): Centre for Reviews and Dissemination (UK); 1995-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK72798/>
- 26 Journal of Applied Pharmaceutical Science Vol. 8(07), pp 132-140, July, 2018 Available online at <http://www.japsonline.com> DOI: 10.7324/JAPS.2018.8721 ISSN 2231-3354
- 27 Ivanov, I. G., Vrancheva, R. Z., Petkova, N. T., Tumbarski, Y., Dincheva, I. N., & Badjakov, I. K. (2019). Phytochemical compounds of anise hyssop (*Agastache foeniculum*) and antibacterial, antioxidant, and acetylcholinesterase inhibitory properties of its essential oil. *Journal of Applied Pharmaceutical Science*, 9(2), 072-078
- 28 Barnes LA, Leach M, Anheyer D, Brown D, Carè J, Lauche R, Medina DN, Pinder TA, Bugarcic A, Steel A. The effects of *Hedera helix* on viral respiratory infections in humans: A rapid review. *Adv Integr Med*. 2020 Dec;7(4):222-226. doi: 10.1016/j.aimed.2020.07.012. Epub 2020 Aug 13. PMID: 32837900; PMCID: PMC7424313
- 29 Stansbury, J., Saunders, P. R., & Zampieron, E. (2013). The use of lobelia in the treatment of asthma and respiratory illness. *Journal of Restorative Medicine*, 2(1), 94-100
- 30 Pizzorno, J. E., Murray, M. T., & Joiner-Bey, H. (2016). *Bronchitis and pneumonia. The Clinician's Handbook of Natural Medicine*, 147
- 31 Salem, S., Leghouchi, E., Soulimani, R., & Bouayed, J. (2021). Reduction

of paw edema and liver oxidative stress in carrageenan-induced acute inflammation by *Lobaria pulmonaria* and *Parmelia caperata*, lichen species, in mice. International journal for vitamin and nutrition research. Internationale Zeitschrift für Vitamin- und Ernährungsforschung. Journal international de vitaminologie et de nutrition, 91(1-2), 143–151. <https://doi.org/10.1024/0300-9831/a000620>

32 Lu, Y., & Foo, L. Y. (2002). Polyphenolics of *Salvia*—a review. *Phytochemistry*, 59(2), 117–140. [https://doi.org/10.1016/s0031-9422\(01\)00415-0](https://doi.org/10.1016/s0031-9422(01)00415-0)

33 Bazarnova, Y., Politaeva, N., & Lyskova, N. (2018). Research for the lichen *Usnea barbata* metabolites. *Zeitschrift für Naturforschung. C, Journal of biosciences*, 73(7-8), 291–296. <https://doi.org/10.1515/znc-2017-0177>

34 Khan, A. U., & Gilani, A. H. (2011). Blood pressure lowering, cardiovascular inhibitory and bronchodilatory actions of *Achillea millefolium*. *Phytotherapy research: PTR*, 25(4), 577–583. <https://doi.org/10.1002/ptr.3303> <https://pubmed.ncbi.nlm.nih.gov/20857434>

35 Benedek, B., Rothwangl-Wiltschnigg, K., Rozema, E., Gjoncaj, N., Reznicek, G., Jurenitsch, J., Kopp, B., & Glasl, S. (2008). Yarrow (*Achillea millefolium* L. s.l.): pharmaceutical quality of commercial samples. *Die Pharmazie*, 63(1), 23–26

36 Wieland, L. S., Piechotta, V., Feinberg, T., Ludeman, E., Hutton, B., Kanji, S., Seely, D., & Garritty, C. (2021). Elderberry for prevention and treatment of viral respiratory illnesses: a systematic review. *BMC complementary medicine and therapies*, 21(1), 112. <https://doi.org/10.1186/s12906-021-03283-5>

37 Wieland, L. S., Piechotta, V., Feinberg, T., Ludeman, E., Hutton, B., Kanji, S., Seely, D., & Garritty, C. (2021). Elderberry for prevention and treatment of viral respiratory illnesses: a systematic review. *BMC complementary medicine and therapies*, 21(1), 112. <https://doi.org/10.1186/s12906-021-03283-5>

38 Wieland LS, Piechotta V, Feinberg T, Ludeman E, Hutton B, Kanji S, Seely D, Garritty C. Elderberry for prevention and treatment of viral respiratory illnesses: a systematic review. *BMC Complement Med Ther*. 2021 Apr 7;21(1):112. doi: 10.1186/s12906-021-03283-5. PMID: 33827515; PMCID: PMC8026097

39 Kronbichler A, Effenberger M, Eisenhut M, Lee KH, Shin JI. Seven recommendations to rescue the patients and reduce the mortality from COVID-19 infection: An immunological point of view. *Autoimmun Rev*. 2020 Jul;19(7):102570. doi: 10.1016/j.autrev.2020.102570. Epub 2020 May 3. PMID: 32376397; PMCID: PMC7252097

40 Zhou, L-K., et al. (2020) Absorbed plant MIR2911 in honeysuckle decoction inhibits SARS-CoV-2 replication and accelerates the negative conversion of infected patients. *Cell Discovery*. doi.org/10.1038/s41421-020-00197-3

41 M.S. Nair, Y. Huang, D.A. Fidock, S.J. Polyak, J. Wagoner, M.J. Towler, P.J. Weathers, *Artemisia annua* L. extracts inhibit the in vitro replication of SARS-CoV-2 and two of its variants. *Journal of Ethnopharmacology*, Volume 274, 2021, 114016, ISSN 0378-8741

42 Carr, A. C., & Maggini, S. (2017). Vitamin C and Immune Function. *Nutrients*, 9(11), 1211. <https://doi.org/10.3390/nu9111211>

43 Bucher A, White N. Vitamin C in the Prevention and Treatment of the Common Cold. *Am J Lifestyle Med*. 2016 Feb 9;10(3):181-183. doi: 10.1177/1559827616629092. PMID: 30202272; PMCID: PMC6124957. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6124957/>

44 Hughes, D. A., & Norton, R. (2009). Vitamin D and respiratory health. *Clinical and experimental immunology*, 158(1), 20–25. <https://doi.org/10.1111/j.1365-2249.2009.04001.x>

45 Yu H, Su X, Lei T, Zhang C, Zhang M, Wang Y, Zhu L, Liu J. Effect of Omega-3 Fatty Acids on Chronic Obstructive Pulmonary Disease: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *Int J Chron Obstruct Pulmon Dis*. 2021 Sep 23;16:2677-2686. doi: 10.2147/COPD.S331154. PMID: 34588776; PMCID: PMC8476109

46 Mehrabi, S., Moradi, M. M., Khodamoradi, Z., & Nazarinia, M. A. (2020). Effects of N-acetylcysteine on Pulmonary Functions in Patients with Systemic Sclerosis: A Randomized Double Blind, Placebo Controlled Study. *Current rheumatology reviews*, 16(2), 149–157. <https://doi.org/10.2174/1573397115666191212092608>

47 Secor ER Jr, Szczepanek SM, Castater CA, Adami AJ, Matson AP, Rafti ET, Guernsey L, Natarajan P, McNamara JT, Schramm CM, Thrall RS, Silbart LK. Bromelain Inhibits Allergic Sensitization and Murine Asthma via Modulation of Dendritic Cells. *Evid Based Complement Alternat Med*. 2013;2013:702196. doi: 10.1155/2013/702196. Epub 2013 Dec 5. PMID: 24381635; PMCID: PMC3870104 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3870104/>

48 Secor, E. R., Jr Shah, S. J., Guernsey, L. A., Schramm, C. M., & Thrall, R. S. (2012). Bromelain limits airway inflammation in an ovalbumin-induced murine model of established asthma. *Alternative therapies in health and medicine*, 18(5), 9–17

49 Cermelli, C., Fabio, A., Fabio, G., & Quaglio, P. (2008). Effect of eucalyptus essential oil on respiratory bacteria and viruses. *Current microbiology*, 56(1), 89–92. <https://doi.org/10.1007/s00284-007-9045-0>

50 Nogueira de Melo, G. A., Grespan, R., Fonseca, J. P., Farinha, T. O., Silva, E. L., Romero, A. L., Bersani-Amado, C. A., & Cuman, R. K. (2011). *Rosmarinus officinalis* L. essential oil inhibits in vivo and in vitro leukocyte migration. *Journal of medicinal food*, 14(9), 944–946. <https://doi.org/10.1089/jmf.2010.0159>

51 Sienkiewicz, M., Wasiela, M., & Głowacka, A. (2012). Aktywność przeciwbakteryjna olejku oreganowego (*Origanum heracleoticum* L.) wobec szczepów klinicznych *Escherichia coli* i *Pseudomonas aeruginosa* [The antibacterial activity of oregano essential oil (*Origanum heracleoticum* L.) against clinical strains of *Escherichia coli* and *Pseudomonas aeruginosa*]. *Medycyna doświadczalna i mikrobiologia*, 64(4), 297–307

52 Li, M., Zhu, L., Liu, B., Du, L., Jia, X., Han, L., & Jin, Y. (2016). Tea tree oil nanoemulsions for inhalation therapies of bacterial and fungal pneumonia. *Colloids and surfaces. B, Biointerfaces*, 141, 408–416. <https://doi.org/10.1016/j.colsurfb.2016.02.017>

53 Park, J. H., Seo, M. W., Jung, H. C., Song, J. K., & Lee, J. M. (2021). Association

between Health-Related Physical Fitness and Respiratory Diseases in Adolescents: An Age- and Gender-Matched Study. *International journal of environmental research and public health*, 18(12), 6655. <https://doi.org/10.3390/ijerph18126655>

54 <https://www.lung.org/lung-health-diseases/wellness/breathing-exercises>

Sinusitis Protocol

- Rosenfeld RM, Piccirillo JF, Chandrasekhar SS, Brook I, Kumar KA, Kramper M, Orlandi RR, Palmer JN, Patel ZM, Peters A, Walsh SA, Corrigan MD (April 2015). "Clinical practice guideline (update): Adult Sinusitis Executive Summary". *Otolaryngology–Head and Neck Surgery*. 152 (4): 598–609. doi:10.1177/0194599815574247. PMID 25833927. S2CID 206469424
- Reh DD, Higgins TS, Smith TL. Impact of tobacco smoke on chronic rhinosinusitis: a review of the literature. *Int Forum Allergy Rhinol*. 2012 Sep-Oct;2(5):362-9. doi: 10.1002/alr.21054. Epub 2012 Jun 13. PMID: 22696460; PMCID: PMC3443524
- Brewer JH, Thrasher JD, Hooper D. Chronic illness associated with mold and mycotoxins: is naso-sinus fungal biofilm the culprit? *Toxins* (Basel). 2013 Dec 24;6(1):66-80. doi: 10.3390/toxins6010066. PMID: 24368325; PMCID: PMC3920250
- Arca KN, Halker Singh RB. Dehydration and Headache. *Curr Pain Headache Rep*. 2021 Jul 15;25(8):56. doi: 10.1007/s11916-021-00966-z. PMID: 34268642; PMCID: PMC280611
- Egan M, Hickner J. Saline irrigation spells relief for sinusitis sufferers. *J Fam Pract*. 2009 Jan;58(1):29-32. PMID: 19141268; PMCID: PMC3183918
- Vairo GL, Miller SJ, McBrier NM, Buckley WE. Systematic review of efficacy for manual lymphatic drainage techniques in sports medicine and rehabilitation: an evidence-based practice approach. *J Man Manip Ther*. 2009;17(3):e80-9. doi: 10.1179/jmt.2009.17.3.80E. PMID: 20046617; PMCID: PMC2755111
- Gowrishankar S, Muthumanickam S, Kamaladevi A, Karthika C, Jothi R, Boomi P, Maniazhagu D, Pandian SK. Promising phytochemicals of traditional Indian herbal steam inhalation therapy to combat COVID-19 - An in silico study. *Food Chem Toxicol*. 2021 Feb;148:111966. doi: 10.1016/j.fct.2020.111966. Epub 2021 Jan 4. PMID: 33412235; PMCID: PMC7780060
- Ben-Arye, E., Dudai, N., Eini, A., Torem, M., Schiff, E., & Rakover, Y. (2011). Treatment of upper respiratory tract infections in primary care: a randomized study using aromatic herbs. *Evidence-based complementary and alternative medicine: eCAM*, 2011, 690346. <https://doi.org/10.1155/2011/690346>
- Prall S, Bowles EJ, Bennett K, Cooke CG, Agnew T, Steel A, Hausser T. Effects of essential oils on symptoms and course (duration and severity) of viral respiratory infections in humans: A rapid review. *Adv Integr Med*. 2020 Dec;7(4):218-221. doi: 10.1016/j.aimed.2020.07.005. Epub 2020 Aug 3. PMID: 32837899; PMCID: PMC7398051
- Kuo, Y. C., Tsai, W. J., Shiao, M. S., Chen, C. F., & Lin, C. Y. (1996). *Cordyceps sinensis* as an immunomodulatory agent. *The American journal of Chinese medicine*, 24(2), 111–125. <https://doi.org/10.1142/S0192415X96000165>
- Hudson J, Vimalanathan S. Echinacea—A Source of Potent Antivirals for Respiratory Virus Infections. *Pharmaceuticals* (Basel). 2011 Jul 13;4(7):1019–31. doi: 10.3390/ph4071019. PMCID: PMC4058675
- Wieland, L. S., Piechotta, V., Feinberg, T., Ludeman, E., Hutton, B., Kanji, S., Seely, D., & Garritty, C. (2021). Elderberry for prevention and treatment of viral respiratory illnesses: a systematic review. *BMC complementary medicine and therapies*, 21(1), 112. <https://doi.org/10.1186/s12906-021-03283-5>
- Zakay-Rones, Z., Thom, E., Wollan, T., & Wadstein, J. (2004). Randomized study of the efficacy and safety of oral elderberry extract in the treatment of influenza A and B virus infections. *The Journal of international medical research*, 32(2), 132–140. <https://doi.org/10.1177/147323000403200205>
- Bayan L, Kouliand PH, Gorji A. Garlic: a review of potential therapeutic effects. *Avicenna J Phytomed*. 2014 Jan;4(1):1-14. PMID: 25050296; PMCID: PMC4103721
- Mashhadi NS, Ghiasvand R, Askari G, Hariri M, Darvishi L, Mofid MR. Anti-oxidative and anti-inflammatory effects of ginger in health and physical activity: review of current evidence. *Int J Prev Med*. 2013 Apr;4(Suppl 1):S36-42. PMID: 23717767; PMCID: PMC3665023
- Chu M, Ding R, Chu ZY, Zhang MB, Liu XY, Xie SH, Zhai YJ, Wang YD. Role of berberine in anti-bacterial as a high-affinity LPS antagonist binding to TLR4/MD-2 receptor. *BMC Complement Altern Med*. 2014 Mar 6;14:89. doi: 10.1186/1472-6882-14-89. PMID: 24602493; PMCID: PMC3946165
- Süleyman, H., Odabasoglu, F., Aslan, A., Cakir, A., Karagoz, Y., Gocer, F., Hali-ci, M., & Bayir, Y. (2003). Anti-inflammatory and antiulcerogenic effects of the aqueous extract of *Lobaria pulmonaria* (L.) Hoffm. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 10(6-7), 552–557. <https://doi.org/10.1078/094471103322331539>
- Turker, A. U., & Gurel, E. (2005). Common mullein (*Verbascum thapsus* L.): recent advances in research. *Phytotherapy research: PTR*, 19(9), 733–739. <https://doi.org/10.1002/ptr.1653>
- Saeed, S., & Tariq, P. (2009). Antibacterial activity of oregano (*Origanum vulgare* Linn.) against gram positive bacteria. *Pakistan journal of pharmaceutical sciences*, 22(4), 421–424
- Lin Z. B. (2005). Cellular and molecular mechanisms of immuno-modulation by *Ganoderma lucidum*. *Journal of pharmacological sciences*, 99(2), 144–153. <https://doi.org/10.1254/jphs.crj05008x>
- Satoh T, Trudler D, Oh C-K, Lipton SA. Potential Therapeutic Use of the Rosemary Diterpene Carnosic Acid for Alzheimer's Disease, Parkinson's Disease, and Long-COVID through Nrf2 Activation to Counteract the NLRP3 Inflammasome. *Antioxidants*. 2022; 11(1):124. <https://doi.org/10.3390/antiox11010124>
- Bakhshae M, Mohammad Pour AH, Esmaeili M, Jabbari Azad F, Alipour Taleh G,

Salehi M, Noorollahian Mohajer M. Efficacy of Supportive Therapy of Allergic Rhinitis by Stinging Nettle (*Urtica dioica*) root extract: a Randomized, Double-Blind, Placebo-Controlled, Clinical Trial. Iran J Pharm Res. 2017 Winter;16(Suppl):112-118. PMID: 29844782; PMCID: PMC5963652

23 Wu, S., & Xiao, D. (2016). Effect of curcumin on nasal symptoms and airflow in patients with perennial allergic rhinitis. *Annals of allergy, asthma & immunology: official publication of the American College of Allergy, Asthma, & Immunology*, 117(6), 697–702.e1. <https://doi.org/10.1016/j.anai.2016.09.427>

24 Pagano, C., Ceccarini, M. R., Calarco, P., Scuota, S., Conte, C., Primavilla, S., Ricci, M., & Perioli, L. (2019). Bioadhesive polymeric films based on usnic acid for burn wound treatment: Antibacterial and cytotoxicity studies. *Colloids and surfaces. B, Biointerfaces*, 178, 488–499. <https://doi.org/10.1016/j.colsurfb.2019.03.001>

25 African Journal of Pharmacy and Pharmacology Vol. 6(41), pp. 2895-2899, 8 November, 2012 Available online at <http://www.academicjournals.org/AJPP> DOI: 10.5897/AJPP12.397 ISSN 1996-0816 © 2012 Academic Journals

26 Fischer, W., Currais, A., Liang, Z., Pinto, A., & Maher, P. (2019). Old age-associated phenotypic screening for Alzheimer's disease drug candidates identifies sterubin as a potent neuroprotective compound from Yerba santa. *Redox biology*, 21, 101089. <https://doi.org/10.1016/j.redox.2018.101089>

27 Carr, A. C., & Maggini, S. (2017). Vitamin C and Immune Function. *Nutrients*, 9(11), 1211. <https://doi.org/10.3390/nu9111211>

28 Aranow C. Vitamin D and the immune system. *J Investig Med*. 2011 Aug;59(6):881-6. doi: 10.2310/JIM.0b013e31821b8755. PMID: 21527855; PMCID: PMC3166406

29 Yan F, Polk DB. Probiotics and immune health. *Curr Opin Gastroenterol*. 2011 Oct;27(6):496-501. doi: 10.1097/MOG.0b013e32834baa4d. PMID: 21897224; PMCID: PMC4006993

30 Nakamura, S., Hashimoto, Y., Mikami, M., Yamanaka, E., Soma, T., Hino, M., Azuma, A., & Kudoh, S. (2003). Effect of the proteolytic enzyme serrapeptase in patients with chronic airway disease. *Respirology (Carlton, Vic.)*, 8(3), 316–320. <https://doi.org/10.1046/j.1440-1843.2003.00482.x>

31 Sivaramakrishnan, G., & Sridharan, K. (2018). Role of Serratiopeptidase After Surgical Removal of Impacted Molar: A Systematic Review and Meta-analysis. *Journal of maxillofacial and oral surgery*, 17(2), 122–128. <https://doi.org/10.1007/s12663-017-0996-9>

Osteoporosis Protocol

1 Compston JE, McClung MR, Leslie WD. Osteoporosis. *Lancet*. 2019;393(10169):364-376. doi:10.1016/S0140-6736(18)32112-3

2 Poursmaeilif, Kamalidehghan B, Kamarehei M, Goh YM. A comprehensive overview on osteoporosis and its risk factors. *Ther Clin Risk Manag*. 2018;14:2029-2049. Published 2018 Nov 6. doi:10.2147/TCRM.S138000

3 Panula J, Pihlajamäki H, Mattila VM, et al. Mortality and cause of death in hip fracture patients aged 65 or older: a population-based study. *BMC Musculoskelet Disord*. 2011;12:105. Published 2011 May 20. doi:10.1186/1471-2474-12-105

4 Negrete-Corona J, Alvarado-Soriano JC, Reyes-Santiago LA. Fractura de cadera como factor de riesgo en la mortalidad en pacientes mayores de 65 años. Estudio de casos y controles [Hip fracture as risk factor for mortality in patients over 65 years of age. Case-control study]. *Acta Ortop Mex*. 2014;28(6):352-362

5 Mundi S, Pindiprolu B, Simunovic N, Bhandari M. Similar mortality rates in hip fracture patients over the past 31 years. *Acta Orthop*. 2014;85(1):54-59. doi:10.3109/17453674.2013.878831

6 Katsoulis M, Benetou V, Karapetyan T, et al. Excess mortality after hip fracture in elderly persons from Europe and the USA: the CHANCES project. *J Intern Med*. 2017;281(3):300-310. doi:10.1111/joim.12586

7 Bijelic R, Milicevic S, Balaban J. Risk Factors for Osteoporosis in Postmenopausal Women. *Med Arch*. 2017;71(1):25-28. doi:10.5455/medarh.2017.71.25-28

8 Kelsey JL. Risk factors for osteoporosis and associated fractures. *Public Health Rep*. 1989;104 Suppl(Suppl):14-20

9 Management of osteoporosis in postmenopausal women: the 2021 position statement of The North American Menopause Society. *Menopause*. 2021;28(9):973-997. doi:10.1097/GME.0000000000001831

10 Maurel DB, Boisseau N, Benhamou CL, Jaffre C. Alcohol and bone: review of dose effects and mechanisms. *Osteoporos Int*. 2012;23(1):1-16. doi:10.1007/s00198-011-1787-7

11 Abukhadir SS, Mohamed N, Mohamed N. Pathogenesis of alcohol-induced osteoporosis and its treatment: a review. *Curr Drug Targets*. 2013;14(13):1601-1610. doi:10.2174/13894501113146660231

12 Sampson HW. Alcohol's harmful effects on bone. *Alcohol Health Res World*. 1998;22(3):190-194

13 Li H, Wallin M, Barregard L, et al. Smoking-Induced Risk of Osteoporosis Is Partly Mediated by Cadmium From Tobacco Smoke: The MrOS Sweden Study. *J Bone Miner Res*. 2020;35(8):1424-1429. doi:10.1002/jbmr.4014

14 Mędrela-Kuder E, Szymura K. Selected anti-health behaviours among women with osteoporosis. *Rocz Panstw Zakl Hig*. 2018;69(4):397-403. doi:10.32394/rpzh.2018.0046

15 Teucher B, Fairweather-Tait S. Dietary sodium as a risk factor for osteoporosis: where is the evidence?. *Proc Nutr Soc*. 2003;62(4):859-866. doi:10.1079/PNS2003300

16 Kim Y, Kim HY, Kim JH. Associations Between Reported Dietary Sodium Intake and Osteoporosis in Korean Postmenopausal Women: The 2008-2011 Korea National Health and Nutrition Examination Survey. *Asia Pac J Public Health*. 2017;29(5):430-439. doi:10.1177/1010539517712759

17 DiNicolantonio JJ, Mehta V, Zaman SB, O'Keefe JH. Not Salt But Sugar As Aetiological In Osteoporosis: A Review. *Mo Med*. 2018;115(3):247-252

18 Bonjour JP. Protein intake and bone health. *Int J Vitam Nutr Res*. 2011;81(2-3):134-142. doi:10.1024/0300-9831/a000063 Wengreen HJ, Munger RG, West NA, et al. Dietary protein intake and risk of osteoporotic hip fracture in elderly residents of Utah. *J Bone Miner Res*. 2004;19(4):537-545. doi:10.1359/JBMR.040208

19 Cuenca-Sánchez M, Navas-Carrillo D, Orenes-Piñero E. Controversies surrounding high-protein diet intake: satiating effect and kidney and bone health. *Adv Nutr*. 2015;6(3):260-266. Published 2015 May 15. doi:10.3945/an.114.007716

20 Kerstetter JE, O'Brien KO, Insogna KL. Dietary protein, calcium metabolism, and skeletal homeostasis revisited. *Am J Clin Nutr*. 2003;78(3 Suppl):584S-592S. doi:10.1093/ajcn/78.3.584S

21 Hallström H, Byberg L, Glynn A, Lemming EW, Wolk A, Michaëlsson K. Long-term coffee consumption in relation to fracture risk and bone mineral density in women. *Am J Epidemiol*. 2013;178(6):898-909. doi:10.1093/aje/kwt062

22 Hasling C, Søndergaard K, Charles P, Mosekilde L. Calcium metabolism in postmenopausal osteoporotic women is determined by dietary calcium and coffee intake. *J Nutr*. 1992;122(5):1119-1126. doi:10.1093/jn/122.5.1119

23 1994;271(4):280-283. doi:10.1001/jama.1994.03510280042030

24 Douglas P, Kiel, David T. Felson, Marian T. Hannan, Jennifer J. Anderson, Peter W. F. Wilson, Caffeine and the Risk of Hip Fracture: The Framingham Study, *American Journal of Epidemiology*, Volume 132, Issue 4, October 1990, Pages 675–684, <https://doi.org/10.1093/oxfordjournals.aje.a115709>

25 Shea B, Wells G, Cranney A, et al. Meta-analyses of therapies for postmenopausal osteoporosis. VII. Meta-analysis of calcium supplementation for the prevention of postmenopausal osteoporosis. *Endocr Rev*. 2002;23(4):552-559. doi:10.1210/er.2001-7002

26 Weaver CM, Alexander DD, Boushey CJ, et al. Calcium plus vitamin D supplementation and risk of fractures: an updated meta-analysis from the National Osteoporosis Foundation [published correction appears in *Osteoporos Int*. 2016 Aug;27(8):2643-6]. *Osteoporos Int*. 2016;27(1):367-376. doi:10.1007/s00198-015-3386-5

27 Cano A, Chedraui P, Goulis DG, et al. Calcium in the prevention of postmenopausal osteoporosis: EMAS clinical guide. *Maturitas*. 2018;107:7-12. doi:10.1016/j.maturitas.2017.10.004

28 Holick, M.F. Optimal Vitamin D Status for the Prevention and Treatment of Osteoporosis. *Drugs Aging* 24, 1017–1029 (2007). <https://doi.org/10.2165/00002512-200724120-00005>

29 Cranney A, Horsley T, O'Donnell S, et al. Effectiveness and safety of vitamin D in relation to bone health. *Evid Rep Technol Assess (Full Rep)*. 2007;(158):1-235

30 Castiglioni S, Cazzaniga A, Alibetti W, Maier JA. Magnesium and osteoporosis: current state of knowledge and future research directions. *Nutrients*. 2013;5(8):3022-3033. Published 2013 Jul 31. doi:10.3390/nu5083022

31 Sharma R, Sharma P, Kumar P, Gupta G (2016) Role of magnesium in post-menopausal women with osteoporosis and osteopenia. *Asian J Pharm Clin Res* 9:198-199

32 Mederle OA, Balas M, Ioanoviciu SD et al (2018) Correlations between bone turnover markers, serum magnesium and bone mass density in postmenopausal osteoporosis. *Clin Interv Aging* 13:1383-1389

33 Okyay E, Ertugrul C, Acar B et al (2013) Comparative evaluation of serum levels of main minerals and postmenopausal osteoporosis. *Maturitas* 76:320-325. <https://doi.org/10.1016/j.maturitas.2013.07.015>

34 Hayhoe RPG, Lentjes MAH, Luben RN et al (2015) Dietary magnesium and potassium intakes and circulating magnesium are associated with heel bone ultrasound attenuation and osteoporotic fracture risk in the EPIC-Norfolk cohort study. *Am J Clin Nutr* 102:376-384

35 Kunutsor SK, Whitehouse MR, Blom AW, Laukkanen JA (2017) Low serum magnesium levels are associated with increased risk of fractures: a long-term prospective cohort study. *Eur J Epidemiol* 32:593-603. <https://doi.org/10.1007/s10654-017-0242-6>

36 Abraham GE, Grewal H (1990) A total dietary program emphasizing magnesium instead of calcium. Effect on the mineral density of calcaneous bone in postmenopausal women on hormonal therapy. *J Reprod Med* 35:503-507

37 Stendig-Lindberg G, Tepper R, Leichter I (1993) Trabecular bone density in a two-year controlled trial of peroral magnesium in osteoporosis. *Magnes Res* 6:155-163

38 Liam Fouhy, Kelsey Mangano, Xiyuan Zhang, Brittany Adelman, Katherine Tucker, Sabrina Noel, Dietary Calcium and Magnesium Are Associated with Lower Odds of Osteoporosis Among Puerto Rican Older Adults When Dietary Calcium to Magnesium Ratio Is Considered, *Current Developments in Nutrition*, Volume 5, Issue Supplement_2, June 2021, Page 15, https://doi.org/10.1093/cdn/nzab033_015

39 Rondanelli, M., Faliva, M.A., Tartara, A. et al. An update on magnesium and bone health. *Biomaterials* 34, 715-736 (2021). <https://doi.org/10.1007/s10534-021-00305-0>

40 Tsugawa N, Shiraki M. Vitamin K Nutrition and Bone Health. *Nutrients*. 2020;12:1909. doi: 10.3390/nu12071909

41 Finnes T.E., Lofthus C.M., Meyer H.E., Sogaard A.J., Tell G.S., Apalset E.M., Gjesdal C., Grimnes G., Schei B., Blomhoff R., et al. A combination of low serum concentrations of vitamins PK and D is associated with increased risk of hip fractures in elderly Norwegians: A NOREPOS study. *Osteoporos. Int*. 2016;27:1645-1652. doi: 10.1007/s00198-015-3435-0

42 Moore A.E., Kim E., Dulnoan D., Dolan A.L., Voong K., Ahmad I., Gorska R., Harrington D.J., Hampson G. Serum vitamin K. *Bone*. 2020;115630. doi: 10.1016/j.bone.2020.115630

43 Capozzi A, Scambia G, Lello S. Calcium, vitamin D, vitamin K2, and magnesium supplementation and skeletal health. *Maturitas*. 2020;140:55-63. doi:10.1016/j.maturitas.2020.05.020

44 Fusaro M, Cianciolo G, Brandi ML, et al. Vitamin K and Osteoporosis. *Nutrients*. 2020;12(12):3625. Published 2020 Nov 25. doi:10.3390/nu12123625

45 Tusar K Giri, David Newton, Opal Chaudhary, Elena Deych, Nicola Napoli, Reina Villareal, Kathy Diemer, Paul E Milligan, and Brian F Gage, International Journal for Vitamin and Nutrition Research 2020 90:1-2, 42-48

46 Brzezińska O, Łukasik Z, Makowska J, Walczak K. Role of Vitamin C in Osteoporosis Development and Treatment-A Literature Review. *Nutrients*. 2020;12(8):2394. Published 2020 Aug 10. doi:10.3390/nu12082394

47 Finck H, Hart AR, Jennings A, Welch AA. Is there a role for vitamin C in preventing osteoporosis and fractures? A review of the potential underlying mechanisms and current epidemiological evidence. *Nutr Res Rev*. 2014;27(2):268-283. doi:10.1017/S0954422414000195

48 Malmir H, Shab-Bidar S, Djafarian K. Vitamin C intake in relation to bone mineral density and risk of hip fracture and osteoporosis: a systematic review and meta-analysis of observational studies. *Br J Nutr*. 2018;119(8):847-858. doi:10.1017/S0007114518000430

49 Kim MH, Lee HJ. Osteoporosis, vitamin C intake, and physical activity in Korean adults aged 50 years and over. *J Phys Ther Sci*. 2016;28(3):725-730. doi:10.1589/jpts.28.725

50 Ruiz-Ramos M, Vargas LA, Fortoul Van der Goes TI, Cervantes-Sandoval A, Mendoza-Nunez VM. Supplementation of ascorbic acid and alpha-tocopherol is useful to preventing bone loss linked to oxidative stress in elderly. *J Nutr Health Aging*. 2010;14(6):467-472. doi:10.1007/s12603-010-0099-5

51 Kołodziejaska B, Stępień N, Kolmas J. The Influence of Strontium on Bone Tissue Metabolism and Its Application in Osteoporosis Treatment. *Int J Mol Sci*. 2021;22(12):6564. Published 2021 Jun 18. doi:10.3390/ijms22126564

52 Maria S, Swanson MH, Enderby LT, et al. Melatonin-micronutrients Osteopenia Treatment Study (MOTS): a translational study assessing melatonin, strontium (citrate), vitamin D3 and vitamin K2 (MK7) on bone density, bone marker turnover and health related quality of life in postmenopausal osteopenic women following a one-year double-blind RCT and on osteoblast-osteoclast co-cultures. *Aging (Albany NY)*. 2017;9(1):256-285. doi:10.18632/aging.101158

53 Jugdaohsingh R. Silicon and bone health. *J Nutr Health Aging*. 2007;11(2):99-110

54 Spector TD, Calomme MR, Anderson S, Swaminathan R, Jugdaohsingh R, Vanden-Berge DA, Powell JJ. Effect of bone turnover and BMD of low dose oral silicon as an adjunct to calcium/vitamin D3 in a randomized placebo-controlled trial. *Journal of Bone Mineral Research*. 2005;20:S172

55 Schiano A, Eisinger F, Detolle P, Laponche AM, Brisou B, Eisinger J. Silicium, tissu osseux et immunité [Silicon, bone tissue and immunity]. *Rev Rhum Mal Osteoartic*. 1979;46(7-9):483-486

56 Price CT, Koval KJ, Langford JR. Silicon: a review of its potential role in the prevention and treatment of postmenopausal osteoporosis. *Int J Endocrinol*. 2013;2013:316783. doi:10.1155/2013/316783

57 Gaffney-Stomberg, E. The Impact of Trace Minerals on Bone Metabolism. *Biol Trace Elem Res* 188, 26–34 (2019). <https://doi.org/10.1007/s12011-018-1583-8>

58 Khaliq, H., Juming, Z. & Ke-Mei, P. The Physiological Role of Boron on Health. *Biol Trace Elem Res* 186, 31–51 (2018). <https://doi.org/10.1007/s12011-018-1284-3>

59 Rondanelli M, Faliva MA, Peroni G, et al. Pivotal role of boron supplementation on bone health: A narrative review. *J Trace Elem Med Biol*. 2020;62:126577. doi:10.1016/j.jtemb.2020.126577

60 Sharma T, Mandal CC. Omega-3 fatty acids in pathological calcification and bone health. *J Food Biochem*. 2020;44(8):e13333. doi:10.1111/jfbc.13333

61 Kim BJ, Yoo HJ, Park SJ, et al. Association of blood n-3 fatty acid with bone mass and bone marrow TRAP-5b in the elderly with and without hip fracture. *Osteoporos Int*. 2019;30(5):1071-1078. doi:10.1007/s00198-019-04881-3

62 Kajarabille N, Díaz-Castro J, Hijano S, López-Frías M, López-Aliaga I, Ochoa JJ. A new insight to bone turnover: role of ω-3 polyunsaturated fatty acids. *ScientificWorldJournal*. 2013;2013:589641. Published 2013 Nov 4. doi:10.1155/2013/589641

63 Abdelhamid A, Hooper L, Sivakaran R, Hayhoe RPG, Welch A, PUFHA Group. The Relationship Between Omega-3, Omega-6 and Total Polyunsaturated Fat and Musculoskeletal Health and Functional Status in Adults: A Systematic Review and Meta-analysis of RCTs. *Calcif Tissue Int*. 2019;105(4):353-372. doi:10.1007/s00223-019-00584-3

64 Rondanelli M, Tartara A, Fossari F, et al. Adequate intake and supplementation of B vitamins, in particular folic acid, can play a protective role in bone health [published online ahead of print, 2021 Oct 4]. *Curr Aging Sci*. 2021;10.2174/1874609814666211005101730. doi:10.2174/1874609814666211005101730

65 Yazdanpanah N, Zillikens MC, Rivadeneira F, et al. Effect of dietary B vitamins on BMD and risk of fracture in elderly men and women: the Rotterdam study. *Bone*. 2007;41(6):987-994. doi:10.1016/j.bone.2007.08.021

66 Puel C, Coxam V, Davicco MJ. Régime méditerranéen et ostéoporose [Mediterranean diet and osteoporosis prevention]. *Med Sci (Paris)*. 2007;23(8-9):756-760. doi:10.1051/medsci/20072389756

67 Romero Pérez A, Rivas Velasco A. Adherence to Mediterranean diet and bone health. *Nutr Hosp*. 2014;29(5):989-996. Published 2014 May 1. doi:10.3305/nh.2014.29.5.7332

68 Puel C, Coxam V, Davicco MJ. Régime méditerranéen et ostéoporose [Mediterranean diet and osteoporosis prevention]. *Med Sci (Paris)*. 2007;23(8-9):756-760. doi:10.1051/medsci/20072389756

69 Fernández-Real JM, Bulló M, Moreno-Navarrete JM, et al. A Mediterranean diet enriched with olive oil is associated with higher serum total osteocalcin levels in elderly men at high cardiovascular risk. *J Clin Endocrinol Metab*.

2012;97(10):3792-3798. doi:10.1210/jc.2012-2221

70 Atmaca A, Kleerekoper M, Bayraktar M, Kucuk O. Soy isoflavones in the management of postmenopausal osteoporosis. *Menopause*. 2008;15(4 Pt 1):748-757. doi:10.1097/gme.0b013e31815c1e7f

71 Chen YM, Ho SC, Lam SS, Ho SS, Woo JL. Soy isoflavones have a favorable effect on bone loss in Chinese postmenopausal women with lower bone mass: a double-blind, randomized, controlled trial. *J Clin Endocrinol Metab*. 2003;88(10):4740-4747. doi:10.1210/jc.2003-030290

72 Zhang Y, Chen WF, Lai WP, Wong MS. Soy isoflavones and their bone protective effects. *Inflammopharmacology*. 2008;16(5):213-215. doi:10.1007/s10787-008-8018-7

73 Zhang Y, Chen WF, Lai WP, Wong MS. Soy isoflavones and their bone protective effects. *Inflammopharmacology*. 2008;16(5):213-215. doi:10.1007/s10787-008-8018-7

74 Anupama DS, Norohna JA, Acharya KK, Ravishankar, George A. Effect of exercise on bone mineral density and quality of life among postmenopausal women with osteoporosis without fracture: A systematic review. *Int J Orthop Trauma Nurs*. 2020;39:100796. doi:10.1016/j.ijotn.2020.100796

75 Bonaiuti D, Shea B, Iovine R, et al. Exercise for preventing and treating osteoporosis in postmenopausal women. *Cochrane Database Syst Rev*. 2002;(3):CD000333. doi:10.1002/14651858.CD000333

76 Howe TE, Shea B, Dawson LJ, et al. Exercise for preventing and treating osteoporosis in postmenopausal women. *Cochrane Database Syst Rev*. 2011;(7):CD000333. Published 2011 Jul 6. doi:10.1002/14651858.CD000333.pub2

77 Moreira LD, Oliveira ML, Lirani-Galvão AP, Marin-Mio RV, Santos RN, Lazaretti-Castro M. Physical exercise and osteoporosis: effects of different types of exercises on bone and physical function of postmenopausal women. *Arq Bras Endocrinol Metabol*. 2014;58(5):514-522. doi:10.1590/0004-2730000003374

78 Jolly JJ, Chin KY, Alias E, Chua KH, Soelaiman IN. Protective Effects of Selected Botanical Agents on Bone. *Int J Environ Res Public Health*. 2018;15(5):963. Published 2018 May 11. doi:10.3390/ijerph15050963

79 Zhang G, Qin L, Shi Y. Epimedium-derived phytoestrogen flavonoids exert beneficial effect on preventing bone loss in late postmenopausal women: a 24-month randomized, double-blind and placebo-controlled trial. *J Bone Miner Res*. 2007;22(7):1072-1079. doi:10.1359/jbmr.070405

80 Pastorino G, Cornara L, Soares S, Rodrigues F, Oliveira MBPP. Liquorice (Glycyrrhiza glabra): A phytochemical and pharmacological review. *Phytother Res*. 2018;32(12):2323-2339. doi:10.1002/ptr.6178

81 Rho TW, Lee SY, Han SY, et al. Glycyrrhizae Radix Inhibits Osteoclast Differentiation by Inhibiting c-Fos-Dependent NFATc1 Expression. *Am J Chin Med*. 2017;45(2):283-298. doi:10.1142/S0192415X17500185

82 Lin Y, Murray MA, Garrett IR, et al. A targeted approach for evaluating preclinical activity of botanical extracts for support of bone health. *J Nutr Sci*. 2014;3:e13. Published 2014 May 13. doi:10.1017/jns.2014.5

83 Galanis D, Soultanis K, Lelovas P, et al. Protective effect of Glycyrrhiza glabra roots extract on bone mineral density of ovariectomized rats. *Biomedicine (Taipei)*. 2019;9(2):8. doi:10.1051/bmcdcn/2019090208

84 Joyce KM, Wong CP, Scriven IA, et al. Isoliquiritigenin Decreases Bone Resorption and Osteoclast Differentiation. *Mol Nutr Food Res*. 2022;66(11):e2100974. doi:10.1002/mnfr.202100974

85 Somjen D, Katzbarg S, Vaya J, Kaye A. M., Hendel D., Posner, G. H. & Tamir, S. (2004). Estrogenic activity of glabridin and glabrene from licorice roots on human osteoblasts and prepubertal rat skeletal tissues. *Journal of Steroid Biochemistry and Molecular Biology*, 91(4–5), 241–246

86 Nieves, J.W. Skeletal effects of nutrients and nutraceuticals, beyond calcium and vitamin D. *Osteoporos Int* 24, 771–786 (2013). <https://doi.org/10.1007/s00198-012-2214-4>

87 Lambert MNT, Thybo CB, Lykkeboe S, et al. Combined bioavailable isoflavones and probiotics improve bone status and estrogen metabolism in postmenopausal osteopenic women: a randomized controlled trial. *Am J Clin Nutr*. 2017;106(3):909-920. doi:10.3945/ajcn.117.153353

88 Chen, Y. M., Wang, I. L., Zhu, X. Y., Chiu, W. C., & Chiu, Y. S. (2021). Red Clover Isoflavones Influence Estradiol Concentration, Exercise Performance, and Gut Microbiota in Female Mice. *Frontiers in nutrition*, 8, 623698. <https://doi.org/10.3389/fnut.2021.623698>

89 Adhikari, B. M., Bajracharya, A., & Shrestha, A. K. (2015). Comparison of nutritional properties of Stinging nettle (*Urtica dioica*) flour with wheat and barley flours. *Food science & nutrition*, 4(1), 119–124. <https://doi.org/10.1002/fsn3.259>

Pain Management Protocol

1 Centers for Disease Control and Prevention. (2019, September 16). *Prevalence of chronic pain and high-impact chronic pain among adults - United States, 2016*. Centers for Disease Control and Prevention. Retrieved August 31, 2022, from <https://www.cdc.gov/mmwr/volumes/67/wr/mm6736a2.htm?suggestedcitation>

2 Wu, A., March, L., Zheng, X., Huang, J., Wang, X., Zhao, J., Blyth, F. M., Smith, E., Buchbinder, R., & Hoy, D. (2020). Global low back pain prevalence and years lived with disability from 1990 to 2017: Estimates from the global burden of disease study 2017. *Annals of Translational Medicine*, 8(6), 299–299. <https://doi.org/10.21037/atm.2020.02.175>

3 Raffaelli, W., & Arnaudo, E. (2017). Pain as a disease: An overview. *Journal of Pain Research, Volume 10*, 2003–2008. <https://doi.org/10.2147/jpr.s138864>

4 Sheng, J., Liu, S., Wang, Y., Cui, R., & Zhang, X. (2017). The link between depression and chronic pain: Neural mechanisms in the brain. *Neural Plasticity*, 2017, 1–10. <https://doi.org/10.1155/2017/9724371>

- 5 Centers for Disease Control and Prevention. (2022, June 2). *Death Rate Maps & Graphs*. Centers for Disease Control and Prevention. Retrieved August 31, 2022, from <https://www.cdc.gov/drugoverdose/deaths/index.html>
- 6 MediLexicon International. (n.d.). *What is pain, and how do you treat it?* Medical News Today. Retrieved August 30, 2022, from <https://www.medicalnewstoday.com/articles/145750>
- 7 Dubin, A. E., & Patapoutian, A. (2010). Nociceptors: The sensors of the pain pathway. *Journal of Clinical Investigation*, 120(11), 3760–3772. <https://doi.org/10.1172/jci42843>
- 8 4 trends in opioid use, harms, and treatment - national center for ... (n.d.). Retrieved August 31, 2022, from <https://www.ncbi.nlm.nih.gov/books/NBK458661/>
- 9 Yu, G., Xiang, W., Zhang, T., Zeng, L., Yang, K., & Li, J. (2020). Effectiveness of boswellia and Boswellia extract for osteoarthritis patients: A systematic review and meta-analysis. *BMC Complementary Medicine and Therapies*, 20(1). <https://doi.org/10.1186/s12906-020-02985-6>
- 10 Siddiqui M. Z. (2011). Boswellia serrata, a potential antiinflammatory agent: an overview. *Indian journal of pharmaceutical sciences*, 73(3), 255–261. <https://doi.org/10.4103/0250-474X.93507>
- 11 Prabhavathi, K., Chandra, U. S. J., Soanker, R., & Rani, P. U. (2014). A randomized, double blind, placebo controlled, cross over study to evaluate the analgesic activity of Boswellia serrata in healthy volunteers using mechanical pain model. *Indian Journal of Pharmacology*, 46(5), 475. <https://doi.org/10.4103/0253-7613.140570>
- 12 Noguchi, K., & Okubo, M. (2011). Leukotrienes in nociceptive pathway and neuropathic/inflammatory pain. *Biological & Pharmaceutical Bulletin*, 34(8), 1163–1169. <https://doi.org/10.1248/bpb.34.1163>
- 13 Moncivaiz, A. (2018, October 6). *Boswellia: Uses, dosage, side effects, and more*. Healthline. Retrieved August 29, 2022, from <https://www.healthline.com/health/boswellia#side-effects>
- 14 Porter, B., Marie, B. S., Milavetz, G., & Herr, K. (2021). Cannabidiol (CBD) use by older adults for acute and chronic pain. *Journal of Gerontological Nursing*, 47(7), 6–15. <https://doi.org/10.3928/00989134-20210610-02>
- 15 Caruso, C. (2022, July 15). *Building a better painkiller*. Building a Better Painkiller | Harvard Medical School. Retrieved August 29, 2022, from <https://hms.harvard.edu/news/building-better-painkiller>
- 16 Wade, D. T., Robson, P., House, H., Makela, P., & Aram, J. (2003). A preliminary controlled study to determine whether whole-plant cannabis extracts can improve intractable neurogenic symptoms. *Clinical Rehabilitation*, 17(1), 21–29. <https://doi.org/10.1191/0269215503cr581oa>
- 17 Russo, E. (2008). Cannabinoids in the management of difficult to treat pain. *Therapeutics and Clinical Risk Management*, Volume 4, 245–259. <https://doi.org/10.2147/term.s1928>
- 18 Shafik Boyaji, M. D. (2020, September 23). *CBD for chronic pain: The science doesn't match the marketing*. Harvard Health. Retrieved August 29, 2022, from <https://www.health.harvard.edu/blog/cbd-for-chronic-pain-the-science-doesnt-match-the-marketing-2020092321003>
- 19 Iffland, K., & Grotenhermen, F. (2017). An update on safety and side effects of Cannabidiol: A review of Clinical Data and relevant animal studies. *Cannabis and Cannabinoid Research*, 2(1), 139–154. <https://doi.org/10.1089/can.2016.0034>
- 20 Peng, Y., Ao, M., Dong, B., Jiang, Y., Yu, L., Chen, Z., Hu, C., & Xu, R. (2021). Anti-inflammatory effects of curcumin in the inflammatory diseases: Status, limitations and countermeasures. *Drug Design, Development and Therapy*, Vol.15, 4503–4525. <https://doi.org/10.2147/dddt.s327378>
- 21 Pari, L., Tewas, D., & Eckel, J. (2008). Role of curcumin in health and disease. *Archives of Physiology and Biochemistry*, 114(2), 127–149. <https://doi.org/10.1080/13813450802033958>
- 22 Wang, Z., Singh, A., Jones, G., Winzenberg, T., Ding, C., Chopra, A., Das, S., Danda, D., Laslett, L., & Antony, B. (2021). Efficacy and safety of turmeric extracts for the treatment of knee osteoarthritis: A systematic review and meta-analysis of Randomised Controlled Trials. *Current Rheumatology Reports*, 23(2). <https://doi.org/10.1007/s11926-020-00975-8>
- 23 Sun, J., Chen, F., Braun, C., Zhou, Y.-Q., Rittner, H., Tian, Y.-K., Cai, X.-Y., & Ye, D.-W. (2018). Role of curcumin in the management of pathological pain. *Phytomedicine*, 48, 129–140. <https://doi.org/10.1016/j.phymed.2018.04.045>
- 24 Zhu, X., Li, Q., Chang, R., Yang, D., Song, Z., Guo, Q., & Huang, C. (2014). Curcumin alleviates neuropathic pain by inhibiting p300/CBP histone acetyltransferase activity-regulated expression of BDNF and cox-2 in a rat model. *PLoS ONE*, 9(3). <https://doi.org/10.1371/journal.pone.0091303>
- 25 Hasriadi, Dasuni Wasana, P. W., Vajragupta, O., Rojsitthisak, P., & Towiwat, P. (2021). Mechanistic insight into the effects of curcumin on neuroinflammation-driven chronic pain. *Pharmaceuticals*, 14(8), 777. <https://doi.org/10.3390/ph14080777>
- 26 Hartung, J. E., Eskew, O., Wong, T., Tchivileva, I. E., Oladosu, F. A., O'Buckley, S. C., & Nackley, A. G. (2015). Nuclear factor-kappa B regulates pain and COMT expression in a rodent model of inflammation. *Brain, Behavior, and Immunity*, 50, 196–202. <https://doi.org/10.1016/j.bbi.2015.07.014>
- 27 Pareek, A., Suthar, M., Rathore, G. S., & Bansal, V. (2011). Feverfew (tanacetum parthenium L.): A systematic review. *Pharmacognosy Reviews*, 5(9), 103. <https://doi.org/10.4103/0973-7847.79105>
- 28 Mannelli, L. D.-C., Tenci, B., Zanardelli, M., Maidecchi, A., Lugli, A., Mattoli, L., & Ghelardini, C. (2015). Widespread pain reliever profile of a flower extract of Tanacetum parthenium. *Phytomedicine*, 22(7-8), 752–758. <https://doi.org/10.1016/j.phymed.2015.05.006>
- 29 *Feverfew*. Mount Sinai Health System. (n.d.). Retrieved August 29, 2022, from <https://www.mountsinai.org/health-library/herb/feverfew>
- 30 Bhardwaj, Neha et al. "Suppression of inflammatory and allergic responses by pharmacologically potent fungus Ganoderma lucidum." Recent patents on inflammation & allergy drug discovery vol. 8,2 (2014): 104-17
- 31 Zhi-Bin Lin. "Cellular and Molecular Mechanisms of Immuno-modulation by Ganoderma lucidum [italic]." J Pharmacol Sci 99, 144-153. 2005
- 32 Eugene R. Zampieron, ND*; Ellen J. Kamhi, PhD, RNb, Natural Support for Autoimmune and Inflammatory Disease, Journal of Restorative Medicine, 2012 Volume 1, Issue 1, pages 38-47, DOI: <https://doi.org/10.14200/jrm.2012.1.1003>
- 33 Cai, Zhe et al. "Anti-inflammatory activities of Ganoderma lucidum (Lingzhi) and San-Miao-San supplements in MRL/lpr mice for the treatment of systemic lupus erythematosus." Chinese Medicine vol. 11 23. 29 Apr. 2016. 8. Zhi-Bin Lin. "Cellular and Molecular Mechanisms of Immuno-modulation by Ganoderma lucidum." J Pharmacol Sci 99, 144-153. 2005
- 34 Li, E. K., Tam, L. S., Wong, C. K., Li, W. C., Lam, C. W., Wachtel-Galor, S., Benzie, I. F., Bao, Y. X., Leung, P. C., & Tomlinson, B. (2007). Safety and efficacy of Ganoderma lucidum (lingzhi) and San Miao San supplementation in patients with rheumatoid arthritis: a double-blind, randomized, placebo-controlled pilot trial. *Arthritis and rheumatism*, 57(7), 1143–1150. <https://doi.org/10.1002/art.22994>
- 35 Sajjad Ahmadi-Renani, Mahdi Fasihi-Ramandi, Kazem Ahmadi. "Ganoderma lucidum: A promising anti-inflammatory medicinal plant.", J HerbMed Pharmacol. 2014; 3(1):67-68
- 36 H; P. B. N. (n.d.). St. John's wort. National Center for Biotechnology Information. Retrieved June 17, 2022, from <https://pubmed.ncbi.nlm.nih.gov/32491397/>
- 37 Butterweck, V. (2003). Mechanism of action of St John's wort in depression. *CNS Drugs*, 17(8), 539–562. <https://doi.org/10.2165/00023210-200317080-00001>
- 38 Galeotti, N., Vivoli, E., Bilia, A. R., Vincieri, F. F., & Ghelardini, C. (2010). St. John's wort reduces neuropathic pain through a hypericin-mediated inhibition of the protein kinase C and ϵ activity. *Biochemical Pharmacology*, 79(9), 1327–1336. <https://doi.org/10.1016/j.bcp.2009.12.016>
- 39 Finnerup, N. B., Kuner, R., & Jensen, T. S. (2021). Neuropathic pain: From mechanisms to treatment. *Physiological Reviews*, 101(1), 259–301. <https://doi.org/10.1152/physrev.00045.2019>
- 40 Sanna, M. D., Ghelardini, C., & Galeotti, N. (2016). St. John's wort potentiates anti-nociceptive effects of morphine in mice models of neuropathic pain. *Pain Medicine*. <https://doi.org/10.1093/pm/pnw241s>
- 41 Nicolussi, S., Drewe, J., Butterweck, V., & Meyer zu Schwabedissen, H. E. (2020). Clinical relevance of St. John's wort drug interactions revisited. *British Journal of Pharmacology*, 177(6), 1212–1226. <https://doi.org/10.1111/bph.14936>
- 42 Markowitz, J. S. (2003). Effect of st John's wort on drug metabolism by induction of cytochrome P450 3A4 enzyme. *JAMA*, 290(11), 1500. <https://doi.org/10.1001/jama.290.11.1500>
- 43 Dannawi, M. (2002). Possible serotonin syndrome after combination of buspirone and St John's wort. *Journal of Psychopharmacology*, 16(4), 401–401. <https://doi.org/10.1177/026988110201600420>
- 44 Lévesque H, Lafont O. L'aspirine à travers les siècles: rappel historique [Aspirin throughout the ages: a historical review]. *Rev Med Interne*. 2000 Mar;21 Suppl 1:8s-17s. French. doi: 10.1016/s0248-8663(00)88720-2. PMID: 10763200
- 45 Sneider, W. (2000). The discovery of aspirin: A reappraisal. *BMJ*, 321(7276), 1591–1594. <https://doi.org/10.1136/bmj.321.7276.1591>
- 46 Wood, J. N. (2015). From Plant Extract to Molecular Panacea: A commentary on stone (1763) 'an account of the success of the bark of the willow in the cure of the agues'. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 370(1666), 20140317. <https://doi.org/10.1098/rstb.2014.0317>
- 47 Vlachojannis, J., Magora, F., & Chrubasik, S. (2011). Willow species and aspirin: Different mechanism of actions. *Phytotherapy Research*, 25(7), 1102–1104. <https://doi.org/10.1002/ptr.3386>
- 48 Chrubasik, S., Eisenberg, E., Balan, E., Weinberger, T., Luzzati, R., & Conrath, C. (2000). Treatment of low back pain exacerbations with willow bark extract: A randomized double-blind study. *The American Journal of Medicine*, 109(1), 9–14. [https://doi.org/10.1016/s0002-9343\(00\)00442-3](https://doi.org/10.1016/s0002-9343(00)00442-3)
- 49 Chrubasik, S., Weiser, T., & Beime, B. (2010). Effectiveness and safety of topical capsaicin cream in the treatment of Chronic Soft Tissue pain. *Phytotherapy Research*, 24(12), 1877–1885. <https://doi.org/10.1002/ptr.3335>
- 50 Derry, S., Lloyd, R., Moore, R. A., & McQuay, H. J. (2009). Topical capsaicin for chronic neuropathic pain in adults. *Cochrane Database of Systematic Reviews*. <https://doi.org/10.1002/14651858.cd007393.pub2>
- 51 Silva, G. L., Luft, C., Lunardelli, A., Amaral, R. H., Melo, D. A., Donadio, M. V., Nunes, F. B., de Azambuja, M. S., Santana, J. C., Moraes, C. M., Mello, R. O., Cassel, E., Pereira, M. A., & de Oliveira, J. R. (2015). Antioxidant, analgesic and anti-inflammatory effects of lavender essential oil. *Anais da Academia Brasileira de Ciências*, 87(2 Suppl), 1397–1408. <https://doi.org/10.1590/0001-3765201520150056>
- 52 Sasanejad, P., Saeedi, M., Shoeibi, A., Gorji, A., Abbasi, M., & Foroughpour, M. (2012). Lavender essential oil in the treatment of migraine headache: A placebo-controlled clinical trial. *European Neurology*, 67(5), 288–291. <https://doi.org/10.1159/000335249>
- 53 Göbel, H., Heinze, A., Heinze-Kuhn, K., Göbel, A., & Göbel, C. (2016). Oleum menthae piperitae (pfefferminzöl) in der Akuttherapie des Kopfschmerzes vom Spannungstyp. *Der Schmerz*, 30(3), 295–310. <https://doi.org/10.1007/s00482-016-0109-6>
- 54 Davies, S. J., Harding, L. M., & Baranowski, A. P. (2002). A novel treatment of postherpetic neuralgia using peppermint oil. *The Clinical Journal of Pain*, 18(3), 200–202. <https://doi.org/10.1097/00002508-200205000-00011>

- 55 Göbel, H., Schmidt, G., & Soyka, D. (1994). Effect of peppermint and eucalyptus oil preparations on neurophysiological and experimental algometric headache parameters. *Cephalalgia*, 14(3), 228–234. <https://doi.org/10.1046/j.1468-2982.1994.014003228.x>
- 56 Zhou, W. B., Meng, J. W., & Zhang, J. (2021). Does low grade systemic inflammation have a role in chronic pain? *Frontiers in Molecular Neuroscience*, 14. <https://doi.org/10.3389/fnmol.2021.785214>
- 57 Ricker, M. A., & Haas, W. C. (2017). Anti-inflammatory diet in clinical practice: A Review. *Nutrition in Clinical Practice*, 32(3), 318–325. <https://doi.org/10.1177/0884533617700353>
- 58 Brain, K., Burrows, T. L., Bruggink, L., Malfiet, A., Hayes, C., Hodson, F. J., & Collins, C. E. (2021). Diet and chronic non-cancer pain: The state of the art and Future Directions. *Journal of Clinical Medicine*, 10(21), 5203. <https://doi.org/10.3390/jcm10215203>
- 59 Guo, R., Chen, L.-H., Xing, C., & Liu, T. (2019). Pain regulation by gut microbiota: Molecular mechanisms and therapeutic potential. *British Journal of Anaesthesia*, 123(5), 637–654. <https://doi.org/10.1016/j.bja.2019.07.026>
- 60 Mayer, E. A. (2011). Gut feelings: The emerging biology of Gut–Brain Communication. *Nature Reviews Neuroscience*, 12(8), 453–466. <https://doi.org/10.1038/nrn3071>
- 61 Dworsky-Fried, Z., Kerr, B. J., & Taylor, A. M. W. (2020). Microbes, microglia, and pain. *Neurobiology of Pain*, 7, 100045. <https://doi.org/10.1016/j.ynpai.2020.100045>
- 62 Arfailasufandi, R., & Andiarna, F. (2018). The influence of dark chocolate to reduce menstrual pain in primary dysmenorrhea. *Journal of Health Science and Prevention*, 2(1), 27–35. <https://doi.org/10.29080/jhsp.v2i1.107>
- 63 De Feo, M., Paladini, A., Ferri, C., Carducci, A., Del Pinto, R., Varrassi, G., & Grassi, D. (2020). Anti-inflammatory and anti-nociceptive effects of cocoa: A review on future perspectives in treatment of pain. *Pain and Therapy*, 9(1), 231–240. <https://doi.org/10.1007/s40122-020-00165-5>
- 64 De Feo, M., Paladini, A., Ferri, C., Carducci, A., Del Pinto, R., Varrassi, G., & Grassi, D. (2020). Anti-inflammatory and anti-nociceptive effects of cocoa: A review on future perspectives in treatment of pain. *Pain and Therapy*, 9(1), 231–240. <https://doi.org/10.1007/s40122-020-00165-5>
- 65 Terry, R., Posadzki, P., Watson, L. K., & Ernst, E. (2011). The use of ginger (zingiber officinale) for the treatment of pain: A systematic review of Clinical Trials. *Pain Medicine*, 12(12), 1808–1818. <https://doi.org/10.1111/j.1526-4637.2011.01261.x>
- 66 *Acupuncture: What is it, how it works, treatments*. Cleveland Clinic. (n.d.). Retrieved September 1, 2022, from <https://my.clevelandclinic.org/health/treatments/4767-acupuncture#:~:text=Acupuncture%20is%20a%20treatment%20that,its%20effectiveness%20for%20some%20conditions>
- 67 Patil, S., Sen, S., Bral, M., Reddy, S., Bradley, K. K., Cornett, E. M., Fox, C. J., & Kaye, A. D. (2016). The role of acupuncture in pain management. *Current Pain and Headache Reports*, 20(4). <https://doi.org/10.1007/s11916-016-0552-1>
- 68 Mayo Foundation for Medical Education and Research. (2022, April 30). *Acupuncture*. Mayo Clinic. Retrieved September 1, 2022, from <https://www.mayoclinic.org/tests-procedures/acupuncture/about/pac-20392763#s>
- 69 Goto, Y., Hayasaka, S., Kurihara, S., & Nakamura, Y. (2018). Physical and mental effects of bathing: A randomized intervention study. *Evidence-Based Complementary and Alternative Medicine*, 2018, 1–5. <https://doi.org/10.1155/2018/9521086>
- 70 Petersen, B. W., Arbuckle, H. A., & Berman, S. (2015). Effectiveness of saltwater baths in the treatment of epidermolysis bullosa. *Pediatric Dermatology*, 32(1), 60–63. <https://doi.org/10.1111/pde.12409>
- 71 Bai, R., Li, C., Xiao, Y., Sharma, M., Zhang, F., & Zhao, Y. (2019). Effectiveness of spa therapy for patients with chronic low back pain. *Medicine*, 98(37). <https://doi.org/10.1097/md.00000000000017092>
- 72 Gröber, U., Werner, T., Vormann, J., & Kisters, K. (2017). Myth or reality—transdermal magnesium? *Nutrients*, 9(8), 813. <https://doi.org/10.3390/nu9080813>
- 73 Rath, L. (n.d.). *Epsom salts bath: How to take one, what it does*. WebMD. Retrieved June 21, 2022, from <https://www.webmd.com/a-to-z-guides/epsom-salt-bath#s>
- 74 Muanjai, P., Mickevicius, M., Kamandulis, S., Snieckus, A., & Jones, D. A. (2019). The relationship between stiffness and pain following unaccustomed eccentric exercise: The effects of gentle stretch and repeated bout. *European Journal of Applied Physiology*, 119(5), 1183–1194. <https://doi.org/10.1007/s00421-019-04108-7>
- 75 Law, L. F., & Sluka, K. A. (2016). How does physical activity modulate pain? *Pain*, 158(3), 369–370. <https://doi.org/10.1097/j.pain.0000000000000792>
- 76 Lee, M. S., Pittler, M. H., & Ernst, E. (2007). External qigong for pain conditions: A systematic review of randomized clinical trials. *The Journal of Pain*, 8(11), 827–831. <https://doi.org/10.1016/j.jpain.2007.05.016>
- 77 Zeidan, F., & Vago, D. R. (2016). Mindfulness meditation-based pain relief: A mechanistic account. *Annals of the New York Academy of Sciences*, 1373(1), 114–127. <https://doi.org/10.1111/nyas.13153>
- 78 Brandel, M. G., Lin, C., Hennen, D., Khazen, O., Pilitsis, J. G., & Ben-Haim, S. (2022). Mindfulness meditation in the treatment of chronic pain. *Neurosurgery Clinics of North America*, 33(3), 275–279. <https://doi.org/10.1016/j.nec.2022.02.005>
- 79 *Mindfulness meditation to Control Pain*. Harvard Health. (2019, June 15). Retrieved August 31, 2022, from <https://www.health.harvard.edu/pain/mindfulness-meditation-to-control-pain>
- 80 Nunez, K. (2020, July 20). *Why do we sleep?* Healthline. Retrieved August 20, 2022, from <https://www.healthline.com/health/why-do-we-sleep#>
- 81 Van Cauter, E., & Plat, L. (1996). Physiology of growth hormone secretion during sleep. *The Journal of Pediatrics*, 128(5). [https://doi.org/10.1016/s0022-3476\(96\)70008-2](https://doi.org/10.1016/s0022-3476(96)70008-2)
- 82 Boggero, I. A., Krietsch, K. N., Pickerill, H. M., Byars, K. C., Homan, K. J., Williams, S. E., & King, C. D. (2021). Improvements in sleep correlate with improvements in clinical outcomes among adolescents undergoing intensive interdisciplinary pain treatment. *The Clinical Journal of Pain*, 37(6), 443–453. <https://doi.org/10.1097/ajp.0000000000000936>
- 83 Staffe, A. T., Bech, M. W., Clemmensen, S. L., Nielsen, H. T., Larsen, D. B., & Petersen, K. K. (2019). Total sleep deprivation increases pain sensitivity, impairs conditioned pain modulation and facilitates temporal summation of pain in healthy participants. *PLOS ONE*, 14(12). <https://doi.org/10.1371/journal.pone.0225849>
- 84 Whale, K., & Gooberman-Hill, R. (2022). The importance of sleep for people with chronic pain: Current insights and evidence. *JBM Plus*, 6(7). <https://doi.org/10.1002/jbm4.10658>
- 85 Habib, A. M., Nagi, K., Thillaiappan, N. B., Sukumaran, V. K., & Akhtar, S. (2020). Vitamin D and its potential interplay with pain signaling pathways. *Frontiers in Immunology*, 11. <https://doi.org/10.3389/fimmu.2020.00820>
- 86 Shipton, E. E., & Shipton, E. A. (2015). Vitamin D deficiency and pain: Clinical evidence of low levels of vitamin D and supplementation in chronic pain states. *Pain and Therapy*, 4(1), 67–87. <https://doi.org/10.1007/s40122-015-0036-8>
- 87 Mayo Clinic. (2009, March 27). Inadequate Vitamin D Levels Linked to High Use of Narcotic Medication by Patients in Chronic Pain. ScienceDaily. Retrieved August 30, 2022 from www.sciencedaily.com/releases/2009/03/090320112114.htm
- 88 Habib, A. M., Nagi, K., Thillaiappan, N. B., Sukumaran, V. K., & Akhtar, S. (2020). Vitamin D and its potential interplay with pain signaling pathways. *Frontiers in Immunology*, 11. <https://doi.org/10.3389/fimmu.2020.00820>
- 89 Wartenberg, L. (2022, March 10). *Vitamin D deficiency: Symptoms, treatments, and causes*. Healthline. Retrieved August 22, 2022, from <https://www.healthline.com/nutrition/vitamin-d-deficiency-symptoms>
- 90 Bjarnadottir, A. (2021, February 18). *How much vitamin D should you take for optimal health?* Healthline. Retrieved June 22, 2022, from <https://www.healthline.com/nutrition/how-much-vitamin-d-to-take>
- 91 Zhang, M., Han, W., Hu, S., & Xu, H. (2013). Methylcobalamin: A potential vitamin of Pain Killer. *Neural Plasticity*, 2013, 1–6. <https://doi.org/10.1155/2013/424651>
- 92 Leishear, K., Boudreau, R. M., Studenski, S. A., Ferrucci, L., Rosano, C., Rekeneire, N., Houston, D. K., Kritchevsky, S. B., Schwartz, A. V., Vinik, A. I., Hogervorst, E., Yaffe, K., Harris, T. B., Newman, A. B., & Strotmeyer, E. S. (2012). Relationship between vitamin B12 and sensory and motor peripheral nerve function in older adults. *Journal of the American Geriatrics Society*, 60(6), 1057–1063. <https://doi.org/10.1111/j.1532-5415.2012.03998.x>
- 93 Maroon, J. C., & Bost, J. W. (2006). Ω-3 fatty acids (fish oil) as an anti-inflammatory: An alternative to nonsteroidal anti-inflammatory drugs for discogenic pain. *Surgical Neurology*, 65(4), 326–331. <https://doi.org/10.1016/j.surneu.2005.10.023>
- 94 Rajaei, E., Mowla, K., Ghorbani, A., Bahadoram, S., Bahadoram, M., & Dargahi-Malamir, M. (2015). The effect of omega-3 fatty acids in patients with active rheumatoid arthritis receiving dmards therapy: Double-blind randomized controlled trial. *Global Journal of Health Science*, 8(7), 18. <https://doi.org/10.5539/gjhs.v8n7p18>
- 95 Swanson, D., Block, R., & Mousa, S. A. (2012). Omega-3 fatty acids EPA and DHA: Health benefits throughout life. *Advances in Nutrition*, 3(1), 1–7. <https://doi.org/10.3945/an.111.000893>
- 96 Schwalfenberg, G. K., & Genies, S. J. (2017). The importance of magnesium in clinical healthcare. *Scientifica*, 2017, 1–14. <https://doi.org/10.1155/2017/4179326>
- 97 Tarleton, E. K., Kennedy, A. G., Rose, G. L., & Littenberg, B. (2020). Relationship between magnesium intake and chronic pain in U.S. adults. *Nutrients*, 12(7), 2104. <https://doi.org/10.3390/nu12072104>
- 98 Kreutzweiser, D., & Tawfik, Q. A. (2019). Expanding role of NMDA receptor antagonists in the management of pain. *CNS Drugs*, 33(4), 347–374. <https://doi.org/10.1007/s40263-019-00618-2>
- 99 Park, R., Ho, A. M.-H., Pickering, G., Arendt-Nielsen, L., Mohiuddin, M., & Gilron, I. (2020). Efficacy and safety of magnesium for the management of chronic pain in adults: A systematic review. *Anesthesia & Analgesia*, 131(3), 764–775. <https://doi.org/10.1213/ane.0000000000004673>
- 100 Pickering, G., Mazur, A., Troussellard, M., Bienkowski, P., Yaltsewa, N., Amessou, M., Noah, L., & Pouteau, E. (2020). Magnesium status and stress: The Vicious Circle Concept revisited. *Nutrients*, 12(12), 3672. <https://doi.org/10.3390/nu12123672>
- 101 Roman, P., Abalo, R., Marco, E. M., & Cardona, D. (2018). Probiotics in digestive, emotional, and pain-related disorders. *Behavioural Pharmacology*, 29(2 and 3), 103–119. <https://doi.org/10.1097/fbp.0000000000000385>
- 102 Cannarella, L. A., Mari, N. L., Alcántara, C. C., Iryioda, T. M., Costa, N. T., Oliveira, S. R., Lozovoy, M. A., Reiche, E. M., Dichi, I., & Simão, A. N. (2021). Mixture of probiotics reduces inflammatory biomarkers and improves the oxidative/nitrosative profile in people with rheumatoid arthritis. *Nutrition*, 89, 111282. <https://doi.org/10.1016/j.nut.2021.111282>
- 103 Carr, A. C., & McCall, C. (2017). The role of Vitamin C in the treatment of Pain: New insights. *Journal of Translational Medicine*, 15(1). <https://doi.org/10.1186/s12967-017-1179-7>
- 104 Zelfand E. Vitamin C, Pain and Opioid Use Disorder. Integr Med (Encinitas). 2020 Jun;19(3):18-29. PMID: 33132774; PMCID: PMC7572147.
- 105 Evangelou A, Kalfakakou V, Georgakas P, Koutras V, Vezyraki P, Iliopoulou L, Vadalouka A. Ascorbic acid (vitamin C) effects on withdrawal syndrome of heroin abusers. In Vivo. 2000 Mar-Apr;14(2):363-6. PMID: 10836211.
- 106 Ballaz, S. J., & Rebec, G. V. (2019). Neurobiology of vitamin C: Expanding the focus from antioxidant to endogenous neuromodulator. *Pharmacological Research*, 146, 104321. <https://doi.org/10.1016/j.phrs.2019.104321>

- 107 Queremel Milani DA, Davis DD. Pain Management Medications. [Updated 2022 Jul 4]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK560692/>
- 108 Lee, W. M. (2004). Acetaminophen and the U.S. acute liver failure study group: Lowering the risks of hepatic failure. *Hepatology*, 40(1), 6–9. <https://doi.org/10.1002/hep.20293>
- 109 McEvoy, L., Carr, D. F., & Pirmohamed, M. (2021). Pharmacogenomics of NSAID-induced upper gastrointestinal toxicity. *Frontiers in Pharmacology*, 12. <https://doi.org/10.3389/fphar.2021.684162>
- 110 Ramic, E., Prasko, S., Gavran, L., & Spahic, E. (2020). Assessment of the antidepressant side effects occurrence in patients treated in primary care. *Materia Socio Medica*, 32(2), 131. <https://doi.org/10.5455/msm.2020.32.131-134>
- 111 Walia, K. S., Khan, E. A., Ko, D. H., Raza, S. S., & Khan, Y. N. (2004). Side effects of antiepileptics- A Review. *Pain Practice*, 4(3), 194–203. <https://doi.org/10.1111/j.1533-2500.2004.04304.x>
- 112 Wiegel, M., Gottschaldt, U., Hennebach, R., Hirschberg, T., & Reske, A. (2007). Complications and adverse effects associated with continuous peripheral nerve blocks in orthopedic patients. *Anesthesia & Analgesia*, 104(6), 1578–1582. <https://doi.org/10.1213/01.ane.0000261260.69083.f3>
- 113 Benyamin, R., Trescot, A. M., Datta, S., Buenaventura, R., Adlaka, R., Sehgal, N., Glaser, S. E., & Vallejo, R. (2008). Opioid complications and side effects. *Pain physician*, 11(2 Suppl), S105–S120.
- 114 U.S. Department of Health and Human Services. (2022, July 21). *Overdose death rates*. National Institutes of Health. Retrieved August 31, 2022, from <https://nida.nih.gov/research-topics/trends-statistics/overdose-death-rates#>
- 115 Seaman, D. R. (2002). The diet-induced proinflammatory state: *Journal of Manipulative and Physiological Therapeutics*, 25(3), 168–179. <https://doi.org/10.1067/mmt.2002.122324>
- 116 Shin, D., Hong, S. J., Lee, K. W., Shivappa, N., Hebert, J. R., & Kim, K. (2022). Pro-inflammatory diet associated with low back pain in adults aged 50 and older. *Applied Nursing Research*, 66, 151589. <https://doi.org/10.1016/j.apnr.2022.151589>
- 117 Kiecolt-Glaser, J. K. (2010). Stress, food, and inflammation: Psychoneuroimmunology and nutrition at The Cutting Edge. *Psychosomatic Medicine*, 72(4), 365–369. <https://doi.org/10.1097/psy.0b013e3181dbf489>
- 118 NHS. (n.d.). NHS choices. Retrieved August 3, 2022, from <https://www.nhs.uk/live-well/eat-well/how-to-eat-a-balanced-diet/what-are-processed-foods/>
- 119 Geiselman, J. F. (2019). The clinical use of IGG food sensitivity testing with migraine headache patients: A literature review. *Current Pain and Headache Reports*, 23(11). <https://doi.org/10.1007/s11916-019-0819-4>
- 120 Ohtsuka, Y. (2015). Food intolerance and mucosal inflammation. *Pediatrics International*, 57(1), 22–29. <https://doi.org/10.1111/ped.12546>
- 121 Lunde, C. E., & Sieberg, C. B. (2020). Walking the tightrope: A proposed model of chronic pain and stress. *Frontiers in Neuroscience*, 14. <https://doi.org/10.3389/fnins.2020.00270>
- 122 Baum, A. (1990). Stress, intrusive imagery, and chronic distress. *Health Psychology*, 9(6), 653–675. <https://doi.org/10.1037/0278-6133.9.6.653>
- 123 Hannibal, K. E., & Bishop, M. D. (2014). Chronic stress, cortisol dysfunction, and pain: A psychoneuroendocrine rationale for stress management in pain rehabilitation. *Physical Therapy*, 94(12), 1816–1825. <https://doi.org/10.2522/ptj.20130597>
- 124 Green, B. N., Johnson, C. D., Snodgrass, J., Smith, M., & Dunn, A. S. (2016). Association between smoking and back pain in a cross-section of adult Americans. *Cureus*. <https://doi.org/10.7759/cureus.806>
- 125 LaRowe, L. R., & Ditte, J. W. (2020). Pain, nicotine, and tobacco smoking: Current state of the science. *Pain*, 161(8), 1688–1693. <https://doi.org/10.1097/j.pain.0000000000001874>
- 126 Taylor, D. H., Hasselblad, V., Henley, S. J., Thun, M. J., & Sloan, F. A. (2002). Benefits of smoking cessation for longevity. *American Journal of Public Health*, 92(6), 990–996. <https://doi.org/10.2105/ajph.92.6.990>
- 127 Baradaran Mahdavi, S., Riahi, R., Vahdatpour, B., & Kelishadi, R. (2021). Association between sedentary behavior and low back pain; a systematic review and meta-analysis. *Health Promotion Perspectives*, 11(4), 393–410. <https://doi.org/10.34172/hpp.2021.50>
- 128 Murtagh, E. M., Murphy, M. H., & Boone-Heinonen, J. (2010). Walking: The first steps in cardiovascular disease prevention. *Current Opinion in Cardiology*, 25(5), 490–496. <https://doi.org/10.1097/hco.0b013e31832833ce972>
- 129 Hamilton, M. T., Healy, G. N., Dunstan, D. W., Zderic, T. W., & Owen, N. (2008). Too little exercise and too much sitting: Inactivity physiology and the need for new recommendations on sedentary behavior. *Current Cardiovascular Risk Reports*, 2(4), 292–298. <https://doi.org/10.1007/s12170-008-0054-8>
- 130 Goldberg, D. S., & McGee, S. J. (2011). Pain as a global public health priority. *BMC Public Health*, 11(1). <https://doi.org/10.1186/1471-2458-11-770>
- 131 Centers for Disease Control and Prevention. (2022, June 2). *Death Rate Maps & Graphs*. Centers for Disease Control and Prevention. Retrieved August 31, 2022, from <https://www.cdc.gov/drugoverdose/deaths/index.html>
- and *Investigational Dermatology*, Volume 9, 241–248. <https://doi.org/10.2147/ccid.s114830>
- 4 American Chemical Society. (2011, May 11). The skinny on how shed skin reduces indoor air pollution. *ScienceDaily*. Retrieved May 23, 2022 from www.sciencedaily.com/releases/2011/05/110509114034.htm
- 5 Bowe, Whitney. The effect of probiotics on immune regulation, acne, and photoaging. *International Journal of Women's Dermatology*, 1.2 (2015): 85–89
- 6 https://www.avogel.ca/download/Studies/Echinacea_potential_effect_on_acne_2010.pdf
- 7 Sharma, M., Schoop, R., Suter, A., & Hudson, J. B. (2010). The potential use of echinacea in acne: Control of Propionibacterium acnes growth and inflammation. *Phytotherapy Research*, 25(4), 517–521. <https://doi.org/10.1002/ptr.3288>
- 8 Cohen, M. M. (2014). Tulsi - Ocimum sanctum: A herb for all reasons. *Journal of Ayurveda and Integrative Medicine*, 5(4), 251. <https://doi.org/10.4103/0975-9476.146554>
- 9 erqueira, R. O., Frey, B. N., Leclerc, E., & Brietzke, E. (2017). Vitex Agnus Castus for premenstrual syndrome and premenstrual dysphoric disorder: A systematic review. *Archives of Women's Mental Health*, 20(6), 713–719. <https://doi.org/10.1007/s00737-017-0791-0>
- 10 Shoemaker, S. V. (2020, May 27). Black cohosh: Benefits, dosage, side effects, and more. Healthline. Retrieved June 7, 2022, from <https://www.healthline.com/health/food-nutrition/black-cohosh>
- 11 Jung, J., Kwon, H., Hong, J., Yoon, J., Park, M., Jang, M., & Suh, D. (2014). Effect of dietary supplementation with omega-3 fatty acid and gamma-linolenic acid on acne vulgaris: A randomised, double-blind, controlled trial. *Acta Dermato Venereologica*, 94(5), 521–525. <https://doi.org/10.2340/00015555-1802>
- 12 Kim, Y.-G., Lee, J.-H., Park, S., & Lee, J. (2022). The anticancer agent 3,3'-diindolyl-methane inhibits multispecies biofilm formation by acne-causing bacteria and candida albicans. *Microbiology Spectrum*, 10(1). <https://doi.org/10.1128/spectrum.02056-21>
- 13 Grant, P., & Ramasamy, S. (2012). An update on plant derived anti-androgens. *International Journal of Endocrinology and Metabolism*, 10(2), 497–502. <https://doi.org/10.5812/ijem.3644>
- 14 Schmid-Wendtner, M.-H., & Korting, H. C. (2006). The ph of the skin surface and its impact on the barrier function. *Skin Pharmacology and Physiology*, 19(6), 296–302. <https://doi.org/10.1159/000094670>
- 15 Lee, M.-hee, Nam, T. G., Lee, I., Shin, E. J., Han, A.-ram, Lee, P., Lee, S.-Y., & Lim, T.-G. (2018). Skin anti-inflammatory activity of Rose petal extract (*rosa gallica*) through reduction of MAPK signaling pathway. *Food Science & Nutrition*, 6(8), 2560–2567. <https://doi.org/10.1002/fsn.3.870>
- 16 Tadić, V., Arsić, I., Zvezdanović, J., Žugić, A., Cvetković, D., & Pavkov, S. (2017). The estimation of the traditionally used Yarrow (*achillea millefolium* L. asteraceae) oil extracts with anti-inflammatory potential in topical application. *Journal of Ethnopharmacology*, 199, 138–148. <https://doi.org/10.1016/j.jep.2017.02.002>
- 17 Greive, K., Heinicke, I., Barnes, T., & Adams, D. (2015). Evaluation of a topical treatment for the relief of sensitive skin. *Clinical, Cosmetic and Investigational Dermatology*, 405. <https://doi.org/10.2147/ccid.s87509>
- 18 Cha, H. M., Kim, S.-K., Kook, M. C., & Yi, T.-H. (2020). Lactobacillus paraplantarum THG-G10 as a potential anti-acne agent with anti-bacterial and anti-inflammatory activities. *Anaerobe*, 64, 102243. <https://doi.org/10.1016/j.anaerobe.2020.102243>
- 19 Nasri, H., Bahmani, M., Shahinfard, N., Moradi Nafchi, A., Saberianpour, S., & Rafieian Kopaei, M. (2015). Medicinal plants for the treatment of acne vulgaris: A review of recent evidences. *Jundishapur Journal of Microbiology*, 8(11). <https://doi.org/10.5812/jjm.25580>
- 20 Bassett, I. B., Barnetson, R. S., & Pannowitz, D. L. (1990). A comparative study of tea-tree oil versus benzoylperoxide in the treatment of acne. *Medical Journal of Australia*, 153(8), 455–458. <https://doi.org/10.5694/j.1326-5377.1990.tb126150>
- 21 Shah, R., & Peethambaran, B. (2018). Anti-inflammatory and anti-microbial properties of achillea millefolium in acne treatment. *Immunity and Inflammation in Health and Disease*, 241–248. <https://doi.org/10.1016/b978-0-12-805417-8.00019-6>
- 22 Pappas, A. (2009). The relationship of diet and acne. *Dermato-Endocrinology*, 1(5), 262–267. <https://doi.org/10.4161/derm.1.5.10192>
- 23 Bowe, Whitney. The effect of probiotics on immune regulation, acne, and photoaging. *International Journal of Womens Dermatology*, 1.2 (2015): 85–89
- 24 Meixiong, J., Ricco, C., Vasavda, C., & Ho, B. K. (2022). Diet and acne: A systematic review. *JAAD International*, 7, 95–112. <https://doi.org/10.1016/j.jdin.2022.02.012>
- 25 Bowe, Whitney. The effect of probiotics on immune regulation, acne, and photoaging. *International Journal of Women's Dermatology*, 1.2 (2015): 85–89
- 26 Cha, H. M., Kim, S.-K., Kook, M. C., & Yi, T.-H. (2020). Lactobacillus paraplantarum THG-G10 as a potential anti-acne agent with anti-bacterial and anti-inflammatory activities. *Anaerobe*, 64, 102243. <https://doi.org/10.1016/j.anaerobe.2020.102243>
- 27 Makrantonaki, E., Ganceviciene, R., & Zouboulis, C. C. (2011). An update on the role of the sebaceous gland in the pathogenesis of acne. *Dermato-Endocrinology*, 3(1), 41–49. <https://doi.org/10.4161/derm.3.1.13900>
- 28 Mayo Foundation for Medical Education and Research. (2020, October 14). *How much water do you need to stay healthy?* Mayo Clinic. Retrieved June 7, 2022, from <https://www.mayoclinic.org/healthy-lifestyle/nutrition-and-healthy-eating/in-depth/water/art-20044256#>
- 29 Grant, P. (2009). Spearmint herbal tea has significant anti-androgen effects in polycystic ovarian syndrome. A randomized controlled trial. *Phytotherapy Research*, 24(2), 186–188. <https://doi.org/10.1002/ptr.2900>

Acne Protocol

- 1 Mukherjee, J. S. Global Health and the Global Burden of Disease. Oxford Scholarship Online. 10.1 (2017); 93
- 2 Prescott, S. et al. The skin microbiome: impact of modern environments on skin ecology, barrier integrity, and systemic immune programming. *World Allergy Organization Journal*, 10. 29 (2017): 63-72
- 3 Elsaie, M. (2016). Hormonal treatment of acne vulgaris: An update. *Clinical, Cosmetic*

- 30 Kim, Y.-G., Lee, J.-H., Park, S., & Lee, J. (2022). The anticancer agent 3,3'-diindolyl-methane inhibits multispecies biofilm formation by acne-causing bacteria and candida albicans. *Microbiology Spectrum*, 10(1). <https://doi.org/10.1128/spectrum.02056-21>
- 31 Carey, E. (2019, August 27). *10 low-glycemic fruits for diabetes*. Healthline. Retrieved May 31, 2022, from <https://www.healthline.com/health/diabetes/low-glycemic-fruits-for-diabetes>
- 32 Aghasi, M., Golzarand, M., Shab-Bidar, S., Aminianfar, A., Omidian, M., & Taheri, F. (2019). Dairy intake and acne development: A meta-analysis of observational studies. *Clinical Nutrition*, 38(3), 1067–1075. <https://doi.org/10.1016/j.clnu.2018.04.015>
- 33 Sethi, S., Tyagi, S. K., & Anurag, R. K. (2016). Plant-based milk alternatives an emerging segment of Functional Beverages: A Review. *Journal of Food Science and Technology*, 53(9), 3408–3423. <https://doi.org/10.1007/s13197-016-2328-3>
- 34 Cervantes, J., Eber, A. E., Perper, M., Nascimento, V. M., Nouri, K., & Keri, J. E. (2017). The role of zinc in the treatment of acne: A review of the literature. *Dermatologic Therapy*, 31(1). <https://doi.org/10.1111/dth.12576>
- 35 Sturgeon, S. R., Heersink, J. L., Volpe, S. L., Bertone-Johnson, E. R., Puleo, E., Stanczyk, F. Z., Sabelawski, S., Wähälä, K., Kurzer, M. S., & Bigelow, C. (2008). Effect of dietary flaxseed on serum levels of estrogens and androgens in postmenopausal women. *Nutrition and Cancer*, 60(5), 612–618. <https://doi.org/10.1080/01635580801971864>
- 36 Maruyama, H., Sakamoto, T., Araki, Y., & Hara, H. (2010). Anti-inflammatory effect of bee pollen ethanol extract from cistus sp. of Spanish on carrageenan-induced rat hind paw edema. *BMC Complementary and Alternative Medicine*, 10(1). <https://doi.org/10.1186/1472-6882-10-30>
- 37 Zari, S., & Alrahmani, D. (2017). The association between stress and acne among female medical students in Jeddah, Saudi Arabia. *Clinical, Cosmetic and Investigational Dermatology*, Volume 10, 503–506. <https://doi.org/10.2147/ccid.s148499>
- 38 El-akawi, Z., Abdel-Latif, N., & Abdul-Razzak, K. (2006). Does the plasma level of vitamins A and E affect acne condition? *Clinical and Experimental Dermatology*, 31(3), 430–434. <https://doi.org/10.1111/j.1365-2230.2006.02106.x>
- 39 Kubala, J. (2022, April 6). Vitamin A. Healthline. Retrieved June 7, 2022, from <https://www.healthline.com/nutrition/vitamin-a#toxicity-dosage>
- 40 Walocko, F. M., Eber, A. E., Keri, J. E., AL-Harbi, M. A., & Nouri, K. (2017). The role of nicotinamide in acne treatment. *Dermatologic Therapy*, 30(5). <https://doi.org/10.1111/dth.12481>
- 41 Illinois, U. of. (2018, March 13). Niacin: Side effects, dosage, uses, and more. Healthline. Retrieved June 7, 2022, from <https://www.healthline.com/health/drugs/niacin-oral-tablet>
- 42 Lim, S.-K., Ha, J.-M., Lee, Y.-H., Lee, Y., Seo, Y.-J., Kim, C.-D., Lee, J.-H., & Im, M. (2016). Comparison of vitamin D levels in patients with and without acne: A case-control study combined with a randomized controlled trial. *PLOS ONE*, 11(8). <https://doi.org/10.1371/journal.pone.0161162>
- 43 Bjarnadottir, A. (2021, February 18). How much vitamin D should you take for optimal health? Healthline. Retrieved June 7, 2022, from <https://www.healthline.com/nutrition/how-much-vitamin-d-to-take>
- 44 Cervantes, J., Eber, A. E., Perper, M., Nascimento, V. M., Nouri, K., & Keri, J. E. (2017). The role of zinc in the treatment of acne: A review of the literature. *Dermatologic Therapy*, 31(1). <https://doi.org/10.1111/dth.12576>
- 45 Link, R. (2022, January 18). Zinc supplements: Benefits, dosage, and side effects. Healthline. Retrieved June 7, 2022, from <https://www.healthline.com/nutrition/zinc-supplements>
- 46 National Institute of Arthritis, Musculoskeletal Diseases, and Skin Diseases, Seen on: <https://www.niams.nih.gov/health-topics/acne#tab-treatment>
- 47 Blasiak, R. C., Stamey, C. R., Burkhart, C. N., Lugo-Somolinos, A., & Morrell, D. S. (2013). High-dose isotretinoin treatment and the rate of retreat, relapse, and adverse effects in patients with acne vulgaris. *JAMA Dermatology*, 149(12), 1392. <https://doi.org/10.1001/jamadermatol.2013.6746>
- 48 Prescott, S. et al. The skin microbiome: impact of modern environments on skin ecology, barrier integrity, and systemic immune programming. *World Allergy Organization Journal*, 10. 29 (2017): 63-72

Kidney Protocol

- 1 <https://www.niddk.nih.gov/health-information/kidney-disease/kidneys-how-they-work>
- 2 <https://www.niddk.nih.gov/health-information/kidney-disease/chronic-kidney-disease-ckd/causes>
- 3 Centers for Disease Control and Prevention. Chronic Kidney Disease in the United States, 2019. Atlanta, GA: US Department of Health and Human Services. Centers for Disease Control and Prevention; 2019
- 4 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5537833/>
- 5 <https://www.med.umich.edu/1libr/urology/LowOxalateDietGuidelines.pdf>
- 6 <https://www.uclahealth.org/core-kidney/electrolytes>
- 7 Zhang, X., Nie, Q., Zhang, Z., Zhao, J., Zhang, F., Wang, C. ... Song, G. (2021). Resveratrol affects the expression of uric acid transporter by improving inflammation. *Molecular Medicine Reports*, 24, 564. <https://doi.org/10.3892/mmr.2021.12203>
- 8 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4330481/>
- 9 <https://pubmed.ncbi.nlm.nih.gov/31378102/>
- 10 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6966103/>
- 11 Reinhart KM, Coleman CI, Teevan C, Vachhani P, White CM. Effects of garlic on blood pressure in patients with and without systolic hypertension: A meta-analysis. *Ann Pharmacother*. 2008;42:1766–71 <https://journals.sagepub.com/doi/abs/10.1345/aph.1L319>

- 12 <https://pubmed.ncbi.nlm.nih.gov/11807965/>
- 13 Tassell, M. C., Kingston, R., Gilroy, D., Lehane, M., & Furey, A. (2010). Hawthorn (Crataegus spp.) in the treatment of cardiovascular disease. *Pharmacognosy reviews*, 4(7), 32–41. <https://doi.org/10.4103/0973-7847.65324> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3249900/>
- 14 Tabassum, N., & Ahmad, F. (2011). Role of natural herbs in the treatment of hypertension. *Pharmacognosy reviews*, 5(9), 30–40. <https://doi.org/10.4103/0973-7847.79097> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3210006/>
- 15 <https://www.ncbi.nlm.nih.gov/labs/pmc/articles/PMC3210006>
- 16 <https://pubmed.ncbi.nlm.nih.gov/23042598/>
- 17 <https://www.health.harvard.edu/heart-health/meditation-and-a-relaxation-technique-to-lower-blood-pressure>
- 18 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3679769/>
- 19 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5655940/>
- 20 <https://pubmed.ncbi.nlm.nih.gov/9894438/>
- 21 <https://www.cdc.gov/diabetes/managing/diabetes-kidney-disease.html>
- 22 <https://pubmed.ncbi.nlm.nih.gov/842585/>
- 23 <https://pubmed.ncbi.nlm.nih.gov/19800084/>
- 24 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4027280/>
- 25 <https://pubmed.ncbi.nlm.nih.gov/21480806/>
- 26 <https://pubmed.ncbi.nlm.nih.gov/19839001/>
- 27 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6943541/>
- 28 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5376420/>
- 29 Ma H.-T., Hsieh J.-F., Chen S.-T. Anti-diabetic effects of Ganoderma lucidum. *Phytochemistry*. 2015;114:109–113. doi: 10.1016/j.phytochem.2015.02.017
- 30 Xiao C, Wu Q, Zhang J, Xie Y, Cai W, Tan J. Antidiabetic activity of Ganoderma lucidum polysaccharides F31 down-regulated hepatic glucose regulatory enzymes in diabetic mice. *J Ethnopharmacol*. 2017;196:47-57
- 31 <https://pubmed.ncbi.nlm.nih.gov/18813412/>
- 32 <https://pubmed.ncbi.nlm.nih.gov/19114224/>
- 33 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4829575/>
- 34 <https://www.verywellhealth.com/>
- 35 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5637251/>
- 36 <https://pubmed.ncbi.nlm.nih.gov/32824103/>
- 37 Zhang, H. W., Lin, Z. X., Tung, Y. S., Kwan, T. H., Mok, C. K., Leung, C., & Chan, L. S. (2014). Cordyceps sinensis (a traditional Chinese medicine) for treating chronic kidney disease. *The Cochrane database of systematic reviews*, (12), CD008353. <https://doi.org/10.1002/14651858.CD008353.pub2>
- 38 Zhang, H. W., Lin, Z. X., Tung, Y. S., Kwan, T. H., Mok, C. K., Leung, C., & Chan, L. S. (2014). Cordyceps sinensis (a traditional Chinese medicine) for treating chronic kidney disease. *The Cochrane database of systematic reviews*, (12), CD008353. <https://doi.org/10.1002/14651858.CD008353.pub2>
- 39 <https://pubmed.ncbi.nlm.nih.gov/32856245/>
- 40 <https://pubmed.ncbi.nlm.nih.gov/15638071>
- 41 <https://pubmed.ncbi.nlm.nih.gov/11746861/>
- 42 <https://www.niddk.nih.gov/health-information/urologic-diseases/kidney-infection-pyelonephritis/treatment>
- 43 Mahdavi, R., Khabbazi, T., & Safa, J. (2019). Alpha lipoic acid supplementation improved antioxidant enzyme activities in hemodialysis patients. *International journal for vitamin and nutrition research*, 89(3-4), 161–167 <https://doi.org/10.1024/0300-9831/a000552>
- 44 Hsu, S. P., Chiang, C. K., Yang, S. Y., & Chien, C. T. (2010). N-acetylcysteine for the management of anemia and oxidative stress in hemodialysis patients. *Nephron. Clinical practice*, 116(3), c207–c216 <https://doi.org/10.1159/000317201>
- 45 Albertoni, G., & Schor, N. (2015). Resveratrol plays important role in protective mechanisms in renal disease--mini-review, 37(1), 106–114. <https://doi.org/10.5935/0101-2800.20150015>

Anti-Aging Protocol

- 1 <https://www.pewresearch.org/fact-tank/2016/04/21/worlds-centenarian-population-projected-to-grow-eightfold-by-2050/>
- 2 Lombard DB, Chua KF et al.: DNA Repair, Genome Stability, and Aging. *Cell* 2005;120 (4): 497-512
- 3 Dizdaroglu, M., Jaruga, P., Birincioglu, M., & Rodriguez, H. (2002). Free radical-induced damage to DNA: mechanisms and measurement. *Free radical biology & medicine*, 32(11), 1102–1115. [https://doi.org/10.1016/s0891-5849\(02\)00826-2](https://doi.org/10.1016/s0891-5849(02)00826-2)
- 4 Lagouge M, Larsson NG: The Role of Mitochondrial DNA Mutations and Free Radicals in Disease and Aging. *J Intern. Med*. 2013;273(6):529-543
- 5 <https://www.ox.ac.uk/news/2022-07-26-genetic-study-provides-evidence-alcohol-accelerates-biological-aging>
- 6 Aubert, G. and P. M. Lansdorp, Telomeres and aging. *Physiol Rev*. 2008. 88(2): p. 557-579
- 7 López-Otín C, Blasco MA et al.: The hallmarks of aging. *Cell* 2013;153 (6):1194-1217
- 8 Yamada M, Naito K, et al.: Prevalence of atherosclerosis in relation to atomic bomb

- radiation exposure: An RERF Adult Health Study. *International Journal of Radiation Biology* 2005; 81:821-6
- 9 Morita A. (2007). Tobacco smoke causes premature skin aging. *Journal of dermatological science*, 48(3), 169–175. <https://doi.org/10.1016/j.jdermsci.2007.06.015>. <https://pubmed.ncbi.nlm.nih.gov/17951030/>
- 10 Crimmins EM. Lifespan and Healthspan: Past, Present, and Promise. *Gerontologist*. 2015 Dec;55(6):901-11. doi: 10.1093/geront/gnv130. Epub 2015 Nov 10. PMID: 26561272; PMCID: PMC4861644. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4861644/>
- 11 https://www.rerf.or.jp/en/programs/roadmap_e/health_effects-en/late-en/aging/
- 12 <https://www.ox.ac.uk/news/2022-07-26-genetic-study-provides-evidence-alcohol-accelerates-biological-aging>
- 13 Thornton MJ. Estrogens and aging skin. *Dermatoendocrinol*. 2013 Apr 1;5(2):264-70. doi: 10.4161/derm.23872. PMID: 24194966; PMCID: PMC3772914. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3772914/>
- 14 Morley, John E., Diabetes and Aging: Epidemiologic Overview, *Clinics in Geriatric Medicine*, Volume 24, Issue 3, 2008, Pages 395-405, ISSN 0749-0690, <https://doi.org/10.1016/j.cger.2008.03.005>
- 15 Mirian Ayumi Kurauti, Gabriela Moreira Soares, Carine Marmentini, Gabriela Alves Bronczek, Renato Chaves Souto Branco, Antonio Carlos Boscherio, Insulin and aging, Editor(s): Gerald Litwack, Vitamins and Hormones, Academic Press, Vol 115, 2021, Pages 185-219, ISSN 0083-6729, ISBN 9780323855488, <https://doi.org/10.1016/bs.vh.2020.12.010>
- 16 Goronzy JJ, Weyand CM. Immune aging and autoimmunity. *Cell Mol Life Sci*. 2012 May;69(10):1615-23. doi: 10.1007/s00018-012-0970-0. Epub 2012 Apr 1. PMID: 22466672; PMCID: PMC4277694. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4277694/> and <https://my.clevelandclinic.org/health/diseases/21624-autoimmune-diseases>
- 17 Hammond B.R., Fletcher L.M., Roos F., Wittwer J., Schalch W. A double-blind, placebo-controlled study on the effects of lutein and zeaxanthin on photostress recovery, glare disability, and chromatic contrast. *Investig. Ophthalmol. Vis. Sci*. 2014;55:8583-8589. doi: 10.1167/jovs.14-15573
- 18 Khachik F. Distribution and metabolism of dietary carotenoids in humans as a criterion for development of nutritional supplements. *Pure Appl. Chem*. 2006;78:1551-1557. doi: 10.1351/pac200678081551. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6539799/>
- 19 Prasad K.N., Wu M., Bondy S.C. Telomere shortening during aging: attenuation by antioxidants and anti-inflammatory agents. *Mech. Aging Dev*. 2017;164:61-66. doi: 10.1016/j.mad.2017.04.004
- 20 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3650504/#:~:text=Three%20major%20classes%20of%20phytochemicals,flavonoid%20within%20the%20polyphenol%20family>. <https://lpi.oregonstate.edu/mic/dietary-factors/phytochemicals/chlorophyll-metallo-chlorophyll-derivatives> <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7146365/>
- 21 Herrera-Arellano A, Flores-Romero S, Chávez-Soto M.A., Tortoriello J. Effectiveness and tolerability of a standardized extract from *Hibiscus sabdariffa* in patients with mild to moderate hypertension: A controlled and randomized clinical trial. *Phytomedicine*. 2004;11:375-382. doi: 10.1016/j.phymed.2004.04.001
- 22 Zafra-Stone S, Yasmin T, Bagchi M, Chatterjee A, Vinson J.A., Bagchi D. Berry anthocyanins as novel antioxidants in human health and disease prevention. *Mol. Nutr. Food Res*. 2007;51:675-683. doi: 10.1002/mnfr.200700002.
- 23 Feitelson M.A., Arzumanyan A., Kulathinal R.J., Blain S.W., Holcombe R.F., Mahajna J., Marino M., Martinez-Chantar M.L., Nawroth R., Sanchez-Garcia I., et al. Sustained proliferation in cancer: Therapeutic targets. *Semin. Cancer Biol*. 2016;35:25-54. doi: 10.1016/j.semcancer.2015.02.006
- 24 Parletta N., Milte C.M., Meyer B.J. Nutritional modulation of cognitive function and mental health. *J. Nutr. Biochem*. 2013;24:725-743
- 25 Janice K. Kiecolt-Glaser, Martha A. Belury, Rebecca Andridge, William B. Malarkey, Beom Seuk Hwang, Ronald Glaser: Omega-3 supplementation lowers inflammation in healthy middle-aged and older adults: A randomized controlled trial. *Brain, Behavior, and Immunity*, 2012; 26 (6) Ohio State University. (2012, October 1). Omega-3 supplements may slow a biological effect of aging. *ScienceDaily*
- 26 USDA, US Department of Agriculture, Agricultural Research Service <https://fdc.nal.usda.gov/fdc-app.html#/food-details/175168/nutrients>
- 27 Zasada, Malwina et al: "Retinoids: active molecules influencing skin structure formation in cosmetic and dermatological treatments", August 30 2019, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6791161/>
- 28 Pullar, J. M., Carr, A. C., & Vissers, M. (2017). The Roles of Vitamin C in Skin Health. *Nutrients*, 9(8), 866. <https://doi.org/10.3390/nu9080866>.
- 29 Goodwill, A. M., & Szoek, C. (2017). A Systematic Review and Meta-Analysis of The Effect of Low Vitamin D on Cognition. *Journal of the American Geriatrics Society*, 65(10), 2161-2168. <https://doi.org/10.1111/jgs.15012> <https://pubmed.ncbi.nlm.nih.gov/28758188/>
- 30 Mark, K. A., Dumas, K. J., Bhaumik, D., Schilling, B., Davis, S., Oron, T. R., Sorensen, D. J., Lukanic, M., Brem, R. B., Melov, S., Ramanathan, A., Gibson, B. W., & Lithgow, G. J. (2016). Vitamin D Promotes Protein Homeostasis and Longevity via the Stress Response Pathway Genes skn-1, ire-1, and xbp-1. *Cell reports*, 17(5), 1227-1237. <https://doi.org/10.1016/j.celrep.2016.09.086>. <https://pubmed.ncbi.nlm.nih.gov/27783938/>
- 31 Meehan M, Penckofer S. The Role of Vitamin D in the Aging Adult. *J Aging Gerontol*. 2014 Dec;2(2):60-71. doi: 10.12974/2309-6128.2014.02.02.1. PMID: 25893188; PMCID: PMC4399494. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4399494/>
- 32 Meydani SN, Lewis ED, Wu D. Perspective: Should Vitamin E Recommendations for Older Adults Be Increased? *Adv Nutr*. 2018 Sep 1;9(5):533-543. doi: 10.1093/advances/nmy035. PMID: 30107519; PMCID: PMC6140432. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6140432/>
- 33 Schagen SK, Zampeli VA, Makrantonaki E, Zouboulis CC. Discovering the link between nutrition and skin aging. *Dermatoendocrinol*. 2012 Jul 1;4(3):298-307. doi: 10.4161/derm.22876. PMID: 23467449; PMCID: PMC3583891. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3583891/>
- 34 Emami, M. R., Safabakhsh, M., Alizadeh, S., Asbaghi, O., & Khosroshahi, M. Z. (2019). Effect of vitamin E supplementation on blood pressure: a systematic review and meta-analysis. *Journal of human hypertension*, 33(7), 499-507. <https://doi.org/10.1038/s41371-019-0192-0>. <https://pubmed.ncbi.nlm.nih.gov/30846828/> Browne D, McGuinness B, Woodside JV, McKay GJ. Vitamin E and Alzheimer's disease: what do we know so far? *Clin Interv Aging*. 2019 Jul 18;14:1303-1317. doi: 10.2147/CIA.S186760. PMID: 31409980; PMCID: PMC6645610. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6645610/>
- 35 Prasad, Ananda S: "Discovery of Human Zinc Deficiency: Its Impact on Human Health and Disease", March 6 2013, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3649098/>
- 36 Park H, Kim CW, Kim SS, Park CW. The therapeutic effect and the changed serum zinc level after zinc supplementation in alopecia areata patients who had a low serum zinc level. *Ann Dermatol*. 2009 May;21(2):142-6. doi: 10.5021/ad.2009.21.2.142. Epub 2009 May 31. PMID: 20523772; PMCID: PMC2861201. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2861201/>
- 37 Ángel Julio Romero Cabrera: Pathobiol Aging Age Relat Dis. 2015; 5: Published online 2015 Feb 5. doi: 10.3402/pba.v5.25592 Zinc, aging, and immunosenescence: an overview
- 38 Dang Y, An Y, He J, Huang B, Zhu J, Gao M, Zhang S, Wang X, Yang B, Xie Z. Berberine ameliorates cellular senescence and extends the lifespan of mice via regulating p16 and cyclin protein expression. *Aging Cell*. 2020 Jan;19(1):e13060. doi: 10.1111/acel.13060. Epub 2019 Nov 26. PMID: 31773901; PMCID: PMC6974710
- 39 Qiu, S., Sun, H., Zhang, A. H., Xu, H. Y., Yan, G. L., Han, Y., & Wang, X. J. (2014). Natural alkaloids: basic aspects, biological roles, and future perspectives. *Chinese journal of natural medicines*, 12(6), 401-406. [https://doi.org/10.1016/S1875-5364\(14\)60063-7](https://doi.org/10.1016/S1875-5364(14)60063-7). <https://pubmed.ncbi.nlm.nih.gov/24969519/>
- 40 Zhang, C., Sheng, J., Li, G., Zhao, L., Wang, Y., Yang, W., Yao, X., Sun, L., Zhang, Z., & Cui, R. (2020). Effects of Berberine and Its Derivatives on Cancer: A Systems Pharmacology Review. *Frontiers in pharmacology*, 10, 1461. <https://doi.org/10.3389/fphar.2019.01461>
- 41 Haijing Zhang, Chunhui Zhao, Guoqiong Cao, Limin Guo, Shuai Zhang, Yuexia Liang, Chunxia Qin, Ping Su, Hang Li, Wensheng Zhang. Berberine modulates amyloid- β peptide generation by activating AMP-activated protein kinase. *Neuropharmacology*, Vol 125, 2017, Pages 408-417, ISSN 0028-3908. <https://doi.org/10.1016/j.neuropharm.2017.08.013>
- 42 Wang, H., Zhu, C., Ying, Y., Luo, L., Huang, D. and Luo, Z. (2017). Metformin and berberine, two versatile drugs in treatment of common metabolic diseases. *Oncotarget*, [online] 9(11), pp.10135-10146. doi:10.18632/oncotarget.20807
- 43 Hoffman, Richard: "Micronutrient deficiencies in the elderly - could ready meals be part of the solution?", January 12 2017, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5465850/>
- 44 Lupu, Mihaela-Adi et al: "Beneficial effects of food supplements based on hydrolyzed collagen for skin care (Review)", July 2020, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7271718/>
- 45 Bolke L, Schlippe G, Gerß J, Voss W. A Collagen Supplement Improves Skin Hydration, Elasticity, Roughness, and Density: Results of a Randomized, Placebo-Controlled, Blind Study. *Nutrients*. 2019 Oct 17;11(10):2494. doi: 10.3390/nu11102494. PMID: 31627309; PMCID: PMC6835901. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6835901/>
- 46 Herskovits AZ, Guarente L. Sirtuin deacetylases in neurodegenerative diseases of aging. *Cell Res*. 2013 Jun;23(6):746-58. doi: 10.1038/cr.2013.70. Epub 2013 May 21. PMID: 23689277; PMCID: PMC3674397
- 47 Tarantini, S., et al (2019). Nicotinamide mononucleotide (NMN) supplementation rescues cerebrovascular endothelial function and neurovascular coupling responses and improves cognitive function in aged mice. *Redox Biology*, [online] 24, p.101192. doi:10.1016/j.redox.2019.101192.
- 48 Picciotto, N.E., Gano, L.B., Johnson, L.C., Martens, C.R., Sindler, A.L., Mills, K.F., Imai, S. and Seals, D.R. (2016). Nicotinamide mononucleotide supplementation reverses vascular dysfunction and oxidative stress with aging in mice. *Aging Cell*, 15(3), pp.522-530. doi:10.1111/acel.12461
- 49 Zhou, D. D., Luo, M., Huang, S. Y., Saimaiti, A., Shang, A., Gan, R. Y., & Li, H. B. (2021). Effects and Mechanisms of Resveratrol on Aging and Age-Related Diseases. *Oxidative medicine and cellular longevity*, 2021, 9932218. <https://doi.org/10.1155/2021/9932218>
- 50 Petruk, Ganna et al: "Antioxidants from Plants Protect against Skin Photoaging", August 2 2018, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6098906/>
- 51 Li Juan et al: "A comparative study of anti-aging properties and mechanism: resveratrol and caloric restriction", September 12 2017, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5630366/>
- 52 Madeo, F., Carmona-Gutierrez, D., Kepp, O. and Kroemer, G. (2018). Spermidine delays aging in humans. *Aging*, 10(8), pp.2209-2211. doi:10.18632/aging.101517
- 53 Eisenberg, T., Abdellatif, M., Schroeder, S. et al. Cardioprotection and lifespan extension by the natural polyamine spermidine. *Nat Med* 22, 1428-1438 (2016). <https://doi.org/10.1038/nm.4222>
- 54 Mishra, Shrikant et al: "The effect of curcumin (turmeric) on Alzheimer's disease: An overview", January 2008, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2781139/>

55 Shen, Li Rong et al: "Curcumin-supplemented diets increase superoxide dismutase activity and mean lifespan in *Drosophila*", August 2013, pubmed.ncbi.nlm.nih.gov/22653297/

56 Morse, Robert: "Curcumin's Anti-Aging Properties", October 2021, lifeextension.com/magazine/2021/ss/curcumin-anti-aging

57 Shalini V.K., Srinivas L. Fuel smoke condensate induced DNA damage in human lymphocytes and protection by turmeric (*Curcuma longa*) Mol. Cell. Biochem. 1990;95:21–30. doi: 10.1007/BF00219526. [PubMed] [CrossRef] [Google Scholar]

58 Chen X., Wang J., Fu Z., Zhu B., Wang J., Guan S., Hua Z. Curcumin activates DNA repair pathway in bone marrow to improve carboplatin-induced myelosuppression. Sci. Rep. 2017;7:1–11. doi: 10.1038/s41598-017-16436-9. [PMC free article] [PubMed] [CrossRef] [Google Scholar]

59 Jiao, C., Xie, Y., Yun, H., Liang, H., He, C., Jiang, A., Wu, Q., & Yang, B. B. (2020). The effect of *Ganoderma lucidum* spore oil in early skin wound healing: interactions of skin microbiota and inflammation. Aging, 12(14), 14125–14140. <https://doi.org/10.18632/aging.103412> <https://pubmed.ncbi.nlm.nih.gov/32692722/>

60 Wang J, Cao B, Zhao H, Feng J. Emerging Roles of *Ganoderma lucidum* in Anti-Aging. Aging Dis. 2017 Dec 1;8(6):691-707. doi: 10.14336/AD.2017.0410. PMID: 29344411; PMCID: PMC5758346

61 Kumar R., Gupta K., Saharia K., Pradhan D., Subramaniam J.R. *Withania somnifera* root extract extends lifespan of *Caenorhabditis elegans*. Ann. Neurosci. 2013;20:13. doi: 10.5214/ans.0972.7531.200106. [PMC free article] [PubMed] [CrossRef] [Google Scholar]

62 Akhoun B.A., Pandey S., Tiwari S., Pandey R. Withanolide A offers neuroprotection, ameliorates stress resistance and prolongs the life expectancy of *Caenorhabditis elegans*. Exp. Gerontol. 2016;78:47–56. doi: 10.1016/j.exger.2016.03.004. [PubMed] [CrossRef] [Google Scholar]

63 Raguram V., Subramaniam J. *Withania somnifera* Root Extract Enhances Telomerase Activity in the Human HeLa Cell Line. Adv. Biosci. Biotechnol. 2016;7:199–204. doi: 10.4236/abb.2016.74018. [CrossRef] [Google Scholar]

64 <https://pubmed.ncbi.nlm.nih.gov/18803231/> Ji, D. B., Ye, J., Li, C. L., Wang, Y. H., Zhao, J., & Cai, S. Q. (2009). Antiaging effect of *Cordyceps sinensis* extract. Phytotherapy research: PTR, 23(1), 116–122. <https://doi.org/10.1002/ptr.2576>

<https://www.hindawi.com/journals/ecam/2012/273435/>, <https://europepmc.org/article/med/15506292>, <https://www.sciencedirect.com/science/article/abs/pii/S0141813020330221> <https://pubs.rsc.org/en/content/articlelanding/2016/ra/c6ra23612j/unaauth>

65 Lai, P. L., Naidu, M., Sabaratnam, V., Wong, K. H., David, R. P., Kuppusamy, U. R., Abdullah, N., & Malek, S. N. (2013). Neurotrophic properties of the Lion's mane medicinal mushroom, *Herichium erinaceus* (Higher Basidiomycetes) from Malaysia. International journal of medicinal mushrooms, 15(6), 539–554. <https://doi.org/10.1615/intjmed-mushrv15.i6.30> <https://pubmed.ncbi.nlm.nih.gov/24266378/>

66 Mori, K., Inatomi, S., Ouchi, K., Azumi, Y., & Tuchida, T. (2009). Improving effects of the mushroom Yamabushitake (*Herichium erinaceus*) on mild cognitive impairment: a double-blind placebo-controlled clinical trial. Phytotherapy research : PTR, 23(3), 367–372. <https://doi.org/10.1002/ptr.2634> <https://pubmed.ncbi.nlm.nih.gov/18844328/>

67 Xu, Hui, et al. "Chemical Analysis of *Herichium Erinaceum* Polysaccharides and Effect of the Polysaccharides on Derma Antioxidant Enzymes, MMP-1 and TIMP-1 Activities." *International Journal of Biological Macromolecules*, vol. 47, no. 1, 1 July 2010, pp. 33–36, pubmed.ncbi.nlm.nih.gov/20380848/, 10.1016/j.ijbiomac.2010.03.024. Accessed 18 May 2022.

68 Noh, Hyung Jun, et al. "Chemical Constituents of *Herichium Erinaceum* Associated with the Inhibitory Activity against Cellular Senescence in Human Umbilical Vascular Endothelial Cells." *Journal of Enzyme Inhibition and Medicinal Chemistry*, vol. 30, no. 6, 13 Feb. 2015, pp. 934–940, 10.3109/14756366.2014.995181. Accessed 1 Sept. 2022.

69 de Oliveira Zanuso, B., de Oliveira Dos Santos, A. R., Miola, V., Guissoni Campos, L. M., Spilla, C., & Barbalho, S. M. (2022). Panax ginseng and aging related disorders: A systematic review. *Experimental gerontology*, 161, 111731. <https://doi.org/10.1016/j.exger.2022.111731>

70 Naveed, Akhtar & Zaman, Shahiq uz & Khan, Barkat & Khan, Haji M. shoaib & Ahmad, Mahmood & Rasool, Fatima & Mahmood, Tariq & Rasul, Akhtar. (2011). Evaluation of various functional skin parameters using a topical cream of *Calendula officinalis* extract. African journal of pharmacy and pharmacology. 5. 199-206

71 Poblócka-Olech, L., Inkielewicz-Stepniak, I., & Krauze-Baranowska, M. (2019). Anti-inflammatory and antioxidative effects of the buds from different species of *Populus* in human gingival fibroblast cells: Role of bioflavonones. *Phytomedicine*, 56, 1-9

72 Warsito, M. F., & Kusumawati, I. (2019). The impact of herbal products in the prevention, regeneration and delay of skin aging. *Reviews on Biomarker Studies in Aging and Anti-Aging Research*, 155-174

73 <https://www.ncbi.nlm.nih.gov/labs/pmc/articles/PMC5485961/> Mármol, I., Sánchez-de-Diego, C., Jiménez-Moreno, N., Ancin-Azpilicueta, C., & Rodríguez-Yoldi, M. J. (2017). Therapeutic Applications of Rose Hips from Different Rosa Species. *International journal of molecular sciences*, 18(6), 1137. <https://doi.org/10.3390/ijms18061137>

74 <https://www.health.harvard.edu/staying-healthy/exercise-and-aging-can-you-walk-away-from-father-time>

Exercise and aging Maria Antoinette Fiatarone Singh, MD, FRACP,a,* aUniversity of Sydney, East Street, Lidcombe, NSW 2141, Australia bHebrew Rehabilitation Center for Aged, 1200 Center Street, Roslindale, MA 02131, USA

Clin Geriatr Med 20 (2004) 201 – 221

[1] American College of Sports Medicine, Pollock M, Gaesser G, et al. ACSM Position Stand on the recommended quantity and quality of exercise for developing and maintaining cardiorespiratory and muscular fitness, and flexibility in healthy adults. Med Sci Sports Exerc 1998;30(6): 975 – 91.

[2] American Geriatrics Society, British Geriatrics Society, American Academy of Orthopaedic Surgeons Panel on Falls Prevention. Guideline for the prevention of falls in older persons. J Am Geriatr Soc 2001;49:664 – 72.

[3] American Diabetic Association. Diabetes mellitus and exercise: position statement. Diabetes Care 1990;13(7):804 – 5.

[4] Drinkwater B, Grimson S, Cullen-Raab D, Harter-Snow C. ACSM position stand on osteoporosis and exercise. Med Sci Sports Exerc 1995;27(4):i – vii

[5] Pate RR, Pratt M, Blair SN, et al. Physical activity and public health: a recommendation from the Centers for Disease Control and Prevention and the American College of Sports Medicine. JAMA 1995;273(5):402 – 7.

[6] Bouchard C, Shephard R, Stephen T, et al. Exercise, fitness and health: a consensus of current knowledge. Champaign, IL: Human Kinetic Publisher; 1990.

[7] Bouchard C. Physical Activity, Fitness, and Health: International Proceedings and Consensus Statement. Human Kinetics. Champaign (IL): 1994.

[8] US Department of Health and Human Services. Physical activity and health: a report of the Surgeon General. Atlanta: US Department of Health and Human Services, Centers for Disease 212 M.A. Fiatarone Singh / Clin Geriatr Med 20 (2004) 201–221 Control and Prevention, National Center for Chronic Disease Prevention and Health Promotion; 1996

[9] Mazzeo R, Cavanaugh P, Evans W, et al. Exercise and physical activity for older adults. Med Sci Sports Exerc 1998;30(6):992 – 1008

[10] American Geriatrics Society Panel on Exercise and Osteoarthritis. Exercise prescription for older adults with osteoarthritis pain: consensus practice recommendations. A supplement to the AGS Clinical Practice Guidelines on the Management of Chronic Pain in Older Adults. J Am Geriatr Soc 2001;49:808 – 23

[11] US Department of Health and Human Services. Healthy people 2010 [draft for public comment]. Washington, DC: US Department of Health and Human Services; 1999.

[12] Christmas C, Andersen R. Exercise and older patients: guidelines for the clinician. J Am Geriatr Soc 2000;48:318 – 24

75 <https://www.endocrine.org/topics/edc/what-edcs-are/common-edcs>

76 Samaras, Nikolaos, et al. "Off-Label Use of Hormones as an Antiaging Strategy: A Review." *Clinical Interventions in Aging*, July 2014, p. 1175, www.ncbi.nlm.nih.gov/pmc/articles/PMC4116364/, 10.2147/cia.s48918. Accessed 18 Aug. 2019

77 Satriyasa, Bagus Komang. "Botulinum Toxin (Botox) a for Reducing the Appearance of Facial Wrinkles: A Literature Review of Clinical Use and Pharmacological Aspect." *Clinical, Cosmetic and Investigational Dermatology*, vol. Volume 12, no. 12, Apr. 2019, pp. 223–228, www.ncbi.nlm.nih.gov/pmc/articles/PMC6489637/, 10.2147/ccid.s202919. Hyaluronan-based hydrogels as dermal fillers: The biophysical properties that translate into a "volumetric" effect a

78 Saccon, T. D., Nagpal, R., Yadav, H., Cavalcante, M. B., Nunes, A., Schneider, A., Gesing, A., Hughes, B., Yousefzadeh, M., Tchkonja, T., Kirkland, J. L., Niedernhofer, L. J., Robbins, P. D., & Mastermak, M. M. (2021). Senolytic Combination of Dasatinib and Quercetin Alleviates Intestinal Senescence and Inflammation and Modulates the Gut Microbiome in Aged Mice. The journals of gerontology. Series A, Biological sciences and medical sciences, 76(11), 1895–1905. <https://doi.org/10.1093/gerona/ab002>

79 Klimova, Blanka, et al. "Anti-Aging Drugs – Prospect of Longer Life?" *Current Medicinal Chemistry*, vol. 25, no. 17, 22 May 2018, pp. 1946–1953, 10.2174/0929867325666171129215251. Accessed 14 Mar. 2021

Energy Protocol

1 Hills RD Jr, Pontefract BA, Mishcon HR, Black CA, Sutton SC, Theberge CR. Gut Microbiome: Profound Implications for Diet and Disease. *Nutrients*. 2019 Jul 16;11(7):1613. doi: 10.3390/nu11071613. PMID: 31315227; PMCID: PMC6682904 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6682904/>

2 Meredith Genevive R., Rakow Donald A., Eldermire Erin R. B., Madsen Cecelia G., Shelley Steven P., Sachs Naomi, *Minimum Time Dose in Nature to Positively Impact the Mental Health of College-Aged Students, and How to Measure it: A Scoping Review*, *Frontiers in Psychology*, 2020, doi=10.3389/fpsyg.2019.02942

3 Tandon, N., & Yadav, S. S. (2020). Safety and clinical effectiveness of *Withania Somnifera* (Linn.) Dual root in human ailments. *Journal of ethnopharmacology*, 255, 112768. <https://doi.org/10.1016/j.jep.2020.112768>

4 Friedman M. (2015). Chemistry, Nutrition, and Health-Promoting Properties of *Herichium erinaceus* (Lion's Mane) Mushroom Fruiting Bodies and Mycelia and Their Bioactive Compounds. *Journal of agricultural and food chemistry*, 63(32), 7108–7123. <https://doi.org/10.1021/acs.jafc.5b02914>

5 Tang, W., Gao, Y., Chen, G., Gao, H., Dai, X., Ye, J., Chan, E., Huang, M., & Zhou, S. (2005). A randomized, double-blind and placebo-controlled study of a *Ganoderma lucidum* polysaccharide extract in neurasthenia. *Journal of medicinal food*, 8(1), 53–58. <https://doi.org/10.1089/jmf.2005.8.53>

6 Lin B, Li S. *Cordyceps* as an Herbal Drug. In: Benzie IFF, Wachtel-Galor S, editors. *Herbal Medicine: Biomolecular and Clinical Aspects*. 2nd edition. Boca Raton (FL): CRC Press/Taylor & Francis; 2011. Chapter 5. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK92758/>

7 Bach HV, Kim J, Myung SK, Cho YA. Efficacy of Ginseng Supplements on Fatigue and Physical Performance: A Meta-analysis. *J Korean Med Sci*. 2016 Dec;31(12):1879-1886. doi: 10.3346/jkms.2016.31.12.1879. PMID: 27822924; PMCID: PMC5102849

8 Lekontseva, Y., Zhukova, I., & Wacker, A. (2017). *Rhodiola rosea* in Subjects with Prolonged or Chronic Fatigue Symptoms: Results of an Open-Label Clinical Trial. *Complementary medicine research*, 24(1), 46–52. <https://doi.org/10.1159/000457918>

9 Kahnum, F. (2006, November 20). *Rhodiola rosea*: A Versatile Adaptogen. Comprehensive Reviews. Retrieved February 7, 2022, from <https://ift.onlinelibrary.wiley.com/doi/10.1111/j.1541-4337.2005.tb00073.x>

10 Yepes-Pérez AF, Herrera-Calderon O, Sánchez-Aparicio JE, Tiessler-Sala L, Maréchal

JD, Cardona-G W. Investigating Potential Inhibitory Effect of Uncaria tomentosa (Cat's Claw) against the Main Protease 3CLpro of SARS-CoV-2 by Molecular Modeling. Evid Based Complement Alternat Med. 2020 Sep 30;2020:4932572. doi: 10.1155/2020/4932572. PMID: 33029165; PMCID: PMC7532411

11 Kaur G, Invally M, Sanzagiri R, Buttar HS. Evaluation of the antidepressant activity of Moringa oleifera alone and in combination with fluoxetine. J Ayurveda Integr Med. 2015 Oct-Dec;6(4):273-9. doi: 10.4103/0975-9476.172384. PMID: 26834427; PMCID: PMC4719488

12 Moss, M., Hewitt, S., Moss, L., & Wesnes, K. (2008). Modulation of cognitive performance and mood by aromas of peppermint and ylang-ylang. The International journal of neuroscience, 118(1), 59–77 https://doi.org/10.1080/00207450601042094

13 Tardy AL, Pouteau E, Marquez D, Yilmaz C, Scholey A. Vitamins and Minerals for Energy, Fatigue and Cognition: A Narrative Review of the Biochemical and Clinical Evidence. Nutrients. 2020 Jan 16;12(1):228. doi: 10.3390/nu12010228. PMID: 31963141; PMCID: PMC7019700

14 DeLoughery T. G. (2017). Iron Deficiency Anemia. The Medical clinics of North America, 101(2), 319–332. https://doi.org/10.1016/j.mcna.2016.09.004

15 Al Alawi AM, Majoni SW, Falhammar H. Magnesium and Human Health: Perspectives and Research Directions. Int J Endocrinol. 2018 Apr 16;2018:9041694. doi: 10.1155/2018/9041694. PMID: 29849626; PMCID: PMC5926493

16 Ayers, J., Cook, J., Koenig, R. A., Sisson, E. M., & Dixon, D. L. (2018). Recent Developments in the Role of Coenzyme Q10 for Coronary Heart Disease: A Systematic Review. Current atherosclerosis reports, 20(6), 29. https://doi.org/10.1007/s11883-018-0730-1

Hair Loss Protocol

1 Hagenaars, S. P., Hill, W. D., Harris, S. E., Ritchie, S. J., Davies, G., Liewald, D. C., Gale, C. R., Porteous, D. J., Deary, I. J., & Marioni, R. E. (2017). Genetic prediction of male pattern baldness. *PLoS genetics*, 13(2), e1006594 https://doi.org/10.1371/journal.pgen.1006594

2 Li R, Grosschmidt FF, Kiefer AK, Stefansson H, Nyholt DR, Song K, Vermeulen SH, Kanoni S, Black D, Medland SE, Dimitriou M, Waterworth D, Tung JY, Geller F, Heilmann S, Hillmer AM, Bataille V, Eigelshoven S, Hanneken S, Moebus S, Herold C, den Heijer M, Montgomery GW, Deloukas P, Eriksson N, Heath AC, Becker T, Sulem P, Mangino M, Vollenweider P, Spector TD, Dedoussis G, Martin NG, Kiemeny LA, Mooser V, Stefansson K, Hinds DA, Nöthen MM, Richards JB. Six novel susceptibility Loci for early-onset androgenetic alopecia and their unexpected association with common diseases. *PLoS Genet*. 2012 May;8(5): e1002746. doi: 10.1371/journal.pgen.1002746. Epub 2012 May 31. PMID: 22693459; PMCID: PMC3364959

3 Cranwell W, Sinclair R. Male Androgenetic Alopecia. [Updated 2016 Feb 29]. In: Feingold KR, Anawalt B, Boyce A, et al., editors. Endotext [Internet]. South Dartmouth (MA); MDText.com, Inc.; 2000

4 Johnson, J. J., Bailey, H. H., & Mukhtar, H. (2010). Green tea polyphenols for prostate cancer chemoprevention: a translational perspective. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 17(1), 3–13. https://doi.org/10.1016/j.phymed.2009.09.011

5 Harvey C. J. (2020). Combined Diet and Supplementation Therapy Resolves Alopecia Areata in a Paediatric Patient: A Case Study. *Cureus*, 12(11), e11371. https://doi.org/10.7759/cureus.11371

6 Pham, C. T., Romero, K., Almhanna, H. M., Griggs, J., Ahmed, A., & Tosti, A. (2020). The Role of Diet as an Adjuvant Treatment in Scarring and Nonscarring Alopecia. *Skin appendage disorders*, 6(2), 88–96. https://doi.org/10.1159/000504786

7 Morinaga, H., Mohri, Y., Grachtchouk, M., Asakawa, K., Matsumura, H., Oshima, M., Takayama, N., Kato, T., Nishimori, Y., Sorimachi, Y., Takubo, K., Suganami, T., Iwama, A., Iwakura, Y., Dlugosz, A. A., & Nishimura, E. K. (2021). Obesity accelerates hair thinning by stem cell-centric converging mechanisms. *Nature*, 595(7866), 266–271. https://doi.org/10.1038/s41586-021-03624-x

8 David, L. A., Maurice, C. F., Carmody, R. N., Gootenberg, D. B., Button, J. E., Wolfe, B. E., Ling, A. V., Devlin, A. S., Varmuza, F., Fischbach, M. A., Biddinger, S. B., Dutton, R. J., & Turnbaugh, P. J. (2014). Diet rapidly and reproducibly alters the human gut microbiome. *Nature*, 505(7484), 559–563. https://doi.org/10.1038/nature12820

9 Duan, Y., Zeng, L., Zheng, C., Song, B., Li, F., Kong, X., & Xu, K. (2018). Inflammatory Links Between High Fat Diets and Diseases. *Frontiers in immunology*, 9, 2649. https://doi.org/10.3389/fimmu.2018.02649

10 Hasegawa, S., Ichijima, T., Sonaka, I., Ohsaki, A., Okada, S., Wakiguchi, H., Kudo, K., Kittaka, S., Hara, M., & Furukawa, S. (2012). Cysteine, histidine and glycine exhibit anti-inflammatory effects in human coronary arterial endothelial cells. *Clinical and experimental immunology*, 167(2), 269–274. https://doi.org/10.1111/j.1365-2249.2011.04519.x

11 Brzezińska-Wcisła L. (2001). Ocena skuteczności witaminy B6 i pantotenianu wapniowego na wzrost włosów w aspekcie klinicznym i trichologicznym w hysieniu rozlanym kobiet [Evaluation of vitamin B6 and calcium pantothenate effectiveness on hair growth from clinical and trichographic aspects for treatment of diffuse alopecia in women]. *Wiadomości lekarskie* (Warsaw, Poland: 1960), 54(1-2), 11–18.

12 Stamatiadis, D., Bulteau-Portois, M. C., & Mowszowicz, I. (1988). Inhibition of 5 alpha-reductase activity in human skin by zinc and azelaic acid. *The British journal of dermatology*, 119(5), 627–632. https://doi.org/10.1111/j.1365-2133.1988.tb03474.x

13 Kalkan, G., Yigit, S., Karakus, N., Ates, O., Bozkurt, N., Ozdemir, A., & Pancar, G. S. (2013). Methylenetetrahydrofolate reductase C677T mutation in patients with alopecia areata in Turkish population. *Gene*, 530(1), 109–112. https://doi.org/10.1016/j.gene.2013.08.016

14 Glynis A. (2012). A Double-blind, Placebo-controlled Study Evaluating the Efficacy of an Oral Supplement in Women with Self-perceived Thinning Hair. *The Journal of clinical and aesthetic dermatology*, 5(11), 28–34.

15 Varani, J., Dame, M. K., Rittie, L., Fligel, S. E., Kang, S., Fisher, G. J., & Voorhees, J. J. (2006). Decreased collagen production in chronologically aged skin: roles of age-dependent alteration in fibroblast function and defective mechanical stimulation. *The American*

journal of pathology, 168(6), 1861–1868. https://doi.org/10.2353/ajpath.2006.051302

16 Wu, G. (2009). Amino acids: metabolism, functions, and nutrition. *Amino acids*, 37(1), 1–17. https://doi.org/10.1007/s00726-009-0269-0

17 Meeraarani, P., Ramadass, P., Toborek, M., Bauer, H. C., Bauer, H., & Hennig, B. (2000). Zinc protects against apoptosis of endothelial cells induced by linoleic acid and tumor necrosis factor alpha. *The American journal of clinical nutrition*, 71(1), 81–87. https://doi.org/10.1093/ajcn/71.1.81

18 Beoy, L. A., Woei, W. J., & Hay, Y. K. (2010). Effects of tocotrienol supplementation on hair growth in human volunteers. *Tropical life sciences research*, 21(2), 91–99

19 Ablon G. (2015). A 3-month, randomized, double-blind, placebo-controlled study evaluating the ability of an extra-strength marine protein supplement to promote hair growth and decrease shedding in women with self-perceived thinning hair. *Dermatology research and practice*, 2015, 841570. https://doi.org/10.1155/2015/841570

20 Le Floc'h, C., Cheniti, A., Connétable, S., Piccardi, N., Vincenzi, C., & Tosti, A. (2015). Effect of a nutritional supplement on hair loss in women. *Journal of cosmetic dermatology*, 14(1), 76–82. https://doi.org/10.1111/jocd.12127

21 Kwon, Y. (2019). Use of saw palmetto (*Serenoa repens*) extract for benign prostatic hyperplasia. *Food science and biotechnology*, 28(6), 1599–1606. https://doi.org/10.1007/s10068-019-00605-9

22 Opoku-Acheampong, A. B., Penugonda, K., & Lindshield, B. L. (2016). Effect of Saw Palmetto Supplements on Androgen-Sensitive LNCaP Human Prostate Cancer Cell Number and Syrian Hamster Flank Organ Growth. *Evidence-based complementary and alternative medicine: eCAM*, 2016, 8135135. https://doi.org/10.1155/2016/8135135

23 Marks, L. S., Hess, D. L., Dorey, F. J., Luz Macairan, M., Cruz Santos, P. B., & Tyler, V. E. (2001). Tissue effects of saw palmetto and finasteride: use of biopsy cores for in situ quantification of prostatic androgens. *Urology*, 57(5), 999–1005. https://doi.org/10.1016/s0090-4295(00)01052-9

24 Cho, Y. H., Lee, S. Y., Jeong, D. W., Choi, E. J., Kim, Y. J., Lee, J. G., Yi, Y. H., & Cha, H. S. (2014). Effect of pumpkin seed oil on hair growth in men with androgenetic alopecia: a randomized, double-blind, placebo-controlled trial. *Evidence-based complementary and alternative medicine: eCAM*, 2014, 549721. https://doi.org/10.1155/2014/549721

25 Larré, S., Camparo, P., Comperat, E., Boulbés, D., Haddoum, M., Baulande, S., Soularue, P., Costa, P., & Cussenot, O. (2012). Biological effect of human serum collected before and after oral intake of Pygeum africanum on various benign prostate cell cultures. *Asian journal of andrology*, 14(3), 499–504. https://doi.org/10.1038/aja.2011.132

26 Auf'mkolk, M., Ingbar, J. C., Kubota, K., Amir, S. M., & Ingbar, S. H. (1985). Extracts and auto-oxidized constituents of certain plants inhibit the receptor-binding and the biological activity of Graves' immunoglobulins. *Endocrinology*, 116(5), 1687–1693. https://doi.org/10.1210/endo-116-5-1687

27 Johnson, T. A., Sohn, J., Inman, W. D., Bjeldanes, L. F., & Rayburn, K. (2013). Lipophilic stinging nettle extracts possess potent anti-inflammatory activity, are not cytotoxic and may be superior to traditional tinctures for treating inflammatory disorders. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 20(2), 143–147. https://doi.org/10.1016/j.phymed.2012.09.016

28 Dong, J. Z., Lei, C., Ai, X. R., & Wang, Y. (2012). Selenium enrichment on Cordyceps militaris link and analysis on its main active components. *Applied biochemistry and biotechnology*, 166(5), 1215–1224. https://doi.org/10.1007/s12010-011-9506-6

29 Gärtner, R., Gasnier, B. C., Dietrich, J. W., Krebs, B., & Angstwurm, M. W. (2002). Selenium supplementation in patients with autoimmune thyroiditis decreases thyroid peroxidase antibodies concentrations. *The Journal of clinical endocrinology and metabolism*, 87(4), 1687–1691. https://doi.org/10.1210/jcem.87.4.8421

30 Kaplan, D., & Dosiou, C. (2021). Two Cases of Graves' Hyperthyroidism Treated with Homeopathic Remedies Containing Herbal Extracts from *Lycopus* spp. and *Melissa officinalis*. *Journal of the Endocrine Society*, 5(Suppl 1), A971. https://doi.org/10.1210/jendso/bvab048.1984

31 Palacio, C., Masri, G., & Mooradian, A. D. (2009). Black cohosh for the management of menopausal symptoms: a systematic review of clinical trials. *Drugs & aging*, 26(1), 23–36. https://doi.org/10.2165/0002512-200926010-00002

32 van Die, M. D., Burger, H. G., Teede, H. J., & Bone, K. M. (2013). Vitex agnus-castus extracts for female reproductive disorders: a systematic review of clinical trials. *Planta medica*, 79(7), 562–575. https://doi.org/10.1055/s-0032-1327831

33 Atteia, H. H., Alzahrani, S., El-Sherbeen, N. A., Youssef, A. M., Farag, N. E., Mehanna, E. T., Elhawary, K., Ibrahim, G. A., Elmistekawy, A., & Zaitone, S. A. (2020). Evening Primrose Oil Ameliorates Hyperleptinemia and Reproductive Hormone Disturbances in Obese Female Rats: Impact on Estrus Cyclicity. *Frontiers in endocrinology*, 10, 942. https://doi.org/10.3389/fendo.2019.00942

34 Panahi, Y., Taghizadeh, M., Marzoni, E. T., & Sahebkar, A. (2015). Rosemary oil vs minoxidil 2% for the treatment of androgenetic alopecia: a randomized comparative trial. *Skinmed*, 13(1), 15–21.

35 Wachtel-Galor S, Yuen J, Buswell JA, et al. *Ganoderma lucidum* (Lingzhi or Reishi): A Medicinal Mushroom. In: Benzie IFF, Wachtel-Galor S, editors. *Herbal Medicine: Biomolecular and Clinical Aspects*. 2nd edition. Boca Raton (FL): CRC Press/Taylor & Francis; 2011. Chapter 9. Available from: https://www.ncbi.nlm.nih.gov/books/NBK92757/

36 Song, J., Wang, Y., Teng, M., Cai, G., Xu, H., Guo, H., Liu, Y., Wang, D., & Teng, L. (2015). Studies on the Antifatigue Activities of Cordyceps militaris Fruit Body Extract in Mouse Model. *Evidence-based complementary and alternative medicine: eCAM*, 2015, 174616. https://doi.org/10.1155/2015/174616

37 Benson, K. F., Stamets, P., Davis, R., Nally, R., Taylor, A., Slater, S., & Jensen, G. S. (2019). The mycelium of the Trametes versicolor (Turkey tail) mushroom and its fermented substrate each show potent and complementary immune activating properties in vitro. *BMC complementary and alternative medicine*, 19(1), 342. https://doi.org/10.1186/s12906-019-2681-7

38 Medic, G., Wille, M., & Hemels, M. E. (2017). Short- and long-term health

consequences of sleep disruption. *Nature and science of sleep*, 9, 151–161. <https://doi.org/10.2147/NSS.S134864>

- 39 Shinjiyo, N., Waddell, G., & Green, J. (2020). Valerian Root in Treating Sleep Problems and Associated Disorders-A Systematic Review and Meta-Analysis. *Journal of evidence-based integrative medicine*, 25, 2515690X20967323. <https://doi.org/10.1177/2515690X20967323>
- 40 Elsas, S. M., Rossi, D. J., Raber, J., White, G., Seeley, C. A., Gregory, W. L., Mohr, C., Pfankuch, T., & Soumyanath, A. (2010). Passiflora incarnata L. (Passionflower) extracts elicit GABA currents in hippocampal neurons in vitro, and show anxiogenic and anticonvulsant effects in vivo, varying with extraction method. *Phytomedicine: international journal of phytotherapy and phytopharmacology*, 17(12), 940–949. <https://doi.org/10.1016/j.phymed.2010.03.002>
- 41 Scholey, A., Gibbs, A., Neale, C., Perry, N., Ossoukhova, A., Bilog, V., Kras, M., Scholz, C., Sass, M., & Buchwald-Werner, S. (2014). Anti-stress effects of lemon balm-containing foods. *Nutrients*, 6(11), 4805–4821. <https://doi.org/10.3390/nu6114805>
- 42 Harber, V. J., & Sutton, J. R. (1984). Endorphins and exercise. *Sports medicine (Auckland, N.Z.)*, 1(2), 154–171. <https://doi.org/10.2165/00007256-198401020-00004>
- 43 Gawel, M. J., Park, D. M., Alagband-Zadeh, J., & Rose, F. C. (1979). Exercise and hormonal secretion. *Postgraduate medical journal*, 55(644), 373–376. <https://doi.org/10.1136/pgmj.55.644.373>
- 44 Choi, J., Jun, M., Lee, S., Oh, S. S., & Lee, W. S. (2017). The Association between Exercise and Androgenetic Alopecia: A Survey-Based Study. *Annals of dermatology*, 29(4), 513–516. <https://doi.org/10.5021/ad.2017.29.4.513>
- 45 Droste, S. K., Gesing, A., Ulbricht, S., Müller, M. B., Linthorst, A. C., & Reul, J. M. (2003). Effects of long-term voluntary exercise on the mouse hypothalamic-pituitary-adrenocortical axis. *Endocrinology*, 144(7), 3012–3023. <https://doi.org/10.1210/en.2003-0097>
- 46 Salmon P. (2001). Effects of physical exercise on anxiety, depression, and sensitivity to stress: a unifying theory. *Clinical psychology review*, 21(1), 33–61. [https://doi.org/10.1016/s0272-7358\(99\)00032-x](https://doi.org/10.1016/s0272-7358(99)00032-x)
- 47 Broman-Fulks, J. J., & Storey, K. M. (2008). Evaluation of a brief aerobic exercise intervention for high anxiety sensitivity. *Anxiety, stress, and coping*, 21(2), 117–128. <https://doi.org/10.1080/10615800701762675>
- 48 Smits, J. A., Berry, A. C., Rosenfield, D., Powers, M. B., Behar, E., & Otto, M. W. (2008). Reducing anxiety sensitivity with exercise. *Depression and anxiety*, 25(8), 689–699. <https://doi.org/10.1002/da.20411>
- 49 Ströhle, A., Graetz, B., Scheel, M., Wittmann, A., Feller, C., Heinz, A., & Dimeo, F. (2009). The acute antipanic and anxiolytic activity of aerobic exercise in patients with panic disorder and healthy control subjects. *Journal of psychiatric research*, 43(12), 1013–1017. <https://doi.org/10.1016/j.jpsychires.2009.02.004>
- 50 McWilliams, L. A., & Asmundson, G. J. (2001). Is there a negative association between anxiety sensitivity and arousal-increasing substances and activities?. *Journal of anxiety disorders*, 15(3), 161–170. [https://doi.org/10.1016/s0887-6185\(01\)00056-1](https://doi.org/10.1016/s0887-6185(01)00056-1)
- 51 Panahi Y, Taghizadeh M, Marzony ET, Sahebkar A. Rosemary oil vs minoxidil 2% for the treatment of androgenetic alopecia: a randomized comparative trial. *Skinmed*. 2015 Jan-Feb;13(1):15-21. PMID: 25842469. <https://europepmc.org/article/med/25842469>
- 52 Panahi, Y., Taghizadeh, M., Marzony, E. T., & Sahebkar, A. (2015). Rosemary oil vs minoxidil 2% for the treatment of androgenetic alopecia: a randomized comparative trial. *Skinmed*, 13(1), 15–21.
- 53 Gottumukkala, V. R., Annamalai, T., & Mukhopadhyay, T. (2011). Phytochemical investigation and hair growth studies on the rhizomes of Nardostachys jatamansi DC. *Pharmacognosy magazine*, 7(26), 146–150. <https://doi.org/10.4103/0973-1296.80674>
- 54 Araújo, L. A., Addor, F., & Campos, P. M. (2016). Use of silicon for skin and hair care: an approach of chemical forms available and efficacy. *Anais brasileiros de dermatologia*, 91(3), 331–335. <https://doi.org/10.1590/abd1806-4841.20163986>

Hemorrhoids Protocol

- 1 Moesgaard, F., Nielsen, M. L., Hansen, J. B., & Knudsen, J. T. (1982). High-fiber diet reduces bleeding and pain in patients with hemorrhoids: a double-blind trial of Vi-Siblin. *Diseases of the colon and rectum*, 25(5), 454–456. <https://doi.org/10.1007/BF02553653>
- 2 Mott, T., Latimer, K., & Edwards, C. (2018). Hemorrhoids: Diagnosis and Treatment Options. *American family physician*, 97(3), 172–179
- 3 Perez-Miranda, M., Gomez-Cedenilla, A., León-Colombo, T., Pajares, J., & Mate-Jimenez, J. (1996). Effect of fiber supplements on internal bleeding hemorrhoids. *Hepato-gastroenterology*, 43(12), 1504–1507
- 4 Garg, P., & Singh, P. (2017). Adequate dietary fiber supplement and TONE can help avoid surgery in most patients with advanced hemorrhoids. *Minerva gastroenterologica e dietologica*, 63(2), 92–96. <https://doi.org/10.23736/S1121-421X.17.02364-9>
- 5 Sheikh, P., Lohsiriwat, V., & Shelygin, Y. (2020). Micronized Purified Flavonoid Fraction in Hemorrhoid Disease: A Systematic Review and Meta-Analysis. *Advances in therapy*, 37(6), 2792–2812. <https://doi.org/10.1007/s12325-020-01353-7>
- 6 Cospite M. (1994). Double-blind, placebo-controlled evaluation of clinical activity and safety of Daflon 500 mg in the treatment of acute hemorrhoids. *Angiology*, 45(6 Pt 2), 566–573
- 7 Corsale, I., Carrieri, P., Martellucci, J., Piccolomini, A., Verre, L., Rigutini, M., & Panucci, S. (2018). Flavonoid mixture (diosmin, troxerutin, rutin, hesperidin, quercetin) in the treatment of I-III degree hemorrhoidal disease: a double-blind multicenter prospective comparative study. *International journal of colorectal disease*, 33(11), 1595–1600. <https://doi.org/10.1007/s00384-018-3102-y>
- 8 Giannini, I., Amato, A., Basso, L., Tricomi, N., Marranci, M., Pecorella, G., Tafuri, S., Pennisi, D., & Altomare, D. F. (2015). Flavonoids mixture (diosmin, troxerutin, hesperidin) in the treatment of acute hemorrhoidal disease: a prospective, randomized, triple-blind, controlled trial. *Techniques in coloproctology*, 19(6),

339–345. <https://doi.org/10.1007/s10151-015-1302-9>

- 9 Annoni, F., Boccasanta, P., Chiurazzi, D., Mozzi, E., & Oberhauser, V. (1986). Trattamento dei sintomi acuti della malattia emorroidaria con O-(beta-idrossietil)-rutosidea ad alte dosi per via orale [Treatment of acute symptoms of hemorrhoid disease with high-dose oral O-(beta-hydroxyethyl)-rutosides]. *Minerva medica*, 77(37), 1663–1668.
- 10 Chiaretti, M., Fegatelli, D. A., Pappalardo, G., Venti, M., & Chiaretti, A. I. (2020). Comparison of Centella with Flavonoids for Treatment of Symptoms in Hemorrhoidal Disease and After Surgical Intervention: A Randomized Clinical Trial. *Scientific reports*, 10(1), 8009. <https://doi.org/10.1038/s41598-020-64772-0>
- 11 Gravina, A. G., Pellegrino, R., Facchiano, A., Palladino, G., Loguercio, C., & Federico, A. (2021). Evaluation of the Efficacy and Safety of a Compound of Micronized Flavonoids in Combination With Vitamin C and Extracts of Centella asiatica, Vaccinium myrtillus, and Vitis vinifera for the Reduction of Hemorrhoidal Symptoms in Patients With Grade II and III Hemorrhoidal Disease: A Retrospective Real-Life Study. *Frontiers in pharmacology*, 12, 773320. <https://doi.org/10.3389/fphar.2021.773320>
- 12 Garg, P., & Singh, P. (2017). Adequate dietary fiber supplement and TONE can help avoid surgery in most patients with advanced hemorrhoids. *Minerva gastroenterologica e dietologica*, 63(2), 92–96. <https://doi.org/10.23736/S1121-421X.17.02364-9>
- 13 Kecmanović, D., Pavlov, M., Ceranić, M., Sepetkovski, A., Kovacević, P., Stamenković, A., Masirević, V., & Ranković, V. (2004). Plantago ovata (Laxomucil) posle hemoroidektomije [Plantago ovata (Laxomucil) after hemorrhoidectomy]. *Acta chirurgica Iugoslavica*, 51(3), 121–123. <https://doi.org/10.2298/aci0403121k>
- 14 Joksimovic, N., Spasovski, G., Joksimovic, V., Andreevski, V., Zuccari, C., & Omini, C. F. (2012). Efficacy and tolerability of hyaluronic acid, tea tree oil and methyl-sulfonyl-methane in a new gel medical device for treatment of haemorrhoids in a double-blind, placebo-controlled clinical trial. *Updates in surgery*, 64(3), 195–201. <https://doi.org/10.1007/s13304-012-0153-4>
- 15 Amatur, A., Meucci, M., & Mari, F. S. (2020). Treatment of haemorrhoidal disease with micronized purified flavonoid fraction and sucralfate ointment. *Acta bio-medica: Atenei Parmensis*, 91(1), 139–141. <https://doi.org/10.23750/abm.v91i1.9361>
- 16 MacKay D. (2001). Hemorrhoids and varicose veins: a review of treatment options. *Alternative medicine review: a journal of clinical therapeutic*, 6(2), 126–140.
- 17 Ahad, H.A., Kumar, C.S., Reddy, K.K., Kranthi, G., Mahesh, K., Teja, K.S., & Kumar, M.T. (2010). Herbal Treatment for Hemorrhoids. *Journal of Innovative Trends in Pharmaceutical Sciences*. 1(5): 236-244
- 18 Bylka, W., & Kornobis, J. (2005). Rusczyk kolczasty, w leczeniu chorób naczyń żylnych [Butcher's Broom, in the treatment of venous insufficiency]. *Polski merkuriusz lekarski : organ Polskiego Towarzystwa Lekarskiego*, 19(110), 234–236.
- 19 MacKay D. (2001). Hemorrhoids and varicose veins: a review of treatment options. *Alternative medicine review: a journal of clinical therapeutic*, 6(2), 126–140.
- 20 Chiaretti, M., Fegatelli, D. A., Pappalardo, G., Venti, M., & Chiaretti, A. I. (2020). Comparison of Centella with Flavonoids for Treatment of Symptoms in Hemorrhoidal Disease and After Surgical Intervention: A Randomized Clinical Trial. *Scientific reports*, 10(1), 8009. <https://doi.org/10.1038/s41598-020-64772-0>
- 21 Senatore, F., Mścisz, A., Mrugasiewicz, K., & Gorecki, P. (1989). Steroidal constituents and anti-inflammatory activity of the horse chestnut (Aesculus hippocastanum L.) bark. *Bollettino della Società italiana di biologia sperimentale*, 65(2), 137–141.
- 22 Suter, A., Bommer, S., & Rechner, J. (2006). Treatment of patients with venous insufficiency with fresh plant horse chestnut seed extract: a review of 5 clinical studies. *Advances in therapy*, 23(1), 179–190. <https://doi.org/10.1007/BF02850359>
- 23 Sirtori C. R. (2001). Aescin: pharmacology, pharmacokinetics and therapeutic profile. *Pharmacological research*, 44(3), 183–193. <https://doi.org/10.1006/phrs.2001.0847>
- 24 Gravina, A. G., Pellegrino, R., Facchiano, A., Palladino, G., Loguercio, C., & Federico, A. (2021). Evaluation of the Efficacy and Safety of a Compound of Micronized Flavonoids in Combination With Vitamin C and Extracts of Centella asiatica, Vaccinium myrtillus, and Vitis vinifera for the Reduction of Hemorrhoidal Symptoms in Patients With Grade II and III Hemorrhoidal Disease: A Retrospective Real-Life Study. *Frontiers in pharmacology*, 12, 773320. <https://doi.org/10.3389/fphar.2021.773320>
- 25 Mahboubi M. (2017). Effectiveness of Myrtus communis in the treatment of hemorrhoids. *Journal of integrative medicine*, 15(5), 351–358. [https://doi.org/10.1016/S2095-4964\(17\)60340-6](https://doi.org/10.1016/S2095-4964(17)60340-6)
- 26 Panahi, Y., Mousavi-Nayeeni, S.M., Sahebkar, A., Rahimnia, A., Beiraghdar, F., Sol-Tanbakhs, M., Imani, S. (2014). Myrtus communis essential oil for the treatment of hemorrhoids: a randomized double-blind double-dummy parallel-group comparative study. *Turkish journal of pharmaceutical sciences*, 11(1):1-8
- 27 Neamsuvan, O., & Ruangrit, T. (2017). A survey of herbal weeds that are used to treat gastrointestinal disorders from southern Thailand: Krabi and Songkhla provinces. *Journal of ethnopharmacology*, 196, 84–93. <https://doi.org/10.1016/j.jep.2016.11.033>
- 28 Romm, A. (2017). Botanical medicine for Women's health E-book. Elsevier Health Sciences.
- 29 InformedHealth.org [Internet]. Cologne, Germany: Institute for Quality and Efficiency in Health Care (IQWiG); 2006-. Enlarged hemorrhoids: How can you relieve the symptoms yourself? 2014 Jan 29 [Updated 2017 Nov 2]. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK279466/>
- 30 Akram M. (2013). Minireview on Achillea millefolium Linn. *The Journal of membrane biology*, 246(9), 661–663. <https://doi.org/10.1007/s00232-013-9588-x>
- 31 Hashempur, M. H., Khademi, F., Rahmanifard, M., & Zarshenas, M. M. (2017). An Evidence-Based Study on Medicinal Plants for Hemorrhoids in Medieval Persia. *Journal of evidence-based complementary & alternative medicine*, 22(4), 969–981. <https://doi.org/10.1177/2156587216688597>
- 32 Shafik A. (1993). Role of warm-water bath in anorectal conditions. The “thermo-sphincteric reflex”. *Journal of clinical gastroenterology*, 16(4), 304–308. <https://doi.org/10.1177/2156587216688597>

Sleep Protocol

- 1 Kanagasabai, T., & Ardern, C. I. (2015). Contribution of Inflammation, Oxidative Stress, and Antioxidants to the Relationship between Sleep Duration and Cardiometabolic Health. *Sleep*, 38(12), 1905–1912. <https://doi.org/10.5665/sleep.5238>
- 2 Rasch, B., & Born, J. (2013). About sleep's role in memory. *Physiological reviews*, 93(2), 681–766. <https://doi.org/10.1152/physrev.00032.2012>
- 3 Pauer, M. J., Kuchenbecker, J. A., Neitz, M., & Neitz, J. (2012). Changes in the colour of light cue circadian activity. *Animal behaviour*, 83(5), 1143–1151. <https://doi.org/10.1016/j.anbehav.2012.01.035>
- 4 University of Washington Health Sciences/UW Medicine. (2020, February 20). Let there be 'circadian' light: New study describes science behind best lights to affect sleep, mood and learning. *ScienceDaily*. Retrieved September 4, 2022 from www.sciencedaily.com/releases/2020/02/202020141731.htm
- 5 Taranto-Montemurro, L., Messineo, L., Sands, S., Azarbarzin, A., Marques, M., & Wellman, A. (2017). 0394 Effect of background noise on sleep quality. *Sleep*, 40(suppl_1): A146–A147. <https://doi.org/10.1093/sleep/zsx050.393>
- 6 Lunsford-Avery, J.R., Engelhard, M.M., Navar, A.M., & Kollins, S.H. (2018). Validation of the Sleep Regularity Index in Older Adults and Associations with Cardiometabolic Risk. *Scientific Reports*. 8:14158. <https://doi.org/10.1038/s41598-018-32402-5>
- 7 Wehrens, S., Christou, S., Isherwood, C., Middleton, B., Gibbs, M. A., Archer, S. N., Skene, D. J., & Johnston, J. D. (2017). Meal Timing Regulates the Human Circadian System. *Current biology: CB*, 27(12), 1768–1775.e3. <https://doi.org/10.1016/j.cub.2017.04.059>
- 8 Rubio-Arias, J. A., Marín-Cascales, E., Ramos-Campo, D. J., Hernandez, A. V., & Pérez-López, F. R. (2017). Effect of exercise on sleep quality and insomnia in middle-aged women: A systematic review and meta-analysis of randomized controlled trials. *Maturitas*, 100, 49–56. <https://doi.org/10.1016/j.maturitas.2017.04.003>
- 9 Siu, P. M., Yu, A. P., Tam, B. T., Chin, E. C., Yu, D. S., Chung, K. F., Hui, S. S., Woo, J., Fong, D. Y., Lee, P. H., Wei, G. X., & Irwin, M. R. (2021). Effects of Tai Chi or Exercise on Sleep in Older Adults With Insomnia: A Randomized Clinical Trial. *JAMA network open*, 4(2), e2037199. <https://doi.org/10.1001/jamanetworkopen.2020.37199>
- 10 D'Aurea, C., Poyares, D., Passos, G. S., Santana, M. G., Youngstedt, S. D., Souza, A. A., Bícudo, J., Tufik, S., & de Mello, M. T. (2019). Effects of resistance exercise training and stretching on chronic insomnia. *Revista brasileira de psiquiatria (Sao Paulo, Brazil: 1999)*, 41(1), 51–57. <https://doi.org/10.1590/1516-4446-2018-0030>
- 11 Reid, K. J., Baron, K. G., Lu, B., Naylor, E., Wolfe, L., & Zee, P. C. (2010). Aerobic exercise improves self-reported sleep and quality of life in older adults with insomnia. *Sleep medicine*, 11(9), 934–940. <https://doi.org/10.1016/j.sleep.2010.04.014>
- 12 Lindahl, O., & Lindwall, L. (1989). Double blind study of a valerian preparation. *Pharmacology, biochemistry, and behavior*, 32(4), 1065–1066. [https://doi.org/10.1016/0091-3057\(89\)90082-8](https://doi.org/10.1016/0091-3057(89)90082-8)
- 13 Salter, S., & Brownie, S. (2010). Treating primary insomnia - the efficacy of valerian and hops. *Australian family physician*, 39(6), 433–437.
- 14 <https://thesleepdoctor.com/sleep-aids/>
- 15 Abdellah, S. A., Berlin, A., Blondeau, C., Guinobert, I., Guilbot, A., Beck, M., & Duforez, F. (2019). A combination of Eschscholtzia californica Cham. and Valeriana officinalis L. extracts for adjustment insomnia: A prospective observational study. *Journal of traditional and complementary medicine*, 10(2), 116–123. <https://doi.org/10.1016/j.jtcm.2019.02.003>
- 16 Abdullhazadeh, M., Matourypour, P., & Naji, S. A. (2017). Investigation effect of oral chamomilla on sleep quality in elderly people in Isfahan: A randomized control trial. *Journal of education and health promotion*, 6, 53. https://doi.org/10.4103/jehp.jehp_109_15
- 17 Ngan, A., & Conduit, R. (2011). A double-blind, placebo-controlled investigation of the effects of Passiflora incarnata (passiflora) herbal tea on subjective sleep quality. *Phytotherapy research: PTR*, 25(8), 1153–1159. <https://doi.org/10.1002/ptr.3400>
- 18 Maroo, N., Hazra, A., & Das, T. (2013). Efficacy and safety of a polyherbal sedative-hypnotic formulation NSF-3 in primary insomnia in comparison to zolpidem: a randomized controlled trial. *Indian journal of pharmacology*, 45(1), 34–39. <https://doi.org/10.4103/0253-7613.106432>
- 19 Sarris, J., Panossian, A., Schweitzer, I., Stough, C., & Scholey, A. (2011). Herbal medicine for depression, anxiety and insomnia: a review of psychopharmacology and clinical evidence. *European neuropsychopharmacology: the journal of the European College of Neuropsychopharmacology*, 21(12), 841–860. <https://doi.org/10.1016/j.euroneuro.2011.04.002>
- 20 Savage, K., Firth, J., Stough, C., & Sarris, J. (2018). GABA-modulating phytochemicals for anxiety: A systematic review of preclinical and clinical evidence. *Phytotherapy research: PTR*, 32(1), 3–18. <https://doi.org/10.1002/ptr.5940>
- 21 Cases, J., Ibarra, A., Feuillère, N., Roller, M., & Sukkar, S. G. (2011). Pilot trial of Melissa officinalis L. leaf extract in the treatment of volunteers suffering from mild-to-moderate anxiety disorders and sleep disturbances. *Mediterranean journal of nutrition and metabolism*, 4(3), 211–218. <https://doi.org/10.1007/s12349-010-0045-4>
- 22 Ranjbar, M., Firoozabadi, A., Salehi, A., Ghorbanifar, Z., Zarshenas, M. M., Sadegh-niaat-Haghighi, K., & Rezaeizadeh, H. (2018). Effects of Herbal combination (Melissa officinalis L. and Nepeta menthoides Boiss. & Buhse) on insomnia severity, anxiety and depression in insomniacs: Randomized placebo controlled trial. *Integrative medicine research*, 7(4), 328–332. <https://doi.org/10.1016/j.imr.2018.08.001>
- 23 Yao, C., Wang, Z., Jiang, H., Yan, R., Huang, Q., Wang, Y., Xie, H., Zou, Y., Yu, Y., & Lv, L. (2021). Ganoderma lucidum promotes sleep through a gut microbiota-dependent and serotonin-involved pathway in mice. *Scientific reports*, 11(1), 13660. <https://doi.org/10.1038/s41598-021-92913-6>
- 24 Shergis, J. L., Hyde, A., Meaklim, H., Varma, P., Da Costa, C., & Jackson, M. L. (2021). Medicinal seeds Ziziphus spinosa for insomnia: A randomized, placebo-controlled, cross-over, feasibility clinical trial. *Complementary therapies in medicine*, 57, 102657. <https://doi.org/10.1016/j.ctim.2020.102657>
- 25 Cao, Y., Zhen, S., Taylor, A. W., Appleton, S., Atlantis, E., & Shi, Z. (2018). Magnesium Intake and Sleep Disorder Symptoms: Findings from the Jiangsu Nutrition Study of Chinese Adults at Five-Year Follow-Up. *Nutrients*, 10(10), 1354. <https://doi.org/10.3390/nu10101354>
- 26 Abbasi, B., Kimiagar, M., Sadeghniaat, K., Shirazi, M. M., Hedayati, M., & Rashidkhani, B. (2012). The effect of magnesium supplementation on primary insomnia in elderly: A double-blind placebo-controlled clinical trial. *Journal of research in medical sciences: the official journal of Isfahan University of Medical Sciences*, 17(12), 1161–1169.
- 27 Mah, J., & Pitre, T. (2021). Oral magnesium supplementation for insomnia in older adults: a Systematic Review & Meta-Analysis. *BMC complementary medicine and therapies*, 21(1), 125. <https://doi.org/10.1186/s12906-021-03297-z>
- 28 Gandolfi, J. V., Di Bernardo, A., Chanes, D., Martin, D. F., Joles, V. B., Amendola, C. P., Sanches, L. C., Ciorlia, G. L., & Lobo, S. M. (2020). The Effects of Melatonin Supplementation on Sleep Quality and Assessment of the Serum Melatonin in ICU Patients: A Randomized Controlled Trial. *Critical care medicine*, 48(12), e1286–e1293. <https://doi.org/10.1097/CCM.0000000000004690>
- 29 Janjua, I., & Goldman, R. D. (2016). Sleep-related melatonin use in healthy children. *Canadian family physician Medecin de famille canadien*, 62(4), 315–317.
- 30 Riha R. L. (2018). The use and misuse of exogenous melatonin in the treatment of sleep disorders. *Current opinion in pulmonary medicine*, 24(6), 543–548. <https://doi.org/10.1097/MCP.0000000000000522>
- 31 Urrila, A. S., Hakkarainen, A., Castaneda, A., Paunio, T., Marttunen, M., & Lundbom, N. (2017). Frontal Cortex Myo-Inositol Is Associated with Sleep and Depression in Adolescents: A Proton Magnetic Resonance Spectroscopy Study. *Neuropsychobiology*, 75(1), 21–31. <https://doi.org/10.1159/000478861>
- 32 Mashayekh-Amiri, S., Delavar, M. A., Bakouei, F., Faramarzi, M., & Esmaeilzadeh, S. (2022). The impact of myo-inositol supplementation on sleep quality in pregnant women: a randomized, double-blind, placebo-controlled study. *The journal of maternal-fetal & neonatal medicine: the official journal of the European Association of Perinatal Medicine, the Federation of Asia and Oceania Perinatal Societies, the International Society of Perinatal Obstetricians*, 35(18), 3415–3423. <https://doi.org/10.1080/14767058.2020.1818225>
- 33 Kanagasabai, T., & Ardern, C. I. (2015). Contribution of Inflammation, Oxidative Stress, and Antioxidants to the Relationship between Sleep Duration and Cardiometabolic Health. *Sleep*, 38(12), 1905–1912. <https://doi.org/10.5665/sleep.5238>
- 34 Otocka-Kmiecik, A., & Król, A. (2020). The Role of Vitamin C in Two Distinct Physiological States: Physical Activity and Sleep. *Nutrients*, 12(12), 3908. <https://doi.org/10.3390/nu12123908>
- 35 Grandner, M. A., Jackson, N., Gerstner, J. R., & Knutson, K. L. (2014). Sleep symptoms associated with intake of specific dietary nutrients. *Journal of sleep research*, 23(1), 22–34. <https://doi.org/10.1111/jsr.12084>
- 36 Takahashi H., Mizuno H., Yanagisawa A. High-dose intravenous vitamin C improves quality of life in cancer patients. *Pers. Med. Universe*. 2012;1: 49–53. doi: 10.1016/j.pmu.2012.05.008.
- 37 Aspy, D. J., Madden, N. A., & Delfabbro, P. (2018). Effects of Vitamin B6 (Pyridoxine) and a B Complex Preparation on Dreaming and Sleep. *Perceptual and motor skills*, 125(3), 451–462. <https://doi.org/10.1177/0031512518770326>
- 38 Bravaccio, C., Terrone, G., Rizzo, R., Gulisano, M., Tosi, M., Curatolo, P., & Emberti Gialloreti, L. (2020). Use of nutritional supplements based on melatonin, tryptophan and vitamin B6 (Melamil Tripto®) in children with primary chronic headache, with or without sleep disorders: a pilot study. *Minerva pediatrica*, 72(1), 30–36. <https://doi.org/10.23736/S0026-4946.19.05533-6>
- 39 Djokic, G., Vojvodic, P., Korcok, D., Agic, A., Rankovic, A., Djordjevic, V., Vojvodic, A., Vlasovic-Jovicic, T., Peric-Hajzler, Z., Matovic, D., Vojvodic, J., Sijan, G., Wollina, U., Tirant, M., Thuong, N. V., Fioranelli, M., & Lotti, T. (2019). The Effects of Magnesium - Melatonin - Vit B Complex Supplementation in Treatment of Insomnia. *Open access Macedonian journal of medical sciences*, 7(18), 3101–3105. <https://doi.org/10.3889/oamjms.2019.771>
- 40 Kim, S., Jo, K., Hong, K. B., Han, S. H., & Suh, H. J. (2019). GABA and l-theanine mixture decreases sleep latency and improves NREM sleep. *Pharmaceutical biology*, 57(1), 65–73. <https://doi.org/10.1080/13880209.2018.1557698>
- 41 Bravaccio, C., Terrone, G., Rizzo, R., Gulisano, M., Tosi, M., Curatolo, P., & Emberti Gialloreti, L. (2020). Use of nutritional supplements based on melatonin, tryptophan and vitamin B6 (Melamil Tripto®) in children with primary chronic headache, with or without sleep disorders: a pilot study. *Minerva pediatrica*, 72(1), 30–36. <https://doi.org/10.23736/S0026-4946.19.05533-6>
- 42 Shannon, S., Lewis, N., Lee, H., & Hughes, S. (2019). Cannabidiol in Anxiety and Sleep: A Large Case Series. *The Permanente journal*, 23, 18–041. <https://doi.org/10.7812/TPP/18-041>
- 43 Dos Reis Lucena, L., Dos Santos-Junior, J. G., Tufik, S., & Hachul, H. (2021). Lavender essential oil on postmenopausal women with insomnia: Double-blind randomized trial. *Complementary therapies in medicine*, 59, 102726. <https://doi.org/10.1016/j.ctim.2021.102726>
- 44 Nasiri Lari, Z., Hajimonfarednejad, M., Riasatian, M., Abolhassanzadeh, Z., Iraj, A., Vojoud, M., Heydari, M., & Shams, M. (2020). Efficacy of inhaled Lavandula angustifolia Mill. Essential oil on sleep quality, quality of life and metabolic control in patients with diabetes mellitus type II and insomnia. *Journal of ethnopharmacology*, 251, 112560. <https://doi.org/10.1016/j.jep.2020.112560>
- 45 Zhong, Y., Zheng, Q., Hu, P., Huang, X., Yang, M., Ren, G., Du, Q., Luo, J., Zhang, K., Li, J., Wu, H., Guo, Y., & Liu, S. (2019). Sedative and hypnotic effects of compound

Anshen essential oil inhalation for insomnia. *BMC complementary and alternative medicine*, 19(1), 306. <https://doi.org/10.1186/s12906-019-2732-0>

46 Ebbens, M. M., & Verster, J. C. (2010). Clinical evaluation of zaleplon in the treatment of insomnia. *Nature and science of sleep*, 2, 115–126. <https://doi.org/10.2147/nss.s6853>

47 Dang, A., Garg, A., & Rataboli, P. V. (2011). Role of zolpidem in the management of insomnia. *CNS neuroscience & therapeutics*, 17(5), 387–397. <https://doi.org/10.1111/j.1755-5949.2010.00158.x>

Weight Loss Protocol

1 Ranade, Manjiri, and Nikhil Mudgalkar. “A simple dietary addition of fenugreek seed leads to the reduction in blood glucose levels: A parallel group, randomized single-blind trial.” *Ayu* vol. 38,1-2 (2017): 24-27. doi:10.4103/ayu.AYU_209_15 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5954247/>

2 Mathern, Jocelyn R et al. “Effect of fenugreek fiber on satiety, blood glucose and insulin response and energy intake in obese subjects.” *Phytotherapy research: PTR* vol. 23,11 (2009): 1543-8. doi:10.1002/ptr.2795 <https://pubmed.ncbi.nlm.nih.gov/19353539/#>

3 Bae, JiYoung et al. “Fennel (*Foeniculum vulgare*) and Fenugreek (*Trigonella foenum-graecum*) Tea Drinking Suppresses Subjective Short-term Appetite in Over-weight Women.” *Clinical nutrition research* vol. 4,3 (2015): 168-74. doi:10.7762/cnr.2015.4.3.168 <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4525133/>

4 Chevassus, Hugues et al. “A fenugreek seed extract selectively reduces spontaneous fat consumption in healthy volunteers.” *European journal of clinical pharmacology* vol. 65,12 (2009): 1175-8. doi:10.1007/s00228-009-0733-5 <https://pubmed.ncbi.nlm.nih.gov/19809809/>

5 Hiwatashi K, Kosaka Y, Suzuki N, Hata K, Mukaiyama T, Sakamoto K, Shirakawa H, Komai M. Yamabushitake mushroom (*Hericium erinaceus*) improved lipid metabolism in mice fed a high-fat diet. *Biosci Biotechnol Biochem*. 2010;74(7):1447-51

6 Chen Diling, Guo Yinrui, Qi Longkai, Tang Xiaocui, Liu Yadi, Feng Jiaxin, Zhu Xiangxiang, Zeng Miao, Shuai Ou, Wang Dongdong, Xie Yizhen, Burton B. Yang, Wu Qingping. Metabolic regulation of *Ganoderma lucidum* extracts in high sugar and fat diet-induced obese mice by regulating the gut-brain axis. *Journal of Functional Foods*. 2020,65:103639

7 Gao, Yihuai & Lan, Jin & Dai, Xihu & Ye, Jingxian & Zhou, Shufeng. A Phase I/II Study of Ling Zhi Mushroom *Ganoderma lucidum* (W.Curt.:Fr.)Lloyd (Aphyllphoromycetideae) Extract in Patients with Type II Diabetes Mellitus. *International Journal of Medicinal Mushrooms*. 2004;6. 33-40

8 Chang CJ, Lin CS, Lu CC, Martel J, Ko YF, Ojcius DM, Tseng SF, Wu TR, Chen YY, Young JD, Lai HC. *Ganoderma lucidum* reduces obesity in mice by modulating the composition of the gut microbiota. *Nat Commun*. 2015 Jun 23;6:7489. doi: 10.1038/ncomms8489. Erratum in: *Nat Commun*. 2017 Jul 11;8:16130. PMID: 26102296; PMCID: PMC4557287

9 Verpeut, Jessica L et al. “Citrus aurantium and *Rhodiola rosea* in combination reduce visceral white adipose tissue and increase hypothalamic norepinephrine in a rat model of diet-induced obesity.” *Nutrition research (New York, N.Y.)* vol. 33,6 (2013): 503-12. doi:10.1016/j.nutres.2013.04.001

10 Panossian, Alexander et al. “The adaptogens *rhodiola* and *schizandra* modify the response to immobilization stress in rabbits by suppressing the increase of phosphorylated stress-activated protein kinase, nitric oxide and cortisol.” *Drug target insights* vol. 2 (2007): 39-54. <https://pubmed.ncbi.nlm.nih.gov/21901061/>

11 <https://www.healthline.com/health-news/interval-workouts-will-help-you-lose-weight-more-quickly>

12 Sharma S, Agrawal RP, Choudhary M, Jain S, Goyal S, Agarwal V. Beneficial effect of chromium supplementation on glucose, HbA1C and lipid variables in individuals with newly onset type-2 diabetes. *J Trace Elem Med Biol*. 2011;25(3):149-153. doi:10.1016/j.jtemb.2011.03.003

13 Pattar GR, Tackett L, Liu P, Elmendorf JS. Chromium picolinate positively influences the glucose transporter system via affecting cholesterol homeostasis in adipocytes cultured under hyperglycemic diabetic conditions. *Mutat Res*. 2006;610(1-2):93-100. doi:10.1016/j.mrgentox.2006.06.018

14 Askari, M., Mozaffari, H., Jafari, A., Ghanbari, M., & Darooghegi Mofrad, M. (2021). The effects of magnesium supplementation on obesity measures in adults: a systematic review and dose-response meta-analysis of randomized controlled trials. *Critical reviews in food science and nutrition*, 61(17), 2921–2937. <https://doi.org/10.1080/10408398.2020.1790498>

15 Sharma VK, Trakroo M, Subramaniam V, Rajajeyakumar M, Bhavanani AB, Sahai A. Effect of fast and slow pranayama on perceived stress and cardiovascular parameters in young health-care students. *Int J Yoga*. 2013 Jul;6(2):104-10. doi: 10.4103/0973-6131.113400. PMID: 23930028; PMCID: PMC3734635

16 <https://chopra.com/articles/release-toxins-with-kapalabhati-breath>

Anti-Inflammatory Diet

1 Yates, T, Khunti, K, Wilmot, E. G., Brady, E., Webb, D., Srinivasan, B., Henson, J, Talbot, D., & Davies, M. J. (2012). Self-reported sitting time and markers of inflammation, insulin resistance, and adiposity. *American Journal of Preventive Medicine*, 42(1), 1–7. <https://doi.org/10.1016/j.amepre.2011.09.022>

2 Ries, J. (2021, November 10). An anti-inflammatory diet high may decrease your dementia risk. Healthline. Retrieved June 2, 2022, from <https://www.healthline.com/health-news/an-anti-inflammatory-diet-high-in-veggies-may-decrease-your-dementia-risk>

3 Tedeschi, S. K., Frits, M., Cui, J., Zhang, Z. Z., Mahmoud, T., Iannaccone, C., Lin, T. C., Yoshida, K., Weinblatt, M. E., Shadick, N. A., & Solomon, D. H. (2017). Diet and Rheumatoid Arthritis Symptoms: Survey Results from a Rheumatoid Arthritis Registry. *Arthritis care & research*, 69(12), 1920–1925. <https://doi.org/10.1002/acr.23225>

4 Pedersen B. K. (2006). The anti-inflammatory effect of exercise: its role in dia-

betes and cardiovascular disease control. *Essays in biochemistry*, 42, 105–117. <https://doi.org/10.1042/bse0420105>

5 Coussens, L. M., & Werb, Z. (2002). Inflammation and cancer. *Nature*, 420(6917), 860–867. <https://doi.org/10.1038/nature01322>

6 Kolb, H., & Mandrup-Poulsen, T. (2010). The global diabetes epidemic as a consequence of lifestyle-induced low-grade inflammation. *Diabetologia*, 53(1), 10–20. <https://doi.org/10.1007/s00125-009-1573-7> Tilg, H., & Moschen, A. R. (2010). Evolution of inflammation in nonalcoholic fatty liver disease: the multiple parallel hits hypothesis. *Hepatology* (Baltimore, Md.), 52(5), 1836–1846. <https://doi.org/10.1002/hep.24001>

7 Kaluza, J., Håkansson, N., Harris, H. R., Orsini, N., Michaëlsson, K., & Wolk, A. (2019). Influence of anti-inflammatory diet and smoking on mortality and survival in men and women: two prospective cohort studies. *Journal of internal medicine*, 285(1), 75–91. <https://doi.org/10.1111/joim.12823>

8 Szeto, Y. T., Kwok, T. C., & Benzie, I. F. (2004). Effects of a long-term vegetarian diet on biomarkers of antioxidant status and cardiovascular disease risk. *Nutrition* (Burbank, Los Angeles County, Calif.), 20(10), 863–866. <https://doi.org/10.1016/j.nut.2004.06.006>

9 Casas, R., Sacanella, E., Urpí-Sardà, M., Chiva-Blanch, G., Ros, E., Martínez-González, M. A., Covas, M. I., Rosa Ma Lamuela-Raventos, Salas-Salvadó, J., Fiol, M., Arós, F., & Estruch, R. (2014). The effects of the mediterranean diet on biomarkers of vascular wall inflammation and plaque vulnerability in subjects with high risk for cardiovascular disease. A randomized trial. *PLoS one*, 9(6), e100084. <https://doi.org/10.1371/journal.pone.0100084>

10 Zhong, Z., Wheeler, M. D., Li, X., Froh, M., Schemmer, P., Yin, M., Bunzendahl, H., Bradford, B., & Lemasters, J. J. (2003). L-Glycine: a novel antiinflammatory, immunomodulatory, and cytoprotective agent. *Current opinion in clinical nutrition and metabolic care*, 6(2), 229–240. <https://doi.org/10.1097/00075197-200303000-00013>

11 Hasegawa S, Ichiyama T, Sonaka I, Ohsaki A, Okada S, Wakiguchi H, Kudo K, Kittaka S, Hara M, Furukawa S. Cysteine, histidine and glycine exhibit anti-inflammatory effects in human coronary arterial endothelial cells. *Clin Exp Immunol*. 2012 Feb;167(2):269-74. doi: 10.1111/j.1365-2249.2011.04519.x. PMID: 22236003; PMCID: PMC3278693

12 Hochberg, M. C., Martel-Pelletier, J., Monfort, J., Möller, I., Castillo, J. R., Arden, N., Berenbaum, F., Blanco, F. J., Conaghan, P. G., Doménech, G., Henrotin, Y., Pap, T., Richette, P., Sawitzke, A., du Souich, P., Pelletier, J. P., & MOVES Investigation Group (2016). Combined chondroitin sulfate and glucosamine for painful knee osteoarthritis: a multicentre, randomised, double-blind, non-inferiority trial versus celecoxib. *Annals of the rheumatic diseases*, 75(1), 37–44. <https://doi.org/10.1136/annrheumdis-2014-206792>

13 De Stefanis, D., Mastrocola, R., Nigro, D., Costelli, P., & Aragno, M. (2015). Effects of chronic sugar consumption on lipid accumulation and autophagy in the skeletal muscle. *European Journal of Nutrition*, 56(1), 363–373. <https://doi.org/10.1007/s00394-015-1086-8>

14 Hammerling, U., Bergman Laurila, J., Grafström, R., & Ilbäck, N. G. (2016). Consumption of Red/Processed Meat and Colorectal Carcinoma: Possible Mechanisms Underlying the Significant Association. *Critical reviews in food science and nutrition*, 56(4), 614–634. <https://doi.org/10.1080/10408398.2014.972498>

15 Stanhope, K. L., & Havel, P. J. (2009). Fructose consumption: considerations for future research on its effects on adipose distribution, lipid metabolism, and insulin sensitivity in humans. *The Journal of nutrition*, 139(6), 1236S–1241S. <https://doi.org/10.3945/jn.109.106641>

16 Nestel, P. (2014). Trans fatty acids: Are its cardiovascular risks fully appreciated? *Clinical Therapeutics*, 36(3), 315–321. <https://doi.org/10.1016/j.clinthera.2014.01.020>

17 Do dietary lectins cause disease? The evidence is suggestive—and raises interesting possibilities for treatment, David L J Freed, *Allergist, BMJ*. 1999 Apr 17; 318(7190): 1023–1024. <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC1115436/>

18 Gundry, S. R. (2020). Abstract P219: The Vast Majority of People Who Eat “Gluten Free” for IBS, Celiac, or Autoimmune Disease Have Markers of Leaky Gut That Resolve When “Gluten Free” Foods Containing Lectins, Like Corn, Other Grains, Beans, and Night-shades are Removed from Their Diet. *Circulation*, 141(Suppl_1), AP219-AP219

19 Wahls, T. L., Chenard, C. A., & Snetselaar, L. G. (2019). Review of two popular eating plans within the multiple sclerosis community: low saturated fat and modified paleolithic. *Nutrients*, 11(2), 352

20 Mullington, J. M., Simpson, N. S., Meier-Ewert, H. K., & Haack, M. (2010). Sleep loss and inflammation. *Best practice & research. Clinical endocrinology & metabolism*, 24(5), 775–784. <https://doi.org/10.1016/j.beem.2010.08.014>

21 Pedersen B. K. (2006). The anti-inflammatory effect of exercise: its role in diabetes and cardiovascular disease control. *Essays in biochemistry*, 42, 105–117. <https://doi.org/10.1042/bse0420105>

22 Yan Q. (2018). Stress and Systemic Inflammation: Yin-Yang Dynamics in Health and Diseases. *Methods in molecular biology* (Clifton, N.J.), 1781, 3–20. https://doi.org/10.1007/978-1-4939-7828-1_1

Elimination Diet

1 Wood, R. A. (2015). Diagnostic elimination diets and oral food provocation. *Food Allergy: Molecular Basis and Clinical Practice*, 87–95. <https://doi.org/10.1159/000371680>

2 Tuck, C. J., Biesiekierski, J. R., Schmid-Grendelmeier, P., & Pohl, D. (2019). Food intolerances. *Nutrients*, 11(7), 1684. <https://doi.org/10.3390/nu11071684>

3 Adam Rindfleisch, J. (2012). Food intolerance and elimination diet. *Integrative Medicine*. <https://doi.org/10.1016/b978-1-4377-1793-8.00068-6>

4 Raman, R. (2017, July 2). *How to do an elimination diet and why*. Healthline. Retrieved September 6, 2022, from https://www.healthline.com/nutrition/elimination-diet#TOC_TITLE_HDR_2

5 Swain, A. R., Soutter, V. L., & Loblay, R. H. (2011). *Rpah Elimination Diet Handbook: With Food & Shopping Guide*. Allergy Unit, Royal Prince Alfred Hospital

- 6 Kanagaratham, C., El Ansari, Y. S., Lewis, O. L., & Oettgen, H. C. (2020). IGE and IGG antibodies as regulators of mast cell and Basophil functions in food allergy. *Frontiers in Immunology*, 11. <https://doi.org/10.3389/fimmu.2020.603050>
- 7 Tuck, C. J., Biesiekierski, J. R., Schmid-Grendelmeier, P., & Pohl, D. (2019). Food intolerances. *Nutrients*, 11(7), 1684. <https://doi.org/10.3390/nu11071684>
- 8 Jenkins, & Vickers. (1998). Unreliability of IGE/igg4 antibody testing as a diagnostic tool in food intolerance. *Clinical & Experimental Allergy*, 28(12), 1526–1529. <https://doi.org/10.1046/j.1365-2222.1998.00367.x>
- 9 Atkinson, W. (2004). Food elimination based on IGG antibodies in irritable bowel syndrome: A randomised controlled trial. *Gut*, 53(10), 1459–1464. <https://doi.org/10.1136/gut.2003.037697>
- 10 Syrigou, E., Angelakopoulou, A., Zande, M., Panagiotou, I., Roma, E., & Pitsios, C. (2015). Allergy-test-driven elimination diet is useful in children with eosinophilic esophagitis, regardless of the severity of symptoms. *Pediatric Allergy and Immunology*, 26(4), 323–329. <https://doi.org/10.1111/pai.12389>
- 11 Dhar, S., & Srinivas, S. M. (2016). Food allergy in atopic dermatitis. *Indian Journal of Dermatology*, 61(6), 645. <https://doi.org/10.4103/0019-5154.193673>
- 12 Nigg, J. T., & Holton, K. (2014). Restriction and elimination diets in ADHD treatment. *Child and Adolescent Psychiatric Clinics of North America*, 23(4), 937–953. <https://doi.org/10.1016/j.jchc.2014.05.010>
- 13 Alpaly, K., Ertaş, M., Orhan, E. K., Üstay, D. K., Lieners, C., & Baykan, B. (2010). Diet restriction in migraine, based on IGG against foods: A clinical double-blind, randomised, cross-over trial. *Cephalalgia*, 30(7), 829–837. <https://doi.org/10.1177/0333102410361404>
- 14 Stamp, L. K., James, M. J., & Cleland, L. G. (2005). Diet and rheumatoid arthritis: A review of the literature. *Seminars in Arthritis and Rheumatism*, 35(2), 77–94. <https://doi.org/10.1016/j.semarthrit.2005.05.001>
- 15 *Elimination diet - veterans affairs*. (n.d.). Retrieved September 7, 2022, from <https://www.va.gov/WHOLEHEALTHLIBRARY/docs/Elimination-Diet.pdf>
- 16 Swain, A. R., Soutter, V. L., & Loblay, R. H. (2011). RPAH Elimination Diet Handbook: With Food & Shopping Guide. Allergy Unit, Royal Prince Alfred Hospital.
- 17 Raman, R. (2017, July 2). *How to do an elimination diet and why*. Healthline. Retrieved September 7, 2022, from https://www.healthline.com/nutrition/elimination-diet#TOC_TITLE_HDR_5
- 18 Rakel, D. (2018). Chapter 86, “The Elimination Diet.” In *Integrative medicine* (4th ed), Elsevier
- 19 MediLexicon International. (n.d.). *Food intolerance: Causes, types, symptoms, and diagnosis*. Medical News Today. Retrieved September 7, 2022, from <https://www.medicalnewstoday.com/articles/263965>
- 20 Mitchell, N., Hewitt, C. E., Jayakody, S., Islam, M., Adamson, J., Watt, I., & Torgerson, D. J. (2011). Randomised controlled trial of food elimination diet based on IGG antibodies for the prevention of migraine like headaches. *Nutrition Journal*, 10(1). <https://doi.org/10.1186/1475-2891-10-85>

Intermittent Fasting

- 1 Patterson RE, Laughlin GA, LaCroix AZ, Hartman SJ, Natarajan L, Senger CM, Martinez ME, Villaseñor A, Sears DD, Marinac CR, Gallo LC. Intermittent Fasting and Human Metabolic Health. *J Acad Nutr Diet*. 2015 Aug;115(8):1203-12. doi: 10.1016/j.jand.2015.02.018. Epub 2015 Apr 6. PMID: 25857868; PMCID: PMC4516560
- 2 Kim BH, Joo Y, Kim MS, Choe HK, Tong Q, Kwon O. Effects of Intermittent Fasting on the Circulating Levels and Circadian Rhythms of Hormones. *Endocrinol Metab* (Seoul). 2021 Aug;36(4):745-756. doi: 10.3803/EnM.2021.405. Epub 2021 Aug 27. PMID: 34474513; PMCID: PMC8419605
- 3 de Cabo, R., & Mattson, M. P. (2019). Effects of Intermittent Fasting on Health, Aging, and Disease. *The New England journal of medicine*, 381(26), 2541–2551. <https://doi.org/10.1056/NEJMr1905136> <https://pubmed.ncbi.nlm.nih.gov/31881139/>
- 4 Dong TA, Sandesara PB, Dhindsa DS, Mehta A, Arneson LC, Dollar AL, Taub PR, Sperling LS. Intermittent Fasting: A Heart Healthy Dietary Pattern? *Am J Med*. 2020 Aug;133(8):901-907. doi: 10.1016/j.amjmed.2020.03.030. Epub 2020 Apr 21. PMID: 32330491; PMCID: PMC7415631
- 5 Chen Y, Su J, Yan Y, Zhao Q, Ma J, Zhu M, He X, Zhang B, Xu H, Yang X, Duan Y, Han J. Intermittent Fasting Inhibits High-Fat Diet-Induced Atherosclerosis by Ameliorating Hypercholesterolemia and Reducing Monocyte Chemoattraction. *Front Pharmacol*. 2021 Sep 30;12:719750. doi: 10.3389/fphar.2021.719750. PMID: 34658858; PMCID: PMC8517704
- 6 Mattson, M. P., Longo, V. D., & Harvie, M. (2017). Impact of intermittent fasting on health and disease processes. *Ageing research reviews*, 39, 46–58. <https://doi.org/10.1016/j.arr.2016.10.005>
- 7 Cheng CW, Adams GB, Perin L, Wei M, Zhou X, Lam BS, Da Sacco S, Mirisola M, Quinn DI, Dorff TB, Kopchick JJ, Longo VD. Prolonged fasting reduces IGF-1/PKA to promote hematopoietic-stem-cell-based regeneration and reverse immunosuppression. *Cell Stem Cell*. 2014 Jun 5;14(6):810-23. doi: 10.1016/j.stem.2014.04.014. Erratum in: *Cell Stem Cell*. 2016 Feb 4;18(2):291-2. PMID: 24905167; PMCID: PMC4102383. [https://www.cell.com/cell-stem-cell/fulltext/S1934-5909\(14\)00151-9](https://www.cell.com/cell-stem-cell/fulltext/S1934-5909(14)00151-9)
- 8 Hannan, M. A., Rahman, M. A., Rahman, M. S., Sohag, A., Dash, R., Hossain, K. S., Farjana, M., & Uddin, M. J. (2020). Intermittent fasting, a possible priming tool for host defense against SARS-CoV-2 infection: Crosstalk among calorie restriction, autophagy and immune response. *Immunology letters*, 226, 38–45. <https://doi.org/10.1016/j.imlet.2020.07.001>
- 9 Hannan, M. A., Rahman, M. A., Rahman, M. S., Sohag, A., Dash, R., Hossain, K. S., Farjana, M., & Uddin, M. J. (2020). Intermittent fasting, a possible priming tool for host defense against SARS-CoV-2 infection: Crosstalk among calorie restriction, autophagy

and immune response. *Immunology letters*, 226, 38–45. <https://doi.org/10.1016/j.imlet.2020.07.001>

- 10 Dias, G. P., Murphy, T., Stangl, D. et al. Intermittent fasting enhances long-term memory consolidation, adult hippocampal neurogenesis, and expression of longevity gene Klotho. *Mol Psychiatry* 26, 6365–6379 (2021)

- 11 Kai Liu, Bo Liu, Leonie K. Heilbronn, Intermittent fasting: What questions should we be asking?, *Physiology & Behavior*, Volume 218, 2020, 112827, ISSN 0031-9384, <https://doi.org/10.1016/j.physbeh.2020.112827>.

- 12 Lebofsky, J. Should You Do Intermittent Fasting When Having Your Period?. Should You Do Intermittent Fasting When Having Your Period? - WeFast

Ketogenic Diet

- 1 D'Andrea Meira, I., Romão, T. T., Pires do Prado, H. J., Krüger, L. T., Pires, M. E., & da Conceição, P. O. (2019). Ketogenic diet and epilepsy: What we know so far. *Frontiers in Neuroscience*, 13. <https://doi.org/10.3389/fnins.2019.00005>
- 2 Paoli, A., Bosco, G., Camporesi, E. M., & Mangar, D. (2015). Ketosis, ketogenic diet and food intake control: A complex relationship. *Frontiers in Psychology*, 6. <https://doi.org/10.3389/fpsyg.2015.00027>
- 3 Sreenivas, S. (n.d.). *Keto Diet: A beginner's guide*. WebMD. Retrieved September 6, 2022, from <https://www.webmd.com/diet/keto-diet-for-beginners>
- 4 Masood W, Annamaraju P, Uppaluri KR. Ketogenic Diet. [Updated 2022 Jun 11]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK499830/>
- 5 Snetselaar, L. G., de Jesus, J. M., DeSilva, D. M., & Stookey, E. E. (2021). Dietary guidelines for Americans, 2020–2025. *Nutrition Today*, 56(6), 287–295. <https://doi.org/10.1097/nt.0000000000000512>
- 6 Masood W, Annamaraju P, Uppaluri KR. Ketogenic Diet. [Updated 2022 Jun 11]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK499830/>
- 7 Nieman, D. C., Gillitt, N. D., Henson, D. A., Sha, W., Shanely, R. A., Knab, A. M., Cialdella-Kam, L., & Jin, F. (2012). Bananas as an energy source during exercise: A metabolomics approach. *PLoS ONE*, 7(5). <https://doi.org/10.1371/journal.pone.0037479>
- 8 Gomez-Arbelaiz, D., Crujeiras, A. B., Castro, A. I., Martinez-Olmos, M. A., Canton, A., Ordoñez-Mayan, L., Sajoux, I., Galban, C., Bellido, D., & Casanueva, F. F. (2018). Resting metabolic rate of obese patients under very low-calorie ketogenic diet. *Nutrition & Metabolism*, 15(1). <https://doi.org/10.1186/s12986-018-0249-z>
- 9 Volek, J. S., & Westman, E. C. (2002). Very-low-carbohydrate weight-loss diets revisited. *Cleveland Clinic Journal of Medicine*, 69(11), 849–849. <https://doi.org/10.3949/ccjm.69.11.849>
- 10 Noakes, M., Foster, P. R., Keogh, J. B., James, A. P., Mamo, J. C., & Clifton, P. M. (2006). Comparison of isocaloric very low carbohydrate/high saturated fat and high carbohydrate/low saturated fat diets on body composition and cardiovascular risk. *Nutrition & Metabolism*, 3(1). <https://doi.org/10.1186/1743-7075-3-7>
- 11 Daly, M. E., Paisey, R., Paisey, R., Millward, B. A., Eccles, C., Williams, K., Hammersley, S., MacLeod, K. M., & Gale, T. J. (2006). Short-term effects of severe dietary carbohydrate-restriction advice in type 2 diabetes-A randomized controlled trial. *Diabetic Medicine*, 23(1), 15–20. <https://doi.org/10.1111/j.1464-5491.2005.01760.x>
- 12 Yang, M., Bai, W., Jiang, B., Wang, Z., Wang, X., Sun, Y., Liu, F., Cui, G., Song, X., Li, J., & Wu, F. (2022). Effects of a ketogenic diet in women with PCOS with different uric acid concentrations: A prospective cohort study. *Reproductive BioMedicine Online*, 45(2), 391–400. <https://doi.org/10.1016/j.rbmo.2022.03.023>
- 13 Zupec-Kania, B. A., & Spellman, E. (2008). An overview of the ketogenic diet for pediatric epilepsy. *Nutrition in Clinical Practice*, 23(6), 589–596. <https://doi.org/10.1177/0884533608326138>
- 14 Włodarek, D. (2019). Role of ketogenic diets in neurodegenerative diseases (Alzheimer's disease and Parkinson's disease). *Nutrients*, 11(1), 169. <https://doi.org/10.3390/nu11010169>
- 15 Dowis, K., & Banga, S. (2021). The potential health benefits of the ketogenic diet: A narrative review. *Nutrients*, 13(5), 1654. <https://doi.org/10.3390/nu13051654>
- 16 Seidelmann, S. B., Claggett, B., Cheng, S., Henglin, M., Shah, A., Steffen, L. M., Folsom, A. R., Rimm, E. B., Willett, W. C., & Solomon, S. D. (2018). Dietary carbohydrate intake and mortality: A prospective cohort study and meta-analysis. *The Lancet Public Health*, 3(9). [https://doi.org/10.1016/s2468-2667\(18\)30135-x](https://doi.org/10.1016/s2468-2667(18)30135-x)
- 17 Nettleton, J. A., Brouwer, I. A., Geleijnse, J. M., & Hornstra, G. (2017). Saturated fat consumption and risk of coronary heart disease and ischemic stroke: A science update. *Annals of Nutrition and Metabolism*, 70(1), 26–33. <https://doi.org/10.1159/000455681>
- 18 Batch, J. T., Lamsal, S. P., Adkins, M., Sultan, S., & Ramirez, M. N. (2020). Advantages and disadvantages of the Ketogenic Diet: A review article. *Cureus*. <https://doi.org/10.7759/cureus.9639>
- 19 Ma, W., Nguyen, L. H., Song, M., Wang, D. D., Franzosa, E. A., Cao, Y., Joshi, A., Drew, D. A., Mehta, R., Ivey, K. L., Strate, L. L., Giovannucci, E. L., Izard, J., Garrett, W., Rimm, E. B., Huttenhower, C., & Chan, A. T. (2021). Dietary fiber intake, the gut microbiome, and chronic systemic inflammation in a cohort of adult men. *Genome Medicine*, 13(1). <https://doi.org/10.1186/s13073-021-00921-y>
- 20 Watanabe, M., Tuccinardi, D., Ernesti, I., Basciani, S., Mariani, S., Genco, A., Manfrini, S., Lubrano, C., & Gnessi, L. (2020). Scientific evidence underlying contraindications to the ketogenic diet: An update. *Obesity Reviews*, 21(10). <https://doi.org/10.1111/obr.13053>

Mediterranean Diet

- 1 Nestle, M. (1995). Mediterranean diets: Historical and research overview. *The American Journal of Clinical Nutrition*, 61(6). <https://doi.org/10.1093/ajcn/61.6.1313>

- 2 Lăcătușu, C.-M., Grigorescu, E.-D., Floria, M., Onofriescu, A., & Mihai, B.-M. (2019). The Mediterranean diet: From an environment-driven food culture to an emerging medical prescription. *International Journal of Environmental Research and Public Health*, 16(6), 942. <https://doi.org/10.3390/ijerph16060942>
- 3 Lăcătușu, C.-M., Grigorescu, E.-D., Floria, M., Onofriescu, A., & Mihai, B.-M. (2019). The Mediterranean diet: From an environment-driven food culture to an emerging medical prescription. *International Journal of Environmental Research and Public Health*, 16(6), 942. <https://doi.org/10.3390/ijerph16060942>
- 4 Mayo Foundation for Medical Education and Research. (2021, July 23). *Mediterranean diet for heart health*. Mayo Clinic. Retrieved September 14, 2022, from <https://www.mayoclinic.org/healthy-lifestyle/nutrition-and-healthy-eating/in-depth/mediterranean-diet/art-20047801>
- 5 Arija Val, V., Babio, N., Fernández-Ballart, J., & Serra-Majem, L. (2004). La Dieta Mediterránea y el mar: pescados [Mediterranean Diet and the sea: fishes]. *Archivos latinoamericanos de nutrición*, 54(2 Suppl 1), 72–75
- 6 Mazzocchi, A., Leone, L., Agostoni, C., & Pali-Schöll, I. (2019). The secrets of the Mediterranean diet. does [only] olive oil matter? *Nutrients*, 11(12), 2941. <https://doi.org/10.3390/nu11122941>
- 7 Minzer, S., Estruch, R., & Casas, R. (2020). Wine intake in the framework of a Mediterranean diet and chronic non-communicable diseases: A short literature review of the last 5 years. *Molecules*, 25(21), 5045 <https://doi.org/10.3390/molecules25215045>
- 8 Féart, C., Samieri, C., Allès, B., & Barberger-Gateau, P. (2012). Potential benefits of adherence to the Mediterranean diet on Cognitive Health. *Proceedings of the Nutrition Society*, 72(1), 140–152. <https://doi.org/10.1017/s0029665112002959>
- 9 Trichopoulou, A., & Vasilopoulou, E. (2000). Mediterranean diet and longevity. *British Journal Of Nutrition*, 84(6), 205–209. <https://doi.org/10.1079/096582197388554>
- 10 Trichopoulou, A., Martínez-González, M. A., Tong, T. Y. N., Forouhi, N. G., Khandelwal, S., Prabhakaran, D., Mozaffarian, D., & de Lorgeril, M. (2014). Definitions and potential health benefits of the Mediterranean diet: Views from experts around the world. *BMC Medicine*, 12(1). <https://doi.org/10.1186/1741-7015-12-112>
- 11 van den Brandt, P. A. (2011). The impact of a Mediterranean diet and healthy lifestyle on premature mortality in men and women. *American Journal of Clinical Nutrition*, 94(3), 913–920. <https://doi.org/10.3945/ajcn.110.008250>
- 12 Milenkovic, T., Bozhinovska, N., Macut, D., Bjekic-Macut, J., Rahelic, D., Velija Asimi, Z., & Burekovic, A. (2021). Mediterranean diet and type 2 diabetes mellitus: A perpetual inspiration for the scientific world. A Review. *Nutrients*, 13(4), 1307. <https://doi.org/10.3390/nu13041307>
- 13 Spence, J. (2019). Nutrition and risk of stroke. *Nutrients*, 11(3), 647. <https://doi.org/10.3390/nu11030647>
- 14 Magriplis, E., Panagiotakos, D., Kyrou, I., Tsioufis, C., Mitsopoulou, A.-V., Karageorgou, D., Dimakopoulos, I., Bakogianni, I., Chourdakis, M., Micha, R., Michas, G., Ntouroupi, T., Tsaniklidou, S.-M., Argyri, K., & Zampelas, A. (2020). Presence of hypertension is reduced by Mediterranean diet adherence in all individuals with a more pronounced effect in the obese: The Hellenic National Nutrition and Health Survey (HNNHS). *Nutrients*, 12(3), 853 <https://doi.org/10.3390/nu12030853>
- 15 Ghosh, T. S., Rampelli, S., Jeffery, I. B., Santoro, A., Neto, M., Capri, M., Giampieri, E., Jennings, A., Candela, M., Turrioni, S., Zoetendal, E. G., Hermes, G. D., Elodie, C., Meunier, N., Brugere, C. M., Pujos-Guillot, E., Berendsen, A. M., De Groot, L. C., Feskens, E. J., ... O'Toole, P. W. (2020). Mediterranean diet intervention alters the gut microbiome in older people reducing frailty and improving health status: The nu-age 1-year dietary intervention across five European countries. *Gut*, 69(7), 1218–1228. <https://doi.org/10.1136/gutjnl-2019-319654>
- 16 Lourida, I., Soni, M., Thompson-Coon, J., Purandare, N., Lang, I. A., Ukoumunne, O. C., & Llewellyn, D. J. (2013). Mediterranean diet, cognitive function, and dementia. *Epidemiology*, 24(4), 479–489 <https://doi.org/10.1097/ede.0b013e3182944410>
- 17 Mentella, Scaldaferrri, Ricci, Gasbarrini, & Miggiano. (2019). Cancer and mediterranean diet: A Review. *Nutrients*, 11(9), 2059. <https://doi.org/10.3390/nu11092059>
- 18 Ventriglio, A., Sancassiani, F., Contu, M. P., Latorre, M., Di Slavatore, M., Fornaro, M., & Bhugra, D. (2020). Mediterranean diet and its benefits on health and Mental Health: A Literature Review. *Clinical Practice & Epidemiology in Mental Health*, 16(1), 156–164. <https://doi.org/10.2174/1745017902016010156>
- 19 Davis, C., Bryan, J., Hodgson, J., & Murphy, K. (2015). Definition of the Mediterranean diet; a literature review. *Nutrients*, 7(11), 9139–9153. <https://doi.org/10.3390/nu7115459>
- 20 Mayo Foundation for Medical Education and Research. (2021, July 23). *Mediterranean diet for heart health*. Mayo Clinic. Retrieved September 14, 2022, from <https://www.mayoclinic.org/healthy-lifestyle/nutrition-and-healthy-eating/in-depth/mediterranean-diet/art-20047801>

The Wahls Protocol Diet

- 1 Burch, K. (2022, March 14). *Some people see remarkable benefit on the Wahls Protocol*. Verywell Health. Retrieved September 16, 2022, from <https://www.verywellhealth.com/reasons-to-try-the-wahls-protocol-for-ms-5218077>
- 2 Dr. Terry Wahls, MD & Author. (2019, September 27). *My story*. Dr. Terry Wahls, MD & Author. Retrieved September 16, 2022, from <https://terrywahls.com/about/about-terry-wahls/>
- 3 Lindberg, S. (2019, January 25). *The Wahls Diet for autoimmune disorders: 5 tasty recipes*. Healthline. Retrieved September 15, 2022, from <https://www.healthline.com/health/wahls-diet-recipes-for-multiple-sclerosis>
- 4 *The wahls protocol and HD - hopes Huntington's disease information*. (n.d.). Retrieved September 16, 2022, from <https://hopes.stanford.edu/the-wahls-protocol-and-hd/>
- 5 AutoimmuneMom. (2021, August 20). *Wahls protocol: Restoring your cells to regain function and mobility lost from autoimmune*. Autoimmune Association. Retrieved September 16, 2022, from <https://autoimmune.org/wahls-protocol-autoimmune-diet/>
- 6 Pagán, C. N. (n.d.). *Wahls Protocol Diet for ms treatment: Effectiveness and foods*. WebMD. Retrieved September 15, 2022, from <https://www.webmd.com/multiple-sclerosis/wahl-diet-ms>
- 7 Konijeti, G. G., Kim, N. M., Lewis, J. D., Groven, S., Chandrasekaran, A., Grandhe, S., Diamant, C., Singh, E., Oliveira, G., Wang, X., Molparia, B., & Torkamani, A. (2017). Efficacy of the autoimmune protocol diet for inflammatory bowel disease. *Inflammatory Bowel Diseases*, 23(11), 2054–2060. <https://doi.org/10.1097/mib.0000000000001221>
- 8 Khanna, S., Jaiswal, K. S., & Gupta, B. (2017). Managing rheumatoid arthritis with dietary interventions. *Frontiers in Nutrition*, 4. <https://doi.org/10.3389/fnut.2017.00052>
- 9 Abbott, R. D., Sadowski, A., & Alt, A. G. (2019). Efficacy of the autoimmune protocol diet as part of a multi-disciplinary, supported lifestyle intervention for Hashimoto's thyroiditis. *Cureus*. <https://doi.org/10.7759/cureus.4556>
- 10 Wahls, T. L., Chenard, C. A., & Snetselaar, L. G. (2019). Review of two popular eating plans within the multiple sclerosis community: Low saturated fat and modified Paleolithic. *Nutrients*, 11(2), 352. <https://doi.org/10.3390/nu11020352>
- 11 Lindberg, S. (2019, January 25). *The Wahls Diet for autoimmune disorders: 5 tasty recipes*. Healthline. Retrieved September 16, 2022, from <https://www.healthline.com/health/wahls-diet-recipes-for-multiple-sclerosis>
- 12 Burch, K. (2022, March 14). *Some people see remarkable benefit on the Wahls Protocol*. Verywell Health. Retrieved September 16, 2022, from <https://www.verywellhealth.com/reasons-to-try-the-wahls-protocol-for-ms-5218077#citation-3>
- 13 Gohil K. Multiple Sclerosis: Progress, but No Cure. P T. 2015 Sep;40(9):604-5. PMID: 26417181; PMCID: PMC4571850.



*If we don't make the time to focus on our wellness,
we may be forced to take the time in the future to focus on our illness.*

"I believe that there are no coincidences in life, and the fact that you're holding this book in your hands is not one either.

If you're dealing with a chronic disease and have lost hope, I know exactly how you feel. I was there for years. Many people today live in a state of perpetual illness, managing their symptoms with painkillers and other drugs, but not addressing the root cause of their problems.

Together with the help of MDs, NDs, and scientists, I've compiled 45 easy-to-follow science-backed holistic protocols to help you improve the quality of your life. In this book, you'll discover protocols for the following issues: Rheumatoid Arthritis, Depression, Inflammatory Bowel Disease: Crohn's Disease and Ulcerative Colitis, Lyme Disease, Multiple Sclerosis, Lupus, Pain Management, Psoriasis and Eczema, Vertigo and Tinnitus, Leaky Gut, Gastritis, and IBS, GERD and Acid Reflux, Ulcer, Kidney Issues, Liver Issues, Autoimmune Issues, Thyroid Issues (including Graves' Disease and Hashimoto's Thyroiditis), Anxiety and Stress, Adrenal Fatigue, Energy, Prostate, Men's Sexual Health, Women's Sexual Health, Weight Loss, Flu, Viral, and Immune System Issues, Food Poisoning, Fungal Infections, Parasites, Hair Loss, Type-2 Diabetes, Heart Health, Acne, Candida, Alzheimer's and Dementia, Parkinson's, Bacterial Infections, Asthma, Sinusitis, Hemorrhoids, Sleep Issues, Lungs and COPD, Menopause, Osteoporosis, Brain and Nerve Health, Allergies, and Anti-Aging.

Even if you're in good health right now, it's wise to have these on hand as one day they may help you or someone you care about.

This book is written so that anyone may understand and integrate these protocols into their daily life and begin taking charge of their health naturally. This book is a culmination of 30+ years of the author's experience and knowledge, paired with expert input from people in the field of natural medicine.

It is also an excellent desk reference for MDs, NPs, and nurses wanting to provide more integrative care, wellness coaches, herbalists, naturopathic doctors, acupuncturists, and teaching institutions."

Nicole Apelian, Ph.D.

Dr. Nicole Apelian is an herbalist, author, scientist, survival skills consultant, TV celebrity, wilderness skills instructor, wild foods expert, and MS survivor and thriver.

In 2015 she was among the first women ever selected for the History Channel's hit TV Show "Alone". Despite having Multiple Sclerosis, she survived solo for 57 days in the wilderness with little more than her hunting knife and the wild foods and medicines she found there.

Growing and harvesting wild medicinal plants and mushrooms is a way of life Nicole adopted for her personal wellness after being diagnosed with MS. She found that nature's medicine and a holistic lifestyle were more effective than standard treatments for MS. She went from wheelchair to walking; taking back full control over her life.

In this book, Dr. Nicole Apelian pairs her expertise and scientific knowledge with peer-reviewed research – doing the deep-dive into the current research so you don't have to. She has helped thousands of people thrive by following her holistic wellness protocols and by using herbal remedies.

Copyrighted by Global Brother SRL

